

Merged research report for the Surreal project

Polish Models of Omnific Arithmetic

Polish Presburger models for Glazer’s Question 2, Borel presentations and support barriers, continuous Presburger arithmetic, local compactness, Polish series fields, Baire-category rigidity, nonsplit models, a claimed ATR_0 bound for Question 1, locally compact cones, elementary embeddings of lexicographic models, local geometric codes, Polish group completions and class manifolds, orders in one real dimension, the exponent-group threshold, Levi-Civita self-embeddings, Borel Hilbert orders, Borel summability, Borel conjugacy, support complexity, Borel flows and two derivations

Built from twenty-four manuscripts dated October 3 and 4, 2026

Sources: manuscripts 05 (the base text of Part I), 04 (Part I, marked [04]) and 03 (Part II) of batch 86, arrival commit 31fdc6571, placement commit 3d2177df4; pinned to repository commits a11efab09 (05), 3ad5f878c (04) and c5b4881fa (03). Manuscripts 07 (the base text of Part III), 06 (Part III, marked [06]), 08 (Part IV) and 09 (Part V) of batch 86, arrival commit fb8414869, placement commit 0bd0e5527; pinned to 0f95145cb (07), fa2f3e419 (06), 883e0b3b2 (08) and 4af956fe0 (09). Batch-87 manuscripts 03, 01, 04–06 (sources 12, 10, 13–15): arrival fa0a0576e, placement d151b39ca; pins 0f084afa9 (12, 15), 6fef5383b (10, 14), eab47e330 (13). Batch-89 manuscript 02 (source 16, Part X): arrival 7c0f2d9f9, placement 23adb85f9, pin 0f084afa9. Batch-90 manuscripts 01 and 03 (sources 17 and 18, Parts XI and XII): arrival a162e4386, placement 12076b2e8, pins 7c0f2d9f9 (17) and 2b7b388ba (18). Batch-92 manuscripts 01, 02 and 05 (sources 19–21, Parts XIII–XV): arrival 9dc8db274, placement e24ce2ce0, pins a32e1b139 (19) and 9fe62865d (20, 21). Batch-93 manuscripts 03 and 04 (sources 22, 23, Parts XVI, XVII): arrival 2faa3b37a, placement 5e4d8eed8, pin 3c25fbb55. Batch-96 manuscripts 01, 02, 03 and 07 (sources 24–27, Parts XVIII–XXI): arrivals 62846e17a, 7be14aa84, fb9f5884b and 78528873b, placement 54ece48ab, pins 715a716a3 (24) and c39974f12 (25, 26); 27 names README blobs, not a commit. Their titles and contributions are in [section 1.2](#); the provenance of every section is in [appendix E](#), and the non-claims of every source are collected in [appendix F](#).

Abstract

Elliot Glazer's *A Topological Tennenbaum Theorem* shows that an uncountable Polish model of $Q + IOpen$ cannot have continuous addition and Borel multiplication, and asks (Question 2) whether some uncountable Polish space supports a model of Presburger arithmetic with continuous addition. Part I answers that question, as printed, affirmatively. The nonnegative cone $M_{\mathbb{R}} = (\mathbb{R}_{>0} \times \mathbb{Z}) \cup (\{0\} \times \mathbb{N})$ of the lexicographic group $\mathbb{R} \vec{\times} \mathbb{Z}$, with the usual topology on the real coordinate and the discrete one on the integer coordinate, is a perfect, locally compact, uncountable Polish model of the full theory $\text{Th}(\mathbb{N}; 0, 1, +, <)$ whose addition is continuous. Replacing $\mathbb{R}$ by the lexicographic group $\mathbb{Q}^{\mathbb{N}}$ gives a model homeomorphic to Baire space, with Type-2 computable addition and division by standard integers. Every definable relation is Δ_2^0 , and this is optimal for the order; the truncated subtraction and the predecessor are discontinuous exactly on explicit loci; no unital semiring multiplication exists on either model; the self-embeddings are classified, and both models embed additively in Conway's omnific integers $\mathbf{Oz}$ as $r\omega + n$ and $\sum_j a_j \omega^{1/(j+1)} + n$. Glazer speculated that the answer would be negative; the affirmative answer is for Question 2 as printed, and its priority is not established.

In Part II, the well-ordered subsets of a countable linear order L form a Borel set exactly when L is scattered, and a Π_1^1 -complete set otherwise; the same condition decides whether a full real Hahn field $\mathbb{R}((t^{\Gamma}))$ has a standard Borel parametrization with Borel coefficient maps. The finite Puiseux principal-part ring $\mathcal{A}_{\text{fin}}$, of which $M_{\mathbb{R}}$ is one stratum, satisfies open induction and has a Polish presentation on which addition and multiplication are continuous off explicit closed nowhere dense cancellation loci; by Glazer's Corollary 1 no Borel-compatible Polish recoding makes its addition continuous everywhere.

In Part III, a Borel invariant order on a Polish abelian group has Δ_2^0 cones, so every divisible Polish group with Borel positive cone yields a Polish model; the cones $(\mathbb{Q}^{\alpha} \vec{\times} \mathbb{Z})_{\geq 0}$, α countably infinite, give $\aleph_1$ pairwise nonisomorphic models on Baire space, other families give exactly $2^{\aleph_0}$, and one order formula has unbounded oscillation ranks. The cone of $\mathbb{Z} + t\mathbb{R}[t]$ (t infinite) carries a locally compact Polish topology with continuous addition and multiplication and fails open induction at $x^2 < t$; and in every Polish topological cone of a discretely ordered ring with both operations continuous, 0 is isolated, which is Glazer's Theorem 1 with PA weakened to PA^- . Part IV shows that a discretely ordered ring whose additive group is Hausdorff, locally compact and second countable with Borel order is countable and discrete, whatever its multiplication, and classifies the locally compact Borel $\mathbb{Z}$ -groups as $\mathbb{R}^d \times E$. Part V shows that a full Hahn field $k((t^{\Gamma}))$ admits a Polish additive topology with Baire order and Borel scalar maps exactly when k is countable and $\Gamma \cong \mathbb{Z}$; the Puiseux field over a countable field admits none, its Levi-Civita completion exactly one; and integer parts of such fields are countable and closed discrete.

Parts VI–XXI add seventeen later manuscripts: a discretely ordered ring with a Hausdorff Baire ccc additive topology and a Baire-property positive cone is countable and discrete, whatever its multiplication (VI); nonsplit Polish Presburger models exist on every countable core (VII); an unreviewed manuscript claims that Glazer's Theorem 2 is provable in ATR_0 (VIII); locally compact cones of Borel ordered Polish groups are countable sums of lines and closed rays (IX); all self-embeddings of $(\mathbb{Q}^{\alpha} \vec{\times} \mathbb{Z})_{\geq 0}$ are continuous exactly when $\alpha \leq \omega$ and have closed images exactly when $\alpha < \omega \cdot 2$ (X; the counterexamples use choice); one internal geometric code forces discontinuous multipliers in every uncountable interval of its indices (XI); a cone topology extends to its group of differences exactly when partial subtraction is continuous (XII); Borel Presburger orders on $\mathbb{R} \times E$ have countable complete invariants but Borel-complete isomorphism (XIII); the cone of $\mathbb{Z} \oplus \bigoplus_{\gamma > 0} \mathbb{R}T^{\gamma}$ admits continuous addition and Borel multiplication exactly for discrete $\Gamma \leq \mathbb{R}$ (XIV); Levi-Civita fields over $\mathbb{R}_{\text{alg}}$ are co-Hopfian exactly in finite rational rank (XV); Borel orders on Hilbert spaces are ordinal lexicographic (XVI); Borel additive maps of real Levi-Civita fields preserve sums (XVII); and Parts XVIII–XXI classify their rank-one derivations up to Borel conjugacy, the complexity of well-ordered supports, Borel flows over divisible exponent groups, and pairs of derivations.

The report merges twenty-four manuscripts, credits each result to its sources and keeps their limitations; it is an unrefereed AI-assisted draft, and nothing in it is formalized.

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1 Scope, sources and conventions

This report collects twenty-four manuscripts, dated October 3 and 4, 2026, on one subject: where Elliot Glazer’s topological Tennenbaum programme meets the repository’s omnific arithmetic. Their common spine is the question which arithmetic operations of a Polish or standard Borel presentation of a substructure of $\mathbf{Oz}$ can be continuous. Three of them (03, 04, 05) were written first; the four later ones (06, 07, 08, 09) arrived together, and 08 and 09 cite 05 and 03 as earlier companion manuscripts. [Part I](#) (label prefix `pma:pr:`) constructs Polish models of Presburger arithmetic with continuous addition and studies them; its base text is manuscript 05, and the further results of manuscript 04 are printed as marked sections. [Part II](#) (label prefix `pma:bor:`) is manuscript 03: Borel presentations of Hahn fields, support barriers, and the finite principal-part ring on which Glazer’s theorem forbids continuous addition. Parts I and II meet in [section 26](#). [Part III](#) (label prefix `pma:cp:`) merges manuscripts 07 (base) and 06 (marked [06]) on continuous Presburger arithmetic: Borel orders on Polish groups, a general realization theorem, ordinal and continuum families on Baire space, a Polish semiring cone, and the isolated-zero theorem; it meets Parts I and II in [section 42](#). [Part IV](#) (prefix `pma:lc:`) is manuscript 08: local compactness forces arithmetic rigidity. [Part V](#) (prefix `pma:ser:`) is manuscript 09, which continues Part II: Polish topologies on Hahn, Puiseux and Levi-Civita fields and their integer parts.

Five further manuscripts (sources 10 and 12–15, batch 87) continue the report. [Part VI](#) (prefix `pma:br:`) merges sources 12 (base) and 10 (marked [10]), which prove that a Baire-category regularity of the order forces arithmetic rigidity without local compactness. [Part VII](#) (prefix `pma:nsp:`) is source 13: nonsplit Polish Presburger models, exact continuity and optimal computation from streams. [Part VIII](#) (prefix `pma:atr:`) is source 14, which claims an ATR_0 proof of Glazer’s Theorem 2. [Part IX](#) (prefix `pma:lcc:`) is source 15: locally compact cones and the limits of multiplication.

A thirteenth manuscript, source 16 of batch 89, continues Parts I and III. [Part X](#) (prefix `pma:emb:`) is source 16: elementary embeddings and Polish elementary submodels of the ordinal-indexed models $(\mathbb{Q}^\alpha \overrightarrow{\times} \mathbb{Z})_{\geq 0}$, with two sharp ordinal thresholds.

Two manuscripts of batch 90, sources 17 and 18, continue Parts IV–IX. [Part XI](#) (prefix `pma:lgc:`) is source 17: a local form of source 14’s power-of-two obstruction, in which one internally coded geometric progression replaces total exponentiation, and a canonical countable arithmetic core. [Part XII](#) (prefix `pma:gcm:`) is source 18: an intrinsic criterion for a cone topology to extend to the group of differences, which answers the report’s group-completion question, and, in a second division, sethood of connected class manifolds in a class theory without Foundation, with Glazer’s class-manifold question.

Three manuscripts of batch 92, sources 19–21, continue Parts II–V and IX. [Part XIII](#) (prefix `pma:ord:`) is source 19: a normal form and complete countable invariant for the Borel Presburger orders on $\mathbb{R} \times E$, the classification problem Part IV left open, with Borel-complete isomorphism and 2^c abstract against c Borel automorphisms; with Parts IV, IX and XII it classifies every uncountable locally compact Polish Presburger model with continuous partial subtraction ([section 182](#)). [Part XIV](#) (prefix `pma:egt:`) is source 20: the exponent groups $\Gamma \leq \mathbb{R}$ for which the cone of $\mathbb{Z} \oplus \bigoplus_{\gamma > 0} \mathbb{R}T^\gamma$ carries a Polish topology with continuous addition and Borel multiplication, and an obstruction from division and one root axiom, weaker than open induction. [Part XV](#) (prefix `pma:rk:`) is source 21, which continues Part V: the left-finite Levi-Civita field over $\mathbb{R}_{\text{alg}}$ with exponent group Γ is co-Hopfian exactly when Γ has finite rational rank, with a correction of a lemma of Kuhlmann and Serra.

Two manuscripts of batch 93, sources 22 and 23, continue Parts I, III, V, X and XV. [Part XVI](#)

(prefix `pma:hbo:`) is source 22: every Borel translation-invariant total order on a separable real Hilbert space is lexicographic in a unique orthonormal basis indexed by a countable ordinal, whose successor is the oscillation rank of the cone; its flag theorem refines Part III's cone recursion for real Polish vector spaces, and its Hilbert cones give a classified family of Polish Presburger models. [Part XVII](#) (prefix `pma:bsd:`) is source 23, on Part V's Levi-Civita field with real coefficients: Borel additive maps are exactly the real-linear valuation-continuous ones and preserve infinite sums, the additive group is not Polishable in its coefficient Borel structure, and the Borel derivations are classified.

Four manuscripts of batch 96, sources 24–27, continue Parts II and XVII. [Part XVIII](#) (prefix `pma:bct:`) is source 24: on Part XVII's left-finite field in rational rank one, explicit Borel normal forms and complete invariants for conjugacy of derivations, a divisibility threshold (conjugacy is smooth exactly when at most one prime is infinitely divisible in Γ) and an exact criterion for coefficientwise analytic flows. [Part XIX](#) (prefix `pma:spc:`) is source 25: the exact Borel level and Wadge degree of the set of well-ordered supports of every countable order, which answers most of Part II's [question 28.3](#), and closed certificate spaces that make strong summation continuous. [Part XX](#) (prefix `pma:bfl:`) is source 26: for a countable divisible exponent group, the Borel flows are classified by a local-finiteness criterion on exponent windows, strict derivations integrate exactly in finite rational rank, and in infinite rank integrable derivations and their time-one maps are not closed under sums, brackets and composition. [Part XXI](#) (prefix `pma:tdv:`) is source 27: in rational rank one, noncommuting ordered pairs of derivations have an explicit Borel cross-section, and the finite-dimensional Lie algebras of Borel derivations are classified. Together they answer Part XVII's [question 237.4](#) in rational rank one and [question 237.5](#) in rational rank one and for divisible exponent groups.

1.1 Glazer's paper and its two questions

Glazer's paper [1] is an eight-page preprint, dated November 20, 2023 on its first page and submitted to arXiv on November 22, 2023 as arXiv:2311.13699. On October 3, 2026 the arXiv record listed one version (v1) and no journal reference. It answers negatively a question of Enayat, Hamkins and Wcisło [2]: no model of Peano arithmetic on a perfect Polish space has continuous addition and multiplication (its Theorem 1). Its Theorem 2 says that if $(M, +, \cdot) \models Q + I\Delta_0 + \text{Exp}$ lives on an uncountable Polish space with Borel addition and multiplication, then neither operation is continuous. Glazer notes that the axiom Exp is used only to obtain the function $x \mapsto 2^x$, and only for the discontinuity of multiplication; this gives his Corollary 1: *there is no model of $Q + I\text{Open}$ on an uncountable Polish space in which addition is continuous and multiplication is Borel*. The paper ends with two questions. Question 1 asks whether Theorem 2 is provable in ATR_0 . Question 2 is quoted in [section 2](#): does some uncountable Polish space support a model of Presburger arithmetic with continuous addition?

[write] (Batch 87, 3 October 2026.) Two different questions of Glazer carry the number 1 in the repository. His earlier paper *A choiceless box game paradox* (arXiv:2211.10474, 2022) ends its Section 5 with another question: does ZF prove that the box game $G([3], \mathbb{R}, \{0, 1\})$ is winnable, that is, can three players facing continuum many boxes of bits guarantee that at most one of them guesses wrong? Batch 87's manuscript 02 studies that question in the research-report collection's SetTheory/Cardinals/docs/reports/ordinals-and-order-types/measurable-box-games, and answers it only for blind outputs measurable for the cylinder σ -algebra. There it is Research question Q1. It is *not* Question 1 of this report, which asks whether Theorem 2 of the Topological Tennenbaum paper is provable in ATR_0 ; the two

reports share no theorem and no notation.

Glazer’s speculation. Immediately after Question 2, Glazer writes that he speculates that an affirmative answer to Question 1 “would lead to a negative answer to Question 2” [1, p. 8]. His reason is an analogy: removing the exponential function from the proof that multiplication is discontinuous would parallel removing multiplication from a proof that addition is discontinuous. Neither source of [Part I](#) mentions this sentence; it is recorded here because [Part I](#) goes the other way.

How this report reads it [write]. [Part I](#) answers Question 2 *as printed* affirmatively ([theorem 3.1](#)). The question asks only for an uncountable Polish space, a model of Presburger arithmetic and continuous addition; it does not ask that the topology be the order topology, and it mentions no multiplication. The speculation is a heuristic about proofs, not a stated implication, and nothing here bears on Question 1, which stays open; no conclusion about Question 1 is drawn from the affirmative answer. Three facts locate the difference. First, the affirmative models are not an artefact of a missing definability condition: every definable relation in them is Δ_2^0 ([theorem 7.1](#)), so the order and every other definable relation are Borel, as in the Borel models of Glazer’s theorems. Second, what fails in them is multiplication: no unital semiring multiplication exists on them at all ([theorem 10.4](#)), so Corollary 1 does not apply. Third, on a ring that does carry Borel multiplication and open induction, Corollary 1 does forbid continuous addition under every Borel-compatible Polish recoding ([theorem 25.2](#)), and [section 26](#) shows that the real model of [Part I](#) sits inside that ring. We do not know whether Glazer intended Question 2 with a further hypothesis, for instance an expansion to a model of a stronger theory, under which his speculation could hold; that would be a different question.

[write] (Batch 87.) “Question 1, which stays open” was written for Parts I and II; see the paragraph on Parts VI–IX below, where a manuscript’s affirmative answer, not independently reviewed, is recorded.

Parts III–V and the speculation [write]. None of sources 06–09 quotes the sentence either. Source 07 comes closest: after restating Question 1 it observes that, because the affirmative Presburger construction stands, a route from a weaker metatheory to a *negative* Presburger conclusion cannot yield that conclusion as stated ([section 43](#)). Read literally, the outcome the speculation predicts is therefore excluded for Question 2 as printed, whatever the answer to Question 1; only a variant of Question 2 with further hypotheses could carry it. Source 06’s isolated-zero theorem ([theorem 39.1](#)) removes a hypothesis from one of Glazer’s proofs, in the spirit of his analogy, but from his Theorem 1: with Glazer’s subset-sum and zero-one method, and without his injectivity claim, it weakens Peano arithmetic to PA^- in “no perfect Polish space carries a model with continuous addition and multiplication” ([corollary 39.2](#)). It does not bear on Question 1, which concerns his Theorem 2; Glazer himself remarks [1, p. 7] that Theorem 1 and Corollary 1 can be proven in ATR_0 , and source 06’s project 6 asks for the strength of the isolated-zero principle. Sources 08 and 09 concern rings, where multiplication is present, and both state that they answer neither of Glazer’s questions. Priority of source 06’s strengthening is not established.

Parts VI–IX, Question 1 and the speculation [write]. (Batch 87.) Source 14 ([Part VIII](#)) claims an affirmative answer to Question 1: Glazer’s Theorem 2 is provable in ATR_0 , under a stated convention for coding Polish presentations and Borel data ([corollary 111.2](#)), with a stronger,

cofinal conclusion ([theorem 104.1](#)). This report prints that answer as a claim. It is unrefereed and AI-assisted; its ATR_0 bookkeeping rests partly on inputs it declares rather than proves, and its author did not see the Gaifman–Dimitracopoulos chapter whose internal MRDP theorem it uses. It has not been independently reviewed, either here or in the repository’s Hilbert’s-tenth programme, whose review of the archive read its internal-MRDP input and states that it does not certify the claimed resolution ([appendix E](#)). The statements of this report that Question 1 stays open were made for Parts I–V and remain true in the sense that no reviewed answer is known; they are qualified by this paragraph. If source 14’s claim is correct, the outcome Glazer speculated, a negative answer to Question 2 following from an affirmative answer to Question 1, does not occur for Question 2 as printed, which Part I answers affirmatively; as noted above, that outcome was already excluded for Question 2 as printed, whatever the answer to Question 1. Source 14 does not mention the speculation. Sources 10, 12 and 15 state that they do not answer Question 1, and source 13 that its results do not resolve it; their theorems concern signed group topologies (10, 12), inherited cones (15) and the positive models of Part I (13).

Parts XI and XII [write]. (Batch 90.) Neither source answers Glazer’s Question 1 or 2. Source 17’s ATR_0 bound ([theorem 147.1](#)) concerns its local theorem, whose arithmetic input is one internal code rather than total exponentiation; like source 14’s, it is a claimed upper bound and has not been reviewed. Source 18 says that its completion criterion does not answer Question 2 anew. Its second division concerns another question of Glazer, from his Oberwolfach abstract on Global Choice without Foundation ([question 165.6](#)), not from the Topological Tennenbaum paper.

Parts XIII–XV [write]. (Batch 92.) None of sources 19–21 answers Glazer’s Question 1 or 2, and each says so; source 19 calls itself a continuation, not a new claim to the affirmative answer to Question 2. Source 20’s [theorem 183.2](#) weakens the open induction in Glazer’s Corollary 1 to Euclidean division and one integer-root axiom, for discretely ordered rings with Borel multiplication; it is a ZFC theorem with no reverse-mathematical bound, and it neither uses nor certifies Part VIII’s claim. Source 21 concerns ordered fields, which carry no discrete arithmetic.

Parts XVI and XVII [write]. (Batch 93.) Neither source answers Glazer’s Question 1 or 2, and both say so. Source 22 takes the affirmative answer to Question 2 as the repository’s and calls its models an extension of it; its no-ring theorem is narrower than Glazer’s theorems and does not replace them. Source 23 states that its non-Polishability proposition is not a solution of either question and that there is no conflict with Glazer’s results; its questions on logical strength are distinct from Question 1 and use no claim of [Part VIII](#).

Parts XVIII–XXI [write]. (Batch 96.) None of the four sources answers Glazer’s Question 1 or 2, and each says so. Source 24 cites his Theorem 2 for the tension between Borel regularity and continuity of arithmetic operations; its “divisibility threshold” concerns the smoothness of conjugacy of derivations, not Glazer’s theorems or Part XIV’s exponent-group threshold. Source 25 treats Glazer’s theorem as motivation, not as a lemma: Hahn fields are not, merely as ordered fields, his arithmetic models, and its certificate spaces are spaces of names, so its continuity results neither evade nor contradict his theorem. Source 26’s question on foundational strength ([question 283.9](#)) is distinct from his Question 1 and uses no claim of [Part VIII](#). Source 27 claims no named question of Glazer’s solved.

Priority. Priority of the affirmative answer is **not established**. Source 05 searched for the exact question and combinations of its terms and found no earlier answer, and says that this does not prove that none exists; source 04 says that historical priority has not been established. The intake re-checked only that the arXiv record has one version and no journal reference (October 3, 2026). The answer is an elementary construction; an earlier publication of it, or a published variant of the question, may exist.

1.2 Twenty-four manuscripts, one report

Source	Manuscript (delivered pages)	Pin	Printed as
05	<i>Polish Presburger Arithmetic inside the Omnific Integers: An affirmative construction for Glazer’s Question 2, effective Baire-space models, and structural obstructions</i> (24, A4); archive <code>polish_presburger_glazer</code>	a11efab09	base text of Part I ; worked examples in appendix A
04	<i>Polish Presburger Arithmetic: A positive construction for Glazer’s question, an effective Baire-space model, and exact regularity boundaries</i> (25, letter); archive <code>Polish_Presburger_Glazer_ProveIt</code>	3ad5f878c	Part I , every statement marked [04]; second proofs; appendices B and C
03	<i>Borel Presentations and Support Barriers in Omnific Arithmetic: Scattered exponent orders, Hahn fields, and the exact discontinuities of finite normal forms</i> (23, letter); archive <code>glazer_proveit_research</code>	c5b4881fa	Part II and appendix D
07	<i>Continuous Presburger Arithmetic on Polish Spaces: An affirmative construction for Glazer’s question, Baire-space models, and omnific realizations</i> (27, letter); archive <code>polish_presburger_research</code>	0f95145cb	base text of Part III ; appendices G and H
06	<i>Continuous Presburger Arithmetic on Polish Spaces: Lexicographic models, an isolated-zero obstruction, and a boundary between additive and semiring arithmetic</i> (21, letter); archive <code>polish_presburger_article</code>	fa2f3e419	Part III , every passage marked [06]; appendices I and J
08	<i>Local Compactness Forces Arithmetic Rigidity: Borel ordered groups, Presburger models, and multiplication without regularity assumptions</i> (23, A4); archive <code>glazer_proveit_local_compactness</code>	883e0b3b2	Part IV and appendix K
09	<i>Polishability at the Hahn–Puisseux–Levi-Civita Boundary: Automatic order bounds, local Polish hulls, and countable omnific integer parts</i> (30, letter); archive <code>Polish_Hahn_Puisseux_Levi_Civita_Research</code>	4af956fe0	Part V and appendices L and M

The numbers are the manuscript numbers of batch 86 of the repository’s intake

(manuscripts 01 and 02 of that batch are unrelated OEIS reports; manuscript 10, on the standard cut in one infinite interval, was written into the report `Algebra/SurrealNumbers/docs/surreal/discrete-initial-subgroups-and-omnific-normalization`). Manuscripts 03–05 arrived in commit 31fdc6571 and were placed in 3d2177df4; manuscripts 06–09 arrived in fb8414869 and were placed in 0bd0e5527. Every pin is an ancestor of its placement commit, and the repository files the sources inspected are unchanged between their pins and the placements.

Why one report. 04 and 05 prove the same main theorem independently: they share 0.6% of their word 8-grams, essentially only Glazer’s quoted question and his bibliography entry, and have different pins, notation and supplementary theorems. 03 answers the same brief from the other side: its finite principal-part ring contains the real model of 04 and 05 as one stratum (section 26), and its Theorem 10.2 (theorem 25.2) is Glazer’s obstruction on that ring. No earlier report of the collection names Glazer, his questions, Polish models of Presburger arithmetic, or Borel presentations of Hahn fields.

Why Parts III–V are here. Sources 06 and 07 answer Glazer’s Question 2 again and go on from Part I’s models; neither is an edition of the other or of 04 and 05 (at most 0.74% shared word 8-grams among manuscripts 03–10). Source 08 names 05 and 03 as its prior work and asks what local compactness forces on every Borel ordered presentation; source 09 names them too and asks the Polish-group question that follows 03’s Borel classification. No other report of the collection treats Polish topologies on Puiseux or Levi-Civita fields. Source 07 is the base of Part III because its realization theorem has the weakest hypotheses (Part III).

Why base 05. On the shared theorems 05 has the weaker hypotheses: its Polish criterion (theorem 5.2) is an equivalence for every Polish group topology on the coefficient group, where 04’s transfer theorem (theorem 5.4) is a sufficient condition under the stronger hypothesis that the positive cone be open and division continuous; its Δ_2^0 theorem holds under the same weaker hypothesis; and it treats the models explicitly as fragments of **Oz**. Results of 04 not in 05 are printed in full and marked [04]: continuity of fixed division under its hypothesis, the general closed-order obstruction and the invariant-metric obstruction, the reduction of all unital additive maps to the divisible part, the matrix classification of the self-embeddings of the Baire model with automatic continuity, and the Laurent-series model with its growth-order obstruction. Where 04 proves a result of 05 by a different argument, its proof is printed as a marked second proof.

Sources of Parts VI–IX [write]. Five manuscripts of batch 87 arrived in commit fa0a0576e and were placed in d151b39ca. They are numbered here by their file prefixes, which continue the batch-86 numbers 03–09: batch-87 manuscript 01 is source 10, manuscript 03 is source 12, 04 is 13, 05 is 14 and 06 is 15. Batch-87 manuscript 02, on Glazer’s box-game paper, became the report `SetTheory/Cardinals/docs/reports/ordinals-and-order-types/measurable-box-games`; prefix 11 is unused. All five pins are ancestors of the placement commit; at the pins of sources 10, 12, 14 and 15 this report consisted of the delivered base text of Part I, and source 13’s pin is one commit before the report was placed. Sources 10, 12, 13 and 14 describe themselves as prepared with ChatGPT, source 15 as prepared with AI assistance.

Source	Manuscript (delivered pages)	Pin	Printed as
12	<i>Baire-Category Rigidity for Discretely Ordered Rings: Beyond local compactness, without measurable multiplication</i> (23, A4); archive Baire_Category_Arithmetic_Rigidity; batch 87, manuscript 03	0f084afa9	base text of Part VI ; appendices N and O
10	<i>Baire Order Forces Arithmetic Rigidity: Discretely ordered rings with Polish addition and completely unrestricted multiplication</i> (23, A4); archive Glazer_ProveIt_Baire_Rigidity; manuscript 01	6fef5383b	Part VI , every passage marked [10]; appendices P and Q
13	<i>Polish Presburger Arithmetic Beyond the Split Construction: Nonsplit models, exact continuity, and optimal computation from streams</i> (39, A4); archive polish_presburger_frontiers; manuscript 04	eab47e330	Part VII ; appendices R and S
14	<i>Diophantine Definability and Topological Arithmetic: An ATR_0 proof of Glazer's theorem, cofinal unary discontinuities, and nonsplit Presburger models</i> (26, letter); archive glazer_atr0_research; manuscript 05	6fef5383b	Part VIII ; appendices T and U
15	<i>Polish Arithmetic: Multiplication Obstructions and Local Compactness</i> (32, A4); archive polish_arithmetic_package; manuscript 06	0f084afa9	Part IX ; appendix V

Why Parts VI–IX are here. Each of the five answers or continues a question or result of this report. Sources 12 and 10 answer Part IV's [question 57.1](#); source 13 answers four questions of Part I and one of Part III, wholly or in part; source 14 claims to answer Glazer's Question 1, which Part III prints as [question 43.1](#); source 15 classifies the locally compact cones of Part IV without assuming a locally compact ambient group. Sources 12 and 10 prove the same principal theorem by different proofs and are merged as one part, with base 12 for its weaker hypotheses ([Part VI](#)); sources 13, 14 and 15 are parts of their own. One construction, $\mathbb{R} \overrightarrow{\times} K$ with a countable nonsplit $\mathbb{Z}$ -group K , occurs in sources 13, 14 and 15; it is printed once, in [Part VII](#), and flagged in Parts VIII and IX. No two of the five are editions of one another: the placement measured under 0.8% shared word 8-grams between any two of them.

Source of Part X [write]. (Batch 89.) Source 16 is manuscript 02 of batch 89. It arrived in commit 7c0f2d9f9 with three other manuscripts, a fifth arriving in 8ea27d6c0, and all five were placed in 23adb85f9; the other four went to surreal-well-orders, birthday-cut-offs-and-hereditary-sets and the research-report collection's naming-elementary-embeddings. It is numbered by its file prefix 16 (the batch-87 numbers end at 15). Its pin 0f084afa9 is an ancestor of the placement; at the pin this report was the delivered base text of Part I. It names this report's questions on endomorphisms and on elementary-submodel lattices and answers them for its family, so it is a part of this report, not a report of its own. It is not an

edition of any source: the placement measured 1.24% shared word 8-grams with this report, shared Presburger phrasing. It describes itself as prepared for Vladimir Reshetnikov; its delivery README calls it AI-assisted.

Source	Manuscript (delivered pages)	Pin	Printed as
16	<i>Elementary Embeddings and Polish Submodels of Lexicographic Presburger Arithmetic: Matrix normal forms, closed images, and two sharp ordinal thresholds</i> (23, A4); archive Glazer_ProveIt_Embeddings_and_Thresholds; batch 89, manuscript 02	0f084afa9	Part X ; appendices W and X

Sources of Parts XI and XII [write]. (Batch 90.) Sources 17 and 18 are manuscripts 01 and 03 of batch 90. Six archives arrived in commit a162e4386 and were placed in 12076b2e8; the other four went to discrete-initial-subgroups-and-omnific-normalization, birthday-cutoffs-and-hereditary-sets, the new report cantor-families-of-surreal-subfields and the research-report collection's measurable-box-games. The two are numbered by their file prefixes 17 and 18. Their pins 7c0f2d9f9 and 2b7b388ba are ancestors of the placement; at both pins this report had Parts I–V, and both sources read the batch-87 archives of Parts VI–IX directly. Source 17 answers source 14's [question 115.2](#) in part, and source 18 answers the report's group-completion question ([question 57.2](#) and its repetitions), so each is a part of this report. Neither is an edition of any source: the placement measured at most 0.52% shared word 8-grams among the six batch-90 manuscripts; source 17 shares 1.23% with this report and 1.34% with source 14's manuscript (the shared scale and fusion construction), source 18 at most 0.35% with any report. Source 17 describes itself as prepared with ChatGPT, source 18 as prepared with AI assistance.

Source	Manuscript (delivered pages)	Pin	Printed as
17	<i>Local Geometric Codes and Topological Arithmetic: Partial exponentiation, interval discontinuities, and a canonical countable arithmetic core</i> (30, letter); archive local_geometric_codes_research; batch 90, manuscript 01	7c0f2d9f9	Part XI ; appendices Y and Z
18	<i>Polish Group Completions and the Size of Class Manifolds: An intrinsic cancellation criterion and a local rank approach to Glazer's manifold question</i> (31, A4); archive glazer_proveit_topology_research; manuscript 03	2b7b388ba	Part XII ; appendices AA and AB

Sources of Parts XIII–XV [write]. (Batch 92.) Sources 19, 20 and 21 are manuscripts 01, 02 and 05 of batch 92, whose six archives arrived in commits 9dc8db274 and afd7ffabb. The three were placed in e24ce2ce0; the other three (box and hat games) became Parts III and IV of the research-report collection's measurable-box-games (e38f368c2). They are numbered by the file prefixes 19, 20 and 21, which continue this report's local sequence. Source 19 is pinned to a32e1b139, 48 commits before the placement, and sources 20 and 21 to 9fe62865d, 52 commits before it; at those pins this report had Parts I–X and Parts I–IX respectively, and

none of the three read this report's article. Source 19 answers Part IV's order-classification question for one real dimension and Part IX's threshold-order question, source 20 answers Part II's [question 28.7](#), and source 21 names source 09 as its prior report and generalizes its field, so each is a part of this report. None is an edition of any source: the placement measured at most 1.02% shared word 8-grams with this report (source 19, mostly with Part IX and the bibliography), 0.60% and 0.97% for sources 20 and 21, and at most 0.14% among the three. All three describe themselves as prepared with ChatGPT.

Source	Manuscript (delivered pages)	Pin	Printed as
19	<i>One Real Dimension, Arbitrarily Complicated Arithmetic: Order normal forms, Borel-complete classification, and measurable automorphism rigidity for Polish Presburger models</i> (22, A4); archive <code>One_Real_Dimension_Arithmetic</code> ; batch 92, manuscript 01	a32e1b139	Part XIII ; appendices AC and AD
20	<i>The Exponent-Group Threshold for Polish Omnific Arithmetic: Separated grids, an induction-free root obstruction, and finite-support surreal semirings</i> (23, A4); archive <code>Glazer_ProveIt_Exponent_Threshold</code> ; manuscript 02	9fe62865d	Part XIV ; appendices AE and AF
21	<i>Rational Rank and Self-Embeddings of Levi-Civita Fields: A sharp co-Hopfian dichotomy, proper immediate self-copies, and integer-part obstructions</i> (20, A4); archive <code>Levi_Civita_Rank_and_Self_Embeddings</code> ; manuscript 05	9fe62865d	Part XV ; appendix AG

Sources of Parts XVI and XVII [write]. (Batch 93.) Sources 22 and 23 are manuscripts 03 and 04 of batch 93, whose six archives arrived in commits 2faa3b37a (manuscripts 01–05) and de37a66d1 (06). The two were placed in 5e4d8eed8; the other four, batch 93A, go to other reports. They are numbered by the file prefixes 22 and 23, which continue this report's local sequence. Both are pinned to 3c25fbb55, 57 commits before the placement, when this report had Parts I–X written (379 pages, thirteen sources); source 22 read this report's README there, source 23 only the root and surreal READMEs. Source 22 extends Part III's descriptive theory and Part I's Polish Presburger models to Hilbert spaces and answers Part I's [question 14.6](#) for them; source 23 continues Part V's Levi-Civita field with real coefficients and settles the non-claim this report's README made about it; so both are parts of this report. Neither is an edition of any source: the placement measured 0.65% shared word 8-grams of source 22 with this report and 0.14% of source 23, at most 0.24% with any single earlier manuscript (source 23 with source 21), and at most 0.05% between the two. Both describe themselves as prepared with ChatGPT.

Source	Manuscript (delivered pages)	Pin	Printed as
22	<i>Ordinal Rigidity of Borel Hilbert Orders: Canonical flags, exact oscillation ranks, Polish Presburger models, and omnific realizations</i> (22, letter); archive <code>Ordinal_Rigidity_Borel_Hilbert_Orders</code> ; batch 93, manuscript 03	3c25fbb55	Part XVI ; appendices AH and AI
23	<i>Borel Regularity Forces Summability: Automatic continuity and measurable derivations on left-finite series fields</i> (24, letter); archive <code>Borel_Summability_and_Derivations</code> ; manuscript 04	3c25fbb55	Part XVII ; appendices AJ and AK

Sources of Parts XVIII–XXI [write]. (Batch 96.) Sources 24–27 are manuscripts 01, 02, 03 and 07 of batch 96; they arrived in commits 62846e17a, 7be14aa84, fb9f5884b and 78528873b and were placed together in 54ece48ab. They are numbered by the file prefixes 24–27, which continue this report’s local sequence, in manuscript order; batch 96’s other manuscripts go to other reports. Source 24 is pinned to 715a716a3, the parent of its arrival commit, 39 commits before the placement; sources 25 and 26 to c39974f12, 38 commits before it. At both pins this report had Parts I–XVII (603 pages), and its article is unchanged from there to the placement. Source 27 records no commit: it names the blobs a7ebc59ad and e24d8fc8d of the surreal and Baker–Campbell–Hausdorff READMEs, current from ad52ef11e to the placement. None of the four read this report’s article. Source 24 read the root and surreal READMEs and describes Part XVII as an unpublished draft whose Questions 13.4 and 13.5 it follows up (they are [questions 237.4](#) and [237.5](#)); source 25 read source 03’s PDF and a Lean file; source 26 read this README, the root documentation, source 23’s PDF and a Lean file of the BCH project; source 27 cites sources 23 and 24 as earlier manuscripts. Sources 24, 26 and 27 continue Part XVII and source 25 answers a question of Part II, so all four are parts of this report. None is an edition of another source: sources 24 and 27 share 1.8% and 2.5% of their word 8-grams (boilerplate and bibliography), source 26 shares 0.5% with source 24 and 0.6% with source 27, source 25 at most 0.41% with any of the three and 0.67% with this report, source 24 about 1% with source 23 and source 26 0.1% with Part XVII. All four describe themselves as prepared with ChatGPT.

Source	Manuscript (delivered pages)	Pin	Printed as
24	<i>A Divisibility Threshold for Borel Conjugacy: Explicit normal forms, measurable classification, and analytic flows on left-finite series fields</i> (25, A4); archive <code>Borel_Conjugacy_Divisibility_Threshold</code> ; batch 96, manuscript 01	715a716a3	Part XVIII ; appendices AL and AM
25	<i>Exact Support Complexity and Strong Summation in Hahn Fields: An order-obstruction classification and certificates that restore continuity</i> (25, letter); archive <code>glazer_proveit_support_complexity</code> ; manuscript 02	c39974f12	Part XIX ; appendix AN
26	<i>Borel Flows on Left-Finite Series Fields: A local-finiteness criterion, a rational-rank dichotomy, and failure of Lie closure</i> (24, A4); archive <code>Borel_Flows_Glazer_ProveIt</code> ; manuscript 03	c39974f12	Part XX ; appendices AO and AP
27	<i>Two Derivations Remove the Borel Conjugacy Obstruction: Scalar normalizing coordinates, simultaneous normal forms, and finite-dimensional Lie algebras on left-finite series fields</i> (19, A4); archive <code>Two_Derivations_Borel_Conjugacy</code> ; manuscript 07	none (blobs a7ebc59ad, e24d8fc8d)	Part XXI ; appendix AQ

Reading conventions. In [Part I](#), unmarked text is source 05's; text from source 04 is marked [04] (statements in their titles, inserted passages in the sentence that introduces them). In [Part II](#) all unmarked text is source 03's; in [Part III](#) unmarked text is source 07's and text from source 06 is marked [06]; in [Parts IV](#) and [V](#) all unmarked text is that of source 08 and source 09 respectively. In [Part VI](#) unmarked text is source 12's and text from source 10 is marked [10]; in [Parts VII](#) to [IX](#) all unmarked text is that of sources 13, 14 and 15 respectively, in [Part X](#) that of source 16, in [Parts XI](#) and [XII](#) that of sources 17 and 18, in [Parts XIII](#) to [XV](#) that of sources 19, 20 and 21, in [Parts XVI](#) and [XVII](#) that of sources 22 and 23, and in [Parts XVIII](#) to [XXI](#) that of sources 24, 25, 26 and 27. Text written for this report is marked [merge] (a mathematical observation combining the sources, with proof) or [write] (editorial notes: provenance, citations of the collection, corrections). Phrases such as "this manuscript" or "this article" inside a source's text refer to that source. References to the collection's other reports give their labels in typewriter type, for example `odg:thm:floor`; these labels are not hyperlinks.

Labels. Every label carries the prefix `pma:`. Source 05's labels are `pma:pr:<label>`, source 03's are `pma:bor:<label>`. Source 04's labels are `pma:pr:<label>`, except the ten that coincide with labels of 05 (`eq:cone`, `eq:division`, `sec:algebra`, `sec:baire`, `sec:embeddings`, `sec:formalization`, `sec:multiplication`, `sec:questions`, `thm:baire`, `thm:qe`), which take the sub-prefix `pma:pr:s04:`. Of these, `eq:cone` and `eq:division` denote displays identical to 05's and are printed once, under 05's labels; source 04's displays `eq:divisionformula`, `eq:monuslocus`, `eq:surrealR` and `eq:surrealQ` are likewise printed once as 05's (4.2), (8.2), (9.1) and (9.2), and its Definition 2.1 (`def:split`) is 05's setup in [section 3](#). Where 04 states a theorem that 05 also states, 04's label is a second label of 05's statement (for example

pma:pr:thm:monuslocus is [theorem 8.3](#)). Labels written for the merge have the bare prefix pma: or a part prefix. Sources 07 and 06 have labels pma:cp:(label), except six labels of 06 that coincide with labels of 07, which take pma:cp:s06: ([Part III](#)); source 08 has pma:lc:, source 09 pma:ser:. A few labels were added to unlabelled statements of these sources so that notes can cite them. Parts VI–IX use pma:br: (sources 12 and 10; seven labels of source 10 that coincide with labels of source 12 take pma:br:s10:), pma:nsp: (13), pma:atr: (14) and pma:lcc: (15); their introductions list the labels they add. The batch-87 write also labelled fifteen questions of Parts I, III and IV so that notes can cite them: pma:pr:q:zerodim, pma:pr:q:mindchange, pma:pr:q:beyondsplit, pma:pr:q:rigidity, pma:pr:q:discprofiles, pma:pr:q:effectivetruth, pma:pr:q:metrics, pma:cp:q:glazerone, pma:cp:q:spaces, pma:cp:q:nonsplit, pma:cp:q:connected, pma:lc:q:weakenborel, pma:lc:q:sizespectra, pma:lc:q:strength and pma:lc:q:certified. [Part X](#) uses pma:emb: (source 16; its introduction lists the labels added), and the batch-89 write labelled four further questions of [Part I](#): pma:pr:q:coefficientgroups, pma:pr:q:lattices, pma:pr:q:surrealcutoffs and pma:pr:q:embeddingcategories. [Parts XI](#) and [XII](#) use pma:lgc: (source 17) and pma:gcm: (source 18); their introductions list the labels added. The batch-90 write labelled eight questions of other parts so that notes can cite them: pma:bor:q:minimal, pma:br:q:completion, pma:br:s10:q:completion, pma:atr:q:strength, pma:atr:q:weaker, pma:atr:q:parameters, pma:lcc:q:connected and pma:lcc:q:completion. [Parts XIII](#) to [XV](#) use pma:ord: (source 19), pma:egt: (source 20) and pma:rk: (source 21); their introductions list the labels added. The batch-92 write labelled six questions of other parts so that notes can cite them: pma:ser:q:supportcompletions, pma:ser:q:integerparts, pma:lc:q:orders, pma:lcc:q:boundary, pma:lcc:q:threshold and pma:lcc:q:isocomplexity. [Parts XVI](#) and [XVII](#) use pma:hbo: (source 22) and pma:bsd: (source 23); their introductions list the labels added. The batch-93 write labelled three questions of other parts so that notes can cite them: pma:cp:q:closurerank, pma:cp:q:simplicity and pma:ser:q:descriptiveranks. [Parts XVIII](#) to [XXI](#) use pma:bct: (source 24), pma:spc: (source 25), pma:bfl: (source 26) and pma:tdv: (source 27); unlabelled numbered statements carry n(number) and labels written for this report w-, and their introductions list the labels added. The batch-96 write labelled four questions of Part II so that notes can cite them: pma:bor:q:exactcomplexity, pma:bor:q:subfields, pma:bor:q:observability and pma:bor:q:certified.

1.3 Notation

The sources use clashing symbols. The table fixes the convention of this report; renamed symbols are marked $\rightarrow$. Its second block, for Parts III–V, was added with them. The third block, for Parts VI–IX, was added with those, the fourth, for Part X, with it, the fifth, for Parts XI and XII, with them, the sixth, for Parts XIII–XV, with those, the seventh, for Parts XVI and XVII, with them, and the eighth, for Parts XVIII–XXI, with those.

Notion	This report	In the sources
Presburger arithmetic	$\text{Th}(\mathbb{N}; 0, 1, +, <)$	05: the same; 04: $\text{Pr}_{\mathbb{N}} = \text{Th}(\mathbb{N}; 0, 1, +)$, kept in 04's text. The two are definitionally equivalent: $x \leq y$ is $\exists z (x + z = y)$.
Lexicographic group, coefficient group D dominant	$D \overrightarrow{\times} \mathbb{Z} = G_D$	05: the same; 04: $D \oplus_{\text{lex}} \mathbb{Z} = G(D) \rightarrow G_D$

Notion	This report	In the sources
Nonnegative cone	$M_D; M_E$ for a subgroup E	04: $M(D), M(E) \rightarrow M_D, M_E$
Baire coefficient group	$D_B = \mathbb{Q}^{\mathbb{N}}$, lexicographic, each $\mathbb{Q}$ discrete	05: the same; 04: $D_\omega \rightarrow D_B$ (the same group, order and topology)
Baire model	$M_B = (D_B \overrightarrow{\times} \mathbb{Z})_{\geq 0}$	04: $M_\omega \rightarrow M_B$ (the same model)
Laurent model (04)	$M_L = M_{D_L}, D_L = \mathbb{Q}((t))$	04 only
Baire space	$\mathbb{N}^{\mathbb{N}}$	04's macro $\backslash B = \mathbb{N}^{\mathbb{N}}$; 03's $\backslash B = \mathcal{B}$ is defined but never used
Homeomorphisms with Baire space	05: $h : \mathbb{N}^{\mathbb{N}} \rightarrow M_B$ (tag code); 04: $\Psi : M_B \rightarrow \mathbb{N}^{\mathbb{N}}$, charts $\Psi_{\geq}, \Psi_{>}$, integer code $\zeta : \mathbb{Z} \rightarrow \mathbb{N}$	04: $H, H_{\geq}, H_{>}, h \rightarrow \Psi, \Psi_{\geq}, \Psi_{>}, \zeta$. The two codings are different maps.
Lexicographic symbol	$\overrightarrow{\times}$ (05); $\mathbb{Z}_{\text{lex}}^d$ for lexicographic powers (03)	04's subscript lex in $\oplus_{\text{lex}} \rightarrow \overrightarrow{\times}$
Division by a standard m	$q_m(x) \in M$, $r_m(x) \in \{0, \dots, m-1\}$ (05); $Q_m(x) \in M$, $R_m(x) = (0, z \bmod m) \in M$ (04)	kept; R_m is r_m viewed as an element of the model
Congruence predicates	$D_m(t) \iff t \in mG$ (05); $u \equiv_m v \iff u - v \in mG$ (04)	kept; $u \equiv_m v \iff D_m(u - v)$
Coset invariance	" mG -invariant" (05) = " m -residue invariant" (04)	the same notion
Cofinal infinite element	"cofinal infinite additive unit" (05) = "arithmetically infinite order unit" (04)	the same notion; not a multiplicative unit
Truncated subtraction	$T(x, y)$ (05), $x \dot{-} y$ (04)	the same operation; $\text{Disc}(f)$ is the discontinuity set of f
Pivots, filtration (04)	$p(k), F_k$ in theorem 11.4 ; F_n in proposition 10.6	local; p elsewhere is the predecessor; 03's closed pieces are $F_{Z,k}$
Growth classes (04)	$\mathcal{A}(M)$	local to proposition 10.7 ; not the ring $\mathcal{A}_{\text{fin}}$
Finite principal-part ring (03)	$\mathcal{A}_{\text{fin}} = \mathbb{Z} \oplus \bigoplus_{q \in \mathbb{Q}_{>0}} \mathbb{R}X^q$, cone $\mathcal{A}_{\text{fin}, \geq 0}$	03: $\mathcal{A}, \mathcal{A}_{\geq 0} \rightarrow \mathcal{A}_{\text{fin}}, \mathcal{A}_{\text{fin}, \geq 0}$. Not the collection's $\mathbf{Oz}$, which the base of the omnific Diophantine report once called A .
Rational Hahn ring (03)	$\mathcal{H}_{\mathbb{Q}}$, (17.1)	03: $H \rightarrow \mathcal{H}_{\mathbb{Q}}$
Kleene–Brouwer embedding (03)	$\mu_{\text{KB}} : \mathbb{N}^{<\mathbb{N}} \rightarrow \mathbb{Q} \cap (0, 1)$	03: $h \rightarrow \mu_{\text{KB}}$. Not the dyadic coding κ or the midpoint coding μ into (1, 2) of the omnific-notations report.
Growth variable (03)	$X = t^{-1}$, read as ω	03 §2.1: increasing- t convention in sections 18 and 19 , reverse X -convention from section 22 on; $q \mapsto -q$ converts
Well-ordered subsets (03)	$\text{WO}(L) \subseteq 2^L$	not the well-orderings $\text{WO}_\mu(X)$ of type μ of the surreal-well-orders report
Local sets (03)	W (proof of theorem 23.2); $\mathcal{Z}$ (lemma 24.2); $D_+, D_\times$ (theorem 24.3)	03: $D \rightarrow W$ and $\mathcal{Z}$; $D_+, D_\times$ kept. In Part I , D is always the divisible coefficient group.
L	a countable linear order (03)	in 05's and 04's elimination proofs, a lower bound or a common multiple (local)
e_j, e	unit vectors of $\mathbb{Q}^{\mathbb{N}}$ (05, 04); computable enumerations $e, e_+, e_{\geq 0}$ (05, section 6)	as in 05

Notion	This report	In the sources
Omnific integers	$\mathbf{Oz} = \mathbb{Z} \oplus \Pi$, growth notation ω^γ	as in the collection's docs/NOTATION.md
<i>Parts III–V (sources 07, 06, 08, 09)</i>		
The variable t	Part III: t positive <i>infinite</i> (06's $\mathbb{Z} + t\mathbb{R}[t]$, $t \mapsto \omega$); Part V: t positive <i>infinitesimal</i> (09's Hahn convention, $t^q \mapsto \omega^{-q}$)	kept in each part; tempting false reading: 06's t and 09's t are reciprocal, not equal. 03 uses $X = t^{-1}$ (infinite); 08 writes X for 06's t
Polynomial ring $\mathbb{Z} + t\mathbb{R}[t]$	A , cone $P = A_{\geq 0}$, topology τ_P by degree and constant term (06)	$= 08$'s $A_* = \mathbb{Z} + X\mathbb{R}[X]$ = the ring $\backslash\text{Pmod}[X]$ of the omnific Diophantine report (odg:frac:sub:models) and, at $t = \omega$, $\mathbb{Z} + \omega\mathbb{R}[\omega]$; under $t \mapsto X$ the integer-exponent subring of 03's $\mathcal{A}_{\text{fin}}$. Not 03's cone $\mathcal{A}_{\text{fin}, \geq 0}$
Presburger arithmetic	$\text{Th}(\mathbb{N}; 0, 1, +, <)$; 06: Pres	07's macro for $\text{Th}(\mathbb{N}; 0, 1, +, <)$ is expanded; 06's Pres = $\text{Th}(\mathbb{N}, 0, 1, +, <)$ kept. 08: signed language $\mathcal{L}_{\text{Pres}}$
Lexicographic product	$\overrightarrow{\times}$	07's $A \times_{\text{lex}} \mathbb{Z}$, 06's $D \text{ lex } \mathbb{Z}$ and 08's $\mathbb{R}^d \times_{\text{lex}} \mathbb{Z} \rightarrow \overrightarrow{\times}$ (same order: left factor dominant)
Real model	$M_{\mathbb{R}} = M_{\mathbb{R}}$	the same set and topology in 06, 07, 08; 07's second topology $M_{\mathbb{R}}^{\#}$ is a different space on the same algebra
Model families (07)	M_A, G_A (general A); $M_\alpha = (\mathbb{Q}^\alpha \overrightarrow{\times} \mathbb{Z})_{\geq 0}$; M_s with $B_s = \mathbb{Q} + \mathbb{Q}s$, $s \in (1, 2)$ irrational	M_ω is Part I's M_B ; 07's H_s is a convex subgroup
Model families (06)	$M(D), G(D)$; $M_\omega (= M_B)$; M_α (as 07); M'_s, G'_s, H'_s for $s \in 2^{\mathbb{N}}$	06: $M_s, G_s, H_s \rightarrow M'_s, G'_s, H'_s$ (07's M_s is another family); 06's $H = \mathbb{Q}^{\mathbb{N}}$ in section 35.5 is local
Models of Part IV	$G_d = \mathbb{R}^d \overrightarrow{\times} \mathbb{Z}$, M_d (so $M_1 = M_{\mathbb{R}}$); $H_d \subseteq \mathbf{Oz}; \mathbb{R}^d \times E$	08's D in theorem 47.2 is a countable discrete group, not Part I's divisible D
Series fields of Part V	$\mathcal{P}_k$ (Puisseux), $\mathcal{L}_k$ (Levi-Civita), $\mathcal{H}_k = k((t^{\mathbb{Q}}))$, integer part I_k ; hulls F_A, J_C	09's macros $\backslash\text{PP}, \backslash\text{LL}, \backslash\text{HH}, \backslash\text{II}$ renamed in the source only; same symbols printed. $\mathcal{P}_k$ is not 03's cone $\mathcal{A}_{\text{fin}, \geq 0}$; 03's $\mathcal{P}$ is the real-coefficient Puiseux field
Well-founded trees Typography	WF $\mathbf{1}, \dot{-}, \text{IOpen}, \text{ProveIt}$	09's WF $\rightarrow$ 03's WF (same set) 07's $\backslash\text{ind}, \backslash\text{dotminus}$; 08's IOpen; 07's ProveIt $\rightarrow$ these (same meaning)
Local letters	A (a Polish group in 07, a ring in 06 and 08); P (a cone); H, D, B, J	local to the statement or proof where they are introduced
<i>Parts VI–IX (sources 12, 10, 13, 14, 15)</i>		
Ordinary-integer-bounded part (10)	$\text{Fin}(A) = \{x : x \leq \underline{n} \text{ for some ordinary } n\}$	10's macro $\backslash\text{Fin}$ renamed in the source only; printed Fin in roman type. Not Part II's Fin (sans-serif), the set of finite subsets

Notion	This report	In the sources
Principal convex hull (12, 10)	$H_a = \{x : x \leq na \text{ for some ordinary } n\}$	the same in both sources; local to Part VI. Not source 08's finite surreal scales $H_d \subseteq \mathbf{Oz}$, 03's ring $\mathcal{H}_{\mathbb{Q}}$, or source 15's $H = G^0$
Unit-image spectrum (12, 10)	$\mathcal{U}(G)$	10: $\mathcal{S}(G, e) \rightarrow \mathcal{U}(G)$ (the same set)
Cardinal functions (12, 10)	c, d (12); cell, dens, ext, L (10)	kept in each source's text; c = cell is cellularity, d = dens density. $\mathfrak{c}$ is the continuum, written $\mathfrak{c}$ in Part III
Continuous scalars (10)	$\text{Sc}_{\text{cont}}(A)$	not source 08's Borel scalars $\mathcal{B}(A)$
The ring $\mathbb{Z} + t\mathbb{R}[t]$ (12, 10, 15)	12: $A = \mathbb{Z} + t\mathbb{R}[t]$; 10: $A = \mathbb{Z} + X\mathbb{R}[X]$; 15: R_0	the ring of Part III and source 08's A_* ; t and 10's X are positive <i>infinite</i> , as in Part III, not Part V's infinitesimal t
Lexicographic products (12–15)	$\overrightarrow{\times}$	12's and 13's $\oplus_{\text{lex}}$, 13's, 14's and 15's $\text{lex} \rightarrow \overrightarrow{\times}$ (left factor dominant)
Baire model (13)	$M_\omega, D_\omega = \mathbb{Q}^\omega$	kept; Part I's $M_B, D_B, \mathbb{Q}^{\mathbb{N}}$ (the same objects)
Countable core (13, 14, 15)	K (13, general); $K_* = G_{\zeta_*}$ (14); $C = N - N$ (15)	14: $D \rightarrow K_*$ in its Section 10 and $D \rightarrow K$ in the commentary on its question on nonsplit topology, because D is the divisible coefficient group in Parts I and VII. 13's explicit core is $\text{isg}'s G_{\varepsilon_2}$
Divisible kernel (13)	$D = \ker r = \bigcap_m mG$ in section 95	local; r is the canonical residue map, not Part I's remainder r_m
Revision numbers (13)	$\mu(f), \mu_{\text{eff}}(f)$	not Part II's μ_{KB}
Exponential and unit (14)	$e(x) = 2^x$, internal; e_0 the unit of K_*	14's e is not a unit; Part I's e_j are unit vectors
Root scales (14)	$t_n, q_n, X_n, h_{k,n}$	local to section 108 ; unrelated to the variable t of Parts III and V
Cantor space (14)	$2^{\mathbb{N}}$	14's $\mathbb{C} = 2^\omega \rightarrow 2^{\mathbb{N}}$
Surreal embeddings (14, 15)	$\Phi, \Phi_\omega = \omega\Phi$ (14); Φ_L (15)	14's $\Psi \rightarrow \Phi_\omega$, because Ψ is Part I's Baire homeomorphism; 10's Ψ in theorem 79.14 is local, like source 08's in theorem 46.2
Sets of Cantor space (14)	D, D^*, E_n	local to sections 110 and 111
Components (15)	$H = G^0$ and its infinitesimal subgroup C	source 08's C and B_C , letters exchanged; tempting false reading: C is the identity component, as in Part IV
Models on a countable order (15)	M_L, G_L, D_L, V_L (L italic)	not Part I's Laurent model $M_L = M_L$
Locally compact carrier (15)	X_{lc}	15 only
Typography	$\text{IOpen}, \text{Pres}, \text{Provelt}, \dot{-}$	12's roman $\text{IOpen}, \text{Pres}$, 13's $\backslash \text{dotminus}$ and 14's $\text{ProveIt} \rightarrow$ these (same meaning)

Part X (source 16)

Notion	This report	In the sources
Coefficient group (16)	$H_\alpha = \mathbb{Q}^\alpha$ (full product, discrete coordinates, lexicographic), $G_\alpha = H_\alpha \times \mathbb{Z}$ ordered as $H_\alpha \xrightarrow{\times} \mathbb{Z}$, $M_\alpha = (G_\alpha)_{\geq 0}$	kept: H_α is Part III's A_α (source 07), not renamed because source 16's A is a matrix; H_ω is Part I's D_B . Tempting false reading: H_α is not Part VI's principal hull H_a , source 07's convex subgroup H_s , source 08's $H_d \subseteq \mathbf{Oz}$ or source 15's $H = G^0$
Finite-support vectors (16)	$D_B = \mathbb{Q}^\omega$, $D_B^{\text{fin}} = \mathbb{Q}^{(\omega)}$; dual $H_\alpha^\circ = \mathbb{Q}^{(\alpha)}$, pairing $\langle c, h \rangle$, annihilator $V^\perp$	16: $D, D_{\text{fin}} \rightarrow D_B, D_B^{\text{fin}}$ (the same group as Part I's); D is the generic divisible group in Parts I and VII
Valuation, pivots (16)	$v(h)$, the first nonzero index; pivot map p ; pivot subset $S \subseteq \alpha$; $\text{otp}(S)$	v and p as in source 04's theorem 11.4 ; p elsewhere is the predecessor; S is not source 04's shift S (example 11.6)
Matrices and maps (16)	$A = (a_{ji})$, left inverse L, U_V , E_S , quotient map B , projection π	local to Part X ; A is not Part III's group A or A_α , B not the Baire coefficient group
Wild functional, shear (16)	φ (exists by choice); Sh_φ ; graph space W_φ	16: $S_\varphi \rightarrow \text{Sh}_\varphi$, because S is the pivot subset; not Part I's shift σ
Terminal tail (16)	$\alpha = \lambda + n$, $K_\alpha \cong \mathbb{Q}^n$	16's local K in the proof of theorem 134.1 $\rightarrow K_\alpha$ (the same set); not source 13's core K or source 14's K_*
Restriction, lexicographic sum (16)	$x \upharpoonright \omega$; $\overrightarrow{\times}$	16's macro $\backslash \text{res} (\upharpoonright) \rightarrow \backslash \text{rstr}$, because Part I's $\backslash \text{res}$ is res ; 16's roman $\text{lex} \rightarrow \overrightarrow{\times}$
Omnific map (16)	ι_α , exponents $\gamma_i = (i+1)^{-1}$	= Part III's $J_\alpha, \varepsilon_\beta$ (theorem 40.1); Part I's ι_B for $\alpha = \omega$
Local letters (16)	$N, \Gamma = N - N$ in the proof of theorem 131.1	16: $S, K \rightarrow N, \Gamma$, because S is the pivot subset
<i>Parts XI and XII (sources 17, 18)</i>		
Cantor space (17)	$2^{\mathbb{N}}$	17's $\backslash C = 2^\omega \rightarrow 2^{\mathbb{N}}$, as for source 14
Geometric code (17)	base $q \geq 2$; decoder $\beta(u, v, i) = \text{rem}(u, 1 + (i+1)v)$; $E(i) = \beta(u, v, i)$ through C ; $\text{Geo}(q, C, u, v)$; endpoint map e_q	q is the base, not a scale; E is not source 18's E_M ; e_q is partial, not source 14's total exponential e
Scales (17)	$c, t_n, p_n = \lfloor c/t_n \rfloor, X_n, x_n, h_{k,n}, y_n, V_s, Q_j, \delta_n, z_r$	source 14's q_n is 17's p_n ; the other letters play their roles in section 108 ; t_n is unrelated to the variable t of Parts III and V
Local data (17)	width $D = d - b$; domains $D_q, \mathcal{G}_M$; anchor $B = b + 2c$; local logarithm L ; parameter set P_E	local to Part XI ; D is not the divisible group of Parts I and VII nor source 14's Cantor set D ; L is not Part I's Laurent model
Countable core (17)	$N_M = N_M$	Glazer's cut; not source 15's countable model N or its manifold boundary, nor source 16's local N
Polynomial cut (17)	$K = \bigcup_n [0, c^n)^M$	local; not source 13's core K , source 14's K_* or source 18's hull K

Notion	This report	In the sources
Monoid and group (18)	$M, G = M - M$, $E_M = \{(y, x) : x \leq y\}$, partial subtraction δ , $F(x, z) = (x + z, x)$, $B_U = U - U$	local to Part XII ; G is not Part IV's G_d or Part III's G_A ; tempting false reading: δ is not the monus $\dot{-}$ (example 155.4)
Monus (18)	$\dot{-}$	18's $\backslash\text{dotminus} \rightarrow \dot{-}$ (same operation)
Divisions (18)	Divisions I–III	source 18's three $\backslash\text{parts}$; its “Research question n ” is Question 165. n
Class theory (18)	$\text{GB}_{-\text{Fnd}}^{\text{rep}} + \text{AC}$; Ord; power set $\mathcal{P}(A)$; $\mathfrak{c} = 2^{\aleph_0}$	18's own notation, Foundation omitted, not negated; $\mathcal{P}$ is not Part II's Puiseux field $\mathcal{P}$; 18's $\backslash\text{cfrac} \rightarrow \mathfrak{c}$
Class spaces (18)	$X_\alpha, N(x), H(A), O_\alpha$, clopen hull $K, \theta = \kappa^+$, components $C(x)$, closure pattern Φ_A , ray $L = \text{Ord} \times [0, 1)$	local to Division II; $H(A)$ is not Part VI's H_a or Part X's H_α ; Φ_A is not source 14's surreal map Φ
Surreal stages (18)	$\text{No}_{<\beta}$, birthday $b(s)$	local to section 163
Typography	Provelt	17's typewriter ProveIt $\rightarrow$ Provelt; 18's is already sans-serif

Parts XIII–XV (sources 19, 20, 21)

Models with coded scales below $\mathbb{R}$ (19)	$G_L^{\text{lo}} = \mathbb{R} \overrightarrow{\times} V_L \overrightarrow{\times} \mathbb{Z}$, $M_L^{\text{lo}} = (G_L^{\text{lo}})_{\geq 0}$; $G_D^{\text{lo}} = \mathbb{R} \overrightarrow{\times} D \overrightarrow{\times} \mathbb{Z}, M_D^{\text{lo}}$	19: $G_L, M_L, G_D, M_D \rightarrow G_L^{\text{lo}}, M_L^{\text{lo}}, G_D^{\text{lo}}, M_D^{\text{lo}}$. Tempting false readings: Part IX's $G_L = V_L \overrightarrow{\times} \mathbb{R} \overrightarrow{\times} \mathbb{Z}$ puts the coded scales <i>above</i> $\mathbb{R}$; Part I's $G_D = D \overrightarrow{\times} \mathbb{Z}$ has no real coordinate; Part I's M_L is the Laurent model
Coded ordered group (19)	$V_L = \bigoplus_{i \in \mathbb{N}} \mathbb{Q}e_i$, least L -index dominant	19: $D_L \rightarrow V_L$, Part IX's group with the same order; not Part IX's $D_L = V_L \overrightarrow{\times} \mathbb{R}$
Carrier (19)	X_{lc}	19: $X \rightarrow X_{\text{lc}}$, Part IX's space
Normal data (19)	$G(E) = \mathbb{R} \times E, \mathbf{1} = (0, u)$; $A, B = E/A, <_A, <_B$; $C = \mathbb{R} \times \{0\}$, $K = \mathbb{R} \times A, I = \{0\} \times A$; shear S_f ; standard part st	C is Part IV's C and Part IX's $H = G^0$; I is Part IV's B_C and Part IX's C (<i>letters exchanged</i>); A, B, K are local, not Part III's groups or the Baire coefficient group; 19's $\backslash\text{st} \rightarrow \backslash\text{stp}$ (same symbol)
Codes (19)	LO; code space $\mathcal{P}_E$; $H, H_L, h_k^\pm(L), \theta_L, m_L, 0_L, 1_L$	local to Part XIII ; $\mathcal{P}_E$ is not Part II's Puiseux field $\mathcal{P}$ or Part V's $\mathcal{P}_k$
Exponent rings (20)	$A_\Gamma = \mathbb{Z} \oplus \bigoplus_{\gamma \in \Gamma_{>0}} \mathbb{R}T^\gamma$, $P_\Gamma = (A_\Gamma)_{\geq 0}, T$ positive infinite; series field L_Γ	$A_\mathbb{Q}$ is Part II's $\mathcal{A}_{\text{fin}} (T = X)$ and $P_\mathbb{Z}$ Part III's cone of $\mathbb{Z} + t\mathbb{R}[t] (T = t)$; P_Γ is not Part III's P ; L_Γ is Part V's Levi-Civita field mirrored ($T = t^{-1}$), with real coefficients
Positive part (20)	$\text{pp}_{>0}$	20's macro $\backslash\text{PP}$ renamed in the source only (same symbol printed); not the report's $\mathcal{A}_{\text{fin}, \geq 0}$, Part II's cone
Arithmetic axioms (20)	DIV (division with an order-bounded remainder), Root_k ; external cut I ; grid bound C ; $M = R_{\geq 0}$	DIV is not division by standard integers; I, C local, not Part V's I_k or Part IV's C

Notion	This report	In the sources
Levi-Civita fields (21)	$\mathcal{L}^\Gamma, \mathcal{L}^\Delta, \mathcal{H}^\Gamma = k((t^\Gamma)), \mathcal{I}^\Gamma$, surreal image $\mathcal{S}^\Gamma$; $k = \mathbb{R}_{\text{alg}}, t$ positive <i>infinitesimal</i>	21: $\mathcal{L}_\Gamma, \mathcal{L}_\Delta, \mathcal{H}_\Gamma, \mathcal{I}_\Gamma, \mathcal{S}_\Gamma \rightarrow$ exponent group as superscript (macros renamed). Tempting false reading: 21's $\mathcal{L}_\mathbb{Q}$ is not Part V's $\mathcal{L}_k$ with $k = \mathbb{Q}$; $\mathcal{L}^\mathbb{Q}$ is Part V's $\mathcal{L}_{\mathbb{R}_{\text{alg}}}$
Characters (21)	$H_\Gamma = \text{Hom}(\Gamma, 1 + \mathfrak{m})$ with $\star$; multiplier field E_Γ ; metric $d = e^{-v}$	H_Γ is not a Hahn field; Part V's metric is $\min\{1, 2^{-v}\}$ (same topology)
Typography (19)	$\mathfrak{c}$, Provelt	19's $\backslash\text{cfra} \rightarrow \mathfrak{c}$; its typewriter $\backslash\text{repo} \rightarrow \text{Provelt}$
<i>Parts XVI and XVII (sources 22, 23)</i>		
Hilbert family (22)	$H_\theta^{\ell^2} = \ell^2(\theta, \mathbb{R})$, tails $H_{\theta, \alpha}^{\ell^2}$; $G_\theta^{\ell^2} = H_\theta^{\ell^2} \overrightarrow{\times} \mathbb{Z}$ (Hilbert times discrete topology), $M_\theta^{\ell^2} = (G_\theta^{\ell^2})_{\geq 0}$; cone P_θ ; leading index v_θ	22: $H_\theta, H_{\theta, \alpha}, G_\theta, M_\theta \rightarrow H_\theta^{\ell^2}, H_{\theta, \alpha}^{\ell^2}, G_\theta^{\ell^2}, M_\theta^{\ell^2}$. Tempting false readings: Part X's $H_\alpha = \mathbb{Q}^\alpha$ and Part III's G_α, M_α have discrete rational coordinates and the product topology; $M_\omega^{\ell^2}$ is not Baire space; v_θ is Part X's v for real coordinates
Flag (22)	stages $\mathcal{E}_\alpha$, functionals f_α , terminal index θ ("flag rank")	22: E_α , and the flag H_α of theorem 210.2 , $\rightarrow \mathcal{E}_\alpha$; θ is Part III's cone closure rank ρ ; not Part III's cut $E_\alpha = \{x < p\}$
Omnific map (22)	$J_\theta^{\ell^2}(x, n) = n + \sum_\alpha x_\alpha \omega^{\theta-\alpha}$, surreal subtraction in the exponent	22: $J_\theta \rightarrow J_\theta^{\ell^2}$; not Part III's J_α (exponents $1/(\beta+1)$, rational coefficients)
Operators (22)	shift S , diagonal Diag , pivot map φ , entries $t_{\beta\alpha}$; divisible part $D = H_\theta^{\ell^2} \times \{0\}$	22: diagonal $D \rightarrow \text{Diag}$; S is not Part X's pivot subset or Part I's shift σ ; D in section 220 is the divisible part, as in Parts I and VII
Typography (22)	Provelt, $\langle x, y \rangle, \ x\ $, span	22's typewriter $\backslash\text{repo} \rightarrow \text{Provelt}$
Left-finite real fields (23)	$\mathcal{L}_{\mathbb{R}}^\Gamma, \Gamma \leq \mathbb{R}$ countable, real coefficients, t positive <i>infinitesimal</i> ; topologies τ_c, τ_v ; slices $\text{Sl}_S \cong \mathbb{R}^S$	23: $L_\Gamma, F_S \rightarrow \mathcal{L}_{\mathbb{R}}^\Gamma, \text{Sl}_S$. Tempting false readings: Part XIV's L_Γ is this field in the infinite variable $T = t^{-1}$, supports finite above each bound; Part XV's $\mathcal{L}^\Gamma$ has coefficients in $\mathbb{R}_{\text{alg}}$; Part V's $\mathcal{L}_k$ is indexed by the coefficient field ($\mathcal{L}_{\mathbb{R}}^\mathbb{Q} = \mathcal{L}_{\mathbb{R}}$); Part XV's F_S are subfields local; δ is not Part XII's partial subtraction; gain is Part XV's valuation gain; E is the restriction of the Euler operator $\mathfrak{D}$ of the real-vector-space-structure report
Derivations (23)	$\delta(\gamma) = t^{-\gamma} D(t^\gamma), D_\delta$, Euler E, E_i , gain, $\text{Der}_{\text{Bot}}, d_h, B_\delta$	23: $H_N \rightarrow W_N$, because H_α is Part X's $\mathbb{Q}^\alpha$; Φ_s is not Part XV's Φ_u
Flows and groups (23)	$\Phi_s = \exp(sD)$; subgroups W_N	23's $\backslash\text{restr} \rightarrow \backslash\text{rstr}$ (same symbol)
Restriction (23)	$\upharpoonright$	
<i>Parts XVIII–XXI (sources 24–27)</i>		
Rank-one derivations (24)	$\mathcal{L}_{\mathbb{R}}^\Gamma$ with $\mathbb{Z} \leq \Gamma \leq \mathbb{Q}$; $E = t d/dt, D_a = aE$; gain $\alpha = v(a), c = \text{lc}(a)$, residue $\rho = \text{ct}(1/a), \theta = \rho/\alpha; J$; normal form a^{nf}	24: $L_\Gamma \rightarrow \mathcal{L}_{\mathbb{R}}^\Gamma$ (as in Part XVII), $\backslash\mathfrak{m} \rightarrow \mathfrak{m}, \backslash\text{norma} \rightarrow \backslash\text{bctnf}$ (same symbols). ρ is not Part III's cone closure rank, θ not Part XVI's ordinal, J not Part III's J_α

Notion	This report	In the sources
Coordinates (24)	G_Γ (Borel ordered-field automorphisms); $S_h f = f \circ h$, $h = At^r(1+u)$, $S_h S_k = S_{k \circ h}$; $U_\Gamma = \{r \in \mathbb{Q}_{>0} : r\Gamma = \Gamma\}$; “ S_h carries D_a to D_b ” means $S_h^{-1} D_a S_h = D_b$	Part XXI (27) uses the same substitutions but acts by $g \cdot D = g D g^{-1}$, the opposite orientation; conjugating coordinates are inverted between the two parts
Primes (24)	heights $h_p(\Gamma)$, p -adic valuations v_p ; infinitely divisible primes $\omega_\infty(\Gamma)$; exponent representative κ	24: $P_\infty(\Gamma) \rightarrow \omega_\infty(\Gamma)$, because Part XIV’s $P_\infty = P_{\mathbb{Z}[1/p]}$ is a semiring; v_p is not the series valuation v
Bookkeeping (24)	I with $EI = 1$ (a formal log t ; not adjoined to the field)	24: $\ell \rightarrow I$, because Part XVII’s ℓ , ℓ_i are the rational coordinate maps; 24’s functional ℓ in lemma 242.1 is kept, as in Part XVII
Flows (24)	Φ_s ; Euler flow $T_s f = f(e^{cs}t)$; tangent automorphism Ψ , $\Psi^{[z]}$; time change τ	T_s is not Part XIV’s variable T ; Part XVII’s Euler flow is Ψ_s , with $T_s = \Psi_{cs}$ for $b = 1$
Words (24)	“threshold” (divisibility), “smooth” (descriptive set theory)	not Part XIV’s exponent-group threshold; not differentiability
Well-ordered supports and codes (25)	$\text{WO}(L)$; $V(L) = \{a \in \mathbb{R}^L : \text{supp } a \in \text{WO}(L)\}$, $V(\Gamma) = V_\Gamma$	25: $\mathcal{H}_L \rightarrow V(L)$, because $\mathcal{H}$ is Part II’s $\mathcal{H}_\mathbb{Q}$, Part V’s $\mathcal{H}_k$ and Part XV’s $\mathcal{H}^\Gamma$; for groups it is Part II’s V_Γ . Tempting false reading: $V(L)$ is not the coded group V_L of Parts IX and XIII
Benchmarks and embeddings (25)	Fin , $\text{Fin}^\mathbb{N}$; $B = \omega^* \cdot \omega$; $\hookrightarrow$; KB	Fin , $\text{Fin}^\mathbb{N}$ are Part II’s (Part V’s FIN , $\text{FIN}^\mathbb{N}$); 25’s sans-serif $\text{KB} \rightarrow \text{KB}$; B is not Part I’s M_B
Cuts and heights (25)	$\text{Cut}(L)$; final segments F_i ($i \in I$), $c_{i,j}$, $Z_{i,j}$; heights h_i	local; F_i is not Part XV’s F_S or Part XVII’s F_N
Families and sums (25)	$\mathcal{S}_\Gamma$, $U(X)$, $I_\gamma(X)$, strong sum s ; unit arrays $\mathbf{e}_k$	25: $E_k \rightarrow \mathbf{e}_k$, because E , E_i are Part XVII’s Euler derivations; s is not Part XVII’s time
Certificates (25)	$\text{Cert}_\Gamma^\mathbb{N}$, $\text{Cert}_\Gamma^\mathbb{R}$; bounds b , m , least bounds b^* , m^* ; sections $r_\mathbb{N}$, $r_\mathbb{R}$; windows Ω_k ; soft thresholds v_k , w_k ; regularizations R_k	25: $C_\Gamma^\mathbb{N}$, $C_\Gamma^\mathbb{R}$, windows $E_k \rightarrow \text{Cert}_\Gamma^\mathbb{N}$, $\text{Cert}_\Gamma^\mathbb{R}$, Ω_k , because C is Part II’s code set; v_k is not a valuation
Local sets (25)	K , K_i , partitions Q_n ; ideal $\mathcal{J}$; supports S ; sets C , C'	25: E , E_i , $\mathcal{P}_n$, $\mathcal{I}$, A , $B \rightarrow$ these, because $\mathcal{P}$ is Part II’s $\mathcal{P}$ and Part V’s $\mathcal{P}_k$, $\mathcal{I}$ Part V’s $\mathcal{I}_k$ and Part XVII’s support family, A Part II’s ring $\mathcal{A}_{\text{fin}}$
Groups on a coded order (25)	$G = \bigoplus_n \mathbb{Z}e_n$, $G' = G \oplus \mathbb{Z}u$, G_L , G'_L	local; not Part IX’s G_L or Part XIII’s G_L^{lo}
Pointclasses (25)	Σ_2^0 , Π_3^0 , Π_1^1	25’s $\backslash \text{sig}$, $\backslash \text{pie} \rightarrow$ Part V’s macros (same symbols); 25 has no δ (its $\backslash \text{del}$ is unused)
Left-finite fields (26)	$\mathcal{L}_\mathbb{R}^\Gamma$, $\Gamma \leq \mathbb{R}$ countable <i>divisible</i> ; filtration $\mathcal{L}_{\mathbb{R}, \geq a}^\Gamma$; windows $\text{Win}_{a,b} = \mathcal{L}_{\mathbb{R}, \geq a}^\Gamma / \mathcal{L}_{\mathbb{R}, \geq b}^\Gamma$	26: $\mathcal{L}_\Gamma$, $\mathcal{L}_{\Gamma, \geq a}$, $V_{a,b} \rightarrow \mathcal{L}_\mathbb{R}^\Gamma$, $\mathcal{L}_{\mathbb{R}, \geq a}^\Gamma$, $\text{Win}_{a,b}$. Tempting false readings: Part XV’s $\mathcal{L}^\Gamma$ is a surreal image; Part II’s V_Γ and Part XIII’s V_L are groups; a window is countably infinite-dimensional, not finite-dimensional

Notion	This report	In the sources
Flows and generators (26)	Borel field flow Φ_s ; TLF (local finiteness on windows); logarithmic values $\delta(\gamma)$; gain constant ε ; gaps κ_n	26: $d(\gamma), \delta, \varepsilon_n \rightarrow \delta(\gamma), \varepsilon, \kappa_n$, Part XVII's letters; "strict" $(v(Dx) > v(x))$ is weaker than uniform positive gain
Alternating construction (26)	$D_e, D_o, D_{eo} = [D_e, D_o]$, shears $\Phi_s^e, \Phi_s^o, A = \Phi_1^e \Phi_1^o$; generic D_1, D_2 in section 281	26: $E, F, H, P_s, Q_s \rightarrow$ these, because E, E_i are Part XVII's Euler derivations, H_α Part X's group, P_θ Part XVI's cone; A is local, not Part II's $\mathcal{A}_{\text{fin}}$
Parameters (26)	$\mathfrak{P}_B = \mathfrak{m}^B$, locus $\mathfrak{S}_B$; transcendental $\mathfrak{g}$	26: $\mathcal{P}, \mathcal{I}, \theta \rightarrow \mathfrak{P}_B, \mathfrak{S}_B, \mathfrak{g}$; not Part II's $\mathcal{P}$, Part V's $\mathcal{P}_k, \mathcal{I}_k$, or Part XVI's θ ; $\mathfrak{P}_B \subseteq$ Part XVII's $\mathcal{B}_\Gamma$
Left-finite field (27)	$\mathcal{L}_\mathbb{R}^\Gamma, \mathbb{Z} \subseteq \Gamma \subseteq \mathbb{Q}$; G_Γ ordered Borel automorphisms, Θ_y the substitution $t \mapsto y$; $U_\Gamma = \{r \in \mathbb{Q}_{>0} : r\Gamma = \Gamma\}$	27: $K_\Gamma \rightarrow \mathcal{L}_\mathbb{R}^\Gamma$. Tempting false readings: Part II's K_Γ is $\mathbb{R}((t^\Gamma))$, Part XVII's K_Γ the surreal image; Θ_y is Part XVIII's S_y
Scalar data (27)	canonical scalar H_h , set Can_Γ ; coordinate $y_h, \Theta_h, N_h = \Theta_h^{-1}$; signed exponent representative κ	27: $H_h, \mathcal{H}_\Gamma \rightarrow H_h, \text{Can}_\Gamma$, because Part V's $\mathcal{H}_k$ and Part XV's $\mathcal{H}^\Gamma$ are Hahn fields; 27's κ is signed, Part XVIII's is defined on positive exponents
Pairs (27)	NC_Γ , ratio $h = b/a, j = D_a(h)$, normal coefficient A , normal pair (D_A, D_{HA})	27: $\mathcal{P}_\Gamma \rightarrow \text{NC}_\Gamma$ (Part V's $\mathcal{P}_k$ is the Puiseux field); conjugation $g \cdot D = g D g^{-1}$, opposite to Part XVIII (remark 289.2)
Lie algebras (27)	$\mathfrak{a}_p, \mathfrak{s}_p$; $\mathfrak{sl}_2$ -triple X_+, X_-, X_0 ; affine action $\Phi_{a,b}$	27: $e, f, H \rightarrow X_+, X_-, X_0$; $\Phi_{a,b}$ is not Part XV's Φ_u or Part XVII's $\exp(sD)$

1.4 Relation to the repository

The formal project `Logic/PresburgerArithmetic`. Both sources of [Part I](#) inspected this project at their pins, 05 its Lean file `Lean/PresburgerArithmetic/Cooper.lean` and 04 its Rocq file `Coq/Cooper.v`. The project decides Presburger sentences over the ordinary integers. Its periodicity lemmas assume a predicate on $\mathbb{Z}$ invariant under one standard shift; [example 4.4](#) and [proposition 4.6](#) show that this hypothesis cannot be carried verbatim to nonstandard $\mathbb{Z}$ -groups, and [lemma 4.7](#) and [theorem 4.8](#) give the model-uniform replacement. This is a porting requirement, not a defect of the integer theorems. [Section 13.5](#) names the declarations. **Nothing in this report is formalized**, and its placement in the Surreal collection, with a pointer from the formal project, confers no formal status.

The Surreal collection. Results already in the collection are cited, not reprinted as new:

- `exr:prop:division` and the paragraph after it in the `exponential-relations-over-r-omnific-integers` report: $\mathbf{Oz}$ is a $\mathbb{Z}$ -group, hence a model of Presburger arithmetic in its ordered-group presentation. [Proposition 4.2](#) and [theorem 4.9](#) prove the classical facts again for the split groups G_D .
- `isg:cf:main:presburger` in the `discrete-initial-subgroups-and-omnific-normalization` report: a set-sized $\mathbb{Z}$ -group has an initial realization in $\mathbf{Oz}$ if and only if it

splits as $G^{\text{dv}} \xrightarrow{\sim} \mathbb{Z}$ with $G^{\text{dv}} = \bigcap_n nG$. The groups G_D of [Part I](#) are split by construction, with $\bigcap_m mG_D = D \times \{0\}$ ([corollary 4.3](#)), so that theorem applies to them [merge]; whether 05's maps $\iota_{\mathbb{R}}, \iota_{\mathbb{B}}$ are initial realizations in its sense is not checked here.

- `onot:neg:lem:kbtree`, `onot:neg:thm:pionesone` and `onot:neg:rem:secondroute` in the omnific-notations report: the lightface Π_1^1 -completeness of support validity. 03's [lemma 18.2](#) and [theorem 22.1](#) are their boldface counterparts (notes after those statements).
- `odg:thm:floor` in the omnific-diophantine-geometry report, proved in Lean for every surreal: 03's [lemma 23.1](#) is its restriction to the Puiseux field (note after it).
- `dsn:cor:openinduction` in the definable-surreals-and-omnific-integers report, and the Shepherdson paragraph of the omnific-diophantine-geometry report: the route of 03's [theorem 23.2](#) (note after it).

All these labels exist at the repository's current state and already existed at the pins of sources 03–05. Of them only `odg:thm:floor` is formalized (`docs/FORMALIZATION.md`, row “Full `odg:thm:floor`”); the others are recorded there as pending.

Parts III–V. Sources 06 and 07 cite the formal Presburger project (its `Lean Syntax.lean`, `Decision.lean`, `Audit.lean`) and source 06 also `Logic/Interpretability/PAHF/Lean/PAHF/PASyntax.lean`; their descriptions are correct at the current state ([section 41](#)). Source 06 cites `isg:thm:suspension`, `isg:cf:main:presburger` and `isg:cf:thm:PA`, and the omnific Diophantine report's `odg:frac:sub:models`, `odg:frac:prop:intersection` and `odg:frac:ex:workspace` for the ring $\mathbb{Z} + t\mathbb{R}[t]$; the notes of Parts III–V also cite `odg:thm:induction`, `odg:rem:ringinduction`, `odg:thm:floor`, `dsn:cor:openinduction`, `onot:neg:lem:kbtree` and `onot:neg:thm:pionesone`. All exist at the current state; of them only `odg:thm:floor` is formalized. Placement beside the formal Presburger project confers no formal status on Parts III–V either.

Parts VI–IX. Sources 10 and 12 read `Logic/PresburgerArithmetic/README.md` and `Algebra/SurrealNumbers/README.md`; source 13 also the Lean files `Syntax.lean` and `Decision.lean` of the Presburger project, the discrete-initial-subgroups-and-omnific-normalization report and the Hilbert's-tenth programme's scope note; source 14 `Logic/PeanoArithmetic/ListCoding/Lean/PAListCoding/ExponentiationDiophantine.lean` (`naturalPowGraph_diophantine`, `power_polynomial_exists`, proved from Mathlib's `Dioph.pow_dioph`) and its Coq sibling `ExponentiationDiophantine.v`; source 15 the two Presburger Lean files and `OmnificIntegers.lean`, `OmnificPurelyInfiniteModule.lean` and `OmnificAlgebraDiscreteness.lean` in `Algebra/SurrealNumbers/Surreal/Foundations`. Their descriptions are correct at the current state; the two READMEs have changed since the pins only by the catalogue's pointers to this report (2b7b388ba), and the other files are unchanged. Those Lean and Coq files are formal developments, but about the ordinary integers, the standard natural numbers and the omnific coefficient construction; none states a result of Parts VI–IX, and source 14 stresses that the standard exponentiation theorem is not the internal statement it uses. The notes of Parts VII and VIII cite `isg:cf:prop:family`, `isg:cf:thm:familytest`, `isg:cf:lem:noninteger`, `isg:cf:eq:explicitparameter`, `isg:cf:main:presburger` and `isg:cf:cor:unit`; all exist at the current state, and `docs/FORMALIZATION.md` records the results among them as pending. Nothing in Parts VI–IX is formalized, and placement beside these projects confers no formal status.

Part X. Source 16 read this report at its pin (source 05's delivered text), the batch-86 archives then in docs/incoming (now Parts III–V and source 03 of discrete-initial-subgroups-and-omnific-normalization), Logic/PresburgerArithmetic/README.md and Algebra/SurrealNumbers/README.md. Its descriptions are correct at the pin; the two READMEs have changed since only by the catalogue's counts and pointers. Its formalization route ([section 137](#)) proposes modules that do not exist; the Presburger project's Lean and Rocq files are about the ordinary integers. Nothing in Part X is formalized, and docs/FORMALIZATION.md maps none of its labels.

Parts XI and XII. (Batch 90.) Source 17 read this report at its pin (Parts I–V), the batch-87 and batch-89 archives then in docs/incoming (now Parts VI–X) and Logic/PeanoArithmetic/ListCoding, whose Lean and Coq modules prove standard-model theorems on sequence coding and exponentiation, not the internal ID_0 statements it uses; it says so, and the project's README is unchanged since the pin. Source 18 read this report at its pin, the batch-87 archives of sources 13 and 15, Logic/PresburgerArithmetic/README.md, the foundations and birthday-cutoffs-and-hereditary-sets reports and the Lean files SignSequence.lean and SignSequenceTopology.lean in Algebra/SurrealNumbers/Surreal/Foundations. The Presburger README, both Lean files and the foundations report are unchanged since its pin; this report and the birthday-cutoffs report have grown. Its descriptions are correct: the Presburger project's semantics are over the ordinary integers, and SignSequenceTopology.lean proves the small-subset clauses of the foundations report's found:thm:discrete, which [proposition 163.4](#) restates for sets. Nothing in Parts XI and XII is formalized, docs/FORMALIZATION.md maps none of their labels, and placement beside these projects confers no formal status.

Parts XIII–XV. (Batch 92.) Source 19 read Logic/PresburgerArithmetic/README.md and Algebra/SurrealNumbers/README.md; source 20 the root README, the surreal catalogue Algebra/SurrealNumbers/docs/README.md, this report's README and Surreal/Foundations/OmnificFloor.lean; source 21 the surreal README, Surreal/HahnSeries/CharacterTwist.lean and Surreal/HahnSeries/BinomialRoots.lean. Every file they read is unchanged between its pin and the placement, except this report's README; the READMEs have since gained the catalogue's pointers (cd514f61b). Their descriptions are correct. The Lean declarations they name exist and are formal: omnificFloor_spec and existsUnique_omnific_integerPart prove odg:thm:floor , characterTwist and twist_inverse belong to odg:def:lem:twist , and the root-uniqueness lemmas of BinomialRoots.lean are mapped in docs/FORMALIZATION.md; they concern surreals and full Hahn series, not the finite-support or left-finite statements of these parts. Nothing in Parts XIII–XV is formalized, docs/FORMALIZATION.md maps none of their labels, and placement beside these projects confers no formal status.

Parts XVI and XVII. (Batch 93.) Source 22 read this report's README, the surreal catalogue Algebra/SurrealNumbers/docs/README.md and the README of surcomplex/hahn-hilbert-geometry at its pin; source 23 read the root README and Algebra/SurrealNumbers/README.md on the moving default branch. Their descriptions are correct; between the pin and the placement the surreal catalogue changed only by counts and pointers, and this report's README by Parts XI–XV. Source 22 uses ordinary real Hilbert spaces, not the surreal Hilbert geometry of that report. Source 23's statement that no declaration of the repository verifies its theorems is true of its main theorem but incomplete for one consequence: the image clause of its rank-one corollary for $\Gamma = \mathbb{Z}$ and d/dt is

`range_derivative_eq_ker_residue` in `Surreal/Algebra/LaurentResidueChange.lean`, proved over every field of characteristic zero and mapped in `docs/FORMALIZATION.md` to the surcomplex analysis report ([remark 239.5](#)). Nothing else in Parts XVI and XVII is formalized, `docs/FORMALIZATION.md` maps none of their labels, and placement beside these projects confers no formal status.

Parts XVIII–XXI. (Batch 96.) Source 24 read the root README and `Algebra/SurrealNumbers/README.md` at 715a716a3; both are unchanged from there to the placement, and its descriptions of them are accurate. Its statement that no repository declaration verifies its new theorems is true. Two ingredients are formal in special forms: for Hahn series, $\text{ct}(Ef) = 0$ and $\ker E = \mathbb{R}$ (`Surreal.EulerDerivation.coeff_zero_euler`, `euler_eq_zero_iff`, mapped in `docs/FORMALIZATION.md` to the surcomplex differential-equations report), and for $\Gamma = \mathbb{Z}$ the integration criterion (`Surreal.LaurentResidue.exists_derivative_eq_iff`, characteristic zero; [remark 239.5](#)). Source 25’s [proposition 267.1](#) concerns the Lean predicate `Surreal.HahnSeries.PowerSeriesSummable` of `Surreal/HahnSeries/StrongEvaluation.lean`, which it read at its pin (unchanged at the placement) and which `docs/FORMALIZATION.md` maps, as proved, to `thm:exact` of the hahn-evaluation-at-omega report; the proposition’s real $\mathbb{N} \times \mathbb{N}$ instance with the product topology has no Lean statement. Source 26 read `Algebra/BakerCampbellHausdorff/Lean/BCH/Formal/Trunc.lean` at its pin (unchanged at the placement) as an interface, and describes it correctly; the formal BCH theorem `BCH.bch_formula_formal` supplies only the Lie-polynomial input of [proposition 285.3](#). Source 27 read the surreal and BCH READMEs; its affine BCH identity ([294.2](#)) is, in the 2×2 matrix algebra, the Lean theorem `BCH.exp_mul_exp_of_lie_eq_smul` (`Algebra/BakerCampbellHausdorff/Lean/BCH/Eigen.lean`; [remark 294.5](#)). Nothing else in Parts XVIII–XXI is formalized, `docs/FORMALIZATION.md` maps none of their labels, and placement beside these projects confers no formal status.

1.5 The status statements of the sources

The twenty-four manuscripts state their status as follows; these statements apply to the parts of this report taken from them.

Source 05: status and scope

The mathematical assertions designated as theorems are accompanied by proofs. The construction answers the cited question *as written*; independent bibliographic priority has not been established. Classical Presburger elimination and elementary topological-group facts are not claimed as new. This AI-assisted manuscript is not independently refereed or Lean-verified. The accompanying executable tests are finite checks, not a replacement for the proofs.

Source 04: from its abstract

The construction gives an affirmative answer to Question 2 as printed in Elliot Glazer’s 2023 paper *A Topological Tennenbaum Theorem*. The historical priority of this answer has not been established. The new results are supported here by conventional proofs and executable exact tests, not by a claimed Lean or Rocq kernel check.

Source 03: status and provenance

Status and provenance. Complete mathematical arguments are provided for the constructions and deductions, with classical inputs identified. ProveIt's existing coanalytic support lower bound and omnific floor formula are credited, not claimed as new. The main contribution of this report is the presentation dichotomy and its algebraic consequences, together with the exact finite-support cancellation analysis. Global novelty and independent correctness review remain open; no established famous conjecture or Lean verification is claimed.

Source 07: status

Status. The main constructions and their consequences are proved below. Classical model theory and standard descriptive set theory are identified as inputs. No new result in this report has been checked by Lean or Rocq, and no claim of publication priority or external peer review is made.

Source 06: from its abstract

The article connects these constructions to ProveIt's existing Presburger elimination development and its polynomial omnific subrings. All new assertions are proved in ordinary mathematics here; no machine-checked certification or unrestricted publication-priority claim is made.

Source 08: status

Status. The central results below have complete mathematical proofs, with classical inputs identified. Their bibliographic priority has not been established. This is an AI-assisted, unrefereed research manuscript, not a claim of an independently certified breakthrough or a Lean-checked proof. The included exact-arithmetic tests are supplementary checks only.

Source 09: status

Status. The central deductions are proved in this manuscript. Classical category theorems, Hahn and Puiseux constructions, real-closedness inputs, and previous ProveIt support and floor results are explicitly credited. Historical novelty has not been established by an exhaustive literature search. This is an unrefereed research contribution, not a claimed solution of a famous conjecture and not a Lean- or Rocq-verified development.

Source 12: status and scope

The results designated as theorems have mathematical proofs below. Historical priority has not been established. This is an AI-assisted, unrefereed manuscript, not an independently certified breakthrough and not a Lean- or Rocq-checked development. The accompanying exact-arithmetic checks are supplementary, not proofs of the topological theorems.

Source 10: status

Complete mathematical proofs are supplied for the main claims. The proposed contributions have not been independently refereed, and bibliographic priority has not been established. This is not a claim of a Lean- or Rocq-checked theorem. Classical inputs and prior project results are distinguished from the deductions made here.

Source 13: from its abstract

The contributions are conventional proofs, with explicit classical inputs and limited exact computational diagnostics. The earlier existence construction is credited, and no first-publication priority, refereeing, or new proof-assistant verification is claimed.

Source 14: status

Complete mathematical proofs of the stated contributions are supplied. The classical MRDP and coded descriptive-set-theoretic inputs are identified explicitly. The manuscript is not a proof-assistant formalization. Priority beyond the inspected literature and repository materials has not been established.

Source 15: from its abstract

We give full mathematical proofs, an exact comparison with the inspected ProveIt sources, and a research agenda. Historical priority and a machine-checked formalization of these extensions are not claimed.

Source 16: status (from its title page)

The article supplies mathematical proofs and explicitly credits the existing Presburger constructions, Hahn-group theory, and topological linear algebra. It resolves specified structural questions for this family of models; it does not claim a literature-certified breakthrough or a new solution of Glazer's already-addressed Question 2. The accompanying computations check finite rational examples, not the infinite theorems. No new proof-assistant formalization is asserted.

[write] Source 14's claimed answer to Glazer's Question 1 has not been independently reviewed ([Part VIII](#)).

[write] Source 16 did not see source 04's results on the Baire model, which it proves again for $\alpha = \omega$ ([Part X](#)).

Source 17: status (from its title page)

A proposed research extension with proofs and independent in-session mathematical review. The manuscript is not a claim of established publication priority, external peer review, or a completed proof-assistant formalization.

Source 18: from its abstract

The unrestricted higher-dimensional manifold question remains open in this work. Classical cardinal inequalities and earlier Presburger examples are credited. The contribution consists of conventional proofs, sharpness examples, and a scoped repository comparison. Historical priority and proof-assistant verification are not claimed.

[write] Source 17's ATR_0 bound for its local theorem is a claimed sketch, not independently reviewed ([Part XI](#)).

Source 19: scope and status (from its title page)

This is a proof-based continuation, not a new claim to the earlier affirmative construction for Glazer's Question 2. It settles the one-real-dimensional order-classification problem stated in the companion report. Historical priority of the new formulations is not established. The manuscript is unrefereed and not kernel-verified; finite exact tests supplement, but do not certify, its proofs.

Source 20: status (from its title page)

This is an unrefereed research manuscript with full mathematical proofs and executable finite diagnostics. The proposed extensions are distinguished from Glazer's method and from constructions already in ProveIt. No kernel verification or independently established priority is claimed.

Source 21: status and scope (from its title page)

Complete mathematical arguments are supplied for the stated results, with classical inputs identified. Historical priority has not been established. This is an AI-assisted, unrefereed manuscript, not an independently certified breakthrough or a Lean-checked development. The literature correction is explicitly version-specific. The accompanying 2,931 exact finite checks supplement, but do not verify, the infinite proofs.

[write] Source 19's "companion report" is source 08 ([Part IV](#)); the order-classification problem is [question 57.3](#), which source 19 settles for $d = 1$. The propositions of [section 182](#) and [remark 204.2](#) were written for this report.

Source 22: status (from its title page)

A self-contained research draft with conventional proofs and supplemental exact finite tests. It is not independently refereed or proof-assistant verified. The precise classification and its synthesis with oscillation ranks are developed here; priority over all earlier literature is not certified. No participation or endorsement by Elliot Glazer is implied.

Source 23: status and attribution (from its title page)

Complete mathematical proofs are supplied for the stated results. This is an AI-assisted, unrefereed manuscript, not an independently certified priority claim or a Lean-checked development. The exact novelty of the classification remains to be established by specialist review. Neither Elliot Glazer nor the ProveIt maintainer is represented as an author, collaborator, or endorser. No solution of a named remaining open problem in Glazer's paper is claimed.

[write] Source 22's transfinite flag is Part III's cone recursion, which it did not see ([proposition 224.1](#)); [section 224](#), [section 239](#) and [question 222.10](#) were written for this report. Source 23's "Lean-checked" disclaimer is accurate for its theorems; one special case is formalized independently ([remark 239.5](#)).

Source 24: status and attribution (from its title page)

This is an AI-assisted, unrefereed research manuscript with complete arguments for its stated theorems. Classical normal-form and flow mechanisms are explicitly credited. The exact historical novelty of the assembled classification and arithmetic threshold has not been independently established. Neither Elliot Glazer nor the ProveIt maintainer is represented as an author, collaborator, or endorser. The accompanying rational jet checks are not a proof-assistant formalization.

Source 25: status (from its title page)

Complete proofs are supplied for the stated deductions, with classical inputs and prior results identified. The report does not claim historical priority, independent peer review, a solution to a famous published conjecture, or Lean verification of its new arguments. It develops a concrete research direction connecting Glazer's topological arithmetic with ProveIt's Hahn-series and surreal-number program.

Source 26: status and attribution (from its title page)

Full mathematical proofs are supplied, together with exact finite algebraic checks. This is an AI-assisted, unrefereed research manuscript, not a Lean-checked development or an independently certified priority claim. Neither Elliot Glazer nor the repository maintainer is represented as a coauthor or endorser. No remaining numbered problem in Glazer's own paper is claimed to be solved.

Source 27: status (from its title page)

Proofs are supplied for the stated theorems. This is an AI-assisted, unrefereed manuscript, not a certified priority claim or a Lean-checked development. Classical one-dimensional Lie theory is credited explicitly. The new contribution proposed here is the scalar-coordinate and simultaneous-classification package, together with its precise Borel and ramification statements. Neither Elliot Glazer nor the ProveIt maintainer is represented as an author or endorser.

[write] (Batch 96.) Source 24's description of Part XVII as an unpublished draft is stale (note in [appendix AM](#)). Source 25's "Lean verification" disclaimer is accurate; its [proposition 267.1](#) is a statement about the Lean predicate `PowerSeriesSummable`, not a Lean theorem (note there). Source 26's malformed reference to its time-one theorem is restored (introduction of [Part XX](#)). Source 27's "Lean-checked" disclaimer is accurate for its theorems; its affine BCH identity is, in matrix form, a Lean theorem of the BCH project ([remark 294.5](#)). [Sections 255, 270, 285 and 300](#), [question 268.12](#) and [question 283.10](#) were written for this report.

Every limitation and non-claim of the twenty-four sources is also listed, source by source, in [appendix F](#).

Part I

Polish Presburger models

This part answers Glazer’s Question 2 as printed. Its base text is source 05; source 04’s further results and second proofs are marked [04]. Repository pins: 05 inspected commit `a11efab09`, 04 commit `3ad5f878c`; the files they name are unchanged at the placement.

2 The research overlap and the precise target

The relevant intersection is *topological and effective model theory of weak arithmetic*. Glazer’s work supplies a sharply formulated question; ProveIt supplies two concrete mathematical interfaces: an executable Cooper quantifier-elimination development and a normal-form construction of the omnific integers. This makes it possible to connect an external Polish topology, internal induction, and exact surreal coordinates without appealing to the repository’s unreviewed research claims.

In the final section of [1], Glazer asks:

Does some uncountable Polish space support a model of Presburger arithmetic with continuous addition?

The source is Question 2 on page 8 of `arXiv:2311.13699v1`. We use precisely that formulation. The topology need not be the arithmetic order topology, and no multiplication is required. These two distinctions are essential.

Glazer’s obstruction for stronger arithmetic includes the impossibility of an uncountable Polish model of $\mathbb{Q} + I\text{Open}$ with continuous addition and Borel multiplication [1, Corollary 1]. The earlier study of Enayat, Hamkins, and Wcisło [2] motivates the broader problem of finding topological presentations of arithmetic. Our construction does not contradict those results: it supports the additive language, and its additive structure prevents a semiring expansion.

2.1 Repository interfaces actually inspected

The repository snapshot used here is

`a11efab09a2d08f692a09829cb6d3440ddd1795a`.

The Presburger README [14] describes a complete Lean integer quantifier eliminator and an independent Rocq/Coq proof of the normalized one-variable step. The inspected file `PresburgerArithmetic/Cooper.lean` [15] states `periodic_interval` and `periodic_has_residue` for predicates on the ordinary type `Int`. Their hypotheses are entirely appropriate there. They must not be transferred unchanged to a nonstandard ordered group; section 4.2 gives an explicit counterexample and the correct replacement.

The surreal README [16] records an omnific ring whose normal forms have nonnegative exponents and integer constant term. Only this normal-form interface and the standard arithmetic of its additive group are needed below. We do not assume that the present topology or any of this article’s theorems has already been formalized in that repository.

2.2 What is established, and what is not

The principal result is an explicit affirmative construction, rather than a claim that a difficult negative theorem has been overturned. The lexicographic algebra is classical. The contribution developed here is the precise topology and its application to the stated question, together with an effective Baire presentation and a collection of sharp structural consequences.

The literature search verified the question in the source PDF and checked the arXiv record; it did not establish the absence of an earlier solution. Consequently, “an affirmative answer to the stated question” is the appropriate conclusion, not an independently certified first solution. No answer to Glazer’s separate reverse-mathematical Question 1 is claimed.

All constructions and proofs take place in ordinary ZFC with set-sized carriers. The notation **Oz** is used only to identify where explicit set-sized images lie. No topology on an unrestricted proper class, and no set of all surreal numbers, is used.

2.3 Source 04’s framing

Source 04 opens with the same question and the same two repository connections (its Section 1). It adds three points, in its own words; “this article” and “this package” there mean source 04.

The interpretation of the signature matters. Throughout this article, Presburger arithmetic is

$$\text{Pr}_{\mathbb{N}} = \text{Th}(\mathbb{N}; 0, 1, +),$$

with its definable order, and with no multiplication of two variables. Continuity of addition does *not* mean continuity of every definable function. Indeed, the difference between these requirements is central to our result. Adding order and fixed congruence predicates to the language does not change which first-order sets are definable, but the predicates need not be open or closed in the topology.

The characterization of Presburger groups as discretely ordered abelian groups with divisible quotient by the standard integers, and their quantifier elimination after naming congruences, are classical. A convenient modern reference is Jeřábek’s preliminaries [4, Section 2]. We include proofs of the facts needed here so the construction can be audited independently of that characterization.

The specific contribution is the topological application to the printed question, strengthened by an effective model on literal Baire space and an analysis of definable operations and self-embeddings. The finite-residue argument also identifies an exact hypothesis needed when transporting a standard-integer proof to arbitrary models. It is not a claim that the repository’s integer proof is defective.

This is an unrefereed research manuscript. The statements called theorems below have proofs in the manuscript; the Python program is an independent finite diagnostic and a usable stream implementation. No new theorem in this package is represented as machine-checked. The literature check verified the wording of Glazer’s question and the relevant background; it did not establish that the affirmative construction has never appeared elsewhere. Thus an answer to the specified question is claimed at the mathematical level, but first-publication priority is not claimed.

3 The construction in one theorem

We include 0 in $\mathbb{N}$. For an ordered abelian group D , let

$$G_D = D \overrightarrow{\times} \mathbb{Z}$$

mean the direct-product additive group with the D coordinate more significant in the order. Thus

$$(d, n) < (e, m) \iff d < e \text{ or } (d = e \text{ and } n < m). \quad (3.1)$$

Write $\mathbf{1} = (0, 1)$ and let

$$M_D = (G_D)_{\geq 0} = (D_{>0} \times \mathbb{Z}) \cup (\{0\} \times \mathbb{N}). \quad (3.2)$$

The ambient product topology, whenever specified, is separate from (3.2)'s arithmetic order.

Theorem 3.1 (Polish and effective Presburger models). *The following assertions hold.*

- (1) *With $D = \mathbb{R}$ in its usual topology, $M_{\mathbb{R}}$ is an uncountable, locally compact Polish space without isolated points. Coordinatewise addition is continuous, and*

$$(M_{\mathbb{R}}; 0, \mathbf{1}, +, <) \models \text{Th}(\mathbb{N}; 0, 1, +, <).$$

- (2) *With $D = \mathbb{Q}^{\mathbb{N}}$, ordered by the first differing coordinate and with each rational coordinate discrete, the space $M_{\mathbb{B}} = M_D$ is homeomorphic to $\mathbb{N}^{\mathbb{N}}$. There is an explicit such homeomorphism under which addition, successor, and division by each positive standard integer are Type-2 computable.*
- (3) *In both models, every first-order definable relation, with parameters, is Δ_2^0 . The order is neither open nor closed. In the Baire presentation, each formula has a computable bound on the number of changes in a convergent truth approximation, relative to its parameter codes.*
- (4) *Both models embed as ordered additive submonoids of $\mathbf{Oz}$, fixing 0 and 1. Neither admits any unital semiring multiplication with its given addition, zero, and unit.*

The proofs occupy [sections 4 to 7, 9 and 10](#). The first assertion alone answers the source question. The later assertions explain how much regularity is possible and where it necessarily stops.

For orientation, an element $(r, -17)$ with $r > 0$ is a *positive* element of $M_{\mathbb{R}}$. Indeed it exceeds every standard numeral $(0, k)$. Nevertheless,

$$(1/j, 0) \longrightarrow (0, 0) \quad (j \longrightarrow \infty). \quad (3.3)$$

in the chosen topology. Arithmetic size and topological proximity are different notions. The construction would fail if one silently demanded that this topology be the order topology.

Source 04's statement [04]. Source 04 (its Section 1.2) sets up the same real model:

Let $\mathbb{Z}$ have the discrete topology and put

$$M_{\mathbb{R}} = (\mathbb{R}_{>0} \times \mathbb{Z}) \cup (\{0\} \times \mathbb{N}). \quad (3.4)$$

Set $0 = (0, 0)$, $1 = (0, 1)$, and

$$(r, z) + (s, w) = (r + s, z + w).$$

Order the ambient group $\mathbb{R} \times \mathbb{Z}$ lexicographically, with the real coordinate dominant. Then $M_{\mathbb{R}}$ is exactly its nonnegative cone.

Remark 3.2 (Source 04's statement of [theorem 3.1](#) [04]). The structure $(M_{\mathbb{R}}; 0, 1, +)$ is a model of $\text{Pr}_{\mathbb{N}}$. With the topology inherited from $\mathbb{R} \times \mathbb{Z}_{\text{discrete}}$, its universe is an uncountable perfect locally compact Polish space, and addition is jointly continuous. There is also a model of $\text{Pr}_{\mathbb{N}}$ on $\mathbb{N}^{\mathbb{N}}$ with jointly continuous addition. In both models, quotient and remainder by each fixed positive standard integer are continuous, and every first-order definable relation with parameters is Δ_2^0 .

The algebraic assertion is proved in [section 4](#) and [section 4.3](#); the topology and the explicit Baire-space realization are proved in [sections 5.2, 5.4](#) and [6.3](#). The proof uses familiar $\mathbb{Z}$ -groups, not an alteration of what Presburger arithmetic means.

[merge] The two statements agree. Source 04's adds continuity of quotient and remainder by each fixed standard integer, which it proves for every admissible coefficient group ([theorem 5.4](#)) and which 05 proves for its two models ([proposition 8.1](#)); source 05's adds the optimal order rank, the finite-mind-change truth approximation and the omnific realization. Both prove the no-semiring and self-embedding results.

4 Full Presburger arithmetic, not merely a few axioms

4.1 Discretely ordered groups with standard residues

Definition 4.1. A $\mathbb{Z}$ -group in this article is a linearly ordered abelian group G with a distinguished least positive element $\mathbf{1}$, such that for every positive standard integer m and every $x \in G$ there exist $q \in G$ and $r \in \{0, \dots, m-1\}$ satisfying

$$x = mq + r\mathbf{1}. \quad (4.1)$$

Here mq denotes repeated addition by a *standard* integer, not a new multiplication operation on G .

Ordered abelian groups are torsion-free. The quotient and remainder in (4.1) are unique: if $m(q - q') = (r' - r)\mathbf{1}$ and $q > q'$, then $q - q' \geq \mathbf{1}$, whereas $r' - r < m$; the other inequality is similar. We suppress $\mathbf{1}$ in standard integer constants when no confusion is possible.

Proposition 4.2 (The split lexicographic group). *If D is a divisible ordered abelian group, then $D \overrightarrow{\times} \mathbb{Z}$ is a $\mathbb{Z}$ -group. Its standard division algorithm is*

$$(d, n) = m(d/m, \lfloor n/m \rfloor) + (0, n \bmod m). \quad (4.2)$$

If $(d, n) \geq 0$, the displayed quotient is nonnegative.

Proof. Coordinatewise addition preserves the lexicographic order. A positive (d, n) either has $d > 0$, in which case it exceeds $(0, 1)$, or has $d = 0$ and $n \geq 1$. Thus $(0, 1)$ is the least positive element. Divisibility and torsion-freeness of D give the unique element d/m . Ordinary Euclidean division of n now proves (4.2). If $d > 0$, then $d/m > 0$, regardless of the integer quotient. If $d = 0$ and $n \geq 0$, the integer quotient is nonnegative. $\square$

Source 04 proves the same statement (its Lemma 2.2) with the same explicit quotient and the same nonnegativity clause; its uniqueness argument compares coordinates instead of using least positivity.

Second proof (source 04). If $(d, z) > 0$ and $d > 0$, then $(d, z) > (0, 1)$. If $d = 0$, positivity says $z \geq 1$. This proves discreteness. Existence in (4.1) follows from divisibility of D and ordinary integer Euclidean division, including for negative z .

For uniqueness, equality of $mq + r\mathbf{1}$ and $mq' + r'\mathbf{1}$ first gives equality of the D -coordinates of q, q' . Equality of the integer coordinates then gives ordinary uniqueness of quotient and remainder. Finally, if $d > 0$, then $d/m > 0$. If $d = 0$ and $z \geq 0$, then $\lfloor z/m \rfloor \geq 0$. $\square$

Corollary 4.3 ([04]). For $m \geq 1$,

$$mG_D = D \times m\mathbb{Z}, \quad (d, z) \equiv (e, w) \pmod{mG_D} \iff z \equiv w \pmod{m}.$$

Moreover,

$$\bigcap_{m \geq 1} mG_D = D \times \{0\}. \quad (4.3)$$

Proof. Every D -coordinate is an m -multiple. The only ordinary integer divisible by every positive integer is zero. $\square$

The standard subgroup $\mathbb{Z}\mathbf{1}$ is convex, and $G_D/\mathbb{Z}\mathbf{1}$ is the ordered divisible group D . This gives the usual quick recognition of G_D as a Presburger group. For the precise model-theoretic conclusion, we next give a direct elimination argument.

The implication from these axioms to *full* Presburger arithmetic is important enough to prove. We use the classical elimination architecture of Presburger and Cooper; [3] supplies historical and algorithmic context. The proof below makes explicit which algebraic hypotheses are needed in a nonstandard group.

4.2 Coset invariance versus standard-shift periodicity

For $m \geq 1$, write $mG = \{mh : h \in G\}$. A predicate P is *mG -invariant* if

$$y - z \in mG \implies (P(y) \leftrightarrow P(z)). \quad (4.4)$$

This is stronger than invariance under the single standard shift $m\mathbf{1}$.

Example 4.4 (A false generalization of an integer lemma). In $G = \mathbb{R} \overrightarrow{\times} \mathbb{Z}$, let $P(d, n)$ mean $d > 0$. For every positive standard m ,

$$P(x + m\mathbf{1}) \leftrightarrow P(x)$$

holds. Yet P has witnesses and is false at every standard residue $0, 1, \dots, m-1$. Thus

$$(\exists x P(x)) \not\iff \bigvee_{0 \leq r < m} P(r\mathbf{1}).$$

It also fails the naive finite-interval reduction with lower bound 0 and upper bound $(1, 0)$. No flaw in the repository's Int theorem is involved: its domain is the ordinary integers, where every multiple is obtained by a standard finite iteration.

The general counterexample [04]. Source 04 phrases coset invariance as a definition and gives the counterexample for every nonzero divisible D , with the predicate $d \neq 0$ in place of $d > 0$. In its text G is a linearly ordered abelian group with a least positive element $\mathbf{1}$ and unique division by positive standard integers as in (4.1), that is, a $\mathbb{Z}$ -group.

Definition 4.5 ([04]). A predicate $P \subseteq G$ is *m-residue invariant* if

$$u - v \in mG \implies (P(u) \iff P(v)).$$

No definability hypothesis is imposed in this definition.

The condition is stronger than invariance under the single translation $x \mapsto x + m\mathbf{1}$. This distinction is invisible over the ordinary group $\mathbb{Z}$, but is essential over nonstandard groups.

Proposition 4.6 (One-step periodicity is insufficient [04]). *If $D \neq 0$, the external predicate*

$$P(d, z) \iff d \neq 0$$

on G_D satisfies $P(x + m\mathbf{1}) \leftrightarrow P(x)$ for every standard $m \geq 1$. Nevertheless P is nonempty and is false at every standard residue $0, 1, \dots, m - 1$.

Proof. A standard integer translation does not change the D -coordinate. A nonzero D -coordinate exists, whereas every standard integer has D -coordinate zero. In particular P is not m -residue invariant: $(d, 0) - (0, 0) \in mG_D$ for every d . $\square$

Lemma 4.7 (The coset-invariant finite search). *Let G be a $\mathbb{Z}$ -group, $m \geq 1$ standard, and P an mG -invariant predicate. For $L, U \in G$,*

$$\begin{aligned} \exists y (L \leq y \leq U \wedge P(y)) \\ \iff \bigvee_{0 \leq r < m} (L + r \leq U \wedge P(L + r)). \end{aligned} \quad (4.5)$$

With only an upper bound, candidates $U - r$ suffice; with only a lower bound, candidates $L + r$ suffice. With neither bound, the candidates are $0, \dots, m - 1$.

Proof. Given a witness $y \geq L$, write $y - L = mh + r$ with $0 \leq r < m$. One has $h \geq 0$: otherwise $h \leq -1$, and then $mh + r \leq -m + (m - 1) = -1$, contradicting $y - L \geq 0$. Consequently

$$L \leq L + r \leq y \leq U.$$

The difference $y - (L + r) = mh$ lies in mG , so $P(L + r)$ holds. Conversely every displayed candidate is already a witness. The same argument applied to $U - y$ proves the upper-bound version. Without bounds, use the standard remainder of y itself. $\square$

Congruence predicates have exactly the stronger invariance (4.4). If M is a common multiple of finitely many standard moduli, any Boolean combination of congruences with those moduli is MG -invariant. This remains true with arbitrary nonstandard parameters: adding Mh to a variable changes an affine congruence term by an element of each relevant modulus subgroup.

4.3 Second route [04]: finite witnesses at a boundary

Source 04's form of [lemma 4.7](#) works with strict bounds and an explicit finite candidate set. It is the same theorem with a different normalization of the bounds, and it is the step that 04 uses in its elimination proof below. Here G is a $\mathbb{Z}$ -group, and a standard integer r also denotes $r\mathbf{1}$.

Theorem 4.8 (Finite boundary criterion [04]). *Let A, B be finite lists of elements of G , let $m \geq 1$, and let P be m -residue invariant. Put*

$$\Phi(x) := P(x) \wedge \bigwedge_{a \in A} (a < x) \wedge \bigwedge_{b \in B} (x < b).$$

Define a finite candidate set C by

$$C = \begin{cases} \{a_* + 1, \dots, a_* + m\}, & A \neq \emptyset, \quad a_* = \max A, \\ \{b_* - 1, \dots, b_* - m\}, & A = \emptyset, B \neq \emptyset, \quad b_* = \min B, \\ \{0, \dots, m - 1\}, & A = B = \emptyset. \end{cases} \quad (4.6)$$

Then

$$(\exists x \in G) \Phi(x) \iff \bigvee_{c \in C} \Phi(c). \quad (4.7)$$

Proof. The reverse direction is immediate. Suppose first that $A \neq \emptyset$ and x is a witness. Divide $x - a_* - 1$ by m to obtain

$$x - a_* = mq + s, \quad 1 \leq s \leq m.$$

Here q is an element of G , not necessarily a standard integer. If $q < 0$, discreteness gives $q \leq -1$, whence $mq + s \leq 0$, contradicting $x > a_*$. Thus $q \geq 0$ and $c = a_* + s \leq x$. This c is larger than every lower bound and smaller than every upper bound because $c \leq x < b$. Also $x - c = mq \in mG$, so $P(c)$ holds.

If A is empty and B is nonempty, divide $b_* - x - 1$ instead. The result is $b_* - x = mq + s$ with $q \geq 0$ and $1 \leq s \leq m$. The candidate $c = b_* - s$ satisfies $x \leq c < b_* \leq b$ for every upper bound and is in the same residue class as x . Finally, when both lists are empty, take the unique standard representative of x modulo mG . $\square$

This proof uses no induction through a nonstandard interval. Its only finite searches have standard lengths fixed by the formula being eliminated. Weak bounds can be converted to strict ones using discreteness: $a \leq x$ is $a - 1 < x$, and $x \leq b$ is $x < b + 1$.

4.4 A self-contained elimination argument

Expand the ordered-group language by predicates

$$D_m(t) \iff t \in mG \quad (m \geq 2),$$

and regard D_1 as always true. All of these are definable by existential formulas in the group language.

Theorem 4.9 (Classical Presburger elimination over $\mathbb{Z}$ -groups). *In the expanded language*

$$\{0, \mathbf{1}, +, -, \leq, (D_m)_{m \geq 2}\},$$

the axioms of $\mathbb{Z}$ -groups eliminate quantifiers and form a complete theory. In particular every $\mathbb{Z}$ -group is elementarily equivalent to $(\mathbb{Z}; 0, \mathbf{1}, +, -, \leq, (m \mid \cdot)_m)$.

Proof. It suffices to eliminate one existential quantifier from a quantifier-free formula. Rewrite the formula in disjunctive normal form. Negated inequalities are replaced using discreteness, for example $\neg(s \leq t) \leftrightarrow t + 1 \leq s$. An equality is the conjunction of two inequalities; a disequality is a disjunction of strict inequalities. Finally, standard residue exhaustiveness gives

$$\neg D_m(t) \longleftrightarrow \bigvee_{r=1}^{m-1} D_m(t - r).$$

Thus it is enough to eliminate $\exists x$ from a conjunction of affine inequalities and positive divisibility atoms, along with conjuncts not involving x .

Let A be a positive common multiple of every nonzero absolute coefficient of x occurring in that conjunction; take $A = 1$ if there are none. Set $y = Ax$ and add the condition $D_A(y)$. An atom

$$ax + t \leq 0, \quad a \neq 0,$$

can be multiplied by the positive standard integer $k = A/|a|$, giving

$$\text{sgn}(a)y + kt \leq 0.$$

It is now an upper or lower bound on y . For a congruence, torsion-free cancellation gives the equivalence

$$D_m(ax + t) \iff D_{km}(\text{sgn}(a)y + kt). \quad (4.8)$$

Indeed $kz \in kmG$ if and only if $z \in mG$: one direction multiplies a witness, and the other cancels k from $kz = kmh$. The additional atom $D_A(y)$ is exactly what ensures that every candidate y comes from a unique x .

After normalizing signs, we have finitely many lower bounds $L_i \leq y$, finitely many upper bounds $y \leq U_j$, and a conjunction $P(y)$ of congruence conditions. Let M be a positive common multiple of all their moduli, including A . Then P is MG -invariant.

If lower bounds exist, choose their maximum L . By [lemma 4.7](#), a witness exists exactly when one of $L, L + 1, \dots, L + M - 1$ satisfies all the normalized conjuncts. The maximum need not be introduced as a new function symbol: take a finite disjunction over the indices i , require $L_h \leq L_i$ for every h , and substitute $L_i + r$ for y . This produces a quantifier-free formula in the parameters. If there are upper bounds but no lower bounds, choose a minimal U_j and substitute $U_j - r$. If no bounds exist, substitute the standard residues r . Conjuncts independent of x are retained throughout.

This eliminates one existential quantifier. Induction on formula complexity, replacing universal quantification by negated existential quantification, eliminates all quantifiers.

A quantifier-free sentence uses only standard integer multiples of **1**. Their order and their residues are the same in every $\mathbb{Z}$ -group, by least positivity and uniqueness of standard remainders. Therefore all $\mathbb{Z}$ -groups satisfy the same sentences. The ordinary integers are one such group, proving completeness and the final assertion. $\square$

4.5 Second route [04]: normalization and elimination

Source 04 proves [theorem 4.9](#) by a differently organized argument: it keeps negated congruences instead of rewriting them through residues, substitutes away equalities in the eliminated variable, and uses strict bounds with [theorem 4.8](#). Its statement is uniform over all groups G of [section 4.3](#).

Use the group language

$$L_{Pr} = \{0, 1, +, -, <, (\equiv_m)_{m \geq 2}\},$$

where $u \equiv_m v$ means $u - v \in mG$. Affine terms have integer coefficients. The congruence predicates are definable by existential quantification in the underlying additive language.

Theorem 4.10 (Quantifier elimination, with a uniform proof [04]). *Every L_{Pr} formula is equivalent, uniformly in all groups G satisfying the hypotheses above, to a quantifier-free Boolean combination of affine inequalities, affine equalities, and congruences. Consequently every such G is elementarily equivalent to $(\mathbb{Z}; 0, 1, +, -, <)$ in the reduct language.*

Proof. It is enough to eliminate one existential quantifier from a conjunction of literals, after writing the quantifier-free part in disjunctive normal form. Negated equality is split into two strict inequalities, and negated inequalities are converted by discreteness. Negated congruences can be kept as negated congruences.

Let x be the variable to eliminate. If it occurs with no nonzero coefficient, the quantifier is vacuous. Otherwise let L be the least common multiple of the positive absolute values of its nonzero coefficients. Introduce $y = Lx$ and impose $y \in LG$. There is a unique x for every such y because G is torsion-free.

For an atom with term $ax + t$, $a \neq 0$, put $B = L/|a|$. Multiplying an inequality or equality by the positive standard integer B replaces its x -coefficient by $\text{sgn}(a)$ on y . For divisibility use

$$ax + t \in nG \iff \text{sgn}(a)y + Bt \in nBG. \quad (4.9)$$

Indeed the forward implication multiplies a witness by B , and the reverse implication cancels B in the torsion-free group. Atoms with zero x -coefficient remain independent conditions.

The resulting conjunction has only coefficient 1 or -1 on y in its inequalities, equalities, and congruences. An equality involving y permits substitution. In the absence of such an equality, the inequalities form finite strict lower and upper lists. The remaining congruences and negated congruences form a predicate $P(y)$ invariant modulo mG , where m is the least common multiple of their moduli, including L . For example, if $n \mid m$, replacing y by $y + mg$ cannot change membership of $\pm y + t$ in nG .

Apply [theorem 4.8](#). The maximum or minimum appearing there is not an extra function symbol: take a finite disjunction over choices of a largest lower bound, or a smallest upper bound, and add the finitely many inequalities that express that choice. Substitution of the finite candidates leaves a quantifier-free formula in the remaining variables. Negation handles universal quantifiers, so induction on the external formula tree proves elimination.

Finally a closed quantifier-free formula has the same truth value in every such G as in $\mathbb{Z}$. Its terms are standard numerals. Their equality and order are fixed, and their congruences have the usual residues by division. Eliminating all quantifiers therefore proves elementary equivalence. $\square$

Corollary 4.11 (The entire induction scheme). *For every divisible ordered abelian group D ,*

$$(M_D; 0, 1, +, <) \models \text{Th}(\mathbb{N}; 0, 1, +, <).$$

In particular it satisfies induction for every first-order formula of this additive language, with arbitrary parameters from M_D .

Proof. By [proposition 4.2](#) and [theorem 4.9](#), $G_D \equiv \mathbb{Z}$. The nonnegative cone is uniformly definable by $x \geq 0$. Relativizing every quantifier to that cone translates any sentence about

M_D into a sentence about G_D . The same translation in $\mathbb{Z}$ gives the ordinary nonnegative integers. Each universally closed induction instance is therefore true in M_D . $\square$

Proposition 4.12 (Definable least elements). *Every nonempty definable subset of M_D , allowing parameters, has a least element.*

Proof. Eliminate quantifiers and take a finite disjunction of conjunctions as in [theorem 4.9](#). Include $x \geq 0$ explicitly. For each conjunction, the substitution $y = Ax$ introduces a lower bound $y \geq 0$. Hence a nonempty conjunction has a least solution among the finitely many valid candidates at its maximum lower bound; the residue argument shows that one of these candidates lies below every witness. Dividing by A is order-preserving. Finally take the minimum of the least solutions of the nonempty disjuncts. $\square$

This also gives a direct induction check after elimination: a nonempty set of counterexamples to an induction instance would have a least member. It cannot be 0, and its predecessor would contradict the induction step. The external standard part $\{(0, n) : n \in \mathbb{N}\}$ is a proper inductive subset when $D \neq 0$, but is not definable, even with parameters. Induction quantifies over definable sets, not over every external subset.

Source 04 proves [corollary 4.11](#) (its Corollary 3.5, for $\text{Pr}_{\mathbb{N}}$) by the same relativization, and adds [04]:

The same argument shows that every definable subset of M_D^k is a Boolean combination of restrictions of group affine inequalities, equalities, and standard congruences. This observation will control its Borel complexity without an induction over projections of Borel sets.

5 Polishness and a locally compact example

A Polish space is a separable space whose topology is induced by some complete metric. A particular convenient metric may be incomplete without the space ceasing to be Polish. This distinction matters for open rays.

5.1 A useful exact criterion

We first record the standard subspace criterion with a proof, so that the completeness issue is visible.

Lemma 5.1 (Polish subspaces). *A subspace A of a Polish space X is Polish if and only if A is a G_δ subset of X .*

Proof. Fix a compatible complete metric d on X . If $A = \bigcap_j U_j$ with U_j open, consider on A the continuous functions $f_j(x) = 1/d(x, X \setminus U_j)$; when $U_j = X$, set $f_j = 0$. The graph of $x \mapsto (f_j(x))_j$ is closed in $X \times \mathbb{R}^{\mathbb{N}}$. Indeed a limit outside some U_j would force f_j to tend to infinity, not to a real number. Projection identifies this closed graph homeomorphically with A . Countable products and closed subspaces of complete separable metric spaces are complete and separable.

Conversely, let δ be a complete metric inducing the subspace topology on A . For each n , let W_n be the union of all ambient open sets U that meet A and satisfy

$$\text{diam}_d(U) < 2^{-n}, \quad \text{diam}_\delta(U \cap A) < 2^{-n}.$$

These unions cover A . If $x \in \bigcap_n W_n$, choose $x \in U_n$ and $a_n \in U_n \cap A$. Then $a_n \rightarrow x$ in d . For each fixed n , all sufficiently late a_k belong to U_n , so (a_k) is δ -Cauchy. Its δ -limit lies in A , and agreement of the two topologies identifies that limit with x . Thus $A = \bigcap_n W_n$. $\square$

Theorem 5.2 (The positive-cone criterion). *Let D be a divisible ordered abelian group equipped with a Polish topological-group topology, not necessarily its order topology. Give $\mathbb{Z}$ the discrete topology. Then the subspace $M_D \subseteq D \times \mathbb{Z}$ is Polish if and only if $D_{>0}$ is G_δ in D . When this holds, $D_{>0}$ is in fact Δ_2^0 , and addition on M_D is continuous.*

Proof. If $D_{>0}$ is G_δ , so is its union with the closed singleton $\{0\}$. On each clopen integer slice, the carrier (3.2) is either $D_{>0}$ or $D_{\geq 0}$. A countable topological sum of G_δ subspaces is G_δ in the corresponding sum: its complement is a countable union of F_σ subsets of the clopen slices. Apply lemma 5.1.

Conversely, the slice with integer coordinate -1 is closed in M_D and homeomorphic to $D_{>0}$. If M_D is Polish, so is that slice, and lemma 5.1 makes $D_{>0}$ a G_δ subset of D . Negation is a homeomorphism on D , so $D_{<0}$ is G_δ as well. The complement of $D_{>0}$ is $D_{<0} \cup \{0\}$, also G_δ . Thus $D_{>0}$ is both F_σ and G_δ .

Finally the cone is closed under coordinatewise addition. Restricting continuous addition on $D \times \mathbb{Z}$ to a subspace with its image in that subspace gives a continuous map $M_D^2 \rightarrow M_D$. $\square$

5.2 Admissible coefficient groups and continuous division [04]

Source 04 proves a sufficient version of theorem 5.2 under a stronger hypothesis, and obtains in return the continuity of division by standard integers for every such D ; source 05 proves that continuity only for its two explicit models (proposition 8.1). The Polish clause of theorem 5.4 is the case of theorem 5.2 in which $D_{>0}$ is open.

Definition 5.3 ([04]). An *admissible topological divisible group* is a divisible linearly ordered abelian group D with a Polish group topology such that $D_{>0}$ is open and each map $d \mapsto d/m$, $m \geq 1$, is continuous. The specified topology is additional data; it is not silently identified with the order topology of G_D or M_D .

Theorem 5.4 (Transfer [04]). *Let D be an admissible topological divisible group, and give G_D the product topology with discrete $\mathbb{Z}$. Then M_D , with its subspace topology, is Polish. Addition on M_D is jointly continuous. The functions*

$$Q_m(d, z) = (d/m, \lfloor z/m \rfloor), \quad R_m(d, z) = (0, z \bmod m)$$

are continuous maps from M_D to itself for each fixed $m \geq 1$.

Proof. The ambient group is Polish. Its complement of M_D is

$$(D_{<0} \times \mathbb{Z}) \cup (\{0\} \times \mathbb{Z}_{<0}).$$

The first set is open and the second is closed. In a metrizable space every open set is F_σ , so this complement is F_σ and M_D is G_δ . A G_δ subspace of a Polish space is Polish. Equivalently, the integer-coordinate slices are a countable disjoint sum of copies of the Polish spaces $D_{\geq 0}$ and $D_{>0}$.

The set M_D is closed under addition; restriction of the continuous ambient addition is therefore continuous as a map into the subspace. The displayed formulas for Q_m, R_m are continuous in the ambient coordinates: division on D is continuous, and every function on the discrete coordinate is continuous. Their images belong to M_D by proposition 4.2. $\square$

5.3 The real-coordinate model, including a complete metric

For $D = \mathbb{R}$, the positive cone is open. More explicitly, the slice

$$X_n = \{r : (r, n) \in M_{\mathbb{R}}\}$$

is $[0, \infty)$ when $n \geq 0$ and $(0, \infty)$ when $n < 0$. Hence

$$M_{\mathbb{R}} \cong \bigsqcup_{n \geq 0} [0, \infty) \sqcup \bigsqcup_{n < 0} (0, \infty). \quad (5.1)$$

This proves separability, local compactness, and absence of isolated points directly.

For a fully explicit compatible complete metric, define

$$d_n(r, s) = \begin{cases} \min\{1, |r - s|\}, & n \geq 0, \\ \min\{1, |\log r - \log s|\}, & n < 0, \end{cases}$$

and set

$$d((r, n), (s, m)) = \begin{cases} d_n(r, s), & n = m, \\ 2, & n \neq m. \end{cases} \quad (5.2)$$

Every Cauchy sequence is eventually contained in one slice. The closed ray with the first metric is complete; the open ray with the logarithmic metric is isometric, before truncation, to $\mathbb{R}$. Thus (5.2) is complete. It induces exactly the topology in (5.1), which is the original product-subspace topology. Continuity of addition follows from that topology, not from a claim of uniform continuity in this metric.

The map $r \mapsto (r, 0)$ on $r > 0$ establishes uncountability. Together with [corollary 4.11](#), this completes the simplest proof of the affirmative answer.

Remark 5.5 (Why the arithmetic order topology is not used). The least positive element is 1. In G_D , the order interval $(x - 1, x + 1)$ is the singleton $\{x\}$. The positive cone has the same discreteness at its minimum. Consequently its arithmetic order topology is discrete, and an uncountable such topology cannot be Polish. The topology of (5.1) is different, as (5.2) and the convergence in (3.3) explicitly demonstrate.

5.4 The real-coordinate model, source 04's account [04]

Source 04 gives the same slice decomposition and the same complete metric as (5.2), except that distinct slices are at distance 1; it adds two arithmetic illustrations and a remark.

Take $D = \mathbb{R}$ with the usual order and topology. It is admissible, so [corollary 4.11](#) and [theorems 5.4](#) and [7.2](#) apply to (3.4).

Topology, perfection, and a complete metric. For $z \geq 0$, the slice of integer coordinate z is a copy of $[0, \infty)$; for $z < 0$, it is a copy of $(0, \infty)$. Each slice is clopen in $M_{\mathbb{R}}$. Thus $M_{\mathbb{R}}$ is uncountable, locally compact, and has no isolated points. In particular every standard numeral $(0, n)$ is nonisolated.

It is important not to assert that the inherited Euclidean metric is complete: $(0, \infty)$ with that metric is not. Here is a compatible complete metric, with different integer slices distance 1 apart:

$$\rho((r, z), (s, w)) = \begin{cases} 1, & z \neq w, \\ \min\{1, |r - s|\}, & z = w \geq 0, \\ \min\{1, |\log r - \log s|\}, & z = w < 0. \end{cases} \quad (5.3)$$

The last line only occurs for positive r, s . A Cauchy sequence is eventually in one slice. On a nonnegative slice use completeness of $[0, \infty)$; on a negative slice use the homeomorphism $r \mapsto \log r$ onto $\mathbb{R}$. The truncated metrics are metrics, and the inter-slice distance preserves the triangle inequality. Countable dense subsets in the slices give a countable dense subset of the whole space.

Arithmetically huge elements converging to zero. Put $a_j = (1/j, 0)$ for $j \geq 1$. Then

$$a_j > n\mathbf{1} \quad \text{for every standard } n, \quad a_j \longrightarrow 0 \quad \text{topologically.}$$

There is no contradiction. The arithmetic order and the chosen topology are distinct structures. The arithmetic order topology on M_D would be discrete, since every nonzero point has a predecessor and a successor, and 0 is isolated by its immediate successor. That discrete topology is *not* the topology used in the construction.

For a concrete division example, in $M_{\mathbb{R}}$ one has

$$(2, -7) = 3(2/3, -3) + (0, 2).$$

The quotient $(2/3, -3)$ is positive because its real coordinate is positive. This illustrates why restricting the integer coordinate to nonnegative values at all positive real coordinates would destroy the desired arithmetic: the negative integer sheets are necessary.

Remark 5.6 (Why a Cartesian product of two positive cones is not used [04]). The universe is not $\mathbb{R}_{\geq 0} \times \mathbb{N}$. The latter does not have the required predecessor property at points $(r, 0)$ with $r > 0$. By contrast, in $M_{\mathbb{R}}$ every positive (r, z) has predecessor $(r, z - 1)$, except that the ordinary zero remains the least element as required.

6 An explicit Baire-space realization

Let

$$D_B = \mathbb{Q}^{\mathbb{N}}$$

with coordinatewise addition. Each factor $\mathbb{Q}$ is discrete, *not* given its Euclidean subspace topology. Two distinct sequences are compared at the least index at which they differ. This lexicographic order makes D_B a divisible ordered abelian group. Its product topology is Polish and zero-dimensional; for example,

$$d_D(a, b) = 2^{-\min\{j: a_j \neq b_j\}} \quad (a \neq b), \quad d_D(a, a) = 0$$

is a compatible complete metric.

The sets

$$C_{k,q} = \{a : a_0 = \cdots = a_{k-1} = 0, a_k = q\}, \quad k \in \mathbb{N}, q \in \mathbb{Q}_{>0},$$

are clopen cylinders, and

$$(D_B)_{>0} = \bigsqcup_{k,q>0} C_{k,q}. \quad (6.1)$$

No restriction is placed on later coordinates. Each cylinder is homeomorphic to $\mathbb{Q}^{\mathbb{N}}$. The nonnegative cone $(D_B)_{\geq 0}$ is closed, since the negative cone is open. [theorem 5.2](#) already proves that

$$M_B = (D_B \overrightarrow{\times} \mathbb{Z})_{\geq 0}$$

is Polish. We now identify its homeomorphism type without relying on a space-classification theorem.

6.1 The zero branch and its explicit tree code

Fix computable bijections

$$e : \mathbb{N} \rightarrow \mathbb{Q}, \quad e_+ : \mathbb{N} \rightarrow \mathbb{Q}_{>0}, \quad e_{\geq 0} : \mathbb{N} \rightarrow \mathbb{Q}_{\geq 0},$$

with computable inverses. A code $s \in \mathbb{N}^{\mathbb{N}}$ determines a sequence $a \in (D_B)_{\geq 0}$ as follows. As long as all previously chosen coefficients are zero, set $a_j = e_{\geq 0}(s_j)$. Once the first positive coefficient has appeared, use $a_j = e(s_j)$ at all later stages. This is a bijection: the inverse uses the corresponding enumeration at each step. A finite output prefix depends on exactly a finite input prefix, in both directions. Thus it is a homeomorphism

$$\mathbb{N}^{\mathbb{N}} \cong (D_B)_{\geq 0}. \quad (6.2)$$

The all-zero rational sequence is not omitted; it is the branch that continues choosing zero indefinitely.

Theorem 6.1 (Baire-space carrier). *There is an explicit homeomorphism $h : \mathbb{N}^{\mathbb{N}} \rightarrow M_B$. In particular M_B is uncountable, has no isolated points, and is nowhere locally compact.*

Proof. Partition M_B into the following countably many clopen blocks. For each $n \geq 0$, take the entire slice $(D_B)_{\geq 0} \times \{n\}$. For $n < 0$, partition that slice further into $C_{k,q} \times \{n\}$ using (6.1). By (6.2), every nonnegative slice is homeomorphic to Baire space; every negative-slice block is so by coding its unrestricted rational tail. A countable disjoint sum of copies of $\mathbb{N}^{\mathbb{N}}$ is homeomorphic to $\mathbb{N}^{\mathbb{N}}$: use the first natural coordinate to choose a block and the remaining coordinates to choose a point in it.

For completeness, no nonempty basic cylinder of $\mathbb{N}^{\mathbb{N}}$ is relatively compact. Its projection to the next unrestricted discrete coordinate has infinite image, whereas the continuous image of a compact set in a discrete space is finite. Every neighborhood contains such a cylinder. The other two assertions follow from the elementary properties of the product $\mathbb{N}^{\mathbb{N}}$. $\square$

6.2 A total code for every sequence of natural numbers

Here is a fully specified block indexing, also used in the accompanying program. Let $\langle a, b \rangle$ be Cantor's pairing function. The first digit of a Baire code is a tag. An even tag $2n$ specifies the nonnegative integer coordinate n and the tree code (6.2). An odd tag

$$2\langle a, \langle k, b \rangle \rangle + 1 \quad (6.3)$$

specifies integer coordinate $-a - 1$, first positive rational coordinate k , and value $e_+(b)$ there. The remaining digits code the unrestricted rational tail using e .

Every natural tag is of exactly one of these forms. Every infinite natural sequence therefore codes a valid element; no hidden domain test is required. Conversely, a point with negative integer coordinate has a first positive rational coefficient, so its tag and tail are uniquely recoverable. This describes h and h^{-1} explicitly.

Theorem 6.2 (Computable continuous arithmetic). *Under this presentation, the operations*

$$x + y, \quad x + \mathbf{1}, \quad q_m(x) = \lfloor x/m \rfloor, \quad r_m(x) = x \bmod m$$

are Type-2 computable for every positive standard integer m . Here $q_m(x)$ is the unique quotient in the model's Euclidean division by m , not a real-valued floor operation. The remainders have their finite discrete topology.

Proof. A Type-2 computation reads finitely many input digits to produce each requested output digit. Decode the integer coordinates n, z of the two inputs. Each rational coefficient $a_j + b_j$ is computable from finite prefixes. If $n + z \geq 0$, output tag $2(n + z)$ and encode the coefficient sequence by the nonnegative-cone tree. The tree decisions only inspect finite prefixes.

If $n + z < 0$, at least one input has a negative integer coordinate and hence a strictly positive coefficient sequence. Its tag supplies an index k of a positive leading coefficient. Since both coefficient sequences are lex-nonnegative, their sum is strictly positive and its first nonzero coordinate occurs at or before k . Search these finitely many coordinates, output the negative tag (6.3), and then encode the tail. Thus the tag itself is computable; the algorithm does not attempt to decide whether an arbitrary sequence is identically zero.

For division use

$$q_m(a, n) = (a/m, \lfloor n/m \rfloor), \quad r_m(a, n) = n \bmod m.$$

If the quotient's integer coordinate is negative, the input integer coordinate was negative as well, and the same known leading index can be used. Otherwise the nonnegative tree code applies. Constants and successor are special cases of these finite-prefix procedures. $\square$

The adjective “computable” here concerns operations on infinite sequence codes. It does not assert that every element is computable, or that equality and order of arbitrary streams can be decided in finite time. The distinction will be quantified next.

6.3 Second route [04]: explicit charts

Source 04 builds the same space: its M_ω is M_B , with the same group $\mathbb{Q}^{\mathbb{N}}$, the same discrete coordinates and the same lexicographic order. Its charts differ from the tag coding of [theorem 6.1](#): they go from the model to Baire space and code the nonnegative and the negative integer slices separately. Source 04's homeomorphism H , its charts $H_\geq, H_>$ and its integer coding h are written $\Psi, \Psi_\geq, \Psi_>$ and ζ here ([section 1.3](#)).

6.3.1 The lexicographic rational product

Let

$$D_B = \mathbb{Q}^{\mathbb{N}}$$

with coordinatewise addition, the product topology in which *each copy of $\mathbb{Q}$ is discrete*, and lexicographic order by the first unequal coordinate. A nonzero sequence has a least nonzero index, so the order is defined on all sequences, not merely on finite-support ones. It is a linear order compatible with addition. Division by standard integers is coordinatewise.

The metric

$$d(u, v) = \begin{cases} 0, & u = v, \\ 2^{-k}, & k = \min\{i : u_i \neq v_i\}, \end{cases}$$

is a complete compatible ultrametric. Finite-support rational sequences form a countable dense set. The positive cone is

$$(D_B)_{>0} = \bigcup_{k \in \mathbb{N}} \bigcup_{q \in \mathbb{Q}_{>0}} \{d : d_0 = \dots = d_{k-1} = 0, d_k = q\},$$

an open union of clopen cylinders. Hence D_B is admissible. Write

$$M_B = M_{D_B}.$$

It is already an uncountable Polish Presburger model with continuous addition. We now identify its space explicitly with $\mathbb{N}^{\mathbb{N}}$.

6.3.2 Two elementary coding homeomorphisms

Fix bijections

$$c : \mathbb{Q} \longrightarrow \mathbb{N}, \quad b : \mathbb{Q}_{>0} \longrightarrow \mathbb{N}_{>0}, \quad a : \mathbb{N} \times \mathbb{Q}_{>0} \longrightarrow \mathbb{N}, \quad \zeta : \mathbb{Z} \longrightarrow \mathbb{N}.$$

Since their domains and codomains are discrete, these are homeomorphisms between discrete spaces.

For $d \in (D_B)_{\geq 0}$ define $\Psi_{\geq}(d) \in \mathbb{N}^{\mathbb{N}}$ as follows. Send the zero sequence to the zero sequence. If k is the first nonzero index of d , then $d_k > 0$, and set

$$\Psi_{\geq}(d)_i = \begin{cases} 0, & i < k, \\ b(d_k), & i = k, \\ c(d_i), & i > k. \end{cases} \quad (6.4)$$

Every nonzero element of $\mathbb{N}^{\mathbb{N}}$ has a first positive entry, so this coding has an evident inverse. Both it and its inverse preserve information in finite initial segments: to determine the first n coordinates, only the first n coordinates are needed. It is therefore a homeomorphism, including at the zero sequence.

For $d \in (D_B)_{>0}$, let k again be its first nonzero index and put

$$\Psi_{>}(d)_0 = a(k, d_k), \quad \Psi_{>}(d)_{j+1} = c(d_{k+1+j}) \quad (j \geq 0). \quad (6.5)$$

The pair (k, d_k) identifies a clopen positive cylinder; within that cylinder the tail is an arbitrary rational sequence. This is another homeomorphism onto $\mathbb{N}^{\mathbb{N}}$.

Theorem 6.3 (A concrete Baire homeomorphism [04]). *The map*

$$\Psi : M_B \longrightarrow \mathbb{N}^{\mathbb{N}}, \quad \Psi(d, z) = \begin{cases} \zeta(z) \frown \Psi_{\geq}(d), & z \geq 0, \\ \zeta(z) \frown \Psi_{>}(d), & z < 0, \end{cases} \quad (6.6)$$

is a homeomorphism. Here $\zeta(z) \frown u$ means prepend the one entry $\zeta(z)$ to the sequence u .

Proof. The integer slices of M_B are clopen. Their images are exactly the clopen cylinders of $\mathbb{N}^{\mathbb{N}}$ specified by the first coordinate. On each slice the map is one of the two homeomorphisms just proved. These slice maps assemble to a bijection whose map and inverse are continuous. $\square$

Transporting addition by Ψ gives a binary operation on the *literal* set $\mathbb{N}^{\mathbb{N}}$:

$$u \oplus v = \Psi(\Psi^{-1}(u) + \Psi^{-1}(v)). \quad (6.7)$$

This proves the Baire-space assertion of [remark 3.2](#); it does not merely produce a Borel-isomorphic universe. The space is perfect and nowhere locally compact. For the latter assertion, every nonempty basic cylinder in $\mathbb{N}^{\mathbb{N}}$ has infinitely many disjoint next-coordinate cylinders covering it, with no finite subcover. If a compact neighborhood existed, a closed basic cylinder inside it would be compact, a contradiction.

6.3.3 Effective operations on infinite streams

A map on Baire space is *Type-2 computable* here if an algorithm, given oracle access to the input sequences, returns every requested output coordinate after reading finitely many input coordinates. The number of queries may depend on the input. This is a statement about computations with infinite input oracles, not an enumeration of an uncountable model by ordinary natural-number indices.

Choose the bijections above effectively. For example, index positive rationals by the binary tree starting at $1/1$, with children

$$\frac{p}{q} \mapsto \frac{p}{p+q}, \quad \frac{p+q}{q}.$$

Every positive rational in lowest terms has a unique finite path: repeatedly subtract the smaller numerator or denominator from the larger. The sum strictly decreases until $(1, 1)$ is reached. Recording whether the larger entry was the numerator or denominator and reversing the record recovers the path. This proves both bijectivity and effective invertibility. Enumerate $\mathbb{Q}$ by interleaving positive and negative rationals and adding zero. Combine this enumeration with an effective pairing of two natural numbers to obtain a and with a signed-integer coding to obtain ζ .

Theorem 6.4 (Effective strengthening [04]). *The homeomorphism Ψ and its inverse can be chosen Type-2 computable. With this choice, the addition $\oplus$ in (6.7), and quotient and remainder by each fixed positive integer, are Type-2 computable on $\mathbb{N}^{\mathbb{N}}$.*

Proof. For $\Psi_{\geq}$, a requested coordinate i requires only the first $i + 1$ coefficients: either all are zero, the first nonzero is at i , or it occurred earlier. These are precisely the three cases in (6.4). The same finite-prefix procedure computes its inverse. No decision whether an entire infinite sequence is zero is made.

For $\Psi_{>}$, search for the first nonzero coefficient. The search terminates because this chart has a positive, not merely nonnegative, domain. Once found, all output coordinates are computable; its inverse reads the encoded pair in the first coordinate and then the tail. Reading the integer coordinate chooses the appropriate chart of Ψ and Ψ^{-1} .

Coordinatewise rational addition and division are computable, as are integer addition, quotient, and remainder. Compose these algorithms with Ψ and Ψ^{-1} . Whenever the output has negative integer coordinate its D_B -coordinate is strictly positive, so the positive-chart search is guaranteed to terminate. For addition, for instance, a negative sum of integer coordinates requires at least one negative input integer coordinate; that input has strictly positive D_B -coordinate, and the other has nonnegative D_B -coordinate. Their sum is strictly positive. $\square$

The companion `verification.py` implements these maps as memoized indexed streams. A caller supplies a function returning the i th natural number of a sequence. The module exports `baire_add`, `baire_quotient`, and `baire_remainder`. Its round-trip checks include long initial zero blocks, which exercise the boundary chart at the zero coefficient sequence.

7 Definability, effective truth, and the optimal order rank

A subset of a metrizable space is Δ_2^0 when it is both F_σ and G_δ . Finite Boolean combinations of such sets are again Δ_2^0 . Closed sets, open sets, and clopen sets belong to this class.

Theorem 7.1 (All definable relations have low Borel rank). *Under the hypotheses of [theorem 5.2](#), every first-order definable subset of M_D^k , with parameters, is Δ_2^0 .*

Proof. In $G_D = D \times \mathbb{Z}$, the strict positive cone is

$$(D_{>0} \times \mathbb{Z}) \cup (\{0\} \times \mathbb{Z}_{>0}).$$

It is Δ_2^0 , as is its complement. Affine group terms are continuous maps to G_D , so their inequalities have Δ_2^0 truth sets. Equalities have closed truth sets because the ambient group is Hausdorff.

Divisibility is especially simple. Since D is divisible,

$$mG_D = D \times m\mathbb{Z}. \quad (7.1)$$

This is clopen. Thus congruence atoms have clopen truth sets, including when their affine terms contain parameters. Translate a cone formula into the group and apply [theorem 4.9](#). The result is a finite Boolean combination of these atoms. Finally restrict to M_D^k . $\square$

Second route [04]. Source 04 proves [theorem 7.1](#) for its admissible groups ([section 5.2](#)), a special case of the hypotheses of [theorem 5.2](#), by an explicit description of the order relation:

Theorem 7.2 (Low complexity of all definable relations [04]). *For every admissible D , every first-order definable relation with parameters on M_D , in every finite arity, is simultaneously F_σ and G_δ .*

Proof. It suffices by [theorem 4.10](#) to consider affine atoms. Affine group maps are continuous, so an affine equality defines a closed set. A congruence condition depends only on finitely many discrete integer coordinates and is clopen, by [corollary 4.3](#).

For the order, write two group elements as $(d, z), (e, w)$. The strict comparison is

$$(d, z) < (e, w) \iff e - d > 0 \text{ or } (e = d \text{ and } z < w). \quad (7.2)$$

The first condition is open; the second is closed because equality of D -coordinates is closed and the integer comparison is clopen. Thus the order relation is F_σ . Its complement has the identical form with the opposite strict comparison and weak integer comparison, hence is also F_σ . Pulling back by affine maps preserves both classes. Finite Boolean combinations of Δ_2^0 sets remain Δ_2^0 , which proves the claim. $\square$

This is stronger than merely obtaining a Borel presentation. The proof also explains why quantification does not raise the complexity in this particular presentation: its apparent projections are replaced by finite residue searches and finite Boolean operations.

7.1 A formula-by-formula bound on changes of mind

Theorem 7.3 (Effective finite-mind-change truth). *For each first-order Presburger formula $\varphi(\bar{x}; \bar{p})$, one can compute a natural number K_φ and an algorithm producing*

$$A_s(\bar{x}; \bar{p}) \in \{0, 1\} \quad (s \in \mathbb{N})$$

from finite prefixes of the Baire codes, such that

$$\lim_s A_s(\bar{x}; \bar{p}) = \mathbf{1}_{M_B \models \varphi(\bar{x}; \bar{p})}.$$

For every input tuple the displayed sequence changes value at most K_φ times. Parameter codes are supplied as oracles; they are not assumed computable.

Proof. Compute a quantifier-free group formula equivalent to the relativized cone formula, by [theorem 4.9](#). An affine group term has the form (c, n) , where n is an exactly computable integer and each rational coefficient c_j is computable from finite prefixes of the inputs.

To approximate the sign of (c, n) , first use the sign of n . At stage s , inspect $c_0, \dots, c_{s-1}$. If a nonzero coefficient has been found, use the sign of the first such coefficient instead. If $c = 0$, the initial answer was correct. Otherwise the answer becomes correct when the first nonzero coefficient is reached and never changes again. An equality atom can be handled in the same way: it starts with the answer determined by n and becomes permanently false if a nonzero coefficient is found. A congruence atom is decided immediately from n by [\(7.1\)](#).

Evaluate the fixed Boolean combination of these approximations at each stage. Its value can change only when an equality or inequality atom changes. The number of those atom occurrences is a computable bound K_φ . Each stage reads only finitely many digits. The truth set is effectively F_σ because it is

$$\bigcup_N \bigcap_{s \geq N} \{\bar{x} : A_s(\bar{x}; \bar{p}) = 1\},$$

where the stage sets are effectively clopen relative to the parameter codes. The same expression with 0 proves effective G_δ status. $\square$

No effective stopping time is asserted: a later rational coordinate may be the first nonzero one. Deciding sentences and deciding arbitrary parameter instances are different problems. The former is ordinary Presburger decidability; the latter is represented here by a convergent, uniformly bounded-change process.

7.2 Why rank one is impossible for the order

Theorem 7.4 (Open or closed arithmetic order forces discreteness). *Let X carry a model of Presburger arithmetic and a topology in which successor $x \mapsto x + 1$ is continuous. If either of the relations $<$ or $\leq$ is open or closed in X^2 , then X is discrete. Consequently, in any uncountable Polish model with continuous addition, the arithmetic order is neither open nor closed.*

Proof. If $\leq$ is open, then for fixed a both sets $\{x : x \leq a\}$ and $\{x : a \leq x\}$ are open. Their intersection is $\{a\}$, so every point is isolated.

If $<$ is closed, taking the complement and reversing coordinates makes $\leq$ open. If $<$ is open, the discrete-order equivalence

$$x \leq y \iff x < y + 1$$

and continuity of successor make $\leq$ open. Finally, if $\leq$ is closed, then

$$x < y \iff x + 1 \leq y$$

makes $<$ closed, reducing to the previous case. An uncountable discrete space is not separable, hence not Polish. $\square$

Together, [theorems 7.1](#) and [7.4](#) make the Δ_2^0 order bound optimal. This is a statement about the order relation, not about every individual definable set: many congruence sets are clopen. It also rules out a continuous Boolean-valued order test on the Baire model, despite its computable addition.

7.3 A general obstruction from closed order [04]

In a Presburger model M , define truncated subtraction by

$$x \dot{-} y = \begin{cases} z, & y + z = x \text{ for some } z \in M, \\ 0, & x < y. \end{cases}$$

The witness in the first case is unique. Equivalently this is $\max\{x - y, 0\}$ in the ordered group completion.

Theorem 7.5 (Closed order forces discreteness [04]). *Let a model of $\text{Pr}_{\mathbb{N}}$ carry a Hausdorff topology. If its weak order relation is closed in M^2 , then every point of M is isolated. If the topology is also separable, then M is countable.*

Proof. By totality, the complement of the converse weak order is strict order. Thus $<$ is open in M^2 . Fixing either argument gives open order rays. For $x > 0$, let $x - 1$ be its predecessor. Discreteness of the arithmetic order gives

$$\{x\} = \{y : x - 1 < y < x + 1\}.$$

At zero one has $\{0\} = \{y : y < 1\}$. Each singleton is open, so the topology is discrete. A dense subset of a discrete space must contain every point; separability therefore implies countability. $\square$

Corollary 7.6 (No continuous monus on an uncountable separable model [04]). *If truncated subtraction on a Hausdorff Presburger model is jointly continuous, the topology is discrete. In particular no uncountable separable Hausdorff Presburger model has continuous truncated subtraction.*

Proof. The relation $x \leq y$ is exactly $x \dot{-} y = 0$. A singleton is closed in a Hausdorff space, so continuity would make weak order closed. Apply [theorem 7.5](#). $\square$

No continuity hypothesis on addition was needed for this obstruction. Consequently, extending the signature of our examples by their definable monus operation destroys the assertion that all named operations are continuous. Definitional equivalence of first-order theories does not by itself preserve this topological property.

Remark 7.7 (Two incomparable order obstructions [merge]). [Theorem 7.4](#) (source 05) and [theorem 7.5](#) (source 04) both show that the order of a nondiscrete model cannot be too simple, under incomparable hypotheses. 05 assumes that successor is continuous, assumes no separation axiom, and treats four cases ($<$ or $\leq$, open or closed). 04 assumes only that the weak order $\leq$ is closed and the topology Hausdorff, with no continuity at all. Neither implies the other: 04 covers a closed $\leq$ in a topology where successor is discontinuous, 05 covers open orders and topologies that are not Hausdorff. Both are therefore printed. For the metrizable models of this part, where successor is continuous, both give the same conclusion, and so does [corollary 7.6](#) for truncated subtraction, whose exact discontinuity locus is [theorem 8.3](#).

8 Which definable operations are continuous?

The coordinate topology makes several arithmetic operations exceptionally regular, but not every definable operation. This section identifies exact boundaries in the two explicit models.

Proposition 8.1 (Euclidean self-similarity). *In $M_{\mathbb{R}}$ and $M_{\mathbb{B}}$, for each positive standard integer m the map*

$$x \mapsto (q_m(x), r_m(x)) \quad (8.1)$$

is a homeomorphism from the model onto $M \times \{0, \dots, m-1\}$, where the finite factor is discrete. Its inverse is $(q, r) \mapsto mq + r\mathbf{1}$.

Proof. Uniqueness and existence are [proposition 4.2](#). In both chosen coefficient topologies, division of d by m is continuous. The integer maps $n \mapsto \lfloor n/m \rfloor$ and $n \mapsto n \bmod m$ are continuous on a discrete space. Hence (8.1) is continuous into the product. Repeated addition and addition of a fixed standard numeral make its inverse continuous. $\square$

The same formula also explains why all standard congruences ignore the entire coefficient sequence. Two elements have the same remainder modulo every positive standard integer if and only if their integer coordinates agree. The forward direction uses the elementary fact that an ordinary integer divisible by every positive integer is zero. Their infinite coordinates can still be arbitrary and different.

Proposition 8.2 (The total predecessor has exactly one discontinuity). *Define*

$$p(0) = 0, \quad p(x) = x - \mathbf{1} \quad (x > 0).$$

In either explicit model, p is continuous exactly at the points other than 0.

Proof. Away from 0, this is a restriction of translation by $-\mathbf{1}$ in the ambient topological group. Since $\{0\}$ is closed, the nonzero domain is open, and translation has its image in the cone.

In the real model let $t_j = 1/j$; in the Baire model let $t_j = e_j$, the rational sequence with 1 in coordinate j and zeros elsewhere. In either case $t_j > 0$ and $t_j \rightarrow 0$ in D . Then

$$(t_j, 0) \rightarrow 0, \quad p(t_j, 0) = (t_j, -1).$$

The latter sequence does not converge to $p(0) = 0$, because its integer coordinate is always -1 and integer coordinates are discrete. $\square$

Source 04 proves [proposition 8.2](#) (its Proposition 7.4), treating the numerals $(0, n)$, $n \geq 1$, separately:

Second proof (source 04). At points with positive D -coordinate, $p(d, z) = (d, z - 1)$ locally. At an ordinary numeral $(0, n)$ with $n \geq 1$, the same formula holds on its integer-coordinate slice and maps into M_D , so it is continuous there. At zero, the sequence $(s_j, 0) \rightarrow 0$ used above has $p(s_j, 0) = (s_j, -1)$, which cannot converge to zero. $\square$

Theorem 8.3 (Exact discontinuity locus of truncated subtraction). *For $M = M_{\mathbb{R}}$ or $M_{\mathbb{B}}$, let*

$$T(x, y) = \max\{x - y, 0\}$$

using the arithmetic order in the ambient group. Its discontinuity locus is exactly

$$\{((d, n), (d, z)) \in M^2 : n \neq z\}. \quad (8.2)$$

Proof. Both positive and negative cones in the coefficient group D are open for our two examples. If the coefficient difference $d - e$ is nonzero, its sign is locally fixed. Thus T locally agrees either with the continuous ambient difference or with the constant zero map.

Suppose instead that $x = y$. Along inputs sufficiently near this pair, the integer difference is zero. The output is either their ambient difference or zero. For any neighborhood V of zero in the ambient group, both possible outputs lie in V when the inputs are sufficiently near (x, x) . This proves continuity there.

It remains to consider equal coefficient coordinates and $k = n - z \neq 0$. Use the positive null sequence t_j from [proposition 8.2](#). If $k > 0$, replace y by $y + (t_j, 0)$. These inputs converge to (x, y) , but their truncated differences are zero, whereas $T(x, y) = (0, k)$. If $k < 0$, replace x by $x + (t_j, 0)$. The outputs are (t_j, k) , which cannot converge to the original value 0 because their integer coordinate is the nonzero integer k . All perturbed inputs remain in the positive cone. This proves exactly (8.2). $\square$

Source 04 states [theorem 8.3](#) as its Theorem 7.3, for the same two models and with the same locus, written $\text{Disc}(\dot{-}) = \{((d, z), (e, w)) : d = e \text{ and } z \neq w\}$, and adds [04]: *this locus is a closed nowhere dense subset of M^2 , and truncated subtraction is continuous on its dense open complement*. Its proof covers the addition.

Second proof (source 04), with the nowhere-density clause. Suppose first that $d \neq e$. The sign of $d - e$ is locally constant because both strict cones of D are open. On a neighborhood the monus operation is therefore either the continuous ambient difference or the constant zero map. Its values lie in the positive cone, so it is continuous as a map to that subspace.

Next let $d = e$ and $z = w$. In a sufficiently small neighborhood the two integer coordinates stay equal to these fixed integers. Monus has integer coordinate zero and D -coordinate $\max\{u - v, 0\}$. This positive-part map is continuous at zero in both groups under consideration. Over $\mathbb{R}$ this follows from the usual absolute-value estimate. Over D_B , if $u - v$ has its first n coordinates zero, so does its positive part: it is either $u - v$ or zero. These finite-prefix neighborhoods prove continuity at the point.

For the converse, choose $s_j > 0$ in D with $s_j \rightarrow 0$: use $1/j$ in $\mathbb{R}$ or the j th unit vector in D_B . Let $x = (d, z)$, $y = (d, w)$ and $z \neq w$. If $z < w$, replace x by $x_j = (d + s_j, z) \in M_D$. Then $x_j \rightarrow x$ but

$$x_j \dot{-} y = (s_j, z - w), \quad x \dot{-} y = 0.$$

The output integer coordinate is constantly negative and cannot converge to the zero integer coordinate. If $z > w$, instead replace y by $y_j = (d + s_j, w)$. Then $x \dot{-} y_j = 0$ for every j , whereas $x \dot{-} y = (0, z - w) \neq 0$. This proves discontinuity in both cases, including at the boundary $d = 0$.

The set in (8.2) is closed: equality of the D -coordinates is closed and inequality of the discrete integer coordinates is clopen. It has empty interior. At any of its points, perturbing the first D -coordinate upward by a sufficiently small positive element stays in M_D^2 , lies in any prescribed neighborhood, and makes the two D -coordinates unequal. Thus the closed set is nowhere dense. $\square$

The standard numerals form a closed, relatively discrete subset of both models, yet zero is not isolated in the whole space. This causes no conflict with [proposition 8.2](#). Nor does continuity of standard division imply continuity of every definable piecewise operation: the branch boundary is decisive.

Source 04 continues [04]:

The failure is not an indiscriminate loss of regularity: the graph of monus is still Δ_2^0 by [theorem 7.2](#), because monus is definable. Its ordinary predecessor specialization has a smaller exceptional set.

8.1 Polish completeness need not be translation-invariant [04]

Proposition 8.4 (A complete-metric obstruction [04]). *If zero is nonisolated in a metrizable Presburger model, there is no compatible complete metric invariant under all additive translations. More generally, for any compatible complete metric, the successor map cannot be a uniform homeomorphism onto its image.*

Proof. Successor is a bijection from the model onto $M \setminus \{0\}$. If it were a uniform homeomorphism onto this subspace, every Cauchy sequence in the image would pull back to a Cauchy sequence in the complete domain and would therefore converge in the image. Thus the image would be a complete subspace of a metric space, hence closed. Since zero is nonisolated, $M \setminus \{0\}$ is not closed, a contradiction. A translation-invariant metric would make successor an isometry, which is a special case. $\square$

This applies to both main examples. Their Polishness and continuous addition therefore cannot be replaced by a claim of completeness in the uniform structure of an invariant additive metric.

Structure or operation	Real-coordinate model	Baire-coordinate model
Underlying space	Perfect, locally compact, Polish	Homeomorphic to $\mathbb{N}^{\mathbb{N}}$
Addition	Continuous	Type-2 computable
Fixed quotient/remainder	Continuous	Type-2 computable
All definable relations	Δ_2^0	Δ_2^0
Monus	Discontinuous exactly on (8.2)	Same exact locus
Total predecessor	Discontinuous only at 0	Discontinuous only at 0
Arithmetic order topology	Discrete; not used	Discrete; not used

9 Explicit set-sized fragments of the omnific integers

We use Conway normal-form notation

$$x = \sum_{\gamma \in S} r_\gamma \omega^\gamma,$$

where S is reverse well-ordered and all coefficients are real. In the omnific ring the occurring exponents are nonnegative, and the coefficient at exponent zero is an ordinary integer. This is the interface recorded in the repository's surreal development [16]. Addition is coefficientwise, and a nonzero normal form has the sign of its leading coefficient.

Theorem 9.1 (One-scale and countable-scale realizations). *There are injective ordered additive maps, fixing 0 and 1,*

$$\iota_{\mathbb{R}} : \mathbb{R} \overrightarrow{\times} \mathbb{Z} \longrightarrow \mathbf{Oz}, \quad \iota_{\mathbb{R}}(r, n) = r\omega + n, \quad (9.1)$$

$$\iota_{\mathbb{B}} : \mathbb{Q}^{\mathbb{N}} \overrightarrow{\times} \mathbb{Z} \longrightarrow \mathbf{Oz}, \quad \iota_{\mathbb{B}}(a, n) = \sum_{j=0}^{\infty} a_j \omega^{1/(j+1)} + n. \quad (9.2)$$

Their positive images, equipped with the transported coefficient topologies, are respectively the Polish models $M_{\mathbb{R}}$ and $M_{\mathbb{B}}$.

Proof. The first formula is a finite normal form. For the second, the fixed exponents

$$1, \frac{1}{2}, \frac{1}{3}, \dots$$

strictly decrease and are positive. Every nonempty subset of these exponents has a largest element, namely the one with least index. Thus every possible support in (9.2) is reverse well-ordered, and adjoining exponent zero preserves that property. The expression is a legitimate normal form, even when infinitely many coefficients are nonzero. All positive-exponent coefficients are real, and the constant term is the integer n , so the value is omnific.

Uniqueness of normal forms gives injectivity. Addition of two such expressions is coordinatewise on a common fixed support, giving additivity. The leading nonzero coefficient is the first nonzero a_j ; if all are zero, the sign is that of n . Therefore the map preserves and reflects the lexicographic order. The same reasoning proves the one-scale case. Transporting a topology through a bijection onto its image preserves all the asserted topological properties. $\square$

9.1 What this realization does and does not preserve

These are ordered additive embeddings, *not* ring embeddings. Both images contain ω , but neither contains ω^2 . Their multiplicative nonclosure is therefore immediate from the normal forms. The stronger theorem in section 10 will show that replacing surreal multiplication by some other multiplication cannot repair the problem while retaining the same carrier, addition, and unit.

The topologies in theorem 9.1 are precisely specified coefficient topologies. The real coordinate is Euclidean in (9.1); the rational coordinates are discrete and combined by a product topology in (9.2). Neither is the discrete arithmetic order topology. No assertion that they are a canonical topology on all omnific integers is intended.

The ordinary constant-term map becomes simply

$$\text{ct}(\iota(d, n)) = n.$$

It is continuous into the discrete integers for these models. Standard congruence data see only this coordinate, matching (7.1). The standard part itself remains externally specified and nondefinable in the Presburger structure.

9.2 Why no proper-class difficulty arises

The source sets $\mathbb{R} \times \mathbb{Z}$ and $\mathbb{Q}^{\mathbb{N}} \times \mathbb{Z}$ are ordinary sets. The normal-form constructions above are definable assignments on those sets. Replacement therefore gives set-sized images. The set of birthdays of their elements has an ordinal supremum; choosing a larger ordinal places both images inside a set-sized birthday cutoff of **No**. No explicit optimal cutoff is needed for any theorem here.

Accordingly, one can conduct the entire argument in the product-group models and use theorem 9.1 only as a final realization theorem. This is also the most economical formalization strategy. It avoids making a class-sized object the carrier of a Polish space or a first-order structure in ordinary set-based semantics.

Source 04's account [04]. Source 04 proves [theorem 9.1](#) as its Proposition 10.1 with the same proof; its second map is written ι_ω there and ι_B here. It adds the following remarks on the normal-form rule and on the transported topologies.

The construction is purely set-sized, but it has a natural expression in the surreal normal-form language. We use only the classical normal-form rule: a reverse-well-ordered set of surreal exponents with real coefficients defines a surreal sum; addition is coefficientwise, and order is decided by the greatest exponent with nonzero coefficient. Omnific integers have positive growth exponents and an ordinary integer constant term. This is the normal-form description relevant to the ordered-group study in [5] and to the repository's omnific development [19].

We transfer the topologies from the coefficient models to these images. This is not a claim that the whole proper class **Oz** has a Polish topology, nor that a canonical global surreal topology is being restricted. For example, ω/j converges to zero in the transported real-coefficient topology on the first image, despite each term being arithmetically infinite. Likewise $\omega^{1/(k+1)}$ converges to zero in the second coefficient topology.

These are additive subgroups, not subrings. Squaring ω yields ω^2 , which belongs to neither displayed image. The absence of inherited multiplicative closure is consistent with the stronger nonexistence result [corollary 10.5](#) for *any* compatible semiring multiplication on these positive cones.

For foundations, the domains $\mathbb{R} \times \mathbb{Z}$ and $\mathbb{Q}^{\mathbb{N}} \times \mathbb{Z}$ are sets. One may work entirely with their normal-form codes in ZFC and interpret the notation inside a sufficiently large set-sized portion of the surreal construction. Alternatively one may use a class theory for the ambient surreal field while retaining these set-sized domains. None of the proofs quantifies over a set of all surreals, a set of all class maps, or a Polish space whose points form a proper class.

10 An absolute obstruction to multiplication

A Borel obstruction would not rule out a wildly discontinuous multiplication. For the models constructed here, the obstruction is stronger and purely algebraic.

Definition 10.1. In a Presburger model M , a positive element u is a *cofinal infinite additive unit* if

$$u > n\mathbf{1} \quad \text{for every standard } n,$$

and for every $x \in M$ there is a positive standard integer N with $x \leq Nu$. The word “unit” here refers only to cofinality under repeated addition; it is not a multiplicative unit.

Source 04 splits the hypothesis of [theorem 10.4](#) into two notions and proves a positivity lemma first:

Definition 10.2 ([04]). An element h in an ordered additive monoid is an *order unit* if for every x there is a positive standard integer N with $x \leq Nh$. It is *arithmetically infinite* if $h > n\mathbf{1}$ for every standard n .

Lemma 10.3 (Positivity is automatic in a semiring expansion [04]). *Suppose a Presburger monoid is expanded to a unital semiring with the same addition, zero, and one. Then the product of two positive elements is positive, and multiplication by a positive element is strictly increasing on either side.*

Proof. If $x, y > 0$, write $x = 1 + a$, $y = 1 + b$ with $a, b \geq 0$. Distributivity and the unit laws give $xy = 1 + a + b + ab \geq 1$. If $x > y$, write $x = y + c$ with $c > 0$. For $z > 0$, one has

$xz = yz + cz > yz$ and similarly on the other side. Here the order is the additive natural order of the Presburger model, so no separate order axiom for the multiplication has been assumed. $\square$

Theorem 10.4 (Cofinal-infinite-element obstruction). *A Presburger model with a cofinal infinite additive unit cannot be expanded to a unital semiring with the same addition, zero, and multiplicative unit $\mathbf{1}$. No continuity, measurability, commutativity of multiplication, or induction involving multiplication is assumed.*

Proof. Suppose such a multiplication exists. Cofinality applied to u^2 gives a positive standard N such that $u^2 \leq Nu$. Since u exceeds every standard numeral, it has a decomposition

$$u = (N + 1)\mathbf{1} + v \quad (v \in M).$$

Distributivity and the unit law give

$$u^2 = (N + 1)u + vu \geq (N + 1)u > Nu,$$

a contradiction. The first inequality uses only that vu belongs to the nonnegative carrier. The last one follows from $u > 0$ and additive cancellation. In any semiring expansion of this additive monoid, its natural order $x \leq y \leftrightarrow \exists z (x + z = y)$ is already the Presburger order, so no additional order-compatibility axiom was hidden in the argument. $\square$

Corollary 10.5. *Neither $M_{\mathbb{R}}$ nor $M_{\mathbb{B}}$ admits a unital semiring multiplication with the displayed addition and unit.*

Proof. In $M_{\mathbb{R}}$ take $u = (1, 0)$. It exceeds every standard numeral, and an integer $N > r$ gives $(r, n) < Nu$. In $M_{\mathbb{B}}$ take $u = (e_0, 0)$, where e_0 has first coefficient 1 and all later coefficients zero. For any (a, n) choose an integer $N > a_0$; comparison at coordinate zero gives $(a, n) < Nu$. Apply [theorem 10.4](#). $\square$

In their omnific realizations, this cofinal element is ω . The missing larger scale ω^2 is not merely a defect of one chosen multiplication: every possible semiring multiplication would have to produce an element growing faster than every standard multiple of ω , which the additive carrier cannot contain.

For a different uncountable Polish Presburger model that does allow a semiring expansion algebraically, Glazer's cited Borel obstruction for $Q + I\text{Open}$ would still matter. Our argument is not a replacement for that general theorem; it identifies a stronger obstruction in these particular additive structures.

An arithmetically infinite order unit in 04's sense is exactly a cofinal infinite additive unit in 05's sense, so [theorem 10.4](#) is 04's Theorem 9.3 and [corollary 10.5](#) its Corollary 9.4. Source 04's proof routes the strict inequality through [lemma 10.3](#):

Second proof of theorem 10.4 (source 04). Suppose such a multiplication exists and let h be the order unit. Choose a standard $N \geq 1$ with $h^2 \leq Nh$. Since $h > N\mathbf{1}$, [lemma 10.3](#) gives $h^2 > Nh$, a contradiction. $\square$

This algebraic argument is specific to the supplied models and does not replace the general topological theorem about stronger arithmetic. Presburger models need not be additive reducts of models of Peano arithmetic.

10.1 A variation with no largest growth class [04]

Let $D_L = \mathbb{Q}((t))$ be the additive group of formal Laurent series

$$f = \sum_{k \geq k_0} a_k t^k, \quad k_0 \in \mathbb{Z}, \quad a_k \in \mathbb{Q},$$

ordered by the sign of the coefficient at the least nonzero exponent. Use the t -adic topology, with rational coefficients discrete. Its basic neighborhoods at zero are

$$F_n = \{f : v(f) \geq n\}, \quad n \in \mathbb{N},$$

where $v(0) = +\infty$. A bounded compatible metric is $\min\{1, 2^{-v(f-g)}\}$ for $f \neq g$.

Proposition 10.6 (A Laurent-series Polish construction [04]). *The group D_L is admissible, and $M_L = M_{D_L}$ is a Polish Presburger model homeomorphic to Baire space, with continuous addition and fixed quotient and remainder. Unlike the two main models, M_L has no arithmetically infinite order unit.*

Proof. A Cauchy sequence of Laurent series eventually has the same entire negative part, because its pairwise differences eventually lie in F_0 . That common negative part is finite. Every subsequent coefficient stabilizes in turn, giving a Laurent-series limit. Thus the metric is complete. Laurent polynomials with rational coefficients form a countable dense set. The leading sign is determined on a basic neighborhood, so the strict positive cone is open. Addition and division by fixed positive integers are coefficientwise continuous.

For the asserted homeomorphism, decompose D_L into the clopen cosets $p + \mathbb{Q}[[t]]$, where p is a finite negative-power rational polynomial. There are countably many such cosets, and each is a rational product. If $p \neq 0$, the sign on its coset is constant, determined by its leading coefficient. The coset $p = 0$ contributes respectively $(D_B)_{>0}$ or $(D_B)_{\geq 0}$ to the positive or nonnegative cone. Thus both Laurent cones are countable disjoint sums of spaces explicitly homeomorphic to $\mathbb{N}^{\mathbb{N}}$, by the codings in [section 6.3](#). Prepending an index codes such a sum by $\mathbb{N}^{\mathbb{N}}$. The integer slices of M_L are another such sum, proving the claim.

Finally, for $f > 0$ with least exponent k , the element $(t^{k-1}, 0)$ is larger than every standard multiple of (f, z) : its leading exponent is strictly earlier. Standard finite elements also fail to be order units. $\square$

The absence of an order unit does not establish the existence of a semiring expansion. In this example another simple obstruction decides that issue negatively.

10.1.1 A growth-order obstruction

For positive arithmetically infinite elements, define

$$x \asymp y \iff (\exists m, n \in \mathbb{N}_{>0})(x \leq my \wedge y \leq nx).$$

Let $\mathcal{A}(M)$ be the set of these Archimedean growth classes, ordered so that $[x] < [y]$ means $nx < y$ for every positive standard n . These are external set-sized classes of elements of our set-sized model; no first-order definability is being asserted.

Proposition 10.7 (Growth classes must embed into their strict final segments [04]). *If a Presburger monoid M has a unital semiring expansion, then for every arithmetically infinite h , the map*

$$[x] \mapsto [hx]$$

is an order embedding of $\mathcal{A}(M)$ into its strict final segment above $[h]$.

Proof. Multiplication by $h > 0$ preserves and reflects every comparison $x \leq ny$. Preservation follows from distributivity and positivity. For reflection, if $x > ny$ then strict monotonicity gives $hx > n(hy)$. It follows that multiplication induces an order embedding on growth classes. Because x is infinite, $x > n\mathbf{1}$ for every standard n , so $hx > nh$ for every n . Thus its class lies strictly above $[h]$. $\square$

Corollary 10.8 ([04]). *The Laurent-series model M_L also has no unital semiring expansion.*

Proof. Its infinite growth classes are determined exactly by the least Laurent exponent, with increasing growth corresponding to decreasing exponent. Hence $\mathcal{A}(M_L)$ has order type $\mathbb{Z}$. A strict final segment of $\mathbb{Z}$ is well-ordered, while $\mathbb{Z}$ contains an infinite strictly decreasing sequence. There is no order embedding of $\mathbb{Z}$ into such a final segment. Apply [proposition 10.7](#). $\square$

The obstruction is not a statement that every split Presburger group forbids multiplication. For example, take $D = X\mathbb{Q}[X]$ ordered by its highest-degree nonzero coefficient, and let the constant term be an ordinary integer. The ordered ring

$$\mathbb{Z} + X\mathbb{Q}[X]$$

has this split additive group, least positive element 1, and a positive cone closed under its polynomial multiplication. Its growth classes have order type $\mathbb{N}_{>0}$, which does embed in each of its strict final segments. This countable comparison makes the scope of the obstruction explicit.

11 Endomorphisms, elementary submodels, and a contrast

The two models have markedly different self-embedding behavior. All homomorphisms in the next theorem preserve addition, zero, and $\mathbf{1}$; continuity and injectivity are *not* assumed.

11.1 The real model: endomorphisms and elementary submodels

Theorem 11.1 (Rigidity of the real-coordinate model). *Every unital additive endomorphism $f : M_{\mathbb{R}} \rightarrow M_{\mathbb{R}}$ has the form*

$$f(r, n) = (cr, n) \quad (c \in \mathbb{R}_{>0}). \quad (11.1)$$

Conversely each such formula is an endomorphism. All of them are homeomorphisms and Presburger automorphisms. In particular $M_{\mathbb{R}}$ has no proper elementary self-embedding.

Proof. The group completion of the cancellative monoid $M_{\mathbb{R}}$ is $G = \mathbb{R} \overrightarrow{\times} \mathbb{Z}$. Thus f extends uniquely to an additive group map $F : G \rightarrow G$ fixing $\mathbf{1}$. Inside G ,

$$\bigcap_{m \geq 1} mG = \mathbb{R} \times \{0\}.$$

One inclusion follows from real divisibility; the other follows because the integer coordinate of an element of the intersection is divisible by every positive integer. Every additive group map preserves this intersection. Hence

$$F(r, 0) = (h(r), 0), \quad F(0, n) = (0, n)$$

for some additive map $h : \mathbb{R} \rightarrow \mathbb{R}$.

Since f maps the positive cone into itself, $h(r) \geq 0$ whenever $r \geq 0$. Thus h is monotone. If $c = h(1)$, rational comparisons and additivity give $h(r) = cr$ for every real r : sandwich r between rational numbers and use the corresponding inequalities for h . Here $c \geq 0$. But $c = 0$ would send the positive element $(1, -1)$ to $(0, -1)$, outside the cone. Thus $c > 0$.

Formula (11.1) and its inverse with scalar $1/c$ preserve the cone, addition, order, and unit, and are continuous in the product coordinates. This proves all assertions. $\square$

Theorem 11.2 (All elementary submodels of the real model). *The elementary submodels of $M_{\mathbb{R}}$ are exactly*

$$M_E = (E \overrightarrow{\times} \mathbb{Z})_{\geq 0}, \quad E \text{ a } \mathbb{Q}\text{-linear subspace of } \mathbb{R}. \quad (11.2)$$

Among them, the only ones Polish in the inherited topology are the standard copy $\mathbb{N}$ and the whole model $M_{\mathbb{R}}$. Every nonzero proper choice of E gives a dense, non-Polish elementary submodel.

Proof. If E is a $\mathbb{Q}$ -linear subspace, then $E \overrightarrow{\times} \mathbb{Z}$ is a $\mathbb{Z}$ -group. Its inclusion in $\mathbb{R} \overrightarrow{\times} \mathbb{Z}$ preserves and reflects the expanded quantifier-free language, including all congruences, since a standard modulus depends only on the integer coordinate. Quantifier elimination makes this inclusion elementary. Relativizing to positive cones proves $M_E \leq M_{\mathbb{R}}$.

Conversely, let $S \leq M_{\mathbb{R}}$, and let $H = S - S$ be its group completion inside G . If $a, b \in S$ and $a - b \geq 0$, the unique solution of $b + x = a$ lies in S by elementarity. Therefore $S = H \cap M_{\mathbb{R}}$. Since H contains every standard integer, subtracting the integer coordinate from any $(r, n) \in H$ shows that $H = E \times \mathbb{Z}$ for an additive subgroup $E \leq \mathbb{R}$. For $r \in E_{\geq 0}$, the model's unique division of $(r, 0)$ by a positive standard m has remainder zero and quotient $(r/m, 0)$. Elementarity puts that quotient in S . Applying the same observation to negatives in H shows E is divisible, hence $\mathbb{Q}$ -linear.

If $E \neq 0$, it contains all rational multiples of a nonzero real and is dense in $\mathbb{R}$. Thus M_E is dense in every slice of $M_{\mathbb{R}}$. Suppose M_E is also Polish in the inherited topology. Its closed slice of integer coordinate zero is homeomorphic to $E \cap [0, \infty)$. By lemma 5.1, this is G_{δ} in $\mathbb{R}$. Its negative is also G_{δ} , so their union E is G_{δ} in $\mathbb{R}$. For any $r \in \mathbb{R}$, the dense G_{δ} sets E and $r + E$ intersect by the Baire category theorem. Hence $r \in E - E = E$, giving $E = \mathbb{R}$. Finally $E = 0$ gives the relatively discrete standard copy $\mathbb{N}$, which is Polish, and $E = \mathbb{R}$ gives the whole model. $\square$

For example, $E = \mathbb{Q}$ gives a countable dense elementary submodel. Its inherited topology is not Polish; there is no inconsistency between its density and the complete metric on the ambient model.

11.2 Reduction to the divisible part [04]

This section considers embeddings in the arithmetic language containing $0, 1, +$. Naming 1 is explicit. Order is already definable, so its preservation is not an extra independent assumption.

Proposition 11.3 (All unital additive maps [04]). *Let D, E be divisible linearly ordered abelian groups. Every map $f : M_D \rightarrow M_E$ preserving $0, 1, +$ is injective and has the unique form*

$$f(d, z) = (T(d), z), \quad (11.3)$$

where $T : D \rightarrow E$ is a $\mathbb{Q}$ -linear order embedding. Conversely every such T defines a map of this kind. Every one of these maps is an elementary embedding of Presburger models.

Proof. If $x > y$, then $x = y + 1 + u$ for some $u \in M_D$. Thus $f(x) = f(y) + 1 + f(u) > f(y)$, so f strictly preserves order and is injective. Every group element is a difference of nonnegative elements. The rule

$$\bar{f}(x - y) = f(x) - f(y)$$

therefore extends f uniquely to an additive order embedding of group completions. It is well defined because $x - y = x' - y'$ implies $x + y' = x' + y$, and f preserves that equality.

For each $d \in D$ and every standard $m \geq 1$, the element $(d, 0)$ is an m -multiple. Its image is consequently in every mG_E , whose intersection is $E \times \{0\}$ by (4.3). Hence $\bar{f}(d, 0) = (T(d), 0)$. Since $f(1) = 1$, the map fixes every standard integer, giving (11.3). Divisibility and torsion-freeness make T $\mathbb{Q}$ -linear; order preservation and injectivity are inherited. The converse follows directly from the formula.

Such a group map preserves and reflects order, equality, and all congruences, the last because integer coordinates are unchanged. Quantifier elimination in theorem 4.10 makes it elementary on the groups. Relativizing to the nonnegative cones makes f elementary as well. $\square$

Source 04 proves theorem 11.1 (its Theorem 8.2) from proposition 11.3:

Second proof (source 04). By proposition 11.3, it remains to classify increasing additive maps $T : \mathbb{R} \rightarrow \mathbb{R}$. Put $c = T(1) > 0$. For rational q one has $T(q) = cq$. For a real r , compare r with rational numbers from below and above and use monotonicity. The resulting inequalities force $T(r) = cr$. The inverse dilation is $f_{1/c}$, and both dilations are continuous in the specified product topology. $\square$

Here 1 in $T(1)$ denotes the real vector $1 \in D$, not the arithmetic unit $(0, 1)$. The distinction prevents a mistaken conclusion that c must equal one. The model is co-Hopfian for unital additive maps but is not rigid.

11.3 The Baire model: all self-embeddings [04] and clopen self-copies

Source 04 classifies every unital additive self-map of M_B ; source 05's shift theorem follows.

For $d \in D_B \setminus \{0\}$ define

$$v(d) = \min\{k : d_k \neq 0\}.$$

Let e_k be the rational sequence with a single 1 in coordinate k , and put

$$F_k = \{d : d_0 = \cdots = d_{k-1} = 0\}.$$

Thus $F_0 = D_B$; the subgroups F_k form a neighborhood basis at zero.

Theorem 11.4 (Complete matrix classification [04]). *Every $\mathbb{Q}$ -linear order embedding $T : D_B \rightarrow D_B$, without any assumed continuity, has the form*

$$(Td)_i = \sum_{k:p(k) \leq i} a_{ik} d_k, \tag{11.4}$$

where $p : \mathbb{N} \rightarrow \mathbb{N}$ is strictly increasing and

$$a_{ik} = 0 \ (i < p(k)), \quad a_{p(k),k} > 0. \tag{11.5}$$

The coefficients below each pivot, a_{ik} with $i > p(k)$, are arbitrary rationals. Conversely every such matrix defines an order embedding. All the sums in (11.4) are finite. Every such map is continuous and has closed image. It is surjective exactly when $p(k) = k$ for all k .

Proof. Let T first be an arbitrary $\mathbb{Q}$ -linear order embedding. Since $Te_k > 0$, its first nonzero coefficient is positive. Define $p(k) = v(Te_k)$. The inequalities

$$ne_{k+1} < e_k \quad (n = 1, 2, \dots)$$

remain true after applying T . If $p(k+1) < p(k)$, the image on the left already dominates the right for $n = 1$. If the pivots were equal, a large enough standard n would make the positive leading rational coefficient on the left larger than the one on the right. Both are impossible. Thus $p(k+1) > p(k)$, and in particular $p(k) \geq k$.

Every $d \in F_k$ satisfies $|d| < Ne_k$ for some standard integer $N > 0$: choose $N > |d_k|$, with a further unit of margin. Order preservation implies $|Td| < NTe_k$, hence

$$T(F_k) \subseteq F_{p(k)} \subseteq F_k. \quad (11.6)$$

This already proves continuity at zero, and therefore everywhere by additivity. It also excludes noncontinuous linear maps arising from a choice of a vector-space basis.

Let $a_{ik} = (Te_k)_i$. To compute coordinate i of Td , subtract the finite initial sum $\sum_{k=0}^i d_k e_k$. The remainder belongs to F_{i+1} , whose image has zero coordinates through i by (11.6). Thus $(Td)_i = \sum_{k=0}^i a_{ik} d_k$. Terms with $p(k) > i$ vanish, yielding (11.4) and (11.5).

Conversely, a strictly increasing p satisfies $p(k) \geq k$, so each row in (11.4) has finite support. The formula defines a linear continuous map on all rational sequences. If $d \neq 0$ has first nonzero index k , then all rows before $p(k)$ vanish and

$$(Td)_{p(k)} = a_{p(k),k} d_k.$$

Its sign is the sign of d_k . This proves order preservation and injectivity, including comparison of arbitrary pairs by their difference.

To describe the range, for any proposed target sequence y recursively recover

$$d_k = \frac{y_{p(k)} - \sum_{j < k} a_{p(k),j} d_j}{a_{p(k),k}}. \quad (11.7)$$

Each recovered coordinate is a continuous function of finitely many coordinates of y . The target belongs to the range exactly when, for every nonpivot row i ,

$$y_i = \sum_{k:p(k) < i} a_{ik} d_k. \quad (11.8)$$

These are closed, indeed clopen, finite-coordinate conditions. Their intersection is closed. If every row is a pivot, the recovery solves the system for every target. This happens exactly when $p(k) = k$. If a row j is omitted, no image has its first nonzero coordinate equal to j , so surjectivity fails. $\square$

Corollary 11.5 (Automatic continuity and the topology of self-images [04]). *Every unital additive self-map of M_B is elementary, continuous, and a homeomorphism onto a closed image. Its image is clopen when the complement of $p(\mathbb{N})$ is finite, and closed nowhere dense when that complement is infinite. The automorphisms are exactly the infinite lower triangular rational matrices with strictly positive diagonal, acting on the divisible coordinate and fixing the integer coordinate.*

Proof. Combine [proposition 11.3](#) and [theorem 11.4](#). The inverse on the image is continuous by the coordinate-recovery formula. The group image is $T(D_B) \times \mathbb{Z}$, and its intersection with M_B is exactly the monoid image, because T reflects order.

When finitely many rows are not pivots, (11.8) gives a finite intersection of clopen conditions, so the group and monoid images are clopen. When infinitely many rows are not pivots, they occur arbitrarily far out. At an image point whose D_B -coordinate is positive, keep an initial segment containing its leading coordinate and change a later nonpivot coordinate so that its constraint fails. The result remains positive and lies in any sufficiently specified cylinder. At a point $(0, z)$ in the image, necessarily $z \geq 0$. In any given zero-prefix cylinder first insert a positive pivot far beyond that prefix, using a small-in-topology image vector Te_k . Then violate a still later nonpivot constraint. Positivity is unchanged and the resulting point is outside the image. Thus the image has empty interior even at the boundary. It is closed, hence nowhere dense. $\square$

Example 11.6 (Two contrasting proper elementary copies [04]). The shift

$$S(d)_0 = 0, \quad S(d)_{k+1} = d_k$$

has pivots $p(k) = k + 1$ and produces a proper clopen elementary self-copy of M_B . The insertion map

$$E(d)_{2k} = d_k, \quad E(d)_{2k+1} = 0$$

has pivots $p(k) = 2k$ and produces a closed nowhere dense elementary self-copy. Both are homeomorphisms onto their images. Their order-theoretic and model-theoretic behavior is identical at the level of elementary embedding, while their image interiors are different.

Theorem 11.7 (Clopen elementary self-copies of the Baire model). *Let*

$$\sigma((a_0, a_1, a_2, \dots), n) = ((0, a_0, a_1, a_2, \dots), n).$$

Then $\sigma : M_B \rightarrow M_B$ is a proper elementary embedding and a homeomorphism onto a clopen subspace. Its iterates satisfy

$$\bigcap_{k \geq 0} \sigma^k(M_B) = \{(0, n) : n \in \mathbb{N}\}.$$

Thus a descending sequence of proper clopen elementary self-copies has intersection exactly the standard elementary submodel.

Proof. Inserting an initial zero preserves and reflects the lexicographic order, addition, and every standard congruence. It maps the divisible coefficient group into the subgroup with first coordinate zero. Quantifier elimination gives elementarity, first for the groups and then for their cones. The image is exactly the subset with $a_0 = 0$, which is clopen because the first rational coordinate is discrete. Deleting that zero is a continuous inverse on the image. It is proper, since elements with $a_0 > 0$ are omitted. The k -th image has its first k rational coordinates zero. Membership in all images therefore means that every rational coordinate is zero; positivity then forces $n \geq 0$. $\square$

These clopen submodels are external subsets, not Presburger-definable subsets, even with parameters: a proper definable subset containing zero and closed under successor would violate induction. Topological simplicity and internal definability are different properties.

[merge] The map σ of theorem 11.7 (source 05) is the shift S of example 11.6 (source 04), with pivots $p(k) = k + 1$. Source 05 adds the intersection of the iterated images; source 04 adds the insertion map E , whose image is closed and nowhere dense.

[write] (Batch 89.) Source 16 (Part X) proves proposition 11.3, theorem 11.4, and corollary 11.5 again, without having seen them, and extends them and theorem 11.7 to the cones $M_\alpha = (\mathbb{Q}^\alpha \vec{\times} \mathbb{Z})_{\geq 0}$ for every countable ordinal α (theorems 129.1, 130.5, 130.7 and 135.1). Automatic continuity of all self-embeddings fails from $\alpha = \omega + 1$ on, and closedness of all their images from $\alpha = \omega \cdot 2$ on (theorems 133.3 and 134.2; both counterexamples use choice).

12 A compactness obstruction independent of multiplication

The positive answer on Baire space does not extend to every familiar uncountable Polish space. In particular compact carriers are impossible. The underlying fact is the standard compact-cancellative-monoid argument, which we include in its commutative form.

Lemma 12.1 (Compact cancellation produces inverses). *Every compact Hausdorff commutative cancellative topological monoid is a group algebraically.*

Proof. Write the identity as 0 and fix a . Let S be the closure of $\{na : n \geq 1\}$. Continuity of addition makes S a nonempty compact subsemigroup. By the compact intersection property and Zorn's lemma, S has a minimal nonempty compact subsemigroup K .

For $x \in K$, the set $x + K$ is a nonempty compact subset of K and is a subsemigroup: for $u, v \in K$,

$$(x + u) + (x + v) = x + (x + u + v) \in x + K.$$

Minimality gives $x + K = K$. Since $x \in K$, there is $y \in K$ with $x + y = x = x + 0$, and cancellation gives $y = 0$. Thus $0 \in K \subseteq S$. But $a + X$ is compact and hence closed, and contains every positive multiple of a . Therefore it contains S and in particular 0. So $a + b = 0$ for some $b \in X$. $\square$

Corollary 12.2 (No compact positive-cone model). *No model of Presburger arithmetic has a compact Hausdorff topology with continuous addition. In particular Cantor space cannot be a carrier, although Baire space can.*

Proof. Its addition is a commutative cancellative monoid. The lemma would give an additive inverse of 1, but no nonnegative element can sum with 1 to zero. $\square$

This obstruction uses neither Borel definability nor multiplication. It is separate from [theorem 10.4](#): the real model is locally compact but not compact, and the Baire model is nowhere locally compact.

13 Formalization and reproducibility

13.1 A minimal theorem dependency graph

The construction naturally splits into independently checkable layers. The core algebra does not require real analysis or surreal numbers. First define a discretely ordered abelian group with standard residue representatives. Prove the coset-invariant finite-search lemma. Then prove coefficient normalization, elimination, completeness, and the positive-cone interpretation. Only afterward supply the real or rational-sequence carrier and its topology. The omnific realization can be verified last, as an independent bridge.

A practical Lean development can use the following *proposed* modules. These names are a plan, not existing checked files:

Proposed module	Required mathematical content
ZGroup.Basic	Least positive element, standard residue witnesses, uniqueness, and order properties of the quotient.
ZGroup.CosetCooper	Full mG -invariance and the finite-search criterion; the counterexample to weaker periodicity.
ZGroup.Elimination	Affine-term semantics over an arbitrary $\mathbb{Z}$ -group, coefficient scaling, quantifier elimination.
PolishPresburger.Real	The positive cone of $\mathbb{R} \overrightarrow{\times} \mathbb{Z}$ and its complete slice metric.
PolishPresburger.Baire	The rational-sequence cone and the explicit block/tree homeomorphism.
PolishPresburger.Structure	Order-rank obstruction, absence of semiring expansions, and self-embedding theorems.
PolishPresburger.Omnific	The fixed-support normal-form maps and their additive/order laws.

13.2 The nonstandard Cooper boundary must be explicit

The inspected repository theorem `periodic_interval` has an ordinary-integer predicate and a one-step periodicity hypothesis [15]. Its proof can iterate by any ordinary integer quotient. Over a general $\mathbb{Z}$ -group, the quotient in $x - L = mq + r$ is an element of that group, not necessarily a standard numeral. A proof by ordinary natural-number induction cannot supply invariance under that shift for an arbitrary external predicate.

The correct interface is therefore a predicate on the quotient G/mG , or equivalently the hypothesis (4.4). Affine congruence atoms satisfy it by algebra, and Boolean combinations preserve it. This makes the generalized arithmetic lemma reusable and prevents an incorrect formal transfer from being hidden by notation. The original integer theorem should be retained as an instance; it does not need to be weakened or retracted.

The existing decision procedure’s semantics are over integers. Reusing its syntax and transformations is plausible, but proving soundness over all $\mathbb{Z}$ -groups is an additional obligation. The repository’s existence of an executable integer eliminator, by itself, does not certify [corollary 4.11](#) for a new carrier.

13.3 Executable material included with this manuscript

`polish_presburger.py` implements exact finite-rank rational arithmetic and the Baire codec of [section 6](#). The codec uses a computable positive-rational enumeration and Cantor pairing, supplies addition and standard division on infinite-sequence oracles, and gives a one-change comparison approximation. Every Baire input code is valid. The lower-level encoding routine takes a positive-cone promise, documented in its interface; the public arithmetic operations preserve that promise.

`verify.py` performs deterministic tests using exact `Fraction` arithmetic and a fixed seed. The executed run produced **17,930 successful checks**. They cover pairing and rational-enumeration round trips, additive and order identities, Euclidean division, cone closure, coset-invariant finite candidates, Baire-code prefix round trips, arithmetic output prefixes, the comparison change bound, and the predecessor boundary witness. The deliberately false weak-periodicity generalization is tested as a counterexample, not asserted as a theorem.

Reproduction

With Python 3.10 or later, run `python verify.py`. No external Python package is needed. The script writes `verification_results.json`. Compile the article twice with `pdflatex -interaction=nonstopmode -halt-on-error polish_presburger_glazer.tex` to resolve references.

Tests on finite-rank elements do not quantify over arbitrary elements of $\mathbb{Q}^{\mathbb{N}}$, and finite prefixes do not decide equality of infinite streams. Polishness, induction, and universal semantic statements are established by the proofs, not by the test count. No new Lean or Rocq formalization is included in this archive.

[write] The box above gives source 05's delivery names. In this report 05's programs are `code/05-omnific-presburger-polish_presburger.py` and `code/05-omnific-presburger-verify.py`, its recorded run is `data/05-omnific-presburger-verification_results.json`, and its manuscript is merged into `article.tex`. The runner imports `polish_presburger` by its delivery name and writes its record beside itself, so it is rerun on a copy with the delivery names restored (the report's README gives the commands). The intake reran it on such a copy with Python 3.14.4: 17,930 checks passed, and the record differed from the delivered one only in the interpreter version and line endings.

13.4 The formalization interface, source 04's audit [04]

Source 04 inspected the Rocq file of the formal project rather than the Lean file, at its own pin.

13.4.1 A pinned, limited source audit

The repository files used in this manuscript were checked at commit

3ad5f878c26d2d82c092190af2f3edc50d6feb06.

The relevant Presburger source is `Logic/PresburgerArithmetic/Coq/Cooper.v`; the project description is `Logic/PresburgerArithmetic/README.md`. The latter describes the Lean pipeline through syntax, normal form, one-variable elimination, and sentence decision [17]. The present manuscript does not claim to have rebuilt that repository or audited its unrelated mathematical projects.

In the inspected Coq source, the predicate hypothesis is defined over *ordinary* integers by one-step translation invariance. The lemma `periodic_shift` then proves invariance under arbitrary integer multiples, using induction on ordinary positive and negative integer representations. This supports `periodic_has_residue` and the interval criterion [18]. That is sound for its stated standard-integer domain.

For a model-uniform version, replacing the type of integers by an arbitrary Presburger group while leaving the predicate hypothesis unchanged would be invalid: [proposition 4.6](#) is a counterexample. The quotient in $x = mq + r$ may have a nonzero divisible coordinate, and iteration through standard integer shifts does not reach mq . The repair is conceptual, not a missing induction tactic: use [definition 4.5](#). Congruence literals satisfy that stronger property directly, so [theorem 4.8](#) supplies the needed uniform lemma.

13.4.2 A concrete implementation sequence

A formalization can be divided into independently reviewable stages. The following are proposed new components, not assertions about existing repository declarations.

Abstract arithmetic semantics. Define a class of linearly ordered abelian groups with a named least positive unit and a standard finite quotient–remainder principle. Interpret the existing affine syntax in an arbitrary group of this class. Keep the syntactic integers as *coefficients*; they must not be confused with the semantic carrier.

Finite boundary theorem. Formalize m -residue invariance as invariance under differences in mG . Prove [theorem 4.8](#), separating existence of a finite candidate from any decidability hypothesis on its predicate. Then specialize to a Boolean evaluator when the residue conditions are decidable. This keeps the mathematical statement valid even when the carrier has undecidable equality or order in a computational representation.

Model-uniform elimination. Reuse the finite syntax transformations, with a fresh semantic correctness proof for coefficient normalization, cancellation in [\(4.9\)](#), bound selection, and finite substitution. The correctness endpoint should quantify over all groups satisfying the abstract assumptions. Specialization to $\mathbb{Z}$ must recover the current integer semantics rather than replace it.

Split models and topological structure. Construct G_D and its nonnegative cone, prove explicit division, and obtain the complete Presburger theory by relativization. Next formalize the G_δ argument, continuity of restricted addition, and the coordinate formulas for fixed quotient and remainder. For D_B , construct the two coding homeomorphisms rather than invoking a black-box classification of zero-dimensional Polish spaces.

Self-embeddings. First prove extension from the positive cone to the group completion and preservation of the divisible subgroup. For D_B , prove the filtration lemma [\(11.6\)](#) before introducing matrices. This exposes the reason for automatic continuity and avoids an unproved assumption that an arbitrary linear map is determined by its values on the coordinate vectors. Those vectors are *not* a vector-space basis of the full product; the finite-row conclusion requires the order argument.

13.4.3 Trust and proof-status ledger

Component	Status in this package	Dependence
Presburger characterization and QE	Classical; self-contained argument supplied	Ordered groups and finite syntax
Affirmative Polish construction	Conventional proof supplied	Explicit split group and topology
Baire homeomorphism and stream operations	Proof and executable implementation	Effective rational coding
Monus and embedding theorems	Conventional proofs supplied	Earlier lemmas in this manuscript
Finite arithmetic diagnostics	Executed successfully	Python standard library, exact fractions
New Lean/Rocq kernel certification	Not supplied	Proposed work above
Historical priority of the answer	Not established	Further literature and expert review

The distinction is especially important here because the repository already contains machine-checked material. Reusing its mathematical strategy does not automatically confer its certification status on the new semantic carrier or on the topology.

13.5 Relation to the formal project `Logic/PresburgerArithmetic` [write]

At the placement, the project's Lean development (namespace `PresburgerArithmetic`) and its Rocq file are unchanged from the two pins. The declarations the sources discuss are:

- Lean, `Lean/PresburgerArithmetic/Cooper.lean`: `periodic_shift_of_mod`, `periodic_interval`, `periodic_has_residue`, `periodic_unbounded_above`, `periodic_unbounded_below` and `cooper_finite_criterion`, all for a predicate P on Int with the one-step hypothesis $\forall x (P(x + m) \leftrightarrow P(x))$;
- Lean, `Lean/PresburgerArithmetic/Decision.lean`: the decision procedure `PresburgerArithmetic.Formula.presburgerArithmetic_decidable` and its correctness theorem `Formula.decideSentence_eq_true_iff`;
- Rocq, `Coq/Cooper.v`: `periodic`, `periodic_shift`, `periodic_has_residue`, `periodic_interval`, `periodic_unbounded_above`, `periodic_unbounded_below`, `cooper_finite_criterion` and `cooper_step_decidable`, all over $\mathbb{Z}$.

These statements are about the ordinary integers, where they are correct. [Example 4.4](#) and [proposition 4.6](#) show that their periodicity hypothesis cannot be read over a nonstandard $\mathbb{Z}$ -group, and [lemma 4.7](#) and [theorem 4.8](#) state the coset-invariant replacement that a model-uniform version would need. Neither source changed the project, and neither reports a defect in it. **No statement of this report is formalized:** the proposed modules of [section 13](#) and the stages of [section 13.4](#) are plans, and the collection's `docs/FORMALIZATION.md` maps no label of this report.

14 Further research questions

Sources 05 and 04 each end with a list of questions (ten and eleven). Four questions occur in both lists and are merged in [section 14.1](#); the remaining six of 05 and seven of 04 follow, so that this part has seventeen questions. Source 05 introduces its list as follows, and source 04 its own list as in the second paragraph. None of these questions is answered in this report; [Part II](#) bears on [question 14.4](#) and on the last questions of both lists ([section 26](#)).

[write] (Batch 87.) The sentence “None of these questions is answered in this report” was true of Parts I–V. Source 13 ([Part VII](#)) answers [question 14.8](#) for parameter-free formulas, the existence half of [question 14.11](#), [question 14.13](#), and [question 14.16](#) for joint uniform continuity; sources 13 and 15 settle [question 14.3](#) and [question 14.5](#) for topologies inherited from a Polish ordered group ([Part IX](#)). Notes after these questions give the details; the other questions, and the remaining parts of these, stay open.

[write] (Batch 89.) Source 16 ([Part X](#)) answers [questions 14.9](#) and [14.14](#) for the cones $(\mathbb{Q}^\alpha \vec{\times} \mathbb{Z})_{\geq 0}$, α a countable ordinal, and the clause of [question 14.6](#) on continuous order-preserving endomorphisms for the groups $\mathbb{Q}^\alpha$; it asks [question 14.3](#) and the first sentence of [question 14.10](#) again. Notes after these questions say what stays open.

The following questions are proposed directions arising from the proved results. Except where explicitly linked to the source question, their status is not presented as a certification that the literature has no answer.

The construction answers the printed existence question, but it leaves several sharper problems. The questions below distinguish implementation work, structural problems, and possible new obstructions. No solution to them is asserted merely by listing them.

14.1 Questions asked by both sources

Question 14.1 (A formal certificate for the affirmative answer [05, 04]). *Source 05*: Can the one-scale model, the full $\mathbb{Z}$ -group elimination theorem, and its Polish topology be formalized with a compact audit boundary in ProveIt? The key initial milestone is the generic coset-invariant Cooper lemma, rather than merely a finite test of arithmetic identities. *Source 04*: Can the complete proof of [remark 3.2](#), including the Baire homeomorphism and the full induction scheme via model-uniform elimination, be added to ProveIt with no new axioms? The first milestone should be the finite boundary lemma with residue invariance, followed by the semantic specialization checks described in [section 13.4](#).

Question 14.2 (Weak metatheory [05, 04]). *Source 05*: What weak foundational theory suffices to certify the affirmative construction, including the chosen representation of Polishness? The real model has an explicit complete metric, and the Baire model has an explicit sequence code, so a general model-existence theorem is unnecessary. This is distinct from Glazer’s Question 1 concerning the strength needed for his stronger-arithmetic obstruction. *Source 04*: What is the weakest useful subsystem of second-order arithmetic in which the Baire-space construction, its stream algorithms, and its formula-by-formula model correctness can be coded and proved? Distinguish coding all points of Baire space from asserting a set of all such points. This question concerns the positive construction, not a resolution of the separate reverse-mathematical problem posed in Glazer’s Question 1 [1].

Question 14.3 (Connected carriers [05, 04]). *Source 05*: What additional obstruction prevents or permits a continuous-addition model on a connected Polish space? Our locally compact model has countably many ray components, and the arithmetic order cannot be open or

closed. A proof for a prescribed connected carrier must use more than the order-topology observation alone. *Source 04*: Can a connected uncountable Polish space support a Presburger model with continuous addition? Our examples do not decide this: their integer coordinate is a continuous map onto a countable discrete space. Requiring all fixed remainders to be continuous would already force nontrivial clopen partitions, so the connected problem must separate continuity of addition from continuity of these definable functions.

[write] (Batch 87.) Negative for topologies inherited from a Polish ordered group: source 13 shows that every cone with a topology inherited from a Polish topological $\mathbb{Z}$ -group has continuous residues and is therefore disconnected (note after [theorem 95.4](#)), and source 15's [theorem 121.1](#) gives the locally compact case with its exact carrier. For a topology on the cone alone the question stays open ([question 103.4](#)).

[write] (Batch 89.) Source 16 asks it again ([question 138.10](#)); none of its carriers is connected.

[write] (Batch 90.) Narrowed by [Part XII](#): a cone topology with continuous addition is inherited from a Hausdorff group topology on the group of differences exactly when partial subtraction is continuous ([theorem 153.1](#)), and the group is then Polish ([theorem 154.1](#)). So the negative answer above covers exactly the presentations with continuous partial subtraction ([corollary 155.3](#)), and a connected example, if one exists, must have discontinuous partial subtraction. The question stays open.

Question 14.4 (Multiplication and growth orders [05, 04]). *Source 05*: Can one give an equally explicit Polish continuous-addition Presburger model whose underlying additive structure admits a unital semiring expansion? Any candidate must avoid a cofinal infinite additive unit. If one also requires $Q + I\text{Open}$, the multiplication cannot be Borel by the cited obstruction. Separating algebraic expandability from Borel expandability is the central issue. *Source 04*: Find admissible uncountable topological groups D whose infinite growth order embeds into each of its strict final segments, thereby passing [proposition 10.7](#). For their positive cones, determine whether any semiring multiplication exists, and, separately, which descriptive regularity it can have. Glazer's obstruction to Borel multiplication under stronger arithmetic remains relevant, but it does not by itself classify bare semiring expansions.

[write] (Batch 86C.) The first sentence of source 05's part is answered affirmatively in [Part III](#): source 06's cone of $\mathbb{Z} + t\mathbb{R}[t]$ ([theorem 38.1](#)) is an explicit uncountable, locally compact Polish model of $\text{Th}(\mathbb{N}; 0, 1, +, <)$ with continuous addition whose additive structure carries a unital semiring multiplication, continuous for the same topology. It has no cofinal infinite element and fails open induction ([theorem 38.2](#)). Whether such a model can also satisfy $Q + I\text{Open}$ with non-Borel multiplication, and source 04's growth-order question, remain open.

[merge] For [question 14.4](#): the real model is the additive monoid of one stratum of 03's ring $\mathcal{A}_{\text{fin}, \geq 0}$, which does carry a semiring multiplication, Borel and continuous off a closed nowhere dense set ([theorem 24.3](#)); but $\mathcal{A}_{\text{fin}, \geq 0}$ is not a model of Presburger arithmetic, and on it continuous addition is impossible ([theorem 25.2](#)). 03's [question 28.1](#) asks the multiplicative counterpart on $\mathcal{A}_{\text{fin}, \geq 0}$.

14.2 Further questions of source 05

Question 14.5 (Prescribed zero-dimensional locally compact carriers). Which uncountable zero-dimensional locally compact Polish spaces support continuous-addition Presburger models? Compact spaces are excluded and Baire space is realized, but Baire space is nowhere locally compact. This leaves a meaningful topological region not settled by our examples.

[write] (Batch 87.) None, among cones with a topology inherited from a Borel ordered Polish group: source 15's [corollary 121.2](#) shows that every uncountable locally compact such cone is homeomorphic to the countable sum of lines and closed rays, and its [theorem 121.4](#) shows that 0 is isolated in every zero-dimensional locally compact continuous-addition presentation. Presentations not inherited from a Polish group remain open ([question 126.1](#)).

Question 14.6 (Classifying admissible coefficient groups). Which divisible ordered Polish abelian groups have G_δ positive cone? [Theorem 5.2](#) turns this into an exact criterion for the split construction. Can their possible underlying topologies, order ranks, and continuous order-preserving endomorphisms be classified under useful additional hypotheses?

[write] (Batch 89.) For the groups $\mathbb{Q}^\alpha$, α a countable ordinal (discrete coordinates, lexicographic order), the clause on continuous order-preserving endomorphisms is answered by source 16 for the strictly order-preserving ones, the order embeddings: they are the row-finite rational matrices with strictly increasing pivot map and positive pivots ([theorem 130.5](#)); every order embedding of $\mathbb{Q}^\alpha$ into itself is continuous exactly when $\alpha \leq \omega$ ([theorems 133.1](#) and [133.3](#), the discontinuous ones built with choice); and these groups have F_σ , hence G_δ , positive cones ([\(128.3\)](#); [theorem 35.1](#)). Weakly monotone endomorphisms, the discontinuous order embeddings for $\alpha > \omega$ ([question 138.4](#)), other coefficient groups and the rest of the question stay open.

[write] (Batch 93, 4 October 2026.) Answered for separable real Hilbert spaces by [Part XVI](#) (source 22, written without knowledge of this question). Every translation-invariant total order with Borel, equivalently G_δ , cone on such a space is lexicographic in a unique orthonormal basis indexed by a countable ordinal θ ([theorem 210.1](#) and [corollary 213.3](#)); the order ranks are exactly the countable ordinals, all countably infinite ones on one carrier ([theorem 214.4](#) and [corollary 214.5](#)); the continuous order embeddings, the strictly order-preserving continuous endomorphisms, are the bounded operators with strictly increasing positive pivots ([theorem 216.2](#)), the automorphisms the bounded invertible lower-triangular ones with positive diagonal ([corollary 216.3](#)). For the first sentence, a real Polish vector space with zero dual, such as L^p with $0 < p < 1$, carries no such order at all ([theorem 217.2](#), [proposition 224.2](#) [merge]). Weakly monotone endomorphisms, other Polish vector spaces and other divisible groups stay open. The proofs are unrefereed and not formalized.

Question 14.7 (Continuity of a richer definable language). How much of Presburger definable choice can be made continuous in one uncountable Polish presentation? Standard quotients and remainders are continuous here, but total predecessor and truncated subtraction are not everywhere continuous. Is there a natural maximal collection of definable operations admitting simultaneous continuity?

Question 14.8 (Optimal finite-mind-change complexity). For a given Presburger formula, what is the smallest possible uniform change bound in the Baire presentation? The proof gives the number of order/equality atom occurrences after elimination. That bound need not be optimal and may be much larger than necessary. Relating the minimal bound to the finite difference hierarchy or to formula structure would refine the Δ_2^0 result.

[write] (Batch 87.) Answered by source 13 in the rational-stream presentation: for a parameter-free formula the optimum is computable and equals the longest alternating chain of feasible sign patterns, at most the rank of the order and equality forms and at most the arity ([theorems 100.1](#) and [100.4](#)); the arity bound is attained ([theorem 100.2](#)); and no uniform algorithm computes the optimum for a fixed parameter ([theorem 100.5](#)). A bound on revisions is not a bound on running time or on the coordinates read.

Question 14.9 (Polish elementary-submodel lattices). Which lattices of closed or clopen elementary submodels can occur in a Polish continuous-addition Presburger model? The real model has only two Polish elementary submodels with inherited topology, whereas the Baire model has infinitely many proper clopen elementary self-copies with standard intersection. What intermediate behaviors can be realized?

[write] (Batch 89.) Answered by source 16 for the family $M_\alpha = (\mathbb{Q}^\alpha \vec{\times} \mathbb{Z})_{\geq 0}$, α a countable ordinal, which contains the Baire model $M_\omega = M_B$. Its elementary submodels that are Polish in the inherited topology are the M_V for the closed $\mathbb{Q}$ -subspaces V of $\mathbb{Q}^\alpha$ ([theorem 131.1](#)); they form a complete lattice, with meet the intersection and join the closure of the sum, anti-isomorphic to the lattice of all subspaces of the finite-support space $\mathbb{Q}^{(\alpha)}$ ([theorem 132.4](#)); each is clopen exactly when its pivot subset omits finitely many coordinates, and closed nowhere dense otherwise ([theorem 132.1](#)), and is isomorphic to $M_{\text{otp}(S)}$ for its pivot subset S ([corollary 131.4](#)). The clopen self-copies intersect in the terminal tail, the standard copy exactly when α is 0 or a limit, and the closed submodels that are intersections of clopen self-copies are exactly those containing that tail ([theorems 135.1](#) and [135.2](#)). For $\alpha > \omega$ an automorphism can move a clopen elementary submodel onto a dense non-Polish one ([proposition 133.5](#), built with choice). Which lattices occur in other Polish continuous-addition Presburger models stays open.

Question 14.10 (Canonical surreal cutoffs and topologies). Find useful explicit birthday bounds for the two fixed-support omnific images and formalize the coefficient-topology transport inside those cutoffs. More generally, which support sets admit Polish coefficient topologies with continuous addition and full Presburger induction on the resulting positive cone? Such a classification should keep ring closure separate from additive closure.

[write] (Batch 89.) Source 16 asks the first sentence again for the whole family M_α ([question 138.12](#)); its realization is Part III's [theorem 40.1](#).

14.3 Further questions of source 04

The last question below and the last question of [section 14.2](#) (canonical surreal cutoffs and topologies) overlap in their second halves: both ask which set-sized supports or subgroups of **Oz** carry Polish coefficient topologies with continuous addition. They are printed separately because their first halves differ. [Theorem 25.2](#) gives a negative datum for one ring-closed family ([section 26](#)).

Question 14.11 (Beyond split topological models). Which Polish presentations of Presburger groups admit a continuous integer-coordinate retraction and a topological decomposition $D \vec{\times} \mathbb{Z}$ of the kind used here? What can occur when the algebraic extension or the topology does not split? Any characterization must separate the standard subgroup, the divisible subgroup, and the full system of residues modulo standard integers.

[write] (Batch 87.) Source 13's [theorem 95.4](#) characterizes, among Polish group presentations, those with a continuous integer-coordinate retraction and a topological decomposition $D \vec{\times} \mathbb{Z}$: exactly the algebraically split ones, for which the retraction is automatically continuous. Its [theorem 94.1](#) gives nonsplit uncountable perfect Polish models from every countable nonsplit $\mathbb{Z}$ -group, locally compact or homeomorphic to Baire space. What can occur in general is not classified.

[write] (Batch 89.) Source 16's [question 138.9](#) asks which Polish continuous-addition Presburger models have a Polish group completion with a discrete standard integer coordinate; the note after it points here and to [question 57.2](#).

Question 14.12 (Rigidity with continuous addition). Is there an uncountable rigid Presburger model with a Polish topology making addition continuous and order Borel? Rigid models are studied by Jeřábek [4]; our split examples are not rigid, because positive rational scaling of their divisible part gives nontrivial automorphisms. One must therefore move beyond the present easy symmetry.

[write] Source 13 asks this again ([question 103.5](#)); its nonsplit models are not rigid either.

[write] (Batch 92.) Answered negatively for the uncountable locally compact Polish models with continuous partial subtraction: each is a normal model of [Part XIII](#) ([proposition 182.3](#) [merge]), and each such model has a faithful continuous action of $\mathbb{R}_{>0}$ by automorphisms ([corollary 171.3](#)). Otherwise open; source 19 asks it again ([question 180.6](#)).

Question 14.13 (Exact discontinuity profiles of definable functions). For each definable function $f : M_D^k \rightarrow M_D$, can its discontinuity set be computed from a quantifier-free description of its graph? For the two main models, determine whether every such set admits a finite description in terms of equality of divisible-coordinate affine expressions and residue conditions. Keep graph complexity separate from target-valued Baire class; they are not interchangeable statements for arbitrary target spaces.

[write] (Batch 87.) Answered by source 13: the discontinuity set is computed from a quantifier-free description of the graph, as a finite union of locally closed sets inside finitely many nowhere dense affine hyperplanes ([theorem 98.3](#) and [corollary 98.4](#)); a description by equalities and residue conditions alone does not suffice, so the proposed fragment is too small ([section 98](#)); and the target-valued Baire class is determined separately ([section 99](#)).

Question 14.14 (Embedding categories for other coefficient groups). Extend [theorem 11.4](#) to products of ordered coefficient groups other than $\mathbb{Q}$, and to countable support systems with more complicated growth orders. Which hypotheses force every order embedding to be continuous? Which image topologies replace the finite-versus-infinite omitted-pivot dichotomy?

[write] (Batch 89.) Answered by source 16 for products $\mathbb{Q}^\alpha$ indexed by countable ordinals, with rational coefficients. The continuous order embeddings $\mathbb{Q}^\alpha \rightarrow \mathbb{Q}^\beta$ are the row-finite matrices with increasing positive pivots ([theorem 130.5](#)); they have continuous left inverses and closed images ([theorem 130.7](#)), and Borel ones are continuous ([theorem 130.3](#)). Every order embedding of $\mathbb{Q}^\alpha$ into itself is continuous exactly when $\alpha \leq \omega$ ([theorems 133.1](#) and [133.3](#)), and has closed image exactly when $\alpha < \omega \cdot 2$ ([theorems 134.1](#) and [134.2](#)); the failures are built with choice ([remark 133.2](#)). The finite-versus-infinite omitted-pivot dichotomy persists for the images of continuous embeddings and for every closed submodel ([theorem 132.1](#)); from $\omega \cdot 2$ on, discontinuous self-embeddings can have dense non-Borel images. Coefficient groups other than $\mathbb{Q}$ and index orders that are not well ordered stay open ([questions 138.7](#) and [138.8](#)).

[write] (Batch 93.) Answered in part for real ℓ^2 coefficients by [Part XVI](#): the bounded order embeddings $H_\theta^{\ell^2} \rightarrow H_\eta^{\ell^2}$ are the ordinal-echelon operators ([theorem 216.2](#)); Borel order embeddings are bounded, hence continuous ([lemma 216.1](#)), but every order embedding of $H_\theta^{\ell^2}$ into itself is continuous only for $\theta \leq 1$; and the image is Polish exactly when the operator is bounded below, which fails for a bounded diagonal operator from $\theta = \omega$ on ([corollary 216.8](#) and [theorem 218.4](#), [proposition 224.5](#) [merge]). The finite-versus-infinite omitted-pivot dichotomy of $\mathbb{Q}^\alpha$ is replaced by the bounded-below criterion. Other coefficient groups and index orders stay open.

Question 14.15 (Effective truth versus effective operations). The Baire presentation has computable addition and fixed division, but not a computable test for every definable relation on arbitrary input streams. Give sharp oracle bounds for evaluation of formulas in this presentation, including uniform bounds under parameter coding and quantifier elimination. Distinguish the decidability of closed Presburger sentences from the discontinuous evaluation of an atomic relation on arbitrary infinite names.

Question 14.16 (Metrics compatible with arithmetic translations). What useful uniform continuity properties of addition can be retained by compatible complete metrics after the invariant-metric obstruction in [proposition 8.4](#)? Determine whether addition can be made jointly uniformly continuous, or all translations uniformly continuous in one direction, and quantify the unavoidable distortion of their inverses. This would clarify completion arguments for nonclosed positive cones.

[write] (Batch 87.) Addition cannot be made jointly uniformly continuous for any compatible complete metric on $M_{\mathbb{R}}$, $M_{\mathbb{B}}$ or the countable-core models, and not even translations can be uniformly equicontinuous at the identity (source 13's [corollary 101.3](#)). The other parts of the question stay open.

Question 14.17 (Coefficient topologies inside surreal arithmetic). Classify set-sized omnific subgroups carrying Polish coefficient topologies with continuous addition, and determine how those topologies behave under changes of normal-form support. The natural comparison is between coefficient topology, arithmetic order, and model-theoretic elementary embedding, not between a Polish topology and the whole proper class of surreals.

15 Conclusions of Part I

The affirmative construction can be summarized by one change of viewpoint: use the lexicographic order to interpret arithmetic, and a coefficient topology to interpret continuity. Divisibility of the leading-coordinate group supplies all standard residues; quantifier elimination then supplies full Presburger induction. Neither countability of the standard model nor discreteness of the arithmetic order forces the external presentation to be countable or discrete.

The real model gives a locally compact answer, while the rational-sequence model gives a genuinely effective answer on Baire space. Their order relations have the smallest possible nontrivial Borel rank, yet they exhibit different elementary self-embedding behavior. Their realization inside **Oz** connects these results directly to the repository's normal-form arithmetic. Finally, the cofinal-infinite-element argument shows exactly why these additive examples cannot be mistaken for topological models of full arithmetic.

Source 04's conclusion [04]. The separation needed for the construction is simple: place a divisible ordered group above a discrete integer coordinate algebraically, while retaining the ordinary or product topology on that divisible group. The arithmetic becomes discretely ordered without forcing the topology to be discrete. Unique standard residues give the whole Presburger theory, not just a fragment.

The real-coordinate construction supplies a perfect locally compact example; the rational product supplies literal Baire space and effective stream operations. Their regularity is sharply delimited: every definable relation has Borel rank at most two, whereas monus is discontinuous exactly on an explicitly described geometric boundary. Their self-embeddings

range from unavoidable surjective dilations to closed nowhere dense elementary copies, but all are automatically continuous.

For Provelt, the most immediate next step is precise and finite: generalize the Cooper residue lemma with the correct subgroup-invariance hypothesis, then connect the already familiar syntax to these new semantics. That would turn the conventional answer presented here into a kernel-checked one without changing the mathematical construction.

Part II

Borel presentations and support barriers

This part is source 03, pinned to repository commit c5b4881fa. It keeps 03's definitions, in particular coefficient observability ([definition 17.1](#)) and its two exponent conventions: $X = t^{-1}$ is read as ω , and the change $q \mapsto -q$ converts the increasing t -convention into the reverse X -convention ([section 17](#)). Four symbols are renamed ([section 1.3](#)): 03's ring $\mathcal{A}$ is $\mathcal{A}_{\text{fin}}$, its rational Hahn ring H is $\mathcal{H}_{\mathbb{Q}}$, its Kleene–Brouwer embedding h is μ_{KB} , and two local sets called D are W and $\mathcal{Z}$. Several of 03's results restate results already in the collection; they keep their proofs here, because [theorem 25.2](#) needs their exact hypotheses, and a note after each names the collection's statement. In 03's text, "this report" means source 03.

16 The intersection and the questions answered

Glazer's paper *A Topological Tennenbaum Theorem* studies the interaction between arithmetic, Borel definability, and complete separable topologies. The part used here is particularly well matched to omnific arithmetic: an uncountable Polish model of Robinson arithmetic plus open induction cannot have continuous addition and Borel multiplication [[1](#), Corollary 1]. This is weaker than the hypothesis needed in his separate multiplication obstruction; we preserve that distinction throughout.

The relevant Provelt material is the surreal-number project, especially the reports on omnific notations and computer algebra, and its construction of the omnific ring and floor. The inspected repository guides explicitly distinguish research drafts, finite verification programs, and statements mapped to Lean proofs [[20](#), [21](#)]. That distinction is retained here. The repository already reports a Π_1^1 classification of unrestricted support validity, using Kleene–Brouwer coding. Thus merely repeating a well-founded-tree reduction would not constitute a new advance on this material.

Instead, we ask for an exact boundary and a positive arithmetic counterpart. For which countable exponent orders is support validity Borel? Does the answer also determine whether the whole Hahn field can be given a standard Borel presentation with readable coefficients? Can a concrete uncountable integer part have arithmetic that is continuous almost everywhere in the category sense, even though Glazer rules out global continuity? Precisely where does continuity fail?

16.1 Main theorem map

The report proves the following connected package.

1. **Support and field-presentation dichotomy.** For countable L , $\text{WO}(L)$ is Borel if and

only if L contains no copy of $\mathbb{Q}$. Otherwise it is coanalytic-complete (Theorem 18.4). For countable ordered abelian Γ , the same condition characterizes coefficient-observable standard Borel presentations of $\mathbb{R}((t^\Gamma))$ (Theorem 20.2).

2. **Exact benchmark complexities and a root-closure obstruction.** The support predicates for $\mathbb{Z}$ and $\mathbb{Z}_{\text{lex}}^d$, $d \geq 2$, are respectively Σ_2^0 -complete and Π_3^0 -complete. A nontrivial countable exponent group in a square-root-closed full real Hahn field cannot lie on the Borel side (Theorem 19.1, Corollary 20.4).
3. **Uniform bounds and localization.** Every Borel coefficient-observable family of valid series has its support order types bounded below one countable ordinal. Arbitrarily large countable support types, and coanalytic-complete validity, already occur in a fixed 0–1 coefficient slice inside $[X, 2X)$ (Theorems 21.2 and 22.1).
4. **A positive model with exact negative boundaries.** The finite principal-part model $\mathcal{A}_{\text{fin}, \geq 0}$ satisfies open induction and has a Polish presentation with explicit closed nowhere dense discontinuity sets for both operations. Yet no Polish topology with the same Borel sets makes addition continuous everywhere (Theorems 23.2, 24.3, and 25.2).

The order-theoretic dichotomy is deduced from classical Hausdorff theory and well-founded-tree completeness. The uniform bound is an application of classical analytic boundedness, for which we include a proof. The integer-part model is of Shepherdson type, not a newly discovered model of arithmetic. The report's claims of contribution concern the explicit combined classification, the observable-field criterion, the benchmark calculations, and the exact topological analysis. These are proved here; bibliographic priority is not asserted.

16.2 A precise source boundary

The source audit used selected repository guides at commit

c5b4881fa603339c7ff90f5e1098ede0651243d8.

The inspected paths are recorded in Appendix D. This was not an exhaustive audit of the repository, nor a rebuild of its Lean development. No new theorem below depends on an unreviewed repository theorem: the required normal-form and integer-part arguments are supplied independently. The repository is used to identify the research overlap and to avoid claiming its previously recorded results as new.

17 Conventions and the meaning of a presentation

17.1 Two opposite exponent conventions

For an ordered abelian group Γ , a Hahn series is written

$$f = \sum_{\gamma \in \Gamma} a_\gamma t^\gamma, \quad a_\gamma \in \mathbb{R},$$

with $\text{supp}(f) = \{\gamma : a_\gamma \neq 0\}$ well-ordered in the increasing order of Γ . Its sign is the sign of the coefficient at the least exponent. Thus t is infinitesimal when $\Gamma = \mathbb{Q}$ and $t = t^1$.

For the omnific applications we instead write $X = t^{-1}$, so X is positive infinite. In

$$f = \sum_{q \in \mathbb{Q}} b_q X^q,$$

the support must be *reverse* well-ordered: every nonempty subset has a greatest exponent. The change $q \mapsto -q$ converts one convention into the other. All descriptive-set-theoretic statements in Sections 18 and 19 use the increasing convention. Section 22 explicitly reverses it.

The ring

$$\mathcal{H}_{\mathbb{Q}} = \left\{ n + \sum_{q \in S} b_q X^q : n \in \mathbb{Z}, S \subseteq \mathbb{Q}_{>0} \text{ reverse well-ordered} \right\} \quad (17.1)$$

is a set-sized rational-exponent omnific ring. Replacing X^q by the Conway monomial ω^q identifies it with the corresponding normal-form subring of **Oz**. Hahn addition and multiplication agree with those normal-form operations. The normal-form viewpoint and its ordered-algebraic context are standard; see [11, 5].

17.2 Borel spaces, completeness, and observability

A Polish space is a separable topological space admitting a compatible complete metric. A standard Borel space is a measurable space isomorphic to a Borel subset of a Polish space. We use the usual boldface pointclasses: Σ_2^0 consists of countable unions of closed sets, Π_3^0 of countable intersections of Σ_2^0 sets, and Π_1^1 denotes coanalytic sets. An analytic set is a projection of a Borel set in a product of Polish spaces. Completeness is with respect to continuous reductions from zero-dimensional Polish spaces. Standard references are [8, 9].

Definition 17.1. Let K be a set of series over a countable exponent alphabet. A *coefficient-observable standard Borel parametrization* of K consists of a standard Borel space Y and a surjection $\theta : Y \rightarrow K$ such that, for every exponent γ , the map

$$y \mapsto [t^\gamma] \theta(y)$$

is Borel as a map to $\mathbb{R}$. Repetitions of names are allowed.

No Borel structure on K is presupposed in this definition. Since the alphabet is countable, coefficient observability says exactly that the composite code map $Y \rightarrow \mathbb{R}^\Gamma$ is Borel. This is stronger than the mere existence of a set-theoretic bijection with the reals, and different from an abstract Borel presentation that is allowed to make every canonical coefficient unobservable. Our negative results explicitly retain the observability hypothesis.

Definition 17.2. For a countable linear order L , put

$$\text{WO}(L) = \{S \in 2^L : (S, <_L) \text{ is well-ordered}\}.$$

The empty support is allowed. The order L is *scattered* if it contains no order-isomorphic copy of $\mathbb{Q}$.

A subset of 2^L is a set of codes, not a new collection of proper-class objects. In particular, asking whether $\text{WO}(L)$ is Borel is an ordinary set-sized descriptive-set-theoretic question.

18 The exact support dichotomy

18.1 A complete obstruction on a dense countable alphabet

Lemma 18.1. *For every countable linear order L , $\text{WO}(L)$ is coanalytic.*

Proof. A support S fails to be well-ordered exactly when there exists $(x_n)_{n \in \mathbb{N}} \in L^{\mathbb{N}}$ with $x_n \in S$ and $x_{n+1} <_L x_n$ for every n . The relation on $(S, (x_n))$ specified by these countably many coordinate conditions is closed, with L discrete. Its projection is analytic. Its complement is $\text{WO}(L)$. $\square$

Let Tr be the closed space of trees $T \subseteq \mathbb{N}^{<\mathbb{N}}$, where trees are closed under initial segments, and let $\text{WF} \subseteq \text{Tr}$ be the well-founded trees. Recall briefly why WF is coanalytic-complete. If A is coanalytic in Baire space, its complement is the projection of the body of a tree R on pairs of finite sequences. For $x \in \mathbb{N}^{\mathbb{N}}$, let

$$T_x = \{s : (x \upharpoonright |s|, s) \in R\}.$$

Membership of a fixed s depends on finitely many coordinates of x , so $x \mapsto T_x$ is continuous, and $x \in A$ exactly when T_x has no branch. This is the standard universal-tree proof [9, Section 16.B].

Lemma 18.2 (Kleene–Brouwer coding). *There is a fixed injection $\mu_{\text{KB}} : \mathbb{N}^{<\mathbb{N}} \rightarrow \mathbb{Q} \cap (0, 1)$ such that*

$$T \in \text{WF} \iff \mu_{\text{KB}}[T] \in \text{WO}(\mathbb{Q}).$$

Consequently $\text{WO}(\mathbb{Q})$ is Π_1^1 -complete.

Proof. Order the finite sequences by the Kleene–Brouwer order: a proper extension precedes its prefix, and otherwise the first differing entry determines the order, with the ordinary order on $\mathbb{N}$. This is a countable linear order. Embed it in $\mathbb{Q} \cap (0, 1)$ as follows. Enumerate its elements. At each finite stage, put the next element at a rational strictly between its nearest already assigned lower and upper neighbors, using 0 and 1 as external bounds when needed. A midpoint works. The resulting μ_{KB} is one fixed order embedding of the entire tree alphabet, not an embedding selected separately for each T .

If T has a branch, its successively longer prefixes form a strictly descending Kleene–Brouwer sequence. If T is well-founded, well-founded induction over its nodes shows that the Kleene–Brouwer order on each descendant subtree is a well-order: it is the increasing $\mathbb{N}$ -indexed sum of the child-subtree orders, followed by the root. Therefore $\text{KB}(T)$ is a well-order exactly when T is well-founded.

The map $T \mapsto \mu_{\text{KB}}[T]$ is continuous. At an exponent in the range of μ_{KB} , its characteristic function reads exactly one tree-membership bit; outside the range it is constantly zero. Thus it reduces WF to $\text{WO}(\mathbb{Q})$. Lemma 18.1 supplies membership in the pointclass. $\square$

[write] Lemma 18.2 and part (ii) of theorem 18.4 below are boldface counterparts of results already in the collection’s omnific-notations report: `onot:neg:lem:kbtree` (a tree is well-founded exactly when its Kleene–Brouwer order is a well-order), `onot:neg:thm:pionesone` (Π_1^1 -completeness, for recursive trees, of the validity of decidable dyadic supports in $[1, 2)$ with coefficient one) and `onot:neg:rem:secondroute` (the midpoint codings of that report’s sources 08 and 09, which are the construction of μ_{KB} up to the target interval). Source 03 reproves the reduction to make the continuous (boldface) form explicit, as its contribution ledger says (appendix D). What is new in section 18 is the dichotomy between scattered and nonscattered orders.

18.2 Why scattered alphabets are different

We use the classical countable Hausdorff characterization: starting with finite linear orders, closing under $\mathbb{Z}$ -indexed sums and finite sums through countable ordinal stages produces precisely the countable scattered orders. A precise formulation is recalled in [10, Definition 1]. Only this characterization, not its more advanced rank consequences, is needed below.

Lemma 18.3. *Suppose $L = \sum_{i \in \mathbb{Z}} L_i$ is a lexicographic sum and each $\text{WO}(L_i)$ is Borel. Then $\text{WO}(L)$ is Borel.*

Proof. For $S \subseteq L$, let $S_i = S \cap L_i$ and $I(S) = \{i : S_i \neq \emptyset\}$. Then

$$S \in \text{WO}(L) \iff (\forall i \in \mathbb{Z} S_i \in \text{WO}(L_i)) \wedge (\exists N \in \mathbb{Z} \forall i < N S_i = \emptyset). \quad (18.1)$$

Necessity of the second condition follows because unboundedly negative occupied indices yield an infinite descending sequence, choosing one element from each of a decreasing sequence of such indices. Conversely, when the condition holds, S is a well-ordered sum of well-ordered fibers over a subset of an integer tail.

Each restriction map $S \mapsto S_i$ is continuous. The fiber-empty conditions are closed. The right-hand side of (18.1) is therefore Borel. For a finite sum the analogous criterion has only finitely many fiber conditions and no additional obstruction. $\square$

Theorem 18.4 (Support dichotomy). *For every countable linear order L :*

- (i) *if L is scattered, $\text{WO}(L)$ is Borel;*
- (ii) *if L is not scattered, $\text{WO}(L)$ is Π_1^1 -complete.*

In particular, Borelness of $\text{WO}(L)$ is equivalent to scatteredness of L .

Proof. For a finite order the support predicate is the whole finite code space. Induction through the Hausdorff construction, using Lemma 18.3 and the finite-sum observation, proves (i).

For (ii), fix an order embedding $j : \mathbb{Q} \rightarrow L$. Extending a support by zero outside $j[\mathbb{Q}]$ gives a continuous map $S \mapsto j[S]$ from $2^{\mathbb{Q}}$ to 2^L , preserving and reflecting well-orderedness. Compose this map with Lemma 18.2, and use Lemma 18.1. A coanalytic-complete set is not Borel, completing the equivalence. $\square$

Remark 18.5. The quantification here is over *all subsets of one fixed countable order*. It is not a statement about the Borel complexity of the class of all scattered orders, nor about finding a Hausdorff decomposition uniformly from an arbitrary code for L . The fixed decomposition used in the positive proof need not be computable.

[write] (Batch 90, 4 October 2026.) The report Algebra/SurrealNumbers/docs/foundation s-and-computation/cantor-families-of-surreal-subfields (batch-90 manuscript 04; AI-assisted, unrefereed, not formalized) codes subsets $S \subseteq \mathbb{Q}$ into countable real closed subfields $K_S^{(1)}$ of **No** so that $K_S^{(1)}$ embeds in $K_T^{(1)}$ exactly when $(S, <)$ embeds in $(T, <)$ (its Theorem 5.1). For the source field of order type ω^* the embedding locus is $2^{\mathbb{Q}} \setminus \text{WO}(\mathbb{Q})$, and its Theorem 6.3 (csf:thm:fixed-hard) proves that locus analytic-complete by the Kleene–Brouwer argument of Lemma 18.2, which its Lemma 6.2 repeats. By the dichotomy above, once those parameters are restricted to subsets of a fixed countable $L \subseteq \mathbb{Q}$, the locus is Borel exactly when L is scattered; that report records this, in a note after its Theorem 6.3, as bearing on its Question 13.1 without answering it.

[write] (Batch 96.) Source 25's cut test ([theorem 258.2](#)) proves part (i) at the level Π_3^0 without a Hausdorff decomposition, and its [theorem 266.3](#) shows that recognizing scatteredness is Π_1^1 -complete even among the compatible orders of one fixed free abelian group with a prescribed least positive element.

19 Exact benchmarks: integers, finite lexicographic powers, and rationals

The dichotomy gives a qualitative boundary. On useful discrete alphabets one can calculate the precise Borel level.

Theorem 19.1. *With the product topology on the corresponding binary support spaces:*

- (i) $\text{WO}(\mathbb{Z})$ is Σ_2^0 -complete;
- (ii) for every integer $d \geq 2$, $\text{WO}(\mathbb{Z}_{\text{lex}}^d)$ is Π_3^0 -complete;
- (iii) every well-ordered subset of $\mathbb{Z}_{\text{lex}}^d$ has order type at most ω^d , and this bound is attained.

Proof. A subset of $\mathbb{Z}$ is well-ordered if and only if it is bounded below. Hence

$$\text{WO}(\mathbb{Z}) = \bigcup_{N \in \mathbb{Z}} \{S : \forall n < N (n \notin S)\},$$

a countable union of closed sets. The continuous map $A \subseteq \mathbb{N} \mapsto \{-n - 1 : n \in A\}$ reduces the familiar Σ_2^0 -complete set Fin of finite subsets of $\mathbb{N}$ to $\text{WO}(\mathbb{Z})$. This proves (i).

For completeness of the benchmark being used, Fin is countable and dense in $2^{\mathbb{N}}$. Were it G_δ , it would be comeager, while it is also meager; the Baire theorem rules this out. The usual completeness criterion for Borel sets in zero-dimensional Polish spaces then gives its Σ_2^0 -completeness [9, Section 16.A].

For $d = 2$, view the order as $\sum_{i \in \mathbb{Z}} \mathbb{Z}$. Formula (18.1) is the intersection of a Π_3^0 condition (every fiber is bounded below) and a Σ_2^0 condition (the occupied outer indices are bounded below). The result is Π_3^0 . For $d > 2$, the same formula and induction work: countable intersections of Π_3^0 sets remain Π_3^0 .

For hardness, use

$$\text{Fin}^{\mathbb{N}} = \{a \in 2^{\mathbb{N} \times \mathbb{N}} : \forall m \{n : a(m, n) = 1\} \text{ is finite}\}.$$

This set is Π_3^0 -complete. Indeed, any Π_3^0 subset of a zero-dimensional Polish space is a countable intersection of Σ_2^0 sets; reduce each of these sets to Fin and take the product of the reductions. Map a to the support

$$S_a = \{(m, -n - 1) : a(m, n) = 1\} \subseteq \mathbb{Z}_{\text{lex}}^2. \quad (19.1)$$

The outer indices are automatically bounded below by zero. A fiber is well-ordered exactly when its row is finite. Thus (19.1) is a continuous reduction. Embedding $\mathbb{Z}_{\text{lex}}^2$ into $\mathbb{Z}_{\text{lex}}^d$ gives hardness for all $d \geq 2$.

Finally, a well-ordered subset of $\mathbb{Z}$ has type at most ω . Inductively, a well-ordered support in $\mathbb{Z}^d$ is a sum of at most ω fibers of type at most ω^{d-1} , and so has type at most $\omega^{d-1} \cdot \omega = \omega^d$. The subset $\mathbb{N}^d$, with the same lexicographic convention (first coordinate most significant), has type exactly ω^d . $\square$

Exponent alphabet	Valid-support complexity	Support-type bound
$\mathbb{Z}$	Σ_2^0 -complete	$\leq \omega$
$\mathbb{Z}_{\text{lex}}^d, d \geq 2$	Π_3^0 -complete	$\leq \omega^d$
A fixed countable scattered order	Borel	Some countable bound
$\mathbb{Q}$, or any countable nonscattered order	Π_1^1 -complete	Unbounded in ω_1

The last two support-type assertions follow also from Theorem 21.2 below. An important feature of the table is that the finite lexicographic rank can grow while the exact Borel level stays at Π_3^0 . Support order type and Borel complexity measure different things. Conversely, passing from $\mathbb{Z}$ to its divisible hull $\mathbb{Q}$ is a genuine jump out of the entire Borel hierarchy.

[write] (Batch 96.) Parts (i) and (ii) are the cases $\mathbb{Z}$ and $\mathbb{Z}_{\text{lex}}^d$ of source 25's theorem 259.6, which covers every countable order: a scattered, not well-ordered L gives Σ_2^0 or Π_3^0 according as $\omega^* \cdot \omega$ does not or does embed in L , so the observation above holds for all scattered orders.

20 An observable-presentation dichotomy for full Hahn fields

Fix a countable ordered abelian group Γ . Let

$$V_\Gamma = \{a \in \mathbb{R}^\Gamma : \text{supp}(a) \in \text{WO}(\Gamma)\}.$$

As a set, V_Γ is the canonical coefficient presentation of the full Hahn field $K_\Gamma = \mathbb{R}((t^\Gamma))$.

Lemma 20.1. *If Γ is scattered, V_Γ is Borel. If Γ is nonscattered, V_Γ is Π_1^1 -complete.*

Proof. The support map

$$\sigma : \mathbb{R}^\Gamma \rightarrow 2^\Gamma, \quad \sigma(a)(\gamma) = 1 \iff a_\gamma \neq 0$$

is Borel. Thus the scattered case follows from Theorem 18.4. In all cases, failure of validity is witnessed by an infinite descending sequence of nonzero coordinates, an analytic condition, so V_Γ is coanalytic. The binary-coefficient inclusion $2^\Gamma \hookrightarrow \mathbb{R}^\Gamma$ is continuous and reduces $\text{WO}(\Gamma)$ to V_Γ , proving completeness in the remaining case. $\square$

Theorem 20.2 (Observable Hahn-field criterion). *For countable ordered abelian Γ , the following are equivalent:*

- (i) Γ is scattered as a linear order;
- (ii) V_Γ is a Borel subset of $\mathbb{R}^\Gamma$;
- (iii) K_Γ admits a coefficient-observable standard Borel parametrization, allowing repeated names;
- (iv) K_Γ admits a standard Borel presentation with Borel coefficient maps, Borel addition and multiplication, Borel inversion on the nonzero elements, and Borel order.

Proof. Lemma 20.1 gives (i) $\Leftrightarrow$ (ii). Assume (ii), and use V_Γ itself as the standard Borel domain. Addition and negation are coordinatewise and therefore Borel. For $\Gamma = \{0\}$ all assertions

reduce to those for $\mathbb{R}$. Otherwise, for multiplication enumerate $\Gamma = \{\delta_0, \delta_1, \dots\}$ without repetition. At a fixed exponent γ , the product coefficient is

$$(ab)_\gamma = \lim_{N \rightarrow \infty} \sum_{j < N} a_{\delta_j} b_{\gamma - \delta_j}. \quad (20.1)$$

For valid Hahn supports, only finitely many terms are nonzero. To see the finiteness, infinitely many contributing δ would contain a strictly increasing sequence in the well-ordered support of a ; the corresponding $\gamma - \delta$ would be a strictly decreasing sequence in the support of b . This is impossible. Thus the sums in (20.1) eventually stabilize. They are continuous on the ambient product, and their pointwise limit on the Borel domain is Borel.

The standard Hahn-field theorem supplies closure under multiplication and the existence of multiplicative inverses; its well-ordered-support lemma is the usual Neumann lemma [11]. The graph

$$\{(a, b) \in (V_\Gamma \setminus \{0\}) \times V_\Gamma : ab = 1\}$$

is Borel and has exactly one point in each vertical section. The Lusin–Novikov theorem gives Borel inversion. Equivalently, for an open set U the inverse image under inversion is the projection of this graph intersected with $V_\Gamma \times U$, whose sections have at most one point.

Positivity is Borel because it is expressed by

$$\exists \gamma \in \Gamma (a_\gamma > 0 \wedge \forall \delta < \gamma a_\delta = 0).$$

There are only countably many coordinates. Order follows by subtraction. This proves (iv), which immediately implies (iii).

Assume (iii). The coefficient code map from its standard Borel parameter space to $\mathbb{R}^\Gamma$ is Borel, and its image is exactly V_Γ . Consequently V_Γ is analytic. For nonscattered Γ , Lemma 20.1 makes it coanalytic-complete and hence nonanalytic. Thus Γ must be scattered, proving (i). $\square$

Remark 20.3 (What the negative direction does and does not exclude). The theorem excludes even a surjective system of Borel-observable names, not just unique or canonical names. It does not exclude abstract Borel models in which the coefficient functions fail to be Borel. Nor does it exclude useful Borel subfields of K_Γ , including real closed subfields. The word *full* is essential.

[write] (Batch 96.) Source 25’s [theorem 265.2](#) proves (i) $\Leftrightarrow$ (iii) again, with closed Polish name spaces and continuous decoders, and extends it to strongly summable families and to Borel witness relations.

20.1 Root closure forces the non-Borel side for full fields

Corollary 20.4. *Suppose $\Gamma \neq \{0\}$ is countable and every positive element of the full Hahn field $\mathbb{R}((t^\Gamma))$ has a square root in that field. Then Γ is nonscattered, and the field admits no coefficient-observable standard Borel parametrization. In particular, this holds whenever the full Hahn field is real closed.*

Proof. For every $\gamma \in \Gamma$, the positive monomial t^γ has a root f with $f^2 = t^\gamma$. Taking least exponents gives $2 \min \text{supp}(f) = \gamma$. Thus Γ is 2-divisible. For $a < b$ in Γ , the element $a + (b - a)/2$ lies strictly between them. A nontrivial ordered group has no endpoints, so Γ is a countable dense linear order without endpoints and is order-isomorphic to $\mathbb{Q}$. Apply Theorem 20.2. $\square$

This is a necessary-condition argument; it does not use a classification of which Hahn fields are real closed. It also explains the source of the jump: arbitrary well-ordered supports over a densely divisible scale are the problem, not the mere availability of algebraic roots. The Puiseux field in Section 23 supplies the contrasting real closed Borel subfield.

[write] (Batch 96.) Source 25's [corollary 260.3](#) needs only one positive exponent that can be halved indefinitely, and adds that the code set is Π_1^1 -complete.

21 Every observable family has a countable support bound

We now strengthen the nonexistence statement into a restriction on every individual Borel system of names. The essential input is the classical boundedness theorem for analytic families of well-founded trees [8]. The following proof makes the particular form we need explicit.

Lemma 21.1 (Analytic boundedness, with proof). *If $\mathcal{T} \subseteq \text{WF}$ is analytic, there is a countable ordinal β bounding the root ranks of all trees in $\mathcal{T}$.*

Proof. The empty tree is assigned rank zero and can be discarded. If no trees remain, take $\beta = 1$. Otherwise write the remaining analytic family $\mathcal{T}$ as the projection of a closed set $C \subseteq \mathbb{N}^{\mathbb{N}} \times \text{Tr}$. Fix a binary coding of trees using an enumeration of $\mathbb{N}^{<\mathbb{N}}$. Construct a countable tree U whose level- k nodes are triples (s, u, v) with $s, u \in \mathbb{N}^k, v \in 2^k$, for which some $(y, T) \in C$ satisfies

$$u = y \upharpoonright k, \quad v = \text{code}(T) \upharpoonright k, \quad s \in T.$$

Taking initial segments preserves these conditions, so U is a tree.

If U had an infinite branch, its three coordinates would give $b, y \in \mathbb{N}^{\mathbb{N}}$ and a binary code z . For each k choose a witness $(y_k, T_k) \in C$ for the level- k node. The witnesses converge to (y, T) , where T has code z : for every fixed number of coordinates the witnesses agree with (y, z) once k is large. Since C and Tr are closed, $(y, T) \in C$. For every j , the node $b \upharpoonright j$ belongs to T_k for all $k \geq j$, and hence belongs to T . Thus $T \in \mathcal{T}$ has a branch, a contradiction. Therefore U is well-founded.

For each $(y, T) \in C$, the assignment

$$s \mapsto (s, y \upharpoonright |s|, \text{code}(T) \upharpoonright |s|)$$

embeds the extension tree T into U , taking children to children. Induction on node ranks gives $\text{rk}(T) \leq \text{rk}(U)$. A well-founded countable tree has countable rank. Taking $\beta = \text{rk}(U) + 1$ gives a strict bound. $\square$

Theorem 21.2 (Uniform support bound). *Let L be countable, Y standard Borel, and $c : Y \rightarrow \mathbb{R}^L$ Borel. Suppose every $\text{supp}(c(y))$ is well-ordered in L . Then there is $\alpha < \omega_1$ such that*

$$\text{otp}(\text{supp}(c(y))) < \alpha \quad \text{for every } y \in Y.$$

The same conclusion holds for reverse well-ordered supports, with the reverse support order used to compute the type.

Proof. For $S \subseteq L$, form the descending-sequence tree D_S . Its root is the empty sequence, and its other nodes are finite strictly decreasing sequences of members of S . The map $S \mapsto D_S$ is continuous on binary support codes; the composite $y \mapsto D_{\text{supp}(c(y))}$ is Borel, since each

nonzero coefficient test is Borel. Its image is an analytic family of well-founded trees. Apply Lemma 21.1.

For a well-ordered S of type ξ , the root rank of D_S is exactly ξ . Indeed, the rank of a node ending in $s \in S$ is the order type of the initial segment of S below s ; induction verifies the recursion. The root rank is the supremum of these ranks plus one, namely ξ . Reversing L proves the last assertion. $\square$

Corollary 21.3. *For a countable linear order L , the following are equivalent:*

- (i) L is scattered;
- (ii) the order types of all well-ordered subsets of L have a common countable upper bound;
- (iii) $\text{WO}(L)$ is analytic;
- (iv) $\text{WO}(L)$ is Borel.

Proof. Theorem 18.4 gives (i) $\Leftrightarrow$ (iv) and excludes (iii) in the nonscattered case. If (iv) holds, apply Theorem 21.2 to the identity binary-coefficient coding of $\text{WO}(L)$ to obtain (ii). If L is nonscattered, it contains $\mathbb{Q}$, and every countable ordinal embeds in $\mathbb{Q}$ by the finite-stage rational insertion construction used in Lemma 18.2. Its well-ordered suborders therefore have unbounded countable types, contradicting (ii). $\square$

The bound is uniform over the entire parameter space, even if it is uncountable. It is not asserted to be computable from an arbitrary code for the Borel map. In particular, this is a boldface descriptive theorem, not an unqualified statement that every effective system has a computable ordinal bound.

22 The obstruction already lies in one omnific interval

Return to the X -convention. Let $I = \mathbb{Q} \cap (1/3, 2/3)$ and consider the compact space of candidate coefficient arrays

$$C = \left\{ X + \sum_{q \in I} \varepsilon_q X^q : \varepsilon_q \in \{0, 1\} \right\}. \quad (22.1)$$

Here the displayed sums are only *candidate codes*. A member of C denotes a Hahn or surreal normal form only when its nonzero support is reverse well-ordered.

Theorem 22.1 (Bounded-interval localization). *The valid-code subset C_{valid} of (22.1) is Π_1^1 -complete. Every valid value lies in the omnific interval $[X, 2X)$. The reverse support order types in this slice are unbounded in ω_1 .*

Proof. Take the fixed increasing Kleene–Brouwer embedding μ_{KB} from Lemma 18.2 and put

$$e(s) = \frac{2}{3} - \frac{\mu_{\text{KB}}(s)}{3} \in I.$$

This reverses the Kleene–Brouwer order. Define the candidate code

$$z_T = X + \sum_{s \in T} X^{e(s)}. \quad (22.2)$$

The map $T \mapsto z_T$ is continuous in the coefficient product topology. The support below the leading exponent 1 is reverse well-ordered exactly when $\text{KB}(T)$ is well-ordered, exactly when $T \in \text{WF}$. Appending the single larger exponent 1 changes none of these equivalences. Coanalytic membership follows from the descending-sequence characterization applied to the reverse exponent order.

For every valid code, the lower part is nonnegative because all its nonzero coefficients are positive, while $2X - z_T$ has greatest exponent 1 and coefficient 1. Hence $X \leq z_T < 2X$; the lower equality occurs when the lower support is empty.

Every countable ordinal ξ embeds into I in its reverse order, by the same rational-insertion principle. Taking that binary support gives total reverse support type $1 + \xi$. These types are unbounded below ω_1 . $\square$

Corollary 22.2. *There is no coefficient-observable standard Borel parametrization onto C_{valid} , even with repeated names. More generally, no such parametrization can cover a family whose support types are unbounded in ω_1 .*

Proof. The general statement is the contrapositive of Theorem 21.2. Apply it to Theorem 22.1. $\square$

This localization removes several possible explanations for the obstruction. It is not due to large coefficients, an unbounded range of leading degrees, or a need to represent the proper class of all surreals. The leading monomial is fixed, the other coefficients are bits, the remaining exponents lie in one fixed rational band, and all actual values lie in one bounded omnific interval.

Remark 22.3 (Relation to Provel's existing lower bound). The use of well-founded trees, and the fact that general support validation is coanalytic-hard, are already recorded in the omnific-notations report [21]. The present localization is included as an explicit corollary with a fixed coefficient cylinder. The more substantial extensions are the scattered/nonscattered classification and its observable-field and uniform-family formulations. None of these descriptions establishes global priority for a result outside the inspected material.

[write] Theorem 22.1 and the collection's `onot:neg:thm:pionesone` are the same phenomenon in two slices, as the remark above says: there, lightface completeness for recursive trees, with decidable dyadic supports in $[1, 2)$ and coefficient one; here, boldface completeness with the fixed leading monomial X , bits on the band $(1/3, 2/3)$ and values in $[X, 2X)$. The unboundedness of the support types and corollary 22.2 are not in the omnific-notations report.

23 A real closed Borel subfield and its finite omnific integer part

The preceding obstructions do not make exact non-Archimedean algebra impossible. A particularly useful positive model separates finite support, bounded-denominator infinite support, and arbitrary Hahn support.

Let

$$\mathcal{P} = \bigcup_{n \geq 1} \mathbb{R}((t^{1/n})), \quad X = t^{-1}. \quad (23.1)$$

The union is a field: any two denominators have a common multiple. Newton–Puiseux says that the corresponding complex field $\bigcup_n \mathbb{C}((t^{1/n}))$ is algebraically closed [12, 13]. It is exactly $\mathcal{P}(i)$, and $\mathcal{P}$ is ordered by the leading coefficient. Hence $\mathcal{P}$ is real closed.

Define

$$\mathcal{A}_{\text{fin}} = \left\{ n + \sum_{q \in S} a_q X^q : n \in \mathbb{Z}, S \subseteq \mathbb{Q}_{>0} \text{ finite}, a_q \in \mathbb{R} \right\}. \quad (23.2)$$

Zero coefficients may be removed to obtain a unique finite normal form. The notation $\mathbb{Z} \oplus \bigoplus_{q>0} \mathbb{R}X^q$ describes an additive direct sum. It does not mean that $\mathcal{A}_{\text{fin}}$ is an $\mathbb{R}$ -algebra: its constant subring is $\mathbb{Z}$, not $\mathbb{R}$.

Lemma 23.1 (Exact principal-part floor). *The ring $\mathcal{A}_{\text{fin}}$ is an integer part of $\mathcal{P}$. More explicitly, write uniquely*

$$f = p + r + \varepsilon,$$

where p contains the positive powers of X , $r \in \mathbb{R}$ is the constant term, and ε has only negative powers of X . Then p is finite, ε is infinitesimal, and the unique $a \in \mathcal{A}_{\text{fin}}$ with $a \leq f < a + 1$ is

$$a = p + \begin{cases} \lfloor r \rfloor - 1, & r \in \mathbb{Z} \text{ and } \varepsilon < 0, \\ \lfloor r \rfloor, & \text{otherwise.} \end{cases} \quad (23.3)$$

Proof. For $f \in \mathbb{R}((t^{1/n}))$, its exponents are bounded below in the lattice $\frac{1}{n}\mathbb{Z}$. Only finitely many are negative in the t -convention, so the positive- X principal part p is finite. Every remaining nonconstant term has negative X -exponent; if the tail is nonzero, its greatest exponent is negative, so it is smaller in absolute value than every positive real.

When $r \notin \mathbb{Z}$, the fractional part $r - \lfloor r \rfloor$ lies strictly between 0 and 1, and adding an infinitesimal preserves both inequalities. When $r \in \mathbb{Z}$ and $\varepsilon \geq 0$, the remainder is in $[0, 1)$. When $r \in \mathbb{Z}$ and $\varepsilon < 0$, subtracting one from the proposed floor makes the remainder $1 + \varepsilon$, which also lies in $[0, 1)$. This proves the inequalities.

The ring is discretely ordered. A positive element with nonempty positive support exceeds every real, in particular 1; a positive constant is an ordinary positive integer. Thus 1 is its least positive element, and two distinct elements of $\mathcal{A}_{\text{fin}}$ cannot both satisfy the floor inequalities. $\square$

The boundary correction in (23.3) is essential. For example, $\lfloor -X^{-1} \rfloor_{\mathcal{A}_{\text{fin}}} = -1$, not 0. This correction is also explicitly recorded in ProveIt's current omnific-floor description [20]; it is not a new correction discovered by this report.

[write] [Lemma 23.1](#) is the collection's `odg:thm:floor` (omnific-diophantine-geometry report) read in the Puiseux field. Under 03's identification of X with ω (section 17), $\mathcal{P}$ is a subfield of $\mathbf{No}$; p , r and ε are the positive-growth part, the real constant and the infinitesimal tail of `odg:eq:floor`; and $\mathbf{Oz} \cap \mathcal{P} = \mathcal{A}_{\text{fin}}$, because the positive-growth part of a Puiseux series is finite. Formula (23.3) is therefore the same formula, and its negative-infinitesimal correction is the case $r \in \mathbb{Z}$, $\varepsilon < 0$ there. `odg:thm:floor` is proved in Lean for every surreal (`Algebra/SurrealNumbers/Surreal/Foundations/OmnificFloor.lean`: `omnificFloor_spec`, `existsUnique_omnific_integerPart`, `omnificFloor_of_integer_negative_tail`; row "Full `odg:thm:floor`" of `docs/FORMALIZATION.md`). The Lean development does not mention $\mathcal{P}$ or $\mathcal{A}_{\text{fin}}$, so the lemma as stated here, with the finiteness of p and the integer-part property of $\mathcal{A}_{\text{fin}}$ in $\mathcal{P}$, is not formalized. Source 03 credits the Surreal README for the correction.

Theorem 23.2. *The nonnegative cone $\mathcal{A}_{\text{fin}, \geq 0}$ is an uncountable model of Robinson arithmetic plus open induction in the ordered semiring language.*

Proof. A discretely ordered ring gives the elementary semiring and predecessor axioms on its nonnegative cone. In particular, if $x > 0$ then $x - 1 \geq 0$. Uncountability follows already from the distinct elements aX , $a > 0$ real.

We prove open induction directly, using the familiar integer-part direction of Shepherdson's theorem [6, 7]. Let $W \subseteq \mathcal{A}_{\text{fin}, \geq 0}$ be a nonempty set defined by a quantifier-free formula with parameters from $\mathcal{A}_{\text{fin}}$. Interpret the same formula in the real closed field $\mathcal{P}$. Its truth set is a finite union of intervals and points: the finitely many roots of the polynomials appearing in the formula partition the line into regions of constant sign.

Intersect these finitely many cells with $[0, \infty)$ and choose the first one meeting $\mathcal{A}_{\text{fin}}$. A point cell contributes its own element. For an interval cell with lower endpoint $b \in \mathcal{P}$, the integer-part property gives the least integer-part element at or above b , or the least one strictly above b when the endpoint is excluded. This element is nonnegative; if it were above the interval, that interval would contain no element of $\mathcal{A}_{\text{fin}}$ at all. For a cell beginning at 0, the same argument starts with 0 or 1. Thus W has a least element.

If a quantifier-free induction instance failed while its base and successor steps held, the nonempty quantifier-free set of counterexamples would have a least member m . It would satisfy $m > 0$, so $m - 1 \in \mathcal{A}_{\text{fin}, \geq 0}$ would satisfy the formula. The successor step would then force the formula at m , a contradiction. $\square$

[write] The proof of [theorem 23.2](#) is the Shepherdson route of the collection's `dsn:cor:openinduction` (definable-surreals-and-omnific-integers report: open induction for the integer part I_A of a real closed field K_A) and of the paragraph citing Shepherdson's theorem in the subsection "Sums of squares and induction" of the `omnific-diophantine-geometry` report (open induction for the omnific integers of a divisible Hahn workspace). Here it is applied to $\mathcal{A}_{\text{fin}}$ inside the real closed field $\mathcal{P}$; uncountability and the semiring axioms are 03's. `dsn:cor:openinduction` is pending in `docs/FORMALIZATION.md`. As 03 says, $\mathcal{A}_{\text{fin}, \geq 0}$ is a Shepherdson-type model, not a new model of arithmetic.

23.1 Why the Puiseux field itself has Borel codes

In the ambient coefficient space $\mathbb{R}^{\mathbb{Q}}$, the field (23.1) is exactly the union, over $n \geq 1$ and $k \in \mathbb{Z}$, of the closed sets

$$\{a : a_q = 0 \text{ whenever } q \notin \frac{1}{n}\mathbb{Z} \text{ or } q < k/n\}.$$

It is therefore F_σ and standard Borel. The convolution and unique graph arguments of Theorem 20.2 make its field operations Borel. Every such support has increasing order type at most ω .

This gives a useful three-way contrast. Finite principal parts have finite support; the real closed Puiseux field has a uniform support bound ω ; the full rational-exponent Hahn field has supports of unbounded countable type. Real closedness alone does not prevent a Borel coefficient presentation. Full arbitrary-support closure is what makes Corollary 20.4 apply.

24 A Polish finite-normal-form presentation

24.1 The topology and its Borel structure

Write an element of $\mathcal{A}_{\text{fin}, \geq 0}$ uniquely as $(S, n, (a_q)_{q \in S})$, where all $a_q \neq 0$. If $S \neq \emptyset$, positivity means $a_{\max S} > 0$ and the integer n is unrestricted. If $S = \emptyset$, we require $n \geq 0$.

For finite nonempty $S \subseteq \mathbb{Q}_{>0}$ and $n \in \mathbb{Z}$, let

$$U_{S,n} = \{a \in (\mathbb{R} \setminus \{0\})^S : a_{\max S} > 0\}.$$

For $S = \emptyset$, let $U_{\emptyset,n}$ be a singleton when $n \geq 0$. Put on $\mathcal{A}_{\text{fin}, \geq 0}$ the topological disjoint union

$$(\mathcal{A}_{\text{fin}, \geq 0}, \tau_{\text{FS}}) = \coprod_{S,n} U_{S,n}. \quad (24.1)$$

Support and integer constant are locally constant in this topology.

Proposition 24.1. *The topology τ_{FS} is Polish. Its Borel sets are exactly the Borel sets obtained from the canonical finite-support coefficient coding.*

Proof. Each nonsingleton component is open in a finite-dimensional Euclidean space, and hence Polish. There are countably many components. A countable topological disjoint union of Polish spaces is Polish: choose bounded complete metrics on the components and put distance 2 between distinct components. Taking the union of countable dense sets gives separability.

Every coefficient map is continuous for (24.1). Conversely, an exact support condition consists of nonzero tests at finitely many coordinates and zero tests at countably many others, so is Borel in the coefficient product. On each such Borel piece the coefficient coordinates give exactly the usual finite-dimensional Borel structure. Since there are countably many pieces, the two Borel structures agree. $\square$

This is not the order topology. The order topology on a discretely ordered uncountable arithmetic domain is discrete and therefore not separable. Nor is τ_{FS} the ambient coefficient subspace topology, where convergence need not make the exact support eventually constant. The distinction is responsible for the cancellation discontinuities below.

24.2 A general support-normalization lemma

Lemma 24.2. *Let $U \subseteq \mathbb{R}^d$ be nonempty and open, and let $(p_j)_{j \in J}$ be a finite family of real polynomials. Define a map whose output is the finite list of nonzero coordinates $p_j(u)$, with the target topologized as a disjoint union by exact nonzero support. Let $E = \{j : p_j \not\equiv 0\}$. Its discontinuity set is exactly*

$$\mathcal{Z} = \bigcup_{j \in E} \{u \in U : p_j(u) = 0\}.$$

This set is closed and nowhere dense in U . The map is continuous on countably many closed subsets covering U .

Proof. Outside $\mathcal{Z}$, all nonidentically-zero coordinates stay nonzero on a sufficiently small neighborhood, while the others are always zero. The output support is therefore constant and the coordinate formulas are continuous.

At $u \in \mathcal{Z}$, choose $j \in E$ with $p_j(u) = 0$. A nonzero real polynomial cannot vanish on a nonempty open set. Thus arbitrarily near u there are points where this coordinate is nonzero. Their output support differs from the support at u , so their images lie outside the clopen target component containing the image of u . Continuity fails. This also proves that the finite union of zero sets is nowhere dense; closedness is immediate.

For $Z \subseteq E$ and integer $k \geq 1$, use the closed set

$$F_{Z,k} = \{u \in U : p_j(u) = 0 \ (j \in Z), \quad |p_j(u)| \geq 1/k \ (j \in E \setminus Z)\}. \quad (24.2)$$

These sets cover U : at any point choose the actual zero set and then a large enough k for the finitely many remaining coordinates. On each piece, the output support is constant and the map is continuous. For a zero-dimensional component consisting of one point, the same conclusion holds with empty discontinuity set. $\square$

24.3 The exact arithmetic cancellation loci

Fix an input component $U_{S,n} \times U_{T,m}$. Extend the variable coefficient families by $a_q = 0$ for $q \notin S$ and $b_q = 0$ for $q \notin T$. For addition, the output coefficient at $r > 0$ is

$$A_r = a_r + b_r, \quad (24.3)$$

and its constant is $n + m$. For multiplication, the output constant is nm and the coefficient at $r > 0$ is

$$M_r = nb_r + ma_r + \sum_{p+q=r} a_p b_q. \quad (24.4)$$

Only finitely many exponents need to be considered: $S \cup T$ for addition and $S \cup T \cup (S + T)$ for multiplication.

For multiplication define

$$E_{S,T,n,m} = \{r \in S \cup T \cup (S + T) : M_r \neq 0\},$$

where identity means polynomial identity in the independent input coefficient variables after fixing the integers n, m . This removes absent terms and zero-factor components correctly.

Theorem 24.3 (Exact cancellation theorem). *On the Polish presentation (24.1), addition and multiplication are Borel. Their discontinuity sets are given componentwise by*

$$D_+ \cap (U_{S,n} \times U_{T,m}) = \bigcup_{r \in S \cap T} \{a_r + b_r = 0\}, \quad (24.5)$$

$$D_\times \cap (U_{S,n} \times U_{T,m}) = \bigcup_{r \in E_{S,T,n,m}} \{M_r = 0\}. \quad (24.6)$$

Globally, both sets are closed and nowhere dense. Both operations are continuous on countably many closed pieces. In particular, both are continuous at every point of the common dense open set $\mathcal{A}_{\text{fin}, \geq 0}^2 \setminus (D_+ \cup D_\times)$.

Proof. On an input component, the actual normal form is obtained from (24.3) or (24.4) by deleting the zero coefficients. The integer output constant is fixed. Apply Lemma 24.2. For addition, a coefficient occurring in only one input support is constrained to be nonzero, so the possible zero sets are precisely those in (24.5). For multiplication the expected polynomial support is exactly $E_{S,T,n,m}$.

Every input component is clopen, and these components partition the domain. A set whose intersection with each component is closed is globally closed; a set whose intersection with each nonempty open component is nowhere dense has empty interior and, when closed, is nowhere dense globally. This gives the assertions about D_+ and $D_\times$.

The pieces (24.2) are closed relative to a clopen component, hence closed in the full domain. Taking them over countably many components, zero patterns, and integers k gives countable closed covers for both operations. To see Borelness directly, the inverse image of an open set is a countable union of sets open relative to these closed pieces. Such sets are Borel (indeed F_σ in a metric space). $\square$

Example 24.4 (Addition: cancellation below the leading term). For real δ sufficiently near 0, set

$$x = X^2 + X, \quad y_\delta = X^2 + (-1 + \delta)X.$$

All these elements are positive and remain in fixed input components. But

$$x + y_\delta = 2X^2 + \delta X.$$

At $\delta = 0$ the output support loses exponent 1, so the outputs do not converge to $2X^2$ in τ_{FS} . Leading coefficients of positive inputs cannot cancel each other; the lower coefficient is what fails here.

Example 24.5 (Multiplication: a vanished middle coefficient). Let $x = X + 1$ and $y_\delta = (1 + \delta)X - 1$. Near $\delta = 0$ both inputs are positive, their supports and integer constants are fixed, and

$$xy_\delta = (1 + \delta)X^2 + \delta X - 1.$$

The middle term disappears at $\delta = 0$. Thus $(X + 1, X - 1) \in D_\times$. Multiplication by 0, on the other hand, is constant and continuous; this explains the exclusion of identically zero coefficient polynomials.

Remark 24.6 (A measurability qualification). Countable closed-piece continuity is the statement proved here. We do not identify it without further hypotheses with the Baire-one class of maps into this disconnected target. Nor does the dense-open conclusion say that an algebraic cancellation can be ignored by exact computation: deleting a single zero coefficient changes the canonical component.

25 Glazer's obstruction survives every compatible Polish recoding

We use precisely the following published result.

Theorem 25.1 (Glazer, Corollary 1 of [1]). *There is no model of Robinson arithmetic plus IOpen on an uncountable Polish space in which addition is continuous and multiplication is Borel.*

Theorem 25.2 (No removal of the addition obstruction). *No Polish topology on the set $\mathcal{A}_{\text{fin}, \geq 0}$ with the same Borel sets as τ_{FS} makes addition continuous. More generally, there is no Polish presentation Borel-isomorphic to the canonical one in which transported addition is continuous.*

Proof. By Theorem 23.2, the algebraic structure is an uncountable model of the required arithmetic. By Theorem 24.3, multiplication is Borel in the canonical Borel structure. Changing the Polish topology without changing its Borel sets leaves multiplication Borel on the product; finite products of standard Borel spaces have the same product Borel structure. Continuous addition would contradict Theorem 25.1. Transporting the operations through a Borel isomorphism changes none of these hypotheses. $\square$

Corollary 25.3. *The distinction between global continuity and continuity off a closed nowhere dense set is genuine for this explicit model: the latter is attained for both operations, while the former is impossible for addition under any Borel-compatible Polish presentation.*

This is a sharp distinction between an empty discontinuity set and an explicit closed nowhere dense one. It is not a proof that our particular zero loci have minimal cardinality, minimal dimension, or a minimal descriptive rank among all recodings.

Glazer's Theorem 2 also excludes continuous multiplication for uncountable Borel models of the stronger theory $\mathbf{Q} + I\Delta_0 + \text{Exp}$ [1]. We have proved only open induction for $\mathcal{A}_{\text{fin}, \geq 0}$. The multiplication obstruction from that stronger result is therefore *not* transferred to every compatible topology on $\mathcal{A}_{\text{fin}, \geq 0}$. Multiplication fails to be continuous in our displayed topology by Theorem 24.3; whether a different compatible Polish topology can make multiplication continuous remains a separate question here.

26 Where the two parts meet [merge]

The real model of Part I is one stratum of the cone $\mathcal{A}_{\text{fin}, \geq 0}$ of section 24. Read ω as X and let

$$\iota : M_{\mathbb{R}} \longrightarrow \mathcal{A}_{\text{fin}}, \quad \iota(r, n) = rX + n,$$

which is 05's map $\iota_{\mathbb{R}}$ of theorem 9.1 in 03's notation. Put

$$C_{\mathbb{R}} = \{a \in \mathcal{A}_{\text{fin}, \geq 0} : \text{supp } a \subseteq \{1\}\} = \bigcup_{n \in \mathbb{Z}} U_{\{1\}, n} \cup \bigcup_{n \geq 0} U_{\emptyset, n}.$$

Proposition 26.1 (The real model as a stratum [merge]). (i) ι is an injective ordered additive map onto $C_{\mathbb{R}}$, fixing 0 and 1. The set $C_{\mathbb{R}}$ is clopen in $(\mathcal{A}_{\text{fin}, \geq 0}, \tau_{\text{FS}})$ and closed under addition, but not under multiplication.

- (ii) On $\{(r, n) \in M_{\mathbb{R}} : r > 0\}$, ι is a homeomorphism onto $\bigcup_n U_{\{1\}, n}$, for the topology of $M_{\mathbb{R}}$.
- (iii) At the standard numerals the two topologies differ: every ordinary integer $n \geq 0$ is isolated in τ_{FS} , but $(0, n)$ is not isolated in $M_{\mathbb{R}}$. Thus τ_{FS} induces on $C_{\mathbb{R}}$ a topology strictly finer than the one transported from $M_{\mathbb{R}}$.
- (iv) Addition is continuous on $(C_{\mathbb{R}}, \tau_{\text{FS}})$. Consequently $(C_{\mathbb{R}}, \tau_{\text{FS}})$ is another uncountable, locally compact Polish model of $\text{Th}(\mathbb{N}; 0, 1, +, <)$ with continuous addition; it is not perfect, its isolated points being exactly the standard numerals.

Proof. (i) An element of $\mathcal{A}_{\text{fin}}$ with support in $\{1\}$ is $n + aX$; by (24.1) it lies in $\mathcal{A}_{\text{fin}, \geq 0}$ when $a > 0$, or $a = 0$ and $n \geq 0$. These are exactly the values of ι . Addition is coordinatewise and the order is decided by the leading coefficient, as in (3.1). The set $C_{\mathbb{R}}$ is a union of components of the topological sum (24.1), hence open, and its complement is the union of the other components, hence also open. Finally $X \cdot X = X^2 \notin C_{\mathbb{R}}$.

(ii) Both $U_{\{1\}, n}$ and the part $r > 0$ of the slice of integer coordinate n in $M_{\mathbb{R}}$ are the open ray $(0, \infty)$ with its Euclidean topology, and ι is the identity on the coefficient.

(iii) $U_{\emptyset, n}$ is a singleton component of (24.1). In $M_{\mathbb{R}}$, $(1/j, n) \rightarrow (0, n)$ by (5.1).

(iv) On a component $U_{S, n} \times U_{T, m}$ with $S, T \subseteq \{1\}$, (24.5) is a union over $r \in S \cap T \subseteq \{1\}$ of the sets $\{a_1 + b_1 = 0\}$, which are empty because both leading coefficients are positive. So $C_{\mathbb{R}}^2$ does not meet D_+ , and addition maps $C_{\mathbb{R}}^2$ into $C_{\mathbb{R}}$. A clopen subspace of a Polish space is Polish, the components are points and open rays, and the algebraic structure is that of $M_{\mathbb{R}}$, so corollary 4.11 applies. $\square$

The boundary. Glazer’s Corollary 1 concerns a model of $Q+I\text{Open}$ with Borel multiplication. The ring cone $\mathcal{A}_{\text{fin}, \geq 0}$ is such a model (theorems 23.2 and 24.3), so no Borel-compatible Polish topology on $\mathcal{A}_{\text{fin}, \geq 0}$ makes its addition continuous (theorem 25.2). The stratum $C_{\mathbb{R}}$ is not such a model: as an additive monoid it is $M_{\mathbb{R}}$, which carries no unital semiring multiplication at all (corollary 10.5), and the multiplication of $\mathcal{A}_{\text{fin}, \geq 0}$ leaves it. On $C_{\mathbb{R}}$ addition is continuous for τ_{FS} itself; the discontinuities of addition on $\mathcal{A}_{\text{fin}, \geq 0}$ lie on the cancellation loci (24.5), which need two inputs whose supports share an exponent below their leading ones, and so lie outside $C_{\mathbb{R}}^2$. Thus Part I and theorem 25.2 delimit, inside one ring, where Corollary 1 applies: the additive stratum admits continuous addition for both topologies, while the ring, with its Borel multiplication and open induction, admits it for none.

The Baire model. The image $\iota_B(M_B)$ of theorem 9.1 lies in 03’s rational Hahn ring $\mathcal{H}_{\mathbb{Q}}$ of (17.1): its supports are contained in $\{1, \frac{1}{2}, \frac{1}{3}, \dots\} \cup \{0\}$ and have reverse order type at most $\omega + 1$. Its elements with infinitely many nonzero coefficients lie neither in $\mathcal{A}_{\text{fin}}$ nor in the Puiseux field $\mathcal{P}$, whose denominators are bounded. Each coefficient of $\iota_B(a, n)$ is a_j , n or 0, a continuous function on M_B , so the composite of 05’s homeomorphism $h : \mathbb{N}^{\mathbb{N}} \rightarrow M_B$ (theorem 6.1) with ι_B is a coefficient-observable standard Borel parametrization of $\iota_B(M_B)$ in the sense of definition 17.1. This is consistent with theorem 21.2, whose bound can be taken to be $\omega + 2$ here, while the full field $\mathbb{R}((t^{\mathbb{Q}}))$ lies on the non-Borel side of theorem 20.2. The Baire model is one of the restricted support families that section 27 calls for.

A correction to the placement record [write]. The placement commit 3d2177df4 and the intake dossier said that 03’s topology τ_{FS} restricts on the stratum to the topology of 04 and 05. Parts (ii) and (iii) of proposition 26.1 give the exact statement: the two topologies agree on the elements with positive real coefficient and differ at the standard numerals, which τ_{FS} isolates. Both make addition continuous on the stratum, so the conclusion drawn there stands.

27 Consequences for exact notation systems and formalization

27.1 A practical separation of three design goals

The results separate three tasks that can otherwise be conflated. *Arithmetic* asks whether operations can be performed on valid inputs. *Validation* asks whether an arbitrary code is a valid well-based normal form. *Topological presentation* asks whether the operations respect a complete separable topology on the names or values. A positive answer to one is not a positive answer to the others.

In the finite-support model, every code carries an explicit finite list, and arithmetic is polynomial coefficient arithmetic followed by exact zero tests. The canonical domain is Polish, the operations are Borel, and their failure of continuity is exactly the support-changing cancellation calculated above. No general theorem of computability over arbitrary real coefficients is claimed: a real coefficient supplied by an unrestricted approximation oracle need not have decidable equality with zero.

For full Hahn fields on scattered exponent groups, support validation is a Borel predicate. This is not the same as decidability by a finite algorithm. The Π_3^0 examples already involve quantification over infinitely many rows. For dense rational exponents the obstruction is stronger: no standard Borel system of fully observable names covers every valid series.

A useful exact system must therefore limit its admissible support families, relax coefficient observability, or accept a nonstandard measurable domain. The Puiseux field demonstrates that a severe-looking support restriction may still permit real closedness. Theorem 21.2 says every observable Borel system necessarily has some uniform countable support bound, even when that bound is not built into the notation explicitly.

27.2 No paradoxes are needed

All main results are in ZFC and concern sets. For countable Γ , the ambient coefficient space $\mathbb{R}^\Gamma$ is a set of cardinality continuum. Its valid-code subset is a set whether or not it is Borel. The rational Hahn ring $\mathcal{H}_\mathbb{Q}$ and its finite subring $\mathcal{A}_{\text{fin}}$ are also sets. The construction never forms a set of all surreal numbers or quantifies over all class maps as a set.

The optional identification with Conway normal forms can be read as a set-indexed definable embedding into the proper class **No**, or carried out inside a sufficiently large set-sized normal-form universe. Standard Replacement handles the image of a fixed set of codes. No global choice, large cardinal, or class-sized Polish topology is used.

27.3 A formalization plan with an explicit dependency boundary

A Lean development can be organized into independent interfaces rather than starting from the entire proper-class surreal hierarchy.

- Layer 1. Finite algebra and normalization.** Use finitely supported rational-exponent coefficient maps with an integer constant. Establish convolution, leading-exponent sign, discreteness, and the coefficient identities (24.3)–(24.4). The exact cancellation examples are small executable tests at this layer.
- Layer 2. Series and arithmetic.** Use a set-sized Hahn or Puiseux field, prove the principal-part floor with its negative-infinitesimal case, and derive quantifier-free induction via the one-variable sign-cell argument. Newton–Puiseux is a named external dependency until its real-closedness interface is formally supplied.
- Layer 3. Topological support components.** Formalize the countable disjoint union, the polynomial normalization lemma, and the closed-piece covers. None of this requires descriptive ranks or proper-class surreals.
- Layer 4. Descriptive-set-theoretic structure.** Formalize the fixed Kleene–Brouwer embedding, Hausdorff-sum support criterion, analytic boundedness, and observable-field equivalence. The classical Hausdorff characterization and Glazer theorem must be proved or explicitly listed as hypotheses; importing them informally is not a checked proof.

This is a proposed architecture, not a claim that these layers have been implemented or that all required library APIs currently exist. The provided Python artifact checks finite rational instances, not Lean terms or infinitary assertions.

28 Further research questions and concrete next targets

The following questions are proposed here. Unless explicitly attributed, “open” means unresolved in this report, not certified as absent from the entire literature.

Question 28.1 (Continuous multiplication on the finite model). Does $\mathcal{A}_{\text{fin}, \geq 0}$ admit a Polish topology with the canonical Borel sets in which multiplication is continuous?

Theorem 25.2 settles the analogous addition question negatively. Glazer’s stronger multiplication theorem cannot be invoked without establishing its extra arithmetic hypotheses. A positive solution would isolate a concrete asymmetry between the two operations; a negative solution should identify an obstruction valid already for this particular open-induction model.

Question 28.2 (How small can the discontinuity set be?). Among all Borel-compatible Polish presentations of $\mathcal{A}_{\text{fin}, \geq 0}$, what restrictions hold on the discontinuity set of addition? Can it be countable? Can it be contained in a prescribed small-dimensional or otherwise canonical set?

The present model gives a nonempty closed nowhere dense set. Glazer excludes only its complete disappearance, not every possible reduction in complexity. This is a natural refinement of the topological theorem that can first be studied on explicit finite-support components.

Question 28.3 (Exact complexity beyond finite lexicographic rank). For a fixed countable scattered order L , compute the least Borel level of $\text{WO}(L)$ directly from an economical Hausdorff decomposition. Which Wadge degrees occur among these support predicates?

The qualitative dichotomy is settled by Theorem 18.4, while Theorem 19.1 shows that simply increasing finite order rank need not increase Borel level. A satisfactory answer should distinguish ordinal support capacity from descriptive complexity rather than treating them as interchangeable ranks.

[write] (Batch 96, 4 October 2026.) Answered in part by Part XIX (source 25, which calls this “Question 12.3” of source 03). For every countable linear order L , $\text{WO}(L)$ is the whole space if L is well ordered, Σ_2^0 -complete if L is not well ordered and $\mathbf{B} = \omega^* \cdot \omega$ does not embed in L , Π_3^0 -complete if L is scattered and $\mathbf{B}$ embeds, and Π_1^1 -complete if $\mathbb{Q}$ embeds; so the least Borel level of a scattered L is decided by whether $\mathbf{B}$ embeds, and the only nontrivial Borel Wadge degrees are those of Fin and $\text{Fin}^{\mathbb{N}}$ (theorem 259.6). No scattered rank raises the level beyond Π_3^0 , as the commentary above anticipates. Not answered, and open: computing the level “directly from an economical Hausdorff decomposition”. Source 25 claims no effective procedure from decomposition codes; its questions 268.2 and 268.9 carry that clause, and its theorem 266.3 shows that scatteredness itself is Π_1^1 -complete already among the orders of one fixed free abelian group. The proofs are unrefereed and not formalized.

Question 28.4 (Borel, algebraically rich subfields of the rational Hahn field). Classify Borel subfields $F \subseteq \mathbb{R}((t^{\mathbb{Q}}))$ containing all rational monomials and stable under specified truncations or algebraic-root operations. Which countable support bounds can such fields attain?

The Puiseux field is one real closed example with supports of type at most ω . The full Hahn field is excluded. An interesting target is a well-described hierarchy of truncation-closed real closed Borel subfields between them, together with provably sharp support bounds at each stage. No claim is made that every ordinal cutoff is preserved by inversion.

[write] (Batch 96, 4 October 2026.) Asked again, independently, by source 25 as question 268.5, for the truncation-closed subfields of full Hahn fields over every dense exponent group, with the added clause: which support languages support continuous certified arithmetic? It is printed once, here; question 268.5 points back. Source 25’s theorem 260.1 excludes the full field over every such group with a Π_1^1 -complete code set; the question stays open.

Review remark 2 (Density versus root closure). The preceding commentary originally cited [corollary 260.3](#) for every countable dense exponent group. That corollary requires a positive element admitting all dyadic divisions, which density does not imply. For example, $\Gamma = \mathbb{Z} + \sqrt{2}\mathbb{Z}$ is dense, but if $(a + b\sqrt{2})/2^n \in \Gamma$ for every n , uniqueness of the two rational coefficients forces both integers a, b to be divisible by every 2^n , so both are zero. The stated exclusion remains valid by the cited group classification: a countable dense order without endpoints contains a copy of $\mathbb{Q}$. This corrects the supporting reference and retains the missing hypothesis of the original citation.

Question 28.5 (How much coefficient observability is necessary?). If only a selected family of coefficient combinations or truncations is required to be Borel, which versions of Theorem [20.2](#) survive?

The current proof uses all countably many coefficient maps to obtain one Borel map into the full product. This pinpoints a possible escape route and suggests investigating minimal observation families that still recover the binary support cylinder in Theorem [22.1](#).

[write] (Batch 96, 4 October 2026.) Asked again, independently, by source 25 as [question 268.8](#): which observations suffice to force the naming barrier of [theorem 265.2](#), which proves (i) $\Leftrightarrow$ (iii) of [theorem 20.2](#) again and extends it to strongly summable families. Printed once, here; open.

Question 28.6 (Certified support languages). Construct useful exact languages whose valid supports carry explicit well-foundedness certificates, and determine their closure under arithmetic, root extraction, and named truncations.

The repository already contains effective restricted omnific calculi [\[21\]](#). The present results suggest comparing them by their Borel parameter spaces and support bounds as well as by decidable equality. A certificate verifier for a fixed language is not a total validator for all raw rational-exponent supports; Theorem [22.1](#) forbids conflating those objectives.

[write] (Batch 96.) Source 25's closed certificate spaces ([theorems 263.2](#) and [264.2](#)) are certified languages of this kind for strong summation over a fixed scattered exponent order, with continuous summation; their closure under multiplication and inversion is its open [question 268.1](#).

Question 28.7 (A minimal arithmetic hypothesis for the presentation barrier). Which concrete consequences of open induction suffice to rule out continuous addition in the finite principal-part model or related integer parts?

A proof specialized to finite Puiseux principal parts might replace the full open-induction interface with explicitly stated root-approximation or division properties. This would clarify the boundary between algebraic normal-form obstructions and arithmetic induction.

[write] (Batch 92, 4 October 2026.) Answered by [Part XIV](#) (source 20, written without knowledge of this question). In every uncountable Polish cone of a discretely ordered commutative ring with Borel multiplication (only countably many scalar maps are used), Euclidean division with order-bounded remainder and integer k th roots for one fixed $k \geq 2$ already forbid continuous addition ([theorem 183.2](#)); these are the “division properties” and “root-approximation” named above, both consequences of open induction and strictly weaker than it ([proposition 188.4](#)). The cone $\mathcal{A}_{\text{fin}, \geq 0}$ of $\mathcal{A}_{\text{fin}} = A_{\mathbb{Q}}$ satisfies both ([corollary 184.5](#) and [theorem 185.1](#)), and for the rings A_{Γ} density of Γ alone suffices, with no root axiom ([theorem 183.1](#)). The proofs are unrefereed and not formalized.

Question 28.8 (Formalized transfer from support theory to arithmetic). Can the observable Hahn-field criterion and the finite-support cancellation theorem be formalized as reusable

results, with the classical Hausdorff and Glazer inputs exposed as exact dependency interfaces?

The suggested first milestone is the support-sum criterion (18.1) together with the normalization Lemma 24.2. Both are modular and elementary enough to be reviewed independently before the larger descriptive-set-theoretic machinery is introduced.

[merge] Question 28.1 is the multiplicative counterpart, on $\mathcal{A}_{\text{fin}, \geq 0}$, of question 14.4. The question on the size of the discontinuity set of addition concerns all of $\mathcal{A}_{\text{fin}, \geq 0}$; on the clopen stratum $C_{\mathbb{R}}$ of proposition 26.1 that set is empty.

29 Conclusions of Part II

The intersection with Glazer’s research produces a precise hierarchy rather than a blanket impossibility theorem. Full Hahn support is Borel exactly on scattered countable exponent orders. Dense alphabets carry a coanalytic-complete validation problem, and no standard Borel family with observable coefficients can exhaust supports of unbounded countable type. This obstruction persists inside one fixed interval of omnific integers.

At the same time, a real closed Puiseux subfield and its finite principal-part integer part have explicit Borel presentations. The latter even admits a Polish topology where both arithmetic operations are continuous off closed nowhere dense sets, with exact polynomial equations describing every failure. Glazer’s theorem makes the remaining addition discontinuity unavoidable under all compatible Polish recodings. The positive construction and the negative result constrain one another: Borel arithmetic is attainable, globally continuous addition is not, and arbitrary well-based support introduces a separate obstruction before topological continuity can even be asked in the canonical standard-Borel setting.

Part III

Continuous Presburger arithmetic on Polish spaces

[write] This part merges two manuscripts of batch 86 that carry the same title, *Continuous Presburger Arithmetic on Polish Spaces*, but are independent texts: source 07 (subtitle *An affirmative construction for Glazer’s question, Baire-space models, and omnific realizations*, dated October 4, 2026, pinned to 0f95145cb) and source 06 (subtitle *Lexicographic models, an isolated-zero obstruction, and a boundary between additive and semiring arithmetic*, dated October 3, 2026, pinned to fa2f3e419). They share at most 0.74% of their word 8-grams, the longest shared runs being Glazer’s quoted question and bibliography entries, and their theorem sets differ.

Base and reading conventions. Source 07 is the base: its realization theorem (theorem 34.1) needs only a divisible Polish abelian group with a Borel positive cone and covers the coordinate families of source 06. Unmarked text in this part is source 07’s. Text from source 06 is marked [06]: in the titles of its sections, subsections and statements, and in the sentence that introduces an inserted passage. Text written for this report is marked [write] or [merge] as in section 1.2. Section and subsection headings of source 06 were moved one level down where its sections are printed inside source 07’s.

What is printed again, and what is new. Both sources prove Part I's main results once more; these statements are printed with their proofs, because the proofs differ in detail, and each says in its title which statement of Part I it repeats:

- the real model ([theorem 32.1](#) of 07 and [theorem 32.2](#) of 06 are [theorem 3.1\(1\)](#));
- the Baire model and its effective addition ([theorems 35.6](#) and [35.7](#) of 06 are [theorems 6.1](#), [6.2](#) and [6.4](#));
- Δ_2^0 definability ([theorem 37.7](#) of 06 and part (iii) of [theorem 34.1](#) extend [theorem 7.1](#));
- the order obstruction ([proposition 37.1](#) is [theorem 7.4](#)), compactness ([proposition 37.3](#) is [corollary 12.2](#)), and the semiring obstruction ([proposition 37.5](#) and [theorem 37.10](#) are [theorem 10.4](#), the latter without a prescribed unit);
- the omnific realizations ([theorem 40.1](#) extends [theorem 9.1](#) from ω to every countably infinite ordinal);
- the classical elimination theorem ([appendices G](#) and [I](#), compare [theorem 4.9](#) and [corollary 4.11](#)).

The results new to the collection are 07's Borel-cone theorem ([theorem 33.1](#)) and cone closure rank, the general realization theorem and the Borelness of definable sets in every continuous presentation ([theorem 34.1](#) and [proposition 34.3](#)), the ordinal family on Baire space ([theorems 35.1](#) and [35.4](#); also 06's [theorem 35.9](#)), the count of exactly $2^{\aleph_0}$ models on Baire space (07's [theorem 35.5](#), and 06's [theorem 35.8](#) with topology, order, 0 and 1 fixed), the oscillation ranks ([theorem 36.1](#) and [corollary 36.2](#)), the discontinuity of min and max ([corollary 37.2](#)), a second topology with continuous predecessor ([proposition 37.4](#)), and source 06's Polish semiring cone of $\mathbb{Z} + t\mathbb{R}[t]$ with continuous addition and multiplication that fails open induction ([theorems 38.1](#) and [38.2](#)) and its isolated-zero theorem ([theorem 39.1](#)). [Section 42](#) locates these results with respect to Parts I and II.

A consequence for Part I's criterion [merge]. Let D be a divisible ordered abelian group with a Polish group topology. By [theorem 5.2](#), M_D is Polish if and only if $D_{>0}$ is G_δ ; by [theorem 33.1](#), a Borel $D_{>0}$ is already Δ_2^0 . Hence

$$M_D \text{ is Polish} \iff D_{>0} \text{ is Borel} \iff D_{>0} \text{ is } \Delta_2^0.$$

The proof of [theorem 34.1](#) also shows that division by a standard integer is then automatically continuous (open mapping), a hypothesis that source 04's [theorem 5.4](#) assumed.

Labels and notation. Labels of this part are `pma:cp:<label>` for both sources, except the six labels of source 06 that coincide with labels of source 07 (`eq:realmodel`, `sec:baire`, `sec:real`, `thm:baire`, `thm:continuum`, `thm:real`), which take the sub-prefix `pma:cp:s06:`. Source 06's family M_s ($s \in 2^{\mathbb{N}}$) is renamed M'_s , with G'_s , H'_s , because source 07's M_s is a different family; [section 1.3](#) lists this and the other conventions of Parts III–V. In particular t is a positive infinite element in this part (source 06's $\mathbb{Z} + t\mathbb{R}[t]$), whereas in [Part V](#) t is a positive infinitesimal.

30 The question and the research intersection

30.1 An exact question with a concrete answer

Glazer’s *A Topological Tennenbaum Theorem* ends with the question:

Does some uncountable Polish space support a model of Presburger arithmetic with continuous addition?

This is Question 2 of the November 2023 version [1]. Here “Polish” means separable and completely metrizable. Presburger arithmetic is the complete first-order theory of addition on the natural numbers, with multiplication of two variables absent from the language.

The answer supplied by this report is affirmative. Its simplest witness is

$$M_{\mathbb{R}} = ((0, \infty) \times \mathbb{Z}) \cup (\{0\} \times \mathbb{N}), \quad (30.1)$$

with coordinatewise addition, $0 = (0, 0)$, $1 = (0, 1)$, and lexicographic arithmetic order that compares the real coordinate first. The topology is the subspace topology from $\mathbb{R} \times \mathbb{Z}_{\text{discrete}}$. The distinction between this topology and the arithmetic order is essential.

There is an even stronger realization on $\mathbb{N}^{\mathbb{N}}$, the usual Baire space. It is obtained by replacing $\mathbb{R}$ with a countable product of discrete copies of $\mathbb{Q}$, then taking the same lexicographic integer extension. No forcing, nonprincipal ultrafilter, or nonconstructive choice of an arithmetic operation is needed for either coordinate model.

Glazer also proves that certain stronger Borel arithmetics on uncountable Polish spaces cannot have either addition or multiplication continuous. His Corollary 1 already excludes continuous addition and Borel multiplication for $Q + I\text{Open}$ [1]. Our result concerns the additive theory. Section 37 explains the precise boundary, including a stronger algebraic obstruction for our examples.

30.2 Why this is a substantive overlap with ProveIt

The repository snapshot examined is commit

0f95145cb65d5dc279e2053e8707a71bb490bb59.

Its timestamp is October 3, 2026, 18:50:05 in the America/Los_Angeles time zone. Two parts of that snapshot are especially relevant:

1. `Logic/PresburgerArithmetic` contains an executable Cooper quantifier-elimination development for formulas over ordinary integers. The main semantic correctness theorem is implemented in Lean; Rocq independently checks its normalized one-variable core [32].
2. The surreal research report on discrete initial subgroups and omnific normalization studies split ordered groups of the form $D \overrightarrow{\times} \mathbb{Z}$, including their relation to omnific integers [33].

The first supplies an existing computational and formalization framework; the second supplies the relevant algebraic shape. The new work here is to give that shape explicit Polish topologies, prove its model-theoretic and descriptive properties, and connect it to the exact published question.

The main positive theorems do not depend on accepting an unrefereed repository theorem. We prove the required ordered-group facts directly and use the classical characterization of Presburger models. The surreal connection is also given by an explicit normal-form map.

30.3 What is proved, and what is a novelty claim

The existence assertion is a theorem with a short proof. The assertion that it is the first published resolution would be a historical claim requiring different evidence. We make only the former claim. A targeted literature search found the question in Glazer’s paper and found no later resolution; that is not an exhaustive priority search.

The same distinction applies to the general cone theorem, the family of Baire-space models, and the rank computations. They are proved here and are useful extensions of the construction. Their precise place in the literature remains suitable for specialist review. In particular, the underlying lexicographic construction of a $\mathbb{Z}$ -group is classical and is not presented as an invention.

30.4 The question, the overlap, and the contribution [06]

30.4.1 A precise research target [06]

Glazer’s work on topological models of arithmetic gives a particularly direct intersection with ProveIt. His 2023 paper ends with the question [1, Question 2]:

Does some uncountable Polish space support a model of Presburger arithmetic with continuous addition?

Here *Presburger arithmetic* means the complete first-order theory of addition on the natural numbers, rather than full Peano arithmetic. We answer this formulation affirmatively, with explicit models and proofs. The use of a topology different from the arithmetic order topology is essential.

The context is a sharp obstruction for stronger arithmetic. Glazer proves that neither arithmetic operation can be continuous in an uncountable Polish Borel model of $\mathbb{Q} + \text{ID}_0 + \text{Exp}$; his additive obstruction already holds for $\mathbb{Q} + \text{IOpen}$ when multiplication is Borel [1, Theorem 2 and Corollary 1]. The constructions below show why the induction and language hypotheses matter.

30.4.2 What is already in ProveIt [06]

We inspected the repository at commit

`fa2f3e419e592a1e90c065b79d559865ec45d436`.

The following connections are concrete, not merely thematic.

1. `Logic/PresburgerArithmetic` contains a Lean development of Cooper elimination and an independent Coq development of the normalized one-variable step [34]. Its current semantics use ordinary integers. The abstract $\mathbb{Z}$ -group semantics needed for the nonstandard models below are an additional formalization task.

2. The report *Discrete Initial Groups and Omnific Normalization* studies split groups $D \vec{\times} \mathbb{Z}$ and their connection to omnific integers [35]. The algebraic form of our examples thus already has a close repository counterpart.
3. The report *Omnific Integers and Omnific–Diophantine Geometry* explicitly studies $\mathbb{Z} + t\mathbb{R}[t]$ and its realization $\mathbb{Z} + \omega\mathbb{R}[\omega]$ [36]. We equip its nonnegative cone with a useful Polish topology and prove continuity of both operations.

The lexicographic $\mathbb{Z}$ -group construction and its model theory are classical [4]; rational polynomial Presburger examples also appear in established lecture notes [24]. The contribution here is the explicit topological realization, its consequences for the stated question, and the accompanying obstruction and abundance results. We use the repository reports as documented points of contact, rather than as unverified premises for our main proofs.

30.4.3 A map of the results [06]

Model	Topology	Continuous operations	Arithmetic boundary
$(\mathbb{R} \vec{\times} \mathbb{Z})_{\geq 0}$	Perfect, locally compact Polish	Addition; fixed standard division	Presburger; no unital semiring expansion
$(\mathbb{Q}^{\mathbb{N}} \vec{\times} \mathbb{Z})_{\geq 0}$	Baire space	Addition, with an effective stream presentation	Presburger; no unital semiring expansion
$(\mathbb{Z} + t\mathbb{R}[t])_{\geq 0}$	Locally compact Polish; isolated standard points	Addition and multiplication	PA^- and Presburger; fails lOpen

The general obstruction in Section 39 says that the last row cannot be made perfect while retaining Polishness and continuity of both operations. The abundance theorem in Section 35.6 gives the exact possible cardinality of the collection of isomorphism types in the Baire-space setting: $2^{\aleph_0}$.

Scope of the literature check. The cited version of Glazer’s paper is arXiv:2311.13699v1. Searches of the paper, its available version record, related primary papers, and the selected repository materials did not locate a later answer or this topological construction. This is a bounded bibliographic finding, not a proof of universal priority. The mathematical assertion that our model answers the displayed question is independent of that priority issue.

31 Conventions and classical inputs

31.1 Language, order, and induction

We use $\mathbb{N} = \{0, 1, 2, \dots\}$. A model of $\text{Th}(\mathbb{N}; 0, 1, +, <)$ is a model of the *complete first-order theory* of $(\mathbb{N}; 0, 1, +, <)$, including every instance of first-order induction with parameters. In this theory,

$$x \leq y \iff \exists z (x + z = y), \quad x < y \iff \exists z (x + z + 1 = y). \quad (31.1)$$

Thus naming $<$ is a definitional convenience. A topological presentation in this report requires $+$ to be jointly continuous. It does not require the definable relation $<$ to be open or closed.

If G is an ordered abelian group, write $G_{\geq 0}$ and $G_{>0}$ for its nonnegative and positive cones. A distinguished $e \in G$ is a *least positive element* if $0 < e \leq g$ whenever $g > 0$. A $\mathbb{Z}$ -group is an ordered abelian group with a least positive element e such that $G/\mathbb{Z}e$ is divisible.

We use the following classical facts about Presburger arithmetic; a convenient precise reference is Jeřábek's preliminary section [4].

1. A group is a model of $\text{Th}(\mathbb{Z}; 0, 1, +, -, <)$ exactly when it is a $\mathbb{Z}$ -group. Its nonnegative cone is then a model of $\text{Th}(\mathbb{N}; 0, 1, +, <)$.
2. In the language enlarged by fixed-modulus congruence predicates, formulas admit quantifier elimination: they reduce to finite Boolean combinations of integer-linear comparisons and congruences.

These inputs address *all* formulas, not merely the displayed division axioms. That is why our construction satisfies full first-order induction. We do not confuse it with second-order induction over every external subset.

31.2 Topological and descriptive notation

An F_σ set is a countable union of closed sets; a G_δ set is a countable intersection of open sets. A Δ_2^0 set is both. In metrizable spaces, open and closed sets are Δ_2^0 , finite Boolean combinations of Δ_2^0 sets are Δ_2^0 , and continuous inverse images preserve this class.

We use three standard descriptive-set-theoretic inputs: a subspace of a Polish space is Polish when it is G_δ ; a Borel injective map between standard Borel spaces has Borel image and inverse onto its image; and a nonempty zero-dimensional Polish space in which every compact subset has empty interior is homeomorphic to $\mathbb{N}^{\mathbb{N}}$. References and proofs of these standard tools can be found in [9, 8].

All metamathematics is carried out in ordinary ZFC. Every model constructed below is a set. Countable ordinals are actual ordinals of this metatheory, not nonstandard ordinal codes inside an arithmetic model.

31.3 Algebraic and topological preliminaries [06]

31.3.1 First-order arithmetic and induction [06]

Let

$$\text{Pres} = \text{Th}(\mathbb{N}, 0, 1, +, <).$$

Adding $0, 1, <$ to the addition-only language is harmless here because they are definable. A model of Pres satisfies every first-order induction instance in this additive language, including instances with parameters. It need not satisfy induction for formulas involving a subsequently added multiplication symbol.

We use PA^- in its standard algebraic sense: the theory of the nonnegative cones of discretely ordered commutative rings, with their inherited $0, 1, +, \cdot, <$. This convention agrees with that used in the topological-arithmetic literature [2]. We will describe all concrete models directly, so the results do not depend on choosing a particular printed list of PA^- axioms.

A discretely ordered abelian group is a linearly ordered abelian group with a least positive element, denoted by 1. The subgroup generated by 1 is a convex copy of $\mathbb{Z}$: the only elements between consecutive standard integers are the endpoints. A $\mathbb{Z}$ -group is such a group for which $G/\mathbb{Z}$ is divisible.

Lemma 31.1 (Classical Presburger characterization [06]; = [theorem 4.9](#) and [corollary 4.11](#)). *If G is a $\mathbb{Z}$ -group, then*

$$(G, 0, 1, +, -, <) \equiv (\mathbb{Z}, 0, 1, +, -, <), \quad G_{\geq 0} \models \text{Pres.}$$

Every formula in the group language is equivalent to a Boolean combination of integer-linear inequalities and congruences modulo positive standard integers.

This characterization is recalled in [4, Section 2]. Appendix I supplies a proof of the elimination step and of the transfer to the nonnegative cone. In particular, our use of this lemma includes full first-order additive induction; it does not merely check a few semiring-like equations.

Lemma 31.2 (Split construction [06]; the setup of [theorem 3.1](#)). *Let D be a divisible ordered abelian group. Give*

$$G(D) = D \times \mathbb{Z}$$

coordinatewise addition and lexicographic order with the D coordinate dominant. Then $G(D)$ is a $\mathbb{Z}$ -group with $1 = (0, 1)$, and

$$M(D) = G(D)_{\geq 0} = (D_{>0} \times \mathbb{Z}) \cup (\{0\} \times \mathbb{N}) \quad (31.2)$$

models Pres.

Proof. If $(d, z) > 0$, either $d > 0$, in which case $(d, z) > (0, 1)$, or $d = 0$ and $z \geq 1$. Thus $(0, 1)$ is least positive. The subgroup $\{0\} \times \mathbb{Z}$ is convex and the quotient is D , which is divisible. Apply Lemma 31.1.

For additional explicitness, if $n \geq 1$ is a standard integer, write $z = nq + r$ in $\mathbb{Z}$, with $0 \leq r < n$. Then

$$(d, z) = n(d/n, q) + (0, r). \quad (31.3)$$

The quotient is unique. If $(d, z) \geq 0$, then $(d/n, q) \geq 0$ as well: either $d > 0$, or $d = 0$ and $z \geq 0$, so $q \geq 0$. $\square$

31.3.2 The topologies under discussion [06]

A space is *Polish* if it is separable and admits a compatible complete metric. Its displayed metric need not itself be complete. We use the product topology for each product of topological groups, and the subspace topology on a cone unless a different topology is explicitly specified.

We write Δ_2^0 for the sets that are simultaneously F_σ and G_δ . This class is closed under finite Boolean operations and continuous inverse images. Every open or closed subset of a metrizable space belongs to this class. A G_δ subspace of a Polish space is Polish. For reference, if $X = \bigcap_n U_n$ in a complete metric space X_0 , the map

$$x \mapsto \left(x, (d(x, X_0 \setminus U_n))^{-1} \right)_{n \in \mathbb{N}}$$

embeds X as a closed graph in a product of complete spaces, with the empty-complement coordinates omitted. This gives the usual complete metrizability argument.

32 The real-coordinate model

Theorem 32.1 (An affirmative construction; = [theorem 3.1\(1\)](#)). *The structure $M_{\mathbb{R}}$ of (30.1), with the operations and order described in Section 30, is an uncountable model of $\text{Th}(\mathbb{N}; 0, 1, +, <)$. Its topology is Polish, locally compact, and has no isolated points. Addition is jointly continuous. Therefore Question 2 of [1], as stated there, has an affirmative answer.*

Proof. Consider the abelian group

$$G_{\mathbb{R}} = \mathbb{R} \times \mathbb{Z}, \quad (r, n) + (s, m) = (r + s, n + m).$$

Give it the order

$$(r, n) < (s, m) \iff r < s \text{ or } (r = s \text{ and } n < m). \quad (32.1)$$

This is a translation-invariant total order. Its least positive element is $e = (0, 1)$: a positive pair either has positive real coordinate and exceeds every $(0, n)$, or has real coordinate zero and a positive integer coordinate.

The subgroup $\mathbb{Z}e$ is $\{0\} \times \mathbb{Z}$, and $G_{\mathbb{R}}/\mathbb{Z}e \cong (\mathbb{R}, +)$ is divisible. Hence $G_{\mathbb{R}}$ is a $\mathbb{Z}$ -group, and its nonnegative cone is a model of $\text{Th}(\mathbb{N}; 0, 1, +, <)$. Computing that cone gives exactly (30.1).

Topologically, $\mathbb{R} \times \mathbb{Z}_{\text{discrete}}$ is Polish, and

$$M_{\mathbb{R}} = ([0, \infty) \times \mathbb{Z}) \setminus (\{0\} \times \mathbb{Z}_{<0}). \quad (32.2)$$

The first set is closed, and the removed set is closed. Thus $M_{\mathbb{R}}$ is open in a closed Polish subspace, so is Polish. Equivalently, it is the countable topological disjoint sum of the slices

$$M_{\mathbb{R}} \cap (\mathbb{R} \times \{n\}) \cong \begin{cases} [0, \infty), & n \geq 0, \\ (0, \infty), & n < 0. \end{cases}$$

Each slice is locally compact and has no isolated points. Those properties therefore hold for the disjoint sum. The map $r \mapsto (r, 0)$ embeds $[0, \infty)$ into the model, proving uncountability.

Ambient coordinatewise addition is continuous. The sum of two nonnegative pairs is again nonnegative, so its restriction is a continuous map $M_{\mathbb{R}}^2 \rightarrow M_{\mathbb{R}}$. $\square$

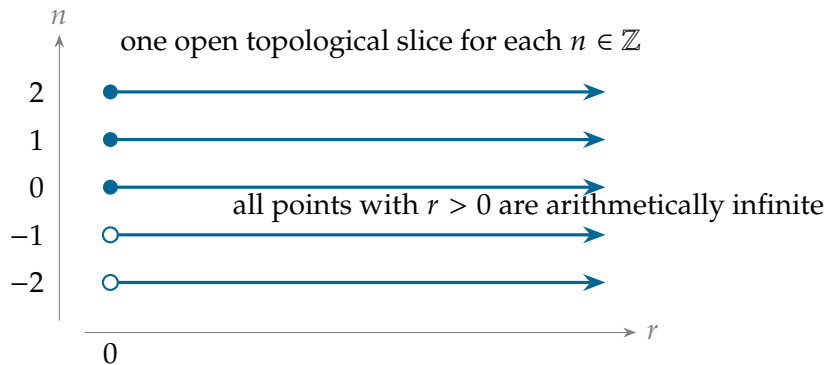


Figure 1: The space $M_{\mathbb{R}}$. Filled endpoints are present; hollow endpoints are absent. Different horizontal slices are topologically separated. Arithmetic order compares the horizontal coordinate before the slice label.

[write] [Theorem 32.1](#) and its proof are those of [theorem 3.1\(1\)](#); source 07 adds the picture in [figure 1](#) and a complete metric that uses $\log r$ on the negative slices ([equation \(32.3\)](#); compare (5.2)). Source 06's account follows in [section 32.3](#).

32.1 A complete metric, rather than an incomplete inherited metric

Being a Polish subspace does not mean that the restriction of the ambient Euclidean product metric is complete. For example, $(1/j, -1)$ approaches a missing endpoint in the ambient space. A compatible complete metric is easy to exhibit.

For the slice with integer coordinate n , put

$$h_n(r) = \begin{cases} r, & n \geq 0, \\ \log r, & n < 0. \end{cases}$$

Define

$$d((r, n), (s, m)) = \begin{cases} 1, & n \neq m, \\ \min\{1, |h_n(r) - h_n(s)|\}, & n = m. \end{cases} \quad (32.3)$$

The triangle inequality holds within each slice and also across slices. A Cauchy sequence is eventually in a single slice. In a nonnegative slice it is complete because $[0, \infty)$ is complete; in a negative slice the logarithm identifies it with a complete copy of $\mathbb{R}$. Thus d is complete and gives precisely the topology used in Theorem 32.1.

32.2 Division and the meaning of nonstandard induction

For a standard integer $d \geq 1$, divide the ordinary integer n as $n = dq + r$, where $0 \leq r < d$. Then

$$(a, n) = d(a/d, q) + (0, r). \quad (32.4)$$

When $(a, n) \in M_{\mathbb{R}}$, its quotient $(a/d, q)$ is also in $M_{\mathbb{R}}$. This remains true for negative n : such an input has $a > 0$, so its positive real coordinate determines positivity of the quotient.

For example,

$$(5/2, -7) = 3(5/6, -3) + (0, 2).$$

The integer coordinate is permitted to be negative in a positive element. Forbidding it would destroy predecessor and division properties and would give the wrong model.

Every (a, n) with $a > 0$ exceeds every standard numeral $(0, k)$. Nevertheless, each induction instance in the first-order additive language holds, by the $\mathbb{Z}$ -group characterization. The external set of standard numerals is proper and successor-closed, but is not definable in this model, even with parameters: a definition would violate the corresponding induction instance. This is the usual distinction between first-order and full second-order induction, manifested in very elementary coordinates.

32.3 A perfect locally compact model [06]

Theorem 32.2 (An affirmative answer to Glazer's question [06]; = [theorem 3.1\(1\)](#)). *With the usual topology on $\mathbb{R}$ and the discrete topology on $\mathbb{Z}$, the space*

$$M_{\mathbb{R}} = ((0, \infty) \times \mathbb{Z}) \cup (\{0\} \times \mathbb{N}) \quad (32.5)$$

is uncountable, perfect, locally compact, and Polish. With coordinatewise addition, $0 = (0, 0)$, $1 = (0, 1)$, and lexicographic order, it is a model of Pres with jointly continuous addition.

Proof. The algebra follows from Lemma 31.2 with $D = \mathbb{R}$. For each integer z , the slice with second coordinate z is clopen and is homeomorphic to

$$I_z = \begin{cases} [0, \infty), & z \geq 0, \\ (0, \infty), & z < 0. \end{cases}$$

Thus $M_{\mathbb{R}}$ is a countable topological disjoint union of locally compact Polish rays. It is second countable and has no isolated points.

One can verify complete metrizability without relying on the disjoint sum assertion. Put $\phi_z(r) = r$ for $z \geq 0$ and $\phi_z(r) = \log r$ for $z < 0$, and define

$$\rho((r, z), (s, w)) = \begin{cases} 2, & z \neq w, \\ \min\{1, |\phi_z(r) - \phi_z(s)|\}, & z = w. \end{cases}$$

This is a compatible complete metric: a Cauchy sequence eventually stays in one slice, where it corresponds to a Cauchy sequence in $[0, \infty)$ or $\mathbb{R}$. Rational points in all slices give a countable dense set.

Finally, $\mathbb{R} \times \mathbb{Z}$ is a topological group under coordinatewise addition, and its nonnegative cone is closed under addition. Restricting the ambient operation gives a continuous map $M_{\mathbb{R}} \times M_{\mathbb{R}} \rightarrow M_{\mathbb{R}}$. $\square$

Example 32.3 (Arithmetic order and topological proximity [06]). Every (r, z) with $r > 0$ is larger, arithmetically, than every standard natural number $(0, n)$. Nevertheless

$$(1/k, 0) \longrightarrow (0, 0).$$

There is no contradiction: the topology is the product topology, not the order topology. In fact, the order topology of any discretely ordered group or its nonnegative cone is discrete, since consecutive integers relative to any point isolate it.

The explicit quotient and remainder maps in (31.3) are continuous for every fixed standard n . In this example they are

$$Q_n(r, z) = (r/n, \lfloor z/n \rfloor), \quad R_n(r, z) = (0, z - n \lfloor z/n \rfloor).$$

The residue classes form a finite clopen partition. These are fixed standard-divisor operations; the assertion does not introduce division by a variable nonstandard element into Presburger arithmetic.

33 Borel orders on Polish abelian groups

The real construction can be extended whenever the positive cone of a divisible ordered Polish group is a Polish subspace. The following theorem shows that Borel definability of the order is already enough. We include the full proof because it is the descriptive engine for the general result.

Theorem 33.1 (Low complexity of an invariant Borel order). *Let A be a Polish abelian group with a translation-invariant total order. If $P = A_{>0}$ is Borel, then all four sign cones are Δ_2^0 . Both order relations $<$ and $\leq$ are Δ_2^0 subsets of A^2 . Moreover, there is a canonically defined countable decreasing chain of closed convex subgroups that decomposes P into countably many locally closed pieces.*

Proof. For a closed subgroup $H \leq A$, write

$$P_H = P \cap H, \quad C(H) = \text{cl}_H(P_H \cup \{0\}), \quad D(H) = C(H) \cap (-C(H)).$$

The set $C(H)$ is a closed additive monoid. Indeed, approximate two of its points by nonnegative points of H and use continuity of addition. Consequently $D(H)$ is a closed subgroup. Totality of the order gives

$$H = C(H) \cup (-C(H)). \quad (33.1)$$

We first show that $D(H) \subsetneq H$ whenever $H \neq \{0\}$. The strict cone P_H is nonmeagre in H . In the nondiscrete case this follows from the Baire category theorem and

$$H = P_H \dot{\cup} (-P_H) \dot{\cup} \{0\},$$

since $\{0\}$ is nowhere dense and negation is a homeomorphism. In the discrete case P_H is nonempty and hence nonmeagre.

As a Borel set, P_H has the Baire property. There is therefore a nonempty open set $V \subseteq H$ on which it is comeagre. For every $t \in V + V$, the set $V \cap (t - V)$ is nonempty and open. Both P_H and $t - P_H$ are comeagre there, so they intersect. Hence

$$V + V \subseteq P_H + P_H \subseteq P_H. \quad (33.2)$$

This is the sumset form of the category argument often associated with Piccard and Pettis; the displayed proof fixes the version needed here.

If $C(H) = H$, then $-P_H \cup \{0\}$ is dense in H . It would meet the nonempty open set $V + V \subseteq P_H$, which is impossible. Thus $C(H) \neq H$, and a fortiori $D(H) \neq H$.

Now form the transfinite recursion

$$H_0 = A, \quad H_{\xi+1} = D(H_\xi), \quad H_\lambda = \bigcap_{\xi < \lambda} H_\xi \quad (\lambda \text{ a limit}). \quad (33.3)$$

Every term is a closed subgroup, with a strict successor decrease whenever the current subgroup is nonzero. This process reaches $\{0\}$ at a countable ordinal η .

To prove the last claim, suppose instead that there were a strict decrease at every $\xi < \omega_1$. Fix a countable base $\mathcal{B}$ of A . For each ξ , choose the first basic set B_ξ meeting H_ξ and disjoint from $H_{\xi+1}$. Such a basic set exists because $H_{\xi+1}$ is closed and properly contained in H_ξ . If $\zeta > \xi$, then B_ξ misses H_ζ , whereas B_ζ meets it. The B_ξ are therefore distinct, contradicting countability.

For $\xi < \eta$, set $C_\xi = C(H_\xi)$ and

$$L_\xi = H_\xi \setminus (-C_\xi).$$

Every member of L_ξ is positive, since a nonpositive member would already lie in $-P_{H_\xi} \cup \{0\} \subseteq -C_\xi$. Equation (33.1) gives the disjoint decomposition

$$H_\xi \setminus H_{\xi+1} = L_\xi \dot{\cup} (-L_\xi).$$

A nonzero element leaves the decreasing chain at a successor stage: its first absent stage cannot be a limit, by the definition of intersection. It follows that

$$P = \bigcup_{\xi < \eta} L_\xi, \quad -P = \bigcup_{\xi < \eta} (-L_\xi). \quad (33.4)$$

Each layer is relatively open in the closed set H_ξ , and hence is F_σ in the ambient metrizable space. Both unions in (33.4) are countable. Thus P and $-P$ are F_σ , and their complements, obtained by adjoining $\{0\}$ to the opposite cone, are also F_σ . This proves all four Δ_2^0 assertions. Applying the continuous difference map $(x, y) \mapsto y - x$ proves the assertions for order relations.

Finally, every H_ξ is order-convex. The limit step is immediate. For a successor step, suppose H is convex, $h \in D(H)$, and $0 \leq x \leq h$. Then $x \in H$ and $x \in C(H)$. Also

$$-x = (-h) + (h - x) \in C(H) + C(H) \subseteq C(H).$$

Hence $x \in D(H)$, proving convexity. $\square$

[write] The sumset step (33.2) is the Pettis argument that source 08 uses in lemmas 47.3 and 53.1 and source 09 states as lemma 61.1. For locally compact second-countable groups, part (1) of theorem 51.2 in Part IV is the special case of theorem 33.1, with a different proof through finite-dimensional cuts.

[write] (Batch 93.) For a real Polish topological vector space with a cone that is Borel, or only hereditarily Baire on closed subspaces, Part XVI's theorem 213.2 sharpens this theorem: the recursion (33.3) is the same (proposition 224.1 [merge]), every step has real codimension one and is the kernel of a continuous functional, and the order is automatically real homogeneous; on a separable Hilbert space this gives a complete classification (theorem 210.1). Source 22 found the recursion independently.

Remark 33.2 (The hypotheses that do the work). The topology must make A a Polish group. A standard Borel group without an established compatible Polish group topology is not enough for this proof. Borelness is used on every closed subgroup; merely knowing that the cone has the Baire property in the ambient space does not automatically give that hereditary regularity.

There is classical related work on measurable orders on locally compact abelian groups and their convex subgroup structure; see Asmar and Montgomery-Smith [22]. We use a direct category proof on Polish groups, including groups that are not locally compact. No historical priority claim for the precise formulation of Theorem 33.1 is needed for the positive construction.

33.1 A countable ordinal invariant of the order and topology

Definition 33.3. The cone closure rank $\rho(A, <)$ is the least η for which the recursion (33.3) reaches $H_\eta = \{0\}$. It is an invariant of the ordered topological group, not just of its abstract group.

Proposition 33.4. For every countable ordinal γ , the product group $\mathbb{R}^\gamma$ with its product topology and lexicographic order has cone closure rank γ . For infinite countable γ , these are different orders on topological groups all isomorphic to $\mathbb{R}^\mathbb{N}$.

Proof. For $\beta \leq \gamma$ put

$$K_\beta = \{x \in \mathbb{R}^\gamma : x_\xi = 0 \text{ for all } \xi < \beta\}.$$

These are closed subgroups. The positive cone is Borel, since it is the countable union, over $\beta < \gamma$, of the sets with zero prefix below β and positive β -coordinate. For $\beta < \gamma$, inside K_β , the closure of the nonnegative cone is exactly

$$\{x \in K_\beta : x_\beta \geq 0\}.$$

The inclusion from left to right follows by coordinate continuity. For the reverse inclusion, points with $x_\beta = 0$ are limits of $x + j^{-1}e_\beta$, which are positive. Intersecting with the negative closure gives $K_{\beta+1}$. At limit stages the intersection is the corresponding K_λ . Thus $H_\beta = K_\beta$ throughout, with the first zero subgroup at γ . Reindexing the countably infinite coordinate set gives the asserted topological group isomorphism. $\square$

Even very low Borel complexity therefore permits arbitrarily long countable closure recursions. The Δ_2^0 conclusion is sharp: the lexicographic positive cone of $\mathbb{R}^2$ is neither open nor closed.

34 A general Polish realization theorem

Theorem 34.1 (Divisible-group realization). *Let A be a divisible Polish abelian group carrying a translation-invariant total order with Borel positive cone. Define*

$$G_A = A \overrightarrow{\times} \mathbb{Z}, \quad M_A = (A_{>0} \times \mathbb{Z}) \cup (\{0\} \times \mathbb{N}). \quad (34.1)$$

Use coordinatewise addition, $0 = (0, 0)$, $1 = (0, 1)$, and the product subspace topology. Then:

- (i) M_A is a Polish model of $\text{Th}(\mathbb{N}; 0, 1, +, <)$, and addition is jointly continuous.
- (ii) Every fixed-standard-divisor quotient and remainder map is continuous. Partial subtraction on its arithmetic domain is continuous.
- (iii) Every relation definable with parameters in a finite Cartesian power of M_A is Δ_2^0 in the product topology.
- (iv) If A is uncountable, so is M_A .

Proof. The order on G_A is lexicographic with A the more significant coordinate. Its least positive element is $(0, 1)$, and its quotient by $\{0\} \times \mathbb{Z}$ is the divisible group A . Thus it is a $\mathbb{Z}$ -group and M_A is its nonnegative cone.

Theorem 33.1 says that $A_{>0}$ and $A_{\geq 0}$ are Δ_2^0 . In the clopen slice $A \times \{k\}$, membership in M_A means $a > 0$ if $k < 0$ and $a \geq 0$ if $k \geq 0$. Both M_A and its complement are therefore countable unions of F_σ sets. In particular M_A is G_δ in $A \times \mathbb{Z}$, so is Polish. Coordinatewise addition restricts continuously to it.

An ordered group is torsion-free. Consequently multiplication by a standard integer $d \geq 1$ is a continuous bijective homomorphism on the divisible Polish group A . Its inverse, division by d , is continuous. Here is a short justification of openness. Given a neighborhood U of zero, choose open V with $V - V \subseteq U$. Countably many translates of V cover A , so countably many translates of dV do as well. The analytic set dV has the Baire property and is nonmeagre. The difference form of the category argument then gives a neighborhood of zero inside $dV - dV \subseteq dU$. Thus multiplication by d is open.

Writing $k = dq + r$, $0 \leq r < d$, the desired maps are

$$Q_d(a, k) = (a/d, q), \quad R_d(a, k) = (0, r). \quad (34.2)$$

They are restrictions of continuous coordinate maps and take their values in M_A . They give the unique decomposition $x = dQ_d(x) + R_d(x)$. Partial subtraction is the restriction of the ambient continuous map $(x, y) \mapsto x - y$ to the subspace where $y \leq x$.

For (iii), consider affine integer-linear expressions on G_A . Their A -coordinate is a continuous affine map into A , and their integer coordinate is a continuous map into the discrete group $\mathbb{Z}$. The nonnegative cone of G_A is Δ_2^0 , by the same slice argument used for M_A . Consequently each linear comparison is Δ_2^0 ; equalities are closed. Since A is divisible, congruence modulo d depends only on the integer coordinate modulo d , so is clopen. Classical Presburger quantifier elimination reduces every parameter-definable relation to a finite Boolean combination of these atoms. Restricting to M_A^k preserves their complexity.

Finally, if A is uncountable, totality and negation show that $A_{>0}$ is uncountable. It embeds into M_A via $a \mapsto (a, 0)$. $\square$

[write] Part (ii) of [theorem 34.1](#) includes, for $A = \mathbb{R}$ and $A = \mathbb{Q}^{\mathbb{N}}$, the continuity statements of [proposition 8.1](#) and of source 04's [theorem 5.4](#); part (iii) extends [theorems 7.1](#) and [7.2](#) to every divisible Polish group with a Borel cone. Source 06 proves part (iii) for its own families in [theorem 37.7](#).

Remark 34.2. Only fixed standard divisors occur in [\(34.2\)](#). A function taking a *variable* divisor as input belongs to a stronger language. The theorem also distinguishes continuous partial subtraction from total truncated subtraction, which will be impossible in an uncountable Polish presentation.

34.1 Definable Borelness for arbitrary continuous presentations

The special Δ_2^0 estimate uses the ambient Polish group. A weaker conclusion holds for every Polish Presburger presentation with continuous addition, whether or not it comes from our construction.

Proposition 34.3. *Let $M \models \text{Th}(\mathbb{N}; 0, 1, +, <)$ have a Polish topology making addition continuous. Every relation definable with parameters is Borel. Every definable partial function between finite Cartesian powers has Borel domain and is Borel measurable on its domain.*

Proof. Equation [\(31.1\)](#) makes $<$ analytic: it is the projection of the closed equality relation $x + z + 1 = y$. Its complement $y \leq x$ is analytic for the same reason. Hence $<$ is Borel by the analytic–coanalytic characterization of Borel sets.

For a fixed $d \geq 1$ and $0 \leq r < d$, the map $x \mapsto dx + r$ is continuous and injective. Its image is Borel by the Borel injective-image theorem. These d residue classes partition M . Congruence relations are therefore Borel, as are integer-linear comparisons after negative coefficients have been moved to the other side. Quantifier elimination now gives every definable relation Borel.

The graph of a definable partial function is a Borel set. Projection from that graph to its input coordinates is Borel and injective. Its image, the domain, is Borel, and its inverse is Borel. Composing that inverse with projection to output coordinates gives the desired function. $\square$

35 A family of models on Baire space

35.1 The countable-ordinal coefficient groups

Fix a countably infinite ordinal α . Let

$$A_\alpha = \mathbb{Q}^\alpha, \quad G_\alpha = A_\alpha \overrightarrow{\times} \mathbb{Z}, \quad M_\alpha = (G_\alpha)_{\geq 0}. \quad (35.1)$$

Here $\mathbb{Q}^\alpha$ is the full Cartesian product, not the finite-support direct sum. Give each copy of $\mathbb{Q}$ the *discrete* topology. On A_α , the least coordinate at which two vectors differ determines their lexicographic comparison. Such a coordinate exists because α is well ordered. Addition and division by a nonzero integer are coordinatewise.

The product topology uses finite coordinate restrictions. In particular, when $\alpha > \omega$, it does not treat an entire transfinite prefix as one open condition. Keeping these two uses of coordinates separate is the main point in the topology proof.

Theorem 35.1 (Continuous arithmetic on Baire space; for $\alpha = \omega$, [theorem 6.1](#)). *For every countably infinite ordinal α , M_α is a continuum-sized model of $\text{Th}(\mathbb{N}; 0, 1, +, <)$ homeomorphic to $\mathbb{N}^\mathbb{N}$. Addition, every quotient and remainder by a fixed positive standard integer, and partial subtraction are continuous. Every parameter-definable relation in every finite arity is Δ_2^0 .*

Proof. The group A_α is divisible, ordered, and Polish: it is a countable product of countable discrete groups. Its topology is zero-dimensional. For each $\beta < \alpha$ define

$$C_\beta^+ = \{a : a_\gamma = 0 \ (\gamma < \beta), \ a_\beta > 0\}, \quad C_\beta^- = \{a : a_\gamma = 0 \ (\gamma < \beta), \ a_\beta < 0\}.$$

Each set is closed. The prefix-zero condition is an intersection of closed coordinate conditions, and the sign condition on a discrete rational coordinate is clopen. Therefore

$$(A_\alpha)_{>0} = \bigcup_{\beta < \alpha} C_\beta^+, \quad (A_\alpha)_{<0} = \bigcup_{\beta < \alpha} C_\beta^-$$

are F_σ , with F_σ complements after adjoining $\{0\}$. Theorem [34.1](#) now gives Polishness and all the arithmetic and definability assertions. Cardinality is $\mathfrak{c}$ because the entire fiber $a_0 = 1$ lies in the positive cone and has countably infinitely many free rational coordinates.

It remains to identify the underlying Polish space. We prove that every nonempty open subset of M_α contains a closed noncompact cylinder. Let U be nonempty and choose $x = (a, n) \in U$. For some finite $F \subseteq \alpha$, the ambient cylinder

$$B = \{(b, n) : b_\gamma = a_\gamma \text{ for } \gamma \in F\}$$

satisfies $x \in B \cap M_\alpha \subseteq U$. Let j be the least natural number not in F . All indices $i < j$ are in F .

If the finite prefix $(a_0, \dots, a_{j-1})$ has a nonzero entry, its first nonzero entry is positive, since $a \geq 0$. Positivity is already forced; in this case additionally set $b_j = 0$. If that prefix is zero, additionally set $b_j = 1$. In both cases the resulting cylinder C fixes only finitely many coordinates, is nonempty, and has strictly positive A_α coordinate. Thus

$$C \subseteq B \cap M_\alpha \subseteq U.$$

The set C is clopen in the ambient product and closed in M_α . Infinitely many rational coordinates remain free. Projection onto any one of them maps C continuously onto the infinite discrete space $\mathbb{Q}$, so C is not compact.

If a compact subset $K \subseteq M_\alpha$ had nonempty interior U , the cylinder just constructed would be a closed subset of K and hence compact, a contradiction. Thus every compact subset has empty interior. The space is nonempty, zero-dimensional, and Polish. The Baire-space characterization gives $M_\alpha \cong \mathbb{N}^\mathbb{N}$. $\square$

[write] For $\alpha = \omega$, M_ω is Part I's Baire model M_B , and [theorem 35.1](#) is [theorem 6.1](#), where an explicit and computable homeomorphism is given ([theorems 6.2](#) and [6.4](#); also source 06's [theorems 35.6](#) and [35.7](#) below). For $\alpha > \omega$ the theorem is new to the collection, and the homeomorphism above comes from the topological characterization of Baire space.

Remark 35.2 (Why the transfinite cases require the full argument). For $\alpha = \omega$, both strict sign cones of A_α are open, since a finite prefix detects each nonzero vector's sign. When $\alpha > \omega$, a vector first nonzero at ω is not an interior point of its sign cone. The closed-piece proof works in both cases. Likewise, the cylinder argument above covers neighborhoods of such transfinite-leading vectors and of zero, not just the dense set of vectors with finite leading coordinate.

Choose a homeomorphism $h : \mathbb{N}^{\mathbb{N}} \rightarrow M_\alpha$ and transport the arithmetic:

$$x \oplus y = h^{-1}(h(x) + h(y)), \quad 0_{\mathbb{N}^{\mathbb{N}}} = h^{-1}(0), \quad 1_{\mathbb{N}^{\mathbb{N}}} = h^{-1}(1).$$

The operation $\oplus$ is jointly continuous. The coefficient model is explicit; the homeomorphism here is supplied by a topological characterization. We do not claim a computable homeomorphism for an arbitrary presentation of a countable ordinal.

35.2 Residue classes as copies of the whole space

Corollary 35.3 (extends [proposition 8.1](#)). *For each standard $d \geq 1$, the map*

$$x \mapsto (Q_d(x), r_d(x))$$

is a homeomorphism $M_A \rightarrow M_A \times \{0, \dots, d-1\}$, where $R_d(x) = r_d(x)1$. Each residue class is clopen and homeomorphic to M_A .

Proof. Continuity of the forward map follows from [Theorem 34.1](#). Its inverse is $(q, r) \mapsto dq + r1$, continuous because the second factor is finite discrete and addition is continuous. Division with remainder proves that the maps are inverse. $\square$

[write] For $M_A = M_{\mathbb{R}}$ and $M_A = M_{\mathbb{B}}$ this is [proposition 8.1](#).

This is an arithmetic explanation of a topological self-similarity. It also applies to $M_{\mathbb{R}}$, whose countably many ray components can be rearranged into any positive finite number of copies.

35.3 An uncountable ordinal spectrum of models

For positive elements g, h of an ordered abelian group, define

$$g \asymp h \iff (\exists k, \ell \in \mathbb{N}_{>0}) (g \leq kh \text{ and } h \leq \ell g).$$

These are the Archimedean equivalence classes. We list the classes in *decreasing magnitude*: a class occurs earlier when its members exceed every standard multiple of a member of the later class.

Theorem 35.4 (An ordinal spectrum on one topological space). *The Archimedean classes of G_α , listed in decreasing magnitude, have order type $\alpha + 1$. Therefore the models M_α , for countably infinite α , are pairwise nonisomorphic, even as additive monoids. There are at least $\aleph_1$ pairwise nonisomorphic continuum-sized Presburger structures continuously presented on $\mathbb{N}^{\mathbb{N}}$. No continuum hypothesis is used.*

Proof. For a positive (a, n) with $a \neq 0$, let $\lambda(a, n)$ be the least $\beta < \alpha$ with $a_\beta \neq 0$. For $a = 0$, let $\lambda(0, n) = \alpha$. If two positive elements have the same leading index $\beta < \alpha$, their leading coefficients are positive rationals. Each can be made larger than the other by multiplication

by a sufficiently large standard integer; lower coordinates do not affect that comparison. Thus they are Archimedean equivalent. The same holds in the integer class.

If $\lambda(g) < \lambda(h)$, then $g > kh$ for every standard positive integer k , since the leading coordinate of g is positive while that coordinate of kh is zero. Every index $\beta \leq \alpha$ occurs. This proves the asserted class order.

An isomorphism of additive positive monoids extends uniquely to an isomorphism of their group completions: represent a group element as a difference of two positives and use cancellation to check independence of the representation. The extension preserves positivity and hence order. It preserves the Archimedean class order. Thus an isomorphism between M_α and M_δ would imply $\alpha + 1 = \delta + 1$, so $\alpha = \delta$.

The set of countably infinite ordinals has cardinality $\aleph_1$. Theorem 35.1 places each corresponding model on Baire space. $\square$

The invariant uses addition. It does not by itself distinguish the pure underlying linear orders; no classification of those pure orders is claimed.

[write] (Batch 89.) Source 16 sharpens this: M_α elementarily embeds into M_β exactly when $\alpha \leq \beta$ (corollary 129.3), and it classifies the embeddings and the Polish elementary submodels of the family (Part X).

35.4 The optimal number of models is the continuum

The ordinal family gives an uncountable hierarchy with a transparent invariant. A different parameter reaches the largest possible number of isomorphism types on a fixed Baire space, without any hypothesis about the value of the continuum.

Theorem 35.5 (Sharp continuum multiplicity). *There are exactly $\mathfrak{c} = 2^{\aleph_0}$ isomorphism types of Presburger models continuously presentable on $\mathbb{N}^{\mathbb{N}}$. The lower bound can be realized with all parameter-definable relations Δ_2^0 and with continuous quotient and remainder by every fixed standard positive integer.*

Proof. For an irrational $s \in (1, 2)$ let $B_s = \mathbb{Q} + \mathbb{Q}s$, ordered as a subgroup of $\mathbb{R}$ but given the discrete topology. This is a countable divisible ordered abelian group. Put

$$A_s = B_s \overrightarrow{\times} \mathbb{Q}^\omega, \quad G_s = A_s \overrightarrow{\times} \mathbb{Z}, \quad M_s = (G_s)_{\geq 0}.$$

The product of the discrete coordinate topologies is Polish. The leading coordinate test makes the sign cones Borel. Theorem 34.1 gives all the arithmetic and definability properties. The finite-cylinder proof of Theorem 35.1 applies with B_s as the first coordinate, so M_s is homeomorphic to Baire space.

The subgroup

$$H_s = \{0\} \times \mathbb{Q}^\omega \times \mathbb{Z}$$

is the unique largest proper convex subgroup of G_s . Indeed it is proper and convex. If a convex subgroup contains an element g with nonzero B_s -coordinate, standard multiples of $|g|$ dominate every element of G_s , because $B_s \subseteq \mathbb{R}$ is Archimedean. Convexity forces such a subgroup to be the whole group. Therefore an isomorphism $M_s \cong M_t$, extended to group completions, induces an ordered additive isomorphism $B_s \cong B_t$.

Every increasing additive isomorphism $f : B_s \rightarrow B_t$ is multiplication by a positive real scalar λ . To see this, let $\lambda = f(1) > 0$. Additivity and divisibility give $f(q) = q\lambda$ for every

rational q . For any $x \in B_s$, comparison with every rational shows that $f(x)/\lambda$ and x have the same rational cut in $\mathbb{R}$, hence are equal. Surjectivity now supplies rational a, b, c, d with

$$1 = \lambda(c + ds), \quad t = \lambda(a + bs), \quad t = \frac{a + bs}{c + ds}.$$

The denominator is nonzero, and $ad - bc \neq 0$, since otherwise t would be rational. Thus for each fixed s only countably many parameters t can yield an isomorphic model. The continuum-sized set of irrational parameters therefore yields continuum many isomorphism types in ZFC.

For the upper bound, fix $X = \mathbb{N}^{\mathbb{N}}$ and a countable dense set $D \subseteq X^2$. Since X is Hausdorff, a continuous map $X^2 \rightarrow X$ is determined by its values on D . There are at most $|X|^{\aleph_0} = \mathfrak{c}$ possible such maps. Choosing finitely many constants does not increase this bound, and all Presburger relations are determined by the addition and constants. Consequently there are at most $\mathfrak{c}$ continuously presented structures and at most $\mathfrak{c}$ isomorphism types. $\square$

[write] Source 06 proves the same count with a different family and adds that the lower bound is attained with the topology, the arithmetic order, 0 and 1 fixed ([theorem 35.8](#)); source 07's statement adds the Δ_2^0 and continuous-division clauses, and [corollary 36.2](#) below adds every oscillation rank. The two refinements are independent, so both statements are printed.

35.5 An explicit model on Baire space [06]

Let $H = \mathbb{Q}^{\mathbb{N}}$, giving every rational coordinate the *discrete* topology. Order H lexicographically: for a nonzero sequence h , its sign is the sign of its first nonzero coordinate. Put

$$G_\omega = H \overrightarrow{\times} \mathbb{Z}, \quad M_\omega = (G_\omega)_{\geq 0}.$$

The coordinatewise group H is divisible. It follows at once that $M_\omega \models \text{Pres}$ and that addition is continuous in the product subspace topology. The remaining issue is to identify that space.

Theorem 35.6 (Baire-space realization [06]; = [theorem 6.1](#)). *The space M_ω is homeomorphic to $\mathbb{N}^{\mathbb{N}}$. Consequently Baire space supports a model of Presburger arithmetic with jointly continuous addition.*

Proof. We give homeomorphisms rather than merely a cardinality argument. The strict cone $H_{>0}$ is the disjoint union of cylinders

$$C_{j,q} = \{h : h(0) = \dots = h(j-1) = 0, h(j) = q\}, \quad j \in \mathbb{N}, q \in \mathbb{Q}_{>0}.$$

Each cylinder is clopen and is homeomorphic to $\mathbb{Q}^{\mathbb{N}}$ by deleting its fixed prefix. A coordinatewise enumeration of $\mathbb{Q}$ identifies $\mathbb{Q}^{\mathbb{N}}$ with $\mathbb{N}^{\mathbb{N}}$. A countable disjoint union of Baire spaces is again Baire space, by using the first coordinate to record the component. Hence $H_{>0} \cong \mathbb{N}^{\mathbb{N}}$.

The closed cone $H_{\geq 0}$ has an equally direct description. Consider the tree of finite rational words whose first nonzero entry, if present, is positive. An all-zero word can be extended by any nonnegative rational; a word already having positive first nonzero entry can be extended by any rational. Every node therefore has countably infinitely many successors. Choose a bijection from $\mathbb{N}$ to each successor set. The resulting rooted-tree isomorphism with $\mathbb{N}^{<\mathbb{N}}$ induces a homeomorphism of branch spaces $H_{\geq 0} \cong \mathbb{N}^{\mathbb{N}}$.

The fixed- z slice of M_ω is clopen, and equals $H_{>0}$ when $z < 0$ and $H_{\geq 0}$ when $z \geq 0$. Their countable disjoint union is therefore Baire space. Transport the arithmetic structure through any of these explicit homeomorphisms. $\square$

35.5.1 A computable stream presentation [06]

Theorem 35.7 (Effective addition [06]; = [theorems 6.2](#) and [6.4](#)). *The homeomorphism in Theorem 35.6 can be chosen computable in both directions for exact rational-coordinate streams. Under this homeomorphism, addition on $\mathbb{N}^{\mathbb{N}}$ is Type-2 computable. The constants 0, 1 and the quotient and remainder maps for each fixed standard divisor are computable as well.*

Proof. Fix computable bijections, with computable inverses, between $\mathbb{N}$ and each of $\mathbb{Q}$, $\mathbb{Q}_{>0}$, $\mathbb{Q}_{\geq 0}$, and $\mathbb{Z}$. Canonical reduced fractions provide such enumerations. Choose the nonnegative enumeration to send 0 to 0. Also fix a computable pairing bijection $\mathbb{N} \leftrightarrow \mathbb{N}^2$.

For $H_{\geq 0}$, decode successive input coordinates using the nonnegative rational enumeration until the first positive output; after that, use the enumeration of all rationals. The inverse uses exactly the same finite-prefix state. Thus each coordinate in either direction requires only finitely much input.

For $H_{>0}$, decode the first input coordinate as (j, k) , emit j zeros and then the k th positive rational, and use the remaining input to produce an arbitrary rational tail. The inverse searches for the first nonzero coordinate. This search terminates on its promised domain $H_{>0}$.

Finally, the first coordinate of a Baire input selects $z \in \mathbb{Z}$; the remaining stream selects a point of $H_{>0}$ or $H_{\geq 0}$ according to the sign of z . This defines a computable homeomorphism

$$F : \mathbb{N}^{\mathbb{N}} \longrightarrow M_{\omega}.$$

Coordinatewise rational addition and integer addition are computable. Therefore

$$a \oplus b = F^{-1}(F(a) + F(b))$$

is Type-2 computable. If the output integer coordinate is negative, the output rational sequence is necessarily strictly positive: a sum of nonnegative ordered-group elements has zero rational part only when both rational parts vanish, in which case both input integers are nonnegative. Thus the search used by F^{-1} always terminates when it is invoked on that branch. Fixed standard division is coordinatewise rational division together with ordinary integer division, so the same argument applies to it and to remainders. $\square$

Here “computable” concerns functions on infinite streams: every requested output coordinate is produced after finitely much input has been inspected. It does not assert a computable enumeration of this uncountable model or decidability of equality between arbitrary streams. It therefore has no conflict with the classical countable-presentation Tennenbaum theorem.

35.6 How many continuous models fit on Baire space? [06]

The existence result has substantial room for variation. The next theorem counts abstract isomorphism types, not merely different coordinate descriptions of one model.

Theorem 35.8 (Sharp abundance [06]; the count of [theorem 35.5](#)). *There are exactly $2^{\aleph_0}$ isomorphism types of Presburger models on Baire space with jointly continuous addition. The lower bound can be realized while keeping the underlying topology, arithmetic order, 0, and 1 fixed and varying only addition.*

Proof. Let $A_0 = \mathbb{Q}$ and $A_1 = \mathbb{Q} + \mathbb{Q}\sqrt{2}$, as additive ordered subgroups of $\mathbb{R}$, and give both groups discrete topology. For $s \in 2^{\mathbb{N}}$ put

$$H'_s = \prod_{n \in \mathbb{N}} A_{s(n)}, \quad G'_s = H'_s \vec{\times} \mathbb{Z}, \quad M'_s = (G'_s)_{\geq 0},$$

using the first differing coordinate for the order on H'_s . Both coefficient groups are countable and divisible. Each has countably infinitely many positive and nonnegative elements. The proof of Theorem 35.6, with its coordinate-dependent enumerations, therefore applies unchanged. Hence each M'_s is a Baire-space model with continuous addition.

For positive elements of an ordered abelian group, say $x \asymp y$ if $x \leq my$ and $y \leq nx$ for some positive standard integers m, n . This is Archimedean equivalence. In G'_s , two positive elements are equivalent exactly when they have the same first nonzero coordinate, or when both are ordinary integers. Indeed, equal-rank real leading coefficients dominate one another after sufficiently large standard integer multiplication; an earlier nonzero coordinate dominates every standard multiple of a later one. The classes, ordered from largest magnitude to smallest, consequently have order type $\omega + 1$.

For each n , define the convex subgroup

$$U_n = \{(h, z) : h(0) = \cdots = h(n-1) = 0\}.$$

It is determined intrinsically by the Archimedean classes of rank at least n , and

$$U_n/U_{n+1} \cong A_{s(n)}. \quad (35.2)$$

Any additive-monoid isomorphism $f : M'_s \rightarrow M'_t$ extends to an ordered-group isomorphism of their difference groups by

$$\tilde{f}(a - b) = f(a) - f(b).$$

This is well-defined since $a - b = c - d$ means $a + d = c + b$; every signed group element is a difference of nonnegative ones. The order is preserved because $a \leq b$ in a cone is equivalent to $\exists c (a + c = b)$.

The ordered set $\omega + 1$ has no nontrivial order automorphism, so $\tilde{f}$ must preserve each rank and send U_n to its counterpart. The quotients in (35.2) then recover $s(n)$: A_0 has rational vector-space dimension 1, whereas A_1 has dimension 2. An additive isomorphism between divisible torsion-free groups is automatically $\mathbb{Q}$ -linear. Thus $M'_s \cong M'_t$ implies $s = t$. This gives $2^{\aleph_0}$ distinct isomorphism types.

For the fixed-order refinement, A_0 and A_1 are countable dense linear orders without endpoints. A back-and-forth construction gives an order isomorphism $\theta : A_0 \rightarrow A_1$ fixing 0: alternately place the next unused element of each enumerated order into the appropriate nonempty interval of the other. Since both topologies are discrete, θ is a homeomorphism. Apply either θ or the identity in coordinate n according to $s(n)$, and leave the integer coordinate unchanged. This gives an order-preserving homeomorphism $M_\omega \rightarrow M'_s$ fixing 0, 1. Pulling back all the additions, and then using one homeomorphism $M_\omega \cong \mathbb{N}^{\mathbb{N}}$, proves the refinement.

Finally, a continuous function $\mathbb{N}^{\mathbb{N}^2} \rightarrow \mathbb{N}^{\mathbb{N}}$ is determined by its values on a fixed countable dense subset. There are at most $(2^{\aleph_0})^{\aleph_0} = 2^{\aleph_0}$ such value assignments. Choices of the two constants do not enlarge this bound, and in Presburger arithmetic the order is determined by addition. Thus no more than continuum many isomorphism types are possible. $\square$

The continuous family in this theorem is not asserted to be an uncountable family of absolutely computable operations. There are only countably many algorithms. The effective

assertion proved above applies to the particular rational ω -indexed model; computable variations can be handled separately, and noncomputable sequences s may serve as oracles.

35.6.1 Countably transfinite Archimedean scales [06]

There is a different useful variation, relevant to the repository's interest in ordered groups and ordinal scales.

Theorem 35.9 (Prescribed well-ordered scales [06]; = [theorems 35.1](#) and [35.4](#)). *For every countable ordinal $\alpha \geq \omega$, the positive cone*

$$M_\alpha = (\mathbb{Q}^\alpha \overrightarrow{\times} \mathbb{Z})_{\geq 0},$$

with discrete rational coordinates and the product subspace topology, is a Baire-space Presburger model with continuous addition. Its Archimedean classes, in decreasing magnitude, have order type $\alpha + 1$. In particular distinct such ordinals give nonisomorphic additive models.

Proof. For $\beta < \alpha$, let

$$C_\beta^\pm = \{h : (\forall \gamma < \beta) h(\gamma) = 0, \pm h(\beta) > 0\}.$$

Each set is closed, including at limit indices, since the preceding zero conditions are closed and the sign condition is clopen in a discrete coordinate. Both

$$M_\alpha = \bigcup_{\beta < \alpha} (C_\beta^+ \times \mathbb{Z}) \cup (\{0\} \times \mathbb{N})$$

and its complement in $\mathbb{Q}^\alpha \times \mathbb{Z}$ are F_σ . Therefore M_α is G_δ in a Polish space, hence Polish and zero-dimensional. Its algebra and addition follow from [Lemma 31.2](#).

Every nonempty open set in M_α contains a closed copy of $\mathbb{Q}^\mathbb{N}$. To see this, choose a point and a neighborhood fixing its integer coordinate and finitely many rational coordinates. If the chosen point first becomes positive at a finite coordinate j , fix all its coordinates through j . If it vanishes on every finite coordinate, choose a finite j beyond all the finitely prescribed finite indices, put zeros below j , and put 1 at j . Keep the other prescribed coordinates. In either case, the resulting finite cylinder is clopen in the ambient product, lies inside the positive cone and the chosen neighborhood, and has countably infinitely many free coordinates. Hence no compact subset of M_α has nonempty interior.

For completeness, a nonempty zero-dimensional Polish space X with this last property is homeomorphic to Baire space. Use a complete compatible metric. Every nonempty clopen set is complete and noncompact, and hence is not totally bounded. It therefore has a countable clopen cover of arbitrarily small diameter with no finite subcover. Removing earlier members from each member disjointifies it into countably infinitely many nonempty clopen sets. Repeat within each piece, with diameters tending to zero. Each branch of the resulting $\mathbb{N}$ -branching tree determines exactly one point by completeness, and each point determines a branch. The pieces are clopen, so the resulting bijection in both directions is continuous. Apply this construction to M_α .

Finally, the same leading-coordinate argument as in [Theorem 35.8](#) identifies the Archimedean class of a noninteger element with its least nonzero index $\beta < \alpha$; the integer class is last. Their ordered quotient is exactly $\alpha + 1$. An additive isomorphism preserves this quotient. Equality of the successor order types implies equality of α and β by removing their final elements. $\square$

The countable ordinals in this theorem describe *classes of magnitude*, not an increasing or decreasing ω_1 -sequence inside one arithmetic order. There is therefore no conflict with the uncountable-monotone-sequence obstruction in [2].

36 Unbounded oscillation at Borel level two

The Borel-level conclusion may suggest that all definable sets in these models have similarly simple topological behavior. A finer invariant shows that they do not. One fixed Presburger formula already has arbitrarily high countable boundary oscillation ranks across the models.

36.1 The derivative and its exact convention

Let X be a space and $E \subseteq X$. For a closed $F \subseteq X$, put

$$D_E(F) = \text{cl}_F(E \cap F) \cap \text{cl}_F(F \setminus E).$$

Starting with $F_0 = X$, set

$$F_{\xi+1} = D_E(F_\xi), \quad F_\lambda = \bigcap_{\xi < \lambda} F_\xi \quad (\lambda > 0 \text{ a limit}). \quad (36.1)$$

The *boundary oscillation rank* used here is the least ξ with $F_\xi = \emptyset$, when such a stage exists; otherwise we assign rank ∞ . For the characteristic function $\mathbf{1}_E$, this is the oscillation derivative at any threshold $0 < \varepsilon \leq 1$. The definition fixes the indexing convention; no identification with all possible conventions for difference-hierarchy ranks is intended.

In A_α , let u be the vector with $u_0 = 1$ and $u_\gamma = 0$ for $\gamma > 0$. Set

$$p = (u, 0), \quad E_\alpha = \{x \in M_\alpha : x < p\}.$$

This is the fixed formula $\varphi(x; y) \equiv x < y$, unary in x , with the indicated parameter p . Its chosen positive zeroth coordinate ensures that all boundary perturbations below stay inside the model.

Theorem 36.1 (Exact boundary oscillation ranks). *Write the countably infinite ordinal α uniquely as*

$$\alpha = \omega \cdot \beta + n, \quad 0 < \beta < \omega_1, \quad n < \omega.$$

For $0 < \mu \leq \alpha$, write

$$C_\mu = \{(a, z) \in M_\alpha : a_\gamma = u_\gamma \text{ for all } \gamma < \mu\}.$$

The recursion (36.1) for E_α satisfies

$$F_\xi = C_{\omega \cdot \xi} \quad (1 \leq \xi \leq \beta), \quad F_{\beta+1} = \emptyset. \quad (36.2)$$

Its rank is exactly $\beta + 1$. Consequently this one formula realizes every countable successor rank at least 2, and its ranks are unbounded below ω_1 .

Proof. Suppose first that $x = (a, z)$ differs from p at a finite first rational coordinate $k < \omega$. Fixing coordinates $0, \dots, k$ decides its comparison with p on a neighborhood of x . Such a point is not in F_1 .

Conversely, suppose that $a_k = u_k$ for every $k < \omega$. Every basic neighborhood of x fixes only finitely many coordinates. Choose an unfixed positive natural index k , change its value

from 0 to -1 or to 1 , and leave everything else unchanged. All earlier coordinates still agree with u . The two resulting points are respectively below and above p , and remain positive because the zeroth coordinate stays 1 . Thus $F_1 = C_\omega$.

For the successor step assume $1 \leq \xi < \beta$ and $F_\xi = C_\mu$, where $\mu = \omega \cdot \xi$. At least ω coordinates remain before α . Inside C_μ , a first residual difference at $\mu + k$, $k < \omega$, is decided by fixing the finite residual prefix $\mu, \dots, \mu + k$. These points leave the derivative. If a point agrees with u throughout the block $[\mu, \mu + \omega)$, every basic relative neighborhood leaves some $\mu + k$ unfixed. Perturbing this coordinate in the two directions gives points on both sides of p , still in C_μ and in M_α . Therefore

$$F_{\xi+1} = C_{\mu+\omega} = C_{\omega \cdot (\xi+1)}.$$

At a nonzero limit $\lambda \leq \beta$, intersecting the prefix conditions and using continuity of ordinal multiplication in its right argument gives

$$F_\lambda = \bigcap_{1 \leq \xi < \lambda} C_{\omega \cdot \xi} = C_{\omega \cdot \lambda}.$$

At stage β , only n rational coordinates and the integer coordinate are free. The resulting fiber is homeomorphic to the discrete space $\mathbb{Q}^n \times \mathbb{Z}$. Every subset of that fiber is relatively clopen, so the next derivative is empty. Each earlier displayed stage contains $\{u\} \times \mathbb{Z}$, proving that no earlier stage is empty.

Taking $\alpha = \omega \cdot \beta$ gives all stated successor ranks. □

Coordinate order α	Boundary oscillation rank of E_α
ω	2
$\omega \cdot 2$	3
ω^2	$\omega + 1$
$\omega \cdot (\omega + 1)$	$\omega + 2$
ω^ω	$\omega^\omega + 1$

The first derivative removes points where a finite prefix decides the comparison. Each later derivative removes one further block of ω coordinates. This is a topological calculation, although the defining formula itself never changes.

36.2 Explicit continuous approximations

There is also a direct approximation of $\mathbf{1}_{E_\alpha}$ by continuous functions. Choose an enumeration $\gamma_0, \gamma_1, \dots$ of α and put $S_N = \{\gamma_0, \dots, \gamma_N\}$. Sort this finite set in its ordinal order. Define $c_N(a, z)$ to be 1 exactly when comparison of the coordinates in S_N , followed by the integer coordinate, places (a, z) below $(u, 0)$.

Each $c_N : M_\alpha \rightarrow \{0, 1\}$ is continuous, because it depends on finitely many discrete coordinates. If $a \neq u$, its true first differing coordinate eventually belongs to S_N . From that point on, every $c_N(a, z)$ is the correct answer; coordinates preceding the true first difference are equal and cannot change it. If $a = u$, every $c_N(a, z)$ already equals $\mathbf{1}_{\{z < 0\}}$. Hence

$$c_N(x) \longrightarrow \mathbf{1}_{E_\alpha}(x)$$

with eventual equality at each point.

Thus these characteristic functions are explicitly Baire class 1, while their boundary oscillation ranks have no countable uniform bound across the family. Low Borel level and finite formula length do not give a uniform bound on this finer topological invariant.

Corollary 36.2 (Optimal multiplicity at every specified rank). *For every countable $\beta > 0$, there are $\mathfrak{c}$ pairwise nonisomorphic continuous Presburger models on Baire space with a parameter p for which the cut $x < p$ has boundary oscillation rank $\beta + 1$. All their definable relations are Δ_2^0 .*

Proof. In Theorem 35.5, replace the tail $\mathbb{Q}^\omega$ by $\mathbb{Q}^{\omega \cdot \beta}$ and choose $p = (1, 0, 0)$. The model remains Polish and homeomorphic to Baire space by the same closed-cylinder proof. An isomorphism still induces an isomorphism of the greatest Archimedean quotients B_s , so the continuum lower bound is unchanged.

For the cut at p , comparison at the discrete B_s -coordinate is locally decided unless that coordinate equals 1. On that fiber, the derivative removes successive blocks of ω rational coordinates, exactly as in Theorem 36.1. The first derivative fixes the B_s -coordinate to 1 and the first ω tail coordinates to zero; the ξ th derivative fixes the first $\omega \cdot \xi$ tail coordinates. The β th derivative is the discrete integer fiber and the next is empty. The rank is $\beta + 1$. $\square$

37 What continuity permits and forbids

37.1 The arithmetic order is not the Polish topology

In any Presburger model, the intrinsic order topology is discrete. For $a > 0$, the arithmetic interval $(a - 1, a + 1)$ is the singleton $\{a\}$; at zero the initial interval $[0, 1)$ is $\{0\}$. An uncountable discrete space is not second countable and hence not Polish. Accordingly, a positive answer to Glazer's question must distinguish continuity of addition from compatibility with the intrinsic order topology.

In $M_{\mathbb{R}}$,

$$x_j = (1/j, 0) \longrightarrow (0, 0), \quad x_j > (0, k) \quad \text{for every standard } k.$$

Topological convergence to zero is entirely compatible with being larger than every standard numeral. In M_α , let v_j be the coordinate unit vector at the natural index j . Then the analogous sequence $(v_j, 0)$ converges to zero and exceeds every standard numeral.

These sequences show explicitly that strict order is neither open nor closed. The ordered pair $(0, 1)$ belongs to $<$ but is approached by $(x_j, 1)$, which do not. The pairs $(1, x_j)$ belong to $<$ and converge to $(1, 0)$, which does not. The following general obstruction explains why this behavior is unavoidable.

Proposition 37.1 (Closed arithmetic order forces discreteness; = theorem 7.4). *Let M be a Hausdorff model of $\text{Th}(\mathbb{N}; 0, 1, +, <)$ with continuous successor. If $\leq$ is closed in M^2 , then the topology of M is discrete. If M is uncountable Polish with continuous addition, neither $<$ nor $\leq$ is open or closed.*

Proof. If $\leq$ is closed, then $<$ is open because its complement is the reverse non-strict order. Sections of $<$ are open. The singleton intervals just described are therefore open, so the topology is discrete.

If $<$ is open, the same singleton argument applies. If $<$ is closed, then $x \leq y \iff x < y + 1$ and continuity of successor make $\leq$ closed. If $\leq$ is open, then $x < y \iff x + 1 \leq y$ makes $<$ open. Each of these four possibilities forces discreteness. An uncountable second-countable space cannot be discrete. $\square$

[write] [Proposition 37.1](#) is [theorem 7.4](#) (source 05), and its first sentence also follows from source 04's [theorem 7.5](#), which needs no continuity of successor. The monus clause of [corollary 37.2](#) is source 04's [corollary 7.6](#); the min and max clauses are new.

Corollary 37.2 (Three individually forbidden operations). *On an uncountable Polish Presburger model with continuous addition, none of the following operations can be continuous:*

$$\min(x, y), \quad \max(x, y), \quad x \dot{-} y = \max\{x - y, 0\}.$$

Proof. In a Hausdorff space equalizers of continuous maps are closed. Each of the three operations, if continuous, would make $\leq$ closed through one of the equivalences

$$x \leq y \iff \min(x, y) = x \iff \max(x, y) = y \iff x \dot{-} y = 0.$$

Apply [Proposition 37.1](#). □

This does not contradict continuity of partial subtraction: its domain is the subspace of pairs with $y \leq x$, and it is not extended across that domain's boundary by assigning the value zero.

37.2 Compactness remains impossible

The following obstruction is established in the earlier work of Enayat, Hamkins, and Wcisło [2]. We include the short jointly continuous case to locate our locally compact example relative to compact spaces.

Proposition 37.3 (= [corollary 12.2](#)). *No infinite compact Hausdorff space supports a Presburger model with jointly continuous addition. In particular, Cantor space does not.*

Proof. The set

$$C = \{(x, y, z) \in M^3 : x + z = y\}$$

is closed and compact. Its projection onto (x, y) is $\leq$, so this relation is compact and closed in the Hausdorff space M^2 . [Proposition 37.1](#) makes M discrete, contradicting compactness and infinitude. □

[write] This is [corollary 12.2](#) of Part I, there with the compact-cancellative-monoid proof.

The model $M_{\mathbb{R}}$ is locally compact but not compact. Its separated ray components and its incomplete ambient-metric ends present no conflict with this obstruction.

37.3 A second topology makes predecessor continuous

In the topology of [Theorem 32.1](#), total truncated predecessor

$$\text{pred}(x) = \begin{cases} 0, & x = 0, \\ x - 1, & x > 0 \end{cases}$$

is discontinuous at zero: $x_j = (1/j, 0) \rightarrow 0$, whereas $\text{pred}(x_j) = (1/j, -1)$ cannot converge to zero. That particular discontinuity is not forced by additive arithmetic.

Proposition 37.4. *The same algebraic model $M_{\mathbb{R}}$ admits a second Polish topology making addition, successor, and truncated predecessor continuous. Its isolated points are exactly the standard numerals.*

Proof. Use the topological disjoint sum

$$M_{\mathbb{R}}^{\sharp} = \mathbb{N}_{\text{discrete}} \sqcup ((0, \infty) \times \mathbb{Z}_{\text{discrete}}),$$

with the same underlying identification $(0, n) \leftrightarrow n$. It is a countable sum of Polish spaces. Standard points are isolated, and nonstandard points lie in open real rays, so have no isolated points.

The product splits into four clopen pieces according to whether each input is standard. On two standard inputs addition is a map on a discrete space. On a mixed piece it is, on each standard slice, translation of the integer coordinate. On two nonstandard inputs it is ordinary coordinate addition on positive real coordinates. Each restriction is continuous, and a sum involving a nonstandard input remains nonstandard. Hence addition is continuous on the whole product.

Successor and truncated predecessor act on the standard discrete part as the familiar maps, and on the nonstandard part as translations by ± 1 of the integer coordinate. They are continuous on the clopen pieces and therefore globally. $\square$

[merge] Under the identification $(a, n) \mapsto aX + n$ of [section 26](#), $M_{\mathbb{R}}^{\sharp}$ is exactly the clopen stratum $(C_{\mathbb{R}}, \tau_{\text{FS}})$ of 03's cone in [proposition 26.1](#), whose parts (iii) and (iv) found the same topology with continuous addition; [proposition 37.4](#) adds that successor and truncated predecessor are continuous. It is also the subspace of polynomials of degree at most one in source 06's cone ([proposition 42.1](#)). Part I does not ask whether the discontinuity of predecessor in [proposition 8.2](#) is forced; this proposition shows that it depends on the topology.

The two presentations are not homeomorphic: one has no isolated points and the other has countably many. Binary truncated subtraction still cannot be continuous in the second presentation, by [Corollary 37.2](#).

37.4 The explicit models have no ordered-semiring expansion

Our examples are subject to an algebraic obstruction stronger than a mere failure of Borel regularity of multiplication.

Proposition 37.5 (Cofinal multiples obstruct multiplication; compare [theorem 10.4](#)). *Suppose $M \models \text{Th}(\mathbb{N}; 0, 1, +, <)$ has an element c such that*

- (i) $c > n1$ for every standard $n \in \mathbb{N}$;
- (ii) the standard multiples $\{nc : n \in \mathbb{N}\}$ are cofinal in M .

Then there is no multiplication, with the given addition and identity 1, making M the nonnegative cone of a discretely ordered ring. More generally, no semiring multiplication can satisfy distributivity, the given identity, and monotonicity under this arithmetic order.

Proof. Suppose such a multiplication exists. By cofinality choose a standard k with $c^2 < kc$; increasing a weak cofinal bound by one gives a strict bound. Since $c > (k+1)1$, additive order gives $c = (k+1)1 + u$ for some $u > 0$. Distributivity and nonnegativity of products give

$$c^2 = (k+1)c + uc \geq (k+1)c > kc,$$

a contradiction. $\square$

Corollary 37.6. *Neither $M_{\mathbb{R}}$ nor any M_{α} has a Peano-arithmetic expansion with the given addition, regardless of the regularity of multiplication.*

Proof. For $M_{\mathbb{R}}$ take $c = (1, 0)$. For M_{α} take $c = (u, 0)$ with $u_0 = 1$ and all other coordinates zero. In each case c exceeds every standard numeral. Given any element, choose a standard integer larger than its leading real or rational coordinate; that multiple of c strictly exceeds the element. Proposition 37.5 applies. $\square$

This proof shows exactly why the additive construction cannot be promoted to a model of full arithmetic by selecting a sufficiently irregular product. It also leaves room for studying other additive models, without the cofinal-multiple property, against Glazer's more general topological obstructions.

[write] Proposition 37.5 is a weaker form of theorem 10.4, which needs neither monotonicity nor commutativity; corollary 37.6 applies it to the whole ordinal family. Source 06's lemma 37.9 and theorem 37.10 remove the remaining hypothesis that the multiplicative unit be the named 1.

37.5 Definable regularity and an algebraic obstruction [06]

37.5.1 All additive definable relations are low in the Borel hierarchy [06]

Theorem 37.7 (Definable relations [06]; extends theorem 7.1). *In $M_{\mathbb{R}}$, in every M_{α} above, and in every M'_s of Theorem 35.8, every first-order Presburger-definable relation with parameters is Δ_2^0 in the corresponding finite product topology. Fixed-standard-modulus residue classes are clopen.*

Proof. In the ambient group $\mathbb{R} \overrightarrow{\times} \mathbb{Z}$, the nonnegative cone and its complement are each a union of an open real-coordinate condition and a closed zero-coordinate condition. Thus the cone is Δ_2^0 . For the countable rational and mixed-coefficient products, the decompositions by first nonzero index used above give the same conclusion. Equality is closed in the ambient Hausdorff topological group.

Every integer-linear affine map from a finite power of the ambient group to that group is continuous. Consequently its comparison with zero defines a Δ_2^0 set. Congruence modulo n depends only on the discrete final integer coordinate, because the other factor is divisible. Such congruences are clopen, including for affine linear combinations and parameters.

Apply the quantifier-elimination conclusion of Lemma 31.1. A formula on the nonnegative cone is translated to the group by restricting its quantified variables to that cone. Elimination produces a finite Boolean combination of the described atomic conditions. Restrict it back to the cone and use closure of Δ_2^0 under finite Boolean operations. $\square$

Remark 37.8 (Graphs versus functions [06]). Low Borel complexity of a definable graph does not imply continuity of its function. In the real example the truncated predecessor

$$p(x) = \begin{cases} 0, & x = 0, \\ x - 1, & x > 0 \end{cases}$$

is definable but discontinuous at zero: $(1/k, 0) \rightarrow 0$, whereas $p(1/k, 0) = (1/k, -1)$ stays in a different integer slice from $p(0)$. In the Baire model, let e_k be the rational sequence with 1 in coordinate k and zeros elsewhere. Then $(e_k, 0) \rightarrow 0$ gives the identical obstruction.

The order relation itself is neither open nor closed in these two basic models. Write $x_k = (e_k, 0)$ in the Baire model and $x_k = (1/k, 0)$ in the real model. In both cases $x_k > 1$ and

$x_k \rightarrow 0$. Thus the pairs $(x_k, 1)$ lie outside the relation $<$ but converge to $(0, 1)$ inside it. The pairs $(1, x_k)$ lie inside $<$ but converge to $(1, 0)$ outside it.

The standard cut $\{0\} \times \mathbb{N}$ is closed in these models and is not definable, even with parameters, in the Presburger language. If it were defined by $\varphi(x, a)$, it would contain zero and be closed under successor, while omitting nonstandard elements. The induction instance for φ would then fail. This supplies a concrete distinction between topological visibility and first-order definability.

37.5.2 Why the lexicographic models cannot acquire multiplication [06]

The failure of multiplication for our first examples is stronger than the absence of a continuous or Borel multiplication.

Lemma 37.9 (The unit is forced [06]). *Suppose a nontrivial Presburger additive monoid M , with least positive element e , is given any unital semiring multiplication, with the same addition and zero. Then its multiplicative identity is e .*

Proof. The natural order $x \leq y \iff \exists z (x + z = y)$ is preserved by multiplication on either side, by distributivity. Write u for the multiplicative identity. It is nonzero, so $e \leq u$ and $0 \leq e^2 \leq eu = e$. By discreteness, e^2 is either 0 or e . If $e^2 = 0$, use division by two in the Presburger monoid to write $u = 2q + re$, where $r = 0$ or 1. Multiplication by e gives $e = ue = 2(qe)$, impossible for the least positive element. Therefore $e^2 = e$. If $u > e$, write $u = e + v$ with $v \geq e$. Then $e = ue = e^2 + ve \geq 2e$, again impossible. Thus $u = e$. $\square$

Theorem 37.10 (No semiring expansion [06]; compare [theorem 10.4](#)). *None of $M_{\mathbb{R}}$, the nontrivial M_{α} , or the M'_s admits a unital semiring multiplication with its prescribed additive monoid. This remains true without imposing any topology or prescribing the multiplicative identity in advance.*

Proof. In each example choose a with leading coordinate 1 in the first divisible coordinate and zero elsewhere. Then $a > n \cdot 1$ for every standard n . On the other hand, for every $b \in M$ there is a standard N with $b < Na$, since an ordinary integer can be chosen larger than the real leading coefficient of b . Thus standard multiples of a are cofinal in M .

Any proposed multiplication has identity 1 by Lemma 37.9. Since $a \geq n \cdot 1$, distributivity and the natural order imply $a^2 \geq na$ for every standard n . Taking $b = a^2$ in cofinality gives $a^2 < Na$ for some standard N , a contradiction. $\square$

The existing repository report contains a more general obstruction to PA expansions of nonstandard split $\mathbb{Z}$ -groups, using residue types [35]. The argument here has a different and stronger conclusion for the displayed groups: their largest Archimedean magnitude class already rules out every unital semiring structure. The next example escapes this obstruction by having polynomial degrees of arbitrarily large magnitude.

38 A Polish semiring with full additive induction [06]

38.1 The ring and its order [06]

Let

$$A = \mathbb{Z} + t\mathbb{R}[t] = \left\{ m + \sum_{i=1}^d a_i t^i : m \in \mathbb{Z}, a_i \in \mathbb{R}, d < \omega \right\}.$$

The indeterminate t is ordered as positive infinite: a nonzero polynomial is positive exactly when its leading coefficient is positive. This is an ordered subring of $\mathbb{R}[t]$ under its leading-term order. It is discrete because a positive constant in A is a positive integer, while every positive nonconstant polynomial exceeds every integer. Put $P = A_{\geq 0}$.

Algebraically, $A/\mathbb{Z} \cong t\mathbb{R}[t]$ is a divisible ordered abelian group. Thus P is a model of Pres as an additive structure, and its inherited polynomial multiplication makes it a model of PA^- . This underlying polynomial construction has classical precedents and occurs explicitly in the repository [24, 36]. Our task is to choose and verify a topology.

38.2 The topology by degree and constant coefficient [06]

For $d \geq 1$ and $m \in \mathbb{Z}$, let

$$P_{d,m} = \{m + a_1t + \cdots + a_d t^d : a_d > 0\}.$$

Give $P_{d,m}$ the coefficient topology $\mathbb{R}^{d-1} \times (0, \infty)$, and make each $n \in \mathbb{N}$ a singleton component. Define P to be the topological disjoint union

$$P = \mathbb{N}_{\text{discrete}} \sqcup \bigsqcup_{d \geq 1, m \in \mathbb{Z}} P_{d,m}. \quad (38.1)$$

In particular, convergence within a nonconstant stratum eventually has fixed degree and constant coefficient.

Theorem 38.1 (A continuous weak arithmetic semiring [06]). *The space in (38.1) is uncountable, locally compact, and Polish. Both polynomial addition and polynomial multiplication are jointly continuous on it. Its additive reduct satisfies full Presburger arithmetic, and its semiring structure satisfies PA^- .*

Proof. Each $P_{d,m}$ is homeomorphic to $\mathbb{R}^d$ by applying log to the positive leading coefficient. The countably many components in (38.1) are locally compact Polish spaces. A complete metric is obtained by bounding complete component metrics by 1 and putting distance 2 between components. Countable dense sets in the components give separability. The algebraic assertions were verified above.

It remains to prove joint continuity rather than only separate continuity. Products of components form an open cover of P^2 . On $P_{d,m} \times P_{e,n}$, positivity of both leading coefficients prevents cancellation in addition, so

$$\deg(p + q) = \max(d, e), \quad (p + q)(0) = m + n.$$

Thus this whole product maps into the fixed component $P_{\max(d,e), m+n}$, and the coefficient map there is linear and continuous.

On the same product,

$$\deg(pq) = d + e, \quad (pq)(0) = mn,$$

and the leading coefficient is a positive product. It maps into the fixed component $P_{d+e, mn}$, with continuous polynomial coefficient maps. If one input is a positive standard integer, its component is a singleton and the corresponding operations are coefficient translation or scalar multiplication. If an input is zero, multiplication is constant on its open singleton component. Pairs of standard integers are handled by singleton products. This proves continuity on every member of an open cover of P^2 , hence joint continuity globally. $\square$

This theorem concerns the nonnegative cone. The analogous topology on all signed polynomials would not make the entire ring a topological ring: cancellation can reduce degree. Even partial subtraction on the cone fails continuity. For instance

$$((1 + 1/k)t, t) \longrightarrow (t, t), \quad (1 + 1/k)t - t = (1/k)t \not\longrightarrow 0,$$

because zero is isolated. This is consistent with the continuity just proved for addition and multiplication of nonnegative elements.

38.3 An exact failed induction instance [06]

Theorem 38.2 (Failure of open induction [06]). *In the polynomial model P , the quantifier-free formula*

$$\varphi(x; t) \equiv x^2 < t$$

defines exactly the standard natural numbers. Consequently P fails !Open , despite satisfying every first-order additive induction instance.

Proof. If $x = n \in \mathbb{N}$, then $n^2 < t$. If x is a positive nonconstant polynomial of degree $d \geq 1$, then x^2 has degree $2d \geq 2$ and positive leading coefficient, so $x^2 > t$. Therefore $\varphi(P; t) = \mathbb{N}$. It contains zero and is closed under successor, but omits t . This falsifies the open-induction instance with parameter t . $\square$

The full omnific integer cone and the polynomial subring are different arithmetic structures. This counterexample concerns the subring $\mathbb{Z} + t\mathbb{R}[t]$ only; it does not transfer to the full omnific integer cone simply because the subring embeds there.

Proposition 38.3 (Additive definability in the polynomial topology [06]). *Every relation definable with parameters in the Presburger reduct of P is Δ_2^0 . Quotient and remainder by each fixed standard divisor are continuous.*

Proof. On a fixed product of coefficient components, an affine integer-linear expression has bounded degree and continuous coefficient functions, with fixed integer constant coefficient. Its positive and negative sign conditions are finite unions of sets where all higher coefficients vanish and a specified coefficient has the required sign; the residual constant case is also included. These sets and their complements are F_σ on each component. There are countably many clopen product components, so the conditions are Δ_2^0 globally. Congruences depend only on the integer constant term and are clopen. Apply Presburger elimination.

For standard $n \geq 1$, division sends the nonconstant coefficients a_i to a_i/n and the constant term m to $\lfloor m/n \rfloor$; the positive leading coefficient preserves degree. Remainder is the standard integer $m \bmod n$. These maps are continuous on each clopen component, hence on all of P . $\square$

39 Zero must be isolated when both operations are continuous [06]

The preceding polynomial example has isolated points, whereas our additive examples are perfect. There is an exact structural reason for this contrast. The essential subset-sum and probability method below comes from Glazer's proof of his first theorem [1]. An audit of its assumptions removes the arithmetic induction hypothesis. We also avoid any need to prove injectivity of the subset-sum limit map.

Theorem 39.1 (Isolated zero [06]). *Let A be a discretely ordered commutative unital ring, and put $M = A_{\geq 0}$. If M carries a Polish topology under which its restricted addition and multiplication are jointly continuous, then 0 is isolated.*

Corollary 39.2 ([06]). *No perfect Polish space supports a topological model of PA^- with continuous addition and multiplication. In particular Baire space supports the additive models constructed above, but cannot support such a weak-arithmetic semiring structure.*

39.1 Borel order from continuous addition [06]

We first record a point that prevents any hidden regularity assumption. For $a, b \in M$,

$$\begin{aligned} a < b &\iff \exists c \in M (b = a + c + 1), \\ a \geq b &\iff \exists c \in M (a = b + c). \end{aligned}$$

The equations define closed subsets of M^3 , because addition is continuous and M is Hausdorff. Their projections are analytic. They are complementary relations, so the analytic/coanalytic separation theorem gives that the order is Borel [8]. This argument uses no order-topology compatibility.

39.2 A uniformly convergent subset-sum map [06]

Suppose, toward a contradiction, that 0 is not isolated. Since $[0, 5] = \{0, 1, \dots, 5\}$ and finite sets are closed, one can choose $x_k > 5$ with $x_k \rightarrow 0$. Fix a complete compatible metric d on M . Set $y_0 = 1$. After choosing $y_0, \dots, y_{n-1}$, choose $k(n)$ so large that, with

$$y_n = x_{k(n)} y_{n-1},$$

we have

$$d(s, s + y_n) < 2^{-n} \quad \text{for every subset sum } s \text{ of } y_0, \dots, y_{n-1}. \quad (39.1)$$

There are only finitely many such sums. For each, continuity implies $s + x_k y_{n-1} \rightarrow s$, so a common choice exists. At the same time, ordered-ring positivity gives

$$y_n > 5y_{n-1} > 0. \quad (39.2)$$

For $r \in 2^{\mathbb{N}}$, define

$$\sigma_N(r) = \sum_{i < N} r(i) y_i.$$

Each σ_N is continuous, since it depends on finitely many bits. For $N \geq 1$ the distance between its next two partial sums is at most 2^{-N} , uniformly in r . Thus σ_N converges uniformly to a continuous function $r \mapsto z_r \in M$. Completeness is used at precisely this point.

If r, s differ only at finitely many positions, continuous addition implies the following algebraic identity in A :

$$z_s - z_r = \sum_i (s(i) - r(i)) y_i. \quad (39.3)$$

For justification, add the finite removed terms to the appropriate sides of the finite partial-sum identity, then pass to the limit using addition only. Rearranging the resulting equality is an algebraic operation in A . No topology on A and no continuity of subtraction are being assumed.

Let $\bar{r}$ denote coordinatewise bit complementation and set

$$\Delta_r = z_r - z_{\bar{r}}$$

as notation for an element of the algebraic ring. Equation (39.3) yields

$$\Delta_s - \Delta_r = 2 \sum_i (s(i) - r(i))y_i \quad (39.4)$$

whenever r, s differ finitely.

39.3 Bounded complementary differences have probability zero [06]

Give $2^{\mathbb{N}}$ fair independent product measure μ . Put $S_j = \sum_{i < j} y_i$. External finite induction, using (39.2), gives

$$4S_j < y_j \quad (j \in \mathbb{N}). \quad (39.5)$$

Indeed it holds for $j = 0$, and $4S_{j+1} = 4S_j + 4y_j < 5y_j < y_{j+1}$.

For fixed n consider

$$E_n = \{r : |\Delta_r| \leq y_n\}.$$

This is a Borel set. To see this without treating Δ as a map into a topological ring, write its definition as the conjunction

$$z_r \leq z_{\bar{r}} + y_n, \quad z_{\bar{r}} \leq z_r + y_n.$$

All terms are continuous M -valued maps and the order is Borel.

Freeze all coordinates except $n+1, \dots, n+k$. If two assignments r, s on these coordinates differ last at $j > n$, then (39.4) and (39.5) imply

$$|\Delta_s - \Delta_r| \geq 2(y_j - S_j) > 2y_n.$$

For the last inequality, $S_j \geq y_n$ and $y_j > 4S_j$. Two values lying in the ordered interval $[-y_n, y_n]$ cannot be this far apart. Thus at most one of the 2^k assignments in each fiber lies in E_n . Integrating over the frozen coordinates gives

$$\mu(E_n) \leq 2^{-k}$$

for every k , and hence $\mu(E_n) = 0$. Therefore

$$C = \bigcap_{n \in \mathbb{N}} \{r : |\Delta_r| > y_n\} \quad (39.6)$$

is Borel and has measure one. In particular, complementary subset sums are unequal almost surely. No injectivity of $r \mapsto z_r$ is needed.

39.4 The sign is a tail event [06]

If r, s differ only at positions at most j , write $b = \Delta_s - \Delta_r$. Equations (39.4) and (39.5) give

$$|b| \leq 2S_{j+1} < 3y_j.$$

For an arbitrary n , take $m = \max(n, j) + 1$. Then

$$y_n + |b| < 4y_{m-1} < y_m.$$

If $r \in C$, it follows that $|\Delta_s| \geq |\Delta_r| - |b| > y_n$ and that Δ_s has the same sign as Δ_r . Since n was arbitrary, C and the sign on C are invariant under every finite change of coordinates.

Consequently the Borel event

$$B = \{r \in C : z_{\bar{r}} < z_r\}$$

belongs to the tail sigma algebra. Such an event has probability zero or one. One quick proof is that it is independent of the first N coordinates for every N ; a monotone-class argument makes it independent of the full sigma algebra, and in particular of itself. Thus $\mu(B) = \mu(B)^2$.

But $r \mapsto \bar{r}$ preserves measure and C , and reverses the sign of Δ_r . Since C has measure one, it follows that $\mu(B) = 1/2$, a contradiction. This proves Theorem 39.1. $\square$

Remark 39.3 (Exactly what the proof uses [06]). The proof uses the ordered-ring laws, discreteness, separability, complete metrizable, continuity, and ordinary external finite induction. It does not use induction *inside* M , standard division in M , internal exponentiation, saturation, or an integer-part theorem. It concludes that zero is isolated. It does not by itself imply that every standard numeral is isolated, since predecessor has not been assumed continuous.

40 Surreal and omnific realizations

40.1 The real model as an omnific subgroup

The most direct connection to the repository is the map

$$J_{\mathbb{R}} : G_{\mathbb{R}} \longrightarrow \mathbf{No}, \quad J_{\mathbb{R}}(r, n) = r\omega + n. \quad (40.1)$$

Here multiplication and addition on the right are surreal field operations. Conway normal forms compare the coefficient of the dominant monomial ω before the constant coefficient. Thus $J_{\mathbb{R}}$ is an ordered additive isomorphism onto $\mathbb{R}\omega + \mathbb{Z}$.

An omnific integer has a Conway normal form with no negative exponents and with integer constant coefficient. This description places $\mathbb{R}\omega + \mathbb{Z}$ inside $\mathbf{Oz}$. The standard normal-form facts used here and below are recalled in [23]; a general treatment of initial ordered subgroups appears in [5].

Transport the topology of $M_{\mathbb{R}}$ through $J_{\mathbb{R}}$. The nonnegative part of $\mathbb{R}\omega + \mathbb{Z}$ becomes an uncountable Polish additive model with continuous addition. This is a topology on a *set-sized substructure*, described by its coefficients. It is not the intrinsic surreal order topology.

For example, in this transported topology,

$$\frac{\omega}{j} \longrightarrow 0,$$

although ω/j exceeds every real number for every positive standard j . A coefficient limit is not a claim about order convergence of surreal numbers.

40.2 Normal forms for the entire ordinal family

Theorem 40.1 (Explicit omnific embedding; extends theorem 9.1). *Fix a countably infinite ordinal α . Regard each ordinal $\beta + 1$ as a surreal number and put*

$$\varepsilon_{\beta} = \frac{1}{\beta + 1} \quad (\beta < \alpha).$$

Then

$$J_\alpha(a, n) = \sum_{\beta < \alpha} a_\beta \omega^{\varepsilon_\beta} + n \quad (40.2)$$

is a well-defined ordered additive embedding of G_α into $\mathbf{Oz}$, sending $(0, 1)$ to 1. Its nonnegative image, with transported coefficient topology, has all the properties of Theorem 35.1 and Theorem 36.1.

Proof. The positive surreal exponents ε_β are strictly decreasing: ordinal inclusion agrees with surreal order, and reciprocation reverses strict order on positive elements of a field. Every subset of $\{\varepsilon_\beta : \beta < \alpha\}$ has a largest element, corresponding to its least index. Thus every support in (40.2) is a set-sized reverse well order, as required for a Conway normal form. The sum denotes this formal normal form, not a limit in an unspecified topology.

Uniqueness of coefficients proves injectivity. Addition of normal forms with a common set of monomials is coefficientwise, proving additivity. For unequal vectors, the least differing β supplies the largest monomial in their difference, so determines its sign. If all rational coordinates agree, the integer constant does. This proves order preservation and reflection.

All displayed exponents are positive, and the constant coefficient is an integer. Hence every value is omnific. The remaining assertions follow by transporting the topology and structure along this isomorphism onto its image. $\square$

[write] For $\alpha = \omega$, J_ω is Part I's map ι_B of theorem 9.1, and $J_\mathbb{R}$ is $\iota_\mathbb{R}$. Source 06's realizations (section 40.4) use the exponents $1/(i+1)$ for $\alpha = \omega$ and $\omega^{-\beta}$ for larger α ; both choices are strictly decreasing sequences of positive surreal exponents. Whether these images are initial in the sense of `isg:cf:main:presburger` is not checked here; the groups are split, so that theorem applies to them abstractly (section 1.4).

Remark 40.2 (Ordinals, reciprocals, and the omega-map). In (40.2), $1/(\beta+1)$ is a reciprocal in the surreal field, not ordinal division. Likewise ω^x denotes Conway's monomial map. It should not be silently replaced by a different surreal exponential convention. For $\alpha = \omega$ the exponents are simply $1, 1/2, 1/3, \dots$

Corollary 40.3 (Omnific realization of the continuum family). For $B_s = \mathbb{Q} + \mathbb{Q}s$ as in Theorem 35.5 and any countably infinite ordinal α , the map

$$J_{s,\alpha}(b, a, n) = b\omega^2 + \sum_{\beta < \alpha} a_\beta \omega^{1/(\beta+1)} + n$$

is an ordered additive embedding of $B_s \times \mathbb{Q}^\alpha \times \mathbb{Z}$ into $\mathbf{Oz}$. Consequently the continuum family, including the models in Corollary 36.2, also has explicit omnific realizations with its topology transported from the coefficient space.

Proof. All tail exponents are positive and at most 1, so the extra exponent 2 lies strictly above them. The same normal-form argument applies, with the real coefficient $b \in B_s$ determining the leading component and the integer n the constant term. $\square$

40.3 Set size and formal foundations

For a fixed α , the domain $\mathbb{Q}^\alpha \times \mathbb{Z}$ is a set, and the exponents in (40.2) form a set. In the sign-sequence presentation of surreal numbers, the normal-form operation assigns a unique surreal set-code to each input. Replacement therefore makes the whole image a set. Its birthdays have an ordinal supremum; after increasing that bound, the entire image lies in a

single set-sized initial segment of the surreal class. The same argument applies to the real model and to each fixed parameter s in the continuum family.

Accordingly, ordinary ZFC suffices to quantify over the model, all its subsets, its topology, and all its finite Cartesian powers. No “set of all surreals” is introduced. The proper class **No** is used as a definable ambient class of set-codes, not as an element of itself or as a domain of unrestricted set comprehension.

The images are additive ordered subgroups. We do not claim that the particular normal-form images are initial under surreal simplicity, closed under surreal multiplication, or closed under arbitrary surreal-analytic operations. Already $(\omega)^2$ is outside the rank-two group $\mathbb{R}\omega + \mathbb{Z}$. Proposition 37.5 also forbids manufacturing a different compatible semiring product internally in these examples.

40.4 Surreal realizations and a formalization route [06]

40.4.1 The same algebra at explicit surreal scales [06]

The connection with the repository’s surreal work is visible directly in Conway normal forms. Using the usual normal-form arithmetic, the map

$$(r, n) \mapsto r\omega + n \quad (40.3)$$

identifies $\mathbb{R} \overrightarrow{\times} \mathbb{Z}$ with an ordered additive subgroup of the omnific integers. The rational-sequence model has the realization

$$((q_i)_{i \in \mathbb{N}}, n) \mapsto n + \sum_{i \in \mathbb{N}} q_i \omega^{1/(i+1)}. \quad (40.4)$$

The exponents are strictly decreasing positive real numbers; every subset of this sequence is reverse well ordered. The expression is therefore a valid set-supported normal form. Its leading coefficient determines the sign, and addition is coefficientwise, exactly as in our lexicographic group. For arbitrary countable α , one can similarly use the strictly decreasing positive surreal exponents $\omega^{-\beta}$, $\beta < \alpha$.

Finally the substitution $t \mapsto \omega$ identifies $\mathbb{Z} + t\mathbb{R}[t]$ with the ring

$$\left\{ n + \sum_{i=1}^d a_i \omega^i : n \in \mathbb{Z}, a_i \in \mathbb{R}, d < \omega \right\},$$

which is explicitly studied in the repository [36]. These identifications use standard surreal normal-form facts, also documented in the cited repository reports [35, 36]; none is needed for the proofs of our Polish models.

Two distinctions matter. First, the subgroup in (40.4) is being used additively; no closure under surreal multiplication is asserted. Second, all Polish topologies here must be *transported* through the displayed identifications. They are not claimed to be inherited from a topology on the proper class **No**. Our constructions are set-sized throughout and need no class of all subsets, global enumeration, or paradox-prone quantification over all surreal numbers.

40.4.2 What the existing formal development certifies [06]

The repository’s Presburger development supplies a promising implementation starting point. In the inspected snapshot, its `Syntax.lean` interprets formulas under valuations `List`

Int. The central results in `Decision.lean`, including `holds_iff_quantifierEliminate`, therefore concern standard integer semantics. They do not already assert that an arbitrary $\mathbb{Z}$ -group satisfies every eliminated formula.

A faithful formalization can separate four layers.

1. **Algebraic semantics.** Define a discretely ordered abelian group with its least positive element and the standard division/remainder schema. Interpret affine terms and formulas in an arbitrary such group. Generalize the Cooper residue and interval lemmas so they no longer mention Int in their semantic valuation type.
2. **Concrete models.** Build the ordered groups $\mathbb{R} \overrightarrow{\times} \mathbb{Z}$, $\mathbb{Q}^{\mathbb{N}} \overrightarrow{\times} \mathbb{Z}$, and the additive group of $\mathbb{Z} + t\mathbb{R}[t]$. Prove the explicit division identity and transfer the generic semantics to their nonnegative cones.
3. **Topology.** Supply the product or disjoint-sum topologies, prove the specified operations continuous, and certify Polishness by the explicit homeomorphisms or complete metrics. The polynomial model requires a topology on the positive cone, not an incompatible claim about a topological signed ring.
4. **Boundary results.** Formalize the cofinal-scale obstruction and the single failed induction instance. The isolated-zero theorem can be a separate later target, adding descriptive-set-theoretic and probability dependencies.

An adjacent interface requires care. The inspected file `PAHF/PASyntax.lean` contains a shallow `PA.Model` interface with induction for every Lean predicate $\alpha \rightarrow \text{Prop}$. Such full external induction excludes the nonstandard objects of interest. The appropriate route uses raw `PreModel` operations, syntactic formulas and satisfaction, and first-order induction instances [37]. For the polynomial construction, one must also keep the additive language distinct from the language with multiplication.

These are proposed formalization steps, not claims that a formalization was run in this project. The deliverable is a self-contained mathematical article; its full proofs make the remaining kernel-checking task precise and reviewable.

41 Formalization scope and a concrete development plan

41.1 What the existing repository actually proves

The repository’s executable development has a specific semantic domain. In the inspected source,

```
Formula.holds : Formula -> List Int -> Prop
```

interprets quantifiers over Lean’s ordinary integer type. In particular, the existential case is an existential over Int. The theorem `holds_iff_quantifierEliminate` proves the computed elimination correct for these integer valuations [32].

This is relevant infrastructure, but does not formally establish that an arbitrary ordered group with a divisible quotient is a Presburger model. It also does not verify a topology on $\mathbb{R} \times \mathbb{Z}$ or $\mathbb{Q}^{\mathbb{N}}$. The source was inspected during this investigation; no Lean or Rocq build was run. The new theorems in this report are ordinary mathematical proofs, not claims of a successful proof-assistant execution.

[write] The claims about the formal project hold at the repository’s current state: `Syntax.lean` defines `holds : Formula -> List Int -> Prop` with an existential over `Int` (lines 192–197), and `holds_iff_quantifierEliminate` is in `Decision.lean` (line 18); the project’s Lean files are unchanged since both pins. [Section 13.5](#) names the declarations that Part I’s porting requirement concerns. **Nothing in this part is formalized.**

41.2 A small first formalization target

The best first target is Theorem [32.1](#), with the algebra and topology proved in separate modules whose assumptions meet in one final statement. A suitable sequence is:

1. Define the coordinate type $\mathbb{R} \times \mathbb{Z}$, its lexicographic order, and the nonnegative subtype $M_{\mathbb{R}}$. Prove the least-positive property and the explicit division identity ([32.4](#)).
2. Prove, or port, the semantic theorem that a $\mathbb{Z}$ -group satisfies the complete Presburger theory. Generalize formula interpretation from `List Int` to valuations in a structure. Avoid assuming the new model satisfies the theory as an unproved axiom.
3. Put the product topology on the coordinates, independently of the lexicographic order instance. Prove closure under addition, restricted continuity, and Polishness using ([32.2](#)) or the explicit metric ([32.3](#)).
4. Combine the semantic and topological theorems. State cardinality through an explicit injection from $\mathbb{R}_{\geq 0}$, so uncountability is not hidden in a typeclass declaration.

The central implementation risk is accidentally selecting the intrinsic order topology when defining a lexicographically ordered type. That topology is discrete. The product topology used here must be installed explicitly and its compatibility claims stated only for the operations that have actually been proved continuous.

41.3 Extending the verified boundary

Once the real-coordinate construction is complete, the $\alpha = \omega$ Baire model is the next manageable target. Its strict coefficient cones are open, avoiding the transfinite Borel decomposition. A coordinate enumeration, cylinder basis, and the noncompact-cylinder lemma can be implemented without any proper-class formalization.

The general cone theorem requires more library infrastructure: Baire category for closed subgroups, Borel sets’ Baire property, countable-base arguments, and transfinite recursion on closed subgroups. The oscillation theorem needs ordinal arithmetic and a general relative-boundary recursion. Those are substantive formalization tasks, not consequences of running the existing integer decision procedure.

The surreal embedding can be isolated still further. One may first work with a set-sized Hahn additive group supported on the fixed exponent set, prove its coordinate order and addition facts, and then use a separately verified normal-form map into a universe-bounded surreal type. This avoids making the core topological result depend on a full surreal field development.

Layer	Mathematical obligation	Present status
Ordinary integer formulas	Executable Cooper elimination and semantic correctness	Existing repository sources inspected
Generic additive models	$\mathbb{Z}$ -group characterization and full theory	Classical theorem used explicitly
Real Polish model	Cone, metric, continuity, uncountability	Proved in this report
Baire ordinal family	G_δ cone, cylinder argument, nonisomorphism	Proved in this report
Oscillation ranks	Relative derivatives and ordinal calculation	Proved in this report
Omnific embedding	Normal-form existence and coefficient transport	Classical normal forms plus proof here
New Lean/Rocq certificates	Kernel-checked versions of the report's new claims	Not supplied

42 Where Part III meets Parts I and II [merge]

Source 06's ring $A = \mathbb{Z} + t\mathbb{R}[t]$, with t positive infinite, becomes under $t \mapsto X$ a subring of 03's finite principal-part ring $\mathcal{A}_{\text{fin}} = \mathbb{Z} \oplus \bigoplus_{q \in \mathbb{Q}_{>0}} \mathbb{R}X^q$ of [Part II](#): the elements with integer exponents. Both rings are ordered by the sign of the leading coefficient, so $P = A_{\geq 0} = A \cap \mathcal{A}_{\text{fin}, \geq 0}$. Write τ_P for source 06's topology ([38.1](#)) on P .

Proposition 42.1 (The stratum of degree one, and two topologies on P [merge]). *The following hold.*

- (i) *The set of elements of degree at most one,*

$$P_{\leq 1} = \{x \in P : \deg x \leq 1\} = \mathbb{N} \cup \bigcup_{m \in \mathbb{Z}} P_{1,m},$$

is clopen in (P, τ_P) . The map $m + at \mapsto (a, m)$ is an isomorphism of ordered additive monoids and a homeomorphism from $P_{\leq 1}$ onto source 07's $M_{\mathbb{R}}^\#$ ([proposition 37.4](#)), and $t \mapsto X$ maps $P_{\leq 1}$ homeomorphically onto the stratum $(C_{\mathbb{R}}, \tau_{\text{FS}})$ of [proposition 26.1](#).

- (ii) *On P , the topology τ_P is strictly coarser than the topology induced by 03's τ_{FS} .*

- (iii) *Addition and multiplication are continuous on (P, τ_P) ([theorem 38.1](#)), but addition is not continuous on P with the topology induced by τ_{FS} .*

Proof. (i) $P_{\leq 1}$ is a union of components of ([38.1](#)). Each $P_{1,m} = \{m + at : a > 0\}$ carries the topology of $(0, \infty)$ in the coefficient a , and the points of $\mathbb{N}$ are isolated. These are exactly the components of $M_{\mathbb{R}}^\#$, namely the rays $(0, \infty) \times \{m\}$ and the isolated standard numerals, and of $C_{\mathbb{R}}$, namely $U_{\{1\},m}$ and $U_{\emptyset,n}$, $n \geq 0$. The algebra is the lexicographic algebra of $\mathbb{R} \overrightarrow{\times} \mathbb{Z}$ in all three cases.

(ii) A component $U_{S,n}$ of τ_{FS} that lies in P has $S \subseteq \{1, \dots, d\}$ with $\max S = d$, so it lies in $P_{d,n}$, and its inclusion there, which inserts zero coefficients, is continuous. Hence every τ_P -open set meets each such component in an open set, and τ_P is coarser. It is strictly coarser: $t^2 + t/j \rightarrow t^2$ in τ_P , inside the component $P_{2,0}$, while $t^2 + t/j$ and t^2 lie in the different components $U_{\{1,2\},0}$ and $U_{\{2\},0}$ of τ_{FS} .

(iii) Put $x_j = t^2 + (1 + 1/j)t$ and $y = t^2 - t$; both lie in P , and $x_j \rightarrow x = t^2 + t$ for τ_{FS} . Then $x_j + y = 2t^2 + t/j$ lies in $U_{\{1,2\},0}$ while $x + y = 2t^2$ lies in $U_{\{2\},0}$, so $x_j + y$ does not converge to $x + y$. The pair (x, y) lies on the cancellation locus ([24.5](#)) of [theorem 24.3](#). $\square$

Glazer’s Corollary 1 inside one ring. Section 26 placed the real model as a stratum of $\mathcal{A}_{\text{fin}, \geq 0}$, which satisfies $Q + \text{IOpen}$ with Borel multiplication and on which no Borel-compatible Polish topology makes addition continuous (theorem 25.2). The integer-exponent subcone P sits between the two: it carries a Polish topology with both operations continuous, and Corollary 1 does not apply to it because P fails open induction at the single formula $x^2 < t$ (theorem 38.2). In $\mathcal{A}_{\text{fin}}$ the same formula is satisfied by $x = \frac{1}{2}X^{1/2}$, which is not a standard integer. Source 06’s isolated-zero theorem (theorem 39.1) explains why τ_P isolates 0, whereas the additive models of Part I are perfect.

Cone topologies versus group topologies. The same ring is source 08’s $A_* = \mathbb{Z} + X\mathbb{R}[X]$ (example 54.1) and, under $t \mapsto \omega$, the ring $\mathbb{Z} + \omega\mathbb{R}[\omega]$ of corollary 55.1 and `odg:frac:prop:` intersection of the omnific Diophantine report. By source 08’s theorem 44.1, no Hausdorff locally compact second-countable additive-group topology on A has a Borel order, so τ_P , a locally compact Polish topology on the cone, is induced by no such group topology. Source 06 already notes that the signed analogue of τ_P is not a ring topology. This is the situation of source 08’s question 57.2 and source 09’s question 73.11, which ask when a cone topology extends to the group of differences; the Hilbert’s-tenth programme’s triage of the arrival (Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/recent_incoming_substrate_triage_20261003.md) makes the same caveat.

The same failed instance elsewhere. The formula $x^2 < X$ defining the standard cut in $\mathbb{Z} + X\mathbb{Q}[X]$ is used by batch 86’s manuscript 10, written into the report Algebra/Surreal1Numbers/docs/surreal/discrete-initial-subgroups-and-omnific-normalization.[write] [Batch-86 reciprocal notes, 3 October 2026.] There it is source 03, Sections 19–22. It is motivated by the same paper of Glazer but is not topological, and its obstruction precedes any topology: the additive group of the difference ring of a nonstandard model of $\text{I}\Delta_0$ has no nonzero homomorphism to $\mathbb{Z}$ (`isg:sc:thm:nochar`), so no such model has a unit-preserving additive map into **Oz** (`isg:sc:cor:omnific`), and by that report’s Presburger criterion the ordered group completion of its additive monoid is not initially realizable (`isg:sc:cor:fragment`). It answers neither of Glazer’s questions. For **Oz** itself the collection records existential failures of induction (`odg:thm:induction`, `odg:rem:ringinduction`); as source 06 says, the failure for P does not transfer to **Oz**.

43 Further research questions and topics

The following program separates an explicitly published question from questions suggested by the constructions. Unless stated otherwise, these are proposed directions for investigation; we do not assert that every formulation is historically new or unresolved in all existing literature.

[06] Source 06 lists twelve numbered projects rather than question environments. Each one that coincides with a question of source 07 is printed after it, with source 06’s number:

43.1 Published reverse-mathematical question

Question 43.1 (Glazer’s Question 1). Can the stronger topological Tennenbaum theorem in [1] be proved over ATR_0 ?

This report does not answer that question. The affirmative Presburger construction shows that a proposed route from a weaker metatheory to a negative Presburger conclusion cannot yield the latter conclusion as stated. It does not determine the metatheoretic strength of the theorem about arithmetic with multiplication.

[write] Source 06's project with the same subject:

6. **Exact strength of the isolated-zero principle.** Formalize the proof of Theorem 39.1 over weak subsystems of second-order arithmetic using codes for Polish spaces and continuous functions. Locate the strength of analytic/coanalytic separation and of the coded measure argument. This is related to, but does not resolve, Glazer's published Question 1 asking whether his full Theorem 2 is provable in ATR_0 [1].

[write] (Batch 87.) Source 14 claims an affirmative answer, under a stated coding convention (Part VIII, corollary 111.2). The claim is unrefereed and has not been independently reviewed; see section 1.1.

43.2 Classification and topology

Question 43.2 (Which spaces and which additive models?). Characterize the uncountable Polish spaces that support continuous Presburger addition, and characterize which continuum-sized Presburger models admit such a presentation.

The real-coordinate model, Baire-space family, and isolated-standard variant give several positive topological types. Compact spaces are excluded. A useful first refinement is to classify locally compact presentations in terms of their components and the position of the standard numerals. A separate algebraic problem asks how much of the existence theory is captured by split $\mathbb{Z}$ -groups.

[write] Source 06's project with the same subject:

4. **Which Polish spaces support additive arithmetic?** Classify the possible underlying spaces. We have a locally compact perfect example and a nowhere locally compact example. Compact Hausdorff spaces are excluded by [2]. A useful next step is to analyze connected spaces and spaces with prescribed sets of isolated points, while specifying whether a group-completion topology is also required.

[write] (Batch 87.) For locally compact cones inherited from a Borel ordered Polish group, source 15 determines the space: it is countable discrete or the countable sum of lines and closed rays (theorem 121.1); on that one space there are exactly $2^{\aleph_0}$ models (theorem 123.1). The general question stays open.

[write] (Batch 92.) The second clause is settled for the uncountable locally compact models with continuous partial subtraction: they are, up to topological and arithmetic isomorphism, the cones of the normal models of Part XIII, classified by countable ordered extensions (proposition 182.3 [merge]). Beyond that class it stays open.

Question 43.3 (Non-split integer extensions). Which ordered extensions $0 \rightarrow \mathbb{Z} \rightarrow G \rightarrow D \rightarrow 0$ with divisible D , but without a group-theoretic splitting, have positive cones with Polish topologies making addition continuous?

The present construction starts with a splitting. General $\mathbb{Z}$ -groups need not possess it. The residue and extension invariants appearing in the repository's omnific research provide

a natural algebraic language for formulating obstructions, but do not automatically decide the topological question.

[write] Source 06's project with the same subject:

2. Which additive groups admit Polish presentations? Find necessary and sufficient conditions for a continuum-sized $\mathbb{Z}$ -group to have a nonnegative cone with a Polish topology and continuous addition. Our split products are a substantial class, but a general $\mathbb{Z}$ -group need not split. In particular, investigate how a nonsplit extension of a divisible ordered group by $\mathbb{Z}$ affects the problem.

[write] (Batch 87.) Existence, for extensions coming from a countable nonsplit core: source 13's [theorem 94.1](#) gives, for every countable $\mathbb{Z}$ -group K , Polish cones of $D \overrightarrow{\times} K$ with continuous addition ($D = \mathbb{R}$ or $\mathbb{Q}^{\mathbb{N}}$), nonsplit when K is, with $2^{\aleph_0}$ nonisomorphic nonsplit examples ([corollary 94.2](#)); sources 14 and 15 give instances. Which extensions admit such topologies is not classified.

Question 43.4 (Connected examples). Does a connected uncountable Polish space support a continuous Presburger presentation?

The examples here have clopen residue classes and hence are disconnected. In an arbitrary presentation, residue classes are Borel by [Proposition 34.3](#); that alone does not make them clopen. The gap between these two regularities is a concrete place to investigate.

[write] (Batch 87.) Negative for cones inheriting a Polish group topology (note after [question 14.3](#)); open otherwise.

[write] (Batch 90.) A connected example must have discontinuous partial subtraction ([corollary 155.3](#); note after [question 14.3](#)); open otherwise.

43.3 Definability and descriptive invariants

Question 43.5 (Ranks for a fixed model). For one fixed continuous presentation, which boundary oscillation ranks occur among parameter-definable sets? How do they depend on arity and on the chosen parameter?

[Theorem 36.1](#) varies the model and uses a parameter with positive zeroth coordinate. A full parameter spectrum requires analyzing cuts based at vectors with a transfinite first nonzero coordinate, where the boundary of the model itself affects the derivative. It should also distinguish the rank of a relation from the rank of a unary section.

Question 43.6 (Uniform definability bounds). Must every Polish Presburger presentation with continuous addition have all definable relations at some uniformly bounded Borel level? Is the Δ_2^0 bound forced, or special to the ambient-group construction?

[Proposition 34.3](#) proves Borelness in complete generality. [Theorem 34.1](#) proves the stronger level-two conclusion under an additional structural hypothesis. Closing that gap would turn the examples into a general regularity theorem.

[write] For the cones of locally compact signed groups with Borel order, source 08's [theorem 52.1](#) gives the uniform Δ_2^0 bound; for an arbitrary Polish presentation the question stays open.

Question 43.7 (Interpreting the cone closure rank). Which structural features of an ordered Polish group are detected by $\rho(A, <)$? How does the rank behave under closed convex subgroups, quotients, products, and order-preserving topological embeddings?

Proposition 33.4 realizes every countable rank on real coordinate groups. A useful next goal is an intrinsic characterization of the first few ranks and precise inequalities for those operations. The rank records the interaction of order with topology; it need not agree with the algebraic Archimedean rank.

[write] (Batch 93.) Answered in part for separable real Hilbert spaces with Borel order, by Part XVI with proposition 224.6 [merge]: there ρ equals the order type of the Archimedean classes, so it does agree with the algebraic Archimedean rank; $\rho + 1$ is the oscillation rank of the cone; a closed convex subgroup is a stage of the flag, of rank η with $\alpha + \eta = \rho$; ρ is monotone under additive order embeddings, continuous or not; and it is a complete invariant. In a real Polish vector space every step of the recursion has codimension one (theorem 213.2). Quotients, products and groups that are not vector spaces stay open.

43.4 Arithmetic expansions and formal verification

Question 43.8 (Additional continuous functions). Which natural expansions of Presburger arithmetic still admit uncountable Polish presentations with all named functions continuous?

Fixed-divisor quotient and remainder are positive examples. Total truncated subtraction, minimum, and maximum are impossible individually. Predecessor depends on the presentation. Expansions by selected unary functions, or restricted coding functions, offer an intermediate program before multiplication is considered.

[write] Source 06's project with the same subject:

9. **Effective comparison and definable functions.** For the explicit computable Baire model, calibrate the effective complexity of its order, equality, minimum, and truncated predecessor. The continuous addition algorithm does not decide an infinite stream's equality to zero. A useful target is an exact classification of the discontinuity or computational strength required by each definable operation.

Question 43.9 (Removing the cofinal-multiple obstruction). Construct and analyze continuous Polish Presburger models in which no nonstandard element has cofinal standard multiples. Which of their additive structures admit an ordered-ring expansion at all?

The explicit family in this report cannot answer the latter question positively. Divisible coefficient groups with no greatest Archimedean class are natural candidates for a next stage. Any resulting multiplication must then be tested against the appropriate arithmetic axioms and Glazer's regularity obstruction.

[write] Source 06's project with the same subject:

3. **Presburger reducts of genuine PA models.** Can the additive reduct of a nonstandard model of PA have an uncountable Polish continuous-addition presentation if its multiplication is allowed to be non-Borel? Glazer's published obstruction rules out a Borel multiplication under stronger arithmetic hypotheses. Our lexicographic examples do not answer this question, because their additive structure already excludes a semiring multiplication.

Question 43.10 (Weak foundations and certified models). What is the weakest useful coded metatheory for the real and Baire constructions? Can the generic Presburger semantic theorem and the Polish topology be joined in a short audited Lean development?

The explicit coordinate formulas suggest a substantially lighter proof than the general transfinite cone theorem. A careful analysis must still specify codes for Borel sets, complete

metrics, countable bases, and satisfaction of first-order formulas. Section 41 gives an implementation order with a concrete first target.

[write] Source 06's project with the same subject:

1. **Formal generic Presburger semantics.** Complete the four-layer route in Section 40.4, beginning with generic $\mathbb{Z}$ -group soundness for the repository's Cooper eliminator. The first deliverable should be a theorem about all models of the division/remainder axioms, followed by the real cone as its first nonstandard instance. This would convert the main answer into a directly auditable formal result.

Question 43.11 (Simplicity and coefficient topologies). Which initially realized omnific additive groups admit coefficient topologies of this kind, and which initiality operations preserve Polishness and continuity of addition?

Theorem 40.1 supplies an explicit ordered embedding without asserting initiality of the displayed image. Combining a proved initial realization theorem with a transported Polish topology is straightforward as an abstract existence argument; preserving a natural coefficient description during that transport is a sharper and more informative goal.

[write] Source 06's project with the same subject:

12. **Surreal topology without class-level assumptions.** Study the transported topologies on set-sized omnific subgroups and subrings such as (40.3) and (40.4). Ask which algebraic embeddings are continuous for these explicit topologies. Keep the support set, coefficient topology, and ambient set theory specified; the proper class of all surreal numbers is not itself an ordinary Polish space.

43.5 Further projects of source 06 [06]

The remaining projects of source 06 have no counterpart among source 07's questions.

5. **How many isolated points are necessary for semirings?** Theorem 39.1 forces zero to be isolated, and the polynomial example isolates every standard numeral. Determine whether a Polish topological PA^- model can have zero as its only isolated point, or classify which patterns of isolated numerals are possible. Requiring continuous predecessor is an additional hypothesis and should be treated separately.
7. **Completeness and separability thresholds.** Find sharp replacements for Polishness in the isolated-zero theorem. Completeness supplies limits of all subset sums, and separability supplies descriptive regularity of the order. One can test these roles separately using incomplete metrizable models and complete nonseparable presentations.
8. **Classification beyond cardinality.** Theorem 35.8 gives the exact number of isomorphism types but no full classification. Determine the descriptive complexity of isomorphism for suitably coded Polish Presburger presentations. Convex quotients and residue extensions provide natural invariants. Compare with the countable-model Scott analysis in [25], without assuming that countable and Polish presentations have identical classification behavior.
10. **Coefficient fields and other support families.** Replace $\mathbb{R}$ in the polynomial construction by an ordered Polish topological field whose positive cone is Polish. The same degree-stratum proof applies whenever coefficient operations are continuous. More ambitiously,

determine which infinite support families allow a Polish positive-cone topology with both operations continuous, and identify where leading-term cancellation or completeness obstructs such a topology.

11. **A bridge from additive induction to open induction.** The formula $x^2 < t$ isolates one exact failure in the polynomial model. Seek a useful classification of additional multiplication principles that remain compatible with a Polish continuous semiring presentation. The class must stop before the hypotheses of Glazer’s additive obstruction, but the interval between additive induction and IOpen contains many distinct theories.

[merge] Relation to Part I’s questions. Source 07’s question on spaces and models and its connected-carrier question, with source 06’s project 4, are [question 14.3](#) extended to all carriers; source 07’s question on weak foundations and source 06’s project 1 continue [questions 14.1](#) and [14.2](#); the cofinal-multiple question and project 3 continue [question 14.4](#). Source 06’s [theorem 38.1](#) answers the first sentence of source 05’s part of that question affirmatively: its cone is an explicit uncountable Polish continuous-addition Presburger model whose additive structure carries a unital semiring multiplication, continuous for the same topology (the note after [question 14.4](#) records this). It fails open induction ([theorem 38.2](#)), so the clause of that question about $Q + \text{IOpen}$, and source 04’s growth-order question, stay open. Project 6 and source 07’s comment on Glazer’s Question 1 bear on the speculation discussed in [section 1.1](#).

Part IV

Local compactness forces arithmetic rigidity

[write] This part is source 08 of batch 86, *Local Compactness Forces Arithmetic Rigidity: Borel ordered groups, Presburger models, and multiplication without regularity assumptions* (dated October 3, 2026; “prepared with ChatGPT”), pinned to repository commit [883e0b3b2](#). All unmarked text is source 08’s; its labels carry the prefix `pma:lc:`. Source 08 cites two earlier manuscripts as “user-library” files: [\[40\]](#) is source 05, the base of [Part I](#), and [\[41\]](#) is source 03, [Part II](#). Its finite checks are shipped as `code/08-local-compactness-verify.py` with the record `data/08-local-compactness-verification_results.json` ([appendix K](#)). Its central theorem is complementary to source 09’s [theorem 62.1](#) in [Part V](#): both make an uncountable discretely ordered ring impossible under a topology, 08 assuming local compactness and no regularity of multiplication, 09 assuming Borel scalar maps and no local compactness. Neither result bears directly on Glazer’s Question 2, which concerns models without multiplication ([section 1.1](#)).

[write] (Batch 87.) [Part VI](#) removes local compactness: [theorem 44.1](#) and source 09’s [theorem 62.1](#) are both special cases of [corollary 78.5](#), which assumes only a Polish (indeed a Hausdorff Baire ccc) additive-group topology and a cone with the Baire property ([section 90](#)).

44 The intersection and the contribution

The point of contact is the topology of arithmetic presentations. In *A Topological Tennenbaum Theorem*, Elliot Glazer rules out continuous addition with Borel multiplication for uncountable Polish models of $Q + \text{IOpen}$; his Question 2 asks for an uncountable Polish Presburger model with continuous addition [[1](#), Corollary 1 and Question 2]. ProveIt’s relevant interfaces are its Cooper-elimination development and its ordered surreal and omnific constructions [[38](#), [39](#)].

These suggest separating three issues: addition, the regularity of order, and the possibility of installing multiplication.

The earlier companion manuscript *Polish Presburger Arithmetic inside the Omnific Integers* already constructs the model

$$M_{\mathbb{R}} = (\mathbb{R}_{>0} \times \mathbb{Z}) \cup (\{0\} \times \mathbb{N}) \quad (44.1)$$

with continuous addition, and gives example-specific multiplication obstructions [40]. Repeating that construction would not be an advance. The question pursued here is structural:

What does local compactness force on every Borel ordered additive-group presentation, and can multiplication evade the obstruction by being completely nonmeasurable?

Our answer is negative for discretely ordered rings. The obstruction occurs before induction, definability of multiplication, or a choice of particular normal forms enters the argument.

[write] Equation (44.1) is Part I's real model (theorem 3.1); source 07's theorem 32.1 and source 06's theorem 32.2 prove it again.

44.1 The central theorem

Theorem 44.1 (Arithmetic rigidity). *Let A be a nonzero commutative unital ring with a total order such that addition preserves order, products of positive elements are positive, and 1 is the least positive element. Suppose that A has a Hausdorff locally compact second-countable topology making $(A, +, -, 0)$ a topological group, and that $A_{>0}$ is Borel. Then the topology is discrete and A is countable. No regularity assumption is imposed on the multiplication map.*

The proof is given in section 48. Its two essential parts are an algebraic growth argument and a classical topological structure argument. The former produces more cosets than the latter permits. In particular, it does not attempt to pass a discontinuous multiplication through a limit.

44.2 Other results proved here

The structural conclusions have independent content even when no ring multiplication exists. For signed Presburger models, we obtain an exact classification of the *pointed topological additive groups*, the local-compactness threshold for their positive cones, and a uniform Δ_2^0 bound on all first-order definable relations. We do not claim that every order itself is the standard product lexicographic order.

There is also a second, distinct obstruction. On these same additive groups, an arbitrary integral-domain multiplication, not required to respect the specified order, has only countably many scalars whose multiplication maps are Borel. For the additive type $\mathbb{R}^d \times \mathbb{Z}$, these scalars are exactly the ordinary integers. This theorem is not vacuous: an explicit algebraic transport construction supplies such ring structures.

44.3 What is classical, and what remains uncertified

Measurable orders on locally compact abelian groups were studied by Hewitt and Koshi and by Asmar and Montgomery-Smith [26, 22]. The compact-group obstruction, finite-dimensional separation arguments, and the structure theorem for locally compact abelian

groups are classical inputs, not discoveries claimed here. We supply the needed proofs except for the principal structure theorem itself. Presburger quantifier elimination is likewise classical [28]; [section 49](#) records the version needed for nonstandard groups.

The contribution proposed for independent review is the amplification and arithmetic-rigidity package, together with its classification, scalar, and omnific consequences. A bounded literature search did not establish priority for these formulations. In particular, lack of a matching search result is not evidence of a first proof. The existing companion constructions are cited as prior work, not relabelled as new.

45 Conventions, topology, and foundations

All groups in this paper are abelian and written additively. An ordered group has a translation-invariant *total* order. The associated strict positive cone is $P = G_{>0}$, and $G_{\geq 0} = P \cup \{0\}$. For a topological group, P is Borel if and only if the binary order relation is Borel: one direction takes a section at zero, and the other uses the continuous difference map $(x, y) \mapsto y - x$.

The phrase *discretely ordered* refers to the *algebraic order*: there is a least positive element, denoted 1. It does not say that the presentation topology is discrete. In a ring, $\underline{n} = n1$ denotes the ordinary integer numeral. Throughout, n, m, N used as moduli, finite indices, or bounds are external ordinary integers, not arbitrary elements of a nonstandard model. We write $|x| = \max(x, -x)$.

A Hausdorff locally compact second-countable topological group is Polish. Here, a Polish space is a separable completely metrizable space. The notation Δ_2^0 means both F_σ and G_δ in the specified topology (usually Polish); Δ_1^0 means clopen. No order topology is silently substituted for the given topology.

Definition 45.1 (Order unit). If V is an ordered additive subgroup, an *order unit* in V is an element $u > 0$ in V such that for every $v \in V$ there exists an ordinary integer $N \geq 1$ with $|v| \leq Nu$.

This is an additive notion. It neither says that u is a ring unit nor that V is closed under multiplication. The distinction is indispensable in the main proof.

All theorems are statements about sets in ZFC. When surreals are mentioned, we work with explicitly set-sized subgroups or subrings of **No** or **Oz**. We never equip the proper class of all surreals with a Polish topology, form a set of all its subsets, or apply Haar measure to a proper class. The algebraic splitting arguments use the ordinary ZFC fact that divisible abelian groups are injective. No assertion about the minimum choice or reverse-mathematical strength of the complete proof is made.

46 An algebraic amplification theorem

The key obstruction can be proved without topology. The strongest useful hypothesis is weaker than divisibility.

Lemma 46.1 (Finite elements are ordinary integers). *In a discretely ordered ring A , every element of $[0, \underline{n}]$, for an ordinary $n \in \mathbb{N}$, is one of $\underline{0}, \dots, \underline{n}$. Consequently, for an additive subgroup $V \subseteq A$ the following are equivalent:*

1. $V \cap \mathbb{Z}1 = \{0\}$;

2. for every $v \in V \setminus \{0\}$ and every ordinary $n \geq 0$, $|v| > \underline{n}$.

Every divisible additive subgroup of A satisfies these conditions.

Proof. The interval assertion follows by external induction on n . If $0 < x \leq \underline{n+1}$, then $x \geq 1$ and $0 \leq x-1 \leq \underline{n}$; the induction hypothesis applies. This proves the equivalence because $|v| \in V$.

For the last assertion, take $v > 0$ in a divisible subgroup and an ordinary $n \geq 0$. Write $v = (n+1)w$ with $w \in V$. Ordered-group torsion-freeness and positivity imply $w > 0$, hence $w \geq 1$. Therefore $v \geq \underline{n+1} > \underline{n}$. Apply the same argument to $-v$ for negative elements. $\square$

Theorem 46.2 (Coset and polynomial amplification). *Let A be a discretely ordered ring, and let $V \subseteq A$ be a nonzero additive subgroup with $V \cap \mathbb{Z}1 = \{0\}$. Suppose $u \in V$ is an order unit. Then:*

1. $uV \cap V = \{0\}$, and the additive map

$$V \longrightarrow A/V, \quad v \longmapsto uv + V \quad (46.1)$$

is injective. In particular, $|A/V| \geq |V|$.

2. The finite-sum map

$$\Psi : \bigoplus_{j \geq 0} V \longrightarrow A, \quad (v_j)_{j \geq 0} \longmapsto \sum_{j \geq 0} u^j v_j \quad (46.2)$$

is an injective additive homomorphism. If the source is ordered by the sign of its highest-index nonzero coefficient, then Ψ is an order embedding.

3. The induced additive map $\bigoplus_{j \geq 1} V \rightarrow A/V$ is injective.

Proof. Suppose $uv \in V$ for $v \neq 0$. Since $u > 0$, the order-unit condition provides an ordinary N with

$$u|v| = |uv| \leq Nu.$$

Cancellation of the positive factor u gives $|v| \leq \underline{N}$, contrary to [lemma 46.1](#). Thus $uv \in V$ implies $v = 0$, which proves the first assertion and injectivity of (46.1).

For the second assertion, take a nonzero finite tuple and let k be its largest nonzero index. If $k = 0$, the result is immediate. For each $j < k$ choose an ordinary $N_j \geq 1$ with $|v_j| \leq N_j u$. Put $N = \sum_{j < k} N_j$. By [lemma 46.1](#), $u > 1$ and $|v_k| > \underline{N}$. The lower-degree terms satisfy

$$\left| \sum_{j < k} u^j v_j \right| \leq \sum_{j < k} u^j |v_j| \leq \sum_{j < k} N_j u^{j+1} \leq Nu^k < u^k |v_k|. \quad (46.3)$$

Therefore the sum in (46.2) is nonzero and has the sign of v_k . Additivity is distributivity. Applying the sign statement to the difference of two tuples proves the claimed order embedding.

Finally, if a finite sum $\sum_{j \geq 1} u^j v_j$ belongs to V , subtract that element as the zeroth coefficient. Injectivity of Ψ makes all coefficients zero, proving the third assertion. $\square$

Remark 46.3 (What the polynomial language does not mean). The domain of Ψ is an ordered additive direct sum. In general V is not a coefficient ring, and the theorem does not define a ring $V[t]$ or an evaluation homomorphism from such a ring. The quotient A/V is also only an additive quotient: V need not be an ideal or convex.

Corollary 46.4 (Divisible-subgroup obstruction). *If a discretely ordered ring contains an uncountable divisible additive subgroup V with an order unit, then A/V is uncountable. More precisely, A/V contains an additive copy of a countable direct sum of copies of V .*

The strength of this result is its insensitivity to the regularity of multiplication. The map (46.1) may be arbitrarily nonmeasurable; its existence as an injection is already enough for a contradiction.

47 Borel orders on locally compact additive groups

This section assembles the classical topological ingredients in the form needed later. The measurable-order literature [26, 22] contains broader structure results. Borelness is useful here because it survives restriction to every closed subgroup.

47.1 A compact subgroup cannot carry the restricted order

Lemma 47.1 (Compact-order obstruction). *A second-countable topological abelian group with a Borel invariant total order has no nontrivial compact subgroup.*

Proof. Let K be a compact subgroup. Ordered groups are torsion-free, so a nontrivial K cannot be finite. Let μ be normalized Haar measure on an infinite K , and set $P_K = K \cap G_{>0}$. Singletons have measure zero: otherwise arbitrarily many disjoint translates of one singleton would have total measure greater than one. Since inversion preserves Haar measure, the partition

$$K = P_K \dot{\cup} (-P_K) \dot{\cup} \{0\}$$

gives $p := \mu(P_K) = 1/2$.

For every $t \in K$, the two upper rays P_K and $P_K + t$ are nested. Their measures are equal by translation invariance. Thus

$$\mu(P_K \cap (P_K + t)) = p \quad (t \in K).$$

On the other hand, the relevant indicator on $K \times K$ is Borel; second countability permits the usual product-measure Fubini theorem. Consequently

$$\begin{aligned} \int_K \mu(P_K \cap (P_K + t)) d\mu(t) &= \int_K \int_K 1_{P_K}(x) 1_{P_K}(x - t) d\mu(x) d\mu(t) \\ &= \int_K 1_{P_K}(x) p d\mu(x) = p^2. \end{aligned}$$

Hence $p = p^2$, contradicting $p = 1/2$. □

47.2 The connected component is an open real vector group

Theorem 47.2 (Topological additive decomposition). *Let G be a Hausdorff locally compact second-countable abelian group with a Borel invariant total order. There are a finite $d \geq 0$ and a countable discrete torsion-free abelian group D such that*

$$G \cong_{\text{top.grp}} \mathbb{R}^d \times D. \quad (47.1)$$

The identity component $C = G^\circ$ is open and is the $\mathbb{R}^d$ factor.

Proof. The principal structure theorem for locally compact abelian groups supplies an open subgroup $H \cong \mathbb{R}^d \times K$, where K is compact; see [27, p. 4] and the classical references recorded there. The image of K is a compact subgroup of G , so lemma 47.1 forces $K = 0$. Thus $H \cong \mathbb{R}^d$ is open and connected. It is also closed, as every open subgroup is closed. Therefore $H = G^\circ = C$.

As an abstract abelian group, C is divisible and hence injective. There is a homomorphism $r : G \rightarrow C$ extending the identity of C . The restriction of r to the open neighborhood C of zero is continuous, so r is continuous at zero and hence everywhere. Let $D = \ker r$. Each coset of C meets D in exactly one point; in particular D is discrete. The maps $(c, e) \mapsto c + e$ and $g \mapsto (r(g), g - r(g))$ are continuous inverses. Second countability makes the family of disjoint open cosets countable. Finally D is torsion-free because it is a subgroup of the ordered group G . $\square$

For completeness, the injectivity fact just used can be obtained by the usual maximal-extension argument. Given a homomorphism from a subgroup B of an abelian group into a divisible group, adjoin one new element x . If a least positive n has $nx \in B$, choose an n th root of its image; otherwise assign any image to x . Each choice extends the homomorphism to $B + \mathbb{Z}x$. Zorn's lemma then gives an extension to the whole group. Its use here is purely algebraic, with continuity recovered from the openness of C .

47.3 A Borel order on the real component is a linear flag

Lemma 47.3 (Finite-dimensional Borel-order lemma). *Every Borel invariant total order on $(\mathbb{R}^d, +)$ is, after an invertible real-linear change of coordinates, the standard lexicographic order. For $d > 0$ it has a positive order unit.*

Proof. Proceed by induction on d . The assertion is trivial for $d = 0$. Let P be the strict positive cone when $d > 0$. Ordered divisibility implies that P is stable under positive rational scaling and rational convex combinations. The sets $P, -P, \{0\}$ cover $\mathbb{R}^d$, so the Baire category theorem implies that P is not meager. Since P is Borel, it has the Baire property and is comeager in some nonempty open set U .

If $z \in U + U$, then $U \cap (z - U)$ is nonempty and open. Both P and $z - P$ are comeager in this intersection, so it contains $x \in P$ with $z - x \in P$. Hence $z \in P + P \subseteq P$. This proves that P has nonempty interior.

Let $F = \overline{P}$. Rational convexity and passage to limits show that F is a closed convex real cone. It is proper: the nonempty open set $-\text{int}(P)$ is disjoint from P , and therefore from its closure. Finite-dimensional separation yields a nonzero real-linear functional ℓ with $\ell(F) \subseteq [0, \infty)$. Totality then gives

$$\{x : \ell(x) > 0\} \subseteq P \subseteq \{x : \ell(x) \geq 0\}. \quad (47.2)$$

Indeed, an element with positive ℓ cannot be zero or negative. The restriction of the order to $\ker \ell$ is again Borel. Apply the induction hypothesis there, and extend its coordinate functionals to $\mathbb{R}^d$, with ℓ as first coordinate. Equation (47.2) and the induction hypothesis give the required lexicographic rule.

In those coordinates, $u = (1, 0, \dots, 0)$ is an order unit: for any $v = (v_1, \dots, v_d)$ choose an ordinary integer $N > |v_1|$. Then $-Nu < v < Nu$. $\square$

Remark 47.4. We did not assume that the given order respects multiplication of vectors by arbitrary positive real numbers. That compatibility is a conclusion of Borelness and finite

dimensionality. Applying an ordered-vector-space classification without first establishing this point would leave a gap.

48 Proof and scope of arithmetic rigidity

Proof of theorem 44.1. Apply theorem 47.2 to the additive group of A . Its identity component C is an open real vector group $\mathbb{R}^d$, and A/C is countable. If $d > 0$, then C is uncountable and divisible. By lemma 47.3, the restricted order has an order unit $u \in C$. The amplification theorem therefore gives an injection

$$C \hookrightarrow A/C, \quad c \mapsto uc + C.$$

This contradicts countability of A/C . Hence $d = 0$. The zero subgroup is then open, so the topology is discrete. A second-countable discrete space is countable. $\square$

[write] Source 09 reaches the same conclusion, discreteness and countability, without local compactness but with Borel scalar maps and a cone with the Baire property (theorem 62.1); the two hypotheses are incomparable. Source 06's theorem 39.1 is a third result of the kind, on the nonnegative cone: if addition and multiplication are both continuous for a Polish topology on the cone, then 0 is isolated.

[write] (Batch 87.) The two hypotheses are no longer incomparable in effect: both conclusions follow from corollary 78.5 of Part VI, which needs neither local compactness nor Borel scalars.

Corollary 48.1 (No compatible ordered multiplication). *Let G be an uncountable discretely ordered abelian group with a Borel order and a locally compact Polish group topology. There is no commutative unital ring multiplication on its underlying set, with unit the named least positive element, which respects the specified order. This remains true when multiplication is allowed to be non-Borel.*

Corollary 48.2 (Positive-cone semirings). *Let G satisfy the hypotheses of corollary 48.1, and let $M = G_{\geq 0}$. There is no commutative unital semiring multiplication on M with the given addition and unit, such that the product of two nonzero elements is nonzero.*

Proof. The cancellative additive monoid M has Grothendieck group G . A commutative semiring multiplication extends by distributivity to this group: $(a-b)(c-d) = ac+bd-ad-bc$. The assumed absence of zero divisors on positive elements makes products of positive elements positive in G . It would therefore yield the ring forbidden by corollary 48.1. $\square$

The group-topology hypothesis is important. A topology on a cancellative positive monoid need not automatically give its Grothendieck group the same desirable separation and local-compactness properties. Thus our result is not a theorem about every possible Polish presentation of a positive arithmetic model. Similarly, the theorem does not replace the general Polish-space result of [1]; it is a different refinement with a stronger topological assumption and much weaker multiplicative assumptions.

The proof uses no standard-residue axiom, no square roots, no induction scheme in the ring, and no compatibility between multiplication and limits. The only induction in lemma 46.1 is ordinary external induction on a finite integer. These distinctions make the theorem suitable for applications to weak arithmetic and integer parts.

49 Signed Presburger arithmetic and a uniform elimination lemma

Definition 49.1 ($\mathbb{Z}$ -group). A $\mathbb{Z}$ -group is an ordered abelian group with least positive element 1 such that for each ordinary $m \geq 1$ and each $x \in G$ there are unique $q \in G$ and $r \in \{0, \dots, m-1\}$ satisfying

$$x = mq + r1. \quad (49.1)$$

Equivalently, $\mathbb{Z}1$ is a convex subgroup and $G/\mathbb{Z}1$ is divisible. For the implication from quotient divisibility, write $x = mq + k1$, perform ordinary Euclidean division on k , and absorb its quotient into q . Uniqueness follows because $|r - s| < m$ cannot be a nonzero multiple of m in a discretely ordered group. The converse follows immediately from (49.1). Convexity of $\mathbb{Z}1$ follows from the finite-interval argument of lemma 46.1, which only needs the group order.

We use the language

$$\mathcal{L}_{\text{Pres}} = \{+, -, 0, 1, <, \equiv_m: m = 2, 3, \dots\}, \quad x \equiv_m y \iff x - y \in mG.$$

The equivalence between $\mathbb{Z}$ -groups and models of signed Presburger arithmetic, and quantifier elimination in this language, are standard [28]. We give the finite-witness argument to identify precisely what transfers from ordinary integers to nonstandard groups.

Lemma 49.2 (Coset-invariant finite witnesses). *Fix an ordinary $m \geq 1$. Let $P(y)$ be a predicate invariant under adding every element of mG . Let a_i, b_j be finite lists of elements of a $\mathbb{Z}$ -group G . Then*

$$\exists y \left(\bigwedge_i a_i < y \wedge \bigwedge_j y < b_j \wedge P(y) \right)$$

is decided by a finite test as follows. If lower bounds occur, let $a = \max_i a_i$ and test $y = a + s1$ for $1 \leq s \leq m$. If only upper bounds occur, let $b = \min_j b_j$ and test $y = b - s1$ for $1 \leq s \leq m$. If no bounds occur, test $y = r1$ for $0 \leq r < m$.

Proof. Only completeness of the candidate list needs proof. Suppose first that $y > a$ is a witness. Euclidean division of $y - a$ gives $y - a = mq + r1$, $0 \leq r < m$. If $r > 0$, then $q \geq 0$; otherwise $q \leq -1$ would imply $y - a \leq -1$. Set $s = r$ in this case. If $r = 0$, positivity implies $q \geq 1$, and we set $s = m$. In either case

$$a < a + s1 \leq y, \quad y - (a + s1) \in mG.$$

Thus $P(a + s1)$ holds and every upper bound remains satisfied. With only upper bounds, apply the reflected argument to $b - y > 0$. With no bounds, use the remainder of y modulo mG . The reverse implications are immediate because every accepted candidate is itself a witness. $\square$

Theorem 49.3 (Classical Presburger elimination, in the needed form). *The axioms of $\mathbb{Z}$ -groups admit quantifier elimination in $\mathcal{L}_{\text{Pres}}$ and form the complete theory of $(\mathbb{Z}, +, -, 0, 1, <, \equiv_m)_{m \geq 2}$. Their nonnegative cones are models of the full first-order theory $\text{Th}(\mathbb{N}; 0, 1, +, <)$, including all its induction instances.*

Proof. After disjunctive normal form, it suffices to eliminate one existential variable from a conjunction. Negated inequalities and equalities become finite combinations of strict inequalities using discreteness. Negated congruences become finite disjunctions of the other ordinary residue classes. Atoms not involving the eliminated variable are carried along.

Normalize the nonzero integer coefficients of the eliminated variable x by taking an ordinary common multiple $L > 0$ of their absolute values and replacing Lx by y , with the additional condition $y \in LG$. For example, with $a \neq 0$ and $k = L/|a|$, an inequality $ax < t$ becomes $\text{sgn}(a)y < kt$, and

$$ax + t \in nG \iff \text{sgn}(a)y + kt \in nkG.$$

The equivalence uses torsion-freeness: $kz \in nkG$ if and only if $z \in nG$. Equalities with coefficient 1 or -1 permit substitution. What remains is a finite collection of upper and lower bounds on y and a Boolean combination $P(y)$ of congruences. This predicate is invariant under mG , where m is a common multiple of the finitely many moduli, including L .

Apply [lemma 49.2](#). Choosing a maximum or minimum among a finite list can itself be expressed by a finite disjunction of comparisons. Thus the output is quantifier-free. Repeating eliminates all quantifiers. A quantifier-free sentence concerns only ordinary integer numerals, their order, and their standard residues, so it has the same truth value in every $\mathbb{Z}$ -group as in $\mathbb{Z}$. This proves completeness.

Finally, the nonnegative cone is definable, and any formula in its language translates to a group formula by restricting its quantifiers to $x \geq 0$. The cone therefore satisfies exactly the first-order sentences true in $\mathbb{N}$. In particular the translated induction instances hold; they are consequences, not additional assumptions of the construction. $\square$

[write] [Lemma 49.2](#) is Part I's coset-invariant finite search, [lemma 4.7](#) (source 05) and source 04's [theorem 4.8](#), and [theorem 49.3](#) is the classical [theorem 4.9](#); source 08 proves them again in its own notation.

Remark 49.4 (The transfer condition is coset invariance). For nonstandard G , invariance under the single ordinary shift $m1$ does not by itself imply invariance under mG . The latter is what the proof needs and what congruence formulas supply. This distinction already appears in the earlier companion manuscript [40]. ProverIt's integer-specific periodicity lemmas must not be generalized merely by replacing the carrier $\mathbb{Z}$ with an arbitrary ordered group.

50 An exact pointed topological-group classification

Theorem 50.1 (Locally compact signed Presburger groups). *Let $(G, +, <, 0, 1)$ be a $\mathbb{Z}$ -group equipped with a locally compact Polish group topology and a Borel order. There exist a finite $d \geq 0$ and a countable $\mathbb{Z}$ -group $(E, +, <_E, 0, 1_E)$ such that*

$$(G, +, 1) \cong_{\text{top.grp},1} (\mathbb{R}^d \times E, +, (0, 1_E)), \quad (50.1)$$

where E is discrete. On the identity component the given order is real-linearly conjugate to the lexicographic order.

Conversely, every such pair (d, E) admits a Borel $\mathbb{Z}$ -group order on the displayed pointed topological group. The theorem classifies the pointed topological additive groups, not all possible full orders on them.

Proof. By [theorem 47.2](#), write $G = C \oplus D$ topologically, with $C \cong \mathbb{R}^d$ and D countable discrete. A divisible subgroup cannot contain a nonzero ordinary integer by [lemma 46.1](#). Hence $C \cap \mathbb{Z}1 = 0$.

Write the named element as $1 = (v, e)$ in this splitting. The element $e \in D$ has infinite order: otherwise a nonzero ordinary multiple of 1 would belong to C . Define a homomorphism on

$\mathbb{Z}e$ by $ne \mapsto nv$ and extend it to $f : D \rightarrow C$ using divisibility of C . Since D is discrete, f is continuous. The shear

$$(c, x) \mapsto (c - f(x), x)$$

is a topological group automorphism taking 1 to $(0, e)$. Transport the order along it and let $E = \{0\} \times D$ with the restricted order and named unit $1_E = e$.

The element 1_E is least positive in E , because it is least positive in the whole group. If $(0, x) = m(c, y) + r(0, e)$ is standard division in G , then $mc = 0$ forces $c = 0$. Thus the quotient lies in E , proving its standard-residue property. The component order is handled by [lemma 47.3](#).

Conversely, put the usual lexicographic order on $\mathbb{R}^d$, and compare pairs first in this vector coordinate and then in E . The element $(0, 1_E)$ is least positive; vector divisibility together with division in E gives (49.1). The order is Borel because the vector comparison is a finite union of coordinate conditions and E is countable discrete. This constructs every asserted topological type. $\square$

Remark 50.2 (Why the order caveat matters). The component G° need not be convex in the full ordered group. For example, take the ordered additive group with coordinates $(r, q, s, n) \in \mathbb{R} \times \mathbb{Q} \times \mathbb{R} \times \mathbb{Z}$, compared in that order, and give $\mathbb{R}^2 \times (\mathbb{Q} \times \mathbb{Z})$ the product topology with the second factor discrete. This is a $\mathbb{Z}$ -group: the first three coordinates are divisible and n supplies the remainder. Its connected component has $q = n = 0$, but a positive element in the q coordinate lies between positive elements at the two real scales. Thus countable discrete scales can be interleaved with real scales.

There is a precise invariant distinguishing this example from a block lexicographic product with all real coordinates first. Define

$$B_C = \{g \in G : |g| < c \text{ for every } c \in C_{>0}\}.$$

In a block product $\mathbb{R}^d \times E$ with the vector coordinate compared first, $B_C = \{0\} \times E$, so $G = C + B_C$. In the displayed interleaved example, comparison with arbitrarily small positive vectors in each real coordinate shows that B_C consists exactly of the integer coordinate. Hence $C + B_C$ misses every element with $q \neq 0$. Both C and B_C are preserved by an ordered topological-group isomorphism, so the example has no such block-order presentation. This does not affect the pointed additive-group decomposition proved above.

51 Positive cones: exact regularity and the dimension threshold

A useful feature of finite-dimensional lexicographic order is that even its arbitrary cuts have low descriptive complexity.

Lemma 51.1 (Cuts in finite lexicographic powers). *Every upper set in $\mathbb{R}^d$ with the lexicographic order is Δ_2^0 in the usual Euclidean topology. No definability hypothesis on the upper set is needed.*

Proof. Induct on d . For $d = 0$ there are only the empty and full sets. For $d > 0$, let U be a nonempty proper upper set. Consider the first coordinates of its elements. They form a nonempty upper subset of $\mathbb{R}$ with a finite lower endpoint a . The endpoint is finite because an excluded first coordinate gives a lower bound and an included one gives an upper bound for the infimum. If all first coordinates occurred, the upper-set condition would make U the whole space, contrary to assumption. For first coordinate $> a$ every point belongs to U , and for first coordinate $< a$ none does. Thus

$$U = \{x : x_1 > a\} \cup (\{a\} \times U'), \quad (51.1)$$

where U' is an upper set in lexicographic $\mathbb{R}^{d-1}$, possibly empty or full. By induction U' is both F_σ and G_δ . Equation (51.1) expresses U as an F_σ set. The corresponding formula for its complement does the same, so U is also G_δ . Empty and full upper sets are immediate. $\square$

Theorem 51.2 (Cone regularity and local compactness). *Let G satisfy theorem 47.2, and write $G^\circ \cong \mathbb{R}^d$. Then:*

1. $G_{\geq 0}$ and the order relation are Δ_2^0 ;
2. $M = G_{\geq 0}$ with its subspace topology is Polish, and addition $M \times M \rightarrow M$ is continuous;
3. M is locally compact if and only if $d \leq 1$.

Proof. The cosets of $C = G^\circ$ are a countable family of clopen subsets of G . Translate each to C and use the linear coordinates of lemma 47.3. In each coset the intersection with M is an upper set for the induced lexicographic order. By lemma 51.1, each such intersection and its relative complement are F_σ in the coset, and hence in G . Taking countably many unions shows that both M and $G \setminus M$ are F_σ . The order relation follows by the continuous difference map. This proves the first assertion.

A G_δ subspace of a Polish space is Polish [8]. The addition on M is the restriction of the continuous group operation, with its image in the subspace M , giving the second assertion.

If $d = 0$, M is discrete. If $d = 1$, each nonempty coset slice is a real line or an open or closed half-line, by the proof of lemma 51.1. Each is locally compact. As the slices are open in M , their union is locally compact.

Suppose now $d \geq 2$. The slice at the identity component is the nonnegative lexicographic cone in $\mathbb{R}^d$. If M had a compact neighborhood K of zero, then for some $\epsilon > 0$ its intersection with C would contain all points of this cone in the Euclidean ball of radius ϵ . In particular, for sufficiently large n ,

$$x_n = (\epsilon/n, -\epsilon/2, 0, \dots, 0) \in K.$$

But in G this sequence converges to

$$x = (0, -\epsilon/2, 0, \dots, 0) < 0.$$

The inclusion $M \hookrightarrow G$ is continuous, so K is compact and therefore closed in the Hausdorff space G . This would put x in $K \subset M$, a contradiction. $\square$

[write] Part (1) is the locally compact case of source 07's theorem 33.1, proved here through lemma 51.1; parts (2) and (3) are new. Proposition 51.4 below is an instance of theorem 7.4 and proposition 37.1.

Corollary 51.3 (The locally compact positive Presburger case). *In the class of positive cones of locally compact Borel signed Presburger-group presentations, an uncountable cone is locally compact exactly when the connected component of its signed group is one-dimensional. Every cone in this class is Polish and has continuous addition.*

This is a classification within the stated presentation class. It does not assume, or prove, that every topological Presburger monoid arises from a locally compact topological group in this manner.

51.1 The order bound is genuinely rank two

Proposition 51.4 (Order is neither open nor closed). *For an uncountable model in [theorem 50.1](#), the strict order on $M = G_{\geq 0}$ is Δ_2^0 but neither open nor closed in M^2 . The same is true of the weak order.*

Proof. Here $d > 0$. Choose $c_n > 0$ in C with $c_n \rightarrow 0$, for example the points $(1/n, 0, \dots, 0)$ in the flag coordinates. The component is divisible, so $c_n > 1$ for every n by [lemma 46.1](#). The pairs $(0, c_n)$ belong to the strict order and converge to $(0, 0)$, which does not; the relation is not closed. The point $(0, 1)$ belongs to the strict order, whereas $(c_n, 1)$ does not and converges to it; the relation is not open. Its complement after exchanging coordinates is the weak order, giving both failures there as well. The upper Δ_2^0 bound follows from [theorem 51.2](#) and restriction to M^2 . $\square$

There is no contradiction between continuous addition and this exact rank-two behavior. Arithmetic comparison is a Borel relation in a topology that is intentionally different from the discrete order topology.

52 Definable relations and continuous standard division

Theorem 52.1 (Uniform definable regularity). *In every model of [theorem 50.1](#), all Presburger-definable subsets of G^k , with parameters and arbitrary finite k , are Δ_2^0 . The same holds for definable subsets of the positive cone M^k in its relative topology. For each ordinary $m \geq 1$, the quotient and remainder maps in [\(49.1\)](#) are continuous, both on G and on M .*

Proof. Use the pointed splitting $G \cong \mathbb{R}^d \times E$. Since the real factor is divisible,

$$mG = \mathbb{R}^d \times mE.$$

This is clopen, because E is discrete. Every congruence condition is therefore the inverse image of a clopen set under a continuous affine map. Every comparison is Δ_2^0 by [theorem 51.2](#). Equality is closed and hence Δ_2^0 in a metrizable space. Affine integer-linear terms are continuous. Quantifier elimination in [theorem 49.3](#) now expresses each definable set as a finite Boolean combination of Δ_2^0 sets, which is again Δ_2^0 .

For positive-cone formulas, restrict quantifiers to $x \geq 0$, apply group quantifier elimination, and then restrict the resulting set to M^k . This preserves the stated relative Borel class.

The remainder map r_m has the finitely many clopen fibers $r1 + mG$, $0 \leq r < m$, so it is continuous into the discrete finite set of ordinary residues. Multiplication by the ordinary integer m is a homeomorphism from G onto mG : in the product splitting it is $(v, e) \mapsto (mv, me)$, whose inverse is continuous on its image. Consequently

$$q_m(x) = (m \cdot)^{-1}(x - r_m(x)1) \tag{52.1}$$

is continuous. If $x \geq 0$, the quotient $q_m(x)$ is nonnegative; otherwise $q_m(x) \leq -1$ would imply $x \leq -m + (m - 1) = -1$. Restricting these maps to M proves the final claim. $\square$

[write] This is the statement of [theorem 7.1](#) and [theorem 34.1\(iii\)](#) for a class that also admits the interleaved, non-block orders of the remark after [theorem 50.1](#). It answers source 07's question on uniform definability bounds ([section 43](#)) within that class.

52.1 A model in every finite real dimension

Take $E = \mathbb{Z}$ and order $\mathbb{R}^d \times \mathbb{Z}$ lexicographically, with the integer coordinate last. Write

$$G_d = \mathbb{R}^d \overrightarrow{\times} \mathbb{Z}, \quad M_d = (G_d)_{\geq 0}. \quad (52.2)$$

For $x = (v, n)$,

$$r_m(v, n) = n \bmod m, \quad q_m(v, n) = (v/m, \lfloor n/m \rfloor). \quad (52.3)$$

Thus standard division is explicit and continuous. Every M_d is a Polish model of full Presburger arithmetic with continuous addition. For $d = 1$, (52.2) is exactly (44.1); for $d \geq 2$ the cone is not locally compact although the signed group is.

The case $d = 1$ therefore provides the same affirmative construction for the question cited in section 44, already given in [40]. The advance here is the classification and dimensional boundary, not a second priority claim for that example.

[write] M_1 is Part I's $M_{\mathbb{R}}$; for $d \geq 2$ the models M_d are new to the collection.

52.2 Why ordinary scalar operations are not ring multiplication

All maps $x \mapsto nx$ for ordinary $n \in \mathbb{Z}$ are continuous group homomorphisms. This remains compatible with the complete absence of an order-compatible ring multiplication in uncountable models. Presburger arithmetic only provides these externally indexed scalar maps, not a binary multiplication of arbitrary elements.

Likewise, low Borel complexity of every definable relation does not make all definable functions continuous. For instance, truncated subtraction on the positive cone can fail to be continuous. A claim of continuity requires an argument such as (52.1), not merely quantifier elimination.

53 Even unordered multiplication has few Borel scalars

The main theorem forbids multiplication preserving the order, without any measurability assumption. We now remove order compatibility instead and ask how much measurable multiplication can survive. In this section the specified Borel $\mathbb{Z}$ -group order and the multiplication may be unrelated; their named additive unit 1 is the same.

53.1 Automatic continuity and component rigidity

Lemma 53.1 (Classical automatic continuity). *A Borel homomorphism between Polish groups is continuous.*

Proof. We give the abelian case needed here. Let $h : G \rightarrow H$ be Borel and additive. For a neighborhood W of zero in H , choose an open neighborhood V with $V - V \subseteq W$. Separability gives countably many translates of V covering H . Their Borel preimages cover the Baire space G , so one preimage B is nonmeager.

A nonmeager set with the Baire property has $B - B$ as a neighborhood of zero. Indeed, it is comeager in some nonempty open set U ; for small t , the set $U \cap (t + U)$ is nonempty open, and the two translated comeager sets meet, putting t in $B - B$. It follows that $h^{-1}(W)$ contains a neighborhood of zero, because $h(B - B) \subseteq V - V$. Thus h is continuous at zero,

and hence everywhere. This is the usual Baire-category automatic-continuity argument; see also [8] for the Polish-group setting. $\square$

[write] [Lemma 53.1](#) is the first half of source 09's [lemma 61.2](#).

Lemma 53.2 (Injective maps cannot move a component into itself from outside). *Let $G \cong \mathbb{R}^d \times E$ with E discrete, and let $C = G^\circ$. Every continuous injective additive endomorphism $h : G \rightarrow G$ satisfies*

$$h(C) = C, \quad h^{-1}(C) = C. \quad (53.1)$$

Proof. Continuity and connectedness give $h(C) \subseteq C$. The restriction to C is a continuous additive map between finite-dimensional real vector groups, hence real-linear: first use rational scalars, then continuity and rational approximation for real scalars. Injectivity makes this linear endomorphism bijective. If $h(g) \in C$, choose $c \in C$ with $h(c) = h(g)$. Injectivity gives $g = c$, proving the preimage statement. $\square$

53.2 The Borel-scalar subring

Definition 53.3. For a ring multiplication on a topological additive group, define

$$\mathcal{B}(A) = \{a \in A : L_a : A \rightarrow A, x \mapsto ax, \text{ is Borel}\}.$$

We call this the *Borel-scalar subring* once its subring property has been proved.

Theorem 53.4 (Countably many Borel scalar multipliers). *Let $(G, +, <, 0, 1)$ be a locally compact Borel Polish $\mathbb{Z}$ -group, and let A be any commutative unital integral-domain multiplication on its underlying additive group, with this same 1. Multiplication need not respect the order. Then $\mathcal{B}(A)$ is a countable closed discrete subring, and the component quotient map is injective on it. In particular:*

1. *no nonzero element of $C = G^\circ$ has a Borel scalar map;*
2. *if the pointed topological group is $\mathbb{R}^d \times \mathbb{Z}$, then $\mathcal{B}(A) = \mathbb{Z}1$;*
3. *if G is uncountable, multiplication cannot be separately Borel in every scalar, and therefore cannot be jointly Borel.*

Proof. We have $C \cap \mathbb{Z}1 = 0$ and $1 \notin C$ by the additive discreteness and divisibility argument used in [theorem 50.1](#). If $a \in C \setminus \{0\}$ and L_a were Borel, then it would be continuous by [lemma 53.1](#). It is injective because A is a domain. But $L_a(1) = a \in C$, contradicting [lemma 53.2](#). This proves $\mathcal{B}(A) \cap C = \{0\}$.

The maps L_0, L_1 are continuous. If $a, b \in \mathcal{B}(A)$, then $L_{a-b} = L_a - L_b$ is Borel, and $L_{ab} = L_a \circ L_b$ is Borel. Thus $\mathcal{B}(A)$ is a subring, and it contains $\mathbb{Z}1$. If two of its elements have the same coset modulo C , their difference belongs to $\mathcal{B}(A) \cap C$ and is zero. Its projection into the countable discrete quotient G/C is therefore injective.

Each open component contains at most one element of $\mathcal{B}(A)$. Within that component a singleton is closed, and its complement is open. Taking the union of these complements over all components proves that $\mathcal{B}(A)$ is closed in G ; the same component neighborhoods make it discrete. It is countable because the quotient is countable.

If the pointed quotient is $\mathbb{Z}$ generated by the image of 1, every $a \in \mathcal{B}(A)$ has the same image as $n1$ for some ordinary n . Injectivity of the projection gives $a = n1$. Finally, an uncountable G has $d > 0$ and nonzero elements in C , whose scalar maps are not Borel. Sections of a jointly Borel multiplication would all be Borel, so joint Borelness is impossible as well. $\square$

Remark 53.5 (Two independent obstructions). Theorem 44.1 uses order-compatible multiplication, but no regularity of its scalar maps. Theorem 53.4 does not require order compatibility, but bounds which scalar maps can be Borel. Neither statement is obtained merely by rephrasing the other.

54 Examples isolating the hypotheses

54.1 The countable conclusion is optimal

Every countable discretely ordered ring has a discrete Polish additive group topology with Borel order and continuous multiplication. Ordinary $\mathbb{Z}$ is the smallest example, but there are non-Archimedean ones, such as $\mathbb{Z}[X]$ ordered by the sign of the highest nonzero coefficient. Thus the conclusion is discreteness and countability, not that the ordered ring must be isomorphic to $\mathbb{Z}$.

54.2 Second countability cannot simply be deleted

Example 54.1 (An uncountable locally compact ordered ring presentation). Let

$$A_* = \mathbb{Z} + X\mathbb{R}[X] \quad (54.1)$$

with the leading-coefficient order, so X is positive and greater than all ordinary integers. This is a discretely ordered domain. Give its additive group the product decomposition

$$A_* = (\mathbb{R}X) \oplus D_*, \quad D_* := \mathbb{Z} \oplus X^2\mathbb{R}[X], \quad (54.2)$$

where $\mathbb{R}X$ has its usual real-line topology and D_* is discrete. Then the additive group is Hausdorff and locally compact, the order is Borel, and the ring is uncountable. The topology is not second-countable or separable.

Proof. The leading-coefficient rule defines an ordered domain, and every positive nonconstant polynomial exceeds all ordinary integers. Positive constants are positive integers, so 1 is least positive. The product in (54.2) is locally compact. Its discrete factor has cardinality $\mathfrak{c}$, making the family of open components uncountable.

For Borelness, a component with nonzero degree-at-least-two part has a constant sign, independent of the X coefficient and integer constant. The union of all such positive components is clopen. On components with no degree-at-least-two part, positivity is the ordinary condition $r > 0$, with the boundary $r = 0$ governed by the integer constant. These conditions and their complements are countable unions of closed sets. Thus the positive cone is in fact Δ_2^0 in this metrizable, nonseparable topology. $\square$

Here $V = \mathbb{R}X$ has the order unit X , and

$$|A_*/V| = |V| = \mathfrak{c}.$$

So the cardinal lower bound of theorem 46.2 is attained. Multiplication by X takes the connected line $\mathbb{R}X$ injectively into the discrete $X^2\mathbb{R}$ coordinate. It is not continuous. It is not even Borel: the inverse images of open sets obtained from arbitrary subsets of that discrete coordinate give arbitrary subsets of the real line, including non-Borel ones. No topological-ring claim is made for this example.

54.3 Borelness of the order is also essential

Example 54.2 (A locally compact Polish additive presentation with wild order). Fix $d \geq 1$. As rational vector spaces, both $\mathbb{R}^d$ and $X\mathbb{R}[X]$ have dimension $\mathfrak{c}$. Choose a $\mathbb{Q}$ -linear isomorphism $\theta : \mathbb{R}^d \rightarrow X\mathbb{R}[X]$, and define

$$\Theta : \mathbb{R}^d \times \mathbb{Z} \longrightarrow A_*, \quad \Theta(v, n) = n + \theta(v).$$

Transport the ring structure and the leading-coefficient order of A_* along Θ , while retaining the usual product topology on $\mathbb{R}^d \times \mathbb{Z}$. This gives an uncountable discretely ordered domain with a locally compact Polish additive-group topology. Its order is necessarily non-Borel by [theorem 44.1](#).

The dimension assertion uses ordinary ZFC basis theory: an infinite dimensional vector space over a countable field has cardinality equal to its basis cardinality, and a countable direct sum of copies of $\mathbb{R}$ still has rational dimension $\mathfrak{c}$. This construction is not effective and is not proposed as a usable notation system.

Independently give the same carrier the *different*, standard Borel lexicographic $\mathbb{Z}$ -group order. The transported multiplication is not compatible with that order, but [theorem 53.4](#) applies to it and gives the exact identity

$$\mathcal{B}(A) = \mathbb{Z}1.$$

Thus the Borel-scalar theorem has concrete algebraic realizations rather than merely a conditional conclusion about nonexistent ring structures.

54.4 Other boundaries

Discreteness of the arithmetic order is indispensable: the ordinary real field is uncountable, locally compact, and has continuous operations and Borel order, but has no least positive element. Totality also matters: our compact-order and finite-dimensional arguments do not apply to a partial order without additional hypotheses.

We do not present a separable non-locally-compact counterexample with all the other hypotheses of [theorem 44.1](#). The general Polish-group problem is left as a research question, rather than being called a settled sharpness result.

[write] (Batch 87.) [Part VI](#) settles the general Polish-group problem: there is no such counterexample ([corollary 78.5](#)). Its examples on the same ring show which further hypotheses cannot be dropped ([section 84](#)).

55 Consequences for surreal and omnific presentations

55.1 A set-sized no-go theorem for actual omnific rings

ProveIt's surreal-number guide describes its construction of the ordered surreal field, its omnific integer ring, the least-positive-element property, and the constant-term map [\[39\]](#). Only elementary finite normal forms and the discrete order are needed below; none of the repository's stronger unrefereed research claims is an input.

Corollary 55.1 (Omnific additive topologies). *Let $A \subseteq \mathbf{Oz}$ be an uncountable set-sized unital subring, with its inherited ordered-ring structure. There is no Hausdorff locally compact second-countable additive-group topology on A for which the inherited order is Borel. This holds without requiring multiplication to have any topological or measurable regularity.*

Proof. Since $1 \in A$ and 1 is least positive in $\mathbf{Oz}$, it is least positive in A . Apply [theorem 44.1](#). $\square$

[merge] Source 06 equips the nonnegative cone of the same ring $\mathbb{Z} + t\mathbb{R}[t]$ with a locally compact Polish topology in which addition and multiplication are continuous ([theorem 38.1](#)). There is no conflict with [corollary 55.1](#): that topology lives on the cone and is induced by no locally compact second-countable group topology on the ring ([section 42](#)).

This is not merely the observation that a Polish space has cardinality at most continuum. The concrete ring

$$\mathbb{Z} + \omega\mathbb{R}[\omega] = \left\{ n + \sum_{j=1}^k r_j \omega^j : n \in \mathbb{Z}, r_j \in \mathbb{R}, k < \omega \right\} \quad (55.1)$$

has cardinality exactly $\mathfrak{c}$, so its cardinality alone permits a Polish coding. Finite surreal normal forms show that it is an ordered copy of A_* in [\(54.1\)](#). The theorem obstructs the combination of Borel order, locally compact additive-group topology, and the actual ordered multiplication even at this set-sized level.

55.2 Finite additive fragments have excellent topology but no ring expansion

Corollary 55.2 (Finite surreal scales). *For every ordinary $d \geq 1$, let*

$$H_d = \{n + r_1\omega + \cdots + r_d\omega^d : n \in \mathbb{Z}, r_j \in \mathbb{R}\} \subseteq \mathbf{Oz}. \quad (55.2)$$

Its inherited ordered additive group, with coefficient topology $\mathbb{R}^d \times \mathbb{Z}$, is a locally compact Borel $\mathbb{Z}$ -group. Its nonnegative cone is Polish with continuous addition and continuous standard quotient and remainder. That cone is locally compact exactly for $d = 1$. All its Presburger-definable relations are Δ_2^0 .

No alternative commutative unital ring multiplication on H_d can respect this addition, order, and named unit. In particular, failure of closure under the actual surreal multiplication cannot be repaired by inventing a different order-compatible multiplication on the same carrier.

Proof. In descending coefficient order $(r_d, \dots, r_1, n)$, addition and order identify H_d with G_d of [\(52.2\)](#). Standard division is [\(52.3\)](#). Apply [theorems 51.2](#) and [52.1](#) and [corollary 48.1](#). Alternatively, the last assertion follows already from algebraic amplification with $V = \sum_{j=1}^d \mathbb{R}\omega^j$ and order unit $u = \omega^d$: $H_d/V \cong \mathbb{Z}$ is countable but V is not. $\square$

55.3 Implications for exact symbolic systems

These results distinguish three legitimate design choices. One may retain finite surreal additive scales and continuous addition, accepting that there is no compatible full multiplication. One may enlarge to a genuine set-sized ordered ring, abandoning local compactness or separability of an additive-group topology. Or one may transport the algebra to a convenient Polish additive space and accept a non-Borel inherited order. The examples in [section 54](#) make the last two alternatives mathematically explicit.

The earlier support-barrier manuscript concerns a different obstruction: which exact Hahn or omnific normal-form families admit Borel coefficient observations [\[41\]](#). Our obstruction requires neither unrestricted supports nor an analysis of support validity. It already appears for the finite polynomial ring [\(55.1\)](#).

56 A formalization plan with explicit trust boundaries

The proof should be formalized in layers. Its algebraic core is much smaller than its locally compact group input, so formalization need not begin with Haar measure or the full theory of real Lie groups.

Layer A: finite-order arithmetic. Formalize that $[0, n1]$ contains exactly the ordinary numerals in a discretely ordered group. Deduce that a subgroup disjoint from $\mathbb{Z}1$ has no nonzero element bounded by a standard integer, and that a divisible subgroup has this property. The induction index is a natural number in the metatheory of the formal development.

Layer B: amplification. For a subgroup V , an element $u \in V$, and the explicit order-unit predicate, prove $uv \in V \Rightarrow v = 0$. Package (46.1) as an injective additive map into the quotient additive group. Then formalize (46.3) for finitely supported coefficient families and derive the ordered direct-sum embedding. No topological assumption, infinite series, or proper class occurs in this layer.

Layer C: the topological interface. State a reusable interface consisting of an open connected subgroup C , a topological additive-group isomorphism $C \cong \mathbb{R}^d$, and a countable quotient. Prove arithmetic rigidity from this interface and the Borel finite-dimensional flag lemma. Separately discharge the interface using the compact-order obstruction and the principal LCA structure theorem. A conditional theorem with this interface is useful, but must not be advertised as an unconditional formalization of the whole locally compact result.

Layer D: Presburger transfer. Prove lemma 49.2 over an abstract $\mathbb{Z}$ -group with a predicate invariant on cosets of mG . Prove the group theory of standard division before connecting to syntax. ProveIt already supplies syntax, coefficient normalization, and an executable integer decision procedure, according to its inspected guide [38]; the semantic transfer must be proved rather than inferred from that guide.

Layer E: topology of presentations. Formalize lexicographic cuts by finite dimension, prove their F_σ and G_δ descriptions, and derive the cone dimension threshold and continuity of the standard quotient and remainder maps. The scalar theorem then reuses an automatic-continuity theorem and finite-dimensional linear algebra.

No Lean files are included as if they had been checked. The companion Python script validates finite exact-arithmetic examples, not proofs in Lean or in any other kernel. In particular it does not test Haar measure, nonmeager sets, arbitrary Borel orders, cardinal injections, or the principal LCA structure theorem.

57 Further research questions and concrete next targets

Question 57.1 (Remove local compactness). Can an uncountable discretely ordered ring have a Polish additive-group topology with Borel order when multiplication has no regularity at all?

The proof here uses local compactness to obtain a finite-dimensional open component with an order unit. A non-locally-compact Polish group can have an infinite-dimensional component or be totally disconnected without being discrete. An extension would need a substitute for that finite-dimensional argument, perhaps an uncountable subgroup to which [theorem 46.2](#) applies and a suitably small quotient. Neither a proof nor a counterexample is claimed here.

[write] Source 09's [theorem 62.1](#) answers this question negatively when every scalar map $x \mapsto ax$ is Borel and the positive cone has the Baire property; without any regularity of multiplication it stays open.

[write] (Batch 87.) Answered negatively, with no regularity of multiplication, by [Part VI](#): sources 12 and 10 prove that a discretely ordered ring with a Hausdorff Baire additive-group topology satisfying the countable chain condition, whose positive cone has the Baire property, is countable and discrete ([corollary 78.4](#)); in particular every Polish additive-group topology with Borel order is discrete on a countable ring ([corollary 78.5](#)). The preceding note's "it stays open" is superseded. The proofs are unrefereed and not formalized.

Question 57.2 (Topological group completion of positive models). Which locally compact Polish cancellative Presburger monoids admit a Hausdorff locally compact Polish topology on their Grothendieck groups that induces the original topology on the positive cone?

A positive answer under usable hypotheses would convert [corollary 51.3](#) into a broader classification of positive arithmetic presentations. The relevant issue is topological embedding of the monoid into its group completion, not the elementary algebraic existence of that completion. Conditions on translations, quotient maps, and neighborhoods of zero are natural starting points.

[write] Source 09 asks the same question as [question 73.11](#). Source 06's cone of $\mathbb{Z} + t\mathbb{R}[t]$ is a data point ([section 42](#)): a locally compact Polish cone with continuous addition, whose group of differences has no locally compact second-countable group topology with Borel order, by [theorem 44.1](#).

[write] (Batch 87.) Sources 12 and 10 ask it again for Baire ccc completions with a Baire-property cone ([Part VI](#), questions on group completion), and source 15 for locally compact Presburger monoids ([Part IX](#)).

[write] (Batch 90.) Answered by [Part XII](#). A locally compact Polish cancellative Presburger monoid with continuous addition admits a Hausdorff group topology on its group of differences inducing its topology if and only if its partial subtraction $\delta(y, x) = y - x$ ($x \leq y$) is continuous ([theorem 153.1](#)); the extending topology is then unique, Polish ([theorem 154.1](#)) and locally compact ([corollary 154.2](#)), and continuity need only be tested at the pairs (x, x) and $(x, 0)$ ([theorem 154.3](#)). Local compactness is not needed for the criterion. Source 06's cone, the data point of the first note, fails it at (t^2, t^2) ([example 155.5](#)). The proofs are unrefereed and not formalized.

Question 57.3 (Classify the orders, not only the pointed groups). For a fixed countable $\mathbb{Z}$ -group E , classify Borel $\mathbb{Z}$ -group orders on $\mathbb{R}^d \times E$ up to continuous additive order isomorphism fixing 1.

The real component carries a finite flag, but discrete coordinates can occur between real scales, as the example following [theorem 50.1](#) shows. The cut description in [lemma 51.1](#) and the continuous separating homomorphisms of the classical measurable-order literature suggest finite flag data together with countable extension data. Determining which data are complete invariants is a concrete next classification problem.

[write] (Batch 92, 4 October 2026.) Answered for $d = 1$ by [Part XIII](#) (source 19, which takes this question as its target): after a continuous shear fixing 1, every Borel (indeed Baire-property) Presburger order on $\mathbb{R} \times E$ is the three-layer order of [theorem 170.2](#), an ordered quotient E/A above $\mathbb{R}$ above an ordered pure subgroup $A \ni 1$, and the complete invariant for continuous isomorphism is the countable ordered extension $(E, 1, A, <_A, <_B)$ ([theorem 171.1](#)); for $E = V \oplus \mathbb{Z}$ with V a $\mathbb{Q}$ -vector space it is a pair of countable ordered divisible groups ([theorem 172.1](#)). Source 19 puts the unit in E ; the shear of [theorem 50.1](#) does that in general ([proposition 182.1](#) [merge]), and abstract isomorphism gives the same classification ([proposition 182.2](#) [merge]). The case $d \geq 2$ stays open ([question 180.1](#)). The proofs are unrefereed and not formalized.

Question 57.4 (Weaken Borel order carefully). Which hereditary measurability assumptions on an invariant order suffice for arithmetic rigidity? In particular, what replaces Borelness if restrictions to every compact subgroup and every closed real subspace are required to be Haar measurable or to have the Baire property?

An ambient Haar-measurable set can have uncontrolled behavior on a null closed subgroup. The present proof intentionally uses Borelness to avoid that problem. A weakening should state restriction hypotheses explicitly rather than treating ambient measurability as automatically hereditary.

[write] (Batch 87.) Answered by [Part VI](#) without any hereditary condition: the Baire property of the cone for the given topology suffices ([lemma 77.2](#) and [corollary 78.4](#)), and on a locally compact group measurability of the cone for completed Haar measure suffices ([theorem 83.3](#)). Both sources of Part VI ask for still weaker regularity.

[write] (Batch 93.) Part XVI's [proposition 217.1](#) gives, on $\mathbb{R}^2$, an order whose cone has the Baire property but is neither Borel nor real homogeneous: for the vector-space conclusions an ambient Baire property is not enough. This does not reopen the answer above, which concerns arithmetic rigidity; source 22's [question 222.5](#) asks the vector-space question.

Question 57.5 (Cardinal and topological size spectra). For nonseparable locally compact additive presentations of discretely ordered rings, what cardinalities of the component quotient can occur relative to the identity component and the weight of the topology?

Example [54.1](#) attains $|A/C| = |C| = \mathfrak{c}$ with a real component. The direct-sum strengthening in [theorem 46.2](#) records more than one injection. It may impose additional ordered-group or module constraints on possible quotients even when it yields no stronger bare infinite-cardinal inequality.

[write] (Batch 87.) For Hausdorff Baire presentations with a Baire-property cone, source 12 proves $c(R) = d(R) = |R|$ ([theorem 78.3](#)) and source 10 adds extent and Lindelöf degree for uncountable rings ([theorem 79.13](#)); [example 54.1](#) attains them. The finer spectra asked for stay open.

Question 57.6 (Realize Borel-scalar subrings). For fixed $\mathbb{R}^d \times E$, which countable subrings can arise as $\mathcal{B}(A)$ for integral-domain multiplications on the additive group?

The projection of $\mathcal{B}(A)$ into E is injective, and it contains the ordinary integers. When $E = \mathbb{Z}$ the answer is exactly $\mathbb{Z}$. For larger E , continuous scalar maps act real-linearly on the component and induce additive maps on E ; compatibility of these two actions and their extension data may give a useful algebraic realization criterion.

Question 57.7 (Effective and reverse-mathematical strength). What is the exact effective content of extracting a finite real flag from a code for a Borel order, and of proving the resulting rigidity statement over weak base theories?

The current ZFC proof uses category, separation, Haar measure, and an LCA structure theorem. It is not an analysis of their strength in second-order arithmetic. Glazer’s separate Question 1 about the strength of his theorem [1] remains outside the results proved here; an argument for the locally compact fragment should not be substituted for an answer to that question.

[write] [Section 1.1](#) discusses Glazer’s Question 1 and the speculation that follows his Question 2.

[write] (Batch 87.) Source 14 ([Part VIII](#)) claims an ATR_0 proof of Glazer’s Theorem 2, not independently reviewed. It is not an analysis of this question, which concerns the locally compact rigidity theorem; sources 12 and 10 ask the corresponding question for Part VI.

Question 57.8 (Certified amplification and notation design). Can the algebraic amplification theorem be kernel-checked first, and then used as an automatic obstruction to proposed finite-scale omnific or non-Archimedean ring interfaces?

A small certificate could consist of a divisible or integer-disjoint additive subgroup, an order unit, and an upper bound on the quotient’s size. This would separate a universally reusable algebraic theorem from application-specific normal-form calculations. The finite-scale examples in [corollary 55.2](#) are a natural first target, followed by the Borel-scalar theorem and the full topological interface.

58 Conclusions of Part IV

Locally compact arithmetic separates cleanly into an additive possibility and a multiplicative obstruction. Uncountable Polish Presburger models with continuous addition are available, and their locally compact signed group presentations have a precise finite-dimensional structure. Their order relations have exact rank-two regularity, and their standard quotient and remainder operations remain continuous.

But a discretely ordered ring cannot use that topology in the uncountable second-countable case. The obstruction does not ask multiplication to be Borel: it uses multiplication only to create an impossible injection from a real component into a countable component quotient. When order compatibility is dropped instead, only countably many scalar maps can remain Borel. This gives a concrete structural connection between Glazer’s arithmetic-presentation questions and ProveIt’s finite surreal, omnific, and Presburger developments, while keeping the unresolved Polish-group and formal-verification questions explicit.

[write] (Batch 87.) The Polish-group question is answered by [Part VI](#).

Part V

Polish topologies at the Hahn–Puisseux–Levi-Civita boundary

[write] This part is source 09 of batch 86, *Polishability at the Hahn–Puisseux–Levi-Civita Boundary: Automatic order bounds, local Polish hulls, and countable omnific integer parts* (dated October 3, 2026; “prepared with AI assistance”), pinned to repository commit `4af956fe0`. All unmarked text is source 09’s; its labels carry the prefix `pma:ser:.` It continues [Part II](#): source 03 asked when Hahn-series codes form a *standard Borel* space with observable coefficients, and source 09 asks

when the additive group of a series field can be made *Polish* with measurable arithmetic. Its “preceding support manuscript” [44] is source 03 (Part II), and its Presburger companion [43] is source 05 (Part I). Its finite checks are shipped as `code/09-hahn-polishability-verify.py` with the record data/`09-hahn-polishability-verification.json`.

Convention [write]. In this part t is a positive infinitesimal, exponents of t that are negative give infinite elements, and the surreal realization is $t^q \mapsto \omega^{-q}$ (equation (71.1)). This is the reverse of Part III, where t is positive infinite, and agrees with source 03’s increasing- t convention; 03’s growth variable is $X = t^{-1}$. Source 09’s macros for its fields are renamed only in the source: the Puiseux field $\mathcal{P}_k$ is printed exactly as before and is not 03’s cone $\mathcal{A}_{\text{fin}, \geq 0}$ (section 1.3).

Reading routes. Sections 61–63 contain the main obstruction and classification. Sections 64–67 explain the Puiseux/Levi-Civita contrast. Sections 68–71 connect the analysis to real closed fields, arithmetic, and ProveIt. The final sections separate proved conclusions from further research.

59 The research intersection and the questions answered

Elliot Glazer’s *A Topological Tennenbaum Theorem* [1] studies Borel arithmetic on uncountable Polish spaces. Its Theorem 2 excludes continuity of either arithmetic operation for Borel models of $Q + I\Delta_0 + \text{Exp}$, and its Corollary 1 excludes continuous addition with Borel multiplication for uncountable Polish models of $Q + \text{IOpen}$. This provides a specific, documented intersection with ProveIt’s work on ordered algebra, exact Hahn normal forms, omnific integers, and formal arithmetic. No inference about Glazer’s present employment or endorsement is needed.

The ProveIt source collection distinguishes formalized constructions from unrefereed research reports. Its surreal documentation describes the normal-form realization of the surreal field and the integer-constant, nonnegative-growth normal forms of the omnific integers [42]. We use those as motivation and as a possible implementation interface, not as substitutes for proofs of the topological assertions below.

Two earlier manuscripts prepared for this research program already address Polish Presburger positive cones and descriptive support barriers [43, 44]. In particular, the preceding support manuscript discusses the coanalytic obstruction for unrestricted rational Hahn supports. Repeating that obstruction would not answer the next question: *even when exact series codes form a standard Borel space, can their additive group be made Polish without losing measurable arithmetic?*

The answer is sensitive to which series are admitted. Fix a countable ordered field k , and use t as a positive infinitesimal. Three natural fields are

$$\mathcal{P}_k = \bigcup_{n \geq 1} k((t^{1/n!})), \quad (59.1)$$

$$\mathcal{L}_k = \left\{ \sum_{q \in \mathbb{Q}} a_q t^q : \text{supp}(a) \cap (-\infty, B] \text{ is finite for every } B \in \mathbb{Q} \right\}, \quad (59.2)$$

$$\mathcal{H}_k = k((t^{\mathbb{Q}})). \quad (59.3)$$

They are, respectively, the Puiseux field, the left-finite Levi-Civita field, and the full rational Hahn field. Their inclusions are strict. The middle field, not the largest or the descriptively

simplest one, admits the desired Polish arithmetic.

59.1 The central conclusions

The term *admissible* is made precise in Definition 60.1. It requires a Polish additive group, a Baire-measurable positive cone, and Borel multiplication by each fixed scalar. It does not initially require jointly continuous multiplication.

Main results. For a nonzero ordered abelian group Γ , the full Hahn field $k((t^\Gamma))$ has an admissible Polish topology if and only if k is countable and $\Gamma \cong \mathbb{Z}$ as an ordered group. For countable ordered k , $\mathcal{P}_k$ has no admissible Polish topology, while $\mathcal{L}_k$ has exactly one: its complete valuation topology. Every discretely ordered ring with an admissible Polish additive group is countable. Every integer part of an admissible Polish ordered field is countable and closed discrete.

There is also a sharp local converse. Every countable family of rational Hahn series lies in a complete separable subfield, and these Polish subfields have no maximal member. Exactly $2^{\aleph_0}$ of them are needed to cover the full Hahn field (Theorem 69.6). Over $\mathbb{R}_{\text{alg}}$ they can be chosen real closed and truncation-closed, with canonical countable omnific integer parts. Thus the Levi-Civita field is not the largest possible positive construction: the global obstruction concerns simultaneous freedom of prefixes, not any one prescribed countable support pattern.

The main advance over a support-code obstruction is quantification over *all* admissible Polish topologies on the abstract field. The Hahn classification is not defeated by abandoning coefficientwise coding. The Puiseux obstruction is stronger still in a different direction: the code domain is already F_σ , but no compatible Polish group topology can support the required arithmetic.

For $k = \mathbb{R}_{\text{alg}}$, the real algebraic numbers, both $\mathcal{P}_k$ and $\mathcal{L}_k$ are real closed and share the same explicitly described countable integer part. Its nonnegative cone satisfies open induction. The integer-part algebra therefore does not distinguish these two sharply different topological situations.

59.2 What is and is not claimed as new

The article gives complete arguments for its central classifications and regularity conclusions. The category tools are classical. The construction and completeness of Levi-Civita fields belong to established non-Archimedean analysis; see, for example, Shamseddine–Berz [31]. Generalized series and Newton–Puiseux theory are classical, with a useful account in Poonen [30]. The coanalytic support obstruction and the negative-infinitesimal floor correction are already present in the preceding project materials.

The contribution developed here is their combination into a presentation-invariant classification, the Puiseux versus Levi-Civita Polishability separation, the local Polish-hull and exact field-cover theorems, and the resulting integer-part and floor-regularity statements. These are proposed research contributions relative to the inspected materials. They are not accompanied by a certified global priority claim. We answer the precisely formulated questions in this article; we do not claim to have newly resolved either of Glazer’s printed open questions.

60 Conventions: four different notions of presentation

All fields, groups, supports, and topological spaces in the main arguments are sets in ZFC. Every ordered field has its specified field-compatible linear order. A *discretely ordered ring* is a commutative ordered unital ring whose least positive element is 1. An *integer part* of an ordered field F is a discretely ordered unital subring $I \subseteq F$ such that each $x \in F$ has an $a \in I$ satisfying $a \leq x < a + 1$.

A Polish space is separable and completely metrizable. A set has the *Baire property* when it differs from an open set by a meager set. Every Borel set has this property. A Polish additive group has jointly continuous addition and continuous negation.

Definition 60.1 (Admissible Polish arithmetic). A topology τ on an ordered field or discretely ordered ring F is *admissible* when:

- (i) $(F, +, \tau)$ is a Polish topological group;
- (ii) $F_{>0}$ has the Baire property in τ ;
- (iii) for each $a \in F$, the map $L_a : F \rightarrow F$, $L_a(x) = ax$, is Borel.

An admissible Polish presentation through another carrier means a transported topology satisfying these conditions. It is therefore equivalent to an admissible topology on the original algebraic carrier.

The following distinctions prevent several false conclusions. A standard Borel presentation supplies measurable codes and operations, but need not supply continuous addition. A *Polishable Borel group* is one whose given Borel structure comes from a Polish group topology. An admissible ordered-field topology may initially be allowed to change the Borel structure; our absolute nonexistence results exclude that possibility too. Finally, a Polish topology on a *positive cone* with continuous addition is only a topological monoid. It is not a Polish topology on the group of differences.

In particular, the ring theorem below cannot be applied by silently adjoining negative elements to a Polish model of Presburger arithmetic. Extending the algebra is easy; extending the topology with all the required properties is a separate problem.

For a Hahn series $f = \sum a_\gamma t^\gamma$, the support is well ordered in Γ , and

$$v(f) = \min \text{supp}(f) \quad (f \neq 0), \quad v(0) = +\infty.$$

The order is determined by the coefficient at the least support exponent. Positive exponents represent small quantities. In particular, if $\alpha < \beta$ and $c \in k_{>0}$, then $ct^\alpha > t^\beta$. We distinguish this formal order from the ordinary absolute value of a coefficient when k happens to be a subfield of $\mathbb{R}$.

61 Category forces local order bounds

The mechanism is a short application of classical category theory. We include the needed proofs to make the hypothesis boundary visible. Pettis's theorem, automatic continuity, and the open mapping theorem are classical; Chen [29] reviews their relationship and develops localic counterparts.

Lemma 61.1 (Pettis, additive form). *Let G be a Polish group and let $A \subseteq G$ have the Baire property and be nonmeager. Then $A - A$ contains a neighborhood of 0.*

Proof. There is a nonempty open set U on which A is comeager. Choose a nonempty open V and a neighborhood W of 0 so small that $V \subseteq U$ and $V - g \subseteq U$ for every $g \in W$. For each such g , both $A \cap V$ and $(g + A) \cap V$ are comeager in V . Their intersection is nonempty because a nonempty open subspace of a Polish space is Baire. An element x of the intersection satisfies $x \in A$ and $x - g \in A$, so $g \in A - A$. $\square$

Lemma 61.2 (Automatic continuity and open mapping). *A Borel homomorphism between Polish groups is continuous. A continuous surjective homomorphism between Polish groups is open.*

Proof. For the first assertion, let $\phi : G \rightarrow H$ be Borel and let U be a neighborhood of 0 in H . Choose an open neighborhood V with $V - V \subseteq U$. Separability gives a countable cover of H by translates of V . Their inverse images are Borel and cover G , so one inverse image A is nonmeager. Then $A - A \subseteq \phi^{-1}(U)$, and Lemma 61.1 gives continuity at 0.

For the second assertion, let U be a neighborhood of 0 in G , and choose an open neighborhood V with $V - V \subseteq U$. Countably many translates of V cover G , hence their images cover H . The set $\phi(V)$ is analytic, and therefore has the Baire property. At least one of its translates is nonmeager. Pettis gives a neighborhood of 0 inside $\phi(V) - \phi(V) \subseteq \phi(U)$. $\square$

The descriptive-set-theoretic facts about analytic sets used here are standard [8]. In particular, continuous images of Polish spaces are analytic, analytic sets have the Baire property, and an analytic set with analytic complement is Borel.

Lemma 61.3 (A Baire order is locally bounded). *Let G be a nontrivial linearly ordered abelian group with a Polish group topology. If $P = \{x : x > 0\}$ has the Baire property, some neighborhood of 0 is contained in a bounded order interval $[-a, a]$, with $a > 0$.*

Proof. For a discrete group, take the neighborhood $\{0\}$ and any $a > 0$. Suppose the topology is not discrete. Then $\{0\}$ is nowhere dense. Inversion is a homeomorphism and takes P to $-P$. If P were meager, then $G = P \cup (-P) \cup \{0\}$ would be meager in itself, contrary to the Baire category theorem. Thus P is nonmeager.

Choose a nonempty open set U on which P is comeager, and choose $p \in P \cap U$. Let V be a symmetric open neighborhood of 0 with $W = p + V \subseteq U$. The reflection $x \mapsto 2p - x$ maps W onto itself. Consequently, both $P \cap W$ and $(2p - P) \cap W$ are comeager in W . The set

$$A = W \cap P \cap (2p - P)$$

has the Baire property and is nonmeager. Every $x \in A$ satisfies $0 < x < 2p$. Therefore $A - A \subseteq (-2p, 2p)$. Pettis gives a neighborhood of 0 inside this interval. Take $a = 2p$. $\square$

Theorem 61.4 (Automatic order-topology refinement). *Every admissible Polish topology on an ordered field refines its order topology. More generally, the same conclusion holds for a Polish additive group topology with Baire positive cone and continuous fixed scalar maps.*

Proof. In the admissible case every L_a is continuous by Lemma 61.2. Choose a neighborhood U of 0 with $U \subseteq [-a, a]$ as in Lemma 61.3. For any $\varepsilon > 0$, put

$$c = \frac{a+1}{\varepsilon} > 0.$$

Then $L_c^{-1}(U)$ is a neighborhood of 0, and its elements satisfy

$$|x| \leq \frac{a}{c} = \varepsilon \frac{a}{a+1} < \varepsilon.$$

Thus every interval $(-\varepsilon, \varepsilon)$ contains a neighborhood of 0. Translating this statement at every point of any order interval proves that the interval is open in the given topology. $\square$

Corollary 61.5 (Uniqueness when the order topology is Polish). *If the order topology of an ordered field is a Polish group topology, then it is the only possible admissible Polish topology on that field.*

Proof. The identity from an admissible topology to the order topology is a continuous bijective group homomorphism between Polish groups. It is open by Lemma 61.2. $\square$

Corollary 61.6 (Order-free hypothesis for real closed fields). *Let F be real closed. If its additive group is Polish and its multiplication is jointly Borel, then its unique field order is Borel. Consequently, Theorem 61.4 applies without separately assuming measurable order.*

Proof. The positive cone is the image of $F \setminus \{0\}$ under the Borel map $x \mapsto x^2$, so is analytic. Its complement is the union of $\{0\}$ with the image of $F \setminus \{0\}$ under $x \mapsto -x^2$, hence is analytic too. The analytic/coanalytic criterion gives Borelness. The scalar maps are sections of multiplication and therefore Borel. $\square$

Remark 61.7. The last corollary uses *joint* Borel multiplication to obtain a Borel square map. Merely postulating Borelness of each fixed scalar map does not by itself justify substituting the variable into both multiplication arguments.

62 Discretely ordered rings and integer parts

Theorem 62.1 (Automatic discreteness). *An admissible Polish topology on a discretely ordered ring is discrete. In particular, the ring is countable.*

Proof. By Lemma 61.3, choose a neighborhood U of 0 contained in $[-a, a]$, where $a > 0$. Multiplication by $b = a + 1$ is a continuous additive homomorphism. Thus $W = L_b^{-1}(U)$ is a neighborhood of 0. If $x \neq 0$, discrete order gives $|x| \geq 1$, whence $|bx| \geq b > a$. Such an x cannot belong to W . Therefore $W = \{0\}$. A discrete separable space is countable. $\square$

[write] Theorem 62.1 and source 08's theorem 44.1 have the same conclusion under incomparable hypotheses: 09 needs Borel scalar maps and a cone with the Baire property but no local compactness, 08 needs local compactness but no regularity of multiplication. Together they answer source 08's question 57.1 under the Borel-scalar hypothesis only. The non-example below is the group of Part I's real model.

[write] (Batch 87.) Part VI answers question 57.1 without the Borel-scalar hypothesis; that hypothesis of theorem 62.1 is redundant, and both theorems are cases of corollary 78.5 (section 90). Source 12's lemma 77.2 is lemma 61.3 for Hausdorff Baire groups.

Corollary 62.2 (A non-Polishable Borel group criterion). *Let R be an uncountable standard Borel discretely ordered ring whose order, addition, negation, and multiplication are Borel. Its additive Borel group is not Polishable.*

Proof. A compatible Polish group topology would preserve the stipulated Borel order and scalar maps and would contradict Theorem 62.1. $\square$

This result needs no induction scheme. That does not make it a theorem with weaker hypotheses than Glazer's arithmetic result: here a Polish group of differences is assumed from the outset. Glazer's theorem concerns the arithmetic domain itself. These are different boundaries.

Theorem 62.3 (All integer parts are countable). *Let F have an admissible Polish topology. Every discretely ordered unital subring $I \subseteq F$ is countable and closed discrete. In particular this holds for every integer part, without any measurability assumption on I .*

Proof. By Theorem 61.4, the interval $(-1, 1)$ is open in F . Its intersection with I is $\{0\}$, so every translate of a sufficiently small order interval meets I in at most one point. For example, two distinct points of I cannot lie in an interval of radius $1/3$, because their nonzero difference has absolute value at least 1.

Each point of I therefore has an open neighborhood containing no other point of I . A countable base supplies distinct base elements for distinct points, proving countability. To see closedness, suppose $x \in \bar{I}$. The interval $(x - 1/3, x + 1/3)$ contains at most one point a of I , and must contain one by closure. If $x \neq a$, a smaller neighborhood of x avoiding a contradicts $x \in \bar{I}$. Hence $x = a \in I$. $\square$

Example 62.4 (An important non-example). The lexicographically ordered group $\mathbb{R} \times \mathbb{Z}$, with its ordinary product group topology, has a Borel discrete order: $(0, 1)$ is its least positive element. It is uncountable and Polish. This does not contradict Theorem 62.1, since no ordered-ring structure with the required scalar maps has been supplied. The scalar $a + 1$ in the proof is a genuine ring-theoretic resource.

63 A complete classification for full Hahn fields

The existence and algebraic operations of full Hahn fields are classical [30]. We use the entire well-ordered-support field, not a smaller convergence-restricted subfield.

Lemma 63.1 (A bounded increasing sequence in a noncyclic group). *Let $\Gamma \neq 0$ be an ordered abelian group. Either Γ is order-isomorphic to $\mathbb{Z}$, or there are*

$$0 < \gamma_0 < \gamma_1 < \cdots < \delta$$

with $\delta \in \Gamma_{>0}$.

Proof. Suppose first that Γ has no least positive element. Choose $0 < \varepsilon_0 < \delta$ and then recursively $0 < \varepsilon_{n+1} < \varepsilon_n$. Put $\gamma_n = \delta - \varepsilon_n$.

Suppose instead that ε is the least positive element. If every positive element is bounded by a positive integer multiple of ε , then for every $x > 0$ choose the least n with $x \leq n\varepsilon$. The difference $x - (n - 1)\varepsilon$ is positive and at most ε , hence equals ε . Thus $x = n\varepsilon$ and $\Gamma = \mathbb{Z}\varepsilon$. Otherwise some δ exceeds every $n\varepsilon$, and $\gamma_n = (n + 1)\varepsilon$ has the required properties. $\square$

Lemma 63.2 (Disjoint intervals in a Hahn field). *Let $H = k((t^\Gamma))$ with $\Gamma \neq 0$. If k is uncountable, its order topology has an uncountable family of pairwise disjoint nonempty open intervals. If Γ is not order-isomorphic to $\mathbb{Z}$, it has such a family of cardinality $2^{\aleph_0}$, even with centers having only 0–1 coefficients.*

Proof. For uncountable k , fix $\delta > 0$. The intervals

$$(c - t^\delta, c + t^\delta), \quad c \in k,$$

are disjoint: every positive nonzero constant dominates $2t^\delta$.

For noncyclic Γ , use Lemma 63.1. For $b \in 2^{\mathbb{N}}$ define

$$f_b = \sum_{n \geq 0} b(n)t^{\gamma_n}.$$

This is a legal Hahn series: each support is a subset of the increasing sequence (γ_n) and hence is well ordered. If $b \neq c$, let m be the first index where they differ. Then $v(f_b - f_c) = \gamma_m < \delta$, so $|f_b - f_c| > 2t^\delta$. The intervals $(f_b - t^\delta, f_b + t^\delta)$ are pairwise disjoint. $\square$

Theorem 63.3 (Full Hahn classification). *For an ordered field k and nonzero ordered abelian group Γ ,*

$$k((t^\Gamma)) \text{ admits an admissible Polish topology} \iff |k| = \aleph_0 \text{ and } \Gamma \cong \mathbb{Z} \text{ as ordered groups.}$$

In the positive case the topology is unique and is the Laurent valuation topology; all field operations, including inversion away from 0, are continuous.

Proof. An admissible topology refines the order topology by Theorem 61.4. A second-countable space has only countably many pairwise disjoint nonempty open sets, since one can choose a distinct basic open set inside each. Lemma 63.2 therefore forces k countable and $\Gamma \cong \mathbb{Z}$.

Conversely identify Γ with $\mathbb{Z}$ and use the metric

$$d(f, g) = \min\{1, 2^{-v(f-g)}\}, \quad 2^{-\infty} = 0.$$

A Cauchy sequence eventually stabilizes all coefficients of exponent at most N , for every integer N . Stabilization for $N = 0$ also gives a common finite negative principal part for its tail. The stabilized coefficients therefore define a Laurent series, and the sequence converges to it. Finite Laurent polynomials form a countable dense set because k is countable.

The valuation identities $v(fg) = v(f) + v(g)$ and $v(f + g) \geq \min\{v(f), v(g)\}$ give continuity of addition and multiplication. For $f \neq 0$, sufficiently small changes do not change $v(f)$, and

$$v(f^{-1} - g^{-1}) = v(f - g) - v(f) - v(g)$$

gives continuity of inversion. The order and valuation topologies agree: a valuation ball sufficiently far above the leading exponent of a positive ε lies inside $(-\varepsilon, \varepsilon)$, while a sufficiently small monomial interval lies inside any prescribed valuation ball. Thus the order is open and the topology is admissible. Uniqueness follows from Corollary 61.5. $\square$

[write] Source 03's theorem 20.2 is the Borel counterpart: $\mathbb{R}((t^\Gamma))$ has a coefficient-observable standard Borel parametrization if and only if Γ is scattered. The two results are consistent: for $\Gamma = \mathbb{Z}^d$, $d \geq 2$, the canonical support domain is Borel (03's theorem 19.1) but no admissible Polish topology exists (the corollary below). For real coefficients, lemma 63.2 excludes an admissible Polish topology on every field containing $\mathbb{R}$ and a nonzero infinitesimal, among them 03's real Puiseux field $\mathcal{P}$, which nevertheless has Borel codes (section 23).

Corollary 63.4 (The obstruction is not a coding artifact). *For any countable ordered field k , the full field $k((t^{\mathbb{Q}}))$ has no admissible Polish topology, regardless of how its elements are encoded. For finite lexicographic powers $\Gamma = \mathbb{Z}^d$, $d \geq 2$, the same conclusion holds even when the canonical support domain is Borel.*

Proof. Both ordered groups are noncyclic. In the second assertion the relevance of Borel support coding is only comparative; the topological conclusion uses Theorem 63.3 alone. $\square$

The excluded case $\Gamma = 0$ is deliberate. It reduces to the coefficient field itself; for example $\mathbb{R}$ with its usual topology is admissible. The nontrivial infinitesimal scale is exactly what separates distinct constants in Lemma 63.2.

For countable k and countable noncyclic Γ , the order density and cellularity of the full Hahn field are both exactly $2^{\aleph_0}$: the lemma gives the lower bound, and $|k^\Gamma| = 2^{\aleph_0}$ gives the upper bound.

64 The Puiseux obstruction: a Borel field that is not Polishable

A second category obstruction does not use an order at all.

Theorem 64.1 (Countable proper-subfield cover). *Let F be an uncountable field whose additive group has a Polish topology. Assume every fixed scalar map is Borel. Then F cannot be the union of countably many Borel proper subfields.*

Proof. Suppose $F = \bigcup_n K_n$, with each K_n a Borel proper subfield. By the Baire category theorem some K_n is nonmeager. As it is an additive subgroup, Pettis shows that $K_n = K_n - K_n$ is open. Choose $a \in F \setminus K_n$. The maps L_a and $L_{a^{-1}}$ are Borel homomorphisms, hence continuous; thus L_a is a homeomorphism and aK_n is open. But

$$K_n \cap aK_n = \{0\}.$$

Indeed, a nonzero $x = ay$ with $x, y \in K_n$ would imply $a = x/y \in K_n$. It follows that $\{0\}$ is open. The Polish group would be discrete and countable, a contradiction. $\square$

The Borel hypothesis on the covering subfields matters. A category argument cannot simply be applied to arbitrary nonmeasurable subfields. In the Puiseux application the subfields will actually be closed under *every* admissible topology, so no measurability assumption is being hidden.

Let k now be a countable ordered field, and set

$$K_n = k((t^{1/n!})), \quad \mathcal{P}_k = \bigcup_{n \geq 1} K_n.$$

The factorial convention makes the chain increasing. Each K_n is a proper subfield of $\mathcal{P}_k$, because rational monomials of denominators not dividing $n!$ occur at later stages.

Lemma 64.2 (Closed stages). *Each K_n is closed in the order topology of $\mathcal{P}_k$, and hence closed under every admissible Polish topology on $\mathcal{P}_k$.*

Proof. If $f \notin K_n$, some nonzero coefficient f_q has $q \notin (1/n!)\mathbb{Z}$. Every g agreeing with f at all exponents at most q still lies outside K_n . The set

$$\{g : v(g - f) > q\}$$

is an order-open neighborhood of f . To check order-openness, a monomial interval of exponent greater than q about any of its points lies inside it. Thus the complement of K_n is open. Theorem 61.4 transfers this closedness to any admissible topology. $\square$

Theorem 64.3 (Absolute Puiseux non-Polishability). *The Puiseux field $\mathcal{P}_k$ over a countable ordered field has no admissible Polish topology. In its canonical standard Borel presentation, even without mentioning the order, no compatible Polish additive-group topology can make all fixed scalar maps Borel.*

Proof. The field is uncountable, since $k((t))$ already contains all binary power series. For an arbitrary admissible topology, Lemma 64.2 makes the proper subfields K_n closed, and Theorem 64.1 applies. For the canonical Borel structure, K_n is defined by the vanishing of all coefficients outside $(1/n!)\mathbb{Z}$, hence is Borel. The same theorem applies directly, without an order hypothesis. $\square$

The ordinary valuation topology on $\mathcal{P}_k$ is separable and metrizable, but is not complete. More strongly, it is meager in itself: each closed stage K_n has empty interior. Given $f \in K_n$ and a valuation ball about f , add a monomial t^q of sufficiently large rational exponent with $q \notin (1/n!)\mathbb{Z}$. The resulting point is in the ball but outside K_n . Thus merely exhibiting an incomplete metric is not the reason for Theorem 64.3; the theorem rules out every admissible replacement.

65 The Levi-Civita completion: a positive and unique presentation

The classical Levi-Civita field is often defined with real coefficients. Here the coefficient field is deliberately *countable*. This changes separability decisively. Our proofs work for every countable ordered k ; the algebraic and completeness arguments also work without an order on k .

Definition 65.1 (Left-finite support). A set $S \subseteq \mathbb{Q}$ is *left-finite* if $S \cap (-\infty, B]$ is finite for every rational B . The field candidate $\mathcal{L}_k$ consists of all functions $f : \mathbb{Q} \rightarrow k$ with left-finite support, written $\sum f_q t^q$.

A nonempty left-finite set has a least element. Indeed, if $b \in S$, the finite nonempty set $S \cap (-\infty, b]$ has a least element, which is least in S . A left-finite support is either finite or an increasing sequence tending to $+\infty$. In particular, it is a legal Hahn support.

Proposition 65.2 (Field operations). *The usual coefficientwise addition and convolution multiplication turn $\mathcal{L}_k$ into a subfield of $k((t^{\mathbb{Q}}))$.*

Proof. A finite union of left-finite supports is left-finite, so addition is defined and preserves the support condition. Let S, T be nonempty left-finite sets with minima s_0, t_0 . If $s + t \leq B$, then $s \leq B - t_0$ and $t \leq B - s_0$. There are only finitely many such pairs. Thus convolution coefficients are finite sums, and the product has left-finite support. The coefficient at the least exponent of a nonzero product is the product of the two leading coefficients, so the product is nonzero.

Write a nonzero series in the form

$$f = at^q(1 + h), \quad a \in k^\times, \quad q \in \mathbb{Q}, \quad h = 0 \text{ or } v(h) = \delta > 0.$$

For $h \neq 0$, define

$$g = \sum_{j \geq 0} (-h)^j.$$

At exponents at most B , only indices j with $j\delta \leq B$ can contribute, apart from the constant term $j = 0$. There are finitely many such indices, and each corresponding power has finite support below B . Therefore g is left-finite and all coefficient manipulations are finite below any prescribed cutoff. The geometric identity gives $(1 + h)g = 1$. Hence $f^{-1} = a^{-1}t^{-q}g \in \mathcal{L}_k$. The case $h = 0$ is immediate. $\square$

Theorem 65.3 (A complete separable field topology). *For countable ordered k , the metric*

$$d(f, g) = \min\{1, 2^{-v(f-g)}\}, \quad 2^{-\infty} = 0, \quad (65.1)$$

makes $\mathcal{L}_k$ an uncountable Polish topological field. Its topology equals the order topology and is the unique admissible Polish topology on $\mathcal{L}_k$.

Proof. The valuation inequality gives the ultrametric inequality for d . Let (f_j) be Cauchy. For each integer $N \geq 0$ there is an index j_N such that all f_j , $j \geq j_N$, have exactly the same coefficients at every exponent at most N . Choose the j_N increasing. These stabilized finite truncations are compatible, and their union defines a coefficient function f . For every bound B choose an integer $N \geq \max\{0, B\}$. The support of f below B is contained in the finite stabilized support below N . Thus $f \in \mathcal{L}_k$. Stabilization also shows $f_j \rightarrow f$.

Finite rational-exponent polynomials over k form a countable set. Truncating any left-finite series above successively larger integers proves density. The binary series $\sum_{n \geq 0} b(n)t^n$ show uncountability.

Addition, multiplication, and inversion are continuous by the valuation identities used in Theorem 63.3. For completeness, continuity of multiplication at (f, g) follows from

$$(f + h)(g + u) - fg = fu + gh + hu$$

and the lower bound on the valuation of a sum. If $v(h) > v(f)$ and $f \neq 0$, then $v(f + h) = v(f)$, and the inverse-difference identity proves continuity of inversion there.

For $\varepsilon > 0$, every series of valuation strictly greater than $v(\varepsilon)$ has absolute ordered value less than ε . Conversely, an interval of radius t^{B+1} lies inside the ball $\{x : v(x) > B\}$. Translation gives equality of the two topologies. The positive cone is open, so the topology is admissible. Uniqueness follows from Corollary 61.5. $\square$

Proposition 65.4 (Completion and strict inclusions). *The valued-field completion of $\mathcal{P}_k$, and also of the rational-function field $k(t^{\mathbb{Q}})$, is $\mathcal{L}_k$. Moreover,*

$$k(t^{\mathbb{Q}}) \subseteq \mathcal{P}_k \subsetneq \mathcal{L}_k \subsetneq \mathcal{H}_k.$$

Proof. Each Puiseux support lies on a discrete rational lattice and is bounded below, so is left-finite. Both $\mathcal{P}_k$ and $k(t^{\mathbb{Q}})$ contain every finite rational-exponent polynomial. These polynomials are dense in the complete field $\mathcal{L}_k$, proving the completion assertion.

For strictness of the first displayed strict inclusion, use

$$x = \sum_{n \geq 2} t^{n+1/n}. \quad (65.2)$$

Its exponents are increasing and unbounded, so $x \in \mathcal{L}_k$. Their reduced denominators are exactly n , since $\gcd(n^2 + 1, n) = 1$. They cannot all divide one fixed factorial, so $x \notin \mathcal{P}_k$. For the other strict inclusion use

$$y = \sum_{n \geq 2} t^{1-1/n}. \quad (65.3)$$

The support is increasing of order type ω , hence well ordered, but is infinite below 1. Thus $y \in \mathcal{H}_k \setminus \mathcal{L}_k$. $\square$

The partial sums of (65.2) are valuation-Cauchy. The partial sums of (65.3) are not: their successive differences have valuations bounded above by 1. A Hahn-summable family need not be a convergent sequence of partial sums in the valuation topology. This is the precise support distinction behind the completion boundary.

Proposition 65.5 (The topology is not locally compact). *No point of $\mathcal{L}_k$ has a compact neighborhood.*

Proof. Every neighborhood of 0 contains a valuation ball $V_B = \{f : v(f) > B\}$. Fix $q > B$ and infinitely many distinct coefficients $c_j \in k$. Then $c_j t^q \in V_B$, and the mutual distances are the same positive number $\min\{1, 2^{-q}\}$. Thus no set containing V_B is totally bounded, and it cannot be compact. Translation deals with other points. $\square$

Remark 65.6 (Why real coefficients do not work). The usual real-coefficient Levi-Civita field is complete in its valuation topology but is not separable in that topology. More strongly, it has no admissible Polish topology at all: the constant-indexed disjoint intervals from Lemma 63.2 lie inside this smaller field as well. The positive construction is therefore not a claim that the conventional real-coefficient field is Polish.

[write] (Batch 93, 4 October 2026.) Part XVII adds two facts about the real-coefficient field, for every countable $\Gamma \neq \{0\}$: its additive group with the coefficient Borel structure is not Polishable (proposition 229.1, source 23), and the argument of this remark extends from $\Gamma = \mathbb{Q}$ to every such Γ and gives a second proof of that statement (proposition 239.1 [merge]). Source 23 also classifies its Borel additive maps and Borel derivations.

66 A prefix criterion for separability

The Hahn and Levi-Civita results can be organized without referring to a particular coding of supports. Let F be any subfield of $k((t^{\mathbb{Q}}))$ that contains k and all rational monomials. For $B \in \mathbb{Q}$, let

$$\pi_B(f) = (f_q)_{q \leq B}, \quad V_B = \{f \in F : v(f) > B\}.$$

The prefix $\pi_B(f)$ may be infinite. It need not itself represent an element of F .

Theorem 66.1 (Countable prefixes characterize order separability). *The following conditions are equivalent:*

- (i) *the valuation topology of F is separable;*
- (ii) *the order topology of F is second countable;*
- (iii) *for every integer $N \geq 0$, the set $\pi_N(F) = \{\pi_N(f) : f \in F\}$ is countable.*

Every admissible Polish topology on F forces these conditions.

Proof. The order and valuation topologies agree by the monomial comparison in Theorem 65.3. The prefix classes at cutoff N are exactly the open cosets of V_N . A dense countable set meets every such coset, proving (i) $\Rightarrow$ (iii).

Conversely, choose one representative of each prefix class for every integer $N \geq 0$. The union D of these countably many countable sets is dense: every valuation neighborhood contains a coset at an integer cutoff and hence one of the chosen representatives. The cosets themselves give a countable base. This proves (iii) $\Rightarrow$ (i),(ii), and (ii) $\Rightarrow$ (i) is immediate.

Finally, in an admissible topology each V_N is open by Theorem 61.4. Its pairwise disjoint cosets are therefore countable in number. This gives (iii) even when the candidate Polish topology is strictly finer than the valuation topology. $\square$

For $\mathcal{L}_k$, each prefix is a finite rational-exponent polynomial, so there are only countably many. The same holds for $\mathcal{P}_k$. For $\mathcal{H}_k$, a binary family supported below a fixed bound gives continuum many different prefixes. Thus the prefix criterion captures the Hahn obstruction but cannot by itself capture the Puiseux obstruction: separability and complete metrizability remain distinct questions.

A useful sufficient condition follows. If F has countably many prefixes at each cutoff and is complete for the valuation metric, then its order topology is Polish and admissible; it is then the unique admissible topology. Whether all admissible intermediate-field presentations must arise this way is left as a sharply stated further question in Section 73.

67 Exact descriptive complexity of the three code domains

In this section k is a countable ordered field with the discrete topology, and all domains are regarded as subsets of the Polish product $k^{\mathbb{Q}}$. The phrase “complete” refers to continuous reductions, with zero-dimensional Polish source spaces. This is a complexity statement about coefficient codes; it is not a statement that their subspace topology is the desired group topology.

Theorem 67.1 (Puiseux codes). *The coefficient-code domain $\mathcal{P}_k \subseteq k^{\mathbb{Q}}$ is Σ_2^0 -complete.*

Proof. For positive integers n and nonnegative integers M , the set of all coefficient functions supported on

$$(1/n!)\mathbb{Z} \cap [-M, \infty)$$

is closed in $k^{\mathbb{Q}}$. Every such support defines a Puiseux series, and every Puiseux series occurs for some n, M . Hence $\mathcal{P}_k$ is F_σ .

Let $\text{FIN} \subseteq 2^{\mathbb{N}}$ be the set of binary sequences with finite support, a standard Σ_2^0 -complete set. Define a continuous map by placing the bit $b(m)$ at exponent $1 - 1/(m + 2)$ and placing zero at every other exponent. Finite support gives a polynomial. Infinite support gives infinitely many nonzero exponents in the bounded interval $[1/2, 1)$, which cannot lie in one discrete rational lattice. The map reduces FIN to $\mathcal{P}_k$. $\square$

Theorem 67.2 (Levi-Civita codes). *The coefficient-code domain $\mathcal{L}_k \subseteq k^{\mathbb{Q}}$ is Π_3^0 -complete.*

Proof. At a fixed integer cutoff $N \geq 0$, having finite support below N is F_σ : take the union over finite $S \subseteq \mathbb{Q} \cap (-\infty, N]$ of the closed condition that all coefficients below N outside S vanish. Intersect over N to obtain a Π_3^0 definition.

The set

$$\text{FIN}^{\mathbb{N}} = \{b \in 2^{\mathbb{N} \times \mathbb{N}} : \text{each row } b(n, \cdot) \text{ has finite support}\}$$

is Π_3^0 -complete. Place the coefficient $b(n, m)$ at

$$q(n, m) = n + 1 - \frac{1}{m + 2}, \quad (67.1)$$

and zero elsewhere. These rational exponents are pairwise distinct. If one row is infinite, its support is infinite below $n + 1$, so the resulting code is not left-finite. If every row is finite, only finitely many rows can contribute below any prescribed bound, and each contributes finitely many terms. The map is continuous and gives the required reduction. $\square$

[write] These are the Borel levels of source 03's [theorem 19.1](#) ($\text{WO}(\mathbb{Z})$ is Σ_2^0 -complete, $\text{WO}(\mathbb{Z}_{\text{lex}}^d)$ is Π_3^0 -complete for $d \geq 2$), attained here by different sets.

For completeness, the usual hardness inputs can be obtained directly. Given an F_σ set $A = \bigcup_n C_n$ in Baire space, take the C_n closed and increasing. At stage s , let $m_s(x)$ be the least $n \leq s$ whose closed-set tree is compatible with the length- s prefix of x , or $s + 1$ if there is none. The values m_s are nondecreasing. Output a 1 exactly when the value increases. This is a continuous map into $2^\mathbb{N}$; it has finitely many 1s exactly when $x \in A$. Coordinatewise application to a countable intersection of F_σ sets proves the completeness of $\text{FIN}^\mathbb{N}$.

Theorem 67.3 (Full rational Hahn codes; classical support obstruction). *The coefficient-code domain $\mathcal{H}_k \subseteq k^\mathbb{Q}$ is Π_1^1 -complete. Consequently, it has no standard Borel parametrization with Borel coefficient observations and exactly one code for each series.*

Proof. A support fails to be well ordered exactly when there is an infinite strictly decreasing sequence of rational exponents with nonzero coefficients. Existence of that sequence is an analytic condition. Thus $\mathcal{H}_k$ is coanalytic.

For hardness use the standard Π_1^1 -complete set WF of well-founded trees on $\mathbb{N}$ [8]. Give $\mathbb{N}^{<\mathbb{N}}$ the Kleene–Brouwer order: proper extensions precede their prefixes, and incompatible sequences are compared at their first differing coordinate. This is a countable linear order, so it embeds into $\mathbb{Q} \cap (0, 1)$. Fix such an embedding e . For a tree T , put coefficient 1 at $e(s)$ exactly when $s \in T$, and put zero elsewhere. The map is continuous. An infinite branch gives a descending chain in the Kleene–Brouwer order. Conversely, well-founded induction on a well-founded tree shows that the order below each node is a countable ordered sum of the child orders, followed by the node itself; it is therefore a well-order. Hence $T \in \text{WF}$ exactly when the resulting support is well ordered.

If a standard Borel space parametrized all Hahn series uniquely and every coefficient were Borel, its coefficient map into $k^\mathbb{Q}$ would be a Borel injection with range exactly $\mathcal{H}_k$. The Lusin–Souslin theorem would make that range Borel, contradicting completeness. $\square$

[write] [Theorem 67.3](#) is a second proof of the non-scattered half of source 03's [theorems 18.4](#) and [20.2](#) (via [lemma 18.2](#)) and of the collection's lightface results `onot:neg:lem:kbtree` and `onot:neg:thm:pionesone`; source 09 credits “the preceding project report” and claims no priority for it.

[write] (Batch 96.) For $k = \mathbb{R}$ it is also the case $\Gamma = \mathbb{Q}$ of source 25's [theorem 260.1](#).

The last theorem is included to complete the comparison, not to reclaim the project's earlier coanalytic support result [44]. It is also logically weaker than the absolute topological obstruction for a different reason: it requires coefficient-observable codes, whereas [Theorem 63.3](#) does not.

Proposition 67.4 (The Polish Levi-Civita topology preserves the Borel codes). *The valuation topology on $\mathcal{L}_k$ has the same Borel sets as the subspace Borel structure inherited from $k^\mathbb{Q}$.*

Proof. Each coefficient map is continuous for the valuation topology: fixing the entire prefix up to its exponent fixes that coefficient. Thus coefficient Borel sets are valuation-Borel. In the other direction, take the countable base consisting of balls centered at finite rational-exponent polynomials with nonnegative integer cutoffs. A basic ball is defined by countably many coefficient equalities. It is therefore Borel in the coefficient structure. Every valuation-open set is a countable union of these basic balls, proving the converse inclusion. $\square$

Both $\mathcal{P}_k$ and $\mathcal{L}_k$ carry Borel field operations in their coefficient structures. For $\mathcal{L}_k$ this follows from Proposition 67.4 and continuous valuation arithmetic. On $\mathcal{P}_k$ the same Borel equality holds for its countable valuation base; its field operations are valuation-continuous. This does not imply that $\mathcal{P}_k$ is completely metrizable as a group.

Field	Exact coefficient-code complexity	Admissible Polish arithmetic
$\mathcal{P}_k$	Σ_2^0 -complete	None; proper-subfield cover obstruction
$\mathcal{L}_k$	Π_3^0 -complete	Exists and is unique; valuation completion
$\mathcal{H}_k$	Π_1^1 -complete	None; uncountably many separated prefixes

Table 2: Countable ordered coefficients and rational exponents throughout. A low Borel rank does not predict Polishability.

68 Real closedness and completion without a defect argument

To connect these fields to integer-part arithmetic, we specialize to a real closed coefficient field. A useful proof avoids importing a full theory of henselian defect. The sole substantial algebraic input is the classical Newton–Puiseux theorem in characteristic zero: for an algebraically closed field K of characteristic zero, $\bigcup_n K((t^{1/n!}))$ is algebraically closed. Poonen gives this as Corollary 7 [30].

Lemma 68.1 (Completion preserves algebraic closedness). *The metric completion of a non-Archimedean valued algebraically closed field is algebraically closed.*

Proof. Write $|\cdot|$ for the multiplicative real-valued absolute value; it is not the truncated metric. Let K be the field and $\widehat{K}$ its completion. Take a monic polynomial

$$P(X) = X^d + \sum_{i < d} a_i X^i \in \widehat{K}[X], \quad d \geq 1.$$

Choose monic polynomials $P_n \in K[X]$ of degree d whose coefficients converge to those of P , and positive real numbers $\varepsilon_n \rightarrow 0$ such that every coefficient of $P_{n+1} - P_n$ has absolute value at most ε_n . There is a common real bound $M \geq 1$ for 1 and for the absolute values of all coefficients of all P_n . Every root z of any P_n satisfies $|z| \leq M$: otherwise the monic term would strictly dominate every lower-degree term.

Choose a root z_0 of P_0 . Having chosen z_n , factor $P_{n+1}(X) = \prod_{j=1}^d (X - w_j)$ over K . Since $P_n(z_n) = 0$,

$$\prod_{j=1}^d |z_n - w_j| = |P_{n+1}(z_n)| \leq \varepsilon_n M^{d-1}.$$

Some root w_j therefore satisfies $|z_n - w_j| \leq (\varepsilon_n M^{d-1})^{1/d}$. Choose it as z_{n+1} . Consecutive distances tend to zero, and the ultrametric inequality then makes (z_n) Cauchy: the distance between any two late terms is at most the largest intervening consecutive distance. Let $z \in \widehat{K}$ be its limit. Continuity of polynomial evaluation and coefficient convergence give $P(z) = 0$. Thus every nonconstant polynomial has a root, which is equivalent to algebraic closedness. $\square$

Theorem 68.2 (Real closed Puiseux and Levi-Civita fields). *If k is a real closed ordered field, then $\mathcal{P}_k$ and $\mathcal{L}_k$ are real closed. In particular, this applies to the countable field $k = \mathbb{R}_{\text{alg}}$.*

Proof. Let $K = k(i)$, which is algebraically closed because k is real closed. Splitting coefficients into their real and imaginary parts gives

$$\mathcal{P}_K = \mathcal{P}_k(i), \quad \mathcal{L}_K = \mathcal{L}_k(i).$$

For the first identity, finitely many denominator bounds can be replaced by one common bound; for the second, a finite union of left-finite supports is left-finite. By Newton–Puiseux, $\mathcal{P}_K$ is algebraically closed. By Proposition 65.4 and Lemma 68.1, so is $\mathcal{L}_K$. The fields $\mathcal{P}_k$ and $\mathcal{L}_k$ are ordered and adjoining i makes them algebraically closed; the standard characterization of real closed fields therefore proves the assertion. $\square$

Corollary 68.3 (A ring-language separation). *For $k = \mathbb{R}_{\text{alg}}$, the field $\mathcal{P}_k$ has no Polish additive-group topology with jointly Borel multiplication, even without a separately specified measurable order. The field $\mathcal{L}_k$ does have such a topology, and it is unique.*

Proof. Real closedness makes the order Borel by Corollary 61.6. Apply Theorems 64.3 and 65.3. $\square$

Real closedness here is a classical algebraic property proved using a classical input; it is not the proposed novelty. Its role is to remove the order-measurability hypothesis from the final field-language contrast.

69 Local Polish hulls, unbounded support ranks, and an exact cover number

The failure of Polishability for the full Hahn field is global, not an obstruction to accommodating any prescribed countable collection of its elements. This section proves that much stronger local positive statement. In particular, the Levi-Civita field is a minimal completion, not a maximal Polish subfield of the Hahn field.

Lemma 69.1 (Metric completeness of the rational Hahn field). *For every field k , $k((t^{\mathbb{Q}}))$ is complete for the metric $d(f, g) = \min\{1, 2^{-v(f-g)}\}$.*

Proof. For a Cauchy sequence, every prefix up to an integer $N \geq 0$ eventually stabilizes, just as in Theorem 65.3. This time a stabilized prefix need not have finite support, but it has well-ordered support because it is a prefix of a Hahn series. The compatible stabilized prefixes define a coefficient function f . Its support is well ordered: given any nonempty subset S of that support, choose $s \in S$ and an integer $N \geq \max\{0, s\}$. The nonempty set $S \cap (-\infty, N]$ has a least element since it lies in the stabilized well-ordered prefix. That element is also least in S , as all remaining points exceed N . Thus f is a Hahn series, and stabilization shows that it is the limit of the original sequence. $\square$

Theorem 69.2 (Countable-parameter Polish hulls). *Let k be countable and ordered, $H = k((t^{\mathbb{Q}}))$, and $A \subseteq H$ countable. Then*

$$F_A = \overline{k(t^{\mathbb{Q}})(A)}^v \subseteq H \tag{69.1}$$

is a complete separable ordered subfield containing $\mathcal{L}_k \cup A$. Its valuation topology is its unique admissible Polish topology. It is the smallest valuation-closed subfield of H containing $k(t^{\mathbb{Q}}) \cup A$. Every valuation-complete separable intermediate field $\mathcal{L}_k \subseteq F \subseteq H$ arises in this way.

Proof. The algebraic field $k(t^{\mathbb{Q}})(A)$ is countable. Its closure is separable and is complete by Lemma 69.1. Closure preserves addition and multiplication by continuity, and preserves inversion at every nonzero limit by the inverse-difference identity. Thus F_A is a subfield. The closure of $k(t^{\mathbb{Q}})$ in H is exactly $\mathcal{L}_k$: it is dense in $\mathcal{L}_k$, and $\mathcal{L}_k$ is complete and hence closed in the metric space H . Therefore $\mathcal{L}_k \subseteq F_A$. The valuation and order topologies agree, and Corollary 61.5 gives uniqueness. Minimality follows directly from taking closure. Finally, for a complete separable intermediate field F , choose a countable valuation-dense subset $A \subseteq F$. Completeness makes F closed in H , and (69.1) then gives $F_A = F$. $\square$

Taking $A = \{y\}$ for the bounded infinite support in (65.3) immediately gives

$$\mathcal{L}_k \subsetneq F_{\{y\}} \subsetneq H.$$

The second strict inclusion follows from Theorem 63.3. Thus bounded infinite supports do not individually prevent Polish field arithmetic. What is impossible is allowing all independently varying bounded prefixes at once.

Theorem 69.3 (A support bound for every separable family). *Let $X \subseteq H = k((t^{\mathbb{Q}}))$ be valuation-separable, and let $D \subseteq X$ be countable and dense. Then*

$$\text{ot}(\text{supp } f) \leq \beta_D := \sup_{d \in D} \text{ot}(\text{supp } d) < \omega_1 \quad (f \in X). \quad (69.2)$$

Here the empty support has order type 0. In particular, every admissibly Polish intermediate field has a uniform countable support-order bound. There is no uniform countable bound valid for all Polish hulls F_A .

Proof. For $f \in X$ and every integer $N \geq 0$, density supplies a $d_N \in D$ with $v(f - d_N) > N$. Their prefixes through N agree. Hence

$$\text{ot}(\text{supp } f \cap (-\infty, N]) \leq \text{ot}(\text{supp } d_N) \leq \beta_D.$$

These prefixes are increasing initial segments whose union is the support of f , so their order types have supremum $\text{ot}(\text{supp } f)$. This proves (69.2). The supremum defining β_D is countable because D and all rational supports are countable.

For an admissible topology, a countable dense set remains valuation-dense by Theorem 61.4, so the preceding argument applies. For the last assertion, let α be any countable ordinal. Embed its linear order into $\mathbb{Q} \cap (0, 1)$ and let h_α have coefficient 1 on that support and 0 elsewhere. It belongs to the Polish hull $F_{\{h_\alpha\}}$ and has support order type α . $\square$

[write] Source 03's theorem 21.2 proves a uniform countable support bound for every Borel coefficient-observable family by analytic boundedness; theorem 69.3 is the elementary counterpart for separable families, as source 09 says below.

This bound uses separability and prefix stabilization, not analytic boundedness or a Borel-domain hypothesis. It is a specialized elementary counterpart of the support bounds in the preceding Borel-support manuscript. It also makes the failure of maximality explicit.

Corollary 69.4 (No maximal admissibly Polish intermediate field). *Every admissibly Polish field F with $k(t^{\mathbb{Q}}) \subseteq F \subseteq H$ is properly contained in another such field. The larger field can have an element whose support order type exceeds all support order types in F .*

Proof. Choose a countable dense set D in the admissible topology of F and let β_D be as in Theorem 69.3. Choose a Hahn series h with support order type $\beta_D + 1$. The hull $F_{D \cup \{h\}}$ contains the valuation closure of D , and hence contains F . It also contains $h \notin F$. Theorem 69.2 proves that it is admissibly Polish. $\square$

69.1 Real-closed, truncation-closed hulls

With $k = \mathbb{R}_{\text{alg}}$, a hull can be required to preserve both real closedness and all rational truncations. Full Hahn real closedness follows from the classical algebraic-closedness theorem for Hahn fields with algebraically closed coefficients and divisible exponent group [30, Corollary 4], applied after adjoining i .

Theorem 69.5 (A strengthened hull with a countable truncation core). *Let $H = \mathbb{R}_{\text{alg}}((t^{\mathbb{Q}}))$ and let $A \subseteq H$ be countable. There exist a countable real closed truncation-closed subfield $B \subseteq H$ and a field $C = \overline{B}^v$ such that*

$$k(t^{\mathbb{Q}}) \cup A \subseteq B \subseteq C,$$

C is real closed and admissibly Polish, and every rational prefix of every $f \in C$ belongs to B :

$$T_{\leq q}(C) \subseteq B \quad (q \in \mathbb{Q}). \quad (69.3)$$

In particular, C is truncation-closed and contains $\mathcal{L}_k \cup A$.

Proof. Start with the countable field $B_0 = k(t^{\mathbb{Q}})(A)$. Given B_n , adjoin $T_{\leq q}b$ for all $b \in B_n$ and all $q \in \mathbb{Q}$, then take its real closure inside the real closed field H to obtain B_{n+1} . Each step is countable: there are countably many truncations and countably many elements algebraic over a countable field. The union $B = \bigcup_n B_n$ is a countable field, is truncation-closed, and is real closed. For the last assertion, any finite list of polynomial coefficients and any positive element occur at one stage, and the requisite odd-degree roots and positive square roots occur at the next stage.

The field $C = \overline{B}^v$ is complete and separable, hence admissibly Polish by the same argument as Theorem 69.2. For $f \in C$ and $q \in \mathbb{Q}$, choose $b \in B$ with $v(f - b) > q$. Then $T_{\leq q}f = T_{\leq q}b \in B$, proving (69.3). To prove real closedness, regard $B(i)$ inside $H(i)$. The coefficientwise valuation satisfies

$$v(a + ib) = \min\{v(a), v(b)\} \quad (a, b \in H).$$

Consequently its completion is $C(i)$. Since B is real closed, $B(i)$ is algebraically closed; Lemma 68.1 makes $C(i)$ algebraically closed. As C is ordered, it is real closed. $\square$

69.2 An exact global invariant

Let $\text{pcov}_k(H)$ be the least cardinality of a family of admissibly Polish intermediate subfields $k(t^{\mathbb{Q}}) \subseteq F \subseteq H$ whose union is H . The topology on a covering field need not initially be the inherited valuation topology.

Theorem 69.6 (The Polish field-cover number). *For every countable ordered coefficient field k ,*

$$\text{pcov}_k(k((t^{\mathbb{Q}}))) = 2^{\aleph_0}.$$

For $k = \mathbb{R}_{\text{alg}}$, the same minimum holds even if all covering fields are required to be real closed, truncation-closed, and Polish in their valuation topology.

Proof. For the upper bound, $|H| = 2^{\aleph_0}$, and the fields $F_{\{h\}}$ from Theorem 69.2, one for each $h \in H$, cover H . For real algebraic coefficients use the stronger hulls from Theorem 69.5 instead.

For the lower bound, use the continuum-sized binary family

$$h_b = \sum_{n \geq 0} b(n)t^{1-1/(n+2)}, \quad b \in 2^{\mathbb{N}}.$$

Distinct centers differ in absolute ordered value by more than $2t$. In any one admissibly Polish intermediate field, the intervals of radius t about the centers it contains are nonempty, open, and pairwise disjoint. It therefore contains only countably many members of this binary family. A union of $\kappa < 2^{\aleph_0}$ such fields contains at most $\kappa \cdot \aleph_0 < 2^{\aleph_0}$ members of the family, and cannot cover H . This cardinal calculation is in ZFC and uses no continuum hypothesis. $\square$

The local and global statements now fit together exactly. Every countable collection of Hahn data is contained in one well-behaved Polish field; no such field is maximal; nevertheless continuum many are necessary to cover all Hahn data. These conclusions are not consequences of the mere cardinality $|H| = 2^{\aleph_0}$: the lower bound uses the fixed-scale separation forced by arithmetic and order.

70 A countable omnific integer part and its exact floor regularity

Throughout this section $k = \mathbb{R}_{\text{alg}} \subseteq \mathbb{R}$. Define

$$\mathcal{I}_k = \mathbb{Z} \oplus \bigoplus_{q \in \mathbb{Q}_{<0}} kt^q. \quad (70.1)$$

Thus an element has finitely many negative-exponent terms and an ordinary integer constant term. The direct sum is literal finite support; replacing it by arbitrary Hahn support would change the ring.

Theorem 70.1 (A common integer part). *The ring $\mathcal{I}_k$ is countable and discretely ordered. It is an integer part of both $\mathcal{P}_k$ and $\mathcal{L}_k$. If*

$$f = p + c + u,$$

where p is the negative principal part, $c = f_0 \in k$, and u has only positive exponents, then

$$\lfloor f \rfloor_{\mathcal{I}_k} = p + \lfloor c \rfloor - \begin{cases} 1, & c \in \mathbb{Z} \text{ and } u < 0, \\ 0, & \text{otherwise.} \end{cases} \quad (70.2)$$

Proof. Finite rational supports with coefficients in a countable field form a countable set. Addition and multiplication preserve finite negative support and integer constant terms, so $\mathcal{I}_k$ is a countable unital subring. If a nonzero element has a negative leading exponent, its absolute ordered value exceeds every positive ordinary integer. Otherwise it is a nonzero ordinary integer. Hence 1 is its least positive element.

For $f \in \mathcal{L}_k$, left-finiteness makes p finite, and every nonzero u is smaller in absolute ordered value than every positive constant of k . If $c \notin \mathbb{Z}$, both $c - \lfloor c \rfloor$ and $\lfloor c \rfloor + 1 - c$ are positive constants; the infinitesimal u cannot change either inequality. If $c \in \mathbb{Z}$ and $u \geq 0$, the floor is $p + c$. If $c \in \mathbb{Z}$ and $u < 0$, the floor is $p + c - 1$. In every case the stated element a satisfies $a \leq f < a + 1$. Uniqueness follows because distinct elements of $\mathcal{I}_k$ differ by at least 1 in absolute ordered value. Since $\mathcal{P}_k \subseteq \mathcal{L}_k$ and contains $\mathcal{I}_k$, the same result holds in $\mathcal{P}_k$. $\square$

[write] With $t^q \mapsto \omega^{-q}$, formula (70.2) is the collection's floor rule `odg:thm:floor`, proved in Lean for every surreal, and source 03's [lemma 23.1](#). For $k = \mathbb{R}_{\text{alg}}$, the ring $\mathcal{I}_k$ is source 03's finite principal-part ring $\mathcal{A}_{\text{fin}}$ with coefficients restricted to $\mathbb{R}_{\text{alg}}$ (with $X = t^{-1}$), a countable subring of it; it is also the Shepherdson ring used by batch 86's manuscript 10 in Algebra/SurrealNumbers/docs/surreal/discrete-initial-subgroups-and-omnific-normalization.

Formula (70.2), particularly its correction for a negative infinitesimal at an integer constant, is a specialization of the omnific normal-form floor rule already documented in ProveIt [42]. It is not claimed as a new floor formula. The following topological consequences are the point of its use here.

Theorem 70.2 (Exact discontinuity set and Baire class). *Give $\mathcal{L}_k$ its unique admissible Polish topology and $\mathcal{I}_k$ the discrete topology. The floor map $\lfloor \cdot \rfloor_{\mathcal{I}_k} : \mathcal{L}_k \rightarrow \mathcal{I}_k$ is Baire class one. Its set of discontinuities is exactly $\mathcal{I}_k$, a countable closed nowhere-dense subset of $\mathcal{L}_k$.*

Proof. Its fibers are the half-open intervals $[a, a + 1)$ for $a \in \mathcal{I}_k$. If $f \notin \mathcal{I}_k$ and $a = \lfloor f \rfloor_{\mathcal{I}_k}$, then $a < f < a + 1$, so a sufficiently small order neighborhood of f lies in the same fiber. The floor is continuous there. At $a \in \mathcal{I}_k$, the sequence $a - t^n$ converges to a , while

$$\lfloor a - t^n \rfloor_{\mathcal{I}_k} = a - 1 \neq a = \lfloor a \rfloor_{\mathcal{I}_k}.$$

Thus the discontinuity set is exactly $\mathcal{I}_k$.

For $n \geq 0$ let $T_n f = \sum_{q \leq n} f_q t^q$, a finite polynomial. The map T_n into its discrete, countable range is locally constant: $v(f - g) > n$ implies $T_n f = T_n g$. Consequently

$$F_n(f) = \lfloor T_n f \rfloor_{\mathcal{I}_k}$$

is a continuous map $\mathcal{L}_k \rightarrow \mathcal{I}_k$. These maps converge pointwise to the floor, and indeed are eventually equal to it at each input. The negative principal part and constant are already fixed for all $n \geq 0$. When the constant is an integer and the positive tail is nonzero, the sign needed for the correction is determined by the first positive exponent, which is included once n is large enough. In all other cases no such correction can change. This proves the Baire class one assertion.

Closed discreteness follows from Theorem 62.3. Its interior is empty because every neighborhood of $a \in \mathcal{I}_k$ contains $a + t^n \notin \mathcal{I}_k$ for large n . Hence it is nowhere dense. $\square$

Example 70.3 (Boundary tests). The formulas include

$$\lfloor -t \rfloor_{\mathcal{I}_k} = -1, \quad \left\lfloor 7 - t^{1/3} \right\rfloor_{\mathcal{I}_k} = 6, \quad \left\lfloor 7 + t^{1/3} \right\rfloor_{\mathcal{I}_k} = 7,$$

and

$$\left\lfloor 3t^{-2} - 5t^{-1/2} + \frac{3}{2} - t^{10} \right\rfloor_{\mathcal{I}_k} = 3t^{-2} - 5t^{-1/2} + 1.$$

A finite-precision procedure that ignores a small negative tail at an integer constant is wrong exactly at the kind of boundary detected by the theorem. The continuous approximants above permit eventual correction; they do not provide an input-uniform stage at which correction is finished.

70.1 Floors in the larger Polish hulls

The strengthened hulls of Theorem 69.5 support the same regularity without restricting their negative principal parts to finite support.

Theorem 70.4 (Canonical integer parts in real closed Polish hulls). *Let $B \subseteq C \subseteq \mathbb{R}_{\text{alg}}((t^{\mathbb{Q}}))$ be as in Theorem 69.5. Then*

$$J_C = \{f \in C : \text{supp } f \subseteq \mathbb{Q}_{\leq 0}, f_0 \in \mathbb{Z}\}$$

is a countable closed discrete integer part of C , and $J_C \subseteq B$. Its floor is Baire class one as a map into discrete J_C , with discontinuity set exactly J_C . Under the surreal normal-form embedding it is a subring of the omnific integers.

Proof. For $f \in C$, the principal part $p = T_{\leq 0}f - f_0$ belongs to B by (69.3). Thus the formula

$$\lfloor f \rfloor_{J_C} = p + \lfloor f_0 \rfloor - \mathbf{1}_{\{f_0 \in \mathbb{Z}, f - T_{\leq 0}f < 0\}}$$

defines an element of C with no positive exponents and integer constant. The same leading-term comparisons as in Theorem 70.1 prove the integer-part inequalities and discrete order. Ring closure follows from convolution: negative exponents cannot contribute to the constant term. Every $j \in J_C$ equals $T_{\leq 0}j$, so lies in the countable set B . Closed discreteness also follows from Theorem 62.3.

For each nonnegative integer n , the map $T_{\leq n} : C \rightarrow B$, with B given the discrete topology, is locally constant. The maps $f \mapsto \lfloor T_{\leq n}f \rfloor_{J_C}$ are therefore continuous and are eventually equal to the floor at each input. Only the first nonzero positive term is needed to decide the possible correction. The interval fibers and the sequence $a - t^n \rightarrow a$ give the exact discontinuity set as before. Finally, replacing t^q by ω^{-q} turns the nonpositive support and integer constant conditions into the omnific normal-form conditions. $\square$

Remark 70.5 (The target topology matters). The Baire class one assertion is about a *discrete* integer-part target, not just its inclusion in the field. It is stronger than merely producing continuous real-valued approximants to the ordinary floor on $\mathbb{R}$. Indeed, the ordinary floor $\mathbb{R} \rightarrow \mathbb{Z}_{\text{discrete}}$ is not Baire class one: every continuous map from connected $\mathbb{R}$ to discrete $\mathbb{Z}$ is constant, and a pointwise convergent sequence of constant maps has a constant limit. The locally constant prefix maps are a genuine non-Archimedean resource.

70.2 The common integer part satisfies open induction

The classical relationship between integer parts of real closed fields and open induction is often called the integer-part characterization of open induction. For our specific ring the needed implication has a short proof.

Proposition 70.6 (Open induction). *The nonnegative cone $\mathcal{I}_{k, \geq 0}$ satisfies $Q + \text{IOpen}$ in the language of ordered arithmetic, with parameters allowed in the induction scheme.*

Proof. The elementary axioms of discretely ordered semiring arithmetic hold in the non-negative cone of any discretely ordered unital ring. Every positive $a \in \mathcal{I}_k$ has predecessor $a - 1 \geq 0$.

A quantifier-free formula in one variable with parameters from $\mathcal{I}_k$ is a Boolean combination of polynomial equations and inequalities. In the real closed field $\mathcal{L}_k$, the real roots of its finitely many nonzero polynomials partition the line into finitely many intervals and points on which the formula has constant truth value. Every nonempty trace of one of these pieces on $\mathcal{I}_{k, \geq 0}$ has a least element: the lower endpoint, when included and already in $\mathcal{I}_k$, or the first integer-part element above that endpoint, with 0 serving as a lower bound when needed. The floor property supplies this first element. Therefore every nonempty quantifier-free definable subset of $\mathcal{I}_{k, \geq 0}$ has a least element.

If a quantifier-free formula held at 0 and were preserved by successor but failed somewhere, its counterexamples would have a least element a . Then $a > 0$ and the formula holds at $a - 1$, contradicting preservation by successor. This is open induction. $\square$

[write] This is the route of `dsn:cor:openinduction` and of source 03's [theorem 23.2](#), which proves open induction for $\mathcal{A}_{\text{fin}}$; the Shepherdson model is classical [6].

Corollary 70.7 (Arbitrary countable support complexity in countable arithmetic). *For every countable ordinal α , there is a real closed Polish truncation-closed field $C \subseteq \mathbb{R}_{\text{alg}}((t^{\mathbb{Q}}))$ whose countable canonical integer part J_C contains an element with support order type α . The nonnegative cone of J_C satisfies $\mathcal{Q} + \text{IOpen}$.*

Proof. Embed α into $\mathbb{Q} \cap (-2, -1)$ and take the Hahn series with coefficient 1 on that support. Put it in a strengthened hull using Theorem 69.5. Its support is negative and its constant is 0, so it belongs to J_C . The proof of Proposition 70.6 applies verbatim with this integer part and its real closed ambient field. $\square$

There is no conflict with Glazer’s result: the arithmetic model $\mathcal{I}_{k, \geq 0}$ is countable. Its uncountable real closed ambient field $\mathcal{L}_k$ is not a discretely ordered arithmetic model. Moreover, $\mathcal{P}_k$ and $\mathcal{L}_k$ share this same integer part even though only the latter supports admissible Polish field arithmetic.

70.3 Why the finite integer part does not extend to the full Hahn field

The element

$$z = t^{-2}y = \sum_{n \geq 2} t^{-1-1/n} \in \mathcal{H}_k,$$

where y is from (65.3), has infinitely many negative terms. For every $a \in \mathcal{I}_k$, at least one negative coefficient of $z - a$ remains nonzero; its leading exponent is negative, so $|z - a| > 1$. Hence $\mathcal{I}_k$ is not an integer part of $\mathcal{H}_k$.

Replacing finite negative support in (70.1) by arbitrary well-ordered negative support gives an integer part of $\mathcal{H}_k$, with the same coefficientwise floor rule. This larger ring is uncountable: binary choices along the support of z already produce continuum many elements. Theorem 62.3 supplies another obstruction to admissible Polish arithmetic on this full Hahn field. Here that argument agrees with, but is less general than, Theorem 63.3.

71 Surreal realization and a formalization interface

71.1 A set-sized embedding, not a topology on a proper class

With $k = \mathbb{R}_{\text{alg}} \subseteq \mathbb{R}$, the normal-form assignment

$$\iota \left(\sum_{q \in \mathbb{Q}} a_q t^q \right) = \sum_{q \in \mathbb{Q}} a_q \omega^{-q} \quad (71.1)$$

realizes the three fields as set-sized subfields of the surreal numbers. Well-ordered support in the q -variable becomes reverse-well-ordered support in the Conway growth exponents $-q$. The normal-form rules preserve addition, multiplication, and order; t corresponds to ω^{-1} . The image of $\mathcal{I}_k$ has only positive growth exponents and an integer constant term, so lies in $\mathbf{Oz}$.

This is a formal normal-form embedding. It does not assert that an arbitrary infinite series can be evaluated at an infinite real or surreal argument by ordinary analytic convergence. In particular, the direction of the exponent change in (71.1) is essential. The full proper class $\mathbf{No}$ is never asserted to be a Polish space. The series carriers have cardinality continuum and

can be constructed as subsets of the set $k^{\mathbb{Q}}$. Only afterwards are they identified with surreal normal forms.

The project documentation is an interface to these structures, not a proof that the new topological statements have already been formalized. The reported kernel checks in Provel concern the repository's declared constructions and theorems. No new kernel run was performed for this article.

71.2 A modular Lean development plan

A formalization need not begin with surreal arithmetic. Most of the proof is independent of it. A suitable division into modules is as follows.

Category and ordered groups. Formalize the Baire-order local-bound lemma over a linearly ordered abelian group, separately from the field specialization. Its interface should return an order bound $a > 0$ and a neighborhood inclusion, not an implicit assertion that the ambient topology is the order topology. Import a proved Pettis lemma and Borel-homomorphism automatic continuity at clearly stated instances.

Ordered rings and fields. Prove automatic discreteness using one scalar $a + 1$, and prove the field refinement theorem using $(a + 1)/\varepsilon$. These are different lemmas: quotient scalars are not available in a general ordered ring. Then prove closed discreteness and countability of any integer-part subring without assuming its carrier is measurable.

Series and topology. Construct left-finite coefficient functions as a subtype of $\mathbb{Q} \rightarrow k$. Give finite-below-cutoff witnesses explicitly. Prove convolution finiteness, the inverse construction, valuation identities, and metric completeness. The countable dense subspace consists of finitely supported functions. A separate theorem identifies the metric and order topologies.

Obstruction certificates. Formalize the binary family from Lemma 63.2; the only infinite support needed for the negative result has order type at most ω . For Puiseux, package the general countable proper-subfield-cover theorem, then prove each lattice-support stage is closed. This separation prevents a proof of incompleteness of one metric from being mistaken for a proof of absolute non-Polishability.

Integer parts and measurable floors. First prove (70.2) in the left-finite series subtype. Then identify this ring with the corresponding set-sized subring of the repository's omnific integers. The exact discontinuity theorem and the pointwise eventually constant sequence (F_n) can be formalized independently of the full surreal field.

Descriptive complexity. The explicit maps in (67.1) and the FIN reduction are the finite combinatorial portion. The well-founded-tree reduction and Lusin–Souslin step belong in a separate descriptive-set-theory layer. They are not required for the core Hahn classification or the positive Levi-Civita construction.

This architecture leaves no proper-class quantification in the main proof. Universe parameters are needed only when identifying the set-sized series carrier with a universe-relative surreal implementation. No additional large cardinal assumption is used.

72 Consequences for exact symbolic arithmetic

An exact coefficient stream and a valuation-Cauchy name provide different information. A coefficient stream reports one coefficient at a time. A valuation-Cauchy name eventually fixes *all* coefficients below a specified cutoff. In a left-finite representation, certifying a completed prefix therefore includes a support bound, not just a list of currently observed nonzero terms.

Theorem 65.3 supports continuous arithmetic in the second representation. Proposition 67.4 says that the measurable information is the same, but it does not turn these representations into the same effective oracle. No computability claim follows merely from Borel equivalence.

For an exact rational-coefficient prototype one can represent finite truncations by dictionaries $q \mapsto a_q$ with $q, a_q \in \mathbb{Q}$. When multiplying to cutoff B , lower bounds on the two supports determine which input prefixes suffice: terms of the first factor above $B - v(g)$, and terms of the second above $B - v(f)$, cannot affect the result. For inversion of $at^q(1 + h)$, a positive lower bound $\delta \leq v(h)$ bounds the number of geometric powers needed below any fixed cutoff. These are mathematically certified finite reductions, not numerical approximations to real coefficients.

Floor is different. Its Baire class one approximants may change once the first nonzero infinitesimal term is revealed at an integer boundary. No continuous exact floor algorithm on the whole field can exist, because the mathematical function is discontinuous there. The theorem pinpoints the necessary limitation instead of attributing it to a poor choice of code.

The accompanying Python program uses rational exponents and coefficients to check finite algebraic identities, truncated geometric inverses, the floor inequalities, and finite instances of the support witnesses. The deterministic run passed 20,116 finite assertions; its output records the counts by test family. These tests catch implementation mistakes in the supplied examples; they cannot establish Baire category, quantification over all Polish topologies, infinite support completeness, or real closedness. The conventional proofs are the basis of those assertions.

73 Further research questions and concrete targets

The following questions are proposed next steps. They are not asserted to be previously published open problems, and no novelty claim is inferred from failure to locate an answer in the inspected sources.

Question 73.1 (Intermediate-field rigidity). Let k be countable and ordered, and let $\mathcal{P}_k \subseteq F \subseteq k((t^{\mathbb{Q}}))$. If F has an admissible Polish topology, must its valuation topology be complete, and must the admissible topology equal the valuation topology?

The prefix criterion gives countably many prefixes, but it does not prove completeness of a coarser topology. A proof must close precisely that gap; one cannot appeal to the open mapping theorem until both topologies are known to be Polish.

Question 73.2 (Classification of Polish hulls). Within a fixed standard Borel family of coefficient-observable parameters, what is the descriptive complexity of equality and inclusion between the countable-parameter hulls F_A ? Can one construct useful canonical dense cores or invariants for their ordered-field isomorphism types?

The existence of proper Polish extensions of $\mathcal{L}_k$ and the absence of maximal ones are settled by Theorem 69.2 and Corollary 69.4. The remaining issue is classification, not existence. The

ambient parameter space must be specified: the full Hahn coefficient domain is not standard Borel.

Question 73.3 (Minimal scalar hypotheses). How few Borel scalar maps suffice to force the order topology in the rational-exponent setting? Can a specified countable set of monomial multiplications replace all scalar maps under additional algebraic or truncation-closure assumptions?

The current proof uses a scalar depending on the local order bound and the target interval. An answer should identify which part of that freedom is essential, rather than dropping scalar regularity altogether.

Question 73.4 (Order-free non-real-closed fields). For countable ordered k that is not real closed, can the Hahn and Puiseux nonexistence statements be derived from a Polish additive group and jointly Borel multiplication alone, without requiring the displayed order to have the Baire property?

Corollary 61.6 settles the real closed case. In general a field may have several orders, and the image-of-squares argument does not identify one positive cone.

Question 73.5 (Other support completions). Extend the left-finite construction and its classification to countable ordered exponent groups of higher rank. Which support restrictions give separable complete field topologies, and which admit a countable cofinal valuation scale but still fail by a proper-subfield cover?

The inverse proof in Proposition 65.2 explicitly uses $n\delta \rightarrow +\infty$ in an Archimedean exponent group. It cannot be copied unchanged into a lexicographic higher-rank group.

[write] (Batch 92, 4 October 2026.) Not answered. Source 21 (Part XV) extends the left-finite construction in the other direction: to every $\mathbb{Q}$ -subspace $\Gamma \leq \mathbb{R}$, of ordered rank one and arbitrary *rational* rank, over $\mathbb{R}_{\text{alg}}$; the field is complete, real closed, and Polish exactly when Γ is countable (propositions 196.2, 196.3 and 197.1), and its self-embeddings are classified (theorems 195.1 and 203.1). Higher *ordered* rank stays open; source 21 asks for it again (question 208.12).

Question 73.6 (Descriptive ranks beyond the trichotomy). Which Borel ranks can occur for coefficient-code domains of admissibly Polish subfields of $k((t^{\mathbb{Q}}))$? Is there a useful algebraic characterization of when a Borel support family yields a Polishable additive group?

Table 2 shows that a smaller rank is not sufficient. A productive refinement should combine support closure, completeness of prefixes, and the proper-subfield obstruction.

[write] (Batch 93.) Not answered; a contrast from Part XVII. With real coefficients the left-finite support family gives a Π_3^0 code domain (proposition 226.4) whose additive group is not Polishable (proposition 229.1), whereas over a countable k the same family gives a Polish field (theorem 65.3). The question is for countable k , so this bears on it without answering it.

Question 73.7 (The least support bound of a one-parameter hull). For $y = \sum_{n \geq 2} t^{1-1/n}$, determine the least countable ordinal bounding all support order types in $F_{\{y\}}$. How does that least bound change when real closure and closure under all rational truncations are imposed before metric completion?

Theorem 69.3 proves existence of a bound but does not compute it. A solution would require quantitative control of support order types under convolution, inversion, algebraic extension, and stabilized-prefix limits. It would turn the qualitative local-hull theorem into an ordinal complexity calculation.

Question 73.8 (Other integer parts and floor discontinuities). Classify integer parts of $\mathcal{L}_k$ and the larger real closed Polish hulls up to field automorphisms. Which floors admit canonical or effective locally constant approximants for presentations not built from truncations?

Countability and closed discreteness hold for every integer part by Theorem 62.3. The discontinuity set of its floor is also its integer part in the order topology. The strengthened hulls retain explicit truncation approximants. The remaining constructive issue is canonical or effective approximation for other presentations, together with a classification of the integer parts.

[write] (Batch 92, 4 October 2026.) Answered in part by Part XV for $k = \mathbb{R}_{\text{alg}}$, where $\mathcal{L}_k$ is source 21's $\mathcal{L}^{\mathbb{Q}}$. Every field automorphism carries the common integer part $\mathcal{I}_k = \mathcal{I}^{\mathbb{Q}}$ to an integer part, and the automorphism $t \mapsto t + t^2$ carries it to a different one, since it sends t^{-1} to $t^{-1} - 1 + t - t^2 + \dots \notin \mathcal{I}_k$ (paragraph after corollary 205.4); every field self-embedding is onto (theorem 199.2), and the image of a proper self-embedding of a field of infinite rational rank contains no integer part of the field (corollary 205.4). The classification up to automorphisms and the effective-approximation question stay open.

Question 73.9 (A uniform effective presentation). For computably presented k , construct a certified Type-2 representation of $\mathcal{L}_k$ with computable arithmetic, explicit precision transfer for products and inverses, and a finite-mind-change floor procedure for $k = \mathbb{R}_{\text{alg}}$.

The representation must include effective support-completion information. Bare enumerations of nonzero coefficients do not suffice to know that a prefix is complete. Any claimed effective embedding into Baire space should state which oracle representation is being used.

Question 73.10 (Formal proof milestones). Can the local order-bound theorem, automatic discreteness, and full Hahn classification be formalized as a small reusable library independent of the full surreal development? What is the minimal imported descriptive-set-theory interface for the exact code-complexity theorems?

These are concrete implementation tasks. The first milestone is the category-to-order bridge; the coanalytic lower bound should not block the rest of the project.

Question 73.11 (Positive cones versus groups of differences). Characterize Borel discretely ordered semirings whose positive cones admit Polish continuous addition but whose groups of differences are not Polishable with measurable multiplication. Which hypotheses force a topology to extend from the cone to its group completion?

This returns directly to the boundary emphasized by Glazer's arithmetic questions and the earlier Presburger constructions. Theorem 62.1 provides an obstruction on the group side, not a classification of monoid presentations.

[write] This is source 08's question 57.2; source 06's cone of $\mathbb{Z} + t\mathbb{R}[t]$ is a data point (section 42).

[write] (Batch 90.) The last sentence is answered by Part XII: when addition is continuous and the natural order of the cone is linear, its topology extends to a (unique, Polish) group topology on the group of differences exactly when partial subtraction is continuous (theorems 153.1 and 154.1); with a semiring multiplication jointly continuous on the cone, the completion is a Polish topological ring (corollary 154.7). For the cones of uncountable discretely ordered rings partial subtraction is never continuous (proposition 155.6). Part XII does not address the first sentence.

Question 73.12 (Weak choice and category). Which portions of the classification survive in weaker choice settings? Separate the dependent choices used to build the bounded increasing

exponent sequence, the Baire theorem, and the selections of representatives of countably many prefix classes.

The present paper works in ZFC and makes no choice-free assertion. A foundational analysis would be a distinct project rather than an automatic consequence of the set-sized constructions.

74 Conclusions of Part V

The main dividing line is not whether exact infinite series can be assigned Borel codes. It is whether their arithmetic, order, and completeness fit within a separable Polish group. A Baire-measurable order and Borel scalar arithmetic force local order bounds and hence the order topology. Full Hahn support then admits too many mutually separated prefixes, except in the countable-coefficient Laurent case. Puiseux support avoids that obstruction but fails a different category test because it is a countable union of proper measurable subfields. Left-finite completion avoids both obstructions and supplies a unique Polish topological field. Countable-parameter hulls go further: arbitrary countable Hahn data fit into a Polish subfield, even with real closedness and truncation closure over real algebraic coefficients. There is no maximal Polish intermediate field, while the exact cover number of the full field is the continuum.

The integer-part consequences connect this topological classification to arithmetic and surreal normal forms, including arbitrary prescribed countable support types inside suitable countable integer parts. A Polish ordered field cannot hide an uncountable discrete integer part, even a nonmeasurable one. The real algebraic-coefficient Levi-Civita field nevertheless supports a countable omnific integer part with open induction, and its floor has exactly controlled discontinuities. These results supply a precise next layer beyond Borel support analysis while leaving topology rigidity on intermediate fields and the broader positive-cone classification open.

Part VI

Baire order forces arithmetic rigidity

[write] This part merges two manuscripts of batch 87 that prove the same principal theorem by different proofs. Source 12 is *Baire-Category Rigidity for Discretely Ordered Rings: Beyond local compactness, without measurable multiplication* (dated October 3, 2026; “research manuscript prepared with ChatGPT for Vladimir Reshetnikov”), pinned to repository commit [0f084afa9](#). Source 10 is *Baire Order Forces Arithmetic Rigidity: Discretely ordered rings with Polish addition and completely unrestricted multiplication* (same date and wording of authorship), pinned to [6fef5383b](#). They share under 0.8% of their word 8-grams; neither is an edition of the other. Both continue [Part IV](#): they answer its [question 57.1](#) negatively, without any regularity of multiplication.

The principal theorem. A discretely ordered ring whose additive group carries a Hausdorff Baire topology satisfying the countable chain condition, and whose strict positive cone has the Baire property, is countable, and the topology is discrete. In particular this holds for every Polish additive-group topology with Borel order. Source 12 proves it as [corollary 78.4](#)

by packing $|R|$ disjoint open translates ([lemma 78.1](#)); source 10 proves it (its Theorem 5.1, [corollary 78.4](#)) by the ordered quotient embedding $A/\text{Fin}(A) \hookrightarrow A/H_a$ of [theorem 79.7](#). Both use an order-bounded neighbourhood of 0 ([lemma 77.2](#)), which is source 09's [lemma 61.3](#) of [Part V](#), extended from Polish to Hausdorff Baire groups.

Base and reading conventions. Source 12 is the base, because on the shared statements it has the weaker hypotheses: its cardinal theorem ([theorem 78.3](#)) needs neither the chain condition nor uncountability; its semiring theorem ([theorem 81.2](#)) does not assume commutativity; its non-Polishability theorem ([theorem 82.2](#)) assumes no regularity of multiplication; and it has a second, Haar-measure route ([theorem 83.3](#)). Unmarked text in this part is source 12's. Text from source 10 is marked [10]: in the titles of its sections, subsections and statements, and in the sentence that introduces an inserted passage. Source 10's sections are printed as subsections of source 12's sections on the same subject, one level down; its section on continuous scalar maps is a section of its own ([section 85](#)). Text written for this report is marked [write] or [merge] as in [section 1.2](#).

What is printed once. The principal theorem ([corollary 78.4](#); source 10's proof is printed as a second proof in [section 79.2](#)); the bounded-neighbourhood lemma (source 10's proof, which avoids Pettis's lemma, is a second proof); the open principal convex subgroup ([proposition 79.1](#), source 10's Proposition 4.3); the absence of an ordered-ring expansion ([corollary 81.1](#), source 10's Corollary 5.3 for Polish topologies); the three topologies on $\mathbb{Z} + t\mathbb{R}[t]$ ([section 84](#); source 10's Examples 9.1, 9.3 and 9.4, whose additional remarks are printed as marked passages); and the completion without a discrete order ([proposition 86.2](#), source 10's Proposition 9.2, with its proof as a second proof). Where source 10 states a sharper or differently scoped version, both are printed: the exact Polish criterion ([corollary 78.6](#)), the unit-image spectrum ([theorem 80.1](#) with an explicit bound, source 10's [theorem 80.4](#) with three further clauses), signed Presburger models ([corollary 81.3](#) with a Borel cone, source 10's [corollary 81.4](#) with a Baire-property cone) and the omnific corollary ([corollary 86.3](#); source 10's [corollary 86.4](#) adds extent and Lindelöf degree).

What is new to the collection. The theorems of this part are new to the collection except where a note names an earlier statement. They make several results of Parts IV and V corollaries, among them [theorems 44.1](#) and [62.1](#); [section 90](#) lists them with the deductions. Neither source claims historical priority.

Labels and notation. Labels of this part are `pma:br:<label>` for both sources, except the seven labels of source 10 that coincide with labels of source 12 (`cor:omnific`, `cor:polish`, `sec:category`, `sec:conventions`, `sec:embeddings`, `sec:questions`, `thm:main`), which take the sub-prefix `pma:br:s10:.` Where source 10 states a result that source 12 also states, source 10's label is a second label of source 12's statement: `pma:br:lem:localbound` is [lemma 77.2](#), `pma:br:prop:openhull` is [proposition 79.1](#), `pma:br:s10:thm:main` is [corollary 78.4](#), `pma:br:cor:noexp` is [corollary 81.1](#), `pma:br:prop:completion` is [proposition 86.2](#), and source 10's examples `ex:incomplete`, `ex:nonccc`, `ex:nonbp` are labels of source 12's [sections 84.1](#) to [84.3](#). Source 10's references to its displays `eq:localbound` and `eq:polynomialring`, which are not printed, point to source 12's [\(77.1\)](#) and [\(84.1\)](#). One label was added to an unlabelled statement of source 12 (`pma:br:cor:signed-presburger`). Source 10's $\text{Fin}(A)$, the ordinary-integer-bounded part, is printed with roman type and is not Part II's set Fin of finite sets; its unit-image spectrum $\mathcal{S}(G, e)$ is printed $\mathcal{U}(G)$, source 12's

name for the same set. [Section 1.3](#) lists these and the other conventions of Parts VI–IX; in particular source 12’s t and source 10’s X are the same positive *infinite* variable of $\mathbb{Z} + t\mathbb{R}[t]$, as in [Part III](#).

75 The research intersection and the question answered

The point of contact is the topology of arithmetic structures: how much arithmetic can coexist with a well-behaved additive topology? Glazer’s *A Topological Tennenbaum Theorem* rules out uncountable Polish models of $\mathcal{Q} + \text{IOpen}$ with continuous addition and Borel multiplication [1, Corollary 1]. His Question 2 asks about continuous addition for uncountable Polish Presburger models. ProveIt contains complementary interfaces: integer Presburger elimination and ordered surreal and omnific constructions [51, 52]. This suggests examining what happens when an additive presentation is required to support a full, order-compatible multiplication.

An earlier project report established a local-compactness obstruction [49]. Its first research question in Section 14 asks whether an uncountable discretely ordered ring can carry a Polish additive-group topology with Borel order when multiplication has no regularity assumption. The answer proved here is *no*. More strongly, Borelness can be weakened to the Baire property of a single cone, and Polishness can be weakened to Baire plus the countable chain condition.

The principal result

Baire-category rigidity. If a discretely ordered ring R has a Hausdorff Baire additive-group topology and a positive cone with the Baire property, then

$$c(R) = d(R) = |R|.$$

If this topology satisfies the countable chain condition, then R is countable and the topology is discrete.

There is no regularity hypothesis on multiplication.

Here $c(R)$ is cellularity and $d(R)$ is density. The cardinal equality is not merely an upper-bound obstruction: the proof explicitly constructs $|R|$ disjoint open translates. It identifies a mechanism that persists far beyond the finite-dimensional locally compact setting of the previous report.

75.1 What advances beyond the earlier work

The local-compactness argument used structural information about locally compact abelian groups. The proof below needs no decomposition into real components, no finite-dimensional flag, and no regularity of scalar multipliers. Its central interface is just an order-bounded neighborhood. This makes the argument both more general and shorter to formalize. The main consequences developed here are as follows.

1. A universal packing theorem gives exact density and cellularity, with both Baire-category and Haar-measure routes to its hypothesis.
2. The images of a least positive element under *all* ordered additive self-embeddings are order-bounded in every uncountable Baire ccc presentation; no regularity of the embeddings is required.

3. An uncountable such ordered group admits no compatible unital ordered-ring multiplication, and its nonnegative cone admits no unital semiring multiplication with the prescribed addition and unit.
4. Polynomial examples explain why Baire category, ccc, and regularity of order cannot simply be deleted. A natural completion destroys discrete order even before topology is considered.

These statements are established within the manuscript. We do not claim that each auxiliary lemma is historically new. Pettis's theorem and its measure counterpart are classical; the contribution here is their use with discrete arithmetic separation, the resulting sharp conclusions, and the resolution of the specified project question.

75.2 The scope is not Glazer's entire setting

A positive arithmetic structure has an additive *monoid*. Our central hypothesis is a topology on its signed additive *group*, with continuous inversion as well as continuous addition. Algebraically taking differences does not, by itself, construct a Polish group topology. Therefore the theorem below is not a replacement for Glazer's theorem under weaker hypotheses on the same objects. The carrier and topology requirements are different.

Glazer's Question 1 concerns provability in ATR_0 [1, Question 1]; we do not settle that question. Nor do we re-present the earlier affirmative Presburger construction [50] as a new answer. The exact open question answered here is the one in the preceding local-compactness report.

[write] Source 12's "earlier project report" [49] is source 08, [Part IV](#); its "first research question in Section 14" is [question 57.1](#), and the question on weakening Borel order is the fourth question of that section ([question 57.4](#)). The "earlier affirmative Presburger construction" [50] is source 05, [Part I](#). Source 12's pin precedes the placement of Parts III–V, so it did not see source 09 ([Part V](#)).

75.3 The intersection and the result as source 10 states it [10]

The topic is the boundary between topological presentations of arithmetic and the algebra of non-Archimedean integer systems. It connects a specific part of Elliot Glazer's published research with two concrete parts of ProveIt, rather than relying on a speculative account of his interests. In *A Topological Tennenbaum Theorem*, Glazer proves, in particular, that an uncountable Polish model of $\mathbb{Q} + \text{IOpen}$ cannot have continuous addition and Borel multiplication [1, Corollary 1]. ProveIt develops Cooper elimination for Presburger arithmetic and an ordered surreal/omnific algebra [54, 55]. These are natural interfaces at which to ask whether an additive presentation can be expanded by multiplication.

Two earlier project manuscripts supply the immediate starting point. *Local Compactness Forces Arithmetic Rigidity* proves a locally compact, second-countable version without regularity assumptions on multiplication, and explicitly asks whether local compactness can be removed [56]. *Polishability at the Hahn–Puisseux–Levi–Civita Boundary* isolates the category argument that makes a Baire order locally bounded [57]. The present article combines that local bound with a different quotient construction. The quotient has a countable kernel even when the open subgroup is neither divisible nor finite dimensional.

Main conclusion. Let A be a nonzero discretely ordered ring. If $(A, +)$ is a Polish topological group and $\{x \in A : x > 0\}$ has the Baire property, then A is countable and its topology is discrete. Multiplication is an arbitrary algebraic operation compatible with the ordered-ring axioms; it is not required to be measurable.

Thus the answer to the general Polish-group question posed in [56] is negative: there is no uncountable example. We actually prove the stronger Baire–ccc statement. Here *ccc* means that every family of pairwise disjoint nonempty open sets is countable.

75.3.1 What changes in the proof

The earlier locally compact argument finds an open real vector component and uses multiplication to move an uncountable divisible subgroup into its quotient. That approach makes finite-dimensional structure appear essential. Instead, for a positive element a , consider

$$H_a = \{x \in A : |x| \leq na \text{ for some ordinary } n \in \mathbb{N}\}.$$

This is a convex additive subgroup. Its elements need not be bounded by a *single* multiple of a ; the ordinary integer n may depend on x . Nevertheless,

$$ax \in H_a \iff |x| \leq n1 \text{ for some ordinary } n.$$

For a discretely ordered ring, the right side says exactly that $x \in \mathbb{Z}1$. Therefore multiplication by a , followed by the additive quotient map, has the countable kernel $\mathbb{Z}1$ on *all* of A . When H_a is open, ccc makes the target countable. No continuity of this map is needed: an algebraic injection is enough.

75.3.2 Scope and contribution ledger

Item	Status in this article
Category local boundedness	Classical category mechanism, already used in [57]; reproved here for Hausdorff Baire groups.
Principal-hull quotient embedding	Main algebraic observation, proved for arbitrary ordered rings in section 79.1 .
Baire–ccc arithmetic rigidity	Main extension beyond [56]; proved without any multiplication regularity in section 79.2 .
Full-cardinality discrete witnesses	Quantitative strengthening for nonseparable presentations; proved in section 79.3 .
Presburger examples	Standard divisible-plus-integer constructions, including examples already used in project reports; not claimed as new.
Universal no-expansion theorem	A consequence for <i>every</i> uncountable Polish signed Presburger group with a Baire positive cone.
Historical priority	Not established by the literature check.
Machine verification	No new Lean/Rocq formalization is claimed; exact-arithmetic tests are supplementary only.

The signed-group hypothesis matters. A Polish topology on a positive arithmetic monoid does not automatically extend to a Polish topology on its group of differences. We return to this distinction in [section 81.4](#). We also do not claim to settle Glazer’s reverse- mathematical Question 1, which concerns an ATR_0 proof of his Theorem 2 [1].

[write] Source 10's [56] is source 08, [Part IV](#), and [57] is source 09, [Part V](#); the category argument it credits to [57] is [lemma 61.3](#). The "general Polish-group question posed in" [56] is [question 57.1](#).

76 Conventions and elementary ordered algebra

All structures and topological spaces in the theorems are sets, and the ambient foundation is ZFC. We use the usual standard natural numbers when writing an external countable union or a sequence indexed by $\mathbb{N}$.

Definition 76.1. A *discretely ordered ring* is a nonzero unital ring R with a total order such that addition preserves order, the product of two strictly positive elements is strictly positive, and 1 is the least strictly positive element. Unless stated otherwise, rings are commutative.

This is the standard discrete-order convention used in weak arithmetic [47]. It is not a statement that a supplied topology is discrete. In fact, distinguishing those two meanings of discreteness is central to the problem.

The ring order makes the additive group ordered and torsion-free. For $x \in R$ put $|x| = \max(x, -x)$. Directly from the order axioms,

$$x \neq y \implies |x - y| \geq 1, \quad |xy| = |x| |y|. \quad (76.1)$$

There are no zero divisors: the absolute value of a product of nonzero elements is positive.

Lemma 76.2 (Ordinary integers form a convex subring). *If R is discretely ordered and $0 \leq x \leq n$ for an ordinary natural number n , then x is one of $0, 1, \dots, n$. Consequently, either $R = \mathbb{Z}$ as an ordered unital ring, or there is $c \in R$ with $c > n$ for every ordinary $n \in \mathbb{N}$.*

Proof. The first assertion follows by external induction on n . If $x > 0$, then $x \geq 1$, so $0 \leq x - 1 \leq n - 1$. For the second assertion choose an element outside the ordinary integer subring and take its absolute value. If it were bounded by any ordinary natural number, the first assertion would put it in that subring. Thus its absolute value is larger than all ordinary natural numbers. $\square$

This uses no induction axiom *inside* the ring. It is an ordinary metatheoretic proof about a finite integer interval.

Definition 76.3. An *additive-group topology* on R is a topology making $(R, +, 0, -)$ a topological group. We always require it to be Hausdorff. We make no compatibility assumption between this topology and the order topology. Write $P = \{x \in R : x > 0\}$.

A set has the *Baire property* if its symmetric difference with an open set is meager. A space is *Baire* if no nonempty open set is meager in the space. These are distinct notions: a positive cone may have the Baire property in a space that is itself meager.

A space satisfies the *countable chain condition*, abbreviated ccc, if every family of pairwise disjoint nonempty open sets is countable. Its cellularity $c(X)$ is the supremum of the cardinalities of such families. Its density $d(X)$ is the least cardinality of a dense subset. For any space,

$$c(X) \leq d(X) \leq |X|. \quad (76.2)$$

Indeed, each member of a disjoint open family meets a dense set, and the chosen intersection points are different. Separability implies ccc. A Polish space is separable and completely metrizable, and is Baire.

For Borelness, regularity of the binary order and regularity of its cone are equivalent in an additive topological group: the cone is a section of the order relation, and

$$x < y \iff y - x \in P.$$

Our category theorem needs only the Baire property of P itself. It does not quantify over the Borel complexity of every definable relation.

76.1 Conventions of source 10 [10]

Throughout the main development, an *ordered ring* is a nonzero commutative unital ring A with a translation-invariant total order such that $x, y > 0$ implies $xy > 0$. This is the strict ordered-domain convention. It implies cancellation of positive factors in inequalities. We do not use the weaker convention allowing positive zero divisors. The noncommutative extension is recorded in [remark 78.9](#).

Write $\underline{n} = n1_A$ for an ordinary integer represented in A . An ordered ring is *discretely ordered* when 1_A is its least positive element. This is an algebraic property, not a statement about the topology. The standard examples and the connection to weak arithmetic are discussed in [\[47\]](#). All uses of induction on an ordinary integer in this paper are external finite mathematical arguments; no first-order induction axiom inside A is assumed.

The topology τ is a topology on the additive group: addition and negation are continuous. Unless stated separately, multiplication is not a topological operation. The topology need not agree with the order topology. We write

$$P = \{x \in A : x > 0\}, \quad |x| = \max(x, -x).$$

The set P has the *Baire property* if it differs from an open set by a meager set. This condition on the positive cone is exactly the regularity used. In particular, a Borel positive cone suffices. We do not silently replace it by a stronger hereditary condition on every closed subspace, nor do we need such a condition.

A space is *Baire* when each countable intersection of open dense sets is dense. In a Baire space, no nonempty open set is meager. Open subspaces of a Baire space are Baire. A Polish space is separable and completely metrizable, so it is Baire and ccc. For the latter fact, a countable dense set meets each nonempty member of a disjoint open family at a different point.

All structures in the theorems are sets, and all arguments take place in ZFC. Quotients such as A/H are quotients of *additive groups*, unless explicitly specified otherwise. No ideal property is presumed. When **No** or **Oz** appears, we apply the result only to specified set-sized subrings or to a fixed universe-bounded formal model. There is no Polish topology on a proper class in the sense used here.

77 From category to an order-bounded neighborhood

The following elementary form of the Pettis lemma is the only category input beyond the definition of a Baire space. We include the proof to make the later dependency boundary explicit [\[45\]](#).

Lemma 77.1 (Pettis difference lemma). *Let G be a Baire topological abelian group. If $A \subseteq G$ has the Baire property and is nonmeager, then $A - A$ contains an open neighborhood of 0.*

Proof. There is a nonempty open set U on which A is comeager. For any $h \in U - U$, the intersection $U \cap (h + U)$ is nonempty and open. Both A and $h + A$ are comeager in that intersection. Since a nonempty open subspace of a Baire space is Baire, they meet there. Thus $h = x - y$ for some $x, y \in A$. We have proved

$$U - U \subseteq A - A.$$

The set $U - U$ is open and contains 0. □

Pettis alone is not enough: differences of an arbitrary positive set need not be order-bounded. A reflection supplies the missing bounded set.

Lemma 77.2 (Bounded-neighborhood lemma). *Let G be a nontrivial linearly ordered abelian group with a Hausdorff Baire group topology. If its strict positive cone P has the Baire property, then there are $a > 0$ and an open neighborhood W of 0 such that*

$$0 \in W \subseteq (-a, a). \tag{77.1}$$

Proof. If the topology is discrete, take $W = \{0\}$ and any $a > 0$. Suppose it is nondiscrete. A Hausdorff topological group with an isolated point is discrete, so $\{0\}$ is closed with empty interior and hence nowhere dense. Because

$$G = P \dot{\cup} \{0\} \dot{\cup} (-P)$$

and inversion is a homeomorphism, P cannot be meager: otherwise G would be meager in itself, contrary to the Baire assumption.

Choose a nonempty open U in which P is comeager, and choose $u \in U \cap P$. Reflection $x \mapsto 2u - x$ is a homeomorphism. Therefore $2u - P$ is comeager in $2u - U$. The open set

$$V = U \cap (2u - U)$$

contains u . On V , both P and $2u - P$ are comeager. Hence

$$A = P \cap (2u - P) = (0, 2u)$$

is nonmeager and has the Baire property. By [lemma 77.1](#), there is an open neighborhood W of 0 with $W \subseteq A - A$. If $x, y \in A$, then $-2u < x - y < 2u$. Thus $W \subseteq (-2u, 2u)$, and $a = 2u$ works. □

[write] [Lemma 77.2](#) is source 09's [lemma 61.3](#), stated there for Polish groups and proved by the same reflection-and-Pettis argument; [lemma 77.1](#) is the Baire-group form of [lemma 61.1](#). Source 12 states both for Hausdorff Baire groups and does not cite source 09, which it had not seen. Source 10 states the lemma as its Lemma 4.2 (pma:br:lem:localbound is a second label of [lemma 77.2](#)), credits the Polish version to source 09, and proves it without Pettis's lemma, as follows.

The category mechanism is elementary and is related to the usual Pettis sumset argument; [29] discusses Pettis's theorem and its connections with automatic continuity. We supply the full argument because its hypotheses are central to the result. The Polish version of the local-bound lemma was already isolated in [57].

Lemma 77.3 (Comeager semigroups have interior [10]). *Let G be a Baire topological group, written additively and assumed abelian here. Suppose a set P is comeager in a nonempty open set O and satisfies $P + P \subseteq P$. Then $O + O \subseteq P$.*

Proof. Take $z \in O + O$. The set $W = O \cap (z - O)$ is nonempty and open. The sets P and $z - P$ are both comeager in W : the second assertion uses that reflection followed by translation is a homeomorphism. Because W is Baire, these two comeager subsets intersect. Choose x in their intersection. Then $x \in P$ and $z - x \in P$, hence $z = x + (z - x) \in P + P \subseteq P$. $\square$

Second proof of lemma 77.2 [10]. Source 10 writes c and U for the bound a and the neighbourhood W .

For a discrete group use $U = \{0\}$ and any $c > 0$. Otherwise every singleton is closed with empty interior and hence nowhere dense. Inversion takes the positive cone P to $-P$. If P were meager, the partition $G = P \cup (-P) \cup \{0\}$ would make G meager in itself, a contradiction. Thus P is nonmeager. Since it has the Baire property, it is comeager in some nonempty open set O .

Choose $p \in O \cap P$. By lemma 77.3, $O + O \subseteq P$. The element $2p$ lies in this open set, so there is a symmetric neighborhood U of 0 with $2p + U \subseteq O + O$. For $u \in U$, both u and $-u$ belong to U . Therefore $2p + u > 0$ and $2p - u > 0$. These inequalities say $-2p < u < 2p$. Set $c = 2p$. $\square$

Remark 77.4. The lemma applies to ordered groups, not just rings. The order may have many Archimedean levels, the group may be infinite-dimensional or non-locally-compact, and its positive cone need not be open, closed, or Borel. Inversion and translation are used; multiplication is absent.

The proof needs one locally comeager part of the positive cone, then a symmetric overlap. This is why there is no need to classify Borel orders or to restrict them to real subspaces or compact subgroups.

78 Algebraic packing and maximal cellularity

Lemma 78.1 (Arithmetic packing). *Let R be a discretely ordered ring with an additive-group topology. Suppose that an open neighborhood W of 0 satisfies $W \subseteq (-a, a)$ for some $a > 0$. Set $b = 2a + 1$. Then*

$$\{br + W : r \in R\} \quad (78.1)$$

is a family of $|R|$ pairwise disjoint nonempty open sets.

Proof. Every translate is open and nonempty. Suppose $r \neq s$. By discrete order, $|r - s| \geq 1$, and therefore

$$|b(r - s)| = b|r - s| \geq b = 2a + 1 > 2a. \quad (78.2)$$

On the other hand, $W - W \subseteq (-2a, 2a)$. If $br + w_1 = bs + w_2$ for some $w_1, w_2 \in W$, then $b(r - s) = w_2 - w_1 \in W - W$, contradicting (78.2). Thus the translates are pairwise disjoint. Since they are nonempty, distinct indices give distinct sets. $\square$

Remark 78.2 (Why no scalar continuity is hidden). It is not asserted that bW is open, that $r \mapsto br$ is continuous, or that its graph is measurable. The open sets are translates of one fixed open set W . The scalars merely choose their centers.

Theorem 78.3 (Maximal cellularity and density). *Let R be a discretely ordered ring with a Hausdorff Baire additive-group topology. If $P = \{r : r > 0\}$ has the Baire property, then there is a family of $|R|$ pairwise disjoint nonempty open subsets of R . In particular,*

$$\boxed{c(R) = d(R) = |R|.} \quad (78.3)$$

Proof. Use [lemma 77.2](#) to obtain $W \subseteq (-a, a)$, and apply [lemma 78.1](#). This proves $|R| \leq c(R)$. Combine this with [\(76.2\)](#). $\square$

Corollary 78.4 (Ccc rigidity). *Under the hypotheses of [theorem 78.3](#), if the topology satisfies ccc, then R is countable and the topology is discrete.*

Proof. The explicit family in [lemma 78.1](#) and ccc imply countability. A countable Hausdorff nondiscrete topological group has every singleton nowhere dense, so it is meager in itself. This contradicts the Baire assumption. Hence the group is discrete. $\square$

Corollary 78.5 (Polish rigidity; answer to the earlier question). *Every discretely ordered ring with a Polish additive-group topology and a positive cone having the Baire property is countable and discrete. In particular, an uncountable discretely ordered ring cannot have a Polish additive-group topology with Borel order, regardless of the regularity of multiplication.*

Proof. A Polish space is Baire and separable; separability implies ccc. Borel sets have the Baire property. Apply [corollary 78.4](#). $\square$

[write] [Corollary 78.4](#) is source 10's main theorem, its Theorem 5.1 (pma:br:s10:thm:main); source 10's proof, through the principal-hull quotient embedding, is printed in [section 79.2](#). With [corollary 78.5](#) it answers [question 57.1](#) of [Part IV](#) negatively. Source 10 states the Polish case as an exact criterion:

Corollary 78.6 (Exact Polish presentation criterion [10]). *A discretely ordered ring admits a Polish additive-group topology with a Baire-property positive cone if and only if its underlying set is countable. Every such topology is discrete.*

Proof. Polish spaces are Baire, Hausdorff, and ccc, so necessity and uniqueness of the possible topology follow from [corollary 78.4](#). Conversely, a countable ring with the discrete topology is a Polish topological ring, and all subsets, in particular its positive cone, are Borel. $\square$

Here “uniqueness” refers to the topology for a fixed underlying ring: it must be the discrete topology. The theorem does not classify countable discretely ordered rings up to isomorphism. There are many nonstandard ones, including $\mathbb{Z}[X]$ ordered with X positive infinite.

Remark 78.7 (Countable does not mean standard). A countable nonstandard discretely ordered ring can carry the discrete Polish topology. The corollary does not assert that the ring is $\mathbb{Z}$; it asserts that its cardinality is countable and its given Baire ccc group topology is discrete.

Corollary 78.8 (A separable category dichotomy). *Let a discretely ordered ring have a separable Hausdorff additive-group topology and a positive cone with the Baire property. Then either it is countable discrete, or it is meager in itself.*

Proof. If the topology is Baire, use [corollary 78.4](#). Otherwise some nonempty open U is meager in the group. Let D be countable and dense. For each x , the nonempty open set $x - U$ meets D , so $x \in D + U$. Thus $G = \bigcup_{d \in D} (d + U)$ is a countable union of meager sets. $\square$

The same packing proof works for noncommutative discretely ordered unital rings, provided the order axioms make products of positives positive: left multiplication by $b > 0$ is order-preserving and satisfies $|bx| = b|x|$. Commutativity of multiplication is not essential to [theorem 78.3](#).

Remark 78.9 (Noncommutative extension [10]). The main proof also applies to an associative unital ring with a total translation-invariant order and strictly positive products, without commutativity. Use left multiplication $x \mapsto ax$. For $a > 0$, $|ax| = a|x|$, positive left multiplication reflects inequalities, and ordinary integer multiples of 1 commute with a . These are all the multiplicative facts used in the kernel computation. The quotient maps remain homomorphisms of the abelian additive groups. We keep the commutative convention elsewhere to match the arithmetic and omnific applications.

79 Principal convex subgroups and a second obstruction

There is a useful purely algebraic form of the argument. It reveals why an open convex subgroup cannot rescue an uncountable ring presentation.

For an ordered abelian group G and $a > 0$, define the principal convex subgroup

$$H_a = \{x \in G : \text{for some ordinary } n \geq 1, |x| \leq na\}. \quad (79.1)$$

It is a subgroup by the triangle inequality, and it is convex by construction. The indexing quantifier is external and ranges over the ordinary natural numbers.

Proposition 79.1 (An open principal convex subgroup). *Every nontrivial ordered Hausdorff Baire topological abelian group whose positive cone has the Baire property has an open subgroup H_a of the form (79.1). If the topology is ccc, G/H_a is countable.*

Proof. Choose $W \subseteq (-a, a)$ using [lemma 77.2](#). Then $W \subseteq H_a$. A subgroup containing a neighborhood of the identity is open, because it is the union of its translates of that neighborhood. Its cosets are disjoint nonempty open sets, so ccc makes the coset space countable. $\square$

[write] Source 10 states [proposition 79.1](#) for ordered rings, with $a \geq 1$ so that $\text{Fin}(A) \subseteq H_a$ (its Proposition 4.3, `pma:br:prop:openhull`), and adds that the open subgroup H_a is also closed, its complement being a union of open cosets. Its proof is the one above. Source 10 continues [10]:

It is important that H_a , rather than the interval $[-a, a]$, is the subgroup used in the proof. The interval need not be a subgroup. Conversely, H_a need not be bounded by one fixed element of A . Only membership in some ordinary multiple of $[-a, a]$ is needed.

Theorem 79.2 (Every principal convex quotient is as large as the ring). *Let $R \neq \mathbb{Z}$ be a discretely ordered ring. For every $a > 0$,*

$$[R : H_a] = |R|. \quad (79.2)$$

More explicitly, choose $c > n$ for every ordinary natural number n , and put $b = (a + 1)c$. The map

$$R \longrightarrow R/H_a, \quad r \longmapsto br + H_a \quad (79.3)$$

is injective.

Proof. Such a c exists by [lemma 76.2](#). For every ordinary $n \geq 1$,

$$b = (a + 1)c > (a + 1)n > an.$$

If $r \neq s$, then $|r - s| \geq 1$, so $|b(r - s)| \geq b > na$ for every ordinary n . Therefore $b(r - s) \notin H_a$, proving injectivity in (79.3). The quotient has at most $|R|$ elements, so its cardinality is exactly $|R|$. $\square$

No ideal is being asserted here. H_a is an additive convex subgroup; R/H_a is used only as an additive quotient. Confusing a convex subgroup with a ring ideal would obscure the purpose of the construction.

Combining [proposition 79.1](#) and [theorem 79.2](#) gives another proof of the countability part of [corollary 78.4](#) for $R \neq \mathbb{Z}$. This route uses an infinite element to make a quotient injection. By contrast, the packing proof uses only the finite expression $2a + 1$, works uniformly for $R = \mathbb{Z}$, and directly attains the full cellularity bound. It is the preferred proof of the main theorem.

Example 79.3 (Polynomial interpretation). In $\mathbb{Z} + t\mathbb{Q}[t]$, ordered by the leading coefficient with t infinite, H_{t^d} consists exactly of polynomials of degree at most d . Multiplication by t^{d+1} sends every nonzero difference outside that subgroup. Thus a principal convex quotient retains at least as much cardinality as the entire ring, despite discarding a complete finite range of growth degrees.

For real rather than rational coefficients, the same description holds: each individual leading coefficient is bounded by an ordinary integer. This observation will also be useful in the sharpness constructions.

79.1 The principal-hull quotient embedding [10]

79.1.1 The ordinary-integer-bounded part

Definition 79.4 ([10]). For an ordered ring A , define

$$F = \text{Fin}(A) = \{x \in A : \exists n \in \mathbb{N} (|x| \leq \underline{n})\}. \quad (79.4)$$

For $a > 0$, define its principal convex additive hull by

$$H_a = \{x \in A : \exists n \in \mathbb{N} (|x| \leq na)\}. \quad (79.5)$$

The quantifiers over $\mathbb{N}$ in these definitions are external ordinary quantifiers, not quantifiers over a nonstandard arithmetic model.

Lemma 79.5 ([10]). *The set F is a convex additive subgroup and a unital subring of A . For every $a > 0$, H_a is a convex additive subgroup, the smallest such subgroup containing a . If $a \geq 1$, then $F \subseteq H_a$.*

Proof. The triangle inequality gives closure under addition, and absolute value gives closure under negation. Bounds by ordinary integers multiply, so $|x| \leq \underline{n}$ and $|y| \leq \underline{m}$ imply $|xy| \leq \underline{nm}$. Thus F is a subring containing 1. If $|z| \leq |x|$ and x belongs to either hull, the same bound puts z in that hull; this proves convexity. Any convex additive subgroup containing a contains all na and all intervals $[-na, na]$, hence contains H_a . Finally, $\underline{n} \leq na$ when $a \geq 1$. $\square$

Lemma 79.6 (Finite elements in a discretely ordered ring [10]). *If A is discretely ordered, then $\text{Fin}(A) = \mathbb{Z}1_A$.*

Proof. For each ordinary $n \geq 0$, we prove externally that $[0, \underline{n}] = \{0, \dots, \underline{n}\}$. At $n = 0$ this is immediate. If $0 < x \leq \underline{n+1}$, discreteness gives $x \geq 1$, and consequently $0 \leq x-1 \leq \underline{n}$. Apply the induction hypothesis. Negating deals with negative finite elements. Conversely, every ordinary integer is bounded by its ordinary absolute value. $\square$

The loss of finiteness in F is precisely what permits uncountable ordered fields such as $\mathbb{R}$ to escape the arithmetic obstruction. The next theorem separates this algebraic issue from topology entirely.

79.1.2 A quotient embedding with exactly known kernel

Theorem 79.7 (Principal-hull amplification [10]). *Let A be an ordered ring and $a > 0$. The additive homomorphism*

$$T_a : A \longrightarrow A/H_a, \quad T_a(x) = ax + H_a \quad (79.6)$$

has kernel exactly $\text{Fin}(A)$. It induces an ordered additive embedding

$$\Phi_a : A/\text{Fin}(A) \hookrightarrow A/H_a, \quad x + \text{Fin}(A) \mapsto ax + H_a. \quad (79.7)$$

In particular,

$$|A/\text{Fin}(A)| \leq |A/H_a|. \quad (79.8)$$

Proof. Distributivity makes T_a additive. Since $a > 0$, absolute values and positive cancellation give the equivalences

$$\begin{aligned} T_a(x) = 0 &\iff ax \in H_a \\ &\iff \exists n \in \mathbb{N} (|ax| \leq na) \\ &\iff \exists n \in \mathbb{N} (a|x| \leq a\underline{n}) \\ &\iff \exists n \in \mathbb{N} (|x| \leq \underline{n}) \\ &\iff x \in \text{Fin}(A). \end{aligned}$$

The induced additive map is therefore well defined and injective.

Both quotient orders are defined because their subgroups are convex. A positive nonzero class $x + \text{Fin}(A)$ has a representative satisfying $x > \underline{n}$ for every ordinary n . Hence $ax > na$ for every n . Every member of H_a is at most some na , so $ax + H_a$ is positive and nonzero. Applying the same reasoning to $-x$ proves that signs, and hence all comparisons of distinct classes, are preserved. $\square$

Remark 79.8 (Why this is not a continuity argument [10]). The map T_a is used only as a homomorphism of abstract additive groups. It may be discontinuous or nonmeasurable. The quotient A/H_a need not be a ring, and multiplication by a need not descend to any endomorphism of A/H_a . What descends is exactly the map from $A/\text{Fin}(A)$ specified in (79.7). No other descent is asserted.

Corollary 79.9 ([10]). *If an ordered ring has an additive order unit $a > 0$, so that $H_a = A$, then $A = \text{Fin}(A)$. A discretely ordered ring with an additive order unit is therefore just its standard integer subring.*

Proof. The target in (79.7) is zero, so the source is zero. Apply lemma 79.6 in the discrete case. $\square$

Remark 79.10 (The change from the earlier amplification theorem [10]). The order-unit theorem in [56] uses a subgroup avoiding nonzero ordinary integers in order to obtain an injective map *before* quotienting its domain. Here there is no reason to avoid those integers. They are precisely the kernel, and their countability is harmless. Taking the principal convex hull makes an order unit available without any local compactness, vector structure, or divisibility assumption.

[write] Lemma 79.6 is the first assertion of lemma 76.2 and of source 08's lemma 46.1. Theorem 79.7 generalizes part (1) of source 08's theorem 46.2: it needs neither an integer-free subgroup nor an order unit, and it holds in every ordered ring.

79.2 Arithmetic rigidity without local compactness [10]

Source 10's main theorem is corollary 78.4; its proof is a second proof of that corollary.

Second proof of corollary 78.4 [10]. By proposition 79.1, some principal convex subgroup H_a is open. Its cosets are pairwise disjoint nonempty open sets, so ccc implies that A/H_a is countable. By theorem 79.7 and lemma 79.6, there is an injection

$$A/\mathbb{Z}1_A \hookrightarrow A/H_a.$$

Thus $A/\mathbb{Z}1_A$ is countable. Each coset of $\mathbb{Z}1_A$ is countable, so A is countable.

A countable Hausdorff Baire group must be discrete. Otherwise it has no isolated point, since any isolated point can be translated to 0 and then to every other point. Every singleton would consequently be nowhere dense, and their countable union would make A meager in itself. This contradicts the Baire hypothesis. $\square$

79.3 The cardinal theorem for general ordered rings [10]

The argument has a useful form before discreteness of the order is imposed. It measures the topological size needed to accommodate the quotient by ordinary-integer-bounded elements.

For an infinite space X , let $\text{cell}(X)$ be the supremum of the cardinalities of disjoint nonempty open families, $\text{dens}(X)$ the least cardinality of a dense subset, and $\text{ext}(X)$ the supremum of the cardinalities of closed discrete subsets. We use the usual infinite-cardinal convention for the Lindelöf degree $L(X)$: every open cover has a subcover of size at most $L(X)$.

Theorem 79.11 (Relative cardinal obstruction [10]). *Let A be an ordered ring with a Hausdorff additive-group topology. Suppose some neighborhood of 0 is order bounded. Then there is an open convex additive subgroup H such that*

$$A/\text{Fin}(A) \hookrightarrow A/H \tag{79.9}$$

is an ordered additive embedding. In particular,

$$|A/\text{Fin}(A)| \leq |A/H| \leq \text{cell}(A) \leq \text{dens}(A). \tag{79.10}$$

Moreover, A has a closed discrete subset of cardinality $|A/H|$. The order-bounded-neighborhood hypothesis follows from a Baire group topology and a Baire-property positive cone.

Proof. Choose $a \geq 1$ so that the neighborhood lies in $[-a, a]$, and put $H = H_a$. The proof of [proposition 79.1](#) shows that H is open without needing Baire once the neighborhood bound is given. Apply [theorem 79.7](#). The cosets form a disjoint nonempty open family, proving the middle bound. A dense set must meet every member of any such family, proving the last bound.

Choose one representative from each coset and call the resulting set T . Each point of T is isolated in T by its coset. To see that T is closed, take $x \notin T$. Its coset contains exactly one selected point t . Since the group is Hausdorff, the coset with t removed is an open neighborhood of x disjoint from T . Thus T is closed and discrete, and $|T| = |A/H|$. The last sentence is [lemma 77.2](#). $\square$

Corollary 79.12 (Cardinality of the finite part [10]). *For a Hausdorff Baire ccc additive presentation of an ordered ring with a Baire-property positive cone,*

$$|A| = |\text{Fin}(A)| = \max(\aleph_0, |[0, 1]_A|). \quad (79.11)$$

Consequently a countable unit interval forces a countable, discrete additive group.

Proof. By [theorem 79.11](#), $A/\text{Fin}(A)$ is countable. The group $\text{Fin}(A)$ contains $\mathbb{Z}1$ and is infinite. A countable union of its cosets has the same cardinality as $\text{Fin}(A)$, giving the first equality.

Every ordinarily bounded element lies in an ordinary integer translate of $[0, 1]$. Indeed, inside a fixed finite interval $[-n, n]$, one can choose the largest ordinary integer k not exceeding that element, using only a finite search. Hence

$$\text{Fin}(A) = \bigcup_{k \in \mathbb{Z}} (k + [0, 1]_A).$$

This proves the upper bound for the second equality; the reverse bound follows from the inclusions of $[0, 1]$ and $\mathbb{Z}1$ in $\text{Fin}(A)$. A countable Hausdorff Baire group is discrete as in [corollary 78.4](#). $\square$

This relative formulation does not exclude the usual real field: its finite part is all of $\mathbb{R}$, and its unit interval already has its full cardinality. Nor does it exclude every non-Archimedean ordered field. Its conclusion is an exact restriction on how much cardinality can occur beyond the finite part in a ccc presentation.

Theorem 79.13 (Maximal topological size [10]). *Let A be an uncountable discretely ordered ring equipped with a Hausdorff Baire additive-group topology and a Baire-property positive cone. Then some open convex additive subgroup H has*

$$|A/H| = |A|. \quad (79.12)$$

There is a closed discrete subset of A of size $|A|$, and

$$\text{cell}(A) = \text{dens}(A) = \text{ext}(A) = L(A) = |A|. \quad (79.13)$$

Proof. Every coset of $\mathbb{Z}1$ is countable. Since A is uncountable, cardinal arithmetic gives $|A/\mathbb{Z}1| = |A|$. By [theorem 79.11](#), an open H satisfies $|A| \leq |A/H| \leq |A|$, proving (79.12) and the existence of the closed discrete transversal.

The cosets witness $\text{cell}(A) \geq |A|$, and no disjoint family can have more members than there are points. A dense set must meet every coset, so $\text{dens}(A) = |A|$. The closed discrete transversal gives the equality for extent. Finally, the coset cover has no proper subcover and has $|A|$ members, while any open cover of a space of size $|A|$ has a subcover of size at most $|A|$, by choosing one covering set per point. This proves the equality for $L(A)$. $\square$

Thus a nonseparable counterexample cannot be merely a little larger than a Polish space in its supply of disjoint open sets. It must have as many disjoint open cosets, and as many points in one closed discrete set, as it has elements altogether. The explicit locally compact example in [section 84.2](#) attains this conclusion.

[write] [Theorem 79.13](#) and source 12's [theorem 78.3](#) overlap: source 12 proves $c(R) = d(R) = |R|$ for every discretely ordered ring with a Hausdorff Baire topology and a Baire-property cone, with no uncountability assumption; source 10 adds, for uncountable rings, an open convex subgroup of full index, a closed discrete set of full size, and the equalities for extent and Lindelöf degree. Source 10's cell and dens are source 12's c and d .

79.4 A complementary bounded-subgroup amplification [10]

The principal-hull theorem is the shortest route to the main result. A different amplification statement retains information about multiple independent additive copies and is useful for algebraic certificates. Here a subgroup is bounded by one element, which is stronger than having an order unit.

Theorem 79.14 (Bounded-subgroup amplification [10]). *Let A be a discretely ordered ring, and let $H \leq (A, +)$ satisfy $|h| \leq c$ for every $h \in H$, where $c \geq 0$. If $b > c$, then $bH \cap H = \{0\}$ and the additive map*

$$H \longrightarrow A/H, \quad h \longmapsto bh + H$$

is injective. With $a = c + 2$, the finite-sum map

$$\Psi : \bigoplus_{j \geq 0} H \longrightarrow A, \quad (h_j) \longmapsto \sum_{j \geq 0} a^j h_j \quad (79.14)$$

is an ordered additive embedding when the source is ordered by the sign of its highest-index nonzero coordinate. Its restriction to the positive-index summands remains injective after quotienting by H .

Proof. If $h \neq 0$, then $|h| \geq 1$, so $|bh| = b|h| \geq b > c$. Thus $bh \notin H$. This proves the first assertion and the coset injection.

For the second, take a nonzero finite tuple and let k be the largest index with $h_k \neq 0$. If $k = 0$, the conclusion is immediate. For $k > 0$, the magnitude of its lower terms is at most

$$B = c \sum_{j < k} a^j.$$

The geometric-sum identity and $a - 1 = c + 1$ give

$$(a - 1)B = c(a^k - 1) < (a - 1)a^k.$$

Indeed, subtracting the middle expression from the right gives $a^k + c > 0$. Cancelling $a - 1 > 0$ yields $B < a^k$. Since $|h_k| \geq 1$, the leading term has magnitude $a^k |h_k| \geq a^k > B$. The sum is therefore nonzero and has the sign of h_k . Distributivity gives additivity; applying the sign conclusion to differences gives the order embedding.

If a positive-index sum belongs to H , subtract that element as the zeroth coefficient in (79.14). Injectivity of Ψ forces every coefficient to vanish. $\square$

No divisibility, convexity, ideal property, or integer-disjointness is assumed of H . The source is an additive direct sum, not a polynomial ring with coefficient ring H . Its ordered embedding gives more structure than a cardinal inequality, although countable direct sums do not increase an infinite cardinal.

79.4.1 Two initial approaches become corollaries

A non-Archimedean topological group has a basis at the identity of open subgroups; this terminology is standard in Polish group theory [53]. If the additive group in [corollary 78.4](#) is non-Archimedean, a bounded neighborhood contains an open bounded subgroup H . Then [theorem 79.14](#) injects H into the countable coset space A/H , yielding countability. This is a valid proof in that case, but the principal-hull argument makes the non-Archimedean assumption unnecessary.

Likewise, an open real topological vector subgroup can be treated by order-unit amplification. If $U \subseteq [-u, u]$ is a neighborhood in that subgroup, then $v/n \rightarrow 0$ implies $|v| \leq nu$ for some ordinary n , making u an order unit. Divisibility makes each nonzero member larger in magnitude than every ordinary integer. Multiplication by u then injects the subgroup into its additive quotient, as in [56]. This works in infinite as well as finite dimension. Again, [corollary 78.4](#) is stronger and needs neither vector structure nor an open subgroup of a prescribed kind.

These two arguments help explain the discovery path, but they should not be mistaken for the endpoint. Arbitrary Polish additive groups, including mixed or otherwise unclassified ones, are covered by the principal-hull proof.

[write] [Theorem 79.14](#) is a variant of parts (2) and (3) of source 08's [theorem 46.2](#), with a bound by one element in place of an integer-free subgroup with an order unit.

80 Arbitrary self-embeddings have bounded unit images

The topological obstruction has an interpretation that does not mention multiplication. It restricts purely algebraic ordered self-embeddings.

Let G be an ordered abelian group with a least positive element e . Write $\text{Emb}_{+,<}(G)$ for the set of injective additive maps $f : G \rightarrow G$ that preserve strict order. The maps are not required to preserve e , nor to be continuous or measurable. Define the unit-image spectrum

$$\mathcal{U}(G) = \{f(e) : f \in \text{Emb}_{+,<}(G)\}.$$

This terminology refers only to a distinguished least positive element; G need not be a ring.

Theorem 80.1 (Bounded unit-image spectrum). *Suppose G is uncountable and has a Hausdorff Baire ccc group topology, an invariant total order, and a positive cone with the Baire property. If e is its least positive element, then $\mathcal{U}(G)$ is bounded above in G . In fact, if $W \subseteq (-a, a)$ is the neighborhood from [lemma 77.2](#), then*

$$f(e) \leq 2a \quad \text{for every } f \in \text{Emb}_{+,<}(G). \quad (80.1)$$

Proof. Suppose $f(e) > 2a$. For distinct $x, y \in G$, discreteness gives $|x - y| \geq e$. Additivity and order preservation imply

$$|f(x) - f(y)| = f(|x - y|) \geq f(e) > 2a.$$

Consequently the sets $f(x) + W$, indexed by all $x \in G$, are pairwise disjoint by the same difference argument as in [lemma 78.1](#). This contradicts ccc because G is uncountable. $\square$

Corollary 80.2 (Cofinal unit movement is impossible). *An uncountable discretely ordered abelian group with a family of ordered additive self-embeddings whose values at the least positive element are cofinal cannot have a Hausdorff Baire ccc group topology with a Baire-property positive cone.*

Proof. A cofinal subset cannot be bounded above in a nontrivial ordered group, since adding a positive element always produces a larger element. Apply [theorem 80.1](#). $\square$

For a discretely ordered ring, the maps $x \mapsto bx$ for $b > 0$ are ordered additive embeddings, and their values at 1 range over the entire positive cone. The main ccc obstruction is therefore a special case of this cofinal-unit-movement obstruction. The formulation is useful for other algebraic structures equipped with an expansive family of endomorphisms, even before a ring structure is supplied.

80.1 A spectrum can be infinite and still bounded

Consider

$$G_{\mathbb{R}} = \mathbb{R} \overrightarrow{\times} \mathbb{Z}, \quad e = (0, 1), \quad (80.2)$$

with the product topology $\mathbb{R}_{\text{usual}} \times \mathbb{Z}_{\text{discrete}}$ and lexicographic order in which the real coordinate is more significant. This is an uncountable Polish ordered group with least positive element e . Its cone is

$$(\mathbb{R}_{>0} \times \mathbb{Z}) \cup (\{0\} \times \mathbb{Z}_{>0}),$$

a Δ_2^0 set.

For each ordinary $k \geq 1$, the map $(r, n) \mapsto (r, kn)$ is an ordered additive embedding. Thus $\mathcal{U}(G_{\mathbb{R}})$ is infinite, although all these images are less than $(1, 0)$. In fact these are exactly the possible images of e . To see this, an additive map from the divisible group $\mathbb{R}$ to $\mathbb{Z}$ is zero. Therefore an ordered additive embedding restricts on $\mathbb{R} \times \{0\}$ to $r \mapsto (\alpha(r), 0)$, where α is an increasing additive injection. It follows that $\alpha(r) = \lambda r$ for some $\lambda > 0$: first prove the identity on rationals, then squeeze an arbitrary real between rationals. Write $f(e) = (b, k)$. Positivity excludes $b < 0$. If $b > 0$, choose $r > 0$ with $\lambda r < b$; then $(r, 0) > e$ but $f(r, 0) < f(e)$, a contradiction. Thus $b = 0$ and $k \geq 1$. Hence

$$\mathcal{U}(G_{\mathbb{R}}) = \{(0, k) : k \in \mathbb{Z}_{>0}\}. \quad (80.3)$$

This illustrates why [theorem 80.1](#) gives an *order bound*, not a finite bound on the number of embeddings or possible unit images.

80.2 All four clauses of the bounded spectrum [10]

Source 10 proves [theorem 80.1](#) as clause (3) of the following theorem; clauses (1), (2) and (4) are additional. Its own section reads [10]:

The proof uses less than a ring multiplication. This gives an independent result about ordered additive groups and explains which part of multiplication is incompatible with the topology.

Definition 80.3 ([10]). For a discretely ordered abelian group $(G, +, <, e)$ with least positive e , define the *unit-image spectrum*

$$\mathcal{U}(G) = \{f(e) : f : G \longrightarrow G \text{ is an additive order embedding}\}.$$

No continuity or measurability is required of f . For $a > 0$, define $H_a = \{x : |x| \leq na \text{ for some } n \in \mathbb{N}\}$.

Theorem 80.4 (Bounded unit-image spectrum [10]). *Let G be an uncountable discretely ordered abelian group with a Hausdorff Baire ccc group topology and a Baire-property positive cone. Then:*

1. If $f : G \rightarrow G$ is an additive order embedding and $a = f(e)$, there is an ordered additive embedding $G/\mathbb{Z}e \hookrightarrow G/H_a$.
2. For every $a \in \mathcal{U}(G)$, the subgroup H_a is not open.
3. There is $b > 0$ such that $0 < a < b$ for every $a \in \mathcal{U}(G)$.
4. Each subgroup H_a with $a \in \mathcal{U}(G)$ is meager.

Proof. For the first assertion use $x \mapsto f(x) + H_a$. Additivity and order reflection imply

$$|f(x)| \leq na \iff f(|x|) \leq f(ne) \iff |x| \leq ne.$$

The ordinarily bounded elements of a discretely ordered abelian group are exactly $\mathbb{Z}e$, by the finite-interval proof of [lemma 79.6](#), which uses no multiplication. The kernel is therefore $\mathbb{Z}e$. The sign proof for the quotient is the same as in [theorem 79.7](#).

If H_a were open, ccc would make G/H_a countable. The injection would make $G/\mathbb{Z}e$, and then G , countable, contrary to hypothesis. For the third assertion, the local-bound lemma provides $b > 0$ with H_b open. If $a \geq b$, then $H_b \subseteq H_a$, so H_a is open. Thus no such a can lie in the spectrum.

Finally, each interval $[-na, na]$ has the Baire property, since it is formed from translates of the positive cone, their complements, and singletons. Therefore H_a , their countable union, has the Baire property. A subgroup with the Baire property that is nonmeager in a Baire group is open: if it is comeager in an open set O , the usual intersection argument shows that $O - O$ is contained in the subgroup. This is an open neighborhood of zero. Since the second assertion excludes openness, H_a is meager. $\square$

The final assertion concerns *each* hull separately. It does not say that the spectrum itself, or the union of all these hulls, is meager. Such a union may have uncountably many terms.

Every positive scalar in an ordered ring gives an additive order embedding by multiplication. Thus its unit-image spectrum contains all positive elements. The bounded-spectrum theorem prohibits this in an uncountable Baire ccc discrete ordered group, independently of any compatibility among the individual embeddings. Associativity and the existence of one binary multiplication are not needed for this reduct obstruction.

This is also a useful self-embedding result for a set-sized integer-like ordered group. A Polish presentation may allow many discontinuous additive order embeddings, but they cannot send the least positive element to arbitrarily large scales. It is not a theorem about field self-embeddings of **No**: those fix 1, and **No** is not discretely ordered in the first place.

[write] [Corollary 80.2](#) contains source 15's [theorem 119.2](#) of [Part IX](#): a positive endomorphism T with $T(e) > 0$ is an ordered additive embedding, so endomorphisms with $T_a(e) = a$ for every $a > 0$ give cofinal unit images, and [Part VI](#) needs only a Baire-property cone and no regularity of the maps (note in [section 119](#)). The exact spectrum [\(80.3\)](#) extends [Part I's theorem 11.1](#) (unital endomorphisms of $M_{\mathbb{R}}$) to all ordered self-embeddings of $G_{\mathbb{R}} = \mathbb{R} \overrightarrow{\times} \mathbb{Z}$, not necessarily unital.

81 Ring expansions, positive semirings, and Presburger arithmetic

Corollary 81.1 (No ordered-ring expansion). *Let G be an uncountable ordered abelian group with least positive element e , a Hausdorff Baire ccc group topology, and a positive cone having the Baire property. No multiplication on the set G makes $(G, +, \cdot, 0, e, <)$ a discretely ordered unital ring.*

Proof. Such a multiplication would satisfy the hypotheses of [corollary 78.4](#), contradicting uncountability. Equivalently, its positive left multipliers would contradict [corollary 80.2](#). $\square$

The multiplication is ruled out algebraically relative to the prescribed addition, order, and unit. The statement is stronger than saying that a particular candidate multiplication is not Borel.

[10] Source 10 states this corollary for Polish topologies (its Corollary 5.3, `pma:br:cor:noexp`) and adds:

This excludes the existence of the multiplication itself, not merely its continuity or Borel measurability. It is therefore stronger than a regularity obstruction on the specified additive presentation. It does not, however, remove the topological-group hypothesis or the Baire condition on order.

Theorem 81.2 (No positive-semiring expansion). *Under the hypotheses of [corollary 81.1](#), let $M = G_{\geq 0}$. There is no associative unital semiring multiplication on M with its given addition, zero, and unit e .*

Proof. Suppose there were such a multiplication. Every $g \in G$ is a difference of two elements of M , and addition on M is cancellative. Extend the multiplication to differences by

$$(a - b)(c - d) = (ac + bd) - (ad + bc), \quad a, b, c, d \in M. \quad (81.1)$$

We verify that it is independent of representatives. If $a - b = a' - b'$, then $a + b' = a' + b$ in M . Multiplying this identity on the right by $c \in M$ and using distributivity gives $ac + b'c = a'c + bc$, so $ac - bc = a'c - b'c$ in G . The same holds with d ; subtracting gives independence of the first difference. Left distributivity proves independence of the second. Expanding finite products proves associativity and both distributive laws on G ; e remains the unit. Thus this is a unital ring extension.

Products of nonnegative elements are nonnegative because they belong to M . For strictly positive $x, y \in M$, the least-positive-element property gives $x = e + u$ and $y = e + v$ with $u, v \in M$. Therefore

$$xy = e + u + v + uv \geq e > 0.$$

The extended ring has the original discrete order. This contradicts [corollary 81.1](#). $\square$

This proof does not assume commutative semiring multiplication: the formula and positivity argument still work, and the main packing theorem also has the required noncommutative version. Additive cancellation is available because M is a cone in a group.

81.1 Application to signed Presburger models

A signed Presburger model is a model of the complete ordered additive theory of the integers. In particular, it is an ordered abelian group with a least positive element 1 and the standard-integer division-with-remainder laws. We use only the ordered-group and least-positive-element properties in the obstruction; quantifier elimination and induction are unnecessary.

Corollary 81.3. *An uncountable signed Presburger model with a Polish additive-group topology and a Borel positive cone has no compatible unital ordered-ring expansion. Its nonnegative cone has no unital semiring expansion with its prescribed addition and unit.*

Proof. Apply [corollary 81.1](#) and [theorem 81.2](#). □

The group $G_{\mathbb{R}}$ in (80.2) is a concrete example of the additive behavior that survives. For a fixed ordinary $m > 0$, divide the real coordinate by m and use the ordinary residue of the integer coordinate. This provides the familiar split signed Presburger example; its positive counterpart was studied in the earlier report [50]. The new general theorem explains the obstruction to multiplication independently of its particular coordinate system.

81.2 The group-completion boundary

For an abstract cancellative commutative monoid, the algebraic Grothendieck group always exists. Given a *topology* on the monoid, several additional questions arise: whether the difference quotient is Hausdorff, whether addition and inversion are continuous, whether the original cone embeds topologically, and whether the resulting group is Baire and ccc. None is supplied by the purely algebraic construction (81.1).

A useful conditional consequence is therefore precise. If a positive arithmetic presentation admits an order-compatible embedding as the nonnegative cone of a Hausdorff Baire ccc topological group with a Baire-property positive cone, and its carrier is uncountable, then no unital semiring multiplication can extend its addition. But without that group-topology hypothesis, this manuscript makes no such general claim. The classical literature on topological semigroup embeddings [48] is relevant to that remaining boundary; no unverified embedding criterion from that literature is being used here.

81.3 Presburger arithmetic: presentations but no expansions [10]

81.3.1 Signed models and the universal obstruction

A $\mathbb{Z}$ -group is a discretely ordered abelian group with named least positive element 1 in which every element has a unique standard remainder modulo each positive ordinary integer m :

$$\forall g \exists!(q, r) \in G \times \{0, 1, \dots, m-1\} : \quad g = mq + r1, \quad (81.2)$$

The usual interpretation is as a model of signed Presburger arithmetic. Cooper elimination reduces formulas to inequalities and congruences; the relevant finite-residue mechanism is the one formalized in ProveIt [54]. For clarity, the transfer argument needed for this abstract characterization is included in [appendix P](#).

Corollary 81.4 (All uncountable Polish signed Presburger groups [10]). *Let G be an uncountable $\mathbb{Z}$ -group with a Polish additive-group topology and a Baire-property positive cone. It has no expansion to a discretely ordered ring with the same addition, order, and named 1. The proposed multiplication may be arbitrary and need not be definable.*

Proof. Apply [corollary 81.1](#). The standard division axioms of a $\mathbb{Z}$ -group are not needed for this obstruction. □

The last observation is useful: the obstruction belongs to the weaker ordered-group reduct. Presburger quantifier elimination explains why many such reducts exist and are well behaved, but the contradiction requires no quantifier elimination or induction.

81.3.2 Divisible-plus-integer examples

Let D be a divisible linearly ordered abelian group. Put

$$G_D = D \overrightarrow{\times} \mathbb{Z}, \quad 1_G = (0, 1), \quad (81.3)$$

where the first coordinate is compared first. The element $(0, 1)$ is least positive. If $g = (d, n)$ and $m \geq 1$ is ordinary, write $n = mq + r$ with $0 \leq r < m$. Then

$$(d, n) = m(d/m, q) + r(0, 1). \quad (81.4)$$

Uniqueness follows from torsion-freeness and the ordinary remainder uniqueness in $\mathbb{Z}$. Thus G_D is a $\mathbb{Z}$ -group.

Three useful choices are

$$D = \mathbb{R}, \quad D = \mathbb{Q}^{\mathbb{N}}, \quad D = \ell^2(\mathbb{N}, \mathbb{R}).$$

Use the usual order on $\mathbb{R}$, and on either sequence space compare at the first nonzero coordinate. Use the ordinary real topology in the first case, the product of discrete rational topologies in the second, and the norm topology in the third. These are divisible ordered groups; the order is defined additively and need not be a lattice order in a functional-analytic sense.

Each G_D is an uncountable Polish additive group with a Borel order. For the sequence cases its strict positive cone is

$$P_D = \bigcup_{j \geq 0} \{d : d_0 = \dots = d_{j-1} = 0, d_j > 0\}. \quad (81.5)$$

In $\mathbb{Q}^{\mathbb{N}}$ each displayed condition is clopen. In ℓ^2 , each term is the intersection of a closed set with an open set and is F_σ , since the space is metrizable. Thus P_D is F_σ . Its complement is $-P_D \cup \{0\}$, also F_σ , so P_D is Δ_2^0 . The corresponding cone in G_D is

$$(P_D \times \mathbb{Z}) \cup (\{0\} \times \mathbb{Z}_{>0}),$$

again Δ_2^0 . The same conclusion is immediate for $D = \mathbb{R}$.

These constructions, including the real-coordinate model, belong to the existing Presburger project context and are not claimed as new here [56]. What [corollary 81.4](#) adds is a theorem covering *all* uncountable Polish signed models with the stated order regularity, not only these examples or the locally compact ones.

81.3.3 Positive cones still give Polish arithmetic monoids

Since the nonnegative cone $G_D^{\geq 0}$ is G_δ in a Polish space, it is Polish with its subspace topology. Addition restricts continuously, and this cone is a model of nonnegative Presburger arithmetic. For $D = \mathbb{R}$ it is explicitly

$$M_{\mathbb{R}} = (\mathbb{R}_{>0} \times \mathbb{Z}) \cup (\{0\} \times \mathbb{N}). \quad (81.6)$$

Thus additive Polish arithmetic is not ruled out. What fails is its ordered-ring expansion on the corresponding signed group.

For these explicit cones, even an ordered semiring multiplication compatible with the given addition and order is impossible when its ring of differences uses the displayed signed ordered group. The extension of such a multiplication to differences is algebraic; the

compatible Polish signed topology is already provided by G_D . This is more specific than asserting the same conclusion for every Polish monoid, a step not justified by the present proof.

[write] [Corollary 81.4](#) assumes a Baire-property cone where source 12's [corollary 81.3](#) assumes a Borel one; the semiring clause of the latter is [theorem 81.2](#). The models G_D with $D = \mathbb{R}$ and $D = \mathbb{Q}^{\mathbb{N}}$ are Part I's G_D ([section 3](#)); their cones are $M_{\mathbb{R}}$ and $M_{\mathbb{B}}$. Source 10 credits these constructions to source 08 [[56](#)]; they are source 05's, [theorem 3.1](#). The case $D = \ell^2(\mathbb{N}, \mathbb{R})$ is new here.

81.4 Positive monoids, group completion, and the exact boundary [10]

The group structure is used twice in the category proof: inversion exchanges P and $-P$, and reflection $x \mapsto z - x$ is a homeomorphism. Neither operation is available on an arbitrary positive monoid. These are substantive uses, not notational conveniences.

For a cancellative commutative monoid M , its algebraic Grothendieck group can be formed from pairs (x, y) , with

$$(x, y) \sim (x', y') \iff x + y' = x' + y,$$

interpreted as formal differences. A semiring multiplication extends algebraically to this group by

$$(x - y)(u - v) = (xu + yv) - (xv + yu).$$

However, none of these formulas proves that the quotient has a compatible Hausdorff Polish group topology, or that the original positive cone embeds with its original topology. Quotient topologies can fail the required separation or completeness properties.

Proposition 81.5 (Conditional completion obstruction [10]). *Let M be an uncountable cancellative ordered semiring whose signed ring of differences A is discretely ordered. Suppose A admits a Polish additive-group topology with a Baire-property positive cone. Then such a semiring cannot exist. In particular this applies whenever a specified topological group completion supplies those properties.*

Proof. The natural map from M to A is injective by cancellation, so A is uncountable. The ring then contradicts [corollary 78.6](#). $\square$

[write] (Batch 90.) The completion hypothesis can be made intrinsic. By [Part XII](#), a Polish cone with continuous addition has such a completion exactly when its partial subtraction is continuous, so [proposition 155.6](#) [merge]: on the nonnegative part of an uncountable discretely ordered ring, every Polish topology making addition continuous makes partial subtraction discontinuous, whatever the multiplication.

For the explicit models $G_D^{\geq 0}$ in [section 81.3](#), the signed topology is already available. For an arbitrary positive model, it is not. Therefore the present proof does not by itself remove the Borel multiplication assumption from Glazer's positive-model theorem. The relationship between those results is a comparison of different hypothesis sets, not an unconditional strengthening in every setting.

Glazer's Question 2 asks for an uncountable Polish Presburger model with continuous addition [1]. The additive examples recalled here belong to the already developed affirmative constructions in the project; they are not presented as a new resolution in this manuscript. Our new conclusion is their universal signed ordered-ring non-expandability under Baire comparison, and the general rigidity mechanism behind it.

82 Borel structures, Polishability, and meager embeddings

Definition 82.1. For this section, a *Polish group presentation of a fixed Borel structure* is a Polish additive-group topology whose Borel sets are exactly the specified Borel sets. This is a requirement, not an automatic consequence of having a standard Borel group.

Theorem 82.2 (No compatible Polish group presentation). *Let an uncountable discretely ordered ring have a specified standard Borel structure in which its positive cone is Borel. There is no Polish additive-group topology generating that Borel structure. No Borelness of multiplication is assumed.*

Proof. In such a topology the cone would remain Borel, so [corollary 78.5](#) would force the carrier to be countable. $\square$

The result distinguishes two failures. A displayed metric may merely fail to be complete; [theorem 82.2](#) rules out *every* compatible Polish group topology preserving the prescribed Borel sets. It does not rule out a Polish topology after the measurable structure and the order's regularity have been changed.

Theorem 82.3 (Borel ordered-ring subgroups are meager when uncountable). *Let K be a Polish abelian group. Suppose $A \subseteq K$ is an uncountable Borel subgroup, equipped algebraically with a discretely ordered ring structure whose addition is the inherited group addition. If its positive cone is Borel as a subset of K , then A is meager in K .*

Proof. If A were nonmeager, [lemma 77.1](#) would show that $A - A = A$ contains a neighborhood of 0. Thus A is open. An open subgroup is also closed, since its complement is a union of open cosets. Consequently A , with the inherited topology, is a Polish group. Its positive cone is Borel, so [corollary 78.5](#) makes A countable, contradicting the hypothesis. $\square$

Corollary 82.4 (Closed subrings and integer parts). *In the setting of [theorem 82.3](#), a closed subgroup admitting a discretely ordered ring structure with Borel cone must be countable and discrete in its subspace topology. More generally, an uncountable Borel integer part of an ordered field with a Polish additive topology and Borel order must be meager in the ambient additive group.*

Proof. A closed subgroup of a Polish group is Polish, so the first statement is [corollary 78.5](#). An integer part is a discretely ordered subring; its positive cone is the intersection of its carrier with the field's positive cone. Thus the second statement follows from [theorem 82.3](#). $\square$

An integer part here means a discretely ordered subring I such that for every field element x there is $i \in I$ with $i \leq x < i + 1$. The approximation clause is not needed for this corollary; only discrete order and subring structure are used. This is another instance where weaker algebraic hypotheses suffice than full arithmetic.

83 A parallel Haar-measure theorem

The bounded-neighborhood mechanism can also be obtained from measure rather than category. This gives a separate improvement of the order regularity hypothesis in the locally compact setting. We use a regular Haar measure and its completion; no assumption about restrictions to null subgroups is imposed.

Lemma 83.1 (Positive-measure differences). *Let G be a Hausdorff locally compact abelian group with regular Haar measure μ . If a compact set K has $\mu(K) > 0$, then $K - K$ contains a neighborhood of 0.*

Proof. Choose an open $U \supseteq K$ of finite measure with $\mu(U) < 2\mu(K)$. Compactness and continuity of addition give a neighborhood W of 0 such that $K + W \subseteq U$. For $h \in W$, the sets K and $K + h$ both lie in U , so

$$\mu(K \cap (K + h)) \geq 2\mu(K) - \mu(U) > 0.$$

Thus they meet, and $h \in K - K$. □

This is the compact-set form of the classical Steinhaus difference-set principle [46]; the displayed argument establishes the locally compact version actually needed here.

Lemma 83.2 (Haar-measurable order gives a bounded neighborhood). *Let G be a nontrivial ordered abelian group with a Hausdorff locally compact group topology. If its positive cone is measurable for the completion of a regular Haar measure, then 0 has an order-bounded open neighborhood.*

Proof. The discrete case is immediate. Otherwise $\mu(\{0\}) = 0$: an atom at 0, together with translation invariance and a compact neighborhood of finite measure, would force that neighborhood to be finite, and the Hausdorff group would then be discrete.

Choose a compact set of positive finite measure. Its partition by P , $-P$, and $\{0\}$ shows that one of the first two pieces has positive measure. Invert the negative piece if necessary. Inner regularity for completed Haar-measurable sets provides a compact $B \subseteq P$ with $0 < \mu(B) < \infty$.

All integrations that follow may be performed in a sigma-compact open subgroup containing B . Such a subgroup exists: take a compact identity neighborhood, adjoin B and $-B$, and take the subgroup generated by their union. It is a countable union of compact sets and contains an open identity neighborhood. Haar measure restricted to it is sigma-finite. Tonelli's theorem and translation invariance give

$$\begin{aligned} \int_G \mu(B \cap (a - B)) d\mu(a) &= \int_G \int_G \mathbf{1}_B(x) \mathbf{1}_B(a - x) d\mu(x) d\mu(a) \\ &= \mu(B)^2 > 0, \end{aligned} \tag{83.1}$$

where the integrands vanish outside that subgroup and its relevant compact sumsets. Hence there is an a such that $K = B \cap (a - B)$ has positive measure. It is compact. For any $x \in K$, both x and $a - x$ are positive, so $0 < x < a$ and in particular $a > 0$. Thus $K \subseteq (0, a)$ and $K - K \subseteq (-a, a)$. Apply lemma 83.1. □

Theorem 83.3 (Haar-measurable arithmetic rigidity). *Let a discretely ordered ring have a Hausdorff locally compact additive-group topology. If its positive cone is measurable for the completion of a regular Haar measure, then*

$$c(R) = d(R) = |R|.$$

If the topology is ccc, it is countable and discrete. In particular, this conclusion holds for second-countable locally compact additive-group topologies with Haar-measurable order.

Proof. Combine lemmas 78.1 and 83.2 with (76.2). Under ccc the ring is countable. Locally compact Hausdorff spaces are Baire, so the same singleton argument as in corollary 78.4 makes it discrete. □

The earlier report asked how to weaken Borel order without assuming that ambient measurability automatically passes to every closed subgroup [49, Section 14]. This proof avoids that issue: it works directly with a positive-measure compact subset of the ambient cone. The sole restriction used for Tonelli is to an *open* sigma-compact subgroup containing

that compact subset. No conclusion about measurability on a null closed subgroup is required.

[write] [Theorem 83.3](#) answers source 08's [question 57.4](#) for ambient Haar measurability, and [lemma 77.2](#) answers it for the Baire property: in both routes the regularity of the cone for the given topology suffices, and no condition on restrictions to compact subgroups or closed real subspaces is needed. [Theorem 83.3](#) also strengthens [theorem 44.1](#) from Borel to Haar-measurable order, and removes second countability from its cardinal part.

84 Sharpness: three topologies on one polynomial ring

The same algebraic example isolates the role of the principal hypotheses. Let

$$A = \mathbb{Z} + t\mathbb{R}[t] = \left\{ z + \sum_{j=1}^N r_j t^j : z \in \mathbb{Z}, r_j \in \mathbb{R}, N \in \mathbb{N} \right\}. \quad (84.1)$$

Order a nonzero polynomial by the sign of its leading coefficient, so that t is larger than every ordinary real constant wherever that comparison is made in $\mathbb{R}[t]$. This is an ordered ring: leading coefficients multiply, and comparison is invariant under translation. If a positive element has degree zero, it is a positive integer; if it has positive degree, it exceeds every ordinary integer. Hence 1 is its least positive element. The cardinality of A is $\mathfrak{c}$.

[write] Source 10 prints the same three topologies on this ring (written $\mathbb{Z} + X\mathbb{R}[X]$) as its Examples 9.1, 9.3 and 9.4; their labels are second labels of the three subsections below. The ring is source 08's A_* ([example 54.1](#)) and source 06's $\mathbb{Z} + t\mathbb{R}[t]$ ([theorem 38.1](#)).

84.1 A separable metrizable ring that is meager in itself

Identify a polynomial with its coefficient sequence and give A the subspace topology from

$$\mathbb{Z}_{\text{discrete}} \times \mathbb{R}^{\mathbb{N}_{>0}}.$$

The higher-coordinate sequence has finite support. The ambient space is metrizable and second countable, so A is separable and metrizable. Addition and inversion are continuous. Multiplication is jointly continuous as well, since its n th coefficient is the finite polynomial expression

$$(fg)_n = \sum_{i=0}^n f_i g_{n-i}. \quad (84.2)$$

Continuity into the product follows coordinate by coordinate; the product lies again in the finite-support subring.

For $N \geq 0$, let A_N be the set of polynomials of degree at most N . It is closed: all higher coefficients vanish. It has empty interior, because a basic open set constrains only finitely many coordinates, and a nonzero term can be inserted at a still higher degree. Thus

$$A = \bigcup_{N \geq 0} A_N \quad (84.3)$$

is meager in itself.

Nevertheless the cone has low Borel complexity. For $d \geq 1$, the condition

$$f_j = 0 \text{ for all } j > d, \quad f_d > 0$$

is the intersection of a closed set and an open set. In a metrizable space every open set is F_σ , so this condition is F_σ . The positive degree-zero polynomials form a countable union of closed singletons. Taking the countable union over d proves that P is F_σ . Its complement is $(-P) \cup \{0\}$, also F_σ . Therefore P is Δ_2^0 .

This example has continuous multiplication, not merely arbitrary multiplication. It shows that the Baire hypothesis cannot be replaced by metrizability or separability. By [theorem 82.2](#), there is also no Polish additive-group topology generating the same Borel structure: the defect is not repaired by choosing another compatible complete group metric.

[10] Source 10 adds:

This example shows why a separable metrizable topology with Borel order is not enough, even when multiplication happens to be continuous. The failure occurs in the Baire hypothesis, not in the algebra. For an even smaller example, $\mathbb{Z}$ with its p -adic subspace topology is a countable, nondiscrete, non-Archimedean topological ring; every subset is Borel, but it is again meager in itself.

84.2 A locally compact Baire ring presentation without ccc

Decompose the additive group as

$$A = \mathbb{R}t \oplus D, \quad D = \mathbb{Z} \oplus \bigoplus_{j \geq 2} \mathbb{R}t^j, \quad (84.4)$$

and give it the product topology with $\mathbb{R}t$ usual and D discrete. This is a Hausdorff locally compact metrizable topological group and is Baire. It is not separable: the distinct sets $\mathbb{R}t + d$, $d \in D$, are $\mathfrak{c}$ disjoint open subsets.

Its cone is again Δ_2^0 . When the high-degree part of d is nonzero, its leading coefficient determines the sign of $rt + d$ independently of r . Such a coset is either wholly positive or wholly negative. When $d = z \in \mathbb{Z}$, positivity is determined by $r > 0$, or by $r = 0$ and $z > 0$. Thus the positive cone is the union of an open set and the closed set of positive integer points at $r = 0$. The negative cone has the analogous form. Metrizability gives the claimed Δ_2^0 complexity.

Choosing rational r in each open coset gives a dense set of size $\mathfrak{c}$. The disjoint cosets give the matching lower bound. Hence

$$c(A) = d(A) = |A| = \mathfrak{c},$$

exactly as [theorem 78.3](#) predicts. The ring multiplication is not continuous in this topology. Indeed, $t/n \rightarrow 0$ whereas multiplication by t sends that sequence to t^2/n , which does not converge to 0 in the discrete D coordinate. This is permitted by the theorem, which assumes nothing about multiplication.

[write] This is source 08's [example 54.1](#). [10] Source 10 adds the extent and the Lindelöf degree: by [theorem 79.13](#), or directly by choosing one point on each line, $\text{cell}(A) = \text{dens}(A) = \text{ext}(A) = L(A) = |A| = \mathfrak{c}$. It notes:

This is not asserted to be a topological ring. The continuity of addition is all the main theorem assumes, and multiplication in this example fails continuity for many fixed scalars; see [section 85](#).

84.3 A Polish additive topology with a non-Baire positive cone

Let $V = t\mathbb{R}[t]$, regarded only as a rational vector space. Its dimension is $\mathfrak{c}$: it contains a copy of the rational vector space $\mathbb{R}$, and its cardinality is $\mathfrak{c}$. The vector space $\mathbb{Q}^{\mathbb{N}}$ also has dimension $\mathfrak{c}$. To justify the latter assertion, its cardinality is $\mathfrak{c}$; over a countable field, an infinite-dimensional vector space has cardinality the maximum of its dimension and $\aleph_0$. A countable-dimensional one would be countable. These facts force the dimension to be $\mathfrak{c}$.

Using bases, choose a rational-linear isomorphism

$$\psi : V \longrightarrow \mathbb{Q}^{\mathbb{N}}.$$

Transport the Polish group topology of $\mathbb{Z}_{\text{discrete}} \times \mathbb{Q}_{\text{discrete}}^{\mathbb{N}}$ to the additive group $A = \mathbb{Z} \oplus V$ through $z + v \mapsto (z, \psi(v))$. The ring and its original leading-coefficient order remain algebraically unchanged. The new topology is Polish and makes addition and inversion continuous. By [corollary 78.5](#), its positive cone does *not* have the Baire property.

This is an existence construction using choice of vector-space bases, not a computable coding. It shows that regularity of order cannot be omitted from the theorem. No claim is made that multiplication is Borel in the transported presentation.

[write] This is a variant of source 08's [example 54.2](#), which transports $\mathbb{R}^d \times \mathbb{Z}$ and calls the order necessarily non-Borel; here the transported topology is that of $\mathbb{Q}^{\mathbb{N}} \times \mathbb{Z}$ and the conclusion is that the cone lacks the Baire property. [10] Source 10's version of the example ends:

The isomorphism is chosen using vector-space bases. It is not an algorithm, and the construction gives no Borel multiplication or regular comparison procedure. Its point is exact: additive Polishability alone does not prohibit uncountable discretely ordered rings. Order regularity is an indispensable part of the theorem.

84.4 Other indispensable distinctions

The ordinary real field is an uncountable Polish ordered ring with continuous operations and open positive cone. It does not contradict our theorem because it has no least positive element. Thus discrete order is essential.

[10] Source 10 adds:

The usual real field is an uncountable Polish topological ring with open positive cone. It escapes [corollary 78.4](#) because it has elements strictly between 0 and 1. In the relative theorem, its finite part is already all of the ring. This is the correct boundary, not a failure of Baire category or of the chain condition.

The group $G_{\mathbb{R}}$ from [\(80.2\)](#) is uncountable, Polish, locally compact, discretely ordered as a group, and has a Δ_2^0 positive cone. It lacks any compatible unital ordered-ring expansion by [corollary 81.1](#). Thus continuous addition and regular discrete order alone are not contradictory. It is the combination with an unrestricted positive multiplier family that creates the packing.

85 Continuous scalar maps beyond the separable setting [10]

Once ccc is imposed, the main theorem already gives discreteness. The following observation is instead meaningful for nondiscrete, necessarily non-ccc Baire presentations such as [section 84.2](#).

Proposition 85.1 (A bounded subring of continuous scalars [10]). *Let A be a discretely ordered ring with a nondiscrete Hausdorff Baire additive-group topology and a Baire-property positive cone. Define*

$$\text{Sc}_{\text{cont}}(A) = \{a \in A : L_a : A \rightarrow A, L_a(x) = ax, \text{ is continuous}\}.$$

Then $\text{Sc}_{\text{cont}}(A)$ is a unital subring containing $\mathbb{Z}1$, and for some $c > 0$ it is contained in $[-c, c]$.

Proof. The identities $L_{a+b} = L_a + L_b$, $L_{-a} = -L_a$, and $L_{ab} = L_a \circ L_b$ prove the subring assertion. Integer scalar maps are continuous because the additive operations are continuous. Choose a neighborhood $U \subseteq [-c, c]$ by local boundedness. If $|a| > c$, then for every nonzero x , discreteness gives $|ax| = |a||x| \geq |a| > c$. Hence $L_a^{-1}(U) = \{0\}$. Continuity of L_a would make $\{0\}$ a neighborhood and the group discrete. Therefore such a cannot belong to $\text{Sc}_{\text{cont}}(A)$. $\square$

Corollary 85.2 ([10]). *A discretely ordered ring with a Hausdorff Baire additive topology, a Baire-property positive cone, and continuous multiplication by every fixed scalar is discrete. No separability or ccc assumption is needed.*

Proof. If the topology were nondiscrete, [proposition 85.1](#) would bound all of A by one c , contradicting the existence of $c + 1$. $\square$

In [section 84.2](#), the continuous-scalar subring is exactly $\mathbb{Z}$. Every integer works. If a has positive degree k and leading coefficient $\alpha \neq 0$, the restriction of L_a to the open line $\mathbb{R}X$ has degree- $(k + 1)$ coefficient $r \mapsto \alpha r$ in the discrete factor D . This is not continuous, so L_a is not continuous. Thus the bounded-scalar conclusion is nonvacuous and can be exact.

We deliberately do not identify Borel scalar maps with continuous ones in this nonseparable discussion. Standard automatic-continuity statements for Borel homomorphisms between Polish groups do not license that substitution for an arbitrary nonseparable target. In the Polish case of the main theorem, every map is already continuous because the topology has been proved discrete.

[write] [Corollary 85.2](#) generalizes source 09's [theorem 62.1](#) in another direction: continuous scalar maps replace Borel ones, and no separability is assumed. Compare source 08's [theorem 53.4](#), which counts the Borel scalars of an integral-domain multiplication unrelated to the order.

86 Completion, units, and omnific applications

86.1 Completion may destroy discrete order algebraically

The additive completion of the coefficient-topology example in [section 84.1](#) is

$$\widehat{A} = \mathbb{Z} + t\mathbb{R}[[t]]. \quad (86.1)$$

The higher coefficients may now have infinite support. The topology is the full product topology, and [\(84.2\)](#) defines a jointly continuous multiplication. Every coefficient sequence is the limit of its finite truncations, so A is dense. Completeness of the product additive group gives the asserted completion.

One might try to extend the leading-degree discrete order to this Polish ring. The obstruction is stronger than failure of Borelness: *there is no discrete ring order on this completion at all.*

Lemma 86.1 (Units in a discretely ordered ring). *Every unit of a discretely ordered ring is 1 or -1 .*

Proof. If $uv = 1$ and $u > 0$, then $v > 0$, and discreteness gives $u, v \geq 1$. If $u > 1$, then $u \geq 2$, and hence $uv \geq 2$, a contradiction. Thus $u = 1$. If $u < 0$, apply the positive case to $-u$ and $-v$. $\square$

Proposition 86.2 (No discrete order on the coefficient completion). *The ring $\widehat{A} = \mathbb{Z} + t\mathbb{R}[[t]]$ admits no unital discrete ring order.*

Proof. The element $1 - t$ is a unit with inverse

$$1 + t + t^2 + \cdots.$$

This identity is a formal coefficient identity, not an analytic convergence assertion. It also follows from the exact finite identity

$$(1 - t) \sum_{j=0}^N t^j = 1 - t^{N+1} \quad (86.2)$$

by passing to the coefficient limit. But $1 - t$ is neither 1 nor -1 , contrary to [lemma 86.1](#). $\square$

Second proof [10]. Truncations show that the polynomial ring is dense in the product coefficient group of all such formal series. This product is completely metrizable, and the coefficientwise ring operations extend by the finite convolution formulas.

In any discretely ordered ring, the only units are 1 and -1 . Indeed, if $uv = 1$, then $|u|, |v| \geq 1$ and $1 = |u||v| \geq |u| \geq 1$, so $|u| = 1$. In $\widehat{A}$, however, the element $1 - X$ is a unit with inverse $\sum_{n \geq 0} X^n$. It is neither 1 nor -1 as a formal series. This contradicts the necessary unit property. $\square$

The example separates three issues: the finite-polynomial ring has a discrete order and a useful incomplete topology; its completion has excellent topological ring operations; and the completion loses the algebraic possibility of discrete order. It would be incorrect to attribute this last failure only to descriptive-set-theoretic complexity.

86.2 Set-sized omnific subrings

ProveIt's surreal development describes the omnific ring as the normal-form elements with nonnegative exponents and integer constant term; it records in particular the least-positive-element property [52]. The present theorem uses only the resulting ordered-ring laws. It does not use the deeper claims in the repository's unrefereed reports.

Corollary 86.3 (Omnific presentation obstruction). *Let B be any uncountable set-sized unital subring of the omnific integers, with its inherited order. It admits no Polish additive-group topology in which the inherited positive cone has the Baire property. More generally, any Hausdorff Baire additive-group topology with such a cone satisfies $c(B) = d(B) = |B|$.*

Proof. An ordered unital subring of a discretely ordered ring is discretely ordered: its positive elements are positive in the larger ring and therefore at least 1. Apply [theorem 78.3](#) and [corollary 78.5](#). $\square$

[10] Source 10 proves the same corollary with the cardinal equalities of [theorem 79.13](#):

The inspected ProveIt surreal README describes the construction of the omnific ring, its least positive element 1, its constant-term homomorphism to $\mathbb{Z}$, and the fact that an omnific integer bounded by an ordinary real is an ordinary integer [55]. The first two properties already make its set-sized unital ordered subrings eligible for the present theorem. We do not need the full normal-form theory, constant-term retraction, or real closedness of the surreal field for the obstruction.

Corollary 86.4 (Set-sized omnific presentations [10]). *Let A be a set-sized unital ordered subring of $\mathbf{Oz}$. If its additive group is given a Polish group topology and the inherited positive cone has the Baire property, then A is countable and the topology is discrete. If instead the topology is Hausdorff and Baire and A is uncountable, a Baire-property positive cone forces a full-cardinality closed discrete subset and the equalities of [theorem 79.13](#).*

Proof. The inherited order is discrete, with least positive element 1. Apply [corollary 78.6](#) and [theorem 79.13](#). $\square$

A frequent coding pattern is a closed additive subgroup of a countable product of countable discrete groups. Such a group is Polish: use a complete product metric and restrict it to the closed subgroup. Its finite-coordinate agreement subgroups are open and form a basis. Thus an uncountable exact-stream representation of a set-sized omnific subring in this format cannot simultaneously give continuous additive group operations and a Baire comparison predicate.

This does not rule out countable notation systems, incomplete spaces of finite expressions, non-Baire comparisons, non-Polish presentation spaces, or purely algebraic formalizations. A kernel proof of field or ring laws does not produce a Polish presentation or a measurable order algorithm. Likewise, being homeomorphic to Baire space as a bare space does not by itself give a non-Archimedean *group* topology.

Finite polynomial expressions

$$z + r_1\omega + r_2\omega^2 + \cdots + r_N\omega^N$$

provide the direct connection with (84.1). Finite normal-form arithmetic identifies their ring and order with $\mathbb{Z} + t\mathbb{R}[t]$ through $t \mapsto \omega$. The sharpness examples are thus also examples on an explicit set-sized omnific subring; one does not need the full proper class of surreal numbers to see the obstruction.

An additive fragment such as $\mathbb{R}\omega + \mathbb{Z}$ behaves differently. It is closed under addition but not under multiplication, since ω^2 lies outside it. Giving that fragment the topology of $G_{\mathbb{R}}$ is compatible with a regular discrete additive order. The theorem explains why expanding it to a ring while retaining that additive presentation is impossible.

86.3 Foundations and universe boundaries

No theorem above treats the proper class $\mathbf{Oz}$ as a Polish space. A Polish space is set-sized. To discuss the entire surreal or omnific class in a class theory, one must separately specify which collections of classes and which class maps are allowed; none of that is needed for [corollary 86.3](#).

In a type-theoretic implementation, fix a universe containing the carrier of the ring or additive group and a universe large enough for its topology and cardinal invariants. The packing map is a set-indexed family within those universes. If a universe-relative construction

of surreals is used, first identify the desired set-sized unital subring and its order. A birthday cutoff or a finite list of permitted scales must not be assumed to be a subring without proving closure.

The finite-polynomial example can instead be defined from scratch, without any surreal ontology: its coefficient sequences are finitely supported functions on the ordinary naturals, with an integer constant term. This is a particularly clean target for a foundationally modest formalization of the examples.

86.3.1 Universe discipline [10]

A proper class such as the full **No** or **Oz** is not an underlying set for the theorems here. In a set-theoretic treatment, first choose a set-sized subring closed under the needed operations. In a dependent- type-theoretic treatment, first specify a universe level and a type carrying the actual ordered ring, rather than a type of arbitrary syntax with an unproved interpretation map. The theorem then concerns that type's underlying set in a metatheoretic model.

The quotient $A/\mathbb{Z}1$ is a set-sized additive quotient. No “set of all class functions”, class-valued probability space, or topology on all proper-class subsets is formed. These restrictions are part of the statement, not a later repair to its proof.

87 Formalization architecture and proof audit

The repository snapshot used for the interface review is

`0f084afa9be466593ffa6aff37749e78c0adc733`.

The inspected Presburger README describes an integer-specific Cooper eliminator with syntax, normalization, elimination, and a decision procedure. The inspected surreal README records the ordered omnific construction and distinguishes proved-library interfaces from research reports [51, 52]. We have not built the repository or independently verified those advertised audit results.

The central theorem can be formalized independently of Cooper elimination, nonstandard-model syntax, or the full surreal construction. Its proof should be split into interfaces that expose exactly what is used.

87.1 The algebraic interface

The first component is a discretely ordered ring and its absolute-value inequalities. The core certificate is the statement

$$r \neq s, a > 0 \implies 2a < |(2a + 1)(r - s)|. \quad (87.1)$$

Next, a set-level lemma says that if $W \subseteq (-a, a)$, the translates $(2a + 1)r + W$ are disjoint for distinct indices. Neither lemma needs Baire category, cardinal arithmetic, topological continuity of scalar multiplication, or properties of a particular ring.

For Lean, a robust design would expose the least-positive-element property explicitly as a hypothesis on the ordered ring, rather than assuming that a convenient name for a discrete-order typeclass matches the required semantics. The proof only needs the implication $0 < x \implies 1 \leq x$. A Rocq development can package precisely the same fields in a record. This paragraph is a design specification, not a claim that any particular library API has been tested.

87.2 The topological and category interface

The second component concerns an ordered additive group with a Hausdorff group topology. It should prove, in order:

1. a nondiscrete Hausdorff topological group has nowhere-dense singletons;
2. a nonmeager Baire-property set has differences containing a neighborhood of zero;
3. reflection of a locally comeager part of the positive cone produces a nonmeager interval $(0, 2u)$;
4. a Baire-property positive cone yields an order-bounded neighborhood.

The last statement should be written for groups rather than rings, so that the self-embedding theorem can reuse it without introducing unneeded multiplication.

The ccc consequence need not formalize general cardinal invariants first. It can directly turn the family of translates into an injection from the carrier into a countable family. The exact cardinal equalities can be added later. This keeps the first kernel-checking target small.

87.3 Separate modules and theorems

A reasonable dependency order is:

Module	Mathematical boundary
DiscreteOrderGap	(76.1) and (87.1)
DisjointTranslates	The packing lemma, with an arbitrary bounded set
BaireOrderedGroup	Pettis and the bounded-neighborhood lemma
ArithmeticRigidity	Baire ccc and Polish conclusions
UnitImageSpectrum	All ordered additive self-embeddings, no regularity
PositiveExpansion	Difference-ring construction and semiring obstruction
PolynomialExamples	Finite-support ring and the completion unit witness
OmnificApplication	An adapter using proved subring and discrete-order laws

The measure theorem belongs in a later module, because it adds Haar measure, compact inner regularity, and Tonelli. It is logically independent of the category route to the principal theorem.

87.4 Trust boundaries of this package

The package contains mathematical proofs and an executable exact arithmetic script. It does not contain a new Lean or Rocq proof, and no such toolchain was run during preparation. The tests check finite polynomial identities in $\mathbb{Z} + t\mathbb{Q}[t]$. They do not certify Baire category, all Borel sets, uncountable cardinalities, arbitrary semiring completions, or a theorem for every ordered ring.

The most important audit questions are therefore mathematical. Does one mistakenly use continuity of $r \mapsto br$? No: only translates of W are used. Does the reflection require multiplication? No: $2u - x$ is formed in the additive group. Is an open convex subgroup assumed to be an ideal? No. Is a positive monoid silently given a Polish group completion? No. Is the proper class of all omnific integers treated as a set? No. The manuscript's statements are formulated to keep each of these distinctions visible.

87.5 A small algebraic core before topology [10]

A sensible implementation order is the following.

Layer A: finite intervals and hulls. Prove finite-interval exhaustion from least positivity of 1; define $\text{Fin}(A)$ and H_a using ordinary natural numbers; prove their additive subgroup and convexity properties. Keep the hull's external ordinary index distinct from an element of a nonstandard model.

Layer B: the exact kernel. Build the additive map $x \mapsto ax + H_a$ and prove that its kernel is $\text{Fin}(A)$. Obtain the quotient injection by a general additive-group quotient theorem. This layer should depend only on ordered algebra, not topology, measurable spaces, or surreal normal forms. The bounded-direct-sum theorem can be implemented independently using finite supports and a largest nonzero index.

Layer C: the category interface. Prove [lemmas 77.2](#) and [77.3](#). The only topological input needed by the algebraic layer is one bounded neighborhood. Explicitly expose that interface before connecting it to the Baire-property positive cone. This makes the theorem reusable with other sources of local boundedness.

Layer D: quotient cardinality and discreteness. Prove that an open subgroup has at most countably many cosets in a ccc group. Combine the injection with countability of $\mathbb{Z}1$. Finish with the elementary countable-Hausdorff-Baire-group lemma. A separate cardinal module can produce the closed discrete transversal and the maximal-cardinality results using choice.

Layer E: syntax and applications. Connect the theorem to the actual omnific ordered ring and to abstract $\mathbb{Z}$ -groups. ProveIt's Presburger development exposes affine syntax, coefficient normalization, a normalized Cooper step, and a full Lean sentence decision procedure, while its Rocq development independently checks the decisive one-variable step [\[54\]](#). Transfer to abstract $\mathbb{Z}$ -group semantics is a separate theorem; an integer implementation alone is not evidence that the transfer has been formalized.

No new source file is labeled "Lean verified" in this package. The repository was inspected at the snapshot recorded in the source audit; its builds were not rerun. The mathematical arguments above do not rely on the repository's unrelated research claims.

88 Further research questions and concrete next targets

The following questions are proposed research directions arising from the proved results. Except for the earlier project questions explicitly identified in [sections 75](#) and [83](#), we do not claim that a complete literature search has established their open status.

[10] Source 10 introduces its twelve questions as follows. Each one that asks the same as a question of source 12 is printed after it; the other four follow in [section 88.1](#).

The general Polish signed-group question is resolved by [corollary 78.4](#); it should no longer appear on the open list. The following questions identify extensions not settled by this paper. Unless attributed explicitly, they are proposed research targets, not assertions that a problem's literature-wide open status has been certified.

[write] The question that both sources answer is [question 57.1](#) of [Part IV](#); their answers are [corollaries 78.4](#) and [78.5](#). Source 08's [question 57.4](#) is answered as stated (note after [theorem 83.3](#)); the questions below on weaker regularity ask for more.

Question 88.1 (Positive monoids beyond Polishable signed groups). Can an uncountable Polish continuous-addition positive arithmetic model admit an algebraically compatible unital semiring multiplication when no compatible Baire ccc signed-group presentation with regular order exists?

The new obstruction shows exactly where a positive example would have to escape: failure of an appropriate group topology or failure of order regularity after passage to differences. Requiring stronger arithmetic, such as $Q + \text{IOpen}$, also invokes Glazer's obstruction if multiplication is Borel. The non-Borel multiplication case and the weaker algebraic case should be kept separate.

[10] Source 10 asks:

Question 88.2 (Positive arithmetic without a signed topology [10]). Can an uncountable Polish positive arithmetic monoid with continuous addition and suitably regular order carry an ordered semiring multiplication when no regularity of that multiplication is assumed?

The task is to recover a replacement for the group reflection argument, or to construct a genuine example escaping it. A result must state its arithmetic axioms and must not silently assume a Polish group completion. Glazer's positive-model work is the closest published starting point, but its Borel multiplication hypothesis changes the available tools.

Question 88.3 (A usable group-completion criterion). Which conditions on a Polish cancellative ordered monoid ensure that its algebraic difference group has a Hausdorff Baire ccc topology with a Baire-property cone and the required compatibility with the original presentation?

The first useful outcome would be a sufficient criterion stated in terms of translations and the difference quotient map. A proof must check the quotient topology rather than appealing only to algebraic cancellation. A converse would identify the precise gap between the signed theorem here and the positive-monoid setting.

[10] Source 10 asks:

Question 88.4 (A usable group-completion criterion [10]). Which Polish cancellative Presburger monoids embed topologically as the positive cone of a Hausdorff Polish topological group?

A promising formulation would use translation embeddings, local neighborhood compatibility, and a complete uniform structure on formal differences. Both the topology on the original monoid and the regularity of the completed order must be tracked. Such a criterion would turn [proposition 81.5](#) into a practical obstruction.

[write] Both are source 08's [question 57.2](#), asked also by source 09 as [question 73.11](#), for Baire ccc completions with a Baire-property cone.

[write] (Batch 90.) Both are answered by [Part XII](#) for monoids with continuous addition and linear natural order. Such a Polish monoid embeds topologically as the positive cone of a Hausdorff Polish topological group exactly when its partial subtraction is continuous ([theorems 153.1](#) and [154.1](#)), which answers source 10's question; the group is then Polish, hence Baire and ccc, and the cone, a G_δ , has the Baire property, which answers source 12's for Polish monoids. For the cones of uncountable discretely ordered rings the criterion always fails ([proposition 155.6](#)).

Question 88.5 (The topology of regular ordered additive groups). For a fixed Polish abelian group, which discrete invariant orders have a Baire-property positive cone, and which open principal convex subgroups can they produce?

[Proposition 79.1](#) provides an open principal convex subgroup with countable quotient. This is necessary, not a classification. One can ask whether recursive data from open convex subgroups and their ordered quotients give complete invariants in natural classes such as non-Archimedean Polish groups.

Question 88.6 (Exact unit-image spectra). Which subsets of the positive cone occur as $\mathcal{U}(G)$ for uncountable Baire ccc discretely ordered groups with regular cone?

The spectrum is bounded by [theorem 80.1](#); [\(80.3\)](#) gives one exact infinite example. Possible next invariants include its cofinality within the convex subgroup it generates, closure under ordinary multiplication of indices when available, and the effects of restricting to continuous or elementary self-embeddings. These restrictions need not produce the same spectrum.

[10] Source 10 asks:

Question 88.7 (The intrinsic unit-image spectrum [10]). For an uncountable Baire ccc discretely ordered group with regular order, how can $\mathcal{U}(G)$ be classified algebraically and descriptively?

The spectrum is order bounded and each of its principal hulls is meager. Determining when the spectrum itself is meager, Borel, analytic, or definable requires additional information. One should first analyze explicit divisible-plus-integer examples, then groups without a global additive order unit.

Question 88.8 (Cardinal and component spectra). For a fixed uncountable discretely ordered ring, which Hausdorff Baire regular-order group topologies realize the forced equality $c(R) = d(R) = |R|$, and what weights and connected-component structures can occur?

The equality itself is settled here. The remaining realization problem is more refined: prescribed local compactness, completeness, metrizability, or topology on selected coefficient subgroups. [Section 84.2](#) shows that local compactness can coexist with exact maximal cellularity on a continuum-sized ring.

[10] Source 10 asks:

Question 88.9 (Optimal nonseparable presentations [10]). Which discretely ordered rings admit nondiscrete Hausdorff Baire additive topologies with Borel order, and which full-cardinality open quotient structures can occur?

The polynomial example realizes maximal cellularity with an open real line. The cardinal lower bound is exact, but it does not classify possible local groups or orders. A classification for rings $\mathbb{Z} + I$ with a specified divisible additive ideal I would make the boundary concrete.

[write] This refines source 08's [question 57.5](#): the equality $c = d = |R|$ is now proved.

Question 88.10 (Regularity weaker than Baire or Haar measurability). What assumptions on an ordered positive cone are sufficient to force an order-bounded neighborhood without requiring the full Baire property or completed Haar measurability?

The proof isolates a concrete intermediate property: some order-bounded set has a difference set containing a neighborhood of zero. A productive weakening should yield that property directly. In non-locally-compact groups, any proposed measure substitute needs a

carefully specified notion of null set and invariance; there is no general locally finite Haar measure to import unchanged.

[10] Source 10 asks:

Question 88.11 (Alternative notions of regular order [10]). Can the category hypothesis be replaced by a useful measure-theoretic or definability hypothesis that still forces a bounded neighborhood?

A positive result needs an explicit substitute for the nonmeager-cone argument. Ambient measurability alone should not be assumed to survive restriction to arbitrary null or meager closed subgroups. The algebraic kernel theorem is already available once a local bound is proved.

[write] Source 08's [question 57.4](#) asked for this with hereditary conditions; Part VI answers it with the Baire property or completed Haar measurability of the cone itself.

Question 88.12 (Definability of integer parts in Polish fields). What descriptive complexities can occur for uncountable integer parts inside ordered fields with Polish additive topology and Borel order?

A Borel integer part is meager by [corollary 82.4](#), and its inherited Borel ordered-group structure is not compatibly Polishable when uncountable. These obstructions leave room for finer questions about Borel ranks, analytic presentations, and the complexity of floor-selection maps. The field, its topology, and its order should be fixed before these complexities are compared.

Question 88.13 (Which completions preserve discrete order?). For natural additive topologies on discretely ordered subrings, which completions avoid new units and retain any discrete ring order?

The unit witness in [proposition 86.2](#) gives an algebraic necessary condition independent of Borel complexity. It is not sufficient: avoiding nontrivial units need not produce an order, and producing an order need not make it regular. A useful theory would separate these three successive obstructions.

[10] Source 10 asks:

Question 88.14 (Completion with controlled units [10]). Which dense discretely ordered subrings admit useful completions without introducing new nontrivial units or other elementary obstructions to discrete order?

The coefficient completion in [proposition 86.2](#) fails before measurability is considered. A systematic completion audit could separate algebraic failures from the category obstruction and could inform the design of exact symbolic versus stream-based omnific interfaces.

Question 88.15 (Reverse mathematics and effective witnesses). What is the exact proof-theoretic strength of the bounded-neighborhood lemma when the order cone is supplied by a Borel code or by an explicit Baire-property witness?

Supplying an open set and a coded meager exceptional set may change the strength of the problem compared with extracting such witnesses from arbitrary Borel codes. The packing stage is elementary once the bound is supplied. An analysis of this signed fragment must not be confused with a solution to Glazer's published ATR_0 question.

[10] Source 10 asks:

Question 88.16 (Reverse-mathematical strength [10]). What weak subsystem suffices for a coded version of [corollary 78.4](#), including Baire-property witnesses and the countability of the relevant coset space?

The main proof uses neither Haar measure nor a principal structure theorem for locally compact groups. Its critical formal issues are category on coded spaces, selecting countably many coset witnesses, and passing from countable quotient and kernel to countable domain. This is a new target; no consequence for Glazer’s distinct ATR_0 question is claimed.

[write] Compare source 08’s [question 57.7](#). Source 14 ([Part VIII](#)) claims an ATR_0 proof of Glazer’s Theorem 2, a different theorem; neither source of Part VI draws on it.

Question 88.17 (Choice and pathological Polish presentations). How much choice is required to construct the Polish additive presentation in [section 84.3](#), and what regularity principles rule out analogous examples over weaker foundations?

The displayed construction uses rational vector-space bases. It provides no effective presentation and no assertion about minimal choice strength. Studying it over ZF with specified choice or regularity axioms would connect the arithmetic obstruction with the foundational side of this research intersection.

Question 88.18 (A small kernel-checked theorem). Can the gap inequality, bounded-neighborhood lemma, and ccc consequence be formalized as a compact independent development, then instantiated at a set-sized omnific subring?

The initial milestone should be the packing lemma with its full hypothesis audit. The next is the reflection argument, not the much larger machinery of arithmetic quantifier elimination. The module boundaries in [section 87](#) make this a concrete implementation project rather than a request to formalize an unspecified entire theory.

[10] Source 10 asks:

Question 88.19 (Certified impossibility interfaces [10]). Can the exact-kernel theorem become a reusable kernel-checked obstruction certificate for proposed arithmetic presentations?

A certificate could expose a positive scale, an open principal hull, a countable kernel, and a bound on the quotient size. This is much smaller than a formalization of the entire surrounding theory and would allow Provel to reject incompatible combinations of algebraic and topological interface claims.

[write] Compare source 08’s last question ([question 57.8](#)).

88.1 Further questions of source 10 [10]

Question 88.20 (Beyond cardinality: quotient order types [10]). What additional restrictions does the ordered embedding $A/\text{Fin}(A) \hookrightarrow A/H_a$ impose on cofinalities, convex-subgroup chains, and ordered ranks?

The main theorem uses only cardinality. The embedding preserves order, so it may yield sharper obstructions for proposed arithmetic reducts whose quotients have small rank even when both cardinalities are large. A useful first target is a uniform rank calculation for iterated polynomial and lexicographic extensions.

Question 88.21 (The bounded continuous-scalar subring [10]). Which subrings can occur as $\text{Sc}_{\text{cont}}(A)$ in a nondiscrete Baire presentation with regular order?

It must contain the ordinary integers and be bounded in the ambient order. The example here realizes exactly $\mathbb{Z}$. Constructing examples with prescribed larger scalar subrings, or finding new algebraic restrictions on them, would sharpen the link with topological algebra. Borel and continuous scalars must be kept separate outside the separable setting.

Question 88.22 (Countable unit intervals in general ordered rings [10]). How restrictive is the relative criterion $|A| = \max(\aleph_0, |[0, 1]_A|)$ when combined with model-theoretic conditions on A ?

The theorem isolates the finite part as the possible location of uncountability. Investigating ordered fields, integer-part extensions, and definable subrings with controlled bounded intervals may yield applications beyond discretely ordered arithmetic.

Question 88.23 (Definability of the transported counterexample [10]). What is the best possible descriptive complexity of an ordered-ring comparison relation after transporting its addition to $\mathbb{Q}^{\mathbb{N}} \times \mathbb{Z}$?

The cone cannot have the Baire property, so it cannot be Borel or analytic in that Polish presentation. The basis-selected construction does not provide any finite projective definition. The problem should be treated with explicit set-theoretic assumptions rather than assuming that a nonconstructive transport is an effective code.

89 Conclusions of Part VI

The decisive topological resource is an order-bounded neighborhood, not local compactness. Category regularity of an invariant positive cone provides such a neighborhood in every Hausdorff Baire group. Discrete ring arithmetic then amplifies the unit gap into a maximal family of disjoint open translates:

$$\text{regular cone} \implies W \subseteq (-a, a) \implies \{(2a + 1)r + W : r \in R\} \text{ disjoint.}$$

This resolves the earlier local-compactness-removal question and gives the sharp equality $c(R) = d(R) = |R|$. A separate measure argument obtains the same conclusion from ambient Haar measurability in locally compact groups.

The consequences are both restrictive and diagnostic. They prohibit arbitrary ordered-ring and positive-semiring expansions of a broad class of Polish additive presentations, constrain all ordered additive self-embeddings, and explain why natural coefficient completions of omnific polynomial fragments must fail somewhere. At the same time, the examples show that uncountable continuous addition, nonseparable Baire ring presentations, and pathological Polish ordered algebra can all exist outside the precise hypothesis boundary.

Source 10's conclusion [10]. The decisive improvement is to retain the ordinary-integer kernel instead of trying to eliminate it before forming a quotient. A Baire order provides an open principal convex additive hull. Multiplication by its defining positive scale sends the quotient by finite elements injectively into its open coset space. In discretely ordered arithmetic, finite elements are exactly the ordinary integers. A countable open coset space therefore leaves no room for an uncountable ring.

This resolves the project's remove-local-compactness question for signed additive groups, strengthens it from Polish to Baire-ccc topologies and from Borel order to the Baire property, and gives an exact cardinal obstruction outside the separable setting. It also identifies a

purely ordered-group limitation on additive self-embeddings. The remaining boundary is not local compactness or infinite dimension; it is the passage from positive monoids to signed topological groups, along with the precise regularity of order.

90 Where Part VI meets Parts I–V [merge]

Part VI was written against Parts I and IV (source 12) and Parts I, IV and V (source 10). Its principal theorem has hypotheses that several earlier results of this report satisfy, so those results become corollaries. Each deduction below is one line; none uses more than is stated.

Proposition 90.1 (Earlier results that follow from Part VI [merge]).

1. *Theorem 44.1* (source 08): a Hausdorff locally compact second-countable group is Polish and a Borel cone has the Baire property, so [corollary 78.5](#) applies. So does [corollary 55.1](#), by [corollary 86.3](#).
2. *Corollaries 48.1 and 48.2* (source 08) are the locally compact cases of [corollary 81.1](#) and [theorem 81.2](#); the latter needs neither commutativity nor the absence of zero divisors.
3. *Theorem 62.1* (source 09): an admissible topology is a Polish additive-group topology with a Baire-property cone, so [corollary 78.5](#) gives the conclusion; the Borel-scalar part of admissibility is not needed. [Corollary 62.2](#) is likewise a case of [theorem 82.2](#), which needs no Borelness of multiplication.
4. Part I's [corollary 10.5](#), for the cones $M_{\mathbb{R}}$ and $M_{\mathbb{B}}$, is a case of [theorem 81.2](#): their groups G_D are uncountable Polish groups with Δ_2^0 cones. Part I's [theorem 10.4](#) itself is purely algebraic and is not a case.

Proof. Each item is the stated specialization; the hypotheses compared are those of the cited statements. □

Not every earlier result becomes a corollary. Source 09's [theorem 62.3](#) concludes that every discretely ordered subring of a field with an admissible topology is countable and closed discrete, with no measurability assumption on the subring; source 12's [corollary 82.4](#) concludes only meagerness, for Borel integer parts, but without Borel scalars. The two are incomparable. Source 08's [theorem 53.4](#) concerns multiplications unrelated to the order, to which Part VI does not apply. The positive-cone results of Part III ([theorems 38.1](#) and [39.1](#)) concern topologies on the cone, not on its group of differences; both sources of Part VI state that their theorems do not apply to a positive monoid without a compatible signed group topology ([sections 81.2](#) and [81.4](#)).

Questions. Part VI answers [question 57.1](#) negatively and [question 57.4](#) as stated; notes after both questions say so. It bears on [question 57.5](#) and asks [question 57.2](#) again. It does not bear on Glazer's questions: both sources state that they answer neither.

Part VII

Polish Presburger arithmetic beyond the split construction

[write] This part is source 13 of batch 87, *Polish Presburger Arithmetic Beyond the Split Construction: Nonsplit models, exact continuity, and optimal computation from streams* (dated October 4, 2026; “research report prepared with ChatGPT”), pinned to repository commit eab47e330, one commit before Parts I and II were placed. All unmarked text is source 13’s; its labels carry the prefix `pma:nsp:`, and nine labels were added to unlabelled statements so that notes can cite them. It continues [Part I](#) and [Part III](#): it starts from the split models of Part I, read as the three batch-86 manuscripts in `docs/incoming`, and answers four of Part I’s questions, wholly or in part: [question 14.8](#) (for parameter-free formulas, exactly), [question 14.11](#) (its existence half), [question 14.13](#) and [question 14.16](#) (for joint uniform continuity, negatively); for countable cores it also answers the existence part of Part III’s [question 43.3](#), which it did not see. Notes after the relevant results say which.

Printed once here. The countable-core transfer ([theorem 94.1](#)) places an arbitrary countable $\mathbb{Z}$ -group K , with the discrete topology, below $\mathbb{R}$ or $\mathbb{Q}^{\mathbb{N}}$. With $D = \mathbb{R}$ and a nonsplit core it is also source 14’s [proposition 113.2](#) and source 15’s [theorem 122.2](#); those are printed in [Parts VIII](#) and [IX](#) as flagged instances (note after the theorem).

What is recalled and what is new. Source 13’s Section 2 ([section 92](#)) recalls Part I. Its nonsplit core K is a member of the factorial family of the discrete-initial-subgroups-and-omnific-normalization report, its algebraic splitting criterion is part of that report’s `isg:cf:main:presburger`, and its omnific criterion is that theorem with `isg:cf:cor:unit`; source 13 says so in each case. New to the collection are the localization presentation of K and its fixed division, the countable-core transfer, automatic continuity of residues and the topological splitting criterion, the continuity compiler and exact discontinuity sets, the target-valued Baire classification, the rank and exact revision theorems, and the closed-translate metric obstruction. Source 13 claims no priority.

Notation. Source 13’s M_ω , D_ω and $\mathbb{Q}^\omega$ are Part I’s M_B , D_B and $\mathbb{Q}^{\mathbb{N}}$; its $G(D)$, $M(D)$ are Part I’s G_D , M_D ; its lexicographic sums $\oplus_{\text{lex}}$ are printed $\overrightarrow{\times}$, and its `\dotminus` as $\dot{-}$. In [section 95](#) the letter D is the divisible kernel $\ker r$, not a coefficient group, and r the canonical residue map; $\mu(f)$ in [section 100](#) is a revision number. [Section 1.3](#) lists these.

Provenance. Source 13’s finite diagnostics are shipped as `code/13-beyond-split-verify.py` with the record `data/13-beyond-split-verification_results.json`, and its source record as `data/13-beyond-split-provenance.json` ([appendices R](#) and [S](#)).

91 The research intersection and the contribution

The relevant intersection is topological and effective model theory of weak arithmetic. Glazer’s paper [\[1\]](#) asks whether an uncountable Polish space can support a model of Presburger arithmetic with continuous addition. It also proves strong obstructions once multiplication

and suitable induction are available. The distinction between continuous basic operations and the regularity of definable operations is therefore mathematically consequential.

Provelt provides two direct connections. Its Presburger project formalizes an executable Cooper elimination procedure over the ordinary integers, with an independent Rocq proof of the finite-residue core [59, 60, 61]. Its surreal project develops the ordered field of surreal numbers and the omnific integers [62]. These connections are established by mathematical sources and actual repository material; the inaccessible LinkedIn profile supplied in the request is not used to infer a current appointment or professional affiliation.

91.1 What was already in the repository

At the inspected revision, the incoming directory contained three related research packages. Two independently develop Polish Presburger models [63, 64]; the third concerns Borel presentations and support barriers in Hahn and omnific arithmetic [65]. The repository’s own triage records a README-level inventory rather than a validation of those proofs [66]. For this article the three LaTeX manuscripts were extracted and read directly.

The first two packages already contain the positive cones

$$M_D = (D_{>0} \times \mathbb{Z}) \cup (\{0\} \times \mathbb{N}), \quad D = \mathbb{R} \text{ or } \mathbb{Q}^\omega,$$

continuous addition and fixed division, low Borel complexity of all definable relations, examples of discontinuous definable operations, and embedding results. The first package even classifies all self-embeddings of its rational-stream model. Repeating these claims would not constitute a further result for the present request.

Instead, the present work addresses their questions about nonsplit presentations, exact discontinuity profiles, optimal finite revisions of truth values, and complete metrics with uniform addition. The connection to omnific arithmetic uses the repository’s existing classification: splitting over the standard integers is exactly the obstruction to a unital additive ordered realization in **Oz** [67]. The new models provide concrete Polish applications of that obstruction.

Topic	Starting point in the inspected reports	Result developed here
Polish Presburger existence	Split real and Baire constructions	Explicit nonsplit Polish models; automatic continuity of the canonical retraction
Definable regularity	Δ_2^0 relations; selected operations	Exact finite continuity compiler for every definable total function
Computation from streams	A bound by the number of atomic tests	A bound by linear rank; optimal arity bound and an exact finite criterion
Complete metrics	No invariant complete metric	No complete metric with uniformly equicontinuous translations at zero

[write] The three “incoming” packages are batch 86’s manuscripts 04 ([63]), 05 ([64]) and 03 ([65]); one commit after source 13’s pin they were placed in this report, and they are now [Part I](#) (sources 05 and 04) and [Part II](#) (source 03). The “repository’s own triage” [66] is a routing note of the Hilbert’s-tenth programme ([appendix E](#)). The “existing omnific splitting criterion” [67] is `isg:cf:main:presburger of the discrete-initial-subgroups-and-omnific-normalization` report. The four questions taken up are [Part I’s question 14.11](#),

question 14.13, question 14.8 and question 14.16; “no invariant complete metric” in the table is proposition 8.4. Source 13 did not see Parts III–VI.

91.2 Scope of the claims

An “open question” from an earlier repository manuscript means a question that that manuscript explicitly left unresolved. It does not assert that the question had independently become a recognized open problem throughout the literature. The present results settle the specified cases with proofs. They do not resolve Glazer’s separate question about the reverse-mathematical strength of his multiplication theorem, and they do not classify arbitrary Polish topological monoids.

Several ingredients are classical: the theory of $\mathbb{Z}$ -groups, Presburger quantifier elimination, elementary facts about Polish spaces, and the ordered-group universality of the surreal numbers. These are cited where used. Their applications and the additional arguments are supplied in full. In particular, a computational diagnostic is never substituted for an infinitary proof.

92 Arithmetic models and their two topologies

We work in ordinary classical mathematics. A Polish space is separable and admits a compatible complete metric. Unless stated otherwise, all groups and parameter tuples are sets. The appearances of **No** and **Oz** will be confined to embeddings with set-sized domain and image.

92.1 Presburger groups and positive cones

Write

$$\text{Pr}_{\mathbb{N}} = \text{Th}(\mathbb{N}; 0, 1, +), \quad \text{Pr}_{\mathbb{Z}} = \text{Th}(\mathbb{Z}; 0, 1, +, -, <).$$

Order is definable in the monoid language by $x \leq y \iff \exists z (x + z = y)$. A $\mathbb{Z}$ -group is a linearly ordered abelian group G with least positive element 1_G such that $G/\mathbb{Z}1_G$ is divisible. The classical characterization says that these are exactly the models of $\text{Pr}_{\mathbb{Z}}$; their nonnegative cones are exactly the associated models of $\text{Pr}_{\mathbb{N}}$ [4, Section 2]. This includes the full first-order induction scheme with parameters, not only the elementary equations for addition.

For every standard integer $m \geq 1$ and every $g \in G$, there are unique $q \in G$ and $r \in \{0, \dots, m-1\}$ such that

$$g = mq + r1_G. \tag{92.1}$$

To see existence, divisibility of the quotient gives $g = mq_0 + k1_G$; ordinary integer division of k gives the asserted remainder. For uniqueness, if $m(q - q') = (r' - r)1_G$, convexity of the standard integers forces $q - q'$ to be standard and the usual remainder argument applies. If $g \geq 0$, then $q \geq 0$: a negative quotient would be at most -1_G and would make $mq + r1_G < 0$.

After congruence predicates are named, quantifier elimination reduces each formula, uniformly over $\mathbb{Z}$ -groups, to a finite Boolean combination of affine inequalities, equalities, and congruences [58, 4]. Transferring this theorem to a positive cone means relativizing each quantified variable to $x \geq 0$ before elimination. No multiplication of two variables is introduced.

The word “uniformly” matters. A subset of a nonstandard group that is invariant under adding a standard integer m need not be invariant under adding every member of mG .

Presburger congruence predicates have the latter, full-coset invariance. The earlier packages correctly isolate that condition when adapting Cooper’s finite-boundary proof [63, 64]. The formal source inspected here evaluates its syntax specifically in $\mathbb{Z}$; a model-uniform formalization remains a separate task.

92.2 The real and stream coefficient groups

Let D be either $\mathbb{R}$ with its usual topology and order, or

$$D_\omega = \mathbb{Q}^{\mathbb{N}}$$

with each rational factor discrete. The latter carries the first-difference lexicographic order: a nonzero sequence is positive exactly when its first nonzero coefficient is positive. Both groups are divisible Polish topological groups. Their strict positive and negative cones are open; for D_ω they are countable unions of finite-prefix cylinders.

Let $G_D = D \times \mathbb{Z}$, with the D coordinate dominant and the integer coordinate discrete. Its positive cone is M_D as above, and $1 = (0, 1)$. The integer quotient in (92.1) is explicit:

$$q_m(u, n) = (u/m, \lfloor n/m \rfloor), \quad r_m(u, n) = n \bmod m. \quad (92.2)$$

These maps, as well as addition, are continuous in the specified topology. We denote the real model by $M_{\mathbb{R}}$ and the stream model by M_ω .

For clarity, the arithmetic order is not the topology being imposed. Every arithmetic element has a successor, while zero is topologically nonisolated. For example,

$$(1/j, 0) \longrightarrow 0 \quad \text{in } M_{\mathbb{R}}, \quad (e_j, 0) \longrightarrow 0 \quad \text{in } M_\omega.$$

Here e_j is the stream with a single coefficient 1 at position j . Each displayed nonzero element is greater in arithmetic order than every standard integer. Convergence of these elements is one reason that arithmetic order and continuous subtraction cannot be silently assumed.

92.3 Polishness, Baire space, and definable relations

The cone M_D is open in the closed Polish subspace $D_{\geq 0} \times \mathbb{Z}$: its complement there is $\{0\} \times \mathbb{Z}_{<0}$, a closed set. Open subspaces of Polish spaces are Polish. In the real model the integer slices are copies of $[0, \infty)$ or $(0, \infty)$, so the space is perfect and locally compact.

The stream model is zero-dimensional and nowhere locally compact. Every neighbourhood contains a closed cylinder with one positive coordinate fixed and infinitely many free rational coordinates. Such a cylinder is homeomorphic to $\mathbb{Q}^{\mathbb{N}}$ and cannot be compact, since a free rational coordinate has an infinite discrete image. The standard characterization of Baire space therefore gives $M_\omega \cong \mathbb{N}^{\mathbb{N}}$ [8]. The earlier packages provide explicit coding homeomorphisms as well [63, 64].

Every parameter-definable relation on M_D is Δ_2^0 in this topology. Indeed, a strict affine inequality is the union of an open leading-sign condition and a closed leading-zero condition with a discrete integer test. Its complement has the same form. Equalities are closed and congruences are clopen, since $mG_D = D \times m\mathbb{Z}$. Quantifier elimination then gives the claim. This background bound concerns sets and graphs; it does not by itself prove a target-valued Baire-class bound for functions. That distinction will become explicit below.

[write] This section recalls Part I, as source 13 says: M_D and its Polishness are [theorems 3.1](#) and [5.2](#), the homeomorphism $M_\omega \cong \mathbb{N}^{\mathbb{N}}$ is [theorem 6.1](#), and the Δ_2^0 bound is [theorems 7.1](#) and [7.2](#). Source 13’s $D_\omega = \mathbb{Q}^{\mathbb{N}}$ and M_ω are Part I’s D_B and M_B (the same group, order and topology); its $\mathbb{Q}^\omega$ is $\mathbb{Q}^{\mathbb{N}}$.

93 A nonsplit countable arithmetic core

The split constructions in the incoming reports do not exhaust the Polish possibilities. Nonsplit countable $\mathbb{Z}$ -groups are classical, and the repository already contains an explicit profinite-parameter family [67, section “An explicit factorial family of nonsplit models”]. The following localization presentation is a particularly elementary member of that family. Its elements and its fixed-division algorithms use ordinary rational arithmetic only. The new topological step below is to transfer arbitrary countable arithmetic cores to perfect Polish universes while preserving their nonsplitting obstruction.

Put

$$A = \mathbb{Z}_{(2)} = \{a/b : a \in \mathbb{Z}, b \in \mathbb{N}_{>0} \text{ odd}\}, \quad B = \mathbb{Z}[1/2] = \{c/2^\ell : c \in \mathbb{Z}, \ell \in \mathbb{N}\},$$

and let $K = A \times B$ with coordinatewise addition. Its distinguished unit is $e = (1, 1)$. Define

$$\pi : K \longrightarrow \mathbb{Q}, \quad \pi(u, v) = v - u.$$

Order K by declaring

$$(u, v) >_K 0 \iff v - u > 0 \text{ or } (u = v \text{ and } u > 0). \quad (93.1)$$

The second case involves an ordinary integer, since $A \cap B = \mathbb{Z}$.

Proposition 93.1 (Explicit nonsplit $\mathbb{Z}$ -group; a member of the family of `isg:cf:prop:family`). *The ordered group $(K, +, <_K, e)$ is a model of the ordered-group version of Presburger arithmetic. The sequence*

$$0 \longrightarrow \mathbb{Z} \xrightarrow{n \mapsto ne} K \xrightarrow{\pi} \mathbb{Q} \longrightarrow 0$$

is exact and does not split. In fact,

$$\text{Hom}(K, \mathbb{Z}) = 0, \quad \text{Div}(K) = 0,$$

where $\text{Div}(K)$ is the largest divisible subgroup.

Proof. The equality $A \cap B = \mathbb{Z}$ follows by writing a rational in lowest terms: its denominator must be both odd and a power of two. Every rational belongs to $A + B$. Indeed, write its denominator as $2^\ell b$ with b odd, and choose integers s, t with $s2^\ell + tb = 1$; then

$$\frac{a}{2^\ell b} = \frac{as}{b} + \frac{at}{2^\ell}.$$

Replacing the first summand by its negative shows that π is onto. Its kernel is exactly $\mathbb{Z}e$.

Formula (93.1) is the lexicographic extension of the usual orders on $\mathbb{Z}e$ and $K/\mathbb{Z}e = \mathbb{Q}$. It therefore defines a translation-invariant linear order. The subgroup $\mathbb{Z}e$ is convex; e is the least positive element; and the quotient is divisible. The standard characterization of $\mathbb{Z}$ -groups now applies: a discretely ordered abelian group is a Presburger group precisely when its quotient by the subgroup generated by its least positive element is divisible [4, §2].

For completeness, the two assertions about homomorphisms and divisibility are elementary. The group A is divisible by 3^j for every j . Consequently, the image of any $a \in A$ under a homomorphism $A \rightarrow \mathbb{Z}$ would be divisible by every 3^j , hence would be zero. The same argument with powers of two applies to B . Thus every homomorphism $K = A \times B \rightarrow \mathbb{Z}$ is zero. A splitting over $\mathbb{Z}e$ would give a retraction taking e to 1, which is impossible.

There is no nonzero infinitely 2-divisible element of A , and no nonzero infinitely 3-divisible element of B . For example, if $a/b \neq 0$ is in lowest terms with b odd, $(a/b)/2^j$ fails to belong to A for all sufficiently large j . Hence a divisible subgroup of either factor is zero. Projection of a divisible subgroup of K to either factor is divisible, so the divisible subgroup itself is zero. $\square$

93.1 Exact relation to the repository's profinite family

Let $\varepsilon_2 \in \widehat{\mathbb{Z}} = \prod_p \mathbb{Z}_p$ have 2-adic coordinate 1 and every odd-prime coordinate 0. The canonical residue homomorphism of K is

$$r_K(u, v)_2 = u, \quad r_K(u, v)_p = v \quad (p \text{ odd}),$$

where each rational is interpreted in the indicated p -adic integer ring. In particular, $r_K(-1, 0) = -\varepsilon_2$.

This identifies K explicitly with the repository family member G_{ε_2} . To verify the identification without any extension classification, let S_n be the ordinary residue of ε_2 modulo $n!$, and put

$$H = \bigcup_{n \geq 1} (\mathbb{Z}(1/n!, S_n/n!) + \mathbb{Z}(0, 1)) \subseteq \mathbb{Q} \overrightarrow{\times} \mathbb{Q}.$$

The map $T(u, v) = (v - u, v)$ is a unital ordered group embedding. Writing $n! = 2^h m$ with m odd, the congruences $S_n \equiv 1 \pmod{2^h}$ and $S_n \equiv 0 \pmod{m}$ show that every displayed generator of H belongs to $T[K]$. Conversely, write $T(u, v) = (k/n!, l/n!)$ for sufficiently large n . The condition $u \in A$ gives $2^h \mid l - k$, and $v \in B$ gives $m \mid l$. Hence $n! \mid l - kS_n$, proving

$$T(u, v) = k(1/n!, S_n/n!) + \frac{l - kS_n}{n!}(0, 1) \in H.$$

Thus $T[K] = H = G_{\varepsilon_2}$. The localization presentation is a transparent specialization of the existing family, rather than an independent claim of algebraic priority.

[write] As source 13 says, K is a member of the factorial family of the discrete-initial-subgroups-and-omnific-normalization report: $K \cong G_{\varepsilon_2}$ in the notation of `isg:cf:prop:family`, and its nonsplitting is the case $\zeta = \varepsilon_2 \notin \mathbb{Z}$ of `isg:cf:thm:familytest`. The localization coordinates and the fixed-division algorithm below are source 13's.

93.2 An exact fixed-division algorithm

Let $n \geq 1$, and write $n = 2^k m$ with m odd. For an input $x = (u, v) \in K$, choose presentations

$$u = \frac{a}{b} \quad (b \text{ odd}), \quad v = \frac{c}{2^\ell}.$$

There is a unique integer $r \in \{0, \dots, n-1\}$ satisfying

$$\begin{aligned} r &\equiv ab^{-1} \pmod{2^k}, \\ r &\equiv c(2^\ell)^{-1} \pmod{m}. \end{aligned} \tag{93.2}$$

Congruences modulo 1 impose no condition. Both indicated inverses exist, and the Chinese remainder theorem gives r . Define

$$q_n^K(u, v) = \left(\frac{u - r}{n}, \frac{v - r}{n} \right), \quad r_n^K(u, v) = r. \tag{93.3}$$

Then $q_n^K(u, v) \in K$ and

$$(u, v) = nq_n^K(u, v) + re.$$

The first congruence in (93.2) removes exactly the forbidden power-of-two factor from the denominator of $(u - r)/n$; the second removes exactly the forbidden odd factor from the denominator of $(v - r)/n$. Conversely, membership of both quotient coordinates in A and B forces these two congruences. This also proves uniqueness of q and r directly.

If $x \geq_K 0$, then $q_n^K(x) \geq_K 0$. If $\pi(x) > 0$, use $\pi(q_n^K(x)) = \pi(x)/n > 0$. If $\pi(x) = 0$, then $x = te$ with an integer $t \geq 0$, and the formula reduces to ordinary Euclidean division of t . Thus this is also the fixed-division algorithm for the nonnegative cone.

94 Transferring every countable core to a perfect Polish universe

The preceding core can be replaced by any countable Presburger group. Let $(K, +, <, e)$ be such a group, equipped with its discrete topology, and let D be either

$D = \mathbb{R}$ with its usual topology, or $D = \mathbb{Q}^{\mathbb{N}}$ with its product of discrete topologies.

In the second case, order D lexicographically at the first differing coordinate. Let

$$G = D \overrightarrow{\times} K, \quad 1_G = (0, e), \quad P = G_{\geq 0},$$

and give G the product topology. Its nonnegative cone is

$$P = (D_{>0} \times K) \cup (\{0\} \times K_{\geq 0}). \quad (94.1)$$

Theorem 94.1 (Countable-core transfer). *The ordered group G is a $\mathbb{Z}$ -group, and P is an uncountable Presburger model with continuous addition. The space P is perfect and Polish. It is locally compact when $D = \mathbb{R}$, and is homeomorphic to Baire space when $D = \mathbb{Q}^{\mathbb{N}}$.*

All fixed standard quotient and remainder functions on P are continuous. Every Presburger-definable relation on any finite power of P is Borel; indeed, it belongs to Δ_2^0 in these two presentations. The map $k \mapsto (0, k)$ embeds K as a closed discrete elementary substructure of G .

There is a unit-preserving homomorphism $G \rightarrow \mathbb{Z}$ if and only if there is one $K \rightarrow \mathbb{Z}$. In particular, taking the core of Proposition 93.1 gives nonsplit models in both topological types, and their largest divisible subgroup is exactly $D \times \{0\}$.

Proof. The element $(0, e)$ is least positive. The quotient by its integer multiples is $D \oplus (K/\mathbb{Z}e)$, which is divisible. This proves the arithmetic claim by the characterization of $\mathbb{Z}$ -groups.

Both choices of D are Polish topological groups. For both, $D_{>0}$ is open and $D_{\geq 0}$ is closed. For rational streams, a nonzero sequence's sign is fixed by its first nonzero coordinate, which proves the open-cone assertion. The slice of P at a fixed $k \in K$ is $D_{\geq 0}$ when $k \geq 0$, and $D_{>0}$ when $k < 0$. It is therefore Polish, and P is a countable topological disjoint union of Polish spaces, hence Polish. Equivalently, P is a G_δ subset of $D \times K$. It need not be closed there: if $k < 0$, positive $d_j \rightarrow 0$ give $(d_j, k) \in P$ converging in G to $(0, k) \notin P$. No completeness claim is made for the restriction of the ambient product metric.

Addition is the restriction of the continuous group addition on G . Every point of either cone $D_{\geq 0}$ or $D_{>0}$ is nonisolated. This is immediate for $\mathbb{R}$; for rational streams, one can change a sufficiently late coordinate while preserving every prescribed initial coordinate and, when necessary, keep the first nonzero coordinate positive. Hence P is perfect and uncountable.

For $D = \mathbb{R}$, the slices are $[0, \infty)$ and $(0, \infty)$, so P is locally compact. For rational streams, the slices, and hence P , are zero-dimensional. No nonempty open subset of P is compact: inside every nonempty basic neighborhood one can find a closed cylinder in which a later rational coordinate is unrestricted; its projection onto the infinite discrete space $\mathbb{Q}$ is onto, contradicting compactness. For clarity at the all-zero stream, first choose a sufficiently late coordinate to be 1, then leave a still later coordinate unrestricted. Thus P is nowhere locally compact. The classical characterization of Baire space as the nonempty zero-dimensional Polish space with no nonempty compact open subsets yields $P \cong \mathbb{N}^{\mathbb{N}}$.

Write $k = nq_n^K(k) + r_n^K(k)e$, with $0 \leq r_n^K(k) < n$. On G set

$$q_n^G(d, k) = (d/n, q_n^K(k)), \quad r_n^G(d, k) = r_n^K(k).$$

These are continuous, since division by n is continuous on D and every map from the discrete factor K is continuous. For a nonnegative input, the quotient remains nonnegative, by the

sign of d/n when $d > 0$ and by the ordinary Presburger division axiom in K when $d = 0$. Restriction gives the desired maps on P .

The strict order on G is

$$(d, k) < (d', k') \iff d < d' \text{ or } (d = d' \text{ and } k < k').$$

The first condition is open; the equality condition is closed; and the condition on the discrete coordinates is clopen. Hence the order is Δ_2^0 . Linear equality is closed, congruence modulo each standard n is clopen, and integer-affine maps are continuous. Presburger quantifier elimination expresses every definable relation as a finite Boolean combination of these atomic relations [4, §2]. Finite Boolean combinations of Δ_2^0 sets remain Δ_2^0 . Restricting to the cone and its finite powers preserves the conclusion. This argument concerns definable sets, and by itself does not assert a target-valued Baire class for every definable function.

The copy $\{0\} \times K$ is closed and discrete. It is elementary because it is itself a unit-containing $\mathbb{Z}$ -subgroup, or directly by the same quantifier elimination. Finally, every homomorphism $D \rightarrow \mathbb{Z}$ is zero because D is divisible. Thus any unit-preserving homomorphism $G \rightarrow \mathbb{Z}$ restricts to one on K , and any such map on K extends by zero on D . For the explicit reduced core, projecting a divisible subgroup of G to K gives zero, while $D \times \{0\}$ is divisible. This proves the last assertion. $\square$

The transfer is algebraically nonsplit over the distinguished standard $\mathbb{Z}$, although it is deliberately a product with a discrete *countable core*. These two statements are compatible. Confusing them would erase exactly the distinction resolved by the construction.

[write] **One construction in three sources.** With $D = \mathbb{R}$ and a countable nonsplit core, [theorem 94.1](#) is the construction of source 14's [proposition 113.2 \(Part VIII\)](#), whose core K_* is the member G_{ζ_*} of the same factorial family, and of source 15's [theorem 122.2 \(Part IX\)](#), whose core is the difference group of an arbitrary countable Presburger model. The construction is printed here, once, in its general form; Parts VIII and IX print those two statements with their own proofs, flagged as instances, because they add the omnific obstruction and the boundary identification respectively. With [theorem 95.4](#) below, the theorem answers the existence half of Part I's [question 14.11](#) and, for countable cores, Part III's [question 43.3](#): every ordered extension $0 \rightarrow \mathbb{Z} \rightarrow G \rightarrow D \oplus (K/\mathbb{Z}) \rightarrow 0$ obtained from a countable nonsplit $\mathbb{Z}$ -group K has a Polish positive cone with continuous addition. Neither question asks only for existence, and neither classification is attempted.

94.1 Continuum many topological examples

Corollary 94.2 (Many nonsplit Polish arithmetic structures). *There are $2^{\aleph_0}$ pairwise nonisomorphic Presburger structures on locally compact perfect Polish spaces having continuous addition and every standard fixed-division operation, with no retraction onto the standard integers. There are likewise $2^{\aleph_0}$ such structures whose underlying spaces are homeomorphic to Baire space.*

Proof. Use the countable cores G_ζ from the repository's factorial family, with $\zeta \in \widehat{\mathbb{Z}} \setminus \mathbb{Z}$, and apply [Theorem 94.1](#). The family has $2^{\aleph_0}$ extension classes modulo endpoint-preserving isomorphism: the parameters ζ, η define the same such class exactly when $\zeta - \eta \in \mathbb{Z}$. Every nonsplit core is reduced, meaning that its largest divisible subgroup is zero [67, section "An explicit factorial family of nonsplit models"].

For either fixed choice of D , the maximal divisible subgroup of $D \overrightarrow{\times} G_\zeta$ is $D \times \{0\}$. Thus a unital group isomorphism between two transferred groups would induce a unital *abstract*

group isomorphism between their reduced quotients G_ζ and G_η . Such an isomorphism induces multiplication by some nonzero rational on the quotient $\mathbb{Q}$ of the cores. There are only countably many possibilities. For each such quotient map, at most one endpoint-marked extension class is possible: comparing two isomorphisms with the same quotient map gives an endpoint-preserving isomorphism between their targets. Consequently, each abstract unital group isomorphism type accounts for at most countably many marked extension classes, leaving $2^{\aleph_0}$ different transferred types. An isomorphism of positive cones would extend to their Grothendieck groups, so the same conclusion holds for the Presburger monoids.

Only an abstract quotient by the maximal divisible subgroup is used in this argument. That subgroup need not be convex, and no ordered quotient by it is asserted. $\square$

[write] By source 15's [theorem 121.1 \(Part IX\)](#), the locally compact models of this corollary all live on the one space $X_{\mathbb{Q}}$; source 15's [theorem 123.1](#) gives $2^{\aleph_0}$ split models on that space with standard boundary. The two continuum families are disjoint: one is nonsplit, the other split.

95 Canonical residues and automatic topological splitting

The algebraic criterion recalled below is already present in Provelt [67]; the residue kernel is standard [4]. The topological consequences are developed separately.

In this section G is a $\mathbb{Z}$ -group: a discretely ordered abelian group with named least positive element 1_G such that $G/\mathbb{Z}1_G$ is divisible. Standard division with remainder gives, for every integer $m \geq 1$, a homomorphism

$$r_m : G \longrightarrow \mathbb{Z}/m\mathbb{Z}, \quad r_m(1_G) = 1,$$

whose kernel is mG . The maps are compatible, giving the canonical residue homomorphism

$$r : G \longrightarrow \widehat{\mathbb{Z}} = \varprojlim_m \mathbb{Z}/m\mathbb{Z}.$$

Let

$$D = \ker r = \bigcap_{m \geq 1} mG.$$

Lemma 95.1 (The divisible kernel). *The group D is the largest divisible subgroup of G , and $D \cap \mathbb{Z}1_G = \{0\}$.*

Proof. The group G is torsion-free. If $x \in D$ and $m \geq 1$, there is a unique $y \in G$ with $my = x$. For every $n \geq 1$, choose $z \in G$ with $mnz = x$. Cancellation gives $y = nz$, so $y \in \bigcap_n nG = D$. Hence D is divisible. Conversely, every divisible subgroup lies in each mG , and therefore in D . If $k1_G \in D$ for $k \in \mathbb{Z}$, then m divides k for every standard positive m , which forces $k = 0$. $\square$

Theorem 95.2 (Algebraic splitting criterion; part of `isg:cf:main:presburger`). *The following conditions are equivalent:*

1. *There is a group homomorphism $\rho : G \longrightarrow \mathbb{Z}$ with $\rho(1_G) = 1$.*
2. *$G = D \oplus \mathbb{Z}1_G$ as abelian groups.*
3. *$r(G)$ is precisely the standard diagonal copy of $\mathbb{Z}$ in $\widehat{\mathbb{Z}}$.*

When these conditions hold, ρ is unique, its kernel is D , and the group decomposition is lexicographic for the induced order on D .

Proof. Assume (1) and put $K = \ker \rho$. The map $k \mapsto k + \mathbb{Z}1_G$ is an isomorphism $K \cong G/\mathbb{Z}1_G$: subtracting $\rho(g)1_G$ from g proves surjectivity, and the intersection is trivial because $\rho(1_G) = 1$. The quotient is divisible, so K is divisible and $K \subseteq D$. Conversely, every homomorphism from a divisible group to $\mathbb{Z}$ is zero, so $D \subseteq K$. Thus $K = D$, proving (2).

Assume (2). Every $d \in D$ has zero residue modulo every m , so $r(d + k1_G) = r(k1_G)$. Since all standard integer elements occur, this proves (3).

Assume (3). For each $g \in G$, there is a unique standard $k \in \mathbb{Z}$ with $r(g) = r(k1_G)$. Define $\rho(g) = k$. Injectivity of the diagonal map $\mathbb{Z} \rightarrow \widehat{\mathbb{Z}}$ and the homomorphism property of r show that ρ is a homomorphism with $\rho(1_G) = 1$. This proves (1).

Every such retraction vanishes on D and fixes the integer summand, so it is unique. Finally, $\mathbb{Z}1_G$ is convex in G . Any nonzero element of D lies outside this convex subgroup, and its sign is unchanged by adding a standard integer. Thus the decomposition has the order of $D \times \mathbb{Z}$. $\square$

[write] [Theorem 95.2](#) is part of `isg:cf:main:presburger`, as the opening sentence of this section says; [lemma 95.3](#) and [theorem 95.4](#) are new.

95.1 Automatic continuity in Polish topological groups

The next theorem assumes a *group* topology: addition and inversion on the entire signed group G are continuous. It is not a theorem about an arbitrary topology on the positive monoid. No Borelness or compatibility assumption for the arithmetic order is needed in this theorem.

Lemma 95.3 (Finite residues are automatically continuous). *If a $\mathbb{Z}$ -group G carries a Polish group topology, every mG is open and closed, every r_m is continuous, r is continuous, and D is a closed Polish subgroup.*

Proof. The map $g \mapsto mg$ is continuous, so its image mG is analytic and has the Baire property. There are exactly m cosets, represented by $0, 1_G, \dots, (m-1)1_G$. The Baire category theorem implies that mG is nonmeagre.

For completeness, a subgroup A of a Polish group which has the Baire property and is nonmeagre is open. There is a nonempty open set V on which A is comeagre. For all sufficiently small u , the open overlap $V \cap (u + V)$ is nonempty. On that overlap, A and $u + A$ are both comeagre, hence intersect. Thus $u \in A - A = A$, proving that A contains a neighbourhood of 0. Apply this fact to $A = mG$.

An open subgroup is also closed, since its other cosets are open. The fibres of r_m are these open cosets, proving continuity. The inverse-limit map r is continuous because all its coordinates are continuous. Its kernel D is closed; a closed subgroup of a Polish group is Polish. $\square$

Theorem 95.4 (Topological splitting criterion). *Let G be a $\mathbb{Z}$ -group carrying a Polish group topology. The conditions of [Theorem 95.2](#) are equivalent to the existence of a continuous unit-preserving retraction $\rho : G \rightarrow \mathbb{Z}$, where $\mathbb{Z}$ is discrete. When they hold, the canonical map*

$$D \times \mathbb{Z} \longrightarrow G, \quad (d, k) \longmapsto d + k1_G$$

is a topological group isomorphism and an isomorphism of the corresponding lexicographically ordered groups.

Proof. Only automatic continuity and the topological assertion require proof. Under algebraic splitting, D is closed by Lemma 95.3, and its cosets $D + k1_G$, $k \in \mathbb{Z}$, cover G . If D had empty interior, every such closed coset would be nowhere dense, contradicting the Baire category theorem. Thus D is open. Its cosets are the fibres of ρ , so ρ is continuous into discrete $\mathbb{Z}$. The displayed map is continuous, and its inverse is

$$g \mapsto (g - \rho(g)1_G, \rho(g)),$$

which is continuous. The order statement is already proved in Theorem 95.2. $\square$

In particular, every positive cone with the topology inherited from a Polish topological $\mathbb{Z}$ -group has continuous fixed residues. Since each residue class occurs, such a cone is disconnected (already reduction modulo 2 gives a nontrivial clopen partition). This does not settle the connected-universe question for arbitrary Polish monoids with continuous addition: one would first need an appropriate Polish group completion.

[write] (Batch 87.) This settles Part I's question 14.3 and Part III's question 43.4 negatively for cones whose topology is inherited from a Polish topological $\mathbb{Z}$ -group: such a cone is disconnected. Source 15's theorem 121.1 gives the locally compact case again, with the exact carrier. For a topology on the cone alone both questions stay open, as source 13 says.

95.2 Compatibility with the nonsplit example

The nonsplit construction with $L = \mathbb{Z}_{(2)} \times \mathbb{Z}[1/2]$ and diagonal unit $e = (1, 1)$ is consistent with all of these conclusions. In $G = \mathbb{Q}^\omega \overrightarrow{\times} L$, with L discrete, the maximal divisible subgroup is $D = \mathbb{Q}^\omega \times \{0\}$. It is clopen. Its quotient is the countable discrete group L , rather than $\mathbb{Z}$. The finite residue maps are continuous, but their coherent values cannot be represented by a single standard integer for every element. The failure of an integer-coordinate retraction is algebraic, not a failure of continuity of individual residues.

96 The existing omnific criterion and its new Polish applications

The equivalence in this section is recalled from [67, Presburger initiality criterion and unit-preserving omnific embeddings], which proves a stronger result about initial images. It is not claimed as a new theorem here. The normal-form proof below is included to make the application independent of that stronger initiality machinery.

The following equivalence is included as background and an independently checkable explanation of the obstruction, not as a new classification. The repository manuscript [67], theorem labelled `isg:cf:main:presburger`, already proves a stronger criterion with *initial* realizations in **No** and **Oz**. Here we need only ordered additive embeddability, and the sufficiency proof uses a common monomial shift instead of initial-realization theory.

An omnific integer has a Conway normal form whose exponents are nonnegative and whose coefficient at exponent zero is an ordinary integer [5, Section 6]. Denote its constant coefficient by

$$c_0 : \mathbf{Oz} \longrightarrow \mathbb{Z}.$$

Normal-form addition is coefficientwise, so c_0 is additive and $c_0(1) = 1$. In fact it is a unital ring homomorphism: a sum of two nonnegative exponents is zero only when both are zero. Only its additive property is needed below. No continuity or measurability of c_0 is assumed.

Lemma 96.1 (Moving a set-sized group into purely infinite surreals). *Every set-sized divisible ordered abelian group D admits an ordered additive embedding into the group of purely infinite omnific integers.*

Proof. Use the standard embedding theorem that every divisible ordered abelian group has an isomorphic copy as an ordered additive subgroup of $\mathbf{No}$; the stronger initial-subgroup result is recalled in [5, Introduction]. Let $i : D \rightarrow \mathbf{No}$ be such an embedding. Since D is a set and every surreal has a set-sized Conway support, the union

$$S = \bigcup_{d \in D} \text{supp}(i(d))$$

is a set of surreal exponents. Choose a surreal β satisfying

$$\beta > 0, \quad \beta > -s \quad (s \in S).$$

For example, the surreal cut $\{\{0\} \cup \{-s : s \in S\} \mid \emptyset\}$ has this property. Multiplication by the positive monomial ω^β shifts every exponent s to $\beta + s > 0$. Thus

$$j(d) = \omega^\beta i(d)$$

has only strictly positive exponents, is an omnific integer, and has zero constant coefficient. Multiplication by a fixed positive surreal preserves addition and strict order, so j is the desired embedding. $\square$

Theorem 96.2 (Omnific criterion, recalled; `isg:cf:main:presburger`, `isg:cf:cor:unit`). *Let G be a set-sized $\mathbb{Z}$ -group with least positive element 1. The following conditions are equivalent.*

1. *There is a unital additive homomorphism $G \rightarrow \mathbf{Oz}$, with no assumption of injectivity or order preservation.*
2. *There is a unital ordered additive embedding $G \hookrightarrow \mathbf{Oz}$.*
3. *There is an additive retraction $\rho : G \rightarrow \mathbb{Z}$ with $\rho(1) = 1$.*
4. *G is isomorphic, as an ordered group with distinguished unit, to $D \vec{\times} \mathbb{Z}$ for some divisible ordered abelian group D .*
5. *The canonical residue homomorphism $r : G \rightarrow \widehat{\mathbb{Z}}$ has image exactly the standard diagonal copy of $\mathbb{Z}$.*

When these conditions hold, the retraction is unique and $\ker \rho$ is the largest divisible subgroup of G . The same conditions characterize the existence of a unital additive homomorphism from $G_{\geq 0}$ to $\mathbf{Oz}$, or of a unital ordered additive embedding from $G_{\geq 0}$ to $\mathbf{Oz}_{\geq 0}$.

Proof. Condition (ii) implies (i). If f witnesses (i), then $\rho = c_0 \circ f$ witnesses (iii).

Suppose (iii). Algebraically,

$$G = \ker \rho \oplus \mathbb{Z}1, \quad g = (g - \rho(g)1) + \rho(g)1.$$

The kernel $D = \ker \rho$ is divisible. One can see this from $D \cong G/\mathbb{Z}1$, or directly as follows. If $d = nq + s1$ with $0 \leq s < n$, then $0 = n\rho(q) + s$, so $s = 0$ and $\rho(q) = 0$. Thus $d/n \in D$.

Every nonzero $d \in D$ is larger in absolute value than every standard integer. Otherwise, discreteness and a finite induction would place d in $\mathbb{Z}1$, contradicting $D \cap \mathbb{Z}1 = 0$. Consequently the sign of $d + n1$ for $d \neq 0$ is the sign of d . This proves that the displayed algebraic decomposition has the lexicographic order required in (iv).

Suppose (iv). Choose $j : D \rightarrow \mathbf{Oz}$ as in Lemma 96.1. Then

$$f(d, n) = j(d) + n$$

is additive and preserves the unit. Its nonzero purely infinite part dominates every integer, so it is an ordered embedding. This proves (ii).

For the residue formulation, recall that for each standard $n > 0$ there is a unique homomorphism $r_n : G \rightarrow \mathbb{Z}/n\mathbb{Z}$ taking 1 to 1; the compatible system forms $r : G \rightarrow \widehat{\mathbb{Z}}$ [4, §2]. If (iii) holds, uniqueness gives $r_n(g) = \rho(g) \bmod n$, so $r(G)$ is exactly diagonal $\mathbb{Z}$. Conversely, under (v), define $\rho(g)$ to be the unique integer whose diagonal residue is $r(g)$. This is the required additive retraction.

Two retractions differ by a homomorphism vanishing on $\mathbb{Z}1$. Their difference therefore factors through the divisible group $G/\mathbb{Z}1$, and every homomorphism from a divisible group to $\mathbb{Z}$ is zero. This proves uniqueness. A divisible subgroup of G is killed by ρ ; since the kernel itself is divisible, it is the largest one.

Finally, G is the Grothendieck group of its nonnegative cone. An additive map from the cone to the additive group of $\mathbf{Oz}$ extends uniquely by $\widetilde{f}(a - b) = f(a) - f(b)$; this is well defined because $a - b = a' - b'$ implies $a + b' = a' + b$. Thus a unital cone homomorphism implies (i). Conversely, the ordered embedding constructed above restricts to an ordered embedding of the cones. $\square$

[write] The equivalence of (i), (ii) and (iii) is `isg:cf:cor:unit`, and $(iii) \Leftrightarrow (iv) \Leftrightarrow (v)$ is contained in `isg:cf:main:presburger`; source 13 recalls them and supplies a direct normal-form proof.

96.1 A two-element certificate excluding even a homomorphism

For the explicit core, the failure is visible without any abstract decomposition. In its nonnegative cone put

$$x = (-1, 0), \quad y = (0, 1) = x + e.$$

Both are positive because $\pi(x) = \pi(y) = 1$. The element x is divisible in the positive cone by 3^j for every j , with quotient $(-3^{-j}, 0)$; the element y is divisible there by 2^j for every j , with quotient $(0, 2^{-j})$.

Suppose a unital additive map $f : K_{\geq 0} \rightarrow \mathbf{Oz}$ existed. The integer $c_0(f(x))$ would be divisible by every 3^j , so it would be zero. Similarly, $c_0(f(y)) = 0$. But $y = x + e$ would give

$$0 = c_0(f(y)) = c_0(f(x)) + c_0(f(e)) = 0 + 1,$$

a contradiction. These same two points, placed in the fiber $D = 0$, prove the obstruction for both perfect Polish models of Theorem 94.1. In particular, neither a different topology nor a loss of injectivity can repair the unital additive embedding problem.

96.2 The precise set/class boundary

The theorem does not quantify over a set of all surreal numbers. It can be read in a conservative class framework such as NBG with the usual surreal construction, while G, D ,

$i[D]$, and S in the proof are sets. In a set-theoretic coding, surreal numbers are set-sized sign sequences and the embedding has a set-sized graph. Replacement bounds the birthdays of its image and the supports occurring in the proof by a suitable ordinal. The cut defining β has set-sized left and right sides, so it is a legitimate surreal cut. After construction, all elements of the final embedded group also belong to a sufficiently large set-sized birthday stage. No assertion that the whole proper class carries a Polish topology is involved.

The set-size hypothesis is used decisively when taking the union of all supports and choosing a common monomial shift. This proof does not provide the same conclusion for arbitrary proper-class groups. It also proves an additive and order-theoretic criterion only. The embedding produced by multiplying a group copy by one monomial is not asserted to preserve a multiplication operation or to have an initial image in the simplicity tree.

96.3 Relation to the incoming research questions

The countable-core transfer answers the existence part of the incoming “beyond split topological models” question. It supplies explicit uncountable Polish examples with continuous fixed divisions for which no integer-coordinate retraction exists at all. The omnific criterion identifies exactly why the incoming additive realizations inside **Oz** cannot cover these examples: preserving the standard unit forces the entire coherent residue system to come from one ordinary integer. Thus the obstruction is algebraic and persists before imposing any regularity on a proposed embedding.

The countable nonsplit algebra and the general omnific criterion have explicit prior counterparts in the repository. The contribution proposed here is their transfer to perfect Polish models, together with elementary CRT coordinates and a short divisibility certificate. The searches recorded for this project do not establish historical priority in the entire literature.

[write] The “incoming ‘beyond split topological models’ question” is Part I’s [question 14.11](#); the existence half is answered here, the characterization asked for there is not.

97 Finite continuous branches for every definable function

Finite affine descriptions of Presburger functions are classical; see [28]. We prove the precise candidate form needed for the topological arguments, directly from quantifier elimination.

Let D be a divisible ordered abelian group, and put $G(D) = D \overrightarrow{\times} \mathbb{Z}$, with positive cone $M(D)$. Write q_m for Euclidean quotient by a positive standard integer m :

$$q_m(d, n) = \left(\frac{d}{m}, \left\lfloor \frac{n}{m} \right\rfloor \right).$$

For the two topological models $D = \mathbb{R}$ and $D = \mathbb{Q}^\omega$, every $q_m : G(D) \rightarrow G(D)$ is continuous. Parameters are held fixed unless they are explicitly made variables.

Theorem 97.1 (Finite continuous branches). *Every Presburger-definable total function $f : M(D)^k \rightarrow M(D)^r$, with a finite parameter tuple a , admits a finite definable partition $C_1, \dots, C_s$ of its domain and candidate maps $F_j : G(D)^k \rightarrow G(D)^r$ such that $f = F_j$ on C_j . Each coordinate of each candidate has the form*

$$q_m \left(c + \sum_i b_i x_i + \sum_t c_t a_t \right), \quad m \in \mathbb{Z}_{>0}, \quad b_i, c_t \in \mathbb{Z}, \quad c \in \mathbb{Z}\mathbf{1}. \quad (\text{B1})$$

The candidates and partition are obtainable by a finite syntactic procedure from a formula for the graph. The same assertion holds with arbitrary ambient inputs $G(D)^k$ provided the output is still in $M(D)^r$. In the two topological models every candidate is continuous into the ambient group. A candidate need not be $M(D)$ -valued outside its own cell.

Proof. First take a single output variable y . Relativize the original formula to the ordered group and eliminate quantifiers. Conjoin $y \geq 0$ and put the resulting formula into disjunctive normal form. Replace an equality by two inequalities, a disequality by a disjunction of strict inequalities, and strict inequalities by weak inequalities with a unit shift. Each disjunct is a conjunction of linear inequalities, congruence conditions, and negated congruence conditions.

For a particular disjunct, take L to be a common positive multiple of the absolute values of all its nonzero coefficients of y , with $L = 1$ if needed. Substitute $u = Ly$ and add $L \mid u$. Multiplying an inequality by a positive standard integer makes its coefficient of u equal to 1 or -1 . A congruence with nonzero coefficient b is transformed using

$$d \mid (by + t) \iff d \frac{L}{|b|} \mid \left(\operatorname{sgn}(b)u + \frac{L}{|b|}t \right). \quad (\text{B2})$$

This equivalence follows by torsion-freeness from $d \mid w \iff cd \mid cw$ for positive standard c . Terms independent of u are left unchanged.

Let m be a common multiple of L and all transformed moduli. Split into the finitely many standard residues $u \equiv r \pmod{m}$. Within one such residue, all congruence literals reduce to conditions on the inputs and parameters. The remaining constraints on u have the form

$$u \geq b_1, \dots, u \geq b_p, \quad u \leq c_1, \dots, u \leq c_q, \quad u \equiv r \pmod{m}, \quad (\text{B3})$$

where all b_i, c_i are affine group terms. The lower-bound list is nonempty because $y \geq 0$ supplied $u \geq 0$.

Fix an input for which this disjunct has a solution. Let $B = \max_i b_i$, and let $s \in \{0, \dots, m-1\}$ be the unique standard integer such that $B + s \equiv r \pmod{m}$. Then $u_0 = B + s$ is the least element of this residue class above B . To check this in an arbitrary nonstandard group, if $u \geq B$ has the same residue, write $u = B + s + mv$. If $v < 0$, discrete order gives $v \leq -1$, hence $u \leq B + s - m < B$, a contradiction. Thus $u \geq u_0$. Because some solution satisfies the upper bounds, so does u_0 . It therefore satisfies the entire disjunct. Since that disjunct lies inside the graph of a function, its solution is unique, and the value of y is exactly $q_L(B + s)$. The expression B is one of the finitely many terms b_i , and s belongs to a finite standard set. Enumerating all such possibilities over all disjuncts gives the candidates (B1).

For vector outputs, apply this argument to each coordinate graph, obtained by existentially quantifying the other coordinates, and take finite products of candidate lists. To make a partition, assign an input to the first candidate which satisfies the original graph. This is a finite definable partition: quotient functions are themselves Presburger definable, so their use here can be eliminated from formulas. Continuity of the candidates follows from continuity of affine group maps and of q_m . $\square$

Audit of the nonstandard step. The proof used the least element of a fixed congruence class above a given bound, explicitly justified from division and discrete order. It did not use an iteration of successor a nonstandard number of times, or infer invariance modulo mG from invariance under one standard translation. This is the distinction highlighted in the archived manuscripts' finite-residue proofs.

98 An exact compiler for continuity loci

Let D denote either $\mathbb{R}$ with its usual topology and order, or $\mathbb{Q}^\omega$ with its product-of-discrete topology and lexicographic order. Put

$$G = D \overrightarrow{\times} \mathbb{Z}, \quad M = G_{\geq 0} = (D_{>0} \times \mathbb{Z}) \cup (\{0\} \times \mathbb{N}).$$

Coordinates of an input tuple are written $x = (u, n) \in M^k$, where $u \in D^k$ and $n \in \mathbb{Z}^k$. All definability below means Presburger definability, with a fixed finite tuple of parameters, unless an expanded coordinate language is explicitly specified. Coordinate projections themselves are not presumed to be Presburger definable.

We use the finite-branch normal form: a definable total map $f : M^k \rightarrow M^r$ admits a finite definable partition $C_1, \dots, C_s$ and maps $F_1, \dots, F_s : G^k \rightarrow G^r$ such that $f|_{C_j} = F_j|_{C_j}$ and each F_j is continuous. Its D -coordinates are rational-affine expressions in u ; its integer coordinates are integer parts of rational-affine expressions in n , with any needed fixed residue partition absorbed into C_j . The extensions F_j need not have values in M away from C_j . This causes no problem for the continuity criterion, which compares limits in the ambient Hausdorff group G^r .

Apply quantifier elimination to the cells. List their affine group forms as

$$L_i(u, n) = (\ell_i(u), z_i(n)), \quad \ell_i(u) = a_i \cdot u + b_i, \quad i = 1, \dots, t.$$

Here $a_i \in \mathbb{Q}^k$, $b_i \in D$ is a rational-linear combination of parameter coordinates, and after clearing denominators z_i is an integer-affine expression. Atomic formulas compare L_i with zero or test an integer congruence. When $\ell_i(u) \neq 0$, the sign of L_i is the sign of $\ell_i(u)$. When $\ell_i(u) = 0$, it is the sign of $z_i(n)$. An equality $L_i = 0$ requires both coordinates to vanish. A congruence depends only on integer coordinates.

98.1 Rational directions describe all local sign possibilities

Fix $x = (u, n) \in M^k$. The integer tuple n is constant on a neighbourhood of x . Let

$$I(x) = \{i : \ell_i(u) = 0\}, \quad B(x) = \{q : u_q = 0\}.$$

For $i \notin I(x)$, the sign of ℓ_i is locally constant. For $i \in I(x)$, a perturbation $\delta \in D^k$ changes its value to $a_i \cdot \delta$. The only local domain restrictions are $\delta_q \geq 0$ for $q \in B(x)$: an input with zero D -coordinate has nonnegative integer coordinate, and may move to zero or positive D -coordinate, but not to negative D -coordinate.

Lemma 98.1 (Rational directions suffice). *A finite pattern of signs of the forms $a_i \cdot \delta$, together with the restrictions $\delta_q \geq 0$ for $q \in B(x)$, is realizable in D^k if and only if it is realizable by a rational vector $v \in \mathbb{Q}^k$.*

Proof. For $D = \mathbb{R}$, the conditions are a finite homogeneous system of rational linear equalities, strict inequalities, and weak inequalities. A nonempty rational polyhedral relatively open region contains a rational point. More explicitly, solve the equality subsystem over $\mathbb{Q}$, restrict to its rational basis, and approximate a real solution closely enough to preserve the finitely many strict signs and the face determined by any vanishing weak inequalities.

For $D = \mathbb{Q}^\omega$, suppose δ realizes the pattern. For each nonzero form, include the coordinate at which its first nonzero entry occurs, and do likewise for each nonzero δ_q appearing in a

domain restriction. Choose N beyond all these finitely many coordinates and put

$$v_q(\varepsilon) = \sum_{j=0}^N \delta_q(j) \varepsilon^j.$$

Every required nonzero linear form in these polynomials has the sign of its first nonzero coefficient when $\varepsilon > 0$ is sufficiently small. Every required zero form is identically zero, since all its sequence coordinates were zero. A sufficiently small positive rational ε therefore gives the required rational vector. Conversely, a rational vector is realized in $\mathbb{Q}^\omega$ by putting it in a single coordinate. $\square$

For the topological use, rational directions are not just witnesses to abstract consistency. In the real model, $\delta = \varepsilon v$ tends to zero as $\varepsilon \downarrow 0$. In the stream model, put $\delta_q^{(N)} = v_q e_N$, where e_N has a single 1 in coordinate N . Then $\delta^{(N)} \rightarrow 0$. These perturbations preserve all inactive nonzero signs eventually and realize the chosen active signs exactly. If all required signs are zero, the constant zero perturbation is allowed; this correctly accounts for a cell which contains x itself.

Theorem 98.2 (Exact local accessibility). *The following are equivalent for a cell C_j :*

1. $x \in \overline{C_j}$, with closure relative to M^k .
2. There is $v \in \mathbb{Q}^k$ satisfying $v_q \geq 0$ for $q \in B(x)$ such that the Boolean formula for C_j is true after: freezing all integer coordinates at n ; retaining the signs of the inactive $\ell_i(u)$; and replacing each active sign by $\text{sgn}(a_i \cdot v)$.

Proof. A rational witness gives arbitrarily close points in the cell by the perturbations just constructed. Conversely choose a point of C_j sufficiently close that the finitely many inactive signs and all integer coordinates agree with those of x . Its displacement realizes the required homogeneous sign pattern. Apply the rational-direction lemma. $\square$

This theorem is an algorithm: enumerate the finitely many assignments of signs $-1, 0, 1$ to the active forms, evaluate the Boolean cell formula, and decide feasibility of the associated rational linear system. Strict inequalities introduce no difficulty; a homogeneous feasible system with finitely many strict positive requirements can be rescaled so that all those positive quantities are at least 1, and ordinary rational linear feasibility then applies. If there are no strict requirements, solve the weak system directly. Rational Fourier–Motzkin elimination is one possible implementation.

98.2 The continuity criterion and its finite output

Theorem 98.3 (The branch-limit criterion). *For $x \in M^k$,*

$$f \text{ is continuous at } x \iff \forall j (x \in \overline{C_j} \implies F_j(x) = f(x)). \quad (\text{C})$$

Proof. Suppose f is continuous and $x \in \overline{C_j}$. Since the domain is metrizable there is a sequence $x_m \in C_j$ tending to x . In G^r , $f(x_m) = F_j(x_m) \rightarrow F_j(x)$ by continuity of F_j , and $f(x_m) \rightarrow f(x)$ by continuity of f . Uniqueness of limits yields the equality. The value $F_j(x)$ is allowed initially to lie outside M^r ; the conclusion rules that possibility out for accessible branches.

Conversely, assume every accessible branch agrees at x . For each inaccessible branch choose a neighbourhood of x avoiding its cell. There are only finitely many such branches. On every

remaining branch, continuity of its ambient extension at x gives the usual neighbourhood estimate around $f(x)$. Intersect the finitely many neighbourhoods. This proves continuity into G^r , hence into its subspace M^r . $\square$

In particular, the discontinuity set is exactly

$$\bigcup_{i,j=1}^s \{x \in C_i : x \in \overline{C_j}, F_i(x) \neq F_j(x)\}. \quad (D)$$

Using rational accessibility to compile the closure conditions produces a finite formula for this set. Its atoms are signs and equalities of rational-affine D -coordinate expressions, and Presburger conditions on integer coordinates. It is crucial to describe this as an *external coordinate formula*. It need not be a formula in the original Presburger structure.

Corollary 98.4 (Exact form of discontinuity sets). *The discontinuity set of every definable $f : M^k \rightarrow M^r$ is a Δ_2^0 set. More strongly, it is a finite union of locally closed sets and is contained in a finite union of closed nowhere dense affine D -coordinate hyperplanes. Thus it is nowhere dense; its closure is nowhere dense as well. It need not itself be closed.*

Proof. In both coefficient models, each strict sign set of a continuous D -affine form is open and each zero set is closed. Integer conditions are clopen because the integer-coordinate space is discrete. Finite Boolean combinations therefore form finite unions of intersections of an open and a closed set. Such locally closed sets and their finite unions are $F_\sigma \cap G_\delta$ in a metrizable space.

If x lies on none of the nonconstant hyperplanes $\ell_i(u) = 0$ associated with the cell partition, all cell truth values are locally constant. Forms constant in the variable u also have locally constant truth values: they either have fixed nonzero sign, or are decided by the locally constant integer tuple. Hence f locally agrees with one continuous candidate and is continuous. Each nonconstant hyperplane is closed and nowhere dense relative to M^k : vary an input coordinate with nonzero coefficient by an arbitrarily small positive perturbation, retaining membership in M . In the stream model such a perturbation may be placed arbitrarily far out in the sequence. A finite union remains closed nowhere dense. $\square$

98.3 Two exact examples and necessary qualifications

Define

$$f(x, y) = \begin{cases} x, & y = 0, \\ 0, & y \neq 0. \end{cases}$$

This map is Presburger definable, and in both models its exact discontinuity set is

$$\text{Disc}(f) = \{(x, 0) : x \neq 0\}.$$

At a point of this set, let the second input approach zero through $(\varepsilon, 0)$ in the real model or through $(e_N, 0)$ in the stream model. The outputs become zero, whereas $f(x, 0) = x \neq 0$. If $y \neq 0$, the function is locally zero since the complement of the closed singleton $\{0\}$ is open. At $(0, 0)$ every output is either zero or the first input, so continuity holds. This discontinuity set is not closed.

For truncated subtraction

$$x \dot{-} y = \max(x - y, 0),$$

write $x = (u, n)$ and $y = (v, m)$. The compiler gives

$$\text{Disc}(\dot{-}) = \{((u, n), (v, m)) : u = v, n \neq m\}. \quad (\text{M})$$

If $u - v \neq 0$, its sign is stable and one of the two continuous branches is selected nearby. If $u = v$ and $n = m$, both ambient branches agree at zero. If $u = v$ and $n \neq m$, arbitrarily small admissible perturbations of the D -coordinates realize both signs of their difference; the branch limits are the distinct points $(0, n - m)$ and zero in G . This proves the formula, including boundary inputs with $u = v = 0$.

The set in (M) is not Presburger definable, even with parameters. If it were, its section at $y = 0$, with the point $x = 0$ added, would define exactly the standard copy of $\mathbb{N}$ in M . That set contains zero and is closed under successor, so the induction scheme with those parameters would force it to be all of M , a contradiction. Thus an exact continuity compiler must be allowed to output the expanded coordinate description, rather than claiming definability in the input language.

Strict signs cannot be removed from the output language. The archived research question proposed descriptions involving equalities of divisible-coordinate affine expressions and residue conditions. That fragment is too small in both models. Choose a parameter $a = (\alpha, c)$ with $\alpha > 0$, and define

$$g_a(x, y) = \begin{cases} 1, & x > a \text{ and } y = 0, \\ 0, & \text{otherwise.} \end{cases}$$

Writing $x = (u, n)$, the exact discontinuity set is

$$\text{Disc}(g_a) = \{(x, 0) : u \geq \alpha\}.$$

For $u > \alpha$ the output at $y = 0$ is one and nearby nonzero y gives zero. For $u < \alpha$ a neighbourhood forces $x < a$, so the map is zero. At $u = \alpha$, approaching from larger divisible coordinate gives one with $y = 0$, while approaching through nonzero y gives zero, regardless of the integer comparison at the point. At $y \neq 0$, the map is locally zero. These observations prove the formula in both models.

In either model restrict to $x = (t\alpha, 0)$ with rational $t \geq 0$, and to $y = 0$. Each affine equality in this remaining scalar variable holds at at most one t or at every t , because D is torsion-free and $\alpha \neq 0$. A finite Boolean combination therefore defines a finite or cofinite subset of $\mathbb{Q}_{\geq 0}$. All residue conditions are constant on this slice. But the resulting threshold $t \geq 1$ is neither finite nor cofinite. Thus the equality-and-residue-only proposal has a negative answer; the additional sign predicates in the compiler are necessary.

[write] (Batch 87.) This answers Part I's [question 14.13](#): the discontinuity set of every definable total function is computed from the graph ([theorem 98.3](#), [theorem 98.5](#)), and the fragment proposed there, equalities of divisible-coordinate affine expressions and residue conditions, is too small, by the example g_a . The formula (M) is Part I's (8.2) ([theorem 8.3](#)). [Corollary 98.4](#) generalizes that theorem to every definable function. The output is a formula in an expanded two-sorted coordinate language, not a Presburger definition, as source 13 stresses (and the Hilbert's-tenth programme's review of the arrival notes, [appendix E](#)).

98.4 Deciding global continuity after coordinate elimination

The output formula (D) lives in a two-sorted language. The first sort is the divisible ordered abelian group D , with rational scalar maps; the second is Presburger integer arithmetic. Each

atomic relation belongs entirely to one of the two sorts. The domain condition is the finite Boolean combination

$$\bigwedge_{q=1}^k (u_q > 0 \vee (u_q = 0 \wedge n_q \geq 0)).$$

The apparent interaction between the sorts occurs only through Boolean combinations. Consequently, after distributing a finite disjunction, existential quantification of all input coordinates separates into ordinary divisible-ordered-group feasibility and ordinary Presburger integer feasibility.

Theorem 98.5 (Effective global continuity criterion). *Given a Presburger definition of a total function, with a finite parameter tuple, one can effectively construct a finite Boolean combination Ψ of:*

- *signs and equalities of rational-linear expressions in the D -coordinates of the parameters; and*
- *Presburger conditions on the integer coordinates of the parameters,*

such that the function is globally continuous exactly when Ψ holds. The same procedure compiles continuity at a named point by treating that point's coordinates as additional parameters.

Proof. Apply (D), add the domain condition, and existentially quantify the inputs. Perform quantifier elimination for divisible ordered abelian groups and for Presburger arithmetic, then negate the result. All preceding transformations are finite syntactic procedures. The rational accessibility feasibility tests depend only on the rational coefficients of the active forms and the selected zero-coordinate face; they therefore may be performed at compilation time for every possible finite sign pattern. $\square$

This yields a genuine ordinary decision algorithm when the needed parameter sign/equality tests are decidable, in particular for rational real parameters and for exactly represented finitely supported rational sequences. With arbitrary infinite input or parameter streams, it yields an oracle reduction, not an ordinary decision procedure. A zero-sequence test is already discontinuous, so that distinction cannot be removed. For the stream model, each finitely many affine sequence forms can be scanned until its first nonzero coordinate is found, treating the form as zero in the meantime; the Boolean answer has a computable eventually stable approximation. A separate sharper rank argument may improve the mind-change bound; it is not needed for the compiler theorem.

99 Exact target-valued Baire classification

Here “Baire class one” means a pointwise limit of continuous maps into the stated target space, with that target's actual topology. It is not being used as a synonym for having F_σ inverse images of open sets. That distinction is decisive for the real-coordinate model.

99.1 The stream model: explicit locally constant approximations

Let $D_\omega = \mathbb{Q}^\omega$ with discrete rational coordinates and lexicographic order. Let $T_N : G(D_\omega) \rightarrow G(D_\omega)$ retain the first N rational coordinates, set all later rational coordinates to zero, and retain the integer coordinate. Its image is the subgroup $G_N \cong \mathbb{Q}^N \times \mathbb{Z}$. This is an additive projection and commutes with every q_m . It need not preserve the positive cone: a positive element whose first nonzero rational coordinate occurs after N , and whose integer coordinate

is negative, is sent to a negative element. The following construction explicitly accounts for that issue.

Theorem 99.1 (Locally constant approximation with eventual exact prefixes). *Every parameter-definable total function $f : M(D_\omega)^k \rightarrow M(D_\omega)^r$ is a pointwise limit of locally constant functions f_N with the same domain and target. Given a graph formula and streams for its parameters, these functions can be computed uniformly: f_N requires only the first N rational coordinates and the integer coordinates of the inputs and parameters. For each fixed input x , there is N_x such that*

$$f_N(x) = T_N f(x) \quad (N \geq N_x). \quad (\text{B4})$$

Proof. Let $R(x, a, y)$ be an ambient group formula for the graph, explicitly including $y \in M^r$. Replace it by the following totalized graph: if there is a unique $y \in M^r$ satisfying $R(x, a, y)$, return that y ; otherwise return the zero tuple. This defines a function $F : G^{k+p} \rightarrow M^r$ for every input and parameter tuple in every $\mathbb{Z}$ -group. It agrees with f at the original parameter tuple and at inputs in the original domain. The definition is a first-order formula using existence and uniqueness; no choice operation is needed.

Evaluate this same totalized formula in the finite-coordinate model G_N on $(T_N x, T_N a)$, and include its output in $M(D_\omega)^r$. Call the resulting output $f_N(x)$. It always lies in the target, even when a truncated input or parameter lies outside the positive cone. Every f_N depends on finite coordinate data and is therefore locally constant. It is effective because arithmetic and order in $\mathbb{Q}^N \times \mathbb{Z}$ are decidable and the finite-branch theorem provides a finite list of possible outputs.

Apply the finite-branch theorem and quantifier elimination to the totalized graph, with the parameters treated as variables. In every atomic cell condition, a nonzero affine rational-coordinate stream has a first nonzero coordinate. After that coordinate is retained, its sign is fixed forever. A zero stream remains zero in every truncation. Integer equalities and congruences are unchanged. Consequently every atomic truth value, and hence every cell truth value, eventually agrees with its value at (x, a) . There are only finitely many atoms. For all sufficiently large N , the selected candidate is therefore the same candidate F_j as at (x, a) . Since affine group terms and fixed quotients commute with T_N ,

$$f_N(x) = F_j(T_N x, T_N a) = T_N F_j(x, a) = T_N f(x).$$

This proves (B4). Coordinate truncations converge in the product topology, proving the asserted pointwise convergence. $\square$

The theorem gives an actual target-valued Baire-one approximation, rather than only a classification of the graph or its coordinate functions. The integer output coordinate stabilizes, and each desired finite output prefix is eventually exact. There is no claimed effective bound on the stabilization stage. A stream which vanishes for a long initial segment may reveal its first nonzero coordinate arbitrarily late.

A dimension bound on branch revisions. Suppose there are k input streams and p parameter streams. Compile the totalized function into a common finite list of quantifier-free cell conditions, and let ρ be the rational rank of the coefficient vectors of all affine D -coordinate forms occurring in their comparisons. Then the selected candidate index in the approximation above changes at most $\rho \leq k + p$ times. The entire output integer tuple likewise changes at most ρ times.

Indeed, the finite-rank truth theorem, Theorem 100.1, applies simultaneously to all the cell atoms. A new stage at which an as-yet-zero leading form becomes nonzero contributes

an independent row functional on the ρ -dimensional span of comparison forms, so there are at most ρ such stages. Between them every atom and hence the selected branch is unchanged. For a fixed candidate, its output integer tuple depends only on the input and parameter integer tuples, which truncation leaves unchanged. This proves both bounds. It does not bound the number of distinct full stream outputs: even a single fixed branch can acquire further nonzero output coordinates at infinitely many stages.

99.2 The real model: a complete criterion and a sharp obstruction

Write $M_{\mathbb{R}} = M(\mathbb{R})$. For $n = (n_1, \dots, n_k) \in \mathbb{Z}^k$, the input slice of fixed integer coordinates is

$$X_n = \prod_{i=1}^k I_{n_i} \times \{n\}, \quad I_t = \begin{cases} [0, \infty), & t \geq 0, \\ (0, \infty), & t < 0. \end{cases} \quad (\text{B5})$$

Each X_n is nonempty, connected, and clopen in $M_{\mathbb{R}}^k$. For a vector-valued map, let $\pi_{\mathbb{Z}}f$ denote the entire tuple of output integer coordinates.

Theorem 99.2 (Exact real-model Baire classification). *For a Presburger-definable total function $f : M_{\mathbb{R}}^k \rightarrow M_{\mathbb{R}}^r$, the following are equivalent:*

1. f is Baire class one with target $M_{\mathbb{R}}^r$.
2. f belongs to some countable Baire class with that target.
3. For every $n \in \mathbb{Z}^k$, the map $\pi_{\mathbb{Z}}f$ is constant on X_n .

Thus every definable map admitted anywhere in the countable target-valued Baire hierarchy already belongs to class one. Other definable maps are outside that hierarchy altogether.

Proof. Necessity. For every continuous $g : X_n \rightarrow M_{\mathbb{R}}^r$, its integer-coordinate map is continuous into the discrete space $\mathbb{Z}^r$, so connectedness forces it to be constant. This property is preserved by pointwise limits. Indeed, suppose g_m have constant integer tuples c_m on X_n and $g_m \rightarrow g$ pointwise. At one fixed point of X_n , convergence in the discrete coordinate forces c_m eventually to equal one tuple c . The same sequence c_m occurs at every point, so $\pi_{\mathbb{Z}}g$ is identically c throughout X_n . Transfinite induction over the usual countable Baire hierarchy proves the claim for every countable Baire class, establishing (2) $\Rightarrow$ (3).

Sufficiency. In both coefficient models every definable cell is Δ_2^0 . For completeness, a group equality is closed; a strict or weak order condition is a finite union of an open strict sign condition on the D -coordinate and a closed zero condition with a clopen integer test. Both it and its complement are F_σ . Congruence predicates are clopen. Quantifier elimination and finite Boolean closure prove the assertion.

On a metric space the real-valued characteristic function of a Δ_2^0 set C is Baire one. To see this directly, choose increasing closed exhaustions $A_m \uparrow C$ and $B_m \uparrow (X \setminus C)$. When both are nonempty, the continuous functions

$$u_m(x) = \frac{d(x, B_m)}{d(x, A_m) + d(x, B_m)}$$

converge to $\mathbf{1}_C$. The denominator is positive because the two closed sets are disjoint. If A_m is empty use $u_m = 0$; if only B_m is empty use $u_m = 1$. These conventions preserve eventual convergence.

Apply this construction to the finite partition in the finite-branch theorem. If H_j is the real-coordinate part of the continuous candidate F_j , then the real-coordinate part h of f is

$$h = \sum_{j=1}^s \mathbf{1}_{C_j} H_j.$$

Replacing each characteristic function by its continuous approximation gives continuous maps h_m into $\mathbb{R}^r$ converging pointwise to h .

Under condition (3), write the output integer tuple as $w(n)$; it is a continuous function of the input because the slices X_n are clopen. For $m \geq 1$ and output coordinate j , replace $h_{m,j}$ by

$$\tilde{h}_{m,j}(x) = \begin{cases} \max\{h_{m,j}(x), 0\}, & w_j(n) \geq 0, \\ \max\{h_{m,j}(x), 1/m\}, & w_j(n) < 0. \end{cases}$$

These are continuous, land in the required real-coordinate half-lines, and converge to h_j . Thus $x \mapsto (\tilde{h}_m(x), w(n))$ are continuous maps into the actual target $M_{\mathbb{R}}^r$ converging pointwise to f . This proves (3) $\Rightarrow$ (1); (1) $\Rightarrow$ (2) is immediate. $\square$

A parameter-free example. The definable function

$$\text{sgn}_M(x) = \begin{cases} 0, & x = 0, \\ 1, & x > 0 \end{cases}$$

has a Δ_2^0 graph and Δ_2^0 inverse images of open sets. On $M_{\mathbb{R}}$ it belongs to no countable target-valued Baire class: its integer output is zero at the endpoint of X_0 and one elsewhere on that connected half-line. On $M(D_\omega)$ it is Baire class one by the explicit approximation theorem. This isolates the precise reason that graph complexity alone cannot settle the target-valued question.

An effective addition to the classification. The criterion (3) can itself be compiled into the external two-sorted coordinate language used by the continuity theorem: assert that there are no two inputs with the same integer tuple but different integer output tuples. Substitute the finite branch candidates, split the finite cell cases, and eliminate the real-coordinate quantifiers by divisible-ordered-group elimination and the integer quantifiers by Presburger elimination. The result is a finite Boolean combination of rational-linear sign tests on the real parts of the parameters and Presburger conditions on their integer parts. Consequently membership in the target-valued Baire hierarchy is decidable for parameter-free definitions, and for parameter presentations with decidable required real sign/equality tests. Arbitrary real streams do not supply those tests as uniformly decidable operations.

99.3 Topological extension to a countable discrete core

Corollary 99.3 (Regularity over every countable discrete core). *Let K be any countable $\mathbb{Z}$ -group with the discrete topology, let $D = \mathbb{R}$ or $D = \mathbb{Q}^\omega$ with the respective topologies used above, and put*

$$G = D \overrightarrow{\times} K, \quad P = G_{\geq 0}.$$

For every parameter-definable total map $f : P^k \rightarrow P^r$, the following topological conclusions continue to hold:

1. *There are finitely many continuous ambient quotient-affine candidates. The rational-direction accessibility criterion and branch-limit continuity criterion remain valid with the core coordinates frozen. Consequently $\text{Disc}(f)$ is a finite union of locally closed sets contained in finitely many closed nowhere dense affine D -coordinate hyperplanes.*
2. *For $D = \mathbb{Q}^\omega$, there are locally constant maps $f_N : P^k \rightarrow P^r$ such that, for each x , eventually $f_N(x) = T_N f(x)$; here T_N truncates only the rational stream and leaves the K -coordinate unchanged. In particular f is target-valued Baire class one.*
3. *For $D = \mathbb{R}$, membership in any countable target-valued Baire class is equivalent to class one, and holds exactly when the output K -tuple is constant on every slice with fixed input K -tuple.*

Proof. The group G is a $\mathbb{Z}$ -group, and its standard quotient maps are

$$q_m(d, b) = (d/m, q_m^K(b)).$$

They are continuous because K is discrete. The finite-branch proof uses only Presburger elimination, discrete order, and standard division, so it applies unchanged. On a neighbourhood of an input, all core coordinates are fixed. Every core affine equality, order test, and congruence therefore has a fixed truth value. The only varying signs are the same rational-affine D -coordinate forms as before. At a zero D -coordinate the domain again allows exactly nonnegative directions. The accessibility proof, branch-limit proof, and nowhere-dense hyperplane argument are consequently unchanged. Core conditions are clopen, so the finite locally closed description follows as well.

In the stream case, totalize the graph on the ambient group and evaluate it on truncated inputs and parameters in $\mathbb{Q}^N \vec{\times} K$. This is again a $\mathbb{Z}$ -group, so the totalized output exists uniquely and lies in its positive cone. The resulting map is locally constant, because it depends only on a finite rational prefix and discrete core data. All core tests remain fixed while the leading-coordinate tests eventually stabilize. Quotient-affine candidates commute with truncation, giving $f_N(x) = T_N f(x)$ eventually, exactly as before.

In the real case the fixed-core slices are products of $[0, \infty)$ and $(0, \infty)$, according to the signs of their core coordinates. They are connected and clopen. Projection to the discrete target K^r therefore gives the same transfinite constancy obstruction. For sufficiency, approximate the leading real coordinates and clamp them to $[0, \infty)$ or to positive values according to the constant output core coordinate. The earlier construction then produces continuous maps into P^r converging pointwise to f . $\square$

This extension is purely topological. No computable presentation of K is assumed. Countability and discreteness alone do not supply an algorithm for evaluating core arithmetic or make the displayed locally constant approximants uniformly computable. Effective versions must separately specify a presentation of K and the available operations and decision procedures.

100 The exact number of changes of mind

The earlier Provett report [64] proved a finite-change truth theorem and bounded the number of changes by the number of equality and order atoms after quantifier elimination. Its question entitled “Optimal finite-mind-change complexity” asked for the smallest possible bound for an individual formula. We first give a bound by linear dimension, which is sharp as a function of the number of input streams. We then give a finite, effective calculation of the optimum for each parameter-free formula.

100.1 Presentation and counting convention

Let

$$D = \mathbb{Q}^{\mathbb{N}}, \quad G = D \overrightarrow{\times} \mathbb{Z}, \quad M = G_{\geq 0},$$

where a nonzero rational stream is positive when its first nonzero coefficient is positive, and the D coordinate precedes the integer coordinate in the order on G . Give G the product topology in which each rational coordinate and the integer coordinate are discrete. An element of M has the form $x = (h, n)$, with

$$h \geq_{\text{lex}} 0, \quad h = 0 \implies n \geq 0.$$

Thus $n < 0$ guarantees that h has a positive first nonzero coefficient. The earlier report provides a computable homeomorphism between this presentation and a total coding by Baire space. A valid coordinate code can also be converted to that Baire code effectively: when $n < 0$, search for its guaranteed positive leading coefficient before producing the corresponding negative-slice tag.

A *truth approximation* for a relation $f : M^d \rightarrow \{0, 1\}$ is a sequence of continuous maps $A_s : M^d \rightarrow \{0, 1\}$ satisfying

$$\lim_{s \rightarrow \infty} A_s(\bar{x}) = f(\bar{x}) \quad \text{for every } \bar{x} \in M^d.$$

Its number of changes on $\bar{x}$ is

$$\#\{s \in \mathbb{N} : A_{s+1}(\bar{x}) \neq A_s(\bar{x})\}.$$

The initial output is not counted as a change. A computable approximation means that the stage maps are uniformly Type-2 computable. Each output must use only finite input information, but its computation may inspect an input-dependent finite number of coordinates. In particular, the definition does not require an output before the integer coordinates have been read.

Write $\mu(f)$ for the least uniform change bound among continuous approximations, and $\mu_{\text{eff}}(f)$ for the corresponding minimum among computable approximations. The constructions below show that these minima exist for the definable relations under consideration.

100.2 The dimension bound

Translate a Presburger formula on M to the ordered group G by relativizing its quantified variables to the positive cone. Effective quantifier elimination for $\mathbb{Z}$ -groups then produces a Boolean combination of atoms

$$t_j(\bar{x}) > 0, \quad t_j(\bar{x}) = 0, \quad m_\ell \mid u_\ell(\bar{x}),$$

where each affine term has integer coefficients. For an order or equality atom write

$$t_j(\bar{x}) = \sum_{i=1}^d c_{ji} x_i + b_j \mathbf{1}, \quad c_j = (c_{j1}, \dots, c_{jd}) \in \mathbb{Z}^d.$$

Repeated occurrences do not need to be duplicated in the following rank. Define

$$U = \text{span}_{\mathbb{Q}} \{c_j : j \text{ is an order or equality atom}\}, \quad r = \dim_{\mathbb{Q}} U.$$

Only these atoms enter r ; congruence coefficients do not.

Theorem 100.1 (Linear dimension bounds truth changes; sharpens [theorem 7.3](#)). *For every Presburger formula with d external stream variables, effective quantifier elimination and rational Gaussian elimination compute a uniform truth approximation with at most*

$$r \leq d$$

changes on every input. Here an external stream variable can be a free variable or a parameter supplied as an oracle. In particular, a parameter-free formula in k free variables requires at most k changes, independently of its length and its quantifier alternation. A formula in k variables with p supplied stream parameters admits the uniform bound $k + p$.

Proof. Write $x_i = (h_i, n_i)$. First read all integer coordinates. The integer coordinate of t_j is the exactly known integer

$$N_j = \sum_i c_{ji} n_i + b_j,$$

and its stream coordinate is $\sum_i c_{ji} h_i$. Congruence atoms are decided immediately from the corresponding integer coordinate, since

$$mG = D \times m\mathbb{Z}.$$

Let

$$v_a = (h_1(a), \dots, h_d(a)) \in \mathbb{Q}^d \quad (a \in \mathbb{N}).$$

At stage s , use the sign of the first nonzero value among $c_j \cdot v_0, \dots, c_j \cdot v_{s-1}$ if there is one. Otherwise use the sign of N_j . This gives an approximation to every order atom. For equality use the answer $N_j = 0$ until a nonzero stream coefficient is seen, at which point the answer is permanently false. Evaluate the fixed Boolean combination at each stage to obtain A_s .

There are finitely many atoms. Every nonzero stream has a first nonzero coefficient, so every atom eventually becomes permanently correct. Consequently A_s converges to the desired truth value.

To count changes, restrict the row functional $c \mapsto c \cdot v_a$ to U , and denote the resulting element of U^* by ℓ_a . Put

$$W_s = \text{span}_{\mathbb{Q}}\{\ell_0, \dots, \ell_{s-1}\} \subseteq U^*.$$

An order or equality atom has no witnessed nonzero stream coefficient exactly when its coefficient vector is annihilated by all members of W_s . If $\ell_s \in W_s$, every such coefficient vector is also annihilated by ℓ_s . Hence no unresolved atom can become resolved at that stage. Resolved atoms never change again, and congruence atoms are already fixed. Thus

$$A_{s+1} \neq A_s \implies \dim W_{s+1} > \dim W_s.$$

The dimension starts at zero and never exceeds r , which proves the bound. The construction is uniform and reads only finite prefixes at each stage. All its preprocessing is effective. $\square$

The proof also gives the input-dependent bound $\dim \text{span}\{\ell_a : a \in \mathbb{N}\}$, which can be strictly smaller than r . It does not provide a computable time at which the last change has occurred. A new independent row can first appear arbitrarily far along the input streams.

Many atoms, only two possible changes. For example, the formula

$$((x + y = 0) \text{ XOR } (x = y)) \wedge \bigwedge_{m=1}^{32} (mx < (m+1)y \vee x = y)$$

has many order atoms, but all their coefficient vectors lie in $\mathbb{Q}^2$. On inputs with integer coordinates zero, take row $v_2 = (1, 1)$, row $v_7 = (1, 0)$, and all other rows zero. The displayed approximation has successive relevant answers 0, 1, 0: the first row resolves $x + y$, while the later row resolves $x - y$. The many inequalities remain true after the first row. Both rank increases can therefore matter to the final Boolean answer.

100.3 Sharpness of the bound by arity

Theorem 100.2 (Exactly d changes can be necessary). *For every $d \geq 1$, the parameter-free Presburger relation*

$$P_d(x_1, \dots, x_d) \iff \text{an odd number of the statements } x_i \neq 0 \text{ hold}$$

satisfies

$$\mu(P_d) = \mu_{\text{eff}}(P_d) = d.$$

Thus the dimension bound is sharp even when no order atoms or quantifiers are used. For any proposed computable approximation with fewer than d changes, a violating input can be chosen computable and with every stream of finite support. For arbitrary continuous stage maps the same lower-bound argument gives an existential violating input.

Proof. The upper bound follows from the preceding theorem. The parity relation is a finite Boolean combination of the equality atoms $x_i = 0$; it is therefore a Presburger formula, with no parity operation added to the language.

For the lower bound restrict to the closed slice on which all integer coordinates are zero. Initially let every stream be zero. Its truth value is zero, so convergence gives a stage t_0 at which the proposed approximation outputs zero there. By continuity, the value at that stage is fixed on some finite-prefix neighborhood of this tuple.

Insert a single positive rational coefficient, sufficiently far out in the first stream to remain in this neighborhood. The tuple still belongs to M^d , its first coordinate is now nonzero, and its truth value has toggled to one. Convergence gives a later stage $t_1 > t_0$ outputting one. Choose a finite-prefix neighborhood that fixes the outputs at both chosen stages. Insert a sufficiently delayed positive coefficient in the second, previously zero, stream. Its truth value toggles again.

Continue in this way through the d streams. At each step preserve all previously selected output values, and choose the next correct output at a strictly later stage. The final tuple has finite support in each stream and satisfies

$$A_{t_j}(\bar{x}) = j \bmod 2 \quad (0 \leq j \leq d).$$

There must therefore be at least one change between every pair of successive selected stages, giving at least d changes altogether.

If the stage maps are computable, the construction is effective relative to their index. At each step the current finite-support input is computable, so search for a stage with its known correct truth value. The computation terminates by convergence. Record the finitely

many input coordinates queried by all the selected computations and place the next positive coefficient beyond them. This preserves their outputs and gives a computable final input. Mere continuity supplies the requisite neighborhoods without an effective procedure for finding them, which is enough for the noneffective lower bound. $\square$

100.4 An exact finite calculation for every parameter-free formula

The rank bound is often not the individual optimum. For example, $x_1 = 0 \wedge \dots \wedge x_d = 0$ has coefficient rank d , but a single change suffices: begin with the answer determined by the integer coordinates and change permanently to false when a nonzero stream coefficient appears. The following finite construction computes the exact optimum instead of just a rank bound.

Fix a parameter-free formula $\varphi(\bar{x})$ and an effectively obtained quantifier-free group formula Ψ as above. Include standard numerals in the affine constants b_j . Put

$$C = \{c_j \neq 0 : c_j \text{ occurs in an order or equality atom}\} \cup \{e_1, \dots, e_d\},$$

where e_i is the i th coordinate vector. Retain just one copy of each vector. Including the coordinate vectors is essential: they record the positivity requirements defining M^d .

Integer regimes. The truth of Ψ at fixed stream signs depends on the integer tuple $\bar{n} \in \mathbb{Z}^d$ only through finitely many data:

1. the bits $\eta_i = [n_i < 0]$;
2. the signs $\alpha_j = \text{sgn}(c_j \cdot \bar{n} + b_j)$ for its order and equality atoms;
3. the truth values β_ℓ of its congruence atoms evaluated on $\bar{n}$.

A choice of all these data is an *integer regime* R . There are finitely many choices. Whether a regime is nonempty is an ordinary Presburger decision problem over $\mathbb{Z}$: ask whether there exists an integer tuple having the stipulated signs and congruences. Discard every infeasible regime.

The finite sign-pattern poset. For a feasible regime R , let $\mathcal{P}_R$ be the set of vectors

$$\sigma(v) = (\text{sgn}(c \cdot v))_{c \in C} \in \{-1, 0, 1\}^C$$

realized by rational vectors $v \in \mathbb{Q}^d$ satisfying

$$v_i \geq 0 \quad \text{for all } i, \quad v_i > 0 \quad \text{whenever } \eta_i = 1.$$

Order these patterns by

$$\sigma \leq \tau \iff \forall c \in C (\sigma_c \neq 0 \implies \tau_c = \sigma_c).$$

In other words, passing upward can replace zeros by signs but cannot change an already nonzero sign. This is a finite partially ordered set. It is computable: enumerate the finitely many proposed sign vectors and decide the corresponding finite systems of rational linear equalities and strict inequalities. Rational linear elimination suffices. Since the systems are homogeneous, one can alternatively replace their strict positive inequalities by inequalities

at least one whenever there is a strict inequality; a common positive rescaling proves equivalence.

Label each $\sigma \in \mathcal{P}_R$ by a bit $\lambda_R(\sigma)$, obtained by evaluating Ψ as follows. For an order atom, use the nonzero stream sign σ_{c_j} when available and otherwise its integer sign α_j . An equality atom holds exactly when both those signs are zero. A congruence atom has the fixed value β_ℓ . For a zero coefficient vector set $\sigma_0 = 0$ by convention.

Let

$$a(R) = \max_{\sigma_0 < \dots < \sigma_q \text{ in } \mathcal{P}_R} \#\{j \in \{1, \dots, q\} : \lambda_R(\sigma_j) \neq \lambda_R(\sigma_{j-1})\}, \quad N_\varphi = \max_{R \text{ feasible}} a(R).$$

Singleton chains are allowed and contribute zero. The finite maximum is effectively computable, for example by dynamic programming on the finite partially ordered sets.

Lemma 100.3 (Finite sign compression). *For any finite $C \subseteq \mathbb{Q}^d$, every sign pattern*

$$(\text{sgn}_{\text{lex}}(\sum_i c_i h_i))_{c \in C}$$

of a tuple of rational streams is realized by a single rational vector. If C contains the coordinate vectors, the rational vector has exactly the same zero and positive coordinate pattern as the streams.

Proof. Choose a finite prefix long enough to witness the first nonzero coefficient of every nonzero linear combination indexed by C . Let its rows be $v_0, \dots, v_{L-1}$. For a sufficiently small positive rational ε , set

$$v = v_0 + \varepsilon v_1 + \dots + \varepsilon^{L-1} v_{L-1}.$$

For each $c \in C$ whose stream is nonzero, the first nonzero coefficient of the polynomial $c \cdot v$ determines its sign for all sufficiently small positive ε . Finitely many such requirements can be satisfied simultaneously. A combination that is identically zero annihilates every row and remains zero. Applying this also to the coordinate vectors proves the final assertion. If all combinations are zero, the zero vector suffices. $\square$

Theorem 100.4 (Exact computable change complexity). *For every parameter-free Presburger formula φ on this presentation,*

$$\mu(\mathbf{1}_\varphi) = \mu_{\text{eff}}(\mathbf{1}_\varphi) = N_\varphi \leq r \leq d.$$

Thus the minimum possible number of changes of mind is computable from the formula. It is the maximum number of truth-value alternations along a chain of feasible sign patterns, maximized over its finitely many feasible integer regimes.

Proof. First, the sign-compression lemma shows that $\mathcal{P}_R$ describes exactly the finite stream-sign patterns occurring on the integer slice of any tuple $\bar{n}$ in regime R . Coordinate signs enforce precisely the necessary conditions: every stream is lexicographically nonnegative, and a stream is nonzero when its integer coordinate is negative. Conversely, a feasible rational vector gives a valid tuple by placing that vector in the first row and making all later rows zero.

For the upper bound, read the integer coordinates and hence their regime. If $n_i < 0$, scan until the positive leading coefficient of h_i has been witnessed. Validity of the input guarantees termination. Choose L beyond all these witnessed indices; if there are no negative integer coordinates, use $L = 0$. This finite preliminary scan is part of the computation of the first output, so it does not itself count as a change.

At stage s truncate all streams after their first $L + s$ coefficients, and let σ_s be their pattern on C . Each truncated tuple is still in the same integer slice of M^d , so by sign compression $\sigma_s \in \mathcal{P}_R$. Revealing a new row cannot change a previously witnessed nonzero leading sign. Therefore

$$\sigma_0 \leq \sigma_1 \leq \sigma_2 \leq \cdots.$$

Output $A_s = \lambda_R(\sigma_s)$. Every relevant linear combination eventually has its correct sign, and only finitely many combinations are involved, so these outputs converge to truth. After repeated patterns are removed, their path is a chain in $\mathcal{P}_R$. Consequently it has at most $a(R) \leq N_\varphi$ changes. All the computations are effective and inspect finite input information.

For the lower bound, choose a feasible regime R and a chain attaining N_φ . Remove consecutive portions with constant label to obtain

$$\sigma_0 < \sigma_1 < \cdots < \sigma_{N_\varphi}, \quad \lambda_R(\sigma_{j+1}) \neq \lambda_R(\sigma_j).$$

Fix an integer tuple $\bar{n}$ in that regime and a one-row rational tuple realizing σ_0 . For any continuous truth approximation, choose a stage t_0 outputting its correct label.

Suppose a finite-support tuple with pattern σ_j has been chosen and the outputs at the earlier selected stages have been fixed. Take a finite-prefix neighborhood preserving all those outputs. At a coordinate beyond both that prefix and the current finite support, append a rational row v realizing σ_{j+1} . Every previously nonzero linear combination retains its old leading coefficient and hence its sign. Every combination previously zero acquires the sign of $c \cdot v$. By the defining extension order, the resulting tuple has exactly pattern σ_{j+1} . The coordinate vectors in C ensure that the tuple remains in M^d with the same integer regime.

Its truth value is the opposite label, so convergence yields a strictly later stage $t_{j+1} > t_j$ with that value. Continue until the end of the chain. The final tuple preserves alternating outputs at $t_0 < \cdots < t_{N_\varphi}$ and therefore forces at least N_φ changes. This lower bound holds even for arbitrary continuous stage maps. With computable stage maps, integer and rational witnesses can be found effectively, and the finite-prefix neighborhoods can be recovered from finite oracle-query traces. The same argument then constructs a computable finite-support witness.

Finally, along any chain of realizable sign patterns, let Z_j be the span of the order/equality coefficient vectors still assigned zero. These subspaces form a descending chain. Whenever one of these forms becomes nonzero, the descent is strict: a rational vector realizing the new pattern annihilates all still-zero forms but does not annihilate the newly nonzero one. A truth-value change requires at least one order/equality form to become nonzero, because the integer regime and all congruence values are fixed. Thus every truth change strictly decreases $\dim Z_j$, which starts at most r . This proves $N_\varphi \leq r \leq d$ and completes the argument. $\square$

100.5 A sharp obstruction when parameters are fixed

Theorem 100.5 (Uniform parameter optimization is impossible). *For a fixed parameter $p \in M$, let $f_p(x) = \mathbf{1}_{x < p}$. Then*

$$\mu(f_p) = \begin{cases} 0, & p = 0, \\ 1, & p > 0. \end{cases}$$

The same values hold for the minimum among approximations computable relative to a supplied name of p . There is no Type-2 computable procedure that obtains this minimum uniformly from the name of p . Even when p is promised computable, there is no algorithm that obtains the minimum from an index computing its stream.

Proof. The relation is empty when $p = 0$, so its characteristic function is constant. For every p , the single inequality $x - p < 0$ has coefficient rank one in the two supplied streams and therefore has a uniform one-change approximation relative to p .

Suppose $p > 0$. The element $x = p - 1$ belongs to M : if the stream of p is positive, subtracting one does not change that fact, and if its stream is zero, its integer coordinate is at least one. Let e_N be the stream whose only nonzero coefficient is a one at coordinate N , and put $\delta_N = (e_N, 0)$. In the product topology, $\delta_N \rightarrow 0$, whereas in the arithmetic order $\delta_N > 1$ for every N . Thus

$$x + \delta_N \longrightarrow x, \quad f_p(x) = 1, \quad f_p(x + \delta_N) = 0.$$

Consequently f_p is not continuous. A zero-change approximation is constant in its stage index and equals its first continuous stage map, so none can approximate this function. Its optimum is exactly one.

The map from p to this optimum is discontinuous at zero, as witnessed by $\delta_N \rightarrow 0$. It therefore cannot be Type-2 computable. For the stronger index statement, given a machine index e , define a computable stream h_e that has coefficient one at its halting time if the machine halts and has all other coefficients zero. Computing its j th coefficient requires only a bounded simulation through step j . The element $p_e = (h_e, 0)$ is always a valid computable parameter, and

$$\mu(f_{p_e}) = 1 \iff \text{machine } e \text{ halts.}$$

A uniform algorithm for the minimum would decide the halting problem. $\square$

Scope of the resolution. This theorem completely answers the earlier report's individual-formula optimization question for parameter-free formulas in the specified rational-stream presentation. A formula with parameters that are allowed to vary as additional input streams is covered after those parameters are made additional free variables. For one fixed arbitrary parameter tuple, the dimension bound remains valid uniformly relative to its supplied names, but the exact finite-poset calculation just proved is not asserted: fixing an infinite parameter stream can prevent the free delayed tail modifications used in the lower bound. The last theorem shows that a uniform algorithm for arbitrary fixed computable parameters is impossible, even for a single unary order formula.

The finite procedure is a decidability result about optimal change counts, not an efficient quantifier-elimination or truth algorithm in the ordinary running-time sense. It can require a large quantifier-free formula, many integer regimes, and many feasible sign patterns. Nor does a bound on changes identify a finite stopping stage for every input. These distinctions are essential to the statement.

Further questions suggested by the theorem. Classify fixed parameter tuples or restricted formulas for which a finite effective replacement for the sign-pattern calculation exists, despite the general obstruction just proved. Study the complexity of computing N_φ from a formula already in quantifier-free form, and compare the finite poset with effective finite difference hierarchies. Seek syntactic fragments for which the integer regimes and sign-pattern chains can be analyzed without full elimination. Finally, formalize the finite-sign compression lemma, rank argument, and delayed-tail lower bound as independently checkable steps in ProveIt.

[write] (Batch 87.) This answers Part I's [question 14.8](#) exactly for parameter-free formulas ([theorem 100.4](#)) and negatively for uniform optimization at a fixed parameter ([theorem 100.5](#)).

The rank bound of [theorem 100.1](#) sharpens the atom-count bound of [theorem 7.3](#). A revision count bounds changes of the output, not running time, the number of stream coordinates read, or arithmetic operations, and no stage after which the answer is final is computed; the Hilbert's-tenth programme's review of the arrival makes the same distinction ([appendix E](#)). Part I's [question 14.15](#) on sharp oracle bounds is related and stays open.

101 A completeness obstruction beyond invariant metrics

Let M be a commutative cancellative monoid with identity 0. Assume that its algebraic order

$$x \leq y \iff (\exists z \in M) x + z = y$$

is a linear order. In particular it is antisymmetric. Define truncated subtraction by

$$x \dot{-} y = \begin{cases} z, & y + z = x, \quad y \leq x, \\ 0, & x < y. \end{cases}$$

Cancellation makes z unique. Let M carry a metrizable topology in which addition is continuous.

Definition 101.1. For a compatible metric d on M , say that translations are *uniformly equicontinuous at the identity* if, for every $\varepsilon > 0$, there is a neighbourhood U of 0 such that

$$d(b + u, b) < \varepsilon \quad (b \in M, u \in U). \quad (\text{UE0})$$

The same neighbourhood U must work for all b . Joint uniform continuity of addition for the product metric $\max\{d(x, x'), d(y, y')\}$ implies (UE0) by comparing (b, u) with $(b, 0)$.

Theorem 101.2 (Closed translates forced by complete uniformity). *Suppose $x \dot{-} y$ is continuous at (a, a) for every $a \in M$. If a compatible complete metric d satisfies (UE0), every translate $c + M$ is closed in M .*

Proof. Fix $c \in M$ and a convergent sequence $x_n = c + b_n \rightarrow x$. We prove that (b_n) is d -Cauchy. Given $\varepsilon > 0$, select U as in (UE0). Continuity of truncated subtraction at (x, x) , where its value is 0, gives a neighbourhood V of x such that

$$s, t \in V \implies s \dot{-} t \in U.$$

For all sufficiently large n, m , both x_n, x_m belong to V . If $b_m \leq b_n$, write $b_n = b_m + z$. Cancellation implies $z = x_n \dot{-} x_m$, so $z \in U$, and

$$d(b_n, b_m) = d(b_m + z, b_m) < \varepsilon.$$

If $b_n \leq b_m$, interchange the indices and use symmetry of d . The order is total, so these cases cover every pair. Thus (b_n) is Cauchy. Completeness gives $b_n \rightarrow b \in M$. Continuity of addition gives $x_n = c + b_n \rightarrow c + b$. Uniqueness of limits yields $x = c + b \in c + M$. Metric spaces are sequential, so $c + M$ is closed. $\square$

Corollary 101.3 (No complete metric with uniformly continuous addition; strengthens [proposition 8.4](#)). *Under the hypotheses of the preceding theorem, suppose some translate $c + M$ is not closed. No compatible complete metric satisfies (UE0); in particular no compatible complete metric makes addition jointly uniformly continuous.*

The theorem uses ordinary continuity at each individual diagonal point. It does not assume a uniform modulus for truncated subtraction. Nor does it require translations to be isometries or their inverses to be uniformly continuous.

101.1 Application to the real and rational-stream cones

Put $G = H \overrightarrow{\times} \mathbb{Z}$, with least positive element $1_G = (0, 1)$, and let $M = G_{\geq 0}$ carry the subspace topology of $H \times \mathbb{Z}$. Here either $H = \mathbb{R}$ with its usual topology, or $H = \mathbb{Q}^\omega$ with discrete rational factors and first-difference lexicographic order.

Lemma 101.4. *Truncated subtraction on M is continuous at every diagonal point.*

Proof. Fix $a = (h, k)$. Near (a, a) both integer coordinates equal k , because the integer factor is discrete. For $x = (u, k)$ and $y = (v, k)$, therefore,

$$x \dot{-} y = ((u - v)^+, 0), \quad w^+ = \max\{w, 0\}.$$

For $H = \mathbb{R}$ the positive-part map is continuous. For $H = \mathbb{Q}^\omega$, it is continuous at 0: if the first N coordinates of w vanish, the first N coordinates of w^+ also vanish. Since $u - v \rightarrow 0$ as $(x, y) \rightarrow (a, a)$, the displayed formula proves the claim in both cases. $\square$

Every nonzero element of M has a predecessor, so $1_G + M = M \setminus \{0\}$. This translate is not closed. Indeed, $((1/n), 0) \rightarrow 0$ in the real example, and $(e_n, 0) \rightarrow 0$ in the rational-stream example, where e_n is the rational stream with a single 1 in coordinate n . All these points are nonzero. Consequently Corollary 101.3 applies.

Both spaces are nevertheless Polish and addition is continuous. The obstruction concerns the simultaneous demands of a compatible complete metric and uniform continuity. The incomplete restriction of an invariant metric from the ambient group has uniformly continuous addition, so completeness is essential.

[write] (Batch 87.) Corollary 101.3 answers Part I's question 14.16 negatively for joint uniform continuity of addition, and for uniform equicontinuity of translations at the identity, in $M_{\mathbb{R}}$, $M_{\mathbb{B}}$ and the countable-core models; it strengthens proposition 8.4. The question's other parts (uniform continuity of all translations in one direction, the distortion of their inverses) stay open.

102 Consequences, examples, and the formalization boundary

The results separate three different forms of regularity. An arithmetic operation can be continuous, uniformly continuous for an incomplete metric, or uniformly continuous for a complete compatible metric. The first two hold for addition in the explicit cones; the third is impossible. A definable map can have a simple graph while failing to be a pointwise limit of continuous maps into its actual target. Finally, a truth value can be eventually correct after finitely many revisions without there being an effective stage at which correctness can be announced permanently.

102.1 A compact comparison of the models

Property	$M_{\mathbb{R}}$	M_{ω}	Nonsplit countable-core models
Perfect Polish carrier	Yes, locally compact	Yes, Baire space	Either type, by choice of coefficient group
Addition and fixed division	Continuous	Continuous	Continuous
All definable relations	Δ_2^0	Δ_2^0	Δ_2^0
Retraction onto standard integers	Unique, continuous	Unique, continuous	Absent for the explicit core
Unital additive map to Oz	Exists	Exists	Impossible for the explicit core
Every definable map target-valued Baire one	No; exact slice criterion	Yes, explicit finite-prefix approximants	Stream: yes; real: exact core-slice criterion
Complete metric with uniform addition	Impossible	Impossible	Impossible for both explicit coefficient choices

The metric conclusion extends to the countable-core cones without any new arithmetic assumption. Near a diagonal point the two discrete core coordinates agree, so truncated subtraction is $((u - v)^+, 0_K)$. The diagonal continuity proof is unchanged. The successor image again omits exactly zero, which is nonisolated.

The no-homomorphism result concerns **Oz**, not all of **No**. For example, the countable core itself has the unital ordered additive embedding

$$(u, v) \mapsto (v - u)\omega + v$$

into **No**. Its finite constant coefficient need not be an integer. This example pinpoints why the omnific restriction matters.

102.2 What the continuity procedure returns

For a graph formula defining f , the procedure has a concrete finite output: a Boolean expression describing $\text{Disc}(f)$ in a language with one divisible ordered group sort and one discrete integer sort. It can also decide global continuity in the absence of arbitrary undecidable parameter comparisons. A full implementation would perform the following transformations.

1. Relativize the graph to the signed group and eliminate its quantifiers; extract a finite list of affine quotient candidates.
2. Partition by the first successful candidate, and eliminate the remaining quantifiers in its cell conditions.
3. For each possible pattern of zero and nonzero leading forms, solve the finite rational directional-feasibility problems for the accessible cells.
4. Compare every accessible candidate with the actual candidate at the point, using the exact continuity criterion.

5. If global continuity is desired, eliminate the input coordinates in the two separate sorts.

This output language is necessary, even for a simple operation. In the monus example the condition that two leading coordinates are equal detects standard finite difference, which is unavailable as a Presburger definition of the standard cut. Thus the compiler is an analysis tool for the presentation, not an additional definability theorem inside the original arithmetic.

102.3 What can be formalized directly in Provelt

The inspected Lean syntax interprets affine expressions in lists of ordinary integers, and its central theorem quantifies over `List Int` [60, 61]. It therefore cannot be cited as an existing proof that every nonstandard $\mathbb{Z}$ -group validates the same elimination algorithm. That semantic generalization has to be implemented and proved.

A useful implementation order is as follows. First define an abstract discretely ordered abelian group with a named least positive unit and standard division with remainder. Prove the full-coset finite-boundary lemma and the model-uniform elimination theorem. These algebraic steps support the finite-candidate theorem and the arithmetic correctness of all the explicit models.

Next formalize rational sign feasibility, finite sign compression, and the rank argument. These are finite algebraic statements plus an elementary eventual-stabilization argument. The exact minimum theorem then separates into a finite poset calculation and the delayed-tail lower-bound construction. The existing integer eliminator can decide the feasibility of each integer regime once its relation to the abstract semantics has been established.

The topological component can be developed independently: the positive-cone construction, the finite-branch continuity lemma, the rational-direction realization, and the closed-translate metric theorem. The last of these is particularly small and has no dependence on surreal numbers, descriptive set-theoretic classification, or Presburger quantifier elimination.

Finally, the omnific application needs the additive constant map and an ordered embedding of a set-sized divisible group into **No**. The explicit two-element contradiction only needs the former. It is therefore a shorter initial formalization target than the general surreal embedding criterion.

103 Further research questions and concrete next steps

The questions below arise after the proved results. They are not presented as already solved, and their scopes are narrower than a blanket request for a new theory of all Polish arithmetic.

Question 103.1 (Fixed-parameter revision complexity). For a fixed parameter tuple, classify the exact degree of difficulty of computing the optimal revision number. The uniform decision problem is already obstructed by the threshold example above. Which representations with certified zero tests, rational dependence data, or finite support restore an effective finite classification?

The next attainable case is a tuple with explicitly supplied finite supports. Its known initial coordinates can be incorporated into the sign-pattern calculation, after which later perturbations see zero parameter tails. The proof must account for the transition between these fixed initial coordinates and the freely variable tail.

Question 103.2 (Complexity of the exact finite algorithms). How hard is it to compute the optimal revision number or the exact continuity condition from a quantifier-free graph? Can the sign-pattern search be compressed using the intersection lattice of the rational hyperplanes, rather than enumerating all ternary patterns?

The theorems prove decidability, not a good running-time bound. It is useful to separate the cost of Presburger elimination from the combinatorial cost of feasible sign patterns and alternating chains. A small verified implementation of the latter would already be a substantive addition to the repository.

Question 103.3 (Effective computation for countable cores). For $D \vec{\times} K$, where K is a countable $\mathbb{Z}$ -group equipped with the discrete topology, classify the oracle information needed for an effective continuity compiler and for finite-prefix evaluation of definable maps. How does the answer depend on the chosen presentation of K and the names of its parameters?

The purely topological continuity and Baire conclusions extend by the same finite-branch proofs, as established above. Computation must distinguish knowing a core element as a discrete topological point from having algorithms for its arithmetic. A countable carrier does not by itself provide decidable operations or order. A useful first case is a computable presentation with a decidable atomic diagram and named standard quotient functions.

Question 103.4 (Connected Polish monoid presentations). Can a connected uncountable Polish space support Presburger addition when no signed Polish group presentation is supplied?

The automatic continuity of standard residues settles the negative answer for cones inheriting a Polish group topology. It does not settle the question for arbitrary monoid topologies. The essential missing step is a suitable group-completion theorem, or an example showing why such a completion cannot be demanded.

[write] This is Part I's [question 14.3](#) and Part III's [question 43.4](#), restricted to monoid topologies with no signed Polish group presentation; see the note after [theorem 95.4](#).

[write] (Batch 90.) The “suitable group-completion theorem” is now [theorems 153.1](#) and [154.1](#): a monoid topology with continuous addition comes from a signed Polish group presentation exactly when partial subtraction is continuous. The question is narrowed to presentations with discontinuous partial subtraction (note after [question 14.3](#)); it stays open.

Question 103.5 (Rigidity with continuous addition). Which rigid Presburger models admit a Polish topology with continuous addition and Borel order? How does the answer change when one demands a Polish topology on the entire signed group?

The split models have nontrivial positive dilations of their divisible part. The nonsplit construction here retains such a divisible coefficient part and is not a rigid example. Jeřábek's structural theory [4] gives algebraic constraints, but these do not automatically produce or exclude a prescribed Polish topology.

[write] Part I's [question 14.12](#) asks the first sentence.

[write] (Batch 92.) Rigid examples are excluded among the uncountable locally compact Polish models with continuous partial subtraction, which are exactly the cones inheriting a Polish group topology on the whole signed group ([proposition 182.3](#) and [corollary 171.3](#); note after [question 14.12](#)).

Question 103.6 (Open divisible subgroups and continuous core decompositions). In which Polish Presburger groups with Borel order is the maximal divisible subgroup $\ker r$ open?

When does the algebraic decomposition admit a continuous splitting with a countable discrete reduced core?

The transfer examples have an open divisible factor and a countable reduced core. The automatic-continuity theorem proves a different, narrower statement: if a retraction onto the standard integers exists algebraically, it is continuous. It does not classify arbitrary countable cores or make every algebraic decomposition topological.

Question 103.7 (Sufficiency in the metric problem). For a cancellative linearly ordered Polish monoid with diagonal continuity of monus, is closedness of all translates part of a useful sufficient criterion for a compatible complete metric with uniformly continuous addition? What additional hypotheses are necessary?

The closed-translate theorem supplies a necessary condition only. One should first study concrete subclasses, such as cones with continuous ambient subtraction or locally compact monoids, and keep the uniform structure as part of the data.

Question 103.8 (Embeddings between different countable cores). Describe continuous elementary embeddings between $D \vec{\times} K$ and $D' \vec{\times} K'$, especially when the cores are reduced and nonsplit. Which information in the coherent residue image is preserved by a topological or ordered embedding?

The automatic continuity theorem shows that all finite residue maps are topologically visible on a Polish signed group. The distinction between their separate continuity and a single integer retraction is therefore a natural invariant to retain in such a classification.

Question 103.9 (Minimal foundational and formalization requirements). What is the weakest convenient subsystem in which the explicit stream model, the rank bound, and the finite-pattern optimization can be proved for coded inputs? Can the mathematical and computational parts be isolated into small reusable proof-assistant modules?

This is a question about the positive construction and its classification algorithms. It is separate from Glazer's published reverse-mathematical question about the strength of the topological Tennenbaum theorem. A proof of one would not be a proof of the other.

[write] Compare Part I's [question 14.2](#). Source 14 ([Part VIII](#)) claims an answer to Glazer's published question; it is a different theorem from the positive construction, as source 13 says.

Part VIII

An ATR_0 bound for Glazer's Question 1 (claimed)

[write] This part is source 14 of batch 87, *Diophantine Definability and Topological Arithmetic: An ATR_0 proof of Glazer's theorem, cofinal unary discontinuities, and nonsplit Presburger models* (dated October 2026, its records October 4; "prepared with ChatGPT"), pinned to repository commit 6fef5383b. All unmarked text is source 14's; its labels carry the prefix `pma:atr:`.

What this part claims, and its status

Source 14 claims that Glazer’s Theorem 2 (no model of $\mathbf{Q} + \mathbf{I}\Delta_0 + \mathbf{Exp}$ on an uncountable Polish space has continuous addition or continuous multiplication when both are Borel) is provable in $\mathbf{ATR}_0$, under the coded-model convention of its Section 2 ([section 105.2](#)): an affirmative answer to Glazer’s Question 1, which this report printed as open ([question 43.1](#)). It proves more: discontinuous translations and discontinuous multiplications by internal powers of two occur cofinally ([theorem 104.1](#)).

This answer is printed **as claimed**. The manuscript is AI-assisted and unrefereed. Parts of its $\mathbf{ATR}_0$ bookkeeping rest on inputs it declares rather than proves (uniform perfect-set and order homogenization, coded Baire approximation, analytic separation, and the evaluation of countable Borel iterations; note in [section 105.4](#)), and its author did not see the Gaifman–Dimitracopoulos chapter whose internal MRDP theorem it relies on, taking the statement from Chen, Müller and Yokoyama. Neither the intake nor the repository’s Hilbert’s-tenth programme, which reviewed the archive’s provenance and read the MRDP input and the automatic-Borel argument, has reviewed the whole proof ([appendix E](#)). It is an upper bound only. Nothing in it is formalized.

The construction it shares with Parts VII and IX. Source 14’s Section 10 ([section 113](#)) builds the locally compact Polish Presburger model $(\mathbb{R} \overrightarrow{\times} K_*)_{\geq 0}$ on a countable nonsplit $\mathbb{Z}$ -group K_* . That group is the member G_{ζ_*} , $\zeta_* = \sum_k k!$, of the factorial family of the discrete-initial-subgroups-and-omnific-normalization report, and the omnific obstruction it proves is that report’s `isg:cf:cor:unit`; source 14 cites neither (notes after [lemma 113.1](#) and [proposition 113.3](#)). The construction is source 13’s countable-core transfer ([theorem 94.1](#)), printed in general in [Part VII](#); [proposition 113.2](#) is printed here with its own proof as a flagged instance.

Notation. Source 14’s countable group D of its Section 10 is printed K_* , and the generic countable $\mathbb{Z}$ -group of the commentary on its question on nonsplit Presburger topology is printed K , because D is the divisible coefficient group in Parts I and VII; its map $\Psi = \omega\Phi$ is printed Φ_ω , because Ψ is Part I’s Baire homeomorphism; its Cantor space 2^ω is printed $2^\mathbb{N}$. In Sections 7 and 8 of source 14 ([sections 110](#) and [111](#)) the letter D is a Borel set of Cantor space, and $e(x) = 2^x$ is the internal exponential, not a unit; t_n , q_n , X_n are its root scales, unrelated to the variable t of Parts III and V. The equation tags (3.1)–(10.6) are source 14’s own: (3.x) are in [section 106](#), (5.x) in [section 108](#), (6.x) in [section 109](#), (7.x) in [section 110](#), (8.x) in [section 111](#) and (10.x) in [section 113](#).

Provenance. Source 14’s finite checks are shipped as `code/14-atr0-tennenbaum-verify_finite_lemmas.py`, which prints its record (`data/14-atr0-tennenbaum-finite_check_results.txt`); its proof audit is `14-atr0-tennenbaum-PROOF_AUDIT.txt`. Its source audit is not shipped, because it contains a third party’s personal profile address; its verified content is summarized in [appendix E](#).

104 The question, the intersection, and the contributions

Elliot Glazer’s research in set theory and formal systems, together with his work on foundations of proof assistants, makes the interface between arithmetic models, definability,

and formalization a direct match for Provelt; an institutional author biography confirms these interests [77]. The particularly useful source is his *A Topological Tennenbaum Theorem* [1]. The relevant part of Provelt is its work on Diophantine exponentiation, arithmetic coding, and Polish models of omnific arithmetic [73, 75].

In ordinary mathematics, Glazer proves that a model of $\mathbb{Q} + \text{I}\Delta_0 + \text{Exp}$ on an uncountable Polish space, with both arithmetic operations Borel, cannot have either operation continuous. His Question 1 asks whether this conclusion is provable in ATR_0 . The unresolved issue identified there concerns the descriptive complexity of exponentiation and the choices used in the multiplicative argument. We give a proof using internal MRDP, Borel regularization, and category.

Here and below, *uncountable* is external to the arithmetic object structure; its precise coded meaning is specified in Section 105.2. Arithmetic comparisons refer to the model's order. A Polish space is a separable space admitting a complete compatible metric. Neither that topology nor its metric is assumed to agree with the arithmetic order topology.

Theorem 104.1 (Main theorem: cofinal unary discontinuity). *Work in ATR_0 , with the coded-presentation convention of Section 105. Let $\mathcal{M}$ have an uncountable Polish carrier M and Borel addition and multiplication.*

- (i) *If $\mathcal{M} \models \mathbb{Q} + \text{IOpen}$, then for every $b \in M$ there is an $a \geq b$ such that*

$$\tau_a : M \longrightarrow M, \quad \tau_a(x) = a + x$$

is discontinuous.

- (ii) *If $\mathcal{M} \models \mathbb{Q} + \text{I}\Delta_0 + \text{Exp}$, then for every $b \in M$ there is an $a \geq b$ such that*

$$\lambda_a : M \longrightarrow M, \quad \lambda_a(x) = 2^a x$$

is discontinuous.

In particular neither addition nor multiplication is separately continuous, and neither is jointly continuous, under the hypotheses of (ii).

The two witnesses a need not be the same. *Cofinal* here means unbounded in the arithmetic order; it makes no assertion about topological density, cardinality, or category of the set of parameters.

Part (i) uses a weak theory already covered by Glazer's additive obstruction. The substantive new connection for Question 1 is the passage from a *provably correct internal* Diophantine definition to a Borel exponentiation map. Cofinality and the unary formulation are additional consequences of the proof supplied here.

A second contribution records a concrete distinction that is useful for the repository's surreal program.

Theorem 104.2 (A nonsplit Presburger boundary; an instance of theorem 94.1). *There is an explicitly described uncountable, locally compact, perfect Polish model of Presburger arithmetic with jointly continuous addition whose standard-integer subgroup is not a direct summand of its Grothendieck group. The model has an ordered additive embedding into the surreal numbers using expressions*

$$x\omega^2 + q\omega + p \quad (x \in \mathbb{R}, p, q \in \mathbb{Q}),$$

but has no additive map to the canonical omnific integers that sends its unit to 1.

The abstract existence of nonsplit $\mathbb{Z}$ -groups is classical [4]; we supply an elementary explicit example, its Polish realization, and the omnific obstruction. We do not claim the first discovery of nonsplit Presburger groups.

104.1 What was already present in the repository

The audit used the snapshot

```
6fef5383b126be343cccc9baf47081ef37ab5afe.
```

The inspected Polish-arithmetic article already gives affirmative constructions for Glazer’s Presburger Question 2, including real and Baire-space carriers, detailed definability results, and omnific realizations. It explicitly does not answer Question 1. The three original incoming Glazer-related bundles were also inspected; none contains the internal-MRDP argument developed here [75, 76].

The existing Lean module on Diophantine exponentiation proves the graph theorem over standard $\mathbb{N}$, with interfaces including `naturalPowGraph_diophantine` and `power_polynomial_exists`. This is relevant mathematical infrastructure, but it is not itself an internal proof over $\text{I}\Delta_0 + \text{Exp}$. Confusing those two statements would leave a real gap in the present argument. Section 114 explains the distinction.

104.2 Contribution and dependency ledger

Statement	Role	Status here
Internal MRDP over $\text{I}\Delta_0 + \text{Exp}$	Correctness in nonstandard models	Classical input [68, 69]
Borel exponentiation	Removes the projective complexity obstacle	Proved from that input
Common Cantor restriction	Regularizes countably many Borel maps together	Proof and coding supplied
Cofinal unary obstructions	Strengthens the continuity conclusion	Full proofs supplied
ATR_0 upper bound	Answers the question in the cited 2023 paper	Proof with explicit coding
Nonsplit Polish model and omnific obstruction	Complementary Presburger application	Explicit construction and proof
An optimal lower bound or Lean verification	Further research	Not claimed

[write] “The inspected Polish-arithmetic article” [75] is this report as it stood at source 14’s pin 6fef5383b: then the delivered text of batch-86 source 05, now the base of Part I. Parts II–V were added later the same day, and none of them answers Question 1 (section 1.1). The “three original incoming Glazer-related bundles” are batch-86 manuscripts 03, 04 and 05, now Parts I and II; [76] is a routing note of the Hilbert’s-tenth programme. The Lean declarations named here exist at the current state in `Logic/PeanoArithmetic/ListCoding/Lean/PAListCoding/ExponentiationDiophantine.lean` and are proved there for the standard natural numbers; as source 14 says, they are not the internal statement it needs.

105 Framework and the established inputs

105.1 Object arithmetic and external mathematics

The object language contains $0, 1, +, \cdot, <$, with the usual discretely ordered semiring conventions. The order may equivalently be recovered from addition:

$$x < y \iff \exists z (x + z + 1 = y).$$

We use IOpen for induction for quantifier-free formulas and $\text{I}\Delta_0$ for induction for bounded formulas. Write

$$T_{\text{exp}} = \mathbf{Q} + \text{I}\Delta_0 + \text{Exp}.$$

The exponentiation axiom is the ordinary first-order assertion of total exponentiation through finite-sequence coding, not an assumption that an external exponential operation has already been supplied as a Borel map. The standard equivalent formulations of elementary arithmetic may be used.

The external theory ATR_0 is second-order arithmetic with arithmetical transfinite recursion. Its natural role is that Borel codes may require transfinite recursion to evaluate. The object theory and the external theory do different jobs: T_{exp} controls the algebra in M , while ATR_0 controls codes for the Polish space and its Borel data.

105.2 Coded-model convention

A coded Polish space is specified by a countable dense metric presentation and its completion; a point is represented by a suitable fast Cauchy name. Equality of names means equality of the represented points. Borel maps are specified by standard Borel codes, including uniform codes for inverse images of a countable basis. All functions and relations must respect equality of point names.

For the reverse-mathematical statement, a coded space or coded Borel subset A is *uncountable* when no coded sequence of point names covers it. In symbols, this is

$$\neg \exists (x_n)_{n \in \mathbb{N}} \forall x (x \in A \longrightarrow \exists n \ x = x_n),$$

where membership is interpreted using the supplied Borel code. The word external does not introduce an ambient cardinality predicate unavailable in second-order arithmetic.

The notation $\mathcal{M} \models T_{\text{exp}}$ is ordinary semantic notation in the mathematical exposition. For the reverse-mathematical claim it is interpreted *instance by instance*: every particular first-order formula needed by the proof is translated directly into second-order arithmetic, with its variables ranging over point names. We do not postulate a truth predicate for all first-order formulas on an uncountable carrier.

Lemma 105.1 (Fixed-proof interpretation). *Let π be a fixed finite first-order derivation of a sentence ψ from a theory T , and let Γ_π be the finite collection of nonlogical axiom instances actually used. For a coded real-domain structure, the direct translations of those instances imply the direct translation of ψ . No uniform satisfaction predicate is required.*

Proof. Translate each of the finitely many formulas in π . Object quantifiers become quantifiers over valid point names, and atomic formulas use the specified equality and operation relations.

Logical axioms remain valid under this interpretation; modus ponens and generalization remain valid as well. Thus the translated finite derivation proves

$$\bigwedge_{\theta \in \Gamma_\pi} \theta^M \longrightarrow \psi^M.$$

This is a fixed syntactic translation. It neither constructs a set of truths nor proves uniform soundness for arbitrary proof codes. The elementary logic involved is available already in the base second-order framework, once the presentation laws are assumed. $\square$

Consequently the main argument can be stated with a sufficient fixed finite arithmetic package, giving an actual second-order sentence. Using the familiar model-theoretic notation is harmless provided this convention is kept explicit.

105.3 The exact internal MRDP input

The theorem of Gaifman and Dimitracopoulos needed here is the following proof-theoretic form of MRDP [68]. Its exact statement is explicitly recorded in the recent research paper of Chen, Müller, and Yokoyama: see its Section 1.4 and Definition 5.2 [69].

Theorem 105.2 (Gaifman–Dimitracopoulos; established input). *For each fixed bounded arithmetic formula $\theta(\bar{x})$, there are fixed polynomial terms $P(\bar{x}, \bar{u})$ and $Q(\bar{x}, \bar{u})$, with natural-number coefficients, such that*

$$T_{\text{exp}} \vdash \theta(\bar{x}) \longleftrightarrow \exists \bar{u} P(\bar{x}, \bar{u}) = Q(\bar{x}, \bar{u}).$$

The same conclusion holds for each fixed Σ_1 formula by including its initial existential variables in $\bar{u}$.

Only one standard instance, for the graph of 2^x , is essential to the multiplicative theorem. No uniform formalization of the whole MRDP theorem inside the external theory is needed. The theorem's *internal correctness* is essential: a polynomial that merely agrees with exponentiation on standard integers would be insufficient.

The original 1982 contribution is a chapter in monograph no. 30 of *L'Enseignement Mathématique*, not an article in journal volume 30. Its bibliography is confirmed by the author's publication list [78]. An accessible scan of the original chapter was not obtained in this audit; the precise input above is corroborated by [69], rather than inferred from a casual summary.

[write] [Theorem 105.2](#) is a declared input. Source 14's author did not see the original chapter [68] and took the statement from [69]; the intake did not check either source.

105.4 Coded descriptive-set-theoretic inputs

We use the following standard results in their coded forms, available in ATR_0 ; the relevant implementation of the perfect-set and order steps is also described in Glazer's Optimization section [1, 70, 71].

1. Disjoint analytic sets admit a Borel separator. In particular, an analytic set with analytic complement has a Borel code.
2. Borel sets have the property of Baire, with an open approximant and a coded countable cover of the error by closed nowhere-dense sets.

3. An uncountable Borel subset of a Polish space admits a continuous injective Cantor map. For a Borel linear order, this may be refined to an increasing continuous Cantor map.
4. The preceding perfect-set/order construction can be carried out for a coded sequence of uncountable Borel sets without adding a countable-choice principle for arbitrary relations.

For the order refinement, pull the arithmetic order back along a Cantor injection, color pairs by agreement with lexicographic order, and use perfect-set homogeneity for a Borel two-coloring of pairs. Reverse the Cantor order if necessary. The uniform version asserts the existence of one real coding the sequence of embeddings; it does not assert an effective or Borel selection operator on all input codes. Apply the pseudohierarchy construction relative to the join of the input codes, then one tagged Baire approximation to the pullback pair-colorings, a common perfect tree avoiding all errors, and the coded open-color homogeneous-tree construction in each section. This is the common coding and perfect-tree construction used in Glazer's optimization.

These are declared inputs, not hidden consequences of elementary metric completeness. The new uses of them, particularly the tagged continuity construction, are proved below.

[write] These four coded inputs, together with [lemma 108.2](#) and the evaluation of countable Borel iterations along a pseudohierarchy in [section 112](#), are where the ATR_0 bookkeeping of source 14 rests on statements it declares rather than proves. Whether each holds in ATR_0 in the uniform form used is part of what an independent review would have to check.

106 Internal Diophantine definitions become Borel maps

106.1 The functional projection principle

Lemma 106.1 (Total analytic functions). *Let X, Y be Polish spaces. Suppose $R \subseteq X \times Y$ is analytic and for every $x \in X$ there is exactly one y with $R(x, y)$. Then the function $f(x) = y$ has a Borel graph and is a Borel map. From the analytic data and the asserted totality and uniqueness, the needed codes can be obtained in ATR_0 .*

Proof. The complement of the graph is analytic because

$$\neg R(x, y) \iff \exists z (R(x, z) \wedge z \neq y).$$

Here totality supplies an output, and uniqueness identifies all other pairs as nongraph pairs. Analytic separation therefore gives a Borel graph.

To obtain a map code explicitly, let $(U_j)_{j \in \mathbb{N}}$ be a countable basis of Y . Form the tagged sets

$$\begin{aligned} A &= \{(j, x) : \exists y (R(x, y) \wedge y \in U_j)\}, \\ B &= \{(j, x) : \exists y (R(x, y) \wedge y \notin U_j)\}. \end{aligned}$$

They are analytic, disjoint, and exhaust $\mathbb{N} \times X$. One separator is therefore exactly A . Its sections give all the codes for $f^{-1}(U_j)$ at once. This avoids a countable sequence of independently chosen separators. $\square$

Theorem 106.2 (Automatic Borel definability). *Suppose $\mathcal{M} \models T_{\text{exp}}$ has a Polish carrier and Borel arithmetic operations. If a fixed Σ_1 formula $\varphi(\bar{x}, y)$ provably defines a total single-valued function*

in T_{exp} , then its interpretation in $\mathcal{M}$ is a Borel map. Also, any relation whose positive and negative definitions are provably Σ_1 over T_{exp} is Borel.

Proof. Use Theorem 105.2 to write

$$\varphi(\bar{x}, y) \longleftrightarrow \exists \bar{u} P(\bar{x}, y, \bar{u}) = Q(\bar{x}, y, \bar{u})$$

inside T_{exp} . Polynomial evaluation in $\mathcal{M}$ is Borel: the polynomial is a fixed finite expression built from Borel operations. Equality of two values is a closed relation on the carrier, so the equality locus is Borel. Projection over the finite tuple $\bar{u}$ makes the displayed relation analytic. Lemma 106.1 applies.

For a relation with positive and negative Σ_1 definitions, apply internal MRDP to both definitions and then analytic separation. The transfer of each fixed arithmetic derivation is justified by Lemma 105.1. $\square$

Corollary 106.3. *Every fixed bounded arithmetic relation is Borel in a Borel presentation of a model of T_{exp} .*

Proof. A bounded formula and its negation are both bounded. Apply the relational part of Theorem 106.2. $\square$

This corollary is not obtained by pretending that an internally bounded quantifier ranges over an externally finite set. In an uncountable nonstandard model, an interval $[0, a]$ can be uncountable. The Diophantine translation is doing substantive work.

106.2 Exponentiation is an instance of the principle

For clarity, one may use the standard remainder-based sequence coding. Let

$$\beta(u, v, i) = u \bmod (1 + (i + 1)v).$$

The assertion $\beta(u, v, i) = r$ can be expanded using a bounded quotient witness and a polynomial equality, so it is bounded arithmetic syntax. The graph formula

$$\begin{aligned} E(x, y) \iff \exists u \exists v (\beta(u, v, 0) = 1 \wedge \beta(u, v, x) = y \\ \wedge \forall i < x \beta(u, v, i + 1) = 2\beta(u, v, i)) \end{aligned} \quad (3.1)$$

is Σ_1 after expanding the remainder notation. All intermediate remainders and quotient witnesses can be bounded by terms in the code parameters.

The usual arithmetic properties, proved internally in T_{exp} , are

$$e(0) = 1, \quad e(1) = 2, \quad e(a + b) = e(a)e(b), \quad a < b \implies e(a) < e(b), \quad (3.2)$$

where $e(x)$ is the unique y satisfying $E(x, y)$. Only fixed proofs of these facts and of totality and uniqueness are used.

Corollary 106.4 (Borel internal exponentiation). *In every Borel Polish presentation of a model of T_{exp} , the internally defined function $e(x) = 2^x$ is Borel. This conclusion, including a Borel-map code, follows in ATR_0 .*

Proof. Apply Theorem 106.2 to the fixed formula E . $\square$

The auxiliary Diophantine witnesses need not be unique. Only the output y must be unique. Nor do we require that those witnesses depend continuously or Borel-measurably on x . The argument concerns projection and the resulting function, not a selection of polynomial solutions.

107 A common Cantor set of continuity

The next lemma explains precisely how the Borel code for e replaces higher-complexity choices.

Lemma 107.1 (Simultaneous Cantor regularization). *Let Y be a Polish space and let $(H_n)_{n \in \mathbb{N}}$ be a uniformly coded sequence of Borel maps $2^{\mathbb{N}} \rightarrow Y$. There is a continuous increasing injection $F : 2^{\mathbb{N}} \rightarrow 2^{\mathbb{N}}$, for the lexicographic order, such that every $H_n \circ F$ is continuous. The construction is available in ATR_0 .*

Proof. Let (U_j) be a countable basis of Y . Apply the Baire-property construction to the single Borel set

$$A = \{(n, j, r) : H_n(r) \in U_j\} \subseteq \mathbb{N} \times \mathbb{N} \times 2^{\mathbb{N}}.$$

It gives open approximants $V_{n,j}$ and, uniformly, a countable family of closed nowhere-dense sets covering every symmetric difference

$$H_n^{-1}(U_j) \Delta V_{n,j}.$$

Let G be the complementary dense G_δ subset of $2^{\mathbb{N}}$. On G , every basis preimage for every H_n is relatively open. Thus every $H_n|_G$ is continuous.

Write $G = \bigcap_\ell W_\ell$, with each W_ℓ dense open. Construct nonempty clopen cylinders V_s , $s \in 2^{<\omega}$, so that the two successors lie inside $V_s \cap W_{|s|}$, are disjoint, occur in left-to-right lexicographic order, and have lengths tending to infinity with $|s|$. Each branch has a unique intersection point. The branch map F is continuous, injective, increasing, and has range in G . Therefore all $H_n \circ F$ are continuous. $\square$

The use of a single tagged set matters in weak metatheory. It produces one coded family of approximations and errors; it does not appeal to unrestricted countable choice. The conclusion also does not say that the original H_n are continuous on all of Cantor space.

108 Arithmetic scales above any prescribed parameter

We next replace informal fractional powers by exact floor operations. This also makes the cofinal strengthening transparent.

108.1 Borel arithmetic order, division, and square roots

Lemma 108.1. *In a Borel presentation of a model of $\mathbf{Q} + \mathbf{IOpen}$, the arithmetic order is Borel. Floor division by a positive element, its remainder, and the function $x \mapsto \lfloor \sqrt{x} \rfloor$ are Borel. Truncated subtraction is also Borel.*

Proof. Both $x < y$ and its complement have existential definitions:

$$x < y \iff \exists z (x + z + 1 = y), \quad x \geq y \iff \exists z (y + z = x).$$

They are analytic, so separation gives a Borel order.

Open induction supplies Euclidean quotient and remainder and the largest q with $q^2 \leq x$. For example, the quantifier-free inequality $q^2 > x$ holds at $q = x + 1$; the least-transition argument supplied by open induction gives a least such q . The division argument uses $dq > x$, with $d > 0$, in the same way.

The characterizing graphs

$$x = dq + r, \quad r < d, \quad \text{and} \quad q^2 \leq x < (q + 1)^2$$

are Borel and define unique total outputs on their stated domains. For coding purposes, extend quotient and remainder by zero at divisor zero, so their domains are the Polish space M^2 . The tagged inverse-open argument of Lemma 106.1 supplies map codes. Truncated subtraction has the Borel graph

$$(y \leq x \wedge y + z = x) \vee (x < y \wedge z = 0),$$

which again gives a unique total output. No continuity of subtraction is asserted. $\square$

Lemma 108.2 (Uniform increasing Cantor maps; established input). *For a Borel linear order on an uncountable Polish space, every uncountable Borel subset contains an increasing continuous Cantor image. Given a coded sequence of such subsets, the images can be supplied by a single real coding a sequence of embeddings in ATR_0 .*

This is the perfect-set/order-homogenization input described in Section 105. Its uniform form is the one used in Glazer's weak-metatheory optimization [1]. No uncountable choice of points is involved.

108.2 The exact scale construction

Fix any threshold $b \in M$. An increasing Cantor image in M supplies a c_0 for which $[0, c_0]$ is uncountable: take its upper endpoint. Enlarge it to

$$c = \max\{c_0, (b + 2)^4\}.$$

Define, for external $n \in \mathbb{N}$,

$$t_0 = c, \quad t_{n+1} = \lfloor \sqrt{t_n} \rfloor, \quad q_n = \lfloor c/t_n \rfloor. \quad (5.1)$$

Every interval $[0, t_n]$ is uncountable. To verify this without induction on an assertion of uncountability, fix an arbitrary external n . At each level $i < n$, quotient and remainder upon division by $t_{i+1} + 1$ inject $[0, t_i]$ into $[0, t_{i+1}]^2$, since

$$t_i < (t_{i+1} + 1)^2.$$

Iteration along the finite binary tree gives an injection $[0, c] \hookrightarrow [0, t_n]^{2^n}$. A coded enumeration of $[0, t_n]$ would enumerate its finite Cartesian power. Reconstruct the tree upward with $x = (t_{i+1} + 1)q + r$, retaining results in $[0, c]$ and using zero otherwise. Every element of $[0, c]$ would occur, contradicting its uncountability. Correctness of these finite calculations uses arithmetic induction relative to their evaluation traces, not induction on a projective uncountability predicate. In particular every t_n exceeds every standard integer, all the divisions in (5.1) are legitimate, and $q_n \geq 1$.

Set

$$X_n = \{q_n i : i < t_{n+2}\}. \quad (5.2)$$

Each X_n is uncountable because multiplication by $q_n > 0$ is injective. It is Borel by the direct description

$$X_n = \{x < q_n t_{n+2} : \text{rem}(x, q_n) = 0\}.$$

Also $X_n \subset [0, c)$, because $q_n t_n \leq c$ and $t_{n+2} < t_n$.

The scale separation has a short exact proof:

$$q_{n+1} \geq q_n t_{n+1} \geq q_n t_{n+2}^2 > 8q_n t_{n+2}. \quad (5.3)$$

The first inequality follows from $c \geq q_n t_n \geq q_n t_{n+1}^2$, and the last uses $t_{n+2} > 8$.

We use the shifted sets X_{n+1} . By Lemma 108.2, choose uniformly increasing continuous injections

$$f_n : 2^{\mathbb{N}} \longrightarrow X_{n+1}.$$

Write $\epsilon_k = \chi_{\{k\}}$, so $\epsilon_k \rightarrow 0^\omega$. For a suitable increasing Cantor restriction F , or simply $F = \text{id}$, put

$$x_n = f_n(F(0^\omega)), \quad h_{k,n} = f_n(F(\epsilon_k)) - x_n. \quad (5.4)$$

The subtraction denotes the unique nonnegative arithmetic difference, not an assumed continuous operation.

Lemma 108.3 (Separated perturbations). *The parameters in (5.4) satisfy*

$$b < h_{k,n}, \quad q_{n+1} \leq h_{k,n} < q_{n+1} t_{n+3}, \quad x_n < c. \quad (5.5)$$

Every choice $h_n = h_{k_n,n}$ satisfies

$$h_{n+1} > 8h_n. \quad (5.6)$$

If $F = \text{id}$, then $x_n + h_{k,n} \rightarrow x_n$. If e is Borel and F is obtained by Lemma 107.1 for $H_n = e \circ f_n$, then

$$e(x_n + h_{k,n}) \longrightarrow e(x_n). \quad (5.7)$$

Proof. The differences are positive multiples of q_{n+1} , because both endpoints lie in X_{n+1} . They are smaller than $q_{n+1} t_{n+3}$. Moreover

$$q_1 \geq t_1 \geq (b+2)^2 > b,$$

and (5.3) makes the subsequent q_n increase. The scale inequality gives (5.6), independently of the choices of k_n . The convergence assertions follow from the specified continuity and from $\epsilon_k \rightarrow 0^\omega$. $\square$

Although t_n decreases and remains internally infinite for every standard n , no paradox is involved. The recursion is externally countable, and internal nonstandard intervals need not behave like standard finite intervals.

109 A quantitative fusion lemma

The following metric estimate will serve both operations. Suppose finite vertices $V_s \in M$, $s \in 2^{<\omega}$, are constructed level by level, with

$$V_{s0} = V_s.$$

Assume vertices at each level are pairwise distinct. Put

$$\Delta_0 = 1, \quad \Delta_n = \min(\{1\} \cup \{d(V_s, V_t) : s, t \in 2^n, s \neq t\}) \quad (n > 0).$$

Zero extensions imply $\Delta_{n+1} \leq \Delta_n$.

Lemma 109.1 (Fusion estimate). *If*

$$d(V_s, V_{s1}) < 2^{-n-4} \Delta_n \quad (s \in 2^n), \quad (6.1)$$

then

$$z_r = \lim_N V_{r|N}$$

exists for every $r \in 2^{\mathbb{N}}$, uniformly in r , and the map $r \mapsto z_r$ is a continuous injection. More precisely,

$$d(V_{r|N}, z_r) \leq 2^{-N-3} \Delta_N, \quad (6.2)$$

and if $r|N \neq s|N$, then

$$d(z_r, z_s) \geq (1 - 2^{-N-2}) \Delta_N > 0. \quad (6.3)$$

Proof. For $i \geq N$, $\Delta_i \leq \Delta_N$. The sum of the tail edge bounds is at most

$$\Delta_N \sum_{i=N}^{\infty} 2^{-i-4} = 2^{-N-3} \Delta_N.$$

Completeness gives the limit and (6.2). Since $\Delta_N \leq 1$, the bound also gives a common Cauchy rate. The finite-level maps $r \mapsto V_{r|N}$ are locally constant, so uniform convergence makes the limit map continuous.

If the prefixes at level N differ, their vertices are at least Δ_N apart. Subtracting the two tail bounds gives (6.3). Thus distinct branches have distinct limits. $\square$

At an implementation level, the search for (6.1) need not compare arbitrary real numbers exactly. Choose rational lower bounds for the finitely many positive separation distances and dovetail a search for certified strict inequalities. The convergence used below guarantees termination at each stage. This is a recursive construction with supplied codes, not an arbitrary sequence of choices.

110 Cofinal discontinuity of additive translations

We first give a full additive proof, including the category lemma that will be reused.

Lemma 110.1 (Category zero–one law for finite changes). *If $D \subseteq 2^{\mathbb{N}}$ has the property of Baire and membership in D is invariant under changing finitely many coordinates, then D is meager or comeager.*

Proof. If D is nonmeager, it is comeager in some nonempty basic cylinder. Finite bit flips take this cylinder to every cylinder of the same length. Invariance makes D comeager in every one of them. There are finitely many such cylinders and they cover Cantor space, so D is comeager. $\square$

Theorem 110.2. *Under the hypotheses of Theorem 104.1(i), discontinuous translations occur cofinally in the arithmetic order.*

Proof. Fix $b \in M$, and suppose for contradiction that every translation τ_a with $a \geq b$ is continuous. Use Section 108 with $F = \text{id}$. For every $m \geq 2c$, the translating constant $m - x_n$ is larger than c , hence larger than b . Therefore

$$m + h_{k,n} = (m - x_n) + (x_n + h_{k,n}) \longrightarrow m. \quad (7.1)$$

Only a fixed translation is used in this passage to the limit.

Choose $h_n = h_{k_n, n}$ recursively and set

$$V_s = 2c + \sum_{i < |s|, s(i)=1} h_i.$$

At stage n , (7.1) allows k_n to make every one of the finitely many new edges satisfy (6.1). The growth $h_{n+1} > 8h_n$ makes all subset sums at a fixed level distinct. For example, induction gives the integer inequality

$$7 \sum_{i < n} h_i < h_n;$$

the largest differing summand dominates all earlier ones. Lemma 109.1 gives a continuous injection $r \mapsto z_r$.

Identify subsets of $\mathbb{N}$ with their characteristic functions. For $j \notin A$, the exact finite-sum identities and continuity of the fixed translation by $h_j > b$ give

$$z_{A \cup \{j\}} = z_A + h_j. \quad (7.2)$$

This identity, not continuity of order or subtraction, controls the limits.

Let r^* denote coordinatewise complementation and define

$$E_n = \{r : z_{r^*} < z_r \leq z_{r^*} + h_n\}, \quad D = \{r : \forall n (z_{r^*} + h_n < z_r)\}. \quad (7.3)$$

These are Borel sets.

Let $r(j) = 0$ and let s be obtained by flipping the j th bit. Equation (7.2) says

$$z_s = z_r + h_j, \quad z_{r^*} = z_{s^*} + h_j. \quad (7.4)$$

If $r, s \in E_n$ and $j > n$, then

$$z_{s^*} + h_j < z_r, \quad z_r + h_j \leq z_{s^*} + h_n.$$

Adding and cancelling yields $2h_j < h_n$, which is impossible. Thus E_n is disjoint from its image under any sufficiently far bit flip. If E_n were nonmeager, it would be comeager in some cylinder. A flip beyond that cylinder's length would give two disjoint comeager subsets of the same cylinder. Hence each E_n is meager.

The set D is invariant under a single bit flip. In the notation (7.4), $r \in D$ immediately implies $s \in D$. Conversely, suppose $s \in D$ and fix n . Choose $k = \max(n, j) + 1$; growth gives

$$h_k > h_n + 2h_j.$$

Then

$$z_{s^*} + h_k < z_s = z_r + h_j$$

implies, after comparison and cancellation,

$$z_{r^*} + h_n = z_{s^*} + h_j + h_n < z_r.$$

Thus $r \in D$. Iterating proves finite-change invariance.

The sets D and $D^* = \{r^* : r \in D\}$ are disjoint. Their union is comeager, because $z_r \neq z_{r^*}$ by injectivity, and

$$2^{\mathbb{N}} \setminus (D \cup D^*) \subseteq \bigcup_n (E_n \cup E_n^*). \quad (7.5)$$

Complementation is a homeomorphism. Lemma 110.1 says D is meager or comeager. In the first case both D and D^* are meager, contradicting their comeager union. In the second case they are disjoint comeager sets, another contradiction. This proves the theorem. $\square$

The proof never takes an arithmetic inequality through a metric limit. In particular it does not assert $z_r \geq 2c$ because every approximating vertex has that property.

111 Cofinal discontinuity of power-of-two multipliers

We now prove the part that resolves the weak-metatheory question.

Theorem 111.1 (A Borel exponential obstruction). *Suppose $\mathcal{M} \models \mathbf{Q} + \mathbf{IOpen}$ has an uncountable Polish carrier and Borel arithmetic operations. Suppose in addition there is a Borel function $e : M \rightarrow M$ satisfying (3.2). Then, for every $b \in M$, some fixed multiplication map $x \mapsto e(a)x$, with $a \geq b$, is discontinuous.*

Proof. Fix b , and assume that multiplication by $e(a)$ is continuous for every $a \geq b$. Construct c, t_n, q_n, X_n, f_n as in Section 108. Apply Lemma 107.1 to all maps $e \circ f_n$, and define $x_n, h_{k,n}$ using the resulting common F . We have (5.7).

For each $m \geq 2c$, the parameter $m - x_n > c > b$. Using only continuity of multiplication by the fixed element $e(m - x_n)$, we obtain

$$\begin{aligned} e(m + h_{k,n}) &= e(m - x_n)e(x_n + h_{k,n}) \\ &\longrightarrow e(m - x_n)e(x_n) = e(m). \end{aligned} \quad (8.1)$$

Choose $h_n = h_{k_n,n}$ recursively, write $y_n = e(h_n)$, and put

$$V_s = e(2c) \prod_{i < |s|, s(i)=1} y_i. \quad (8.2)$$

Each current vertex is $e(m_s)$, with $m_s = 2c + \sum_{i < |s|, s(i)=1} h_i \geq 2c$. Thus (8.1) allows all new edges at stage n to satisfy (6.1).

The arithmetic identities for e give

$$y_0 \geq 2, \quad y_{n+1} > y_n^5. \quad (8.3)$$

Let $Q_n = \prod_{i < n} y_i$. Induction gives $Q_n^4 < y_n$: it holds at $n = 0$, and

$$Q_{n+1}^4 = Q_n^4 y_n^4 < y_n^5 < y_{n+1}.$$

Consequently the largest differing index in two finite products dominates all preceding factors. The vertices at every level are distinct. Lemma 109.1 again produces a continuous injection $r \mapsto z_r$.

For $j \notin A$, continuity of multiplication by $y_j = e(h_j)$, whose exponent is $h_j > b$, gives

$$z_{A \cup \{j\}} = y_j z_A. \quad (8.4)$$

Every z_A is strictly positive. If $z_A = 0$, flip any bit. Equation (8.4), and cancellation of the nonzero factor y_j when the bit is removed, give a second input whose limit is zero. This contradicts injectivity. The carrier is the nonnegative arithmetic domain, so nonzero means positive. No order-closedness assumption is used.

Define direct comparison events

$$E_n = \{r : z_{r^*} < z_r \leq y_n z_{r^*}\}, \quad D = \{r : \forall n (y_n z_{r^*} < z_r)\}. \quad (8.5)$$

They are Borel. With $r(j) = 0$ and s its one-bit flip, (8.4) gives

$$z_s = y_j z_r, \quad z_{r^*} = y_j z_{s^*}. \quad (8.6)$$

If $r, s \in E_n$ and $j > n$, then

$$y_j z_{s^*} < z_r, \quad y_j z_r \leq y_n z_{s^*}.$$

Multiplying the first inequality by y_j , combining, and cancelling $z_{s^*} > 0$, yields $y_j^2 < y_n$. This contradicts (8.3). The same cylinder argument used in Section 110 therefore makes each E_n meager.

The set D is invariant under one-bit flips. If $r \in D$, (8.6) immediately implies $s \in D$. For the converse, suppose $s \in D$, fix n , and put $k = \max(n, j) + 1$. Then

$$y_k > y_n y_j^2.$$

We have

$$y_j z_r = z_s > y_k z_{s^*} > y_n y_j^2 z_{s^*}.$$

Cancelling $y_j > 0$ proves $z_r > y_n z_{r^*}$, as required.

The category contradiction now finishes the proof: D, D^* are disjoint, have a comeager union by

$$2^{\mathbb{N}} \setminus (D \cup D^*) \subseteq \bigcup_n (E_n \cup E_n^*),$$

and are interchanged by a homeomorphism. Finite-change invariance gives the meager-or-comeager dichotomy, which is incompatible with these two facts. $\square$

Corollary 111.2 (The answer to Glazer’s Question 1, as claimed by source 14). *With the coding convention of Section 105, Glazer’s topological Tennenbaum theorem for $\mathbf{Q} + \mathbf{I}\Delta_0 + \mathbf{Exp}$ is provable in $\mathbf{ATR}_0$. In fact Theorem 104.1 holds.*

Proof. Corollary 106.4 supplies the Borel function e and its internally proved algebraic properties. Apply Theorem 111.1 and Theorem 110.2. Joint continuity implies continuity of every unary section, so the original continuity conclusion follows as well. $\square$

Status of this answer [write]

Corollary 111.2 is a **claimed** affirmative answer to Glazer’s Question 1, which this report printed as open (question 43.1; section 1.1). It is an unrefereed, AI-assisted manuscript’s claim, an upper bound under the coded-model convention of section 105.2, and it has **not been independently reviewed**: not by the intake, and not by the Hilbert’s-tenth programme’s scoped review of the same archive, which read the internal-MRDP input and the automatic-Borel argument and states that it does not certify the claimed resolution (appendix E). Its $\mathbf{ATR}_0$ bookkeeping rests partly on declared inputs (note in section 105.4), and its author did not see the Gaifman–Dimitracopoulos chapter it relies on. No reversal is claimed: $\mathbf{ACA}_0$ is not shown insufficient.

[write] Glazer names two places where his proof of Theorem 2 uses more than $\mathbf{ATR}_0$ [1, p. 8]: the comprehension used to define a partial map g from preimages under $\exp \circ f_n$ in his Claim 2, and $\Sigma_2^1\text{-AC}_0$ to choose the points $s_{k,n}$. Source 14 replaces the first by the Borel exponentiation of corollary 106.4 and the common continuity restriction of lemma 107.1, and the second by the finite searches of the fusion estimate (lemma 109.1). Its additive part, theorem 110.2, strengthens Glazer’s Corollary 1, which Glazer already proves in $\mathbf{ATR}_0$ [1, p. 7], from measure to category and to cofinal unary maps.

112 Why the proof stays in ATR_0

The new bridge is short, but its metatheoretic interpretation needs care. The following table states the roles of the ingredients.

Operation	Data or justification	Choice or truth issue
Internal exponentiation formula	One fixed MRDP proof instance	Finite proof translation only
Polynomial evaluation	Finite composition of Borel operations	No solution selector
Exponentiation map code	One tagged analytic separator	No separate separators chosen
Arithmetic scales	Borel floor-root and quotient maps	Coded recursive evaluation
Increasing Cantor maps	Uniform perfect/order construction	Use the coded uniform form
Common continuity restriction	One tagged Baire approximation	One exceptional cover
Metric fusion	Finite searches with a common tail bound	No arbitrary real choices
Tail contradiction	Baire property and finite flips	No measure theory

112.1 Countable iteration of Borel functions

The sequence $t_{n+1} = \lfloor \sqrt{t_n} \rfloor$ should not be justified by an unexamined appeal to dependent choice. Evaluation of the finitely many Borel maps involved can be organized along a common countable well-order α . Successive evaluations can then be performed along $\alpha \cdot \omega$, with each block using the point name produced by the preceding one. This is arithmetical transfinite recursion, precisely the available principle. The rational search used in fusion is handled within the same coded evaluation framework.

112.2 Uniform meager covers

The passage from the individual meagerness of E_n to the meagerness of their union also has a uniform implementation. Apply the Baire-property construction to

$$\{(n, r) : r \in E_n\} \subseteq \mathbb{N} \times 2^{\mathbb{N}}.$$

Because each section is meager, each open approximant section must be empty: a nonempty open subset of Cantor space cannot be meager. The uniformly supplied error cover therefore covers every section. Its sections, together with their complement-images, supply the one countable cover needed in (7.5) and its multiplicative analogue.

112.3 An upper bound, not a reversal

An affirmative answer to the question about provability in ATR_0 is an upper bound on the metatheory required. Nothing here proves that the theorem is equivalent to ATR_0 over RCA_0 , or that ACA_0 cannot prove a suitably restricted version. Those are separate questions, sensitive to how uncountability, the Borel data, and the arithmetic hypotheses are coded.

Similarly, the proof does not construct a global satisfaction predicate for T_{exp} , a set of all Borel functions, or a collection of all models on an uncountable carrier. The cofinality

assertions themselves can be written as second-order formulas about point names and the supplied Borel codes.

112.4 Three tempting shortcuts that are insufficient

First, a Diophantine representation proved only on standard $\mathbb{N}$ does not automatically identify the internally defined function in an arbitrary nonstandard model. Theorem 105.2 is the remedy.

Second, a Borel graph is not the chosen endpoint of the argument. The tagged basis-preimage construction explicitly gives the Borel-map information needed for regularization.

Third, transferring the topology along $e : M \rightarrow e[M]$ does not immediately reduce multiplication to addition. The Borel subspace $e[M]$ need not be Polish in the inherited topology, and a Polish refinement can destroy the continuity one hoped to preserve. The direct metric fusion works in the original complete carrier.

113 A nonsplit Polish Presburger model

This supplementary section is ordinary set-theoretic mathematics. The ATR_0 upper-bound claim concerns the coded topological-arithmetic theorem above; no formalization of the surreal statements in ATR_0 is asserted.

The preceding obstruction needs more arithmetic than addition alone. To make that boundary concrete, we now construct a Presburger model with continuous addition whose algebra differs from the split examples already investigated in Provel.

113.1 A countable nonsplit ordered group

For $n \geq 1$, define

$$s_n = \sum_{k=1}^{n-1} k!, \quad e_0 = (1, 0), \quad a_n = \left(\frac{s_n}{n!}, \frac{1}{n!} \right) \in \mathbb{Q}^2. \quad (10.1)$$

Let K_* be the subgroup generated by $e_0, a_1, a_2, \dots$. Order $\mathbb{Q}^2$ by the second coordinate first and then by the first coordinate:

$$(p, q) > 0 \iff q > 0 \text{ or } (q = 0 \text{ and } p > 0).$$

Lemma 113.1 ($K_* = G_{\zeta_*}$ of `isg:cf:prop:family`). *The group K_* is countable, e_0 is its least positive element, and projection onto the second coordinate induces*

$$K_*/\mathbb{Z}e_0 \cong \mathbb{Q}.$$

The exact sequence $0 \rightarrow \mathbb{Z}e_0 \rightarrow K_ \rightarrow \mathbb{Q} \rightarrow 0$ does not split.*

Proof. Countability is immediate. If $i \leq N$, then

$$a_i = \frac{N!}{i!} a_N - \frac{s_N - s_i}{i!} e_0, \quad (10.2)$$

and both coefficients are integers. Hence every finite linear combination of generators has the form $me_0 + ka_N$ for suitable integers m, k and N . Its second coordinate is zero exactly

when $k = 0$. Thus the kernel of the projection is $\mathbb{Z}e_0$. The image contains $1/n!$ for every n , so it is all of $\mathbb{Q}$. An element with positive second coordinate is larger than e_0 ; an element with zero second coordinate is an integer multiple of e_0 . This proves the least-positive claim.

The generators obey

$$(n+1)a_{n+1} = a_n + e_0. \quad (10.3)$$

If a homomorphism $\rho : K_* \rightarrow \mathbb{Z}$ satisfied $\rho(e_0) = 1$, write $b_n = \rho(a_n)$. Iterating (10.3) gives

$$n!b_n = b_1 + s_n. \quad (10.4)$$

Both s_n and $n! - s_n$ tend to infinity. For the second assertion, $s_n \leq 2(n-1)!$ for $n \geq 3$, so $n! - s_n \geq (n-2)(n-1)!$. For sufficiently large n , (10.4) would therefore put the integer multiple $n!b_n$ strictly between 0 and $n!$. This is impossible. A splitting of the displayed exact sequence would give such a retraction, proving nonsplitting. $\square$

[write] K_* is the group G_{ζ_*} , $\zeta_* = \sum_k k!$, of the discrete-initial-subgroups-and-omnific-normalization report (isg:cf:eq:explicitparameter), with the two coordinates exchanged: source 14's $a_n = (s_n/n!, 1/n!)$ and $e_0 = (1, 0)$ are that report's $x_n = (1/n!, S_n/n!)$ and $e = (0, 1)$, with $S_n = s_n$, and relation (10.3) of lemma 113.1 is its transition relation with every digit equal to 1. This lemma is therefore isg:cf:prop:family with isg:cf:thm:familytest for ζ_* , proved with the same estimate $s_n \leq 2(n-1)!$ (isg:cf:lem:noninteger). Source 14 does not cite that report; its statement that it does not claim the first discovery of nonsplit Presburger groups stands.

This supplies all algebra needed here without invoking the classification of extensions of $\mathbb{Q}$ by $\mathbb{Z}$.

113.2 The Polish carrier

Let

$$G = \mathbb{R} \overrightarrow{\times} K_*$$

as an ordered group, with the real coordinate having priority. Give it the product group topology $\mathbb{R} \times K_{*,\text{discrete}}$. Its distinguished unit is $(0, e_0)$. Let $M = G_{\geq 0}$, with the inherited topology and addition.

Proposition 113.2 (an instance of theorem 94.1). *The structure M is an uncountable, locally compact, perfect Polish model of Presburger arithmetic with jointly continuous addition. Its group completion is G , and the standard-integer subgroup of G is not a direct summand.*

Proof. The group K_* is countable discrete, hence Polish. The slices of M are

$$M \cap (\mathbb{R} \times \{d\}) = \begin{cases} [0, \infty) \times \{d\}, & d \geq 0, \\ (0, \infty) \times \{d\}, & d < 0. \end{cases} \quad (10.5)$$

These slices are clopen in M . They are locally compact Polish spaces without isolated points. A countable topological disjoint union of Polish spaces is Polish: choose bounded complete metrics on the slices and a fixed positive distance between distinct slices. This also proves local compactness and perfectness directly. The carrier is uncountable because any one slice is uncountable. Addition is the restriction of continuous group addition on $\mathbb{R} \times K_{*,\text{discrete}}$.

The least positive element of G is $(0, e_0)$, and

$$G/\mathbb{Z}(0, e_0) \cong \mathbb{R} \oplus \mathbb{Q}$$

is divisible. The standard characterization of Presburger groups as discretely ordered abelian groups with divisible quotient by the subgroup generated by 1 therefore applies [4, Section 2]. Its nonnegative cone is a model of Presburger arithmetic.

Every element of a linearly ordered group is a difference of nonnegative elements, so G is the Grothendieck group of M . A retraction $G \rightarrow \mathbb{Z}$ fixing its unit would restrict on $\{0\} \times K_*$ to the forbidden retraction in Lemma 113.1. Thus the extension remains nonsplit. $\square$

[write] Proposition 113.2 is the case $D = \mathbb{R}$, $K = K_*$ of source 13's theorem 94.1, printed there in general; source 14's proof is its own. Source 15 uses the same construction in theorem 122.2.

In fact every first-order definable relation, with parameters, is Δ_2^0 in this presentation. On each clopen discrete fiber of a finite power of M , a linear inequality is a real open or closed half-space, with its equality boundary decided by the order of K_* . Both the inequality and its complement are F_σ after taking the countable union of fibers. Congruence classes are clopen unions of fibers: $nG = \mathbb{R} \times nK_*$, so divisibility of the real coordinate leaves the real part unrestricted. Presburger quantifier elimination [4], and closure of Δ_2^0 under finite Boolean operations, prove the assertion. Thus this is a Borel arithmetic presentation at the additive level.

113.3 A surreal realization and an omnific obstruction

For $(x, (p, q)) \in G$, set

$$\Phi(x, (p, q)) = x\omega^2 + q\omega + p. \quad (10.6)$$

These are ordinary finite Conway normal forms.

Proposition 113.3 (second clause: `isg:cf:cor:unit`). *The map $\Phi : G \rightarrow \mathbf{No}$ is additive, injective, preserves order, and sends the distinguished unit to 1. There is nevertheless no additive map $M \rightarrow \mathbf{Oz}$ that sends the distinguished unit to 1.*

Proof. Addition of these finite normal forms is coefficientwise. The sign of their difference is determined by its first nonzero coefficient in the order $\omega^2, \omega, 1$. This is exactly the order on G , so Φ is an ordered additive embedding and has the stated unit value.

For the second assertion, recall the defining normal-form property of the canonical omnific integers: their constant coefficient is an integer, and their normal forms have no negative exponents [72]. Taking that coefficient gives an additive map

$$\text{ct} : \mathbf{Oz} \longrightarrow \mathbb{Z}, \quad \text{ct}(1) = 1.$$

Suppose $f : M \rightarrow \mathbf{Oz}$ were additive with $f(1) = 1$. All $(0, a_n)$ lie in M , since their second coordinate inside K_* is positive. Thus $b_n = \text{ct}(f(0, a_n)) \in \mathbb{Z}$ would satisfy

$$(n+1)b_{n+1} = b_n + 1$$

by (10.3), and hence the impossible relation (10.4). No extension of f to the group completion is needed for this contradiction. $\square$

Preservation of the unit is essential to the obstruction. The map

$$\Phi_\omega(x, (p, q)) = \omega\Phi(x, (p, q)) = x\omega^3 + q\omega^2 + p\omega$$

is an ordered additive embedding into $\mathbf{Oz}$, but sends the distinguished unit to ω . Its constant term is zero.

[write] The second assertion of [proposition 113.3](#) is the case $G = K_*$ of `isg:cf:cor:unit` (a $\mathbb{Z}$ -group admits an additive map to **Oz** sending its least positive element to 1 only if it splits), and the embedding Φ_ω is the one that report records after that corollary, $(a, b) \mapsto a\omega^2 + b\omega$ on G_ζ . Source 13 gives a two-element certificate of the same obstruction for its core ([Part VII](#)).

The distinction is algebraic. The topology under discussion is the explicitly given Polish topology on G and M . The image of Φ is discrete in the surreal order topology, since its ordered-group gaps are at least 1. Consequently Φ is not a homeomorphic realization of the perfect Polish carrier in that order topology.

The use of surreals creates no collection of all surreal numbers as a set. Formula (10.6) describes a set-indexed family of finite normal forms; it can be handled within a sufficiently large set-sized stage, or as a class-valued map in a conservative class framework. The nonexistence argument is stronger and more general: it applies to any target additive group that has a unit-preserving additive map to $\mathbb{Z}$.

114 Consequences and a concrete formalization route

114.1 What the result says about representations

In a Borel arithmetic presentation, internal exponentiation can be discontinuous, but it cannot have arbitrary projective complexity: its provable Diophantine definition supplies Borel regularity. This amount of regularity is enough to find a common perfect set on which the required countably many exponential maps are continuous. Completeness then turns small metric perturbations into a Cantor family of limits. Arithmetic growth and finite-bit symmetries make that family impossible if the relevant translations remain continuous.

The result distinguishes three notions that a representation theorem must keep separate:

1. correctness on standard natural numbers;
2. correctness in every model of a specified weak arithmetic;
3. regularity of the interpreted operations in an external topological presentation.

The passage from the second to the third is the main bridge of this article. The first alone does not imply either.

The cofinal strengthening says more than a failure somewhere in a two-variable function. No final arithmetic segment of translation parameters can consist entirely of continuous maps; the same holds for exponents of continuous powers-of-two multipliers. In particular the failure cannot be confined to a bounded set of arithmetic constants.

114.2 Existing interfaces and missing obligations

The relevant Lean file is

```
Logic/PeanoArithmetic/ListCoding/Lean/PAListCoding/
ExponentiationDiophantine.lean.
```

It imports Mathlib's Diophantine development and proves the standard-natural graph statement using `Dioph.pow_dioph` [73]. The corresponding Coq file provides explicit formula and single-equation interfaces [74].

Layer	Existing or established material	Additional obligation
Standard graph	$\mathbb{N}$ -valued exponentiation is Diophantine	Keep the standard-domain scope explicit
Internal graph	Gaifman–Dimitracopoulos theorem	Extract one weak-arithmetic proof for $E(x, y)$
Presentation semantics	Borel arithmetic operations on a Polish space	Interpret the finite object proof on point names
Borel decoding	Theorem 106.2	Formalize tagged analytic separation
Perfect restriction	Lemma 107.1	Formalize uniform Baire approximations and fusion
Continuity obstruction	Sections 110–111	Formalize limits, one-bit identities, and category

A practical development should proceed in this order.

1. **Fix the arithmetic sentence.** Choose one exact coded-sequence definition E and one pair of polynomial terms. Record the finite weak-arithmetic proof obligations connecting them. Do not silently substitute a theorem about $\mathbb{N}$.
2. **Separate interpretation from topology.** Prove the fixed-proof interpretation lemma for the chosen syntax and point-equality relation. This module should have no claims about continuity.
3. **Prove the abstract Borel decoding lemma.** Its inputs are analytic relation codes, totality, and uniqueness. The result should include uniform inverse-basis codes.
4. **Formalize the metric and category core.** The fusion estimate and one-bit contradiction can be verified in ordinary mathematical foundations before reverse-mathematical code bookkeeping is added.
5. **Assemble the exact upper bound.** Audit every use of countable sequences. Tagged constructions should replace implicit choices, and the final statement should name the coding convention.

No Lean or Coq file in this package is presented as a completed formalization of these obligations. The included program checks finite arithmetic estimates and the explicit nonsplit presentation; it does not check the infinite-dimensional semantic theorem.

[write] The two repository files named here are unchanged at the current state. Placement of this report beside them gives Part VIII no formal status; none of its statements is formalized.

115 Further research questions and topics

115.1 Reverse-mathematical strength

Question 115.1. For a fixed, precise coding of Borel arithmetic presentations, is the topological Tennenbaum statement equivalent to ATR_0 over RCA_0 , or does a weaker subsystem suffice?

The present proof provides an upper bound. A useful first step is to separate the contribution of analytic separation from that of the perfect-set/order lemma. Restrictions on the Borel ranks of the supplied operations may allow a sharper statement. A proposed reversal must specify the meaning of uncountability and the available presentation codes; otherwise different formulations can conceal different principles.

115.2 Weaker object arithmetic

Question 115.2. Can the power-of-two multiplier obstruction be established under a strictly weaker arithmetic hypothesis than $\text{I}\Delta_0 + \text{Exp}$, without assuming a total Borel exponential homomorphism as extra data?

Theorem 111.1 isolates exactly the usable external substitute. The remaining task is to obtain enough internally correct exponential behavior in weaker arithmetic. This should not be confused with a claimed proof of MRDP in $\text{I}\Delta_0$; that is a separate longstanding problem [69]. Partial exponentiation cuts are a natural intermediate setting.

[write] (Batch 90.) Answered in part by Part XI, in its explicit-code direction. Over $\text{I}\Delta_0$, one internal geometric code through an externally uncountable interval, with no total exponential, already gives a discontinuous multiplier $x \mapsto E(a)x$ in every uncountable subinterval (theorem 139.1), and uncountable Borel $\text{I}\Delta_0$ -models with such a code in which total exponentiation fails exist (proposition 146.2); they are polynomial cuts of Borel models of PA. The hypotheses are incomparable with those of theorem 111.1: theorem 139.1 assumes $\text{I}\Delta_0$ and a code, theorem 111.1 assumes $\text{Q} + \text{IOpen}$ and a total Borel exponential, so neither theorem strengthens the other. Open: the bare $\text{I}\Delta_0$ case without a code (question 149.4) and the identification of coded with ordinary bounded-definition exponentiation (question 149.1). Unrefereed and not formalized.

115.3 How complicated are the discontinuous parameters?

There is an immediate general upper bound. If $F : P \times X \rightarrow Y$ is Borel between Polish spaces, discontinuity of the section F_p is equivalent to

$$\exists x \exists \varepsilon \in \mathbb{Q}_{>0} \exists (x_k)_{k \in \mathbb{N}} \forall k \left[d_X(x_k, x) < 2^{-k} \wedge d_Y(F(p, x_k), F(p, x)) \geq \varepsilon \right].$$

The displayed witness relation is Borel, so the discontinuous-parameter set is analytic and the continuous-parameter set is coanalytic. The countable sequence of witnesses can be selected in ATR_0 , which already proves $\Sigma_1^1\text{-AC}_0$ [70, Theorem V.8.3]. Apply this to the jointly Borel families $F(a, x) = a + x$ and $F(a, x) = e(a)x$.

Question 115.3. For the arithmetic families above, is the coanalytic upper bound sharp, or must the parameter sets

$$\{a : \tau_a \text{ is continuous}\}, \quad \{a : \lambda_a \text{ is continuous}\}$$

be Borel under some natural additional hypotheses? Can their complements be forced to be nonmeager, or must one settle for arithmetic cofinality?

The theorem proves cofinal discontinuity, which does not settle a topological size question. The outstanding problem is sharper classification for arithmetic families, beyond the general analytic/coanalytic bound just proved.

[write] (Batch 90.) For multipliers $E(a)$ given by one internal geometric code, every convex set of continuous-multiplier indices is countable (proposition 145.5); whether the whole set is Borel, or countable, stays open (question 149.8).

115.4 Borel-rank bounds for internally computable functions

Question 115.4. Given a fixed internal proof that a Σ_1 graph is total and single-valued, what bounds can one give on the Borel complexity of its interpretation in terms of the Borel codes for the primitive arithmetic operations?

The analytic-separation argument gives existence of codes, not a small fixed finite rank. A quantitative version could turn a qualitative impossibility theorem into an analysis of the representation cost of arithmetic decoding. One should distinguish graph ranks from ranks of inverse images of open sets.

115.5 The computational content of Borel decoding

Question 115.5. What is the uniform computational strength of the operation that takes coded Borel arithmetic and an internal Diophantine correctness proof to a Borel exponentiation code?

A precise formulation can compare this operation with analytic separation or transfinite Borel evaluation. The fixed-proof theorem shows that uniform arithmetic truth is not required. Determining whether separation is genuinely necessary for the full code-construction problem would clarify the source of noncomputability.

115.6 Nonsplit Presburger topology

Question 115.6. How can continuous unit-preserving homomorphisms and elementary embeddings between the countable-slice Presburger models $(\mathbb{R} \vec{\times} K)_{\geq 0}$ be classified? Which additional specified topological or definability conditions constrain the extension class or force splitting?

The slice argument already applies to every countable $\mathbb{Z}$ -group K with its discrete topology. In particular, an extension $0 \rightarrow \mathbb{Z} \rightarrow K \rightarrow V \rightarrow 0$ with V countable, divisible, and ordered carries the lexicographic order with convex kernel $\mathbb{Z}$, and yields the same locally compact Polish positive cone and definability bounds. General existence is therefore settled by this construction. Classification of maps, extension classes under extra constraints, and other Polish carriers remains a separate task.

[write] The general existence statement is source 13's [theorem 94.1](#). With it, this answers the existence part of Part III's [question 43.3](#) for countable cores.

115.7 Which arithmetic fragments fit inside the omnific integers?

Question 115.7. For set-sized Presburger groups admitting a retraction to $\mathbb{Z}$, which additional conditions allow unit-preserving omnific embeddings with specified definability, topology, or initiality in the surreal simplicity order?

Proposition [113.3](#) supplies a necessary obstruction: the constant-term map forces such a retraction. The question concerns these extra requirements, rather than bare ordered additive existence. The explicit formula (10.6) also makes clear that an algebraic surreal realization need not carry the desired topology. Any prescribed topology and notion of initiality must be stated before an embedding classification is attempted.

115.8 A proof-assistant milestone with a finite target

Question 115.8. Can one extract and verify one explicit $\text{I}\Delta_0 + \text{Exp}$ -proof of the Diophantine correctness of $E(x, y)$, together with its finite semantic interpretation?

This is more focused than immediately formalizing every part of MRDP or all reverse mathematics of Borel sets. It would close the exact gap between the repository’s standard-natural exponentiation theorem and the principal new application in this article. The tagged Borel lemma and the metric/category core can then be developed independently.

116 Verification, scope, and research status

The mathematical review concentrated on the points where the argument could otherwise fail:

1. internal, rather than merely standard, Diophantine correctness;
2. Borel-map codes in addition to graph codes;
3. uniformity of the perfect restriction and meager covers;
4. positivity of multiplicative limits without order continuity;
5. exact one-bit identities and inclusive comparison endpoints;
6. control of all continuity parameters above an arbitrary threshold;
7. the distinction between a fixed finite proof translation and a global satisfaction construction.

Separate in-session proof audits checked the arithmetic/descriptive bridge and the metric/-category argument. This is a mathematical review record, not a substitute for independent external scrutiny.

The accompanying Python program uses exact integers and rational numbers to check finite instances of the root-scale inequalities, subset-sum separation, product-growth estimates in logarithmic coordinates, factorial identities, and the finite obstruction to a proposed retraction. These computations test concrete algebraic details; they neither simulate a nonstandard model nor prove the infinite topological theorem.

The claim about the question’s status is deliberately precise. The argument answers Question 1 *as stated in the inspected 2023 paper*. The current repository manuscripts inspected for this task leave that question open. Searches did not locate a later arXiv version or a separate indexed solution. This is not an exhaustive bibliographic priority determination.

The final package includes the complete LaTeX source, its compiled PDF, build instructions, the finite-check program and its results, and concise source and proof-audit records.

Part IX

Locally compact cones and the limits of multiplication

[write] This part is source 15 of batch 87, *Polish Arithmetic: Multiplication Obstructions and Local Compactness*, subtitled *Structural extensions of the Presburger constructions at the intersection of Elliot Glazer’s work and ProvelIt* (dated October 2026, its source snapshot and proof audit October 4; “research manuscript prepared for Vladimir Reshetnikov with AI assistance”),

pinned to repository commit `0f084afa9`, before Parts III–V were placed. All unmarked text is source 15’s; its labels carry the prefix `pma:lcc:`. It continues [Part IV](#): it classifies the locally compact cones of Borel ordered Polish groups without assuming a locally compact ambient group.

What is new, and what is a special case of Part VI. New to the collection are the classification of locally compact inherited cones with the exact carrier X_{lc} ([theorems 120.5](#) and [121.1](#)), the compact-open isolated-zero theorem without a group topology ([theorem 121.4](#)), the boundary theorems ([theorems 122.1](#) and [122.2](#)), the exact count of $2^{\aleph_0}$ models on X_{lc} with standard boundary ([theorem 123.1](#)) and its omnific realization ([theorem 124.3](#)). Source 15’s Section 3, the “induction-free obstruction to Borel multiplication” ([section 119](#)), and its [theorems 119.6](#) and [124.1](#) are special cases of [Part VI](#), which source 15 did not see; they are printed, with their different proofs, and flagged in their titles. Its background results repeat Part I and Part III, also flagged. The construction of [theorem 122.2](#) is source 13’s countable-core transfer ([Part VII](#)).

Notation. Source 15’s lexicographic products are printed $\overrightarrow{\times}$. In [theorem 122.1](#) its $H = G^0$ and C are source 08’s C and B_C (letters exchanged; tempting false reading: C is the identity component, as in [Part IV](#)). Its M_L , G_L , D_L , V_L , with an italic countable linear order L , are not Part I’s Laurent model $M_L = M_L$, and its Φ_L is not source 14’s Φ . Its $\mathcal{A}(\cdot)$, the ordered set of Archimedean classes, has the meaning of the same symbol in [proposition 10.7](#). [Section 1.3](#) lists these.

Provenance. Source 15 ships no executable checks, by design ([section 125](#)); its build script is `code/15-polish-arithmetic-build.sh` and its proof audit `15-polish-arithmetic-proof_audit.txt`. Its source manifest is not shipped; see [appendix E](#).

117 The research intersection and the contribution

The intersection chosen here is the topological model theory of weak arithmetic, together with ordered additive groups and set-sized fragments of Conway’s omnific integers. It is supported by Glazer’s own mathematical paper, rather than inferred from a professional profile. In *A Topological Tennenbaum Theorem*, Glazer proves strong nonexistence results for uncountable Polish models of arithmetic and asks whether an uncountable Polish space can support Presburger arithmetic with continuous addition [1]. The ProveIt repository contains both an integer Cooper elimination development and a research manuscript on Polish Presburger models inside the omnific integers [83, 84, 85].

The repository manuscript already supplies an affirmative construction for Glazer’s Question 2 as it is stated in the cited 2023 paper. In particular, the nonnegative cone of $\mathbb{R} \overrightarrow{\times} \mathbb{Z}$ works when the real coordinate has its usual topology and the integer coordinate is discrete. That construction, its Baire-space analogue, and the basic affirmative answer are background for this article. Repeating them would not constitute further progress. The present questions are what their locally compact generalisations look like, whether their additive structures can support multiplication, and how much arithmetic variation survives a fixed underlying topology.

117.1 Five principal results

The following statements give a guide to the proofs. A Borel order below means that the order relation is Borel for the specified topology. It does not mean that the topology is the order topology.

1. **Local compactness classification.** For a Borel ordered abelian Polish group G , local compactness of $G_{\geq 0}$ at zero forces G to be countable discrete or topologically $\mathbb{R} \times A$ with A countable discrete and torsion-free. These conditions are also sufficient. If G has a least positive element and is uncountable, its cone is homeomorphic to

$$X_{lc} = \left(\bigsqcup_{j \in \mathbb{N}} \mathbb{R} \right) \sqcup \left(\bigsqcup_{j \in \mathbb{N}} [0, \infty) \right). \quad (117.1)$$

2. **An absolute multiplication obstruction in this case.** If H is an uncountable Archimedean additive subgroup of a discretely ordered unital ring R , multiplication by any fixed $h > 0$ in H induces an injection $H \hookrightarrow R/H$. The locally compact groups above have an Archimedean real subgroup of countable index. Consequently their positive cones admit no unital semiring multiplication with the same addition and prescribed unit, whether measurable or not.
3. **A Borel multiplication obstruction without local compactness.** In a Polish abelian group with Borel discrete order and least positive element e , positive Borel endomorphisms cannot send e to every positive element unless the group is discrete. In particular, a nondiscrete such cone admits no semiring multiplication for which each fixed right-multiplier map is Borel. This is a group-theoretic extension of Glazer's probabilistic mechanism, with a stronger topological hypothesis and substantially weaker arithmetic assumptions.
4. **Optimal arithmetic diversity on one locally compact space.** There are exactly $2^{\aleph_0}$ isomorphism classes of full Presburger models with continuous addition on the space X_{lc} . An explicit family encodes subsets of $\mathbb{N}$ in its orders of Archimedean classes and admits finite-normal-form realizations in the omnific integers.
5. **Arithmetic on the topological boundary.** For an uncountable locally compact inherited Presburger cone, the endpoints of its closed-ray components form a countable discrete elementary initial submodel. Every countable Presburger model can occur as this boundary. Even when the boundary is fixed to the standard natural numbers, the continuum lower bound in the preceding item persists.

These results settle structural follow-up questions inside a precisely specified class of presentations. They do not settle the existence of all possible connected or zero-dimensional locally compact Polish Presburger presentations. A topology on a positive monoid need not extend to a Polish group topology on its difference group. That distinction is essential and will be demonstrated by a counterexample.

117.2 Existing results and the status of the extensions

The probabilistic construction of rapidly convergent, rapidly growing subseries, the use of complementation, and the final zero-one-law contradiction come from Glazer [1, Theorem 1]. Automatic continuity of Borel homomorphisms is classical [45]. Measurable orders on locally

compact abelian groups have an extensive earlier theory, including the work discussed by Asmar and Montgomery-Smith [22]; Borel ordered-group classification also appears in Christensen, Kanovei, and Reeken [79]. Neither the no-compact-subgroup lemma nor the real-line order lemma is offered as a new discovery.

The contribution developed here is the exact condition that the *cone itself* be locally compact, its application to arithmetic carriers, the cardinal-index obstruction, the boundary interpretation and its realization converse, the optimal family on the classified carrier, and a unified statement of the multiplier barrier at the level of positive endomorphisms. The proofs are supplied in full. A targeted source search did not establish publication priority for these exact combined statements. They should be treated as proved claims in an unrefereed research manuscript, with the standard dependencies explicitly identified, rather than as a certified historical first or an announced breakthrough.

Topic	Earlier material	Present treatment
Continuous Presburger models	Real and Baire coefficient constructions in the repository	Used as established background
Topological Tennenbaum mechanism	Glazer’s zero-one-law proof	Positive-endomorphism formulation and Borel group consequences
Local compactness	A real-coordinate example; follow-up classification questions	Exact classification of inherited cones
Multiplication	Special algebraic obstructions	General Borel obstruction; absolute obstruction for the classified cones
Number of models	Earlier families on Baire space	Sharp continuum family on one locally compact carrier
Manifold boundary	Endpoints visible in the real example	Elementary initial submodel; every countable model realized

117.3 The repository snapshot

All local source claims refer to the inspected ProveIt checkout at commit

```
0f084afa9be466593ffa6aff37749e78c0adc733.
```

The earlier Polish article has Git blob 88f0244d0f1155cf44cd7d4915a3a5dfa3e50d03 at this snapshot. Its actual source was read; descriptive README filenames and page counts were not taken as authoritative for the current combined manuscript. The inspection is a mathematical and source-level audit. No new Lean or Rocq build is asserted, and no repository changes are part of this deliverable. The accompanying source manifest records the files used.

[write] “The earlier Polish article” [84] is this report as it stood at source 15’s pin 0f084afa9: then the delivered text of batch-86 source 05, now the base of Part I. Parts III–V were placed after that pin, so source 15 did not see them; its “earlier families on Baire space” are Part I’s,

and Part III has since added continuum families on Baire space ([theorem 35.5](#)). Its “follow-up classification questions” are Part I’s [questions 14.3](#) and [14.5](#). The source manifest is not shipped, because it contains a third party’s personal profile address; its verified content is summarized in [appendix E](#).

118 Arithmetic and topology: the precise framework

118.1 Discrete order is independent of the chosen topology

Definition 118.1. A *discretely ordered abelian group with distinguished unit* is an ordered abelian group $(G, +, 0, <)$ with a least positive element 1. Its nonnegative cone is the cancellative commutative monoid $M = G_{\geq 0}$.

For every standard n , the interval $[0, n1]$ consists exactly of $0, 1, \dots, n1$. The assertion follows by external induction, using the absence of an element strictly between successive translates of 0 and 1. It does not bound nonstandard elements by standard integers. The order topology of such a group or cone is discrete, since each point has immediate neighbours, with the usual endpoint modification at zero. Thus an uncountable Polish presentation must use a different topology.

A *Polish group* is a topological group whose topology is separable and completely metrizable. An arbitrary compatible complete metric need not be translation invariant. A Borel order can be discontinuous: $\mathbb{R} \overrightarrow{\times} \mathbb{Z}$ with product topology is the basic example. For a topological group, the positive cone is Borel if and only if the entire order relation is Borel, since $x \leq y$ exactly when $y - x$ is in that cone.

We use the phrase *inherited group presentation* to mean exactly that $M = G_{\geq 0}$ has the subspace topology of a specified Polish abelian group G with Borel order. The phrase carries a genuine hypothesis. It is not a theorem that every continuous-addition Polish monoid has such a presentation.

118.2 The coefficient construction and full induction

Let D be a divisible ordered abelian group. Its unique rational division makes it a $\mathbb{Q}$ -vector space. Set

$$G_D = D \overrightarrow{\times} \mathbb{Z}, \quad M_D = (D_{>0} \times \mathbb{Z}) \cup (\{0\} \times \mathbb{N}), \quad 1 = (0, 1). \quad (118.1)$$

The D coordinate is compared first. In particular, a negative integer coordinate is compatible with being positive whenever the D coordinate is positive.

Proposition 118.2 (The standard algebraic certificate; = [corollary 4.11](#)). *For every divisible ordered abelian group D , M_D is a model of the entire first-order theory of $(\mathbb{N}, 0, 1, +, <)$.*

Proof. The group G_D has least positive element $(0, 1)$. For every standard $m \geq 2$, each (d, k) has the unique decomposition

$$(d, k) = m(d/m, \lfloor k/m \rfloor) + (0, r), \quad 0 \leq r < m, \quad (118.2)$$

where r is the ordinary remainder of k modulo m . For positive input, the quotient is nonnegative: it has positive D coordinate if $d > 0$, and an ordinary nonnegative integer coordinate if $d = 0$.

These are the axioms of a $\mathbb{Z}$ -group: a discretely ordered abelian group with exactly the standard residue classes modulo every standard positive integer. The classical quantifier-elimination theorem says that the theory of $\mathbb{Z}$ -groups in the language expanded by these congruence predicates is complete and is the theory of the ordered integers; see [82, Chapter 8] and the elimination method of [58]. Relativizing quantifiers to the nonnegative cone gives the asserted elementary equivalence. In particular every first-order Presburger induction instance holds in M_D . $\square$

This use of quantifier elimination is a classical dependency, not a claim that verifying a few division equations alone proves arbitrary induction. It also exposes the formalization boundary. ProveIt's current executable semantics quantify over Int ; the nonstandard semantic theorem must be supplied before substituting an arbitrary group G_D for that type. In particular, invariance under $x \mapsto x + m$ for standard iterations is weaker than invariance under every shift in the full subgroup mG_D .

118.3 Exactly when the inherited coefficient cone is Polish

Equip D with a Polish group topology and $\mathbb{Z}$ with the discrete topology. Write $P = D_{\geq 0}$. Each fixed-integer slice of M_D is homeomorphic to P or $P \setminus \{0\}$.

Proposition 118.3 (Positive-cone criterion; = [theorem 5.2](#)). *The space M_D is Polish in the inherited product topology if and only if P is G_δ in D . In that case addition on M_D is continuous. If the condition holds, the strict positive cone and the order relation on G_D are also Borel.*

Proof. If M_D is Polish, its clopen slice at integer coordinate zero is a Polish subspace of D equal to P . A subspace of a Polish space is Polish exactly when it is G_δ . Conversely, if P is G_δ , then $P \setminus \{0\}$ is G_δ as well. The countable topological sum of the resulting Polish slices is Polish. For example, choose bounded complete metrics on the slices and set the distance between distinct slices equal to a fixed larger constant. Addition is the restriction of the continuous ambient group operation; the cone is closed under addition. $\square$

For $D = \mathbb{R}$, the slice at $k \geq 0$ is $[0, \infty)$ and the slice at $k < 0$ is $(0, \infty)$. Consequently $M_{\mathbb{R}}$ has the space X_{lc} of [\(117.1\)](#). It is locally compact and has no isolated points. The map $r \mapsto \log r$ identifies an open ray with $\mathbb{R}$ and supplies a compatible complete metric there. This explains how a subspace defined using open rays can still be completely metrizable.

For contrast, on $D = \mathbb{R}^2$ with its Euclidean topology and lexicographic order,

$$P = \{(x, y) : x > 0\} \cup \{(0, y) : y \geq 0\}$$

is G_δ and F_σ , but is not locally compact at zero. Every relative neighbourhood of zero contains a sequence $(1/j, -\varepsilon)$ for some fixed $\varepsilon > 0$ whose ambient limit $(0, -\varepsilon)$ is missing. A compact relative neighbourhood would contain that limit, a contradiction. Thus local compactness of the coefficient group is weaker than local compactness of its cone.

[write] [Propositions 118.2](#) and [118.3](#) are Part I's [theorem 3.1](#) and [corollary 4.11](#) and [theorem 5.2](#), proved again; source 15 calls them background. The lexicographic $\mathbb{R}^2$ example is the step $d \geq 2$ of part (3) of source 08's [theorem 51.2](#).

119 An induction-free obstruction to Borel multiplication

The construction of Polish positive cones with continuous addition raises a natural strengthening question: can their addition be supplemented by a measurable multiplication? When

the topology comes from a Polish additive group, the answer has a particularly strong form. No arithmetic induction axiom is needed, and even separate Borel measurability in one argument is already too much.

The probabilistic mechanism below is adapted from the proof of Glazer's first topological Tennenbaum theorem [1, Theorem 1]. We give every step of the group version. The changes relevant here are that Borel multiplier maps become continuous by automatic continuity, limits are taken in the ambient additive group, and a finite-block probability estimate removes the need to construct an injective map from Cantor space. We present this as a consequence and extension of that mechanism; we do not assert independent historical priority for its ingredients.

[write] **A special case of Part VI.** The obstruction of this section is subsumed by [Part VI](#), which source 15 did not see. A positive additive endomorphism T with $T(e) > 0$ is an ordered additive embedding: $x > 0$ gives $x \geq e$ and $T(x) \geq T(e) > 0$. So the endomorphisms T_n of [theorem 119.2](#) have cofinal unit images, and source 12's [corollary 80.2](#) already excludes an uncountable group with a Baire-property cone, a Hausdorff Baire ccc group topology and such a family, with no regularity of the T_n ; a countable Polish group is discrete. Likewise [corollary 119.3](#) is a case of [theorem 81.2](#), [corollary 119.4](#) of [corollary 78.5](#) (and source 09's [theorem 62.1](#) with right instead of left scalars), and [corollary 119.5](#) of [theorem 82.2](#); none of the Part VI versions needs Borel multiplier maps. [Theorem 119.6](#) is a case of source 10's [theorem 79.7](#) and of part (1) of source 08's [theorem 46.2](#). The section is printed because its proof of [theorem 119.2](#) is different: it adapts Glazer's zero-one-law argument rather than packing open sets, and its proof of [lemma 119.1](#) is source 08's [lemma 53.1](#) and source 09's [lemma 61.2](#) again. Its mechanism is that of source 06's [theorem 39.1](#) in Part III.

119.1 Automatic continuity of individual multiplier maps

Recall that an ordered abelian group is an abelian group equipped with a translation-invariant linear order. An endomorphism $T : G \rightarrow G$ is *positive* if $x \geq 0$ implies $T(x) \geq 0$. If $e > 0$ is the least positive element, then

$$0 \leq x \leq ne \implies x \in \{0, e, \dots, ne\} \quad (n \in \mathbb{N}).$$

This elementary assertion is proved by ordinary, external induction on the standard integer n . It is not an induction scheme imposed on the ordered group as an arithmetic structure.

Lemma 119.1 (Borel homomorphisms are continuous; = [lemma 53.1](#)). *Let G be a Polish group and H a second-countable topological group. Every Borel homomorphism $f : G \rightarrow H$ is continuous.*

Proof. We recall the category argument associated with the Piccard–Pettis theorem [45]. First, if $A \subseteq G$ has the Baire property and is nonmeagre, there is a nonempty open set O in which A is comeagre. For every g sufficiently close to the identity, $O \cap gO$ is nonempty and open. The sets A and gA are comeagre in this intersection, so they meet. Consequently AA^{-1} contains a neighborhood of the identity.

Now let V be an identity neighborhood in H , and choose an open identity neighborhood W such that $WW^{-1} \subseteq V$. Second countability implies that the subspace $f(G)$ is Lindelöf. The cover $\{f(g)W \cap f(G) : g \in G\}$ therefore has a countable subcover, with centers $f(g_j)$. If $A = f^{-1}(W)$, then

$$G = \bigcup_{j \in \mathbb{N}} g_j A.$$

The set A is Borel and cannot be meagre, by the Baire category theorem. Thus AA^{-1} contains an identity neighborhood, and

$$AA^{-1} \subseteq f^{-1}(WW^{-1}) \subseteq f^{-1}(V).$$

This proves continuity at the identity, hence everywhere. $\square$

For a ring whose addition makes it a Polish group, the map $x \mapsto xy$, for any fixed y , is an additive homomorphism. Accordingly, a Borel hypothesis on this one-variable map already gives all the multiplication continuity used below. Joint continuity of multiplication is neither assumed nor needed.

119.2 The positive-endomorphism obstruction

Theorem 119.2 (Borel multiplier obstruction; a case of [corollary 80.2](#)). *Let G be a Polish abelian topological group with a Borel, translation-invariant linear order and a least positive element e . Suppose that for every $a > 0$ there exists a positive Borel additive endomorphism $T_a : G \rightarrow G$ such that*

$$T_a(e) = a.$$

There is no required measurability or algebraic coherence in the parameter a . Then G is discrete and consequently countable.

Proof. Suppose that G is not discrete. Fix a complete metric d inducing its topology. We do not assume that d is translation invariant.

A positive null sequence. There is a sequence of nonzero points $v_j \rightarrow 0$. Put $u_j = |v_j|$, where this absolute value means $\max(v_j, -v_j)$ in the given group order. Since both v_j and $-v_j$ tend to zero, so does u_j , even though the absolute-value map need not be continuous. The finite set $\{e, 2e, \dots, 5e\}$ does not contain zero. After discarding finitely many terms, discreteness of the order therefore gives

$$u_j > 5e, \quad \text{and hence} \quad u_j \geq 6e \quad \text{for every } j.$$

Rapid topological convergence and rapid order growth. We recursively construct positive elements $y_0, y_1, \dots$, beginning with $y_0 = e$, such that

$$y_n > 5y_{n-1} \quad (n \geq 1) \tag{119.1}$$

and

$$d(s, s + y_n) < 2^{-n} \quad \text{for every} \quad s \in \left\{ \sum_{i < n} \varepsilon_i y_i : \varepsilon_i \in \{0, 1\} \right\}. \tag{119.2}$$

Suppose the elements before y_n have been chosen. Lemma [119.1](#) implies that $T_{y_{n-1}}$ is continuous, so $T_{y_{n-1}}(u_j) \rightarrow 0$. For each of the finitely many s in (119.2), continuity of addition gives

$$d(s, s + T_{y_{n-1}}(u_j)) \rightarrow 0.$$

Choose j large enough for all these inequalities and set $y_n = T_{y_{n-1}}(u_j)$. Positivity and additivity imply

$$y_n \geq T_{y_{n-1}}(6e) = 6y_{n-1} > 5y_{n-1}.$$

This completes the recursion.

All binary subseries converge. For $r \in 2^{\mathbb{N}}$, define

$$s_N(r) = \sum_{i < N} r(i)y_i.$$

If $N \geq 1$, consecutive partial sums either agree or differ by the addition of y_N . Thus, uniformly in r ,

$$d(s_N(r), s_L(r)) \leq \sum_{i=N}^{L-1} 2^{-i} \quad (L > N \geq 1).$$

Completeness gives a limit

$$z(r) = \lim_{N \rightarrow \infty} s_N(r) \in G,$$

and the convergence is uniform in r . Since each s_N is continuous on Cantor space, so is z . Write $\bar{r}(i) = 1 - r(i)$, and put

$$D(r) = z(r) - z(\bar{r}).$$

The map $D : 2^{\mathbb{N}} \rightarrow G$ is continuous, and

$$D(\bar{r}) = -D(r). \quad (119.3)$$

If r and s differ on only finitely many coordinates, continuity of the group operations gives the exact identity

$$D(r) - D(s) = 2 \sum_i (r(i) - s(i))y_i. \quad (119.4)$$

To see this, the corresponding identity for finite partial sums is constant once the partial sums include every changed coordinate; then take limits.

Order separation on a finite block. Let $S_m = \sum_{i < m} y_i$. From (119.1), induction gives

$$4S_m < y_m \quad (m \geq 1). \quad (119.5)$$

Indeed $4y_0 < y_1$, and $4S_{m+1} = 4S_m + 4y_m < 5y_m < y_{m+1}$. If r and s differ finitely and m is their greatest coordinate of disagreement, then (119.4) implies

$$|D(r) - D(s)| \geq 2(y_m - S_m) > y_m \quad (m \geq 1). \quad (119.6)$$

Give $2^{\mathbb{N}}$ the fair independent Bernoulli product probability measure. Fix $n \geq 0$ and $k \geq 1$, and consider the block $B = \{n+1, \dots, n+k\}$. Once all coordinates outside B are fixed, at most one of the 2^k assignments on B can satisfy

$$|D(r)| \leq y_n.$$

Otherwise, two such assignments would have greatest disagreement $m \in B$, and (119.6) would give

$$|D(r) - D(s)| > y_m \geq y_{n+1} > 5y_n,$$

whereas the triangle inequality in the ordered group would give $|D(r) - D(s)| \leq 2y_n$. The events are Borel because D is continuous and the order is Borel. Integrating the finite conditional counting bound yields

$$\mathbb{P}(|D(r)| \leq y_n) \leq 2^{-k}.$$

As k is arbitrary, this probability is zero. Countable additivity now gives

$$\mathbb{P}(\forall n \ |D(r)| > y_n) = 1. \quad (119.7)$$

In particular, $\mathbb{P}(D(r) = 0) = 0$. No injectivity assertion about z was necessary.

A tail event with probability one half. Define

$$A = \{r : \forall n \ D(r) > y_n\}.$$

By (119.7), almost every r belongs either to A or to its reflected event $\{r : \forall n \ -D(r) > y_n\}$. Complementation preserves product measure and interchanges these two disjoint events by (119.3). Consequently

$$\mathbb{P}(A) = \frac{1}{2}.$$

We verify directly that A is a tail event. Suppose $r \in A$ and s differs from r only at coordinates at most m . Equations (119.4) and (119.5) give

$$|D(s) - D(r)| \leq 2S_{m+1} < y_{m+1}.$$

For any given n , choose $j > \max(n, m + 1)$. The growth condition implies $y_j > y_n + y_{m+1}$, and therefore

$$D(s) \geq D(r) - |D(s) - D(r)| > y_j - y_{m+1} > y_n.$$

Thus $s \in A$. Reversing r and s proves invariance under every finite coordinate change.

For each N , membership in this Borel set is consequently determined by the coordinates with index at least N : replace the first N coordinates by zero to obtain a Borel test on the remaining tail. Hence A belongs to the tail sigma-algebra. Kolmogorov's zero-one law says that its probability is zero or one. Indeed, a tail event is independent of every event determined by finitely many coordinates. The monotone-class theorem extends this independence to all events in the product Borel sigma-algebra, including A itself. Thus $\mathbb{P}(A) = \mathbb{P}(A)^2$, contradicting $\mathbb{P}(A) = 1/2$. This proves that G is discrete. Every discrete second-countable space is countable, completing the proof. $\square$

119.3 Semiring expansions and the difference group

Corollary 119.3 (No Borel semiring expansion of a nondiscrete Polish cone; a case of theorem 81.2). *Let G be a nondiscrete Polish abelian group with a Borel, translation-invariant linear order and least positive element 1. Put $M = G_{\geq 0}$, with its inherited Borel structure. There is no semiring multiplication*

$$\mu : M \times M \longrightarrow M$$

having the original addition and zero, having 1 as multiplicative identity, and satisfying the following weak regularity condition: for every fixed $a \in M$, the map $x \mapsto \mu(x, a)$ is Borel. In particular, there is no jointly Borel such multiplication. The positive cone M need not itself be Polish.

Proof. The maps

$$x^+ = \max(x, 0), \quad x^- = \max(-x, 0)$$

are Borel, take values in M , and satisfy $x = x^+ - x^-$. For $a > 0$, define

$$T_a(x) = \mu(x^+, a) - \mu(x^-, a).$$

This is a Borel map $G \rightarrow G$. It is precisely the additive extension of $x \mapsto \mu(x, a)$ from M to its difference group $G = M - M$. For completeness, if $b - c = b' - c'$ with $b, c, b', c' \in M$, then $b + c' = b' + c$. Distributivity gives

$$\mu(b, a) + \mu(c', a) = \mu(b', a) + \mu(c, a),$$

so the difference of the two products is independent of the chosen representation. This also proves additivity of the extension. For $x \geq 0$, $T_a(x) = \mu(x, a) \geq 0$, and $T_a(1) = a$. Theorem 119.2 now contradicts nondiscreteness. $\square$

If μ is jointly Borel, its usual extension to the whole group can be written explicitly as

$$\begin{aligned}\mu_G(x, y) &= \mu(x^+, y^+) + \mu(x^-, y^-) \\ &\quad - \mu(x^+, y^-) - \mu(x^-, y^+).\end{aligned}\tag{119.8}$$

Thus no measurable selection theorem is required for the passage to differences. The formula is Borel and biadditive. The usual well-definedness argument above, applied in both variables, proves that it is the unique ring multiplication extending μ . Associativity and the unit law extend from M by distributivity. For the obstruction itself, only the individual T_a are needed.

Corollary 119.4 (Polish discretely ordered rings; a case of [corollary 78.5](#)). *Let R be a unital ordered ring whose additive group is Polish and whose order is Borel. Suppose 1 is its least positive element and, for every fixed $a \in R$, right multiplication $x \mapsto xa$ is Borel. Then the additive topology of R is discrete and R is countable.*

No form of IOpen, bounded induction, division with remainder, roots, or exponentiation occurs in these statements. Indeed, Theorem [119.2](#) does not assume a multiplication operation at all: it assumes only the listed positive additive endomorphisms. The order's least positive element, Polish completeness, measurability of the order, and the multiplier regularity supply the obstruction.

Corollary 119.5 (Non-Polishability of the additive Borel group; a case of [theorem 82.2](#)). *Let R be an uncountable discretely ordered unital ring on a standard Borel space, with Borel addition, negation, order, and individual right-multiplier maps. Its additive Borel group is not Polishable: there is no Polish group topology on $(R, +)$ whose Borel sets are the given ones.*

Proof. In any such topology the order and multiplier maps would remain Borel. Corollary [119.4](#) would force R to be countable. $\square$

119.4 An algebraic cardinality obstruction

For an important subclass, measurability can be removed completely. A subgroup H of an ordered abelian group is called *Archimedean* if for all $h, k > 0$ in H , some standard positive integer n satisfies $k < nh$.

Theorem 119.6 (Archimedean subgroup versus quotient cardinality; a case of [theorem 79.7](#)). *Let R be a discretely ordered unital ring, and let $H \subseteq (R, +)$ be an uncountable Archimedean subgroup, with its induced order. Then there is an injective additive homomorphism*

$$H \longrightarrow (R, +)/H.$$

In particular,

$$|H| \leq |R/H|.$$

Thus an ordered abelian group having an uncountable Archimedean subgroup of countable index cannot be the additive ordered group of a discretely ordered unital ring with the prescribed least positive element as unit.

Proof. Every $a > 0$ in H is larger than every standard integer multiple of 1. Otherwise, $0 < a \leq n1$ for some standard n , and discreteness of the ring order gives $a = k1$ for a positive integer k . Archimedeaness would then imply that each $b > 0$ in H is bounded by

$ma = mk1$ for some standard integer m . Discreteness again gives $b \in \mathbb{Z}1$, and hence $H \subseteq \mathbb{Z}1$, contradicting uncountability.

Fix $h > 0$ in H . For any $a > 0$ in H , positivity of multiplication yields

$$ah > nh \quad \text{for every standard } n.$$

Because H is Archimedean, the multiples nh are cofinal in its positive part. Consequently ah is larger than every element of H , and in particular $ah \notin H$. If $a < 0$, applying this argument to $-a$ likewise gives $ah \notin H$. The additive homomorphism

$$a \mapsto ah + H$$

therefore has trivial kernel. □

For example, suppose a coefficient group is algebraically $D = \mathbb{R} \oplus A$, where A is countable, and its order restricts to an Archimedean order on the displayed copy of $\mathbb{R}$. In $G = D \overrightarrow{\times} \mathbb{Z}$, with the integer summand least significant and its positive generator as the proposed unit, take

$$H = \mathbb{R} \oplus \{0\} \oplus \{0\}.$$

Then H is uncountable and $G/H \cong A \oplus \mathbb{Z}$ is countable. There is no compatible semiring multiplication on the positive cone, even with no regularity requirement whatever. This applies to coefficient decompositions of this form supplied by a classification of *locally compact positive cones*. Local compactness of D alone is a different hypothesis and does not by itself supply this decomposition with a countable quotient.

119.5 Scope and sharpness of the hypotheses

The countable discrete case is possible. The usual ordered ring $\mathbb{Z}$, with its discrete topology, satisfies all the hypotheses and all its operations are continuous. Thus the conclusion cannot be strengthened to nonexistence of every such ring.

Complete metrizability matters. Give the usual ordered ring $\mathbb{Z}$ its 2-adic topology, whose basic neighborhoods are the congruence classes modulo 2^k . Addition, negation, and multiplication are continuous. The usual order is Borel: every subset of the countable T_1 space $\mathbb{Z}^2$ is a countable union of closed singletons. The element 1 remains the least positive element algebraically, while the topology is nondiscrete.

This example is not Polish. It is a countable metrizable space without isolated points, so every singleton is nowhere dense; the Baire category theorem rules out any compatible complete metric. The familiar 2-adic metric is explicitly incomplete: the integer partial sums

$$q_N = \sum_{j=0}^N 4^j = \frac{4^{N+1} - 1}{3}$$

are Cauchy and converge in the 2-adic completion to $-1/3$, which is not an integer. This is the same final-digits phenomenon used in the topological arithmetic constructions of Enayat, Hamkins, and Wcisło [2, Section 2].

A least positive element matters. The ordinary ordered field $\mathbb{R}$, with its usual topology, is an uncountable Polish additive group with continuous multiplication and a Borel order. It has no least positive element. Its null sequences have small order magnitude, so the recursion forcing simultaneous order growth and topological convergence cannot begin.

The ambient group topology is a real hypothesis. The corollaries concern a topology making *both* addition and negation continuous on the full difference group. A Polish topology on a positive cone alone does not automatically have this property. One should therefore not infer a blanket nonexistence theorem for all uncountable Polish models of the purely algebraic theory of nonnegative parts of discretely ordered rings. For any application, the original addition, the distinguished element 1, the Borel order, and the specific compatible Polish group topology must all be identified.

Here is an explicit example demonstrating the distinction. The polynomial construction is included as a boundary example; no novelty is claimed for it. Let

$$R_0 = \mathbb{Z} + t\mathbb{R}[t],$$

and order a nonzero polynomial by the sign of its highest-degree coefficient. This makes R_0 a discretely ordered ring in which t is larger than every integer and 1 is the least positive element. Let $M_0 = (R_0)_{\geq 0}$. Give M_0 the topological disjoint-union decomposition

$$M_0 = \mathbb{N} \sqcup \bigsqcup_{d \geq 1} ((0, \infty) \times \mathbb{R}^{d-1} \times \mathbb{Z}).$$

The factor $(0, \infty)$ records the leading coefficient of a degree- d polynomial, the middle factors its other nonconstant coefficients, and $\mathbb{Z}$ its constant coefficient. The constants have the discrete topology. This is an uncountable Polish space, being a countable topological sum of Polish spaces, and zero is isolated. It is also locally compact. Each stratum is open and locally compact: near a nonconstant polynomial, fix its integer constant coefficient and choose a bounded coefficient box with closure keeping the leading coefficient strictly positive. That closure is compact inside the same stratum. An isolated constant has its singleton as a compact neighborhood.

Addition and multiplication are jointly continuous. On a fixed pair of nonconstant degree strata, addition has degree equal to the larger input degree and multiplication has degree equal to their sum: positive leading coefficients prevent cancellation. Within these fixed strata the coefficient formulas are continuous polynomial functions. Addition of a constant and multiplication by a positive constant have the same property. The remaining cases involve the isolated zero or two isolated constants.

In fact its additive reduct is a model of *full* Presburger arithmetic. The convex subgroup generated by 1 in R_0 is the constant copy of $\mathbb{Z}$, and

$$R_0/\mathbb{Z} \cong t\mathbb{R}[t]$$

is a divisible ordered abelian group. The standard $\mathbb{Z}$ -group characterization therefore applies [4, Section 2]. More explicitly, for each standard integer $m > 0$, write the constant coefficient c of $p = c + \sum_{i=1}^d a_i t^i$ as $c = mq + r$, with $0 \leq r < m$. Then

$$p = m \left(q + \sum_{i=1}^d \frac{a_i}{m} t^i \right) + r$$

is its unique division by m with standard remainder. When $p \geq 0$, its quotient is also nonnegative. These division schemes, together with the discrete ordered-group axioms, give the complete Presburger theory on the nonnegative cone.

Thus M_0 is an uncountable locally compact Polish model of full Presburger arithmetic admitting a continuous semiring multiplication. It is a model of the algebraic theory of

nonnegative parts of discretely ordered rings, but it fails open induction in the language *with multiplication* explicitly. The quantifier-free formula with parameter t ,

$$\varphi(x) \equiv x^2 < t,$$

holds in M_0 exactly for the ordinary nonnegative integer constants. Hence $\varphi(0)$ and $\forall x (\varphi(x) \Rightarrow \varphi(x+1))$ hold, while $\varphi(t)$ does not.

The additive Borel group of R_0 , using the corresponding coefficient Borel structure, cannot carry a compatible Polish group topology. Besides Corollary 119.5, this has a direct category proof. Put

$$G_d = \{p \in R_0 : \deg p \leq d\}, \quad d \geq 0,$$

including zero in every G_d . These are Borel additive subgroups and $R_0 = \bigcup_d G_d$. In a hypothetical Polish group topology with these Borel sets, one G_d would be nonmeagre. The Piccard–Pettis argument makes that subgroup open. A second-countable group has at most countably many cosets of an open subgroup. But

$$G_{d+1}/G_d \cong (\mathbb{R}, +)$$

already supplies uncountably many such cosets, a contradiction. The obstruction therefore concerns the passage from a Polish positive monoid to a Polish difference group, not the existence of the displayed positive semiring itself.

[write] The degree-stratum cone M_0 is batch-86 source 06's cone of $\mathbb{Z} + t\mathbb{R}[t]$ in Part III, with the same topology: [theorems 38.1](#) and [38.2](#), and its group of differences is not compatibly Polishable for the reason given in [section 42](#); source 15 claims no novelty for it, and had not seen Part III. The 2-adic example is new to this report; source 10 makes the same remark for the p -adic topology ([section 84.1](#)).

120 Classification of locally compact coefficient presentations

The positive-cone construction has a surprisingly rigid locally compact case. The result below classifies the topological spaces obtained by this construction, including presentations whose coefficient group was not initially assumed locally compact. Its scope is precisely the inherited product topology; arbitrary Polish presentations of Presburger arithmetic are not covered.

Let D be a divisible abelian Polish group with a translation-invariant linear order $<$ whose positive cone is Borel. Write

$$P = D_{\geq 0}, \quad P^* = D_{> 0}, \quad M_D = (P^* \times \mathbb{Z}) \cup (\{0\} \times \mathbb{N}),$$

and give M_D the topology inherited from $D \times \mathbb{Z}$, with $\mathbb{Z}$ discrete. Arithmetic is inherited from the lexicographic ordered group $D \overrightarrow{\times} \mathbb{Z}$, and the distinguished arithmetic zero is $(0, 0)$. In this section no preliminary assumption that P is G_δ is needed: in the cases classified as locally compact, Polishness follows from the classification itself.

120.1 A measurable-order obstruction to compact subgroups

Lemma 120.1 (compare [lemma 47.1](#)). *Let G be a second-countable Hausdorff topological group with a translation-invariant linear order whose positive cone is Borel. If the order is invariant under left translations, then G has no nontrivial compact subgroup. In particular, this applies to the ordered abelian groups used here.*

Proof. An ordered group is torsion-free, so a nontrivial compact subgroup K would be infinite. Its normalized Haar measure μ is atomless: by translation invariance, one positive point mass would give the same positive mass to every point of the infinite group.

First consider the abelian case, which is all that is needed below. The two sets $K_{>0}$ and $K_{<0}$ have the same Haar measure, because inversion preserves Haar measure on a compact group. They partition $K \setminus \{0\}$, so each has measure $1/2$. Let X, Y, Z be independent Haar-distributed points of K . Ties have probability zero, and exchangeability gives

$$\Pr(X < Z \text{ and } Y < Z) = \frac{1}{3}.$$

Conditioning on $Z = z$, translation invariance makes $X - z$ and $Y - z$ independent Haar-distributed points. The same probability must therefore be $(1/2)^2 = 1/4$, a contradiction.

For a group with a merely left-invariant order, replace $X - z$ and $Y - z$ by $z^{-1}X$ and $z^{-1}Y$; inversion still interchanges the positive and negative cones and preserves normalized Haar measure on a compact group. The identical argument applies. $\square$

The argument isolates the obstruction: a translation-invariant measurable linear order is incompatible with a uniform distribution on a nontrivial compact subgroup. No continuity of the order relation is assumed.

120.2 Local compactness at zero determines the coefficient group

Lemma 120.2. *Let D be a Hausdorff ordered abelian topological group. If its nonnegative cone P is locally compact at zero in the subspace topology, then D is locally compact.*

Proof. There is a compact set $K \subseteq P$ which contains a neighborhood of zero relative to P . Choose an ambient open neighborhood W of zero such that $P \cap W \subseteq K$, and put $U = W \cap (-W)$. Every $x \in U$ belongs to $K \cup (-K)$: if $x \geq 0$, then $x \in P \cap W$, and otherwise $-x \in P \cap W$. Thus $K \cup (-K)$ is a compact neighborhood of zero in D . Translation gives local compactness everywhere. $\square$

Lemma 120.3. *Let C be the nonnegative cone of a translation-invariant linear order on the additive group $\mathbb{R}^n$. If C is locally compact at zero in the Euclidean subspace topology, then $n \leq 1$.*

Proof. Choose a compact $K \subseteq C$ and a Euclidean open neighborhood W of zero such that $C \cap W \subseteq K$. Since K is closed in $\mathbb{R}^n$ and $K \subseteq C$, the set $C \cap W = K \cap W$ is closed relative to W .

The cone C is invariant under multiplication by every positive rational number. Indeed, positive integral multiples preserve sign in an ordered abelian group, and division in $\mathbb{R}^n$ is unique. If $x_j \in C$ converges to x , choose a positive rational q small enough that $qx \in W$. For all sufficiently large j , $qx_j \in C \cap W$. Relative closedness yields $qx \in C$, hence $x \in C$. Thus C is globally closed.

If $n \geq 2$, the sets

$$C \cap S^{n-1} \quad \text{and} \quad (-C) \cap S^{n-1}$$

are nonempty disjoint closed sets covering the connected sphere S^{n-1} . This is impossible. Nonemptiness follows by taking a nonzero vector and its negative; closedness also implies that rescaling a positive vector to the sphere preserves its sign, by rational approximation. $\square$

Lemma 120.4 (= lemma 47.3 for $d = 1$, with the Baire property). *A translation-invariant linear order on the additive group $\mathbb{R}$ whose positive cone has the Baire property is either the usual order or its reverse.*

Proof. Let B be its strict positive cone. The sets B , $-B$, and $\{0\}$ partition $\mathbb{R}$, so B is nonmeager. Since B has the Baire property, it is comeager in some nonempty bounded open interval I . For every $t \in I + I$, the interval $I \cap (t - I)$ is nonempty. The sets B and $t - B$ are both comeager there, so their intersection is nonempty. Consequently

$$I + I \subseteq B + B \subseteq B.$$

The interval $I + I$ does not contain zero, and thus lies entirely in one of the two usual half-lines. Positive rational dilations of a nonempty open interval in $(0, \infty)$ cover $(0, \infty)$; the analogous statement holds on the negative half-line. The cone B is invariant under these dilations. It therefore contains one usual strict half-line, and the partition $B \sqcup (-B) = \mathbb{R} \setminus \{0\}$ prevents it from containing anything else. $\square$

The only structural theorem about locally compact groups needed next is the classical principal structure theorem: a locally compact abelian group has an open subgroup topologically isomorphic to $\mathbb{R}^n \times K$, with $n < \omega$ and K compact. We use the standard statement in Morris [80, Chapter 6, Theorem 25]; it is also explicitly recalled by Gabrielyan and Morris [81, p. 68].

Theorem 120.5 (Locally compact coefficient classification; extends [theorem 51.2\(3\)](#)). *For D and M_D as above, the following conditions are equivalent.*

1. M_D is locally compact at its arithmetic zero $(0, 0)$.
2. M_D is locally compact.
3. As a topological group, D is either a countable discrete $\mathbb{Q}$ -vector space or $\mathbb{R} \times A$, where A is a countable discrete $\mathbb{Q}$ -vector space.

In the first case M_D is a countably infinite discrete space. In the second case M_D is an uncountable locally compact Polish space, and

$$M_D \cong \left(\bigsqcup_{j \in \mathbb{N}} \mathbb{R} \right) \sqcup \left(\bigsqcup_{j \in \mathbb{N}} [0, \infty) \right) \quad (120.1)$$

as a topological space. Both families of connected components are countably infinite.

Proof. The slice with integer coordinate zero is a clopen subspace of M_D homeomorphic to P . Condition (1) therefore makes P locally compact at zero. By [Lemma 120.2](#), D is locally compact. The principal structure theorem supplies an open subgroup $H \cong \mathbb{R}^n \times K$. [Lemma 120.1](#) forces $K = \{0\}$, so $H \cong \mathbb{R}^n$ is open.

An ordered divisible abelian group is naturally a vector space over $\mathbb{Q}$. The subgroup H is a $\mathbb{Q}$ -subspace. Choose an algebraic vector-space complement A , so $D = H \oplus A$. It is discrete in its subspace topology because $H \cap A = \{0\}$ and H is open. There are only countably many H -cosets: a countable dense subset of D meets every open coset. Thus A is countable. The addition map $H \times A \rightarrow D$ is a topological isomorphism, since on each clopen slice it is translation from H onto an open coset.

The intersection $P \cap H$ is locally compact at zero and is the nonnegative cone of the restricted order on $H \cong \mathbb{R}^n$. [Lemma 120.3](#) gives $n \leq 1$. If $n = 0$, D is countable and discrete. If $n = 1$, $D \cong \mathbb{R} \times A$. This proves (1) $\Rightarrow$ (3).

For the converse, the discrete case is immediate. Suppose $D \cong \mathbb{R} \times A$ with A countable discrete. The restricted order on the open subgroup $\mathbb{R} \times \{0\}$ is Borel. By [Lemma 120.4](#), after

changing the orientation of the real coordinate it is the usual order. For each $a \in A$, each section

$$S_a = \{r \in \mathbb{R} : (r, a) > 0\}$$

is an upper set for the usual real order. It is consequently one of

$$\emptyset, \quad \mathbb{R}, \quad (b, \infty), \quad [b, \infty) \quad (b \in \mathbb{R}).$$

For $a = 0$ it is $(0, \infty)$. The corresponding nonnegative section at $a = 0$ is $[0, \infty)$; at $a \neq 0$ the strict and nonstrict sections agree.

The (a, k) -slices of M_D , for $(a, k) \in A \times \mathbb{Z}$, are clopen and have one of these interval types, with the replacement $(0, \infty) \mapsto [0, \infty)$ exactly when $a = 0$ and $k \geq 0$. Every nonempty slice is locally compact and homeomorphic either to $\mathbb{R}$ or to $[0, \infty)$. There are countably many slices. In particular, the slices $a = 0, k < 0$ give infinitely many line components, while $a = 0, k \geq 0$ give infinitely many closed-ray components. This proves (3) $\Rightarrow$ (2), the homeomorphism (120.1), and Polishness. Finally, (2) $\Rightarrow$ (1) is immediate. $\square$

120.3 Consequences and the precise boundary of the result

Corollary 120.6. *The inherited coefficient construction produces no uncountable zero-dimensional locally compact space. Every uncountable locally compact carrier it produces is homeomorphic to the carrier obtained from $D = \mathbb{R}$.*

The classification also excludes connected uncountable carriers inside this construction; the discrete integer coordinate already supplies clopen slices. It gives a stronger conclusion than an existence obstruction for one proposed space: the entire locally compact homeomorphism type is fixed.

It is essential not to infer that the open connected subgroup $\mathbb{R} \times \{0\}$ must be convex in the arithmetic order. For example, put the lexicographic order with real coordinate first on $D = \mathbb{R} \times \mathbb{Q}$ and give $\mathbb{Q}$ the discrete topology. Then $(0, 1)$ lies between $(-1, 0)$ and $(1, 0)$, although it does not lie in that connected subgroup. The coefficient cone remains locally compact, as the theorem predicts.

Likewise, the theorem does not assert that every uncountable Polish model of Presburger arithmetic with continuous addition has the carrier (120.1). A presentation whose topology is not inherited from a Polish coefficient group can evade these hypotheses. The broader inherited-group statement is proved next. Further questions are collected in Section 126.

[write] (Batch 87.) Within inherited coefficient presentations, [corollary 120.6](#) answers Part I's [question 14.5](#) negatively and the remark after it excludes connected carriers ([question 14.3](#)); [theorem 121.1](#) below extends both to every inherited Polish group cone. For presentations not inherited from a Polish ordered group both questions stay open ([question 126.1](#)). [Theorem 120.5](#) extends part (3) of source 08's [theorem 51.2](#) by assuming local compactness of the cone rather than of the coefficient group, and fixes the carrier.

121 The classification extends to arbitrary inherited group cones

The coefficient group need not be divisible, and the ambient ordered group need not split as a coefficient group followed by its standard integer part. The same arguments classify every inherited positive cone in a Borel ordered abelian Polish group. Write X_{lc} for the countable disjoint union of lines and closed rays displayed in (120.1).

Theorem 121.1 (General inherited-cone classification; extends [theorems 47.2](#) and [50.1](#)). *Let G be an abelian Polish group with a translation-invariant linear order whose nonnegative cone P is Borel. The following are equivalent:*

1. P is locally compact at zero in the topology inherited from G ;
2. P is locally compact in that topology;
3. as a topological group, G is either countable discrete, or $G \cong \mathbb{R} \times A$ for a countable discrete torsion-free abelian group A .

If these equivalent local-compactness conditions hold and, in addition, G has a least positive element e and P is uncountable, then P is homeomorphic to X_{lc} .

Proof. By Lemma [120.2](#), local compactness of P at zero gives local compactness of G . The principal structure theorem and Lemma [120.1](#) supply an open subgroup $H \cong \mathbb{R}^n$. The cone $P \cap H$ is locally compact at zero, so Lemma [120.3](#) gives $n \leq 1$.

Here is a direct splitting argument that does not assume divisibility of G . Since an ordered abelian group is torsion-free, it embeds in its rational hull $G_{\mathbb{Q}} = G \otimes_{\mathbb{Z}} \mathbb{Q}$, a $\mathbb{Q}$ -vector space. The subgroup H is divisible and is therefore already a $\mathbb{Q}$ -subspace of this hull. Choose a vector-space complement B to H in $G_{\mathbb{Q}}$ and put $A = G \cap B$. For $g \in G$, its decomposition $g = h + b$ has $h \in H \subseteq G$, so $b = g - h$ lies in A . Hence $G = H \oplus A$. The subgroup A is torsion-free, is discrete because H is open, and is countable because there are only countably many open H -cosets. Addition gives the topological isomorphism $H \times A \cong G$, as in the earlier proof. This proves (1) $\Rightarrow$ (3).

For (3) $\Rightarrow$ (2), the discrete case is immediate. In the other case Lemma [120.4](#) orients the real subgroup in its usual direction. In each clopen coset of that subgroup, the positive cone is an upper interval. Each nonempty such interval is a line or an open or closed ray, and is locally compact. Condition (2) plainly implies (1).

Suppose now that G has a least positive element e and is uncountable. Then $H \cong \mathbb{R}$. For every $h \in H$ with $h > 0$ and every standard $n \geq 0$, divisibility of H and minimality of e give

$$\frac{h}{n+1} \geq e, \quad \text{and consequently} \quad h \geq (n+1)e > ne.$$

In particular, $H \cap \mathbb{Z}e = \{0\}$, so the cosets $H + ke$, $k \in \mathbb{Z}$, are pairwise distinct. In an oriented real coordinate on H , their intersections with P are

$$P \cap (H + ke) \cong \begin{cases} [0, \infty), & k \geq 0, \\ (0, \infty), & k < 0. \end{cases}$$

The dominance inequality just proved determines the sign away from the real origin; the sign of ke determines whether that endpoint belongs. All other coset sections are again empty, lines, or open or closed rays. There are countably many such clopen sections, with infinitely many line components and infinitely many closed-ray components already supplied by the standard integer translates. Thus $P \cong X_{lc}$. $\square$

Corollary 121.2. *Suppose a locally compact Polish model of Presburger arithmetic is the nonnegative cone, with its inherited topology, of a Borel ordered abelian Polish group. If the model is uncountable, its carrier is X_{lc} . In particular, an uncountable zero-dimensional locally compact continuous-addition Presburger model cannot admit such an inherited Polish group presentation.*

The condition concerns a compatible topology on the entire group of differences, not merely the existence of that group algebraically. Every cancellative commutative monoid has an algebraic group of differences; it is precisely the compatible Polish group topology and inherited-cone condition that are additional here. No algebraic splitting of a general Presburger group over its standard $\mathbb{Z}$ -subgroup is assumed.

[write] (Batch 90.) By [Part XII](#), a locally compact Polish Presburger model with continuous addition is such an inherited cone exactly when its partial subtraction is continuous; [proposition 155.7](#) [merge] restates the corollary in that form.

[write] [Theorem 121.1](#) strengthens source 08's [theorems 47.2](#) and [50.1](#): no ambient local compactness is assumed, and the homeomorphism type of the cone is determined. For [Part III's question 43.2](#) it settles the locally compact inherited case.

121.1 A compact-open obstruction without a group topology

One useful obstruction survives even when no ambient topological group is available.

Lemma 121.3 (compare [lemma 12.1](#)). *Let M be a Hausdorff commutative cancellative topological monoid whose only invertible element is zero. Its only nonempty compact subsemigroup is $\{0\}$.*

Proof. Let C be a nonempty compact subsemigroup. For each $a \in C$, the set $a + C$ is a nonempty compact subset of C . These sets have the finite intersection property: for $a_1, \dots, a_m \in C$, their intersection contains $a_1 + \dots + a_m + C$. Choose

$$p \in \bigcap_{a \in C} (a + C).$$

Since $p \in C$, the intersection condition includes $p \in p + C$. Thus $p = p + c$ for some $c \in C$, and cancellation against $p = p + 0$ gives $c = 0$. We have shown that every such C contains zero.

For any $a \in C$, the translate $a + C$ is itself a nonempty compact subsemigroup: the sum of $a + x$ and $a + y$ equals $a + (a + x + y)$, with $a + x + y \in C$. Applying the previous paragraph to $a + C$ gives $0 \in a + C$. Hence a is invertible in M and must be zero. Therefore $C = \{0\}$. $\square$

Theorem 121.4 (Compact-open neighborhoods force isolated zero). *Let M be a Hausdorff commutative cancellative topological monoid with jointly continuous addition and no nonzero invertible elements. If zero has a compact open neighborhood, then zero is isolated. Consequently, zero is isolated in every zero-dimensional locally compact Hausdorff continuous-addition Presburger presentation.*

Proof. Let K be a compact open neighborhood of zero. For each $x \in K$, joint continuity at $(x, 0)$ supplies neighborhoods U_x of x and V_x of zero such that $U_x + V_x \subseteq K$. Finitely many U_x cover K . Intersecting their corresponding V_x , and then intersecting with K , gives an open neighborhood V of zero satisfying

$$V \subseteq K, \quad K + V \subseteq K.$$

If $v \in V$, every positive integer multiple of v lies in K . The closure of this set of multiples is a nonempty compact subsemigroup: K is closed because it is compact in a Hausdorff space, and joint continuity preserves the subsemigroup condition under closure. [Lemma 121.3](#) says that this closure is $\{0\}$. Since it contains v , we have $v = 0$. Hence $V = \{0\}$ is open.

Finally, a zero-dimensional locally compact Hausdorff space has a compact open neighborhood at each point: inside an open neighborhood with compact closure choose a clopen neighborhood, which is then compact. A Presburger monoid satisfies cancellation and has no nonzero additive units, so the first assertion applies. $\square$

This last conclusion concerns the topology at zero only. It does not rule out zero-dimensional locally compact Presburger presentations with isolated zero, and it does not assert that every standard natural number is isolated: continuity of addition alone does not make every translation an open map.

[write] [Theorem 121.4](#) extends Part I's [corollary 12.2](#) (no compact carrier) and [lemma 12.1](#); compare source 06's [theorem 39.1](#), which isolates 0 when addition and multiplication are both continuous.

122 Arithmetic on the manifold boundary

The countably many endpoints of the locally compact carrier have an intrinsic arithmetic meaning. Here $\partial_{\text{man}}P$ denotes the boundary of P as a one-dimensional topological manifold with boundary, rather than its boundary inside the ambient group. On X_{lc} these are exactly the endpoints of the closed-ray components.

Theorem 122.1 (The boundary is an elementary initial submodel). *Let G be a Borel ordered abelian Polish group whose nonnegative cone P , in its inherited topology, is an uncountable locally compact model of Presburger arithmetic. Put $H = G^0$, the connected component of zero, and define*

$$C = \{g \in G : |g| < h \text{ for every } h \in H \text{ with } h > 0\}.$$

Then $H \cong \mathbb{R}$, C is a countable discrete convex subgroup containing the standard integer subgroup, and

$$\partial_{\text{man}}P = C_{\geq 0}.$$

With its inherited arithmetic, this boundary is a countable elementary initial submodel of P and has the discrete subspace topology.

Proof. [Theorem 121.1](#) gives an open subgroup isomorphic to $\mathbb{R}$. An open subgroup is also closed; since this one is connected, it is exactly $H = G^0$.

The set C is a subgroup. If $c, d \in C$ and $h \in H$ is positive, then $|c| < h/2$ and $|d| < h/2$, whence $|c + d| < h$; closure under negation is immediate. It is convex because $0 \leq x \leq |c|$ with $c \in C$ implies $x < h$ for every positive $h \in H$. No nonzero element of H lies in C , so $C \cap H = \{0\}$. The quotient G/H is countable, and the map $C \rightarrow G/H$ is injective. Thus C is countable. It is discrete since H is open and intersects it only at zero. If e is the least positive element, the dominance inequality in [Theorem 121.1](#) shows that every standard multiple of e belongs to C .

Each H -coset meets P in a line or an open or closed ray, or not at all. Suppose $c \in C_{\geq 0}$. For $h \in H$, the sign of $c + h$ is positive when $h > 0$, negative when $h < 0$, and nonnegative when $h = 0$. Therefore

$$P \cap (c + H) = c + H_{\geq 0},$$

and c is the endpoint of this closed ray. Conversely, if $x \in P$ is such an endpoint in its H -coset, then $x - h < 0$ for every positive $h \in H$. Consequently $0 \leq x < h$ for every such h , so $x \in C$. This proves the asserted identification with the intrinsic manifold boundary.

Finally, G is a Presburger ordered group, or $\mathbb{Z}$ -group, and C inherits its least positive element e . Division stays inside C . More explicitly, if

$$c = mq + re, \quad 0 \leq r < m,$$

with $c \in C$ and a standard positive integer m , then $mq = c - re \in C$. Convexity gives $q \in C$, since $|q| \leq m|q| = |mq|$. Thus C is itself a $\mathbb{Z}$ -group and its inclusion into G preserves and reflects every congruence predicate. Quantifier elimination for Presburger arithmetic in the language with congruence predicates makes this inclusion elementary. Restricting to nonnegative cones gives $C_{\geq 0} \leq P$. Convexity makes it an initial submodel. $\square$

The word “initial” is arithmetic: elements below a boundary element are again boundary elements. It does not mean that the endpoint set is a connected interval in the ambient topology; that endpoint set is discrete.

[write] In source 08’s notation (remark after [theorem 50.1](#)) source 15’s $H = G^0$ is C and its C is B_C : the letters are exchanged. The theorem shows that in the locally compact case B_C is a countable discrete convex elementary initial submodel.

Theorem 122.2 (Every countable Presburger model occurs as a boundary; the construction of [theorem 94.1](#)). *For every countable model N of Presburger arithmetic, there is an uncountable locally compact inherited Polish-group presentation P with continuous addition such that*

$$P \cong X_{lc} \text{ topologically,} \quad \partial_{\text{man}} P \cong N \text{ arithmetically.}$$

Proof. Let $C = N - N$ be the countable ordered difference group of N and give C the discrete topology. It is a $\mathbb{Z}$ -group. Set

$$G = \mathbb{R} \overrightarrow{\times} C$$

and give it the product topology, with the real coordinate dominant in the arithmetic order. This is an abelian Polish group with Borel order and least positive element $(0, 1_C)$. Since

$$G/\mathbb{Z}(0, 1_C) \cong \mathbb{R} \overrightarrow{\times} (C/\mathbb{Z}1_C)$$

is divisible, G is a $\mathbb{Z}$ -group and $P = G_{\geq 0}$ is a full Presburger model. Addition is continuous. The general classification gives its carrier X_{lc} . Its connected component is $\mathbb{R} \times \{0\}$, and the infinitesimal subgroup relative to that component is exactly $\{0\} \times C$. The preceding theorem identifies its manifold boundary with $\{0\} \times N$. $\square$

[write] The model in this proof is source 13’s countable-core transfer with $D = \mathbb{R}$ and $K = N - N$ ([theorem 94.1](#)); the identification of its boundary with N is source 15’s, through [theorem 122.1](#).

123 Continuum many models on one locally compact carrier

The topological rigidity in [Theorem 120.5](#) does not imply rigidity of the arithmetic structure. We now exhibit the maximum possible number of pairwise nonisomorphic continuous-addition Presburger models on the single carrier

$$X_{lc} = \left(\bigsqcup_{j \in \mathbb{N}} \mathbb{R} \right) \sqcup \left(\bigsqcup_{j \in \mathbb{N}} [0, \infty) \right).$$

The ingredients are ordinary finite-support ordered groups; no novelty claim is made for the Hahn-group method itself.

123.1 A recoverable continuum family of countable orders

For a subset $S \subseteq \mathbb{N}$, put

$$k_n(S) = \begin{cases} 1, & n \notin S, \\ 2, & n \in S, \end{cases} \quad L_S = \sum_{n \in \mathbb{N}} (\mathbb{Z} + \mathbf{k}_n(S)),$$

where $+$ and $\sum$ denote ordered sums, $\mathbb{Z}$ has its usual two-sided order, and $\mathbf{k}$ is a finite chain of size k . The countable order L_S has neither a least nor a greatest element.

For elements of a linear order, define finite condensation by

$$x \sim_{\text{fin}} y \iff [\min(x, y), \max(x, y)] \text{ is finite.}$$

This is an equivalence relation with convex classes and is preserved by all order isomorphisms. In L_S its classes are, in succession,

$$\mathbb{Z}_0, \mathbf{k}_0(S), \mathbb{Z}_1, \mathbf{k}_1(S), \mathbb{Z}_2, \mathbf{k}_2(S), \dots$$

Indeed, two points in one integer block have a finite interval between them, as do two points in one finite block. An interval crossing from an integer block into the following finite block contains an infinite terminal segment of that integer block. An interval crossing from a finite block into the following integer block contains an infinite initial segment of the latter. Every interval crossing farther contains one of these infinite intervals.

Consequently the condensation quotient has order type ω . Every automorphism of ω is the identity, and an order isomorphism of L_S with L_T must match the corresponding finite classes. Their cardinalities recover $k_n(S)$ for every n , and hence recover S . Thus

$$L_S \cong L_T \iff S = T. \quad (123.1)$$

123.2 Encoding an order into Archimedean classes

For a countable linear order L , let

$$V_L = \bigoplus_{\ell \in L} \mathbb{Q}$$

be the group of finitely supported rational functions on L . For a nonzero $v \in V_L$, let $\lambda(v)$ be the least point in its finite support and declare

$$v > 0 \iff v(\lambda(v)) > 0.$$

Every nonempty finite subset of a linear order has a least element, so this definition requires no well-ordering of L . It gives the usual lexicographic ordered-group structure, with earlier indices dominating later indices. The group V_L is countable and divisible. Equip it with the discrete topology and define

$$D_L = V_L \overrightarrow{\times} \mathbb{R}, \quad G_L = D_L \overrightarrow{\times} \mathbb{Z}, \quad M_L = (G_L)_{\geq 0}.$$

The topology of D_L is the product of the discrete topology on V_L and the usual topology on $\mathbb{R}$; the topology on M_L is inherited from $D_L \times \mathbb{Z}$. The coefficient order is Borel: each nonzero positive V_L -coset is wholly positive, each negative coset is wholly negative, and the zero coset uses the usual real order.

The group G_L is a discretely ordered abelian group with least positive element $1 = (0, 0, 1)$, and its quotient by $\mathbb{Z}1$ is the divisible ordered group D_L . Equivalently, for every standard

$m \geq 2$, division by m has precisely the usual residue representatives $0, \dots, m-1$. In coordinates it is given by

$$(d, z) = m \left(\frac{d}{m}, \left\lfloor \frac{z}{m} \right\rfloor \right) + (0, z \bmod m).$$

For a nonnegative input the quotient is again nonnegative. By the standard ordered-group characterization of Presburger models, M_L is a model of the full first-order theory of $(\mathbb{N}, 0, 1, +, <)$, including its entire induction scheme. Addition is continuous because it is the restriction of the group operation. Theorem 120.5 gives $M_L \cong X_{lc}$ as a topological space.

For completeness, the invariant distinguishing these models can be checked directly. For positive elements of an ordered abelian group, write

$$x \asymp y \iff (\exists m, n \in \mathbb{N}_{>0}) (x \leq my \text{ and } y \leq nx).$$

This is Archimedean equivalence. Order its classes by dominance:

$$[x] \triangleleft [y] \iff x > ny \text{ for every } n \in \mathbb{N}_{>0}. \quad (123.2)$$

Thus a more dominant class is earlier. The condition is independent of the positive representatives and is invariant under ordered-group isomorphisms. In V_L , two positive elements are Archimedean equivalent exactly when they have the same least support index. Equality of the indices allows integer multiples of either positive leading coefficient to dominate the other; inequality of the indices makes the earlier coordinate dominate every positive integer multiple of the later one. Hence

$$\mathcal{A}(V_L) \cong L, \quad \mathcal{A}(D_L) \cong L + 1, \quad \mathcal{A}(G_L) \cong L + 2,$$

where $\mathcal{A}$ denotes the set of positive Archimedean classes with the convention (123.2). The two appended classes come respectively from the real summand and the integer summand.

An isomorphism of Presburger monoids $M_L \rightarrow M_{L'}$ extends uniquely to an isomorphism of their Grothendieck groups, by $x - y \mapsto h(x) - h(y)$. It preserves the positive cones, and therefore preserves the ordered groups and their Archimedean-class orders. For $L = L_S$, the order L_S has no greatest element. In $L_S + 2$, the greatest class and its immediate predecessor are therefore precisely the two appended classes. Removing those two intrinsically specified classes recovers L_S . There is no absorption of an end block. We conclude from (123.1) that

$$M_{L_S} \cong M_{L_T} \text{ as Presburger structures} \iff S = T. \quad (123.3)$$

Theorem 123.1 (Sharp number of models on one locally compact carrier; compare theorem 35.5). *There are exactly $2^{\aleph_0}$ isomorphism classes of full Presburger models with continuous addition whose underlying topological space is X_{lc} . The lower bound is realized by inherited coefficient presentations.*

Proof. The lower bound is (123.3). Choose a homeomorphism from each of the displayed carriers to the fixed space X_{lc} and transport its arithmetic structure.

For the upper bound, a continuous map from X_{lc}^2 to X_{lc} is determined by its restriction to a countable dense subset, because the target is Hausdorff. Since $|X_{lc}| = 2^{\aleph_0}$, there are at most

$$(2^{\aleph_0})^{\aleph_0} = 2^{\aleph_0}$$

such maps. Choosing the constants 0 and 1 does not increase this bound. The order is recovered from addition by $x \leq y \iff (\exists z) x + z = y$, so it contributes no additional choices. Hence there are at most continuum many such structures, and in particular at most continuum many isomorphism classes. $\square$

This separates two classification problems. The locally compact carrier is unique inside the uncountable inherited coefficient construction, but its arithmetic structures achieve the largest multiplicity permitted by continuity on a separable space. Further questions are to determine the topological Presburger automorphism groups of these presentations and to determine the descriptive complexity of isomorphism on the full space of continuous Presburger operations on X_{lc} . Within the displayed S -indexed family the isomorphism relation has already been determined: it is exactly equality of the parameters.

In fact, the family above fixes even the boundary arithmetic. In $G_{L_S} = V_{L_S} \overrightarrow{\times} \mathbb{R} \overrightarrow{\times} \mathbb{Z}$, the subgroup infinitesimal relative to the open real component is precisely the final $\mathbb{Z}$ summand. Thus $\partial_{\text{man}} M_{L_S}$ is the standard model $\mathbb{N}$ for every S . The continuum lower bound therefore persists after prescribing the boundary to be standard arithmetic.

[write] The analogous counts on Baire space are Part III's [theorems 35.5](#) and [35.8](#). Source 13's [corollary 94.2](#) gives $2^{\aleph_0}$ nonsplit locally compact models; by [theorem 121.1](#) they too live on X_{lc} , whereas the family above is split.

[write] (Batch 92.) The two further questions named above are addressed by [Part XIII](#): the continuous Presburger automorphism groups of all inherited locally compact presentations are computed in [corollary 171.2](#) (with [proposition 182.3](#) [merge]), and the descriptive complexity of isomorphism is bounded below as recorded after [question 126.6](#). Source 19's coding family places the coded scales below the real line, the family above places them above it; both recover the coding order from the Archimedean classes.

124 Consequences for multiplication and for omnific arithmetic

124.1 The locally compact obstruction is entirely algebraic

The topological classification and the cardinality theorem combine to give a conclusion stronger than the Borel multiplier obstruction in this particular setting.

Theorem 124.1 (No multiplication on an uncountable locally compact inherited cone; a case of [theorem 81.2](#)). *Let G be an abelian Polish group with a Borel translation-invariant linear order and least positive element 1. Suppose that its positive cone $M = G_{\geq 0}$, with the inherited topology, is uncountable and locally compact. There is no unital semiring multiplication on M having its existing addition, zero, and distinguished unit 1. No continuity or measurability of the proposed multiplication is assumed. Neither Presburger induction nor multiplication commutativity is required for this conclusion.*

Proof. By [Theorem 121.1](#), G has an open subgroup $H \cong \mathbb{R}$ and G/H is countable. The order restricted to H is usual or reversed by [Lemma 120.4](#); therefore H is an uncountable Archimedean subgroup in its induced order.

A proposed multiplication on M extends uniquely by distributivity to its additive group of differences G . The construction is the algebraic version of [\(119.8\)](#), and does not use any Borel hypothesis. Products of nonnegative elements remain nonnegative. Moreover, the product of two strictly positive elements is strictly positive. Indeed, each can be written as $1 + x$ with $x \geq 0$, and

$$(1 + x)(1 + y) = 1 + x + y + xy > 0.$$

Thus the extension makes G a discretely ordered unital ring. [Theorem 119.6](#) then injects the uncountable group H into the countable quotient G/H , a contradiction. $\square$

The topology is used to locate the Archimedean subgroup of countable index. Once that subgroup has been found, the obstruction survives every change of topology. In particular, refining one of these cones' topologies to isolate its standard natural numbers cannot make its *unchanged additive structure* support a semiring multiplication. The polynomial example in Section 119 has a different ordered additive group and therefore does not contradict this theorem.

124.2 Explicit finite normal forms in the omnific integers

The family from Section 123 can be placed inside the omnific integers without invoking a topology on a proper class, infinite support convergence, or a global choice of normal forms. All the normal forms needed for each element are finite. We use the standard convention that an omnific integer has real coefficients at strictly positive surreal exponents and an integer coefficient at exponent zero. The inspected Provel development realizes this as a coefficient pullback inside its surreal normal-form ring [85].

Lemma 124.2. *Every countable linear order L admits an order-reversing embedding $q : L \rightarrow \mathbb{Q} \cap (1, 2)$.*

Proof. Enumerate L . Assign exponents one at a time, maintaining the required reversed inequalities. At each finite stage, the already assigned elements above and below the next element determine an open interval between finitely many rational bounds; use 1 or 2 when one side has no assigned bound. The interval is nonempty because the finite assignment respects the order. Density of $\mathbb{Q}$ supplies a new rational strictly inside it. The union of these finite assignments is the desired injection. $\square$

For $v \in V_L$, $r \in \mathbb{R}$, and $k \in \mathbb{Z}$, define

$$\Phi_L(v, r, k) = \sum_{\ell \in \text{supp}(v)} v(\ell) \omega^{q(\ell)} + r\omega + k. \quad (124.1)$$

The convention is that zero coefficients are omitted. The finite support has a well-defined greatest exponent, and every exponent $q(\ell)$ lies strictly between 1 and 2.

Theorem 124.3 (A set-sized omnific realization; compare theorems 9.1 and 40.1). *The map Φ_L is an injective order-preserving additive homomorphism from*

$$G_L = V_L \overrightarrow{\times} \mathbb{R} \overrightarrow{\times} \mathbb{Z}$$

into the omnific integers, and sends its distinguished unit to 1. Its nonnegative image is an additive Presburger substructure of the nonnegative omnific integers. With the topology transported from M_L , this image is a locally compact Polish model on X_{lc} . For the orders L_S , these images yield continuum many nonisomorphic such arithmetic structures with standard boundary.

Proof. The expression (124.1) has no negative exponent and has integer constant term, so it is an omnific integer. Uniqueness of finite normal forms gives injectivity. Addition is coefficientwise, giving additivity and preservation of the unit.

If $v \neq 0$, let ℓ_0 be its least support index. Since q reverses the order, $q(\ell_0)$ is the greatest exponent appearing in the first sum. It also exceeds the exponent 1 of the real summand and the exponent 0 of the integer summand. The sign of the normal form is therefore the sign of $v(\ell_0)$. If $v = 0$ and $r \neq 0$, its sign is the sign of r ; if $v = r = 0$, its sign is the sign of k . This is exactly the lexicographic order on G_L .

Transport the topology along the resulting bijection between M_L and its positive image. The bijection is then a homeomorphism by definition, and the already proved Presburger axioms, continuity of addition, local compactness, and boundary statements are preserved. Each image is a set of cardinality continuum: its parameters range over the set $V_L \times \mathbb{R} \times \mathbb{Z}$, not over all surreal numbers. $\square$

These are additive substructures, with the full Presburger theory verified by their $\mathbb{Z}$ -group certificate. They are not subrings for the usual omnific multiplication. For example,

$$\omega = \Phi_L(0, 1, 0), \quad \omega \cdot \omega = \omega^2,$$

and uniqueness of normal forms shows that ω^2 is outside the image: every exponent allowed by (124.1) is less than 2. Theorem 124.1 says more than this failure of closure. No alternative multiplication on the same positive image can satisfy the unital semiring laws with its existing addition and unit.

The topology just transported is a coefficient topology. It is distinct from the native order topology of the omnific integers, which is discrete because 1 is least positive. ProveIt's separate discreteness development concerns the native fine order topology; it provides no contradiction to the nondiscrete coefficient topology used here [86]. An assertion of "a Polish topology on the omnific integers" without a set-sized carrier and an explicit topology would obscure both of these distinctions.

125 From mathematical proofs to a ProveIt formalization

The repository offers a concrete starting point for formalizing these results, but the distinction between a source-inspected existing declaration and a proposed future theorem must be maintained. The source audit establishes which declarations are present at the pinned commit. The proofs in this article are mathematical proofs; no new Lean kernel certificate accompanies them.

125.1 Existing formal interfaces

The Presburger files are under the directory

`Logic/PresburgerArithmetic/Lean/PresburgerArithmetic/.`

The actual semantics in `Syntax.lean` use valuations of type `List Int`; quantified variables range over `Int`. In `Decision.lean`, the following declarations occur in the namespace `PresburgerArithmetic.Formula` [83]:

- `quantifierEliminate`: the elimination procedure;
- `holds_iff_quantifierEliminate`: its semantic equivalence;
- `decideSentence_eq_true_iff`: correctness for sentences;
- `presburgerArithmetic_decidable`: the resulting decision instance.

These support the executable integer decision procedure. They do not by themselves instantiate that procedure over an arbitrary nonstandard ordered group.

The omnific files are under `Algebra/SurrealNumbers/Surreal/Foundations/.` The inspected `OmnificIntegers.lean` supplies the following chain of interfaces [85]:

1. `normalFormHahnHom` turns the normal-form map into a ring homomorphism, and `nonnegativeSupportSubring` restricts the permitted exponents.
2. `constantCoeff` extracts the real constant coefficient; `omnificSubring` pulls it back along the integer embedding. The abbreviation `OmnificInteger` is a universe-indexed type, and `omnificToSurreal` embeds it into the actual surreal construction.
3. `omnificToSurreal_injective` records injectivity; `omnificConstantCoeff` extracts the integer constant; `omnificPurelyInfiniteIdeal` is its kernel; and `omnificQuotientEquiv` identifies the quotient with $\mathbb{Z}$.
4. In `OmnificPurelyInfiniteModule.lean`, `omnificPurelyInfiniteAddEquiv` transfers the established purely infinite support module. The real and rational module instances provide coefficient divisibility.

These interfaces fit the finite normal-form construction above. Their existence is source evidence for a formalization route, not evidence that the new Polish topology or classification is already implemented.

[write] The declarations named in this subsection exist at the current state under these names. They are about the ordinary integers and the omnific coefficient construction; none of source 15's theorems is formalized.

125.2 A dependency order for implementation

Stage 1: algebraic certificates and the cardinal obstruction. Introduce a semantic interface for discretely ordered abelian groups with standard residue representatives. Prove closure of positive cones, the unique division statement, and extension of additive monoid maps to differences. Then formalize Theorem 119.6: after the bounded integer-interval lemma, its essential object is the additive map $a \mapsto ah + H$ and the proof that its kernel is trivial. This is the shortest substantial result in the article and needs no measure theory or descriptive set theory.

Stage 2: nonstandard Presburger semantics. Parameterize the interpretation of affine terms and congruences by the ordered group. The elimination proof must use the actual subgroup mG , including nonstandard shifts. For example,

$$x \equiv y \pmod{m} \iff x - y \in mG$$

is the semantic congruence relation. Proving a formula invariant under standard iterations of the translation $x \mapsto x + m$ only covers $m\mathbb{Z}1$, which can be a proper subgroup of mG . Generalized elimination or a formal model-theoretic completeness argument closes this gap. Its output should include elementary embeddings of pure $\mathbb{Z}$ -subgroups, the interface needed by the boundary theorem.

Stage 3: topology and local compactness. Build the coefficient models as explicit types with transported product topologies. Prove the G_δ criterion and the compact-neighborhood reflection lemma. The Euclidean closed-cone argument is a small independent target. The full classification then requires the principal LCA structure theorem and the measurable-order compact-subgroup obstruction. Availability of these results in the particular dependency version must be checked before estimating a formalization budget; this manuscript does not assume that they are already packaged in the required form.

Stage 4: the arithmetic boundary and continuum family. Formalize the infinitesimal subgroup C relative to the open real component, its convexity and countability, and its identification with the endpoints. Prove purity under division and then apply the semantic theorem from Stage 2. For the continuum family, finite support makes the ordered-group construction elementary; the recoverable finite-condensation pattern supplies an explicit nonisomorphism certificate. The final omnific embedding uses finite normal forms and the already inspected coefficient interfaces.

Stage 5: the Borel multiplier theorem. This stage requires Borel automatic continuity, a complete metric, Cantor product probability, and a tail zero-one law. The proof is organized so that each dependency can be isolated. The finite-block estimate only counts 2^k assignments and avoids a measurable inverse or an injective Cantor-space coding theorem. The order and the topology must remain separate structures in the formal statement: assuming an order topology would force discreteness at the outset and erase the content of the result.

125.3 What verification establishes

The source comparison fixes the existing background; the proofs check the new implications; and the counterexamples test the boundaries of the hypotheses. Separate AI-assisted mathematical reviews focused on the rational-hull splitting, endpoint identification, quotient injection, finite-block probability argument, and recovery of the continuum parameters. That review is useful error control, but it is not a substitute for an independent referee or a proof assistant.

Finite numerical checks could illustrate the geometric growth inequality, but they would not verify the uncountable cardinalities, Polish completeness, elementarity, or all induction instances. The article therefore supplies deductive arguments for those claims and does not present finite computation as a mathematical certificate.

126 Further research questions and a proposed program

The following are proposed questions arising from the results. Their status is deliberately local: they are not all asserted to be previously published open problems. Where a question comes from Glazer's paper or the earlier repository manuscript, that origin is stated. Questions already settled above are separated from questions outside the proved hypotheses.

126.1 Questions now settled within the inherited-group framework

For an uncountable Presburger positive cone inherited from a Borel ordered abelian Polish group, the locally compact carrier is exactly X_{lc} . Thus connected and zero-dimensional locally compact carriers are excluded in this framework. Every such cone forbids every unital semiring multiplication on the same addition and unit. The fixed carrier supports the maximum possible number, continuum, of isomorphism classes, even with standard boundary. Its intrinsic boundary is always a countable elementary initial submodel, and the set of possible boundary models is exactly the set of countable Presburger models. These are complete answers to the corresponding restricted classification, expansion, multiplicity, and boundary questions.

[write] (Batch 87.) For Part I's [question 14.5](#) and [question 14.3](#), and Part III's [question 43.2](#) and [question 43.4](#), these settle the inherited-group case only.

126.2 Presentations outside Polish group completions

Question 126.1 (Zero-dimensional locally compact arithmetic). Does any uncountable zero-dimensional locally compact Polish space support a full Presburger model with jointly continuous addition?

This extends a follow-up question in the repository manuscript [84]. The classification proves that any affirmative example cannot inherit its topology from a Borel ordered Polish group. Theorem 121.4 adds that its zero must be isolated. The remaining case is therefore precise: an example with isolated zero and no compatible inherited Polish group presentation. A global nonexistence proof must use information beyond the compact-open neighborhood argument.

[write] This is Part I's question 14.5, re-scoped by corollary 121.2 and theorem 121.4.

Question 126.2 (Connected Polish arithmetic). Does a connected uncountable Polish space admit a full Presburger model with jointly continuous addition? What changes if local compactness or finite covering dimension is imposed?

The inherited locally compact case is excluded above. Connectedness without a group-completion hypothesis remains outside these proofs. One possible approach is to study the successor map and the images of translations as topological embeddings; addition is cancellative, but its inverses on these images need not be continuous automatically. The degree-stratum polynomial example is disconnected and therefore does not answer this question.

[write] Compare Part I's question 14.3, Part III's question 43.4 and source 13's question 103.4.

[write] (Batch 90.) "Without a group-completion hypothesis" now means "with discontinuous partial subtraction": an example must have it (corollary 155.3), and in the locally compact case proposition 155.7 [merge] gives the same. Open.

Question 126.3 (A usable group-completion criterion). Which intrinsic conditions on a locally compact Polish Presburger monoid ensure that its topology is inherited from a Polish group topology on its algebraic group of differences?

A useful answer should be stated using translations, the partial difference map on ordered pairs, and neighborhood structure, so it can be checked without first constructing a group topology. The polynomial example shows that cancellation, joint continuity of addition, local compactness, and a full Presburger reduct do not suffice. This question would identify exactly where the classification can be applied to presentations given without coefficient coordinates.

[write] This is source 08's question 57.2 (= source 09's question 73.11); sources 12 and 10 ask it again in Part VI.

[write] (Batch 90.) Answered by Part XII, in the terms the question asks for: the topology is inherited from a Hausdorff group topology on the group of differences exactly when the partial difference map is continuous (theorem 153.1); it suffices to test it at the pairs (x, x) and $(x, 0)$ (theorem 154.3); and the group is then Polish and locally compact (theorem 154.1 and corollary 154.2). The polynomial example fails it at (t^2, t^2) (example 155.5). With corollary 121.2 this gives proposition 155.7 [merge]. Source 18's question 165.3 asks for a criterion in terms of translations alone.

126.3 Arithmetic invariants beyond the carrier

Question 126.4 (Classification with a prescribed boundary). Fix a countable Presburger model N . Classify the locally compact inherited-group presentations with manifold boundary isomorphic to N , up to arithmetic isomorphism and up to topological arithmetic isomorphism.

Existence is settled by Theorem 122.2, and there are already continuum many classes when $N = \mathbb{N}$. The open classification problem is to find complete invariants. Natural data include the ordered quotient above the infinitesimal subgroup, the placement of the real component, and extension data for the standard integer subgroup. A homeomorphism of carriers alone cannot recover these data.

[write] (Batch 92, 4 October 2026.) Answered by corollary 182.4 [merge], from source 19's theorems 170.2 and 171.1 (Part XIII) with proposition 182.2: the uncountable presentations with boundary N are the cones of the normal models whose ordered pure subgroup $(A, <_A, 1)$ is isomorphic to $N - N$; arithmetic and topological arithmetic isomorphism coincide on them, and the complete invariant is the countable ordered extension $(E, 1, A, <_A, <_B)$, which consists of exactly the data suggested above. This reduces the classification to isomorphism of countable structures; its complexity for a fixed N is not determined. The answer is this report's reading of source 19, which does not state it; it is unrefereed and not formalized.

Question 126.5 (Threshold orders and automorphism groups). For a fixed countable torsion-free abelian group A , classify Borel translation-invariant linear orders on $\mathbb{R} \times A$ whose cones are discretely ordered Presburger models. Determine their groups of continuous Presburger automorphisms.

Each real-coset section is empty, full, or a ray. Its threshold, whether an endpoint is present, and the arithmetic relations between cosets satisfy compatibility conditions under addition. Making these conditions explicit could turn the topological classification into an arithmetic parametrization. The real component need not be convex, so assuming a single lexicographic product would miss part of the problem.

[write] (Batch 92, 4 October 2026.) Answered by Part XIII (source 19, written without knowledge of this question), with proposition 182.1 [merge] to move the unit into the countable factor. Up to a continuous shear the thresholds are removable, and the order is the sign rule (170.1): a pure subgroup containing the unit, ordered below the real line, and an ordered quotient above it (theorem 170.2); two such orders are continuously isomorphic exactly when their countable ordered extensions are (theorem 171.1), and also abstractly (proposition 182.2 [merge]). The continuous Presburger automorphism group is an extension of $\mathbb{R}_{>0} \times \text{Aut}(E, u; A, <_A, <_B)$ by $\text{Hom}(E/A, \mathbb{R})$ (corollary 171.2), and no such order is rigid (corollary 171.3). This question's A is source 19's E . The proofs are unrefereed and not formalized.

Question 126.6 (Complexity of isomorphism). In a precise standard Borel coding of continuous binary operations and distinguished constants on X_{lc} , what is the descriptive complexity of the subset defining Presburger models? On that subset, what is the complexity of arithmetic isomorphism and of topological arithmetic isomorphism?

The family $S \mapsto M_{L_S}$ has its own isomorphism relation completely determined: it is equality of the parameters. Cardinality alone does not determine the complexity of isomorphism on the larger space of presentations. A classification should distinguish continuous changes of coordinates from abstract model isomorphisms.

[write] (Batch 92, 4 October 2026.) Answered in part by Part XIII. Source 19 gives a continuous map $L \mapsto (m_L, 0_L, 1_L)$ from countable linear orders into $C(X_{lc} \times X_{lc}, X_{lc}) \times X_{lc} \times X_{lc}$ whose

values are Presburger structures and which reduces isomorphism of linear orders to arithmetic and to topological arithmetic isomorphism alike ([theorem 174.3](#)); so both relations are at least as complex as isomorphism of countable structures and are not Borel on any Borel set of codes containing the image ([remark 182.5](#) [merge]), while on source 19's normal-form codes topological isomorphism is classifiable by countable structures and Borel complete ([theorem 175.2](#)). The complexity of the set of Presburger codes and an upper bound on the whole space stay open ([question 180.3](#)).

126.4 Strengthening and formalizing the obstructions

Question 126.7 (Smaller endomorphism hypotheses). In Theorem [119.2](#), how far can the assumption that every positive a is $T_a(e)$ be weakened?

The proof actually uses only enough positive endomorphisms to continue a sequence that is rapidly increasing in order while topologically summable. A useful refinement would give an intrinsic condition on the positive endomorphism orbit of e that is both necessary and sufficient for this recursion. Merely replacing the orbit condition by an unexplained cofinality assumption would leave the needed control of neighborhoods unproved.

[write] (Batch 87.) Answered by [Part VI](#): by source 12's [theorem 80.1](#), in an uncountable group with a Hausdorff Baire ccc topology and a Baire-property cone every ordered additive self-embedding f satisfies $f(e) \leq 2a$ for the a of [lemma 77.2](#). So a single positive endomorphism T with $T(e) > 2a$ already forces countability, and no regularity of T is needed.

Question 126.8 (Regularity of the order). Which conclusions survive when a Borel order is replaced by a weaker measurable or category-theoretic regularity assumption?

Different steps use different regularity. The real-line lemma uses the Baire property; the compact-subgroup lemma uses Haar measurability; the probabilistic obstruction needs the inverse images of order intervals under the Cantor limit map to be measurable. A satisfactory weakening should state these requirements on the relevant maps and subgroups, since an arbitrary weakening of "Borel" need not be preserved by restriction or inverse image.

[write] For the multiplier obstruction, the Baire property of the cone suffices ([Part VI](#)).

Question 126.9 (Algebraic obstructions beyond one real subgroup). What other additive invariants prevent a discretely ordered group from admitting any unital ordered-ring multiplication?

The injection $H \hookrightarrow G/H$ compares the size of an Archimedean subgroup with its quotient. One can ask for analogues using a chain of Archimedean layers, ranks of quotients, or several independent multiplier images. A useful classification would explain which failures are purely algebraic and which arise only from measurability. The polynomial ring supplies an example where an infinite hierarchy of larger growth layers permits multiplication.

Question 126.10 (Minimal metatheory and Glazer's Question 1). What is the weakest subsystem of second-order arithmetic that proves the particular obstruction theorems in this article? How does it compare with Glazer's original question about proving his Theorem 2 in ATR_0 ?

Glazer's Question 1 is an explicitly published problem [[1](#), Further Work, Question 1]. This article does not answer it. Its classical proofs use completeness, Baire category, product probability, and, in the classification, algebraic complements and LCA structure. The phrase

“induction-free” in Section 119 refers to the *object arithmetic being modeled*, not to the strength of the metatheory used to prove the theorem. Determining the latter requires a separate reverse-mathematical analysis.

[write] Source 14 (Part VIII) claims an affirmative answer to Glazer’s Question 1, not independently reviewed; source 15’s own statement that it does not answer the question stands.

Question 126.11 (A checked formalization). Can the algebraic cardinality theorem, the generalized Presburger semantic certificate, and the boundary theorem be formalized in ProveIt as a coherent first milestone?

These three components connect the repository’s current integer elimination and omnific coefficient interfaces to the new structural results. They also divide naturally into algebra, model theory, and topology. The probability theorem can follow as a second milestone once the needed measure-theoretic interfaces are explicit.

126.5 Recommended order of work

The most concrete next project is the algebraic obstruction and its formal certificate: the statement is short, the proof has few dependencies, and the conclusion is stronger than a continuity obstruction. Classification of presentations with a prescribed boundary is the next mathematical project, because the boundary supplies an intrinsic invariant that the carrier alone misses. The broad existence questions for connected or zero-dimensional presentations should be pursued together with a group-completion criterion, with the polynomial example serving as a test case for every proposed general theorem.

Part X

Elementary embeddings and Polish submodels of lexicographic Presburger arithmetic

[write] This part is source 16, manuscript 02 of batch 89, *Elementary Embeddings and Polish Submodels of Lexicographic Presburger Arithmetic: Matrix normal forms, closed images, and two sharp ordinal thresholds* (23 pages, A4, dated October 3, 2026; “prepared for Vladimir Reshetnikov”; its delivery calls it AI-assisted, with internal AI-assisted proof reviews). It arrived in commit 7c0f2d9f9, as archive Glazer_ProveIt_Embeddings_and_Thresholds, and was placed in 23adb85f9; it is pinned to 0f084afa9, 70 commits before the placement. Its file prefix in this report is 16-presburger-embeddings-. It continues Parts I and III: it takes the family $M_\alpha = (\mathbb{Q}^\alpha \vec{\times} \mathbb{Z})_{\geq 0}$ of theorems 35.1 and 35.9 (α a countable ordinal, rational coordinates discrete) and classifies its elementary embeddings, its elementary submodels that are Polish in the inherited topology, and two ordinal thresholds, for continuity of all self-embeddings ($\alpha \leq \omega$) and for closedness of all their images ($\alpha < \omega \cdot 2$). All unmarked text in this part is source 16’s; its twelve sections are Sections 127–138 (its Section k is Section 126 + k), and its two appendices are appendices W and X.

What was already here. At its pin this report was the delivered base text of Part I (source 05), and the manuscripts of Parts III–V were archives in docs/incoming; source 16 read both,

and credits the construction of M_α to them. It did not see source 04: that manuscript was placed in 3d2177df4, an ancestor of the pin, but as a member of a merge it was not staged, and its text entered the report only with the write ae7b9ab74, after the pin. So the case $\alpha = \beta = \omega$ of source 16's embedding results proves again, without credit, source 04's [proposition 11.3](#) (all unital additive maps are elementary embeddings $T \times \text{id}$), [theorem 11.4](#) (the increasing-positive-pivot matrices, automatically continuous) and [corollary 11.5](#) (closed images, clopen exactly when finitely many pivots are omitted), and source 05's [theorem 11.7](#) is the simplest case of its [theorem 135.1](#). Its model construction ([proposition 128.3](#)) is [theorem 35.1](#) (source 07) and [theorem 35.9](#) (source 06), and its omnific realization ([section 137](#)) is [theorem 40.1](#). Each of these is printed with source 16's proof and flagged in its title or in a note. New to the report are the transfinite cases: the classification of continuous embeddings and of all Polish elementary submodels for every countable α , the pivot normal form, ambient orbits and the lattice, the two thresholds, the automorphism counts and the intersection of all self-copies. Priority is not established.

Use of choice. The discontinuous and non-Borel maps of [sections 133](#) and [134](#) are built from a nonzero $\mathbb{Q}$ -linear functional on $\mathbb{Q}^\omega$ vanishing on the finite-support vectors, which source 16 obtains with the axiom of choice (a Hamel basis of a quotient). So the negative halves of clauses (5) and (6) of [theorem 127.1](#), [theorem 133.3](#), the value 2^c in [theorem 133.4](#), [proposition 133.5](#) and [theorem 134.2](#) are theorems of ZFC, not of ZF; [remark 133.2](#) [merge] shows that ZF with dependent choice does not prove them unless it is inconsistent.

Questions answered. Source 16 answers, for its family, Part I's [question 14.9](#) (Polish elementary-submodel lattices; its "Research question 8") and [question 14.14](#) (embedding categories beyond [theorem 11.4](#), for ordinal-indexed rational products), and the clause of [question 14.6](#) on continuous order-preserving endomorphisms, for the groups $\mathbb{Q}^\alpha$ (its "Research question 5"). Dated notes after those questions say what is answered. Its questions on connected carriers and on arbitrary Polish presentations repeat questions of Parts I, III, IV, VII and IX that Parts VII and IX answer in part; notes after them point there instead of reopening them.

Labels and notation. Every label of source 16 takes the prefix `pma:emb:`. Labels were added to its unlabelled corollary on orbit counts, two examples and its twelve questions, so that notes can cite them, and `pma:emb:rem:choice` was written for this report; four questions of [Part I](#) were labelled for the notes (`pma:pr:q:coefficientgroups`, `pma:pr:q:lattices`, `pma:pr:q:surrealcutoffs`, `pma:pr:q:embeddingcategories`). The fourth block of the table in [section 1.3](#) lists the symbols. In short: source 16's $H_\alpha = \mathbb{Q}^\alpha$ is Part III's A_α , kept because A is its matrix; its G_α and M_α are Part III's; its $D = \mathbb{Q}^\omega$ and D_{fin} are printed D_B and D_B^{fin} (Part I's Baire coefficient group); its shear S_φ is printed Sh_φ , because S is its pivot subset; its restriction symbol is printed $\restriction$; its lexicographic sum is printed $\vec{\sum}$; in two proofs local letters were renamed (S, K to N, Γ in the proof of [theorem 131.1](#); K to K_α in the proof of [theorem 134.1](#), where it is the terminal tail of [section 135](#)). No normalization changed.

Files. Its finite checks are shipped as `code/16-presburger-embeddings-verify_examples.py` with the recorded run data `data/16-presburger-embeddings-verification_results.json` (11,243 exact checks); its build record, source audit and proof-review record as `data/16-presburger-embeddings-BUILD-REPORT.json`, `16-presburger-embeddings-SOURCE-AUDIT.md` and `16-presburger-embeddings-PROOF-REVIEW.md`. The Hilbert's-tenth

programme routed the archive without a correctness finding (appendix E). Nothing in this part is formalized.

127 The research intersection and the contribution

The useful intersection is topological model theory of arithmetic, with a second connection to ordered Hahn groups and the omnific integers. Glazer’s *A Topological Tennenbaum Theorem* [1] rules out uncountable Polish models of sufficiently strong arithmetic with the indicated regularity of their operations. Its final Question 2 asks whether an uncountable Polish space can support Presburger arithmetic with continuous addition. This is a question about addition and first-order induction in the additive language; multiplication is absent.

The current ProveIt corpus already contains affirmative constructions. The maintained manuscript [92] treats the positive cones of $\mathbb{R} \vec{\times} \mathbb{Z}$ and $\mathbb{Q}^\omega \vec{\times} \mathbb{Z}$. It classifies elementary submodels of the real-coordinate model and constructs shift self-embeddings of the rational-sequence model. Its further-research questions ask for classifications of continuous coefficient-group endomorphisms and of Polish elementary-submodel lattices. A separate incoming manuscript [93] already introduces the countable-ordinal family used here. We therefore take the construction of the models as prior work and study their structure.

This distinction matters. Repeating the positive construction would not constitute a new answer for this repository. Nor does the present article settle Glazer’s separate Question 1 about proving his theorem in ATR_0 . Instead, we solve a precise family of follow-up questions and exhibit two different failures of topological rigidity.

[write] At source 16’s pin 0f084afa9 the “maintained manuscript” [92] was source 05’s delivered text, now the base of Part I (its real-model results are theorems 11.1 and 11.2, the shift theorem 11.7); the “separate incoming manuscript” [93] was source 06, now part of Part III, whose thm:ordinals is theorem 35.9. The two questions it names are Part I’s questions 14.6 and 14.9. The self-embeddings of the rational-sequence model were already classified, by source 04, which source 16 did not see (introduction of this part).

127.1 A statement of the main answer

Fix a countable ordinal α . All products below are full products, not finite-support sums. Set

$$H_\alpha = \mathbb{Q}^\alpha, \quad G_\alpha = H_\alpha \times \mathbb{Z}, \quad \mathbf{1} = (0, 1), \quad M_\alpha = (G_\alpha)_{\geq 0}. \quad (127.1)$$

The first nonzero rational coordinate determines the sign in H_α ; in G_α , the H_α coordinate dominates the integer coordinate. Every rational coordinate and the integer coordinate have the discrete topology. The topology on M_α is the subspace topology from G_α .

Theorem 127.1 (Structural classification and thresholds). *For countable ordinals α, β the following hold.*

- (1) M_α is a Polish model of the full first-order theory of $(\mathbb{N}; 0, 1, +, <)$, with continuous addition. For $\alpha \geq \omega$ its underlying space is homeomorphic to Baire space.
- (2) Every map $M_\alpha \rightarrow M_\beta$ preserving 0, 1, and addition is an elementary embedding. Such a map exists exactly when $\alpha \leq \beta$.
- (3) The Borel embeddings are exactly the continuous embeddings. They are described by row-finite rational matrices with strictly increasing pivot map $p : \alpha \rightarrow \beta$ and positive pivot entries. Every such embedding has closed image.

- (4) The elementary submodels of M_α are precisely $M_V = (V \overrightarrow{\times} \mathbb{Z})_{\geq 0}$ for $\mathbb{Q}$ -linear subspaces $V \leq H_\alpha$. The inherited topology is Polish exactly when V is closed. Closed subspaces have canonical pivot subsets $S \subseteq \alpha$ and are order-topologically isomorphic to $\mathbb{Q}^S$. The corresponding submodel is isomorphic to $M_{\text{otp}(S)}$.
- (5) Every elementary self-embedding of M_α is continuous if and only if $\alpha \leq \omega$.
- (6) Every elementary self-embedding of M_α has closed image if and only if $\alpha < \omega \cdot 2$.

All claims are proved below. Parts (1) and the underlying arithmetic construction are included for completeness and attributed to the existing work. The ordered normal forms, their model-theoretic consequences, and the precise combination of thresholds are the focus of this article.

[write] **What this report already contained.** Clause (1) is [theorem 35.1](#) (source 07) and [theorem 35.9](#) (source 06) for infinite α ; for finite α it is elementary. For $\alpha = \beta = \omega$, the first sentence of clause (2) is source 04's [proposition 11.3](#) (proved there for arbitrary divisible coefficient groups, as [theorem 129.1](#) is here), and clauses (3) and (5) are source 04's [theorem 11.4](#) and [corollary 11.5](#). That the M_α are pairwise nonisomorphic is [theorem 35.4](#). Clause (4) has the real model's [theorem 11.2](#) as its counterpart. The rest, in particular everything about $\alpha > \omega$, is new to the report.

Coordinate rank	All self-embeddings continuous?	All self-embedding images closed?	Number of automorphisms
0	Yes	Yes	1
$0 < \alpha < \omega$	Yes; all are onto	Yes	$\aleph_0$
$\alpha = \omega$	Yes	Yes	$\mathfrak{c}$
$\alpha = \omega + n$, $0 < n < \omega$	No	Yes	$2^{\mathfrak{c}}$
$\alpha \geq \omega \cdot 2$	No	No	$2^{\mathfrak{c}}$

127.2 Relation to established mathematics

Quantifier elimination for Presburger arithmetic is classical [3]; we give the needed uniform argument in [appendix W](#). Ordered Hahn groups and their automorphisms have a substantial literature. Kuhlmann and Serra [87] develop the valuation and skeleton framework and matrix descriptions in relevant Hahn settings. Serra's dissertation [88], especially its discussion of strong additivity, already explains closely related automatic behavior for discrete ranks and failures beyond them. Continuity for discrete-field products and preservation of formal infinite sums are also studied by Freni [89]. The unordered closed-subspace theory belongs to classical linearly compact duality; see, for example, [90].

We do not claim these general mechanisms as new. The proofs here specialize them, make the positive-cone and elementary-embedding interfaces explicit, and extract the exact orbit, lattice, and closed-image consequences. The source audit found no statement of this complete package in the inspected ProveIt reports or cited sources. That limited audit does not establish historical priority.

[write] (Batch 89.) For the repository the audit was incomplete through no fault of its own: the case $\alpha = \beta = \omega$ of the dictionary, the matrix classification, automatic continuity and the image topology was already proved by source 04 ([sections 11.2](#) and [11.3](#)), which had been placed but not yet written into this report at source 16's pin. The package is new for transfinite α and as a package.

127.3 Foundations and conventions

We work in ZFC. Every α is a set ordinal, and every product, matrix, subspace, and collection of submodels appearing in a fixed instance is a set. An elementary submodel always has the *inherited* topology unless another topology is specified. A map called Borel is Borel measurable between the indicated Polish spaces. “Clopen” means both closed and open. We use $\mathfrak{c} = 2^{\aleph_0}$.

The product topology is essential. At transfinite coordinate ranks it can differ sharply from both the lexicographic order topology and the Hahn valuation topology. The discontinuous maps below are discontinuous for the product topology specified in (127.1).

128 The arithmetic and topology of the models

128.1 Why the positive cone satisfies full Presburger arithmetic

Definition 128.1. A $\mathbb{Z}$ -group is an ordered abelian group with least positive element 1, such that for every standard positive integer m and every g there are unique q and $r \in \{0, \dots, m-1\}$ with $g = mq + r1$.

The group G_α is a $\mathbb{Z}$ -group. It is discretely ordered because any element with positive coefficient part is greater than every standard integer, whereas the zero coefficient part leaves the usual order on $\mathbb{Z}$. For division, write $n = mq + r$ in $\mathbb{Z}$. Then

$$(h, n) = m(h/m, q) + (0, r). \quad (128.1)$$

The coefficient group is uniquely divisible. The same formula works in the positive cone: if $h > 0$ then $h/m > 0$, and if $h = 0$ then ordinary nonnegative division applies.

The classical quantifier-elimination theorem says that the theory of $\mathbb{Z}$ -groups eliminates quantifiers after adding the predicates $g \equiv_m h$, for every standard $m \geq 2$. Its complete theory is that of $(\mathbb{Z}; 0, 1, +, -, <)$. Relativizing group formulas to the nonnegative cone gives the complete theory of $(\mathbb{N}; 0, 1, +, <)$. This includes every first-order induction instance in the additive language. It is much stronger than verifying finitely many associativity or division axioms.

In our split groups, congruences have a particularly simple meaning:

$$(h, n) \equiv_m (h', n') \iff n \equiv n' \pmod{m}. \quad (128.2)$$

This identity will make elementary embeddings almost entirely a problem about ordered $\mathbb{Q}$ -vector spaces.

128.2 Polishness and the role of the missing boundary

The group H_α is a countable product of countable discrete Polish spaces. Choose a bijective enumeration (i_k) of an infinite α ; a compatible complete metric is

$$d(h, h') = \sum_{k \geq 0} 2^{-k-1} \mathbf{1}_{h_{i_k} \neq h'_{i_k}}.$$

This enumeration specifies a topology, not the lexicographic priority of coordinates, which remains the ordinal order on α .

For $i < \alpha$, the condition $h_j = 0$ for every $j < i$ is closed. Since each rational coordinate is discrete,

$$H_\alpha^{>0} = \bigcup_{i < \alpha} \{h : h_j = 0 \ (j < i), \ h_i > 0\} \quad (128.3)$$

is F_σ . The negative cone is likewise F_σ , so the positive and nonnegative cones are both G_δ as well. It follows that

$$M_\alpha = (H_\alpha^{>0} \times \mathbb{Z}) \cup (\{0\} \times \mathbb{N}) \quad (128.4)$$

is G_δ in G_α : finite unions of G_δ sets are G_δ in a metric space. A G_δ subspace of a Polish space is Polish [8]. Addition is the restriction of the continuous group operation and preserves the cone.

Remark 128.2. The cone need not be closed in its ambient group. For example, when $\alpha \geq \omega$, $(e_k, -1) \in M_\alpha$ converges to $(0, -1) \notin M_\alpha$, where e_k has one nonzero coordinate at k . Thus completeness is obtained from the G_δ theorem, not by claiming that the displayed ambient metric remains complete on the cone.

Proposition 128.3 (Baire-space carrier; for $\alpha \geq \omega$, [theorems 35.1](#) and [35.9](#)). *If $\alpha \geq \omega$ is countable, M_α is homeomorphic to $\mathbb{N}^\mathbb{N}$. If $\alpha < \omega$, it is countable and discrete.*

Proof. The finite case is immediate. In the infinite case the cone is zero-dimensional and Polish. We show that no compact subset has nonempty interior. Let a basic coefficient cylinder, with integer coordinate fixed and only finitely many rational coordinates prescribed, meet the cone. It contains a smaller cylinder in which positivity is already decided at a finite natural-number coordinate. To see this, use an already prescribed positive first nonzero coordinate if it occurs before the first unprescribed natural coordinate; otherwise put a positive rational at that first unprescribed coordinate and zero at the earlier unprescribed ones. A prescribed negative first nonzero coordinate before this point would contradict the assumed nonempty intersection with the cone.

Choose a later natural coordinate that is still unprescribed. It can take every rational value while all these points remain in the smaller cylinder. The projection of a compact set to a discrete rational coordinate is finite. Therefore the cylinder cannot be contained in a compact set. In particular there are no isolated points and the space is nowhere locally compact. The standard characterization of Baire space as the nonempty zero-dimensional Polish space that is nowhere locally compact now applies [8]. $\square$

[write] The cylinder argument is the one of [theorem 35.1](#) (source 07) and [theorem 35.9](#) (source 06); for $\alpha = \omega$, [theorem 6.1](#) gives an explicit, computable homeomorphism.

129 From arithmetic homomorphisms to ordered linear embeddings

The following observation removes any ambiguity about which arithmetic maps are being classified.

Theorem 129.1 (The homomorphism dictionary; = [proposition 11.3](#) [04]). *Let A, B be divisible ordered abelian groups and let $M_A = (A \overrightarrow{\times} \mathbb{Z})_{\geq 0}$, $M_B = (B \overrightarrow{\times} \mathbb{Z})_{\geq 0}$. Every map $f : M_A \rightarrow M_B$ preserving 0, 1, and addition has the unique form*

$$f(a, n) = (T(a), n), \quad (129.1)$$

where $T : A \rightarrow B$ is a strictly order-preserving $\mathbb{Q}$ -linear embedding. Conversely every such T defines such an f , and f is elementary in the full Presburger language.

Proof. The cancellative positive cone generates its ordered group by differences. Hence f extends uniquely to a group homomorphism $F : A \times \mathbb{Z} \rightarrow B \times \mathbb{Z}$ fixing $(0, 1)$: define $F(u - v) = f(u) - f(v)$ for $u, v \geq 0$. This is well-defined because $u - v = u' - v'$ implies $u + v' = u' + v$.

The maximal divisible subgroup of the split group is

$$\bigcap_{m \geq 1} m(A \times \mathbb{Z}) = A \times \{0\}.$$

A homomorphism sends this subgroup into the analogous subgroup of the target. Consequently $F(a, n) = (T(a), n)$ for an additive map T , which is $\mathbb{Q}$ -linear because divisibility is unique in ordered groups.

If $a > 0$, the element $(a, -1)$ belongs to M_A . Its image $(T(a), -1)$ can belong to M_B only if $T(a) > 0$. Thus T strictly preserves positivity. Applying this to differences proves that T is injective and preserves and reflects order.

Conversely such a T plainly preserves the positive cone and the operations. The extended map preserves and reflects equality, order, and every congruence by (128.2). Quantifier elimination therefore makes the group embedding elementary. Restricting quantified variables to the nonnegative cone proves elementarity of f . $\square$

[write] This is source 04's [proposition 11.3](#), with the same hypotheses and conclusion. Both proofs extend f to the group of differences and use the maximal divisible subgroup $A \times \{0\}$; they obtain strict positivity of T differently (source 04 from $x = y + 1 + u$ for $x > y$, source 16 from the positive element $(a, -1)$).

Corollary 129.2 (for M_B , [corollary 11.5](#) [04]). *For the models M_α , “unital additive homomorphism” and “elementary embedding” describe the same maps. Injectivity, order preservation, and elementarity need not be imposed separately.*

129.1 The valuation spectrum

For $0 \neq h \in H_\alpha$, put

$$v(h) = \min\{i < \alpha : h_i \neq 0\}.$$

Two nonzero elements have the same value exactly when their absolute values are Archimedean equivalent, meaning that each is bounded by a standard positive integer multiple of the other. Indeed their first nonzero rational coefficients have a finite rational ratio. If $v(x) < v(y)$, then $|x| > n|y|$ for every positive integer n .

An order-preserving linear embedding $T : H_\alpha \rightarrow H_\beta$ therefore induces a strictly increasing map

$$p : \alpha \longrightarrow \beta, \quad p(i) = v(T(e_i)), \quad v(T(h)) = p(v(h)). \quad (129.2)$$

The last equality follows from Archimedean equivalence. In particular, an embedding exists only if $\alpha \leq \beta$. If that ordinal inequality holds, inserting zeros outside an initial copy of α gives a continuous ordered linear embedding. Thus elementary embeddability among this family is exactly ordinal comparison.

Corollary 129.3 (An ordinal hierarchy on a fixed topological carrier; extends [theorem 35.4](#)). For $\omega \leq \alpha, \beta < \omega_1$, all the carriers M_α are homeomorphic to Baire space, but

$$M_\alpha \text{ elementarily embeds into } M_\beta \iff \alpha \leq \beta.$$

The same equivalence holds when the embedding is required to be continuous or Borel. Two members are isomorphic exactly when their indices agree.

[write] Its last sentence is [theorem 35.4](#) (source 07; source 06's [theorem 35.9](#)), proved there by the Archimedean classes; the embeddability equivalence is new.

130 Borel maps, finite dependence, and matrix normal forms

130.1 A countable-index Baire lemma

Lemma 130.1. *A subgroup with the Baire property and countable index in a Polish group is open. In particular a Borel subgroup of countable index is open, and a G_δ subgroup of a Polish group is closed.*

Proof. Countably many cosets cover the group. The Baire category theorem implies that the subgroup is nonmeagre. A set with the Baire property that is nonmeagre is comeagre in some nonempty open set. The usual Baire argument then shows that its difference set contains a neighborhood of the identity: sufficiently small translates of that open set overlap, and the two comeagre subsets have a common point there. For a subgroup the difference set is the subgroup itself, so it is open.

For the final assertion, a G_δ subgroup K is dense and comeagre in its closure L , which is a Polish group. Every translate $x + K$, $x \in L$, is also comeagre in L and intersects K . Hence $x \in K - K = K$, so $K = L$. $\square$

Lemma 130.2 (Continuous functionals). *A $\mathbb{Q}$ -linear functional $\ell : \mathbb{Q}^I \rightarrow \mathbb{Q}$, where I is countable and all factors are discrete, is continuous if and only if it depends on finitely many coordinates. If it is Borel, it is continuous.*

Proof. For continuity, the inverse image of zero contains a basic open subgroup in which finitely many coordinates vanish. The functional therefore factors through that finite coordinate projection. Conversely every finite rational linear combination of coordinates is continuous. If ℓ is Borel, its kernel is a Borel subgroup of countable index, since $\mathbb{Q}$ is countable. Apply [lemma 130.1](#). $\square$

Theorem 130.3 (Borel equals continuous for arithmetic embeddings). *For an elementary embedding $f : M_\alpha \rightarrow M_\beta$ and its coefficient map T , the following are equivalent:*

$$f \text{ is Borel}; \quad T \text{ is Borel}; \quad T \text{ is continuous}; \quad f \text{ is continuous}.$$

Proof. The coefficient positive and negative cones are Borel by [\(128.3\)](#). If f is Borel, define $T(h)$ by taking the coefficient part of $f(h, 0)$ when $h \geq 0$, and the negative of the coefficient part of $f(-h, 0)$ otherwise. This is a Borel definition on two Borel pieces. Every coordinate of a Borel T is a Borel linear functional, hence continuous by [lemma 130.2](#). Product-topology continuity follows. A continuous T restricts to a continuous f , and any continuous f is Borel. $\square$

130.2 The exact matrices

Definition 130.4. A matrix $A = (a_{ji})_{j < \beta, i < \alpha}$ is row-finite if for each j only finitely many a_{ji} are nonzero. It has increasing positive pivots if there is a strictly increasing $p : \alpha \rightarrow \beta$ with

$$a_{ji} = 0 \quad (j < p(i)), \quad a_{p(i),i} > 0. \quad (130.1)$$

Theorem 130.5 (Matrix classification; for $\alpha = \beta = \omega$, [theorem 11.4](#) [04]). *The continuous elementary embeddings $M_\alpha \rightarrow M_\beta$ are exactly*

$$(h, n) \mapsto (Ah, n), \quad (Ah)_j = \sum_{i < \alpha} a_{ji} h_i, \quad (130.2)$$

where A is row-finite with increasing positive pivots. Each displayed sum is finite. The pivot map is uniquely determined by the embedding.

Proof. For a continuous T , [lemma 130.2](#) supplies each matrix row. The columns are $T(e_i)$. Their first nonzero entries obey (130.1) by (129.2), and positivity gives positive pivots.

Conversely let $h \neq 0$ have first nonzero coordinate i . All coordinates of Ah before $p(i)$ vanish: columns indexed below i are multiplied by zero, while those indexed at least i have no entry that early. At the pivot itself,

$$(Ah)_{p(i)} = a_{p(i),i} h_i.$$

Its sign is the sign of h_i . Thus A is a strictly ordered linear embedding, and row-finiteness gives continuity. Apply [theorem 129.1](#). $\square$

[write] For $\alpha = \beta = \omega$ this is source 04's [theorem 11.4](#), which is stronger there: it assumes no continuity ([theorem 133.1](#) below) and adds that the map is onto exactly when p is the identity.

130.3 Why the images are closed even at limit ranks

Lemma 130.6 (Finite ancestry). *Suppose each vertex i of a well-ordered index set has finitely many designated predecessors, all strictly below i . Then each vertex has only finitely many ancestors obtained by repeatedly taking designated predecessors.*

Proof. Form the tree of finite predecessor paths starting at the vertex. It is finitely branching. An infinite such tree would have an infinite branch by König's lemma, giving an infinite descending sequence in a well-order. Thus the tree and its set of labels are finite. $\square$

Theorem 130.7 (A continuous left inverse and a closed image; for $\alpha = \beta = \omega$, [corollary 11.5](#) [04]). *Every matrix embedding in [theorem 130.5](#) admits a continuous $\mathbb{Q}$ -linear left inverse $L : H_\beta \rightarrow H_\alpha$. Its group image is closed, and its cone image is a closed elementary submodel with inherited Polish topology.*

Proof. At row $p(i)$, columns indexed above i vanish. Define recursively

$$(Ly)_i = \frac{1}{a_{p(i),i}} \left(y_{p(i)} - \sum_{k < i} a_{p(i),k} (Ly)_k \right). \quad (130.3)$$

Each sum is finite. Although the recursion is indexed by an ordinal, [lemma 130.6](#) shows that each output coordinate ultimately depends on only finitely many input coordinates. Thus L is a continuous row-finite linear map. Transfinite induction in [\(130.3\)](#) gives $LA = \text{id}$.

Now AL is a continuous projection and

$$\text{im } A = \{y : ALy = y\} = \ker(\text{id} - AL),$$

which is closed in the Hausdorff group H_β . The cone image is $(\text{im } A \times \mathbb{Z}) \cap M_\beta$, because order is reflected. It is closed in the Polish space M_β , and L restricts to its continuous inverse parametrization. $\square$

The finite-ancestry argument is important. A naive sequence of transfinite row subtractions could alter a limit coordinate infinitely often. Row-finiteness alone does not justify that naive procedure. Equation [\(130.3\)](#) and the well-founded dependency argument supply the missing control.

[write] For $\alpha = \beta = \omega$ the recursion [\(130.3\)](#) is source 04's recovery formula [\(11.7\)](#), where no ancestry lemma is needed because every pivot row has finitely many predecessors, and the closed image is part of [corollary 11.5](#).

131 All elementary submodels and their Polish topology

Theorem 131.1 (Elementary submodels are coefficient subspaces; extends [theorem 11.2](#)). *The elementary submodels of M_α are precisely*

$$M_V = (V \overrightarrow{\times} \mathbb{Z})_{\geq 0}, \quad V \leq_{\mathbb{Q}} H_\alpha.$$

The model M_V is Polish in the inherited topology if and only if V is closed in H_α .

Proof. If V is a rational linear subspace, $V \overrightarrow{\times} \mathbb{Z}$ is a $\mathbb{Z}$ -group and its inclusion preserves and reflects congruences as well as order. Quantifier elimination makes its cone elementary.

Conversely let $N \leq M_\alpha$ and set $\Gamma = N - N$ inside G_α . If $a, b \in N$ and $a - b \geq 0$, the equation $b + x = a$ has a unique solution in the ambient cone. Elementarity puts that solution in N . Thus $N = \Gamma \cap M_\alpha$. The subgroup Γ contains $(0, n)$ for every standard integer n . Subtracting integer coordinates yields $\Gamma = V \times \mathbb{Z}$ for an additive subgroup $V \leq H_\alpha$. The unique standard division of $(v, 0)$ has remainder zero and quotient $(v/m, 0)$. For $v \geq 0$ elementarity puts this quotient in N ; negatives follow by subtraction. Hence V is divisible and therefore a rational linear subspace.

If V is closed, M_V is closed in M_α and is Polish. Conversely, if M_V is Polish, its integer-coordinate-zero slice is clopen in M_V and is homeomorphic to $V^{\geq 0}$. Thus $V^{\geq 0}$ is a Polish subspace of the Polish space H_α , so it is G_δ . Its negative is also G_δ , and

$$V = V^{\geq 0} \cup (-V^{\geq 0})$$

is G_δ . By [lemma 130.1](#) the subgroup V is closed. $\square$

Polishness here cannot be replaced by merely admitting some abstract Polish presentation. A nonclosed V might be algebraically isomorphic to a closed space, but the inherited topology is the issue being classified.

[write] For the real model this is source 05's [theorem 11.2](#), with the same proof; there only two elementary submodels are Polish, here one for each closed subspace of H_α .

131.1 The finite-support annihilator

Write $H_\alpha^\circ = \mathbb{Q}^{(\alpha)}$ for the finite-support dual, with pairing

$$\langle c, h \rangle = \sum_{i < \alpha} c_i h_i.$$

For $V \leq H_\alpha$ define $V^\perp = \{c \in H_\alpha^\circ : \langle c, v \rangle = 0 \text{ for all } v \in V\}$.

Lemma 131.2 (Cylinder separation). *For every linear subspace $V \leq H_\alpha$,*

$$\text{cl}(V) = (V^\perp)^\perp.$$

Proof. Each finite-support functional is continuous, so its kernel contains $\text{cl}(V)$ whenever it contains V . For the reverse inclusion, if $h \notin \text{cl}(V)$ then a finite coordinate set F separates h from V : $h \upharpoonright F \notin \pi_F(V)$. Finite-dimensional linear algebra supplies a functional on $\mathbb{Q}^F$ vanishing on $\pi_F(V)$ but not on $h \upharpoonright F$. Extend it by zero outside F . $\square$

131.2 Canonical ordered normal form

Theorem 131.3 (Pivot normal form of a closed subspace). *Let $V \leq H_\alpha$ be closed. There is a unique subset $S \subseteq \alpha$, its valuation spectrum*

$$S = \{v(x) : 0 \neq x \in V\},$$

with the following description. For each $i \notin S$ there are unique finitely many coefficients $b_{is} \in \mathbb{Q}$, $s \in S$, $s < i$, such that

$$V = \left\{ x : x_i = \sum_{\substack{s \in S \\ s < i}} b_{is} x_s \text{ for every } i \notin S \right\}. \quad (131.1)$$

The coordinates $(x_s)_{s \in S}$ are arbitrary. The resulting parametrization $\mathbb{Q}^S \rightarrow V$ is an ordered topological linear isomorphism.

Proof. Let ℓ_i be the coordinate functional restricted to V . Scan the ordinal α in increasing order, selecting i precisely when ℓ_i is not in the algebraic span of the previously selected functionals. Let S be the selected set. The selected functionals are linearly independent. Every unselected ℓ_i is a finite linear combination of selected ℓ_s , $s < i$; the coefficients are unique by independence. This proves the necessary equations in (131.1).

Given arbitrary $c \in \mathbb{Q}^S$, define $x_s = c_s$ at selected coordinates and define the remaining x_i by those finite equations. We claim $x \in V$. Any finite-support functional annihilating V becomes, after substituting these equations, a finite linear combination of the independent selected functionals which vanishes on V . Every coefficient in that combination is therefore zero, so the original functional also vanishes at x . Since V is closed, lemma 131.2 gives $x \in V$.

All coordinates of the parametrization are finite linear combinations of the c_s , so it is continuous; its inverse is coordinate restriction. If c first becomes nonzero at $s \in S$, all earlier unselected coordinates are combinations of earlier, zero, selected coordinates. Thus the first nonzero coordinate of x is s and has value c_s . The parametrization preserves order. It also shows that the selected set is exactly the valuation spectrum, making S intrinsic. $\square$

This proof uses finite linear duality to handle limit coordinates. It does not select successive vectors and hope that an infinite elimination process converges at the limit.

Corollary 131.4 (Isomorphism types). *Every inherited-Polish elementary submodel of M_α is order-topologically and arithmetically isomorphic to a unique M_β , $\beta \leq \alpha$. All such β occur. For $\alpha = \omega$, a Polish elementary submodel is either a countable discrete M_k for $k < \omega$, or an elementary self-copy of M_ω .*

Proof. Use $\beta = \text{otp}(S)$ in [theorem 131.3](#) and [theorem 129.1](#). Uniqueness follows from the Archimedean rank of the coefficient group. A coordinate subspace supported on an initial segment β realizes every $\beta \leq \alpha$. $\square$

132 Orbits, quotients, and the complete submodel lattice

Theorem 132.1 (Ambient conjugacy and quotient topology). *Let V, W be closed coefficient subspaces of H_α .*

- (1) *A continuous ordered linear automorphism sends V to W if and only if their pivot subsets agree. The same criterion holds for arbitrary ambient arithmetic automorphisms of M_α sending M_V onto M_W .*
- (2) *If S is the pivot subset of V , then as topological vector spaces*

$$H_\alpha/V \cong \mathbb{Q}^{\alpha \setminus S}.$$

- (3) *M_V is clopen in M_α exactly when $\alpha \setminus S$ is finite. Otherwise it is closed and nowhere dense.*

Proof. Let E_S be the coordinate subspace supported on S . Using the coefficients in [\(131.1\)](#), define

$$(U_V x)_s = x_s \ (s \in S), \quad (U_V x)_i = x_i + \sum_{s < i, s \in S} b_{is} x_s \ (i \notin S).$$

This is row-finite lower triangular with diagonal one. Its inverse subtracts the same finite combinations, since selected coordinates are fixed. It is a continuous ordered automorphism and sends E_S onto V . If V, W have the same S , use $U_W U_V^{-1}$.

Every ordered automorphism induces an order automorphism of the ordinal rank α , and this must be the identity. Therefore it preserves valuation spectra even if no regularity is assumed. This proves the converse assertion about orbits.

Conjugating by U_V reduces the quotient to H_α/E_S , which is the product on the complementary coordinates. The subspace E_S is open precisely when only finitely many coordinates are required to vanish. If it is not open it has empty interior, as does any closed subgroup which is not open.

For the positive cone, openness of V implies openness of M_V . Conversely suppose M_V is open. At $(0, 0)$ it contains a neighborhood specified by finitely many coefficient coordinates; thus the nonnegative half of the corresponding basic open subgroup is contained in V . Negating shows that the entire subgroup lies in V , so V is open. To prove nowhere density when the missing set is infinite, conjugate to E_S . Given a point of M_{E_S} and a neighborhood prescribing finitely many coordinates F , choose $j \in (\alpha \setminus S) \setminus F$. Adding the positive coefficient vector e_j preserves the cone and the prescribed coordinates, but leaves E_S . Thus no point of M_{E_S} is interior, and the nonopen closed submodel is nowhere dense. $\square$

[write] For the image of a self-embedding of M_ω , the dichotomy of (3) is source 04's [corollary 11.5](#); here it holds for every closed elementary submodel and every countable α .

Remark 132.2. If $\alpha \setminus S$ is infinite countable, the quotient is a *full product* of copies of $\mathbb{Q}$. Its algebraic dimension over $\mathbb{Q}$ is $\mathfrak{c}$, not $\aleph_0$. Counting the missing pivots as though they were a Hamel basis would give an incorrect codimension.

Corollary 132.3. *For $\alpha = \omega$ there are exactly $\mathfrak{c}$ ambient automorphism orbits of Polish elementary submodels, and already $\mathfrak{c}$ orbits of uncountable elementary self-copies. In the latter case they are all abstractly isomorphic, but need not be conjugate inside the ambient model.*

Proof. The pivot subsets are exactly the subsets of ω ; the infinite ones give self-copies. There are $\mathfrak{c}$ of each. All automorphisms of M_ω will be shown continuous in [section 133](#), so the same count holds without a topological qualification. $\square$

Theorem 132.4 (The lattice). *The inherited-Polish elementary submodels of M_α form a complete lattice. Under $V \mapsto M_V$ its operations are*

$$\bigwedge_t M_{V_t} = M_{\bigcap_t V_t}, \quad \bigvee_t M_{V_t} = M_{\text{cl}(\sum_t V_t)}.$$

The lattice is order anti-isomorphic to the lattice of all linear subspaces of the finite-support space $\mathbb{Q}^{(\alpha)}$, by taking annihilators.

Proof. Intersections of closed linear subspaces are closed, and the smallest closed linear subspace containing a family is the closure of its algebraic sum. The correspondence with elementary submodels follows from [theorem 131.1](#).

The annihilator map reverses inclusion and is injective on closed subspaces by [lemma 131.2](#). If $W \leq \mathbb{Q}^{(\alpha)}$, then $W^\perp$ is closed. If $c \notin W$, ordinary linear algebra on $\mathbb{Q}^{(\alpha)}/W$ gives a functional which vanishes on W but not on c . Every algebraic functional on the finite-support space is evaluation against a point of the full product $\mathbb{Q}^\alpha$. Thus $(W^\perp)^\perp = W$, proving surjectivity and the claimed anti-isomorphism. $\square$

This answers the repository's elementary-submodel-lattice question for the displayed family. It does not classify the lattices of arbitrary Polish presentations of arbitrary Presburger models.

[write] The question is Part I's [question 14.9](#) (source 05's); a dated note after it records this answer.

133 The first threshold: automatic continuity fails at $\omega + 1$

Theorem 133.1 (Automatic continuity through ω ; for $\alpha = \omega$, [theorem 11.4](#) [04]). *Every elementary self-embedding of M_α is continuous when $\alpha \leq \omega$. If $\alpha < \omega$, every such embedding is an automorphism. For $\alpha = \omega$, all embeddings are exactly the increasing-positive-pivot matrices; row-finiteness follows automatically from their pivot condition.*

Proof. Finite products are discrete. A strictly increasing self-map of a finite ordinal is the identity, so the corresponding finite triangular matrix has nonzero diagonal and is invertible.

Now let $T : H_\omega \rightarrow H_\omega$ be any ordered linear embedding, without a regularity assumption. Its pivot map $p : \omega \rightarrow \omega$ is strictly increasing and hence $p(n) \geq n$. If h vanishes in its first n coordinates, then either $h = 0$ or $v(h) \geq n$. Equation (129.2) implies that $T(h)$ vanishes in its first $p(n)$ coordinates. These finite-coordinate kernels form the neighborhood basis of zero, proving continuity. The matrix description follows from [theorem 130.5](#). In any output row j only columns with $p(i) \leq j$ can contribute; there are finitely many of them. $\square$

[write] For $\alpha = \omega$ this is the automatic continuity of source 04's [theorem 11.4](#) and [corollary 11.5](#), proved there by the same filtration argument ([11.6](#)); the finite case is new to the report.

133.1 A functional invisible on every finite-support vector

Let

$$D_B = \mathbb{Q}^\omega, \quad D_B^{\text{fin}} = \mathbb{Q}^{(\omega)}.$$

The finite-support subspace is dense and proper. Choose a nonzero linear functional

$$\varphi : D_B \longrightarrow \mathbb{Q}, \quad \varphi(D_B^{\text{fin}}) = 0. \quad (133.1)$$

Such a functional is obtained by choosing a nonzero functional on D_B/D_B^{fin} and composing with the quotient map. This is an algebraic use of Choice. A continuous functional vanishing on a dense subspace would vanish everywhere, so φ is discontinuous. By [lemma 130.2](#) it is not Borel.

[write] **Choice.** The functional φ is the only nonconstructive object of this part. Everything built from it, [theorem 133.3](#), the value 2^c in [theorem 133.4](#), [proposition 133.5](#), [theorem 134.2](#), [example 136.4](#) and the negative halves of clauses (5) and (6) of [theorem 127.1](#), is proved in ZFC. The next remark shows that choice cannot be removed altogether.

Remark 133.2 (Discontinuous self-embeddings need a set without the Baire property [merge]). Work in ZF with dependent choice. If some elementary embedding $f : M_\alpha \rightarrow M_\beta$ between members of the family (α, β countable) is not continuous, then some subset of a Polish space does not have the Baire property. Consequently, in a model of ZF with dependent choice in which every set of reals has the Baire property, which exists if ZF is consistent [96], every elementary embedding between members of the family is continuous, given by a matrix as in [theorem 130.5](#) and has closed image ([theorem 130.7](#)); there the thresholds of clauses (5) and (6) of [theorem 127.1](#) do not occur.

Proof. By [theorem 129.1](#), $f = (T, \text{id})$ with T an ordered $\mathbb{Q}$ -linear embedding, and f is continuous if T is. If T is not continuous for the product topology, some coordinate functional $\ell = \pi_j \circ T : H_\alpha \rightarrow \mathbb{Q}$ is not continuous. Its kernel is a subgroup of countable index of the Polish group H_α , because $\mathbb{Q}$ is countable. If the kernel had the Baire property it would be open by [lemma 130.1](#), whose proof uses only countably many choices, and then ℓ , constant on the cosets of an open subgroup, would be continuous. Hence the kernel lacks the Baire property, and H_α is homeomorphic to Baire space or countable discrete. In the model described, every coordinate functional is therefore continuous, and [theorems 130.5](#) and [130.7](#) apply. $\square$

This bears on source 16's [question 138.3](#): a non-Borel ordered automorphism of $M_{\omega+1}$ yields a set of reals without the Baire property. Whether a weaker hypothesis than choice produces one, and whether the converse holds, is not settled here. The remark was written for this report and has not been independently reviewed.

Theorem 133.3 (A non-Borel elementary automorphism). *For every countable $\alpha > \omega$, define*

$$(\text{Sh}_\varphi x)_i = \begin{cases} x_\omega + \varphi(x \upharpoonright \omega), & i = \omega, \\ x_i, & i \neq \omega. \end{cases} \quad (133.2)$$

This is an ordered linear automorphism of H_α , with inverse $\text{Sh}_{-\varphi}$. Its induced automorphism of M_α is neither continuous nor Borel. It fixes every finite-support coefficient vector.

Proof. Linearity and the inverse formula are immediate. If the first nonzero coordinate of x is below ω , that coordinate is unchanged and still decides the sign. If all coordinates below ω vanish, the added functional is zero, so every coordinate of x is unchanged. Thus the first nonzero coordinate and its sign are preserved in every case.

The coordinate- ω functional of Sh_φ contains the noncontinuous summand $\varphi(x \upharpoonright \omega)$, so Sh_φ is discontinuous. [Theorem 130.3](#) makes the induced cone automorphism non-Borel. Finally [\(133.1\)](#) proves the finite-support assertion. $\square$

Together [theorems 133.1](#) and [133.3](#) prove the exact first threshold. The shear preserves the Hahn valuation v , and is continuous for the associated finer valuation topology. There is no contradiction: in the product topology, no neighborhood can demand that all the infinitely many coordinates below ω vanish.

133.2 The size of the discontinuous automorphism group

Theorem 133.4 (Exact cardinalities). *For countable α ,*

$$|\text{Aut}(M_\alpha)| = \begin{cases} 1, & \alpha = 0, \\ \aleph_0, & 0 < \alpha < \omega, \\ \mathfrak{c}, & \alpha = \omega, \\ 2^{\mathfrak{c}}, & \alpha > \omega. \end{cases}$$

For every infinite countable α , exactly $\mathfrak{c}$ automorphisms are continuous, equivalently Borel.

Proof. Finite-rank automorphisms are finite triangular rational matrices with positive diagonal, giving a countably infinite set in nonzero rank. At rank ω , matrices supply at most $\mathfrak{c}$ maps and arbitrary positive rational diagonal sequences supply at least $\mathfrak{c}$.

The quotient $D_{\mathbb{B}}/D_{\mathbb{B}}^{\text{fin}}$ has cardinality $\mathfrak{c}$: $D_{\mathbb{B}}$ has that cardinality and each coset is countable. Over a countable field an infinite-dimensional vector space has the cardinality of its Hamel basis, so this quotient has dimension $\mathfrak{c}$. Its linear functionals to $\mathbb{Q}$ number $\aleph_0^{\mathfrak{c}} = 2^{\mathfrak{c}}$. Distinct functionals give distinct shears. There are at most $\mathfrak{c}^{\mathfrak{c}} = 2^{\mathfrak{c}}$ maps on a set of cardinality $\mathfrak{c}$, which proves equality for $\alpha > \omega$.

Continuous maps on a separable space into a Hausdorff space of cardinality $\mathfrak{c}$ are determined by their values on a countable dense subset, so there are at most $\mathfrak{c}$ of them. Positive rational diagonal sequences again attain this bound. Use [theorem 130.3](#) for the Borel assertion. $\square$

[write] For $\alpha = \omega$, source 04's [corollary 11.5](#) identifies the automorphisms as the infinite lower triangular rational matrices with positive diagonal; the count $\mathfrak{c}$ and the counts for $\alpha > \omega$ are new. The value $2^{\mathfrak{c}}$ uses choice (note above). The dimension count for $\mathbb{Q}^\omega/\mathbb{Q}^{(\omega)}$ is the rational, full-product counterpart of the collection's `rvs:eu:thm:fullcodim` and `rvs:eu:cor:continuum` (real-vector-space-structure: the finite-support part of a real Hahn field has codimension equal to its dimension), and that report's subsection "Product topology versus algebraic dimension" contrasts, as the paragraph after [theorem 133.3](#) does here, the coefficientwise topology, in which the finite-support vectors are dense, with the valuation topology.

133.3 A clopen submodel can become dense under an algebraic automorphism

Proposition 133.5. *For $\alpha > \omega$, the shear Sh_φ sends the clopen coefficient subspace $\{x : x_\omega = 0\}$ to*

$$W_\varphi = \{y : y_\omega = \varphi(y \upharpoonright \omega)\}, \quad (133.3)$$

which is dense, proper, and non-Borel. The corresponding elementary cone submodel is also dense and non-Borel, and is not Polish in its inherited topology.

Proof. The image equation is immediate. A nonzero rational functional onto $\mathbb{Q}$ is surjective. Every fiber of φ is dense, since it contains a translate of the dense subspace D_B^{fin} . Thus any finite cylinder and any prescribed value of coordinate ω meet W_φ . It is proper because the value at ω is determined by the preceding block.

The subgroup W_φ has countable index: it is the kernel of $y \mapsto y_\omega - \varphi(y \upharpoonright \omega)$, onto $\mathbb{Q}$. If it were Borel, [lemma 130.1](#) would make it open and hence closed, contradicting its being dense and proper.

For density in the cone, any finite cylinder meeting the cone contains one whose positivity is decided in the first ω coordinates, as in [proposition 128.3](#). The prescribed functional value can still be attained there: first choose a vector in the desired fiber, then change finitely many coordinates to meet the cylinder and force a positive first free coordinate. These changes do not change φ . If the cone submodel were Borel, its integer-zero slice and the negative of that slice would make W_φ Borel. Finally apply [theorem 131.1](#). $\square$

For $\alpha = \omega + 1$, this does *not* yet give a nonclosed image of a self-embedding: the coordinate hyperplane has rank ω , not $\omega + 1$. That distinction leads to the second threshold.

134 The second threshold: nonclosed self-images start at $\omega \cdot 2$

Theorem 134.1 (Closed images below the second infinite block). *If $\alpha < \omega \cdot 2$, every elementary self-embedding of M_α has closed image, even when the embedding is non-Borel.*

Proof. For $\alpha \leq \omega$ use [theorems 130.7](#) and [133.1](#). Otherwise $\alpha = \omega + n$ with $0 < n < \omega$. Let T be the ordered coefficient embedding and p its increasing rank map. The last n positions must be fixed by p : an increasing self-map of an ordinal sends i to at least i , and there is no room to move any of the final finitely many positions to the right. Also every finite rank maps below ω , since infinitely many increasing finite-rank images cannot fit in a finite terminal block.

Let $K_\alpha = \{x : x \upharpoonright \omega = 0\} \cong \mathbb{Q}^n$. The restriction $T \upharpoonright K_\alpha$ is finite triangular with nonzero diagonal, hence maps K_α onto itself. The quotient map induced by T on $H_\alpha/K_\alpha \cong H_\omega$ is an ordered embedding B . By the first-threshold theorem, B is continuous and has closed image. Moreover

$$\text{im } T = \pi^{-1}(\text{im } B),$$

where $\pi : H_\alpha \rightarrow H_\omega$ is the coordinate projection. One inclusion is immediate; for the other, realize the desired quotient coordinate by an image under T and correct the remaining terminal coordinates using the surjectivity of $T \upharpoonright K_\alpha$. Thus $\text{im } T$ is closed. Its cone is closed in M_α . $\square$

Theorem 134.2 (A dense proper non-Borel elementary self-copy). *For every countable $\alpha \geq \omega \cdot 2$ there is an elementary self-embedding of M_α whose image is dense, proper, and non-Borel. In particular its image is not Polish in the inherited topology.*

Proof. Use the functional φ from (133.1). Define $T : H_\alpha \rightarrow H_\alpha$ by

$$\begin{aligned} (Tx)_k &= x_k & (k < \omega), \\ (Tx)_\omega &= \varphi(x \upharpoonright \omega), \\ (Tx)_{\omega+k+1} &= x_{\omega+k} & (k < \omega), \\ (Tx)_i &= x_i & (\omega \cdot 2 \leq i < \alpha). \end{aligned} \tag{134.1}$$

This is linear and injective. If the first nonzero input coordinate lies in the first block, its position and sign are unchanged. If it is $\omega + k$, the first block is zero and the new functional coordinate is zero, so the first nonzero output is at $\omega + k + 1$, with the same coefficient. If it lies at or after $\omega \cdot 2$, all earlier output coordinates vanish and that coordinate is unchanged. Therefore T is an ordered embedding.

Its image is exactly the graph space W_φ of (133.3), with all coordinates other than ω unconstrained. Indeed the shift of the second block can be reversed on every coordinate after ω , and the later blocks are fixed. Proposition 133.5 proves all the asserted topological and Borel properties. The cone map is elementary by theorem 129.1. $\square$

The two thresholds answer different questions. At $\omega + 1$, an automorphism can destroy the topological regularity of a submodel while still mapping the entire ambient structure onto itself. At $\omega \cdot 2$, there is enough space after the first limit coordinate to shift an infinite tail and make the graph itself a self-copy. Theorems 134.1 and 134.2 establish that this extra space is necessary.

135 The intersection of self-copies and clopen approximation

Every ordinal α has a unique expression

$$\alpha = \lambda + n,$$

where $n < \omega$ and λ is zero or a limit ordinal. Define its terminal coefficient tail

$$K_\alpha = \{x \in H_\alpha : x_i = 0 \text{ for } i < \lambda\} \cong \mathbb{Q}^n.$$

Theorem 135.1 (The unavoidable terminal tail; extends theorem 11.7). *The intersection of the images of all elementary self-embeddings of M_α is exactly*

$$(K_\alpha \overrightarrow{\times} \mathbb{Z})_{\geq 0}.$$

The same intersection is obtained using only continuous embeddings with clopen image. Thus the intersection is the standard copy of $\mathbb{N}$ exactly when α is zero or a nonzero limit ordinal. For $0 < \alpha < \omega$ it is the entire model.

Proof. The final n ranks are fixed by every increasing self-map of α , as in theorem 134.1. The restriction of any coefficient embedding to K_α is finite triangular and onto, so every image contains this tail.

If $s < \lambda$, deleting the coordinate s from α leaves an ordered set of order type α . Explicitly, $s + \omega \leq \lambda$; shift $s + k$ to $s + k + 1$ for $k < \omega$ and fix all coordinates outside $[s, s + \omega)$. This gives an order isomorphism from α onto $\alpha \setminus \{s\}$. Repeated deletion proves the corresponding finite-removal assertion. The coordinate hyperplane $x_s = 0$ is therefore a clopen ordered self-copy of H_α , and its cone is a clopen elementary self-copy. Intersect these hyperplanes over all $s < \lambda$. Exactly K_α remains. If $\lambda = 0$ there are no such coordinates, and the preceding surjectivity argument already gives the assertion. $\square$

[write] For $\alpha = \omega$ the terminal tail is 0 and the intersection is the standard copy, as in source 05's [theorem 11.7](#), which obtains it from the iterates of one shift; the theorem extends that to all self-embeddings and every countable α .

Theorem 135.2 (Exactly which closed submodels are intersections of clopen self-copies). *Let $V \leq H_\alpha$ be closed. Then M_V is an intersection of clopen elementary self-copies of M_α if and only if $K_\alpha \subseteq V$. When this condition holds, a descending countable sequence of such self-copies suffices.*

Proof. Necessity follows from [theorem 135.1](#). For sufficiency, enumerate the countably many coordinates and let F_m be the first m of them. Put

$$V_m = \pi_{F_m}^{-1}(\pi_{F_m}(V)).$$

These are descending clopen linear subspaces and $\bigcap_m V_m = V$ by closedness. They contain K_α . Each V_m has only finitely many missing pivots, because it contains the subgroup vanishing on F_m . None of the missing pivots is in the terminal tail, since that tail is contained in V_m . Deleting finitely many nonterminal positions leaves the order type α unchanged. By [theorem 131.3](#), each V_m is therefore a continuous ordered self-copy of H_α . Their cones have the required properties. $\square$

In particular every Polish elementary submodel of M_ω is an intersection of descending clopen self-copies. This includes finite-rank submodels and the standard copy. At $\omega + 1$, the standard copy cannot arise this way: every self-copy contains the last rational scale.

136 Concrete examples and what finite calculations can certify

Example 136.1 (A clopen shift and a nowhere-dense self-copy; = [example 11.6](#) [04]). In H_ω , the shift

$$(Tx)_0 = 0, \quad (Tx)_{j+1} = x_j$$

has pivot map $p(j) = j + 1$ and image $x_0 = 0$, which is clopen. The embedding

$$(Ux)_{2j} = x_j, \quad (Ux)_{2j+1} = 0$$

has pivot map $p(j) = 2j$ and closed nowhere-dense image. Both induce elementary self-copies of the same Baire-space arithmetic model. The difference is the finiteness or infinitude of the missing pivot set.

Example 136.2 (A non-coordinate closed submodel). Let

$$V = \{x \in \mathbb{Q}^\omega : x_{2j+1} = x_{2j} + 2x_{2j-2} \ (j \geq 0)\},$$

where $x_{-2} = 0$. Its pivot set is the even coordinates. The map

$$c \mapsto x, \quad x_{2j} = c_j, \quad x_{2j+1} = c_j + 2c_{j-1}, \quad c_{-1} = 0,$$

is a continuous ordered parametrization. The model M_V is a nowhere-dense elementary self-copy. Its quotient coefficient space is another full rational product. Replacing the right-hand side by zero gives a conjugate submodel, because the pivot set is unchanged.

Example 136.3 (A finite projection normal form). Consider the equations in five rational coordinates

$$x_1 = 2x_0, \quad x_3 = x_0 - x_2,$$

with x_0, x_2, x_4 free. The pivot subset is $\{0, 2, 4\}$. The ambient triangular automorphism

$$(x_0, x_1, x_2, x_3, x_4) \mapsto (x_0, x_1 + 2x_0, x_2, x_3 + x_0 - x_2, x_4)$$

sends the corresponding coordinate subspace to this graph. Its inverse subtracts the two indicated expressions. This is the finite-coordinate version of [theorem 132.1](#).

Example 136.4 (Two independent infinite stages). At rank $\omega + 1$, the shear $(x, z) \mapsto (x, z + \varphi(x))$ is onto, although non-Borel. At rank $\omega \cdot 2$, write the two blocks as (x, y) . The map

$$(x, (y_0, y_1, \dots)) \mapsto (x, (\varphi(x), y_0, y_1, \dots))$$

is an elementary self-embedding onto a dense graph. The extra infinite block makes room for insertion; one terminal coordinate does not.

The package includes an exact rational script checking finite instances of the triangular inverse, the pivot parametrization, order preservation, and standard quotient and remainder. Every calculation uses rational arithmetic. These checks are useful for detecting indexing or sign errors in examples.

They cannot verify the existence of φ , the transfinite normal-form theorem, Baire category arguments, or quantifier elimination on infinite structures. In fact every wild shear above agrees with the identity on all finite-support vectors. No test suite confined to such vectors can distinguish it from the identity. The distinction is established by the proof and its hypotheses.

137 Realization in the omnific integers and a formalization route

137.1 Set-sized fixed-support normal forms

The ProveIt surreal development gives an additional concrete interpretation of the coefficient models [\[95\]](#). For $i < \alpha$, regard the ordinal $i + 1$ as a positive surreal and put

$$\gamma_i = (i + 1)^{-1}.$$

Then $\gamma_i > 0$ and $\gamma_i > \gamma_j$ whenever $i < j$. The exponent set is reverse well-ordered. Conway normal forms therefore give a well-defined map

$$\iota_\alpha(h, n) = \sum_{i < \alpha} h_i \omega^{\gamma_i} + n. \quad (137.1)$$

Zero coefficients are omitted. Normal-form addition is coefficientwise, and the leading term decides the order. Thus ι_α is an additive ordered embedding. Every exponent in its nonconstant part is positive and the constant term is an ordinary integer, so its values are omnific integers in the standard normal-form description [\[91, 95\]](#).

[write] For infinite α the map ι_α is J_α of [theorem 40.1](#) (source 07), with the same exponents $1/(i + 1)$, and for $\alpha = \omega$ it is Part I's ι_B ([theorem 9.1](#)); the set-size argument of the next paragraph is source 07's ([section 40](#)). Source 16 adds the finite α . [\[95\]](#) is the Surreal project's README.

The coefficient topology is transported along ι_α . It is not being identified with a topology on the entire proper class **No**. Every fixed image is a set: its domain $\mathbb{Q}^\alpha \times \mathbb{Z}$ is a set, and Replacement collects its normal-form values. Taking the supremum of their birthdays

supplies a set ordinal bounding the image. No collection of all surreal subclasses or class-sized matrix algebra is required.

This realization preserves the additive ordered structure used in the paper. It does not assert ring closure of the fixed exponent support. Matrix maps and wild shears should accordingly be read as additive arithmetic maps of these fragments.

137.2 Why the limit-coordinate obstruction is meaningful here

At rank $\omega + 1$, the monomial indexed by ω lies after all the monomials with finite indices. The shear changes its coefficient by a linear functional of the entire earlier infinite coefficient block. Every finite sum of monomials is fixed, yet some infinite normal forms move. This is an explicit instance of the distinction between ordinary linearity and strong additivity in Hahn-type settings [87, 88, 89].

At rank ω , every coordinate has only finitely many predecessors. The combination of order preservation and the full product structure forces the finite dependence proved above. At rank $\omega + 1$, valuation preservation alone no longer forces product-topology continuity.

137.3 A tractable proof-assistant decomposition

The repository's existing Presburger decision procedure works over ordinary integers [94]. Its correctness should not be silently treated as a formalization of arbitrary nonstandard $\mathbb{Z}$ -groups. A suitable extension has the following modules.

1. Define $\mathbb{Z}$ -groups with standard division and prove generic Presburger quantifier elimination. The periodic predicates in the Cooper step must be invariant under complete congruence classes, not merely under a single standard successor shift.
2. Prove the positive-cone/group-completion dictionary and the maximal divisible subgroup identity. These isolate the arithmetic content of [theorems 129.1](#) and [131.1](#).
3. Formalize lexicographic products on a well-ordered index type, their valuation spectrum, and row-finite linear maps. The main algebraic certificate is the positive pivot condition.
4. Formalize finite-support dual separation and the greedy coordinate-functional construction. This avoids uncontrolled transfinite row operations.
5. Add the Polish G_δ and Baire-category lemmas, then derive the topology of the cone and closed subgroup results.
6. Treat the non-Borel constructions in a separate module whose explicit hypothesis is a nonzero functional annihilating the finite-support subspace. Prove the shear and graph theorems from that hypothesis before proving its existence using a basis theorem.

This route keeps the constructive matrix calculations separate from the Choice-dependent existence of wild maps. The present package contains ordinary proofs and executable examples, not completed Lean or Rocq implementations of these modules.

[write] Module 1 is the porting requirement of [section 13.5](#), already named by sources 05 and 04: the repository's Cooper lemmas assume invariance under one standard shift and are stated over the integers. No module of this list exists in the repository.

138 Further research questions and proposed next steps

The following questions are deliberately more precise than a request for a general classification of all Polish arithmetic. Each states what has been settled here and what is still missing. Except where the original source is named, they are proposed questions, not certifications that the literature contains no answer.

Question 138.1 (Effective normal forms). For computable row-finite presentations of closed subspaces of $\mathbb{Q}^\omega$, under which effective hypotheses can the pivot set, the reduced coefficients, and a continuous left inverse be computed? The proof uses finite-dimensional linear dependence, but an arbitrary closed-set code need not provide the needed dependence information. Determine the exact oracle or limit-computation complexity.

Question 138.2 (Computable ordinal ranks). For computable ordinals $\alpha > \omega$, give effective versions of the finite-ancestry argument from a row-finite triangular matrix code. A mathematical well-order guarantees termination of the dependency recursion; a useful certified implementation should expose that well-foundedness and quantify the cost of inverse rows.

Question 138.3 (Weak choice and the first threshold). What weak choice principle suffices to produce a nonzero $\mathbb{Q}$ -linear functional on $\mathbb{Q}^\omega / \mathbb{Q}^{(\omega)}$? Does the existence of a non-Borel ordered automorphism of $M_{\omega+1}$ characterize a familiar principle over a choiceless base? The shear theorem reduces one direction to a very specific linear-functional existence statement. This connects directly to Glazer's interest in choice principles.

[write] [Remark 133.2](#) [merge] gives one direction: a non-Borel ordered automorphism of $M_{\omega+1}$ yields a set of reals without the Baire property, so ZF with dependent choice does not prove that one exists, if ZF is consistent. The rest of the question is open.

Question 138.4 (All algebraic endomorphisms at higher ranks). For $\alpha > \omega$, classify the noncontinuous ordered embeddings by successor and limit blocks. The continuous maps are completely described here, and the two thresholds are exact, but the complete algebraic endomorphism monoid at ranks such as ω^2 remains to be analyzed in a usable normal form.

[write] This is what remains of the endomorphism clause of Part I's [question 14.6](#) for the groups $\mathbb{Q}^\alpha$: the continuous order-preserving endomorphisms are classified ([theorem 130.5](#)), the discontinuous ones for $\alpha > \omega$ are not.

Question 138.5 (Which dense elementary subspaces are self-images?). At $\omega \cdot 2$ one dense graph is realized as a self-image. For a general $\mathbb{Q}$ -subspace $V \leq H_\alpha$, characterize when V is order-isomorphic to H_α . Closedness allows the finite-support annihilator theorem; without it, the valuation spectrum alone no longer controls the space.

Question 138.6 (Definability complexity of images). Can one prescribe the descriptive complexity of a proper elementary self-image at a fixed higher rank? Every Borel embedding has a closed image, but an image of a non-Borel embedding can itself be closed. Determine which nonclosed Borel, analytic, or projective coefficient subspaces can arise as algebraic self-images.

Question 138.7 (Other coefficient groups). Replace the rational factors by other ordered Archimedean groups, or mix discrete rational and usual real factors. Which versions of the orbit classification and the two thresholds survive? The rational proof uses unique divisibility, a countable discrete codomain for every coordinate, and finite-dimensional linear duality; each has a different role.

[write] This is the part of Part I's [question 14.14](#) (source 04) that source 16 leaves open; it answers the rest for countable ordinal index sets with rational coefficients (note after that question).

[write] (Batch 93.) Answered in a variant by Part XVI: for real coefficients with the Hilbert topology, not the product topology this question has in mind, every order embedding of $H_\theta^{\ell^2}$ into itself is continuous exactly when $\theta \leq 1$, and every one, Borel or not, has closed image exactly when $\theta < \omega$ ([proposition 224.5](#) [merge], from source 22's [theorem 216.2](#) and [corollary 216.8](#)); both thresholds move down from ω and $\omega \cdot 2$. The orbit classification and the product-topology version stay open.

Question 138.8 (Orders beyond countable ordinals). For a countable linear index order that is not well-ordered, work with a genuine Hahn product of well-ordered supports. The full product is no longer the appropriate carrier. Determine when it is a standard Borel or Polish space and how rank automorphisms alter the classification. Existing Hahn-group theory provides the natural starting point.

Question 138.9 (Arbitrary Polish Presburger presentations). Which Polish continuous-addition Presburger models admit a Polish group completion with a discrete standard integer coordinate? The split family here has both properties by construction. A general answer would identify the reach of the subspace method and the precise additional phenomena in nonsplit models.

[write] Answered in part. Among Polish group presentations, source 13's [theorem 95.4](#) shows that a continuous integer-coordinate retraction exists exactly for the algebraically split ones, where it is automatic, and its [theorem 94.1](#) gives nonsplit Polish models; this is Part I's [question 14.11](#) (note there). Whether a given Polish cone has a Polish group completion at all is source 08's [question 57.2](#), asked again in Parts V, VI and IX; it is open.

[write] (Batch 90.) No longer open: [Part XII](#) answers [question 57.2](#) (note there). A Polish continuous-addition Presburger model has a Polish group completion inducing its topology exactly when its partial subtraction is continuous. Whether that completion has a discrete standard integer coordinate is then the splitting question settled by [theorem 95.4](#).

Question 138.10 (Connected carriers). Can a connected uncountable Polish space support a continuous-addition Presburger model? None of the rational-product carriers here is connected. The matrix classification does not settle this broader question from the repository.

[write] This is Part I's [question 14.3](#), Part III's [question 43.4](#) and source 13's [question 103.4](#). It is answered negatively for cones whose topology is inherited from a Polish topological $\mathbb{Z}$ -group (note after [question 14.3](#)) and is open for other monoid topologies.

[write] (Batch 90.) The "other monoid topologies" are now those with discontinuous partial subtraction (note after [question 14.3](#)).

Question 138.11 (Formalization as the next deliverable). Implement the rational- ω case first: the homomorphism dictionary, automatic continuity, positive-pivot matrices, closed images, and the closed-submodel correspondence. This is a finite collection of theorem interfaces and gives the largest immediate gain before introducing transfinite coordinate ranks.

[write] Compare Part I's [question 14.1](#); at $\alpha = \omega$ the listed interfaces are those of [proposition 11.3](#), [theorems 11.2](#) and [11.4](#), and [corollary 11.5](#).

Question 138.12 (Sharp surreal birthday bounds). For the fixed support in [\(137.1\)](#), replace the abstract Replacement bound by useful explicit birthday estimates uniform in α . Then formalize the transported coefficient topology inside one specified set-sized surreal cutoff.

[write] For $\alpha = \omega$ this is the first sentence of Part I's [question 14.10](#) (source 05).

Part XI

Local geometric codes and a countable arithmetic core

[write] This part is source 17, manuscript 01 of batch 90, *Local Geometric Codes and Topological Arithmetic: Partial exponentiation, interval discontinuities, and a canonical countable arithmetic core* (30 pages, letter, dated October 3, 2026; “prepared with ChatGPT”). It arrived in commit a162e4386, as archive local_geometric_codes_research, and was placed in 12076b2e8; it is pinned to 7c0f2d9f9, the batch-89 arrival, 122 commits before the placement. Its file prefix in this report is 17-local-codes-. All unmarked text in this part is source 17's; its twelve sections are Sections [139–150](#) (its Section k is Section $138 + k$, with statement numbers otherwise unchanged), and its two appendices are [appendices Y and Z](#).

What this part claims, and its status

Let $M \models \mathsf{I}\Delta_0$ have a Polish carrier with Borel addition and multiplication, and let one pair of internal parameters u, v code, through the remainder decoder $\beta(u, v, i) = \text{rem}(u, 1 + (i + 1)v)$, a geometric progression $E(0) = 1, E(i + 1) = qE(i)$ ($q \geq 2$) for all $i < C$. Then every externally uncountable interval $[b, d] \subseteq [0, C]$ contains an a for which $x \mapsto E(a)x$ is discontinuous ([theorem 139.1](#)). With the Harrington–Marker–Shelah theorem on Borel orders, the elements with countable initial segments form a countable Δ_0 -elementary initial submodel N_M ([proposition 145.4](#)), and under separately continuous multiplication every geometric-code length, for every base, lies in it ([theorem 145.6](#)); uncountable Borel $\mathsf{I}\Delta_0$ -models with a long code but without total exponentiation exist ([proposition 146.2](#)).

Relation to Part VIII. [Theorem 139.1](#) localizes source 14's [theorem 111.1](#) to any uncountable interval of a coded exponential. The hypotheses are *incomparable*, so neither theorem strengthens the other: source 17 assumes $\mathsf{I}\Delta_0$ and one internal code, where [theorem 111.1](#) assumes only $\mathsf{Q} + \mathsf{I}\text{Open}$ but a total Borel exponential e given as extra data; [theorem 111.1](#) concludes cofinal discontinuity above every b , [theorem 139.1](#) discontinuity inside every uncountable subinterval of the code's domain. Its ATR_0 upper bound ([theorem 147.1](#)) is, like source 14's, a metamathematical sketch with declared coding interfaces resting on Glazer's uniform perfect-set “Optimization”; it is printed **as claimed** and has not been independently reviewed. The global countability results are ordinary ZFC theorems. Unrefereed, AI-assisted, and not formalized.

What was already here. At its pin this report had Parts I–V only. Source 17 read the batch-87 archives directly, as it says, among them source 14's glazer_atr0_research: its “earlier project manuscript” [103] is [Part VIII](#), whose Theorem 8.1 is [theorem 111.1](#) and whose Question 12.2 is [question 115.2](#). Source 17 proves again, in its own setting ($\mathsf{I}\Delta_0$, standard Borel spaces), several lemmas of [Part VIII](#): its [lemma 140.1](#) is [lemma 106.1](#); its [lemma 140.2](#) is [lemma 108.1](#) for $\mathsf{I}\Delta_0$ in place of $\mathsf{Q} + \mathsf{I}\text{Open}$; its [lemma 140.3](#) is [lemma 108.2](#); its scales and perturbations ([lemmas 142.1 to 142.3](#)) adapt [section 108](#) and [lemma 108.3](#); its [lemma 143.1](#) is [lemma 109.1](#); its [lemma 143.3](#) is [lemma 110.1](#); and its [lemma 143.4](#) isolates the category

argument of the proof of [theorem 111.1](#). These are printed with source 17's proofs and flagged in notes. New to the report are the interval localization with its buffer (4.11)–(4.12) and the shifted cancellation of Step 3 in [section 144](#), which avoids continuity of the small multipliers $E(h_j)$ (source 14 arranges all exponents above b); the countable core and its consequences ([propositions 145.4](#), [145.5](#) and [145.7](#) and [theorem 145.6](#)); the polynomial cuts ([proposition 146.2](#)); and the recoding criterion ([proposition 149.2](#)). The cut of elements with countable initial segments is Glazer's, and source 17's audit credits it to him. [Corollary 146.1](#) is a second route to the multiplicative half of Glazer's Theorem 2, and so of [theorem 104.1\(ii\)](#), for the internally coded formulation of total exponentiation. Priority is not established.

Questions answered. Source 17 answers [question 115.2](#) of [Part VIII](#) *in part*, in its explicit-code direction: one internal geometric code through an uncountable interval already forces a discontinuous multiplier, and [proposition 146.2](#) shows that this hypothesis is strictly weaker than total exponentiation. The bare $\text{I}\Delta_0$ version stays open ([question 149.4](#)), and the coded domain is not identified with ordinary bounded-definition exponentiation ([question 149.1](#)). [Proposition 145.5](#) bears on [question 115.3](#). Dated notes after both questions say so. Glazer's Questions 1 and 2 are not answered by this part: its ATR_0 bound concerns the local theorem only.

Labels and notation. Every label of source 17 takes the prefix `pma:lgc:`; labels were added to its eleven unlabelled questions and to its two appendices, and `pma:part:lgc` to the part. The fifth block of the table in [section 1.3](#) lists the symbols. In short: its Cantor space 2^ω is printed $2^{\mathbb{N}}$; its base q is not a scale, and its scales p_n are source 14's q_n ; $c, t_n, X_n, x_n, h_{k,n}, y_n, V_s, Q_j, \delta_n, z_r$ play the roles they play in [Part VIII](#); its $D = d - b$, D_q, K and $N_M = N_M$ are local and are not the groups D, K, K_* or the countable model N of other parts. Its equation tags (1.1)–(11.1) are printed as delivered and refer to its own sections: (2.x) are in [section 140](#), (3.x) in [section 141](#), (4.x) in [section 142](#), (5.x) in [section 143](#), (6.x) in [section 144](#), (8.x) in [section 146](#), (9.x) in [section 147](#) and (11.x) in [section 149](#). No normalization changed; three `\Needspace` layout commands were dropped.

Files. Its finite diagnostics are shipped as `code/17-local-codes-verify_local_arithmetic.py` with the recorded run `data/17-local-codes-local_arithmetic_diagnostics.json` (14,953 assertions, seed 20261004); its proof-status record and source audit as `17-local-codes-PROOF_STATUS.txt` and `17-local-codes-SOURCE_AUDIT.txt`. The Hilbert's-tenth programme read parts of the manuscript for routing, without a correctness finding ([appendix E](#)). Nothing in this part is formalized.

139 The research question and the contribution

139.1 Why this is a substantive Glazer–ProveIt intersection

Glazer's *A Topological Tennenbaum Theorem* studies arithmetic structures whose underlying sets are Polish spaces. His Theorem 2 rules out continuity of either operation in an uncountable Borel presentation of $\mathbb{Q} + \text{I}\Delta_0 + \text{Exp}$; the multiplication argument uses a globally defined exponential function [1]. The surrounding subject was developed by Enayat, Hamkins, and Wcisło [2]. It asks which algebraic and logical properties of nonstandard arithmetic are compatible with topological regularity.

The relevant part of Provelt includes Polish models of Presburger and omnific arithmetic, Diophantine exponentiation, and explicit arithmetic sequence coding [102, 104]. A more recent project manuscript, *Diophantine Definability and Topological Arithmetic*, already develops a total-exponentiation argument and proposes partial exponentiation as a next step [103]. Its Question 12.2 asks for a power-of-two multiplier obstruction under weaker arithmetic hypotheses, without a total Borel exponential supplied as extra data.

The present work addresses that direction. The replacement input is a single tuple of elements of M , certifying one geometric progression through one internal length. Total exponentiation is not an axiom of the main theorem. The resulting conclusion is local enough to apply to every uncountable subinterval of the certified domain.

139.2 A precise source boundary

The repository was inspected at commit

7c0f2d9f92c3d51ec85703bed1022924a4b7359b,

dated 2026-10-04 at 04:36:43 UTC. The main consolidated Polish-arithmetic document and its README did not yet contain all of the latest placed material. Accordingly, the audit also read the retained arrival manuscripts, including the total-exponentiation ATR_0 article [103]. This matters: neither the earlier Presburger construction nor that earlier total-exponentiation argument is presented here as a new result.

The main new proof obligation is the localization to a bounded code. Its especially useful step is a shifted cancellation identity: continuity of two allowed large multipliers recovers a finite-change identity involving a smaller multiplier whose continuity is unavailable. The global countability conclusions then combine this local result with an established theorem on Borel linear orders [99].

The literature check was targeted. It did not identify the local interval statement or the full coding-domain countability conclusion in the inspected sources. That finding supports treating this as a proposed extension; it does not certify bibliographic priority.

[write] At the pin 7c0f2d9f9 this report had Parts I–V. The “retained arrival manuscripts” are the batch-87 archives, written into Parts VI–IX in 83926d574; the total-exponentiation article [103] is source 14, now [Part VIII](#), whose Theorem 8.1 and Question 12.2 are [theorem 111.1](#) and [question 115.2](#). Source 17’s audit also read the batch-89 archive of source 16 (now [Part X](#)), which it found unrelated to its theorem.

139.3 A first statement of the results

All intervals and arithmetic expressions below are interpreted in M . Countability and continuity are external notions. The precise code is defined in Section 141.

Theorem 139.1 (Local geometric-code obstruction). *Let $M \models \text{I}\Delta_0$ have a Polish topology, and suppose addition and multiplication are Borel. Fix $q \geq 2$, $C, u, v \in M$, and put*

$$E(i) = \text{rem}(u, 1 + (i + 1)v).$$

Assume

$$E(0) = 1, \quad \forall i < C \quad E(i + 1) = qE(i). \tag{1.1}$$

For every $b \leq d \leq C$, if $[b, d]^M$ is externally uncountable, there is $a \in [b, d]^M$ such that

$$\lambda_{E(a)} : M \longrightarrow M, \quad \lambda_{E(a)}(x) = E(a)x \tag{1.2}$$

is discontinuous.

The theorem only uses continuity of unary maps in a contradiction hypothesis. Joint continuity of multiplication is unnecessary. Neither continuity of addition nor compatibility with the arithmetic order topology is assumed.

[write] Compare source 14's [theorem 111.1](#). That theorem assumes only $\mathbf{Q} + \mathbf{IOpen}$ but a total Borel function e satisfying its exponential law, given as extra data, and concludes that $x \mapsto e(a)x$ is discontinuous for some $a \geq b$, for every b . [Theorem 139.1](#) assumes $\mathbf{I}\Delta_0$ and one internal code instead of a total exponential, and concludes discontinuity inside every uncountable subinterval of the code's domain. The hypotheses are incomparable, so neither theorem strengthens the other.

Corollary 139.2 (Countable coded-geometric domain). *Work in ordinary ZFC. Under the hypotheses on M in [Theorem 139.1](#), suppose every map $x \mapsto ax$, $a \in M$, is continuous. Then the set*

$$\mathcal{G}_M = \{C \in M : \text{some } q \geq 2, u, v \text{ satisfy (1.1) through } C\} \quad (1.3)$$

is externally countable.

For an uncountable M , this confines all such index lengths to a proper countable initial submodel. The bases and code parameters need not belong to that submodel.

Corollary 139.3 (Regularity of code certificates). *Under the hypotheses of [Corollary 139.2](#), the relation on M^4*

$$\text{Geo}(q, C, u, v) \iff q \geq 2 \wedge E(0) = 1 \wedge \forall i < C [E(i+1) = qE(i)] \quad (1.4)$$

is Borel. For each fixed $q \geq 2$, the corresponding partial exponential graph is countable and hence F_σ .

The universal quantifier in (1.4) may range over an externally uncountable interval in an arbitrary presentation. Its collapse to a countable family of tests is a consequence of the theorem, not an assumption in its proof.

139.4 What the result resolves

The result gives a proved answer to the repository's partial-code version of the weaker-arithmetic question. It also supplies an alternative route to the multiplicative part of the known total-exponentiation obstruction. There remain two separate problems: removing the explicit code hypothesis entirely, and transferring the countability conclusion to every usual bounded-definition exponential domain in $\mathbf{I}\Delta_0$. The manuscript does not identify those problems with the one it proves.

[write] (Batch 90.) "The repository's partial-code version of the weaker-arithmetic question" is [question 115.2](#) of [Part VIII](#), which a dated note there now marks as answered in part. The "alternative route" is [corollary 146.1](#), for the multiplicative half of [theorem 104.1\(ii\)](#) and of Glazer's Theorem 2 under their coded formulation of total exponentiation.

140 Conventions and descriptive foundations

140.1 Two levels of arithmetic

The underlying structure is

$$\mathcal{M} = (M, 0, 1, +, \cdot, <) \models \mathbf{I}\Delta_0.$$

The order is the usual arithmetic order of a model of bounded induction. The structure includes only nonnegative elements. A notation such as $d - b$, with $b \leq d$, means the unique $r \in M$ such that $b + r = d$.

External indices n, k, j range over the ordinary natural numbers. An element $c \in M$ may exceed every such standard integer. A recurrence indexed externally by $n < \omega$ can therefore descend through internally huge numbers while remaining above every standard bound at each external stage. No internal assertion that an infinite decreasing sequence exists is being made.

A Polish topology is a separable topology induced by a complete metric. Fix one complete compatible metric d_M , replaced by $\min(d_M, 1)$ when convenient. Arithmetic intervals need not be topologically closed. In particular, a topological limit of positive elements cannot be assumed positive without proof.

140.2 Elementary arithmetic available in $\mathsf{I}\Delta_0$

We use cancellation, the ordered-semiring laws, division with remainder, and integer square roots. For $z \in M$ and $s > 0$, there are unique a, r with

$$z = as + r, \quad r < s. \quad (2.1)$$

For $z \in M$, there is a unique t satisfying

$$t^2 \leq z < (t+1)^2. \quad (2.2)$$

These follow by bounded induction, or equivalently by the bounded least-number principle applied to the relevant polynomial inequalities. All searches can be bounded by $z + 1$. We write $\lfloor z/s \rfloor = a$, $\text{rem}(z, s) = r$, and $\lfloor \sqrt{z} \rfloor = t$.

The proof never requires the totality of a variable-height power function. Expressions with fixed standard exponents, such as t^2 , y^5 , and y^8 , are polynomial terms.

140.3 Borel arithmetic order and remainder

Lemma 140.1 (Unique analytic evaluation). *Let X, Y be standard Borel spaces. If a relation $R \subseteq X \times Y$ is analytic and is the graph of a total single-valued function, then that function is Borel.*

Proof. For a Borel $U \subseteq Y$, the set $\{x : \exists y \in U R(x, y)\}$ is analytic. Its complement is $\{x : \exists y \notin U R(x, y)\}$, which is also analytic by totality and uniqueness. Analytic-coanalytic separation therefore makes the preimage Borel. Apply this to a countable basis for a Polish realization of Y . $\square$

The totality qualification is essential. An analytic graph of a partial function can have a non-Borel domain. We use this lemma only after verifying totality on its stated Borel domain.

Lemma 140.2 (Borel elementary operations). *If addition and multiplication on $M \models \mathsf{I}\Delta_0$ are Borel, then its arithmetic order, quotient and remainder functions, integer square-root function, and nonnegative difference on $\{(x, y) : x \leq y\}$ are Borel.*

Proof. The relation $x \leq y$ has the analytic definition $\exists z (x + z = y)$. Its complement has the analytic definition $\exists z (y + z + 1 = x)$. Thus order is Borel.

For $s > 0$, the graph of the pair-valued map $(z, s) \mapsto (a, r)$ in (2.1) is Borel, hence analytic, and the map is total and unique. Extend it by $(0, 0)$ at $s = 0$ to obtain a total Borel function. The square-root graph is given by (2.2), and again totality and uniqueness apply. The difference graph is $x + r = y$ on the Borel domain $x \leq y$; the same argument applies there. $\square$

This argument does not say that all arithmetically bounded quantifiers preserve Borelness in an uncountable model. An internally bounded quantifier can still quantify over uncountably many points of the presentation.

140.4 Increasing Cantor sets

Lemma 140.3 (Perfect-order input). *If A is an uncountable Borel subset of a Polish space equipped with a Borel linear order, there is a continuous injection $f : 2^{\mathbb{N}} \rightarrow A$ increasing from lexicographic order on $2^{\mathbb{N}}$ to that linear order.*

Proof. The Borel perfect-set theorem gives a continuous injection $g : 2^{\mathbb{N}} \rightarrow A$. Color a pair $s < t$ by whether $g(s) < g(t)$. The coloring is Borel. Galvin's perfect-set theorem for pairs gives a perfect homogeneous subset. It contains the image of an increasing continuous Cantor embedding. If the color is reversed, precompose with the order-reversing homeomorphism $r \mapsto 1 - r$. $\square$

The perfect-set partition input is established background; see the treatment used by Glazer and the related theorem of Blass [1, 71]. Section 147 separates its coded uniform use from the later classical Harrington–Marker–Shelah argument.

[write] The three lemmas of this section are source 14's lemma 106.1 (lemma 140.1), lemma 108.1 (lemma 140.2, there for $\mathbf{Q} + \mathbf{I}\mathbf{Open}$ with truncated subtraction, here for $\mathbf{I}\Delta_0$ with the difference on $\{x \leq y\}$) and lemma 108.2 (lemma 140.3, there in its uniform coded form), proved again with source 17's own proofs.

141 One code supplies all the local regularity

141.1 A fixed decoder

Definition 141.1. For $u, v, i \in M$, define

$$\beta(u, v, i) = \text{rem}(u, 1 + (i + 1)v). \quad (3.1)$$

A geometric code of base $q \geq 2$ through C is a pair (u, v) such that

$$\beta(u, v, 0) = 1, \quad \forall i < C \quad \beta(u, v, i + 1) = q\beta(u, v, i). \quad (3.2)$$

Write $\text{Geo}(q, C, u, v)$ for this assertion.

The modulus in (3.1) is always positive. By Lemma 140.2, the decoder is jointly Borel in (u, v, i) , before any recurrence condition is imposed. Once a valid code is fixed, let $E(i) = \beta(u, v, i)$. We restrict its exponential interpretation to $i \leq C$. The decoder itself is defined everywhere, but its values beyond C need not have any exponential meaning.

The code condition is a bounded arithmetic formula: remainders can be expressed using witnesses bounded by u and the moduli. Consequently, $\mathbf{I}\Delta_0$ permits bounded induction over statements involving this fixed decoder. This is an internal use of bounded formulas; Borelness of the decoder was established separately by unique evaluation.

141.2 Local exponential identities

Lemma 141.2 (Algebra inside a code). *Assume $\text{Geo}(q, C, u, v)$. For $i, j \leq C$:*

(i) $E(i) > 0$, and $i < j$ implies $E(i) < E(j)$.

(ii) If $i + j \leq C$, then

$$E(i + j) = E(i)E(j). \quad (3.3)$$

(iii) If $r \in \mathbb{N}$ is fixed and $ri \leq C$, then

$$E(ri) = E(i)^r. \quad (3.4)$$

(iv) Two codes of the same base agree throughout their common certified interval.

Proof. Every remainder $E(i)$ is at most u . Bounded induction on $i \leq C$, using $q \geq 2$ and the recurrence, proves positivity and $E(i + 1) > E(i)$. A further bounded induction proves strict increase between arbitrary indices.

Fix $i \leq C$. Induct on $j \leq C - i$. The base case is $E(i) = E(i)E(0)$. For the step,

$$E(i + j + 1) = qE(i + j) = qE(i)E(j) = E(i)E(j + 1).$$

The equality being proved is a bounded formula: decoder values are bounded by u , and their displayed products by u^2 . Thus this induction is available in $\text{I}\Delta_0$. It proves (3.3).

For each fixed standard r , repeated application of (3.3) proves (3.4); no variable-height exponentiation is used. Finally compare two codes by bounded induction through the smaller length. The entries start at 1 and obey the same recurrence, so their entries coincide. $\square$

Remark 141.3 (No hidden recoding assumption). The lemma compares or evaluates codes that already exist. It does not construct a new code, extend one past its certified length, or prove that two coded lengths have a coded sum. These distinctions will matter for the global domain of coded exponentiation.

141.3 Borel images without a global exponential

Lemma 141.4 (Local logarithmic decoder). *Fix a geometric code through C . There is a total Borel map $L : M \rightarrow [0, C]$ such that*

$$L(E(i)) = i \quad (i \leq C). \quad (3.5)$$

Consequently $E(X)$ is Borel for every Borel $X \subseteq [0, C]$.

Proof. Set $L(0) = 0$. For $z \geq 1$, let $L(z)$ be the largest $i \leq C$ with $E(i) \leq z$. Such an i exists since $E(0) = 1$, and the largest one exists by bounded induction. More explicitly, its graph for $z \geq 1$ is

$$i \leq C \wedge E(i) \leq z \wedge (i = C \vee z < E(i + 1)). \quad (3.6)$$

This graph is Borel, because it uses only the fixed Borel decoder and Borel order. It is total and single-valued, so Lemma 140.1 applies. Strict increase of E proves (3.5).

For Borel $X \subseteq [0, C]$,

$$E(X) = \{z \geq 1 : E(L(z)) = z \wedge L(z) \in X\}. \quad (3.7)$$

The right side is Borel. $\square$

The role of L is purely local. It is a bounded inverse for the values of this one code; it supplies no logarithm or exponentiation theorem outside the certified interval.

141.4 A more general semantic formulation

The local obstruction also holds for a Borel function $E : [0, C] \rightarrow M$ supplied directly, provided it is strictly increasing, $E(0) = 1$, $E(1) \geq 2$ when $C \geq 1$, and (3.3) holds wherever $i + j \leq C$. In this formulation Borel images of E follow from the injective Borel image theorem, or from a supplied local inverse. The code formulation is stronger as an application to weak arithmetic: the extra regularity and the needed identities follow from the finite tuple of code parameters.

142 Exact scales inside an arbitrary interval

Fix a code through C and $b \leq d \leq C$ such that $[b, d]$ is uncountable. Put

$$D = d - b, \quad c = \lfloor \sqrt{D} \rfloor. \quad (4.1)$$

Translation gives a bijection $[0, D] \rightarrow [b, d]$. Moreover,

$$c^2 \leq D < (c + 1)^2. \quad (4.2)$$

Lemma 142.1 (Uncountability survives integer roots). *The interval $[0, c]$ is uncountable. Define externally*

$$t_0 = c, \quad t_{n+1} = \lfloor \sqrt{t_n} \rfloor \quad (n < \omega). \quad (4.3)$$

Then every $[0, t_n]$ is uncountable. In particular every t_n exceeds every standard natural number.

Proof. Division of $z \leq D$ by $c + 1$ gives

$$z = (c + 1)a + r, \quad a, r \leq c,$$

by (4.2). Thus $[0, D]$ injects into $[0, c]^2$. If the latter interval were countable, so would be $[0, D]$, a contradiction.

At each later root step, the same argument injects $[0, t_n]$ into $[0, t_{n+1}]^2$. For each fixed external n , compose the finitely many radix maps. Countability of $[0, t_n]$ would imply countability of $[0, c]$. Finally an interval bounded by a standard integer is externally finite. $\square$

Define

$$p_n = \left\lfloor \frac{c}{t_n} \right\rfloor, \quad X_n = \{p_n i : i < t_{n+2}\}. \quad (4.4)$$

The symbol p_n denotes a scale, and is unrelated to the fixed base q .

Lemma 142.2 (Separated Borel grids). *Each X_n is an uncountable Borel subset of $[0, c]$. Furthermore*

$$p_{n+1} \geq p_n t_{n+1} \geq p_n t_{n+2}^2 > 8p_n t_{n+2}. \quad (4.5)$$

Proof. All t_n are positive and $p_n \geq 1$. Multiplication by p_n injects $[0, t_{n+2})$ into X_n , so X_n is uncountable. Its direct description

$$X_n = \{x < p_n t_{n+2} : \text{rem}(x, p_n) = 0\}$$

makes it Borel. Since $p_n t_n \leq c$ and $t_{n+2} < t_n$, every point of X_n is below c .

The inequalities $c \geq p_n t_n \geq p_n t_{n+1}^2$ give $p_{n+1} \geq p_n t_{n+1}$ after division by t_{n+1} . Apply $t_{n+1} \geq t_{n+2}^2$, and then $t_{n+2} > 8$, to obtain (4.5). $\square$

The sets $E(X_{n+1})$ are uncountable and Borel, by strict increase and Lemma 141.4. Choose increasing continuous injections

$$g_n : 2^{\mathbb{N}} \longrightarrow E(X_{n+1}) \quad (4.6)$$

using Lemma 140.3. Write $\epsilon_k = \chi_{\{k\}}$, so $\epsilon_k \rightarrow 0^\omega$, and define

$$x_n = L(g_n(0^\omega)), \quad h_{k,n} = L(g_n(\epsilon_k)) - x_n. \quad (4.7)$$

All differences are nonnegative arithmetic differences. The maps L and arithmetic subtraction need not be continuous.

Lemma 142.3 (Perturbations and the local buffer). *For all external k, n ,*

$$p_{n+1} \leq h_{k,n} < p_{n+1}t_{n+3} < c, \quad x_n < c, \quad (4.8)$$

and

$$E(x_n + h_{k,n}) \longrightarrow E(x_n) \quad (k \rightarrow \infty). \quad (4.9)$$

Every choice $h_n = h_{k_n,n}$ satisfies

$$h_{n+1} > 8h_n. \quad (4.10)$$

Let $B = b + 2c$. For every external n , every $S \subseteq \{0, \dots, n-1\}$, and every k ,

$$m_S = B + \sum_{i \in S} h_i, \quad m_S + h_{k,n} \text{ belong to } [b, d]. \quad (4.11)$$

Also $m_S - x_n \in [b, d]$.

Proof. Both inverse images in (4.7) lie in X_{n+1} , and the second is strictly greater than the first. Their difference is a positive multiple of p_{n+1} , strictly less than $p_{n+1}t_{n+3}$. The image lies below c , and (4.9) follows directly from continuity of g_n . Apply (4.5) at $n+1$ to obtain (4.10).

For the buffer, at most $n+1$ increments occur, and each is below c . Hence

$$m_S + h_{k,n} < b + (n+3)c < b + c^2 \leq b + D = d, \quad (4.12)$$

because c exceeds the standard number $n+3$. The same bound holds without the new increment. Finally $m_S \geq b+2c$ and $x_n < c$, so $m_S - x_n > b+c \geq b$, while $m_S - x_n \leq m_S \leq d$. $\square$

The buffer is the central localization estimate. Every finite exponent needed in the construction remains inside the one existing code. The external process has infinitely many stages, but no stage evaluates E at an index beyond d .

[write] The scales c, t_n , the grids X_n and the perturbations $x_n, h_{k,n}$ are those of source 14's section 108 and lemma 108.3, with source 14's scales $q_n = \lfloor c/t_n \rfloor$ written p_n here because q is the base of the code; source 14 works above a threshold, with $b < h_{k,n}$ in its (5.5). The localization to $[b, d]$, with the anchor $B = b + 2c$ and the buffer (4.11)–(4.12), is new.

142.1 A finite illustration of the scales

For intuition, take the ordinary integer $c = 2^{4096}$. For the first five stages the root calculations are exact. The possible perturbation ranges in (4.8) satisfy:

Stage n	Lower bound for $h_{k,n}$	Strict upper bound
0	2^{2048}	2^{2560}
1	2^{3072}	2^{3328}
2	2^{3584}	2^{3712}
3	2^{3840}	2^{3904}
4	2^{3968}	2^{4000}

The intervals grow rapidly while remaining below c . An ordinary finite c cannot support this indefinitely. In the proof, uncountability ensures that every externally specified root remains internally infinite.

143 The complete-metric and category mechanisms

143.1 A quantitative binary construction

Let $V_s \in M$, for $s \in 2^{<\omega}$, be finite vertices with $V_{s0} = V_s$, and suppose distinct vertices at the same level are distinct points. Set $\delta_0 = 1$, and for $n > 0$ set

$$\delta_n = \min(\{1\} \cup \{d_M(V_s, V_t) : s, t \in 2^n, s \neq t\}). \quad (5.1)$$

These numbers are positive. The zero-extension convention implies $\delta_{n+1} \leq \delta_n$.

Lemma 143.1 (Uniform limits and separation). *Suppose*

$$d_M(V_s, V_{s1}) < 2^{-n-4}\delta_n \quad (s \in 2^n). \quad (5.2)$$

Then for every $r \in 2^{\mathbb{N}}$ the limit

$$z_r = \lim_{n \rightarrow \infty} V_{r \upharpoonright n} \quad (5.3)$$

exists, and $r \mapsto z_r$ is a continuous injection. For every N ,

$$d_M(V_{r \upharpoonright N}, z_r) \leq 2^{-N-3}\delta_N. \quad (5.4)$$

If $r \upharpoonright N \neq s \upharpoonright N$, then

$$d_M(z_r, z_s) \geq (1 - 2^{-N-2})\delta_N > 0. \quad (5.5)$$

Proof. For $i \geq N$, the edge bound is at most $2^{-i-4}\delta_N$. Summing gives

$$\sum_{i=N}^{\infty} 2^{-i-4}\delta_N = 2^{-N-3}\delta_N.$$

The finite vertices along each branch are Cauchy, with a bound uniform in the branch. Completeness gives (5.3) and (5.4). Finite-level maps are locally constant, so their uniform limit is continuous. For different prefixes at level N , the vertices are at distance at least δ_N . Subtract the two tail bounds to obtain (5.5). $\square$

This lemma identifies the precise role of completeness. It also makes the proof suitable for an explicit formalization: only finitely many separations are considered at a stage, and all later movement is bounded by a summable series.

143.2 Two category facts, with proofs

For $j < \omega$, write $F_j : 2^{\mathbb{N}} \rightarrow 2^{\mathbb{N}}$ for the homeomorphism that flips bit j . Write $r^* = 1 - r$ for the homeomorphism that complements all bits. A cylinder $[s]$ specifies a finite initial string.

Lemma 143.2 (Late-flip meagerness). *Let $A \subseteq 2^{\mathbb{N}}$ have the Baire property. Suppose there is n such that*

$$A \cap F_j(A) = \emptyset \quad (j > n). \quad (5.6)$$

Then A is meager.

Proof. If A were nonmeager, the Baire property would make it comeager in some nonempty cylinder $[s]$. Choose $j > \max(n, |s|)$. Then F_j preserves $[s]$, so $F_j(A)$ is also comeager there. The intersection of two comeager subsets of the nonempty Baire space $[s]$ is nonempty, contradicting (5.6). $\square$

Lemma 143.3 (Finite-change zero-one law). *If $A \subseteq 2^{\mathbb{N}}$ has the Baire property and is invariant under every finite set of bit flips, then A is meager or comeager.*

Proof. Suppose A is nonmeager. It is comeager in some cylinder $[s]$. Finite flips of the first $|s|$ bits carry $[s]$ onto every other cylinder of that length and preserve A . Thus A is comeager in each member of a finite clopen partition of $2^{\mathbb{N}}$. Its complement is therefore meager in $2^{\mathbb{N}}$. $\square$

143.3 An abstract ordered-product contradiction

Lemma 143.4 (No lacunary Cantor product). *Let M be an ordered cancellative arithmetic semiring, with a Polish topology and Borel order and multiplication. Suppose $y_n \in M$ satisfy*

$$y_0 \geq 2, \quad y_{n+1} > y_n^5. \quad (5.7)$$

There cannot be a continuous injection $r \mapsto z_r$ from $2^{\mathbb{N}}$ to M such that, for every $A \subseteq \omega$ and $j \notin A$,

$$z_{A \cup \{j\}} = y_j z_A. \quad (5.8)$$

Here subsets of ω are identified with their characteristic functions.

Proof. First all z_A are positive. If $z_A = 0$, flip any one bit. Formula (5.8) gives another zero value when the bit is added; when it is removed, cancellation of the nonzero y_j gives another zero value. Either alternative contradicts injectivity.

Define the Borel sets

$$\begin{aligned} A_n &= \{r : z_{r^*} < z_r \leq y_n z_{r^*}\}, \\ T &= \{r : \forall n \quad y_n z_{r^*} < z_r\}. \end{aligned} \quad (5.9)$$

They are Borel because z is continuous, the arithmetic operations and order are Borel, and the intersection defining T is externally countable.

Fix $j > n$, and let $r(j) = 0$, $s = F_j(r)$. Then

$$z_s = y_j z_r, \quad z_{r^*} = y_j z_{s^*}. \quad (5.10)$$

If both r and s belonged to A_n , their defining inequalities would imply

$$y_j z_{s^*} < z_r, \quad y_j z_r \leq y_n z_{s^*}. \quad (5.11)$$

Multiply the first inequality by y_j , combine with the second, and cancel the positive z_{s^*} . This gives $y_j^2 < y_n$, contrary to (5.7). Thus A_n has no pair related by a flip at any $j > n$. Lemma 143.2 makes every A_n meager.

Next T is invariant under each bit flip. If $r \in T$ and $r(j) = 0$, then for every n ,

$$z_s = y_j z_r > y_j y_n z_{r^*} = y_n y_j^2 z_{s^*} \geq y_n z_{s^*},$$

so $s \in T$. Conversely suppose $s \in T$, and fix n . For $k = \max(n, j) + 1$, (5.7) implies

$$y_k > y_n y_j^2. \quad (5.12)$$

Consequently

$$y_j z_r = z_s > y_k z_{s^*} > y_n y_j^2 z_{s^*}.$$

Cancel $y_j > 0$ and use (5.10) to obtain $z_r > y_n z_{r^*}$. Since n was arbitrary, $r \in T$.

The sets T and $T^* = \{r : r^* \in T\}$ are disjoint. Their union is comeager. Indeed, $z_r \neq z_{r^*}$ by injectivity. If, for example, $z_{r^*} < z_r$ but $r \notin T$, then $r \in A_n$ for some n . The other orientation gives $r \in A_n^*$ for some n . Hence

$$2^{\mathbb{N}} \setminus (T \cup T^*) \subseteq \bigcup_n (A_n \cup A_n^*), \quad (5.13)$$

a meager set.

By Lemma 143.3, T is either meager or comeager. Its homeomorphic image T^* has the same status. If both are meager their union cannot be comeager in $2^{\mathbb{N}}$; if both are comeager they cannot be disjoint. This is the contradiction. $\square$

Notice that this argument does not use continuity of division, subtraction, order, or the small multipliers y_j . It uses the algebraic identities (5.8), Borel comparisons, and a continuous injective parametrization. That separation is what permits the interval version.

[write] Lemma 143.1 is source 14's lemma 109.1 and lemma 143.3 its lemma 110.1; lemma 143.4 isolates, as an abstract statement, the category argument inlined in source 14's proof of theorem 111.1. Each is proved again here.

144 Proof of the local interval theorem

Proof of Theorem 139.1. Fix $b \leq d \leq C$ with $[b, d]$ uncountable. Suppose for contradiction that

$$\lambda_{E(a)} \text{ is continuous for every } a \in [b, d]. \quad (6.1)$$

Use the scales, grids, x_n , and $h_{k,n}$ of Section 142, and let $B = b + 2c$.

Step 1: transfer convergence using only allowed indices. At stage n , let

$$m = B + \sum_{i \in S} h_i, \quad S \subseteq \{0, \dots, n-1\}.$$

By Lemma 142.3, the exponent $m - x_n$ belongs to $[b, d]$. The local exponential identity and (4.9) give

$$\begin{aligned} E(m + h_{k,n}) &= E(m - x_n) E(x_n + h_{k,n}) \\ &\longrightarrow E(m - x_n) E(x_n) = E(m). \end{aligned} \quad (6.2)$$

The only continuity used here is the one unary map $\lambda_{E(m-x_n)}$, allowed by (6.1). All arguments of E in the equalities lie at most d , as verified in (4.12).

Step 2: choose the finite vertices. Choose k_n recursively, put $h_n = h_{k_n, n}$, and define

$$y_n = E(h_n), \quad V_s = E(B) \prod_{\substack{i < |s| \\ s(i)=1}} y_i. \quad (6.3)$$

Repeated use of Lemma 141.2 identifies V_s with $E(B + \sum_{s(i)=1} h_i)$. The index remains within the code by the local buffer.

The perturbations give $y_0 \geq 2$ and

$$y_{n+1} > y_n^5. \quad (6.4)$$

Indeed $h_{n+1} > 8h_n > 5h_n$, and $5h_n < h_{n+1} < c \leq C$, so $y_n^5 = E(5h_n) < E(h_{n+1})$.

Put $Q_j = \prod_{i < j} y_i$. Ordinary external induction gives

$$Q_j^4 < y_j. \quad (6.5)$$

The base case is $1 < y_0$; the step is

$$Q_{j+1}^4 = Q_j^4 y_j^4 < y_j^5 < y_{j+1}. \quad (6.6)$$

Thus the largest differing index in two finite products dominates all their smaller factors. After cancellation of shared higher factors, one side is at least y_j and the other is at most $Q_j < y_j$. The vertices at each level are distinct.

At stage n , convergence (6.2) holds at all the finitely many current vertices. Choose k_n large enough that all new edges satisfy (5.2). This choice can be made after computing the positive finite separation δ_n . Lemma 143.1 then yields a continuous injection $r \mapsto z_r$.

Step 3: recover the finite-change identity by cancellation. Fix $j \notin A \subseteq \omega$. For a finite truncation long enough to include j , the finite products satisfy

$$E(b)V_{(A \cup \{j\}) \upharpoonright N} = E(b + h_j)V_{A \upharpoonright N}. \quad (6.7)$$

This follows by factoring $y_j = E(h_j)$ and using

$$E(b + h_j) = E(b)E(h_j). \quad (6.8)$$

The two exponents b and $b + h_j$ belong to $[b, d]$. Therefore both outer multiplication maps in (6.7) are continuous by (6.1). Passing to the limits gives

$$E(b)z_{A \cup \{j\}} = E(b + h_j)z_A = E(b)E(h_j)z_A. \quad (6.9)$$

Since $E(b) > 0$, cancellation yields

$$z_{A \cup \{j\}} = y_j z_A. \quad (6.10)$$

This step does not assume continuity of $\lambda_{E(h_j)}$. Also, the product on either side of (6.7) is an ordinary product in M ; it is not evaluated by applying E to an index such as $2b + 2c + \sum h_i$, which might exceed C . Only (6.8) requires a new exponential identity, and $b + h_j \leq d \leq C$.

Step 4: apply the category contradiction. Equations (6.4) and (6.10), together with the continuous injection $r \mapsto z_r$, contradict Lemma 143.4. Thus (6.1) fails, which proves the theorem. $\square$

[write] Step 3 is the step without a counterpart in Part VIII: source 14 keeps every multiplier index above its threshold, while here the increments h_j may lie below b , and the identity (6.10) is recovered from the two allowed multipliers $E(b)$ and $E(b + h_j)$ by cancellation.

144.1 A domain ledger for the proof

The following table records the bounds at the points where total exponentiation could otherwise be inserted silently.

Argument or expression	Reason it is legitimate
$x_n, x_n + h_{k,n}$	Both belong to $X_{n+1} \subset [0, c)$.
$h_n, 5h_n$	$h_n < c$ and $5h_n < h_{n+1} < c$.
$B + \sum_{i \in S} h_i$	At stage n , at most $b + (n + 2)c < d$.
$m + h_{k,n}$	Below $b + (n + 3)c < b + c^2 \leq d$.
$m - x_n$	Strictly above $b + c$, and at most $m \leq d$.
$b + h_j$	Below $b + c \leq d$.
$E(b)V_s$	An ordinary product; no combined exponential index is needed.

In particular, the theorem applies when b is much larger than $d - b$. No comparison between the lower endpoint and the interval's width is assumed.

145 The countable core and coded geometric domains

The conclusions in this section are external theorems in ordinary classical set theory. They combine Theorem 139.1 with a classical structure theorem for Borel orders. No reverse-mathematical upper bound for this additional structure theorem, or for the resulting global countability statements, is claimed here.

Throughout, $M \models \text{IA}_0$ has a Polish carrier, and its arithmetic operations are Borel. Its arithmetic order is therefore Borel, as established in the decoding discussion. An interval such as $[a, b]^M$ always denotes the set of *all* elements between its endpoints in the model; its countability is an external property.

145.1 A classical restriction on Borel linear orders

Theorem 145.1 (Harrington–Marker–Shelah, established theorem). *Every Borel linear order on a standard Borel space admits a Borel order embedding into $(2^\alpha, <_{\text{lex}})$ for some countable ordinal α .*

This is the linear-order specialization of [99, Theorem 3.1]. The exclusion of ω_1 -chains is also stated explicitly in [99, Corollary 3.2]. For clarity, we give the elementary order-theoretic arguments needed to use the representation.

Lemma 145.2. *For a countable ordinal α , the lexicographic order on 2^α contains neither a strictly increasing nor a strictly decreasing sequence indexed by ω_1 .*

Proof. Suppose $(s_\xi)_{\xi < \omega_1}$ is strictly increasing. By induction on $\beta \leq \alpha$, we show that the restrictions $s_\xi \upharpoonright \beta$ are constant on a tail of ω_1 . The assertion for $\beta = 0$ is immediate. Once all coordinates below β are fixed, lexicographic monotonicity makes the β -coordinates nondecreasing in $\{0, 1\}$. They therefore stabilize on a further tail, proving the successor step. At a limit β , take the supremum of the earlier stabilization stages. This is a supremum of countably many countable ordinals and hence is still below ω_1 . At $\beta = \alpha$, the full sequences are constant on a tail, a contradiction. The proof for decreasing sequences reverses the inequalities. $\square$

Corollary 145.3. *Every subset of a Borel linear order has a countable cofinal subset and a countable coinital subset. The subset need not be Borel, analytic, or otherwise definable. In particular, every initial segment of a Borel linear order is Borel.*

Proof. If a nonempty subset A had no countable cofinal subset, transfinite recursion would produce a strictly increasing sequence $(a_\xi)_{\xi < \omega_1}$ in A : at each stage the previously chosen elements form a countable set which is not cofinal, so some element of A is strictly above all of them. Theorem 145.1 and Lemma 145.2 exclude this. Reverse the order for coinitality; the empty subset causes no difficulty. Finally, if I is an initial segment and $\{a_n : n < \omega\} \subseteq I$ is cofinal, then $I = \bigcup_{n < \omega} (-\infty, a_n]$. Principal segments are Borel sections of the order relation, so I is Borel. $\square$

145.2 The largest countable initial submodel

Define the *countable core* by

$$N_M = \{a \in M : [0, a]^M \text{ is externally countable}\}.$$

Proposition 145.4. *The set N_M is countable and is the largest countable initial segment of M . It contains the standard natural numbers and is closed under successor, addition, and multiplication. With its inherited arithmetic,*

$$N_M <_{\Delta_0} M \quad \text{and} \quad N_M \models \mathsf{I}\Delta_0.$$

If M is uncountable, N_M is a proper cut with external cofinality ω , and its complement has external coinitality ω . Thus its boundary is an external (ω, ω) -cut.

Proof. Downward closure is immediate. By Corollary 145.3, choose a countable cofinal subset $\{a_n : n < \omega\}$ of N_M . Then

$$N_M = \bigcup_{n < \omega} [0, a_n]^M$$

is countable. If J is any countable initial segment and $a \in J$, then $[0, a]^M \subseteq J$, so $a \in N_M$. This proves maximality.

Every standard initial interval is finite. Moreover, $[0, a + 1]^M = [0, a]^M \cup \{a + 1\}$, so the core is successor closed. For $a, b \in N_M$,

$$[0, a + b]^M \subseteq [0, a]^M \cup (a + [0, b]^M),$$

which proves closure under addition. For multiplication, the case $a = 0$ is immediate. If $a > 0$, division with remainder, available in $\mathsf{I}\Delta_0$, gives

$$[0, ab]^M \subseteq \{aq + r : q \in [0, b]^M, r \in [0, a)^M\}.$$

The latter is an image of a product of two countable sets, proving closure under multiplication.

These closure properties and downward closure give absoluteness of bounded formulas between N_M and M : term values stay in the core, and each bounded quantifier has the same range in the two structures. To check bounded induction in the core, suppose its base case and every successor step hold there, and fix $a \in N_M$. Bounded absoluteness transfers the base case and all the steps below a to M . Bounded induction in M therefore gives the assertion at a . As a was arbitrary, the core satisfies the induction instance.

For uncountable M , the core is proper and, by successor closure, has no maximum. Its complement has no minimum either: if b were its minimum, then $[0, b]^M = N_M \cup \{b\}$ would be countable, contrary to $b \notin N_M$. Corollary 145.3 now gives countable cofinal and coinital subsets of the two sides. The absence of endpoints excludes finite such subsets, so both respective characters are ω . $\square$

In particular, N_M is an F_σ subset of the Polish carrier, because it is countable. No continuity assumption on arithmetic was used in Proposition 145.4. The notation $<_{\Delta_0}$ asserts bounded elementarity only, not full elementarity or preservation of unbounded existential coding assertions.

[write] The cut N_M of elements with countable initial segments is Glazer's; source 17's audit (17-local-codes-SOURCE-AUDIT.txt) credits it to him and claims only its arithmetic properties and their applications. It is not source 15's manifold boundary (Part IX), which is defined topologically.

145.3 Convex families of continuous coded multipliers

Fix one valid geometric code through C , with base $q \geq 2$, and write $E(i) = \beta(u, v, i)$. Put

$$P_E = \{a \in [0, C]^M : x \mapsto E(a)x \text{ is continuous on } M\}.$$

Proposition 145.5. *Every convex subset of $[0, C]^M$ contained in P_E is countable. Consequently every maximal convex component of P_E is countable.*

Proof. Let $K \subseteq P_E$ be convex. If $a, b \in K$ and $a \leq b$, then $[a, b]^M \subseteq K$. Theorem 139.1 implies that this interval is countable, since all its indexed multipliers are continuous. Choose countable subsets $A, B \subseteq K$ which are respectively coinital and cofinal, using Corollary 145.3. For nonempty K ,

$$K \subseteq \bigcup_{\substack{a \in A, b \in B \\ a \leq b}} [a, b]^M.$$

This is a countable union of countable intervals. The assertion is immediate if K is empty. $\square$

Neither K nor P_E had to be Borel. The proposition does not imply that P_E itself is countable: an uncountable set can have uncountably many singleton convex components. As in Theorem 139.1, continuity is continuity of the multiplier on the entire Polish carrier.

[write] (Batch 90.) This bears on source 14's question 115.3, which asks whether the set of continuous multiplier parameters is Borel and how large its complement is; source 17 does not decide that question (question 149.8).

145.4 All coded geometric lengths lie in the core

For $q, C, u, v \in M$, define

$$\begin{aligned} \text{Geo}(q, C, u, v) \iff & q \geq 2 \wedge \beta(u, v, 0) = 1 \\ & \wedge \forall i < C \beta(u, v, i+1) = q\beta(u, v, i). \end{aligned}$$

This is one internal bounded arithmetic condition. Define

$$D_q = \{C \in M : \exists u, v \in M \text{ Geo}(q, C, u, v)\}, \quad \mathcal{G}_M = \bigcup_{q \geq 2} D_q.$$

The union ranges over all internal bases, including nonstandard ones.

Theorem 145.6. *Suppose every unary map $x \mapsto ax$, $a \in M$, is continuous. Then*

$$\mathcal{G}_M \subseteq N_M.$$

Hence $\mathcal{G}_M$ and every D_q are countable F_σ sets. In particular, this conclusion holds if multiplication is separately continuous, and therefore also if it is jointly continuous.

Proof. If $C \in \mathcal{G}_M$, take a witnessing base and code. Every multiplier indexed by $[0, C]^M$ is continuous. Theorem 139.1, applied to this interval, therefore makes $[0, C]^M$ countable. Thus $C \in N_M$; now apply Proposition 145.4. Countable subsets of a Polish space are F_σ . $\square$

The domains D_q and $\mathcal{G}_M$ are downward closed, because the same code witnesses every shorter length. The geometric domain contains every standard natural number, witnessed already by ordinary finite base-2 codes. For uncountable M , every point outside N_M strictly bounds $\mathcal{G}_M$.

Fix $q \geq 2$. Two certificates through the same C have equal entries at every index $i \leq C$: apply bounded induction to the equality of their entries, using the common initial value and recurrence. Consequently

$$e_q(C) = \beta(u, v, C) \quad (C \in D_q)$$

is well defined, independently of the certificate. Bounded induction inside a single code shows that e_q is strictly increasing and that

$$e_q(a + b) = e_q(a)e_q(b) \quad \text{whenever } a + b \in D_q.$$

For the identity, use a code through $a + b$ and induct on $t \leq b$ in $E(a + t) = E(a)E(t)$. Each induction formula is bounded after expanding the β notation. Under the continuity hypothesis of Theorem 145.6, the graph of e_q is countable, and hence F_σ in M^2 ; its range is countable as well.

Proposition 145.7. *Under the hypotheses of Theorem 145.6, the full relation $\text{Geo} \subseteq M^4$ is Borel.*

Proof. For a fixed $C \in N_M$, let

$$B_C = \{(q, u, v) : \text{Geo}(q, C, u, v)\}.$$

The evaluation map $(u, v, i) \mapsto \beta(u, v, i)$ is Borel, by Borel division-with-remainder decoding. For each fixed i , the recurrence equality is therefore Borel in (q, u, v) . Since $[0, C]^M$ is countable, the defining universal condition for B_C is a countable intersection of Borel conditions. Thus B_C is Borel. Every valid certificate has $C \in N_M$, by Theorem 145.6, so, after the harmless permutation of coordinates,

$$\text{Geo} = \bigcup_{C \in N_M} (\{C\} \times B_C).$$

The union is countable and hence Borel. $\square$

This last proposition is a *consequence* of the continuity obstruction. The local proof does not assume that the certificate relation is Borel. An internally bounded quantifier can range over an externally uncountable set, so its being bounded alone would not justify the countable-intersection argument.

Coding boundaries. No successor, addition, or multiplication closure of D_q or $\mathcal{G}_M$ is asserted here. Such closure requires a code-extension or recoding lemma; the conditional identity for $e_q(a + b)$ does not supply one. Likewise, existence of an endpoint according to a chosen Δ_0 definition of exponentiation has not been identified with existence of the full β -certificate. Finally, membership of C in the core does not place its code parameters or its endpoint $e_q(C)$ in the core. In particular, the initial submodel need not recognize every certificate available in M .

146 The scope is strictly beyond total exponentiation

146.1 Recovering the known total-exponentiation obstruction

Corollary 146.1. *Suppose $M \models \text{I}\Delta_0$ has an uncountable Polish carrier with Borel arithmetic, and every $C \in M$ admits a base-two geometric code through C . Then multiplication is not separately continuous.*

Proof. If multiplication were separately continuous, Theorem 145.6 would make M countable. $\square$

There is also a direct local proof, avoiding the classical global countability argument. Apply Lemma 140.3 to the carrier, and take the upper endpoint of an increasing Cantor image. The interval below that endpoint is uncountable. Choose one code through it and apply Theorem 139.1.

Under the internally coded formulation of total exponentiation used in Glazer's paper, this recovers the multiplication obstruction [1]. The earlier project manuscript obtains global Borel exponentiation by an internal Diophantine argument [103]. The local proof needs only evaluation of the one chosen code.

[write] The earlier argument is source 14's corollary 106.4, Borel internal exponentiation from internal MRDP, used in theorem 111.1 and theorem 104.1(ii). Corollary 146.1 is a second route to the multiplicative conclusion, under the coded formulation of total exponentiation; it does not need internal MRDP, and it does not give source 14's cofinal form.

146.2 Actual models without total exponentiation

The weakening is not merely a change in the list of hypotheses. There are uncountable Borel models of $\text{I}\Delta_0$ in which total exponentiation fails, but one coded exponential interval is externally uncountable. The new theorem applies to them.

Proposition 146.2 (Polynomial cuts retaining a long code). *There is an uncountable standard Borel model $K \models \text{I}\Delta_0$ with Borel arithmetic such that:*

- (i) *a base-two geometric code exists through some $a \in K$ for which $[0, a]^K$ is uncountable;*
- (ii) *there is a $c \in K$ through which no such code exists;*
- (iii) *in the usual bounded-definition sense, total exponentiation also fails in K .*

The standard Borel space K admits a Polish realization. For every Polish topology with this same Borel structure, the local obstruction rules out separate continuity of multiplication.

Proof. Start with an uncountable standard Borel model $M \models \text{PA}$ whose operations are Borel. Existence is an established Borel-model theorem [97, 98], used here in the form explicitly stated in [2, Remark 17(a)] and in the introduction of [1]. An increasing Cantor set supplies $a \in M$ such that $[0, a]^M$ is uncountable. Since $M \models \text{PA}$, choose a code (u, v) for doubling through a .

Choose

$$c > \max(a, u, v, 2)$$

in M . It is nonstandard because a bounds an uncountable interval. Define externally

$$K = \bigcup_{n \in \mathbb{N}} [0, c^n]^M. \quad (8.1)$$

Only fixed standard powers occur in this definition. Each interval is Borel, so K is a Borel subset of M . It is uncountable, since it contains $[0, a]^M$.

It is downward closed. If $x < c^r$ and $y < c^s$ for standard r, s , then $xy < c^{r+s}$. Also $x + y < 2c^{\max(r,s)} \leq c^{\max(r,s)+1}$, because $c \geq 2$. The same estimate shows successor closure. Thus K is an initial subsemiring and $K <_{\Delta_0} M$. The bounded-induction argument of Proposition 145.4 proves $K \models \text{I}\Delta_0$.

All of a, u, v belong to K . The code condition is bounded, so it is absolute between K and M . This proves (i).

The element c itself belongs to K , since $c < c^2$. Suppose $u', v' \in K$ coded doubling through c inside K . Bounded absoluteness would give the same code in M . Uniqueness of the coded sequence in $M \models \text{PA}$ would then give

$$2^c = \beta(u', v', c) \leq u' < c^n \quad (8.2)$$

for some standard n . For every fixed standard n , PA proves that $2^x > x^n$ for all sufficiently large x , with a standard bound. The nonstandard c exceeds that bound, contradicting (8.2). This proves (ii).

For the usual internally correct bounded exponentiation graph, an endpoint at c in K would transfer by bounded absoluteness to the same unique endpoint 2^c in M [100, Lemma 2.1]. But 2^c exceeds every c^n with standard n , so it does not belong to K . This proves (iii).

Finally, a Borel subset of a standard Borel space is a standard Borel space. Equip K with a Polish topology realizing that structure. The restricted operations remain Borel. The retained code and its uncountable interval are unchanged, so Theorem 139.1 applies in every such Polish realization. $\square$

Remark 146.3. The inherited subspace topology on K is not asserted to be Polish. The construction first produces a standard Borel arithmetic structure, then chooses a Polish realization of the same measurable structure. This distinction is needed for the completeness hypothesis of the theorem.

146.3 Elementary boundary examples

Example 146.4 (Countable discrete models). Every countable model of PA, with the discrete topology, is Polish and has continuous arithmetic. Its coded exponentiation domain is its entire carrier, which is externally countable. Thus countability of the coded domain cannot be replaced by finiteness or by containment in the standard natural numbers.

Example 146.5 (Why the base must exceed one). For the constant progression of base 1, the parameters $u = v = 1$ give

$$\beta(1, 1, i) = 1 \bmod (i + 2) = 1$$

for every $i \in M$. Hence arbitrarily long constant progressions have the same simple code. The growth assumption $q \geq 2$ is essential; the obstruction concerns geometric growth, not the mere existence of sequence codes.

Example 146.6 (Borelness in the global step). The linear order ω_1 has countable principal initial segments and is uncountable. It illustrates why intervalwise countability alone does not imply global countability. The Borel-order theorem supplies the additional structure used in Section 145. This order example is not being presented as an arithmetic model.

147 A calibrated reverse-mathematical statement

147.1 What is being coded

The classical theorem has a direct interpretation for ordinary Polish spaces and set-sized structures. A reverse-mathematical statement requires a precise presentation convention. Use a code for a complete separable metric space, codes for Borel addition and multiplication, distinguished point codes for the arithmetic parameters, and the usual equality of represented points. The structure satisfies the required arithmetic axioms in that coded interpretation.

Uncountability of a Borel set means that no real codes a sequence exhausting all its points. The proof supplies the code for a continuous Cantor injection when the perfect-set input is applied. Continuity of a unary map can be expressed metrically using rational radii and the given point and function codes.

This is a scheme for coded presentations with the indicated finite arithmetic proof obligations. It is not a claim that full uncountable-model satisfaction is a single elementary arithmetical predicate.

Theorem 147.1 (Local upper bound, with coded interfaces). *With the preceding coded-presentation convention, Theorem 139.1 is provable in ATR_0 , using the uniform perfect-set/order construction in Glazer's optimization as the established descriptive-set-theoretic input.*

Proof. The fixed decoder (3.1) is obtained from the given Borel operations by unique analytic evaluation and analytic separation. The same applies to the elementary functions and the local logarithmic decoder. Their graphs and inverse images can be constructed as Borel codes.

The root sequence is generated by arithmetic recursion with the Borel square-root evaluator. For each external n , compose the finitely many radix maps to obtain an explicit injection

$$[0, D] \hookrightarrow [0, t_n]^{2^{n+1}}. \quad (9.1)$$

If a sequence enumerated the last interval, its finite Cartesian power, followed by the radix reconstruction, would enumerate $[0, D]$. This establishes the required uncountability for each stage without induction on a projective uncountability predicate.

The Borel sets $E(X_{n+1})$ form one coded sequence, by (3.7). Glazer's optimization provides the sequence of increasing continuous Cantor embeddings uniformly for a coded sequence of uncountable Borel sets [1, Optimization]. This replaces an unexplained appeal to countable choice for perfect-set witnesses.

At stage n of the metric construction, take positive rational lower bounds for the finitely many nonzero separation distances. Search for a k_n with certified strict edge inequalities. Convergence (6.2) makes this search terminate. The uniform tail estimate supplies a code for the limit map.

The comparison events (5.9) are Borel codes. The proof of late-flip meagerness uses Baire approximations, and the finite-change zero-one argument is an elementary cylinder argument in the coded category theory available in ATR_0 [70]. All arithmetic equalities involving E use fixed bounded-induction proofs with the code parameters held fixed. None requires total exponentiation or a Diophantine representation of its global graph. $\square$

This is a metamathematical proof with stated interfaces, not a machine-checked formalization. The new localization introduces no higher projective choice step: it evaluates one already chosen code.

[write] [Theorem 147.1](#) is printed **as claimed**. Its proof is a sketch with declared coding interfaces, and its perfect-set step rests on Glazer’s uniform “Optimization” as an input, as do the ATR_0 claims of [Part VIII \(section 105.4\)](#). It has not been independently reviewed, here or in the Hilbert’s-tenth programme, whose routing review says it does not certify the reverse-mathematical argument ([appendix E](#)). It is an upper bound for the local theorem only; no reversal is claimed.

147.2 The global conclusions are kept separate

The proof that the whole set $\mathcal{G}_M$ is countable uses Harrington–Marker–Shelah and arbitrary-subset countable cofinality. The present manuscript treats those as ordinary ZFC consequences. It does not assign an ATR_0 upper bound to all the global countability, submodel, or certificate regularity conclusions.

In particular, there are three different assertions:

1. A local interval obstruction is proved for a fixed code.
2. Its coded proof admits the stated ATR_0 upper bound.
3. A classical Borel-order theorem converts the family of local obstructions into global countability.

The third assertion should not be silently absorbed into the metatheory claimed for the first two.

147.3 No reversal is claimed

An upper bound in ATR_0 is not a reversal showing that ATR_0 is necessary. The precise strength may depend on how uncountability, the operation graphs, and the order-perfect-set witnesses are supplied. A restricted theorem with a continuous Cantor witness as part of the input could have a different calibration from the unrestricted Borel formulation.

148 Consequences for ProveIt and surreal arithmetic

148.1 An exact addition to the research program

The earlier total-exponentiation manuscript leaves a weaker-arithmetic question open [103, Question 12.2]. The present theorem gives an answer with an explicit, checkable additional condition: one code on an externally uncountable interval suffices. Proposition 146.2 shows that this condition is realized in Borel $\mathsf{I}\Delta_0$ -models where total exponentiation fails.

The proposed result is therefore useful even when internal MRDP has not been established for the object theory. It isolates the arithmetic information used by the topology: a Borel evaluator for a finite tuple, a long interval of correct recurrence, and bounded algebraic identities.

[write] The question cited is [question 115.2](#); the answer is partial (note there).

148.2 Relation to the surreal side

The repository's study of Polish arithmetic inside the omnific integers supplies a natural environment for asking these questions [102]. The results here concern set-sized presentations. To apply them to a structure realized by surreal expressions, one must first specify a set-sized carrier, its arithmetic operations, its topology, and the induction axioms it satisfies.

No topology on the proper class **No** or **Oz** is used in this paper. Nor does the existence of an integer-part structure by itself justify all of $\mathsf{I}\Delta_0$. Once a concrete set-sized presentation has Borel arithmetic and bounded induction, however, its geometric codes are subject to the same local obstruction. An embedding into a surreal structure does not remove that obstruction.

148.3 What can be formalized independently

The proof naturally divides into modules with small interfaces:

Module	Exact output needed by the next module
Remainder evaluation	Borel, total, unique quotient and remainder on a coded arithmetic presentation.
Local geometric code	Positivity, strict increase, and $E(a + b) = E(a)E(b)$ when $a + b \leq C$.
Integer scales	The inequalities (4.5), (4.8), and the local buffer (4.12).
Metric construction	A continuous injective limit map with the explicit estimates (5.4)–(5.5).
Shifted cancellation	The limit identity (6.10) using only multipliers indexed in $[b, d]$.
Category contradiction	Impossibility of the ordered Cantor product in Lemma 143.4.
Classical Borel order	Countability and structure of N_M , followed by code-domain consequences.

The finite algebra and metric estimates can be formalized before the full descriptive-set-theoretic environment. The repository's existing coding modules provide relevant syntax and arithmetic infrastructure [104], but a theorem about standard $\mathbb{N}$ should not be substituted for an internal theorem about all models of $\mathsf{I}\Delta_0$.

148.4 Descriptive complexity: what changes

The following table summarizes the conclusions. Projective upper bounds refer to the given standard Borel presentation.

Object	Without a continuity assumption	With separately continuous multiplication
Fixed decoder $\beta(u, v, i)$	Borel	Borel
Valid-code relation	Coanalytic upper bound	Borel
$\text{Geo}(q, C, u, v)$	Σ_2^1 upper bound	Countable, hence F_σ
D_q	Σ_2^1 upper bound	Countable, hence F_σ
Fixed-base endpoint graph	Σ_2^1 upper bound	Contained in the countable N_M
Union $\mathcal{G}_M$ of all length domains	Σ_2^1 upper bound	Contained in the countable N_M
Countable core N_M	Countable and F_σ , by classical Borel-order theory	Same conclusion

For the coanalytic upper bound, failure of a recurrence is witnessed by an $i < C$ satisfying a Borel condition. Existential quantification over the code parameters then gives the Σ_2^1 upper bounds. These are upper bounds, not completeness claims. The variable-base endpoint relation is a projection of the Borel certificate relation under the continuity hypothesis, so it is analytic; no general Borelness assertion for that projection is made here.

149 Further research: precise targets and priorities

The questions below are research proposals. None is presented as resolved by the preceding theorems. The first four would most directly improve the arithmetic threshold of the result.

149.1 Relating ordinary exponentiation to explicit codes

Question 149.1 (Highest priority: the ordinary exponential domain). Under Borel arithmetic and separately continuous multiplication, must the domain of the usual internally correct bounded exponentiation graph also be externally countable in every $\text{I}\Delta_0$ -model?

The standard partial exponential graph has internal uniqueness and recursion properties [100, 101]. Those facts do not by themselves provide a Borel evaluator on an externally uncountable domain. A possible route is to obtain an explicit sequence code on a sufficiently large polynomial root of every available exponential index.

Proposition 149.2 (A sufficient recoding criterion). *Suppose M satisfies the hypotheses of Theorem 145.6, and let $J \subseteq M$. Assume there is a fixed positive standard integer r such that for every $A \in J$ there is a geometric-code length C with*

$$A < (C + 1)^r. \quad (11.1)$$

Then $J \subseteq N_M$, and hence J is countable.

Proof. Every such C belongs to N_M . Successor and multiplication closure imply $(C+1)^r \in N_M$. Downward closure then gives $A \in N_M$. Equivalently, an r -digit radix expansion injects $[0, A]$ into $[0, C]^r$. $\square$

This proposition identifies a concrete arithmetic lemma that would settle [Question 149.1](#). The proposed recoding implication itself is not proved here. A proof must bound the construction of the new code inside the weak model.

Question 149.3 (Coding domains and closure). For the exact remainder decoder (3.1), what successor, additive, and multiplicative closure properties of D_q are provable in $\text{I}\Delta_0$? Which require additional coding or collection principles?

The issue is not agreement of existing codes: that follows from bounded induction. The issue is construction of a new code whose size may be much larger. A useful answer would compare explicit encodings and state the object-theory strength needed for each conversion.

149.2 The arithmetic threshold for continuous multiplication

Question 149.4 (Highest priority: bare bounded induction). Does an uncountable Polish space support a Borel model of $\text{I}\Delta_0$ in which multiplication is separately continuous, or jointly continuous?

The present theorem gives a strong necessary condition: all nontrivial geometric-code lengths must lie in the canonical countable core. It does not give a construction satisfying that condition, nor show that $\text{I}\Delta_0$ alone forces a code outside the core.

[write] This is the part of source 14's [question 115.2](#) that stays open (note there).

Question 149.5 (Open induction and one regular operation). Which natural models of IOpen admit a Polish Borel presentation with continuous multiplication? Can the finite Puiseux integer-part examples studied in [Provelt](#) be classified for this purpose?

The additive obstruction already has very weak hypotheses in Glazer's work. The multiplication-only problem should be investigated separately. Examples failing open induction cannot establish sharpness at $\text{I}\Delta_0$ or IOpen .

[write] Compare Part II's [question 28.7](#). The finite Puiseux principal-part ring of Part II satisfies $\text{Q} + \text{I}\text{Open}$ ([theorem 23.2](#)) and admits no Borel-compatible Polish recoding with continuous addition ([theorem 25.2](#)); Part V's integer part $\mathcal{I}_k$ satisfies open induction ([proposition 70.6](#)); the example failing open induction is Part III's cone of $\mathbb{Z} + t\mathbb{R}[t]$ ([theorems 38.1](#) and [38.2](#)).

149.3 Structure of the surviving countable part

Question 149.6 (Realizing the core and coding domain). Given a countable model $N \models \text{I}\Delta_0$ and a downward-closed subset $D \subseteq N$, when can they be realized as N_M and D_2 of an uncountable Borel arithmetic model with separately continuous multiplication?

This question combines an existence problem with an invariant problem. Any proposed realization must respect bounded elementarity of the core, agreement of overlapping codes, and whatever code-extension laws are available in the object theory.

Question 149.7 (Values versus index lengths). Can a coded geometric progression have an index $C \in N_M$ and endpoint $E(C) \notin N_M$ in a model with separately continuous multiplication? What relations are possible between the cofinalities of the coded index domain and its range?

The present proof controls index lengths. It does not place the certificate parameters or outputs inside the core. The distinction could be central to possible constructions.

149.4 Regularity of continuity and certificate parameters

Question 149.8 (Size of the continuous-parameter set). For one code E , how complicated can

$$P_E = \{a \leq C : \lambda_{E(a)} \text{ is continuous}\}$$

be? Can it be uncountable while every convex component is countable? Must it be Borel under natural additional hypotheses?

A jointly Borel family of functions has an analytic set of discontinuous parameters, by coding a point and a convergent sequence witnessing failure of continuity. Thus P_E has a general coanalytic upper bound, as also observed in the earlier project manuscript [103]. Proposition 145.5 adds an arithmetic restriction to that general descriptive bound. It does not settle its sharpness.

[write] For a coded exponential this refines source 14's [question 115.3](#); the coanalytic bound is the one proved before that question.

Question 149.9 (Variable-base endpoint relations). Under separately continuous multiplication, is the variable-base endpoint relation

$$\{(q, C, z) : \exists u, v [\text{Geo}(q, C, u, v) \wedge z = \beta(u, v, C)]\}$$

necessarily Borel?

The certificate relation is Borel, and its projection is analytic. Uniqueness of z does not by itself make a partial analytic graph Borel. An answer requires additional structure in the domain of bases for a fixed nonstandard length.

149.5 Metatheory and generalization

Question 149.10 (Exact reverse-mathematical strength). What is the exact strength, over RCA_0 , of the local interval theorem with fixed Borel ranks for the arithmetic operations? What is the additional strength of global countability of $\mathcal{G}_M$?

The two questions should be formalized separately. A useful intermediate target supplies ordered Cantor embeddings as part of the data, isolating the metric/category argument from the perfect-set construction.

[write] Compare source 14's [question 115.1](#), for total exponentiation.

Question 149.11 (Beyond full Polish completeness). Which weaker completeness assumptions suffice for the local obstruction? Is completeness only along the constructed binary system enough in a useful intrinsic class of topological arithmetic structures?

Lemma 143.1 uses a specific family of uniformly Cauchy sequences. Replacing full Polish completeness would require both a replacement for that limit existence and an adequate Borel or Baire-property framework.

Question 149.12 (Other coded multiplicative dynamics). Which internally coded recurrences, beyond $E(i+1) = qE(i)$, admit an analogous local obstruction? Can a general theorem be stated for Borel evaluators with bounded composition identities and sufficiently separated growth?

Lemma 143.4 already separates the topological contradiction from the origin of the factors y_n . The remaining work is arithmetic: construct separated factors and enough transfer identities from the chosen finite code.

Question 149.13 (A full formalization). Can the local geometric-code theorem be formalized in Lean or another proof assistant with the object theory and metatheory kept explicit?

A practical first milestone is the shifted cancellation lemma and complete-metric construction. The next is the fixed-code bounded-induction proof. The descriptive-set-theoretic interfaces should be represented as established lemmas with precise hypotheses, followed by their own formalization.

[write] No module of this plan exists. The ListCoding project cited for it proves standard-model theorems, not internal ones (section 1.4).

150 Verification, provenance, and interpretation

150.1 Mathematical verification

The main proof was checked through independent in-session mathematical reviews. The reviews concentrated on the nontrivial failure modes:

1. Borel evaluation must come from the fixed code, not merely from bounded arithmetic syntax.
2. Every argument of E must remain within its certified interval.
3. The small factor $E(h_j)$ must not be assumed continuous when $h_j < b$.
4. Products such as $E(b)V_s$ must not be replaced by an out-of-domain exponential call.
5. Limits must be shown nonzero without assuming order is closed.
6. Finite-change invariance must hold in both directions.
7. Global countability must use an actual Borel-order theorem rather than an invalid union of countable initial intervals.

The proofs in the manuscript explicitly address each point. The reviews are not independent external peer review and are not proof-assistant certification.

150.2 Finite diagnostics

The accompanying standard-library Python program checks the finite algebraic parts susceptible to rounding or buffer mistakes. With the recorded seed, it passed 14,953 assertions:

Diagnostic	Scope
24 explicit code cases	Chinese-remainder constructions for bases 2, 3, 5, 7, through lengths at most 9; decoder agreement and local identities.
24 scale and buffer scenarios	Exact integer roots with c near $2^{512}, 2^{1024}, 2^{2048}, 2^{4096}$, including irregular roots and endpoint cases.
2,856 vertex bounds	All tested finite vertices remain within the selected local interval; 16 scenarios have the anchor larger than its width.
36 shifted identities	Actual integer product equalities; 18 cases have a combined product exponent outside the code interval.

The last group tests the distinction in (6.7): the equality of ordinary products remains legitimate although combining them into one exponential call would exceed the domain. The program's construction of ordinary finite CRT codes is a diagnostic only. It is not a claim that the corresponding variable-length recoding theorem has been proved in $\mathsf{I}\Delta_0$.

No finite computation can validate external uncountability, a nonstandard model, the infinite category argument, or an ATR_0 provability claim. Those conclusions rest on the written proofs and their stated established inputs.

[write] The program and its record are shipped as `code/17-local-codes-verify_local_arithmetic.py` and `data/17-local-codes-local_arithmetic_diagnostics.json`. Run without an argument, the program writes its record beside itself; rerun it on a copy with an explicit output path (the report's README).

150.3 Claims and their status

Claim	Status in this manuscript
Glazer's total-exponentiation theorem	Established prior work.
Prior project's total-exponentiation ATR_0 argument	Existing research manuscript; credited and used as context.
Fixed-code interval obstruction	Main proved extension, including arbitrary bases $q \geq 2$.
Countability of all geometric-code lengths	Proved classical consequence of the local theorem and established HMS theory.
Borelness of all valid geometric certificates	Proved consequence under separate continuity of multiplication.
Models with a long local code and failure of total exponentiation	Proved polynomial-cut construction from established Borel PA models.
Exact ATR_0 equivalence	Open; only a local upper bound is given.
Countability of every usual partial-exponentiation domain	Open here; an explicit recoding criterion is isolated.
Uncountable Polish $\mathsf{I}\Delta_0$ models with continuous multiplication	Existence remains unresolved here.

150.4 Reading the result at its proper scale

The theorem shows that topological regularity constrains how much internally finite computation can be encoded over an externally large interval. Its proof is local in the arithmetic

resources and in the multiplier parameters, while its countability consequences are global in the presentation.

This provides a concrete advance in the weaker-arithmetic direction identified by Provelt. The remaining work is sharply defined: determine how far explicit coding follows from ordinary partial exponentiation, and determine whether the surviving countable-domain condition can actually coexist with uncountable continuous-multiplication models.

Part XII

Polish group completions and the size of class manifolds

[write] This part is source 18, manuscript 03 of batch 90, *Polish Group Completions and the Size of Class Manifolds: An intrinsic cancellation criterion and a local rank approach to Glazer's manifold question* (31 pages, A4, dated October 4, 2026 UTC; "prepared with AI assistance"). It arrived in commit a162e4386, as archive glazer_proveit_topology_research, and was placed in 12076b2e8; it is pinned to 2b7b388ba, 135 commits before the placement. Its file prefix in this report is 18-polish-completions-. All unmarked text in this part is source 18's; its sixteen sections are Sections 151–166 (its Section k is Section $150 + k$, with statement numbers otherwise unchanged), and its two appendices are [appendices AA](#) and [AB](#). Source 18 divides itself into three parts; to avoid a clash with this report's Parts they are printed as Divisions I–III, and its twelve "Research questions" are Questions 165.1–165.12.

What this part claims, and its status

Division I. Let M be a Hausdorff space with jointly continuous, commutative, cancellative addition whose natural order ($x \leq y$ iff $x + z = y$ for some z) is linear. The topology of M extends to a Hausdorff group topology on $G = M - M$ if and only if partial subtraction $\delta(y, x) = y - x$ on $\{x \leq y\}$ is continuous; the extension is unique, with zero neighbourhoods $U - U$ ([theorem 153.1](#)); it is Polish when M is ([theorem 154.1](#)) and locally compact when M is ([corollary 154.2](#)); continuity need only be tested at the pairs (x, x) and $(x, 0)$ ([theorem 154.3](#)). A Polish Presburger model with continuous addition and partial subtraction has clopen residue classes and is disconnected ([corollary 155.3](#)).

Division II. In a class theory with Foundation omitted (not negated), without Global Choice, a connected Hausdorff class space of bounded tightness on sets with an ordinal exhaustion or a specified neighbourhood operator is a set ([theorem 160.2](#)); connected real class manifolds with an exhaustion, a specified class metric, or a coding in or from $\mathbf{No}^m$ are sets of size $\mathfrak{c}$ ([theorems 161.3, 161.5 and 163.2](#)). Glazer's unrestricted question whether Global Choice suffices is **not** resolved ([question 165.6](#)).

Unrefereed, AI-assisted, internally reviewed in AI-assisted passes, and not formalized; historical priority is not established. Source 18 says that the completion criterion "should not be described as the resolution of a longstanding community-wide problem without further evidence".

Questions answered. [Theorems 153.1 and 154.1](#) and [corollary 154.2](#) answer exactly source 08's [question 57.2](#) (the locally compact Polish Presburger monoids with a locally compact Polish extending group are those with continuous partial subtraction), source 10's

question [question 88.4](#) (which Polish cancellative Presburger monoids embed topologically as the positive cone of a Hausdorff Polish group) and source 15's [question 126.3](#) (an intrinsic condition stated with the partial difference map, as that question asks); the sentence of source 09's [question 73.11](#) asking which hypotheses force a topology to extend from the cone to its group completion; and, for Polish monoids, source 12's [question 88.3](#) (the extending group is Polish, hence Baire and ccc, and the cone is G_δ , hence has the Baire property). Throughout, the monoid is assumed to have continuous addition, as every one of these questions does. Part X's note that this completion question "is open" (after [question 138.9](#)) is superseded. The questions on connected carriers ([questions 14.3, 43.4, 103.4, 126.2](#) and [138.10](#)) are narrowed, not answered: a connected example must have discontinuous partial subtraction. Dated notes after each question say what is answered. Source 18 does not answer Glazer's Question 2 anew, nor his Question 1.

What was already here, and what is new. At its pin this report had Parts I–V; source 18 read it there and the batch-87 archives of sources 13 and 15 (now Parts VII and IX). Its [proposition 155.1](#) is a special case of Part III's [proposition 34.3](#), which it credits; its [example 155.4](#) is Part I's model $M_{\mathbb{R}}$ ([theorem 3.1](#)), whose monus is discontinuous by [theorem 8.3](#) and [corollary 37.2](#); its [example 155.5](#) is Part III's cone of $\mathbb{Z} + t\mathbb{R}[t]$ with the topology τ_P ([theorem 38.1](#)), which it credits to source 15; its [corollary 155.3](#) recovers, by an intrinsic hypothesis, the disconnectedness of cones inheriting a Polish group topology already noted after [question 14.3](#); and [proposition 163.4\(ii\)–\(iii\)](#) restate, for sets, clauses of the foundations report's `found:thm:discrete` that are proved in Lean for the real sign carrier (note after the proposition). New to the report are the criterion and its upgrades ([theorems 153.1, 154.1](#) and [154.3](#), [corollaries 154.2](#) and [154.7](#) and [propositions 154.4](#) and [154.6](#)), the whole of Division II, and Glazer's class-manifold question itself, which no earlier file of the repository mentions (note after [question 165.6](#)). Two consequences written for this report, [propositions 155.6](#) and [155.7](#) [merge], combine the criterion with Parts VI and IX.

Labels and notation. Every label of source 18 takes the prefix `pma:gcm:`; labels were added to its twelve questions and its terminology appendix, `pma:part:gcm` to the part, and `pma:gcm:prop:rings` and `pma:gcm:prop:lc` to the two merge propositions. The fifth block of the table in [section 1.3](#) lists the symbols. In short: its `\dotminus` is printed $\dot{-}$ and its `c` macro is the report's c ; δ is partial subtraction, not the monus (tempting false reading: continuity of δ is not continuity of $\dot{-}$; [example 155.4](#)); its $G = M - M$, E_M , F , B_U and, in Division II, X_α , $N(x)$, $H(A)$, O_α , K , $C(x)$, L and Φ_A are local; its base theory $\text{GB}_{\text{-Fnd}}^{\text{rep}} + \text{AC}$ is its own notation, defined in [section 157](#). No normalization changed.

Files. Source 18 ships no executable checks. Its build script and proof audit are shipped as `code/18-polish-completions-build.sh` and `18-polish-completions-PROOF_AUDIT.txt`. Its source record is not shipped, because it contains a third party's personal profile address; its verified content is summarized in [appendix E](#). The Hilbert's-tenth programme routed the archive from its README only, without a correctness finding ([appendix E](#)). Nothing in this part is formalized.

151 The research intersection and the status of the results

Glazer's own papers provide the relevant mathematical connection. His work includes topological models of arithmetic, the behavior of Global Choice over theories weaker than

ZFC, and reflection and Collection in the presence of urelements [1, 105, 106, 107]. His public research biography also identifies set theory, formal systems, and choice principles as central interests [108]. The topic selection here uses these authored sources, rather than inferring research interests from activity on a professional profile.

The repository provides two concrete starting points. Its Presburger project implements integer quantifier elimination, while its research corpus studies uncountable Polish Presburger presentations and their relation to the omnific integers [109, 110]. Its surreal-foundation reports distinguish classes, birthday cutoffs, set-sized workspaces, and type universes [113, 114]. The present article connects these threads through a precise question: when does local topological data extend to a legitimate global object?

There are two different global objects in this question. For an additive monoid, the desired object is a topological group containing its formal differences. For a class space, the issue is whether the entire carrier is a set. The arguments are independent; neither theorem is used to prove the other. Both isolate a missing hypothesis that informal constructions can easily overlook.

151.1 The arithmetic question actually answered

The inspected incoming manuscript *Polish Arithmetic: Multiplication Obstructions and Local Compactness* [111] explicitly asks for intrinsic conditions ensuring that a locally compact Polish Presburger monoid inherits its topology from a Polish group of differences. It suggests translations, partial subtraction, and neighborhood structure as the appropriate language. The pinned merged report also asks for a locally compact extending group [110, Part IV]; [corollary 154.2](#) supplies that conclusion when the monoid is locally compact.

[Theorems 153.1](#) and [154.1](#) answer this question: continuous partial subtraction is necessary and sufficient, and local compactness is unnecessary. [Theorem 154.3](#) gives two local families of tests in place of continuity on every ordered pair. The theorem applies to every monoid satisfying the stated algebraic hypotheses, not only to models of Presburger arithmetic.

The earlier existence question from Glazer’s paper is not being answered anew. His Question 2 asks for an uncountable Polish Presburger model with continuous addition [1, Further work]. The repository already supplies such models, and its later manuscripts investigate nonsplitting, regularity, and local compactness [110, 112, 111]. Those are background, not discoveries of the present article.

[write] (Batch 90.) The “inspected incoming manuscript” [111] is source 15, now [Part IX](#); its question is [question 126.3](#). The question of “the pinned merged report” in its Part IV is source 08’s [question 57.2](#); the same question is asked as source 09’s [question 73.11](#) and source 12’s and source 10’s [questions 88.3](#) and [88.4](#). Dated notes after each say what [theorems 153.1](#) and [154.1](#) answer, and the note after [question 138.9](#) that the question “is open” is superseded.

151.2 The manifold question partially addressed

In his 2025 Oberwolfach abstract, Glazer discusses classification of connected class curves over ZFC with Foundation omitted, and raises the higher-dimensional sethood question with Global Choice available [106, pp. 106–107]. We retain *connectedness* from that discussion: allowing arbitrary disconnected manifolds would make the proposed sethood statement false immediately, as $\text{Ord} \times \mathbb{R}^n$ with separate Euclidean components shows.

Our result gives sufficient hypotheses for sethood in all finite dimensions. A connected

Hausdorff real class manifold is a set whenever its carrier admits an ordinal exhaustion by sets, a class injection into such a carrier, or a class surjection from such a carrier. Surreal sign sequences provide a particularly relevant exhaustion. We also give a criterion using a specified neighborhood operator and apply it to class metrics and topological groups.

This is a restriction on possible counterexamples, not a proof that Global Choice alone supplies the missing coding or neighborhood operator. Glazer already observes that a global bijection between the universe and the ordinals suffices for the relevant structural consequences of Foundation. Here the rank hypothesis is confined to the carrier or its parametrization, and its exact use is exposed.

[write] (Batch 90.) Glazer’s abstract [106, pp. 106–107] reports that the classification of connected one-dimensional manifolds, not necessarily second countable or set-sized, is not a theorem of ZFC without Foundation but is provable with Global Choice added, and continues that “a natural generalization is that $\text{ZFC} - \text{Fnd} + \text{GC}$ proves that every topological manifold is a set”, for which the one-dimensional methods do not seem to suffice in dimension two; he notes that a global bijection of the universe with Ord is enough. His sentence does not say “connected”: connectedness is source 18’s reading, and [example 162.1](#) shows it is needed for a non-trivial statement. Glazer works in the one-sorted theory $\text{ZFC} - \text{Fnd}$ with a Global Choice symbol, source 18 in its own class theory $\text{GB}_{-\text{Fnd}}^{\text{rep}} + \text{AC}$ ([section 157](#)). The question is new to the repository: no earlier report or README mentions it. It is printed as [question 165.6](#).

151.3 Contribution and attribution

Statement	Role in this article	Status
Partial subtraction criterion and local tests	Resolves the inspected repository’s completion question	Proved here; priority unverified
Automatic Polishness of the extending group	Removes the local compactness restriction	Proved from classical metrization and category
Clopen set hull from bounded tightness and local rank	Supplies sethood without Foundation	Proved with explicit class assumptions
Surreal coding and class metric corollaries	Restrict possible counterexamples	Deductions proved here
Cardinality of a connected Hausdorff real manifold	Quantitative conclusion after sethood	Classical; attributed to Nyikos
Uncountable Polish Presburger examples	Background and diagnostic examples	Already in the repository
General manifold question under Global Choice	Remaining frontier	Not resolved here

Table 3: The claims made and the claims left open.

The repository comparison is pinned to commit 2b7b388ba81a (full identifier in [section 166](#)). Only the specified relevant files and archive members were inspected. Nothing here certifies the rest of the repository. The proofs below are mathematical arguments, not Lean or Rocq certificates.

[write] At this pin the report had Parts I–V; the batch-87 archives it cites are now Parts VII and IX. Its [110, Part III] and [110, Part IV] are this report’s Parts III and IV.

Division I of source 18: An intrinsic criterion for Polish group completion

152 Algebraic differences and the topological obstruction

Definition 152.1. A linearly naturally ordered cancellative monoid is a commutative monoid $(M, +, 0)$ with cancellation, such that

$$x \leq_M y \iff (\exists z \in M) x + z = y$$

is a linear order.

Antisymmetry is part of the hypothesis. In particular, the only invertible element of the monoid is 0. A nonnegative cone in a linearly ordered abelian group is the principal example. The additive reduct of a Presburger model satisfies these conditions.

The algebraic group $G = M - M$ is the quotient of $M \times M$ by

$$(a, b) \sim (c, d) \iff a + d = b + c.$$

Write the class of (a, b) as $a - b$, and identify M with the injective image $a \mapsto a - 0$. Totality gives

$$G = M \cup (-M), \quad M \cap (-M) = \{0\}. \quad (152.1)$$

Indeed, either $a = b + z$ or $b = a + z$, so $a - b = z$ or $a - b = -z$.

If M has a topology, algebraic group completion does not by itself make this embedding a topological embedding. Cancellation says that an equation has at most one solution; it says nothing about continuity of the operation that recovers this solution. Define

$$E_M = \{(y, x) \in M^2 : x \leq_M y\}, \quad \delta : E_M \longrightarrow M, \quad \delta(y, x) = y - x. \quad (152.2)$$

The domain always carries the subspace topology from M^2 . Neither openness nor closedness of E_M is assumed.

Remark 152.2. The operation δ is *partial subtraction*. It is different from total truncated subtraction $y \dot{-} x = \max(y - x, 0)$. Continuity of the latter may fail when one crosses between order regions. Our criterion concerns only the region where the algebraic difference belongs to M .

153 The exact Hausdorff criterion

Theorem 153.1 (Intrinsic group-completion criterion). *Let M be a Hausdorff space with jointly continuous addition, whose additive algebra is a linearly naturally ordered cancellative monoid. The following are equivalent:*

1. $\delta : E_M \rightarrow M$ is continuous.
2. $G = M - M$ admits a Hausdorff topological group topology whose restriction to M is its original topology.
3. The map

$$F : M \times M \longrightarrow M \times M, \quad F(x, z) = (x + z, x),$$

is a topological embedding.

When these conditions hold, the extending group topology is unique. A neighborhood base at zero consists of

$$B_U = U - U, \quad (153.1)$$

where U ranges over the open neighborhoods of zero in M . No metrizability, completeness, or local compactness assumption is needed.

Proof. Restricting continuous subtraction in an extending topological group gives $(2) \Rightarrow (1)$. The map F is a continuous bijection onto E_M , with inverse

$$F^{-1}(y, x) = (x, \delta(y, x)).$$

Thus (1) and (3) are equivalent. We construct the topology for $(1) \Rightarrow (2)$.

Step 1: the group neighborhood base. Each B_U contains zero and is symmetric, and $B_{U \cap V} \subseteq B_U \cap B_V$. Continuity of addition at $(0, 0)$ gives an open neighborhood V of zero with $V + V \subseteq U$. Then

$$B_V + B_V = (V + V) - (V + V) \subseteq B_U.$$

If $g = u - v \in B_U$, continuity supplies an open zero neighborhood W with $u + W \subseteq U$ and $v + W \subseteq U$, whence $g + B_W \subseteq B_U$. The translates of these sets form a base for a group topology: the last inclusion gives internal openness, symmetry gives continuous inversion, and the sum inclusion gives continuous addition. Commutativity removes the conjugation condition.

Step 2: Hausdorffness. Let $g = a - b \neq 0$, so $a \neq b$ in M . Choose disjoint open neighborhoods P of a and Q of b . Choose a zero neighborhood U with $a + U \subseteq P$ and $b + U \subseteq Q$. If $g = u - v$ for $u, v \in U$, then $a + v = b + u$, contradicting $P \cap Q = \emptyset$. Thus $g \notin B_U$. The intersection of the zero neighborhoods is $\{0\}$, so the group topology is Hausdorff. This step uses neither a chosen net nor a countable local base.

Step 3: the original topology is at least as fine. Fix $x \in M$ and a zero neighborhood U in M . Continuity of δ at (x, x) gives a product neighborhood whose intersection with E_M is sent into U . Intersect its two coordinate neighborhoods to get an original neighborhood W of x . For every $y \in W$, totality gives either $y - x \in U$ or $x - y \in U$. Since $0 \in U$,

$$W \subseteq (x + (U - U)) \cap M.$$

Every group-induced neighborhood thus contains an original one.

Step 4: the induced topology is at least as fine. Let O be an original open neighborhood of $x \in M$. Continuity of δ at $(x, 0)$ gives neighborhoods P of x and Q of zero such that

$$\delta((P \times Q) \cap E_M) \subseteq O.$$

Choose a zero neighborhood $U \subseteq Q$ with $x + U \subseteq P$. For $y \in (x + (U - U)) \cap M$, write $y = x + u - v$, $u, v \in U$. The equality $x + u = v + y$ shows that $(x + u, v) \in E_M$, so

$$y = \delta(x + u, v) \in O.$$

Thus $(x + B_U) \cap M \subseteq O$. The two topologies on M are equal.

Step 5: uniqueness. Let τ be any extending group topology. Given $U \ni 0$ open in M , take a symmetric τ -neighborhood V with $V \cap M \subseteq U$. By (152.1),

$$V \subseteq U \cup (-U) \subseteq U - U.$$

Conversely, given a τ -neighborhood W of zero, choose a symmetric τ -neighborhood V with $V - V \subseteq W$, and put $U = V \cap M$. Then $U - U \subseteq W$. This identifies its neighborhood filter with the constructed one. $\square$

Remark 153.2 (Why totality matters). Step 3 and uniqueness use $G = M \cup (-M)$, a stronger assertion than generation by M . No analogous theorem for arbitrary partially ordered cones is asserted here.

154 Polishness, local tests, and the universal property

Theorem 154.1 (Automatic Polish upgrade). *Under the hypotheses of theorem 153.1, suppose M is Polish. Whenever an extending Hausdorff group topology exists, it is Polish.*

Proof. A countable zero-neighborhood base (U_n) in M gives the countable group base $(U_n - U_n)$. The Hausdorff group is metrizable by the classical group metrization theorem [117]. Since it is abelian, a compatible translation-invariant metric exists.

If $D \subseteq M$ is countable dense, then $D - D$ is dense in G , by continuity and surjectivity of the difference map on M^2 . Complete a compatible invariant metric to obtain a separable complete metric group $\widehat{G}$. Addition and inversion extend by their uniform continuity.

The embedded M is completely metrizable in its subspace topology. The classical subspace theorem makes it a G_δ in $\widehat{G}$ [9]; the same holds for $-M$. A finite union of G_δ 's is G_δ , since

$$\left(\bigcap_n O_n\right) \cup \left(\bigcap_m V_m\right) = \bigcap_{n,m} (O_n \cup V_m).$$

Thus $G = M \cup (-M)$ is a dense G_δ subgroup of $\widehat{G}$. For every $h \in \widehat{G}$, the dense G_δ 's G and $h + G$ intersect by Baire category. Hence $h \in G - G = G$, and $G = \widehat{G}$. $\square$

The result preserves the *specified* topology of M ; it is stronger than existence of an unrelated Polish presentation. The finite union is essential to this argument. An arbitrary countable union of completely metrizable subspaces need not be a G_δ .

Corollary 154.2. *If M satisfies the completion criterion and is locally compact, its canonical Hausdorff group completion is locally compact.*

Proof. Take a compact neighborhood K of zero in M , containing an open zero neighborhood U . The set $K - K$ is compact in G , as a continuous image of $K \times K$, and contains the group neighborhood $U - U$. Thus G is locally compact at zero, and translation gives local compactness everywhere. $\square$

Theorem 154.3 (Two local tests suffice). *In theorem 153.1, full continuity of δ is equivalent to continuity at every (x, x) and every $(x, 0)$. If M is metrizable, these conditions are equivalent to the following implications for every $x \in M$ and every pair of sequences satisfying the indicated assumptions:*

$$a_n, b_n \rightarrow x, \quad b_n \leq_M a_n \implies a_n - b_n \rightarrow 0; \quad (154.1)$$

$$a_n \rightarrow x, \quad b_n \rightarrow 0, \quad b_n \leq_M a_n \implies a_n - b_n \rightarrow x. \quad (154.2)$$

Proof. The construction uses continuity of δ only in Steps 3 and 4, at precisely these pairs. Under these restricted assumptions it still gives an extending group topology, whose subtraction then restricts to a continuous δ everywhere. In metrizable E_M , continuity at a point is equivalent to preserving limits of sequences, giving the displayed formulations. $\square$

These tests use only convergent sequences in the original carrier and differences defined there. They can be checked before any group topology is constructed.

Proposition 154.4 (Universal property). *If the criterion holds, every continuous monoid homomorphism $f : M \rightarrow H$ to a Hausdorff abelian topological group extends uniquely to a continuous group homomorphism*

$$\tilde{f} : G \rightarrow H, \quad \tilde{f}(a - b) = f(a) - f(b).$$

Proof. Algebraic group completion gives well-definedness and uniqueness. Given a zero neighborhood W in H , choose $V \ni 0$ with $V - V \subseteq W$, and a zero neighborhood U in M with $f(U) \subseteq V$. Then $\tilde{f}(U - U) \subseteq W$, proving continuity at zero, and hence everywhere. $\square$

Corollary 154.5. *Continuous homomorphisms between monoids satisfying the criterion extend uniquely to their canonical group completions. A homeomorphic monoid isomorphism extends to a topological group isomorphism.*

Proof. Apply [proposition 154.4](#), and in the isomorphism case apply it also to the inverse. $\square$

154.1 Biadditive maps and topological rings

Proposition 154.6. *Suppose M satisfies the completion criterion, with canonical completion G . Every jointly continuous biadditive map $b : M \times M \rightarrow H$, where H is a Hausdorff abelian topological group, extends uniquely to a jointly continuous biadditive map $B : G \times G \rightarrow H$.*

Proof. Biadditivity gives a well-defined algebraic extension

$$B(a_1 - a_2, c_1 - c_2) = b(a_1, c_1) - b(a_1, c_2) - b(a_2, c_1) + b(a_2, c_2).$$

Given a zero neighborhood O in H , choose a symmetric zero neighborhood T with $T + T + T + T \subseteq O$. Continuity of b at $(0, 0)$ supplies open zero neighborhoods U, V in M such that $b(U \times V) \subseteq T$. The formula gives

$$B((U - U) \times (V - V)) \subseteq O,$$

proving joint continuity at $(0, 0)$.

For fixed $g = a_1 - a_2$, the map $x \mapsto b(a_1, x) - b(a_2, x)$ is a continuous monoid homomorphism on M . The universal property makes its extension $B(g, \cdot)$ continuous. The other variable is handled in the same way. Finally,

$$B(g + u, h + v) - B(g, h) = B(g, v) + B(u, h) + B(u, v)$$

combines separate continuity and continuity at $(0, 0)$ to give joint continuity everywhere. Additive generation proves uniqueness. $\square$

Corollary 154.7. *If M is a topological semiring satisfying the additive completion criterion, its algebraic ring completion is a Hausdorff topological ring. If M is Polish, this ring is Polish.*

Proof. Apply [proposition 154.6](#) to $b(x, y) = i(xy)$ with target the completed group G , where $i : M \hookrightarrow G$. The extension is the usual multiplication of formal differences. Distributivity and additive generation preserve associativity and any multiplicative identity. Use [theorem 154.1](#) for Polishness. $\square$

Thus multiplication introduces no further continuity obstruction once joint multiplication on M and the additive completion criterion both hold.

155 Borel regularity, residues, and examples

155.1 Borel subtraction comes for free; continuity does not

The following is a direct special case of the automatic-Borel definability principle already proved in the merged report [[110](#), Part III]. We include the short proof to identify the regularity that is automatic before the completion criterion is applied.

Proposition 155.1. *Suppose M is Polish and satisfies the algebraic hypotheses of [theorem 153.1](#), with continuous addition. Then the order relation E_M is Borel in M^2 , and its partial subtraction $\delta : E_M \rightarrow M$ is Borel measurable.*

Proof. The map $F(x, z) = (x + z, x)$ is continuous and injective. The Lusin–Souslin theorem says that its image is Borel and its inverse, with domain that image, is Borel [[9](#)]. Its image is E_M ; the second coordinate of $F^{-1}(y, x)$ is $\delta(y, x)$. $\square$

Thus Borel measurability is not the missing completion hypothesis. It already follows from cancellation, continuity of addition, and Polishness. The issue is the topology of that Borel inverse.

[write] [Proposition 155.1](#) is the case of order and partial subtraction of Part III’s [proposition 34.3](#), which source 18 credits.

Lemma 155.2. *Every analytic subgroup of finite index in a Polish group is open.*

Proof. Analytic sets have the Baire property [[9](#)]. A subgroup H of finite index cannot be meager, since its finitely many cosets cover the Baire space G . Choose a nonempty open O in which H is comeager. Take a nonempty open $V \subseteq O$ and a zero neighborhood W with $W + V \subseteq O$; for example first choose a point of O , then sufficiently small translated neighborhoods. For $h \in W$, both $H \cap V$ and $(H - h) \cap V$ are comeager in V . Their intersection is nonempty, so $h \in H - H = H$. Consequently $W \subseteq H$, and H is open. For a nonabelian group the same argument uses products and inverses. $\square$

Corollary 155.3. *If a Polish model M of Presburger arithmetic has continuous addition and partial subtraction, then each congruence class modulo every integer $n \geq 2$ is clopen in M . In particular, M is disconnected.*

Proof. Its ordered group completion G is a $\mathbb{Z}$ -group: division with remainder gives $G/nG \cong \mathbb{Z}/n\mathbb{Z}$. By [theorem 154.1](#), G is Polish. The subgroup nG is the continuous image of G under multiplication by n , hence analytic. It has finite index, so [lemma 155.2](#) makes it open; its complement is a union of open cosets, so it is also closed. Restrict the cosets to M . All n residues occur because the standard numerals $0, \dots, n - 1$ occur in M . $\square$

The finite-index argument is classical and its Presburger use is already present in the repository's group-based analysis [110, 112]. The contribution here is an exact intrinsic condition for bringing that argument to bear. A connected Polish Presburger presentation, if one exists, must fail one of the two sequence tests in [theorem 154.3](#). The corollary does not settle existence of such a presentation.

[write] (Batch 90.) The notes after Part I's [question 14.3](#) and source 13's [theorem 95.4](#) already answered the connected question negatively for cones whose topology is inherited from a Polish topological $\mathbb{Z}$ -group. By [theorems 153.1](#) and [154.1](#), a cone topology with continuous addition is so inherited exactly when partial subtraction is continuous, so [corollary 155.3](#) is that answer with an intrinsic hypothesis. The connected-carrier questions of the report are narrowed accordingly (notes after them).

155.2 Partial subtraction is weaker than continuous monus

In a linearly ordered monoid the *monus* is the total operation

$$a \dot{-} b = \begin{cases} a - b, & b \leq a, \\ 0, & a < b. \end{cases}$$

Continuity of this total operation implies continuity of partial subtraction, but the converse is false.

Example 155.4. Let $G = \mathbb{R} \times \mathbb{Z}$ have its product topology and lexicographic order, with the real coordinate dominant. Its nonnegative cone is

$$M_{\mathbb{R}} = (\mathbb{R}_{>0} \times \mathbb{Z}) \cup (\{0\} \times \mathbb{N}).$$

This is a locally compact Polish Presburger monoid with continuous partial subtraction and discontinuous monus.

Proof. In the slice $\mathbb{R} \times \{k\}$, the cone is $[0, \infty)$ for $k \geq 0$ and $(0, \infty)$ for $k < 0$. It is therefore a countable topological sum of locally compact Polish spaces. Addition and partial subtraction are restrictions of the corresponding continuous group operations.

The least positive element is $(0, 1)$. For $n \geq 1$, write $k = nq + s$, $0 \leq s < n$. If $r > 0$, then

$$(r, k) = n(r/n, q) + (0, s);$$

if $r = 0$ and $k \geq 0$, the ordinary division algorithm gives a nonnegative quotient. These are the usual ordered-group conditions for a Presburger cone.

Put $a_j = (1/j, 0)$ and $b = (0, 1)$. Then $a_j \rightarrow 0$ and $a_j > b$, whereas

$$a_j \dot{-} b = (1/j, -1)$$

does not converge to $0 = (0, 0)$: its integer coordinate is constantly -1 . But $0 \dot{-} b = 0$. Hence monus is discontinuous. $\square$

Variants of this lexicographic model occur in Provelt [110]. Here it identifies why continuity of a total truncated operation would impose an unnecessarily strong hypothesis.

[write] This is Part I's model $M_{\mathbb{R}}$ ([theorem 3.1](#)), with the same topology. Its partial subtraction is continuous because it is the restriction of the group subtraction, which is also the case $A = \mathbb{R}$ of Part III's [theorem 34.1](#); the discontinuity of the monus is Part I's [theorem 8.3](#) and Part III's [corollary 37.2](#).

155.3 The existing polynomial obstruction, localized

Example 155.5. Let

$$A = \mathbb{Z} + t\mathbb{R}[t]$$

be the additive ordered group of polynomials with integer constant coefficient, ordered by their leading coefficient, with t positive and larger than every integer. Let $M = A_{\geq 0}$.

Give each nonnegative constant polynomial an isolated point. For each integer constant coefficient m and each degree $d \geq 1$, give the positive polynomials in that stratum their coefficient topology

$$\mathbb{R}^{d-1} \times (0, \infty),$$

and give M the topological sum of all these strata and the isolated constants. Then M is a locally compact Polish Presburger monoid with continuous addition, but its topology extends to no Hausdorff group topology on A .

Proof. There are countably many strata, each locally compact Polish. Addition is continuous on every pair of strata: positive leading coefficients do not cancel, so the degree of a sum is fixed there and its coefficients are polynomial, indeed linear, expressions in the input coefficients. A pair involving an isolated constant is treated in the same way. Since the strata are open, this proves joint continuity globally. The same observation proves continuity of multiplication if it is included in the language.

The least positive element is 1. Given $p \in M$ and $n \geq 1$, divide all nonconstant coefficients by n , and divide the integer constant coefficient with remainder $s \in \{0, \dots, n-1\}$. For a nonconstant positive polynomial the quotient still has positive leading coefficient; for constants use ordinary nonnegative division. Thus the cone is a Presburger model.

Now set

$$a_j = t^2 + t/j, \quad b_j = t^2.$$

Both converge to t^2 in the same degree-two stratum, and $b_j \leq a_j$. But

$$\delta(a_j, b_j) = t/j$$

cannot converge to the isolated point 0. Partial subtraction fails at (t^2, t^2) , so [theorem 153.1](#) rules out any extending Hausdorff group topology. $\square$

This example and its group-completion obstruction are already in the incoming arithmetic manuscript [111]; they are reproved for a self-contained hypothesis check. The present criterion locates the failure at the diagonal. Since 0 is isolated, the zero-axis sequence test is automatic in this example: a sequence $b_j \rightarrow 0$ is eventually 0. Thus the zero-axis test by itself cannot replace the full criterion.

Another view of the same obstruction is that translation by t^2 is not a topological embedding. The sequence t/j does not converge to 0, although its translated sequence converges to the translate of 0. Cancellation is an algebraic statement and supplies no continuity for this inverse translation.

[write] The cone of [example 155.5](#) is Part III's cone of $\mathbb{Z} + t\mathbb{R}[t]$ with source 06's topology τ_p ([theorem 38.1](#)), which source 15 studies again as its degree-stratum cone ([Part IX](#)). The report already knew that its group of differences has no locally compact second-countable group topology with Borel order (Part IV's [theorem 44.1](#), note after [question 57.2](#)) and no Polish group topology compatible with its coefficient Borel structure (source 15). [Example 155.5](#) shows more: no Hausdorff group topology on A extends the topology of the cone, and the

failure sits at the diagonal pair (t^2, t^2) . Since that cone also has continuous multiplication, the example is a case of [proposition 155.6](#) below.

Proposition 155.6 (Uncountable discretely ordered rings [merge]). *Let R be a discretely ordered ring whose nonnegative part $M = R_{\geq 0}$ is uncountable and carries a Polish topology in which addition is continuous. Then partial subtraction $\delta : E_M \rightarrow M$ is discontinuous. No hypothesis on multiplication is made.*

Proof. The natural order of the monoid $(M, +, 0)$ is the order of R : for $x, y \in M$ there is $z \in M$ with $x + z = y$ exactly when $y - x \geq 0$. So M is a Hausdorff, cancellative, linearly naturally ordered monoid with continuous addition, and $M - M = R$. If δ were continuous, [theorems 153.1](#) and [154.1](#) would give a Polish group topology on $(R, +)$ inducing the topology of M , and the proof of [theorem 154.1](#) shows that M is a G_δ in it. So the positive cone of R would be Borel, and in particular would have the Baire property. By source 10's [corollary 78.6](#), R would be countable, contrary to the uncountability of M . $\square$

So the hypothesis of a “specified topological group completion” in source 10's [proposition 81.5](#) can be replaced by continuity of partial subtraction on the cone, which is intrinsic. The proposition applies to the nonnegative part of any uncountable discretely ordered ring, for instance of models of PA^- , IOpen , $\text{I}\Delta_0$ or PA , with a Polish topology making addition continuous on the cone; Part III's cone ([theorem 38.1](#)) shows that such topologies exist, with continuous multiplication as well.

Proposition 155.7 (Locally compact cones [merge]). *Let M be an uncountable locally compact Polish model of Presburger arithmetic with continuous addition and continuous partial subtraction. Then M is homeomorphic to source 15's carrier X_{lc} . In particular, in an uncountable zero-dimensional locally compact Polish Presburger model with continuous addition, partial subtraction is discontinuous.*

Proof. By [theorems 153.1](#) and [154.1](#), $G = M - M$ has a Polish group topology inducing the topology of M , and M is a G_δ in it, hence Borel. Since $G = M \cup (-M)$, G is an ordered abelian group with positive cone M , and its order $\{(x, y) : y - x \in M\}$ is Borel. So M is the nonnegative cone, with its inherited topology, of a Borel ordered abelian Polish group, and source 15's [corollary 121.2](#) gives the carrier X_{lc} , which is not zero-dimensional. $\square$

156 What the completion theorem does and does not classify

The result supplies a recognition theorem, a canonical construction, and a uniqueness statement. Given the original topology on M , one can determine whether it extends by checking partial subtraction; if it extends, the neighborhoods $U - U$ determine the answer.

This is different from classifying all possible topologies on a given algebraic Presburger model. Nor does the theorem classify the resulting Polish groups, their locally compact factors, or the possible interaction of multiplication with those topologies. The repository's structural work remains relevant to these separate tasks.

The hypotheses also have distinct roles. Cancellation is needed for unambiguous differences. Antisymmetry of the natural order identifies the embedded cone correctly. Totality lets every group element be positive or negative; it is used in the topology comparison, uniqueness, and especially the two-piece Polishness argument. Hausdorffness is needed to separate a nonzero difference. No first countability enters the general construction; metrizability enters the sequence formulation, and Polishness enters the completeness upgrade.

Division II of source 18: Local rank, class spaces, and manifolds

157 Foundational conventions

157.1 A precise base theory

For this part we work in a two-sorted theory denoted by $\mathbf{GB}_{\text{-Fnd}}^{\text{rep}} + \mathbf{AC}$. Its set axioms are those of ZFC with Foundation *omitted*; this notation does not assert the negation of Foundation. Its class axioms include extensionality, elementary class comprehension with set and class parameters, and class Replacement: the image of a set under a specified class function is a set. Comprehension formulas quantify over sets, not over arbitrary classes. We use the resulting Separation schemes with class parameters. An equivalent reading is as theorem schemes for specified definable classes and functions in a corresponding one-sorted setting.

Set Choice is available. Global Choice is not assumed. We do not assume a full Collection scheme or recursion along the whole proper class of ordinals. The recursions below have a *set ordinal* as their domain and use a specified functional rule, so ordinary set recursion with class parameters and class Replacement suffice.

This distinction is essential without Foundation. In ZFC, a rank-minimization argument often turns a proper-class search for a witness into a search inside a set. That argument cannot be imported unchanged into the present setting. Similarly, the existence of a witness for each element of a set does not itself specify a class function choosing those witnesses.

These conventions are chosen to make every proof checkable; they should not be silently identified with every notation used for weak set or class theories in the literature. Glazer's work on Global Choice and the Glazer–Yao work on reflection in ZFU explain why such distinctions matter [105, 106, 107]. The ZFU results are context, not axioms imported into our proofs.

[write] Glazer's nonconservativity of Global Choice over Zermelo set theory [105] is treated in the surreal-well-orders report (its Part XV, `swo:gcz:thm:main`); classes, birthday cutoffs and set-sized workspaces are the subject of the foundations and birthday-cutoffs-and-hereditary-sets reports. None of them works without Foundation in the sense of this division.

157.2 How a topology on a proper class is represented

A class cannot be an element of another class in this framework. Accordingly, we never form a class whose members are all open subclasses of a proper class X .

Definition 157.1. A *locally small class space* consists of a class X and a class $\mathcal{B}$ of sets $B \subseteq X$, such that $\mathcal{B}$ covers X and satisfies the usual basis refinement condition. A subclass $O \subseteq X$ is open if each $x \in O$ belongs to some $B \in \mathcal{B}$ with $B \subseteq O$. Closures are computed with these neighborhoods.

All references to class spaces below use this representation. For example, closure membership is given by the elementary formula

$$\text{cl } A = \{x \in X : \forall B \in \mathcal{B}, (x \in B \implies B \cap A \neq \emptyset)\}.$$

For a set $S \subseteq X$, the induced topology on S is a set, by Separation inside $\mathcal{P}(S)$. Hausdorffness has its usual pointwise meaning. A space is connected if it has no separation into two

nonempty open subclasses. In fact, our sethood proofs only need the absence of a nonempty proper *set* that is both open and closed.

A class C with an injection into a set is a set, provided the injection is a specified class function: form its image by Separation inside the target, then apply class Replacement to its unique inverse. We will use this small but useful principle repeatedly.

Definition 157.2. An *ordinal exhaustion* of X is a specified class function $\alpha \mapsto X_\alpha$, for $\alpha \in \text{Ord}$, such that every X_α is a set, the stages are increasing, and

$$X = \bigcup_{\alpha \in \text{Ord}} X_\alpha.$$

No continuity at limit stages is required.

Every set $A \subseteq X$ is bounded in an exhaustion. For each $a \in A$, choose the *least* $\alpha(a)$ with $a \in X_{\alpha(a)}$. This is a specified class function, so Replacement and an ordinal supremum bound all its values. There is no choice from a proper class of competing witnesses in this argument.

Definition 157.3. A *neighborhood operator* is a specified class function $N : X \rightarrow V$ such that $N(x)$ is an open set containing x . An *open enlargement operator* is a specified class function H , defined on all sets $A \subseteq X$, such that $H(A)$ is an open set containing A .

Local smallness says suitable neighborhoods exist individually. It does not assert that they have been selected by a class function. The extra content of these operators is their uniform specification.

157.3 Tightness on set subsets

Definition 157.4. For an infinite cardinal κ , say that X has *tightness at most κ on sets* if, whenever $A \subseteq X$ is a set and $x \in \text{cl } A$, some $B \subseteq A$ satisfies $|B| \leq \kappa$ and $x \in \text{cl } B$.

This is deliberately a condition on set subsets, which are precisely the subsets used in the construction. First countability implies tightness at most $\aleph_0$: choose one point of A in each member of a countable local base. Only a set-indexed choice is used.

A uniform local cardinal bound also suffices. If each point has an open set neighborhood of size at most λ , fix such a neighborhood U for the single point under consideration. If $x \in \text{cl } A$, then

$$B = A \cap U, \quad |B| \leq \lambda, \quad x \in \text{cl } B.$$

Every neighborhood of x meets $A \cap U$, since its intersection with U is again a neighborhood. This proof makes no simultaneous choice of neighborhoods for all points.

158 The closure of a set in a Hausdorff class space

The following is the foundation-free form of a classical cardinal inequality. The inequality is not new. Our purpose is to specify the class-theoretic inference from a bound to actual sethood.

Lemma 158.1 (Closure coding). *In the class theory above, with no use of Choice, the closure of every set A in a Hausdorff class space is a set. There is a specified class injection*

$$\text{cl } A \longrightarrow \mathcal{P}(\mathcal{P}(A)).$$

Proof. For $x \in \text{cl } A$, define the closure pattern

$$\Phi_A(x) = \{B \subseteq A : x \in \text{cl } B\}.$$

This is a set by Separation in $\mathcal{P}(A)$. If $x, y \in \text{cl } A$ and $x \neq y$, take disjoint open neighborhoods $U \ni x$ and $V \ni y$. Put $B = A \cap U$. For every neighborhood $W \ni x$, the neighborhood $W \cap U$ meets A , so $x \in \text{cl } B$. But $V \cap B = \emptyset$, so $y \notin \text{cl } B$. Thus $\Phi_A(x) \neq \Phi_A(y)$.

The image

$$I = \{p \in \mathcal{P}(\mathcal{P}(A)) : \exists x \in \text{cl } A (\Phi_A(x) = p)\}$$

is a set by Separation with class parameters. Each $p \in I$ has a unique preimage. Replacement applied to that inverse yields $\text{cl } A$ as a set. $\square$

The closure-pattern proof is explicitly recorded by YCor [118]; the same cardinal estimate is used in Anschütz's lecture notes [119, Lemma 3.21(iii)]. The preceding proof does not use hereditary ranks, Foundation, first countability, or Global Choice.

With Choice it gives the familiar inequality

$$|\text{cl } A| \leq 2^{2^{|A|}}.$$

For the manifold application a smaller classical estimate is useful.

Lemma 158.2. *If X is first countable and Hausdorff, and A is an infinite set, then*

$$|\text{cl } A| \leq |A|^{\aleph_0}.$$

Proof. The closure is a set by lemma 158.1. Every point in it is the limit of a sequence in A , by taking a decreasing countable neighborhood base and choosing a point of A in each member. Hausdorffness makes each convergent sequence have a unique limit. The subset of $A^{\mathbb{N}}$ consisting of convergent sequences therefore maps onto $\text{cl } A$. Set Choice turns this surjection into the asserted cardinal bound. $\square$

Corollary 158.3. *A Hausdorff class space with a set-sized dense subset is a set.*

Proof. If A is such a subset, then $X = \text{cl } A$. $\square$

Another useful test is a set-indexed basis in a T_0 space. If $(B_i)_{i \in I}$ is a basis and I is a set, then

$$x \mapsto \{i \in I : x \in B_i\}$$

injects the carrier into $\mathcal{P}(I)$, so the carrier is a set. These two observations resolve many class-size concerns before more elaborate topology is needed.

159 Constructing neighborhoods from local rank

Lemma 159.1. *Every ordinal-exhausted locally small class space has a specified neighborhood operator and a specified open enlargement operator.*

Proof. Fix an exhaustion $(X_\alpha)_{\alpha \in \text{Ord}}$. For $x \in X$, some open set neighborhood U exists by local smallness, and this set is contained in an exhaustion stage. Let $\alpha(x)$ be the least ordinal for which there is an open set U with

$$x \in U \subseteq X_{\alpha(x)}.$$

Define

$$N(x) = \bigcup \{U \in \mathcal{P}(X_{\alpha(x)}) : U \text{ is open and } x \in U\}.$$

The displayed family is a set, and is nonempty. Its union is a set, is open, and contains x . This gives a specified class function without choosing a member of the family.

For every set $A \subseteq X$, put

$$H(A) = \bigcup_{x \in A} N(x).$$

Class Replacement makes $\{N(x) : x \in A\}$ a set, and Union makes $H(A)$ a set. It is open and contains A . $\square$

The union in this proof need not be a coordinate chart, have small diameter, or preserve a prescribed cardinal bound. It is an open set neighborhood; that is exactly what the next theorem needs. Cardinality estimates will be recovered only after sethood is known.

160 A clopen set hull and the sethood theorem

Theorem 160.1 (Clopen set hull). *Let X be a Hausdorff locally small class space of tightness at most an infinite cardinal κ on sets. Suppose that X has a specified open enlargement operator H . Then every set $A \subseteq X$ is contained in a clopen subset of X that is itself a set.*

Proof. Let $\theta = \kappa^+$, which is regular under Set Choice. Define a sequence of open sets of length θ by

$$\begin{aligned} O_0 &= H(A), \\ O_{\alpha+1} &= H(\text{cl } O_\alpha), \\ O_\lambda &= \bigcup_{\alpha < \lambda} O_\alpha \quad (\lambda < \theta \text{ a nonzero limit}). \end{aligned}$$

The closure at each successor stage is a set by [lemma 158.1](#). At limit stages Replacement and Union give sets. Hence this is a legitimate set-length recursion. The sets are open and increasing, with $\text{cl } O_\alpha \subseteq O_{\alpha+1}$. Put

$$K = \bigcup_{\alpha < \theta} O_\alpha.$$

It is an open set containing A .

If $x \in \text{cl } K$, tightness gives $B \subseteq K$, $|B| \leq \kappa$, with $x \in \text{cl } B$. For each $b \in B$, take the least $\alpha(b) < \theta$ for which $b \in O_{\alpha(b)}$. Regularity of θ bounds these indices below some $\alpha < \theta$. Thus $B \subseteq O_\alpha$, and

$$x \in \text{cl } B \subseteq \text{cl } O_\alpha \subseteq O_{\alpha+1} \subseteq K.$$

There is always a successor $\alpha + 1 < \theta$, since an infinite initial successor cardinal is a limit ordinal. Therefore K is closed as well as open. $\square$

Theorem 160.2 (Connected carrier theorem). *Let X be a connected Hausdorff locally small class space. If its tightness on sets is bounded by a cardinal and it has either an ordinal exhaustion or a specified neighborhood operator, then X is a set.*

Proof. An exhaustion produces a neighborhood operator by [lemma 159.1](#). A neighborhood operator produces the enlargement $H(A) = \bigcup_{x \in A} N(x)$. If $X = \emptyset$, the conclusion is immediate. Otherwise apply [theorem 160.1](#) to a singleton. The resulting nonempty clopen set must be all of the connected space X . $\square$

Corollary 160.3 (Uniform local size). *An ordinal-exhausted connected Hausdorff locally small class space is a set if there is one cardinal λ such that every point has an open set neighborhood of size at most λ .*

Proof. Uniform local size implies tightness at most $\max(\lambda, \aleph_0)$, as shown in [section 157](#). Apply [theorem 160.2](#). $\square$

Corollary 160.4. *Suppose X is ordinal-exhausted, locally small, Hausdorff, and has bounded tightness on sets. Every specified connected subclass of X , with its subspace topology, is a set.*

Proof. For a subclass Y , intersect the exhaustion and the set neighborhoods with Y . These are sets by Separation. Hausdorffness passes to the subspace. For a set $A \subseteq Y$, $\text{cl}_Y A = Y \cap \text{cl}_X A$; hence a tightness witness in X is also a witness in Y . Thus bounded tightness on sets passes to Y , and [theorem 160.2](#) applies. $\square$

160.1 An exact equivalence in the connected case

Corollary 160.5. *For a connected Hausdorff locally small class space whose tightness on sets is bounded by a cardinal, the following are equivalent:*

- (i) X is a set;
- (ii) X has an ordinal exhaustion;
- (iii) X has a specified neighborhood operator;
- (iv) X has a specified open enlargement operator.

Proof. If X is a set, use the constant exhaustion $X_\alpha = X$ and the constant neighborhood assignment $N(x) = X$. The implications (ii) $\Rightarrow$ (iii) and (iii) $\Rightarrow$ (iv) were proved above. Finally (iv) $\Rightarrow$ (i) follows from [theorem 160.1](#) and connectedness. $\square$

This equivalence identifies the exact missing uniformity. It does not turn pointwise neighborhood existence into a neighborhood operator. Nor does it establish that Global Choice, in every foundation-free formulation, provides such an operator.

161 Components, manifolds, metrics, and groups

161.1 Defining components without a class of proper classes

Proposition 161.1. *Let X be an ordinal-exhausted, locally connected Hausdorff class space whose tightness on sets is bounded by a cardinal. There is a specified class function $x \mapsto C(x)$ such that $C(x)$ is a connected clopen set and is the connected component of x . Consequently the collection of components is a legitimate class of sets.*

Proof. Modify the construction of [lemma 159.1](#). At x , use the least exhaustion stage containing some *connected* open set neighborhood of x , and let $N(x)$ be the union of all such connected open neighborhoods in that stage. Local connectedness and local smallness ensure that this family is nonempty. Every member contains x , so its union is connected. The assertion that a set U is connected is elementary: test separations inside the set topology on U .

Apply the recursion in [theorem 160.1](#) starting with $A = \{x\}$ and this N . The initial set $N(x)$ is connected. The closure of a connected set is connected. The union of the sets $N(y)$ for y in that closure is connected, because each connected $N(y)$ meets the same connected closure, and their union contains that closure. At limits, an increasing union of nonempty connected sets is connected. Thus the terminal set $C(x)$ is connected, and the previous theorem makes it clopen.

A connected subclass meeting $C(x)$ cannot also meet its complement, because that would give a separation by the two clopen pieces. Therefore $C(x)$ is maximal connected. Its construction is a specified set-length recursion, so its graph is given by an elementary formula with the fixed class parameters. We may form the class $\{C(x) : x \in X\}$, whose members are now known to be sets. $\square$

The order of this argument matters: we first construct each component as a set. We do not begin by postulating a collection whose members might have been proper classes.

161.2 Real class manifolds

Definition 161.2. A real class n -manifold, for $1 \leq n < \omega$, is a Hausdorff class space locally homeomorphic to open subsets of $\mathbb{R}^n$. No global second-countability condition is imposed.

Chart domains are sets under our conventions. Even if a chart is initially described as a class bijection from a class domain onto a set-sized Euclidean open, class Replacement applied to its inverse makes the domain a set; Replacement also makes the graph of the restricted chart a set. Rational balls within charts supply a class of set basic opens. Such manifolds are first countable, locally connected, and have local neighborhoods of cardinal at most $\mathfrak{c} = 2^{\aleph_0}$.

Theorem 161.3 (Manifold sethood). *A connected real class manifold with an ordinal exhaustion or a specified neighborhood operator is a set. A nonempty such n -manifold has cardinal exactly $\mathfrak{c}$.*

Proof. First countability supplies countable tightness, so [theorem 160.2](#) proves sethood.

We include the classical cardinal argument to keep the conclusion self-contained. Once X is a set, candidate chart domains form nonempty families inside $\mathcal{P}(X)$, so ordinary Set Choice selects, for each $x \in X$, an open chart neighborhood U_x of size at most $\mathfrak{c}$. For a set $A \subseteq X$ of size at most $\mathfrak{c}$, [lemma 158.2](#) gives

$$|\text{cl } A| \leq \mathfrak{c}^{\aleph_0} = \mathfrak{c}$$

when A is infinite; finite sets are closed in a Hausdorff space. Thus both taking a closure and covering that closure by the chosen U_x 's preserve cardinal at most $\mathfrak{c}$.

Starting from one chart, perform the closure-and-enlargement construction through ω_1 . Every stage has size at most $\mathfrak{c}$, and their union does too, because $\aleph_1 \leq \mathfrak{c}$. Countable tightness and regularity of ω_1 make the union closed, just as in [theorem 160.1](#). It is nonempty and open, so connectedness makes it all of X . This proves $|X| \leq \mathfrak{c}$. A nonempty Euclidean chart has size $\mathfrak{c}$, giving equality. $\square$

The cardinal bound for connected Hausdorff manifolds is classical: Baillif and Gabard attribute it to Nyikos [120, 121], and an explicit proof is discussed by Van Name [122]. We claim no new inequality here. The new work in this article is the preceding control of class size and of the witness-selection step when Foundation is absent.

Corollary 161.4. *For an ordinal-exhausted real class manifold, every connected component is a set of cardinal $\mathfrak{c}$. The whole manifold is a set if and only if its class of components is a set.*

Proof. Use [proposition 161.1](#) and then [theorem 161.3](#) on each component. If the components form a set, their union is a set. If X is a set, Replacement applied to $x \mapsto C(x)$ makes the component collection a set. $\square$

161.3 A metric supplies its own neighborhood operator

Theorem 161.5 (Metric criterion). *A connected real class manifold whose topology is induced by a specified real-valued class metric is a set, without an exhaustion hypothesis.*

Proof. Let d be the metric. At each x , choose for this single existence argument a set chart neighborhood U . Some positive real ball about x lies in U ; hence there is $n \in \mathbb{N}$ such that

$$B_d(x, 1/(n+1)) \subseteq U.$$

This ball is a set by Separation inside U . Let $n(x)$ be the *least* natural number for which the ball $B_d(x, 1/(n+1))$, defined as a class, is a set. Sethood here has the elementary expression that there exists a set with exactly the points satisfying the ball inequality.

The assignment

$$N(x) = B_d(x, 1/(n(x)+1))$$

is therefore a specified neighborhood operator. Apply [theorem 160.2](#), using first countability. $\square$

The metric is required to be one specified class function inducing the topology. Pointwise existence of chart metrics does not provide this hypothesis.

161.4 A parallel criterion for class topological groups

Proposition 161.6. *Let G be a connected Hausdorff class topological group with tightness bounded by a cardinal on set subsets. If the identity has one open neighborhood U that is a set, then G is a set.*

Proof. Translations give the specified neighborhood operator $N(g) = gU$; class Replacement makes each translate a set. Translations also supply a class of set basic opens, by translating the set open subsets of U . Apply [theorem 160.2](#). $\square$

In particular a connected real class Lie group is a set. This is a direct connection between the two parts of the article: for groups, translation makes the neighborhood assignment uniform. For a general class manifold, local charts do not by themselves supply that assignment in the weak foundational setting.

162 Why the hypotheses cannot simply be removed

162.1 Connectedness: a proper class of ordinary components

Example 162.1. The topological sum

$$X = \text{Ord} \times \mathbb{R}^n$$

with one Euclidean component for each ordinal is a proper-class, Hausdorff, first-countable real class manifold. It has the exhaustion $X_\beta = \beta \times \mathbb{R}^n$, and uniform local size c . It is disconnected.

Each point has a coordinate chart contained in its Euclidean component. The map $\alpha \mapsto (\alpha, 0)$ shows that the carrier cannot be a set. This example explains why the connectedness qualification must be retained when interpreting the manifold problem.

162.2 Hausdorffness: a proper class of origins

Example 162.2. Take one shared copy of $\mathbb{R} \setminus \{0\}$ and one origin o_α for each ordinal α . A basic neighborhood of o_α is

$$\{o_\alpha\} \cup ((-\varepsilon, \varepsilon) \setminus \{0\}), \quad \varepsilon > 0;$$

away from the origins use ordinary real intervals avoiding 0. This is a connected, first-countable, locally Euclidean proper-class space with an ordinal exhaustion and local size c . It is not Hausdorff.

Proof. For each α , the shared punctured line together with o_α is an open subspace homeomorphic to $\mathbb{R}$. These connected sheets all intersect, so their union is connected. There are a proper class of distinct origins. Distinct origins cannot have disjoint neighborhoods, because both neighborhoods contain sufficiently small positive real points. The exhaustion consists of the entire shared punctured line and the origins with indices below β .

The sequence $1/j$ converges to every origin. Consequently the closure of the countable set $\{1/j : j \geq 1\}$ is a proper class. This explicitly identifies the failure of [lemma 158.1](#). $\square$

The line with multiple origins is a classical construction. The proper-class indexing above is a hypothesis test, not a claim to have invented its underlying topology.

162.3 Bounded tightness: the ordinal class ray

Example 162.3. Let

$$L = \text{Ord} \times [0, 1)$$

have the lexicographic order topology. It is a connected, ordinal-exhausted, locally small Hausdorff proper-class space. Its tightness on sets is not bounded by any cardinal.

Proof. Each point has a bounded set-sized interval neighborhood. The stages $\beta \times [0, 1)$ give an exhaustion. The carrier is proper because it contains the ordinal-indexed points $(\alpha, 0)$.

For connectedness, every closed interval $[p, q]$ is a set-sized linear continuum. It is densely ordered: within a real block use real density, and between different blocks choose a larger real coordinate in the earlier block. It is order complete within its endpoints. For a nonempty bounded subset, take the supremum of its ordinal coordinates. If that coordinate is unattained, the supremum is the start of its block. If it is attained, take the supremum of the real coordinates in that block; a supremum of 1 moves to the start of the next block. The upper endpoint ensures that the resulting point lies in the interval. The standard supremum argument therefore makes $[p, q]$ connected. Any separation of L would separate an interval joining one point of each side, which is impossible.

Let λ be any infinite regular cardinal, and put

$$p_\lambda = (\lambda, 0), \quad A_\lambda = \{(\alpha, 0) : \alpha < \lambda\}.$$

Then $p_\lambda \in \text{cl } A_\lambda$. For $B \subseteq A_\lambda$ of cardinal less than λ , regularity bounds all ordinal coordinates occurring in B strictly below λ . An interval immediately below and around p_λ avoids B , so $p_\lambda \notin \text{cl } B$. Thus tightness is at least λ . Regular cardinals are unbounded, so there is no uniform bound. $\square$

This space is not a real class manifold at points of uncountable cofinality. The example separates the general class-space theorem from the finite-dimensional manifold corollary.

Example	Connected	Hausdorff	Exhausted	Missing condition
$\text{Ord} \times \mathbb{R}^n$, separate components	No	Yes	Yes	Connectedness
Proper class of origins	Yes	No	Yes	Hausdorffness
Ordinal class ray	Yes	Yes	Yes	A uniform bound on tightness or local cardinality

Table 4: The three counterexamples have different obstructions. None is a counterexample to the stated manifold theorem.

162.4 Why an unnamed global chart choice is not a proof

An attempted proof may say: take a set of points, choose a chart at each point, then take the union of the charts. In the foundational setting of [section 157](#), the candidates at each point form a potentially proper class. Set Choice does not directly choose from that family, because it is not yet a set of nonempty sets. A global choice function on nonempty *sets* cannot simply be applied to a proper class of candidate charts.

Our exhaustion argument repairs this step concretely. It first finds a least stage containing a candidate open neighborhood, collects all such open neighborhoods within the power set of that stage, and takes their union. The metric and group arguments repair the same step by giving an explicit functional assignment. These are sufficient mechanisms; we have not proved that they exhaust every possible route from Global Choice to sethood.

163 Surreal codes and bounded birthdays

163.1 Sign sequences are exhausted without Foundation

Represent surreal numbers by sign sequences

$$s : \alpha \longrightarrow \{-, +\}, \quad \alpha \in \text{Ord}.$$

The ordinal α is the birthday $b(s)$. For every ordinal β ,

$$\mathbf{No}_{<\beta} = \bigcup_{\alpha < \beta} \{-, +\}^\alpha$$

is a set: each function space is a set, and Replacement followed by Union collects the β -indexed family. These increasing stages exhaust the proper class of sign sequences.

Only ordinary recursion and constructions on ordinals enter this argument. No claim is made that every set in the ambient foundation-free universe has a hereditary rank. Finite powers $\mathbf{No}^m$, $1 \leq m < \omega$, have the exhaustion $(\mathbf{No}_{<\beta})^m$.

The repository develops bounded birthday fragments and an ordinal-length sign-sequence carrier, with explicit universe boundaries [113, 114, 115]. That is the concrete reason surreal coding fits the theorem.

Lemma 163.1 (Transport of an exhaustion). *Let C have an ordinal exhaustion (C_α) . A class X has an ordinal exhaustion if either:*

- (i) *there is a specified class injection $j : X \rightarrow C$; or*
- (ii) *there is a specified class surjection $q : C \rightarrow X$.*

No topology on either map is required.

Proof. In case (i), form the set $C_\alpha \cap j[X]$ by Separation. The inverse of j on this set is unique, so Replacement makes

$$X_\alpha = j^{-1}[C_\alpha]$$

a set. These sets increase and cover X . In case (ii), use $X_\alpha = q[C_\alpha]$, a set by Replacement. Surjectivity gives the covering property. $\square$

Theorem 163.2 (Surreal coding obstruction). *Let X be a connected real class manifold. If, for some fixed finite $m \geq 1$, there is a class injection $X \rightarrow \mathbf{No}^m$ or a class surjection $\mathbf{No}^m \rightarrow X$, then X is a set. If nonempty, it has cardinal c .*

Proof. Use the sign-sequence exhaustion, lemma 163.1, and theorem 161.3. $\square$

The result does not require the coding or decoding map to be continuous, injective and surjective at once, or compatible with surreal arithmetic. A specified map with set-valued outputs is what makes Replacement applicable.

A presentation as a quotient needs care. A class function assigning an actual set point to every code is covered by the theorem. Writing a quotient whose alleged points are themselves proper equivalence classes does not create a class of sets in our framework and is not covered by that notation alone.

Corollary 163.3 (Birthday bound for a coded carrier). *If a connected real class manifold has carrier $X \subseteq \mathbf{No}^m$, then there is an ordinal β such that*

$$X \subseteq (\mathbf{No}_{<\beta})^m.$$

Proof. The carrier is a set by theorem 163.2. Apply Replacement to the finitely many coordinate birthdays of its points, take their ordinal supremum, and increase the bound if necessary. $\square$

This is a bound for each particular carrier. It supplies no fixed birthday cutoff valid for all possible manifolds or all possible codings. It also bounds the points, not every auxiliary parameter used in a chosen description of their topology.

[write] The same Replacement bound on birthdays opens the proof of the birthday-cutoffs-and-hereditary-sets report's `hset:kw:thm:coding`, for sets injected into $\mathbf{No}$; source 18 applies it after proving that the carrier is a set.

163.2 Fine surreal topology is a different geometric choice

The coding theorem does not impose the native order topology of $\mathbf{No}$ on the manifold. That distinction matters.

For the following background observations, use the standard surreal ordered-field construction and its *set-cut property*: if L and R are sets of surreals with every element of L less than every element of R , there is a surreal strictly between them. Only set-sized cuts are used. This property is also the mechanism behind the repository's sign-sequence topology results [116]; the field arithmetic used in the elementary proofs below is not asserted to be formalized there.

Proposition 163.4. *For the native order topology of the surreal field:*

- (i) *Every nonempty open interval is a proper class.*
- (ii) *Every set of surreals is closed and discrete.*
- (iii) *A net indexed by a set converges only if it is eventually equal to its limit.*
- (iv) *Every continuous map from a connected set space into $\mathbf{No}^m$, with its finite product of native order topologies, is constant.*

Proof. For (i), suppose (a, b) were a set. The positive distances $b - z$ for $z \in (a, b)$, together with $b - a$, form a set. By the set-cut property choose $\varepsilon > 0$ smaller than every one of these distances. Then $b - \varepsilon \in (a, b)$, but its distance from b is ε , a contradiction.

For (ii), let A be a set and $x \notin A$. The distances $\{|a - x| : a \in A\}$ are a set of positive surreals. Choose a positive ε smaller than all of them. The interval $(x - \varepsilon, x + \varepsilon)$ misses A , proving closedness. Apply the same argument to $A \setminus \{x\}$ at a point $x \in A$ to obtain discreteness.

For (iii), the range of a set-indexed net together with its proposed limit is a set. By (ii) a neighborhood of that limit meets this set only at the limit. Convergence is precisely eventual membership in every such neighborhood, so the net is eventually equal to it.

For (iv), a set subset of $\mathbf{No}^m$ is likewise closed and discrete: use the maximum of the finitely many coordinate distances from a fixed point and repeat the small- ε argument. A continuous image of a connected set space is a connected set. A connected discrete set has at most one point. $\square$

In particular the ordinary copy of $\mathbb{R}$ inside $\mathbf{No}$, with its induced fine topology, is discrete. Its induced topology is not its usual Euclidean topology. Ordinary real chart geometry must therefore be specified independently of this ambient topology.

[write] Clauses (ii) and (iii) are, for sets, clauses 1 and 2 of the foundations report's `found.thm:discrete`, which `docs/FORMALIZATION.md` records as proved in Lean for the real sign carrier, for lower-universe-small subsets and index types (`Surreal.Foundations.SignSequence.isClosed_of_small`, `isDiscrete_of_small`, `tendsto_nhds_iff_eventually_eq_of_small_range`, `tendsto_nhds_iff_eventually_eq` in `Algebra/SurrealNumbers/Surreal/Foundations/SignSequenceTopology.lean`, the file source 18 cites). Clauses (i) and (iv) and [proposition 163.5](#) are not formalized as stated.

There is also a separation statement for proper classes, which does not rely on a set-sized image.

Proposition 163.5. *The fine surreal line is totally separated: any two distinct points are separated by a clopen subclass. The same holds for each finite power $\mathbf{No}^m$.*

Proof. Take $x < y$, let ω be the simplest positive infinite surreal, and put $d = (y - x)/\omega > 0$. Define

$$H = \{z : \exists n \in \mathbb{N}, z < x + nd\}.$$

This is open, contains x , and excludes y , since $nd < \omega d = y - x$ for every finite n . If $z \notin H$, then $q = (z - x)/d$ is at least every natural number, hence is positive infinite. So $q/2$ also exceeds every natural number. With $w = x + (q/2)d$, we have $w < z$, and every point above w lies outside H . The complement of H is open, proving that H is clopen. For finite powers, use a differing coordinate and take the inverse image of such a clopen separator under its projection. $\square$

The fine topology in this subsection is not locally small in the sense of [definition 157.1](#): no nonempty interval is a set. It is represented instead by interval parameters in $\mathbf{No}^2$; openness and closure have their ordinary interval definitions. It is used only for these comparison statements, not as an unmentioned hypothesis in the earlier locally small class-space theorems.

Division III of source 18: Research program, verification, and sources

164 What has been resolved

Question	Outcome and exact scope
Intrinsic group-completion criterion for Polish Presburger monoids	Resolved by continuous partial subtraction, equivalently the two local tests. The extending topology is unique and Polish. Local compactness is unnecessary.
Connectedness when partial subtraction is continuous	Ruled out for Polish Presburger models: all finite residue classes are clopen.
Sethood of ordinal-exhausted connected class manifolds	Proved without Foundation or Global Choice in the explicitly specified base theory.
Manifolds carried by surreal codes	An injection into, or a surjection from, a finite power of sign-sequence surreals suffices for sethood.
Metric and group cases	A specified compatible real metric, or a specified group structure with a set identity neighborhood, supplies the needed neighborhood assignment.
Unrestricted Glazer manifold problem	Not resolved. The article does not derive the necessary assignment from Global Choice alone.
Historical priority of the main criteria	Not established. The results are proved and compared with the inspected repository, with classical inputs credited.

Table 5: A scoped research outcome, distinguishing proved statements from questions still outside the argument.

The group-completion result addresses an explicit question in a research draft. It should not be described as the resolution of a longstanding community-wide problem without further evidence. Likewise, the manifold theorem is a precise sufficient-condition result in the direction of Glazer's question, not a general answer in all the weak theories his work considers.

The strongest unifying lesson is mathematical rather than rhetorical. Formal differences need continuity of the inverse cancellation operation. A class union of local pieces needs a specified functional way of collecting those pieces. Algebraic cancellation and pointwise existence are each weaker than the corresponding topological or foundational requirement.

165 Further research questions and topics

The following are proposed projects. Except where a source is named, they are questions left unanswered by this article, not claims that their status in the entire literature has been settled. Each project includes a concrete next step.

165.1 Polish arithmetic and topological completion

Question 165.1 (Connected Polish Presburger presentations). Does an uncountable connected Polish model of Presburger arithmetic with continuous addition exist?

By [corollary 155.3](#), any such model must have discontinuous partial subtraction. A useful first target is to classify the possible diagonal failures in [theorem 154.3](#). The polynomial obstruction shows that such failures exist in locally compact models, but its stratified topology is highly disconnected and does not settle this question. Start by determining which connectivity assumptions force a Borel inverse of F to become continuous.

[write] This is Part I's [question 14.3](#), asked again as [questions 43.4](#), [103.4](#), [126.2](#) and [138.10](#); by [corollary 155.3](#), and in the locally compact case by [proposition 155.7](#), a connected example must have discontinuous partial subtraction.

Question 165.2 (Can the diagonal test stand alone?). Under the hypotheses of [theorem 153.1](#), does continuity of δ at every (x, x) imply its continuity at every $(x, 0)$? If not, can a Polish counterexample be constructed?

The reverse one-test reduction fails by [example 155.5](#). No independence assertion in the other direction is proved here. An answer would either shorten the exact recognition criterion or identify a second, genuinely different cancellation obstruction.

Question 165.3 (Translation-based recognition). Which hypotheses on the translations $T_a(x) = a + x$ imply the two local tests? In particular, what do properness, openness onto the image, or uniform control of inverse translations contribute in locally compact Polish monoids?

Individual translations being embeddings only controls a fixed parameter a . Partial subtraction controls an inverse while both the sum and the subtrahend vary. The first task is to formulate this parameter dependence as a precise local condition and test whether local compactness upgrades pointwise control to joint control.

[write] Source 15's [question 126.3](#) asked for an intrinsic condition using “translations, the partial difference map on ordered pairs, and neighborhood structure”; continuity of the partial difference map answers it ([theorem 153.1](#)). This question asks for the translation half alone.

Question 165.4 (Completion beyond total cones). What is the best version of the theorem for cancellative commutative monoids whose natural order is only partial?

The identity $G = M \cup (-M)$ is then unavailable. One must separately analyze whether $U - U$ induces the original cone topology, whether the extending topology is unique, and whether Polishness survives. Examples should distinguish failure of existence from failure of completeness. The finite union of two completely metrizable subspaces in the present proof cannot be replaced without argument by an arbitrary union.

165.2 Choice principles and class manifolds

Question 165.5 (The exact strength of manifold neighborhood selection). Over the base theory of [section 157](#), which additional principles imply that every connected real class manifold has a specified open set neighborhood at each point?

By [corollary 160.5](#), this is an exact reformulation of sethood for such manifolds. Compare explicit class uniformization, appropriate Collection schemes, Global Choice in fixed formulations, and carrier-specific rank principles. Any claimed implication must show how the candidate chart family is handled, rather than treating a proper class as the input to a set choice function.

Question 165.6 (The unrestricted higher-dimensional problem). Does the Global Choice assumption in Glazer’s Oberwolfach question suffice for sethood of connected Hausdorff real class manifolds of dimension at least two?

This is the source problem [\[106\]](#), and is not solved here. The present work restricts a possible counterexample: it cannot admit an ordinal exhaustion, a surreal encoding or decoding of the stated kind, a compatible real class metric, or a suitable transitive group structure supplying neighborhoods.

A focused next step is the surface case. One may ask which parts of classical surface arguments construct local data by explicit formulas and which rely on class-wide selections that become nontrivial without Foundation.

[write] (Batch 90.) The source problem, in Glazer’s words [\[106, p. 106\]](#), is whether “ZFC – Fnd + GC proves that every topological manifold is a set”; he adds that the methods for the one-dimensional case do not seem to suffice in dimension two. It enters the repository with this part (note in [section 151](#)). The qualification “connected” and the class-theoretic setting are source 18’s. [Theorems 161.3, 161.5 and 163.2](#) and [proposition 161.6](#) restrict a counterexample; they do not answer the question.

Question 165.7 (Path and chart-chain collection). Which hypotheses guarantee that the points reachable from a fixed point by finite chart chains form a set, when only pointwise chart existence is assumed?

For an exhausted manifold our component construction already answers this. The research issue is to remove the exhaustion without inserting an equivalent neighborhood selection assumption. A rigorous analysis should distinguish the set of possible real coordinate descriptions from the possibly proper class of their realizations on the carrier.

Question 165.8 (Weaker separation and closure control). Can useful manifold-like classes replace Hausdorffness by weaker separation assumptions while retaining set-sized closures of sets?

The multiple-origins example shows that local Euclidean structure alone is insufficient. The hull construction itself only needs a specified set-valued closure operation and bounded tightness; Hausdorffness supplies closure coding. Investigate which weaker uniqueness or separation conditions supply that coding in the chosen foundational theory, and which are preserved under gluing.

165.3 Descriptive complexity, coding, and formalization

Question 165.9 (Complexity of the cancellation obstruction). How complicated can the Borel map $\delta : E_M \rightarrow M$ be for a Polish Presburger monoid with continuous addition?

[Proposition 155.1](#) gives Borel measurability but no uniform Baire-class bound. Study the rank of inverse images of open sets, the descriptive complexity of the discontinuity locus, and whether local compactness or additional arithmetic operations impose a sharper bound. The explicit degree-stratified model is a useful initial test case.

Question 165.10 (Effective birthday bounds under constrained coding). For natural classes of geometric descriptions by surreal sign sequences, can one compute a birthday bound from the description and its operations?

[Corollary 163.3](#) is existential and permits arbitrary codings. An effective result must specify a syntax, allowed coordinate operations, and how gluing changes birthdays. The repository's bounded birthday workspaces supply a concrete starting point [114, 115]. An arbitrary recoding can raise birthdays, so a meaningful theorem must constrain the presentation.

Question 165.11 (A formal proof of the completion criterion). Can [theorems 153.1](#), [154.1](#) and [154.3](#) be formalized as reusable theorems for topological monoids?

A sensible order is to formalize the algebraic positive-cone decomposition, the $U - U$ neighborhood construction, the two topology comparisons, and then the Polish upgrade. The last step should explicitly distinguish complete metrizability of a subspace from completeness of one inherited metric. Only after this infrastructure is in place should one attach arbitrary- $\mathbb{Z}$ -group semantics to the repository's Presburger development.

[write] Nothing of this exists in the repository; the formal Presburger project's semantics are over the ordinary integers ([section 1.4](#)).

Question 165.12 (A formal class-theoretic proof audit). Can the closure-coding and clopen-hull arguments be checked in a formal theory that distinguishes class Replacement, Collection, Global Choice, and Foundation?

A type-universe encoding alone does not establish the same foundation-free theorem. One needs an explicit interpretation of classes and set-valued functions, or a syntactic proof in the specified class theory. The first milestones are the inverse-image sethood lemma, the rank-bounded neighborhood operator, and the set-length recursion. Such a development would directly connect Glazer's foundational interests with the repository's universe and sign-sequence infrastructure.

165.4 Suggested order of work

The most contained projects are the diagonal-test question, the descriptive complexity of subtraction, and a formal proof of the general completion criterion. They have explicit input objects and small proof interfaces.

The neighborhood-selection and surface questions are more ambitious. They require a fixed foundational theory from the beginning. The coding and proof-audit projects provide useful intermediate deliverables: they make the scope of a claimed class construction inspectable even before the unrestricted manifold problem is settled.

166 Verification, limitations, and reproducibility

166.1 Mathematical review performed

The two main proof chains were independently reread in separate review passes. The group-completion review checked the neighborhood axioms, Hausdorff separation, both

directions of topology agreement, uniqueness, and the category argument for Polishness. The class-space review checked the type of every collection used, the use of Replacement, the absence of a hidden Global Choice step, and the regular-cardinal closure argument. The actual manuscript text was also checked against the examples and stated hypotheses.

These are AI-assisted mathematical reviews, not external peer review. No proof-assistant verification of the new results is claimed. No finite numerical experiment could verify these infinite topological and class-theoretic assertions; the evidence offered is the written proof and its dependency audit.

166.2 What was inspected in the repository

The repository snapshot is

```
2b7b388ba81a3355b19a7c2d2fe797e92f572355.
```

The relevant research reports, incoming archive sources, and selected formal-source interfaces were inspected. The archive containing the explicit completion question is

```
docs/incoming/polish_arithmetic_package.zip,  
member polish_arithmetic/polish_arithmetic.tex.
```

The frontiers report is in

```
docs/incoming/polish_presburger_frontiers.zip,  
member polish_presburger_frontiers/article.tex.
```

[write] Both archives were placed in d151b39ca and retired from docs/incoming; they remain at their arrival commit fa0a0576e. They are sources 15 and 13, [Parts VII](#) and [IX](#).

The inspected Presburger formalization has ordinary-integer semantics: formulas are interpreted over $\mathbb{Z}$, and the theorem for quantifier elimination uses integer environments. An arbitrary- $\mathbb{Z}$ -group semantic theorem is not claimed. The repository itself states that the Polish-model research report is not formalized [\[109, 110\]](#).

The inspected sign-sequence files establish order and birthday infrastructure and small-subset topology results at explicit type universes. In particular, the topology file proves closedness and discreteness of lower-universe-small subsets, and eventual equality for convergent maps with small range [\[115, 116\]](#). Those results should not be misreported as a formalization of the class-manifold theorem or as a proof that every external set is small in a chosen universe.

166.3 Novelty and source scope

The criterion is a resolution of a question explicitly posed in the inspected incoming manuscript. The class-space construction is presented as a proved extension and application of classical closure-cardinality methods. The literature checks located earlier sources for the closure estimate and the manifold cardinal bound; they are credited in the relevant proofs.

No search can by itself certify that a theorem has never appeared elsewhere. Historical priority, publication suitability, and claims of a major breakthrough require a broader specialist review. The article therefore states the exact proved results instead of assigning them an unsupported historical status.

The ZIP includes the complete standalone source and PDF, a build script, a source record identifying the inspected snapshot, and a concise proof audit. It contains no copied repository archive and requires no network access to rebuild the PDF.

[write] In this report the source is merged into this part, and the PDF, the delivery README, the checksum manifest and the source record are not shipped; the source record contains a third party's personal profile address, and its verified content is in [appendix E](#). The build script and the proof audit are shipped as `code/18-polish-completions-build.sh` and `18-polish-completions-PROOF_AUDIT.txt`.

Part XIII

One real dimension: order normal forms, Borel-complete classification and automorphisms

[write] This part is source 19, manuscript 01 of batch 92, *One Real Dimension, Arbitrarily Complicated Arithmetic: Order normal forms, Borel-complete classification, and measurable automorphism rigidity for Polish Presburger models* (22 pages, A4, dated 4 October 2026; “research manuscript prepared with ChatGPT for Vladimir Reshetnikov”). It arrived in commit 9dc8db274, as archive `One_Real_Dimension_Arithmetic`, and was placed in e24ce2ce0; it is pinned to a32e1b139, 48 commits before the placement. Its file prefix in this report is `19-one-real-dimension-`. All unmarked text in this part is source 19's; its fifteen sections are Sections [167–181](#) (its Section k is Section $166 + k$, with statement numbers otherwise unchanged), its nine “Research questions” share the report's counter and are Questions 180.1–180.9, and its two appendices are [appendices AC](#) and [AD](#). [Section 182](#) [merge] was written for this report.

What this part claims, and its status

Let (E, u) be a countable torsion-free group with $E/\mathbb{Z}u$ torsion-free and divisible, and give $\mathbb{R} \times E$ the product of the usual and the discrete topology, with unit $\mathbf{1} = (0, u)$. After a continuous additive shear fixing $\mathbf{1}$, every Borel (equivalently Baire-property, equivalently Δ_2^0) Presburger order on $\mathbb{R} \times E$ has three layers: an ordered quotient $B = E/A$ above the real line, the usual real order, and an ordered pure subgroup $A \ni u$ below it ([theorem 170.2](#) and [corollary 170.6](#)). The complete invariant for continuous isomorphism is the countable ordered extension $(E, u, A, <_A, <_B)$; real cut offsets carry no invariant ([theorem 171.1](#)); the continuous automorphism group is computed ([corollary 171.2](#)), and no model in the class is rigid ([corollary 171.3](#)). All the nonnegative cones are the same real-analytic manifold X_{lc} with analytic addition ([theorem 173.1](#)), yet a continuous family of these additions on X_{lc} encodes isomorphism of countable linear orders, so isomorphism in the explicit normal-form codes is Borel complete for countable structures and not Borel ([theorems 174.3](#) and [175.2](#) and [corollary 175.3](#)). On $\mathbb{R} \vec{\times} D \vec{\times} \mathbb{Z}$, $D \neq 0$ countable divisible, there are 2^c abstract and c Borel automorphisms ([theorem 176.1](#)). [Section 182](#) [merge] adds that abstract and topological isomorphism agree in the whole class, and that, with Parts IV, IX and XII, this classifies every uncountable locally compact Polish Presburger model with continuous addition and continuous partial subtraction. Unrefereed, AI-assisted, not formalized; historical priority is not established. The finite checks supplement the proofs and do not certify them.

Questions answered. Source 19 states that it answers no published question other than the classification target it took from source 08. This report reads it as follows (dated notes after each question). Part IV's [question 57.3](#) ("Classify the orders, not only the pointed groups") is answered for $d = 1$ by [theorems 170.2](#), [171.1](#) and [172.1](#); $d > 1$ stays open ([question 180.1](#)). Part IX's [question 126.5](#) is answered, after Part IV's shear moves the unit into E ([proposition 182.1](#) [merge]), by [theorems 170.2](#) and [171.1](#) and [corollary 171.2](#); Part IX's [question 126.4](#) is answered within inherited-group presentations by [corollary 182.4](#) [merge], for arithmetic and for topological arithmetic isomorphism alike; Part IX's [question 126.6](#) is answered in part (lower bounds, [theorems 174.3](#) and [175.2](#) and [remark 182.5](#) [merge]). [Corollary 171.3](#) is a negative answer, inside this class, to Part I's [question 14.12](#) and source 13's [question 103.5](#). Source 19's own question on group completions ([question 180.2](#)) is Part IV's [question 57.2](#), already answered by Part XII. Source 19 answers neither of Glazer's two questions and says so.

What was already here, and what is new. Source 19 read the Presburger and surreal READMEs at its pin, not this report, and so did not know Part IX (written at 83926d574, before its pin) or Part XII (written after it). Its [lemma 169.1](#) is Part IX's [lemma 120.4](#) (= Part IV's [lemma 47.3](#) for $d = 1$), with essentially the same proof; its [lemma 177.1](#) is Part IX's [lemma 124.2](#), same statement and proof; its [theorem 177.2](#) is a mirrored variant of Part IX's [theorem 124.3](#); the carrier in [theorem 173.1](#) is Part IX's X_{lc} ([theorems 120.5](#) and [121.1](#)); [lemma 173.2](#) is a special case of Part XII's [theorem 153.1](#), by a different proof; [proposition 173.3](#) is Part IV's [theorem 52.1](#) and [proposition 51.4](#); [lemma 174.1](#) is Part IX's computation $\mathcal{A}(V_L) \cong L$; and [appendix AC](#) is one more write-up of the classical elimination already in the report ([lemma 4.7](#) and [theorem 4.9](#), [appendix I](#), [theorem 49.3](#), [appendix W](#)). Each is printed with its proof and flagged in a note. New to the report are the normal form and its Baire upgrade, the isomorphism, split and automorphism theorems, the coding

theorems, the Borel-completeness theorem and the abstract versus Borel automorphism count.

Labels and notation. Every label of source 19 takes the prefix `pma:ord:`; labels were added to its subsection on the inspected snapshot, its two examples, its conclusion and its nine questions, `pma:part:ord` to the part, and to [remark 176.3](#) and [section 182](#), which were written for this report. The sixth block of the table in [section 1.3](#) lists the symbols. In short: source 19's $G_L = \mathbb{R} \overrightarrow{\times} D_L \overrightarrow{\times} \mathbb{Z}$, M_L and $G_D = \mathbb{R} \overrightarrow{\times} D \overrightarrow{\times} \mathbb{Z}$, M_D are printed $G_L^{\text{lo}}, M_L^{\text{lo}}, G_D^{\text{lo}}, M_D^{\text{lo}}$ (superscript “lo”: the coded scales lie *below* the real line), because Part IX's $G_L = V_L \overrightarrow{\times} \mathbb{R} \overrightarrow{\times} \mathbb{Z}$ and $D_L = V_L \overrightarrow{\times} \mathbb{R}$ put them above it and Part I's $G_D = D \overrightarrow{\times} \mathbb{Z}$ has no real coordinate (tempting false readings); source 19's D_L is Part IX's V_L , the same ordered group, and is printed V_L ; its fixed manifold X is printed X_{lc} , Part IX's carrier; its `\cfрак`, `\st` and typewriter `\repo` are printed `c`, `st` and `Provelt`. Its $C = \mathbb{R} \times \{0\}$ is Part IV's C and Part IX's $H = G^0$; its I is Part IV's B_C and Part IX's C (letters again exchanged); A, B, K, H_L, H and $\mathcal{P}_E$ are local. No normalization changed.

Files. Source 19's finite checks, build script, claim list and source manifest are shipped as `code/19-one-real-dimension-checks-verify_finite.py`, `code/19-one-real-dimension-build.sh`, `19-one-real-dimension-CLAIMS.md` and `data/19-one-real-dimension-SOURCE_MANIFEST.json`, with the recorded run data `data/19-one-real-dimension-checks-results.json`. The Hilbert's-tenth programme routed the archive from its README only, without a correctness finding ([appendix E](#)). Nothing in this part is formalized.

167 The research intersection and the question actually answered

Elliot Glazer's *A Topological Tennenbaum Theorem* studies the interaction between descriptive set theory, topology, and nonstandard arithmetic. Its final Question 2 asks whether an uncountable Polish space can carry Presburger arithmetic with continuous addition [[1](#), Question 2, p. 8]. Provelt supplies two directly relevant interfaces: its formal Cooper-elimination development, and its work on surreal and omnific additive structures [[126](#), [127](#)]. These are substantive mathematical connections, rather than an inference from a biographical profile.

The earlier companion manuscripts already give the model

$$(\mathbb{R}_{>0} \times \mathbb{Z}) \cup (\{0\} \times \mathbb{N}) \quad (167.1)$$

with coordinatewise addition and the topology inherited from $\mathbb{R} \times \mathbb{Z}_{\text{discrete}}$ [[128](#)]. Repeating that construction would not answer a new research question. The subsequent report *Local Compactness Forces Arithmetic Rigidity* classifies certain *pointed topological groups*, while expressly leaving classification of their full orders open [[129](#), Section 7 and Research Question 3].

The precise question pursued here is:

For a fixed countable pointed group (E, u) , classify the Borel Presburger orders on $\mathbb{R} \times E$, up to continuous additive order isomorphism fixing $(0, u)$. Determine whether the corresponding positive models admit a manageable isomorphism test.

We give a complete structural answer in this one-real-dimensional presentation class, and a sharp obstruction to a simple Borel classification. We do not assume or prove that every locally compact Polish Presburger *monoid* admits a locally compact topological-group completion. Nor do we settle Glazer’s separate reverse-mathematical Question 1 about ATR_0 .

[write] (Batch 92.) The “earlier companion manuscripts” [128] are source 05, now the base of [Part I](#); [display \(167.1\)](#) is Part I’s $M_{\mathbb{R}}$ ([theorem 3.1](#)). The report [129] is source 08, now [Part IV](#): its “Section 7” is Section 50, whose theorem is [theorem 50.1](#), and its “Research Question 3” is [question 57.3](#). The question in the quotation above is that question for $d = 1$, with the unit already in E ; [proposition 182.1](#) [merge] shows that this costs no generality. The group-completion problem set aside in the last paragraph is solved, in general, by Part XII ([theorem 153.1](#)), which source 19 did not see.

167.1 A guide to the contributions

The normal-form and isomorphism theorems are [theorems 170.2](#) and [171.1](#). The Baire-property weakening is settled by [corollary 170.6](#). Their split specialization is [theorem 172.1](#). The fixed-manifold theorem is [theorem 173.1](#), and the continuous coding theorem is [theorem 174.3](#). The exact abstract/Borel automorphism distinction is [theorem 176.1](#). All of these statements are proved below.

The classical inputs are identified separately. Presburger elimination is classical; [appendix AC](#) includes the group-uniform argument needed here. The universality of countable linear-order isomorphism is the Friedman–Stanley theorem, not a new result [123, 124]. The general theory of Presburger automorphisms and rigid models is represented by Jeřábek’s work [4]. Our arguments use ordinary convex-subgroup methods, elementary real analysis, and countable coding; they do not assert that these methods themselves are new.

167.2 The inspected repository snapshot

The repository interfaces were read at the pinned commit

a32e1b139db296512a767ea78296ef3db642578e.

The Presburger README describes an executable integer decision procedure and explicitly distinguishes its ordinary-integer semantics from a future theorem uniform over nonstandard groups [126]. The surreal README separates unrefereed reports from mapped formal results and records the omnific normal-form and constant-term interfaces [127]. We use those interfaces, not the truth of unrelated research claims. No repository file was modified, and no repository-wide Lean build was run.

[write] Both READMEs are unchanged between the pin and the placement `e24ce2ce0` (blobs `51fbd3d11` and `18478d586`); the catalogue commit `cd514f61b` has since added pointers to this report. Both descriptions are correct: the Presburger project’s semantics are stated over `List Int` valuations ([section 1.4](#)). Source 19 did not read this report.

168 Arithmetic and topological conventions

All groups are abelian. An ordered group has a translation-invariant *total* order. A subgroup $A \leq E$ of a torsion-free group is *pure* if $ne \in A$ implies $e \in A$ for every ordinary integer $n \geq 1$. Convex subgroups of ordered groups are pure.

Definition 168.1. A *countable Presburger datum* is a countable torsion-free group E with a specified nonzero $u \in E$, such that $E/\mathbb{Z}u$ is torsion-free and divisible. Put

$$G(E) = \mathbb{R} \times E, \quad \mathbf{1} = (0, u),$$

with coordinatewise addition and the product of the usual real topology and the discrete topology on E .

The quotient condition is algebraic; it does not initially specify an order on E . It is the standard characterization of an unordered $\mathbb{Z}$ -group [4, Section 2]. The real component of $G(E)$ is

$$C = \mathbb{R} \times \{0\}.$$

It is both the identity component and an open subgroup. A topological isomorphism throughout means an algebraic isomorphism that is a homeomorphism; continuity of its inverse is included. A Borel order means its positive cone is Borel, equivalently its binary comparison relation is Borel.

A *Presburger order* on $G(E)$ means a group order in which $\mathbf{1}$ is the least positive element. The quotient $G(E)/\mathbb{Z}\mathbf{1} \cong \mathbb{R} \oplus (E/\mathbb{Z}u)$ is then divisible. A discretely ordered group with divisible quotient by its integer subgroup is called a $\mathbb{Z}$ -group. Its nonnegative cone is a model of

$$\text{Pr}_{\mathbb{N}} = \text{Th}(\mathbb{N}; 0, 1, +, <).$$

This includes the entire first-order induction scheme in the additive language. It does not supply multiplication of two variables.

Lemma 168.2 (Standard residues). *For every $e \in E$ and ordinary $n \geq 1$ there are unique $r \in \{0, \dots, n-1\}$ and $d \in E$ with $e = ru + nd$. The corresponding statement holds in $G(E)$.*

Proof. Divisibility of $E/\mathbb{Z}u$ gives $e = nd_0 + ku$. Divide the ordinary integer k by n to obtain existence. If $(r - r')u = n(d' - d)$, then $d' - d + \mathbb{Z}u$ is torsion in $E/\mathbb{Z}u$, hence is zero. Thus $r - r'$ is divisible by n , so $r = r'$; torsion-freeness gives $d = d'$. For $(x, e) \in G(E)$ the quotient is $(x/n, d)$. $\square$

Proposition 168.3 (Arithmetic transfer). *Every Presburger order on $G(E)$ makes it elementarily equivalent to the ordered additive integers. Its nonnegative cone satisfies $\text{Pr}_{\mathbb{N}}$. Every unital additive order embedding between such groups is elementary.*

Proof. The usual elimination in the language of affine inequalities and fixed congruences uses precisely discreteness and [lemma 168.2](#); a full finite-witness proof is given in [appendix AC](#). Ground formulas then have their ordinary-integer truth values. Restricting quantifiers to the nonnegative cone transfers the corresponding assertion to $\mathbb{N}$. An embedding preserves inequalities and, by the uniqueness of the standard residue, both preserves and reflects every congruence. Apply quantifier elimination once more. $\square$

We work in ZFC. All groups, parameter spaces, and manifolds in the main theorems are sets. The final use of **No** is only an embedding of an explicit set-sized family, with no quantification over a set of all proper-class surreal structures.

169 Borel orders on a real component

The first observation excludes hidden nonmeasurable orders on the real component. Its proof is short enough to include.

Lemma 169.1 (Baire real-order rigidity). *A translation-invariant total order on the additive group $\mathbb{R}$ whose positive cone has the Baire property is either the usual order or its reverse. In particular this holds for every Borel order.*

Proof. Let P be its strictly positive cone. Then $P + P \subseteq P$, $P \cap (-P) = \emptyset$, and $P \cup (-P) = \mathbb{R} \setminus \{0\}$. Consequently P is not meager. Since P has the Baire property, it is comeager in some nonempty open interval I . For every $t \in I + I$, the open set $I \cap (t - I)$ is nonempty, and the two sets P and $t - P$ are comeager there. Thus $t \in P + P \subseteq P$. So P contains a nonempty open interval J .

No such interval can contain zero. Shrinking J , suppose first that $J \subset (0, \infty)$ in the usual order. Group-order compatibility implies $qP = P$ for every positive rational q . The intervals qJ , with $q > 0$ rational, cover $(0, \infty)$, so all usual positive reals lie in P . Totality excludes all usual negative reals. If J lies in the negative half-line, the same argument gives the reverse order. $\square$

[write] (Batch 92.) This is Part IX's [lemma 120.4](#), the case $d = 1$ of Part IV's [lemma 47.3](#) with the Baire property in place of Borelness, with essentially the same proof (an interval of comeagerness, $I + I \subseteq P$, rational dilations). Source 19 did not see Part IX.

Henceforth orient C so its order is usual; replacing x by $-x$ is a continuous pointed-group automorphism and achieves this when necessary.

Lemma 169.2 (The arithmetic unit lies below every real scale). *In a Presburger order on $G(E)$ with the usual order on C ,*

$$n\mathbf{1} < (r, 0) \quad (r > 0, n \in \mathbb{N}).$$

Proof. For $n \geq 1$, the element $(r/(n+1), 0)$ is positive and therefore at least $\mathbf{1}$. It follows that $(r, 0) \geq (n+1)\mathbf{1} > n\mathbf{1}$. The case $n = 0$ is immediate. $\square$

[write] The same dominance inequality is the group form of Part IV's [lemma 46.1](#) (stated there for discretely ordered rings and divisible subgroups) and appears in the proof of Part IX's [theorem 121.1](#).

This is an order-theoretic infinitesimal assertion, not a statement about topological smallness. The sequence $(1/k, 0)$ tends to zero topologically although every term exceeds every ordinary positive integer in the arithmetic order.

170 The three-layer normal form

Definition 170.1 (Normal data). For a countable Presburger datum (E, u) , *normal order data* consist of a pure subgroup $A \leq E$ containing u , a group order $<_A$ on A with least positive element u , and a group order $<_B$ on $B = E/A$. Define an order on $G(E)$ by the sign rule

$$(r, e) > 0 \iff \begin{cases} e + A >_B 0, & e \notin A, \\ r > 0 \text{ or } (r = 0 \text{ and } e >_A 0), & e \in A. \end{cases} \quad (170.1)$$

The first clause is equivalently valid for all e , since it is false when $e \in A$.

Theorem 170.2 (Complete normal form). *Every Borel Presburger order on $G(E)$ is carried by a continuous additive automorphism fixing $\mathbf{1}$ to an order of the form (170.1). Conversely, every choice of normal data gives a Borel Presburger order. In this normal form,*

$$K = \mathbb{R} \times A, \quad I = \{0\} \times A \quad (170.2)$$

are intrinsically the convex hull of C and the elements infinitesimal relative to C , respectively.

170.1 Convex hull and relative standard part

Proof of the forward direction. Orient C by lemma 169.1. Define

$$K = \{g \in G(E) : |g| < (r, 0) \text{ for some } r > 0\}.$$

Here $|g| = \max(g, -g)$ is formed in the given group order. The subgroup K is convex, contains C , and is its convex hull: any convex subgroup containing C contains every element bounded by an element of C . Since $C \subseteq K$, there is a subgroup $A \leq E$ with $K = \mathbb{R} \times A$. Convexity implies purity of K and then purity of A . By lemma 169.2, $u \in A$.

For $g \in K$, define its standard part relative to C by the ordinary real supremum

$$\text{st}(g) = \sup\{t \in \mathbb{R} : (t, 0) \leq g\}. \quad (170.3)$$

The set is nonempty and bounded above. For every $\varepsilon > 0$,

$$(\text{st}(g) - \varepsilon, 0) < g < (\text{st}(g) + \varepsilon, 0). \quad (170.4)$$

The left inequality follows from the defining supremum, and the right one from its upper-bound property. Adding two such inequalities and letting ε decrease shows

$$\text{st}(g + h) = \text{st}(g) + \text{st}(h).$$

Also $\text{st}(r, 0) = r$. Thus $\text{st} : K \rightarrow \mathbb{R}$ is an order-preserving surjective homomorphism. Its kernel is exactly

$$I = \{g \in G(E) : -(r, 0) < g < (r, 0) \text{ for all } r > 0\}.$$

This kernel is convex.

Put $\lambda(e) = \text{st}(0, e)$ for $e \in A$. Then

$$\text{st}(r, e) = r + \lambda(e), \quad I = \{(-\lambda(e), e) : e \in A\}.$$

The map $e \mapsto (-\lambda(e), e)$ transports the order on I to a group order $<_A$ on A . Since $\lambda(u) = 0$ by lemma 169.2, u is its least positive element. The quotient order on $G(E)/K$ identifies with a group order $<_B$ on E/A .

For $e \notin A$, the sign of (r, e) is the sign of its quotient $e + A$. For $e \in A$ with $r + \lambda(e) \neq 0$, its sign is that of $r + \lambda(e)$ by (170.4). On the remaining boundary it is exactly the sign of e in $<_A$. This already describes the original order completely, except for removing λ . $\square$

170.2 Why cut offsets are not extra invariants

Lemma 170.3 (Countable extension of real homomorphisms). *If A is a subgroup of a countable abelian group E , every homomorphism $\lambda : A \rightarrow \mathbb{R}$ extends to a homomorphism $f : E \rightarrow \mathbb{R}$.*

Proof. Enumerate E and adjoin one element x at a time to the current domain H . If no positive multiple of x lies in H , assign $f(x) = 0$ and extend on $H + \mathbb{Z}x$. Otherwise the subgroup $\{n \in \mathbb{Z} : nx \in H\}$ is $m\mathbb{Z}$ for a least $m > 0$. Assign $f(x) = f(mx)/m$. The relation $mx \in H$ ensures that $f(h + kx) = f(h) + kf(x)$ is well defined. The union of these countably many extensions is the desired homomorphism. $\square$

Extend the λ above to $f : E \rightarrow \mathbb{R}$. The shear

$$S_f(r, e) = (r + f(e), e) \quad (170.5)$$

is a continuous additive automorphism; its inverse is S_{-f} . It fixes $\mathbf{1}$ because $f(u) = \lambda(u) = 0$. The description after (170.4) shows that S_f carries the original order to (170.1). It also gives (170.2). This completes the forward direction of [theorem 170.2](#).

Proof of the converse. The order on $\mathbb{R} \times A$ is the lexicographic extension of the usual real order by $<_A$. Extend it to $G(E)$ using the ordered quotient E/A . Explicitly this is (170.1); the positive cone is closed under addition, disjoint from its negative, and partitions all nonzero elements by sign. Hence it defines a translation-invariant total order.

If a positive element has nonzero quotient in E/A , it exceeds every element of $\mathbb{R} \times A$ and in particular $\mathbf{1}$. If its quotient is zero but its real coordinate is positive, it again exceeds $\mathbf{1}$. On the zero-real-coordinate boundary, least positivity is precisely that of u in A . Therefore $\mathbf{1}$ is the least positive element. On each clopen real fiber its positive cone is empty, the entire fiber, an open half-line, or a closed half-line. Countably many fibers imply Borelness. Finally, the definitions of convex hull and infinitesimals recover exactly the two subgroups in (170.2). $\square$

[write] (Batch 92.) In the notation of other parts: C is Part IV's identity component C and Part IX's $H = G^0$; I is Part IV's B_C (remark after [theorem 50.1](#)) and Part IX's infinitesimal subgroup C ([theorem 122.1](#)), so the manifold boundary of the cone is $I_{\geq 0} = \{0\} \times A_{\geq 0}$; K is the convex hull of the identity component, which that remark shows need not be the whole group. In Part IV's interleaved example the real component has dimension two; the one-dimensional interleaving is source 19's [example 172.2](#) below. Part VII's construction ([theorem 94.1](#) with $D = \mathbb{R}$) is the case $A = E$, $B = 0$ of (170.1), with $<_A$ the order of its countable core.

Remark 170.4 (An ordered extension need not split). We retain $A \leq E$ and the quotient E/A as *extension data*. Nothing here asserts $E \cong A \oplus (E/A)$. A divisible quotient need not be a direct summand. This is why the general normal form is a sign rule on $\mathbb{R} \times E$, not an unjustified direct-product decomposition. The split specialization below has an additional vector-space reason for splitting.

Lemma 170.5. *For normal data, (A, u) is itself a countable Presburger datum and E/A is a divisible torsion-free group.*

Proof. Torsion-freeness of E/A is purity. Since $E/\mathbb{Z}u$ is divisible, so is its quotient E/A . To divide $a + \mathbb{Z}u$ in $A/\mathbb{Z}u$, write $a = ne + ku$ in E . Then $ne \in A$, so purity gives $e \in A$. Torsion-freeness of $A/\mathbb{Z}u$ follows by its inclusion in $E/\mathbb{Z}u$. $\square$

Corollary 170.6 (Baire regularity upgrades to Borel rank two). *For a Presburger order on $G(E)$, the following are equivalent: its positive cone has the Baire property; it is Borel; it is Δ_2^0 . The normal form of [theorem 170.2](#) therefore holds under the weaker Baire-property assumption.*

Proof. The real component C is open. Restricting an ambient set with the Baire property to this open subspace preserves the Baire property. Thus [lemma 169.1](#) orients the restricted

order as usual. Every remaining step in the normal-form proof is algebraic or uses ordinary real completeness; no further measurability is needed. In normal form the cone and its complement are countable unions of F_σ half-lines in clopen fibers, so the cone is Δ_2^0 . A homeomorphism preserves that class. The converse implications are immediate. $\square$

[write] The implication from a Borel cone to a Δ_2^0 cone is, for these groups, a case of Part III's [theorem 33.1](#) and of part (1) of Part IV's [theorem 51.2](#); the upgrade from the Baire property is new in this class. Compare Part VI, where a Baire-property cone already forces rigidity of discretely ordered rings ([lemma 77.2](#)).

171 Complete isomorphism invariants

Theorem 171.1 (All continuous isomorphisms). *Let $(E, u, A, <_A, <_B)$ and $(E', u', A', <_{A'}, <_{B'})$ be normal data. Every topological unital additive order isomorphism $G(E) \rightarrow G(E')$ has the unique form*

$$F(r, e) = (ar + f(e), \varphi(e)), \quad (171.1)$$

where $a > 0$, $\varphi : E \rightarrow E'$ is a group isomorphism with $\varphi(u) = u'$ and $\varphi(A) = A'$, the restrictions and induced maps

$$\varphi|_A : (A, <_A) \longrightarrow (A', <_{A'}), \quad \bar{\varphi} : (E/A, <_B) \longrightarrow (E'/A', <_{B'})$$

are order isomorphisms, and $f : E \rightarrow \mathbb{R}$ is a homomorphism vanishing on A . Conversely, every such formula is an isomorphism.

In particular, an isomorphism exists if and only if such a countable ordered-extension isomorphism φ exists.

Proof. A topological-group isomorphism preserves the identity component C . Its restriction there is a continuous additive bijection of $\mathbb{R}$, hence $r \mapsto ar$ for some nonzero real a . Order preservation gives $a > 0$. The induced map on the component quotient is a group isomorphism $\varphi : E \rightarrow E'$. Writing $F(0, e) = (f(e), \varphi(e))$ gives (171.1), with f additive. Fixing the unit gives $\varphi(u) = u'$ and $f(u) = 0$.

Since K is the convex hull of the identity component, it is preserved. Thus $\varphi(A) = A'$, and the induced order on the outer quotient is preserved. The infinitesimal subgroup I relative to C is also intrinsic. In normal form $I = \{0\} \times A$, so $f|_A = 0$ and $\varphi|_A$ preserves its order. These establish necessity.

Conversely, (171.1) is a continuous group isomorphism: solve for e using φ^{-1} and then for r using a^{-1} . It preserves signs in the outer quotient. On $\mathbb{R} \times A$ it is $(r, e) \mapsto (ar, \varphi(e))$, preserving both the real comparison and the boundary order. It fixes the unit. Choosing $a = 1$ and $f = 0$ proves the last assertion. $\square$

[write] (Batch 92.) With Part IV's shear ([proposition 182.1](#) [merge]), [theorems 170.2](#) and [171.1](#) classify, for every countable torsion-free E , the Borel Presburger orders on $\mathbb{R} \times E$ up to continuous isomorphism; with [corollary 171.2](#) below, this answers Part IX's [question 126.5](#) (note there). For $E = \mathbb{Z}u$ the model is Part I's $M_{\mathbb{R}}$ and the corollary gives $\text{Aut}_{\text{top}} = \mathbb{R}_{>0}$, Part I's [theorem 11.1](#). [Proposition 182.2](#) [merge] shows that the word “topological” can be dropped.

Let $\text{Aut}(E, u; A, <_A, <_B)$ denote the countable-data automorphism group appearing in this theorem. It is a subgroup of the permutation group of the countable set E .

Corollary 171.2 (Topological automorphism group). *The normal model has a split exact sequence*

$$1 \longrightarrow \text{Hom}(E/A, \mathbb{R}) \longrightarrow \text{Aut}_{\text{top}}(G(E), <, \mathbf{1}) \longrightarrow \mathbb{R}_{>0} \times \text{Aut}(E, u; A, <_A, <_B) \longrightarrow 1.$$

Under the natural splitting, (a, φ) acts on a shear h by $h \mapsto ah\bar{\varphi}^{-1}$.

Proof. Project (171.1) to (a, φ) . Its kernel consists precisely of f factoring through E/A . The section sets $f = 0$. Conjugating $(r, e) \mapsto (r + h(e + A), e)$ gives the stated action. Equivalently, direct composition is

$$(a, f, \varphi)(b, g, \psi) = (ab, ag + f \circ \psi, \varphi \circ \psi).$$

□

Corollary 171.3 (No rigid model in this presentation class). *Every Borel Presburger order on $G(E)$ admits a faithful action of $\mathbb{R}_{>0}$ by continuous unital additive order automorphisms. The same holds on its nonnegative cone.*

Proof. In normal form use $D_a(r, e) = (ar, e)$, for $a > 0$; conjugate by the normalizing shear in the original coordinates. If that shear is S_f , the formula in the original coordinates is

$$S_f^{-1}D_aS_f(r, e) = (ar + (a - 1)f(e), e).$$

The action is faithful already on positive elements of C .

□

This is compatible with the existence of rigid Presburger models of cardinality continuum in the purely algebraic classification [4]. Those models cannot have a Borel ordered presentation of the particular form classified here. We are not asserting that all uncountable Polish Presburger models are nonrigid.

[write] (Batch 92.) This answers Part I's [question 14.12](#) and source 13's [question 103.5](#) negatively inside this class, and, by [proposition 182.3](#) [merge], for every uncountable locally compact Polish Presburger model with continuous addition and partial subtraction (notes after those questions); source 19's [question 180.6](#) asks the rest again. Part I already observed that its split examples are not rigid.

172 A sharper answer for split countable groups

The general theorem becomes particularly transparent for the pointed group most useful in the coding construction.

Theorem 172.1 (Two countable ordered vector spaces). *Let V be a countable-dimensional, countable $\mathbb{Q}$ -vector space, equipped with the discrete topology, and let*

$$E = V \oplus \mathbb{Z}, \quad u = (0, 1).$$

Every Borel Presburger order on $\mathbb{R} \times E$ is topologically, additively, and unital-order isomorphic to

$$B \overrightarrow{\times} \mathbb{R} \overrightarrow{\times} W \overrightarrow{\times} \mathbb{Z}, \tag{172.1}$$

where B, W are countable divisible ordered groups and $\dim_{\mathbb{Q}} B + \dim_{\mathbb{Q}} W = \dim_{\mathbb{Q}} V$. The topology is the usual topology on the real coordinate and the discrete topology on all other coordinates. Two such models are topologically order isomorphic if and only if their ordered groups B are isomorphic and their ordered groups W are isomorphic.

Proof. Apply [theorem 170.2](#). Since A contains all of $\mathbb{Z}u$, an element (v, z) belongs to A if and only if $(v, 0)$ does. Thus $A = W \oplus \mathbb{Z}$ for a subgroup $W \leq V$. Purity makes W a $\mathbb{Q}$ -linear subspace: if $w \in W$, then $n(w/n) \in W$, hence $w/n \in W$. Choose a vector-space complement $V = B_0 \oplus W$. The order on V/W transfers to B_0 , producing B .

The integer subgroup is convex in $(A, <_A)$ because u is its least positive element. For completeness, if $0 \leq a \leq nu$ for an ordinary n , a finite induction using least positivity shows that a is one of $0, u, \dots, nu$. Consequently the order on A is exactly the lexicographic extension of its quotient order on W by the standard order on $\mathbb{Z}$. This proves [\(172.1\)](#).

Necessity of the pair invariant follows intrinsically: $B = G/K$ and $W = I/\mathbb{Z}1$. Conversely, two order isomorphisms of the respective B and W factors, together with the identity on $\mathbb{R}$ and $\mathbb{Z}$, give a continuous pointed order isomorphism. The coordinate rearrangement between V and $B_0 \oplus W$ is automatically continuous because these factors are discrete. $\square$

Example 172.2 (The same pointed group, different placement of scales). For $V = \mathbb{Q}$, both

$$\mathbb{R} \overrightarrow{\times} \mathbb{Q} \overrightarrow{\times} \mathbb{Z} \quad \text{and} \quad \mathbb{Q} \overrightarrow{\times} \mathbb{R} \overrightarrow{\times} \mathbb{Z}$$

have underlying pointed topological group $\mathbb{R} \times (\mathbb{Q} \oplus \mathbb{Z})$. They are not topologically order isomorphic: in the first model the convex hull of the identity component is the entire group, while in the second its quotient is $\mathbb{Q}$. This is exactly the information omitted by a classification of the pointed topological group alone.

Example 172.3 (A removable tilted cut). In the first model, replace the comparison of (r, q, z) by that of $(r + \sqrt{2}q, q, z)$ in lexicographic order. The nonnegative boundary in the (q, z) fiber is now at $r = -\sqrt{2}q$ rather than at zero. The continuous shear $(r, q, z) \mapsto (r + \sqrt{2}q, q, z)$ removes the tilt. No irrational slope survives as an isomorphism invariant.

[write] (Batch 92.) The case $V = 0$ of [theorem 172.1](#) is Part I's $M_{\mathbb{R}}$; the case $B = 0$ is Part VII's [theorem 94.1](#) with $D = \mathbb{R}$ and a split core $W \overrightarrow{\times} \mathbb{Z}$. [Example 172.2](#) is the one-dimensional counterpart of the interleaved example after Part IV's [theorem 50.1](#), which has two real scales.

173 All the positive cones are one fixed manifold

Let $M = G(E)_{\geq 0}$ with its subspace topology. Define the fixed space

$$X_{lc} = (\mathbb{N} \times [0, \infty)) \sqcup (\mathbb{N} \times \mathbb{R}), \quad (173.1)$$

where each copy is an open-and-closed component. It is a real-analytic one-dimensional manifold with boundary. It is not connected and not compact. It is perfect, locally compact, second countable, and Polish. For example, use the ordinary complete metrics on its components, truncate them at 1, and put distance 2 between distinct components.

Theorem 173.1 (Fixed carrier and analytic addition). *For every order in [theorem 170.2](#), its nonnegative cone M is real-analytically diffeomorphic to X_{lc} , and addition $M \times M \rightarrow M$ is real analytic. In particular it is jointly continuous. The smooth structure here is the fiberwise one inherited from $\mathbb{R} \times E$; products of manifolds with boundary are understood as manifolds with corners.*

Proof. Normalize the order. If $e + A >_B 0$, its entire real fiber belongs to M ; if $e + A <_B 0$, its fiber is absent. For $e \in A$ the fiber is

$$\begin{cases} [0, \infty), & e \geq_A 0, \\ (0, \infty), & e <_A 0. \end{cases} \quad (173.2)$$

Both cases occur countably infinitely often, since $nu \geq_A 0$ and $-nu <_A 0$ for every positive ordinary n . Thus there are countably infinitely many closed-half-line components and countably infinitely many open-line components; any additional full real fibers only enlarge a countably infinite list. Use the identity on half-lines and the logarithm on $(0, \infty)$ to obtain an analytic diffeomorphism to X_{lc} .

On any fixed pair of component labels, addition is simply $(r, s) \mapsto r + s$ into the component with label $e + f$. These formulas are real analytic, including at the boundary in the sense of local extensions to open subsets of Euclidean spaces. Because the component labels are discrete and addition preserves M , the formulas patch to an analytic map. In unnormalized coordinates the shear is itself fiberwise affine and analytic, so the conclusion remains valid. $\square$

[write] (Batch 92.) The topological part is Part IX's classification: X_{lc} is the carrier of [theorems 120.5](#) and [121.1](#) (source 19's display [\(173.1\)](#) defines the same space), and every uncountable inherited locally compact Presburger cone lives on it ([corollary 121.2](#)). The real-analytic structure and the analyticity of addition are new. Local compactness of these cones is the case $d = 1$ of Part IV's [theorem 51.2\(3\)](#).

173.1 Topological monoid isomorphisms recover the signed group

Lemma 173.2 (Local extension to the group completion). *A topological unital additive monoid isomorphism between two cones in [theorem 173.1](#) extends uniquely to a continuous unital additive order isomorphism of their signed groups. Every such monoid isomorphism is therefore real analytic, with real-analytic inverse.*

Proof. Algebraically the group is the Grothendieck group of its nonnegative cone. Extend f by $F(a - b) = f(a) - f(b)$. Cancellativity makes this well defined, and the inverse monoid isomorphism gives its inverse. The cone recovers the order, so F is an order isomorphism.

We check continuity rather than assume it for an arbitrary group completion. In normalized coordinates put $c = (1, 0) \in C$. Every $h \geq c$ is an interior point of M : for sufficiently small real t , one has $h + (t, 0) \geq 0$, and this is an open interval in its clopen fiber. Given $g \in G$, choose $b = \max(-g, 0) + c \in M$. Then $g + b \geq c$, so a neighborhood U of g satisfies $U + b \subseteq M$. On U ,

$$F(x) = f(x + b) - f(b),$$

a continuous expression in the target topological group. Thus F is continuous everywhere, and the same argument applies to its inverse. By [theorem 171.1](#), it is affine on each real fiber; hence it and its inverse are analytic. Normalizing shears do not affect the argument. $\square$

This lemma is deliberately restricted to the cones constructed here. It does not solve the separate existence problem for topological-group completions of arbitrary Polish cancellative monoids.

[write] (Batch 92.) That existence problem is solved by Part XII, which source 19 did not see: a cone topology with continuous addition and a linear natural order extends to a Hausdorff group topology on the group of differences exactly when partial subtraction is continuous, and the extension is unique ([theorem 153.1](#)), Polish ([theorem 154.1](#)) and locally compact when the cone is ([corollary 154.2](#)). [merge] The continuity part of [lemma 173.2](#) also follows from that uniqueness: if $f : M \rightarrow M'$ is a topological monoid isomorphism and $F : G \rightarrow G'$ its algebraic extension, the topology that F pulls back from G' is a Hausdorff group topology

on G inducing the topology of M , so it is the topology of G , and F is a homeomorphism. Source 19's proof by interior translates is different and stays in its own class.

173.2 Low complexity of definability does not simplify isomorphism

Proposition 173.3. *All first-order definable relations, with parameters, in these signed groups and positive cones are Δ_2^0 . For each ordinary $n \geq 1$, quotient and remainder on division by n are continuous. The strict arithmetic order on M^2 is nevertheless neither open nor closed.*

Proof. In normal form each fiber of the positive cone is empty, full, or a half-line with either endpoint convention. Both the cone and its complement are countable unions of F_σ sets in clopen fibers. Thus the cone is Δ_2^0 . Affine group terms are continuous; their comparisons are preimages of this cone. The subgroup $nG = \mathbb{R} \times nE$ is clopen, so congruences are clopen. Quantifier elimination from [appendix AC](#), and closure of Δ_2^0 under finite Boolean combinations, give the assertion. Restriction to M^k preserves the same relative Borel class.

For division, use the decomposition $e = ru + nd$ from [lemma 168.2](#). The maps are

$$q_n(x, e) = (x/n, d), \quad \rho_n(x, e) = r\mathbf{1}.$$

They are continuous on every clopen fiber. If $(x, e) \geq 0$, its quotient cannot be negative: then it would be at most $-\mathbf{1}$, implying $nq_n + \rho_n \leq -n\mathbf{1} + (n-1)\mathbf{1} < 0$. Thus q_n restricts to M .

Finally $c_k = (1/k, 0) \rightarrow 0$, while $c_k > \mathbf{1}$ by [lemma 169.2](#). The pairs $(0, c_k)$ lie in $<$ but tend to $(0, 0)$, so $<$ is not closed. The pairs $(c_k, \mathbf{1})$ lie outside $<$ but tend to $(0, \mathbf{1})$, so it is not open. $\square$

[write] (Batch 92.) For these groups this is Part IV's [theorem 52.1](#) and [proposition 51.4](#), proved there for every locally compact Borel $\mathbb{Z}$ -group; the order obstruction is also Part I's [theorem 7.4](#).

174 Encoding arbitrary countable linear orders

Let LO be the closed subspace of $2^{\mathbb{N} \times \mathbb{N}}$ consisting of strict linear orders on $\mathbb{N}$. For $L \in \text{LO}$, let

$$V_L = \bigoplus_{i \in \mathbb{N}} \mathbb{Q}e_i$$

with the order determined by the *least L -index* in the finite support of a nonzero vector: the vector is positive exactly when its coefficient at that index is positive. The group operation and its underlying countable set do not depend on L . Set

$$G_L^{\text{lo}} = \mathbb{R} \overrightarrow{\times} V_L \overrightarrow{\times} \mathbb{Z}, \quad M_L^{\text{lo}} = (G_L^{\text{lo}})_{\geq 0}. \quad (174.1)$$

Use the product topology $\mathbb{R}_{\text{usual}} \times (V_L \times \mathbb{Z})_{\text{discrete}}$. These are normal models with $A = E = V_L \oplus \mathbb{Z}$ and no scales above $\mathbb{R}$.

174.1 Recovering the order of infinitesimal scales

For positive elements in an ordered group, define Archimedean equivalence by

$$x \asymp y \iff x \leq ny \text{ and } y \leq mx \text{ for some ordinary } m, n \geq 1.$$

Write $x \gg y$ when $x > ny$ for every ordinary $n \geq 1$.

Lemma 174.1 (The Archimedean spine). *For positive $x, y \in V_L$, one has $x \asymp y$ exactly when their least support indices coincide. If those indices are i, j , then $i <_L j$ exactly when $x \gg y$. Consequently the order type L can be reconstructed from V_L as an ordered group.*

Proof. If the indices are different, the more significant coordinate in every $x - ny$ is the earlier one, independent of n . This gives the stated dominance. If the indices coincide, choose ordinary integers n, m strictly exceeding the respective ratios of the positive leading coefficients. The leading coefficient of $ny - x$, and respectively $mx - y$, is then positive, regardless of all later coordinates. Thus the two vectors are Archimedean equivalent. Every index occurs as the leading index of e_i , so no classes are missing. $\square$

[write] (Batch 92.) V_L is Part IX's group V_L , with the same order (source 19 calls it D_L), and the lemma is Part IX's computation $\mathcal{A}(V_L) \cong L$ in [section 123](#), by the same argument. Tempting false reading: Part IX places the coded scales above the real line ($G_L = V_L \overrightarrow{\times} \mathbb{R} \overrightarrow{\times} \mathbb{Z}$), source 19 below it ($G_L^{\text{lo}} = \mathbb{R} \overrightarrow{\times} V_L \overrightarrow{\times} \mathbb{Z}$).

Theorem 174.2 (Exact isomorphism coding). *For $L, K \in \text{LO}$, the following conditions are equivalent:*

- (i) $L \cong K$ as countable linear orders;
- (ii) $V_L \cong V_K$ as ordered groups;
- (iii) $G_L^{\text{lo}} \cong G_K^{\text{lo}}$ as abstract unital ordered groups;
- (iv) $M_L^{\text{lo}} \cong M_K^{\text{lo}}$ as abstract unital additive monoids;
- (v) $M_L^{\text{lo}} \cong M_K^{\text{lo}}$ by a topological unital additive isomorphism;
- (vi) $M_L^{\text{lo}} \cong M_K^{\text{lo}}$ by a real-analytic unital additive isomorphism.

Proof. An order isomorphism of index sets permutes finite-support coordinates, giving (i) $\Rightarrow$ (ii) and an analytic isomorphism of the models. Conversely, (ii) $\Rightarrow$ (i) follows from [lemma 174.1](#).

In G_L^{lo} , the subgroup $H_L = \{0\} \times V_L \times \mathbb{Z}$ is the unique largest proper convex subgroup. It is convex; and any convex subgroup containing an element with nonzero real coordinate contains every element of G_L^{lo} , since a sufficiently large ordinary multiple of that element dominates it. Therefore any abstract ordered-group isomorphism preserves H_L . It also preserves the integer subgroup generated by $\mathbf{1}$. Passing to $H_L/\mathbb{Z}\mathbf{1}$ recovers V_L . This proves (iii) $\Rightarrow$ (ii).

An abstract monoid isomorphism extends algebraically to the signed groups and preserves their cones, hence their orders; so (iv) $\Rightarrow$ (iii). The other implications follow from the explicit coordinate permutation and from forgetting regularity. $\square$

[write] (Batch 92.) Part IX proves the corresponding statement for its subfamily $L = L_S$ and the opposite placement of scales ([\(123.3\)](#)); the theorem here covers every countable L . [Proposition 182.2](#) [merge] extends the equivalence of abstract and topological isomorphism to every model of [theorem 170.2](#).

174.2 A continuous family of analytic additions on a fixed space

It is important that the carrier can be fixed without an arbitrary nonmeasurable choice of homeomorphisms. Choose once and for all an explicit enumeration $(h_j)_{j \in \mathbb{N}}$ without repetitions of $H = \mathbb{Q}^{(\mathbb{N})} \oplus \mathbb{Z}$. For each L , enumerate the sets

$$H_L^+ = \{h : h \geq_L 0\}, \quad H_L^- = \{h : h <_L 0\}$$

in increasing *enumeration index*, writing $h_k^+(L), h_k^-(L)$. Each selected element depends locally constantly on L : to locate the k th eligible index, only finitely many sign decisions are needed, and every sign decision examines only a finite support.

Use the labels $+$ and $-$ for the two parts of X_{lc} in (173.1). Define

$$\begin{aligned} \theta_L(+, k, t) &= (t, h_k^+(L)), & t \geq 0, \\ \theta_L(-, k, s) &= (e^s, h_k^-(L)), & s \in \mathbb{R}. \end{aligned} \quad (174.2)$$

This is a real-analytic diffeomorphism $X_{lc} \rightarrow M_L^{\text{lo}}$. Transport addition and the constants by θ_L :

$$m_L(x, y) = \theta_L^{-1}(\theta_L(x) + \theta_L(y)), \quad 0_L = \theta_L^{-1}(0), \quad 1_L = \theta_L^{-1}(1). \quad (174.3)$$

Theorem 174.3 (Continuous fixed-manifold reduction). *The map*

$$L \mapsto (m_L, 0_L, 1_L)$$

is continuous into $C(X_{lc} \times X_{lc}, X_{lc}) \times X_{lc} \times X_{lc}$, where the function space has the compact-open topology. Every m_L is real analytic. Moreover

$$L \cong K \iff (X_{lc}, m_L, 0_L, 1_L) \cong (X_{lc}, m_K, 0_K, 1_K),$$

whether isomorphism means abstract, topological, or real analytic.

Proof. Fix L and a pair of input component labels. On a neighborhood of L , the two selected h labels are constant. Their sum in H is constant, and its sign and output enumeration index are also locally constant. For the real coordinate, addition is followed either by the identity or by the logarithm. A logarithm is never applied to zero: if the output label is negative, the sum has positive real coordinate; both input real coordinates could be zero only for two nonnegative labels, whose sum is nonnegative. Thus $(L, x, y) \mapsto m_L(x, y)$ is jointly continuous. The same finite-decision argument gives continuity of 0_L and 1_L .

Since $X_{lc} \times X_{lc}$ is locally compact, joint continuity implies continuity into the compact-open function space. Directly, for a compact set of inputs and an open target condition, neighborhoods of each input point provide a finite cover; intersecting the finitely many parameter neighborhoods gives the required compact-open neighborhood. Analyticity for fixed L follows from the displayed formulas or [theorem 173.1](#). The isomorphism assertion is [theorem 174.2](#) transported through (174.2). $\square$

This is stronger than merely producing many models. Every individual operation is analytically regular, the underlying manifold is fixed, and the family varies continuously, but identifying its isomorphism types retains the full complexity of countable linear orders.

175 The exact descriptive-set-theoretic classification statement

The phrase *Borel complete* needs a specified parameter space and a specified comparison relation. It must not be confused with completeness for all analytic equivalence relations.

Definition 175.1. An equivalence relation $\mathcal{E}$ on a standard Borel space is *Borel complete for isomorphism of countable structures* if every isomorphism relation on a Borel class of countable structures in a countable language Borel reduces to $\mathcal{E}$. A reduction F satisfies $x \cong y$ if and only if $F(x) \mathcal{E} F(y)$.

The classical Friedman–Stanley theorem says that isomorphism of countable linear orders is Borel complete in this sense [123, 124]; see also the precise distinction between equivalence relations and isomorphism sets in [125, Section 4]. We use that theorem only after proving the new coding map.

175.1 A standard Borel space of all normal data

Fix (E, u) . Code a subgroup A by a point of 2^E , its positive order by a subset of A , and the quotient order by a subset of E which is a union of A -cosets. Require the subgroup, purity, positivity, totality, translation, and least-unit axioms from [definition 170.1](#). Every requirement involves countably many coordinates, so the admissible code space $\mathcal{P}_E$ is a Borel subset of a countable product of two-point spaces. Codes may equivalently use the corresponding binary order relations.

To include unnormalized presentations explicitly, append a homomorphism $f : E \rightarrow \mathbb{R}$ with $f(u) = 0$ and use the sign rule obtained by replacing r with $r + f(e)$ on $\mathbb{R} \times A$. The homomorphism equations are countably many closed conditions in $\mathbb{R}^E$. By [theorem 170.2](#) and [lemma 170.3](#), this enlarged parameterization represents every Borel Presburger order on $G(E)$ after, if necessary, reversing the real coordinate. The extra parameter does not affect isomorphism type.

The countable structure associated with a code consists of the fixed group E , the constant u , the unary predicate A , its internal order, and the quotient comparison relation pulled back to E . Its language is countable (in fact finite after using group operations as function symbols). By [theorem 171.1](#), two normal codes give topologically isomorphic ordered groups exactly when these countable structures are isomorphic. This association is Borel. For their positive cones the same assertion follows from [lemma 173.2](#).

Theorem 175.2 (Sharp complexity in normal-form coordinates). *For the fixed pointed group $E = \mathbb{Q}^{(\mathbb{N})} \oplus \mathbb{Z}$, topological isomorphism of the models parameterized by $\mathcal{P}_E$ is Borel complete for countable structures and is itself classifiable by countable structures. The same is true for the enlarged parameterization including real shears, and for the corresponding positive cones.*

Proof. The preceding countable-structure construction gives the upper reduction. For the lower reduction, send $L \in \text{LO}$ to the code with $A = E$ and the order $V_L \vec{\times} \mathbb{Z}$ on E . A sign of a fixed finite-support vector depends on finitely many coordinates of L , so this code map is continuous. By [theorem 174.2](#), it reduces linear-order isomorphism to topological isomorphism of the models. Apply Friedman–Stanley. Shears have no effect on equivalence by [theorem 170.2](#). $\square$

Corollary 175.3 (No Borel test and no complete real invariant). *There is no Borel test that decides, for arbitrary $L, K \in \text{LO}$, whether the analytic monoids $(X_{\text{lc}}, m_L, 0_L, 1_L)$ and $(X_{\text{lc}}, m_K, 0_K, 1_K)$ are*

isomorphic. There is also no Borel assignment of a single real number, or a countable sequence of real numbers, whose equality characterizes isomorphism in this family.

Proof. Such a test would make isomorphism of countable linear orders Borel, contrary to its classical non-Borelness. An equality-based invariant in any standard Borel space would give just such a Borel test, since its equality relation is Borel. A countable sequence of reals is again a standard Borel code. Apply [theorem 174.3](#). $\square$

Remark 175.4 (What is, and is not, ruled out). The theorem does not say that there are no complete mathematical invariants. It supplies one: a countable ordered extension, or in the split case a pair of countable ordered vector spaces. What fails is a Borel *equality-based* choice of complete codes. Countable structures are compared up to isomorphism, not by literal equality of their presentations. Nor is there a claimed Borel-complete classification of all possible raw codes for arbitrary Borel relations on uncountable spaces; the standard Borel parameterization used above is explicit.

[write] (Batch 92.) Part IX's [question 126.6](#) asks for the complexity of the set of Presburger structures and of arithmetic and topological isomorphism in a standard Borel coding of continuous operations on X_{lc} ; the sentence after its [theorem 123.1](#) asks for the descriptive complexity of isomorphism on that space. [Theorems 174.3](#) and [175.2](#) and [corollary 175.3](#) answer the lower-bound half for the natural coding $C(X_{lc} \times X_{lc}, X_{lc}) \times X_{lc} \times X_{lc}$ ([remark 182.5](#) [merge]); the upper bound on the whole coding space and the complexity of the set of Presburger structures remain open ([question 180.3](#)).

176 Abstract automorphisms versus measurable automorphisms

The coding family admits an especially clean calculation. For any countable divisible ordered group D , put

$$G_D^{lo} = \mathbb{R} \overrightarrow{\times} D \overrightarrow{\times} \mathbb{Z}, \quad M_D^{lo} = (G_D^{lo})_{\geq 0},$$

with the same product topology. This includes every V_L above.

Theorem 176.1 (All automorphisms and the measurable boundary). *Every abstract unital ordered-group automorphism of G_D^{lo} has a unique form*

$$F(r, d, z) = (ar, \beta(d) + h(r), z), \quad (176.1)$$

where $a > 0$, $\beta \in \text{Aut}(D, <)$, and $h : \mathbb{R} \rightarrow D$ is an arbitrary $\mathbb{Q}$ -linear map. Every such formula is an automorphism. The same classification holds for abstract unital additive monoid automorphisms of M_D^{lo} .

For either the signed group or the positive cone, the following are equivalent for an automorphism: it is Borel measurable; it is continuous; it is real analytic; its parameter h is zero. If $D \neq 0$, then

$$|\text{Aut}_{\text{abstract}}(M_D^{lo})| = 2^c, \quad |\text{Aut}_{\text{Borel}}(M_D^{lo})| = |\text{Aut}_{\text{top}}(M_D^{lo})| = c, \quad c = 2^{\aleph_0}. \quad (176.2)$$

Proof. The unique largest proper convex subgroup of G_D^{lo} is $H = \{0\} \times D \times \mathbb{Z}$, by the same cofinality argument used in [theorem 174.2](#). Every ordered automorphism preserves it and induces an ordered additive automorphism of $G_D^{lo}/H \cong \mathbb{R}$. An order-preserving additive bijection of the usual reals is $r \mapsto ar$ with $a > 0$: determine it first on rational multiples of 1 and then squeeze each real between rational bounds.

Inside H , the subgroup $D \times \{0\}$ is its largest divisible subgroup, equivalently $\bigcap_{n \geq 1} nH$. It is characteristic. The unit is fixed, so the restriction of F to H must be $(d, z) \mapsto (\beta(d), z)$, with

β an ordered-group automorphism. Write $F(r, 0, 0) = (ar, h(r), k(r))$. The map $k : \mathbb{R} \rightarrow \mathbb{Z}$ is zero because the image of a divisible group is divisible and $\mathbb{Z}$ has no nonzero divisible subgroup. Thus h is additive, hence $\mathbb{Q}$ -linear. This yields (176.1) and uniqueness.

Conversely, the formula is bijective with inverse obtained by first solving for r , then for d . It preserves the sign of every element with nonzero real coordinate, regardless of h . On the zero-real boundary it preserves the lexicographic order by β and fixes $\mathbb{Z}$. Hence it is an ordered automorphism. Group completion transfers the classification to M_D^{lo} .

If $h = 0$, the formula and its inverse are fiberwise linear and analytic. If F is Borel on the group, its restriction to $(r, 0, 0)$ shows that h is Borel into the countable discrete group D . If F is Borel only on M_D^{lo} , the section $(r, 0, 0)$ for $r > 0$ first shows h is Borel on $(0, \infty)$. Additivity extends this Borelness to all of $\mathbb{R}$, using $h(-r) = -h(r)$ and $h(0) = 0$.

A Borel additive map from $\mathbb{R}$ to a countable discrete group is zero. Indeed, its kernel N is Borel and has countably many cosets. By the Baire theorem, N is not meager. A nonmeager subgroup with the Baire property contains an interval around zero: choose an interval where it is comeager, then intersect it with a sufficiently small translate of it to see that every sufficiently small difference lies in N . Such a subgroup is all of $\mathbb{R}$. Therefore $h = 0$, proving all the regularity equivalences.

Finally $\dim_{\mathbb{Q}} \mathbb{R} = \mathfrak{c}$. One way to see this is that a vector space over a countable field with infinite basis of size κ has cardinality $\max(\kappa, \aleph_0)$; countable dimension would make $\mathbb{R}$ countable. Assigning an arbitrary value in nonzero countable D to each element of a Hamel basis gives $|D|^{\mathfrak{c}} = 2^{\mathfrak{c}}$ maps h . They give distinct automorphisms. No family of maps on a set of size $\mathfrak{c}$ can exceed $\mathfrak{c}^{\mathfrak{c}} = 2^{\mathfrak{c}}$, giving the first equality. There are at most $\mathfrak{c}$ automorphisms of countable D , and exactly $\mathfrak{c}$ positive real dilations. The regular automorphisms therefore number exactly $\mathfrak{c}$. $\square$

Example 176.2 (A discontinuous automorphism preserving every arithmetic formula). Choose $0 \neq d_0 \in D$ and a Hamel basis of $\mathbb{R}$ over $\mathbb{Q}$ containing 1 and $\sqrt{2}$. Define $h(1) = 0$, $h(\sqrt{2}) = d_0$, and set $h = 0$ on all other basis elements. For rationals $q_n \rightarrow \sqrt{2}$,

$$r_n = \sqrt{2} - q_n \rightarrow 0, \quad h(r_n) = d_0.$$

The automorphism $F(r, d, z) = (r, d + h(r), z)$ is therefore discontinuous in the product topology. Taking $q_n < \sqrt{2}$ gives the same witness inside the positive cone. Yet F preserves addition, order, every standard congruence, and every first-order arithmetic formula. This shows why Δ_2^0 definability of individual relations does not force all *external* symmetries to be measurable.

The elementary algebra behind shears has precedents in the general automorphism analysis of $\mathbb{Z}$ -groups [4, Section 4]. The result established here is the exact regularity distinction for the specified Polish presentations, together with its cardinal calculation.

[write] (Batch 92.) For $D = 0$ the theorem is Part I's [theorem 11.1](#): only positive dilations. Part X's [theorem 133.4](#) finds $2^{\mathfrak{c}}$ automorphisms of $M_{\alpha} = (\mathbb{Q}^{\alpha} \overrightarrow{\times} \mathbb{Z})_{\geq 0}$ for countable $\alpha > \omega$ by a different family (a choice-built functional on $\mathbb{Q}^{\alpha}$, not a Hamel map on $\mathbb{R}$); its [remark 133.2](#) shows that those need a set without the Baire property. The same holds here:

Remark 176.3 (The $2^{\mathfrak{c}}$ automorphisms need a set without the Baire property [merge]). Work in ZF with dependent choice. The classification (176.1) uses no choice. If $h : \mathbb{R} \rightarrow D$ is additive and its kernel N has the Baire property, then N is not meager, because its countably many cosets cover $\mathbb{R}$, and the argument in the proof above gives $N = \mathbb{R}$, so $h = 0$. Hence in a model of ZF with dependent choice in which every set of reals has the Baire property, which exists

if ZF is consistent [96], every abstract automorphism of G_D^{lo} and of M_D^{lo} is continuous, and $|\text{Aut}_{\text{abstract}}(M_D^{\text{lo}})| = \mathfrak{c}$. The count $2^{\mathfrak{c}}$ in (176.2) therefore uses the axiom of choice, as source 19's ledger (appendix AD) says of the Hamel shears.

The remark was written for this report and has not been independently reviewed.

177 A finite-normal-form realization inside the omnific integers

This section gives the second concrete link to ProveIt. The omnific normal-form interface records sums with nonnegative exponents and integer constant coefficient [127]. Only finite sums are needed, so no delicate infinite-support summability is involved.

Lemma 177.1. *Every countable linear order L admits an order-reversing embedding $i \mapsto q_i$ into $(1, 2) \cap \mathbb{Q}$.*

Proof. Enumerate the elements of L . At each finite stage choose a rational strictly between the finitely many already assigned lower and upper bounds, reversing the prescribed order. The endpoint bounds 1 and 2 ensure a nonempty rational interval at every stage. The union is an injective order-reversing map. $\square$

[write] (Batch 92.) This is Part IX's lemma 124.2, with the same statement and proof.

Theorem 177.2 (Omnific realization of the coding family). *For such a choice of (q_i) , the map*

$$\iota_L(r, d, z) = r\omega^3 + \sum_{i \in \text{supp}(d)} d_i \omega^{q_i} + z \quad (177.1)$$

is a unital ordered additive embedding $G_L^{\text{lo}} \hookrightarrow \mathbf{Oz}$. Consequently M_L^{lo} is an additive submonoid of the nonnegative omnific integers. Transporting its topology through ι_L gives exactly the Polish analytic presentation studied above.

Proof. The exponents are positive and distinct, and z is an ordinary integer, so each displayed finite Conway normal form is omnific. Coefficientwise addition proves additivity. Uniqueness of normal form proves injectivity. If $r \neq 0$, the exponent 3 dominates every q_i , so the sign is that of r . If $r = 0$ and $d \neq 0$, its least L -support index has the largest q_i , so the sign is the sign of that coefficient. If $r = d = 0$, the sign is that of z . These are exactly the lexicographic comparisons in G_L^{lo} . The unit maps to 1. $\square$

Remark 177.3 (The topology is external, and the image is not a ring). No claim is made that this topology is a natural surreal order topology. The sequence ω^3/n tends to zero in the transported presentation, but every term is greater than 1, so it does not tend to zero in the surreal order topology. Likewise, the image need not be closed under multiplication: it contains ω^3 but not ω^6 . The theorem is about additive arithmetic, not a topological model of full arithmetic and not a counterexample to Glazer's theorems.

All expressions in (177.1) are indexed by the set $\mathbb{R} \times \mathbb{Q}^{(\mathbb{N})} \times \mathbb{Z}$. Their image is a set by Replacement for this explicitly defined construction. In a class-theoretic formulation, its birthdays have an ordinal bound. One may therefore work in a fixed set-sized surreal stage large enough to contain the image; no set of all surreal numbers or all surreal embeddings is used.

[write] (Batch 92.) **Theorem 177.2** is a mirrored variant of Part IX's theorem 124.3: there the real scale $r\omega$ lies *below* the coded exponents in $(1, 2)$, here the real scale $r\omega^3$ lies above

them, matching the two placements of scales; compare Part I's [theorem 9.1](#) and Part III's [theorem 40.1](#). Part IX's remark after its theorem, that the native order topology of $\mathbf{Oz}$ is discrete and the transported topology is a coefficient topology, applies here too.

178 Exact checks and an implementation boundary

The archive contains a dependency-free Python program `checks/verify_finite.py`. It uses exact rational arithmetic and finite-support vectors. Its tests cover the following finite shadows of the proofs: lexicographic positivity and translation invariance; standard quotient-remainder identities, including negative constant terms; preservation of signs under rational shears; the corresponding inverse maps; permutation transport of a finite index order; and integer-multiple witnesses for equal-leading-index Archimedean comparisons.

It also exhausts all placements of a real scale among the coordinates of a three-dimensional rational vector space, together with all coordinate orders and a range of rational offsets. This checks the *sign-rule implementation* of the normal form. It is not an exhaustive classification of orders on $\mathbb{Q}^3$: ordered rational vector spaces have many orders not given by coordinate flags.

The fixed-carrier transport is tested through a formal real-coordinate encoding rather than floating-point logarithms. The logarithm in [\(174.2\)](#) is only a change of chart; exact tests need only the positive real coordinate before applying that chart. Analyticity and joint continuity in L are proved in [theorem 174.3](#), not established by numerical sampling.

The tests do *not* decide any infinite isomorphism problem, construct a nonmeasurable Hamel basis, prove Borel completeness, quantify over all real numbers, or replace a theorem-prover kernel. Their result is recorded in `checks/results.json`: all 181,071 checks passed, both in ordinary execution and with Python optimization enabled. The document includes no claim that these new theorems have already been formalized in Lean or Rocq.

[write] In this report the program is `code/19-one-real-dimension-checks-verify_finite.py` and its record `data/19-one-real-dimension-checks-results.json`; run without an argument it writes `results.json` beside itself, so the README gives a rerun on a copy. The intake reran it on a copy: 181,071 checks in 23 families pass (seed 20261004), also under -0 , and the record equals the shipped one after removing the carriage returns that Windows text mode adds.

179 Formalization plan tied to the repository

There are three independently useful layers. The first is a generic $\mathbb{Z}$ -group interface: torsion-freeness, the least positive unit, and the unique standard-residue property. The finite-witness lemma in [appendix AC](#) should be proved for predicates invariant on cosets of mG . This is the precise generalization of the ordinary-integer Cooper interface described in the repository [\[126\]](#). Single-shift periodicity is not the appropriate replacement hypothesis in a nonstandard group.

The second layer is the order-classification argument. A minimal sequence of formal modules would establish the Borel-order lemma on $\mathbb{R}$, convex hull K , its relative standard-part homomorphism, extension of the countable real homomorphism, and the sign criterion. The isomorphism theorem then uses only preservation of connected components, convex hulls, and the infinitesimal kernel. Splitting a countable rational vector space yields the sharper pair invariant.

The third layer is the coding and surreal interface. Finite-support rational vectors can be represented without any arbitrary infinite series. Leading-support comparisons prove [lemma 174.1](#) and the forward embedding ([177.1](#)). The Borel and compact-open parameterization should be kept separate from the algebraic embedding: it involves topological function spaces, not merely order homomorphisms. Friedman–Stanley can first be an explicitly stated classical dependency; no new arithmetic axiom or opaque oracle should be introduced to hide that dependency.

A meaningful milestone is a kernel-checked proof of [theorem 170.2](#) under explicit topological hypotheses, followed by the finite-support isomorphism reduction. Compiling the exact-arithmetic Python tests is not a substitute for this milestone. No speculative .lean file is supplied with unverified syntax or concealed sorry declarations.

[write] None of these modules exists. The Presburger project’s Lean and Rocq files are about the ordinary integers ([section 1.4](#)), and docs/FORMALIZATION.md maps no label of this part.

180 Further research questions and concrete next targets

These questions arise from the proved results. Except for the explicitly cited earlier problem, they are proposed research directions, not claims that no answer exists anywhere in the literature.

Question 180.1 (Finite real dimension and interleaved flags). Classify all Borel Presburger orders on $\mathbb{R}^d \times E$ for fixed finite $d > 1$ and countable pointed E . Can the one-dimensional construction be iterated along the ordered flag of the real component to produce a complete finite list of countable extension data and removable cut maps?

The main issue is not the real flag alone: countable scales can be inserted between successive real scales. A successful statement must retain the extension data at each level rather than assume a global lexicographic product. The $d = 1$ theorem provides the base case.

[write] This is the part of Part IV’s [question 57.3](#) left open by this part. For $d \geq 2$ the cones are not locally compact (Part IV’s [theorem 51.2\(3\)](#)), and the interleaved example after [theorem 50.1](#) is a test case.

Question 180.2 (Which positive models have good group completions?). Give necessary and sufficient conditions for a locally compact Polish cancellative Presburger monoid to embed topologically as the nonnegative cone of a locally compact Polish group.

The local interior-translate argument in [lemma 173.2](#) suggests a usable sufficient condition: enough translates must enter a common interior region where differences acquire compatible local charts. The missing step is existence of a Hausdorff group topology, not the algebraic Grothendieck construction.

[write] (Batch 92.) This is Part IV’s [question 57.2](#), asked again by sources 09, 10, 12 and 15, and it was answered by Part XII before this part was written: a locally compact Polish cancellative Presburger monoid with continuous addition embeds topologically as the nonnegative cone of a locally compact Polish group exactly when its partial subtraction is continuous, and the group topology is then unique ([theorems 153.1](#) and [154.1](#) and [corollary 154.2](#); note after [question 57.2](#)). Source 19’s suggested condition through interior translates is a sufficient condition of that kind for its own cones, where partial subtraction is the restriction of the group subtraction. The question is not reopened here.

Question 180.3 (Exact complexity beyond normal-form codes). For a natural standard Borel coding of *all* Polish continuous-addition Presburger presentations on a prescribed carrier, what is the complexity of topological isomorphism?

Our normal-form class is exactly classifiable by countable structures and Borel complete there. The same upper bound need not hold for all Polish presentations. A precise coding of operations and admissible orders should precede any complexity claim.

[write] (Batch 92.) This repeats the open part of Part IX's [question 126.6](#), whose lower bound this part supplies ([remark 182.5](#) [merge]). For the codes with continuous partial subtraction, [proposition 182.3](#) [merge] shows that every such structure is isomorphic to a normal model, so its isomorphism type is determined by countable data; what is missing for an upper bound is a Borel map from the codes to those data, that is, a Borel choice of the group completion, the shear and the normal form, and a computation of the complexity of the set of codes with continuous partial subtraction.

Question 180.4 (Restricted countable scales and tractable classification). Determine the isomorphism complexity when the groups above and below $\mathbb{R}$ have finite rational dimension, finitely many Archimedean classes, or prescribed Archimedean components.

Finite dimension does not imply that each component is a rational coordinate: finite-rank ordered subgroups of $\mathbb{R}$ can contain irrational ratios. The pair invariant from [theorem 172.1](#) cleanly isolates the remaining countable classification problem.

Question 180.5 (Effective interpretation versus isomorphism reduction). How much of the countable order L can be uniformly interpreted in the analytic monoid presentation M_L^{lo} , and at what infinitary complexity?

The proof reconstructs a largest proper convex subgroup and its Archimedean spine, but this does not automatically give a first-order or uniform effective interpretation. The distinctions demonstrated by Knight, Soskova, and Vatev for graph-to-order coding [[124](#)] are directly relevant. An isomorphism reduction should not be mistaken for a stronger interpretability theorem.

Question 180.6 (Rigidity outside the locally compact one-scale class). Can a rigid uncountable Presburger model admit a Polish presentation with continuous addition? What additional hypotheses force a nontrivial continuous dilation action?

[Corollary 171.3](#) excludes rigidity for the whole class treated here. Jeřábek's algebraically rigid models [[4](#)] provide a separate source of test objects. A result would have to confront the regularity of their orders, rather than rely only on cardinality.

[write] (Batch 92.) The first sentence is Part I's [question 14.12](#) and source 13's [question 103.5](#). With [proposition 182.3](#) [merge], [corollary 171.3](#) excludes rigid examples among the uncountable locally compact Polish models with continuous partial subtraction; a rigid example would have to be non-locally-compact or have discontinuous partial subtraction.

Question 180.7 (Regularity when real components are not open). For Polish ordered groups whose real identity component is not open, which hereditary Baire or measurability hypotheses suffice for a normal form? When do such hypotheses automatically bound the Borel rank of the full order?

[Corollary 170.6](#) settles the Baire-property weakening completely in our product class: the order is automatically Δ_2^0 . The extension beyond this class is not formal. An ambient

Baire-property or measurable set can restrict badly to a closed meager or null subgroup. A meaningful extension must state its restriction hypotheses, rather than silently inherit them. [write] Compare Part IV's [question 57.4](#), whose rigidity half Part VI answers with the Baire property of the cone alone (note there).

Question 180.8 (Non-Borel symmetries and named topology). Classify which expansions by selected continuous functions or predicates eliminate the Hamel shears of [theorem 176.1](#) without collapsing the countable-scale isomorphism complexity.

Naming a predicate for the real axis kills the shear h in the simple family, but retaining only arithmetic-definable predicates cannot do so: every abstract arithmetic automorphism already preserves those. A small expansion with a sharp automorphism theorem would distinguish arithmetic information from externally supplied topology.

Question 180.9 (A formal theorem package). Formalize the normal form, the countable-extension invariant, and the finite-support coding in the ProveIt ecosystem, with a machine-readable ledger separating classical dependencies from newly checked statements.

The first useful certificate should state exactly which hypotheses about the real component and the topology are used. The Borel-complete conclusion can then be built on a separately identified formalization or assumption of Friedman–Stanley.

181 Conclusion

The one-real-dimensional classification problem has a precise answer. A Borel Presburger order on $\mathbb{R} \times E$ consists, up to continuous shear, of countable scales above the real line and a countable ordered infinitesimal group below it. The apparent continuum of real cut offsets is coordinate freedom, not additional invariant data. In the split case, the answer is exactly a pair of countable divisible ordered groups.

Topological simplicity is not classification simplicity. All positive cones have the same analytic one-dimensional carrier, every addition is analytic, and the coding family varies continuously. Nevertheless its isomorphism problem contains arbitrary countable linear orders. Within each nontrivial coding model, there is a complementary regularity boundary: only continuum many automorphisms are Borel, while 2^c abstract automorphisms exist.

These are the results proposed for independent checking. Their proofs do not depend on the correctness of the earlier companion report's other theorems. They advance its stated order-classification question in the $d = 1$ case, without presenting the original positive answer to Glazer's Question 2 as a second new discovery, and without claiming a result on his separate reverse-mathematical question.

182 Where Part XIII meets Parts IV, IX and XII [merge]

This section was written for this report. It combines source 19's theorems with Part IV's shear, Part IX's classification of inherited locally compact cones and Part XII's completion criterion. Its proofs use only results proved in those parts and the elementary algebra of convex subgroups; they have not been independently reviewed. Throughout, $G(E) = \mathbb{R} \times E$ carries the product of the usual and the discrete topology, as in [definition 168.1](#).

Proposition 182.1 (The unit can be moved into E). *Let E be a countable torsion-free abelian group, and let $<$ be a Borel translation-invariant linear order on $G(E)$ in which $G(E)$ is a $\mathbb{Z}$ -group with least positive element $\mathbf{1} = (v, e)$. Then e has infinite order, (E, e) is a countable Presburger datum in the sense of [definition 168.1](#), and for every homomorphism $f : E \rightarrow \mathbb{R}$ with $f(e) = v$ the shear $S(r, x) = (r - f(x), x)$ is a topological group automorphism of $G(E)$ with $S(\mathbf{1}) = (0, e)$. The order transported by S is a Borel Presburger order on $G(E)$ with unit $(0, e)$, to which [theorems 170.2](#) and [171.1](#) apply.*

Proof. Suppose $ne = 0$ for an ordinary $n \geq 1$. Then $n\mathbf{1} = (nv, 0)$ is positive, and $w = (nv/(n+1), 0)$ satisfies $(n+1)w = n\mathbf{1} > 0$, so $w > 0$ and therefore $w \geq \mathbf{1}$. Hence $n\mathbf{1} = (n+1)w \geq (n+1)\mathbf{1} > n\mathbf{1}$, a contradiction. So e has infinite order, the assignment $ke \mapsto kv$ is a homomorphism on $\mathbb{Z}e$, and it extends to a homomorphism $f : E \rightarrow \mathbb{R}$, for instance by [lemma 170.3](#) or by divisibility of $\mathbb{R}$. Every homomorphism on the discrete group E is continuous, so S is a topological group automorphism, and $S(\mathbf{1}) = (v - f(e), e) = (0, e)$. The transported order is Borel because S is a homeomorphism, and $(0, e)$ is its least positive element.

In the transported order $G(E)$ is still a $\mathbb{Z}$ -group, so $G(E)/\mathbb{Z}(0, e) \cong \mathbb{R} \oplus (E/\mathbb{Z}e)$ is divisible, and hence so is $E/\mathbb{Z}e$. For torsion-freeness, let $nx = ke$ with $x \in E$ and ordinary $n \geq 1$, and write $k = nq + s$ with $0 \leq s < n$. Then $y = (0, x) - q(0, e)$ satisfies $ny = s(0, e)$. If $s > 0$, then $y > 0$, so $y \geq (0, e)$ and $ny \geq n(0, e) > s(0, e)$, a contradiction. Hence $s = 0$, $ny = 0$, and $y = 0$ by torsion-freeness of the ordered group; that is, $x = qe \in \mathbb{Z}e$. $\square$

This is the shear in the proof of Part IV's [theorem 50.1](#), made explicit for $d = 1$ without assuming local compactness of the group. Together with [theorems 170.2](#) and [171.1](#) and [corollary 171.2](#) it answers Part IX's [question 126.5](#): for a fixed countable torsion-free E , the Borel Presburger orders on $\mathbb{R} \times E$ are classified, up to continuous unital additive order isomorphism, by the countable ordered extensions $(E, e, A, <_A, <_B)$, and their continuous Presburger automorphism groups are the extensions displayed in [corollary 171.2](#). By the next proposition, abstract arithmetic isomorphism gives the same classification.

Proposition 182.2 (Abstract isomorphism is topological). *Let G and G' be the normal models of normal data $(E, u, A, <_A, <_B)$ and $(E', u', A', <_{A'}, <_{B'})$, with C, K, I and C', K', I' as in [theorem 170.2](#). Every unital ordered-group isomorphism $F : G \rightarrow G'$, with no regularity assumed, maps I onto I' and K onto K' , and there is a group isomorphism $\varphi : E \rightarrow E'$ with $\varphi(u) = u'$ and $\varphi(A) = A'$ that induces order isomorphisms $(A, <_A) \rightarrow (A', <_{A'})$ and $(E/A, <_B) \rightarrow (E'/A', <_{B'})$. Consequently the following are equivalent:*

- (i) $G \cong G'$ as abstract unital ordered groups;
- (ii) $G_{\geq 0} \cong G'_{\geq 0}$ as abstract unital additive monoids;
- (iii) $G \cong G'$ by a topological unital ordered-group isomorphism;
- (iv) $G_{\geq 0} \cong G'_{\geq 0}$ by a topological unital monoid isomorphism;
- (v) the ordered extensions $(E, u, A, <_A, <_B)$ and $(E', u', A', <_{A'}, <_{B'})$ are isomorphic.

The same holds for arbitrary Borel Presburger orders on $G(E)$ and $G(E')$, after normalization.

Proof. Step 1: I and K are order-theoretic. The convex subgroups of an ordered group are linearly ordered by inclusion. The subgroup $I = \{0\} \times A$ is countable and convex. Let J be a convex subgroup with $J \not\subseteq I$, so $J \supsetneq I$. If $J \subseteq K$, its image under the order-preserving surjection $st : K \rightarrow \mathbb{R}$ with kernel I (proof of [theorem 170.2](#)) is a nonzero convex subgroup

of $\mathbb{R}$, hence all of $\mathbb{R}$, and so $J = K$; otherwise $J \supsetneq K$. Either way J is uncountable. Hence I is the largest countable convex subgroup of G , and K is the least convex subgroup strictly containing I . An order isomorphism maps convex subgroups to convex subgroups and preserves cardinality, so $F(I) = I'$ and $F(K) = K'$.

Step 2: straightening $F(C)$. We have $C \cap I = 0$ and $C + I = K$, so $X = F(C)$ satisfies $X \cap I' = 0$ and $X + I' = K'$. Let $\pi : K' \rightarrow C'$ be the projection along I' , $\pi(r, a) = (r, 0)$. Its restriction to X is a group isomorphism onto C' , so $X = \{c + h(c) : c \in C'\}$ for the homomorphism $h = (\text{id} - \pi) \circ (\pi|_X)^{-1} : C' \rightarrow I'$. Write $h(r, 0) = (0, \eta(r))$, with $\eta : \mathbb{R} \rightarrow A'$ additive, and put $\vartheta(r, e) = (r, e + \eta(r))$ on G' . Then ϑ is a group automorphism with inverse $(r, e) \mapsto (r, e - \eta(r))$, it fixes I' pointwise, $\vartheta(g) - g \in I'$ for every g , and $\vartheta(C') = X$. The map $F_1 = \vartheta^{-1} \circ F$ is a group isomorphism with $F_1(C) = C'$, $F_1(K) = K'$, $F_1 = F$ on I , and $F_1(g) - F(g) \in I'$ for every g . Neither ϑ nor F_1 need be continuous or order preserving.

Step 3: the extension data. Since $F_1(C) = C'$, F_1 induces a group isomorphism $\varphi : E \cong G/C \rightarrow G'/C' \cong E'$, determined by $F_1(0, x) \in \mathbb{R} \times \{\varphi(x)\}$. As $\mathbf{1} \in I$, $F_1(\mathbf{1}) = F(\mathbf{1}) = \mathbf{1}'$, so $\varphi(u) = u'$. As $F_1(K) = K'$ and $K/C \cong A$, $K'/C' \cong A'$, we get $\varphi(A) = A'$. For $a \in A$, $F_1(0, a) = F(0, a) \in I'$, so $F_1(0, a) = (0, \varphi(a))$; since the order of $I = \{0\} \times A$ is $<_A$ and F is an order isomorphism, $\varphi|_A$ is an order isomorphism onto $(A', <_{A'})$. Finally F_1 and F agree modulo $I' \subseteq K'$, so they induce the same map $G/K \rightarrow G'/K'$, which is an order isomorphism because F is one and K, K' are convex; under $G/K \cong (E/A, <_B)$ and $G'/K' \cong (E'/A', <_{B'})$ this map is the map $\bar{\varphi}$ induced by φ .

Step 4: the equivalences. Steps 1–3 give (i) $\Rightarrow$ (v). By the converse part of [theorem 171.1](#), (v) gives the topological unital ordered isomorphism $(r, e) \mapsto (r, \varphi(e))$, which is (iii); (iii) gives (i), (iv) and (ii) by restriction. A unital monoid isomorphism of the cones extends to a unital isomorphism of the groups of differences that preserves the cones, hence the orders, so (ii) $\Rightarrow$ (i). For unnormalized Borel orders compose with the normalizing shears, which are topological unital order isomorphisms. $\square$

The automorphisms of [theorem 176.1](#) with $h \neq 0$ show that the isomorphism F itself need not be continuous; only the existence of a continuous one is asserted. [Theorem 174.2](#) is the case of the coding family.

Proposition 182.3 (Locally compact models with continuous partial subtraction). *Let M be an uncountable locally compact Polish space carrying a model $(M, 0, 1, +)$ of Presburger arithmetic whose addition is jointly continuous and whose partial subtraction $\delta(y, x) = y - x$, defined for $x \leq y$, is continuous. Then there are normal data $(E, u, A, <_A, <_B)$ such that M is topologically and arithmetically isomorphic to the nonnegative cone of the normal model on $G(E)$. Conversely, every such cone is an uncountable locally compact Polish model of Presburger arithmetic with continuous addition and partial subtraction, homeomorphic to X_{lc} . Two models M, M' of this class are isomorphic as unital additive monoids if and only if they are topologically isomorphic, if and only if their normal data are isomorphic ordered extensions.*

Proof. The natural order of the monoid, $x \leq y$ if and only if $x + z = y$ for some z , is the order of the model, which is linear; the monoid is cancellative and Hausdorff. By Part XII's [theorems 153.1](#) and [154.1](#), the group of differences $G = M - M$ carries a Polish group topology inducing the topology of M . As in the proof of [proposition 155.7](#), M is completely metrizable, hence a G_δ subset of G and Borel, and $G = M \cup (-M)$ is an ordered abelian Polish group with Borel order and nonnegative cone M . Since M is a model of Presburger arithmetic, G is a $\mathbb{Z}$ -group with least positive element 1 (proof of [corollary 155.3](#)). The cone M is locally compact and uncountable, so Part IX's [theorem 121.1](#) gives a topological group isomorphism $G \cong \mathbb{R} \times E$ with E countable, discrete and torsion-free. Transport the order; by

[proposition 182.1](#) and [theorem 170.2](#), a further topological unital order isomorphism brings it to a normal model, and its restriction to M is the required isomorphism of models.

Conversely, [proposition 168.3](#) and [theorem 173.1](#) give the Presburger axioms, local compactness, Polishness and continuous addition, and partial subtraction is the restriction of the continuous subtraction of the group. For the last assertion, a topological monoid isomorphism of two cones extends to a topological group isomorphism ([lemma 173.2](#)), so [theorem 171.1](#) applies, and [proposition 182.2](#) removes the topology. $\square$

The hypothesis on partial subtraction cannot be dropped: Part III's cone of $\mathbb{Z} + t\mathbb{R}[t]$ with the topology of [theorem 38.1](#) is an uncountable locally compact Polish model with continuous addition, and its partial subtraction is discontinuous ([example 155.5](#)); it is not homeomorphic to X_{lc} , since it has components $\mathbb{R}^{d-1} \times (0, \infty)$ with $d \geq 2$. With [corollary 171.3](#) the proposition shows that no model of this class is rigid, which answers Part I's [question 14.12](#) for the class.

Corollary 182.4 (Prescribed boundary). *Fix a countable model N of Presburger arithmetic. The uncountable locally compact inherited Polish-group presentations P of Part IX whose manifold boundary $\partial_{\text{man}}P$ is isomorphic to N are, up to topological and arithmetic isomorphism, exactly the cones of the normal models whose data satisfy $(A, <_A, u) \cong (N - N, <, 1)$ as pointed ordered groups. Two of them are arithmetically isomorphic if and only if they are topologically arithmetically isomorphic, if and only if their ordered extensions $(E, u, A, <_A, <_B)$ are isomorphic. In particular the two classifications asked for in Part IX's [question 126.4](#) coincide, and both are the isomorphism classification of countable ordered extensions of a countable divisible torsion-free ordered group $B = E/A$ by the fixed pointed $\mathbb{Z}$ -group $N - N$.*

Proof. Let $P = G_{\geq 0}$, with G a Borel ordered abelian Polish group and P , in the inherited topology, an uncountable locally compact Presburger model. Then $G = P - P$, and the inclusion $P \subseteq G$ is a topological embedding, so the partial subtraction of P is continuous and [proposition 182.3](#) applies; by the uniqueness clause of [theorem 153.1](#), the group topology used in its proof is the given topology of G . So P is isomorphic, topologically and arithmetically, to the cone of a normal model, by an isomorphism that extends to a topological isomorphism of the ordered groups. Such an isomorphism maps the identity component onto the identity component and hence Part IX's infinitesimal subgroup ([theorem 122.1](#), defined there through G^0) onto source 19's $I = \{0\} \times A$; so $\partial_{\text{man}}P \cong I_{\geq 0} \cong (A, <_A, u)_{\geq 0}$ by that theorem, and $(A, <_A, u)_{\geq 0} \cong N$ exactly when $(A, <_A, u) \cong (N - N, <, 1)$, since an isomorphism of cones extends to the groups of differences. Conversely the cone of every normal model is such a presentation ([theorem 173.1](#)), with boundary $(A, <_A, u)_{\geq 0}$ by the same theorem. The equivalences are [proposition 182.3](#). $\square$

Existence for every N is Part IX's [theorem 122.2](#) (normal data with $E = A = N - N$ and $B = 0$); its continuum of models with standard boundary ([theorem 123.1](#)) consists of normal models with $E = V_{L_S} \oplus \mathbb{Z}u$, $A = \mathbb{Z}u$ and $B = E/A \cong V_{L_S}$, the coded scales placed above the real line. For a fixed N , the complexity of the remaining classification of countable ordered extensions is not determined here.

Remark 182.5 (Lower bounds for Part IX's [question 126.6](#)). Give $C = C(X_{lc} \times X_{lc}, X_{lc}) \times X_{lc} \times X_{lc}$ the product of the compact-open topology and the topology of X_{lc} ; it is Polish, by the standard fact that $C(Y, Z)$ with the compact-open topology is Polish when Y is locally compact and second countable and Z is Polish. Let $\mathcal{P} \subseteq C$ be the set of codes $(m, 0, 1)$ for which $(X_{lc}, m, 0, 1)$ is a model of Presburger arithmetic. By [theorem 174.3](#), $L \mapsto (m_L, 0_L, 1_L)$ is a continuous map from LO into $\mathcal{P}$ that reduces isomorphism of countable linear orders to arithmetic isomorphism and to topological arithmetic isomorphism of the coded structures. By the

Friedman–Stanley theorem [123], every isomorphism relation of a Borel class of countable structures therefore Borel reduces to each of these two relations, and neither relation is Borel on any Borel subset of C containing the image of LO. The complexity of $\mathcal{P}$ itself and an upper bound for the two relations are not determined here (question 180.3).

Part XIV

The exponent-group threshold for Polish omnific arithmetic

[write] This part is source 20, manuscript 02 of batch 92, *The Exponent-Group Threshold for Polish Omnific Arithmetic: Separated grids, an induction-free root obstruction, and finite-support surreal semirings* (23 pages, A4, dated 4 October 2026; “prepared for Vladimir Reshetnikov”, “developed with ChatGPT”). It arrived in commit 9dc8db274, as archive Glazer_Prove It_Exponent_Threshold, and was placed in e24ce2ce0; it is pinned to 9fe62865d, 52 commits before the placement. Its file prefix in this report is 20-exponent-threshold-. All unmarked text in this part is source 20’s; its twelve sections are Sections 183–194 (its Section k is Section $182 + k$, with statement numbers otherwise unchanged), its twelve “Research questions” are Questions 193.1–193.12, and its two appendices are appendices AE and AF. Its title page and abstract are not printed separately (its status statement is in section 1.5); its conventions paragraph follows.

What this part claims, and its status

For a subgroup $\Gamma \leq \mathbb{R}$ let $A_\Gamma = \mathbb{Z} \oplus \bigoplus_{\gamma \in \Gamma_{>0}} \mathbb{R}T^\gamma$, with T positive *infinite* and the leading-term order, and $P_\Gamma = (A_\Gamma)_{\geq 0}$. Then P_Γ has a Polish topology with continuous addition and Borel multiplication if and only if $\Gamma = 0$ or $\Gamma = \alpha\mathbb{Z}$, and then both operations can be continuous in a locally compact topology (theorem 183.1). Independently, in any uncountable Polish cone of a discretely ordered commutative ring, Euclidean division with order-bounded remainder (DIV) and integer k th roots for one fixed $k \geq 2$ (Root_k) already forbid continuous addition with Borel multiplication (theorem 183.2); $\text{DIV} + \text{Root}_2$ is strictly weaker than open induction (proposition 188.4). Every P_Γ satisfies DIV, and Root_k exactly when $\Gamma = k\Gamma$ (corollary 184.5 and theorem 185.1). Both negative theorems go through an abstract separated-grid obstruction adapted from Glazer’s fusion and zero-one method (theorem 186.4).

Unrefereed, AI-assisted, not formalized; historical priority is not established. The finite checks test algebraic identities only.

Questions answered. Part II’s question 28.7 asks which concrete consequences of open induction suffice to rule out continuous addition in the finite principal-part model or related integer parts, suggesting “root-approximation or division properties”. Theorem 183.2 answers it: DIV and one Root_k suffice, with multiplication Borel (only countably many scalar maps are used), in every uncountable Polish cone; Part II’s ring $\mathcal{A}_{\text{fin}}$ is source 20’s $A_\mathbb{Q}$ (with $X = T$), which satisfies DIV and every Root_k ; and for the rings A_Γ density of Γ alone suffices (theorem 183.1). Theorem 183.1 also extends the boundary between the integer-exponent subcone and the whole cone of $\mathcal{A}_{\text{fin}}$, found in Part III’s section 42, from $\mathbb{Z}$ versus $\mathbb{Q}$ to every subgroup of $\mathbb{R}$. Source 20’s question on multiplication-only presentations (question 193.1)

generalizes Part II's [question 28.1](#) and Part XI's [question 149.5](#) without answering them. Source 20 answers neither of Glazer's questions and says so.

What was already here, and what is new. At its pin the report had Parts I–IX; source 20 read this report's README there, not the article. Its [lemma 184.3](#) is Part V's [proposition 65.2](#) mirrored ($T = t^{-1}$, real coefficients, any $\Gamma \leq \mathbb{R}$; in-batch it is source 21's [proposition 196.2](#)); its [theorem 184.4](#) generalizes Part II's [lemma 23.1](#) and Part V's [theorem 70.1](#), and is the restriction of the collection's `odg:thm:floor`, proved in Lean; [proposition 185.2](#) generalizes Part III's [theorem 38.2](#); [proposition 187.2](#) is Part III's [theorem 38.1](#) reparametrized, as source 20 says; for $\Gamma = \mathbb{Q}$, [proposition 187.3](#) is Part II's [theorem 25.2](#) (Glazer's Corollary 1 with [theorem 23.2](#)); [proposition 190.1](#) and the example after it generalize Part II's [theorem 24.3](#) and repeat its example; [lemma 186.2](#) and [theorem 186.4](#) adapt the method that Part VIII ([lemmas 107.1](#) and [109.1](#)), Part IX's [theorem 119.2](#) and Part XI also adapt from Glazer; and [corollary 186.6](#) proves in ZFC, under its own hypotheses, a cofinal-translation conclusion of the kind of Part VIII's claimed [theorem 110.2](#). Each is flagged in a note. New to the report are the threshold for every $\Gamma \leq \mathbb{R}$, the one-root obstruction, the root spectrum, the separated-grid interface, the root-tower transition and the monus locus of [proposition 190.3](#).

Labels and notation. Every label of source 20 takes the prefix `pma:egt:`; labels were added to its first and last sections, its twelve questions, and `pma:part:egt` to the part. The sixth block of the table in [section 1.3](#) lists the symbols. In short: source 20's macro `\PP` for the positive part $pp_{>0}$ is renamed in the source only, because the report's $\mathcal{A}_{\text{fin}, \geq 0}$ is Part II's cone; T is positive *infinite*, as Part II's X and Part III's t , and not Part V's infinitesimal t ; A_Γ with $\Gamma = \mathbb{Q}$ is Part II's $\mathcal{A}_{\text{fin}}$; P_Γ is not Part III's cone P or Part II's Puiseux field $\mathcal{P}$; L_Γ (italic, T infinite, real coefficients) is the mirror image of Part V's $\mathcal{L}_k$ and of source 21's $\mathcal{L}^\Gamma$; the external cut I , the grid bound C and $M = R_{\geq 0}$ are local. No normalization changed.

Files. Source 20's checks, recorded run, build script, proof audit and source record are shipped as `code/20-exponent-threshold-verify.py`, `data/20-exponent-threshold-verification_results.json`, `code/20-exponent-threshold-build.sh`, `20-exponent-threshold-PROOF_AUDIT.md` and `20-exponent-threshold-SOURCES.md`. The Hilbert's-tenth programme routed the archive from its README only ([appendix E](#)). Nothing in this part is formalized.

Conventions. All proofs use ordinary external ZFC. The word “induction-free” describes the axioms imposed on an arithmetic structure, not the metatheory used to prove the theorems. All topological spaces in the main statements are sets.

183 The research intersection and the questions resolved

183.1 Why this topic fits both sides

Glazer studies continuity obstructions in Borel models of arithmetic. His addition obstruction applies under open induction; his stronger arithmetic theorem treats the two operations separately [1]. The historical setting is the study of topological presentations of arithmetic by Enayat, Hamkins, and Wcisło [2]. These papers make the distinction between a Borel operation and a continuous one mathematically substantive.

ProveIt contains a complementary collection of ordered arithmetic models inside the omnific integers. In particular, its report *Polish Models of Omnific Arithmetic* records a continuous semiring topology on the nonnegative cone of $\mathbb{Z} + T\mathbb{R}[T]$, and a contrasting obstruction for a rational-exponent principal-part ring [130]. The first example is already in the repository: the present paper does not claim it as a new construction. We ask for the exact boundary between these examples, rather than repeating either one.

Our source snapshot is the repository revision

9fe62865dc30d71256183466b744b98ebf7e64d6.

The catalogue and selected Lean sources were inspected through the GitHub connector. We did not audit the entire repository or read the complete 349-page merged topology report. The precise scope of inspection is recorded in Appendix AE and in the companion source ledger.

[write] (Batch 92.) The report is this one, which at the pin had Parts I–IX and 349 pages. Its continuous semiring topology on the cone of $\mathbb{Z} + T\mathbb{R}[T]$ is Part III’s [theorem 38.1](#) (source 06), and the obstruction for the rational-exponent principal-part ring is Part II’s [theorem 25.2](#) (source 03). The source ledger is shipped as 20-exponent-threshold-SOURCES.md.

183.2 The classification problem

Fix an additive subgroup $\Gamma \leq \mathbb{R}$. Introduce a positive infinite formal symbol T . A finite sum

$$z + \sum_{\gamma \in S} c_\gamma T^\gamma, \quad z \in \mathbb{Z}, \quad S \subseteq \Gamma_{>0} \text{ finite}, \quad c_\gamma \in \mathbb{R}, \quad (183.1)$$

is positive when its highest nonzero coefficient is positive. This makes T behave like an infinite magnitude; for example, $T^{\gamma'} > nT^\gamma$ whenever $\gamma' > \gamma$ and n is an ordinary positive integer. Let A_Γ be the ring of these sums and let $P_\Gamma = (A_\Gamma)_{\geq 0}$.

The question is not whether the familiar coefficient topology works. We allow *any* Polish topology on this underlying set and ask whether addition can be continuous while multiplication is Borel. This freedom makes the negative direction much stronger than a failure of a particular coordinate representation.

Theorem 183.1 (Exponent-group threshold). *For every subgroup $\Gamma \leq \mathbb{R}$, the following conditions are equivalent.*

- (i) P_Γ has a Polish topology in which addition is continuous and multiplication is Borel.
- (ii) P_Γ has a locally compact, σ -compact Polish topology in which addition and multiplication are both continuous.
- (iii) $\Gamma = \{0\}$ or $\Gamma = \alpha\mathbb{Z}$ for some real $\alpha > 0$.

For every nonzero Γ , the cone P_Γ is uncountable.

Theorem 183.1 is proved in Section 187. In the cyclic case it reuses, with a self-contained proof, the repository’s polynomial-cone construction. Its main extension is the presentation-independent negative direction for *every dense real exponent subgroup*.

183.3 A strictly weaker arithmetic obstruction

For a discretely ordered commutative ring R , put $M = R_{\geq 0}$. We use two arithmetic properties:

$$\text{DIV} : \quad \forall a \geq 0 \forall b > 0 \exists q, r \geq 0 (a = bq + r \wedge r < b), \quad (183.2)$$

$$\text{Root}_k : \quad \forall a \geq 0 \exists r \geq 0 (r^k \leq a < (r+1)^k), \quad k \geq 2 \text{ fixed.} \quad (183.3)$$

Here and below “discretely ordered” means that 1 is the least positive ring element. In particular, $x < y$ is equivalent to $x + 1 \leq y$.

Theorem 183.2 (One-root obstruction). *Suppose $M = R_{\geq 0}$ satisfies DIV and Root_k for one fixed ordinary integer $k \geq 2$. There is no uncountable Polish topology on M in which addition is continuous and multiplication is Borel.*

Section 188 proves this theorem without invoking an induction scheme in M . The separation is genuine: $P_{\mathbb{Z}[1/2]}$ satisfies DIV + Root_2 but fails open induction. Thus this is not merely a rephrasing of an open-induction hypothesis. The dense-exponent theorem goes further in a different direction: it applies to $\mathbb{Z} + \sqrt{2}\mathbb{Z}$, which fails Root_k for every $k \geq 2$.

[write] (Batch 92.) [Theorem 183.2](#) answers Part II’s [question 28.7](#), which asks for such consequences of open induction (note there). For $\Gamma = \mathbb{Z}$ the cone P_Γ is Part III’s cone of $\mathbb{Z} + t\mathbb{R}[t]$ ($T = t$), and for $\Gamma = \mathbb{Q}$ the ring A_Γ is Part II’s finite principal-part ring $\mathcal{A}_{\text{fin}}$ ($T = X$); [theorem 183.1](#) at $\Gamma = \mathbb{Q}$ is Part II’s [theorem 25.2](#), and at $\Gamma = \mathbb{Z}$ it is Part III’s [theorem 38.1](#).

183.4 What is new here, and what is not claimed

The principal proposed contributions are the exact subgroup threshold, the explicit one-root weakening, and the connection between them through separated arithmetic grids. The exact root spectrum and the root-tower transition clarify their algebraic meaning.

The perfect-set tools, the probabilistic fusion mechanism, generalized-series arithmetic, and the polynomial positive example have antecedents. They are used with attribution, not relabelled as discoveries. These results solve classification and weakening problems formulated precisely in this paper; we do not claim to have settled a separately identified published conjecture. In particular, this paper neither proves Glazer’s reverse-mathematical Question 1 nor reproduces the repository’s affirmative work on his Presburger Question 2.

The mathematical arguments below are intended as proofs, not heuristic claims. Their publication priority and independent correctness review remain separate matters. The finite programs test examples and algebraic estimates; they do not certify the infinite descriptive-set-theoretic argument.

184 Finite normal forms and their arithmetic

184.1 The ordered ring

The direct sum in A_Γ is an additive direct sum: only finitely many positive exponents occur in each element. Multiplication is convolution, with $T^\gamma T^\delta = T^{\gamma+\delta}$. Because the sum of two positive exponents is positive, the constant coefficient of a product is the product of the constant coefficients, hence an integer.

For $0 \neq f \in A_\Gamma$, let $\deg f$ be its greatest exponent, with $\deg z = 0$ for a nonzero integer z . Let $\text{lc}(f)$ denote its coefficient there. Declare $f > 0$ if $\text{lc}(f) > 0$.

Proposition 184.1. *The ring A_Γ is a linearly ordered integral domain, 1 is its least positive element, and*

$$\deg(fg) = \deg f + \deg g, \quad \text{lc}(fg) = \text{lc}(f)\text{lc}(g)$$

for nonzero f, g . If $f, g \in P_\Gamma$ are nonzero, then

$$\deg(f + g) = \max(\deg f, \deg g).$$

Proof. Uniqueness of the largest exponent makes the sign comparison total. The highest coefficient of a product cannot vanish, proving both the domain property and multiplicative positivity. A sum of two positive elements has positive highest surviving coefficient; when the degrees agree, their positive leading coefficients add. Translation invariance follows by comparing differences. A positive nonconstant element exceeds every ordinary integer, since subtracting an integer does not change its highest term. A positive constant is a positive integer. Consequently no element lies strictly between 0 and 1. $\square$

Proposition 184.2 (Standard division and the additive reduct). *For every ordinary integer $n \geq 1$ and every $f \in A_\Gamma$, there are unique $q \in A_\Gamma$ and $r \in \{0, \dots, n-1\}$ with $f = nq + r$. Consequently A_Γ is a $\mathbb{Z}$ -group and its nonnegative additive reduct is a model of Presburger arithmetic.*

Proof. Write $f = z + p$ with p having no constant term. Divide $z = nz_0 + r$ in $\mathbb{Z}$ and take $q = z_0 + p/n$. Uniqueness follows from discreteness. The usual characterization of $\mathbb{Z}$ -groups by discrete order and standard division then identifies their theory with that of the ordered additive integers; passing to the definable nonnegative cone gives the stated reduct. This last identification is the classical Presburger characterization, also used in the repository report [130]; the displayed division formula is all that subsequent proofs require. $\square$

Standard division is not the same as DIV: the divisor in (183.2) may itself be an infinite generalized polynomial. We next establish that stronger property.

184.2 A convenient set-sized series field

We work with a standard generalized-series construction and give the needed details to make the hypotheses explicit. Let L_Γ consist of formal sums $f = \sum_{\gamma \in \Gamma} c_\gamma T^\gamma$ such that

$$\text{for every } b \in \mathbb{R}, \quad \{\gamma \in \text{supp } f : \gamma \geq b\} \text{ is finite.} \quad (184.1)$$

Thus supports are finite above every real cutoff; they may extend to $-\infty$. This is a left-finite, or Levi–Civita-type, support condition in the present positive-growth convention. It is stronger than the reverse-well-order condition for a general Hahn series.

Lemma 184.3. *Termwise addition and convolution make L_Γ an ordered field containing A_Γ and $\mathbb{R}$. Every nonzero series has a largest exponent. If all exponents of ε are negative, then ε is infinitesimal: $|\varepsilon| < a$ for every real $a > 0$.*

Proof. Choose an exponent in any nonempty support. There are only finitely many support elements above it, so a largest one exists. This defines the leading-term order.

For a fixed exponent u , a pair of terms contributing to a product must satisfy $\gamma + \delta = u$. If a and b are the largest exponents of the factors, then $\gamma \geq u - b$ and $\delta \geq u - a$. There are only finitely many possibilities. The same argument with $\gamma + \delta \geq v$ shows that a product satisfies (184.1). Associativity and distributivity follow by rearranging finite coefficient sums.

Factor a nonzero series as $cT^a(1 + \varepsilon)$, where $c \neq 0$ and ε has strictly negative exponents. If $\varepsilon \neq 0$, let $-\eta < 0$ be its largest exponent. Then the largest exponent of ε^n is at most $-n\eta$. Therefore

$$(1 + \varepsilon)^{-1} = \sum_{n \geq 0} (-\varepsilon)^n$$

is coefficientwise well-defined and again satisfies (184.1): only finitely many powers can contribute above any cutoff, and each such power has finite support above that cutoff. The formal geometric identity proves it is an inverse. The leading coefficient of a product is the product of the leading coefficients, so the order is a field order. Finally, subtracting any positive real from a series with negative leading exponent gives a series with negative constant leading coefficient. Apply the same argument to its negative to obtain the infinitesimal bound. $\square$

No completeness of a real-valued metric is being asserted here. The convergence used in the geometric identity is coefficientwise local finiteness, not an analytic limit in a Polish field. [write] (Batch 92.) Under $T = t^{-1}$, L_Γ is the Levi-Civita field with real coefficients and exponent group Γ , in Part V's convention (t positive infinitesimal, supports finite below every bound). Lemma 184.3 is Part V's proposition 65.2 with real coefficients and arbitrary $\Gamma \leq \mathbb{R}$, and source 21's proposition 196.2 with real algebraic coefficients; source 21's proposition 196.3 supplies the valuation topology that this part does not use. Part V's field $\mathcal{L}_k$ has countable coefficients, so L_Γ is not Part V's Polish field; source 20 makes no topological claim about it.

184.3 Exact integer parts and full division

Write $f \in L_\Gamma$ uniquely as

$$f = p + c + \varepsilon, \quad p = \text{pp}_{>0}(f), \quad c \in \mathbb{R}, \quad (184.2)$$

where p is the finite positive-exponent part and ε has only negative exponents. Put

$$\lfloor f \rfloor_\Gamma = p + \begin{cases} c - 1, & c \in \mathbb{Z} \text{ and } \varepsilon < 0, \\ \lfloor c \rfloor, & \text{otherwise.} \end{cases} \quad (184.3)$$

The first branch is an integer constant because of its hypothesis.

Theorem 184.4 (Integer-part formula). *For every $f \in L_\Gamma$, formula (184.3) is the unique $q \in A_\Gamma$ such that $q \leq f < q + 1$. In particular A_Γ is an integer part of L_Γ .*

Proof. If c is not an integer, both gaps $c - \lfloor c \rfloor$ and $\lfloor c \rfloor + 1 - c$ are positive reals. The infinitesimal tail is smaller in absolute value than both. If c is an integer and the tail is nonnegative, the inequalities are immediate from $0 \leq \varepsilon < 1$. If the tail is negative, they follow from $-1 < \varepsilon < 0$ and the constant correction $c - 1$. Adding p preserves the inequalities. For uniqueness, two distinct candidates differ in absolute value by at least 1, which is incompatible with membership in the same half-open unit interval. $\square$

The correction at a negative infinitesimal is essential. For example, $T^2 - \frac{1}{2}T^{-1}$ has integer part $T^2 - 1$, not T^2 . This is the same integer-part convention as the omnific floor; the repository contains an explicit Lean implementation and proof of its bounds [131]. The theorem above additionally checks that, under (184.1), the integer part remains *finite-support*.

[write] (Batch 92.) Formula (184.3) is the collection's `odg.thm:floor`, proved in Lean for every surreal (`omnificFloor_spec`, `existsUnique_omnific_integerPart` and

`omnificFloor_of_integer_negative_tail` in `Algebra/SurrealNumbers/Surreal/Foundations/OmnificFloor.lean`), restricted along the embedding (191.1) extended to L_Γ ; Part II's lemma 23.1 is the case $\Gamma = \mathbb{Q}$ on the Puiseux field, Part V's theorem 70.1 the mirrored case with countable coefficients, and source 21's proposition 205.2 the case of algebraic coefficients. The finite-support clause is not formalized: source 20's section 191 names it as an open formalization obligation.

Corollary 184.5 (Euclidean division in every exponent group). *For every $\Gamma \leq \mathbb{R}$, the cone P_Γ satisfies DIV.*

Proof. For $a \geq 0$ and $b > 0$, use the ordered field L_Γ and set $q = \lfloor a/b \rfloor_\Gamma$. Since $a/b \geq 0$, its integer part cannot be negative: otherwise discreteness would give $q + 1 \leq 0 \leq a/b$. Let $r = a - bq$. Multiplying $q \leq a/b < q + 1$ by b gives $0 \leq r < b$. Both q and r belong to A_Γ . $\square$

Remark 184.6. This is division with an *order-bounded remainder*. It does not say that the ring is Euclidean for a well-founded Euclidean norm. The ordered intervals in question can be uncountable and externally infinite.

185 The exact integer-root spectrum

Theorem 185.1 (Root spectrum). *Let $k \geq 2$ be an ordinary integer. Then*

$$P_\Gamma \models \text{Root}_k \iff \Gamma = k\Gamma.$$

Proof. Suppose first that Root_k holds and take $\gamma > 0$ in Γ . Choose $q \in P_\Gamma$ with

$$q^k \leq T^\gamma < (q + 1)^k.$$

The element q cannot be constant because every ordinary constant is smaller than T^γ . Thus $\delta = \deg q > 0$, and $q + 1$ has the same degree. If $k\delta < \gamma$, the upper inequality fails; if $k\delta > \gamma$, the lower one fails. Hence $\gamma = k\delta$. Applying this to positive elements and their negatives proves $\Gamma = k\Gamma$.

Conversely suppose $\Gamma = k\Gamma$. Every positive $f \in L_\Gamma$ has a factorization $aT^\gamma(1 + \varepsilon)$ with $a > 0$ real and ε strictly negative in exponent. The series

$$g = a^{1/k} T^{\gamma/k} \sum_{n \geq 0} \binom{1/k}{n} \varepsilon^n \tag{185.1}$$

lies in L_Γ . Indeed $\gamma/k \in \Gamma$, and the support estimate in Lemma 184.3 justifies every coefficient and every cutoff. The formal binomial identity, applied coefficientwise, gives $g^k = f$. For completeness, its coefficients are characterized recursively by $b_0 = 1$ and $b_{n+1} = b_n(1/k - n)/(n+1)$; the resulting formal power series has k th power $1 + X$. Substituting $X = \varepsilon$ is valid by the same local-finiteness estimate. The leading coefficient of g is positive.

Now take $f \in P_\Gamma$. For $f = 0$, use $q = 0$. Otherwise apply Theorem 184.4 to its positive root g and obtain $q \leq g < q + 1$ with $q \geq 0$. The k th-power map is strictly increasing on the nonnegative part of an ordered field, so $q^k \leq f < (q + 1)^k$. $\square$

Proposition 185.2 (A direct open-induction failure). *If $\Gamma \neq k\Gamma$ for some $k \geq 2$, then P_Γ fails open induction in the ordered-semiring language.*

Proof. Choose $\gamma > 0$ with $\gamma/k \notin \Gamma$. The quantifier-free formula $\varphi(x) : x^k < T^\gamma$ holds at 0. It is closed under successor. For a constant x , this is because $(x+1)^k$ remains constant. For a nonconstant positive x of degree δ , the inequality $x^k < T^\gamma$ forces $k\delta < \gamma$: equality is impossible by the choice of γ . Adding 1 does not change degree, so $(x+1)^k < T^\gamma$ as well. Nevertheless $\varphi(T^\gamma)$ is false. This is an explicit failed instance of induction with parameter T^γ . $\square$

Example 185.3 (Square roots without open induction). For $\Gamma = \mathbb{Z}[1/2]$, multiplication by 2 is onto Γ but multiplication by 3 is not. Thus P_Γ satisfies $\text{DIV} + \text{Root}_2$ and fails Root_3 . More explicitly, $x^3 < T$ defines a proper successor-closed initial segment. For every $m \geq 1$, integer 2^m th roots exist; universal integer k th roots fail whenever k has an odd prime divisor.

Example 185.4 (Dense exponents without any universal root axiom). For $\Gamma = \mathbb{Z} + \sqrt{2}\mathbb{Z}$, the element $1/k$ does not belong to Γ for any $k \geq 2$: an equality $1/k = a + b\sqrt{2}$ with integers a, b forces $b = 0$ and then an impossibility. Consequently every Root_k fails. This group is nevertheless dense, so Theorem 183.1 excludes a continuous-addition Polish presentation with Borel multiplication.

186 Descriptive preliminaries and a separated-grid criterion

186.1 Recovering a Borel order from addition

Lemma 186.1. *Let $M = R_{\geq 0}$ be the nonnegative cone of a discretely ordered ring. If a Polish topology on M makes addition Borel, then its arithmetic order is Borel, regardless of the regularity of multiplication.*

Proof. In M , the relation $x < y$ has the existential description $\exists z \in M (y = x + z + 1)$, whereas its complement has the description $\exists z \in M (x = y + z)$. Their witness relations are Borel, hence both relations are analytic. The analytic/coanalytic separation theorem therefore makes each relation Borel. $\square$

We use three standard descriptive facts: uncountable Borel subsets of Polish spaces contain Cantor sets; injective Borel images of standard Borel spaces are Borel; and analytic sets whose complements are analytic are Borel. These are classical descriptive set theory [8]. The additional order-homogenization input is Galvin's perfect-set theorem for Borel colorings of pairs, the pair case of the partition framework in Blass [71].

Lemma 186.2 (Increasing Cantor extraction). *If X is an uncountable Borel subset of a Polish space carrying a Borel linear order, then there is a continuous injective map $f : 2^{\mathbb{N}} \rightarrow X$ that is strictly increasing for the lexicographic order on $2^{\mathbb{N}}$.*

Proof. Start with a continuous injection of Cantor space into X . Color each unordered pair according to whether its lexicographic orientation agrees with the target order. This is a Borel two-coloring. Galvin's theorem provides a perfect homogeneous subset. A perfect subset of Cantor space contains an order-preserving Cantor copy: recursively choose two disjoint ordered cylinders meeting it in perfect sets, with lengths tending to infinity. The restricted map is either increasing or decreasing. In the latter case precompose with coordinatewise bit complementation. $\square$

[write] (Batch 92.) This is Part VIII's lemma 108.2 (without its ATR_0 uniformity clause) and Part XI's lemma 140.3; the three sources prove or cite it independently.

186.2 The abstract obstruction

Definition 186.3 (Separated grids). A *bounded separated-grid system* in $M = R_{\geq 0}$ consists of an element $C \in M$, uncountable Borel sets $X_n \subseteq [0, C]$, and positive $a_n, b_n \in M$ such that

$$a_n \leq y - x \leq b_n \quad (x, y \in X_n, x < y), \quad a_{n+1} > 5b_n. \quad (186.1)$$

The Borel qualification always refers to the proposed topology on M . The subtraction in the bounds is ring subtraction, not an assumed continuous operation.

Theorem 186.4 (Separated-grid obstruction). A Polish topology on $M = R_{\geq 0}$ with Borel arithmetic order and continuous addition cannot admit a bounded separated-grid system.

This criterion isolates the part of the argument that needs no particular arithmetic induction axiom. The fusion-and-tail-event method is adapted from Glazer [1]; the hypotheses here are an explicit reusable grid interface. We give all metric and finite-perturbation details.

Proof. **Step 1: positive shifts that disappear after a large translation.** Use Lemma 186.2 to choose increasing continuous injections $f_n : 2^{\mathbb{N}} \rightarrow X_n$. Let 0^∞ denote the all-zero sequence and e_j the sequence with its only 1 in position j . Set

$$x_n = f_n(0^\infty), \quad h_{n,j} = f_n(e_j) - x_n.$$

Then $a_n \leq h_{n,j} \leq b_n$. For every $m \geq C$, the element $m - x_n$ is nonnegative, and continuity of addition gives

$$m + h_{n,j} = (m - x_n) + f_n(e_j) \longrightarrow (m - x_n) + x_n = m. \quad (186.2)$$

This does *not* assert $h_{n,j} \rightarrow 0$. No cancellation of a continuous map is used to deduce continuity of its inverse.

Step 2: a metrically convergent, arithmetically lacunary fusion. Fix a compatible complete metric d . We recursively choose $y_n = h_{n,j_n}$. Having chosen y_i for $i < n$, put

$$s_\sigma = C + \sum_{i < n} \sigma(i) y_i \quad (\sigma \in 2^n).$$

Every two different such sums are distinct. Indeed, at the last differing index the corresponding y_i exceeds the sum of all earlier terms, since consecutive terms have ratio greater than 5. Define

$$d_0 = 1, \quad d_n = \min(\{1\} \cup \{d(s_\sigma, s_\tau) : \sigma \neq \tau \in 2^n\}) > 0 \quad (n \geq 1),$$

and

$$\delta_n = 2^{-n-3} \min_{j \leq n} d_j. \quad (186.3)$$

There are only finitely many s_σ at this stage. Equation (186.2) allows us to choose j_n so that

$$d(s_\sigma, s_\sigma + y_n) < \delta_n \quad (\sigma \in 2^n). \quad (186.4)$$

By (186.1), $y_{n+1} > 5y_n$ independently of how large j_n must be.

For $r \in 2^{\mathbb{N}}$ define

$$F(r) = \lim_{N \rightarrow \infty} \left(C + \sum_{i < N} r(i) y_i \right). \quad (186.5)$$

The limit exists uniformly in r : the n th increment is either zero or has metric length less than δ_n , and $\sum_n \delta_n < \infty$. Each finite-prefix map is continuous, so F is continuous.

If r and s have different prefixes of length $n \geq 1$, their prefix sums are at least d_n apart. For either sequence the remaining metric tail is bounded by

$$\sum_{k \geq n} \delta_k \leq d_n \sum_{k \geq n} 2^{-k-3} = 2^{-n-2} d_n \leq d_n/8.$$

Consequently $d(F(r), F(s)) \geq 3d_n/4 > 0$. Thus F is injective.

Step 3: algebra survives finite bit changes. If r, s differ only on a finite set, equality of their sufficiently long prefix sums, after adding fixed finite sums on the two sides, and continuity of addition imply

$$F(s) - F(r) = \sum_i (s(i) - r(i))y_i. \quad (186.6)$$

For clarity, separate indices with $r(i) = 0, s(i) = 1$ from those with the opposite change. Add the first finite sum to each r -prefix and the second to each s -prefix, then take limits. Only after obtaining the limiting equality in M do we subtract in R .

Let $\bar{r}$ be coordinatewise complementation and set $D(r) = F(r) - F(\bar{r}) \in R$. Injectivity implies $D(r) \neq 0$ for every r . For finitely different r, s , (186.6) yields

$$D(s) - D(r) = 2 \sum_i (s(i) - r(i))y_i. \quad (186.7)$$

No topology on R is required for this identity.

Step 4: bounded differences have probability zero. Give $2^{\mathbb{N}}$ fair independent bits, with product probability μ . For fixed n , let

$$E_n = \{r : 0 < D(r) \leq y_n\}.$$

This is Borel: rewrite its conditions as $F(\bar{r}) < F(r) \leq F(\bar{r}) + y_n$, using the Borel order and continuous addition. Condition on all bits outside $B = \{n+1, \dots, n+k\}$. At most one assignment of the k remaining bits can lie in E_n .

To see this, take two assignments and let j be their largest differing index. The lacunarity gives

$$4 \sum_{i < j} y_i < y_j. \quad (186.8)$$

One can prove this directly by induction: if it holds at j , then $4 \sum_{i \leq j} y_i < 5y_j < y_{j+1}$. Now (186.7) gives

$$|D(s) - D(r)| \geq 2 \left(y_j - \sum_{i < j} y_i \right) > y_n,$$

where $j \geq n+1$. Two elements of $(0, y_n]$ cannot have such a difference. The conditional probability is therefore at most 2^{-k} , and integration over the outside bits gives $\mu(E_n) \leq 2^{-k}$. Since k is arbitrary, $\mu(E_n) = 0$. Complementation gives the same conclusion for $-y_n \leq D(r) < 0$.

Step 5: a tail event of probability one half. Define

$$E = \{r : D(r) > y_n \text{ for every } n\}.$$

This is Borel. Since D never vanishes and all bounded positive or negative cases form a countable union of null sets, E and its image under complementation partition a conull set. Symmetry gives $\mu(E) = 1/2$.

The event E is unchanged by any finite modification of bits. Such a modification changes D by an element whose absolute value is at most $2 \sum_{i \leq m} y_i$ for some m . Given any n , choose $N > \max(n, m)$ large enough that $y_N > y_n + 2 \sum_{i \leq m} y_i$. If $r \in E$, then $D(r) > y_N$, so the modified value still exceeds y_n . The converse follows by reversing the modification.

A Borel event invariant under finite bit changes belongs to every tail σ -field: fix the first m coordinates to zero to obtain a Borel function of the remaining coordinates defining the same event. Hence it is independent of each finite-coordinate cylinder. Approximation by finite-coordinate events then makes it independent of itself, so its probability equals its square. This is the zero-one law, contradicting $\mu(E) = 1/2$. $\square$

[write] (Batch 92.) The same fusion and zero-one mechanism, also adapted from Glazer, is used by Part VIII (lemmas 109.1 and 110.1), Part IX's Borel multiplier obstruction (theorem 119.2) and Part XI (lemma 143.1). The separated-grid formulation, which isolates the arithmetic input as a family of bounded, increasingly separated uncountable Borel sets, is source 20's.

186.3 The arithmetic-grid interface

Corollary 186.5 (Borel scalar version). *Suppose M is an uncountable Polish arithmetic cone with continuous addition. Assume there are positive a_n, u_n and C such that*

$$[0, u_n] \text{ is uncountable, } \quad a_n u_n \leq C, \quad a_{n+1} > 5a_n u_n, \quad (186.9)$$

and each map $i \mapsto a_n i$ is Borel. Then a contradiction follows. In particular, jointly Borel multiplication suffices.

Proof. The order is Borel by Lemma 186.1. The image $X_n = \{a_n i : 0 \leq i < u_n\}$ is Borel by injectivity and the Borel-image theorem. If $i < j < u_n$, discreteness gives $j - i \geq 1$, so the positive difference of their images lies between a_n and $a_n u_n$. The hypotheses give the required separated-grid system. $\square$

Only countably many specified scalar maps need regularity. The theorem assumes neither continuity of these scalar maps nor measurability of arbitrary other multipliers. This localizes precisely where multiplication enters the addition obstruction.

186.4 Cofinally many individual translations fail

The same argument can localize the negative conclusion without assuming joint continuity of addition.

Corollary 186.6 (Cofinal translation obstruction). *Suppose a Polish arithmetic cone has Borel addition and a bounded separated-grid system. Then for every $d \in M$ there is $b \geq d$ such that the translation $\tau_b(x) = b + x$ is discontinuous.*

Proof. Suppose instead that every τ_b with $b \geq d$ is continuous. The arithmetic order is Borel by Lemma 186.1. Repeat the proof of Theorem 186.4, changing the base of every prefix sum from C to $C + d$. For $m \geq C + d$ we have $m - x_n \geq d$, so continuity of τ_{m-x_n} gives (186.2). This supplies the same metric fusion.

To pass finite-flip identities to limits, suppose the two fixed positive correction sums are A and B . Add d to both sides of the finite-prefix identity and use continuity of τ_{A+d} and τ_{B+d} . The limiting equality can then be cancelled algebraically to recover (186.6). The

remaining proof is unchanged: the events are Borel because addition is Borel, and the zero-one contradiction follows. $\square$

Consequently, in any Borel Polish presentation of P_Γ with dense Γ , discontinuous translations occur above every parameter. The same conclusion holds for any uncountable Borel Polish cone satisfying DIV + Root $_k$, by the grids constructed below. This is a localization of our obstructions, not an assertion that every translation fails: τ_0 is always the identity. Related cofinal discontinuity claims under other arithmetic hypotheses appear in the repository catalogue [130]; the additional point here is the explicit grid and one-root hypothesis boundary.

[write] (Batch 92.) Those claims are Part VIII's [theorem 104.1\(i\)](#) and [theorem 110.2](#) (cofinal discontinuous translations in Borel presentations of $\mathbb{Q} + \text{IOpen}$), which source 14 proves with its own fusion lemma ([lemma 109.1](#)) and states in ATR $_0$; Part VIII is a claimed, unreviewed part. [Corollary 186.6](#) is an independent ZFC proof, under the different hypothesis of a bounded separated-grid system; with [section 188](#) it covers uncountable Polish cones with Borel addition and multiplication satisfying DIV and one Root $_k$, a class containing the nonnegative parts of uncountable discretely ordered rings with open induction ([proposition 188.4](#)). It does not review or certify source 14's ATR $_0$ bookkeeping.

187 Proof of the exponent-group threshold

187.1 Discrete versus dense subgroups

Lemma 187.1. *An additive subgroup Γ of $\mathbb{R}$ is either trivial, of the form $\alpha\mathbb{Z}$ with $\alpha > 0$, or dense in $\mathbb{R}$.*

Proof. If positive group elements can be arbitrarily small, every real open interval contains an integer multiple of a sufficiently small positive group element; hence the group is dense. Otherwise let $\alpha = \inf \Gamma_{>0} > 0$. If this infimum were not attained, two distinct positive group elements sufficiently close to α would have a positive difference less than α , impossible. Division of any positive group element by the real number α leaves a remainder in $\Gamma \cap [0, \alpha)$, which must be zero. Thus $\Gamma = \alpha\mathbb{Z}$. $\square$

187.2 The positive topology

Assume $\Gamma = \alpha\mathbb{Z}$ and put $U = T^\alpha$. The positive nonconstant elements have the unique form

$$z + c_1U + \cdots + c_dU^d, \quad d \geq 1, z \in \mathbb{Z}, c_d > 0.$$

For each fixed pair (d, z) , give this stratum the usual topology of $\mathbb{R}^{d-1} \times (0, \infty)$. Declare the nonnegative integer constants isolated and take the topological disjoint sum of all strata.

Proposition 187.2 (Degree-and-constant topology). *This topology is locally compact, σ -compact and Polish, and both semiring operations are jointly continuous.*

Proof. There are countably many strata. Each nonconstant stratum is homeomorphic to $\mathbb{R}^d$, via the logarithm of the positive leading coefficient, and each constant stratum is a singleton. Thus each is locally compact, Polish and σ -compact. The countable topological sum has the same properties. For an explicit complete metric, transform each stratum to Euclidean space, use its distance truncated at 1 within a stratum, and use distance 2 between distinct strata. A

Cauchy sequence is eventually in one stratum, proving completeness; a countable union of countable dense sets proves separability.

On the product of two fixed strata, addition has a fixed output degree and a fixed constant coefficient. If the degrees agree and are positive, the leading coefficients add and cannot cancel. If the degrees differ, the larger one survives. The other coefficients are linear functions of the inputs.

For multiplication of two nonzero elements the output degree is the sum of the degrees and its leading coefficient is the product of two positive leading coefficients. The constant coefficient is fixed on each input stratum pair, and all other coefficients are polynomial functions. Multiplication by a constant zero is continuous because that input is isolated and the output is constant. Since the input stratum pairs are open and partition the product space, these checks prove joint continuity. $\square$

This is the existing polynomial-cone construction from the repository, reparameterized by $U = T^\alpha$ [130, Theorem 38.1 as indexed in the catalogue]. When $\Gamma = 0$, the cone is simply $\mathbb{N}$ with its discrete topology.

[write] (Batch 92.) Theorem 38.1 is Part III's [theorem 38.1](#) (source 06), with source 06's topology τ_P ; its partial subtraction is discontinuous at (t^2, t^2) , so the topology does not extend to the group of differences (Part XII's [example 155.5](#)), and its monus is discontinuous exactly on the locus of [proposition 190.3](#) below.

187.3 Dense groups supply bounded separated grids

Proposition 187.3. *If $\Gamma \leq \mathbb{R}$ is dense, no Polish topology on P_Γ makes addition continuous and multiplication Borel.*

Proof. Assume such a topology exists. Choose $\gamma_* \in \Gamma_{>0}$ and a strictly increasing sequence $\alpha_n \in \Gamma$ with $0 < \alpha_n < \gamma_*$ for all n . Using density, choose

$$0 < \beta_n < \alpha_{n+1} - \alpha_n, \quad \beta_n \in \Gamma.$$

Set

$$a_n = T^{\alpha_n}, \quad u_n = T^{\beta_n}, \quad b_n = T^{\alpha_n + \beta_n}, \quad C = T^{\gamma_*}. \quad (187.1)$$

Then $b_n < C$ and $a_{n+1} > 5b_n$, by the strict exponent inequalities. Every interval $[0, u_n)$ is uncountable: choose $\eta_n \in \Gamma$ with $0 < \eta_n < \beta_n$, and observe that all distinct elements rT^{η_n} , $r \in \mathbb{R}_{>0}$, lie in this interval. The scalar maps $i \mapsto a_n i$ are Borel sections of multiplication. Corollary 186.5 gives the contradiction. $\square$

Proof of Theorem 183.1. Proposition 187.2 gives (iii) $\Rightarrow$ (ii), and (ii) $\Rightarrow$ (i) is immediate. Lemma 187.1 and Proposition 187.3 give (i) $\Rightarrow$ (iii). For any nonzero Γ , choose $\gamma > 0$. The elements rT^γ with $r > 0$ real are distinct positive elements, so the cone is uncountable. $\square$

[write] (Batch 92.) For $\Gamma = \mathbb{Q}$, the negative direction is Part II's [theorem 25.2](#): there Glazer's Corollary 1 applies because the cone satisfies Q + IOpen ([theorem 23.2](#)), and it is stated for Borel-compatible recodings; [proposition 187.3](#) needs neither open induction nor compatibility with a given Borel structure. Part III's [proposition 42.1](#) found the boundary inside $\mathcal{A}_{\text{fin}}$ between its integer-exponent subcone ($\Gamma = \mathbb{Z}$, both operations continuous) and the whole cone ($\Gamma = \mathbb{Q}$, no continuous addition); [theorem 183.1](#) places every subgroup of $\mathbb{R}$ on one side of it.

Remark 187.4 (Strength and exact scope). The negative proof does not require a coefficient-observable coding, a Borel set of exponents, or a countable exponent group. It starts with an arbitrary proposed Polish topology on the cone and recovers its arithmetic order from addition. On the other hand it does require Borel regularity of the displayed scalar maps. It does not prove that every Polish topology on the underlying additive monoid makes addition discontinuous when multiplication is allowed to be completely non-Borel.

188 Euclidean division plus one root axiom

188.1 The external countability cut

The next argument takes place in an arbitrary uncountable Polish cone, not just in P_Γ .

Lemma 188.1. *Let $M = R_{\geq 0}$ have a Polish topology with Borel order and satisfy DIV. Define the external set*

$$I = \{a \in M : [0, a] \text{ is countable}\}.$$

Then I is a proper initial segment containing every ordinary numeral and closed under addition and multiplication. It need not be definable in M .

Proof. Discreteness makes the interval below each ordinary numeral finite, and downward closure is immediate. If $a, b \in I$, every element of $[0, a + b]$ lies either in $[0, a]$ or in $a + [0, b]$, so the interval is countable.

For multiplication, the case $b = 0$ is trivial. If $b > 0$, divide each $x \in [0, ab]$ as $x = bq + r$ with $0 \leq r < b$. Then $q \leq a$; otherwise $bq > ab$. The pair (q, r) belongs to the product of the two countable sets $[0, a]$ and $[0, b]$, and determines x . Thus $[0, ab]$ is countable.

Finally apply Lemma 186.2 to M . The greatest point of the resulting increasing Cantor copy bounds uncountably many elements, and therefore does not lie in I . $\square$

[write] Here M is uncountable, as the sentence before the lemma says; properness of I uses it, and for countable M one has $I = M$.

188.2 Slowly descending roots and rapidly ascending quotients

Lemma 188.2. *Assume in addition Root_k for a fixed $k \geq 2$. For any $c \in M \setminus I$, there are elements $r_n \in M \setminus I$ with $r_0 = c$ and*

$$r_{n+1}^{k^2} \leq r_n < (r_{n+1} + 1)^{k^2}. \quad (188.1)$$

Every r_n exceeds every ordinary integer.

Proof. Take two successive integer k th roots. If s is an integer k th root of r_n and r one of s , then

$$r^{k^2} \leq s^k \leq r_n < (s + 1)^k \leq (r + 1)^{k^2}.$$

For the final inequality use $s < (r + 1)^k$, hence $s + 1 \leq (r + 1)^k$ by discreteness. Put $r_{n+1} = r$. If r lay in I , closure of I under successor and multiplication would put $(r + 1)^{k^2}$, and then r_n , in I , a contradiction. Elements outside I exceed every numeral because I is initial and contains the numerals. $\square$

Proposition 188.3. *Under the hypotheses of Lemma 188.2, define $a_n = \lfloor c/r_n \rfloor$ using DIV and put $u_n = r_{n+1}$. Then*

$$a_n \geq 1, \quad a_n u_n \leq c, \quad a_{n+1} > 5a_n u_n,$$

and every interval $[0, u_n)$ is uncountable.

Proof. The root chain is nonincreasing, so $1 \leq r_n \leq c$ and $a_n \geq 1$. Moreover $a_n r_n \leq c$ and $r_{n+1} \leq r_n$, giving $a_n u_n \leq c$. Since $u_n \notin I$, its closed interval is uncountable; deleting the single endpoint preserves uncountability.

Write $r = r_{n+1}$. Division gives

$$c < (a_{n+1} + 1)r.$$

If $a_{n+1} \leq 5a_n r$, then $a_n r \geq 1$ implies

$$c < (a_{n+1} + 1)r \leq 6a_n r^2.$$

But the other quotient inequality and (188.1) give

$$c \geq a_n r_n \geq a_n r^{k^2}.$$

These inequalities are incompatible, since $a_n > 0$ and $r^{k^2} > 6r^2$. For example, $r > 3$ and $k^2 \geq 4$ already imply that last strict inequality. Hence $a_{n+1} > 5a_n r$. $\square$

Proof of Theorem 183.2. If the alleged topology existed, its order would be Borel by Lemma 186.1. Construct I, c, r_n, a_n, u_n as above. Borel multiplication supplies the Borel scalar maps in Corollary 186.5; Proposition 188.3 supplies all its other hypotheses. This yields a contradiction. $\square$

The proof uses only an external sequence of roots, not an internally coded sequence of nonstandard length. It also does not assume that root or quotient functions are continuous. This avoids both an induction assumption and a hidden regularity assumption.

188.3 Why the weakening is strict

Proposition 188.4. *For nonnegative cones of discretely ordered commutative rings, open induction implies DIV and every fixed Root_k . Conversely, DIV + Root_2 does not imply open induction.*

Proof. Fix $a \geq 0$ and $b > 0$. The open predicate $bx \leq a$ holds at 0 and fails at $x = a + 1$, since $b \geq 1$. Open induction therefore supplies q with $bq \leq a$ but $b(q + 1) > a$; otherwise the predicate would be successor closed and hold everywhere. Let $r = a - bq$. This proves DIV. The identical argument with $x^k \leq a$, which also fails at $a + 1$, proves Root_k .

For the converse use Example 185.3 and Corollary 184.5. The ring $A_{\mathbb{Z}[1/2]}$ is discretely ordered, has full division and integer square roots, but its cube-cut predicate violates open induction. $\square$

Within the ordered-ring-cone setting, Theorem 183.2 therefore lowers the arithmetic assumptions of the addition obstruction. It says nothing by itself about the continuity of multiplication when addition is merely Borel, and it makes no reverse-mathematical claim.

189 Finite root extensions and the limit transition

189.1 Every finite stage is topologically tame

Fix an ordinary integer $p \geq 2$. Set

$$\Gamma_m = p^{-m}\mathbb{Z}, \quad A_m = A_{\Gamma_m}, \quad P_m = (A_m)_{\geq 0}.$$

Each P_m has the topology of Proposition 187.2 with $U_m = T^{1/p^m}$.

Proposition 189.1. *The natural inclusion $P_m \hookrightarrow P_{m+1}$ is a closed topological embedding and a semiring homomorphism.*

Proof. The relation $U_m = U_{m+1}^p$ sends a degree- d polynomial to a degree- pd polynomial and inserts zeros in all coefficient positions whose indices are not divisible by p . Its constant coefficient is unchanged. On each stratum this is a closed linear-coordinate embedding with positive leading coordinate. In a target stratum whose degree is not divisible by p , the image is empty. The source and target are topological sums of these strata and the isolated constants; thus the whole image is closed and the induced topology is exactly the source topology. The semiring identities hold algebraically. $\square$

Corollary 189.2 (Failure at the infinite root tower). *The algebraic union*

$$P_\infty = \bigcup_{m \geq 0} P_m = P_{\mathbb{Z}[1/p]}$$

is a union of closed, continuously embedded locally compact Polish semirings. Nevertheless no Polish topology on P_∞ makes addition continuous and multiplication Borel.

Proof. The group $\mathbb{Z}[1/p]$ has arbitrarily small positive elements, so it is dense. Apply Theorem 183.1. $\square$

This statement excludes *all* candidate Polish topologies on the union, not only a naive direct-limit topology. It does not assert that a particular final topology has jointly continuous multiplication; that is a separate topological issue.

189.2 Rational versus incommensurable exponent adjunction

Adjoining finitely many rational exponents to $\mathbb{Z}$ gives $\Gamma = N^{-1}\mathbb{Z}$ for some N , so the positive topology survives. Adjoining an irrational exponent θ and then closing the exponent set under the additive group operations gives the dense group $\mathbb{Z} + \theta\mathbb{Z}$, so it does not survive.

This qualification matters. The ordinary polynomial subring generated by T and T^θ has only nonnegative combinations of 1 and θ as exponents. It is generally smaller than $A_{\mathbb{Z} + \theta\mathbb{Z}}$, which allows every positive group combination, including combinations formed using negative integers. Our theorem concerns the latter ring and does not silently identify these two constructions.

Exponent group	Universal integer roots	Polish continuous arithmetic
$\alpha\mathbb{Z}, \alpha > 0$	No $\text{Root}_k, k \geq 2$	Both operations continuous
$\mathbb{Z}[1/2]$	Exactly $k = 2^m, m \geq 1$	Continuous + with Borel · impossible
$\mathbb{Z} + \sqrt{2}\mathbb{Z}$	No $\text{Root}_k, k \geq 2$	Same obstruction, despite no root axiom
$\mathbb{Q}$	Every Root_k	Same obstruction
$\mathbb{R}$	Every Root_k	Same obstruction, arbitrary coding

All rows have full division. The table summarizes results proved above; it is not a classification of multiplication-only presentations.

190 Borel presentations and generic continuity

The negative theorem must not be misread as an obstruction to Borel arithmetic itself. For countable exponent groups there are simple explicit Polish representations with Borel operations.

190.1 Exact-support charts

Assume Γ is countable. For each nonempty finite $S = \{\gamma_1 < \dots < \gamma_d\} \subseteq \Gamma_{>0}$ and $z \in \mathbb{Z}$, use the chart

$$\left\{ z + \sum_{i=1}^d c_i T^{\gamma_i} : c_1 \cdots c_d \neq 0, c_d > 0 \right\}. \quad (190.1)$$

Give it its coefficient topology and take the topological sum of these charts and the isolated nonnegative integers. It is Polish: splitting by the signs of the nonleading coordinates and applying logarithms to their absolute values gives countably many Euclidean components.

Proposition 190.1. *In the exact-support topology, addition and multiplication are Borel. Each operation is continuous away from a closed nowhere dense subset of P_Γ^2 .*

Proof. Fix two input charts. There are finitely many possible positive output exponents: their union for addition, and the pairwise sums together with terms arising from multiplication by constants for multiplication. Each output coefficient is a polynomial in the finitely many real input coordinates; the output constant is fixed. Partition the input chart pair according to which output coefficients vanish and the signs of the others. Each piece is Borel and has a fixed output chart. On that piece the coordinate formula is continuous, proving Borel measurability.

Discard candidate coefficients whose polynomial is identically zero. For each remaining coefficient, its zero locus is closed and nowhere dense in the open Euclidean input domain. The finite union of these loci is a closed nowhere dense exceptional set. Outside it the output support is locally constant, so the operation is continuous. The countable input chart pairs form a partition into clopen sets; the union of their relative closed exceptional sets is closed in the entire product and has empty interior there. Constant-zero multiplication charts cause no problem because the operation is constant on them. $\square$

This construction and its cancellation mechanism generalize the rational-support presentation already discussed in the repository [130]. We use it here as a companion to the universal obstruction, not as a replacement for that obstruction.

Corollary 190.2. *In these presentations the quotient–remainder map furnished by DIV is Borel. Whenever $\Gamma = k\Gamma$, the integer k th-root function is Borel as well.*

Proof. The graphs are Borel, expressed by the finite algebraic relations and inequalities in (183.2) and (183.3). Their values are unique by discrete order. The projection from either graph to its domain is a Borel bijection between standard Borel spaces, so its inverse is Borel. Composing with the value projection proves the assertion. $\square$

190.2 A visible discontinuity and why it is not the main theorem

Suppose the group contains exponents $0 < \beta < \alpha$. In exact-support charts, let

$$f = T^\alpha + T^\beta, \quad g_s = T^\alpha + (-1 + s)T^\beta, \quad |s| < \frac{1}{2}.$$

Both inputs are positive and stay in fixed charts, with $g_s \rightarrow g_0$. But $f + g_s = 2T^\alpha + sT^\beta$ changes its exact support at $s = 0$. The output chart containing $2T^\alpha$ is open, so the sums do not converge to it.

For cyclic groups the degree topology repairs this defect by allowing intermediate zero coefficients within one chart. For dense groups, Theorem 183.1 says that *no* Polish repair can make addition continuous while preserving Borel multiplication. A coordinate example alone could never establish that presentation-independent conclusion.

[write] (Batch 92.) The rational-support presentation is Part II’s exact-support topology τ_{FS} on $\mathcal{A}_{\text{fin}}$ (proposition 24.1), and proposition 190.1 is its theorem 24.3 for every countable Γ . The example above, with $\alpha = 2$ and $\beta = 1$, is Part II’s example “Addition: cancellation below the leading term” after that theorem.

190.3 An elementary boundary for subtraction

There is also a useful distinction even in the positive cyclic topology. Define truncated subtraction $p \dot{-} q = \max(p - q, 0)$.

Proposition 190.3. *In the degree-and-constant topology, truncated subtraction is discontinuous exactly at pairs of nonconstant polynomials having the same degree and the same leading coefficient.*

Proof. If the degrees differ, the sign and degree of $p - q$ are fixed locally. If the degrees agree but their leading coefficients differ, the same is true on a sufficiently small neighborhood. The output is then either constant zero or a fixed-degree polynomial with continuously varying coefficients. Constant inputs are isolated, and the remaining mixed cases have fixed nonconstant degree, so they are continuous too.

Now suppose the common degree is $d > 0$ and the leading coefficients agree. The value of $p \dot{-} q$ has degree less than d , or is zero. Increase the leading coefficient of p by a positive real $s \rightarrow 0$, keeping every other coefficient fixed. The resulting truncated difference has degree d for all $s > 0$. Since distinct degree strata are clopen, these outputs cannot converge to the original value. $\square$

Thus continuous semiring arithmetic is entirely compatible with noncontinuous truncated subtraction. Group-topology rigidity results must not be applied to these cones without first checking their different hypotheses.

[write] (Batch 92.) For $\Gamma = \mathbb{Z}$ this is Part III's cone; the remark after [theorem 38.1](#) shows that even partial subtraction is discontinuous there, at $((1 + 1/k)t, t) \rightarrow (t, t)$, a pair of the locus above, and Part XII's [example 155.5](#) that this failure, at (t^2, t^2) , is exactly why the topology does not extend to the group of differences ([theorem 153.1](#)). The exact discontinuity locus of the monus is new.

191 Surreal realizations and foundational scope

191.1 The finite-support embedding

Conway normal-form monomials satisfy $\omega^\gamma \omega^\delta = \omega^{\gamma+\delta}$, and finite normal forms are compared by their greatest differing exponent. It follows that

$$\iota_\Gamma \left(z + \sum_{\gamma \in S} c_\gamma T^\gamma \right) = z + \sum_{\gamma \in S} c_\gamma \omega^\gamma \quad (191.1)$$

is an ordered ring embedding into the omnific integers: all its exponents are positive except for an integer constant term. This is the standard finite normal-form interpretation used throughout the repository's surreal developments [132].

[write] (Batch 92.) For $\Gamma = \mathbb{Z}$ this is the map $t \mapsto \omega$ of Part III ([theorem 40.1](#)), and for $\Gamma = \mathbb{Q}$ Part II's reading $X = \omega$; compare Part I's [theorem 9.1](#).

The classification therefore applies to a concrete family of *set-sized subrings* of **Oz**. In particular, the negative result for dense Γ is an obstruction already inside small, finite-support parts of omnific arithmetic. It does not depend on uncountable supports or on the proper-class size of all surreals.

191.2 No topology on a proper class is smuggled in

Everything needed to define A_Γ is a set: a set of real exponents, finite supports, finite real coefficient lists, and an integer constant. The surreal expression in (191.1) is a definable interpretation of this set. Replacement collects its image as a set in an ordinary set-theoretic presentation of surreal numbers. Equivalently, one may work in a sufficiently large type-theoretic universe containing the required set of sign sequences.

Neither **No** nor **Oz** as a whole is declared to be a Polish space. Nor does the proof quantify over a set of all topologies on a proper class. All topological quantifiers concern the set P_Γ . The obstruction survives transport along any bijective presentation of that set because it is expressed using its arithmetic operations and a proposed topology, not a privileged code.

191.3 Relation to the existing formal floor

The inspected file `Surreal/Foundations/OmnificFloor.lean` defines the positive-growth truncation, the infinitesimal tail, and the corrected integer constant. It contains declarations named `omnificFloor_spec` and `existsUnique_omnific_integerPart` [131]. These are appropriate infrastructure for the integer-part step here.

There remains a genuine new formalization obligation: prove that the positive-growth truncations of the particular generalized series used in this paper have finite support and exponents in the chosen subgroup. An existing theorem that every surreal has an omnific floor does not, by itself, prove that this floor belongs to A_Γ . Our support estimate in (184.1) discharges this obligation mathematically.

192 Hypothesis audit and reproducible checks

192.1 The dependency structure

The two negative results share a short logical interface:

$$\begin{aligned} &\text{continuous addition} \implies \text{Borel arithmetic order,} \\ &\text{bounded Borel separated grids} \implies \text{Cantor fusion} \implies \text{zero-one contradiction.} \end{aligned}$$

Dense real exponent groups provide the grids by monomials. Division and one root axiom provide them by a countability cut and quotient estimates. Neither application needs continuity of subtraction.

Hypothesis or input	Exact use
Polish topology	Complete fusion metric, Borel perfect-set theorem, and standard Borel image facts
Continuous addition	Translation limits and passage of finite-sum identities to limits
Borel multiplication	Only the selected maps $i \mapsto a_n i$ need it
Discrete arithmetic order	Distinct grid indices differ by at least 1; quotient and integer-root inequalities
DIV in the root theorem	Closure of the countability cut under products and definition of $a_n = \lfloor c/r_n \rfloor$
One fixed Root_k	Two successive roots give k^2 -power descent
$\Gamma \leq \mathbb{R}$	Discrete/dense dichotomy and real cutoff support estimates
Real coefficients	Uncountably many elements in the required monomial intervals; the positive Euclidean charts

192.2 Sanity checks and non-conclusions

The countability hypothesis cannot be dropped. If one uses a countable coefficient field and a countable exponent group, the finite-support ring is countable; its discrete topology makes every operation continuous. The proof fails at exactly the intended point: the uncountable grids are absent. For example a countable real-closed coefficient field with dyadic exponents supplies a countable DIV + Root_2 example.

[merge] The last sentence is not proved in source 20; here is the proof. Let $K \subseteq \mathbb{R}$ be a countable real closed field and replace $\mathbb{R}$ by K in (183.1) and (184.1), with $\Gamma = \mathbb{Z}[1/2]$. The proofs of [proposition 184.1](#) and [lemma 184.3](#) use only that the coefficients form an ordered field; the proof of [theorem 184.4](#) uses only that the constant c lies in the Archimedean field $K \subseteq \mathbb{R}$, so

that $\lfloor c \rfloor \in \mathbb{Z}$ and the gaps are positive elements of K , which dominate infinitesimals; so DIV holds as in [corollary 184.5](#). In the proof of [theorem 185.1](#) the leading coefficient $a > 0$ has $a^{1/2} \in K$ because K is real closed, $\gamma/2 \in \Gamma$, and the binomial coefficients are rational, so Root₂ holds. The ring is countable.

The negative statements do not say that dense-exponent arithmetic has no standard Borel representation. Section 190 proves the opposite when the exponent group is countable. Nor do they say that multiplication is necessarily discontinuous when only addition is Borel. That is an explicit research question below.

We do not use or certify the repository report's separately marked claim about an ATR_0 proof of Glazer's full theorem. A proof in ZFC using perfect-set and measure arguments does not automatically give an ATR_0 proof. Any such refinement needs a coding convention and an internal construction of every object used.

192.3 What the executable companion checks

The accompanying `verify.py` uses only Python's standard library and exact integer or rational arithmetic. It checks finite generalized polynomial operations, degree and sign identities, polynomial division with the corrected constant term, exact integer-root inequalities, lacunary finite-bit perturbations, and explicit dyadic exponent grids. It writes a machine-readable record to `verification_results.json`.

These tests are deliberately aligned with vulnerable algebraic steps. They do not enumerate a Polish topology, decide Borelness, construct an actual infinite fused Cantor set, or prove a measure-zero assertion. Theorems about those objects are justified by the written proofs, not by the test count.

[write] In this report the program is `code/20-exponent-threshold-verify.py` and its record `data/20-exponent-threshold-verification_results.json`. The program writes `verification_results.json` beside itself (the path is fixed in the code), so the README gives a rerun on a copy. The intake reran it on a copy: 13,034 assertions in 20 groups pass, and the record equals the shipped one except for the interpreter version (3.14.4 for the recorded 3.13.5) and, on Windows, line endings.

192.4 A practical formalization sequence

A formal development should first encode A_Γ by finitely supported functions and prove its ordered-ring and degree laws. The next layer is the support-local-finiteness lemma for geometric and binomial substitution, followed by the restricted floor, DIV, and the root-spectrum theorem. The inequalities in Proposition 188.3 form a compact arithmetic module that can be formalized independently of topology.

The topological positive construction should then be encoded as a countable coproduct of finite-dimensional coefficient charts. Its proofs need no advanced descriptive set theory. The difficult final layer is the separated-grid theorem: Borel order recovery, increasing Cantor extraction, complete-metric fusion, finite-flip identities, and the product-measure zero-one argument. Each imported descriptive theorem needs an explicit statement and an axiom audit. The whole project should expose theorem dependencies rather than treat numerical diagnostics as proof certificates.

No Lean or Rocq source purporting to prove these new theorems is included in this package. The formalization sequence is a plan, not a completed formalization.

[write] Of the inputs named here only the omnific floor is formalized in the repository

(odg: thm: floor, note after [theorem 184.4](#)); docs/FORMALIZATION.md maps no label of this part.

193 Further research questions and concrete next targets

Question 193.1 (Multiplication-only presentations). For dense countable Γ , can P_Γ have a Polish presentation with continuous multiplication and merely Borel addition?

The present grids target addition. A multiplicative analogue would need an appropriate family of multiplicatively separated positive increments or a definable exponential map. The dyadic example makes this especially interesting: its arithmetic is much weaker than the exponential theories in the known two-operation obstruction. A useful first target is $\Gamma = \mathbb{Z}[1/2]$.

[write] (Batch 92.) For $\Gamma = \mathbb{Q}$ this is Part II's [question 28.1](#) (there with the canonical Borel sets), and Part XI's [question 149.5](#) asks it for natural models of IOpen ; all three stay open.

Question 193.2 (Is the grid criterion close to necessary?). Among uncountable Borel ordered arithmetic cones satisfying DIV, does failure of every bounded separated-grid system allow a continuous-addition Polish recoding? If not, identify an explicit independent obstruction.

The criterion is presently sufficient for impossibility, not a characterization. Polynomial cones show that full division alone does not force a contradiction.

Question 193.3 (Weaker quotient hypotheses). Can the one-root theorem replace DIV by division only for the particular descending root sequence, or by a weaker condition making the external countability cut multiplicatively closed?

The proof isolates two uses of division. Separating them could identify a substantially weaker first-order arithmetic theory with the same topological obstruction. One should state the restricted divisor class inside the structure, rather than silently using an external schema.

Question 193.4 (Replacing powers by one polynomial). For which eventually increasing integer polynomials p does the axiom $\forall a \geq 0 \exists r \geq 0 (p(r) \leq a < p(r+1))$, together with suitable division, force the same obstruction?

The quotient argument needs a descent that dominates a quadratic term after finitely many iterations. A theorem in terms of explicit lower and upper polynomial growth bounds would generalize the k^2 estimate. Linear polynomials should be excluded, since ordinary division supplies them in the positive polynomial example.

Question 193.5 (General coefficient fields). Classify the analogous cones with coefficient field $K \leq \mathbb{R}$ in place of $\mathbb{R}$.

For uncountable K , dense exponent groups still have the required uncountable intervals, so the negative proof persists. For countable K and countable Γ , the discrete topology settles the question positively. The remaining coefficient-exponent combinations need both cardinality and descriptive regularity analyses; the Euclidean-chart proof for $K = \mathbb{R}$ cannot simply be copied to an arbitrary subfield.

Question 193.6 (Higher-rank exponent groups). What replaces Theorem [183.1](#) when Γ is an arbitrary ordered abelian group rather than a subgroup of $\mathbb{R}$?

There may be both discrete layers and dense Archimedean quotients. A first test is a lexicographic sum of two cyclic groups. The correct criterion may involve bounded increasing scales together with the sizes of the intervals they create, rather than the mere existence of a least positive exponent.

Question 193.7 (Optimal regularity of arithmetic codes). For a fixed dense countable Γ , what is the best simultaneous Baire class of addition, multiplication, quotient, and the available root functions among Polish presentations inducing the canonical Borel structure?

Generic continuity from Proposition 190.1 does not automatically imply that a target-valued function is Baire class one. The codomain's disconnected support charts make this distinction important. An explicit upper bound and a matching lower bound would sharpen the qualitative threshold.

Question 193.8 (Classifying positive cyclic topologies). Classify locally compact Polish semiring topologies on $P_{\alpha\mathbb{Z}}$. Is the degree-and-constant topology characterized by a natural additional condition on coefficient maps or multiplication by monomials?

The existence theorem does not establish uniqueness. A practical first problem is to classify which strata must be open when 0 and the ordinary numerals are isolated, without presupposing continuity of degree.

Question 193.9 (Topology of the infinite root tower). Describe the final topology on $\bigcup_m P_m$ from Section 189: is addition jointly continuous there, is multiplication jointly continuous, and which completeness or countability properties fail?

The universal Polish obstruction already shows that joint continuity of both operations cannot coexist with Polishness. It does not identify the precise failure of this particular final topology. Sequential and compact-generation methods may separate the alternatives.

Question 193.10 (Non-Polish presentations). How far do the obstructions extend to completely metrizable nonseparable spaces, or to separable spaces lacking complete metrizability?

The finite-fusion estimates themselves use completeness, while the Borel perfect-set extraction uses Polish hypotheses. Counterexamples should isolate these two roles, rather than changing several assumptions at once.

Question 193.11 (Reverse mathematics of the grid interface). Under an explicit code for the ordered cone, scalar maps and grid system, what subsystem of second-order arithmetic proves Theorem 186.4? Which additional strength is needed to manufacture the grids from $\text{DIV} + \text{Root}_k$?

This is intentionally narrower than the full multiplication theorem. It requires uniform codes for the Cantor extractions and a careful replacement of the external countability cut. No upper or lower bound is claimed here.

Question 193.12 (End-to-end formal verification). Can the restricted floor theorem, exact root spectrum, positive topology, and separated-grid contradiction be joined in a single kernel-checked Lean or Rocq development with no admitted descriptive-set-theoretic lemmas?

The natural milestones are the arithmetic grid inequalities, a fully checked metric fusion, and then the perfect-set and probability inputs. The repository's existing floor infrastructure gives a concrete starting point, but the descriptive layer is likely the major library-development cost.

194 Conclusion

The finite-support setting already contains a sharp topological boundary. With real coefficients and a real exponent subgroup, discrete exponents permit locally compact Polish semiring arithmetic; dense exponents obstruct even continuous addition when multiplication remains Borel. The obstruction does not require full open induction: Euclidean division and one fixed integer-root operation suffice, and the weakening is witnessed by an explicit model failing cubic induction.

The organizing idea is not an informal claim that infinite arithmetic is incompatible with topology. It is a precise conflict between bounded uncountable arithmetic grids, increasingly separated positive increments, and a complete separable topology in which finite additions pass to limits. The exponent classification and the root theorem give two independent ways to create that conflict. They also mark what remains unsettled: multiplication-only continuity, weaker arithmetic axioms, higher-rank exponents, optimal descriptive regularity, and full formal verification.

Part XV

Rational rank and self-embeddings of Levi-Civita fields

[write] This part is source 21, manuscript 05 of batch 92, *Rational Rank and Self-Embeddings of Levi-Civita Fields: A sharp co-Hopfian dichotomy, proper immediate self-copies, and integer-part obstructions* (20 pages, A4, dated 4 October 2026; “research manuscript prepared with ChatGPT for Vladimir Reshetnikov”). It arrived in commit 9dc8db274, as archive `Levi_Civita_Rank_and_Self_Embeddings`, and was placed in `e24ce2ce0`; it is pinned to `9fe62865d`, 52 commits before the placement. Its file prefix in this report is `21-levi-civita-rank-`. All unmarked text in this part is source 21’s; its fifteen sections are Sections 195–209 (its Section k is Section $194 + k$, with statement numbers otherwise unchanged), its twelve “Research questions” are Questions 208.1–208.12, and its appendix is [appendix AG](#). [Remark 204.2](#) was written for this report.

What this part claims, and its status

Let $k = \mathbb{R}_{\text{alg}}$, let $0 \neq \Gamma \leq \mathbb{R}$ be a $\mathbb{Q}$ -subspace, and let $\mathcal{L}^\Gamma$ be the field of series over k with left-finite supports in Γ (t positive *infinitesimal*). Every unital field embedding $\mathcal{L}^\Gamma \rightarrow \mathcal{L}^\Delta$ fixes k , scales the valuation by some $\lambda > 0$ with $\lambda\Gamma \subseteq \Delta$, and is continuous with closed image ([theorem 198.1](#)); embeddings correspond to triples (λ, c, u) ([theorem 198.2](#)). Every self-embedding of $\mathcal{L}^\Gamma$ is onto if and only if $\dim_{\mathbb{Q}} \Gamma$ is finite ([theorem 195.1](#)): finite rank gives a uniform valuation gain and a Neumann inverse ([theorem 199.2](#)), and infinite rank a proper isometric immediate self-embedding by a bounded-exponent shift ([theorem 200.2](#)), whose image has an unattained distance from one monomial ([theorem 201.1](#)); for countable Γ of infinite rank there are $2^{\aleph_0}$ closed immediate self-copies in a continuous Cantor family ([theorem 202.1](#)). $\mathcal{L}^\Gamma \hookrightarrow \mathcal{L}^\Delta$ exactly when $\lambda\Gamma \subseteq \Delta$ for some $\lambda > 0$ ([theorem 203.1](#)). A closed subfield with cofinal value group containing an integer part of $\mathcal{L}^\Gamma$ is all of $\mathcal{L}^\Gamma$ ([theorem 205.3](#)). Lemma 4.2.4 of the Kuhlmann–Serra preprint (arXiv:2107.03362v3) fails as stated ([proposition 204.1](#), [remark 204.2](#)). Unrefereed, AI-assisted, not formalized; historical priority is not established. The correction of the preprint is version-specific. The finite checks test finite identities only.

Questions answered. Source 21 states that it resolves no printed question of Glazer’s and answers no question of the report it continues. This report reads it as follows (dated notes after each question). Part V’s [question 73.5](#) (other support completions, of higher *ordered* rank) is not answered: source 21 extends the left-finite construction to every $\mathbb{Q}$ -subspace $\Gamma \leq \mathbb{R}$, of ordered rank one and arbitrary *rational* rank, and its [question 208.12](#) asks the higher-rank question again. Part V’s [question 73.8](#) is answered in part: transporting integer parts by automorphisms gives integer parts other than the canonical one, already for the automorphism $t \mapsto t + t^2$ of Part V’s own field $\mathcal{L}_k$ with $k = \mathbb{R}_{\text{alg}}$ ([corollary 205.4](#) and the paragraph after it); the classification up to automorphisms stays open. For $\Gamma = \mathbb{Q}$, source 21’s $\mathcal{L}^\Gamma$ is Part V’s $\mathcal{L}_k$ with $k = \mathbb{R}_{\text{alg}}$, so [theorem 199.2](#) shows that every field self-embedding of that field is onto, which is new to the report.

What was already here, and what is new. Source 21 names Part V’s source 09 as its prior report, uses no theorem of it, and re-derives its basic layer for general Γ : [proposition 196.2](#) is Part V’s [proposition 65.2](#), [proposition 196.3](#) extends [theorem 65.3](#) and [proposition 65.5](#) (Polish exactly when Γ is countable), [proposition 197.1](#) extends [theorem 68.2](#) with a different proof, [proposition 205.2](#) extends [theorems 62.3](#) and [70.1](#), and [proposition 206.1](#) is Part V’s surreal map ([71.1](#)). In batch 92, source 20’s [lemma 184.3](#) and [theorem 184.4](#) are the same field and floor with real coefficients and $T = t^{-1}$. The valuation part of [theorem 198.1](#) is the Archimedean case of the surreal self-embeddings report’s `sse:prop:valueaction`, and its continuity part an instance of that report’s `sse:an:top:criterion`; the coefficient twist of [theorem 198.2](#) is the repository’s Lean `characterTwist`. Each is flagged in a note. New to the report are the rank dichotomy, the multiplier field, the bounded-exponent shift, the distance gap, the Cantor family, the classification of embeddings between exponent groups, the integer-part criterion and the correction of the preprint.

Labels and notation. Every label of source 21 takes the prefix `pma:rk:`; labels were added to its conclusion, its appendix, its twelve questions, the remark `pma:rk:rem:ks` and `pma:part:rk`. The sixth block of the table in [section 1.3](#) lists the symbols. In short: source 21 indexes its fields by the *exponent group*, $\mathcal{L}_\Gamma$, $\mathcal{H}_\Gamma$, $\mathcal{I}_\Gamma$, while Part V indexes them by the

coefficient field, $\mathcal{L}_k, \mathcal{H}_k, \mathcal{I}_k$. To prevent the false reading of source 21's $\mathcal{L}_{\mathbb{Q}}$ as Part V's $\mathcal{L}_k$ with $k = \mathbb{Q}$, source 21's fields are printed $\mathcal{L}^{\Gamma}, \mathcal{L}^{\Delta}, \mathcal{H}^{\Gamma}, \mathcal{I}^{\Gamma}$ and $\mathcal{S}^{\Gamma}$, with the exponent group as a superscript; so $\mathcal{L}^{\mathbb{Q}}$ is Part V's $\mathcal{L}_{\mathbb{R}_{\text{alg}}}$. Its t is positive infinitesimal, as in Part V, and its metric $d = e^{-v}$ is unbounded where Part V's is $\min\{1, 2^{-v}\}$ (same topology). $H_{\Gamma} = \text{Hom}(\Gamma, 1 + \mathfrak{m})$ is a character monoid, not a Hahn field, and E_{Γ} is the multiplier field. No normalization changed.

Files. Source 21's checks, recorded run, Makefile, build summary, source record and proof audit are shipped as `code/21-levi-civita-rank-code-check_identities.py`, `data/21-levi-civita-rank-audit-checks.json`, `code/21-levi-civita-rank-Makefile`, `data/21-levi-civita-rank-audit-build-summary.json`, `data/21-levi-civita-rank-audit-sources.json` and `21-levi-civita-rank-audit-proof-audit.md`. The Hilbert's-tenth programme routed the archive from its README only and did not check the correction of the preprint, which this report has checked ([remark 204.2](#)). Nothing in this part is formalized.

195 The research intersection and the main conclusions

Glazer's *A Topological Tennenbaum Theorem* studies the incompatibility between uncountable Polish presentations and sufficiently strong arithmetic with continuous operations [1]. ProveIt's surreal-number development provides a complementary setting: ordered fields built from generalized series, monomial maps, and omnific integer parts [136, 137, 138]. The intersection suggests a structural question on the positive side of the topological boundary:

When an uncountable ordered field does admit a complete separable arithmetic topology, how rigid are its self-embeddings, and what can its integer parts detect?

The answer below is an exact finite-versus-infinite rational-rank dichotomy. The obstruction is neither nonmeasurability nor a discontinuous choice of coordinates: in the flexible case the proper self-embeddings are isometries.

The earlier project report *Polishability at the Hahn–Puisseux–Levi-Civita Boundary* [135] studies existence of Polish presentations and countable integer parts. The present article changes the question from existence of a topology to surjectivity of arbitrary field self-embeddings. No theorem from that unrefereed report is used as a proof input here.

[write] (Batch 92.) The earlier project report [135] is source 09, now [Part V](#). Source 21 read the surreal README and two Lean files at its pin, not this report.

195.1 The precise field and the dichotomy

Throughout the main results,

$$k = \mathbb{R}_{\text{alg}} = \{a \in \mathbb{R} : a \text{ is algebraic over } \mathbb{Q}\}, \quad 0 \neq \Gamma \leq_{\mathbb{Q}} \mathbb{R}.$$

The field $\mathcal{L}^{\Gamma}$ has coefficient field k and *left-finite* supports ([section 196](#)). This is a generalized Levi-Civita field; the usual real-coefficient Levi-Civita field is not being silently substituted for this algebraic-coefficient variant. An embedding always preserves 1. A field is *co-Hopfian* when every such self-embedding is onto.

Theorem 195.1 (Rational-rank dichotomy). *The following are equivalent:*

1. $\dim_{\mathbb{Q}} \Gamma < \infty$;
2. $\mathcal{L}^\Gamma$ is co-Hopfian as an abstract field;
3. every isometric field self-embedding of $\mathcal{L}^\Gamma$ is onto.

If $\dim_{\mathbb{Q}} \Gamma = \infty$, there is a proper isometric self-embedding $\Phi : \mathcal{L}^\Gamma \longrightarrow \mathcal{L}^\Gamma$ fixing k and inducing the identity on both Γ and the residue field. Its image is a closed, immediate, real-closed elementary subfield of $\mathcal{L}^\Gamma$.

Here *immediate* means that inclusion changes neither the value group nor the residue field. It does not mean dense. The proof of the finite-rank implication is in [section 199](#); the counterexample and its exact approximation obstruction are in [sections 200](#) and [201](#).

Theorem 195.2 (Topology and abundance). *The valuation topology of $\mathcal{L}^\Gamma$ is completely metrizable and is Polish if and only if Γ is countable. For countably infinite rational rank, $\mathcal{L}^\Gamma$ has $2^{\aleph_0}$ distinct closed proper immediate subfields isomorphic to itself. They can be obtained as images of a jointly continuous family of isometric elementary self-embeddings parametrized by Cantor space.*

The rank in these statements is *rational rank*, not the rank of an ordered group. Every nonzero $\Gamma \subseteq \mathbb{R}$ is Archimedean and has ordered-group rank one. Its dimension over $\mathbb{Q}$ can nevertheless be any finite positive integer or be infinite.

195.2 Contribution ledger and relation to the literature

The established literature supplies generalized-series algebra, valuation theory, and automorphism decompositions; Kuhlmann–Serra is especially close to the monomial-character description [\[133\]](#). The arguments here must separate an embedding from an automorphism. This distinction is material: [section 204](#) gives a counterexample to the as-written converse in that preprint’s Lemma 4.2.4, and then proves a valid monoid formulation for our complete left-finite fields.

The proposed contributions are the exact dichotomy, its quantitative finite-rank proof, the bounded-exponent shift and distance obstruction, the parametrized family of proper immediate self-copies, and the integer-part criterion for surjectivity. Classical real closedness, Presburger constructions from previous reports, and ordinary character twists are not claimed as new. Neither of Glazer’s printed open questions is claimed to be newly resolved by this article. These qualifications concern attribution and scope, not a replacement of proofs by conjecture.

196 Left-finite series and their complete topology

Definition 196.1. A set $S \subseteq \Gamma$ is *left finite* if

$$\#(S \cap (-\infty, B]) < \infty \quad \text{for every } B \in \mathbb{R}.$$

Define

$$\mathcal{L}^\Gamma = \left\{ \sum_{\gamma \in \Gamma} a_\gamma t^\gamma : a_\gamma \in k, \text{ supp}(a) \text{ is left finite} \right\}.$$

For nonzero x , set $v(x) = \min \text{supp}(x)$ and $\text{lc}(x) = a_{v(x)}$; put $v(0) = +\infty$. The order is determined by $x > 0$ if and only if $\text{lc}(x) > 0$.

A nonempty left-finite support has a least element: choose any point of the support and take the minimum of the finite nonempty prefix ending there. It is either finite or has increasing enumeration tending to $+\infty$. In particular, it is countable, even when Γ is uncountable. The field sits in the full Hahn field

$$k[\Gamma] \subseteq k(\Gamma) \subseteq \mathcal{L}^\Gamma \subseteq k((t^\Gamma)) = \mathcal{H}^\Gamma.$$

Here $k[\Gamma]$ is the finite-support group algebra and $k(\Gamma) = \text{Frac}(k[\Gamma])$. The rightmost inclusion is generally strict. A bounded increasing infinite support is well ordered and therefore allowed in $\mathcal{H}^\Gamma$, but forbidden in $\mathcal{L}^\Gamma$. This small distinction will be the entire obstruction in the infinite-rank proof. The classical Levi-Civita literature discusses the left-finite construction and its valuation topology [31].

Proposition 196.2 (Field operations). *Coefficientwise addition and convolution multiplication make $\mathcal{L}^\Gamma$ an ordered field. For nonzero elements,*

$$v(xy) = v(x) + v(y), \quad v(x + y) \geq \min\{v(x), v(y)\}.$$

If $v(x) \neq v(y)$, equality holds in the second formula.

Proof. Unions of finitely many left-finite supports are left finite. If S, T are nonempty such supports, with minima s_0, t_0 , then every pair with $s + t \leq B$ satisfies $s \leq B - t_0$ and $t \leq B - s_0$. There are only finitely many such pairs. Thus convolution coefficients and their left-finite support are well defined. The leading product term is unique and has nonzero coefficient, proving the valuation formula and absence of zero divisors. All ring laws can be checked on these finite coefficient sums.

For $x \neq 0$, write $x = at^\gamma(1 + \varepsilon)$, where $a \in k^\times$ and either $\varepsilon = 0$ or $\delta = v(\varepsilon) > 0$. The expression

$$x^{-1} = a^{-1}t^{-\gamma} \sum_{n \geq 0} (-\varepsilon)^n$$

is left finite. Below any fixed bound, only finitely many powers contribute, since $v(\varepsilon^n) = n\delta$, and each contributing power has left-finite support. Finite geometric identities give the inverse. Finally, a product of positive leading coefficients is positive; the leading-term order is compatible with addition and multiplication. $\square$

Proposition 196.3 (Completion and Polishness). *With $e^{-\infty} = 0$, the formula*

$$d(x, y) = e^{-v(x-y)}$$

defines a complete ultrametric on $\mathcal{L}^\Gamma$. The finite-support group algebra $k[\Gamma]$ is dense. Field operations are continuous, and this topology is also the order topology. It is separable exactly when Γ is countable.

Proof. The valuation laws prove the ultrametric inequality. Truncation at the integer bounds $0, 1, 2, \dots$ approximates every element by finite sums. If (x_n) is Cauchy, then for each integer $B \geq 0$ all coefficients at exponents at most B eventually stabilize simultaneously. The stabilized prefix is a finite prefix of one x_n . These compatible prefixes define a series x with left-finite support, and $v(x_n - x) \rightarrow +\infty$.

Addition is continuous. For multiplication, use $x_n y_n - xy = (x_n - x)y_n + x(y_n - y)$ and the fact that a convergent sequence has valuations bounded below. At $x \neq 0$, sufficiently close y has $v(y) = v(x)$, and

$$v(x^{-1} - y^{-1}) = v(x - y) - 2v(x).$$

This proves continuity of inversion.

For the topology comparison, an order interval $(-r, r)$ with $r > 0$ contains a valuation ball $v(x) > B$ once $B > v(r)$. Conversely, given $B \in \mathbb{R}$, choose $h \in \Gamma$ with $h > B$. Then $(-t^h, t^h) \subseteq \{x : v(x) > B\}$. Such an h exists because the nonzero subgroup Γ is unbounded.

When Γ is countable, so is $k[\Gamma]$. Conversely, the nonempty sets $\{x : v(x) = \gamma\}$, for $\gamma \in \Gamma$, are pairwise disjoint open sets. A separable metric space has only countably many disjoint nonempty open sets, so separability implies countability of Γ . $\square$

Remark 196.4. The ultrametric is intentionally not bounded by 1; this makes the exact scaling and distance identities simpler. Replacing it by $\min\{1, d\}$ would give the same topology and Cauchy sequences. The topology on the constant field k induced by d is discrete, not its usual real topology.

For countable nonzero Γ , the field has cardinality continuum: the upper bound follows by coding coefficient functions on a countable group, and the lower bound follows from binary series $\sum_{n \geq 0} \epsilon_n t^{nh}$ for a fixed $h > 0$. It has no isolated points and is nowhere locally compact. Indeed every neighborhood contains a ball $v(x) > B$; choosing $h > B$, the points mt^h , $m \in \mathbb{N}$, lie in that ball and have constant pairwise distance e^{-h} . A compact closure would be totally bounded, which is impossible. Thus the examples genuinely lie outside locally compact field rigidity phenomena.

[write] (Batch 92.) For $\Gamma = \mathbb{Q}$ these are Part V's [proposition 65.2](#), [theorem 65.3](#) (with the metric $\min\{1, 2^{-v}\}$, the same topology) and [proposition 65.5](#), for Part V's countable coefficient fields; Part V also proves that this topology is the only admissible Polish one. The “Polish exactly when Γ is countable” direction for general Γ is new to the report. Source 20's [lemma 184.3](#) is the same field construction with real coefficients and $T = t^{-1}$.

197 Real closedness and the intrinsic valuation

The following standard valuation-theoretic input is used explicitly. A henselian valued field of residue characteristic zero is defectless: for each finite extension with its unique extended valuation,

$$[E : F] = (vE : vF)[Ev : Fv]. \quad (197.1)$$

This is the residue-characteristic-zero consequence of Ostrowski's lemma; see Kuhlmann [134, Corollary 2.4]. No analogous assertion in arbitrary positive residue characteristic is being assumed.

Proposition 197.1. *The field $\mathcal{L}^\Gamma$ is real closed. Its valuation ring and maximal ideal are*

$$\mathcal{O} = \{x : v(x) \geq 0\}, \quad \mathfrak{m} = \{x : v(x) > 0\},$$

its residue field is k , and v is its natural Archimedean valuation.

Proof. First, completeness for a real-valued valuation gives the simple-root form of Hensel's lemma directly. If $f \in \mathcal{O}[X]$ and $a \in \mathcal{O}$, $f(a)$ has positive value, and $f'(a)$ has value zero, Newton iteration

$$a_{n+1} = a_n - \frac{f(a_n)}{f'(a_n)}$$

keeps $f'(a_n)$ a unit and at least doubles the positive value of $f(a_n)$ at each nonterminal step. Taylor expansion proves these assertions. The corrections have valuations tending to

$+\infty$, so completeness produces the required root. The same argument applies to complex coefficients.

Let $F = \mathcal{L}^\Gamma(i)$. Splitting coefficients into real and imaginary parts identifies F with the left-finite series field over $k(i)$: the union of two left-finite supports is left finite. Thus F is complete and henselian; its residue field $k(i)$ is algebraically closed and its value group Γ is divisible. For a finite extension E/F , the residue degree is one. The ramification index is also one. Indeed, if $n\eta \in \Gamma$ for an element of a finite-index ordered-group extension, divisibility supplies $\gamma \in \Gamma$ with $n\gamma = n\eta$, and torsion-freeness forces $\eta = \gamma$. Equation (197.1) gives $[E : F] = 1$. Hence F is algebraically closed. Since $\mathcal{L}^\Gamma$ is ordered, it is real closed.

The residue map on $\mathcal{O}$ is the coefficient at exponent zero. The coefficient field k is Archimedean. Consequently $v(x) > 0$ exactly when $|x| < 1/n$ for every positive integer n , and $v(x) \geq 0$ exactly when $|x|$ is bounded by an ordinary integer. This identifies v with the natural valuation, with the displayed value-group coordinates. $\square$

[write] (Batch 92.) For $\Gamma = \mathbb{Q}$ this is Part V's [theorem 68.2](#), proved there by Newton–Puiseux and completion; the defect argument here works for every divisible $\Gamma \leq \mathbb{R}$.

Lemma 197.2 (Principal rational powers). *The multiplicative group $1 + \mathfrak{m}$ is uniquely divisible. If $u \in 1 + \mathfrak{m}$, $q \in \mathbb{Q}$, and $v(u - 1) \geq \delta > 0$, then*

$$v(u^q - 1) \geq \delta. \quad (197.2)$$

Proof. For $u = 1 + \varepsilon$, the binomial series

$$(1 + \varepsilon)^q = \sum_{n \geq 0} \binom{q}{n} \varepsilon^n$$

converges in $\mathcal{L}^\Gamma$ because $v(\varepsilon) > 0$. Its finite formal identities prove the rational-power identities; alternatively these follow from uniqueness of the positive roots in an ordered field. It belongs to $1 + \mathfrak{m}$ and satisfies (197.2). For uniqueness, if $n \geq 1$, $y, z \in 1 + \mathfrak{m}$, and $y^n = z^n$, then $(y - z) \sum_{j=0}^{n-1} y^j z^{n-1-j} = 0$. The second factor has residue $n \neq 0$, hence $y = z$. $\square$

[write] The uniqueness half is formalized for Hahn series in `Algebra/SurrealNumbers/Surreal/HahnSeries/BinomialRoots.lean` (`pow_injective_near_one`, `binomialPower_rat_root_unique`, `exists_unique_root_near_one`); the left-finite form and the bound (197.2) are not, as the next paragraph says.

The root-uniqueness mechanism already appears in `ProveIt's BinomialRoots.lean` [138]. The additional requirement here is that every constructed series remain left finite; a theorem about a full Hahn field alone does not supply that support condition.

198 All embeddings: automatic continuity and monomial data

Theorem 198.1 (Automatic valuation scaling). *Every field embedding $\phi : \mathcal{L}^\Gamma \rightarrow \mathcal{L}^\Delta$, where $0 \neq \Gamma, \Delta \leq \mathbb{Q} \mathbb{R}$, fixes k pointwise. There is a unique $\lambda > 0$ with $\lambda\Gamma \subseteq \Delta$ such that*

$$v(\phi x) = \lambda v(x), \quad d(\phi x, \phi y) = d(x, y)^\lambda. \quad (198.1)$$

In particular, ϕ is continuous and its image is closed.

Proof. A field embedding between real-closed fields preserves order because positive elements are precisely nonzero squares. It fixes the prime field. A real algebraic number is specified among the roots of its polynomial by a rational isolating interval, so it is fixed as well.

An order embedding fixing the integers preserves and reflects the Archimedean comparison of any two nonzero elements: the assertions $|x/y| \leq n$ for ordinary n are preserved and reflected individually. It therefore induces an injective order-preserving additive map $h : \Gamma \rightarrow \Delta$ with $h(vx) = v(\phi x)$. Choose $\beta \in \Gamma_{>0}$ and set $\lambda = h(\beta)/\beta$. For every $\gamma \in \Gamma$ and $q \in \mathbb{Q}$,

$$q\beta < \gamma \iff qh(\beta) < h(\gamma).$$

The real numbers γ/β and $h(\gamma)/h(\beta)$ have the same rational lower cut and are equal. Thus $h(\gamma) = \lambda\gamma$. This proves (198.1) and uniqueness.

Finally, a convergent sequence of image points is Cauchy; the second identity in (198.1) makes its preimages Cauchy. Completeness of $\mathcal{L}^\Gamma$ gives a preimage for the limit. In a metric space this proves that the image is closed. $\square$

The conclusion is stronger than Borel automatic continuity: no regularity hypothesis is made on the field homomorphism in the first place. Its source is order definability together with the Archimedean coefficient field.

[write] (Batch 92.) The value-group part is the Archimedean case of `sse:prop:valueaction` in the surreal self-embeddings report (for self-embeddings of **No**), which the Cantor-families report proves for arbitrary ordered fields (`csf:lem:valuation-functor`); the continuity part is an instance of that report's `sse:an:top:criterion` (an embedding of ordered fields with cofinal image is continuous for the order topologies), here with the order topology equal to the valuation topology (proposition 196.3). None of these labels is formalized.

Theorem 198.2 (Complete embedding parametrization). *Field embeddings $\phi : \mathcal{L}^\Gamma \rightarrow \mathcal{L}^\Delta$ correspond exactly to triples*

$$\lambda > 0, \quad \lambda\Gamma \subseteq \Delta, \quad c \in \text{Hom}(\Gamma, k_{>0}^\times), \quad u \in \text{Hom}(\Gamma, 1 + \mathfrak{m}_\Delta).$$

The correspondence is

$$\phi(t^\gamma) = c(\gamma)t^{\lambda\gamma}u(\gamma), \quad \phi\left(\sum a_\gamma t^\gamma\right) = \sum a_\gamma c(\gamma)t^{\lambda\gamma}u(\gamma). \quad (198.2)$$

Every triple in the display defines an embedding; surjectivity is a separate condition.

Proof. For a given embedding, take $c(\gamma)$ to be the leading coefficient of $\phi(t^\gamma)$ and divide off its leading monomial to define $u(\gamma)$. Positivity and multiplication of leading coefficients show that c and u are characters. Continuity and density of finite sums give the second formula.

Conversely, the first formula defines a unital k -algebra homomorphism on $k[\Gamma]$. Distinct monomials have distinct image valuations, so a nonzero finite sum has image valuation exactly λ times its original valuation. Thus this homomorphism is injective and extends uniquely to the completion. Addition and multiplication pass to limits, giving a field embedding. There is no hidden support difficulty: below an output bound B only source exponents $\gamma \leq B/\lambda$ can contribute; there are finitely many, and each corresponding summand is left finite. This also proves the displayed series formula directly. $\square$

The coefficient character is an ordinary twist. `ProveIt's characterTwist` constructs that factor with its reciprocal-character inverse [137]. The principal-unit character is fundamentally

different: substitution acts on its coefficients as series, and taking pointwise reciprocals is not generally a compositional inverse.

[write] `characterTwist`, `support_twist` and `twist_inverse` are in `Algebra/SurrealNumbers/Surreal/HahnSeries/CharacterTwist.lean`, which is unchanged since the `pin; docs/FORMALIZATION.md` maps them to `odg:def:lem:twist` of the omnific Diophantine report, for full Hahn series. The left-finite statements of this part are not formalized.

199 Finite rational rank: a uniform gap forces an inverse

Lemma 199.1 (Quantitative Neumann lemma). *Let $T : \mathcal{L}^\Gamma \rightarrow \mathcal{L}^\Gamma$ be additive. Suppose that for some $\delta > 0$,*

$$v(Tx) \geq v(x) + \delta \quad (x \neq 0). \quad (199.1)$$

Then $\text{id} + T$ is bijective, with inverse

$$Sx = \sum_{n=0}^{\infty} (-1)^n T^n x. \quad (199.2)$$

For $S_N x = \sum_{n=0}^N (-1)^n T^n x$,

$$v(Sx - S_N x) \geq v(x) + (N + 1)\delta. \quad (199.3)$$

Proof. Iteration gives $v(T^n x) \geq v(x) + n\delta$, so the series converges by completeness and has the stated tail bound. Condition (199.1) also gives continuity of T . Finite telescoping yields

$$(\text{id} + T)S_N x = x + (-1)^N T^{N+1} x, \quad S_N(\text{id} + T)x = x + (-1)^N T^{N+1} x.$$

Taking limits proves both inverse identities. Only additivity is needed for the finite sums and their signs. $\square$

Theorem 199.2 (Finite-rank surjectivity). *If $\dim_{\mathbb{Q}} \Gamma = d < \infty$, every field self-embedding of $\mathcal{L}^\Gamma$ is surjective. After normalization of its leading monomials, its inverse has the quantitative form (199.2)–(199.3).*

Proof. Let (λ, c, u) be the parameters of an embedding ϕ . The injective $\mathbb{Q}$ -linear map $\gamma \mapsto \lambda\gamma$ on the finite-dimensional space Γ is onto. Consequently the pure leading-monomial map

$$M\left(\sum a_\gamma t^\gamma\right) = \sum a_\gamma c(\gamma) t^{\lambda\gamma}$$

is an automorphism: its inverse rescales exponents by λ^{-1} and divides their coefficients by c . Set $\tau = M^{-1}\phi$. It fixes k , preserves values and leading coefficients, and has the form $\tau(t^\gamma) = t^\gamma w(\gamma)$ with $w(\gamma) \in 1 + \mathfrak{m}$.

Choose a $\mathbb{Q}$ -basis $b_1, \dots, b_d$ of Γ . If all $w(b_i) = 1$, then $\tau = \text{id}$. Otherwise let

$$\delta = \min\{v(w(b_i) - 1) : w(b_i) \neq 1, 1 \leq i \leq d\} > 0.$$

For $\gamma = \sum_i q_i b_i$, unique divisibility gives $w(\gamma) = \prod_i w(b_i)^{q_i}$. By lemma 197.2, every factor is 1 modulo valuation at least δ , and the same is true of the product. Thus

$$v(\tau(t^\gamma) - t^\gamma) \geq \gamma + \delta.$$

For a left-finite sum x , the termwise formula from theorem 198.2 gives

$$v((\tau - \text{id})x) \geq v(x) + \delta.$$

Apply lemma 199.1 with $T = \tau - \text{id}$. Then τ is bijective, so $\phi = M\tau$ is bijective, with inverse SM^{-1} . $\square$

199.1 The scalar multiplier field

Proposition 199.3. *For finite positive rational rank, the ring*

$$E_\Gamma = \{a \in \mathbb{R} : a\Gamma \subseteq \Gamma\}$$

is a number field embedded in $\mathbb{R}$, and $[E_\Gamma : \mathbb{Q}]$ divides $\dim_{\mathbb{Q}} \Gamma$. The possible valuation scalings of self-embeddings are precisely $(E_\Gamma)_{>0}$.

Proof. The set E_Γ is a subring containing $\mathbb{Q}$. Each nonzero element acts bijectively on the finite-dimensional space Γ , so its inverse belongs to E_Γ . For fixed $b \neq 0$ in Γ , the map $a \mapsto ab$ embeds E_Γ linearly into Γ ; hence its extension degree over $\mathbb{Q}$ is finite. Regard Γ as a vector space over this field to obtain the divisibility of dimensions. The statement about scalings follows from [theorem 198.2](#), using $c = u = 1$ for sufficiency. $\square$

This need not be a totally real number field: the assertion concerns its chosen embedding in $\mathbb{R}$. Examples include $E_{\mathbb{Q}} = \mathbb{Q}$ and $E_{\mathbb{Q} + \mathbb{Q}\sqrt{2}} = \mathbb{Q}(\sqrt{2})$. If α is transcendental and $\Gamma = \mathbb{Q} + \mathbb{Q}\alpha$, then $E_\Gamma = \mathbb{Q}$: writing $a = r + s\alpha$ and requiring $a\alpha \in \mathbb{Q} + \mathbb{Q}\alpha$ forces $s = 0$.

199.2 A concrete inverse and its precision

For $\Gamma = \mathbb{Q}$, consider $\tau(t) = t + t^2 = t(1 + t)$. Unique positive rational powers determine its values on t^q , and completion determines it everywhere. Here one may take $\delta = 1$. In particular,

$$\tau^{-1}(t) = \frac{\sqrt{1+4t} - 1}{2} = t - t^2 + 2t^3 - 5t^4 + 14t^5 - 42t^6 + \dots$$

The displayed square root has constant coefficient 1. More generally, to determine the normalized inverse of x to error valuation greater than a prescribed B , it is enough to take N with $v(x) + (N + 1)\delta > B$ in [\(199.2\)](#). This is a certificate of required iteration depth, not a blanket computational complexity claim: computing a single substitution still depends on how its series and exponents are represented.

200 Infinite rational rank: a proper isometric self-embedding

Lemma 200.1 (Independent exponents approaching a finite bound). *If $\dim_{\mathbb{Q}} \Gamma = \infty$, there are $\beta \in \Gamma_{>0}$ and $\gamma_0 < \gamma_1 < \dots < \beta$ in Γ such that $\gamma_n \rightarrow \beta$ in $\mathbb{R}$ and $\{\beta, \gamma_0, \gamma_1, \dots\}$ is $\mathbb{Q}$ -linearly independent.*

Proof. Fix any $\beta > 0$. At stage n , choose a nonzero element outside the finite-dimensional span of $\beta, \gamma_0, \dots, \gamma_{n-1}$. Change its sign if necessary and multiply it by a positive rational number so that the resulting γ_n lies in

$$(\beta(1 - 2^{-n-1}), \beta(1 - 2^{-n-2})).$$

Rational multiples of a positive real meet every positive open interval. Scaling does not destroy the required independence. The chosen intervals are successive and converge to β . $\square$

Extend the independent set to a $\mathbb{Q}$ -basis $\mathcal{B}$ of Γ . Define a character $u : \Gamma \rightarrow 1 + \mathfrak{m}$ on that basis by

$$u(\gamma_n) = 1 + t^{\gamma_{n+1} - \gamma_n}, \quad u(b) = 1 \quad (b \in \mathcal{B} \setminus \{\gamma_n : n \in \mathbb{N}\}). \quad (200.1)$$

Unique divisibility extends it to all of Γ . By [theorem 198.2](#), it defines an isometric field self-embedding

$$\Phi(t^{\gamma_n}) = t^{\gamma_n} + t^{\gamma_{n+1}}, \quad \Phi(t^b) = t^b \quad (b \text{ in the remaining basis}). \quad (200.2)$$

In particular $\Phi(t^\beta) = t^\beta$.

Theorem 200.2 (Bounded-exponent shift). *The embedding Φ of (200.2) is not surjective: $t^{\gamma_0} \notin \Phi(\mathcal{L}^\Gamma)$. It fixes the coefficient field and induces the identity on the residue field and value group.*

Proof. Set

$$s_N = \sum_{j=0}^N (-1)^j t^{\gamma_j}.$$

A finite telescoping calculation, with its boundary term retained, gives

$$\Phi(s_N) = t^{\gamma_0} + (-1)^N t^{\gamma_{N+1}}. \quad (200.3)$$

If $\Phi(x) = t^{\gamma_0}$, isometry would imply

$$v(x - s_N) = \gamma_{N+1} \quad \text{for every } N.$$

Thus for every j , the coefficient of x at γ_j would be $(-1)^j$: choose any $N \geq j$ and compare below γ_{N+1} . This forces infinitely many nonzero coefficients below β , contradicting left finiteness. The assertions about invariants follow from the leading monomial t^γ in every $\Phi(t^\gamma)$. $\square$

Corollary 200.3. [Theorem 195.1](#) holds. The image $F = \Phi(\mathcal{L}^\Gamma)$ is a closed proper immediate real-closed subfield. It is elementary in $\mathcal{L}^\Gamma$ in the language of ordered fields, and

$$\mathcal{L}^\Gamma \supsetneq \Phi(\mathcal{L}^\Gamma) \supsetneq \Phi^2(\mathcal{L}^\Gamma) \supsetneq \dots$$

is a strictly decreasing chain of closed immediate elementary self-copies.

Proof. Finite rank implies co-Hopfianity by [theorem 199.2](#); co-Hopfianity implies surjectivity of isometric self-embeddings; infinite rank violates the latter by [theorem 200.2](#). Closedness follows from [theorem 198.1](#); immediate and real closed follow from the construction and isomorphism. The elementary-subfield conclusion uses the classical quantifier elimination for real-closed ordered fields (the associated model-completeness facts are recalled in [134, §2]): quantifier-free formulas with parameters in F are absolute between F and $\mathcal{L}^\Gamma$. If two consecutive iterated images were equal, injectivity of the preceding iterate would imply $\Phi(\mathcal{L}^\Gamma) = \mathcal{L}^\Gamma$, a contradiction. $\square$

200.1 Why the formal inverse does not converge

Put $T = \Phi - \text{id}$. For every nonzero x , $v(Tx) > v(x)$, since Φ preserves its leading term. Nevertheless,

$$T^n(t^{\gamma_0}) = t^{\gamma_n}, \quad v(T^n(t^{\gamma_0})) = \gamma_n \longrightarrow \beta < \infty.$$

The formal Neumann inverse of t^{γ_0} is therefore $\sum_{n \geq 0} (-1)^n t^{\gamma_n}$. Its support is well ordered, so it is an element of the full Hahn field, but it is not an element of $\mathcal{L}^\Gamma$ and its partial sums are not Cauchy in the valuation metric. Completeness never promises limits for non-Cauchy sequences.

The operator T strictly decreases every nonzero distance but has no uniform contraction factor less than 1. The missing hypothesis in a Banach-type inverse argument is exactly the uniform δ in (199.1). We do not assert that the substitution Φ extends to a well-defined map on every series of the full Hahn field; that would require additional summability work.

201 An exact approximation gap and no nearest image point

Theorem 201.1 (Unattained distance). *For $F = \Phi(\mathcal{L}^\Gamma)$ and $y = t^{\gamma_0}$,*

$$v(y - z) < \beta \quad (z \in F), \quad \sup_{z \in F} v(y - z) = \beta. \quad (201.1)$$

Consequently

$$\inf_{z \in F} d(y, z) = e^{-\beta} > 0,$$

and no $z \in F$ attains that distance.

Proof. Suppose $z = \Phi(x)$ and $v(y - z) \geq \beta$. Since $\gamma_{N+1} < \beta$, subtracting (200.3) from z shows that $v(z - \Phi(s_N)) = \gamma_{N+1}$. Isometry again forces $v(x - s_N) = \gamma_{N+1}$ for every N , and the same coefficient argument contradicts left finiteness. Thus all values in (201.1) are strictly below β . On the other hand, $z_N = \Phi(s_N)$ has $v(y - z_N) = \gamma_{N+1} \rightarrow \beta$. Taking exponentials proves the metric assertions and nonattainment. $\square$

This supplies a complete closed subfield with no nearest point to a specified outside element. It also separates three notions that can otherwise be confused: equal valuation invariants, immediate extension, and valuation density. The image is immediate because it can match any leading term. Matching arbitrarily long valuation prefixes is a stronger requirement, and (201.1) is its exact failure.

Corollary 201.2 (An order-scale gap). *Let $y' = t^{\gamma_0 - 2\beta}$. For every $z \in F$, the difference $y' - z$ has negative valuation and is unbounded in absolute value by ordinary integers. In particular, no element of F is within order distance 1 of y' .*

Proof. The element $t^{2\beta}$ is fixed by Φ , so multiplication by it permutes F . Hence

$$v(y' - z) = v(y - t^{2\beta}z) - 2\beta < -\beta < 0.$$

The natural-valuation characterization from [proposition 197.1](#) finishes the proof. $\square$

The support obstruction can also be expressed as a nested-ball obstruction. The closed balls centered at s_N with radii $e^{-\gamma_{N+1}}$ are nested and have empty intersection. Their radii approach the positive number $e^{-\beta}$. This does not contradict metric completeness, which concerns radii tending to zero. Nor is failure of spherical completeness itself the rank criterion: bounded rational-exponent supports already witness that failure in rank one. The independent exponent chain is what turns it into a field self-embedding obstruction.

202 Continuum many closed immediate self-copies

Assume now that Γ is countable and has infinite rational rank. A diagonal version of [lemma 200.1](#) supplies an independent family

$$\{\beta\} \cup \{\gamma_{n,j} : n, j \in \mathbb{N}\}$$

such that, for each fixed n , the sequence $\gamma_{n,j}$ increases to β . To construct it, enumerate pairs (n, j) in an order in which $(n, j-1)$ precedes (n, j) , and at each finite stage choose outside the span of all previous choices in the interval used for index j . Extend this family to a basis.

For $S \subseteq \mathbb{N}$, shift the chain with first index n exactly when $n \in S$, and fix all other basis elements. Explicitly,

$$\Phi_S(t^{\gamma_{n,j}}) = \begin{cases} t^{\gamma_{n,j}} + t^{\gamma_{n,j+1}}, & n \in S, \\ t^{\gamma_{n,j}}, & n \notin S. \end{cases}$$

All other basis monomials are fixed. Principal rational powers and [theorem 198.2](#) define the resulting isometric field embedding.

Theorem 202.1 (A Cantor-parametrized family). *Let $F_S = \Phi_S(\mathcal{L}^\Gamma)$. Then*

$$t^{\gamma_{n,0}} \in F_S \iff n \notin S. \quad (202.1)$$

Thus all the fields F_S are distinct; they are closed, immediate, real closed, and isomorphic to $\mathcal{L}^\Gamma$. Every nonempty S gives a proper subfield. The map

$$2^{\mathbb{N}} \times \mathcal{L}^\Gamma \longrightarrow \mathcal{L}^\Gamma, \quad (S, x) \longmapsto \Phi_S(x)$$

is jointly continuous. The range map $S \mapsto F_S$ is Borel in the Effros space of closed subsets and is continuous for the lower Vietoris topology.

Proof. If $n \notin S$, the indicated monomial is fixed. If $n \in S$, the finite telescoping argument along that chain, together with isometry, forces a forbidden bounded infinite support for any putative preimage. Other chains do not affect this argument. This proves (202.1), hence distinctness and properness. The remaining algebraic properties follow as before.

For continuity, fix (S, x) and an error threshold. Approximate x by a finite-support element p to an error with sufficiently large valuation. Each exponent in p is a finite rational combination of basis elements, so $\Phi_{S'}(p) = \Phi_S(p)$ whenever S' agrees with S on finitely many chain indices. Isometry bounds the images of $x - p$ uniformly in S' . Together with $d(\Phi_{S'}(x'), \Phi_{S'}(x)) = d(x', x)$, this proves joint continuity.

Let $D = k[\Gamma]$, a countable dense set. For an open subset $U \subseteq \mathcal{L}^\Gamma$,

$$\{S : F_S \cap U \neq \emptyset\} = \bigcup_{p \in D} \{S : \Phi_S(p) \in U\}.$$

Each set on the right is open. These are precisely the hit-open tests for the lower Vietoris topology and the generators of the Effros Borel structure. $\square$

Restricting to all S containing 0 leaves continuum many proper images, proving the abundance assertion in [theorem 195.2](#). No continuity claim for the full Vietoris topology is needed or made. No assertion about mutual inclusion of F_S and F_T is inferred merely from inclusion of S and T .

203 Embeddings between different exponent groups

Theorem 203.1 (Embedding and isomorphism classification). *For nonzero $\mathbb{Q}$ -subspaces $\Gamma, \Delta \subseteq \mathbb{R}$,*

$$\begin{aligned}\mathcal{L}^\Gamma \hookrightarrow \mathcal{L}^\Delta &\iff \exists \lambda > 0 (\lambda \Gamma \subseteq \Delta), \\ \mathcal{L}^\Gamma \cong \mathcal{L}^\Delta &\iff \exists \lambda > 0 (\lambda \Gamma = \Delta).\end{aligned}$$

The symbols on the left refer to abstract unital field maps; all such maps automatically have the topological regularity of [theorem 198.1](#). If Γ and Δ have the same finite rational rank, every embedding between their fields is onto.

Proof. Necessity follows from the intrinsic valuation map. For an isomorphism, that map is onto. Sufficiency is given by pure monomial rescaling with $c = u = 1$. In the equal finite-rank case, any injective scalar map $\Gamma \rightarrow \Delta$ is onto; normalize the field embedding by its leading-monomial isomorphism and apply the finite-basis proof of [theorem 199.2](#). $\square$

Example 203.2 (Continuum many co-Hopfian Polish fields). For irrational α , set $\Gamma_\alpha = \mathbb{Q} + \mathbb{Q}\alpha$. Two such groups are equivalent up to positive scalar multiplication exactly when their parameters are related by an invertible rational fractional-linear transformation:

$$\alpha' = \frac{a + b\alpha}{c + d\alpha}, \quad a, b, c, d \in \mathbb{Q}, \quad ad - bc \neq 0.$$

Indeed, a rescaled basis $\{1, \alpha'\}$ is a basis of $\mathbb{Q} + \mathbb{Q}\alpha$ precisely when its two columns form an invertible rational matrix. A negative scale can be changed to positive because $-\Gamma_\alpha = \Gamma_\alpha$. Each orbit is countable, so there are continuum many isomorphism types among the rank-two fields $\mathcal{L}^{\Gamma_\alpha}$. Every one is a co-Hopfian Polish real-closed field. They have the same first-order theory but need not be isomorphic.

Example 203.3 (Two independent failures of surjectivity). Let $\alpha > 0$ be transcendental. On $\Gamma = \mathbb{Q}[\alpha]$, $\gamma \mapsto \alpha\gamma$ has proper image, so $t^\gamma \mapsto t^{\alpha\gamma}$ is a proper self-embedding already detected on the value group. By contrast, for $\Gamma = \mathbb{Q}(\alpha)$, any nonzero multiplier preserving Γ is in that field and acts onto it. Nevertheless [theorem 200.2](#) still gives a proper *immediate* self-embedding. Thus co-Hopfianity of the value group under scalar maps alone is not the required criterion.

204 One-sided summability is not surjectivity

This section records a version-specific literature issue because it bears directly on the logical step needed in the preceding proofs. In arXiv:2107.03362v3, Definition 4.2.3 calls a unit character u K -summable when each monomial expansion, after twisting by u , is a summable family with sum in K . Lemma 4.2.4 identifies all such characters with ones arising from strongly additive internal *automorphisms* [133, p. 24]. The following elementary example shows that this one-sided condition, as written, is insufficient.

Proposition 204.1 (Rational-function counterexample). *Take $K = k(t) \subseteq k((t^\mathbb{Z}))$ in characteristic zero and set $u(n) = (1 + t)^n$. For every $f \in K$, its Laurent expansion after the termwise replacement $t^n \mapsto t^n(1 + t)^n$ is Hahn summable and has sum in K . The resulting map*

$$\sigma : K \longrightarrow K, \quad f(t) \longmapsto f(t + t^2)$$

is a valuation-preserving embedding, but is not onto.

Proof. A rational function has a Laurent expansion with integer exponents bounded below. The substituted n th monomial has least exponent n . Thus each coefficient receives finitely many contributions, and the union of supports is still bounded below in $\mathbb{Z}$. Formal composition with $t + t^2$, of value one, is therefore well defined. Applying finite polynomial identities to numerator and denominator shows that its sum is exactly $f(t + t^2) \in k(t)$. Its leading term is unchanged.

The automorphism ρ of the rational function field given by $\rho(t) = -1 - t$ fixes $t + t^2$ but does not fix t . Every element of $k(t + t^2)$ is fixed by ρ , so $t \notin k(t + t^2)$. This is exactly the image of σ , proving non-surjectivity. $\square$

The example is also truncation closed, and the usual coefficient and exponent lifting maps preserve $k(t)$. Thus the issue is not repaired by those elementary restrictions. Its completion $k((t))$ does admit the inverse from [section 199](#). More significantly, the infinite-rank example in [theorem 200.2](#) is already complete and real closed: even those additional hypotheses do not make one-sided summability sufficient in arbitrary rational rank.

This is a correction to the cited preprint statement under its displayed definition, not an audit of the final journal text or a claim that all other results of the paper fail. No external confirmation of the correction or exhaustive search for prior corrections is asserted.

Remark 204.2 (Lemma 4.2.4 of the Kuhlmann–Serra preprint fails as stated [write]). This report checked [proposition 204.1](#) against the text of arXiv:2107.03362v3 (11 April 2022) [133] on 4 October 2026. There k is a field, G a totally ordered abelian group, and a *Hahn field* is any field K with $k(G) \subseteq K \subseteq k((G))$ (Definition 2.1.1). A family of series is *summable* if the union of its supports is well ordered and each exponent lies in only finitely many of the supports (Definition 4.0.1), and *K -summable* if moreover its sum lies in K (Definition 4.0.3); $U = U_K$ is the unit group of the valuation ring of K . By Definition 4.2.3, $u \in \text{Hom}(G, U)$ is *K -summable* if for every $a = \sum a_g t^g \in K$ the family $\{a_g u(g) t^g : g \in \text{supp } a\}$ is K -summable; these form $\text{Hom}^+(G, U)$. The map S of (4.4) sends an internal, strongly additive k -automorphism σ (Definition 4.0.5: σ and σ^{-1} are K -summable maps) to $u_\sigma(g) = t^{-g} \sigma(t^g)$. Lemma 4.2.4 (printed p. 24) states $\text{im } S = \text{Hom}^+(G, U)$. Its proof of “ $\supseteq$ ” defines $\sigma(\sum a_g t^g) = \sum a_g u(g) t^g$ and asserts that σ is a well-defined strongly additive valued k -automorphism, without proving that σ is onto or that σ^{-1} is K -summable. The inclusion “ $\subseteq$ ” is correct.

Counterexample. Let k be any field, $G = \mathbb{Z}$ and $K = k(t) = k(G)$, a Hahn field in this sense, which moreover has the preprint’s first lifting property (its Example 3.1.4(ii)). Put $u(n) = (1+t)^n \in U$. For $f \in k(t)$ with Laurent expansion $\sum_{n \geq n_0} a_n t^n$, the family $\{a_n (1+t)^n t^n\}$ has supports in $[n, \infty)$, so the union of its supports is bounded below and each exponent m lies only in the supports with $n_0 \leq n \leq m$: the family is summable. Its sum is $f(t + t^2)$, because substitution of $t + t^2$, of value one, is a continuous k -algebra endomorphism of $k((t))$ for the t -adic topology. As $f(t + t^2) \in k(t)$, $u \in \text{Hom}^+(\mathbb{Z}, U)$. If $u = u_\sigma$ for some σ in the domain of S , then $\sigma(t) = u(1)t = t + t^2$, so σ and $f \mapsto f(t + t^2)$ are k -homomorphisms of $k(t)$ that agree on t , hence everywhere, and the image of σ is $k(t + t^2)$. But the k -automorphism ρ of $k(t)$ with $\rho(t) = -1 - t$ fixes $t + t^2$, since $(-1 - t) + (-1 - t)^2 = t + t^2$, hence fixes $k(t + t^2)$ pointwise, while $\rho(t) - t = -1 - 2t \neq 0$ in every characteristic. So $t \notin k(t + t^2)$ (indeed $[k(t) : k(t + t^2)] = 2$), σ is not onto, and $u \in \text{Hom}^+(\mathbb{Z}, U) \setminus \text{im } S$. Thus $\text{im } S \neq \text{Hom}^+(G, U)$ for this Hahn field. [Proposition 204.1](#) is this example, stated by source 21 in characteristic zero; the argument does not use the characteristic.

A complete, real closed counterexample. For $k = \mathbb{R}_{\text{alg}}$, a $\mathbb{Q}$ -subspace $\Gamma \leq \mathbb{R}$ of infinite dimension and $K = \mathcal{L}^\Gamma$, which is a Hahn field in the same sense, the character u of (200.1) is also in $\text{Hom}^+(\Gamma, U)$: for $a \in \mathcal{L}^\Gamma$ only finitely many $\gamma \in \text{supp } a$ lie below any bound, and each contributes finitely many exponents below it, so the family $\{a_\gamma u(\gamma) t^\gamma\}$ has left-finite, hence

well-ordered, support union with finite fibres, and its sum $\Phi(a)$ lies in $\mathcal{L}^\Gamma$ (theorem 198.2); but Φ is not onto (theorem 200.2).

Scope. The preprint's Proposition 4.2.6 and Corollary 4.2.7 cite Lemma 4.2.4 for the bijectivity of S ; which of their consequences survive is not determined here. The journal version (*J. Algebra* 605 (2022)) was not checked, and nothing here says that the lemma fails for the maximal Hahn field $k((G))$ or under further hypotheses. Six other reports of the collection cite the preprint (the surreal self-embeddings, omnific-preserving automorphisms, surcomplex field automorphisms, entire-functions, hidden-negative-Hermitian and large-cardinal reports); every citation was read, and none uses the inclusion " $\supseteq$ " of Lemma 4.2.4: they cite the decomposition of given automorphisms, the definitions, or the general theory. The remark was written for this report and has not been independently reviewed.

204.1 A valid substitution monoid

For $\mathcal{L}^\Gamma$, let

$$H_\Gamma = \text{Hom}(\Gamma, 1 + \mathfrak{m}),$$

and write $\Phi_u(t^\gamma) = t^\gamma u(\gamma)$. By theorem 198.2, every element of H_Γ is forward summable and defines an embedding. Define

$$(u \star w)(\gamma) = u(\gamma) \Phi_u(w(\gamma)). \quad (204.1)$$

Proposition 204.3 (Monoid, not automatically group). *The operation $\star$ makes H_Γ a monoid, and $u \mapsto \Phi_u$ is an isomorphism onto the monoid of field self-embeddings fixing k and preserving every leading monomial including its coefficient. Its units correspond exactly to automorphisms. The monoid is a group if and only if $\dim_{\mathbb{Q}} \Gamma < \infty$.*

Proof. The right side of (204.1) is a character with values in $1 + \mathfrak{m}$. On monomials,

$$(\Phi_u \circ \Phi_w)(t^\gamma) = t^\gamma u(\gamma) \Phi_u(w(\gamma)) = \Phi_{u \star w}(t^\gamma).$$

Density proves the identity everywhere. The parametrization is injective, so associativity and the identity character follow from composition. If Φ_u is onto, its inverse also fixes coefficients and preserves leading monomials, hence is Φ_w for a unique $w \in H_\Gamma$. Conversely a two-sided monoid inverse gives a field inverse. The last assertion follows from the finite-rank theorem and the nonunit character in (200.1). $\square$

Thus an automorphism decomposition must use the invertible substitution characters, or supply an actual inverse theorem. The uniform-gap criterion is one such theorem; forward summability alone is not.

205 Integer parts detect proper closed images

Definition 205.1. An *integer part* of an ordered field L is a discretely ordered subring $J \subseteq L$, with least positive element 1, such that for every $x \in L$ there is $j \in J$ satisfying $j \leq x < j + 1$.

205.1 The canonical left-finite integer part

Consider

$$\mathcal{I}^\Gamma = \mathbb{Z} + \left\{ \sum_{\gamma < 0}^{\text{finite}} a_\gamma t^\gamma : a_\gamma \in k \right\}. \quad (205.1)$$

Its construction is classical in this series setting; the prior project report and ProveIt's omnific floor development already discuss the negative-infinitesimal boundary correction [135, 136]. We include the short proof to fix the convention used in the new image-detection theorem.

Proposition 205.2. *The ring $\mathcal{I}^\Gamma$ is an integer part of $\mathcal{L}^\Gamma$. It is closed and discrete in the valuation topology, and is countable when Γ is countable.*

Proof. Finite negative supports are closed under sum and convolution, so (205.1) is a subring. A nonconstant nonzero element has negative valuation and is infinite in absolute value; an element with no negative terms is an ordinary integer. Hence 1 is the least positive element.

Write uniquely

$$x = p_- + a_0 + \varepsilon, \quad p_- \text{ a finite negative-exponent sum}, \quad a_0 \in k, \quad v(\varepsilon) > 0$$

with $\varepsilon = 0$ allowed. Let $m = \lfloor a_0 \rfloor$ in the usual Archimedean sense. Then

$$\lfloor x \rfloor_{\mathcal{I}^\Gamma} = \begin{cases} p_- + m - 1, & a_0 \in \mathbb{Z} \text{ and } \varepsilon < 0, \\ p_- + m, & \text{otherwise.} \end{cases} \quad (205.2)$$

If a_0 is nonintegral, its positive distances from the two adjacent integers dominate the infinitesimal ε . At an integral constant, its sign determines the correction. These observations prove $j \leq x < j + 1$, and discreteness proves uniqueness.

Distinct elements of $\mathcal{I}^\Gamma$ have difference of valuation at most zero, so their metric distance is at least 1. This proves discreteness; a convergent sequence in $\mathcal{I}^\Gamma$ is eventually constant, proving closedness. Countability follows from finite support over two countable sets. $\square$

[write] (Batch 92.) For $\Gamma = \mathbb{Q}$ this is Part V's common integer part $\mathcal{I}_k$ with $k = \mathbb{R}_{\text{alg}}$ (theorem 70.1), which is isg's Shepherdson ring; its floor is the collection's `odg:thm:floor`, proved in Lean, and source 20's theorem 184.4 is the mirrored statement with real coefficients. For countable Γ , Part V's theorem 62.3 already gives that every integer part of the Polish field $\mathcal{L}^\Gamma$ is countable and closed discrete.

205.2 A general closed-subfield obstruction

Theorem 205.3 (Integer parts force density). *Let $E \subseteq \mathcal{L}^\Gamma$ be a subfield whose value group is cofinal in Γ . If E contains an integer part J of all of $\mathcal{L}^\Gamma$, then E is valuation dense in $\mathcal{L}^\Gamma$. In particular, if E is closed, then $E = \mathcal{L}^\Gamma$.*

Proof. Fix $x \in \mathcal{L}^\Gamma$ and a desired threshold $B \in \mathbb{R}$. Cofinality and closure under negation in the value group give $a \in E_{>0}$ with $v(a) < -B$. Choose $j \in J$ with $j \leq ax < j + 1$. Then $0 \leq ax - j < 1$, so $v(ax - j) \geq 0$. Consequently

$$v\left(x - \frac{j}{a}\right) = v(ax - j) - v(a) \geq -v(a) > B,$$

where the zero-error case is interpreted in the usual way. Since $j/a \in E$, this gives arbitrary valuation precision. $\square$

Corollary 205.4 (Surjectivity detected by transported integer parts). *For every field self-embedding ϕ of $\mathcal{L}^\Gamma$, the following are equivalent:*

1. ϕ is surjective;

2. $\phi(I^\Gamma)$ is an integer part of the whole field $\mathcal{L}^\Gamma$;
3. $\phi(\mathcal{L}^\Gamma)$ contains some integer part of the whole field $\mathcal{L}^\Gamma$.

Proof. Surjectivity transports the integer-part property, so (1) implies (2), which implies (3). The image is closed by [theorem 198.1](#); its value group $\lambda\Gamma$ is cofinal because it is a nonzero subgroup of $\mathbb{R}$. Apply [theorem 205.3](#) to obtain (1). $\square$

In the infinite-rank construction, $\Phi(I^\Gamma)$ is always an integer part of $F = \Phi(\mathcal{L}^\Gamma)$, but it is not an integer part of $\mathcal{L}^\Gamma$. [Corollary 201.2](#) gives an explicit witness to the failure. This is stronger than noting that one monomial is absent from the image: no choice of a different integer part lying inside F can repair the problem.

These statements do not say that every embedding preserves the *canonical* integer part as a named predicate. Even the rank-one automorphism $t \mapsto t + t^2$ sends t^{-1} to $t^{-1} - 1 + t - t^2 + \dots$, which is not in the ring ([205.1](#)). Its transported integer part is another integer part of the whole field. Naming the canonical ring therefore creates a different embedding problem.

[write] (Batch 92.) This bears on Part V's [question 73.8](#), which asks to classify the integer parts of $\mathcal{L}_k$ up to field automorphisms: for $k = \mathbb{R}_{\text{alg}}$ and $\Gamma = \mathbb{Q}$, the field is $\mathcal{L}^\Gamma$, every automorphism carries I^Γ to an integer part, the automorphism $t \mapsto t + t^2$ carries it to one different from I^Γ , and by [corollary 205.4](#) a subfield image of a proper self-embedding contains no integer part of the field. The classification itself is not attempted (note after that question).

206 Surreal realizations and the arithmetic boundary

The usual Conway normal-form calculus identifies a series of the present kind with a surreal number by

$$\iota\left(\sum a_\gamma t^\gamma\right) = \sum a_\gamma \omega^{-\gamma}. \quad (206.1)$$

The real exponents on the right are viewed as surreal numbers. The support $\{-\gamma : a_\gamma \neq 0\}$ is reverse well ordered, and the Conway sum is legitimate. Its leading-term order and convolution arithmetic agree with those on the left. These normal forms and the normal-form characterization of omnific integers are standard; see Ehrlich–Kaplan [[5](#), §§3,6.2].

Proposition 206.1. *The map ι is an ordered-field embedding with set-sized image $\mathcal{L}^\Gamma \subseteq \mathbf{No}$. It sends I^Γ into $\mathbf{Oz}$. Conjugating the embeddings of [theorems 195.1](#) and [202.1](#) by ι gives the corresponding self-embeddings of $\mathcal{L}^\Gamma$.*

Proof. Uniqueness and arithmetic of normal forms prove the field and order assertions. In ([205.1](#)), negative t -exponents become positive ω -exponents, and the constant coefficient is integral; this is exactly the required omnific normal-form condition. The source $\mathcal{L}^\Gamma$ is a set, so its image under the displayed map is a set. Conjugation transports all the asserted properties, using the transported valuation metric. $\square$

[write] (Batch 92.) For $\Gamma = \mathbb{Q}$ this is Part V's surreal map ([71.1](#)) in [section 71](#); source 20's ([191.1](#)) is the mirrored map $T^\gamma \mapsto \omega^\gamma$ on finite sums.

The topological claim concerns the transported metric on this *set-sized subfield*, not a Polish topology on the entire proper class $\mathbf{No}$. It does not assert that the ambient order topology of all surreal numbers, restricted to the subfield, is the same topology: ambient infinitesimal neighborhoods can be finer. Nor does the construction extend automatically to a class

self-embedding of **No**. The transported ring $\Phi(I^\Gamma)$ need not lie in the original canonical **Oz** copy, as the preceding rank-one example already illustrates.

All objects used in the proof can be handled in ZFC. A basis is chosen for the set-sized vector space Γ ; no global choice on **No** is used. In a class-theoretic presentation, (206.1) is simply a set-indexed normal-form map. In a type-theoretic implementation, one fixes a universe for Γ and the coefficient field, proves the left-finite series results there, and only then transports them into a suitable universe-indexed surreal construction. Treating all surreal numbers as one small type without such a boundary would not be a legitimate formalization.

Finally, there is no conflict with Glazer’s arithmetic obstruction. The fields here are densely ordered, and their displayed integer parts are countable when the fields are Polish. A field structure, an integer part, and a model of full Peano arithmetic are different requirements. Neither uncountable Polish PA models nor continuity of an uncountable discrete arithmetic expansion is produced. The new conclusion concerns self-embeddings inside an explicit permitted family of Polish ordered fields.

207 Proof audit and a concrete ProveIt formalization plan

207.1 Repository interfaces actually inspected

The repository snapshot is

9fe62865dc30d71256183466b744b98ebf7e64d6.

The following source files were inspected through the GitHub connector, rather than inferred solely from the repository’s overview:

Source under	Relevant content and limitation
Algebra/SurrealNumbers/	
README.md	Reports the constructed surreal field, omnific floor, and the separation between research reports and formal coverage.
Surreal/HahnSeries/	characterTwist, support_twist, and twist_inverse;
CharacterTwist.lean	coefficient characters have explicit inverses.
Surreal/HahnSeries/	pow_injective_near_one,
BinomialRoots.lean	binomialPower_rat_root_unique, and
	exists_unique_root_near_one; root uniqueness uses
	nonzero residue characteristic-zero integer factors.

These are useful interfaces, not a pre-existing formal proof of the co-Hopfian theorem. The repository was not rebuilt for this manuscript, and the finite checks shipped here do not enlarge its kernel-checked coverage.

[write] (Batch 92.) The three files are unchanged between the pin and the placement e24ce2ce0, and the declarations named exist; the surreal README has since gained the catalogue’s pointers (cd514f61b). None of the modules proposed below exists, and docs/FORMALIZATION.md maps no label of this part.

207.2 Recommended dependency order

A practical development should introduce a left-finite predicate on existing Hahn series and prove closure under field operations. It must then prove the truncation/completion

equivalence for real-valued exponent groups. This is the first substantial new layer: an arbitrary Hahn-series support theorem does not imply left finiteness.

The next layer should separate three reusable statements. First, prove valuation scaling for an order embedding between these Archimedean-value groups, initially with coefficient-fixing as an explicit hypothesis. Second, prove the extension of monomial characters to the completion. Third, prove the additive uniform-gap Neumann lemma independently of field multiplication. This separation avoids making invertibility part of the definition of a substitution and then accidentally using it as a conclusion.

For finite rank, the essential finite object is a basis together with positive lower bounds on the valuations of its unit corrections. The support estimate for rational powers can reuse the near-one root interface, but the stronger bound (197.2) must be stated and proved for the left-finite subtype. Real closedness must also be proved for that subtype; it cannot be inherited merely by inclusion in a real-closed Hahn field.

The infinite-rank algebraic core is particularly suitable for independent formalization. First prove the finite telescoping identity for an abstract sequence of monomials. Then prove the coefficient-forcing lemma:

$$(\forall N, v(x - s_N) = \gamma_{N+1}) \implies (\forall j, x_{\gamma_j} = (-1)^j).$$

The contradiction uses only infinitude below one bound and the left-finite predicate. The existence of an independent bounded chain can be formalized separately with standard basis-extension machinery. Finally, transport the maps and integer parts to the already constructed surreal field.

207.3 External inputs and proof obligations

The only substantial classical algebraic input not reproved in full is the residue-characteristic-zero defectless theorem used for real closedness. Real-closed quantifier elimination is used for the model-theoretic labels and the comparison of first-order theories; the embedding, topological, support, and integer-part arguments do not depend on it. The surreal realization uses the usual normal-form arithmetic, independently of the new abstract-field proofs.

Ordinary set-theoretic choice is used for basis extension and, in the continuum-many-isomorphism-types example, cardinal arithmetic of countable equivalence classes. No saturation assumption, continuum hypothesis, large cardinal, or class-sized choice function is used. The main finite-rank proof and the explicit shift after a basis is specified require none of those additional set-theoretic principles.

208 Further research questions and concrete next targets

The questions below are proposed continuations of the results proved here. They are not all claimed to be previously published open problems. Their purpose is to identify exact next theorems rather than broadly request more work on series fields.

Question 208.1 (Which infinite-rank unit characters are invertible?). Find a usable necessary-and-sufficient condition on $u \in \text{Hom}(\Gamma, 1 + \mathfrak{m})$ for Φ_u to be onto. Uniform positive valuation gain is sufficient but may not be necessary. A first target is a condition expressed in terms of the supports of $u(b) - 1$ on a chosen basis and stable under change of basis.

The shift shows that strict gain at each element is insufficient. A correct criterion must rule out correction paths whose cumulative valuation gain has a finite upper bound, or otherwise

show how those paths cancel. A criterion that only checks the first correction of individual basis monomials will miss this obstruction.

[write] (Batch 93.) Bears, for real coefficients and countable Γ : on Part XVII's field $\mathcal{L}_{\mathbb{R}}^{\Gamma}$ a monomial character with uniform positive gain gives an automorphism that is the exponential of a Borel derivation ([remark 239.4](#) [merge], from source 23's [theorem 234.2](#)). Necessity is not addressed.

Question 208.2 (Weighted dependency graphs). For substitutions in which each basis monomial maps to itself plus finitely many later monomials, characterize invertibility by the weighted dependency graph. Does divergence of every infinite path's cumulative exponent gain suffice under an appropriate finite-branching and local-finiteness hypothesis?

The finite-rank Neumann proof corresponds to a uniform lower edge bound. The present counterexample is one infinite path of finite total weight. A graph theorem would convert this contrast into an algorithmic inverse criterion, but branching and cancellations require separate control.

Question 208.3 (Closed immediate self-copies). Classify closed immediate self-copies up to equality and conjugacy. When do two internal unit characters yield the same subfield? Which invariants of the corresponding approximation cuts distinguish conjugacy classes of ranges?

Every isomorphism onto such a subfield has the parameters in [theorem 198.2](#). Immediacy forces $\lambda\Gamma = \Gamma$; precomposing with the inverse of its monomial automorphism therefore gives an internal substitution with the same range. The remaining issue is to organize and compare ranges, rather than just embeddings. The exact distance cut in [theorem 201.1](#) provides an invariant worth testing.

Question 208.4 (Descriptive complexity of surjectivity). Fix a countable effective presentation of Γ and a reasonable code space for unit characters. What is the descriptive complexity of

$$\{u : \Phi_u \text{ is surjective}\}?$$

Can the bounded-path obstruction be used to prove a sharp lower bound?

The Cantor family supplies a continuous family of embeddings and an exact membership code, but it is not by itself a completeness proof for this decision set. A reduction must specify its domain topology and control every possible preimage, not merely selected monomials.

Question 208.5 (Larger coefficient fields). For an Archimedean real-closed coefficient field $k \supsetneq \mathbb{R}_{\text{alg}}$, classify all field self-embeddings of the left-finite field without requiring them to fix k . Is there a replacement for the rational-rank dichotomy involving the transcendence structure of the coefficient section?

The present proofs extend to coefficient-fixing embeddings over such k , but an arbitrary embedding into a non-Archimedean field need not preserve the selected coefficient section. The automatic fixing of real algebraic constants is therefore an essential stated hypothesis of the abstract-field classification.

[write] The extension of the proofs to coefficient-fixing embeddings over a larger Archimedean real closed k is asserted here, not written out; checking it is part of this question.

[write] (Batch 93.) For $k = \mathbb{R}$, countable Γ and Borel embeddings: every Borel field embedding of $\mathcal{L}_{\mathbb{R}}^{\Gamma}$ is real-linear and so fixes $\mathbb{R}$ ([theorem 228.2](#), [remark 239.4](#) [merge]), which reduces this question for them to the coefficient-fixing case. Embeddings that are not Borel are untouched.

[write] (Batch 96.) For $k = \mathbb{R}$, $\Gamma = \mathbb{Q}$ and automorphisms the coefficient-fixing classification is now written out: every field automorphism of $\mathcal{L}_{\mathbb{R}}^{\mathbb{Q}}$ that fixes $\mathbb{R}$ is a substitution $t \mapsto At^r(1+u)$ with $A > 0$, $r \in \mathbb{Q}_{>0}$ and $v(u) > 0$, and these are the automorphisms among the embeddings of [theorem 198.2](#) ([theorem 242.4](#), whose proof uses Borelness only to fix $\mathbb{R}$, and [remark 255.3](#) [merge]). Embeddings, higher rational rank and other k remain as stated, and so do automorphisms that do not fix $\mathbb{R}$.

Question 208.6 (Full Hahn extensions and spherical completions). For the bounded-exponent shift, determine exactly when the monomial assignment extends to a map of the full Hahn field or a chosen maximal immediate extension. When it does, is the extension onto, and how is the formal alternating inverse represented?

The alternating series is an allowed Hahn element, but its existence does not establish that every other Hahn series is substitutable. A proof must verify the required summability families globally.

[write] Whether Kuhlmann–Serra’s Lemma 4.2.4 holds for the maximal Hahn field, where every summable family is summable in the field, is the same kind of question; [remark 204.2](#) leaves it open.

Question 208.7 (Integer-part pairs). Classify embeddings of the pair $(\mathcal{L}^{\Gamma}, \mathcal{I}^{\Gamma})$ that send the canonical integer part into itself, and classify the automorphisms that preserve it exactly. Do these conditions force the principal-unit character to be trivial under additional natural hypotheses?

Even a finite-rank field automorphism can move the canonical integer part. Conversely a monomial embedding with proper value-group image can preserve the ring setwise in the one-sided sense. The pair problem therefore has information not present in field co-Hopfianity.

Question 208.8 (Intrinsic invariants of the image gap). For closed immediate self-copies, study the cuts $\sup_{z \in F} v(x - z)$ and whether those cuts are attained. Which patterns can occur simultaneously for a countable family of elements? Can this distinguish nonconjugate proper self-copies?

[Theorem 201.1](#) gives a finite cut in the value group with no best approximation. Multiple independent chains provide a controlled starting family. Conjugacy questions must allow the scalar change of valuation that any automorphism may induce.

Question 208.9 (Positive and mixed characteristic). Find the correct analogue over perfect coefficient fields of positive characteristic. Which parts survive Frobenius and which are altered by algebraic defect or failure of principal unique divisibility?

The additive Neumann estimate remains valid in suitable complete valued groups. The real-order argument, the characteristic-zero rational binomial series, and the real-closedness proof do not transfer unchanged. A meaningful analogue should first fix the exponent category and specify whether only valuation-preserving embeddings are considered.

Question 208.10 (Effective inverse certificates). Develop a verified procedure that accepts finite basis data and certified support bounds, constructs inverse prefixes, and outputs proofs of the error estimate ([199.3](#)). Determine how coefficient arithmetic, rational denominators, and ordered-group comparison affect its running time.

The iteration depth has already been bounded. Turning it into a usable complexity theorem requires an explicit input model; arbitrary real exponents and arbitrary infinite series are not finite algorithmic inputs.

Question 208.11 (Minimal formalization of the correction). Formalize the rational-function counterexample and the internal substitution monoid before attempting the full real-closed field construction. Can the resulting interfaces prevent other automatic promotion of forward summability to invertibility?

The reflection $t \mapsto -1 - t$ gives a short algebraic certificate of properness. It is an attractive first contribution to ProveIt because it does not depend on complete valuations or surreal foundations.

[write] [Remark 204.2](#) restates the counterexample in the preprint's own terms, for every coefficient field. Nothing of it is formalized in the repository.

Question 208.12 (Beyond Archimedean value groups). Replace $\Gamma \subseteq \mathbb{R}$ by a non-Archimedean ordered $\mathbb{Q}$ -vector space and choose a precise support/completion convention. Which combination of rational rank, ordered rank, and cofinality controls co-Hopfianity?

The formula e^{-v} is no longer directly available, metrizable depends on cofinality, and a fixed positive valuation increment need not be cofinal. Thus the proof of [lemma 199.1](#), not just the exponent classification, must be reconsidered. This is a natural point where the set-theoretic cofinality questions motivating the project become decisive.

[write] (Batch 92.) This is Part V's [question 73.5](#) (higher-rank exponent groups) asked for self-embeddings; source 20's [question 193.6](#) asks the analogous question for its threshold.

209 Conclusion

For algebraic real coefficients and an Archimedean divisible exponent group, rational rank exactly controls field co-Hopfianity. Finite rank converts all principal-unit corrections into one uniform valuation estimate, forcing an inverse. Infinite rank permits an independent bounded chain, and its telescoping substitution forces any inverse to carry a forbidden infinite bounded prefix.

The resulting proper images preserve both classical valuation invariants, remain complete and closed, and are elementary real-closed self-copies. Nevertheless they omit an entire order-scale neighborhood around a specified element and cannot contain any integer part of the whole field. The same mechanism identifies why one-sided summability cannot be substituted for a proof of surjectivity. These are compatible with, and sharpen the structural questions around, the Polish arithmetic boundary that connects Glazer's work to ProveIt's surreal and generalized-series program.

Part XVI

Ordinal rigidity of Borel Hilbert orders

[write] This part is source 22, manuscript 03 of batch 93, *Ordinal Rigidity of Borel Hilbert Orders: Canonical flags, exact oscillation ranks, Polish Presburger models, and omnific realizations* (22 pages, US letter, dated 4 October 2026; "prepared for Vladimir Reshetnikov, research developed and

written with ChatGPT"). It arrived in commit 2faa3b37a, as archive `Ordinal_Rigidity_Borel_Hilbert_Orders`, and was placed in 5e4d8eed8; it is pinned to 3c25fbb55, 57 commits before the placement. Its file prefix in this report is 22-hilbert-orders-. All unmarked text in this part is source 22's; its fourteen sections are Sections 210–223 (its Section k is Section $209 + k$, with statement numbers otherwise unchanged; its equations, numbered consecutively in the source, are numbered by section here), its nine "Research questions", numbered 1–9 in the source, share the report's counter and are Questions 222.1–222.9, and its two appendices are appendices AH and AI. Question 222.10 and section 224 [merge] were written for this report.

What this part claims, and its status

Every translation-invariant total order with Borel positive cone on a separable real Hilbert space is lexicographic in a unique positively oriented orthonormal basis indexed by a unique countable ordinal θ , and is therefore real homogeneous (theorem 210.1). On a fixed infinite-dimensional space the indices are exactly the countably infinite ordinals, embeddability is ordinal comparison, and there are exactly $\aleph_1$ orders up to isomorphism (theorem 214.4 and corollary 214.5). All cones are Δ_2^0 , and the oscillation rank of the cone's characteristic function is exactly $\theta + 1$, a complete invariant of the ambient pair (theorem 215.1 and corollary 215.2). In a general real Polish vector space a hereditarily Baire cone has a canonical countable flag of closed hyperplanes (theorem 213.2). The Borel order embeddings are the bounded operators with strictly increasing positive pivots (theorem 216.2); the image is Polish exactly when the operator is bounded below (corollary 216.8). An order on $\mathbb{R}^2$ whose cone has the Baire property need not be Borel (proposition 217.1), and L^p , $0 < p < 1$, carries no Borel order (corollary 217.3). The cones $M_\theta^{\ell^2} = (H_\theta^{\ell^2} \vec{\times} \mathbb{Z})_{\geq 0}$ are Polish models of $\text{Th}(\mathbb{N}; 0, 1, +, <)$ with continuous addition, classified by θ , embedded additively in $\mathbf{Oz}$, and admitting no compatible discretely ordered ring (theorems 218.1, 218.3, 219.1 and 220.2).

Section 224 [merge] identifies the flag with Part III's cone recursion, extends the L^p theorem to every nonatomic measure, proves the characterization source 22 asserts for invariant orders, and computes the analogues of Part X's two thresholds.

Unrefereed, AI-assisted, not formalized; priority is not established, in particular against Küçük–Soyertem–Küçük (2012), whose full text neither source 22 nor this report has read. The finite checks test finite algebra only.

Questions answered. Source 22 answers no question it names: it takes Glazer's Question 2 as already answered in the repository and calls itself a continuation. This report reads it as follows (dated notes after each question). Part I's question 14.6 is answered for separable real Hilbert spaces: every Borel order is lexicographic in an orthonormal basis, a G_δ cone is the same as a Borel one, the order ranks are exactly the countable ordinals (all countably infinite ones on one carrier), and the continuous order embeddings are the bounded ordinal-echelon operators; it adds that L^p , $0 < p < 1$, carries no such order at all. Part I's question 14.14 is answered in part for real ℓ^2 coefficients. Part III's question 43.7 is answered in part for Hilbert spaces (proposition 224.6 [merge]), and Part X's question 138.7 is answered in a variant, real coefficients with the Hilbert topology rather than the product topology: both of Part X's thresholds move, to $\theta \leq 1$ for automatic continuity and to $\theta < \omega$ for closed images (proposition 224.5 [merge]). Source 22's question 222.8 repeats questions of Parts I, III and X, and its question 222.4 is partly answered by Part III; notes after them say so. Source 22 answers neither of Glazer's questions and says so.

What was already here, and what is new. Source 22 read this report's README at its pin, not its article, and so did not see that the canonical transfinite extraction it presents as its key point is Part III's [theorem 33.1](#), in the tree since `0bd0e5527`, before its pin. Its [theorem 213.2](#) and the Δ_2^0 half of [corollary 213.3](#) are that theorem for real Polish vector spaces, with the same recursion, termination argument and layers ([proposition 224.1](#)); its [lemma 212.1](#) is the sumset step of the same proof; the finite-dimensional case of [theorem 210.1](#) is Part IV's [lemma 47.3](#), refined to an orthonormal basis; its [theorem 218.1](#) is a case of Part III's [theorem 34.1](#); the algebraic half of [theorem 218.3](#) is Part X's [theorem 129.1](#); [theorem 220.2](#) is weaker than Part I's [theorem 10.4](#); and [theorem 219.1](#) is a variant of Part III's [theorem 40.1](#). Each is printed with source 22's proof and flagged in a note. New to the report are the classification of Borel orders on separable Hilbert spaces, the codimension-one flag with continuous functionals and automatic real homogeneity in every real Polish vector space, the regularity collapse, the exact oscillation rank $\theta + 1$ and the ambient classification, the convex-subgroup classification, the pivot normal form for bounded operators and the closed-range criterion, the Baire-but-not-Borel cone, the theorem for vector spaces with zero dual, and the Hilbert family of Presburger models with its embedding and Polish-image theorems.

Labels and notation. Every label of source 22 takes the prefix `pma:hbo:`; labels were added to its remarks, corollary and examples without labels, its conclusion and its nine questions, and to `pma:part:hbo,` [question 222.10](#) and the statements of [section 224](#), which were written for this report. The seventh block of the table in [section 1.3](#) lists the symbols. In short: source 22's $H_\theta = \ell^2(\theta, \mathbb{R})$, its tails $H_{\theta,\alpha}$, and its $G_\theta = H_\theta \overrightarrow{\times} \mathbb{Z}$, $M_\theta = (G_\theta)_{\geq 0}$ and omnific map J_θ are printed $H_\theta^{\ell^2}$, $H_{\theta,\alpha}^{\ell^2}$, $G_\theta^{\ell^2}$, $M_\theta^{\ell^2}$ and $J_\theta^{\ell^2}$ (superscript ℓ^2 : real square-summable coordinates with the Hilbert topology), because Part X's $H_\alpha = \mathbb{Q}^\alpha$, Part III's $G_\alpha, M_\alpha = (\mathbb{Q}^\alpha \overrightarrow{\times} \mathbb{Z})_{\geq 0}$ and J_α use the same letters and ordinal subscripts for discrete rational coordinates with the product topology (tempting false readings: $M_\omega^{\ell^2}$ is not Baire space, and its oscillation ranks differ); its flag stages E_α , and the flag (H_α) of its [theorem 210.2](#), are printed $\mathcal{E}_\alpha$, because Part III's E_α is the cut $\{x < p\}$ of [theorem 36.1](#); its diagonal operator D is printed `Diag`, because D is the divisible group of Parts I and VII and, in [lemma 220.1](#), source 22's own divisible part. Its "flag rank" θ is Part III's cone closure rank ρ . Its typewriter `\repo` is printed `Provelt`. No normalization changed.

Files. Source 22's program, recorded run, build script, build record, proof review and source record are shipped as `code/22-hilbert-orders-verify.py`, `data/22-hilbert-orders-verification_results.json`, `code/22-hilbert-orders-build.sh`, `data/22-hilbert-orders-BUILD_REPORT.json`, `22-hilbert-orders-PROOF_REVIEW.md` and `22-hilbert-orders-SOURCES.md`. Source 22 calls its pin a "tree identifier"; it is a commit (note in [appendix AI](#)). The Hilbert's-tenth programme routed the archive from its README only, without a correctness finding ([appendix E](#)). Nothing in this part is formalized.

210 The intersection and the main results

Why this problem

Glazer's *A Topological Tennenbaum Theorem* studies the obstruction to continuous arithmetic on uncountable Polish spaces. Its concluding Question 2 asks whether some such space supports a model of Presburger arithmetic with continuous addition [1]. The question is about *additive* arithmetic: multiplication is not in the language. The relevant part of `Provelt`

already contains affirmative constructions and a substantial collection of extensions, under `polish-models-of-omnific-arithmetic` [139]. Consequently, merely constructing another copy of $\mathbb{R} \overrightarrow{\times} \mathbb{Z}$ would not be a substantive continuation.

The next structural problem is to classify regular additive orders on a fixed, genuinely infinite-dimensional carrier. Hilbert space is a useful meeting point: its topology is Polish, its orthogonality makes closed hyperplane flags canonical, and its coordinates can be interpreted as coefficients of surreal normal forms. The repository also develops Hilbert geometry at surreal scales [140]. We work with ordinary real Hilbert spaces here, not with a putative positive-definite proper-class Hilbert space over **No**.

Precisely what is established

Put

$$H_\theta^{\ell^2} = \ell^2(\theta, \mathbb{R}) = \left\{ x : \theta \rightarrow \mathbb{R} : \sup_{F \subseteq \theta \text{ finite}} \sum_{\alpha \in F} |x_\alpha|^2 < \infty \right\},$$

with its Hilbert norm. For $x \neq 0$, let

$$v_\theta(x) = \min\{\alpha < \theta : x_\alpha \neq 0\}, \quad x >_\theta 0 \iff x_{v_\theta(x)} > 0.$$

The ordinal is countable, so this is a separable Hilbert space. Infinite sums in $H_\theta^{\ell^2}$ mean unconditional norm sums, not an assertion that θ has order type ω .

Theorem 210.1 (Hilbert classification). *Let H be a separable real Hilbert space and $<$ a translation-invariant total order whose strict positive cone P is Borel. There are a unique countable ordinal θ and a unique ordered orthonormal basis $(e_\alpha)_{\alpha < \theta}$ of H such that*

$$x > 0 \iff \langle x, e_\alpha \rangle > 0 \text{ at the least } \alpha \text{ for which } \langle x, e_\alpha \rangle \neq 0. \quad (210.1)$$

In particular the order is compatible with multiplication by every positive real number, even though this was not assumed. If $\dim H = n < \infty$, then $\theta = n$. If H is infinite-dimensional, its possible indices are exactly $\omega \leq \theta < \omega_1$.

Theorem 210.2 (Rank and regularity). *Every cone in Theorem 210.1 is Δ_2^0 . Its successive relative boundary spaces form a canonical decreasing flag $(E_\alpha)_{\alpha \leq \theta}$, with $E_\theta = \{0\}$. For every $0 < \varepsilon \leq 1$, the oscillation derivative of $\mathbf{1}_P$ starting from H first becomes empty at stage $\theta + 1$. Thus the characteristic functions are Baire class 1, with exact oscillation ranks unbounded in ω_1 on a single fixed infinite-dimensional Hilbert space.*

Theorem 210.3 (Embeddings and arithmetic). *For countable ordinals θ, η , an additive order embedding $H_\theta^{\ell^2} \rightarrow H_\eta^{\ell^2}$ exists if and only if $\theta \leq \eta$; when one exists, one can choose a linear isometric embedding. The same criterion holds for unit-preserving additive embeddings between*

$$M_\theta^{\ell^2} = (H_\theta^{\ell^2} \overrightarrow{\times} \mathbb{Z})_{\geq 0}. \quad (210.2)$$

Each $M_\theta^{\ell^2}$ is a Polish model of full Presburger arithmetic with continuous addition, and all its definable relations with parameters are Δ_2^0 . Its unit-preserving additive embeddings are elementary. For $\theta > 0$, it admits an explicit additive order embedding into $\mathbf{Oz}_{\geq 0}$, but no compatible discretely ordered ring expansion exists on its group completion.

These are classification statements about a specified class, not about all Polish Presburger models. In particular they do not cover the nonsplit or discrete-direction constructions in the

repository. They give exactly $\aleph_1$ isomorphism types on a fixed carrier with Hilbert divisible part of countably infinite Hilbert dimension; this cardinality statement does not assume the continuum hypothesis.

[write] (Batch 93.) In this report's terms: the "flag rank" θ is Part III's cone closure rank ρ ([proposition 224.1](#) [merge]), and the models $M_\theta^{\ell^2}$ have real coordinates with the Hilbert topology, not the discrete rational coordinates of Part III's and Part X's $M_\alpha = (\mathbb{Q}^\alpha \overrightarrow{\times} \mathbb{Z})_{\geq 0}$. The "nonsplit or discrete-direction constructions" are Parts VII, III and X. Part XIII classifies the Borel Presburger orders on $\mathbb{R} \times E$, E countable and discrete, a class this part does not contain.

Prior work and novelty boundary

Lexicographic cones constructed from orthogonal or Schauder bases are established mathematics. Küçük, Soyertem, and Küçük study total ordering cones in separable Hilbert spaces [141]; Tan constructs lexicographic orders from Schauder bases and studies their order ideals [142]. Our starting example at index ω is therefore not new. The point of the present argument is the converse from *Borel additive order alone*, canonical transfinite extraction, exact ordinal classification, and its agreement with an oscillation rank. The last invariant belongs to the general tradition of ordinal ranks for Baire class 1 functions [143]; we compute it directly rather than importing a rank comparison theorem.

The full text of Tan's paper was inspected, but the 2012 paper was available in this session only through its indexed abstract and bibliographic information. No claim about a gap in that paper is made. A potentially equivalent older representation theorem would affect priority, not the proofs supplied below. Appendix AI records the bounded scope of the source audit.

[write] (Batch 93.) The novelty boundary stated above was written without this report's article (source 22 read its README only). The canonical transfinite extraction from a Borel cone, with countable termination, Δ_2^0 layers and the same proof, is Part III's [theorem 33.1](#), in the tree since 0bd0e5527, before source 22's pin; its finite-dimensional form is Part IV's [lemma 47.3](#). [Theorem 213.2](#) is therefore printed as a refinement of [theorem 33.1](#) for real Polish vector spaces ([proposition 224.1](#) [merge]); what is new is listed in the introduction of this part. The priority question against Küçük–Soyertem–Küçük is open and is recorded as [question 222.10](#), with [remark 224.3](#) [merge].

211 Conventions and classical inputs

A *real Polish topological vector space* is a real vector space with a Hausdorff, separable, completely metrizable topology for which addition and scalar multiplication are continuous. Local convexity is *not* included in this definition. Closed real linear subspaces are again Polish. A nonzero such space has no isolated points.

For a translation-invariant total order on an additive group, its strict positive cone satisfies

$$P + P \subseteq P, \quad P \cap (-P) = \emptyset, \quad P \cup (-P) = E \setminus \{0\}. \quad (211.1)$$

Conversely, these identities define the order by $x < y$ if $y - x \in P$. On a real vector space they imply $qP = P$ for every $q \in \mathbb{Q}_{>0}$. Indeed, a sum of positive elements is positive, and $nx > 0$ forces $x > 0$ by trichotomy. Real scalar invariance does not follow purely algebraically; it will be a consequence of regularity.

A set has the *Baire property* if it differs from an open set by a meager set. A Δ_2^0 set is both F_σ and G_δ . In a metrizable space, relatively open subsets of closed sets are F_σ . Finite Boolean

combinations of Δ_2^0 sets are Δ_2^0 . We use the classical facts that a subspace of a Polish space is Polish exactly when it is G_δ , and that an analytic and coanalytic set is Borel; see [8]. A characteristic function of a Δ_2^0 set is Baire class 1. One direct proof is included in Section 215.

We use the geometric Hahn–Banach separation principle in the following form: a proper closed convex subset with nonempty interior in a real Hausdorff topological vector space can be separated from an exterior point by a nonzero continuous real linear functional. Nonempty interior is essential when local convexity is absent. Appendix AH gives the reduction to the algebraic Hahn–Banach theorem.

For the arithmetic application we use the classical quantifier elimination theorem for Presburger arithmetic in the ordered group language with $0, 1, +, -$ and congruence predicates modulo every positive standard integer. Its algebraic characterization says that a discretely ordered abelian group with least positive element 1 is a model of the integer version precisely when it has division with remainder by every positive standard integer. Its nonnegative cone is then a model of $\text{Th}(\mathbb{N}, 0, 1, +, <)$. This is the standard Presburger theorem, not a new result of this report [144, 58].

An *additive order embedding* is an injective additive map that preserves and reflects strict order. Unless explicitly specified, the word does not assert continuity or that the map is a topological embedding onto its image. A *unit-preserving additive embedding* of our nonnegative cones preserves $0, 1, +$ and is injective; for these cones it automatically preserves and reflects the definable order.

212 A category lemma forces a closed halfspace

Lemma 212.1 (Interior from additive positivity). *Let $E \neq \{0\}$ be a real Polish topological vector space and let P satisfy (211.1). If P has the Baire property in E , then $\text{int}_E(P) \neq \emptyset$.*

Proof. If P were meager, so would be $-P$. The singleton $\{0\}$ is nowhere dense because E is nonzero. Equation (211.1) would make E meager in itself, contrary to the Baire category theorem. Thus there is a nonempty open set U on which P is comeager.

Fix $z \in U + U$. The set $V = U \cap (z - U)$ is nonempty and open. Both $P \cap V$ and $(z - P) \cap V$ are comeager in V . Since V is a Baire space, choose $p \in P \cap (z - P) \cap V$. Then $z = p + q$ for some $q \in P$, so $z \in P + P \subseteq P$. Hence the nonempty open set $U + U$ is contained in P . $\square$

[write] (Batch 93.) This is the sumset step (33.2) of Part III’s theorem 33.1, the argument of Part IV’s lemma 47.3 and Part IX’s lemma 120.4, and Part V’s lemma 61.1; source 22’s proof is its own.

Lemma 212.2 (Canonical first quotient). *Under the assumptions of Lemma 212.1, there is a nonzero continuous real linear functional $f : E \rightarrow \mathbb{R}$ for which*

$$\text{cl}_E(P) = \{x : f(x) \geq 0\}, \quad \{x : f(x) > 0\} \subseteq P \subseteq \{x : f(x) \geq 0\}. \quad (212.1)$$

Consequently

$$\text{cl}_E(P) \cap (-\text{cl}_E(P)) = \ker f = \partial_E P. \quad (212.2)$$

The kernel is a canonical closed real hyperplane, and f is unique up to multiplication by a positive real constant.

Proof. Set $C = \text{cl}_E(P)$. Positive rational scalar invariance and closure under addition imply that C is a convex cone: approximate real coefficients by positive rational coefficients and use continuity of the vector operations. Also $0 \in C$, for $p/n \rightarrow 0$ when $p \in P$.

The set $-\text{int}_E(P)$ is a nonempty open set disjoint from P , and therefore disjoint from C . Thus C is proper and has nonempty interior. Separation supplies a nonzero continuous linear functional f nonnegative on C , after reversing its sign if necessary. The nonnegativity follows from the cone property: a functional bounded below on a cone cannot take a negative value on it, since that value could be multiplied by arbitrarily large positive scalars.

If $f(x) > 0$, trichotomy leaves $x \in P$ or $-x \in P$. The latter would give $f(-x) \geq 0$, a contradiction. Hence $\{f > 0\} \subseteq P \subseteq C \subseteq \{f \geq 0\}$. The closure of the open halfspace $\{f > 0\}$ is $\{f \geq 0\}$: for $f(x) = 0$ and $f(u) > 0$, the sequence $x + u/n$ approaches x from that halfspace. This proves (212.1).

Every point of $\ker f$ is approached by points with positive and negative f -value, so it is a boundary point of P . Points outside $\ker f$ lie in the interior of P or of its complement. This proves (212.2). If g gives the same closed halfspace, then $\ker g = \ker f$. Since $E/\ker f$ has real dimension one, $g = cf$; the common orientation forces $c > 0$. $\square$

[write] (Batch 93.) For $E = \mathbb{R}$ this is Part IX's lemma 120.4 and Part XIII's lemma 169.1; for $E = \mathbb{R}^d$, Part IV's lemma 47.3 iterates it. The proof uses neither separability nor local convexity, only complete metrizability and the separation of appendix AH; proposition 224.2 [merge] uses this.

Remark 212.3. The lemma does not say that P is an open halfspace. Its intersection with the kernel contains the entire remaining order. The need to analyze that trace is exactly why a single application of Baire category does not finish the classification.

213 Canonical countable flags in Polish vector spaces

Definition 213.1. A positive cone $P \subseteq E$ is *hereditarily Baire on closed linear subspaces* if $P \cap F$ has the Baire property in F for every closed real linear subspace $F \subseteq E$.

Every Borel cone has this property. Its usefulness is that it explicitly states the regularity required at the later boundary stages.

Theorem 213.2 (Polish flag theorem). *Let E be a real Polish topological vector space, and let P satisfy (211.1) and be hereditarily Baire on closed linear subspaces. Define, until zero is reached,*

$$\mathcal{E}_0 = E, \tag{213.1}$$

$$\mathcal{E}_{\alpha+1} = \text{cl}_{\mathcal{E}_\alpha}(P \cap \mathcal{E}_\alpha) \cap (-\text{cl}_{\mathcal{E}_\alpha}(P \cap \mathcal{E}_\alpha)), \tag{213.2}$$

$$\mathcal{E}_\lambda = \bigcap_{\alpha < \lambda} \mathcal{E}_\alpha \quad \text{for nonzero limit } \lambda. \tag{213.3}$$

There is a countable ordinal θ such that $\mathcal{E}_\theta = \{0\}$, where every earlier $\mathcal{E}_\alpha$ is nonzero. Each successor step has real codimension one. There is a continuous real linear functional $f_\alpha : \mathcal{E}_\alpha \rightarrow \mathbb{R}$, unique up to positive scale, with kernel $\mathcal{E}_{\alpha+1}$ and

$$P = \bigcup_{\alpha < \theta} \{x \in \mathcal{E}_\alpha : f_\alpha(x) > 0\}. \tag{213.4}$$

The cone is Δ_2^0 and is invariant under positive real scalar multiplication. The construction is determined by P and the topology.

Proof. Whenever $\mathcal{E}_\alpha \neq 0$, it is a closed real linear subspace and hence Polish. The trace $P \cap \mathcal{E}_\alpha$ satisfies the cone axioms there and has the Baire property. Lemma 212.2 shows that (213.2) is

a proper closed real hyperplane in $\mathcal{E}_\alpha$ and provides f_α . Limit intersections remain closed real linear subspaces.

We prove countable termination rather than assuming that a countable underlying basis permits only ω steps. Suppose the process continued nontrivially at every $\alpha < \omega_1$. For each such α , choose $x_\alpha \in \mathcal{E}_\alpha \setminus \mathcal{E}_{\alpha+1}$. From a fixed countable base of E , choose B_α containing x_α and disjoint from $\mathcal{E}_{\alpha+1}$. If $\alpha < \beta$, then $\mathcal{E}_\beta \subseteq \mathcal{E}_{\alpha+1}$, so $B_\alpha \cap \mathcal{E}_\beta = \emptyset$, whereas B_β meets $\mathcal{E}_\beta$. Thus all B_α are distinct, contradicting countability of the base. Some countable stage reaches zero.

For $x \neq 0$, the least stage at which x fails to belong cannot be a limit stage, by (213.3). It is uniquely $\alpha + 1$, with $x \in \mathcal{E}_\alpha \setminus \mathcal{E}_{\alpha+1}$. Equation (212.1) in $\mathcal{E}_\alpha$ says precisely that the sign of x is the sign of $f_\alpha(x)$. This proves (213.4) and positive real homogeneity.

Each set in the union is relatively open in the closed set $\mathcal{E}_\alpha$, so it is F_σ in E . There are countably many layers. Hence P is F_σ . Its complement is $-P \cup \{0\}$, also F_σ ; therefore P is G_δ . The displayed recursion, including all relative closures, proves canonicity. $\square$

[write] (Batch 93.) The recursion is Part III's (33.3) for $A = (E, +)$, with the same countable-base termination argument and the same layer decomposition (33.4); its length θ is Part III's cone closure rank (proposition 224.1 [merge]). New here are the codimension-one steps with continuous functionals, the automatic real homogeneity and the weakening of Borel to "hereditarily Baire" (Part III's proof uses Borelness only through the Baire property of each trace, as its remark after theorem 33.1 says).

Corollary 213.3 (Regularity collapse). *For a translation-invariant total order on a real Polish topological vector space, the following are equivalent: its cone is Borel; its cone is Δ_2^0 ; its cone is hereditarily Baire on closed linear subspaces. An analytic or coanalytic cone also satisfies these conditions. The strict order relation is Δ_2^0 in $E \times E$.*

Proof. The first three implications follow from Theorem 213.2 and restriction of Borel sets. If P is analytic, its complement $-P \cup \{0\}$ is analytic, so P is Borel by the analytic/coanalytic criterion. The coanalytic case is identical after complementation and reflection. The order relation is the inverse image of P under the continuous difference map $(x, y) \mapsto y - x$. $\square$

Proposition 213.4 (Converse flag construction). *Suppose a countable decreasing flag of closed real subspaces starts at E , is continuous at limit ordinals, ends in zero, and has a continuous nonzero linear functional f_α on each $\mathcal{E}_\alpha$ with $\ker f_\alpha = \mathcal{E}_{\alpha+1}$. Then (213.4) defines a Borel translation-invariant real-vector-space order on E . Its canonical flag is the given flag.*

Proof. Every nonzero vector has a first departing layer, so trichotomy holds. For positive x, y , if their layers differ, the earlier layer determines the sign of their sum; if the layers agree, the two positive leading functional values add. This proves closure under addition. Positive real homogeneity follows from linearity. The Δ_2^0 argument is the one in Theorem 213.2. Inside $\mathcal{E}_\alpha$ the closure of the positive cone is exactly $\{f_\alpha \geq 0\}$, which recovers the next subspace and hence the entire flag. $\square$

[write] (Batch 93.) Part III's proposition 33.4 is the instance $\mathbb{R}^\gamma$ with the product topology and the coordinate flag.

Remark 213.5 (Existence does not require a Schauder basis). Every separable real Banach space has a countable family of continuous linear functionals separating its points. For each nonzero point choose one separating functional; the open sets on which these functionals are nonzero cover $E \setminus \{0\}$, and second countability gives a countable subcover. Lexicographic comparison by this sequence of functionals gives a Borel order. Redundant coordinates can

be discarded. This uses neither a Schauder basis nor finite support. It does not classify the resulting orders up to bounded linear isomorphism.

[write] (Batch 93.) The sketch is complete as stated: with the countable separating sequence (f_n) , the cone is the countable union of the sets $\{f_1 = \cdots = f_{n-1} = 0 < f_n\}$, each the intersection of a closed and an open set, and it is total and real homogeneous because the f_n are linear and separate points. Nothing is claimed for Polish vector spaces whose dual separates points but which are not Banach.

214 Hilbert rigidity and all convex subgroups

Proof of Theorem 210.1. Apply Theorem 213.2. For each $\alpha < \theta$, the subspace

$$\mathcal{E}_\alpha \cap \mathcal{E}_{\alpha+1}^\perp$$

is one-dimensional. Choose its unique unit vector e_α that is positive in the quotient, equivalently $f_\alpha(e_\alpha) > 0$. Then

$$\mathcal{E}_{\alpha+1} = \mathcal{E}_\alpha \cap e_\alpha^\perp.$$

Induction, including intersections at limits, gives

$$\mathcal{E}_\alpha = (\text{cl span}\{e_\beta : \beta < \alpha\})^\perp. \quad (214.1)$$

The vectors form an orthonormal family, and $\mathcal{E}_\theta = 0$ makes it complete. On $\mathcal{E}_\alpha$, the functionals f_α and $x \mapsto \langle x, e_\alpha \rangle$ have the same kernel and orientation, so they differ by a positive scalar. The lexicographic formula follows.

The canonical recursion fixes $\mathcal{E}_\alpha$; its oriented one-dimensional orthogonal complement fixes e_α . Thus both the ordinal and the ordered orthonormal basis are unique. In finite dimension, exactly n one-dimensional orthogonal directions are removed. In infinite separable dimension, the index is countably infinite. Conversely an orthonormal basis can be indexed in any prescribed countably infinite ordinal type, and Proposition 213.4 constructs the corresponding order. $\square$

[write] (Batch 93.) In finite dimension this is Part IV's lemma 47.3 ("after an invertible real-linear change of coordinates, the standard lexicographic order"), refined to an orthonormal basis; for $n = 1$ it is Part IX's lemma 120.4.

Definition 214.1. For positive elements in an ordered abelian group, write

$$x \asymp y \iff x \leq my \text{ and } y \leq nx \text{ for some } m, n \in \mathbb{N}_{>0},$$

and write $y \ll x$ if $ny < x$ for every $n \in \mathbb{N}_{>0}$. A subgroup is *convex* if a, b in the subgroup and $a \leq z \leq b$ imply that z is in the subgroup.

Lemma 214.2 (Archimedean ranks). For positive $x, y \in H_\theta^{\ell^2}$,

$$x \asymp y \iff v_\theta(x) = v_\theta(y), \quad y \ll x \iff v_\theta(x) < v_\theta(y).$$

Proof. If the leading indices agree, choose integers large enough that the appropriate leading real coefficient dominates the other one strictly. These integer multiples dominate regardless of all later coordinates. If the indices differ, any positive vector of the earlier rank dominates every integer multiple of a positive vector of the later rank. These alternatives exhaust all pairs. $\square$

Theorem 214.3 (Complete convex-subgroup classification). *The convex additive subgroups of $H_\theta^{\ell^2}$ are exactly*

$$H_{\theta,\alpha}^{\ell^2} = \{x : x_\beta = 0 \text{ for every } \beta < \alpha\}, \quad 0 \leq \alpha \leq \theta. \quad (214.2)$$

In particular, they are all closed real linear subspaces, including when the subgroup was not assumed Borel or real linear.

Proof. Every tail in (214.2) is plainly convex. Conversely let C be a nonzero convex subgroup. The set of leading indices of its nonzero elements has a least element α . Choose positive $y \in C$ of that rank. By minimality, $C \subseteq H_{\theta,\alpha}^{\ell^2}$. For any $x \in H_{\theta,\alpha}^{\ell^2}$, Lemma 214.2 gives an integer $n > 0$ with $-ny < x < ny$. Convexity then gives $x \in C$. Thus $C = H_{\theta,\alpha}^{\ell^2}$. The zero subgroup is the tail at θ . $\square$

Theorem 214.4 (Exact isomorphism and embedding types). *For countable ordinals θ, η :*

- (i) $H_\theta^{\ell^2}$ and $H_\eta^{\ell^2}$ are isomorphic as ordered additive groups if and only if $\theta = \eta$.
- (ii) An additive order embedding $H_\theta^{\ell^2} \rightarrow H_\eta^{\ell^2}$ exists if and only if $\theta \leq \eta$.
- (iii) Whenever such an embedding exists, a linear isometric one exists.

No measurability hypothesis is needed for the nonexistence directions.

Proof. An additive order embedding preserves $\asymp$ and $\ll$ because these relations are characterized by inequalities involving standard integer multiples. Applying it to the positive basis vectors gives a strictly increasing map from their source ranks θ into the target ranks η . An ordinal embeds increasingly in another ordinal exactly when it is at most that ordinal. An isomorphism gives an order isomorphism of the rank sets, and hence equality of the ordinals.

For $\theta \leq \eta$, extend a coefficient vector by zero outside the initial segment $\theta \subseteq \eta$. This is a real linear isometry and preserves the first nonzero coordinate. Equality of the ordinals gives an isometric isomorphism. $\square$

Corollary 214.5. *On a fixed infinite-dimensional separable real Hilbert space there are exactly $\aleph_1$ Borel total orders up to additive order isomorphism, and their embeddability types form the chain of countably infinite ordinals. No member of this class is universal under additive order embeddings.*

Proof. Theorems 210.1 and 214.4 give the exact indexing. Given an index η , the model of index $\eta + 1$ cannot embed into it. $\square$

[write] (Batch 93.) Part III's [proposition 33.4](#) already gives $\aleph_1$ distinct cone closure ranks on the one topological group $\mathbb{R}^\mathbb{N}$ with the product topology, and its [theorem 35.9](#) $\aleph_1$ pairwise nonisomorphic models on Baire space; here the carrier is a Hilbert space and the classification is complete.

Example 214.6 (A final coordinate after infinitely many coordinates). On $\ell^2(\mathbb{N}, \mathbb{R})$ compare the usual basis order $e_0, e_1, e_2, \dots$ with the order $e_1, e_2, e_3, \dots, e_0$. Their ranks are ω and $\omega + 1$. In the second order, a nonzero positive vector at the last coordinate is infinitesimal relative to every vector whose leading coordinate is one of $e_1, e_2, \dots$. The intersection of the first n successive hyperplane tails remains $\mathbb{R}e_0$ at stage ω , so one more step is necessary. Re-enumerating a countable set does not preserve its specified well-order type.

Example 214.7 (Infinitely many infinite blocks). Index the standard basis by $\mathbb{N} \times \mathbb{N}$ through any bijection with $\mathbb{N}$, and compare coordinates in block order $(0, 0), (0, 1), \dots, (1, 0), (1, 1), \dots$. The resulting rank is ω^2 . At stage ωm , all blocks before m have been removed. This gives an explicit cone not additively order-embeddable into any cone of rank ωn for finite n .

215 The exact oscillation rank

The Δ_2^0 bound does not make the order structure finite-depth. We now measure the remaining complexity by a standard type of Baire-class 1 rank, with conventions stated explicitly.

For a real function g on a metrizable space X , a closed $F \subseteq X$, and $\varepsilon > 0$, define

$$D_{g,\varepsilon}(F) = \left\{ x \in F : \begin{array}{l} \text{for every neighborhood } U \text{ of } x, \text{ there are } y, z \in U \cap F \\ \text{with } |g(y) - g(z)| \geq \varepsilon \end{array} \right\}. \quad (215.1)$$

This is closed in F . Starting from $F^0 = X$, iterate the derivative at successors and take intersections at limits. Its rank is the least stage at which the iterated closed set is empty. This is the usual oscillation-derivative convention; compare [143]. For a characteristic function and $0 < \varepsilon \leq 1$, the derivative is simply the set of points where both values occur arbitrarily nearby within F .

Theorem 215.1 (Boundary rank equals ordinal flag rank). *Let P satisfy the hypotheses of Theorem 213.2, with terminal index θ . For $g = \mathbf{1}_P$ and every $0 < \varepsilon \leq 1$,*

$$F^\alpha = \mathcal{E}_\alpha \quad (\alpha \leq \theta), \quad F^{\theta+1} = \emptyset.$$

Thus the oscillation rank is exactly $\theta + 1$. The same holds for the characteristic function of $P \cup \{0\}$.

Proof. For $\alpha < \theta$, the trace of P on $\mathcal{E}_\alpha$ contains the open positive halfspace of f_α and is disjoint from the negative halfspace. A point where $f_\alpha \neq 0$ has a neighborhood in $\mathcal{E}_\alpha$ with constant sign. At a point $x \in \mathcal{E}_{\alpha+1}$, choose $u \in \mathcal{E}_\alpha$ with $f_\alpha(u) > 0$. The points $x + tu$ and $x - tu$, for arbitrarily small $t > 0$, have opposite membership in P . Therefore

$$D_{\mathbf{1}_P,\varepsilon}(\mathcal{E}_\alpha) = \mathcal{E}_{\alpha+1}.$$

Induction and limit intersections give the asserted iterates. At stage θ the remaining set is the singleton $\{0\}$; the restriction of either characteristic function to a singleton is constant, so the next derivative is empty. Every earlier stage is nonempty. $\square$

[write] (Batch 93.) This is Part III's boundary oscillation rank (36.1), with the same indexing. Part III's theorem 36.1 computes it for a different set, the cut $x < p$ in $M_\alpha = (\mathbb{Q}^\alpha \overrightarrow{\times} \mathbb{Z})_{\geq 0}$ with discrete rational coordinates, and obtains $\beta + 1$ for $\alpha = \omega \cdot \beta + n$; here every real coordinate costs one derivative step, and the rank is $\theta + 1$. The two results do not compare the same objects.

For completeness, the Baire-class 1 assertion has an elementary metric proof. Write $P = \bigcup_n A_n$ and $X \setminus P = \bigcup_n B_n$ with increasing closed sets A_n, B_n . The two sets at a fixed stage are disjoint. After allowing the evident constant functions if one of the sets is empty, put

$$g_n(x) = \frac{d(x, B_n)}{d(x, A_n) + d(x, B_n)}.$$

The denominator is positive at every point. The functions are continuous; for $x \in P$ they are eventually 1, and for $x \notin P$ they are eventually 0. Hence $g_n \rightarrow \mathbf{1}_P$ pointwise. This construction claims pointwise convergence, not a uniform rate.

Corollary 215.2 (Ambient topological classification). *There is a homeomorphism $h : H_\theta^{\ell^2} \rightarrow H_\eta^{\ell^2}$ satisfying $h(P_\theta) = P_\eta$ if and only if $\theta = \eta$. No additive or linear assumption on h is needed for necessity.*

Proof. The closed sets in (215.1) are carried to the corresponding closed sets by a homeomorphism, so the oscillation ranks are equal. Ordinal successor is injective, hence $\theta = \eta$. The converse is supplied by a basis isometry. $\square$

This gives three descriptions of the same invariant: the order type of Archimedean classes, the length of the canonical closed hyperplane flag, and the predecessor of the oscillation rank. The descriptive hierarchy is therefore visible in both ordered algebra and ambient topology. An order may be Δ_2^0 while requiring arbitrarily long countable boundary recursion. There is no conflict with second countability: each *individual* recursion has only countably many steps, but there is no common countable bound for all cones on ℓ^2 .

216 Embedding operators and the closed-image boundary

Lemma 216.1 (Automatic linear continuity). *A Borel additive map between separable real Hilbert spaces is continuous and real linear.*

Proof. This is the Borel automatic-continuity theorem for Polish groups [45]; here is the needed argument. Let $T : H \rightarrow K$ be Borel. Given a neighborhood V of zero in K , choose a symmetric open neighborhood W with $W - W \subseteq V$. The separable metric subspace $T(H)$ is covered by countably many translates $T(x_j) + W$ with centers in $T(H)$. Thus H is covered by the corresponding translates of $A = T^{-1}(W)$. The Borel set A is nonmeager, since otherwise this would contradict Baire category. A nonmeager set with the Baire property has $A - A$ containing a neighborhood of zero: choose an open set on which A is comeager and intersect it with sufficiently small translates, exactly as in Lemma 212.1. Since $T(A - A) \subseteq W - W \subseteq V$, T is continuous at zero and hence everywhere. Additivity gives rational linearity; continuity and rational approximation then give real linearity. $\square$

[write] (Batch 93.) The continuity half is Part V's lemma 61.2; the same Pettis argument appears in Parts IV, VI and X.

Write $t_{\beta\alpha} = \langle Te_\alpha, e_\beta \rangle$ for a bounded real linear operator $T : H_\theta^{\ell^2} \rightarrow H_\eta^{\ell^2}$, with the source and target canonical bases understood.

Theorem 216.2 (Ordinal pivot normal form). *A bounded real linear operator $T : H_\theta^{\ell^2} \rightarrow H_\eta^{\ell^2}$ is an additive order embedding if and only if there is a strictly increasing map $\varphi : \theta \rightarrow \eta$ such that, for every $\alpha < \theta$,*

$$t_{\beta\alpha} = 0 \quad (\beta < \varphi(\alpha)), \quad t_{\varphi(\alpha),\alpha} > 0. \quad (216.1)$$

For such an operator,

$$v_\eta(Tx) = \varphi(v_\theta(x)) \quad (x \neq 0). \quad (216.2)$$

Together with Lemma 216.1, this classifies all Borel additive order embeddings by bounded ordinal-echelon operators.

Proof. If T is an order embedding, the positive vector Te_α has a leading index $\varphi(\alpha)$ and a positive leading coefficient. Lemma 214.2 forces the leading indices to be strictly increasing. This proves necessity.

Conversely, suppose (216.1). Let $x \neq 0$ have first nonzero coordinate α . Its Hilbert expansion uses only e_γ with $\gamma \geq \alpha$. For $\beta < \varphi(\alpha)$, every vector Te_γ in this expansion has zero β -coordinate. Continuity of T and of coordinate evaluation passes this statement from finite partial sums to Tx . At coordinate $\varphi(\alpha)$, all terms with $\gamma > \alpha$ vanish and the value is $t_{\varphi(\alpha),\alpha}x_\alpha$. This proves (216.2), injectivity, and preservation and reflection of sign. Translation then gives preservation and reflection of order. $\square$

[write] (Batch 93.) This is the real Hilbert analogue of Part X's [theorem 130.5](#) (row-finite rational matrices with increasing positive pivots on $\mathbb{Q}^\alpha$) and of Part I's [theorem 11.4](#): row-finiteness is replaced by boundedness. [Proposition 224.5](#) [merge] compares the two thresholds of Part X.

Corollary 216.3 (Automorphisms). *The Borel additive order automorphisms of $H_\theta^{\ell^2}$ are exactly the bounded invertible real linear operators whose matrices are lower triangular in the ordinal order and have strictly positive diagonal.*

Proof. An automorphism induces an order automorphism of the ordinal of ranks, which must be the identity. Theorem 216.2 gives the claimed matrix form. Conversely, a bounded invertible operator of that form preserves leading rank and sign; its bijectivity makes it an order automorphism. Boundedness of the inverse is the Banach inverse theorem. $\square$

Remark 216.4 (What the matrix conditions do not say). Positive diagonal and formal triangularity are not substitutes for operator boundedness or surjectivity. The theorem starts with a bounded operator, and the automorphism corollary explicitly requires invertibility. Formal infinite matrices without these hypotheses are not covered.

Example 216.5 (Three very different maps at rank ω). The unilateral shift $S(e_n) = e_{n+1}$ is an isometric order embedding with proper closed range. The diagonal map $\text{Diag}(e_n) = (n+1)^{-1}e_n$ is a bounded order embedding with dense proper range, not an automorphism and not a topological embedding for the subspace norm topology on its image. Finally, for $0 < c < 1$, the operator $I + cS$ is an order automorphism: its diagonal is positive, and its bounded inverse is the norm-convergent Neumann series $\sum_{k \geq 0} (-cS)^k$.

Example 216.6 (An algebraic automorphism need not be Borel). Let $a : \mathbb{R} \rightarrow \mathbb{R}$ be a discontinuous additive function. On $H_\omega^{\ell^2}$, the map

$$(x_0, x_1, x_2, \dots) \longmapsto (x_0, x_1 + a(x_0), x_2, \dots)$$

is an additive order automorphism, with the evident inverse obtained by subtracting $a(x_0)$. It preserves sign because the first coordinate is unchanged and the correction vanishes when that coordinate is zero. It is not Borel by Lemma 216.1. Thus ordinal classification under arbitrary additive isomorphisms does not imply that every such isomorphism is a bounded linear map.

[write] (Batch 93.) This is the real-coefficient form of Part X's choice-built maps ([remark 133.2](#) and [theorem 133.3](#)) and of the abstract automorphisms of Part XIII's [theorem 176.1](#). It uses only the coordinates 0 and 1, so it lives on $H_\theta^{\ell^2}$ for every $\theta \geq 2$ ([proposition 224.5](#) [merge]).

Lemma 216.7 (Polish subgroups in their subspace topology). *A subgroup A of a Polish group G is Polish in its subspace topology if and only if it is closed.*

Proof. The reverse implication is standard completeness. For the forward implication, A is G_δ in G , hence is dense G_δ in its closure K . It is comeager in K . For any $k \in K$, both A and kA are comeager in K and therefore meet. Such a meeting implies $k \in A$. Hence $A = K$ is closed. $\square$

[write] (Batch 93.) This is the last clause of Part X's [lemma 130.1](#).

Corollary 216.8. *For a Borel additive order embedding $T : H_\theta^{\ell^2} \rightarrow H_\eta^{\ell^2}$, its image is Polish in the inherited Hilbert topology exactly when its range is closed, equivalently when there is $c > 0$ such that $\|Tx\| \geq c \|x\|$ for all x .*

Proof. Use Lemma [216.7](#) and the bounded inverse theorem on the closed range. Conversely, the displayed lower bound makes the range complete and hence closed. $\square$

[write] (Batch 93.) Compare Part X: every continuous order embedding of $\mathbb{Q}^\alpha$ has closed image ([theorem 130.7](#)), whereas here Diag already fails at $\theta = \omega$ ([example 216.5](#), [proposition 224.5](#) [merge]).

217 Sharp regularity and a nonlocally convex obstruction

The hereditary hypothesis in Theorem [213.2](#) cannot be replaced by the Baire property of the whole cone. The issue already occurs in two real dimensions.

Proposition 217.1 (A Baire cone that is not Borel). *There is a translation-invariant total order on the additive group $\mathbb{R}^2$ whose cone has the Baire property but is not Borel. It is not compatible with multiplication by all positive real scalars.*

Proof. Choose a Hamel basis of $\mathbb{R}$ over $\mathbb{Q}$ containing 1 and $\sqrt{2}$. Let a be the additive automorphism which fixes 1 and every other basis vector except $\sqrt{2}$, and sends $\sqrt{2}$ to $-\sqrt{2}$. Set $Q = \{t : a(t) > 0\}$. It defines an additive total order on $\mathbb{R}$. It is not the usual order or its reverse, since $1 \in Q$ and $\sqrt{2} \notin Q$.

If Q had the Baire property, Lemma [212.2](#) in dimension one would identify it with one of the two open rays. Thus Q does not even have the Baire property, and in particular is not Borel. Now put

$$P = \{(s, t) : s > 0\} \cup (\{0\} \times Q).$$

This is a total additive cone. It differs from the open halfspace $\{s > 0\}$ by a subset of a nowhere dense line, so it has the Baire property in $\mathbb{R}^2$. If it were Borel, its trace on that line would make Q Borel, a contradiction. Finally, the positive vector $(0, 1)$ becomes negative after multiplication by the positive real $\sqrt{2}$. $\square$

[write] (Batch 93.) This realizes the warning in the remark after Part III's [theorem 33.1](#) and bears on Part IV's [question 57.4](#); it concerns the vector-space conclusions, not the arithmetic rigidity that Part VI derives from an ambient Baire property. It is consistent with Part XIII's [corollary 170.6](#): on $\mathbb{R} \times E$ with E discrete the real slices are open, so the ambient Baire property is inherited by them, which fails for the meager line of $\mathbb{R}^2$.

The example explains why intersecting an arbitrary Baire-property set with a closed subspace is not a valid step: on a meager subspace the trace can be completely irregular. Borel regularity avoids precisely this failure.

Theorem 217.2 (No Borel order when the continuous dual vanishes). *If a nonzero real Polish topological vector space E has no nonzero continuous real linear functional, then its additive group has no total order whose cone has the Baire property. In particular it has no Borel additive total order.*

Proof. Lemma 212.2 would produce a nonzero continuous linear functional. Its proof uses separation of a convex set with nonempty interior, and therefore does not require local convexity of E . $\square$

Corollary 217.3 (Nonatomic L^p). *For $0 < p < 1$, the real space $L^p([0, 1])$, with its usual complete quasi-norm topology, admits no Borel translation-invariant additive total order.*

Proof. The metric $d(f, g) = \int_0^1 |f - g|^p$ defines a separable complete vector topology. Let L be a continuous real linear functional. Homogeneity and continuity give a constant C such that $|L(f)| \leq C \|f\|_p$ for all f . For a measurable set A , divide A into n measurable pieces of equal measure, which is possible for Lebesgue measure. Then

$$|L(\mathbf{1}_A)| \leq \sum_{j=1}^n |L(\mathbf{1}_{A_j})| \leq Cn \left(\frac{\mu(A)}{n} \right)^{1/p} = C\mu(A)^{1/p} n^{1-1/p}.$$

Since $1 - 1/p < 0$, the right side tends to zero. Thus L vanishes on indicators and hence on simple functions. Their density and continuity imply $L = 0$. Theorem 217.2 now applies. $\square$

[write] (Batch 93.) Source 22's abstract announces "a nonexistence theorem for nonatomic L^p spaces with $0 < p < 1$ "; the corollary proves it for $L^p([0, 1])$. Proposition 224.2 [merge] proves the announced statement for every nonatomic measure (with $L^p \neq 0$, separable or not), so the abstract's claim stands, with that proof.

Abstract additive orders still exist by algebraic choice arguments. The obstruction is regularity in the given topology, not orderability of a torsion-free abelian group in isolation.

218 A classified hierarchy of Polish Presburger models

For a countable ordinal θ , give

$$G_\theta^{\ell^2} = H_\theta^{\ell^2} \times \mathbb{Z}$$

its product topology, with $\mathbb{Z}$ discrete, and its lexicographic group order, with $H_\theta^{\ell^2}$ the leading component. Its least positive element is $1 = (0, 1)$. Explicitly,

$$M_\theta^{\ell^2} = (P_\theta \times \mathbb{Z}) \cup (\{0\} \times \mathbb{N}). \quad (218.1)$$

Theorem 218.1 (Topology, arithmetic, and definability). *For every countable ordinal θ :*

- (i) $M_\theta^{\ell^2}$ is Polish in its subspace topology and addition is jointly continuous.
- (ii) $M_\theta^{\ell^2} \models \text{Th}(\mathbb{N}, 0, 1, +, <)$.
- (iii) Every definable relation with parameters in $M_\theta^{\ell^2}$ is Δ_2^0 in the appropriate finite Cartesian power.

For $\theta > 0$, the model is uncountable and has no isolated points.

Proof. By Theorem 213.2, P_θ and $P_\theta \cup \{0\}$ are Δ_2^0 . Formula (218.1), or its complement, shows that $M_\theta^{\ell^2}$ is Δ_2^0 in $G_\theta^{\ell^2}$. In particular it is G_δ in a Polish space and therefore Polish. The group addition restricts to a jointly continuous operation on the cone.

The least positive element assertion is immediate: any positive vector component dominates every integer component. For any standard integer $m > 0$ and $(x, n) \in G_\theta^{\ell^2}$, write $n = mq + r$ with $0 \leq r < m$. Then

$$(x, n) = m(x/m, q) + (0, r). \quad (218.2)$$

The remainder is unique. When $(x, n) \geq 0$, the quotient in this formula is also nonnegative: if $x > 0$, then $x/m > 0$, and if $x = 0$ then $n \geq 0$ implies $q \geq 0$. Thus $G_\theta^{\ell^2}$ is a $\mathbb{Z}$ -group and its nonnegative cone is a full Presburger model by the classical characterization stated in Section 211.

To prove the complexity assertion, use Presburger quantifier elimination. Equality of affine integer-linear terms is closed, and its complement is open, so it is Δ_2^0 . Order comparison is the inverse image of the Δ_2^0 group cone under a continuous affine integer-linear map. Congruence modulo m depends only on the discrete integer component, since $H_\theta^{\ell^2}$ is divisible; its interpretation is clopen. A quantifier-free formula is a finite Boolean combination of these sets and is therefore Δ_2^0 . Restriction to powers of $M_\theta^{\ell^2}$ preserves this property. Fixed parameters only translate the affine maps.

For $\theta > 0$, the map $t \mapsto (te_0, 0)$ for $t > 0$ gives uncountably many elements. For any $(x, n) \in M_\theta^{\ell^2}$ and $t > 0$, $(x + te_0, n)$ is another element of the cone tending to (x, n) as $t \downarrow 0$. Hence no point is isolated. $\square$

[write] (Batch 93.) A special case of Part III's [theorem 34.1](#) (every divisible Polish abelian group with Borel positive cone), with the same division formula; $M_1^{\ell^2}$ is Part I's $M_{\mathbb{R}}$.

Remark 218.2 (Not the order topology). The topology just used is the inherited Hilbert-product topology, not the order topology of the discrete ordered group. Every nonempty slice $M_\theta^{\ell^2} \cap (H_\theta^{\ell^2} \times \{n\})$ is a clopen connected component: it is P_θ or $P_\theta \cup \{0\}$, both convex and connected. For $\theta > 0$ these components contain nontrivial paths. In particular this presentation is not homeomorphic to the totally disconnected Baire space, even though both are Polish settings for additive arithmetic.

Theorem 218.3 (All unit-preserving maps). *Every unit-preserving additive embedding $F : M_\theta^{\ell^2} \rightarrow M_\eta^{\ell^2}$ extends uniquely to an additive order embedding $\widehat{F} : G_\theta^{\ell^2} \rightarrow G_\eta^{\ell^2}$, and has the form*

$$\widehat{F}(x, n) = (T(x), n) \quad (218.3)$$

for an additive order embedding $T : H_\theta^{\ell^2} \rightarrow H_\eta^{\ell^2}$. If F is Borel, then T is bounded real linear, and F is continuous. Every such F is an elementary embedding of Presburger structures. Consequently $M_\theta^{\ell^2}$ embeds in $M_\eta^{\ell^2}$ in this sense exactly when $\theta \leq \eta$, and the models are isomorphic exactly when their indices agree.

Proof. Every group element is a difference of two nonnegative elements, and cancellation makes the extension $\widehat{F}(a - b) = F(a) - F(b)$ well defined and unique. Injectivity of F makes the group extension injective. Since positivity is preserved, a negative nonzero element cannot have positive image: its negative has positive nonzero image. Thus order is reflected as well.

The elements divisible by every positive standard integer in $G_\theta^{\ell^2}$ are exactly $H_\theta^{\ell^2} \times \{0\}$: the integer coordinate of such an element must be divisible by every positive integer. An additive map carries these elements into the corresponding subgroup of $G_\eta^{\ell^2}$. Preservation of $(0, 1)$ then proves (218.3).

If F is Borel, choose the positive and negative parts of $g \in G_\theta^{\ell^2}$ by its Borel sign comparison. This is a Borel map $g \mapsto (g^+, g^-)$ into $M_\theta^{\ell^2}$, and $\widehat{F}(g) = F(g^+) - F(g^-)$ is Borel. Its restriction T is continuous and real linear by Lemma 216.1; in a Hilbert space continuous linear means bounded. Formula (218.3) gives continuity of the original F .

The extension preserves and reflects all order and equality relations of affine terms. It also preserves and reflects congruences, since the integer coordinate is fixed. Quantifier elimination therefore makes it elementary, and the same follows for its nonnegative cone. The exact ordinal criteria follow from Theorem 214.4. $\square$

[write] (Batch 93.) The algebraic half is Part X's theorem 129.1 and Part I's proposition 11.3 (any divisible ordered groups), with the same proof through the subgroup $\bigcap_m mG$; the Borel clause is the ℓ^2 analogue of Part X's theorem 130.3.

Theorem 218.4 (When an elementary image is Polish). *For a Borel unit-preserving additive embedding F as in Theorem 218.3, the image $F(M_\theta^{\ell^2})$ is Polish in the subspace topology of $M_\eta^{\ell^2}$ exactly when $T(H_\theta^{\ell^2})$ is closed, equivalently when T is bounded below.*

Proof. If the image cone is Polish, it is G_δ in the Polish ambient group $G_\eta^{\ell^2}$ as well as in $M_\eta^{\ell^2}$. Its negative is also G_δ . A finite union of G_δ sets is G_δ , so

$$\widehat{F}(G_\theta^{\ell^2}) = F(M_\theta^{\ell^2}) \cup (-F(M_\theta^{\ell^2}))$$

is a Polish subgroup of $G_\eta^{\ell^2}$. By Lemma 216.7 it is closed. Formula (218.3) then makes the Hilbert range closed. Conversely, on a closed range times $\mathbb{Z}$ the nonnegative cone is G_δ , hence Polish. The bounded-below criterion is Corollary 216.8. $\square$

[write] (Batch 93.) The analogue of Part X's theorem 131.1 (M_V Polish exactly when V is closed).

For example the shift gives a proper closed Polish elementary submodel of $M_\omega^{\ell^2}$; the diagonal map Diag of Section 216 gives a proper elementary submodel whose inherited topology is not Polish. Isomorphism with an abstract Polish model alone does not make a subspace Polish.

219 Omnific realization without a proper-class topology

We use the standard normal form of a surreal number as a series

$$\sum_{\gamma \in S} r_\gamma \omega^\gamma,$$

where S is a set that is reverse well ordered. Omnific integers are the series with no negative exponent and with integer constant coefficient [145, 72]. Only this normal-form description and coefficientwise addition are needed; no factorization result is used.

Fix $0 < \theta < \omega_1$. Regard ordinals below or equal to θ as their canonical surreal representatives, and put

$$\gamma_\alpha = \theta - \alpha \quad (\alpha < \theta). \quad (219.1)$$

The subtraction in (219.1) is surreal subtraction, not an ordinal subtraction operation. These exponents are positive and strictly decreasing as α increases.

Theorem 219.1 (Explicit omnific coordinates). *The map*

$$J_{\theta}^{\ell^2} : G_{\theta}^{\ell^2} \longrightarrow \mathbf{Oz}, \quad J_{\theta}^{\ell^2}(x, n) = n + \sum_{\alpha < \theta} x_{\alpha} \omega^{\gamma_{\alpha}} \quad (219.2)$$

is a well-defined injective unit-preserving additive order embedding. Every subset of θ occurs as the support index set of the nonconstant part of some element in its image. In particular there are image elements whose nonconstant normal form has support order type exactly θ in decreasing-exponent order.

Proof. For every subset $A \subseteq \theta$, the set $\{\gamma_{\alpha} : \alpha \in A\}$ is reverse well ordered: every nonempty subset has a largest exponent corresponding to its least index. Thus the series in (219.2) is a legitimate set-supported surreal normal form. All exponents are positive and n is an integer, so its value is omnific.

Distinct coordinates give distinct normal forms. Addition is coefficientwise on these fixed supports; the union of two supports is still a subset of the same reverse well-ordered set. The leading nonzero coefficient has the sign of the first nonzero Hilbert coordinate, unless the Hilbert part vanishes, in which case the integer coefficient decides. This proves all algebraic assertions.

For the support assertion, choose an injection $b : \theta \rightarrow \mathbb{N}$ and put $x_{\alpha} = 2^{-b(\alpha)-1}$ on A , and $x_{\alpha} = 0$ elsewhere. Then $\sum_{\alpha} x_{\alpha}^2 \leq \sum_{n \geq 0} 4^{-n-1} < \infty$, so $x \in H_{\theta}^{\ell^2}$ and its support index set is exactly A . $\square$

[write] (Batch 93.) A variant of Part III's [theorem 40.1](#) ($J_{\alpha}(a, n) = \sum_{\beta} a_{\beta} \omega^{1/(\beta+1)} + n$, rational coefficients) and Part I's [theorem 9.1](#): the exponents $\theta - \alpha$ and the real square-summable coefficients differ, so $J_{\theta}^{\ell^2}$ is printed with the superscript ℓ^2 ([section 1.3](#)).

The image is a set of surreals, obtained as the image of the set $H_{\theta}^{\ell^2} \times \mathbb{Z}$. All quantification needed for its arithmetic is set-sized. One may formalize the normal-form interpretation in an ordinary set/class foundation such as ZFC with definable classes, or in a set-sized Hahn environment containing the relevant exponent group. No set of all surreal numbers is formed, and no probability measure or topology on the entire proper class is invoked.

Proposition 219.2 (Three topologies must not be confused). *For infinite θ , the image of $G_{\theta}^{\ell^2}$ can be given a Polish additive-group topology by transporting the Hilbert-product topology through $J_{\theta}^{\ell^2}$. This is neither the inherited coefficient-product topology nor the topology induced from the surreal order topology. The inherited coefficient-product topology is not Polish; the inherited surreal order topology is discrete.*

Proof. Transport of topology gives the first claim directly. For the coefficient-product topology, the Hilbert coefficient space is $\ell^2(\theta)$ as a subgroup of $\mathbb{R}^{\theta}$ with its product topology. It contains all finite-support sequences and hence is dense, but it is proper because the constant sequence 1 is not square-summable. Lemma [216.7](#) shows that this subgroup is not Polish in its subspace topology. Adding the discrete integer factor does not change this conclusion.

For the order-topology assertion, two distinct image elements differ in absolute value by at least 1, because the image is discretely ordered with least positive element 1. The surreal interval $(z - 1/2, z + 1/2)$ therefore meets the image in the singleton $\{z\}$. Every image point is isolated in that inherited topology. $\square$

[write] (Batch 93.) The same distinction is made after Part I's [theorem 9.1](#) and in Part III's [section 40](#).

In particular, saying that a Polish Presburger model is realized “inside the omnific integers” in this report is an *algebraic* statement plus an explicitly transported topology. It is not a claim that the ordinary surreal order topology has an uncountable separable discrete subspace that is Polish.

220 Why multiplication cannot be added

The next obstruction is algebraic and does not assume continuity or measurability of multiplication.

Lemma 220.1 (The divisible part is an ideal). *Suppose the additive group of a unital ring is $G_\theta^{\ell^2}$, with its specified unit. Then $D = H_\theta^{\ell^2} \times \{0\}$ is a two-sided ideal.*

Proof. It equals $\bigcap_{m \geq 1} mG_\theta^{\ell^2}$. If $d \in D$ and r is any ring element, write $d = md_m$ for each positive integer m . Then $rd = m(rd_m)$ and $dr = m(d_mr)$, so both products are divisible by every positive integer. They therefore belong to D . $\square$

Theorem 220.2 (No compatible ordered ring). *For every $\theta > 0$, no multiplication on the additive group $G_\theta^{\ell^2}$ with identity $(0, 1)$ makes its given lexicographic order into a discretely ordered ring order. Commutativity of the proposed multiplication is not needed.*

Proof. Let $u = (e_0, 0)$. It is larger than every standard integer. For every $d \in D$, there exists a positive standard integer n with $d < nu$: choose n strictly larger than the leading real coefficient at coordinate zero if that coordinate is present; later coordinates cannot change that comparison.

In a compatible ordered ring, $u > 0$ and $u - n > 0$ for every positive standard integer n . Positivity of products would yield

$$u^2 - nu = u(u - n) > 0, \quad \text{hence} \quad u^2 > nu \quad \text{for all } n.$$

But Lemma 220.1 puts u^2 in D , where every element is bounded above by some nu . This is a contradiction. $\square$

The same argument rules out a unital semiring multiplication on $M_\theta^{\ell^2}$ that is distributive and has strictly positive products of strictly positive elements: its cancellative additive monoid has the usual ring completion, and that completion would satisfy the contradictory inequalities above.

The explicit omnific image also displays the failure directly: $J_\theta^{\ell^2}(e_0, 0) = \omega^\theta$, whose square has exponent $\theta + \theta$ (surreal addition), larger than every allowed exponent in (219.1). Thus the image is not closed under the ambient omnific multiplication. More strongly, Theorem 220.2 excludes *any* replacement multiplication compatible with the same ordered addition and unit.

This conclusion is narrower than Glazer’s general topological Tennenbaum theorem [1], and its hypotheses are different. It explains why the present positive constructions for additive arithmetic do not extend to ordered ring arithmetic; it does not replace the general theorem or settle all weak-arithmetic topology questions.

[write] (Batch 93.) Theorem 220.2 is weaker than Part I’s theorem 10.4, which from a cofinal infinite additive unit, here $u = (e_0, 0)$, excludes every unital semiring multiplication, with no order axiom on products; compare also Part III’s proposition 37.5 and theorem 37.10.

Source 22's route through the divisible ideal is different. The remark on semirings with positive products is the content of Part I's [lemma 10.3](#).

221 Verification, formalization, and proof dependencies

The accompanying `verify.py` uses exact rational arithmetic. It tests the finite-dimensional algebra that underlies the operator normal form and arithmetic lift: totality and translation invariance of lexicographic comparison, closure of positive cones, rectangular pivot matrices, triangular automorphisms, division with remainder, fixed-support Hahn addition, and positive/negative perturbation witnesses at finite flag boundaries. It also checks small counterexamples to dropping increasing pivots or positive pivot coefficients. Its machine-readable output is `verification_results.json`. The delivered run passes all 136,562 explicit assertions in seven test families, using the fixed random seed 231113699. No floating-point approximation is used in these checks.

[write] In this report the program is `code/22-hilbert-orders-verify.py`, its recorded run data `data/22-hilbert-orders-verification_results.json`, the build script `code/22-hilbert-orders-build.sh` and the build record data `data/22-hilbert-orders-BUILD_REPORT.json`. The intake reran the program on a copy (Python 3.14.4): 136,562 assertions in seven families pass, also under `-O`, and the record equals the shipped one except for the interpreter version (3.14.4 for the recorded 3.13.5) and the carriage returns that Windows text mode adds.

These computations do *not* prove transfinite termination, measurability, the Baire category lemma, completeness of the Hilbert basis, the normal-form theorem for surreals, or Presburger quantifier elimination. The proofs of the new structural statements are the arguments in this article. The computational role is to catch sign, indexing, and remainder mistakes in executable finite instances.

The dependency structure is short. The category argument and geometric separation give the halfspace lemma. Iteration and second countability give the canonical flag. Orthogonal complements identify it with an ordered orthonormal basis. Leading coordinates then give Archimedean ranks, embeddings, and matrix forms. The same boundary flag computes the oscillation derivative. The arithmetic layer adds only the classical $\mathbb{Z}$ -group characterization; the surreal layer adds only normal forms. The ring obstruction is independent of automatic continuity and of descriptive-set-theoretic rank theory.

A useful proof-assistant development would separate four interfaces: (1) strict additive cones and their rational homogeneity; (2) the category/halfspace interface on real Polish vector spaces; (3) a countable closed flag and the Hilbert orthogonal extraction; (4) the algebraic Presburger and normal-form interpretations. The countability proof should be formalized through the explicit injection of successor stages into a countable topological base. For Hilbert sums, it is important to use unconditional summability rather than pretend that the ordinal basis is enumerated in order by $\mathbb{N}$. No Lean or Rocq proof is claimed in this package.

222 Further research questions

The following are proposed continuation problems. Unless otherwise specified, they are questions left open by *this report*, not assertions that the entire literature has established their novelty.

Question 222.1 (Effective flag extraction). Given an effective code for a Borel additive cone on a computable Hilbert space, what information is needed to recover its canonical ordinal flag and oriented basis? Which pointclasses of codes permit hyperarithmetic extraction, and how does the answer depend on a supplied ordinal notation?

The closure operation, the selection of separating functionals, and well-founded iteration should be analyzed separately. A finite matrix checker does not address any of these issues.

Question 222.2 (Uniform families and bounded ranks). Under which precise coding hypotheses does a Borel or analytic family of Borel Hilbert orders have bounded flag rank below ω_1 ? Is the family of valid order codes equipped with a natural coanalytic rank whose value is the flag index?

The exact oscillation calculation makes general boundedness theorems for Baire-class 1 ranks a plausible tool, but a uniform code-to-function map must first be supplied. Merely observing that each section is Δ_2^0 is not a proof of uniform boundedness.

[write] (Batch 93.) The sentence on Baire-class-one boundedness is a heuristic, as source 22 says; no boundedness theorem is claimed.

Question 222.3 (Banach-space classification). For a fixed separable Banach space E , which countable ordinals arise as flag ranks, and when do two flags yield bounded-linearly isomorphic ordered groups? Can a useful complete invariant be expressed by the successive quotient maps and their norm geometry?

The Hilbert proof eliminates this extra geometry by orthogonality. On general Banach spaces a sequence of hyperplanes need not be encoded by an orthonormal basis, and boundedness of a prospective coordinate map is an additional problem.

Question 222.4 (Beyond vector spaces). Classify Borel total orders on Polish divisible abelian groups that are not supplied with a compatible real scalar action. Which groups admit a boundary recursion with Archimedean quotients, and what replaces the one-dimensional continuous real quotient?

This is the natural point of contact with the repository's nonsplit and discrete-direction Presburger models. Our $\aleph_1$ classification should not be extrapolated to that larger class.

[write] (Batch 93.) Partly answered before source 22: Part III's [theorem 33.1](#) already gives the canonical boundary recursion, with closed convex subgroups and Δ_2^0 layers, on every Polish abelian group with a Borel order. What remains is the analysis of the successive quotients, which need not be real lines; Parts IV, VII, IX and XIII treat locally compact, nonsplit and one-real-dimensional cases.

Question 222.5 (Measurability weaker than Borel). Which universally measurable additive cones on real Polish vector spaces must be Borel? Does an appropriate hereditary measurability condition imply the same flag theorem when Baire category is replaced by a measure-theoretic argument?

Proposition [217.1](#) excludes a naive replacement by the Baire property in the whole space. Any proposed weaker condition must control traces on closed hyperplanes.

[write] (Batch 93.) Compare Part IV's [question 57.4](#), which asks the same for arithmetic rigidity and is answered there (notes after it); [proposition 217.1](#) shows that for the vector-space conclusions the ambient Baire property is not enough.

Question 222.6 (Prescribed symmetries). For a given countable family of bounded invertible operators on H , when is there a Borel additive total order preserved by all of them? Can this be characterized by a common countable well-ordered invariant flag with positive one-dimensional quotient actions?

Necessity follows from the canonical flag; sufficiency follows if the flag is complete. Finding such a flag from the operator family is the substantive task, especially at limit stages. [write] (Batch 93.) Source 22's sentence "Necessity follows from the canonical flag; sufficiency follows if the flag is complete" is asserted without proof. [Proposition 224.4](#) [merge] proves both directions, for any family of topological linear automorphisms of a real Polish vector space. The question that remains is the one source 22 calls substantive: to decide, from the operator family, whether such a flag exists.

Question 222.7 (Topology of the arithmetic cones). Determine the homeomorphism type of $M_\theta^{\ell^2}$ as a bare space, forgetting its addition and order. Which distinct indices become homeomorphic after the cone is separated from its ambient Hilbert space? Can its algebraic rank be recovered from a minimal enrichment of the topological space by named maps?

Corollary [215.2](#) settles the ambient-pair version, not the bare-space version. Conflating these equivalence relations would erase an essential distinction.

[write] (Batch 93.) Related to Part III's [question 43.2](#); $M_\theta^{\ell^2}$ for $\theta > 0$ is not zero-dimensional and has connected clopen slices ([remark 218.2](#)), but it answers none of the report's questions on connected carriers, since its components are the slices.

Question 222.8 (Optimal omnific realizations). Can the exponent systems in ([219.1](#)) be chosen with useful minimal birthday or initiality properties while retaining the Hilbert topology on the coefficient space? Which prescribed families of bounded embedding operators admit compatible ambient surreal maps?

The present construction is additive and set-sized. It makes no simplicity-initiality claim, and the no-ring theorem excludes a multiplicative upgrade on the same carrier.

[write] (Batch 93.) The first sentence repeats, for this family, Part I's [question 14.10](#), Part III's [question 43.11](#) and Part X's [question 138.12](#).

Question 222.9 (Foundational strength and audited formalization). What subsystem of second-order arithmetic proves the Hilbert flag classification for coded separable spaces and coded Borel cones? How much transfinite recursion is needed for the exact rank theorem, and can its proof be made compatible with a machine-checked Presburger lift?

This is a separate problem from Glazer's Question 1 concerning his own theorem. The repository lists an unreviewed claimed answer to that question; the present report neither relies on nor audits it [[139](#)]. The two reverse-mathematical questions should not be identified merely because both mention transfinite methods.

[write] (Batch 93.) The "unreviewed claimed answer" is [Part VIII](#); Part XI also claims an ATR_0 bound for its local theorem. Neither is used here.

Further questions and research [write]

(Batch 93.) Under the repository's rule that unproved claims are kept as questions, the write adds the following. The other unproved statements found in source 22 were proved for this report: the abstract's L^p claim ([proposition 224.2](#)) and the characterization asserted after [question 222.6](#) ([proposition 224.4](#)).

Question 222.10 (Priority against Küçük–Soyertem–Küçük [write]). Does the 2012 paper [141] contain [theorem 210.1](#), or its finite-dimensional or real-homogeneous cases, and under which hypotheses?

Source 22 read only the indexed abstract and could not fetch the full text; it makes no exclusive priority claim and asserts no gap in that paper. The write read the abstract on 4 October 2026 (the bibliography entry gives the record): the paper constructs a total ordering cone of a separable Hilbert space from an orthogonal base and states that every total order can be represented by such a cone. Read without a regularity hypothesis that sentence would be contradicted by [proposition 217.1](#) and [remark 224.3](#) [merge], so the paper presumably carries hypotheses that the abstract does not show. What is missing is a reading of the full text, to compare its hypotheses (topological, measurable or none) with Borelness and to see whether transfinite index types occur.

223 Conclusion

The classification is governed by an ordinal, not by cardinality or by Hilbert dimension alone. A Borel additive order successively exposes a continuous one-dimensional real quotient until a countable flag reaches zero. In Hilbert space, orthogonality turns this flag into a canonical well-ordered basis. The ordinal completely determines additive isomorphism and embeddability, and its successor is exactly the oscillation rank of the cone's characteristic function.

Thus one fixed Hilbert topology supports an ω_1 -long hierarchy of additive order types although every cone is Λ_2^0 . Adding the least positive integer gives Polish Presburger models whose embeddings are elementary and whose inherited Polish submodels have an exact closed-range criterion. Surreal normal forms realize every level of the hierarchy additively, while a purely algebraic bound in the divisible part prevents compatible ordered multiplication. These are concrete structural extensions of the Glazer–Provelt intersection, with a clear boundary between proved classification, established inputs, and further research.

224 Where Part XVI meets Parts III, IV and X [merge]

This section was written for this report. It identifies source 22's flag with Part III's cone recursion, proves the two statements of source 22 that were asserted without proof (the L^p theorem of its abstract and the characterization after [question 222.6](#)), and records what source 22's theorems say about questions of Parts I, III and X. The proofs use only results proved in source 22 and in those parts, and classical facts named where they are used; they have not been independently reviewed. Throughout, E is a real Polish topological vector space, not necessarily locally convex, with a translation-invariant total order whose strict positive cone is P , and $(\mathcal{E}_\alpha)$ is the recursion (213.1)–(213.3).

Proposition 224.1 (The flag is Part III's cone recursion). *Suppose P is hereditarily Baire on closed linear subspaces, for instance Borel. Then $\mathcal{E}_\alpha = H_\alpha$ for every $\alpha \leq \theta$, where (H_α) is Part III's recursion (33.3) for the Polish group $A = (E, +)$; hence the terminal index θ of [theorem 213.2](#) is Part III's cone closure rank $\rho(E, <)$, and the layers $\{x \in \mathcal{E}_\alpha : f_\alpha(x) > 0\}$ are Part III's layers L_α of (33.4).*

Proof. Part III defines $C(H) = \text{cl}_H(P_H \cup \{0\})$ and $D(H) = C(H) \cap (-C(H))$ for a closed subgroup H , with $P_H = P \cap H$. If $H = \mathcal{E}_\alpha \neq \{0\}$, then $P_H \neq \emptyset$ by totality, and for $p \in P_H$ the

points p/n lie in P_H (positive rational homogeneity, [section 211](#)) and tend to 0. So $0 \in \text{cl}_H(P_H)$, $C(H) = \text{cl}_H(P_H)$, and $D(H)$ is the right side of [\(213.2\)](#). Both recursions start at E and intersect at limits, so induction gives $\mathcal{E}_\alpha = H_\alpha$ up to the first stage at which $\{0\}$ is reached, which is therefore the same ordinal. Part III's $L_\alpha = H_\alpha \setminus (-C(H_\alpha))$ is $\mathcal{E}_\alpha \setminus \{f_\alpha \leq 0\}$ by [\(212.1\)](#). $\square$

So [theorem 213.2](#) and the implication from Borel to Δ_2^0 in [corollary 213.3](#) are Part III's [theorem 33.1](#) in the vector-space case, by the same proof; what source 22 adds is that every step has real codimension one with a continuous functional, that real homogeneity follows, and that “hereditarily Baire” suffices. Part III's proof uses Borelness only through the Baire property of each trace, so it gives the same strengthening for Polish groups.

Proposition 224.2 (No Baire order without a dual; every nonatomic L^p).

- (i) Let $F \neq \{0\}$ be a real topological vector space whose topology is completely metrizable, separable or not, and on which every continuous real linear functional vanishes. Then no translation-invariant total order on $(F, +)$ has a positive cone with the Baire property; in particular none has a Borel cone.
- (ii) Let $0 < p < 1$, and let μ be a nonatomic measure (every set of positive measure contains a set of strictly smaller positive measure) with $L^p(\mu) \neq \{0\}$. Then $L^p(\mu)$, with the metric $d(f, g) = \int |f - g|^p d\mu$, satisfies the hypotheses of (i).

Proof. (i) The proofs of [lemmas 212.1](#) and [212.2](#) use of the space only the Baire category theorem, in the space and in its nonempty open subsets, and the separation of [appendix AH](#), which needs a Hausdorff topological vector space and nothing more. Both hold in a completely metrizable space. So a total order with a Baire-property cone would give a nonzero continuous linear functional by [lemma 212.2](#), which does not exist. This is [theorem 217.2](#) without separability.

(ii) The metric d is complete and translation invariant, and scalar multiplication is continuous, by the usual Riesz–Fischer argument, which works for $0 < p < 1$. Let L be a continuous linear functional. Choose $\delta > 0$ with $|L(f)| < 1$ whenever $N(f) = \int |f|^p d\mu < \delta$. For $N(f) > 0$, the multiple cf with $c^p N(f) = \delta/2$ gives $|L(f)| \leq C N(f)^{1/p}$ with $C = (2/\delta)^{1/p}$, and this also holds when $N(f) = 0$. Let $0 < \mu(A) < \infty$. By Sierpiński's theorem a nonatomic measure takes every value between 0 and $\mu(A)$ on subsets of A , so for each n the set A splits into n measurable pieces A_j of measure $\mu(A)/n$, and

$$|L(\mathbf{1}_A)| \leq \sum_{j=1}^n |L(\mathbf{1}_{A_j})| \leq C n \left(\frac{\mu(A)}{n} \right)^{1/p} = C \mu(A)^{1/p} n^{1-1/p} \rightarrow 0.$$

So L vanishes on the indicators of sets of finite measure. Every $f \in L^p(\mu)$ is a pointwise limit of simple functions s_n with $|s_n| \leq |f|$; each s_n is supported on a set of finite measure, because $|s_n|^p \leq |f|^p$ is integrable, and dominated convergence gives $d(s_n, f) \rightarrow 0$. Hence $L = 0$. $\square$

Part (ii) is the statement of source 22's abstract (“nonatomic L^p spaces”), of which [corollary 217.3](#) proves the case $L^p([0, 1])$. When $L^p(\mu)$ is separable it is Polish and [theorem 217.2](#) applies as stated; part (i) removes the separability. Sierpiński's theorem is classical and is used only to split A into equal pieces.

Remark 224.3 (Real homogeneity does not replace Borelness). On every infinite-dimensional separable real Hilbert space H there is a translation-invariant total order whose cone is closed under multiplication by positive reals (a total *convex* cone) but is not Borel, so it is not lexicographic in any orthonormal basis. Choose a discontinuous linear functional φ

(a Hamel basis, so the axiom of choice), and a Borel order with cone P' , for instance the lexicographic one in some orthonormal basis. Put

$$P = \{\varphi > 0\} \cup (\ker \varphi \cap P').$$

The three cone identities (211.1) and positive real homogeneity are checked case by case, using that P' is real homogeneous (theorem 213.2). If P were Borel, lemma 212.2 would give a nonzero continuous f_0 with $P \subseteq \{f_0 \geq 0\}$. The kernel of the discontinuous φ is not closed, and a hyperplane that is not closed is dense; so for x_0 with $\varphi(x_0) > 0$ the set $x_0 + \ker \varphi \subseteq P$ is dense, and the closed set $\{f_0 \geq 0\}$ would be all of H , which is impossible. Every cone that is lexicographic in an orthonormal basis is Borel (proposition 213.4).

This bears on question 222.10: the abstract of Küçük–Soyertem–Küçük states that “every total order can be represented by such a cone”, a cone built from an orthogonal base; in infinite dimension that needs a hypothesis beyond real homogeneity. The full text, and so the hypothesis it uses, was not read; no gap in that paper is asserted.

Proposition 224.4 (Invariant orders and invariant flags). *Let $\mathcal{T}$ be any family of continuous linear bijections $E \rightarrow E$ with continuous inverses. There is a translation-invariant total order on E with Borel cone that every $T \in \mathcal{T}$ preserves ($x < y \Rightarrow Tx < Ty$) if and only if there is a decreasing family $(\mathcal{F}_\alpha)_{\alpha \leq \eta}$ of closed linear subspaces, with η a countable ordinal, such that $\mathcal{F}_0 = E$, $\mathcal{F}_\eta = \{0\}$, $\mathcal{F}_\lambda = \bigcap_{\alpha < \lambda} \mathcal{F}_\alpha$ at limits, each $\mathcal{F}_{\alpha+1}$ has codimension one in $\mathcal{F}_\alpha$, every $T \in \mathcal{T}$ satisfies $T(\mathcal{F}_\alpha) = \mathcal{F}_\alpha$ for all α , and on each one-dimensional quotient $\mathcal{F}_\alpha/\mathcal{F}_{\alpha+1}$ every T acts by a positive scalar. For a Borel invariant order, its canonical flag is such a family.*

Proof. Suppose a Borel order with cone P is preserved by every T , so $T(P) \subseteq P$. Then $T(P) = P$: if $p \in P$ and $T^{-1}p \in -P$, then $p \in T(-P) \subseteq -P$, which is impossible, and $T^{-1}p \neq 0$. Each T is a linear homeomorphism of E , so by induction $T(\mathcal{E}_\alpha) = \mathcal{E}_\alpha$ for the canonical flag of theorem 213.2: at successors $T(\text{cl}_{\mathcal{E}_\alpha}(P \cap \mathcal{E}_\alpha)) = \text{cl}_{\mathcal{E}_\alpha}(P \cap \mathcal{E}_\alpha)$, and images of intersections are intersections of images because T is bijective. Since T maps $\mathcal{E}_\alpha$ and $\mathcal{E}_{\alpha+1}$ onto themselves, it induces a linear map of the one-dimensional quotient, so $f_\alpha \circ T = c f_\alpha$ on $\mathcal{E}_\alpha$ for some real $c \neq 0$. For $x \in \mathcal{E}_\alpha$ with $f_\alpha(x) > 0$ we have $x \in P$ by (212.1), so $Tx \in P \cap \mathcal{E}_\alpha \setminus \mathcal{E}_{\alpha+1}$, and $f_\alpha(Tx) > 0$ by the same display; hence $c > 0$.

Conversely, given such a family, each quotient $\mathcal{F}_\alpha/\mathcal{F}_{\alpha+1}$ is a one-dimensional Hausdorff topological vector space, hence linearly homeomorphic to $\mathbb{R}$, so there is a continuous nonzero g_α on $\mathcal{F}_\alpha$ with kernel $\mathcal{F}_{\alpha+1}$. Proposition 213.4 makes $P = \bigcup_\alpha \{x \in \mathcal{F}_\alpha : g_\alpha(x) > 0\}$ the cone of a Borel order. Let $x \in P$ lie in the layer α and $T \in \mathcal{T}$. Then $Tx \in \mathcal{F}_\alpha$, and, since T preserves $\mathcal{F}_{\alpha+1}$, $g_\alpha(Tx) = c g_\alpha(x) > 0$ with the positive scalar c of T on that quotient. So $Tx \in P$, and T preserves the order. $\square$

For bounded invertible operators on a Hilbert space the continuity of the inverse is the bounded inverse theorem, so this is the characterization asserted after question 222.6; countability of the family is not needed. What it does not provide is a way to find such a flag from the family.

Proposition 224.5 (The analogues of Part X’s two thresholds). *Let θ be a countable ordinal.*

- (i) *Every additive order embedding $H_\theta^{\ell^2} \rightarrow H_\theta^{\ell^2}$ is continuous if and only if $\theta \leq 1$.*
- (ii) *For finite $\theta = n$, every additive order embedding $H_n^{\ell^2} \rightarrow H_n^{\ell^2}$ is onto; in particular its image is closed.*
- (iii) *For infinite θ , the diagonal operator $\text{Diag}_b(e_\alpha) = (b(\alpha) + 1)^{-1}e_\alpha$, for any injection $b : \theta \rightarrow \mathbb{N}$, is a bounded order embedding whose image is not closed.*

Hence every order embedding of $H_\theta^{\ell^2}$ into itself, and also every Borel one, has closed image exactly when $\theta < \omega$.

Proof. (i) For $\theta = 0$ there is nothing to prove. An additive map $g : \mathbb{R} \rightarrow \mathbb{R}$ that preserves strict order satisfies $g(q) = qg(1)$ for rational q and is monotone, hence $g(x) = xg(1)$; this is the case $\theta = 1$. For $\theta \geq 2$ the shear of [example 216.6](#), which changes only the coordinate 1, is an additive order automorphism of $H_\theta^{\ell^2}$ that is not Borel, hence not continuous.

(ii) Induction on n ; the cases $n \leq 1$ are covered by (i). Let T be an additive order embedding of $H_n^{\ell^2}$, $n \geq 2$. As in the proof of [theorem 214.4](#), T preserves $\asymp$ and $\ll$ and induces a strictly increasing map of n into itself, the identity; so $T(e_0)$ has leading index 0, and by [lemma 214.2](#) T maps the tail $H_{n,1}^{\ell^2}$, the elements infinitesimal with respect to e_0 , into itself. The tail is order isomorphic to $H_{n-1}^{\ell^2}$, so $T(H_{n,1}^{\ell^2}) = H_{n,1}^{\ell^2}$ by induction. The map $s \mapsto (T(se_0))_0$ is additive and positive for $s > 0$, hence $s \mapsto cs$ with $c > 0$, and $(Tx)_0 = cx_0$ for all x . Given y , put $s = y_0/c$; then $y - T(se_0) \in H_{n,1}^{\ell^2} = T(H_{n,1}^{\ell^2})$, so $y \in T(H_n^{\ell^2})$.

(iii) Diag_b is bounded by 1, injective, and lower triangular with positive diagonal, so it is an order embedding by [theorem 216.2](#) with the identity pivot map. Since b is injective and θ infinite, $\|\text{Diag}_b e_\alpha\|$ takes arbitrarily small values, so Diag_b is not bounded below and its range is not closed ([corollary 216.8](#)). The last sentence combines (ii), (iii) and [lemma 216.1](#). $\square$

For Part X's groups $\mathbb{Q}^\alpha$ with the product topology the corresponding thresholds are $\alpha \leq \omega$ for automatic continuity ([theorems 133.1](#) and [133.3](#)) and $\alpha < \omega \cdot 2$ for closed images ([theorems 134.1](#) and [134.2](#)). With real coordinates both move down: the real line itself already carries discontinuous additive maps, and a bounded operator can shrink infinitely many coordinates. The proof uses the Hilbert topology; Part X's [question 138.7](#), which has the product topology in mind, stays open for it.

Proposition 224.6 (The cone closure rank of a Hilbert order). *Let $<$ be a Borel order of index θ on a separable real Hilbert space H , with cone P , and let $\rho = \rho(H, <)$ be Part III's cone closure rank.*

- (i) $\rho = \theta$, and θ is the order type of the set of Archimedean classes of positive elements; here the cone closure rank agrees with the algebraic Archimedean rank.
- (ii) The oscillation rank of $\mathbf{1}_P$ is $\rho + 1$.
- (iii) Every convex subgroup C is a closed subspace, the stage $\mathcal{E}_\alpha$ for some $\alpha \leq \theta$, and $\rho(C, <)$ is the ordinal η with $\alpha + \eta = \theta$.
- (iv) If $(H, <)$ has an additive order embedding, continuous or not, into another such space $(H', <')$, then $\rho(H, <) \leq \rho(H', <')$.
- (v) ρ is a complete invariant of $(H, <)$ up to additive order isomorphism, and of the pair (H, P) up to homeomorphism.

Proof. (i) is [proposition 224.1](#), [theorem 210.1](#) and [lemma 214.2](#); (ii) is [theorem 215.1](#); (iv) and (v) are [theorem 214.4](#) and [corollary 215.2](#). For (iii), [theorem 214.3](#) and (214.1) give $C = \mathcal{E}_\alpha$. The canonical recursion of C is $C_\beta = \mathcal{E}_{\alpha+\beta}$: successor steps are computed from $P \cap C_\beta$ inside C_β in both recursions, and at a limit λ , $\bigcap_{\beta < \lambda} \mathcal{E}_{\alpha+\beta} = \mathcal{E}_{\alpha+\lambda}$ because $\alpha + \lambda = \sup_{\beta < \lambda} (\alpha + \beta)$. Since $\mathcal{E}_\gamma = \{0\}$ exactly for $\gamma = \theta$, the first β with $C_\beta = \{0\}$ is the η with $\alpha + \eta = \theta$. $\square$

This answers part of Part III's [question 43.7](#) for separable Hilbert spaces: ρ detects the Archimedean structure exactly, behaves as stated under closed convex subgroups and order

embeddings, and is a complete invariant. Quotients and products are not treated, and for general ordered Polish groups the question stays open.

Part XVII

Borel summability and derivations of left-finite fields

[write] This part is source 23, manuscript 04 of batch 93, *Borel Regularity Forces Summability: Automatic continuity and measurable derivations on left-finite series fields* (24 pages, US letter, dated 4 October 2026; “research manuscript prepared with ChatGPT”; its title page calls it “a research direction connecting Elliot Glazer’s work with the surreal-series program in ProveIt”). It arrived in commit 2faa3b37a, as archive `Borel_Summability_and_Derivations`, and was placed in 5e4d8eed8; it is pinned to 3c25fbb55, the pin of source 22, 57 commits before the placement, and also records the later commit 4f03443e0. Its file prefix in this report is 23-borel-summability-. All unmarked text in this part is source 23’s; its fourteen sections are Sections 225–238 (its Section k is Section $224 + k$, with statement and equation numbers otherwise unchanged), its nine questions are Questions 237.1–237.9, and its two appendices are [appendices AJ](#) and [AK](#). [Section 239](#) [merge] was written for this report.

What this part claims, and its status

Let $\Gamma, \Lambda \leq \mathbb{R}$ be countable, and let $\mathcal{L}_{\mathbb{R}}^{\Gamma}$ be the field of real-coefficient series with left-finite support in Γ (t positive *infinitesimal*), with the Borel structure of its coefficients in $\mathbb{R}^{\Gamma}$. An additive map $\mathcal{L}_{\mathbb{R}}^{\Gamma} \rightarrow \mathcal{L}_{\mathbb{R}}^{\Lambda}$ is Borel exactly when it is real-linear and valuation-continuous, exactly when it is given by a matrix with left-finite columns and a uniform cutoff; such a map preserves every left-finitely summable family and is of Baire class one ([theorem 228.2](#)). For $\Gamma \neq 0$ the additive group carries no Polish group topology with these Borel sets ([proposition 229.1](#)). The Borel derivations correspond exactly to the additive $\delta : \Gamma \rightarrow \mathcal{L}_{\mathbb{R}}^{\Gamma}$ with a common lower valuation bound, and kill $\mathbb{R}$ ([theorem 230.2](#)); in rational rank $r < \infty$ they are $\bigoplus_i \mathcal{L}_{\mathbb{R}}^{\Gamma} E_i$, in rank one aE with an exact sequence for primitives ([theorem 231.1](#) and [corollary 231.3](#)). Coefficient continuity is characterized ([theorem 232.1](#)); there are derivations killing $\mathbb{R}[\Gamma]$ that are not Borel ([theorem 233.3](#), with choice); and the positive-gain Borel derivations are the logarithms of the uniformly tangent-to-identity automorphisms ([theorem 234.2](#)).

[Section 239](#) [merge] adds a second route to the non-Polishability through Part V, proves two statements source 23 left as sketches (Baire measurable slices suffice; the Euler derivations have coefficientwise flows), and records the Lean formalization of the rank-one image for $\Gamma = \mathbb{Z}$.

Unrefereed, AI-assisted, not formalized beyond that special case; priority is not established. The finite checks test finite identities only.

Questions answered. Source 23 answers no question it names, and states that it solves no remaining question of Glazer’s. It settles a boundary this report had left as a non-claim: the README said that “the real-coefficient Levi-Civita field is not claimed Polish”; [proposition 229.1](#), with Part V’s remark and [proposition 239.1](#) [merge], shows that it is not Polish in either of two senses (notes there and after Part V’s remark). It bears, without answering

them, on Part V's [question 73.6](#) (a real-coefficient support family that is not Polishable) and on Part XV's [questions 208.1](#) and [208.5](#) ([remark 239.4](#) [merge]). Its [question 237.1](#) is a companion of Part V's [question 73.5](#).

What was already here, and what is new. Source 23 read the root and surreal READMEs, not this report or any Lean file. Its [proposition 226.2](#) re-proves, for real coefficients, the field layer of Part V ([proposition 65.2](#) and [theorem 65.3](#)), Part XIV ([lemma 184.3](#), mirrored) and Part XV ([proposition 196.2](#)); [proposition 226.4](#) is the upper half of Part V's [theorem 67.2](#) with real coefficients; [lemma 227.1](#) rests on Part V's [lemma 61.2](#); the logarithm proof of [theorem 234.2](#) is that of the omnific-preserving-automorphisms report; its flow $t \mapsto t/(1-st)$ is the surcomplex-field-automorphisms report's T_s ; and the image clause of [corollary 231.3](#) for $\Gamma = \mathbb{Z}$ is formalized in Lean. Each is flagged in a note. New to the report are the automatic-summability and matrix theorem, the generic-support lemma, the non-Polishability with the coefficient Borel structure, the classification of Borel derivations and its rational-rank normal form, the coefficient continuity criterion, the explicit wild derivations and the exponential and logarithm correspondence for positive gain.

Labels and notation. Every label of source 23 takes the prefix `pma:bsd:`; labels were added to its unlabelled definitions, examples, corollaries and three subsections, its conclusion, its nine questions and its two appendices, to `pma:part:bsd`, and to the statements of [section 239](#). The seventh block of the table in [section 1.3](#) lists the symbols. In short: source 23's field L_Γ is printed $\mathcal{L}_\mathbb{R}^\Gamma$, with the coefficient field $\mathbb{R}$ as a subscript, as in Part V's $\mathcal{L}_k$, and the exponent group as a superscript, as in Part XV's $\mathcal{L}^\Gamma$, because Part XIV's L_Γ is the same field written in the infinite variable $T = t^{-1}$ (tempting false reading: Part XIV's L_Γ has supports finite *above* each bound) and Part XV's $\mathcal{L}^\Gamma$ has coefficients in $\mathbb{R}_{\text{alg}}$; so $\mathcal{L}_\mathbb{R}^\mathbb{Q}$ is Part V's $\mathcal{L}_\mathbb{R}$ and Part XV's $\mathcal{L}^\Gamma$ is the subfield of $\mathcal{L}_\mathbb{R}^\Gamma$ with algebraic coefficients when Γ is a countable $\mathbb{Q}$ -subspace. Its coefficient slices F_S are printed Sl_S , because Part XV's F_S are the subfields of its Cantor family; its subgroups H_N in the proof of [proposition 229.1](#) are printed W_N , because H_α is Part X's $\mathbb{Q}^\alpha$; and its macro `\restr` is the report's `\rstr` (same symbol). The letters E, E_i (Euler derivations), $D, T, \delta, \Phi_s, K, F, F_N$ and Λ are local to this part; δ is not Part XII's partial subtraction, Φ_s not Part XV's Φ_u , D not the divisible group of Parts I and VII. No normalization changed.

Files. Source 23's program, recorded run and source and proof audit are shipped as `code/23-borel-summability-verification.py`, `data/23-borel-summability-verification_results.json` and `23-borel-summability-SOURCE_AND_PROOF_AUDIT.md`. Source 23 calls its pin a "tree identifier, not a commit identifier"; it is a commit (note in [appendix AK](#)). The Hilbert's-tenth programme routed the archive from its README only ([appendix E](#)).

225 The research intersection and the questions settled here

225.1 From topological arithmetic to measurable differential algebra

Elliot Glazer's *A Topological Tennenbaum Theorem* studies the tension between Borel definability and continuity of arithmetic operations on uncountable Polish spaces [1]. The specific connection used here is not a biographical guess from a professional profile: it is the mathematical

problem of determining how a mild descriptive regularity assumption constrains an algebraic operation.

The relevant part of ProveIt is its surreal-number project. Its documentation describes a normal-form-based field development, strong summation, valuations, and fine calculus; it also distinguishes research reports from the statements actually mapped to formal declarations [149, 150]. These are natural settings in which to ask whether respecting infinite sums has to be an axiom imposed on a derivation or can follow from an independent regularity condition.

The existing generalized-series literature already develops derivations that commute with formal sums. Kuhlmann and Matusinski explicitly build strong linearity and an appropriate Leibniz principle into the notion of a series derivation [146]. We do not claim to introduce that notion. Our direction is instead to *derive* strong summation from coefficient-Borel measurability, on a restricted but substantial support class. The restriction to left-finite supports includes the usual Levi–Civita field; background on that field is given by Shamseddine and Berz [31].

225.2 Three precise questions

The article answers the following questions for real coefficients and countable exponent subgroups of $\mathbb{R}$.

- Q1. Does a Borel additive map on a left-finite series field automatically commute with the field's admissible infinite sums?
- Q2. Which prescriptions for the derivatives of monomials extend to a Borel derivation on the entire completed field, and is such an extension unique?
- Q3. Which of these derivations are genuinely continuous in the coefficient product topology, rather than merely Borel or valuation-continuous?

The answers are affirmative automatic summability for Q1; a uniform lower bound on logarithmic monomial derivatives, with uniqueness, for Q2; and a finite-support plus diagonal condition for Q3. These are the classification problems formulated and settled in this manuscript. They are not being attributed to Glazer as previously published open questions.

225.3 Main theorem in informal form

A coefficient can be inspected with a Borel operation. An infinite series has only finitely many terms below any fixed exponent. The main result says that an additive Borel operation cannot secretly extract information from arbitrarily high input exponents and move it into one bounded output range. It must have a finite valuation cutoff for every such range.

For derivations, additivity of logarithmic monomial values makes these cutoffs much more rigid: one global lower bound suffices. In the ordinary Levi–Civita field $\mathcal{L}_{\mathbb{R}}^{\mathbb{Q}}$, every Borel derivation has the form

$$D = aE, \quad E\left(\sum_{q \in \mathbb{Q}} c_q t^q\right) = \sum_{q \in \mathbb{Q}} q c_q t^q, \quad a \in \mathcal{L}_{\mathbb{R}}^{\mathbb{Q}}. \quad (225.1)$$

Thus one field element classifies every Borel derivation, without assuming in advance that constants are fixed or that infinite sums are preserved.

225.4 What is and is not new

The Baire-category automatic-continuity lemma for Polish groups is classical; we give the needed proof. The general phenomenon is discussed in Rosendal and Suarez [147]. Formal Laurent-series manipulations, Euler derivations, and filtered exponential–logarithm arguments also have a long history. The proposed contribution is the exact automatic-summability theorem on this non-Polishable Borel field, its uniform-bound consequence for arbitrary countable rational rank, and the resulting continuity classification.

A targeted literature search found the cited background but did not establish priority for these exact statements. Accordingly, “proved here” means that an argument is present in this manuscript; it does not mean “first proved in the literature.” The paper does not infer a theorem’s correctness from a repository claim or from numerical experiments.

[write] (Batch 93.) Source 23 read the root and surreal READMEs, not this report. Its field $\mathcal{L}_{\mathbb{R}}^{\mathbb{Q}}$ is Part V’s Levi-Civita field $\mathcal{L}_k$ with $k = \mathbb{R}$, which Part V sets aside as not Polish (remark before section 66); Part XV’s $\mathcal{L}^{\Gamma}$ is the same construction over $\mathbb{R}_{\text{alg}}$, and Part XIV’s L_{Γ} is $\mathcal{L}_{\mathbb{R}}^{\Gamma}$ mirrored ($T = t^{-1}$). Its automatic-continuity input is Part V’s lemma 61.2. The notes below give the overlaps; section 239 [merge] proves the two statements source 23 left as sketches and gives a second route to proposition 229.1.

226 Left-finite fields and their two topologies

Throughout, $\Gamma, \Lambda \leq (\mathbb{R}, +)$ are countable additive subgroups. A subgroup may be $\{0\}$ unless explicitly excluded. Coefficients carry the usual real topology when descriptive set theory is used.

Definition 226.1. A set $S \subseteq \Gamma$ is *left-finite* if

$$\#(S \cap (-\infty, B]) < \infty \quad (B \in \mathbb{R}).$$

Define

$$\mathcal{L}_{\mathbb{R}}^{\Gamma} = \left\{ f = \sum_{\gamma \in \Gamma} f_{\gamma} t^{\gamma} : f_{\gamma} \in \mathbb{R}, \text{ supp } f = \{\gamma : f_{\gamma} \neq 0\} \text{ is left-finite} \right\}.$$

For $f \neq 0$, put $v(f) = \min \text{supp } f$, and put $v(0) = +\infty$. The coefficient of t^{γ} is denoted $[t^{\gamma}]f$.

An infinite left-finite set has a strictly increasing enumeration tending to $+\infty$. In particular it is well ordered. The converse fails: for example, $\{1 - 1/n : n \geq 2\}$ is well ordered but not left-finite. This distinction is essential; $\mathcal{L}_{\mathbb{R}}^{\mathbb{Q}}$ is not the full Hahn field $\mathbb{R}((t^{\mathbb{Q}}))$.

Proposition 226.2. Coefficientwise addition and convolution multiplication make $\mathcal{L}_{\mathbb{R}}^{\Gamma}$ a field containing $\mathbb{R}[\Gamma]$, the finite-support group algebra. The valuation satisfies

$$v(fg) = v(f) + v(g), \quad v(f + g) \geq \min\{v(f), v(g)\}.$$

The field is complete for the valuation topology, and $\mathbb{R}[\Gamma]$ is dense in that topology.

Proof. A nonempty left-finite set has a least element: choose any of its elements and take the minimum of the finite nonempty part below it. If S, T are nonempty left-finite sets with minima s_0, t_0 , every pair satisfying $s + t \leq B$ lies in

$$(S \cap (-\infty, B - t_0]) \times (T \cap (-\infty, B - s_0]).$$

This is finite. Consequently convolution has finite coefficient sums and left-finite support. Associativity and distributivity reduce, below any fixed cutoff, to ordinary finite sums. The product's least exponent is $s_0 + t_0$, and its coefficient there is the product of two nonzero real numbers. This proves the multiplicative valuation formula and the absence of zero divisors.

Write a nonzero element as $f = ct^\alpha(1 + u)$ with $c \neq 0$ and either $u = 0$ or $v(u) > 0$. If $u \neq 0$, the family $((-u)^n)_{n \geq 0}$ is left-finitely summable: only finitely many n can have $nv(u) \leq B$, and each of their supports is left-finite. Therefore

$$f^{-1} = c^{-1}t^{-\alpha} \sum_{n \geq 0} (-u)^n$$

is defined and the finite geometric identity proves that it is an inverse.

For completeness, let (f_n) be valuation-Cauchy. For each real B there is N_B such that all f_n , $n \geq N_B$, agree at every exponent at most B . Each coefficient consequently stabilizes. Define f by these stabilized values. Its support below B agrees with that of one f_n below B , hence is finite. Thus $f \in \mathcal{L}_{\mathbb{R}}^{\Gamma}$ and $f_n \rightarrow f$ in valuation. Finally the finite truncations $\sum_{\gamma \leq N} f_{\gamma} t^{\gamma}$ converge to f as integer $N \rightarrow \infty$. $\square$

[write] (Batch 93.) A re-proof, for real coefficients and countable Γ , of Part V's [proposition 65.2](#) and [theorem 65.3](#) (countable k , $\Gamma = \mathbb{Q}$), Part XV's [proposition 196.2](#) (over $\mathbb{R}_{\text{alg}}$, every $\mathbb{Q}$ -subspace Γ) and Part XIV's [lemma 184.3](#) (real coefficients, $T = t^{-1}$).

Definition 226.3. The *coefficient topology* τ_c on $\mathcal{L}_{\mathbb{R}}^{\Gamma}$ is the subspace topology from $\mathbb{R}^{\Gamma}$. The *coefficient Borel structure* is its Borel σ -algebra. Unless qualified, “Borel” always refers to this structure.

The *valuation topology* τ_v has neighborhoods of zero $\{f : v(f) > B\}$, $B \in \mathbb{R}$. It is metrized, for example, by $d_v(f, g) = \min\{1, e^{-v(f-g)}\}$, where $e^{-\infty} = 0$.

These topologies are different. On the constant subfield, τ_c is the usual real topology, while τ_v is discrete. If $a > 0$ belongs to Γ , then $t^{-na} \rightarrow 0$ coefficientwise but does not converge to zero in valuation. Valuation convergence does imply coefficient convergence, indeed eventual stabilization of each coefficient. No assertion below silently replaces one topology by the other.

Proposition 226.4. $\mathcal{L}_{\mathbb{R}}^{\Gamma}$ is a Borel subset of $\mathbb{R}^{\Gamma}$, hence a standard Borel space. Addition, multiplication, and inversion on nonzero elements are Borel.

Proof. Writing $\Gamma_{\leq N} = \Gamma \cap (-\infty, N]$, the defining support condition is

$$\mathcal{L}_{\mathbb{R}}^{\Gamma} = \bigcap_{N \in \mathbb{N}} \bigcup_{\substack{F \subseteq \Gamma_{\leq N} \\ F \text{ finite}}} \bigcap_{\gamma \in \Gamma_{\leq N} \setminus F} \{f : f_{\gamma} = 0\}. \quad (226.1)$$

Every union displayed is countable, so this is a Borel set (of class at most Π_3^0 in the ambient product). Addition is coefficientwise continuous. For multiplication, enumerate Γ and take finite partial sums of $f_{\gamma} g_{\lambda-\gamma}$ for each output coefficient λ . For any fixed pair (f, g) , only

finitely many summands are nonzero. The coefficient is therefore a pointwise limit of Borel functions. This proves Borel multiplication.

The leading exponent is Borel on the nonzero field: the event $v(f) = \alpha$ says $f_\alpha \neq 0$ and $f_\gamma = 0$ for all $\gamma < \alpha$. Normalizing by that exponent and coefficient is consequently Borel. The geometric inverse in Proposition 226.2 is a coefficientwise limit of Borel finite sums, so inversion is Borel as well. $\square$

[write] (Batch 93.) The upper bound Π_3^0 is that of Part V's theorem 67.2 (Levi-Civita codes over a countable field, where it is exact), here with real coefficients; exactness is not claimed. Part V's question 73.6 asks when a support family yields a Polishable group: over a countable k the left-finite family does (theorem 65.3), over $\mathbb{R}$ it does not (proposition 229.1).

Definition 226.5. A family $(f_i)_{i \in I}$ is *left-finitely summable* if the union of its supports is left-finite and every exponent belongs to the supports of only finitely many f_i . Its sum is given by the resulting finite coefficient sums.

For countable $I = \mathbb{N}$, this condition is equivalent to saying that, for every B , only finitely many i satisfy $v(f_i) \leq B$. The forward implication uses finitely many exponents below B , each with finitely many contributors. The reverse implication uses finitely many individual left-finite supports below B . Since Γ is countable, an arbitrary left-finitely summable family has at most countably many nonzero members. Thus no genuinely uncountable summation principle is needed here.

227 Polish slices and the generic-support argument

For each left-finite $S \subseteq \Gamma$, define the coefficient slice

$$\text{Sl}_S = \{f \in \mathcal{L}_{\mathbb{R}}^\Gamma : \text{supp } f \subseteq S\}.$$

Extension by zero identifies Sl_S with the Polish real vector space $\mathbb{R}^S$. Its role is crucial: the full field need not carry a compatible Polish group topology, as Section 229 will show.

Lemma 227.1 (Finite-coordinate automatic continuity). *Let S be countable. Every Borel additive map $\ell : \mathbb{R}^S \rightarrow \mathbb{R}$ is a continuous real-linear functional and has the form*

$$\ell(x) = \sum_{s \in F} c_s x_s$$

for a finite set $F \subseteq S$.

Proof. We recall the Baire-category argument to specify the exact regularity used. A nonmeager Borel subset A of a Polish group has $A - A$ containing a neighborhood of zero. Indeed A is comeager in some nonempty open set U . For a sufficiently small translation h , the intersection $U \cap (h + U)$ contains a nonempty open set. In that set, both A and $h + A$ are comeager, so they meet. Hence $h \in A - A$.

Fix $\varepsilon > 0$. Cover $\mathbb{R}$ by countably many open intervals of length less than ε . Their Borel inverse images under ℓ cover $\mathbb{R}^S$. One inverse image A is nonmeager, because $\mathbb{R}^S$ is a Baire space. On $A - A$ the values of ℓ lie in $(-\varepsilon, \varepsilon)$. The preceding paragraph gives continuity at zero, and additivity gives continuity everywhere. Rational homogeneity follows from additivity; real homogeneity follows by approximating a real scalar by rationals and using continuity.

Continuity at zero supplies a basic product neighborhood on which $|\ell| < 1$. Such a neighborhood restricts only finitely many coordinates, say those in F . Every vector vanishing on F , and every real multiple of that vector, lies in this neighborhood. Homogeneity forces ℓ to vanish on all such vectors. Therefore ℓ depends only on the coordinates in F , where ordinary finite-dimensional linear algebra gives the displayed form. $\square$

[write] (Batch 93.) The continuity half is Part V's [lemmas 61.1](#) and [61.2](#); the finite-coordinate form is the real analogue of Part X's [lemma 130.2](#) for $\mathbb{Q}^I$ with discrete coordinates. Only the Baire property of the preimages is used, which [proposition 239.2](#) [merge] exploits.

Lemma 227.2 (Generic simultaneous noncancellation). *Let $T : \mathcal{L}_{\mathbb{R}}^{\Gamma} \rightarrow \mathcal{L}_{\mathbb{R}}^{\Lambda}$ be Borel and additive, and let $S \subseteq \Gamma$ be left-finite. Put*

$$U_S = \bigcup_{\gamma \in S} \text{supp } T(t^{\gamma}).$$

Then U_S is left-finite. Moreover, for each $\lambda \in \Lambda$ the coefficient functional on Sl_S is a finite-coordinate real-linear functional.

Proof. Compose T with the coefficient projection at λ and the inclusion $\mathbb{R}^S \simeq \text{Sl}_S \hookrightarrow \mathcal{L}_{\mathbb{R}}^{\Gamma}$. The resulting map ℓ_{λ} is Borel and additive. [Lemma 227.1](#) makes it finite-coordinate and real-linear. Its coefficient on input coordinate γ is exactly $[t^{\lambda}]T(t^{\gamma})$. Thus ℓ_{λ} is nonzero precisely when $\lambda \in U_S$.

For each $\lambda \in U_S$, the set $\{x \in \mathbb{R}^S : \ell_{\lambda}(x) \neq 0\}$ is open and dense. It is dense because the kernel of a nonzero continuous real-linear functional cannot contain an open set. There are only countably many such λ , so the Baire theorem gives an x in their intersection. For the corresponding $f_x \in \text{Sl}_S$, all these output coefficients are nonzero; all output coefficients outside U_S are zero. Consequently

$$\text{supp } T(f_x) = U_S.$$

The left side is left-finite because $T(f_x) \in \mathcal{L}_{\mathbb{R}}^{\Lambda}$. This proves the claim. $\square$

The generic choice is a proof device, not an algorithm for calculating T . Its importance is that cancellation cannot conceal infinitely many forbidden output exponents for *every* possible choice of input coefficients.

228 The automatic-summability and matrix theorem

Definition 228.1. A real matrix $C = (c_{\lambda\gamma})_{\lambda \in \Lambda, \gamma \in \Gamma}$ is *cutoff-admissible* if it satisfies both conditions below:

(M1) For each γ , the column $\sum_{\lambda} c_{\lambda\gamma} t^{\lambda}$ belongs to $\mathcal{L}_{\mathbb{R}}^{\Lambda}$.

(M2) For every $B \in \mathbb{R}$ there is $A \in \mathbb{R}$ such that

$$\lambda \leq B, \quad \gamma > A \quad \implies \quad c_{\lambda\gamma} = 0.$$

Condition (M2) is uniform in *all* output exponents at most B , not just one output coefficient at a time.

Theorem 228.2 (Automatic summability and exact matrix form). *For an additive map $T : \mathcal{L}_{\mathbb{R}}^{\Gamma} \rightarrow \mathcal{L}_{\mathbb{R}}^{\Lambda}$, the following are equivalent:*

- (i) T is Borel for the coefficient Borel structures.
- (ii) T is real-linear and continuous for the valuation topologies.
- (iii) There is a cutoff-admissible matrix C such that

$$[t^\lambda]T(f) = \sum_{\gamma \in \Gamma} c_{\lambda\gamma} f_\gamma \quad (f \in \mathcal{L}_{\mathbb{R}}^\Gamma). \quad (228.1)$$

Each sum in (228.1) has only finitely many nonzero summands for the given f . The matrix is unique and is $c_{\lambda\gamma} = [t^\lambda]T(t^\gamma)$. Every such map preserves all left-finitely summable families and is of Baire class one from $(\mathcal{L}_{\mathbb{R}}^\Gamma, \tau_c)$ to $(\mathcal{L}_{\mathbb{R}}^\Lambda, \tau_c)$.

Proof. Step 1: Borel additivity implies real linearity and a slice formula. For any f , take the left-finite set $S = \text{supp } f$. The coefficient functionals of T on Sl_S are real-linear by Lemma 227.1. Hence $T(af) = aT(f)$ for every real scalar a . Additivity then makes T real-linear on the full field. The same lemma gives (228.1) on each slice, with only finitely many nonzero matrix coefficients in each row restricted to that slice. Condition (M1) follows from $T(t^\gamma) \in \mathcal{L}_{\mathbb{R}}^\Lambda$.

Step 2: Borel additivity implies the uniform cutoff. Suppose (M2) fails for some B . Then

$$\{\gamma \in \Gamma : v(T(t^\gamma)) \leq B\}$$

is unbounded above. Choose distinct $\gamma_n \rightarrow +\infty$ in this set, with $\gamma_n > n$. The set $S = \{\gamma_n : n \geq 1\}$ is left-finite. Lemma 227.2 says that $U_S \cap (-\infty, B]$ is finite. Each chosen column has a nonzero entry in that finite set of rows. One row therefore has nonzero entries in infinitely many of the chosen columns, contrary to the finite-coordinate assertion on Sl_S . Thus (M2) holds.

Step 3: An admissible matrix defines a valuation-continuous map. Fix $f \in \mathcal{L}_{\mathbb{R}}^\Gamma$ and a bound B , and choose A from (M2). Only input exponents in the finite set $\text{supp } f \cap (-\infty, A]$ can contribute below B . Each corresponding column has finite support below B by (M1). Consequently the formula has finite coefficient sums and gives an output with left-finite support. It is real-linear. Moreover,

$$v(f) > A \implies v(Tf) > B, \quad (228.2)$$

which is precisely continuity at zero in valuation; linearity supplies continuity everywhere.

Step 4: Valuation-continuous real-linear maps are Borel. Enumerate Γ without repetition as $\gamma_1, \gamma_2, \dots$ (with the obvious finite modification for $\Gamma = \{0\}$), and define

$$T_m(f) = \sum_{j=1}^m f_{\gamma_j} T(t^{\gamma_j}). \quad (228.3)$$

This is a finite-rank, coefficient-topology continuous map into $\mathcal{L}_{\mathbb{R}}^\Lambda$: each output coefficient is a finite real-linear combination of the input coordinates. For every fixed f , the finite coordinate truncations $\sum_{j \leq m} f_{\gamma_j} t^{\gamma_j}$ converge in valuation to f . Indeed, below any bound there are only finitely many nonzero input terms, and the enumeration eventually contains all of them. Valuation continuity gives $T_m(f) \rightarrow T(f)$ in valuation and hence coefficientwise. Therefore T is a pointwise limit of continuous maps for τ_c , and in particular is Borel. This also proves the claimed Baire-class-one bound.

The implications already prove equivalence. Uniqueness of the matrix follows by evaluating on monomials. Alternatively, valuation continuity in (ii) immediately gives (M2) on monomials through (228.2), and the finite truncation limit reconstructs (228.1).

Step 5: Preservation of sums. Let (f_i) be left-finitely summable, and discard its zero members. They form an at most countable family. For each output bound B , choose A as in (228.2). Only finitely many i have $v(f_i) \leq A$, and all other $T(f_i)$ have valuation greater than B . Thus $(T(f_i))$ is left-finitely summable. Finite partial sums of the input converge in valuation to $\sum_i f_i$. Continuity and additivity give

$$T\left(\sum_i f_i\right) = \sum_i T(f_i).$$

□

[write] (Batch 93.) New to the report. Related by theme are Part X's matrix classification [theorem 130.5](#) for $\mathbb{Q}^\alpha$, the automatic strongness of valued automorphisms in the reports [omnific-preserving-automorphisms](#) and [surcomplex-field-automorphisms](#) (an algebraic route), and the strong K -linear maps of the [three-duals-of-hahn-vector-spaces](#) report (`duals:thm:strong`).

Corollary 228.3 (Monomial determination). *Two Borel additive maps $\mathcal{L}_\mathbb{R}^\Gamma \rightarrow \mathcal{L}_\mathbb{R}^\Lambda$ that agree on every monomial are equal. In particular, there is no nonzero Borel additive map that vanishes on the finite-support group algebra.*

Proof. Their matrices coincide, so Theorem 228.2 applies. Real linearity is automatic, and therefore agreement on monomials implies agreement on the entire finite-support group algebra as well. □

Proposition 228.4 (Coefficient-continuous part). *A map satisfying Theorem 228.2 is continuous for τ_c if and only if every row of its matrix has finite support in Γ .*

Proof. Finite rows make every output coefficient a continuous finite-coordinate linear functional. Conversely, if one output coefficient is continuous at zero, a basic neighborhood controlling finitely many input coordinates maps into $(-1, 1)$. Arbitrary real multiples of any monomial outside those coordinates still lie in the neighborhood. The matrix coefficient of that monomial must therefore vanish. This proves finiteness of the row. □

Example 228.5 (The Baire-class-one bound is sharp). For nonzero Γ , define

$$T(f) = \left(\sum_{\gamma \leq 0} f_\gamma \right) 1.$$

The sum is finite for each f . The matrix has one nonzero row, supported on all exponents at most zero. It satisfies (M1)–(M2), so T is Borel, valuation-continuous, and Baire class one. But for $a > 0$ in Γ , $t^{-na} \rightarrow 0$ coefficientwise while $T(t^{-na}) = 1$. It is not coefficient-continuous.

Example 228.6 (Why pointwise row conditions do not suffice). In $\Gamma = \Lambda = \mathbb{Q}$, set $c_{-1/n, n} = 1$ for $n \geq 1$, and set all other entries to zero. Every row is finite, and every column is a monomial. Yet the candidate image of $\sum_{n \geq 1} t^n$ has support $\{-1/n : n \geq 1\}$, which is not left-finite. Thus this matrix does not even define a map into $\mathcal{L}_\mathbb{R}^\mathbb{Q}$. The missing hypothesis is the uniform bounded-output cutoff (M2).

Corollary 228.7 (The coefficient-Borel dual). *A Borel additive functional $\ell : \mathcal{L}_{\mathbb{R}}^{\Gamma} \rightarrow \mathbb{R}$ is exactly an expression*

$$\ell(f) = \sum_{\gamma \leq A} c_{\gamma} f_{\gamma}$$

for some real bound A and arbitrary real weights c_{γ} . It is coefficient-continuous exactly when only finitely many weights are nonzero.

Proof. Use $\mathcal{L}_{\mathbb{R}}^{\{0\}} = \mathbb{R}$ as the target in Theorem 228.2 and then apply Proposition 228.4. The valuation topology on this target is discrete. $\square$

229 Why the full additive group is not Polishable

A standard Borel space always admits some Polish topology with the same Borel sets. The additional requirement that a given group operation be continuous is much stronger. It fails here.

Proposition 229.1. *If $\Gamma \neq \{0\}$, there is no Polish group topology on $(\mathcal{L}_{\mathbb{R}}^{\Gamma}, +)$ whose Borel sets are exactly the coefficient Borel sets.*

Proof. Fix $a > 0$ in Γ and, for integers $N \geq 0$, let

$$W_N = \{f \in \mathcal{L}_{\mathbb{R}}^{\Gamma} : [t^{-na}]f = 0 \text{ for every integer } n > N\}.$$

Each W_N is a coefficient-Borel subgroup. Left-finiteness implies $\mathcal{L}_{\mathbb{R}}^{\Gamma} = \bigcup_{N \geq 0} W_N$. Its index is uncountable: the elements $rt^{-(N+1)a}$, for $r \in \mathbb{R}$, lie in distinct cosets modulo W_N .

Suppose a compatible Polish group topology existed. By the Baire theorem some W_N would be nonmeager. Because W_N is Borel, the argument in Lemma 227.1 makes $W_N - W_N = W_N$ a neighborhood of zero. It is then an open subgroup. Distinct cosets of an open subgroup are disjoint nonempty open sets, and a separable space has only countably many such sets: each contains a different member of a fixed countable dense set. This contradicts the uncountable index of W_N . $\square$

The proof of Theorem 228.2 is therefore not a direct invocation of automatic continuity for homomorphisms between two Polish groups. The coefficient slices are Polish; the whole additive group is not Polishable in the relevant Borel coding. Moreover, the continuity obtained first is for a different, nonseparable valuation topology. This is the precise reason the slice and generic-support steps matter.

There is no contradiction with Glazer's results. We do not produce a Polish model of arithmetic with the forbidden continuous operations. We study a Borel field and prove a different regularity statement about its additive operators. In particular, this proposition is not a solution of either of the remaining questions stated in Glazer's paper [1].

[write] (Batch 93.) Part V already shows that the real-coefficient Levi-Civita field has no *admissible* Polish topology (remark before section 66); proposition 239.1 [merge] extends that to every $\Gamma \neq 0$ and derives proposition 229.1 from it, a second route. Together they settle, negatively, the boundary that this report's README recorded as a non-claim ("the real-coefficient Levi-Civita field is not claimed Polish"): no Polish group topology with the coefficient Borel sets, and no admissible Polish topology with any Borel sets.

230 Classification of Borel derivations

A *field derivation* is an additive map $D : \mathcal{L}_{\mathbb{R}}^{\Gamma} \rightarrow \mathcal{L}_{\mathbb{R}}^{\Gamma}$ satisfying $D(fg) = fD(g) + gD(f)$. We do not initially require that D vanish on $\mathbb{R}$, be continuous, or commute with infinite sums. From now until explicitly noted, $\Gamma \neq \{0\}$.

Definition 230.1. An additive map $\delta : \Gamma \rightarrow \mathcal{L}_{\mathbb{R}}^{\Gamma}$ is *uniformly valuation-bounded below* if some $m \in \mathbb{R}$ satisfies

$$v(\delta(\gamma)) \geq m \quad (\gamma \in \Gamma). \quad (230.1)$$

The zero map is allowed. For a real-linear operator T on $\mathcal{L}_{\mathbb{R}}^{\Gamma}$, define its *valuation gain* by

$$\text{gain}(T) = \inf_{f \neq 0} (v(Tf) - v(f)),$$

with $v(0) - v(f) = +\infty$. Thus $\text{gain}(0) = +\infty$; the infimum for a general operator can be $-\infty$.

Theorem 230.2 (Bounded logarithmic-velocity classification). *There is a bijection between Borel field derivations D on $\mathcal{L}_{\mathbb{R}}^{\Gamma}$ and uniformly valuation-bounded additive maps $\delta : \Gamma \rightarrow \mathcal{L}_{\mathbb{R}}^{\Gamma}$. The correspondence is*

$$\delta(\gamma) = t^{-\gamma} D(t^{\gamma}) \quad (230.2)$$

and its inverse is

$$D_{\delta} \left(\sum_{\gamma} f_{\gamma} t^{\gamma} \right) = \sum_{\gamma} f_{\gamma} t^{\gamma} \delta(\gamma). \quad (230.3)$$

Every Borel derivation vanishes on $\mathbb{R}$. Formula (230.3) is a left-finitely summable expression, and its optimal gain is

$$\text{gain}(D_{\delta}) = \inf_{\gamma \in \Gamma} v(\delta(\gamma)) > -\infty. \quad (230.4)$$

For the zero derivation both sides equal $+\infty$.

Proof. Necessity: linearity, constants, and monomials. Theorem 228.2 makes a Borel derivation real-linear and valuation-continuous. The Leibniz rule gives $D(1) = 2D(1)$, hence $D(1) = 0$. Real linearity now gives $D(r) = 0$ for $r \in \mathbb{R}$. Applying Leibniz to $t^{\gamma+\eta} = t^{\gamma} t^{\eta}$ proves

$$\delta(\gamma + \eta) = \delta(\gamma) + \delta(\eta).$$

Automatic preservation of the monomial expansion gives (230.3).

Necessity: one lower bound for all logarithmic values. This is the extra rigidity supplied by the derivation law. Valuation continuity supplies $A \in \mathbb{R}$ with

$$v(f) > A \implies v(Df) > 0.$$

Fix $a > 0$ in Γ . Given any $\gamma \in \Gamma$, choose an integer n so that $\eta = \gamma + na$ lies in $(A, A + a]$. Then

$$v(\delta(\eta)) = v(D(t^{\eta})) - \eta > -\eta \geq -(A + a).$$

Additivity gives $\delta(\gamma) = \delta(\eta) - n\delta(a)$. Multiplication by a nonzero real constant does not change valuation. The ultrametric inequality therefore yields the common bound

$$v(\delta(\gamma)) \geq \min\{-(A + a), v(\delta(a))\}. \quad (230.5)$$

The same inequality holds when $n = 0$ or a logarithmic value is zero. Notice that this argument uses integer translations only. It does not assume that Γ is divisible or has finite rational rank.

Sufficiency. Suppose δ is additive and satisfies (230.1). The term $f_\gamma t^\gamma \delta(\gamma)$, when nonzero, has valuation at least $\gamma + m$. For a fixed bound B , only the finitely many $\gamma \in \text{supp } f \cap (-\infty, B - m]$ can contribute at or below B . Each of their series is left-finite. Thus (230.3) is well defined, real-linear, and satisfies

$$v(D_\delta f) \geq v(f) + m. \quad (230.6)$$

It is valuation-continuous and therefore Borel by Theorem 228.2.

On monomials the Leibniz rule is exactly additivity of δ . By real linearity, it holds on the finite-support group algebra. Approximate arbitrary f, g by finite-support truncations f_N, g_N . Multiplication is continuous in the valuation topology, and so is D_δ . Passing to the limit in

$$D_\delta(f_N g_N) = f_N D_\delta(g_N) + g_N D_\delta(f_N)$$

proves the rule on the entire field. This establishes the correspondence and uniqueness.

Optimal gain. Put $m_* = \inf_\gamma v(\delta(\gamma))$. The same termwise argument as above gives $v(D_\delta f) \geq v(f) + m_*$ whenever m_* is finite. On a monomial,

$$v(D_\delta t^\gamma) - v(t^\gamma) = v(\delta(\gamma)).$$

Taking infima gives equality. The zero case follows directly from the formula. $\square$

[write] (Batch 93.) The valuation gain is Part XV's "valuation gain" (lemma 199.1). New to the report.

Corollary 230.3 (Exact extension criterion). *Let $d : \mathbb{R}[\Gamma] \rightarrow \mathcal{L}_\mathbb{R}^\Gamma$ be an $\mathbb{R}$ -derivation, with the codomain viewed as an $\mathbb{R}[\Gamma]$ -module through inclusion. Then d extends to a Borel derivation of $\mathcal{L}_\mathbb{R}^\Gamma$ if and only if*

$$\inf_{\gamma \in \Gamma} v(t^{-\gamma} d(t^\gamma)) > -\infty.$$

When the extension exists, it is unique and is given by (230.3).

Proof. The logarithmic values of d are additive by Leibniz. Necessity and sufficiency follow from Theorem 230.2; uniqueness also follows from Corollary 228.3. $\square$

A generic valuation-continuous real-linear map need not have finite gain: its cutoff may depend nonlinearly on the desired output bound. The derivation identity upgrades arbitrary continuity moduli to the single translation bound (230.6). This upgrade is not an assumption hidden in the word "Borel."

Example 230.4 (Additive operators with unboundedly negative gain). On $\mathcal{L}_\mathbb{R}^\mathbb{Z}$, let $T(t^n) = t^{\lfloor n/2 \rfloor}$ and extend real-linearly by the matrix theorem. Columns are monomials and the image exponents tend to $+\infty$ with n , so (M2) holds. Thus T is Borel and valuation-continuous. But $v(T(t^n)) - v(t^n) = \lfloor n/2 \rfloor - n \rightarrow -\infty$ as $n \rightarrow +\infty$. This cannot be a derivation.

231 Rational-rank normal forms and exact integration

Let $V = \text{span}_\mathbb{Q} \Gamma \subseteq \mathbb{R}$, and let $(b_i)_{i \in I}$ be a $\mathbb{Q}$ -basis of V chosen from Γ . Here I is finite or countably infinite. Write $\ell_i : \Gamma \rightarrow \mathbb{Q}$ for its coordinate maps, so each γ has the finite expression $\gamma = \sum_i \ell_i(\gamma) b_i$. The number $|I|$ is the *rational rank*; it is not the ordered-group rank. Every nonzero subgroup of $\mathbb{R}$ is Archimedean as an ordered group, even when its rational rank is infinite.

Define the coordinate Euler derivations

$$E_i(f) = \sum_{\gamma \in \Gamma} \ell_i(\gamma) f_\gamma t^\gamma. \quad (231.1)$$

The values $\ell_i(\gamma)$ are constants, so the logarithmic map is bounded below by zero. Theorem 230.2 proves that each E_i is Borel with gain zero. Its matrix is diagonal, hence it is also coefficient-continuous.

Theorem 231.1 (Normal form in every countable rational rank). *Borel derivations are parameterized by families $(a_i)_{i \in I}$ of elements of $\mathcal{L}_\mathbb{R}^\Gamma$ satisfying*

$$\inf_{i \in I} v(a_i) > -\infty. \quad (231.2)$$

The logarithmic map and derivation are

$$\delta(\gamma) = \sum_{i \in I} \ell_i(\gamma) a_i, \quad D = D_\delta,$$

where the first sum is finite for every γ . The parameters are unique, $a_i = \delta(b_i)$, and

$$\text{gain}(D) = \inf_{i \in I} v(a_i). \quad (231.3)$$

In finite rational rank r , every tuple in $\mathcal{L}_\mathbb{R}^{\Gamma^r}$ is allowed and

$$\text{Der}_{\text{Bor}}(\mathcal{L}_\mathbb{R}^\Gamma) = \mathcal{L}_\mathbb{R}^\Gamma E_1 \oplus \cdots \oplus \mathcal{L}_\mathbb{R}^\Gamma E_r. \quad (231.4)$$

Proof. Any additive map $\delta : \Gamma \rightarrow \mathcal{L}_\mathbb{R}^\Gamma$ extends uniquely to a $\mathbb{Q}$ -linear map on V : clear denominators in each finite rational relation, and use that the additive group of $\mathcal{L}_\mathbb{R}^\Gamma$ is uniquely divisible. Thus it has the stated coordinate form. A common lower bound for the a_i is a lower bound for every finite rational combination, by the valuation inequality. Conversely, a bound for all $\delta(\gamma)$ bounds $\delta(b_i) = a_i$. This proves the parameterization using Theorem 230.2. The same reasoning, with basis elements giving the reverse inequality, gives (231.3).

If I is finite, every tuple is bounded below by the minimum of its valuations. On every monomial the finite sum $\sum_i a_i E_i$ agrees with D_δ , and the matrix theorem gives equality everywhere. Uniqueness can also be seen by evaluating on t^{b_j} and dividing by that monomial. $\square$

For infinite I , the notation $\sum_i a_i E_i$ should *not* be read as an unqualified assertion that an arbitrary operator sum is summable on every input. The precise definition is the logarithmic map followed by (230.3). This avoids imposing an unnecessary global summability condition on the tuple (a_i) .

Proposition 231.2 (A standard Borel parameter space). *With the chosen basis, the parameter set*

$$\mathcal{B}_\Gamma = \bigcup_{m \in \mathbb{Z}} \bigcap_{i \in I} \{(a_j)_{j \in I} : v(a_i) \geq m\} \subseteq \mathcal{L}_\mathbb{R}^{\Gamma^I}$$

is standard Borel. The evaluation map $\mathcal{B}_\Gamma \times \mathcal{L}_\mathbb{R}^\Gamma \rightarrow \mathcal{L}_\mathbb{R}^\Gamma$, $(a, f) \mapsto D_a(f)$, is Borel.

Proof. The condition $v(a_i) \geq m$ is the countable conjunction that all its coefficients below m vanish, so the displayed set is Borel in the standard Borel product. For fixed γ , the expression $\delta_a(\gamma) = \sum_i \ell_i(\gamma) a_i$ is a finite sum and is Borel in the parameters. Enumerate Γ . Finite partial sums of $f_\gamma t^\gamma \delta_a(\gamma)$ are Borel jointly in (a, f) , and the common valuation bound makes them converge coefficientwise for every $(a, f) \in \mathcal{B}_\Gamma \times \mathcal{L}_\mathbb{R}^\Gamma$. Their limit is Borel. $\square$

231.1 Rank one: every Borel derivation and every antiderivative

If Γ has rational rank one, choose a nonzero basis element $b \in \Gamma$. Then $\ell(\gamma) = \gamma/b \in \mathbb{Q}$, and E denotes the corresponding Euler derivation. For $\Gamma = \mathbb{Q}$ we take $b = 1$.

Corollary 231.3 (One parameter, one integration obstruction). *In rational rank one, every Borel derivation is uniquely $D = aE$. If $a \neq 0$, then*

$$\ker D = \mathbb{R}, \quad \text{im } D = \{g \in \mathcal{L}_{\mathbb{R}}^{\Gamma} : \text{ct}(g/a) = 0\},$$

where $\text{ct}(h) = [t^0]h$. Equivalently,

$$0 \longrightarrow \mathbb{R} \longrightarrow \mathcal{L}_{\mathbb{R}}^{\Gamma} \xrightarrow{D} \mathcal{L}_{\mathbb{R}}^{\Gamma} \xrightarrow{g \mapsto \text{ct}(g/a)} \mathbb{R} \longrightarrow 0 \quad (231.5)$$

is an exact sequence of real vector spaces and Borel additive maps. When $D(f) = g$ is solvable, its solutions are precisely

$$f = c + \sum_{\substack{\gamma \in \Gamma \\ \gamma \neq 0}} \frac{[t^{\gamma}](g/a)}{\ell(\gamma)} t^{\gamma}, \quad c \in \mathbb{R}. \quad (231.6)$$

No nonzero Borel derivation in rational rank one is surjective.

Proof. Because $\ell(\gamma) \neq 0$ for $\gamma \neq 0$, the kernel of E consists exactly of constants. Its image consists exactly of the series with zero constant coefficient: division of the coefficient at $\gamma \neq 0$ by $\ell(\gamma)$ changes no support. Multiplication by nonzero a proves the kernel, image, and primitive formulas. The final map in (231.5) is onto, since it takes ar to r . All maps in the sequence are Borel by the field operations and the matrix theorem. Surjectivity of D fails because a is not in its image. $\square$

For $\Gamma = \mathbb{Q}$ and $D = d/dt = t^{-1}E$, the obstruction is the coefficient of t^{-1} in g . For $D = E$, the obstruction is the constant coefficient. These familiar special cases are included to identify the normalization; the classification says there are no other Borel derivations outside this one-parameter family.

[write] (Batch 93.) For $\Gamma = \mathbb{Z}$ the field $\mathcal{L}_{\mathbb{R}}^{\mathbb{Z}}$ is the Laurent field $\mathbb{R}((t))$, and the image clause for $D = d/dt$ is formalized in Lean over every field of characteristic zero ([remark 239.5 \[merge\]](#)); the classification of Borel derivations is not.

231.2 The finite/infinite rational-rank threshold

Finite rational rank imposes no extra bound on a finite list of prescribed logarithmic values. In infinite rational rank, the bound is genuine. Fix $a > 0$ in Γ and a countably infinite basis $(b_n)_{n \geq 1}$. The prescription

$$\delta(b_n) = t^{-n^2 a}$$

defines an additive map on Γ , hence an $\mathbb{R}$ -derivation on $\mathbb{R}[\Gamma]$, but its logarithmic valuations have no common lower bound. Corollary 230.3 rules out every Borel extension to $\mathcal{L}_{\mathbb{R}}^{\Gamma}$. The algebraic extension argument in Section 233 nevertheless gives field derivations extending this prescription. Thus the obstruction is regularity, not the algebraic possibility of extending a derivation.

232 Exactly which derivations are coefficient-continuous?

The matrix theorem gives an abstract finite-row criterion. For derivations it can be rewritten entirely in terms of the logarithmic map. For $h \in \Gamma$ put

$$d_h(\gamma) = [t^h]\delta(\gamma). \quad (232.1)$$

Each $d_h : \Gamma \rightarrow \mathbb{R}$ is additive.

Theorem 232.1 (Finite-support and diagonal criterion). *Let D_δ be a Borel derivation. It is continuous for the coefficient topology if and only if both of the following hold:*

(C1) *Every logarithmic value $\delta(\gamma)$ has finite support, that is, $\delta(\Gamma) \subseteq \mathbb{R}[\Gamma]$.*

(C2) *The diagonal exceptional set*

$$B_\delta = \{h \in \Gamma : d_h(h) \neq 0\}$$

is finite.

In finite rational rank, these conditions are equivalent to every parameter a_i in Theorem 231.1 having finite support.

Proof. The output coefficient of $D(t^\gamma)$ at exponent λ is

$$c_{\lambda\gamma} = d_{\lambda-\gamma}(\gamma).$$

For fixed λ , reindex its row by $h = \lambda - \gamma$. Additivity gives

$$c_{\lambda, \lambda-h} = d_h(\lambda - h) = d_h(\lambda) - d_h(h). \quad (232.2)$$

At $\lambda = 0$, this row is $-d_h(h)$, so its support is exactly B_δ . If all rows are finite, B_δ is finite. For arbitrary λ , (232.2) then implies that only finitely many h have $d_h(\lambda) \neq 0$. This is exactly (C1).

Conversely, (C1) makes the support of $h \mapsto d_h(\lambda)$ finite for each λ , and (C2) makes the support of $h \mapsto d_h(h)$ finite. Their difference in (232.2) has finite support. Every row is therefore finite, and Proposition 228.4 applies.

In finite rank, (C1) implies finite support of each $a_i = \delta(b_i)$. Conversely, finitely many finite-support a_i have a finite union of supports F . Every $\delta(\gamma)$ is supported in F , and d_h is identically zero for $h \notin F$. Both (C1) and (C2) follow. $\square$

Example 232.2 (A Borel derivation with a discontinuous coefficient). On $\mathcal{L}_{\mathbb{R}}^{\mathbb{Q}}$ set

$$a = \frac{1}{1-t} = \sum_{k \geq 0} t^k, \quad D = aE.$$

It is a Borel derivation of gain zero. For positive integers n put $f_n = -t^{-n}/n$. Then $f_n \rightarrow 0$ coefficientwise but

$$D(f_n) = \frac{t^{-n}}{1-t}, \quad [t^0]D(f_n) = 1.$$

Thus it is not coefficient-continuous. Its failure is exactly the infinite support of the parameter a , not a failure of valuation continuity or summability.

Example 232.3 (Infinite rank permits infinitely many bounded shifts). Choose $\mathbb{Q}$ -linearly independent real numbers $b_0, b_1, \dots \in (1, 2)$ and let $\Gamma = \bigoplus_{n \geq 0} \mathbb{Q}b_n$. Such a sequence exists by induction, since the rational span of finitely many real numbers is countable. Define

$$\delta(b_n) = t^{b_{n+1}}.$$

The valuations have common lower bound 1, so this defines a Borel derivation. Each individual $\delta(\gamma)$ is a finite sum. The only possibly nonzero coefficient maps are $d_{b_{n+1}} = \ell_n$, and

$$d_{b_{n+1}}(b_{n+1}) = \ell_n(b_{n+1}) = 0.$$

Thus B_δ is empty and Theorem 232.1 makes D coefficient-continuous.

Nevertheless, the union of the supports of all $\delta(b_n)$ is the infinite bounded set $\{b_{n+1} : n \geq 0\}$, which is not left-finite. This does not contradict the theorem: the logarithmic values need not form a summable family as γ ranges over all exponents. Summability is required after the input monomial factors and the left-finite input support have been taken into account.

233 Wild algebraic derivations: an explicit obstruction to Borelness

The regularity hypothesis is not vacuous. We construct a nonzero derivation that kills every real coefficient and every monomial. The final extension uses a transcendence basis, but the element witnessing the obstruction is an explicit lacunary series.

Lemma 233.1 (Lacunary transcendence over the monomial field). *Fix $a > 0$ in Γ , and put*

$$F = \sum_{n \geq 1} t^{n!a}, \quad K = \text{Frac}(\mathbb{R}[\Gamma]).$$

Then $F \in \mathcal{L}_{\mathbb{R}}^\Gamma$ is transcendental over K .

Proof. Membership follows because $n!a \rightarrow +\infty$. Suppose a nonzero polynomial over K vanishes at F . Clear denominators to obtain

$$P(X) = \sum_{k=0}^d p_k X^k \in \mathbb{R}[\Gamma][X], \quad P(F) = 0,$$

with $d \geq 1$. Choose real bounds C_- and C_+ containing every exponent in the finite union of the supports of the nonzero p_k .

Let $F_N = \sum_{n=1}^N t^{n!a}$. These are distinct field elements. Since a nonzero degree- d polynomial has at most d roots, $P(F_N) \neq 0$ for all but at most d values of N . Its support has no exponent above $C_+ + dN!a$. Therefore

$$v(P(F_N)) \leq C_+ + dN!a \tag{233.1}$$

whenever $P(F_N) \neq 0$.

On the other hand, $v(F - F_N) = (N+1)!a$ and $v(F) = v(F_N) = a > 0$. Factor each difference $F_N^k - F^k$ by $F_N - F$. All factors in the remaining sum have nonnegative valuation. Because every p_k has valuation at least C_- , the equation $P(F) = 0$ gives

$$v(P(F_N)) = v(P(F_N) - P(F)) \geq (N+1)!a + C_-. \tag{233.2}$$

For sufficiently large N , the lower bound in (233.2) exceeds the upper bound in (233.1), since $((N+1) - d)N!a \rightarrow +\infty$. Choose such an N with $P(F_N) \neq 0$, a contradiction. $\square$

Lemma 233.2 (Algebraic extension of a derivation). *Let $K \subseteq L$ be fields of characteristic zero. Every derivation $d : K \rightarrow K$ extends to a derivation $L \rightarrow L$.*

Proof. Choose a transcendence basis B of L/K . Set the derivatives of its elements to zero and extend d to $K(B)$ by the polynomial and quotient rules. The extension $L/K(B)$ is algebraic and separable. If $\alpha \in L$ has minimal polynomial $P(X) = \sum_k c_k X^k$ over the current base, its derivative is forced to be

$$d(\alpha) = -\frac{\sum_k d(c_k) \alpha^k}{P'(\alpha)}. \quad (233.3)$$

The denominator is nonzero by separability. Defining the derivative of α by this formula makes differentiation vanish on the ideal generated by P , so it defines a derivation on the simple algebraic extension and then on its fraction field. Repeating on finite subextensions is consistent by uniqueness. Their union gives the derivation on L . $\square$

Theorem 233.3 (Algebraic derivations invisible on monomials). *For every nonzero countable $\Gamma \leq \mathbb{R}$, there exists a derivation $D : \mathcal{L}_{\mathbb{R}}^{\Gamma} \rightarrow \mathcal{L}_{\mathbb{R}}^{\Gamma}$ such that*

$$D \upharpoonright \mathbb{R}[\Gamma] = 0, \quad D(F) = 1$$

for the explicit series F in Lemma 233.1. Every such derivation is non-Borel and fails valuation continuity.

Proof. By transcendence, $K(F)$ is a rational-function field in F over K . Define its derivation by $D(K) = 0$ and $D(F) = 1$, and apply Lemma 233.2 to extend it to $\mathcal{L}_{\mathbb{R}}^{\Gamma}$. If it were Borel, Corollary 228.3 would force it to be zero, contradicting $D(F) = 1$. Also $F_N \rightarrow F$ in valuation, while $D(F_N) = 0$ for every N and $D(F) = 1$. Thus it is not valuation-continuous. $\square$

This is not an algorithmically explicit formula for a wild derivation on all of $\mathcal{L}_{\mathbb{R}}^{\Gamma}$. Its extension uses choice through a transcendence basis. What is explicit is the lacunary input and the finite-support approximants that certify failure of continuity and, by the main theorem, failure of Borelness. The distinction prevents an existence proof from being confused with an effective construction.

The unbounded monomial prescription from Section 231 also takes values in $\mathbb{R}[\Gamma]$, extends by the quotient rule to K , and then extends algebraically by Lemma 233.2. Every extension is non-Borel by the boundedness obstruction. This is a second mechanism for wildness: there the obstruction is already visible on monomials, whereas in Theorem 233.3 the monomial data are identically zero.

234 Positive-gain derivations and tangent-to-identity automorphisms

The classification also gives a controlled integration theorem. This section concerns formal operator exponentials in the valuation topology, not an ordinary real-analytic convergence assertion.

Definition 234.1. An $\mathbb{R}$ -linear operator U has *uniformly positive gain* if some $\varepsilon > 0$ satisfies

$$v(Uf) \geq v(f) + \varepsilon \quad (f \neq 0).$$

An $\mathbb{R}$ -algebra automorphism Φ is *uniformly tangent to the identity* if $\Phi - \text{id}$ has uniformly positive gain.

Every positive-gain operator is valuation-continuous and hence Borel by the matrix theorem. If U has gain at least ε , then U^n has gain at least $n\varepsilon$. Consequently, for any real sequence (c_n) , the operator series $\sum_{n \geq 0} c_n U^n$ is well defined on every input: for a fixed output cutoff only finitely many n contribute. Real coefficients, however large, do not change valuation. All manipulations below can therefore be checked coefficient by coefficient using finite sums.

Theorem 234.2 (Exponential–logarithm correspondence). *Let D be a Borel derivation with gain at least $\varepsilon > 0$. For each $s \in \mathbb{R}$ the formula*

$$\Phi_s = \exp(sD) = \sum_{n \geq 0} \frac{s^n D^n}{n!} \quad (234.1)$$

defines a Borel $\mathbb{R}$ -algebra automorphism of $\mathcal{L}_{\mathbb{R}}^{\Gamma}$, and

$$\Phi_s \Phi_r = \Phi_{s+r}, \quad \Phi_0 = \text{id}, \quad \Phi_s^{-1} = \Phi_{-s}.$$

It preserves valuation and leading coefficients, and $\text{gain}(\Phi_s - \text{id}) \geq \varepsilon$.

Conversely, for an $\mathbb{R}$ -algebra automorphism Φ with $\text{gain}(\Phi - \text{id}) \geq \varepsilon > 0$,

$$\log \Phi = \sum_{n \geq 1} \frac{(-1)^{n+1}}{n} (\Phi - \text{id})^n \quad (234.2)$$

is a Borel derivation of gain at least ε . Exponential and logarithm are inverse correspondences between positive-gain derivations and uniformly tangent-to-identity $\mathbb{R}$ -algebra automorphisms.

Proof. Exponentiation. The preceding gain estimate proves summability of (234.1). Iterated Leibniz, proved by induction, says

$$D^n(fg) = \sum_{k=0}^n \binom{n}{k} D^k(f) D^{n-k}(g).$$

Substitution in the exponential gives $\Phi_s(fg) = \Phi_s(f)\Phi_s(g)$. For a coefficient at most B , the terms on the product side have valuation at least $v(f) + v(g) + (k+j)\varepsilon$, so only finitely many pairs (k, j) are relevant. This justifies regrouping without an analytic convergence assumption. The same finite-cutoff argument and the binomial identity give $\Phi_s \Phi_r = \Phi_{s+r}$. Constants are fixed because D kills $\mathbb{R}$. Thus Φ_s is an $\mathbb{R}$ -algebra automorphism with the stated inverse.

For nonzero f , every term of $\Phi_s(f) - f$ has valuation at least $v(f) + \varepsilon$, so the leading exponent and coefficient of f survive. The map is real-linear and valuation-continuous, hence Borel.

Logarithm and the derivation rule. Put $U = \Phi - \text{id}$. The series (234.2) has gain at least ε and is a real-linear Borel operator. To prove that it is a derivation, for each real z define

$$\Psi_z = \sum_{n \geq 0} \binom{z}{n} U^n.$$

Each output coefficient of $\Psi_z(f)$ is a polynomial in z , since only finitely many n contribute to it. The same holds for each coefficient of $\Psi_z(f)\Psi_z(g)$. For a nonnegative integer z , the binomial sum is finite and equals Φ^z , so

$$\Psi_z(fg) = \Psi_z(f)\Psi_z(g).$$

For any fixed output coefficient the difference of the two sides is a real polynomial vanishing at every nonnegative integer. It vanishes identically. Differentiate this coefficient identity at $z = 0$. Since $\Psi_0 = \text{id}$ and

$$\left. \frac{d}{dz} \binom{z}{n} \right|_{z=0} = \frac{(-1)^{n-1}}{n} \quad (n \geq 1),$$

the resulting identity is exactly

$$(\log \Phi)(fg) = f(\log \Phi)(g) + g(\log \Phi)(f).$$

Thus the logarithm is a derivation.

Inverse identities. The scalar formal identities $\exp(\log(1+X)) = 1+X$ and $\log(\exp X) = X$ hold in $\mathbb{Q}[[X]]$; they follow, for example, by differentiating the two sides and comparing constant terms. Substitution of U or D is legitimate because, at a fixed output cutoff on a fixed input, sufficiently high powers vanish from consideration. Every coefficient identity reduces to its finite formal truncation. These identities give the two inverse correspondences. $\square$

[write] (Batch 93.) The proof that the logarithm is a derivation, by a polynomial identity in z vanishing on the natural numbers and differentiated at 0, is the proof of the exponential–logarithm correspondence in the `omnific-preserving-automorphisms` report (`opa:hs:thm:fulllie`), there for an s -adic filtration and here for a valuation-gain filtration. The inverse of $\text{id} + U$ for uniformly positive gain is Part XV’s Neumann mechanism ([lemma 199.1](#)); for real coefficients, [remark 239.4](#) [merge] applies the theorem to Part XV’s monomial characters.

Corollary 234.3 (Joint Borel flows). *For fixed positive-gain D , the map $(s, f) \mapsto \exp(sD)(f)$ from $\mathbb{R} \times \mathcal{L}_{\mathbb{R}}^{\Gamma}$ to $\mathcal{L}_{\mathbb{R}}^{\Gamma}$ is Borel. For fixed f and a fixed exponent, its coefficient is a polynomial in s .*

Proof. Finite exponential partial sums are jointly Borel because each D^n is Borel and real scalar multiplication is coefficient-continuous. They converge coefficientwise. For a fixed input and output coefficient, the gain estimate bounds the number of contributing powers independently of s . $\square$

[write] (Batch 96.) Part XX proves the exponential half again ([corollary 276.3](#)), extends it from a uniform positive gain to window-local nilpotence ([theorem 276.2](#)), extends the logarithm half in the same way ([corollary 279.6](#)), and shows that without a common gain the time-one maps of integrable derivations need not be closed under composition ([theorem 279.1](#)); under a common positive gain they are ([proposition 285.3](#) [merge]).

234.1 Euler normal forms and a concrete flow

In finite rational rank, write $D_a = \sum_i a_i E_i$ and $D_b = \sum_i b_i E_i$. The coordinate Euler operators commute, since they act on each coefficient by commuting scalar multipliers. Expanding the commutator and cancelling the double-derivative terms gives

$$[D_a, D_b] = \sum_j \left(\sum_i a_i E_i(b_j) - b_i E_i(a_j) \right) E_j. \quad (234.3)$$

For any two bounded-gain operators, composition adds gain lower bounds, and taking a difference cannot lower their minimum. In particular, $\text{gain}([D_a, D_b]) \geq \text{gain}(D_a) + \text{gain}(D_b)$ when both displayed gains are finite.

On $\mathcal{L}_{\mathbb{R}}^{\mathbb{Q}}$, take $D = tE = t^2 d/dt$, of gain 1. For $q \in \mathbb{Q}$,

$$D^n(t^q) = (q)_n t^{q+n}, \quad (q)_n = q(q+1) \cdots (q+n-1), \quad (q)_0 = 1.$$

The exponential gives the classical formal binomial expression

$$\Phi_s(t^q) = t^q (1-st)^{-q} = \sum_{n \geq 0} \frac{(q)_n}{n!} s^n t^{q+n}, \quad \Phi_s(t) = \frac{t}{1-st}. \quad (234.4)$$

The right side defines the generalized binomial power; no branch convention for ordinary complex powers is involved. The operator theorem proves that this monomial prescription extends to every left-finite series and that the parameters add under composition.

[write] (Batch 93.) The same flow $t \mapsto t/(1-st)$ is the fixed-shift flow T_s of the `surcomplex-field-automorphisms` report (`saut:thm:shiftflow`, its derivation $\mathcal{D}(t^g) = g t^{g+1}$ being tE).

Positive gain is a sufficient condition for valuation-formal exponentiation, not a necessary condition for every possible flow notion. For example, E itself has gain zero but admits the coefficientwise flow $\sum f_\gamma t^\gamma \mapsto \sum e^{s\gamma} f_\gamma t^\gamma$ in the normalization $\Gamma = \mathbb{Q}$, $b = 1$. Here the exponential is interpreted using ordinary real coefficient convergence rather than eventual valuation stabilization of the operator series. These two integration mechanisms must not be conflated.

[write] (Batch 93.) Source 23 asserts this flow without proof; [proposition 239.3](#) [merge] proves it, for every coordinate Euler derivation.

235 Surreal interpretation and a formalization route

235.1 A set-sized subfield, not a Borel structure on a proper class

The Conway normal-form description presents surreal numbers as generalized real-coefficient series; modern treatments of surreal subfields explicitly use this viewpoint [148]. For our chosen $\Gamma \subseteq \mathbb{R}$, define

$$\iota_\Gamma : \mathcal{L}_{\mathbb{R}}^\Gamma \longrightarrow \mathbf{No}, \quad \sum_\gamma f_\gamma t^\gamma \mapsto \sum_\gamma f_\gamma \omega^{-\gamma}. \quad (235.1)$$

The right side is a Conway normal form: a left-finite support is well ordered, so its negative exponents are reverse well ordered. Normal-form uniqueness makes this map injective. Normal-form addition and multiplication, together with $\omega^x \omega^y = \omega^{x+y}$, show that it is a field homomorphism. All the coefficient sums needed for the image of a product are already finite by Proposition 226.2.

Let $K_\Gamma = \iota_\Gamma(\mathcal{L}_{\mathbb{R}}^\Gamma)$. We give this *set-sized* subfield the Borel structure transported from $\mathcal{L}_{\mathbb{R}}^\Gamma$. The main theorem and the classification transfer verbatim to K_Γ . In particular, with $t = \omega^{-1}$, every Borel derivation on this copy of $\mathcal{L}_{\mathbb{R}}^{\mathbb{Q}}$ is determined by one logarithmic monomial value.

[write] (Batch 93.) ι_Γ is Part V's surreal map (71.1) and Part XV's [proposition 206.1](#), with real coefficients. Under it E (for $\Gamma = \mathbb{Q}$, $b = 1$) is the restriction of the Euler operator $\mathfrak{D}(t^\gamma) = \gamma t^\gamma$ of the `real-vector-space-structure` report, which in the ω -convention reads $\omega^\beta \mapsto -\beta \omega^\beta$. The transfer to K_Γ is a transport of structure and needs no further proof.

This does not equip all of $\mathbf{No}$ with a standard Borel structure. It does not assert that every set-sized surreal subfield is a left-finite field, that the usual surreal exponential preserves K_Γ , or that a particular surreal derivation restricts to it. Compatibility with a chosen exponential,

logarithm, or distinguished surreal derivation is an additional problem. Borel regularity selects a classified family of derivations; it does not select a canonical one.

All the main proofs can be carried out in ordinary ZFC on the set $\mathcal{L}_{\mathbb{R}}^{\Gamma} \subseteq \mathbb{R}^{\Gamma}$, without any class quantification. The surreal interpretation can be added afterwards through the normal-form map. A universe-polymorphic formalization can keep the countable exponent type and its coefficient functions in an explicit universe and use a larger universe for an ambient surreal construction, avoiding any purported set of all surreals.

235.2 How this fits the repository

The inspected ProveIt documentation describes a distinction between research reports, theorems mapped to formal declarations, and executable computations outside the trusted theorem boundary [149, 150]. The present article belongs to the first category. No declaration in that repository has been claimed to verify Theorem 228.2 or its consequences.

[write] (Batch 93.) True for [theorem 228.2](#). For its consequences it is incomplete: the image clause of [corollary 231.3](#) for $\Gamma = \mathbb{Z}$ and $D = d/dt$ is the Lean theorem `range_derivative_eq_ker_residue` ([remark 239.5](#)). Source 23 did not read the Lean files.

A useful formalization order is as follows.

1. Define the left-finite support predicate and the subtype of coefficient functions. Prove convolution finiteness, the field laws, valuation completeness, and convergence of finite truncations. These facts form a self-contained set-sized layer, independent of surreal proper classes.
2. Formalize coefficient slices and the finite-coordinate automatic continuity lemma. The analytical dependencies are the Baire theorem for countable products of $\mathbb{R}$ and the Baire property of Borel sets.
3. Prove the generic-support lemma and cutoff-admissible matrix theorem. This is the critical descriptive-set-theoretic layer; testing finite matrices cannot replace it.
4. Establish the logarithmic map, the integer-translation uniform-bound argument ([230.5](#)), and the normal form of derivations. Prove the exact finite-row criterion before specializing to coefficient continuity of derivations.
5. Add rational-rank parameterization, the rank-one integration exact sequence, and the filtered exponential–logarithm theorem. Transfer to an actual surreal subfield only after the normal-form embedding is available.

This is a dependency plan, not a list of existing Lean theorem names. It allows the new result to be checked without assuming unrelated research claims elsewhere in the repository.

236 Finite truncation, verification artifacts, and proof audit

236.1 An exact truncation law

Write $\pi_{\leq B}(f) = \sum_{\gamma \leq B} f_{\gamma} t^{\gamma}$. If δ has a known lower bound m , the proof of Theorem [230.2](#) gives the exact identity

$$\pi_{\leq B} D_{\delta}(f) = \pi_{\leq B} D_{\delta}(\pi_{\leq B-m}(f)). \quad (236.1)$$

Only finitely many input terms are required. For each such input exponent γ , only the finite truncation $\pi_{\leq B-\gamma}\delta(\gamma)$ is needed. This is a finite evaluation algorithm whenever the input and logarithmic values are supplied by procedures returning their finite truncations.

For a generic Borel additive map, (M2) guarantees existence of the cutoff but does not compute it from an arbitrary Borel code. The proof's Baire-category step is not an extraction algorithm. A certified implementation should accept a cutoff modulus as part of its data, or separately prove a way to obtain one for a restricted representation language.

236.2 What the executable checks cover

The accompanying `verification.py` uses only Python's standard library and exact rational arithmetic. It implements finite rational-exponent series, Euler derivations, multiplication, and controlled truncation. Its 1,640 deterministic exact checks passed in the supplied run. The tests check the Leibniz rule for aE , the commutator identity (234.3) in rank one, the exact truncation identity, the binomial flow formula (234.4), and flow composition at finite cutoffs. It also checks the rank-one primitive formula on exact finite examples.

[write] In this report the program is `code/23-borel-summability-verification.py` and its recorded run data `data/23-borel-summability-verification_results.json`; the program prints its record to standard output. The intake reran it on a copy (Python 3.14.4): 1,640 checks in 13 groups pass, also under `-0`, and the printed record equals the shipped one after removing the carriage returns that Windows text mode adds.

These tests can expose sign, normalization, and cutoff errors in formulas. They are not a proof of the infinite-dimensional matrix theorem, not a mechanized proof of the Baire argument, and not evidence of literature priority. The mathematical justification for passing from finite truncations to the entire field is supplied in the preceding sections.

236.3 The delicate steps and their checks

No hidden Polish topology.

All uses of automatic continuity take place on $\mathrm{Sl}_5 \simeq \mathbb{R}^5$. Proposition 229.1 explicitly rules out applying a Polish group theorem to the full additive field in this coding.

No assumed infinite linearity.

The slice functional is obtained from Borel additivity. Infinite summation is not interchanged with T until the matrix cutoff has been proved.

No cancellation loophole.

The generic-support lemma constructs one coefficient vector for which every nonzero output row on a slice survives simultaneously. Countability of the target exponent group is used exactly here.

No divisibility assumption on the exponent group.

The uniform-bound proof moves an arbitrary exponent into one bounded interval by an integer multiple of a single positive group element. It works for nondivisible subgroups as well as $\mathbb{Q}$ -vector spaces.

No global summability assumption on logarithmic values.

Only a common lower valuation bound is required. The infinite-rank example in Section 232 demonstrates why requiring the union of all logarithmic supports to be left-finite would be an unjustified strengthening.

No analytic rearrangement in exponential or logarithm.

Uniform positive gain turns every fixed output coefficient calculation into a finite calculation. The logarithm's Leibniz rule is proved through polynomial identities in an auxiliary scalar parameter, rather than assumed.

No effective claim about wild derivations.

The lacunary witness is explicit, but extension to the whole field uses a transcendence basis. The article does not provide a computable global wild map.

237 Further research questions and concrete next targets

The following questions go beyond the theorems proved here. They are proposed research directions, not assertions that every formulation is a previously published open problem.

237.1 Support classes beyond left-finiteness

Question 237.1. Which Borel families $\mathcal{I}$ of allowed supports on a countable ordered exponent group have the property that every coefficient-Borel additive map between their series spaces automatically preserves their formal sums?

A first target is a necessary-and-sufficient condition on $\mathcal{I}$ that replaces the “unbounded bad columns admit a left-finite subsequence” step. The product-slice lemma works whenever arbitrary coefficient assignments on an allowed support remain in the domain. The unresolved part is translating simultaneous noncancellation into a useful global matrix condition. Full Hahn support classes require particular care: well ordering alone permits bounded infinite supports, and their descriptive complexity need not match (226.1).

[write] (Batch 93.) A companion of Part V's [question 73.5](#), which asks which support classes give Polish field topologies; neither answers the other.

[write] (Batch 96.) Part XIX makes the last sentence of the commentary exact: for countable $\Gamma \leq \mathbb{R}$ the code set of all well-ordered supports is the whole space for $\Gamma = \{0\}$, Σ_2^0 -complete for nonzero cyclic Γ , and Π_1^1 -complete for noncyclic Γ ([proposition 270.1](#)), against the Π_3^0 left-finite domain (226.1). The question, on automatic summation, stays open.

Review remark 1 (Trivial-group correction). The earlier Batch 96 commentary above, and the guide's summary of [proposition 270.1](#), said “ Σ_2^0 -complete for cyclic $\Gamma \neq 0$ and Π_1^1 -complete otherwise”. The latter claim is false at $\Gamma = \{0\}$: every support is a subset of a singleton, so the code set is the whole coefficient space. Every continuous preimage of that set is the whole domain, which precludes Π_1^1 -completeness. [Proposition 270.1](#) already states this exception correctly. Its distinct strong-family predicate is Σ_2^0 -complete at $\Gamma = \{0\}$; that does not change validity of a single support.

237.2 Coefficient fields other than the reals

Question 237.2. What is the exact analogue of Theorem [228.2](#) for left-finite series with coefficients in a specified non-Archimedean local field or in a finite field, equipped with its natural Polish coefficient topology?

The real proof uses more than additive automatic continuity: continuous real-linear functionals on a product have finite coordinate support, since unrestricted coordinates can be scaled arbitrarily. For other coefficient fields, continuous additive maps may have different scalar linearity and infinite-coordinate behavior. In characteristic p , Frobenius is an immediate warning that additivity does not force linearity over the full coefficient field. An exact answer should identify both the allowed scalar endomorphisms and the support cutoff replacing (M2).

237.3 Higher-rank constants and antiderivatives

Question 237.3. For finite rational rank at least two, classify the constant fields and images of Borel derivations $D = \sum_i a_i E_i$ under explicit hypotheses on the parameters a_i .

The rank-one exact sequence shows that there is a single real-linear obstruction to integration. In higher rank, a coordinate Euler derivation already has all series on its coordinate-kernel subgroup among its constants. Variable parameters can interact and create further cancellations. A concrete first project is the two-dimensional rational span with finite-support parameters: determine when its constant field is exactly $\mathbb{R}$, and whether the cokernel of D is finite-dimensional, countably dimensional, or larger.

[write] (Batch 96.) Not answered. For $\Gamma = \mathbb{Z} + \mathbb{Z}\sqrt{2}$, Part XXI's [example 295.5](#) shows that scalar rigidity and the three-dimensional bound for Lie algebras of Borel derivations both fail; Part XVIII's [question 253.2](#) is the conjugacy counterpart of this question.

237.4 Borel classification up to conjugacy

Question 237.4. What invariants classify Borel derivations under conjugacy by Borel $\mathbb{R}$ -algebra automorphisms of the same left-finite field?

The coefficient tuple is a complete invariant only after a monomial basis is fixed, not under changes of field coordinates. Even the rational-rank-one case asks when aE and bE are related by a field automorphism. The positive-gain subgroup from Theorem [234.2](#) supplies a controlled starting point, while residue obstructions and gain estimates give testable necessary conditions. A classification should distinguish formal conjugacy from coefficient-Borel conjugacy rather than assuming they coincide.

[write] (Batch 96, 4 October 2026.) Answered in rational rank one by [Part XVIII](#) (source 24), for conjugacy by Borel ordered-field automorphisms, which are all the Borel automorphisms when Γ is 2-divisible (for instance $\Gamma = \mathbb{Q}$); every countable $\Gamma \leq \mathbb{R}$ of rational rank one becomes a group $\mathbb{Z} \leq \Gamma \leq \mathbb{Q}$ after a scaling of exponents. The automorphisms are the substitutions $t \mapsto At^r(1+u)$ with $r\Gamma = \Gamma$ ([theorem 242.4](#)). For $a, b \neq 0$, aE and bE are conjugate exactly when $v(a)$ and $v(b)$ have the same sign and the invariants of [theorem 245.1](#) agree: for nonzero gain, the orbit of $|v(a)|$ under $U_\Gamma = \{r \in \mathbb{Q}_{>0} : r\Gamma = \Gamma\}$, the sign of the leading coefficient and, for positive gain, the residue ratio $\text{ct}(1/a)/v(a)$; for gain zero, the leading coefficients modulo U_Γ . Conjugating coordinates can be chosen Borelly on the relation ([theorem 245.2](#)), and the relation is smooth exactly when at most one prime is infinitely divisible in Γ ([theorem 246.4](#)). On the distinction asked for in the commentary: conjugacy by field automorphisms that fix $\mathbb{R}$ and preserve the order, Borel or not, is the same relation, because such automorphisms are substitutions and hence Borel ([remark 255.3](#) [merge]). For pairs, [Part XXI](#) (source 27) answers the simultaneous version for noncommuting ordered pairs in rational rank one, under ordered Borel automorphisms, with an explicit Borel cross-section, the complete invariant

(H, A) , trivial stabilizers and unique Borel conjugators ([theorem 291.2](#) and [corollary 291.3](#)); commuting pairs are no better than single derivations ([theorem 292.2](#)). Open: higher rational rank ([questions 253.2](#) and [298.1](#)), unordered automorphisms when Γ is not 2-divisible ([questions 253.1](#) and [298.2](#)), and conjugacy by automorphisms that do not fix $\mathbb{R}$. The proofs are unrefereed and not formalized.

237.5 Flows at zero or negative gain

Question 237.5. Which Borel derivations admit one-parameter Borel automorphism flows when uniform positive gain is absent, and what additional parameter regularity selects a useful notion of flow?

The gain-zero Euler flow shows that positive gain is not necessary. However, its coefficientwise analytic construction is different from valuation-formal operator exponentiation. A first precise version can require that each coefficient of $\Phi_s(f)$ be real-analytic in s , and that its derivative at zero be the given derivation. The resulting existence and uniqueness problem should explicitly record whether the coefficient functions are merely analytic, entire, or polynomial.

[write] (Batch 93.) [Proposition 239.3](#) [merge] proves the gain-zero example used here, with entire coefficient functions; the question is otherwise open.

[write] (Batch 96, 4 October 2026.) The “otherwise open” of the preceding note is re-scoped: Parts XVIII and XX answer the question in rational rank one and for countable divisible Γ , and agree where both apply. *Rational rank one* ([Part XVIII](#), source 24; $\mathbb{Z} \leq \Gamma \leq \mathbb{Q}$ after a scaling of exponents; one-parameter groups of Borel ordered-field automorphisms, which are all the Borel automorphisms when Γ is 2-divisible). A Borel derivation aE generates a coefficientwise real-analytic flow exactly when $a = 0$ or $v(a) \geq 0$; the flow is unique, its coefficients are polynomials in time for positive gain and finite sums of exponentials for gain zero, all entire in complex time; and for negative gain not even a curve through the identity with one coefficient differentiable at 0 has derivative aE ([theorem 248.3](#)). Borel dependence on time of the coefficients of $\Phi_s(t)$ alone, or their differentiability at 0, forces such a flow ([theorem 248.8](#)), while without time regularity a discontinuous additive change of time gives other homomorphisms. So positive gain is replaced by nonnegative gain, and the regularity that selects the notion is Borel measurability in time of the single coordinate $\Phi_s(t)$; the time-one maps of these flows are the automorphisms that preserve the valuation ([proposition 255.1](#) [merge]). *Divisible Γ of every rational rank* ([Part XX](#), source 26; jointly Borel flows). Every orbit coefficient is automatically an entire function of time, so no analyticity has to be imposed, and the Borel derivations that generate flows are exactly the valuation-nondecreasing ones that are locally finite on every bounded exponent window ([theorem 275.2](#) and [corollary 275.3](#)); negative gain never integrates ([corollary 275.4](#)); in finite rational rank exactly the valuation-nondecreasing Borel derivations integrate ([corollary 277.3](#)); in infinite rank strictness does not suffice, and integrable derivations are not closed under sums or brackets ([theorems 277.2](#) and [278.2](#)). Borel dependence on time of the images of the monomials of one rational basis already makes an action a Borel flow ([proposition 285.1](#) [merge]). At $\Gamma = \mathbb{Q}$ the two answers coincide ([remark 255.2](#), [corollary 285.2](#) [merge]). [Part XXI](#) (source 27) adds an explicit global action of $\text{Aff}^+(\mathbb{R})$ whose one-parameter subgroups are the flows of E/p and $t^p E/p$ ([theorem 294.4](#)), proves again that negative gain is impossible ([remark 294.3](#)), and, in rational rank one, classifies the actions of connected Lie groups whose one-parameter subgroups have Borel time dependence ([theorem 300.3](#) [merge]). Open: nondivisible Γ of rational rank at least two ([question 283.5](#)), and unordered automorphisms when Γ is not 2-divisible ([question 253.1](#)). The proofs are unrefereed and not formalized.

237.6 Effective moduli from symbolic descriptions

Question 237.6. For which effective description languages for Borel additive operators can a cutoff modulus $B \mapsto A(B)$ be computed, or certified, from the description?

The main theorem is existential. A symbolic language generated by finite matrices, exponent substitutions with known growth, coefficient multipliers, and bounded-gain derivations might admit compositional modulus extraction. The target is a proved compiler producing a finite truncation algorithm, not a heuristic that tests finitely many monomials and guesses a global bound. For derivations, an explicit bound on the logarithmic tuple is already a complete certificate.

[write] (Batch 96.) An effective boundary for normal forms, not for cutoff moduli: on an explicit family of coefficient programs over $\mathbb{Z}$ the normalizer of Part XXI is computable but equality of its invariant is co-c.e.-complete ([remark 300.4](#), a deduction of the Hilbert's-tenth review); for single derivations, decidable membership in Γ does not decide conjugacy ([remark 255.4](#)).

237.7 Logical strength of the slice argument

Question 237.7. What is the reverse-mathematical or choice-theoretic strength of suitable coded versions of the automatic-summability theorem?

The proof isolates two regularity inputs: the Baire property of the relevant measurable sets and a countable intersection of dense open sets in a Polish product. A coded theorem must first specify which Borel codes, exponent presentations, and maps are quantified over; different choices may lead to different strengths. This is a natural connection to the foundational questions around topological arithmetic, but it is not a claimed answer to Glazer's question about the strength of his own theorem.

[write] (Batch 93.) Compare Part VIII, whose claimed answer to Glazer's Question 1 is not reviewed, and Part VI's question on the minimal metatheory; source 23 uses neither.

237.8 A minimal regularity hypothesis

Question 237.8. How does ambient coefficient-Borelness compare with weaker hypotheses imposed only on the restrictions of an additive map to left-finite coefficient slices?

The proof gives one useful observation immediately: if every coordinate functional on every slice is Baire measurable, Lemma [227.1](#) and the rest of the argument still work, so the map is in fact globally Borel. What remains is to find smaller, explicitly specified collections of slices that already force the conclusion. For example, can a natural cofinal or effectively generated family of supports test all operators in a useful symbolic class? Any proposed test family must detect the escaping-column configuration in Step 2 of Theorem [228.2](#).

[write] (Batch 93.) The "useful observation" of the second paragraph is stated without proof; [proposition 239.2](#) [merge] proves it. The question of smaller test families is open.

237.9 A formalized proof with a transparent trust boundary

Question 237.9. Can the full slice-to-matrix theorem and bounded logarithmic-velocity classification be formalized in Lean with no theorem-specific axioms or unverified executable reflection?

The decisive deliverable would be a theorem about actual coefficient-function subtypes, followed by a checked transport to an actual surreal subfield. An especially useful intermediate milestone is an independent formal proof of Lemma 227.2: it contains the infinite-dimensional noncancellation step and is the part least faithfully represented by finite computations.

[write] (Batch 93.) One special case is already formal: the rank-one image clause for $\Gamma = \mathbb{Z}$ (remark 239.5).

238 Conclusion

On a countable-exponent left-finite real series field, coefficient-Borel additivity is much stronger than its definition first suggests. It forces real linearity, a uniform matrix cutoff at every bounded output range, valuation continuity, preservation of admissible infinite sums, and a Baire-class-one coefficient presentation. None of these conclusions requires a Polish topology on the full additive group.

For derivations, the cutoff collapses to a single lower bound on logarithmic monomial values. This yields a complete parameterization in every countable rational rank, an exact criterion for coefficient continuity, and a complete rank-one antiderivative test. Wild algebraic derivations and the noncontinuous Borel examples show that the distinctions are genuine. The positive-gain part integrates into an explicit family of Borel automorphisms.

The result connects descriptive regularity with the strong summation used in non-Archimedean calculus: on this support class, strong summation is a theorem forced by measurability rather than an extra axiom. Independent proof review, priority checking, and a formalization of the slice argument are the next steps needed to turn this manuscript into an established research contribution.

239 Where Part XVII meets Parts V, XIV and XV [merge]

This section was written for this report. It connects source 23's non-Polishability with Part V's refinement theorem, proves the two statements source 23 asserts without proof (that Baire measurable slices suffice in theorem 228.2, and that the Euler derivations have coefficientwise flows), and records what source 23's theorems say about Part XV and about the repository's Lean files. The proofs use only results proved in source 23 and in Parts V and XV; they have not been independently reviewed. Throughout, $\Gamma, \Lambda \leq \mathbb{R}$ are countable, and $\mathcal{L}_{\mathbb{R}}^{\Gamma}$ carries the field order in which $f > 0$ when its coefficient at $v(f)$ is positive (t positive infinitesimal, as in Part V and the surreal map (235.1)).

Proposition 239.1 (No admissible Polish topology; a second route). *Let $\Gamma \neq \{0\}$.*

- (i) *The ordered field $\mathcal{L}_{\mathbb{R}}^{\Gamma}$ has no admissible Polish topology in the sense of Part V's definition 60.1, whatever its Borel sets.*
- (ii) *Consequently no Polish group topology on $(\mathcal{L}_{\mathbb{R}}^{\Gamma}, +)$ has exactly the coefficient Borel sets; this is proposition 229.1.*

Proof. (i) Suppose τ is admissible. By Part V's theorem 61.4, τ refines the order topology. Fix $a > 0$ in Γ . For real r the order intervals $I_r = (r - t^a, r + t^a)$ are nonempty and open, hence τ -open, and for $r < r'$ they are disjoint, because $(r' - t^a) - (r + t^a) = (r' - r) - 2t^a$ has positive coefficient $r' - r$ at its least exponent 0. Uncountably many disjoint nonempty open sets contradict separability.

(ii) Let τ be a Polish group topology whose Borel sets are the coefficient Borel sets. The positive cone is the union over $\alpha \in \Gamma$ of the sets $\{f : f_\alpha > 0, f_\gamma = 0 (\gamma < \alpha)\}$, a countable union of coefficient-Borel sets, so it is τ -Borel and has the Baire property. Each $L_a(x) = ax$ is coefficient-Borel by [proposition 226.4](#), hence τ -Borel. So τ is admissible, contrary to (i). $\square$

For $\Gamma = \mathbb{Q}$, part (i) is Part V's remark before [section 66](#), whose argument it repeats. The proof of part (ii) uses the order and the multiplication, which source 23's proof of [proposition 229.1](#) does not need; the two are different routes to the same statement.

Proposition 239.2 (Baire measurable slices suffice). *Let $T : \mathcal{L}_\mathbb{R}^\Gamma \rightarrow \mathcal{L}_\mathbb{R}^\Lambda$ be additive, and suppose that for every left-finite $S \subseteq \Gamma$ and every $\lambda \in \Lambda$ the additive map $\mathbb{R}^S \cong \text{Sl}_S \rightarrow \mathbb{R}$, $f \mapsto [t^\lambda]T(f)$, is Baire measurable (preimages of open sets have the Baire property in $\mathbb{R}^S$). Then T is Borel, and so all conclusions of [theorem 228.2](#) hold.*

Proof. In the proof of [lemma 227.1](#) the preimages are used only through the Baire property: one of the countably many preimages covering $\mathbb{R}^S$ is nonmeager, and for a nonmeager set with the Baire property $A - A$ contains a neighbourhood of zero (Part V's [lemma 61.1](#)). So the lemma holds for Baire measurable ℓ , and [lemma 227.2](#), which uses only the lemma's conclusion and the Baire category theorem in $\mathbb{R}^S$, holds for T . Steps 1 and 2 of the proof of [theorem 228.2](#) use only these two lemmas: they show that the matrix $c_{\lambda\gamma} = [t^\lambda]T(t^\gamma)$ satisfies (M1) and (M2) and that, for each f , with $S = \text{supp } f$, $[t^\lambda]T(f) = \sum_{\gamma \in S} c_{\lambda\gamma} f_\gamma$, the sum having finitely many nonzero terms. So T is the map defined by a cutoff-admissible matrix, which is Borel by the implication (iii) $\Rightarrow$ (i) of [theorem 228.2](#). $\square$

This is the observation stated in the commentary on [question 237.8](#); the question of smaller families of test slices stays open.

Proposition 239.3 (Coefficientwise flows of the Euler derivations). *Let (b_i) be a rational basis as in [section 231](#), E_i a coordinate Euler derivation ([231.1](#)), and for $s \in \mathbb{R}$ put*

$$\Psi_s \left(\sum_{\gamma} f_{\gamma} t^{\gamma} \right) = \sum_{\gamma} e^{s \ell_i(\gamma)} f_{\gamma} t^{\gamma}.$$

- (i) *Each Ψ_s is an $\mathbb{R}$ -algebra automorphism of $\mathcal{L}_\mathbb{R}^\Gamma$ that preserves the valuation, the order and every support, with $\Psi_0 = \text{id}$ and $\Psi_s \Psi_r = \Psi_{s+r}$.*
- (ii) *The map $(s, f) \mapsto \Psi_s(f)$ is continuous from $\mathbb{R} \times (\mathcal{L}_\mathbb{R}^\Gamma, \tau_c)$ to $(\mathcal{L}_\mathbb{R}^\Gamma, \tau_c)$; in particular each Ψ_s is Borel.*
- (iii) *For fixed f and γ , the coefficient $[t^\gamma]\Psi_s(f)$ is an entire function of s with derivative $[t^\gamma]E_i(f)$ at $s = 0$, and it is the sum of the coefficientwise series $\sum_n s^n [t^\gamma]E_i^n(f)/n!$.*
- (iv) *If $s \neq 0$ and $\ell_i(\gamma) \neq 0$, the partial sums of $\sum_n s^n E_i^n(t^\gamma)/n!$ do not converge in the valuation topology.*

Proof. The map $\chi_s(\gamma) = e^{s \ell_i(\gamma)}$ is a homomorphism from Γ to the positive reals. Multiplying coefficients by $\chi_s(\gamma)$ keeps every support, so Ψ_s maps $\mathcal{L}_\mathbb{R}^\Gamma$ to itself; it is additive and real-linear, and Ψ_{-s} is its inverse. For a product,

$$[t^\lambda]\Psi_s(fg) = \chi_s(\lambda) \sum_{\gamma+\eta=\lambda} f_{\gamma} g_{\eta} = \sum_{\gamma+\eta=\lambda} \chi_s(\gamma) f_{\gamma} \chi_s(\eta) g_{\eta} = [t^\lambda](\Psi_s(f)\Psi_s(g)),$$

the sums being finite by [proposition 226.2](#); and $\Psi_s(1) = 1$. The least exponent and the sign of its coefficient are unchanged, so valuation and order are preserved, and $\chi_s \chi_r = \chi_{s+r}$ gives

the group law. This proves (i). Each output coefficient $\chi_s(\gamma)f_\gamma$ is a continuous function of (s, f_γ) , which is (ii) for the product topology. Since $[t^\gamma]E_i^n(f) = \ell_i(\gamma)^n f_\gamma$, (iii) is the exponential series of $e^{s\ell_i(\gamma)}$. For (iv), the difference between $\Psi_s(t^\gamma)$ and the N -th partial sum is $(e^x - \sum_{n \leq N} x^n/n!)t^\gamma$ with $x = s\ell_i(\gamma) \neq 0$, a nonzero multiple of t^γ by Taylor's formula with remainder, so its valuation is γ for every N . $\square$

For $\Gamma = \mathbb{Q}$ and $b = 1$ this is the flow $\sum f_\gamma t^\gamma \mapsto \sum e^{s\gamma} f_\gamma t^\gamma$ that source 23 asserts after (234.4). It is not of the form of [theorem 234.2](#): $\Psi_s - \text{id}$ has gain 0, and the valuation-formal exponential of sE_i does not converge, as (iv) shows. So positive gain is not necessary for a Borel flow whose coefficients are analytic in s ; [question 237.5](#) asks what replaces it in general.

[write] (Batch 96.) Part XX's [theorem 275.2](#) answers this for divisible Γ ; its [example 275.5](#) gives the same flows for every additive $\lambda : \Gamma \rightarrow \mathbb{R}$, and its [corollary 277.3](#) integrates every derivation of gain zero in finite rational rank. In rational rank one Part XVIII's [theorem 248.3](#) gives the flow of every derivation of gain zero.

Remark 239.4 (Borel embeddings fix $\mathbb{R}$; Part XV's questions). Every Borel unital field embedding $\sigma : \mathcal{L}_\mathbb{R}^\Gamma \rightarrow \mathcal{L}_\mathbb{R}^\Lambda$ is real-linear by [theorem 228.2](#), so $\sigma(r) = r\sigma(1) = r$ for real r : it fixes the coefficient field, and it is valuation-continuous and given by a cutoff-admissible matrix. For a countable $\mathbb{Q}$ -subspace Γ and the coefficient field $\mathbb{R}$, this disposes, for Borel embeddings, of the clause of Part XV's [question 208.5](#) that asks for embeddings not required to fix k ; what remains for them is the coefficient-fixing classification, which the note after that question says extends to larger k without a written proof. Embeddings that are not Borel are untouched. For Part XV's [question 208.1](#): if $u : \Gamma \rightarrow 1 + \mathfrak{m}$ is a homomorphism with $\inf_\gamma v(u(\gamma) - 1) \geq \varepsilon > 0$, then $\Phi_u(f) = \sum u(\gamma)f_\gamma t^\gamma$ on $\mathcal{L}_\mathbb{R}^\Gamma$ is well defined (only finitely many terms reach below any bound), satisfies $v((\Phi_u - \text{id})f) \geq v(f) + \varepsilon$, is multiplicative on monomials and hence, being valuation-continuous, everywhere, is onto by the Neumann series $\sum_n (\text{id} - \Phi_u)^n$, and is therefore a Borel $\mathbb{R}$ -algebra automorphism uniformly tangent to the identity; by [theorem 234.2](#) it is $\exp D$ for a unique Borel derivation D of gain at least ε . This adds a logarithm to the sufficient condition of that question and says nothing about necessity.

Remark 239.5 (The rank-one image for $\Gamma = \mathbb{Z}$ is formalized). A left-finite subset of $\mathbb{Z}$ is a subset bounded below, so $\mathcal{L}_\mathbb{R}^\mathbb{Z}$ is the Laurent field $\mathbb{R}((t))$, which is Mathlib's `LaurentSeries = HahnSeries  $\mathbb{Z}$   $\mathbb{R}$` . For $D = d/dt = t^{-1}E$, [corollary 231.3](#) says that g has a primitive exactly when $\text{ct}(gt)$, the coefficient of t^{-1} in g , vanishes. In the namespace `Surreal.LaurentResidue` that is the Lean theorem `exists_derivative_eq_iff`, with `range_derivative_eq_ker_residue` (the range of the derivative is the kernel of the residue), `residue_derivative`, `residue_surjective` and the termwise primitive, in `Algebra/SurrealNumbers/Surreal/Algebra/LaurentResidueChange.lean`, and instantiated at the surcomplex numbers. The first two are proved over every field of characteristic zero (the file's `CharZero` section); the last three over every field; `docs/FORMALIZATION.md` maps them to `b:formalprimitive` of the surcomplex analysis report. They formalize this algebraic clause for one derivation. Borelness, the classification of Borel derivations and the other values of Γ are not formalized, and no label of this part is mapped.

Part XVIII

Borel conjugacy and flows of left-finite derivations

[write] This part is source 24, manuscript 01 of batch 96, *A Divisibility Threshold for Borel Conjugacy: Explicit normal forms, measurable classification, and analytic flows on left-finite series fields* (25 pages, A4, dated 4 October 2026; “Research manuscript prepared with ChatGPT”; its title page calls it “a research direction connecting Elliot Glazer’s work with the surreal-series program in ProveIt”). It arrived in commit 62846e17a, as archive Borel_Conjugacy_Divisibility_Threshold, and was placed in 54ece48ab; it is pinned to 715a716a3, the parent of the arrival commit and 39 commits before the placement, when this report already had Parts I–XVII. Its file prefix in this report is 24-borel-conjugacy-. All unmarked text in this part is source 24’s; its fifteen sections are Sections 240–254 (its Section k is Section 239 + k , with statement and equation numbers otherwise unchanged), its ten questions are Questions 253.1–253.10, and its two appendices are appendices AL and AM. Section 255 [merge] was written for this report.

What this part claims, and its status

Let $\mathbb{Z} \leq \Gamma \leq \mathbb{Q}$ (every countable group of rational rank one has this form after a scaling of exponents), and let $\mathcal{L}_{\mathbb{R}}^{\Gamma}$ be Part XVII’s field of real-coefficient series with left-finite support, with the Borel structure of its coefficients. Every Borel derivation is $D_a = aE$ with $E = t \, d/dt$, and the Borel ordered-field automorphisms are exactly the substitutions $t \mapsto At^r(1+u)$ with $A > 0$, $r\Gamma = \Gamma$ and $v(u) > 0$; when Γ is 2-divisible every Borel field automorphism is ordered (theorem 242.4). A Borel tangent coordinate puts every nonzero D_a into an explicit normal form determined by its gain $v(a)$, its leading coefficient and its residue $\text{ct}(1/a)$ (theorem 244.1). Complete invariants of conjugacy are given in each gain stratum, and a conjugating coordinate can be chosen Borelly on the relation (theorems 245.1 and 245.2). Conjugacy is smooth exactly when at most one prime is infinitely divisible in Γ , and all nonsmoothness lies in gain zero (theorem 246.4). The nonzero centralizers are determined (theorem 247.2). D_a generates a coefficientwise real-analytic flow exactly when $a = 0$ or $v(a) \geq 0$, and then uniquely (theorem 248.3); a one-parameter group of these automorphisms whose image of t has coefficients Borel in time is such a flow (theorem 248.8).

Section 255 [merge] characterizes the time-one maps of these flows, compares the part with Parts XV and XX, records a reviewer’s lemma on effective presentations, and reports the check of an external citation.

Unrefereed, AI-assisted, not formalized; priority is not established. The finite jet checks test finite identities only.

Questions answered. Source 24 answers, in rational rank one, for real coefficients and Borel ordered automorphisms (all Borel automorphisms when Γ is 2-divisible), the two questions of Part XVII it calls Questions 13.4 and 13.5 of an earlier manuscript: question 237.4 (complete invariants, the smoothness threshold and Borel witnesses) and question 237.5 (the exact flow criterion, and Borel dependence of the image of t on time as the regularity that selects it). Dated notes after both questions record this; the earlier note’s “otherwise open” at the second is re-scoped there. Its theorem 242.4 also writes out, for $k = \mathbb{R}$, $\Gamma = \mathbb{Q}$ and automorphisms,

the coefficient-fixing case of Part XV's [question 208.5](#) that the note after that question asserts without proof ([remark 255.3](#) [merge]; dated note there). Source 24 says that it settles no remaining published question of Glazer's.

What was already here, and what is new. Source 24 read the root and surreal READMEs at its pin, not this report or any Lean file, and it describes Part XVII as an unpublished draft. Its [lemmas 241.2](#) and [242.1](#) and [proposition 241.4](#) re-prove for $\mathbb{Z} \leq \Gamma \leq \mathbb{Q}$ Part XVII's [propositions 226.2](#) and [226.4](#) and [lemma 227.1](#); [theorem 242.2](#) and the integration obstruction after ([243.3](#)) are the rank-one cases of [theorem 231.1](#) and [corollary 231.3](#), whose instance $\Gamma = \mathbb{Z}$ is formalized in Lean, as are two clauses for the Euler derivation of Hahn series; [proposition 247.1](#) and the logarithm in [lemma 248.6](#) re-prove the two halves of [theorem 234.2](#); the zero-gain flow of [theorem 248.3](#) extends [proposition 239.3](#) [merge] to every gain-zero derivation; the flow $t \mapsto t/(1-st)$ and the surreal map are Part XVII's. Each is flagged in a note. New to the report are the residue covariance, the explicit Borel normal forms, the complete invariants with Borel witness selection, the prime-height threshold, the centralizers, the analytic-flow criterion, automatic analyticity in time and its corollary for $\mathbb{R}^n$ -actions. Parts XX and XXI, written from sources of the same batch, overlap this part in places that the notes name: source 27 proves [theorem 242.4](#) and the zero-gain classification again, and source 26's flow theorems contain those of [section 248](#) at $\Gamma = \mathbb{Q}$ under incomparable hypotheses ([remark 255.2](#) [merge]).

Labels and notation. Every label of source 24 takes the prefix `pma:bct:.`. Its unlabelled numbered statements are labelled `pma:bct:nk.j` by their source numbers $k.j$ (the ten questions are `pma:bct:n14.1`–`pma:bct:n14.10`); labels were also added to its conclusion and its two appendices, to `pma:part:bct`, and to the statements of [section 255](#). The eighth block of the table in [section 1.3](#) lists the symbols. In short: source 24's L_Γ is printed $\mathcal{L}_\Gamma^\Gamma$, as in [Part XVII](#) (tempting false readings: Part XIV's L_Γ is this field in the infinite variable $T = t^{-1}$, Part II's K_Γ is the full Hahn field and Part XVII's K_Γ the surreal image). Its set $P_\infty(\Gamma)$ of infinitely divisible primes is printed $\omega_\infty(\Gamma)$, because Part XIV's $P_\infty = P_{\mathbb{Z}[1/p]}$ is a semiring ([corollary 189.2](#)); its bookkeeping symbol ℓ with $E\ell = 1$ is printed l , because Part XVII's ℓ and ℓ_i are the rational coordinate maps of Γ ; its slice F_S is printed Sl_S , as in [Part XVII](#); its macros `\m` and `\norma` are the report's `\mm` and `\bctnf` (same symbols). *Direction of conjugacy:* here S_h carries D_a to D_b when $S_h^{-1}D_aS_h = D_b$, with $S_hS_k = S_{k \circ h}$ ([definition 243.1](#)); Part XXI lets automorphisms act by $g \cdot D = gDg^{-1}$, the opposite orientation (note after [definition 243.1](#)). The letters $E, D_a, J, H, K, A, M, \alpha, c, \rho, \theta, \kappa, h_p, v_p, S_h, \Phi_s, T_s, \Psi, \tau$ and λ are local to this part; ρ is not Part III's cone closure rank, θ not Part XVI's ordinal, J not Part III's map J_α , and T_s not Part XIV's variable T . The word “threshold” refers to divisibility, not to Part XIV's exponent-group threshold, and “smooth” has its descriptive-set-theoretic meaning. No normalization changed.

Files. Source 24's program and recorded run are shipped as `code/24-borel-conjugacy-verify_jets.py` and `data/24-borel-conjugacy-verification_report.json`; the program writes its record to `verification_report.json` in the working directory unless `-output` is given (rerun note in [section 251](#)). Not shipped: the manuscript (printed here), its PDF, its delivery README and its checksum manifest (five of five files verified at placement). The Hilbert's-tenth programme reviewed the archive in full in commit `8dffc7af0` (`Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/review_new_borel_62846e17a.md`), found no correction, and added the lemma printed as [remark 255.4](#) [merge].

240 The intersection and the precise questions settled

240.1 Why this topic

Glazer's *A Topological Tennenbaum Theorem* investigates how Borel regularity constrains arithmetic operations on uncountable Polish spaces, and proves that neither addition nor multiplication can be continuous in the models under consideration [1]. The connection used here is the interaction between algebraic structure and descriptive regularity, not a biographical inference from a professional profile.

ProveIt's surreal-number project develops ordered fields, normal-form series, strong summation, and differential constructions, while distinguishing its formalized library from unrefereed research reports [153]. These materials suggest a concrete meeting point: classify measurable differential structures on a set-sized series subfield of the surreals. A previous manuscript in the user's research collection, *Borel Regularity Forces Summability*, asks in Questions 13.4 and 13.5 for conjugacy invariants of Borel derivations and for flows outside the strictly positive-gain case [152]. The present paper answers precise rank-one, order-preserving versions of those questions. It does not claim to settle the remaining published questions in Glazer's paper.

There is a particularly useful tension. Every individual derivation can be put into a short normal form by a Borelly chosen change of variables. Yet a Borel assignment of complete invariants for the *full* conjugacy relation can fail. The failure has an exact cause: the scaling symmetries of the exponent group. Two infinitely divisible primes are enough, and one is not.

240.2 What is classical and what is developed here

Levi–Civita analysis provides the standard background for left-finite rational-exponent series [31]. Strongly additive automorphisms of generalized series fields have an established structural theory [133]. Formal one-variable normal forms and flow embeddings are classical; the power-log extension of Mardešić, Resman, Rolin, and Županović treats closely related normalization and embedding mechanisms [151].

The contribution developed here is the *joint* statement consisting of an explicit Borel normalization on the entire left-finite rank-one field, exact conjugacy invariants for every intermediate group $\mathbb{Z} \subseteq \Gamma \subseteq \mathbb{Q}$, a sharp prime-height criterion for smoothness, and a Borel selection of conjugating witnesses even in the nonsmooth cases. The analytic-flow theorem is formulated for automorphisms of the whole field, with no common support assumed in advance: a real-analytic support lemma supplies it. A further group-theoretic argument shows that mere Borel dependence of the image of t on time already forces analyticity on every field element.

The elementary coefficient-slice argument needed to identify Borel derivations is included, so no theorem of the earlier unrefereed manuscript is required as an unproved input. A targeted literature check found the stated antecedents, but did not certify priority for the exact combined results. Throughout, “proved here” means that the proof is supplied, not that no earlier proof exists.

240.3 A preview of the threshold

A prime p is *infinitely divisible* in Γ when $p^{-n} \in \Gamma$ for every $n \geq 0$. Let $\omega_\infty(\Gamma)$ be the set of such primes. The central conclusion is

$$\text{Borel ordered-field conjugacy of Borel derivations on } \mathcal{L}_\mathbb{R}^\Gamma \text{ is smooth} \iff |\omega_\infty(\Gamma)| \leq 1.$$

Here *smooth* is a descriptive-set-theoretic term: a Borel map into a standard Borel space assigns equal values exactly to conjugate objects. It has nothing to do with differentiability of a function of t .

For $\Gamma = \mathbb{Q}$, the positive-gain part is classified by a sign and one real number. In contrast, the zero-gain part already contains

$$cE \sim dE \iff d/c \in \mathbb{Q}_{>0},$$

which is nonsmooth. For $\Gamma = \mathbb{Z}[1/2]$, the analogous scaling group is $2^\mathbb{Z}$, and a real fundamental interval gives smooth classification. The threshold distinguishes these situations exactly.

[write] (Batch 96.) The “previous manuscript in the user’s research collection” is source 23, printed in this report as [Part XVII](#); it was already there at source 24’s pin 715a716a3, which source 24 did not know (it read only the root and surreal READMEs). Its “Questions 13.4 and 13.5” are [questions 237.4](#) and [237.5](#); the dated notes after them record what this part answers. Glazer’s discontinuity theorem cited above is his Theorem 2, for Borel operations on models of $\mathbb{Q} + I\Delta_0 + \text{Exp}$ ([section 1.1](#)). The “threshold” of this part is a divisibility condition on Γ for smoothness of conjugacy; it is unrelated to Part XIV’s exponent-group threshold ([theorem 183.1](#)). “Smooth” is used in the sense of descriptive set theory, as in Part XIII’s [corollary 175.3](#), whose relation is not even Borel; the relation here is Borel ([theorem 245.1](#)). The field $\mathcal{L}_\mathbb{R}^\Gamma$ is [Part XVII](#)’s field for the groups $\mathbb{Z} \leq \Gamma \leq \mathbb{Q}$; every countable group of rational rank one is of this form after a scaling of exponents, which is an order-preserving, coefficient-Borel isomorphism of the fields.

241 Left-finite fields and support calculus

Fix an additive subgroup $\mathbb{Z} \subseteq \Gamma \subseteq \mathbb{Q}$. The assumption $1 \in \Gamma$ fixes a coordinate t . Every nonzero subgroup of $\mathbb{Q}$ can be put in this form by rescaling its exponents.

Definition 241.1. A subset $S \subseteq \Gamma$ is *left-finite* if $S \cap (-\infty, B]$ is finite for every real B . Define

$$\mathcal{L}_\mathbb{R}^\Gamma = \left\{ \sum_{q \in \Gamma} f_q t^q : f_q \in \mathbb{R}, \text{ supp } f = \{q : f_q \neq 0\} \text{ is left-finite} \right\}.$$

For $f \neq 0$, set $v(f) = \min \text{supp } f$ and $\text{lc}(f) = f_{v(f)}$. Set $v(0) = +\infty$, $\text{ct}(f) = f_0$, and $\mathfrak{m} = \{f : v(f) > 0\}$.

An infinite left-finite set has an increasing enumeration tending to $+\infty$. It is well ordered, but well ordering alone is insufficient: $\{1 - 1/n : n \geq 2\}$ is well ordered and is not left-finite. Thus $\mathcal{L}_\mathbb{R}^\mathbb{Q}$ is the Levi–Civita field, not the full Hahn field with arbitrary well-ordered rational supports. It is also larger than the union of fixed-denominator Laurent fields. For distinct primes p_n , a series supported on $n + 1/p_n$, $n \geq 1$, has left-finite support and no common denominator.

Lemma 241.2 (Support calculus). *The usual addition and convolution product make $\mathcal{L}_{\mathbb{R}}^{\Gamma}$ an ordered field, with positivity determined by the leading coefficient. It is complete for the valuation topology, and finite-support series are valuation-dense. If $S \subseteq \Gamma_{>0}$ is left-finite, the additive monoid generated by S is left-finite. Every element has only finitely many ordered representations as a sum of a fixed or varying number of elements of S , apart from the empty representation of zero.*

Proof. If nonempty left-finite sets S, T have minima s_0, t_0 , a pair with $s + t \leq B$ belongs to the finite product

$$(S \cap (-\infty, B - t_0]) \times (T \cap (-\infty, B - s_0]).$$

This proves finiteness of coefficient convolutions, left-finiteness of products, and $v(fg) = v(f) + v(g)$. If $S \subseteq \Gamma_{>0}$ and $\delta = \min S$, a sum at most B uses at most $\lfloor B/\delta \rfloor$ terms, each from the finite set $S \cap (0, B]$. This proves the monoid assertion, including finiteness of ordered representations.

Write $f = ct^{\alpha}(1+u)$ with $c \neq 0$ and $u \in \mathfrak{m}$. The inverse is $c^{-1}t^{-\alpha} \sum_{n \geq 0} (-u)^n$. The preceding support argument makes this a valid series, and the finite geometric identity proves the inverse law. The leading-coefficient order is compatible with both field operations.

For a valuation-Cauchy sequence, coefficients below each fixed bound eventually stabilize simultaneously. The stabilized coefficients define a series whose support below that bound equals the support of a single term of the sequence, so it is finite. This proves completeness. Finite truncations below integer bounds converge in valuation to the original series. $\square$

For $u \in \mathfrak{m}$ and $b \in \mathbb{Q}$, the series

$$(1+u)^b = \sum_{n \geq 0} \binom{b}{n} u^n, \quad \log(1+u) = \sum_{n \geq 1} \frac{(-1)^{n+1}}{n} u^n, \quad \exp u = \sum_{n \geq 0} \frac{u^n}{n!}$$

are therefore defined in $\mathcal{L}_{\mathbb{R}}^{\Gamma}$. Their formal identities follow by checking finitely many terms below every bound. This use of $\log(1+u)$ does not adjoin $\log t$ to the field.

Lemma 241.3 (Unit estimates). *For $H, K \in 1 + \mathfrak{m}$ and $b \in \mathbb{Q}$, one has*

$$v(\log H - \log K) \geq v(H - K), \quad v(H^b - K^b) \geq v(H - K).$$

The second difference is zero when $b = 0$. Formal differentiation, the product rule, and the chain rule hold for these unit operations.

Proof. Write $H/K = 1 + w$, with $v(w) = v(H - K) > 0$. The differences factor as $\log(1+w)$ and $K^b((1+w)^b - 1)$, respectively. Every nonzero term has valuation at least $v(w)$. Differentiation follows by finite truncation of the formal power-series identities; the supports remain in the monoids of Lemma 241.2. $\square$

241.1 Two topologies, and a Borel coding

The coefficient topology τ_c is inherited from the countable product $\mathbb{R}^{\Gamma}$. The valuation topology τ_v has zero-neighborhoods $\{f : v(f) > B\}$. They must not be confused. On constants, τ_c is the usual real topology, while τ_v is discrete. The monomials t^{-n} tend to zero coefficientwise but not in valuation.

Proposition 241.4. $\mathcal{L}_{\mathbb{R}}^{\Gamma}$ is a standard Borel space in its coefficient coding. Its field operations, v on nonzero elements, leading coefficient, unit powers, unit logarithm, and unit exponential are Borel on their stated domains.

Proof. In $\mathbb{R}^\Gamma$, the left-finite condition is

$$\bigcap_{N \in \mathbb{N}} \bigcup_{\substack{F \subseteq \Gamma \cap (-\infty, N] \\ F \text{ finite}}} \bigcap_{q \in (\Gamma \cap (-\infty, N]) \setminus F} \{f : f_q = 0\}.$$

All unions are countable, so the domain is Borel. Convolution coefficients are limits of finite coordinate sums; for each input pair only finitely many summands contribute. The event $v(f) = q$ specifies that $f_q \neq 0$ and all coefficients below q vanish, so it is Borel. Normalize a nonzero series and apply its geometric inverse to obtain Borel inversion. The unit operations are coefficientwise limits of finite polynomial operations. The same reasoning shows that rational unit powers are jointly Borel when the rational exponent is a parameter in its countable discrete space. $\square$

[write] (Batch 96.) For $\mathbb{Z} \leq \Gamma \leq \mathbb{Q}$ this section re-proves Part XVII's [proposition 226.2](#) (with the order, and the left-finite monoid generated by a left-finite set of positive exponents), its topologies ([definition 226.3](#)) and [proposition 226.4](#); the displayed domain is (226.1). Those are in turn the real-coefficient forms of Part V's field layer (note at the beginning of [Part XVII](#)).

242 Borel derivations and all allowed coordinate changes

Throughout this paper, *Borel* means Borel for the coefficient coding. A derivation is initially only an additive map satisfying the Leibniz rule; linearity over $\mathbb{R}$ is a conclusion, not an assumption.

Lemma 242.1 (Polish coefficient slices). *Let S be countable. Every Borel additive map $\ell : \mathbb{R}^S \rightarrow \mathbb{R}$ is a continuous real-linear functional with finite coordinate support. Consequently, two Borel additive maps $T_1, T_2 : \mathcal{L}_{\mathbb{R}}^\Gamma \rightarrow \mathcal{L}_{\mathbb{R}}^\Gamma$ that agree on every monomial are equal.*

Proof. We recall the automatic-continuity argument; the broader subject is surveyed in [147]. A nonmeager Borel set A in a Polish group is comeager in some nonempty open set, and $A - A$ contains a neighborhood of zero: sufficiently small translates of that open set overlap, and the two corresponding comeager subsets must meet. Cover $\mathbb{R}$ by countably many intervals of arbitrarily small length. One inverse image under ℓ is nonmeager, so its difference set establishes continuity of ℓ at zero. Additivity yields continuity everywhere, and rational homogeneity plus real limits yields real linearity.

A basic product neighborhood on which $|\ell| < 1$ restricts only finitely many coordinates, say those in F . Every real multiple of a vector vanishing on F lies in that neighborhood, so ℓ vanishes on that vector. Thus ℓ depends on finitely many coordinates.

For the final assertion, fix a left-finite $S \subseteq \Gamma$. The series supported in S form a coefficient slice $\text{Sl}_S \simeq \mathbb{R}^S$. Compose each output coefficient of $T_1 - T_2$ with this inclusion. The resulting functional has finite coordinate support, and it vanishes on every coordinate vector by the monomial hypothesis. It is therefore zero. Every input lies in such a slice. $\square$

Theorem 242.2 (Derivation parameterization). *Every Borel derivation on $\mathcal{L}_{\mathbb{R}}^\Gamma$ has the unique form*

$$D_a = aE, \quad E\left(\sum_{q \in \Gamma} f_q t^q\right) = \sum_{q \in \Gamma} q f_q t^q, \quad a \in \mathcal{L}_{\mathbb{R}}^\Gamma.$$

Conversely, every D_a is a Borel derivation. If $a \neq 0$, its exact uniform valuation gain is $\alpha = v(a)$:

$$v(D_a f) \geq v(f) + \alpha,$$

with equality on every nonconstant monomial.

Proof. For a Borel derivation D , each coefficient of its restriction to $\mathbb{R}$ is a Borel additive map $\mathbb{R} \rightarrow \mathbb{R}$. It is real-linear. As $D(1) = 0$, all these maps vanish; hence D kills $\mathbb{R}$. Put $a = D(t)/t$.

For $q = m/n \in \Gamma$, $n > 0$, differentiate $(t^q)^n = t^m$. The integer power rule, including negative integer powers, gives

$$n(t^q)^{n-1}D(t^q) = mt^{m-1}D(t), \quad D(t^q) = qat^q.$$

The Euler operator E is coefficientwise continuous and a derivation, since $q + r$ is the exponent of a product monomial. Thus D_a is Borel and has the same monomial values as D . Lemma 242.1 proves equality. Uniqueness follows from $D_a(t)/t = a$. The valuation inequality and its equality case follow directly from the Euler formula. $\square$

We henceforth use $a \in \mathcal{L}_{\mathbb{R}}^{\Gamma}$ as the standard Borel parameter for D_a . This avoids any unsupported claim that an arbitrary space of functions has a preferred standard Borel coding. [write] (Batch 96.) Lemma 242.1 is Part XVII's lemma 227.1 (its slices Sl_S , printed so here for source 24's F_S) with corollary 228.3, and rests on Part V's lemmas 61.1 and 61.2. Theorem 242.2 is the rational-rank-one case of Part XVII's theorems 230.2 and 231.1 and the first sentence of corollary 231.3, with $b = 1$; source 24's proof needs only lemma 242.1, not Part XVII's generic-support lemma, because a derivation is pinned down by $D(t)$ in rank one.

Definition 242.3. Set

$$U_{\Gamma} = \{r \in \mathbb{Q}_{>0} : r\Gamma = \Gamma\}.$$

Let G_{Γ} be the group of Borel ordered-field automorphisms of $\mathcal{L}_{\mathbb{R}}^{\Gamma}$. An automorphism with coordinate $h = \Phi(t)$ is *tangent* if $h = t(1 + u)$ for $u \in \mathfrak{m}$.

Theorem 242.4 (Coordinate classification). *The elements of G_{Γ} are exactly the substitutions*

$$S_h f = f \circ h = \sum_{q \in \Gamma} f_q h^q, \quad h = At^r(1 + u), \quad A > 0, \quad r \in U_{\Gamma}, \quad u \in \mathfrak{m}, \quad (242.1)$$

where $h^q = A^q t^{rq}(1 + u)^q$. They fix $\mathbb{R}$ and satisfy

$$v(S_h f) = rv(f) \quad (f \neq 0).$$

The coordinate space, evaluation $(h, f) \mapsto S_h f$, composition, and inversion are Borel. In particular,

$$[t^{\lambda}]S_h f \text{ depends, on a fixed support } S, \text{ only on } S \cap (-\infty, \lambda/r]. \quad (242.2)$$

If Γ is 2-divisible, the order restriction is automatic: every Borel field automorphism belongs to G_{Γ} .

Proof. Let $\Phi \in G_{\Gamma}$. The slice argument on constants and $\Phi(1) = 1$ give $\Phi(x) = x$ for $x \in \mathbb{R}$. Since t is positive and smaller than every positive real, so is $h = \Phi(t)$. Thus $h = At^r(1 + u)$ with $A > 0$, $r > 0$, and $u \in \mathfrak{m}$. Positive n th roots are unique in an ordered field. Applying Φ to the rational power relations gives $\Phi(t^q) = h^q$ for every $q \in \Gamma$.

If f has valuation q and leading coefficient c , its absolute value lies between $(|c|/2)t^q$ and $2|c|t^q$. The same inequalities after applying Φ show that $v(\Phi f) = rq$. Surjectivity implies

$r\Gamma = \Gamma$. This scaling law implies valuation continuity; taking finite truncations of f gives (242.1).

Conversely, the family in (242.1) is left-finitely summable: below any bound only finitely many input exponents can contribute, and each individual image has left-finite support. Rational binomial laws show that S_h is an ordered-field embedding. To prove surjectivity, first consider $k = t(1 + w)$. If $w \neq 0$ and $\delta = v(w) > 0$, the operator $N = S_k - \text{id}$ satisfies $v(Nf) \geq v(f) + \delta$. Hence $\sum_{n \geq 0} (-N)^n$ is a two-sided linear inverse to S_k . The inverse of an algebra isomorphism is multiplicative. A general h factors as $h = M \circ k$, where $M(t) = At^r$ and $k = t(1 + u)^{1/r}$; the monomial substitution is invertible because $r\Gamma = \Gamma$.

Borelness of evaluation follows from coefficient limits and the countable partition according to r ; real powers A^q are continuous in $A > 0$. For tangent coordinates, the inverse just constructed is a coefficientwise limit of Borel expressions. General inversion follows from the factorization. Composition is evaluation. Formula (242.2) follows from the least exponent rq of h^q . Notice the convention $S_h S_k = S_{k \circ h}$.

Finally, if Γ is 2-divisible, every positive $ct^q(1 + u)$ has square root $\sqrt{c} t^{q/2}(1 + u)^{1/2}$ in $\mathcal{L}_{\mathbb{R}}^{\Gamma}$. Thus the positive elements are exactly the nonzero squares, a field-definable set preserved by every field automorphism. $\square$

Remark 242.5. In particular, no additional order hypothesis is needed for $\Gamma = \mathbb{Q}, \mathbb{Z}[1/2]$, or $\mathbb{Z}[1/6]$. For a general intermediate group, the present paper classifies the stated ordered automorphisms, not all algebraic automorphisms. For example, $t \mapsto -t$ is an unordered automorphism of $\mathbb{R}((t))$.

[write] (Batch 96.) That Borel automorphisms fix $\mathbb{R}$ is Part XVII's [remark 239.4](#) [merge]. [Theorem 242.4](#) is the real-coefficient, rank-one analogue of Part XV's [theorems 198.1](#) and [198.2](#) (coefficients $\mathbb{R}_{\text{alg}}$, all embeddings); [remark 255.3](#) [merge] compares them and records what the proof gives without Borelness. Source 27 proves the same theorem independently, through the value-group automorphism ([theorem 289.1](#)). The convention $S_h S_k = S_{k \circ h}$ fixes the direction of conjugacy below; Part XXI uses the opposite one (note after [definition 243.1](#)).

243 Residue, primitives, and the transformation law

For $a \neq 0$, define the three numerical parameters

$$\alpha = v(a), \quad c = \text{lc}(a), \quad \rho = \text{ct}(1/a). \quad (243.1)$$

The coefficient ρ is the formal residue of $dt/(ta(t))$. If $\alpha < 0$, then $v(1/a) > 0$, so $\rho = 0$. If $\alpha = 0$, then $\rho = 1/c$. Only the positive-gain stratum has an independent residue parameter.

Definition 243.1 (Direction of conjugacy). A coordinate h carries D_a to D_b if

$$D_a S_h = S_h D_b, \quad \text{equivalently} \quad D_a h = h(b(h)). \quad (243.2)$$

Thus $S_h^{-1} D_a S_h = D_b$. We write $a \sim_{\Gamma} b$ when such a coordinate exists. The equivalence of the two conditions follows from the chain rule on monomials and valuation-continuous extension to every series.

[write] (Batch 96.) *The direction of conjugacy in this report.* In this part the substitution S_h carries D_a to D_b when $S_h^{-1} D_a S_h = D_b$, and substitutions compose contravariantly, $S_h S_k = S_{k \circ h}$. Part XXI (source 27) uses the same substitutions ($\Theta_y = S_y$) but lets the group act by $g \cdot D = g D g^{-1}$ ([\(289.4\)](#)); so “ S_h carries D_a to D_b ” here is “ S_h sends D_b to D_a ” there. For

example, the coordinate $t^{c/d}$ carries dE to cE here ((245.2) with $c_a = d$, $c_b = c$), and Part XXI's [proposition 292.1](#) says that it sends cE to dE . The relation $\sim_\Gamma$ is symmetric, so its invariants do not depend on the convention; formulas for conjugating coordinates and witnesses do, and are inverted between the two parts.

Let

$$Jf = \sum_{q \neq 0} \frac{f_q}{q} t^q. \quad (243.3)$$

Then $EJf = f - \text{ct}(f)$ and $\ker E = \mathbb{R}$. In particular, $EH = f$ is solvable in $\mathcal{L}_\mathbb{R}^\Gamma$ exactly when $\text{ct}(f) = 0$, and Jf is the unique solution with zero constant coefficient. This is the elementary integration obstruction behind the normal forms.

[write] (Batch 96.) This is Part XVII's [corollary 231.3](#) for $a = 1$, with (231.6) giving J . Two clauses are in Lean for Hahn series $R((t^\Gamma))$ with an additive exponent map: $\text{ct}(Ef) = 0$ over every commutative ring (`Surreal.EulerDerivation.coeff_zero_euler`), and $\ker E$ consists of the constants when R has no zero divisors and the exponent map vanishes only at 0 (`euler_eq_zero_iff`), in `Algebra/SurrealNumbers/Surreal/HahnSeries/EulerDerivation.lean`, which `docs/FORMALIZATION.md` maps to the surcomplex differential-equations report (`diff:rs:lem:euler`); $\mathcal{L}_\mathbb{R}^\Gamma$ is an E -stable subfield of $\mathbb{R}((t^\Gamma))$, so they restrict to it. For $\Gamma = \mathbb{Z}$, writing $f = tg$, the solvability criterion is the Lean theorem `Surreal.LaurentResidue.exists_derivative_eq_iff`, proved over every field of characteristic zero ([remark 239.5](#)). The criterion for other Γ , the operator J and everything below are not formalized.

Theorem 243.2 (Covariance). *Suppose $h = At^r(1 + u)$ carries D_a to D_b , with $a, b \neq 0$. Then*

$$\alpha_b = \frac{\alpha_a}{r}, \quad c_b = rc_a A^{-\alpha_b}, \quad \rho_b = \frac{\rho_a}{r}. \quad (243.4)$$

Consequently, $\text{sgn}(\alpha)$ and $\text{sgn}(c)$ are conjugacy invariants. When $\alpha \neq 0$, so is ρ/α .

Proof. Since $EH/h = r + E \log(1 + u)$ has leading coefficient r and valuation zero, (243.2) gives

$$a \frac{Eh}{h} = b(h).$$

Comparison of leading exponents and coefficients gives $\alpha_a = r\alpha_b$ and $rc_a = c_b A^{\alpha_b}$.

Taking reciprocals in the appropriate form gives

$$\frac{1}{a} = \frac{Eh}{h} \frac{1}{b(h)}.$$

Write $1/b = \rho_b + EH_b$, where $H_b = J(1/b)$. The chain rule yields

$$\frac{1}{a} = \rho_b \frac{Eh}{h} + E(H_b(h)).$$

The constant coefficient of the second term is zero. The constant coefficient of Eh/h is r . Therefore $\rho_a = r\rho_b$, as required. $\square$

It is useful to keep a *bookkeeping* symbol l with $El = 1$ and to write

$$F_a = H_a + \rho l, \quad H_a = J(1/a).$$

Then $EF_a = 1/a$. Under a tangent coordinate $h = tH$, substitute $l \mapsto l + \log H$. All occurrences of l cancel in the equations used below. This is a concise notation for an identity in $\mathcal{L}_{\mathbb{R}}^{\Gamma}$; it does not enlarge the coordinate group by adding $\log t$.

[write] (Batch 96.) Source 24 writes this symbol ℓ ; it is printed l because Part XVII's ℓ and ℓ_i are the rational coordinate maps of Γ (section 231). The letter ℓ for a functional in lemma 242.1 is kept, as in Part XVII's lemma 227.1.

244 Explicit tangent normal forms and Borel normalizers

Theorem 244.1 (Tangent normalization). *Every nonzero D_a is carried by a tangent coordinate $h_a = tH$, $H \in 1 + \mathfrak{m}$, to $D_{a^{\text{nf}}}$, where*

$$a^{\text{nf}} = \begin{cases} ct^{\alpha}, & \alpha < 0, \\ c, & \alpha = 0, \\ \frac{ct^{\alpha}}{1 + c\rho t^{\alpha}}, & \alpha > 0. \end{cases} \quad (244.1)$$

The coordinate h_a can be chosen Borelly in a . Two derivations are conjugate by a tangent coordinate exactly when their corresponding parameters in (244.1) agree: (α, c) in negative gain, c in zero gain, and (α, c, ρ) in positive gain.

Proof. Necessity of the parameters follows from Theorem 243.2 with $r = A = 1$. We construct the coordinate explicitly in each case.

Negative gain. Here $\rho = 0$ and $H_a = J(1/a)$ has leading term $-t^{-\alpha}/(\alpha c)$. Put

$$K = -\alpha ct^{\alpha} H_a \in 1 + \mathfrak{m}, \quad h_a = tK^{-1/\alpha}. \quad (244.2)$$

This is a binomial power of a unit multiplied by t ; it does not require a new exponent outside Γ . It satisfies

$$-\frac{h_a^{-\alpha}}{\alpha c} = H_a.$$

Apply E to obtain $h_a^{-\alpha-1} E h_a / c = 1/a$, which is precisely $D_a h_a = h_a c h_a^{\alpha}$.

Zero gain. Now $c/a - 1 \in \mathfrak{m}$. Define

$$h_a = t \exp(J(c/a - 1)). \quad (244.3)$$

Then $E h_a / h_a = 1 + E J(c/a - 1) = c/a$, so $D_a h_a = c h_a$.

Positive gain. Define $H_a = J(1/a)$ and

$$K = -\alpha ct^{\alpha} H_a \in 1 + \mathfrak{m}.$$

We seek a unit $H \in 1 + \mathfrak{m}$ satisfying

$$H^{-\alpha} - \alpha c \rho t^{\alpha} \log H = K. \quad (244.4)$$

Set $H_0 = 1$ and recursively

$$H_{n+1} = \mathcal{T}(H_n), \quad \mathcal{T}(H) = (K + \alpha c \rho t^{\alpha} \log H)^{-1/\alpha}. \quad (244.5)$$

The quantity in parentheses is in $1 + \mathfrak{m}$. Lemma 241.3 implies

$$v(\mathcal{T}(H) - \mathcal{T}(G)) \geq v(H - G) + \alpha \quad (H, G \in 1 + \mathfrak{m}). \quad (244.6)$$

When $\rho = 0$ the left side is infinite, so the assertion remains valid. The successive differences have valuations tending to infinity. Completeness gives a limit $H \in 1 + \mathfrak{m}$, and the estimates make $\mathcal{T}$ continuous, so H satisfies (244.4). Two distinct fixed points would violate (244.6), proving uniqueness for this normalized equation.

With $h_a = tH$, equation (244.4) says

$$-\frac{h_a^{-\alpha}}{\alpha c} + \rho \log H = H_a. \quad (244.7)$$

Add the formal bookkeeping term ρl to both sides and differentiate, or differentiate (244.7) directly. The result is

$$\left(\frac{h_a^{-\alpha}}{c} + \rho \right) \frac{E h_a}{h_a} = \frac{1}{a}.$$

Since $1/a^{\text{nf}}(h_a) = h_a^{-\alpha}/c + \rho$, this is $D_a h_a = h_a a^{\text{nf}}(h_a)$.

Borelness and completeness. The parameters α, c, ρ , the operator J , and every operation in the formulas are Borel. Partitioning the parameter space according to the countably many values of α , every iterate in (244.5) is Borel; each coefficient of its limit is a pointwise limit of Borel functions. Thus $a \mapsto h_a$ is Borel. Finally, two derivations with the same parameters both normalize to the same derivation, and composing one normalizer with the inverse of the other gives a tangent conjugacy. $\square$

244.1 Why unbounded denominators cause no missing limit

The positive-gain construction does not assume that the support lies in a finitely generated rational grid. Let $S = \text{supp}(K - 1) \cup \{\alpha\}$, discarding zero coefficients. This is a left-finite positive set. Every correction in the iteration is supported in the additive monoid generated by S , which is left-finite by Lemma 241.2. Hence the construction cannot accumulate infinitely many exponents below a fixed bound, even when their denominators are unbounded. The same argument applies to the two explicit unit formulas in the other strata.

If $K \neq 1$ and $\delta = v(K - 1)$ in the positive case, then

$$v(H - H_n) \geq \delta + n\alpha.$$

Thus a finite output cutoff has a certified iteration bound. If $K = 1$, the chosen solution is simply $H = 1$.

244.2 A normalization choice is not yet a full orbit selector

For $\alpha > 0$, equation (244.7) chooses the integration constant to be zero. Other tangent conjugacies to the same normal derivation are obtained by time translations, as proved below. The Borel map $a \mapsto h_a$ selects a coordinate for each input, but *not* a canonical representative of every full orbit: the normal parameters still transform under At^r . This distinction is essential in the nonsmooth cases.

Remark 244.2 (Dependence on the coordinate group). The residue is an invariant of the log-free group specified in Theorem 242.4. It must not be advertised as an invariant under

arbitrary logarithmic transseries changes. The power-log theory in [151], in particular its discussion of residual terms and Example 6.3, illustrates that enlargement of a coordinate group can eliminate a classical residual parameter. Our unit logarithms do not make that enlargement.

[write] (Batch 96.) The placement left this citation unchecked; the write checked Example 6.3 in the arXiv version and records what it shows in [remark 255.5](#) [merge].

245 Full conjugacy and a Borel choice of conjugating coordinates

Fix once and for all an enumeration of $\Gamma_{>0}$. For $\beta > 0$ in Γ , let $\kappa(\beta)$ be the first member *in this enumeration* of

$$\{\beta/r : r \in U_\Gamma\}.$$

An orbit need not have a least member in the ordinary numerical order. The enumeration convention avoids that error. Since its domain is countable, κ is a Borel function and is a complete invariant for this exponent scaling relation.

Theorem 245.1 (Complete invariants). *Zero is a conjugacy class by itself. For nonzero a, b , different signs of $v(a)$ and $v(b)$ never mix. Within the three strata, full conjugacy is characterized as follows:*

$$\alpha_a, \alpha_b < 0 : a \sim_\Gamma b \iff (\kappa(-\alpha_a), \operatorname{sgn} c_a) = (\kappa(-\alpha_b), \operatorname{sgn} c_b), \quad (245.1)$$

$$\alpha_a = \alpha_b = 0 : a \sim_\Gamma b \iff c_b/c_a \in U_\Gamma, \quad (245.2)$$

$$\alpha_a, \alpha_b > 0 : a \sim_\Gamma b \iff (\kappa(\alpha_a), \operatorname{sgn} c_a, \rho_a/\alpha_a) = (\kappa(\alpha_b), \operatorname{sgn} c_b, \rho_b/\alpha_b). \quad (245.3)$$

In particular, $\sim_\Gamma$ is Borel, and its positive- and negative-gain restrictions are smooth.

Proof. The covariance laws prove necessity. For sufficiency, normalize a and b using Theorem 244.1. In the nonzero-gain cases, put

$$r = \alpha_a/\alpha_b \in U_\Gamma, \quad A = \left(\frac{rc_a}{c_b} \right)^{1/\alpha_b} > 0. \quad (245.4)$$

The real power is defined because c_a, c_b have the same sign. The normal forms are carried to one another by $M(t) = At^r$: substitution directly gives the leading-coefficient law, and, in positive gain, the denominator matches because $\rho_a = r\rho_b$. In negative gain both residues are zero.

In zero gain, set $r = c_b/c_a \in U_\Gamma$ and $A = 1$. Then $M(t) = t^r$ carries $c_a E$ to $c_b E$. Composing with the tangent normalizers proves sufficiency. The displayed conditions are Borel: the numerical parameters are Borel, U_Γ is countable, and equality in a countable product of discrete and real spaces is Borel. The tuples in the nonzero-gain cases give smoothness. $\square$

For an explicit representative in nonzero gain, put $k = \kappa(|\alpha|)$, $\varepsilon = \operatorname{sgn} c$, and, when $\alpha > 0$, $\theta = \rho/\alpha$. The representatives are

$$\varepsilon t^{-k} E \quad (\alpha < 0), \quad \frac{\varepsilon t^k}{1 + \varepsilon k \theta t^k} E \quad (\alpha > 0). \quad (245.5)$$

Indeed choose $r = |\alpha|/k$ and $A = (r|c|)^{1/(\operatorname{sgn}(\alpha)k)}$ in the monomial step.

Theorem 245.2 (Borel witness selection). *There is a Borel function on $\{(a, b) \in \mathcal{L}_{\mathbb{R}}^{\Gamma^2} : a \sim_{\Gamma} b\}$ assigning a coordinate $w(a, b)$ that carries D_a to D_b . This remains true when $\sim_{\Gamma}$ is nonsmooth.*

Proof. Set $w(0, 0) = t$. For a nonzero conjugate pair, use the normalizers h_a, h_b and the monomial M from the proof of Theorem 245.1, and define

$$w(a, b) = h_b^{-1} \circ M \circ h_a. \quad (245.6)$$

The first coordinate carries a to its normal form, the middle one carries that normal form to the normal form of b , and the last one returns to the b coordinate. This proves the conjugacy equation. All operations, including compositional inversion, are Borel by Theorem 242.4. $\square$

A witness is chosen only *after both conjugate inputs have been supplied*. This is not a Borel choice of one representative from every orbit. No transversal or orbit-invariant normalization follows from Theorem 245.2.

[write] (Batch 96.) When $\sim_{\Gamma}$ is not smooth (theorem 246.4) there is no Borel transversal at all. If T were a Borel set meeting every class in exactly one point, the map sending a to the point of T conjugate to it would have the Borel graph $\{(a, b) : b \in T, a \sim_{\Gamma} b\}$ (the relation is Borel by theorem 245.1), hence would be Borel (a map between standard Borel spaces with Borel graph is Borel [8]), and it would be a Borel complete invariant. Part XXI's theorem 291.2 gives, for ordered pairs of noncommuting derivations, a Borel cross-section with unique witnesses; the obstruction here concerns single derivations.

246 The prime-height theorem and the exact smoothness threshold

Definition 246.1. For a prime p , define its height in Γ by

$$h_p(\Gamma) = \sup\{n \in \mathbb{N} : p^{-n} \in \Gamma\} \in \mathbb{N} \cup \{\infty\}.$$

Let $\omega_{\infty}(\Gamma) = \{p : h_p(\Gamma) = \infty\}$. Ordinary prime-adic valuations of nonzero rational numbers are denoted v_p in this section; they are distinct from the series valuation v .

Lemma 246.2 (Denominator characterization). *For $x \in \mathbb{Q}$, membership in Γ is determined by the heights:*

$$x \in \Gamma \iff v_p(x) \geq -h_p(\Gamma) \text{ for every prime of finite height,}$$

with zero included automatically. Consequently,

$$U_{\Gamma} = \left\{ \prod_{p \in F} p^{n_p} : F \subseteq \omega_{\infty}(\Gamma) \text{ finite, } n_p \in \mathbb{Z} \right\}. \quad (246.1)$$

Proof. If $m/n \in \Gamma$ is reduced and $n > 0$, Bézout's identity gives integers u, v with $um + vn = 1$, hence $1/n = u(m/n) + v \in \Gamma$. Every reciprocal prime power dividing n then belongs to Γ , which proves necessity of the bounds.

Conversely, suppose $n = \prod p_i^{e_i}$ has only allowed prime-power exponents. The elements $1/p_i^{e_i}$ belong to Γ . If a, b are coprime and $1/a, 1/b \in \Gamma$, choose $u, v \in \mathbb{Z}$ with $ub + va = 1$; then $u/a + v/b = 1/(ab) \in \Gamma$. Induction gives $1/n \in \Gamma$, and integer multiples give all rationals satisfying the bounds.

For a finite height h_p , the set of values $v_p(x)$ for nonzero $x \in \Gamma$ has minimum $-h_p$. Multiplication by r preserves Γ only if it preserves this minimum, forcing $v_p(r) = 0$. Conversely, if $v_p(r) = 0$ at every finite-height prime, multiplying or dividing by r preserves every denominator bound. Unique prime factorization now gives (246.1). $\square$

Lemma 246.3 (Dense translation obstruction). *If H is a countable dense additive subgroup of $\mathbb{R}$, the equivalence relation $x \equiv_H y$ defined by $y - x \in H$ is not smooth.*

Proof. Every H -invariant Borel set $A \subseteq \mathbb{R}$ is meager or comeager. Indeed, if A is nonmeager, the Baire property makes it comeager in a nonempty open interval I . Invariance makes it comeager in every $I + h$, $h \in H$. These countably many intervals cover $\mathbb{R}$ by density, so A is comeager in $\mathbb{R}$.

Suppose there were a Borel complete invariant map into a standard Borel space. Using a countable Borel separating family, it may be regarded as a Borel map $f : \mathbb{R} \rightarrow 2^{\mathbb{N}}$ with equality of values exactly on the H -orbits. For each coordinate, the set on which that bit is one is H -invariant, hence meager or comeager. Choose the comeager alternative for each bit. The countable intersection is comeager, and f is constant there. But every fiber is a single countable orbit and therefore meager. This contradicts the Baire category theorem. $\square$

Theorem 246.4 (Divisibility threshold). *For every $\mathbb{Z} \subseteq \Gamma \subseteq \mathbb{Q}$, the following are equivalent:*

- (i) *Conjugacy $\sim_{\Gamma}$ of Borel derivations under G_{Γ} is smooth.*
- (ii) *Its zero-gain restriction is smooth.*
- (iii) *At most one prime is infinitely divisible in Γ .*

If two or more primes are infinitely divisible, nonsmoothness already occurs on the one-parameter family $\{cE : c > 0\}$. The positive- and negative-gain strata remain smooth in every case.

Proof. Theorem 245.1 reduces the only possible obstruction to multiplication of $c \in \mathbb{R}^{\times}$ by U_{Γ} .

If $\omega_{\infty}(\Gamma)$ is empty, $U_{\Gamma} = \{1\}$ and c itself is a complete invariant. If $\omega_{\infty}(\Gamma) = \{p\}$, then $U_{\Gamma} = p^{\mathbb{Z}}$. A complete invariant is

$$\left(\operatorname{sgn} c, |c|p^{-\lfloor \log_p |c| \rfloor} \right) \in \{-1, 1\} \times [1, p). \quad (246.2)$$

This is Borel and gives equal values exactly when the ratio is an integral power of p . Combining it with the smooth invariants on the two other strata and a separate symbol for zero gives smoothness of the entire relation.

Now suppose distinct primes p, q are in $\omega_{\infty}(\Gamma)$. The ratio $\log p / \log q$ is irrational, since a rational relation would imply equality of distinct prime powers. The additive group generated by $\log p, \log q$ is dense. For completeness, pigeonholing the fractional parts of $j \log p / \log q$, $0 \leq j \leq N$, gives nonzero group elements of arbitrarily small absolute value; integer multiples of such elements approximate every real number. Thus $\log U_{\Gamma}$ is a countable dense subgroup of $\mathbb{R}$.

The Borel map $x \mapsto e^x E$ carries the orbit relation for $\log U_{\Gamma}$ exactly to conjugacy on positive Euler coefficients, by (245.2). Lemma 246.3 excludes smoothness there, and hence on the full parameter space. The smoothness assertions on the other strata were already proved in Theorem 245.1. $\square$

246.1 Concrete exponent groups

For a localization $\Gamma = \mathbb{Z}[P^{-1}]$, the prime-height lemma gives $\omega_{\infty}(\Gamma) = P$. The exponent invariant k can be chosen to be the positive integer obtained by removing all prime factors in P from a positive rational exponent. For arbitrary height groups the fixed enumeration definition of κ is preferable.

Exponent group	Scaling group U_Γ	Convenient k for nonzero gain	Full conjugacy
$\mathbb{Z}$	$\{1\}$	Positive integer	Smooth
$\mathbb{Z}[1/2]$	$2^{\mathbb{Z}}$	Positive odd integer	Smooth
$\mathbb{Z}[1/6]$	$2^{\mathbb{Z}}3^{\mathbb{Z}}$	Positive integer coprime to 6	Nonsmooth
$\mathbb{Q}$	$\mathbb{Q}_{>0}$	1	Nonsmooth

The density of Γ in $\mathbb{R}$ is not the criterion. For instance, $\Gamma = \mathbb{Z}[1/2]$ is dense and has smooth conjugacy. Another dense example with trivial U_Γ is the group generated by $1/p$ for all primes p : every prime has height one, and the positive reciprocals tend to zero, yet no prime is infinitely divisible. Thus its zero-gain coefficient is an exact real invariant.

Corollary 246.5 (Classical Levi–Civita field). *On $\mathcal{L}_{\mathbb{R}}^{\mathbb{Q}}$, under all coefficient-Borel field automorphisms, every nonzero derivation is conjugate to exactly one of the following parameter types:*

$$\varepsilon \frac{d}{dt} \quad (\alpha < 0), \quad cE \quad (\alpha = 0), \quad \frac{\varepsilon t}{1 + \varepsilon \theta t} E \quad (\alpha > 0),$$

where $\varepsilon \in \{-1, 1\}$, $\theta \in \mathbb{R}$, and in the middle family c is taken modulo multiplication by $\mathbb{Q}_{>0}$. In the first and third families the displayed sign and real parameter are genuine complete invariants. There is no Borel selection of complete invariants for the middle family.

Proof. Here $U_\Gamma = \mathbb{Q}_{>0}$, every positive rational exponent scales to 1, and order is automatic by 2-divisibility. Apply Theorems 245.1 and 246.4. $\square$

[write] (Batch 96.) Theorems 245.1, 245.2 and 246.4 and this corollary answer Part XVII’s question 237.4 in rational rank one for ordered coordinate changes (all of them when Γ is 2-divisible); see the note after that question. Part XXI (source 27) proves the zero-gain classification again (proposition 292.1, in the opposite direction of conjugacy) and strengthens the “if” direction of theorem 246.4: when $\log U_\Gamma$ is dense, no complete invariant is even Baire measurable on the positive Euler line, also for commuting pairs (theorem 292.2); by (246.1) density of $\log U_\Gamma$ means $|\omega_\infty(\Gamma)| \geq 2$. Smoothness is not decidability: for a suitable Γ with decidable membership, conjugacy of the Euler fields E and pE is co-c.e.-complete in the prime p (remark 255.4 [merge], a reviewer’s lemma).

247 Positive-gain exponentials and exact centralizers

The next result is a filtered exponential construction, included to fix the precise summation convention. It is not asserted as a new exponential principle; related formal-flow constructions occur in [151, 152].

Proposition 247.1 (Positive-gain time maps). *If $\alpha = v(a) > 0$, then for every $s \in \mathbb{R}$,*

$$\Phi_s = \exp(sD_a) = \sum_{n \geq 0} \frac{s^n}{n!} D_a^n \quad (247.1)$$

defines an element of G_Γ . These automorphisms form a one-parameter group and satisfy

$$v((\Phi_s - \text{id})f) \geq v(f) + \alpha.$$

For each fixed f and output exponent, its coefficient in $\Phi_s(f)$ is a polynomial in s . The map $(s, f) \mapsto \Phi_s(f)$ is jointly Borel.

Proof. The gain estimate gives $v(D_a^n f) \geq v(f) + n\alpha$. Below every output bound only finitely many n can contribute, so (247.1) is a valid left-finite sum, and each individual output coefficient is a polynomial in s . The iterated Leibniz identity

$$D_a^n(fg) = \sum_{j=0}^n \binom{n}{j} D_a^j(f) D_a^{n-j}(g)$$

proves multiplicativity after regrouping finite sums below each cutoff. The same finite coefficient argument and the binomial theorem give $\Phi_s \Phi_r = \Phi_{s+r}$ and inverse Φ_{-s} . The automorphism fixes constants, and its difference from the identity has positive gain, so it preserves the leading term and the order. Each finite operator sum is jointly Borel, and its coefficient limits establish the final assertion. $\square$

[write] (Batch 96.) For $\mathbb{Z} \leq \Gamma \leq \mathbb{Q}$ this is the exponential half of Part XVII's [theorem 234.2](#), with [corollary 234.3](#); the “earlier manuscript” cited above is Part XVII. Its gain-one instance $D = tE$ is Part XVII's flow ([234.4](#)).

Theorem 247.2 (Centralizers). *For $a \neq 0$, the centralizer of D_a in G_Γ is, up to the tangent normalizing coordinate, exactly*

$\alpha = v(a)$	Centralizing coordinates in normal form	Group
$\alpha < 0$	t	$\{1\}$
$\alpha = 0$	$At, A > 0$	$(\mathbb{R}_{>0}, \times)$
$\alpha > 0$	$\exp(sD_{a^{\text{nf}}})(t), s \in \mathbb{R}$	$(\mathbb{R}, +)$.

(247.2)

In positive gain, the centralizer in the original coordinate is precisely $\{\exp(sD_a) : s \in \mathbb{R}\}$.

Proof. A normalizer induces an isomorphism of centralizers, so first assume a is in the normal form ([244.1](#)). If $h = At^r(1+u)$ centralizes D_a , the covariance laws with $b = a$ show that $r = 1$ whenever $\alpha \neq 0$, and then $A^\alpha = 1$, so $A = 1$. In the zero-gain case, the leading-coefficient law $c = rc$ still gives $r = 1$, while A is unrestricted.

If $\alpha < 0$, the centralizing equation implies

$$E \left(-\frac{h^{-\alpha} - t^{-\alpha}}{\alpha c} \right) = 0.$$

The expression is a constant, but since $h = t(1+u)$ and $-\alpha > 0$, any nonzero value of the expression has strictly positive valuation. Therefore it is zero, and $h = t$.

If $\alpha = 0$, the equation is $cEh = ch$, hence $Eh = h$. Only exponent 1 can occur, and $h = At$ with $A > 0$. Every such coordinate centralizes cE .

If $\alpha > 0$, write $h = tH$. The centralizing equation is equivalent to

$$-\frac{t^{-\alpha}}{\alpha c}(H^{-\alpha} - 1) + \rho \log H = s \quad \text{for some } s \in \mathbb{R}, \quad (247.3)$$

because $\ker E = \mathbb{R}$. For a fixed s , this equation has a unique solution $H \in 1 + \mathfrak{m}$: rewrite it as

$$H^{-\alpha} - \alpha c \rho t^\alpha \log H = 1 - \alpha c s t^\alpha$$

and use the same contraction as in ([244.5](#)). It remains to identify the solution with the exponential time map.

By Proposition 247.1, $g_s = \exp(sD_a)(t)$ is a tangent centralizing coordinate. Its first possible correction is

$$g_s = t + sct^{1+\alpha} + \text{terms of valuation greater than } 1 + \alpha.$$

Substituting g_s/t into the left side of (247.3) shows that its constant coefficient is s . The whole expression is constant by centralization, so it equals s . Uniqueness in (247.3) identifies every centralizer element with one time map. Distinct s give distinct first correction coefficients; the group law is addition of times. Conjugating the exponential back preserves the operator power series, proving the assertion in the original coordinate. $\square$

In particular, a negative-gain derivation is rigid under the allowed automorphism group even though it has a simple primitive. The ordinary intuition of translating that primitive by a nonzero real time fails because such a translation leaves the positive infinitesimal coordinate domain.

248 Automatic analyticity and the exact flow criterion

The topology used for the time parameter is the ordinary topology of $\mathbb{R}$. No valuation-continuity in real time is imposed or claimed.

Definition 248.1. A *coefficientwise real-analytic flow* is a family $(\Phi_s)_{s \in \mathbb{R}}$ in G_Γ with $\Phi_0 = \text{id}$ and $\Phi_{s+r} = \Phi_s \Phi_r$, such that for every $f \in \mathcal{L}_\mathbb{R}^\Gamma$ and $q \in \Gamma$,

$$s \mapsto [t^q] \Phi_s(f)$$

is real-analytic on $\mathbb{R}$. Its generator is D_a when the derivative at $s = 0$ of every coefficient is the corresponding coefficient of $D_a f$.

This definition initially imposes no common support for the entire time family. Analyticity supplies one.

Lemma 248.2 (Analytic support rigidity). *Let I be a nonempty connected open interval, and let $s \mapsto f_s \in \mathcal{L}_\mathbb{R}^\Gamma$ have real-analytic coefficient functions. Then*

$$S = \bigcup_{s \in I} \text{supp } f_s$$

is left-finite. For any fixed substitution S_h , coefficient differentiation of $S_h(f_s)$ can be passed through S_h whenever a derivative is taken.

Proof. For each q for which the analytic coefficient function is not identically zero, its zero set is discrete in I , hence countable. There are only countably many coordinates. Choose $s_0 \in I$ outside the union of these countable zero sets. Then $\text{supp } f_{s_0}$ equals the full union S , so S is left-finite. Every coefficient derivative is also supported in S .

If $v(h) = r > 0$, formula (242.2) shows that an output coefficient at λ of $S_h(f_s)$ uses only the finite set $S \cap (-\infty, \lambda/r]$, uniformly in $s \in I$. Differentiation is therefore differentiation of a fixed finite real-linear combination. $\square$

Theorem 248.3 (Analytic-flow criterion). *A Borel derivation D_a on $\mathcal{L}_\mathbb{R}^\Gamma$ generates a coefficientwise real-analytic flow in G_Γ if and only if*

$$a = 0 \quad \text{or} \quad v(a) \geq 0. \tag{248.1}$$

The flow is unique in this class. For positive gain its coefficients are polynomials in time. For zero gain they are finite sums of real exponentials in time. Both constructions are jointly Borel in time and the field input, and their coefficient functions extend to entire functions of a complex time variable.

If $v(a) < 0$, even a coefficientwise differentiable curve of ordered Borel automorphisms through the identity cannot have derivative D_a there; the obstruction does not require a group law.

Proof. Existence in positive gain. Use Proposition 247.1. Finite coefficient sums also show directly that its derivative at zero is D_a .

Existence in zero gain. Choose the tangent normalizer h_a with $S_{h_a}^{-1}D_aS_{h_a} = cE$. The Euler flow is

$$T_s f = f(e^{cs}t) = \sum_{q \in \Gamma} f_q e^{cq_s} t^q.$$

Define

$$\Phi_s = S_{h_a} T_s S_{h_a}^{-1}. \quad (248.2)$$

For fixed f , write $g = S_{h_a}^{-1}f$. The support of $T_s g$ is the fixed left-finite support of g . Every output coefficient after S_{h_a} is a finite linear combination of the functions e^{cq_s} , by (242.2). Hence it is an exponential polynomial. The group law is immediate, and finite coefficient differentiation gives generator $S_{h_a}(cE)S_{h_a}^{-1} = D_a$. Joint Borelness follows from the Borel substitution operations. When $a = 0$, take the identity flow.

Negative-gain obstruction. Suppose $\alpha = v(a) < 0$, let $q = -\alpha > 0$, and write $c = \text{lc}(a)$. For any ordered automorphism fixing $\mathbb{R}$, the element $\Phi_s(t^q)$ is positive infinitesimal. In particular,

$$\text{ct}(\Phi_s(t^q)) = 0$$

for every s . Its constant-coefficient derivative at zero is therefore zero. But

$$\text{ct}(D_a(t^q)) = \text{ct}(qat^q) = qc \neq 0.$$

This contradicts the generator condition. Only differentiability of this one coefficient was needed.

Differentiating a general analytic flow. For fixed s and f , differentiate $\Phi_{s+h}(f) = \Phi_s(\Phi_h(f))$ at $h = 0$. Lemma 248.2 justifies passing the derivative through the fixed substitution Φ_s . Differentiating also the commuted expression $\Phi_h(\Phi_s(f))$ gives

$$\frac{d}{ds} \Phi_s(f) = \Phi_s(D_a f) = D_a(\Phi_s(f)) \quad (248.3)$$

coefficient by coefficient. Thus every time map centralizes the generator.

Uniqueness. If $a = 0$, equation (248.3) says that every coefficient has zero derivative everywhere, so the flow is the identity. If $\alpha > 0$, Theorem 247.2 gives $\Phi_s = \exp(\tau(s)D_a)$. The coefficient of $t^{1+\alpha}$ in $\Phi_s(t) - t$ is $c\tau(s)$, so τ is analytic. The group law makes τ additive, hence $\tau(s) = ks$; the generator condition gives $k = 1$.

If $\alpha = 0$, after the normalizing coordinate the centralizer consists of positive dilations. Write their scalar as $A(s) > 0$. Tangent normalization does not change the coefficient of t of a coordinate of valuation one, so $A(s) = [t^1]\Phi_s(t)$ is analytic. The group law gives $A(s+r) = A(s)A(r)$ and $A(0) = 1$, hence $A(s) = e^{ks}$. Its derivative at zero is the leading coefficient c of $D_a(t)$, so $k = c$. This recovers exactly (248.2). $\square$

[write] (Batch 96.) In rational rank one the zero-gain clause generalizes Part XVII's [proposition 239.3](#) [merge], written for the Euler derivations themselves, to every derivation of gain zero; at $a = c$ the flow (248.2) is that proposition's Ψ_{cs} . The negative-gain clause is proved again in Part XX (source 26, [corollary 275.4](#), for jointly Borel flows of a divisible Γ of any rank) and in Part XXI (source 27, [remark 294.3](#), for coefficient-differentiable actions with a Borel generator); the version here assumes only that one coefficient is differentiable at 0. With [theorem 248.8](#) the theorem answers Part XVII's [question 237.5](#) in rational rank one (note after that question); [remark 255.2](#) [merge] compares it with Part XX at $\Gamma = \mathbb{Q}$.

Remark 248.4 (Three different meanings of convergence). For $\alpha > 0$, the exponential in (247.1) is locally finite below each valuation bound. For $\alpha = 0$, the scalar exponentials are ordinary convergent real series, while the dependence on t remains formal. Neither case implies that the given series $a(t)$ or its normalizing coordinate converges for real or complex t . Also, even a nontrivial positive-gain flow usually fails to be continuous in real time for the valuation topology: its first nonzero correction can retain fixed valuation as $s \rightarrow 0$.

Remark 248.5 (Why the analytic hypothesis is useful). A merely smooth coefficient family need not have one common left-finite support: smooth functions can have disjoint time supports and vanish on open intervals. The real-analytic zero-set argument rules this out. The analytic support argument alone does not settle merely C^1 flows. The group-theoretic argument below does: it proves automatic analyticity from much weaker time regularity. The negative-gain obstruction already applies to differentiability at zero.

248.1 A stronger result: Borel time dependence forces analyticity

The analytic support lemma justifies the preceding differential argument. The group law permits a stronger conclusion: analyticity need not be assumed at all. We first establish the two automorphism facts needed for this improvement.

Lemma 248.6 (Logarithms of tangent automorphisms). *Every nonidentity tangent automorphism $\Psi \in G_\Gamma$ has a unique positive-gain derivation D with $\Psi = \exp D$. Its centralizer in G_Γ is $\{\exp(sD) : s \in \mathbb{R}\}$.*

Proof. Put $U = \Psi - \text{id}$. If $\Psi(t) = t(1 + u)$ and $\delta = v(u) > 0$, then U has gain at least δ . Thus

$$D = \log \Psi = \sum_{n \geq 1} \frac{(-1)^{n+1}}{n} U^n$$

is a Borel real-linear operator with positive gain. To verify the Leibniz rule rather than assume it, define $\Psi^{[z]} = \sum_{n \geq 0} \binom{z}{n} U^n$ for real z . For fixed f every coefficient is a polynomial in z , and the union of its supports over z is left-finite. The same holds for products of two such outputs. For every nonnegative integer $z = m$, this operator is Ψ^m , hence multiplicative. Polynomial identity therefore gives $\Psi^{[z]}(fg) = \Psi^{[z]}(f)\Psi^{[z]}(g)$ coefficientwise for all z . Differentiating at zero proves the Leibniz rule for D , since $\frac{d}{dz} \binom{z}{n} \Big|_{z=0} = (-1)^{n+1}/n$ for $n \geq 1$.

The formal identities between exponential and logarithm apply below every cutoff to finite operator computations, so $\exp D = \Psi$ and the logarithm is unique. Moreover, $D(t)$ has the same leading correction as $U(t)$ because all terms $U^n(t)$ for $n \geq 2$ have strictly greater valuation; hence $D \neq 0$. If an allowed automorphism commutes with Ψ , it commutes with U and therefore with D : its valuation scaling permits passage through the convergent operator sum. Conversely, commuting with D implies commuting with $\exp D$. The centralizer theorem for derivations now applies. $\square$

[write] (Batch 96.) The first two sentences are the logarithm half of Part XVII's [theorem 234.2](#), with the same polynomial identity in z (the proof the note after that theorem traces to the omnific-preserving-automorphisms report): in rank one a tangent automorphism $\Psi(t) = t(1 + u)$ is uniformly tangent to the identity in Part XVII's sense, with gain at least $v(u)$. The centralizer statement is new here.

Lemma 248.7 (Linearization of a nonunit dilation). *If $\Psi(t) = At(1 + u)$ with $A > 0$, $A \neq 1$, and $u \in \mathfrak{m}$, there is a tangent coordinate k with*

$$k(\Psi(t)) = Ak(t), \quad S_k^{-1}\Psi S_k = S_{At}.$$

The centralizer of S_{At} in G_Γ consists exactly of the positive dilations S_{Bt} , $B > 0$.

Proof. There is nothing to construct if $u = 0$. Otherwise set $\delta = v(u) > 0$ and work on the real vector space $V = \{f : v(f) > 1\}$. Let $L_0 = S_{At} - A \text{id}$ and $R = \Psi - S_{At}$. On a coefficient of exponent $q > 1$, L_0 multiplies by $A^q - A$, which is nonzero. Thus L_0^{-1} exists on V and preserves valuation. The operator R has gain at least δ on V . Solving $k = t + f$, $f \in V$, amounts to

$$(L_0 + R)f = -Atu.$$

The solution is

$$f = \sum_{n \geq 0} (-L_0^{-1}R)^n L_0^{-1}(-Atu).$$

It converges in valuation, lies in V , and is unique by the positive-gain estimate. Hence k is a tangent coordinate with the required property. All operators used here are coefficient-Borel; small real denominators do not affect their valuation estimates.

A coordinate $w = Bt^r(1 + v)$ centralizing S_{At} satisfies $w(At) = Aw(t)$. Leading terms give $A^r = A$, hence $r = 1$. Each nonzero coefficient of exponent q then requires $A^q = A$, hence $q = 1$. Thus $w = Bt$, and every such dilation centralizes S_{At} . $\square$

Theorem 248.8 (Automatic analyticity in time). *Let $s \mapsto \Phi_s$ be a group homomorphism from $(\mathbb{R}, +)$ to G_Γ . Either of the following assumptions suffices to force a unique analytic flow of the type in Theorem 248.3:*

- (i) *Every coefficient of $\Phi_s(t)$ is a Borel function of s .*
- (ii) *Every coefficient of $\Phi_s(t)$ is differentiable at $s = 0$.*

In particular, under either assumption there is a unique $a \in \mathcal{L}_{\mathbb{R}}^\Gamma$ with $a = 0$ or $v(a) \geq 0$ such that (Φ_s) is the analytic flow generated by D_a . No initial time regularity for the images of other field elements is required. All their coefficient functions are automatically entire in time.

Proof. Write $\Phi_s(t) = A(s)t^{r(s)}(1 + u_s)$. The exponent law gives $r(s + r') = r(s)r(r')$. For every positive integer n , $r(s) = r(s/n)^n$. A positive rational admitting rational n th roots for every n is 1, by unique prime factorization. Hence $r(s) = 1$ for every s , without any regularity hypothesis. The leading coefficient satisfies $A(s + r') = A(s)A(r')$.

Suppose first that $A(s_0) \neq 1$ for some s_0 . Linearize Φ_{s_0} by Lemma 248.7. Every other time map commutes with it, so in the same tangent coordinate every Φ_s is a positive dilation. Tangent conjugation preserves its leading coefficient, which is $A(s)$. The additive function $\lambda(s) = \log A(s)$ is Borel under (i), or differentiable at zero under (ii). In the first case the additive automatic-continuity lemma, and in the second the elementary differentiable Cauchy equation, give $\lambda(s) = cs$. Thus the original family is a fixed tangent conjugate of $S_{e^{cs}t}$. It is the zero-gain flow already constructed.

Now suppose $A(s) = 1$ for all s . All time maps are tangent. If all are the identity, take $a = 0$. Otherwise choose a nonidentity Φ_{s_0} and let $D = \log \Phi_{s_0}$ be its positive-gain derivation. Every time map centralizes Φ_{s_0} , so Lemma 248.6 gives $\Phi_s = \exp(\tau(s)D)$ for a unique real $\tau(s)$. Faithfulness of this exponential family makes τ additive.

Let $D = D_b$, $\beta = v(b) > 0$, and $d = \text{lc}(b)$. Its first correction coefficient gives

$$\tau(s) = \frac{1}{d} [t^{1+\beta}] \Phi_s(t).$$

Thus τ is Borel under (i), or differentiable at zero under (ii). In either case $\tau(s) = ks$, and the generator is kD . The flow is nontrivial, so $k \neq 0$; its gain remains positive. Proposition 247.1 gives entire coefficient functions on every input. Uniqueness of the generator follows by coefficientwise differentiation at zero, and uniqueness of its analytic flow was proved in Theorem 248.3. $\square$

[write] (Batch 96.) For $\Gamma = \mathbb{Q}$ this theorem under hypothesis (i), and the criterion of theorem 248.3, are Part XX's corollary 285.2 [merge], which derives them from source 26's theorem 275.2 and corollary 277.3 with proposition 285.1; for every countable divisible $\Gamma \leq \mathbb{R}$, Borel time dependence of the images of the monomials of one rational basis already gives a jointly Borel flow. Differentiability at zero (hypothesis (ii)) in higher rank is question 283.10. Part XXI extends the theorem to actions of connected Lie groups (theorem 300.3 [merge]).

Corollary 248.9 (Finite-dimensional real parameter groups). *Every homomorphism $\Phi : (\mathbb{R}^n, +) \rightarrow G_\Gamma$ for which the coordinate of $\Phi_x(t)$ is Borel in x factors through a single real linear functional $\lambda : \mathbb{R}^n \rightarrow \mathbb{R}$ and a one-parameter analytic flow. Thus such an action has an image of dimension at most one in this explicit sense.*

Proof. The trivial action factors through the zero functional. Otherwise, divisibility again makes the exponent character trivial. If some image element has leading coefficient different from one, choose such an element; its centralizer is conjugate to positive dilations. If all image elements are tangent, choose a nonidentity one; its centralizer is an exponential family. In either case the whole image lies in this centralizer by commutativity. Extracting its scalar parameter gives a Borel additive map $\mathbb{R}^n \rightarrow \mathbb{R}$, which is real-linear by Lemma 242.1. $\square$

Without time regularity there are genuinely different parameterizations: compose a nontrivial analytic flow with a discontinuous additive function $\tau : \mathbb{R} \rightarrow \mathbb{R}$. Each individual time map remains Borel on $\mathcal{L}_\mathbb{R}^\Gamma$, but the first relevant coefficient is not a Borel function of time. This construction uses a nontrivial Hamel-basis additive map and explains why some time regularity is substantive. The group law alone still forces the two centralizer forms appearing in the proof.

[write] (Batch 96.) The same two lemmas determine which elements of G_Γ are time-one maps of such flows: exactly those with $r = 1$, and then the flow is unique (proposition 255.1 [merge]). For an action of a Lie group by elements of G_Γ in which every coefficient of the image of t is a Borel function on the group, theorem 248.8 applies to every one-parameter subgroup $s \mapsto \exp(sX)$, which is continuous; Part XXI (source 27) uses this for the one-parameter half of its question on joint regularity, and it answers this part's question 253.3 at the level of Lie algebras (note after that question).

249 Worked examples

249.1 A residue hidden beyond the leading term

In $\mathcal{L}_{\mathbb{R}}^{\mathbb{Q}}$, take

$$a = 2t^{1/2}(1 + t^{1/6}), \quad D_a = aE.$$

Put $z = t^{1/6}$. Then

$$\frac{1}{a} = \frac{1}{2}t^{-1/2}(1 - z + z^2 - z^3 + z^4 - \dots),$$

so $\alpha = 1/2$, $c = 2$, $\rho = -1/2$, and $\theta = \rho/\alpha = -1$. The full normal representative is therefore

$$\frac{t}{1-t}E.$$

The parameter -1 could not have been obtained from the leading term of a alone; it arises at the constant term of the reciprocal. The tangent normalizer selected by (244.5) begins

$$\frac{h_a}{t} = 1 + 3z + \frac{3}{4}z^2 - \frac{27}{2}z^3 - \frac{315}{16}z^4 + \frac{621}{16}z^5 + \frac{8255}{64}z^6 + O(z^7).$$

Here $O(z^7)$ is a formal coefficient-order statement. These coefficients are reproduced by the exact verification script. The subsequent monomial coordinate $t \mapsto t^{1/2}$ carries the tangent normal form to the displayed full representative. The same example lies in $\mathcal{L}_{\mathbb{R}}^{\mathbb{Z}[1/6]}$ and has exponent representative $k = 1$ there as well.

249.2 Zero gain with an explicit global time family

For $a = 3(1 + t^{1/2})$ in a field containing $1/2$ in its exponent group,

$$h_a = t(1 + t^{1/2})^{-2}$$

normalizes D_a to $3E$. Solving $h_a(g_s) = e^{3s}h_a(t)$ gives

$$g_s = \Phi_s(t) = e^{3s}t \left(1 + (1 - e^{3s/2})t^{1/2}\right)^{-2}. \quad (249.1)$$

Indeed, writing $z = \sqrt{t}$ gives $\sqrt{h_a(t)} = z/(1+z)$; scaling this by $e^{3s/2}$ and solving for the new square root yields (249.1). Differentiation at zero gives $3t(1 + \sqrt{t}) = D_a(t)$. Every coefficient is a finite sum of exponentials, even though the formal series has infinitely many terms.

249.3 A rectifiable derivation with no automorphism flow

Consider

$$a = t^{-1}(1 + t^{1/2}), \quad D_a = (1 + t^{1/2})\frac{d}{dt}.$$

Its tangent normalizer is

$$\begin{aligned} h_a &= \sum_{n \geq 0} \frac{(-1)^n}{1 + n/2} t^{1+n/2} \\ &= t - \frac{2}{3}t^{3/2} + \frac{1}{2}t^2 - \frac{2}{5}t^{5/2} + \frac{1}{3}t^3 - \dots \\ &= 2\sqrt{t} - 2\log(1 + \sqrt{t}). \end{aligned}$$

It satisfies $D_a h_a = 1$, so the derivation is conjugate to d/dt . Nevertheless it cannot generate an ordered automorphism flow: $D_a(t) = 1 + \sqrt{t}$ has constant coefficient one, while every image of the positive infinitesimal t has constant coefficient zero. Formal rectification and existence of a global real-time action on the field are different questions.

249.4 Positive gain without residue

For $D_a = ct^\alpha E$ with $\alpha > 0$, the time coordinate is explicitly

$$g_s = t(1 - \alpha c s t^\alpha)^{-1/\alpha}.$$

It follows either from the primitive $-t^{-\alpha}/(\alpha c)$ or from the operator exponential. Its coefficients are polynomials in s , and $g_s \circ g_r = g_{s+r}$ follows by taking the $-\alpha$ power. Every coefficient is valid formally for every real s ; no real-variable radius of convergence is involved.

[write] (Batch 96.) For $\alpha = c = 1$ this is Part XVII's flow (234.4), $g_s = t/(1 - st)$, which is the fixed-shift flow T_s of the surcomplex-field-automorphisms report (saut:thm:shiftflow); the two-parameter affine action of Part XXI (theorem 294.4) contains this family as its translation part.

249.5 The same Euler fields in different exponent groups

On $\mathcal{L}_{\mathbb{R}}^{\mathbb{Q}}, E$ and $2E$ are conjugate by $t \mapsto t^2$, whereas E and $\sqrt{2}E$ are not conjugate. On $\mathcal{L}_{\mathbb{R}}^{\mathbb{Z}[1/2]}, E$ and $2E$ remain conjugate, but E and $3E$ are not. On $\mathcal{L}_{\mathbb{R}}^{\mathbb{Z}[1/6]}$, the latter pair becomes conjugate. The equivalence relation on all positive Euler coefficients is smooth in the $\mathbb{Z}[1/2]$ case and nonsmooth in the $\mathbb{Z}[1/6]$ case. An explicit test for a given pair is thus compatible with a genuine obstruction to Borel complete classification of all pairs at once.

250 Surreal interpretation and a formalization route

250.1 A set-sized subfield, not a Polish proper class

The normal-form interpretation of surreal numbers gives the embedding

$$\iota_{\Gamma} : \mathcal{L}_{\mathbb{R}}^{\Gamma} \longrightarrow \mathbf{No}, \quad \sum_{q \in \Gamma} f_q t^q \mapsto \sum_{q \in \Gamma} f_q \omega^{-q}. \quad (250.1)$$

The exponents $-q$ on the right form a reverse well-ordered set. Uniqueness of surreal normal forms gives injectivity; the normal-form sum and product laws give an ordered-field embedding. These are the normal-form structures described in ProveIt's surreal-field documentation [153]. The image has cardinality exactly $2^{\aleph_0}$: it contains $\mathbb{R}$ and is coded by a subset of $\mathbb{R}^{\Gamma}$ with Γ countable.

Transport the coefficient Borel structure, the derivations, and the allowed coordinate changes through ι_{Γ} . Every theorem then holds on this explicit surreal subfield. No Borel structure on the proper class of all surreals is needed. No quantification over a set of all proper-class maps is used. The transported Euler derivation sends ω^{-q} to $q\omega^{-q}$; in particular, it sends ω to $-\omega$. It is the derivation specified here, not an unidentified canonical surreal derivation. The paper does not assert that these automorphisms or derivations extend to the whole surreal field, preserve surreal exponentiation, or preserve birthday and simplicity structure.

[write] (Batch 96.) ι_Γ is Part XVII's map (235.1) for $\mathbb{Z} \leq \Gamma \leq \mathbb{Q}$, hence Part V's surreal map (71.1) with real coefficients; its image is Part XVII's K_Γ , not Part II's Hahn field of the same name. The cited documentation is the surreal README at the pin, which describes these normal-form structures. No statement of this part about the transported derivations or automorphisms is formalized.

250.2 A realistic proof-assistant decomposition

The mathematical dependency graph separates into three manageable layers. The first is algebraic and valuation-theoretic: left-finite supports, finite convolution, unit powers, Euler differentiation, and compositional inversion. The second concerns countable coefficient coding and Baire category. The third combines those layers to prove normalization, conjugacy, and flow statements. This separation is important for ProvelIt's stated trust boundary: an executable calculation is not a formal theorem until a proved checker connects it to the intended semantics [153].

A concrete implementation plan is as follows. Represent Γ as an additive subgroup of $\mathbb{Q}$, and a series as a coefficient function accompanied by its left-finiteness proof. Reuse a Hahn-series construction only after proving that the left-finite subtype is closed under the required operations. Prove an explicit cutoff lemma for every operation, rather than allowing arbitrary infinite coefficient sums. Implement unit powers first on finite truncations, then use the support and completeness lemmas to connect them to full series.

The positive-gain normalizer should expose its numerical error certificate $v(H - H_n) \geq \delta + n\alpha$. A verified finite-jet implementation can then return a truncation together with a proved cutoff bound. The residue covariance and prime-height theorem are comparatively elementary targets once the core algebra is available. The Borel witness map requires countable partitions and pointwise limits, whereas the nonsmoothness proof needs the Baire-property argument for invariant sets. These descriptive-set-theoretic obligations should not be replaced by an undocumented assumption of automatic continuity of the whole coefficient space.

No Lean or Rocq file is supplied with this manuscript, and no declaration in the repository is claimed to verify the new theorems. The repository was used as mathematical context, not as a substitute for checking the proofs above.

[write] (Batch 96.) True of the new theorems. Two ingredients are formal in special forms: the kernel of E and the vanishing of $\text{ct}(Ef)$, for Hahn series, and the integration criterion for $\Gamma = \mathbb{Z}$ (note after (243.3); remark 239.5). Nothing of this part is mapped in docs/FORMALIZATION.md, and its placement beside the formal project confers no formal status.

251 Exact finite-jet verification and reproducibility

The accompanying `verify_jets.py` uses only Python's standard library and `Fraction`. It works in $\mathbb{Q}[z]/(z^{N+1})$, where $z = t^{1/d}$. The integer d records the grid of the particular test and is not imposed on the general theorems.

251.1 The recurrence and an independent residual

Write $a = ct^\alpha U(z)$ with $U(0) = 1$ and $\alpha d \in \mathbb{Z}$. Let $W = U^{-1}$. For $\alpha \neq 0$, the normalized primitive unit K has coefficients

$$K_0 = 1, \quad K_j = \frac{\alpha W_j}{\alpha - j/d} \quad (j > 0, j/d \neq \alpha), \quad K_{\alpha d} = 0 \quad (\alpha > 0).$$

When $\alpha > 0$, the residue is $\rho = W_{\alpha d}/c$. The script computes $H = h_a/t$ by the explicit unit formula or fixed-point iteration as appropriate. It then checks the original coordinate equation, not merely the equation used to generate the coefficients. After cancelling the common monomial and scalar factors, the residual is

$$U(H + EH)(1 + c\rho t^\alpha H^\alpha) - H^{1+\alpha} \quad (\alpha > 0), \quad (251.1)$$

and $U(H + EH) - H^{1+\alpha}$ for $\alpha \leq 0$.

For a unit U and rational b , the script computes $V = U^b$ by the recurrence

$$V_0 = 1, \quad V_n = \frac{1}{n} \sum_{k=1}^n ((b+1)k - n) U_k V_{n-k}.$$

This follows from $UV' = bU'V$, with derivatives in z . Logarithms are computed by integrating U'/U , and exponentials by their own derivative recurrence. Thus the checks do not rely on floating-point approximations or cancellation tolerances.

The positive-flow tests use

$$D_a^n(t) = c^n t^{1+n\alpha} F_n, \quad F_0 = 1, \quad F_{n+1} = U((1 + n\alpha)F_n + EF_n),$$

and verify both composition of time maps and inversion. Separate pullback tests under $t^r H$ check the residue law. The explicit zero-gain time family is checked with rational values for $e^{3s/2}$, so that this additional check also remains exact.

251.2 Recorded run

The supplied report records the following run:

```
python verify_jets.py --degree 24 --seed 20261004 \
  --random-cases 48 --output verification_report.json
```

Exact check family	Number of checks
Normal-form coordinate residuals	51
Residue covariance under pullback	24
Positive-flow composition laws	8
Positive-flow inverse laws	8
Unit exponential–logarithm identities	48
Explicit zero-gain flow identities	4
Total, modulo z^{25}	143

All 143 checks passed. The maximum fixed-point stabilization count observed was 24. The JSON report contains the exact rational coefficients of the worked normalizers. An earlier smaller run at degree 18 also passed, but only the specified degree-24 report is included as the principal reproducibility record.

[write] (Batch 96.) The program and its record are shipped as `code/24-borel-conjugacy-verify_jets.py` and `data/24-borel-conjugacy-verification_report.json`; the degree-18 run is not shipped. Without `-output` the program writes `verification_report.json` in the current directory, so it should be run on a copy with an explicit output path. Rerun so on 4 October 2026 (Python 3.14.4, also under `-0`; the checks raise exceptions rather than use `assert`), it passed the same 143 checks in about five seconds, and its record equals the shipped one after removing carriage returns. As the source says, the checks are regression tests on finite jets, not proofs.

These finite tests can detect sign, exponent, residue, and conjugation-direction errors. They cannot prove the treatment of arbitrary unbounded-denominator supports, Borel uniformity, the nonsmoothness theorem, or the no-flow obstruction. Those assertions depend on the mathematical proofs, not on the test count.

252 Proof audit and scope of the conclusions

Claim	Decisive argument	Main hypothesis
Every Borel derivation is aE	Finite-coordinate Polish slices and rational power identities	Borel coding; rank one
Allowed automorphisms are substitutions	Constants fixed; order comparison; valuation density	Order preservation
Borel tangent normalizer exists	Unit formulas and gain- α contraction	Left-finite support
Complete full conjugacy invariants	Residue covariance and monomial witness	$r\Gamma = \Gamma$
Two-prime smoothness threshold	Prime heights; dense translation zero-one law	Countable exponent group
Centralizer classification	Primitive difference is constant	Same coordinate group
Analytic-flow uniqueness	Analytic support rigidity and centralizers	Real-analytic coefficients
Automatic time analyticity	Automorphism centralizers and the additive Cauchy equation	Borel time coordinate
No negative-gain flow	Constant coefficient of a positive infinitesimal	Order; differentiability at zero

Several possible overextensions are excluded explicitly. Left-finite support is stronger than well-ordered support. A rank-one subgroup of $\mathbb{Q}$ is not an arbitrary countable ordered abelian group. A Borel normalizer need not select full orbits. A log-free normal form need not remain invariant under logarithmic coordinate enlargement. Coefficientwise analyticity in time is not analytic convergence in t . Borelness is not computability. The surreal interpretation is set-sized and does not establish a theorem about all proper-class self-maps of **No**.

The two specific earlier questions answered are therefore: complete conjugacy under Borel ordered-field coordinates in the present rank-one setting, and existence and uniqueness of coefficientwise real-analytic flows for every Borel derivation in that setting. In the 2-divisible cases, including the classical Levi–Civita field, the order restriction is automatic. The general higher-rank and unordered problems remain separate research targets. For real

one-parameter actions, the weaker-time-regularity question is resolved here by automatic analyticity; nonabelian parameter groups provide a further extension.

[write] (Batch 96.) The two earlier questions are Part XVII's [questions 237.4](#) and [237.5](#); the dated notes after them state what remains open. The nonabelian extension is [question 253.3](#), which Part XXI answers at the level of Lie algebras.

253 Further research questions and concrete next projects

The following are proposed extensions of the present work. They are not all asserted to be previously unposed or open in every form in the literature. Each formulation records a boundary that this manuscript does not cross.

253.1 Unordered Borel automorphisms

Question 253.1. For rank-one groups that are not 2-divisible, classify Borel field automorphisms without an order-preservation requirement, and determine the resulting conjugacy invariants of derivations.

The map $t \mapsto -t$ over integer exponents shows that the group can enlarge. A first step is to identify allowed sign characters on Γ and prove valuation control from Borel regularity rather than from ordered comparisons. One should determine whether the two-prime threshold survives or must be modified by additional finite symmetries.

[write] (Batch 96.) Open. Part XXI's question on unordered automorphisms and sign characters ([question 298.2](#)) is the same problem for its cross-section of pairs; the flow results of [section 248](#) are also proved only for ordered automorphisms.

253.2 Higher rational rank

Question 253.2. For a finite-rank subgroup of $\mathbb{R}$ with rational rank at least two, obtain normal forms and conjugacy invariants for Borel derivations expressed in a basis of Euler operators.

There are then several independent monomial logarithmic derivatives. Constants, resonances, and integration obstructions can be infinite-dimensional. A concrete initial problem is rank two with finitely supported coefficient functions: identify conditions yielding a finite-dimensional normal form and determine whether a dense scaling action is still the only source of nonsmoothness.

[write] (Batch 96.) Open. It is the conjugacy counterpart of Part XVII's [question 237.3](#). Two data points are in this report: in rational rank two the scalar rigidity behind Part XXI's classification fails ([example 295.5](#)); and for divisible Γ of any rational rank Part XX decides which Borel derivations integrate to Borel flows ([theorem 275.2](#) and [corollary 277.3](#)), which bears on the flow half of the problem but not on conjugacy invariants.

253.3 Nonabelian real parameter groups

Question 253.3. Classify coefficient-Borel actions of connected nonabelian Lie groups on $\mathcal{L}_{\mathbb{R}}^{\Gamma}$ by ordered-field automorphisms, including the restrictions on their infinitesimal Lie algebras.

Theorem 248.8 settles the weaker-time-regularity problem for one-parameter actions: even Borel time dependence forces analyticity. Corollary 248.9 shows that real vector-group actions factor through one real parameter. Noncommuting generators can behave differently; Euler and positive-gain derivations already give explicit nonzero brackets. The next task is to identify the possible finite-dimensional Lie algebras and construct the corresponding global ordered actions.

[write] (Batch 96.) *Answered at the level of Lie algebras by Part XXI* (source 27, which takes this question as its starting point). Every finite-dimensional real Lie algebra of Borel derivations of $\mathcal{L}_{\mathbb{R}}^{\Gamma}$ has dimension at most three, abelian ones dimension at most one (theorem 293.2); the two- and three-dimensional ones are classified up to conjugacy by a ramification invariant (theorem 293.3), with Borel smooth classification of the subalgebras (proposition 293.4); the positive affine algebra integrates to an explicit global action of $\text{Aff}^+(\mathbb{R})$ (theorem 294.4), and a coefficient-differentiable action with a Borel generator has only generators of nonnegative gain (remark 294.3); the valuation-nondecreasing algebras are 0, a line or conjugate to $\mathfrak{a}_p$ with $p > 0$ (corollary 294.1). *Answered at the level of groups* for ordered automorphisms in this rank-one setting, by Part XXI's theorem 300.3 [merge], which uses theorem 248.8: if the one-parameter subgroups of an action of a connected Lie group by elements of G_{Γ} move t Borelly in time (in particular if the action is jointly coefficient-Borel, or coefficient-differentiable with Borel generators), then the generator map is a Lie algebra homomorphism, its image has dimension at most two, and the action factors, on the universal cover, through source 27's affine action or through one analytic flow; connected semisimple groups act trivially. The one-parameter half, that every one-parameter subgroup of such an action is an analytic flow with a generator of nonnegative gain, is theorem 248.8 itself, and it is the regularity step that answers Part XXI's question on joint regularity (question 298.4) in this setting. Part XXI's theorem was written for this report and checked independently once; it is unrefereed. *Open*: unordered automorphisms, higher rational rank, and actions whose one-parameter subgroups have no time regularity, where additive changes of time without regularity give homomorphisms with no generator.

253.4 Effective normalizers and exponent presentations

Question 253.4. For computably presented Γ , which representations make U_{Γ} , the exponent representative κ , and the conjugating coordinate effectively computable from a symbolic derivation?

The present proofs establish Borelness, not an algorithm for arbitrary real coefficients or arbitrary height presentations. For a finite localization $\mathbb{Z}[P^{-1}]$ with computable coefficient data and certified support bounds, the formulas suggest a practical exact normalizer. The contraction estimate provides a direct verification target for a proof-producing implementation.

[write] (Batch 96.) *Open*. A negative boundary is known: decidable membership in Γ does not suffice even to decide whether E and pE are conjugate (remark 255.4 [merge], a reviewer's lemma), so any positive answer needs effective information on the infinitely divisible primes. The question is a relative of Part XVII's question 237.6; Part XXI's question on effective normalizers is its analogue for pairs, and for pairs over $\mathbb{Z}$ the same review shows that equality of the invariant can be co-c.e.-complete although the normalizer is computable (remark 300.4).

253.5 Logarithmic extensions of the coordinate group

Question 253.5. For a carefully specified countable monomial language containing logarithmic scales, determine how the residue invariant and the descriptive complexity of conjugacy change.

The coordinate group must be fixed before an invariant is proposed. The classical power-log literature already shows that residual parameters can disappear under enlargement [151]. A useful extension here would preserve a standard Borel coding and describe the resulting orbit relation, rather than assuming that the current residue survives unchanged.

[write] (Batch 96.) Open. The cited elimination of a residual parameter was checked (remark 255.5 [merge]); it concerns normal forms of maps, not of derivations.

253.6 Full Hahn supports

Question 253.6. On which Borel-coded subspaces of a full Hahn field do the normalization and Borel witness-selection arguments remain valid?

Arbitrary well-ordered supports can have infinitely many terms below a finite cutoff, so the finite-coordinate argument must be replaced or strengthened. A rigorous project should first settle the descriptive complexity of its chosen support domain and the exact meaning of summation; a formal change from “left-finite” to “well ordered” is not sufficient.

[write] (Batch 96.) Open, with one input now in this report: Part XIX (source 25) determines the exact complexity of the full Hahn domain, which for a noncyclic $\Gamma \leq \mathbb{Q}$ (not scattered) is Π_1^1 -complete in $\mathbb{R}^\Gamma$ (theorem 260.1), against the Borel domain (226.1) of $\mathcal{L}_\mathbb{R}^\Gamma$. So the full Hahn field itself is not a Borel-coded subspace, and the question concerns its Borel subspaces. Part XXI’s question on full Hahn support domains and Part XVII’s question 237.1 are its companions.

253.7 Effective complexity of the zero-gain quotient

Question 253.7. For natural effective presentations of Γ , construct explicit finite or finitely generated approximations to the zero-gain orbit relation and quantify the complexity of their codes and witnesses.

The abstract orbit relation is already explicit: multiplication by the countable group generated by the infinitely divisible primes. The research issue is an effective and uniform construction from exponent-group data, not an unsupported assertion that general countable abelian orbit theory is unknown. The Borel conjugator in Theorem 245.2 gives a useful starting object to analyze.

[write] (Batch 96.) Open. Remark 255.4 [merge] shows that the zero-gain relation can be co-c.e.-complete on finite rational inputs for a Γ with decidable membership, so the “effective presentation” must include information about the infinitely divisible primes.

253.8 Surreal extension constraints

Question 253.8. Which of the automorphisms and derivations constructed on $\iota_\Gamma(\mathcal{L}_\mathbb{R}^\Gamma)$ extend to specified larger surreal subfields while preserving strong sums, exponentiation, or a chosen derivation?

Different requirements may force very different extension groups. Any such project should remain inside a set-sized universe or use a stated class theory, and distinguish field extension from preservation of simplicity and birthday. The present ω^{-q} model gives explicit test cases without making a proper-class Borel claim.

253.9 Complex coefficients and analytic realization

Question 253.9. Develop the corresponding classification over complex coefficients, and then compare formal Borel conjugacy with convergent or sectorial analytic conjugacy for series that admit analytic realizations.

Roots of unity and coordinate branches alter the automorphism and centralizer analysis. Formal time maps need not produce analytic germs in t , and sectorial classification may require Stokes data beyond the formal invariants. These are additional analytic problems, not consequences of the present coefficientwise entire dependence on time.

253.10 Certified implementation

Question 253.10. Formalize the support monoid lemma, the primitive residue law, and the fixed-point cutoff certificate, then connect an exact jet normalizer to the full-series theorem in Lean or Rocq.

This is a concrete route from the supplied regression script to a verified algorithm. The first milestones can be purely algebraic. Descriptive-set-theory and analytic-support results can be added as separate modules, keeping their axiom and library dependencies visible.

[write] (Batch 96.) Open; a companion of Part XVII's [question 237.9](#). The Lean statements that already cover special cases of the primitive residue law are listed in the note after [\(243.3\)](#). [Questions 253.8](#) and [253.9](#) are open as stated.

254 Conclusion

The conjugacy problem separates into two mechanisms. Primitive integration and one positive-gain contraction remove all nonessential coefficients, leaving explicit order, sign, and residue data. The exponent group then acts on those normal forms. In nonzero gain that action admits elementary complete invariants; in zero gain it becomes positive real scaling by U_{Γ} .

Prime heights identify this group exactly. With no infinitely divisible prime it is trivial; with one it has a discrete logarithm; with two it has a dense logarithm. This proves the precise smoothness threshold, while also explaining why Borelly choosing a conjugacy for a supplied pair does not require a Borel orbit selector. The same normal forms determine all nonzero centralizers and, with analytic support rigidity, give the exact existence and uniqueness theorem for real-time analytic automorphism flows. The automorphism centralizers strengthen this to an automatic-regularity theorem: Borel dependence on time of the single coordinate t forces entire coefficient dependence for every field element.

255 Where Part XVIII meets Parts XV, XVII and XX [merge]

This section was written for this report. It characterizes the time-one maps of the flows of [section 248](#), a consequence of source 24's lemmas that it does not state; compares source 24's results with Part XX at $\Gamma = \mathbb{Q}$ and with Part XV; prints a lemma of the Hilbert's-tenth programme's review on effective presentations; and records the check of source 24's one unverified citation. The proofs use only results proved in source 24 and in the parts cited; they have not been independently reviewed. Throughout, $\mathbb{Z} \leq \Gamma \leq \mathbb{Q}$, and for $\sigma \in G_\Gamma$ we write $\sigma(t) = A_\sigma t^{r_\sigma}(1 + u_\sigma)$ as in [\(242.1\)](#).

Proposition 255.1 (Time-one maps). *Let $\sigma \in G_\Gamma$.*

- (i) *If $\sigma = \Phi_1$ for some homomorphism $s \mapsto \Phi_s$ from $(\mathbb{R}, +)$ to G_Γ , with no regularity assumed, then $r_\sigma = 1$.*
- (ii) *If $r_\sigma = 1$, there is a homomorphism $s \mapsto \Phi_s$ from $(\mathbb{R}, +)$ to G_Γ with $\Phi_1 = \sigma$ such that every coefficient of $\Phi_s(t)$ is a Borel function of s . It is the analytic flow of [theorem 248.3](#) generated by a derivation D_a with $a = 0$ if $\sigma = \text{id}$, with $v(a) = 0$ if $A_\sigma \neq 1$, and with $v(a) > 0$ if $A_\sigma = 1$ and $\sigma \neq \text{id}$.*
- (iii) *The homomorphism in (ii) is unique among homomorphisms with $\Phi_1 = \sigma$ for which every coefficient of $\Phi_s(t)$ is a Borel function of s .*

In particular, if no prime is infinitely divisible in Γ , so that $U_\Gamma = \{1\}$, every element of G_Γ is the time-one map of exactly one such flow.

Proof. (i) Since $v(\Phi_s f) = r_{\Phi_s} v(f)$, the map $s \mapsto r_{\Phi_s}$ is a homomorphism into the multiplicative group $U_\Gamma \subseteq \mathbb{Q}_{>0}$. As in the proof of [theorem 248.8](#), $r_{\Phi_1} = (r_{\Phi_{1/n}})^n$ has a rational n th root for every n , so it is 1.

(ii) If $\sigma = \text{id}$, take $\Phi_s = \text{id}$. If $A_\sigma \neq 1$, [lemma 248.7](#) gives a tangent k with $S_k^{-1} \sigma S_k = S_{A_\sigma t}$. Put $\Phi_s = S_k S_{A_\sigma^s t} S_k^{-1}$. Since $S_{Bt} S_{Ct} = S_{BCt}$, this is a homomorphism with $\Phi_1 = \sigma$; by [\(242.2\)](#) each coefficient of $\Phi_s(f)$ is a finite combination of functions A_σ^{qs} , so the family is a coefficientwise real-analytic flow. Its generator is $S_k(\lambda E) S_k^{-1}$ with $\lambda = \log A_\sigma \neq 0$, a Borel derivation, hence D_a for some a by [theorem 242.2](#); since k carries D_a to D_λ , [theorem 243.2](#) with $r = 1$ gives $v(a) = v(\lambda) = 0$. By the uniqueness clause of [theorem 248.3](#), Φ is the flow constructed there. If $A_\sigma = 1$ and $\sigma \neq \text{id}$, then σ is tangent and [lemma 248.6](#) gives a Borel derivation D of positive gain with $\sigma = \exp D$; by [theorem 242.2](#) $D = D_a$ with $v(a) > 0$, and $\Phi_s = \exp(s D_a)$ ([proposition 247.1](#)) has polynomial coefficients and $\Phi_1 = \sigma$.

(iii) Let Φ be a homomorphism with $\Phi_1 = \sigma$ whose image of t has Borel coefficients. Every Φ_s commutes with σ .

If $A_\sigma \neq 1$, the centralizer of σ is S_k times the positive dilations times S_k^{-1} ([lemma 248.7](#)), so $\Phi_s = S_k S_{B(s)t} S_k^{-1}$ with B a homomorphism from $(\mathbb{R}, +)$ to $(\mathbb{R}_{>0}, \times)$ and $B(1) = A_\sigma$. Tangent conjugation preserves the leading coefficient, so $B(s) = [t^1] \Phi_s(t)$ is Borel; $\log B$ is a Borel additive function, hence linear ([lemma 242.1](#) with S a single point), and $B(s) = A_\sigma^s$.

If $A_\sigma = 1$ and $\sigma \neq \text{id}$, write $\sigma = \exp D$ with $D = D_b$, $\beta = v(b) > 0$ and $d = \text{lc}(b)$. By [lemma 248.6](#) the centralizer of σ is $\{\exp(\tau D) : \tau \in \mathbb{R}\}$, so $\Phi_s = \exp(\tau(s) D)$. The coefficient of $t^{1+\beta}$ in $\exp(\tau D)(t)$ is τd , so the exponential family is faithful, τ is additive, $\tau(s) = d^{-1} [t^{1+\beta}] \Phi_s(t)$ is Borel, hence $\tau(s) = \tau(1)s$, and $\exp(\tau(1) D) = \sigma = \exp D$ gives $\tau(1) = 1$.

If $\sigma = \text{id}$, [theorem 248.8](#) makes Φ the analytic flow of some D_a with $a = 0$ or $v(a) \geq 0$. If $v(a) = 0$, then Φ_1 is a tangent conjugate of S_{c^t} with $c = \text{lc}(a) \neq 0$, so the leading coefficient

of $\Phi_1(t)$ is $e^c \neq 1$. If $v(a) > 0$, then $\Phi_1 = \exp(D_a)$, and the coefficient of $t^{1+v(a)}$ in $\Phi_1(t)$ is $\text{lc}(a) \neq 0$. Both contradict $\Phi_1 = \text{id}$, so $a = 0$ and $\Phi_s = \text{id}$. In each case Φ is the homomorphism of (ii).

The last assertion follows from (i)–(iii) and (246.1), since $r_\sigma \in U_\Gamma$. $\square$

Source 24 does not state this proposition; it is assembled from its [lemmas 248.6](#) and [248.7](#) and [theorem 248.8](#) and the uniqueness clause of [theorem 248.3](#). When some prime is infinitely divisible, the elements of G_Γ with $r_\sigma \neq 1$, for example the monomial substitutions $t \mapsto t^p$ with $p \in \omega_\infty(\Gamma)$, lie in no one-parameter subgroup of G_Γ at all, by (i).

Remark 255.2 (Part XX at $\Gamma = \mathbb{Q}$). The only divisible group between $\mathbb{Z}$ and $\mathbb{Q}$ is $\mathbb{Q}$. There, every Borel field automorphism of $\mathcal{L}_\mathbb{R}^\mathbb{Q}$ lies in $G_\mathbb{Q}$ ([theorem 242.4](#)), and a Borel field flow in the sense of Part XX (a jointly coefficient-Borel action of $(\mathbb{R}, +)$ by field automorphisms) has Borel coefficients of $\Phi_s(t)$. So [theorem 248.8](#) applies to it and makes it the analytic flow of a D_a with $a = 0$ or $v(a) \geq 0$; conversely these flows are jointly Borel ([theorem 248.3](#)). Since $v(aEf) = v(a) + v(Ef) \geq v(a) + v(f)$, with equality on nonconstant monomials, the condition $a = 0$ or $v(a) \geq 0$ says exactly that D_a does not decrease valuation (for $Ef \neq 0$). Thus, at $\Gamma = \mathbb{Q}$, [theorems 248.3](#) and [248.8](#) contain Part XX's [corollary 277.3](#) (a Borel derivation generates a Borel field flow exactly when it does not decrease valuation) and the description of the flows in its [theorem 275.2](#) (entire coefficient functions, the generator obtained by differentiating at time zero), under the weaker hypothesis that only the image of t depends Borelly on time; Part XX's local-finiteness characterization is not needed in rank one. Similarly, [proposition 255.1](#) at $\Gamma = \mathbb{Q}$ says that the time-one maps are the σ with $v(\sigma f) = v(f)$, which is condition (i) of Part XX's [theorem 279.5](#); its window condition (ii) holds automatically here, because the iterates of a vector supported in a finite set F are supported in $F + M$, where M is the additive monoid generated by $\text{supp } u_\sigma$, which is left-finite ([lemma 241.2](#)) and so meets every bounded window in a finite set. Away from $\Gamma = \mathbb{Q}$ the hypotheses are incomparable: source 24 allows nondivisible Γ of rank one, source 26 divisible Γ of every countable rational rank. Both are kept.

Remark 255.3 (Part XV's coefficient-fixing case for $k = \mathbb{R}$). The proof of [theorem 242.4](#) uses Borelness only to show that the automorphism fixes $\mathbb{R}$. Hence, for every $\mathbb{Z} \leq \Gamma \leq \mathbb{Q}$, every ordered field automorphism of $\mathcal{L}_\mathbb{R}^\Gamma$ that fixes $\mathbb{R}$ pointwise is a substitution ([242.1](#)), and when Γ is 2-divisible every field automorphism fixing $\mathbb{R}$ is. For $\Gamma = \mathbb{Q}$ this is the description of Part XV's [theorem 198.2](#) with $k = \mathbb{R}$, restricted to automorphisms: S_h with $h = At^r(1+u)$ has

$$\lambda = r, \quad c(q) = A^q, \quad u(q) = (1+u)^q,$$

and every pair of characters of $\mathbb{Q}$ into $\mathbb{R}_{>0}$ and into $1 + \mathfrak{m}$ is of this form, being determined by its value at 1 through uniqueness of positive roots; every such S_h is onto. So source 24 writes out, for $k = \mathbb{R}$, the rank-one $\mathbb{Q}$ -subspace and automorphisms, the coefficient-fixing classification that the note after Part XV's [question 208.5](#) says extends to larger Archimedean real closed k without a written proof. Embeddings, higher rational rank and other k are not treated.

Remark 255.4 (A reviewer's lemma: decidable exponents do not make conjugacy decidable). The following is due to the Hilbert's-tenth programme's review of source 24 (`Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/review_new_borel_62846e17a.md`, commit `8dffc7af0`); it is not a result of source 24. Fix effective enumerations $p_0, p_1, \dots$ of the primes and $M_0, M_1, \dots$ of Turing machines with a c.e.-complete halting set, every halting run taking at least one step, and let H_e be the halting time of M_e . Put $h_{p_e} = H_e - 1$ if M_e halts and $h_{p_e} = \infty$ otherwise, and let Γ consist of the rationals m/n

in lowest terms with $v_p(n) \leq h_p$ for every prime p . Then $\mathbb{Z} \leq \Gamma \leq \mathbb{Q}$, membership in Γ is decidable, and

$$E \sim_{\Gamma} p_e E \iff M_e \text{ never halts,}$$

so the set of e with $E \sim_{\Gamma} p_e E$ is co-c.e.-complete. Indeed, Γ is a group because denominators of sums divide least common multiples; to decide $m/n \in \Gamma$, factor n and, for each $p_e^j \parallel n$, run M_e for j steps, rejecting exactly when it has halted; the height of p_e is exactly h_{p_e} . By [lemma 246.2](#), $p_e \in U_{\Gamma}$ exactly when $h_{p_e} = \infty$, and by [\(245.2\)](#) $E \sim_{\Gamma} p_e E$ exactly when $p_e \in U_{\Gamma}$. The write checked this argument. It bears on [questions 253.4](#) and [253.7](#): an effective theory of normalizers must be given the infinitely divisible primes, not only a membership test. The programme's review of source 26 (commit 3d87c3d6b) notes that the construction gives a nondivisible Γ , so it says nothing about the divisible setting of Part XX.

Remark 255.5 (The cited Example 6.3 of Mardešić, Resman, Rolin and Županović). [Remark 244.2](#) cites Example 6.3 of [\[151\]](#); the placement left the citation unchecked. The write read it on 4 October 2026 in arXiv:1505.05929v2 (10 March 2016, 42 pages; its running title is the journal title). The example takes a parabolic formal diffeomorphism $f(x) = x + x^{k+1} + o(x^{k+1})$ in $\mathbb{R}[[x]]$, recalls that its standard formal normal form in $\mathbb{R}[[x]]$ is $x + x^{k+1} + bx^{2k+1}$ with a residual coefficient b , and proves that in the paper's larger group of power-log changes of variables the normal form is $x + x^{k+1}$: one logarithmic elementary change removes the term x^{2k+1} , and the later terms are removed by changes in that group. So the example supports the remark: enlarging the coordinate group by logarithmic changes eliminates a classical residual parameter. It concerns normal forms of maps rather than of derivations, which is why the remark says "illustrates". The journal version (Adv. Math. **303**) was not consulted, so its numbering of the example is not checked.

Part XIX

Exact support complexity and strong summation in Hahn fields

[write] This part is source 25, manuscript 02 of batch 96, *Exact Support Complexity and Strong Summation in Hahn Fields: An order-obstruction classification and certificates that restore continuity* (25 pages, US letter, dated 4 October 2026; "research report prepared with ChatGPT for Vladimir Reshetnikov"). It arrived in commit 7be14aa84, as archive glazer_proveit_support_complexity, and was placed in 54ece48ab; it is pinned to c39974f12, 38 commits before the placement, when this report already had Parts I–XVII written. Its file prefix in this report is 25-support-complexity-. All unmarked text in this part is source 25's; its fourteen sections are Sections [256–269](#) (its Section k is the k th of them, with statement numbers otherwise unchanged), its eleven research questions are Questions [268.1–268.11](#), and its appendix is [appendix AN. Section 270](#) [merge] was written for this report.

What this part claims, and its status

For a fixed countable linear order L , the set $\text{WO}(L) \subseteq 2^L$ of well-ordered supports and the set $V(L) \subseteq \mathbb{R}^L$ of real coefficient arrays with well-ordered support are the whole space when L is well ordered, Σ_2^0 -complete when L is not well ordered and $B = \omega^* \cdot \omega$ does not embed in L , Π_3^0 -complete when L is scattered and B embeds, and Π_1^1 -complete when $\mathbb{Q}$ embeds; the binary predicates have exactly two nontrivial Borel Wadge degrees, those of Fin and $\text{Fin}^{\mathbb{N}}$ (theorem 259.6). For countable ordered abelian groups the Σ_2^0 case is exactly $\Gamma \cong \mathbb{Z}$ (theorem 260.1). Strongly summable countable families form a Π_3^0 -complete set already over $\mathbb{Z}$, and a Π_1^1 -complete one over non-scattered groups (theorem 261.2); strong summation is Baire class 1 and nowhere continuous in the raw product topology (theorem 262.1). Closed Polish spaces of integer or real support and incidence certificates make summation continuous; integer certificates admit a Σ_2^0 -measurable section but no Baire-class-1 section, real ones an explicit Baire-class-1 section obtained from continuous regularizations that converge exactly on legal inputs (theorems 263.2, 263.3, 264.2 and 264.3). Non-scattered groups admit no exact standard Borel coefficient-observable naming system (theorem 265.2), and scatteredness of the compatible orders on one fixed free abelian group with a prescribed least positive element is Π_1^1 -complete (theorem 266.3). Proposition 267.1 transfers the family results to the Lean predicate `PowerSeriesSummable`, as a statement about it, not a Lean theorem. Section 270 [merge] identifies the results of Parts II and V that source 25 proves again, specializes the classification to countable subgroups of $\mathbb{R}$ (bearing on Part XVII's question 237.1), proves the effectivity sentence of section 264.2, and records two computability observations of the Hilbert's-tenth programme's review of the archive. Unrefereed, AI-assisted, not formalized; priority is not established, and the Π_3^0 upper bound for scattered orders is prior art (Guevara-Uzcátegui) by source 25's own attribution. The finite checks test finite formulas only.

Questions answered. Source 25 answers Part II's question 28.3 (its "Question 12.3" of "the earlier project report"; source 03's question, printed Question 28.3 here) in its Borel-level and Wadge-degree parts: for a fixed countable scattered L the least Borel level of $\text{WO}(L)$ is Σ_2^0 or Π_3^0 according as B does not or does embed in L (the whole space for well-ordered L), and the nontrivial Wadge degrees are exactly those of Fin and $\text{Fin}^{\mathbb{N}}$. The question's first sentence asks to compute the level "directly from an economical Hausdorff decomposition"; that clause is not answered (source 25 says so, and its questions 268.2 and 268.9 carry it), and a dated note after the question re-scopes it. Two of source 25's research questions repeat Part II's: question 268.5 is question 28.4 and question 268.8 is question 28.5; each is merged by a pointer below. Through proposition 270.1 [merge] source 25 bears on Part XVII's question 237.1 without answering it.

What was already here, and what is new. Source 25 knew Part II only as source 03's PDF in the requester's library (by its ledger, the title, the abstract and Question 12.3 were retrieved), did not read this report, and read the Lean file `Algebra/SurrealNumbers/Surreal/HahnSeries/StrongEvaluation.lean`. Its lemma 257.2 is classical and proved in Parts II and V; its cut test (theorem 258.2) sharpens part (i) of Part II's theorem 18.4 from "Borel" to Π_3^0 by a different proof; lemma 259.4 and proposition 259.5 are Part II's lemma 18.2, part (ii) of theorem 18.4 and the non-scattered half of lemma 20.1; parts (i) and (ii) of Part II's theorem 19.1 are the cases $\mathbb{Z}$ and $\mathbb{Z}_{\text{lex}}^d$ of theorem 259.6, and Part V's theorem 67.3 with $k = \mathbb{R}$ is the case $\Gamma = \mathbb{Q}$

of [theorem 260.1](#); [corollary 260.3](#) is Part II's [corollary 20.4](#) with completeness added; and the equivalence of scatteredness with exact naming in [theorem 265.2](#) is (i) $\Leftrightarrow$ (iii) of Part II's [theorem 20.2](#). Each is flagged in a note and printed as a second route. New to the report are the forbidden-order criterion for the Σ_2^0 case and the exact four-way classification, the group classification, the strong-family classification, nowhere-continuity of strong summation, the certificate spaces and the selector and regularization theorems, the strong-family and witness-relation clauses of the naming barrier, the fixed-group uniformity barrier, and the statement about `PowerSeriesSummable`.

Labels and notation. Every label of source 25 takes the prefix `pma:spc:`. Labels were added to its unlabelled numbered statements (`pma:spc:n<number>`), its unlabelled conclusion (`pma:spc:n14`), its eleven research questions (`pma:spc:nq<its question number>`), the subsections cited in notes (`pma:spc:nsec<number>`), and to `pma:part:spc`; statements written for this report carry `pma:spc:w-`. The write also labelled four questions of Part II so that notes can cite them: `pma:bor:q:exactcomplexity`, `pma:bor:q:subfields`, `pma:bor:q:observability` and `pma:bor:q:certified`. The block of the table in [section 1.3](#) for Parts XVIII–XXI lists the symbols. In short: source 25's $\mathcal{H}_L$ is printed $V(L)$, because for groups it is Part II's V_Γ ($V(\Gamma) = V_\Gamma$), while $\mathcal{H}$ already names Part II's $\mathcal{H}_Q$, Part V's $\mathcal{H}_k$ and Part XV's $\mathcal{H}^\Gamma$ (tempting false reading: $V(L)$ is not the coded group V_L of Parts IX and XIII); its certificate spaces $C_\Gamma^{\mathbb{N}}$, $C_\Gamma^{\mathbb{R}}$ are printed $\text{Cert}_\Gamma^{\mathbb{N}}$, $\text{Cert}_\Gamma^{\mathbb{R}}$, because C is Part II's code set; its unit arrays E_k (proof of [theorem 262.1](#)) are printed $\mathbf{e}_k$ and its finite windows E_k ([section 264.1](#)) are printed Ω_k , because E , E_i are Part XVII's Euler derivations; the Σ_2^0 set $E = \bigcup_j F_j$ and the sets E_i in the proof of [lemma 257.2](#) are printed K , K_i ; its supports $A \subseteq L$ are printed S , as in Part II, and its other sets A , B (the Wadge definition in [section 257](#) and [definition 265.1](#)) are printed C , C' , because $\mathcal{A}_{\text{fin}}$ is Part II's ring; its clopen partitions $\mathcal{P}_n$ are printed $\mathcal{Q}_n$, because $\mathcal{P}$ is Part II's Puiseux field and $\mathcal{P}_k$ Part V's; and its ideal $\mathcal{I}$ is printed $\mathcal{J}$, because $\mathcal{I}_k$ is Part V's integer part and $\mathcal{I}$ Part XVII's support family. Its sans-serif KB is printed with the report's roman KB, and its macros `\sig`, `\pie` are the report's Σ_2^0 , Π_3^0 , Π_1^1 (same symbols). The letters F_i , $Z_{i,j}$, h_i , s , v_k , w_k , R_k , G , G' , G_L , G'_L are local to this part: s is the strong sum, not Part XVII's time; v_k is a soft threshold, not a valuation; G_L is not Part IX's G_L . Source 25 has no symbol δ (its `\del` is defined and unused). No normalization changed; its equations, numbered consecutively in the source, are numbered by section, and its “Research questions” share the report's counter.

Files. Source 25's program, recorded run, Makefile and source ledger are shipped as `code/25-support-complexity-checks.py`, `data/25-support-complexity-checks.txt`, `code/25-support-complexity-Makefile` and `25-support-complexity-sources.md`. The ledger calls Part II “a project artifact . . . not bundled in this package”; it was in the repository at the pin (note in [section 256](#)). The program checks with bare assert statements, so under `python -0` it prints the same “all passed” record while checking nothing ([appendix AN](#)). The Hilbert's-tenth programme reviewed the archive in full ([section 270](#)).

256 The intersection and the question answered

Elliot Glazer's *A Topological Tennenbaum Theorem* studies how arithmetic interacts with Borel definability and Polish topology. Its abstract states that in Borel models of arithmetic on an uncountable Polish space neither addition nor multiplication is continuous [1]. This is a documented research connection, rather than an inference from a job title or an inaccessible profile. On the other side, *ProveIt* contains a substantial Hahn-series and surreal-number

development; its `StrongEvaluation.lean` file expresses strong summability through a well-founded union of supports and finite coefficient incidence, and proves preservation of source sums by admissible evaluation [156].

These themes meet at a precise representation problem: before asking whether an infinite algebraic operation is continuous, how difficult is it to recognize its legal inputs, and what additional information makes the operation continuous? The answer depends dramatically on the exponent order and on whether the representation contains explicit support-completion information.

The earlier project report *Borel Presentations and Support Barriers in Omnific Arithmetic* asks, in Question 12.3, for the exact Borel levels and Wadge degrees of $\text{WO}(L)$ for countable scattered L , beyond finite lexicographic rank [158]. We answer those two classification questions. An effective procedure extracting certificates from an arbitrary supplied Hausdorff decomposition is a separate issue, not claimed here.

[write] (Batch 96, 4 October 2026.) Correction. The “earlier project report” is source 03 of this report, printed as [Part II](#), and its Question 12.3 is Part II’s [question 28.3](#). Source 25 read it as a Library PDF and its ledger (`25-support-complexity-sources.md`) calls it a project artifact “not bundled in this package”; but source 03 was placed in this repository in `3d2177df4` and written into this report well before source 25’s pin `c39974f12`, at which the report had Parts I–XVII. Source 25 did not read this report’s article. The answer is partial: the dated note after [question 28.3](#) records the answered parts and re-scopes the rest. The shipped ledger keeps its wording.

256.1 The principal statements

Write $B = \omega^* \cdot \omega$ for an increasing sequence of reversed natural-number blocks. Its concrete order is

$$(n, m) <_B (n', m') \iff n < n' \text{ or } (n = n' \text{ and } m > m'). \quad (256.1)$$

Thus a subset of B is well ordered exactly when it is finite in every block. Throughout, $\hookrightarrow$ denotes an order embedding, not necessarily a convex embedding.

For every countable linear order L , both binary support validity $\text{WO}(L) \subseteq 2^L$ and real coefficient validity $V(L) \subseteq \mathbb{R}^L$ have the following exact pointclass:

Order condition	Complexity
L is well ordered	the entire ambient space
L is not well ordered and $B \not\hookrightarrow L$	Σ_2^0 -complete
L is scattered and $B \hookrightarrow L$	Π_3^0 -complete
$\mathbb{Q} \hookrightarrow L$	Π_1^1 -complete.

For binary supports this gives exactly two nontrivial Borel Wadge degrees: those of Fin and $\text{Fin}^{\mathbb{N}}$.

For a nontrivial countable ordered abelian group Γ , the single-series Σ_2^0 case is exactly $\Gamma \cong \mathbb{Z}$. Strongly summable *families*, however, already form a Π_3^0 -complete set over $\mathbb{Z}$. The extra complexity comes from independently bounding the number of participating rows at every exponent, not from increasing the order rank.

The constructive part of the article produces two kinds of certified names. Both form closed Polish spaces and support continuous strong summation. Integer-valued bounds and real-valued bounds certify exactly the same legal arrays, but have different selector behavior.

The real-valued construction admits continuous finite-support approximations whose limit exists precisely on the legal domain. This also explains why all finite tests can succeed on an illegal infinite input.

256.2 What is already known, and what is proved here

Guevara and Uzcátegui’s Theorem 4.13 identifies the Borel restrictions of the well-order ideal with scattered orders; their Theorem I places the resulting ideals in $F_{\sigma\delta}$ [154]. Consequently the general scattered-order Π_3^0 ceiling is *not new*. Their classification up to ideal isomorphism is finer than the coarse pointclass classification considered here. We give an independent cut-based proof because it yields the height data needed for the certificate constructions.

The report’s proof package consists of the explicit forbidden-order criterion for the F_σ case, its applications to coefficient codes and exponent groups, the exact strong-summability classification, the certificate and selector theorems, and the fixed-group uniformity barrier. All are proved below. Whether some or all formulations already occur elsewhere has not been settled by the source search; “proved here” must not be read as “first proved in the literature.”

Hall, Knight, and Lange study complexity of order-type restrictions and operations on well-ordered subsets of ordered abelian groups [155]. Their question is distinct from recognizing *arbitrary* proposed supports as well ordered. We do not claim to resolve an open problem in that paper.

[write] (Batch 96.) Within this report, the qualitative dichotomy (Borel exactly for scattered L) is Part II’s [theorem 18.4](#), proved there by induction through the Hausdorff construction; the benchmark levels for $\mathbb{Z}$ and $\mathbb{Z}_{\text{lex}}^d$ are Part II’s [theorem 19.1](#), attained by different sets in Part V’s [theorem 67.2](#). The write did not read the Guevara–Uzcátegui or Hall–Knight–Lange papers; the attributions are source 25’s ([question 268.12](#)).

256.3 Scope and the connection to Glazer

All objects below are sets in ZFC. We use countable exponent orders, real coefficients with their usual topology, and countable products. No topology on the proper class of all surreal numbers is formed. A suitable surreal realization can reverse the exponent convention, identifying t^γ with $\omega^{-\gamma}$, but none of the proofs needs that realization.

Glazer’s arithmetic obstruction is motivation and context, not a lemma used to prove the support classification. Hahn fields are not, merely by being ordered fields, the arithmetic models in his theorem. Moreover, redundant certificate spaces are spaces of *names*, not arithmetic universes with unique representatives. Our positive continuity constructions therefore do not evade or contradict his theorem.

257 Conventions and descriptive benchmarks

For a countable linear order L , define

$$\text{WO}(L) = \{S \subseteq L : (S, <) \text{ is well ordered}\}, \quad (257.1)$$

$$V(L) = \{a \in \mathbb{R}^L : \text{supp}(a) \in \text{WO}(L)\}, \quad \text{supp}(a) = \{\ell : a_\ell \neq 0\}. \quad (257.2)$$

We include the empty support. A linear order is *scattered* when it contains no copy of $(\mathbb{Q}, <)$. In a countable linear order, failure of well-ordering is equivalent to the existence of an infinite

strictly decreasing sequence. Subsets and finite unions of well-ordered subsets are well ordered. Thus $\text{WO}(L)$ is an ideal, with the whole power set allowed as an improper ideal.

When $L = \Gamma$ is an ordered abelian group, $V(\Gamma)$ is the coefficient space of the full real Hahn field $\mathbb{R}((t^\Gamma))$. We use its usual formal series interpretation. Nothing in the support proofs depends on analytic convergence of t or on a valuation topology. The raw topology is always the subspace topology inherited from the coefficient product.

[write] $\text{WO}(L)$ is Part II's set of the same name, and $V(\Gamma)$ is Part II's V_Γ (section 20), the canonical coefficient presentation of $K_\Gamma = \mathbb{R}((t^\Gamma))$.

The symbols Σ_2^0 , Π_3^0 , and Π_1^1 mean, respectively, F_σ , $F_{\sigma\delta}$, and coanalytic. Complexity is *boldface*: no computability of L , its cuts, or its presentations is assumed. Completeness means membership in the indicated pointclass and continuous hardness for subsets of zero-dimensional Polish spaces. This convention is important when the target ambient space is $\mathbb{R}^L$ rather than a Cantor cube.

For sets $C \subseteq X$ and $C' \subseteq Y$, $C \leq_W C'$ means that some continuous $f : X \rightarrow Y$ satisfies $C = f^{-1}(C')$. Wadge equivalence is mutual reducibility. The classical facts used without full rederivation are the perfect-set theorem for uncountable closed subsets of a Polish space, strictness of the Borel hierarchy, coanalytic completeness of well-founded trees, and the analytic-image theorem for standard Borel spaces [8].

Definition 257.1 (The two Borel benchmarks). Let

$$\text{Fin} = \{u \in 2^{\mathbb{N}} : \{n : u(n) = 1\} \text{ is finite}\}, \quad (257.3)$$

$$\text{Fin}^{\mathbb{N}} = \{u \in 2^{\mathbb{N} \times \mathbb{N}} : \forall i \{j : u(i, j) = 1\} \text{ is finite}\}. \quad (257.4)$$

The second notation means independent row finiteness; it is not a Fubini product modulo finitely many bad rows.

Lemma 257.2. Fin is Σ_2^0 -complete and $\text{Fin}^{\mathbb{N}}$ is Π_3^0 -complete.

Proof. The upper bounds follow from

$$u \in \text{Fin} \iff \exists N \forall n \geq N \ u(n) = 0, \quad u \in \text{Fin}^{\mathbb{N}} \iff \forall i \exists N \forall n \geq N \ u(i, n) = 0.$$

Here and below a universally quantified collection of coordinate zero conditions is closed.

For completeness of Fin , let $K = \bigcup_j F_j$ in a zero-dimensional Polish space X , where the closed sets F_j increase with j . Choose countable clopen partitions Q_n refining one another so that the cells $C_n(x)$ containing x form a neighborhood base. Define

$$r_n(x) = \min(\{j \leq n : C_n(x) \cap F_j \neq \emptyset\} \cup \{n+1\}).$$

Each r_n is locally constant, and $r_{n+1} \geq r_n$. Membership in some F_j makes this sequence bounded; conversely, if it is bounded then it eventually has a constant value j , and every sufficiently small cell meets F_j , so closedness gives $x \in F_j$. Therefore

$$f(x)(n) = 1 \iff r_{n+1}(x) > r_n(x)$$

defines a continuous reduction of K to Fin .

For $K = \bigcap_i K_i$ with each K_i in Σ_2^0 , reduce K_i to the i th row of $\text{Fin}^{\mathbb{N}}$. The product of these coordinate maps is continuous. $\square$

[write] (Batch 96.) These are classical. In this report Part II proves them inside the proof of [theorem 19.1](#) (by the Baire category theorem and the completeness criterion for Fin , and by products of reductions for $\text{Fin}^{\mathbb{N}}$), and Part V proves them again after [theorem 67.2](#) by the same monotone-index construction as here, there in Baire space with closed-set trees. The proof above is a third route.

Remark 257.3 (A topology distinction that must not be suppressed). The support map $a \mapsto \text{supp}(a)$ from $\mathbb{R}^L$ to 2^L is Borel, but generally not continuous at a zero coefficient. Upper bounds for $V(L)$ will therefore be proved by direct zero-coordinate formulas, not by silently composing a continuous support map. Also, $V(L)$ and $\text{WO}(L)$ having the same complete pointclass does not assert Wadge equivalence across these ambient spaces. If L is nonempty, $\mathbb{R}^L$ is connected and every continuous map from it to a Cantor cube is constant. Our Wadge-degree answer concerns the binary predicates.

258 Countable cuts and a third-level upper bound

A cut is a downward-closed subset of L , with the empty and full cuts included. We regard $\text{Cut}(L)$ as a closed subspace of 2^L : any violation of downward closure is witnessed by two coordinates.

Lemma 258.1 (Countable cuts). *If L is a countable scattered linear order, then $\text{Cut}(L)$ is countable.*

Proof. Suppose instead that $\text{Cut}(L)$ is uncountable. The perfect-set theorem gives a nonempty perfect closed subset P of $\text{Cut}(L)$. Recursively construct nonempty relatively clopen perfect subsets P_s for $s \in 2^{<\mathbb{N}}$ and splitting coordinates $\ell_s \in L$. Start with $P_\emptyset = P$. Since P_s is not a singleton, choose a coordinate on which two of its cuts differ and set

$$P_{s0} = \{D \in P_s : \ell_s \notin D\}, \quad P_{s1} = \{D \in P_s : \ell_s \in D\}.$$

Both sets are nonempty, relatively clopen, and perfect.

Every splitting coordinate in the left subtree of s is less than ℓ_s : a cut in that subtree contains the new splitting coordinate but omits ℓ_s , and downward closure forces the inequality. Similarly every splitting coordinate in the right subtree is greater than ℓ_s . Consequently the map $s \mapsto \ell_s$ embeds the full binary tree with its in-order linear order into L . The in-order on the finite binary strings is countable, dense, and has no endpoints: left and right descendants supply points on both sides, and a suitable descendant lies between any two ordered nodes. It is therefore a copy of $\mathbb{Q}$, contradicting scatteredness. $\square$

List the nonempty final segments of L that have no least element as $(F_i)_{i \in I}$, where I is a finite or countable index set. For each one choose a strictly decreasing coinital sequence

$$c_{i,0} > c_{i,1} > c_{i,2} > \cdots \quad \text{in } F_i. \quad (258.1)$$

Such a sequence is obtained by enumerating F_i and repeatedly choosing a point below the current point and the next enumerated point. Put

$$Z_{i,j} = F_i \cap (-\infty, c_{i,j}). \quad (258.2)$$

For fixed i these sets decrease with j , and their intersection is empty.

Theorem 258.2 (The cut test). *For a countable scattered L and $S \subseteq L$,*

$$S \in \text{WO}(L) \iff \forall i \in I \exists j \in \mathbb{N} \quad S \cap Z_{i,j} = \emptyset. \quad (258.3)$$

In particular, $\text{WO}(L)$ and $V(L)$ belong to Π_3^0 .

Proof. If S is well ordered and $S \cap F_i$ is nonempty, let a be its least element. Choose j with $c_{i,j} \leq a$. Then $S \cap Z_{i,j}$ is empty. If $S \cap F_i$ is empty, any j works.

Conversely, suppose $d_0 > d_1 > \dots$ lies in S . The upward closure

$$F = \{x \in L : \exists n \, d_n \leq x\}$$

is a nonempty final segment with no least element. It is therefore one of the F_i . For every $c \in F$, some $d_n \leq c$, whence $d_{n+1} < c$. Thus S meets $F \cap (-\infty, c_{i,j})$ for every j , contradicting the right-hand side of (258.3).

For binary supports, $S \cap Z_{i,j} = \emptyset$ is closed in 2^L . For real codes its counterpart is

$$\forall \ell \in Z_{i,j} \quad a_\ell = 0,$$

which is closed in $\mathbb{R}^L$. A countable intersection of countable unions of these closed sets is Π_3^0 . $\square$

This proves directly the upper bound supplied by the prior ideal-theoretic classification. More importantly for later sections, it provides numerical height functions.

[write] (Batch 96.) A second route to, and a sharpening of, part (i) of Part II's [theorem 18.4](#): Part II proves only that $\text{WO}(L)$ is Borel, by induction through the Hausdorff construction with [lemma 18.3](#), and remarks that its decomposition need not be computable; the cut test gives the level Π_3^0 directly, and for real codes it avoids the support map, which Part II's [lemma 20.1](#) uses as a Borel map. The cut test uses countability of $\text{Cut}(L)$ instead of a Hausdorff decomposition.

Definition 258.3 (Cut heights). For $i \in I$ and $\ell \in L$, define

$$h_i(\ell) = \min\{j \in \mathbb{N} : \ell \notin Z_{i,j}\}. \quad (258.4)$$

This is always finite, by coinitality; it is zero outside F_i .

Since $Z_{i,j}$ decreases with j ,

$$S \in \text{WO}(L) \iff \forall i \in I \quad \sup(\{h_i(\ell) : \ell \in S\} \cup \{0\}) < \infty. \quad (258.5)$$

This is a family of ordinary integer bounds, rather than an ordinal-valued rank certificate. Its existence is nonuniform in the fixed order L .

Example 258.4 (Two transparent height tables). For $L = \mathbb{Z}$, the only nonempty final segment without a least element is L itself. Taking $c_j = -j$ gives $h(\ell) = \max(0, -\ell)$: legal supports are exactly the subsets bounded below.

For $L = \mathbb{B}$, take F_i to be the union of all blocks numbered at least i and $c_{i,j} = (i, j)$. Then $Z_{i,j} = \{(i, m) : m > j\}$ and $h_i(n, m) = m$ if $n = i$, while it is zero otherwise. The cut test is exactly finiteness in each reversed block.

259 The forbidden-order classification

259.1 When one countable list of supports suffices

An ideal $\mathcal{J} \subseteq 2^L$ is *countably generated* if there are $W_0, W_1, \dots \in \mathcal{J}$ such that every member of $\mathcal{J}$ is contained in some W_j . Closing an ordinary list of ideal generators under finite unions gives this equivalent form.

Lemma 259.1 (Well-ordered interval classes). *Define $x \sim y$ when the closed interval with endpoints x, y is well ordered. Then $\sim$ is an equivalence relation with convex classes. If C is one such class, each upper ray $C_{\geq c}$, $c \in C$, is well ordered.*

Proof. Reflexivity and symmetry are immediate. The interval between x and z is contained in the union of the intervals between x, y and between y, z ; finite unions and subsets preserve well-ordering, proving transitivity. Convexity follows in the same way.

If $C_{\geq c}$ contained a decreasing sequence $d_0 > d_1 > \dots$, that sequence would lie in $[c, d_0]$, a well-ordered interval because $c \sim d_0$. This is impossible. $\square$

Lemma 259.2 (The obstruction creates generators). *If $B \not\rightarrow L$, then $\text{WO}(L)$ is countably generated. Consequently both $\text{WO}(L)$ and $V(L)$ are F_σ .*

Proof. Let $J = L/\sim$ be the ordered set of convex classes from lemma 259.1. If J had an increasing sequence $C_0 < C_1 < \dots$, choose $x_k \in C_k$. Each interval $[x_{2n}, x_{2n+1}]$ is not well ordered, since its endpoints belong to distinct classes. Choose a decreasing ω -sequence in each such interval. These sequences are separated and increase as blocks, so their union is a copy of B . Therefore J has no increasing ω -sequence: it is reverse well ordered.

A well-ordered $S \subseteq L$ meets only finitely many classes. Indeed, an infinite collection of occupied classes in a reverse well-order contains a strictly decreasing sequence of classes, and one point of S from each would be a decreasing sequence in S . In each occupied class C , let a_C be the least point of $S \cap C$. Then

$$S \subseteq \bigcup_{C \text{ occupied}} C_{\geq a_C}.$$

Every such finite union is well ordered by lemma 259.1. There are only countably many possible finite unions, because L is countable. Enumerate them as (W_j) , including the empty set.

It follows that

$$\text{WO}(L) = \bigcup_j 2^{W_j}, \quad V(L) = \bigcup_j \{a \in \mathbb{R}^L : \forall \ell \notin W_j \ a_\ell = 0\}. \quad (259.1)$$

Each displayed piece is closed in the appropriate ambient space. $\square$

259.2 The two Borel lower bounds

Lemma 259.3. *If L is not well ordered, $\text{WO}(L)$ and $V(L)$ are Σ_2^0 -hard. If $B \hookrightarrow L$, they are Π_3^0 -hard.*

Proof. For the first assertion choose $d_0 > d_1 > \dots$ in L . Send $u \in 2^{\mathbb{N}}$ to the support $\{d_n : u(n) = 1\}$, or to its 0–1 coefficient array. A subset of this reversed sequence is well ordered exactly when it is finite. Both maps are continuous and reduce Fin .

For the second assertion, fix an embedding $e : B \hookrightarrow L$. Send a binary matrix u to $\{e(i, j) : u(i, j) = 1\}$. This support is well ordered exactly when each row is finite: finite subsets of the reversed blocks, summed in increasing ω order, form a well-order; an infinite row is a decreasing sequence. Again the characteristic-coefficient version is continuous. Apply lemma 257.2. $\square$

259.3 The coanalytic lower bound

We recall the reduction in detail to specify exactly which unrestricted-input problem is hard. Let Tr be the closed space of prefix-closed subsets of $\mathbb{N}^{<\mathbb{N}}$. Equip the finite sequences with the Kleene–Brouwer order: a proper extension precedes its prefix, and otherwise comparison is by the first unequal natural-number entry.

Lemma 259.4 (A fixed order carries all tree obstructions). *For $T \in \text{Tr}$, the order KB restricted to T is well ordered if and only if T has no infinite branch.*

Proof. The successive prefixes of an infinite branch form a strictly decreasing Kleene–Brouwer sequence. In the other direction, assume T is well founded under proper extension. By well-founded induction on nodes, the subtree above a node has well-ordered Kleene–Brouwer order: it is the increasing natural-number-indexed sum of the child subtrees, followed by the node itself. A well-ordered sum of well-orders is well ordered. Applying this at the root proves the assertion; the empty tree is harmless. $\square$

Every countable linear order embeds in $\mathbb{Q}$. To see this, enumerate its elements and successively place each new image between its finitely many already placed lower and upper neighbors; the density and lack of endpoints of $\mathbb{Q}$ always leave an available rational. In particular, fix an embedding of the *entire* Kleene–Brouwer order on $\mathbb{N}^{<\mathbb{N}}$ into $\mathbb{Q}$.

Proposition 259.5. *If $\mathbb{Q} \hookrightarrow L$, both $\text{WO}(L)$ and $V(L)$ are Π_1^1 -complete.*

Proof. Compose the fixed Kleene–Brouwer embedding with an embedding of $\mathbb{Q}$ into L . Sending T to the image of its nodes is a continuous reduction of the coanalytic-complete set $\text{WF} \subseteq \text{Tr}$ to $\text{WO}(L)$, by [lemma 259.4](#). Using coefficients 0 and 1 gives the same reduction to $V(L)$.

For the upper bound, nonvalidity is witnessed by a sequence $\ell_0 > \ell_1 > \dots$ of active coordinates. The relation between an array and such a sequence is Borel: it consists of countably many order comparisons and nonzero-coordinate conditions. Its projection is analytic, so validity is coanalytic. The same argument applies to binary supports. $\square$

[write] (Batch 96.) [Lemma 259.4](#) and [proposition 259.5](#) prove again Part II’s [lemma 18.2](#) (whose fixed embedding is μ_{KB} into $\mathbb{Q} \cap (0, 1)$), part (ii) of [theorem 18.4](#), the non-scattered half of [lemma 20.1](#) and, for $L = \mathbb{Q}$ and real coefficients, Part V’s [theorem 67.3](#); the note after [lemma 18.2](#) lists the earlier lightface versions in the collection. The proofs are the same reduction; the argument is printed here because the classification below uses it.

259.4 The exact theorem and the answer to Question 12.3

Theorem 259.6 (Exact support classification). *For every countable linear order L , the four cases stated in [section 256](#) hold. In addition, the following are equivalent:*

$$\begin{aligned} B \not\hookrightarrow L; \quad & \text{WO}(L) \text{ is countably generated;} \\ \text{WO}(L) \in \Sigma_2^0; \quad & V(L) \in \Sigma_2^0. \end{aligned} \tag{259.2}$$

For scattered L that is not well ordered, the Wadge degree of $\text{WO}(L)$ is exactly that of Fin or of $\text{Fin}^{\mathbb{N}}$, with the latter occurring precisely when $B \hookrightarrow L$.

Proof. If L is well ordered, all of its subsets are well ordered. Otherwise [lemma 259.3](#) supplies the Σ_2^0 lower bound. When $B \not\hookrightarrow L$, [lemma 259.2](#) supplies the matching upper bound. When L is scattered and contains B , [theorem 258.2](#) and [lemma 259.3](#) give Π_3^0 completeness. In the non-scattered case use [proposition 259.5](#).

A countable generating list gives the closed-piece formulas ([259.1](#)). Conversely, if $B \hookrightarrow L$, the continuous $\text{Fin}^{\mathbb{N}}$ reduction prevents either validity set from being F_σ , by strictness of the Borel hierarchy. This proves ([259.2](#)). Finally, completeness and the zero-dimensional ambient Cantor cubes give the stated mutual continuous reductions for the binary predicates. $\square$

The answer is not an unbounded progression of Borel levels mirroring Hausdorff rank. There are only two nontrivial Borel degrees. High scattered-order rank and large ordinal support capacity may still matter for algebra or for finer ideal structure, but neither forces a higher validity pointclass. This settles the exact-level and Wadge-degree portions of the earlier project question; it does not provide an effective scatteredness test on arbitrary input orders.

[write] (Batch 96.) Parts (i) and (ii) of Part II's [theorem 19.1](#) are the cases $L = \mathbb{Z}$ and $L = \mathbb{Z}_{\text{lex}}^d$, $d \geq 2$, of this theorem: $\mathbb{Z}$ is not well ordered and contains no copy of B (a decreasing sequence in $\mathbb{Z}$ is unbounded below, so no block can lie below a later one), while $(n, m) \mapsto (n, -m - 1, 0, \dots, 0)$ embeds B in $\mathbb{Z}_{\text{lex}}^d$ (first coordinate most significant), which is the reduction ([19.1](#)) of Part II. Part (iii) of [theorem 19.1](#), on support order types, is not covered here. The theorem confirms and generalizes Part II's comment after that theorem that increasing finite order rank need not raise the Borel level: no countable order raises it beyond Π_3^0 without leaving the Borel sets.

260 Consequences for exponent groups and roots

Theorem 260.1 (The ordered-group specialization). *Let Γ be a countable ordered abelian group. Then*

Γ	$V(\Gamma)$ in $\mathbb{R}^\Gamma$
$\{0\}$	the entire ambient space
$\Gamma \cong \mathbb{Z}$	Σ_2^0 -complete
Γ scattered and noncyclic	Π_3^0 -complete
Γ non-scattered	Π_1^1 -complete.

Proof. Every nontrivial ordered group contains $u, 0, -u, -2u, \dots$ for some $u > 0$, so it is not well ordered. If a nontrivial ordered group has no least positive element, then between $a < b$ one can choose $a + c$ with $0 < c < b - a$. It is dense and contains a copy of $\mathbb{Q}$. Consequently a nontrivial scattered Γ has a least positive element u .

If $\Gamma = \mathbb{Z}u$, it has order type $\mathbb{Z}$; its well-ordered subsets are exactly those bounded below, giving the second case. Otherwise choose $a > 0$ outside $\mathbb{Z}u$. Then $a > nu$ for every natural number n . Indeed, if $a \leq nu$ and $n \geq 1$ is least with this property, then $0 < a - (n - 1)u \leq u$, forcing equality with u and hence $a = nu$, a contradiction.

Now

$$e(n, m) = (n + 1)a - (m + 1)u \quad (260.1)$$

embeds B in Γ : increasing m decreases the point within its block, while one extra a dominates every finite multiple of u and separates the blocks. Apply [theorem 259.6](#). $\square$

Exponent group	Single-series validity	Reason
$\mathbb{Z}$	Σ_2^0 -complete	One lower support bound suffices.
$\mathbb{Z}^d$, lexicographic, $d \geq 2$	Π_3^0 -complete	Scattered, noncyclic; contains B.
Any countable noncyclic scattered group	Π_3^0 -complete	Rank does not raise the Borel level.
$\mathbb{Q}$ or $\mathbb{Z}[1/2]$	Π_1^1 -complete	The exponent order is dense.
$\mathbb{Q} \times_{\text{lex}} \mathbb{Z}$	Π_1^1 -complete	Discrete, but contains $\mathbb{Q} \times \{0\}$.

[write] (Batch 96.) The case $\Gamma = \mathbb{Q}$ is Part V's [theorem 67.3](#) for $k = \mathbb{R}$, and the first two rows of the table are Part II's [theorem 19.1](#) transferred to real codes. For countable subgroups of $\mathbb{R}$ only the first, second and fourth rows occur ([proposition 270.1](#) [merge]).

Remark 260.2 (Discrete does not mean scattered). In $\mathbb{Q} \times_{\text{lex}} \mathbb{Z}$, the element $(0, 1)$ is least positive, but the suborder $\{(q, 0) : q \in \mathbb{Q}\}$ is dense. Replacing the scatteredness hypothesis by the existence of a least positive element would therefore make both the validity theorem and the certificate construction false.

Corollary 260.3 (A full-support root-closure barrier). *Suppose $\Gamma \neq \{0\}$ is countable and contains $u/2^n$ for every $n \in \mathbb{N}$ for some $u > 0$. Then $V(\Gamma)$ is Π_1^1 -complete. In particular, if the full Hahn field $\mathbb{R}((t^\Gamma))$ is real closed and $\Gamma \neq \{0\}$, its raw coefficient-validity set is Π_1^1 -complete.*

Proof. The subgroup generated by $\{u/2^n : n \in \mathbb{N}\}$ is an ordered copy of the dyadic rationals. It is a countable dense order without endpoints, so contains $\mathbb{Q}$.

For the last assertion, each positive monomial t^γ has a square root y . The least exponent of a product is the sum of the least exponents: the leading coefficient is the nonzero product of the two leading coefficients. Thus $y^2 = t^\gamma$ implies $2 \text{ val}(y) = \gamma$. Consequently Γ is 2-divisible and any $u > 0$ can be halved repeatedly. Apply the first assertion. $\square$

[write] (Batch 96.) A second route to Part II's [corollary 20.4](#), with two differences: the hypothesis needs only one $u > 0$ that can be halved indefinitely (Part II assumes square roots of all positive elements, hence 2-divisibility of Γ), and the conclusion adds Π_1^1 -completeness of the code set to Part II's nonexistence of a coefficient-observable standard Borel parametrization (which follows by [theorem 265.2](#) or by Part II's [theorem 20.2](#)).

The adjective *full* is essential. A real closed subfield whose elements obey additional support restrictions need not inherit the complexity of all well-ordered supports on its exponent alphabet. Puiseux-type restrictions and other bounded-support languages remain possible; the corollary is not an impossibility theorem for them.

261 The exact complexity of strong summability

Let Γ be a countable ordered abelian group. A raw family is an array $X = (x_{n,\gamma}) \in \mathbb{R}^{\mathbb{N} \times \Gamma}$. Its union support and incidence sets are

$$U(X) = \{\gamma : \exists n \ x_{n,\gamma} \neq 0\}, \quad I_\gamma(X) = \{n : x_{n,\gamma} \neq 0\}. \quad (261.1)$$

Definition 261.1 (Strongly summable family). The strong-summability domain is

$$\mathcal{S}_\Gamma = \{X : U(X) \in \text{WO}(\Gamma) \text{ and } \forall \gamma \in \Gamma \ I_\gamma(X) \text{ is finite}\}. \quad (261.2)$$

For $X \in \mathcal{S}_\Gamma$, its strong sum is the coefficient array

$$s(X)_\gamma = \sum_{n \in I_\gamma(X)} x_{n,\gamma}. \quad (261.3)$$

The union condition implies that each row is a legal Hahn series, and the support of the sum is a subset of $U(X)$. Thus $s(X) \in V(\Gamma)$. The two requirements in (261.2) are precisely the order-theoretic support and finite-incidence conditions used in the standard formal theory of Hahn summable families [157]. They are not analytic convergence conditions for a sequence of real numbers at each exponent.

Theorem 261.2 (Exact strong-summability classification). *In the raw product space $\mathbb{R}^{\mathbb{N} \times \Gamma}$,*

Γ	$\mathcal{S}_\Gamma$
$\{0\}$	Σ_2^0 -complete
$\Gamma \neq \{0\}$ scattered	Π_3^0 -complete
Γ non-scattered	Π_1^1 -complete.

Proof. For the trivial group, (261.2) says that a real sequence is eventually zero. This is F_σ , and its binary restriction is Fin.

Suppose Γ is scattered. Apply the cut test to $U(X)$. The union-support condition is

$$\forall i \in I \exists j \in \mathbb{N} \forall n \in \mathbb{N} \forall \gamma \in Z_{i,j} \quad x_{n,\gamma} = 0. \quad (261.4)$$

The incidence condition is

$$\forall \gamma \in \Gamma \exists N \in \mathbb{N} \forall n \geq N \quad x_{n,\gamma} = 0. \quad (261.5)$$

Both formulas are Π_3^0 with closed innermost conditions, giving the upper bound.

For the lower bound, choose $u > 0$ in a nontrivial Γ . Given a binary matrix $b(i, j)$, define a raw array by

$$x_{j,iu} = b(i, j), \quad x_{j,\gamma} = 0 \quad (\gamma \notin \{iu : i \in \mathbb{N}\}). \quad (261.6)$$

The union support is always contained in the well-order $\{0, u, 2u, \dots\}$. Thus it is automatically well ordered, and strong summability holds exactly when every row of b is finite. The map is continuous, so $\text{Fin}^{\mathbb{N}}$ reduces to $\mathcal{S}_\Gamma$.

If Γ is non-scattered, the union-support condition is coanalytic, using a decreasing sequence and a row index witnessing activity at each exponent. The incidence condition is Borel. Their intersection is coanalytic. Conversely, place any proposed Hahn coefficient array in row zero and make all other rows zero. This continuous map reduces $V(\Gamma)$ to $\mathcal{S}_\Gamma$, yielding coanalytic completeness. $\square$

Corollary 261.3 (The complexity jump already occurs over the integers). *For $\Gamma = \mathbb{Z}$, the single-series domain is Σ_2^0 -complete, whereas the strong-family domain is Π_3^0 -complete. This increase persists even on families whose union support is contained in the fixed well-order $\mathbb{N}u$.*

This distinguishes two sources of difficulty. One is finding a well-founded union of exponents. The other is finding an independent finite stopping point for each coefficient. Eliminating the first source does not eliminate the second.

262 Strong summation is nowhere continuous

Theorem 262.1 (Baire class one, but no continuity points). *For every countable ordered abelian group Γ , the strong-sum map $s : \mathcal{S}_\Gamma \rightarrow V(\Gamma)$ is Baire class 1 and is nowhere continuous in the raw coefficient topologies. Furthermore, $\mathcal{S}_\Gamma$ is meager in itself and is not Polish in its raw subspace topology.*

Proof. For $N \in \mathbb{N}$, the finite partial sum

$$s_N(X)_\gamma = \sum_{n < N} x_{n,\gamma}$$

is continuous as a map to $\mathbb{R}^\Gamma$. It takes $\mathcal{S}_\Gamma$ into $V(\Gamma)$, because a finite union of row supports is well ordered; hence it is also continuous with that subspace as target. At each fixed exponent, finite incidence implies eventual equality $s_N(X)_\gamma = s(X)_\gamma$. Therefore $s_N(X) \rightarrow s(X)$ in the coefficient product, proving Baire class 1.

Fix an exponent γ_0 , an array $X \in \mathcal{S}_\Gamma$, and let $\mathbf{e}_k$ be the array with coefficient 1 at (k, γ_0) and zero elsewhere. Then $X + \mathbf{e}_k$ remains strongly summable: its union support is contained in $U(X) \cup \{\gamma_0\}$ and every incidence set has changed by at most one index. Yet

$$X + \mathbf{e}_k \longrightarrow X, \quad s(X + \mathbf{e}_k) = s(X) + t^{\gamma_0}.$$

The output at γ_0 changes by exactly 1 for every k . Thus s is not continuous at X , and X was arbitrary.

For meagerness, let

$$D_N = \{X \in \mathcal{S}_\Gamma : \forall n \geq N \ x_{n,\gamma_0} = 0\}.$$

Each D_N is relatively closed and $\mathcal{S}_\Gamma = \bigcup_N D_N$. Given any basic relative neighborhood of a point of D_N , choose a row $k \geq N$ beyond all its observed coordinates and add $\mathbf{e}_k$. This stays in that neighborhood but leaves D_N , proving empty relative interior. Thus $\mathcal{S}_\Gamma$ is a nonempty space meager in itself, and cannot be Polish by the Baire category theorem. $\square$

There is no conflict with continuity-point theorems for Baire-class-1 functions on Polish or Baire domains. The legal domain here, with its raw topology, fails the necessary Baire property. It may be standard Borel in the scattered case without being Polish in this particular topology.

[write] (Batch 96.) It is standard Borel exactly in the scattered case ([proposition 270.1](#) [merge]).

The discontinuity witness is stronger than a numerical instability example: any finite set of observed input coordinates can remain exactly unchanged while a chosen output coefficient changes by 1. A topology that observes only more and more raw coefficients cannot infer that no further contribution will arrive. The next sections supply the missing information explicitly.

Lemma 262.2 (Connectedness of the legal family space). *$\mathcal{S}_\Gamma$ is a real vector subspace of $\mathbb{R}^{\mathbb{N} \times \Gamma}$ and is path connected.*

Proof. For $X, Y \in \mathcal{S}_\Gamma$, $U(X + Y) \subseteq U(X) \cup U(Y)$, a well-ordered set, and every incidence set of $X + Y$ is contained in a union of two finite sets. Scalar multiplication also preserves legality. The path $t \mapsto (1 - t)X + tY$ is coordinatewise continuous and stays inside $\mathcal{S}_\Gamma$. $\square$

263 Closed integer-certificate spaces

Assume in this and the following section that Γ is scattered. Fix once and for all the cut heights $h_i : \Gamma \rightarrow \mathbb{N}$ from [definition 258.3](#). A support bound records how far a support may penetrate each final-segment test. An incidence bound records how far down the row index a given exponent can still receive a contribution.

We use *inclusive* row bounds: m_γ permits rows $0, \dots, m_\gamma$. In particular a zero bound permits row zero, and an empty incidence set may also receive bound zero. This convention avoids a special negative value for empty sets.

Definition 263.1 (Integer-certified names). Let $\text{Cert}_\Gamma^\mathbb{N}$ be the set of triples

$$(X, b, m) \in \mathbb{R}^{\mathbb{N} \times \Gamma} \times \mathbb{N}^I \times \mathbb{N}^\Gamma$$

satisfying

$$x_{n,\gamma} \neq 0 \implies h_i(\gamma) \leq b_i \quad \text{for every } i, n, \gamma, \quad (263.1)$$

$$x_{n,\gamma} \neq 0 \implies n \leq m_\gamma \quad \text{for every } n, \gamma. \quad (263.2)$$

Give the natural numbers the discrete topology. Write $p(X, b, m) = X$ for the raw-array projection.

Theorem 263.2 (A closed Polish naming space). $\text{Cert}_\Gamma^\mathbb{N}$ is a closed Polish space and $p[\text{Cert}_\Gamma^\mathbb{N}] = \mathcal{S}_\Gamma$. Strong summation lifts to the continuous map

$$\tilde{s}(X, b, m)_\gamma = \sum_{n \leq m_\gamma} x_{n,\gamma} \quad \text{from } \text{Cert}_\Gamma^\mathbb{N} \text{ to } V(\Gamma). \quad (263.3)$$

Proof. The ambient product is Polish. A violation of (263.1) or (263.2) is open: it fixes a discrete bound and asks that one real coordinate be nonzero. The complement of $\text{Cert}_\Gamma^\mathbb{N}$ is a countable union of these open violations, so the name space is closed.

Any certified array has finite incidence by (263.2) and a well-ordered union support by (258.5), so it is strongly summable. Conversely, for a legal array define

$$b_i^*(X) = \max(\{h_i(\gamma) : x_{n,\gamma} \neq 0 \text{ for some } n\} \cup \{0\}), \quad (263.4)$$

$$m_\gamma^*(X) = \max(I_\gamma(X) \cup \{0\}). \quad (263.5)$$

Both maxima are finite. For the first, the set of heights is a bounded subset of $\mathbb{N}$; for the second use finite incidence. These bounds give a name for X , proving exactness of the projection.

In (263.3), each fixed m_γ is locally constant in the name space. Thus each output coefficient is locally a fixed finite sum of input coordinates. This proves continuity into the product, and hence into its subspace $V(\Gamma)$. $\square$

The certificates are infinite sequences of finite bounds, not a single finite proof string. The closedness theorem says that a violation is finitely observable *relative to the fixed height tables*. It does not make all validity questions decidable, or make the height tables effective for an arbitrary supplied order.

263.1 A Borel selector and two sharp obstructions

A section of p is a map $r : \mathcal{S}_\Gamma \rightarrow \text{Cert}_\Gamma^\mathbb{N}$ with $p \circ r = \text{id}$. The pointwise least bounds define

$$r_\mathbb{N}(X) = (X, b^*(X), m^*(X)). \quad (263.6)$$

Theorem 263.3 (Integer selectors). The section $r_\mathbb{N}$ is Σ_2^0 -measurable: preimages of open sets are F_σ relative to $\mathcal{S}_\Gamma$. Every section of p is nowhere continuous. In fact, there is no Baire-class-1 section with target $\text{Cert}_\Gamma^\mathbb{N}$.

Proof. For each fixed i, j , the condition $b_i^*(X) \leq j$ is closed relative to the legal domain: it says that all coordinates with $h_i(\gamma) > j$ vanish. Likewise $m_\gamma^*(X) \leq j$ is a closed condition. Consequently the fiber $b_i^* = j$ is a closed set minus a closed set (with the evident convention at $j = 0$), and the same is true of $m_\gamma^* = j$. Such fibers are locally closed and therefore F_σ in a metric space. A basic open set of the name space restricts an open set of raw coordinates and finitely many discrete bound coordinates. Its preimage is F_σ . A countable base gives the assertion for all open sets.

For any section r , the identity $s = \tilde{s} \circ r$ holds. Continuity of r at even one point would imply continuity of s there, contradicting [theorem 262.1](#).

Suppose nevertheless that a section r were a pointwise limit of continuous maps $r_k : \mathcal{S}_\Gamma \rightarrow \text{Cert}_\Gamma^\mathbb{N}$. Fix an exponent γ_0 . The m_{γ_0} coordinate of each r_k is a continuous map from the connected space $\mathcal{S}_\Gamma$ to discrete $\mathbb{N}$, and hence is constant. A convergent sequence of such constant maps has a constant limit. Thus $m_{\gamma_0}(r(X))$ would have one fixed value M for all X . But the legal array with coefficient 1 only at $(M + 1, \gamma_0)$ cannot have a certificate with incidence bound M . This contradiction rules out a Baire-class-1 section. $\square$

The distinction between Σ_2^0 -measurability and Baire class 1 is deliberate. They need not coincide for maps into disconnected targets. Treating these two notions as interchangeable would incorrectly strengthen the positive selector statement.

264 Real bounds and continuous regularization

The obstruction in [theorem 263.3](#) concerns the topology of the certificate values, not their logical sufficiency. We now allow bounds to move continuously through nonnegative real values. This changes the available approximating maps while preserving exact input legality.

Definition 264.1 (Real-certified names). Let $\text{Cert}_\Gamma^\mathbb{R}$ consist of triples

$$(X, b, m) \in \mathbb{R}^{\mathbb{N} \times \Gamma} \times [0, \infty)^I \times [0, \infty)^\Gamma$$

obeying the same implications (263.1)–(263.2). Equivalently, with $(t)_+ = \max(t, 0)$,

$$(h_i(\gamma) - b_i)_+ x_{n,\gamma} = 0 \quad \text{for every } i, n, \gamma, \quad (264.1)$$

$$(n - m_\gamma)_+ x_{n,\gamma} = 0 \quad \text{for every } n, \gamma. \quad (264.2)$$

Theorem 264.2 (Real bounds retain exactness and continuity). $\text{Cert}_\Gamma^\mathbb{R}$ is a closed Polish space with $p[\text{Cert}_\Gamma^\mathbb{R}] = \mathcal{S}_\Gamma$. The strong sum lifts to a continuous map $\tilde{s} : \text{Cert}_\Gamma^\mathbb{R} \rightarrow V(\Gamma)$. Every section of its projection is nowhere continuous.

Proof. Each equation in (264.1)–(264.2) is the zero set of a continuous real-valued function. Their countable intersection is closed in a Polish product. Any finite real bound on integer heights or row indices gives an integer bound, proving soundness; the least integer bounds (263.4)–(263.5) prove surjectivity.

Fix a name and an exponent γ . Choose an integer $M > m_\gamma$. On the relative neighborhood where the new m_γ remains less than M , every row with index at least M must have zero coefficient at γ . Therefore throughout that neighborhood the strong-sum coefficient is the fixed finite sum $\sum_{n < M} x_{n,\gamma}$. This proves coordinatewise, hence product, continuity. The section obstruction again follows from $s = \tilde{s} \circ r$ and [theorem 262.1](#). $\square$

Although bounds need not be locally constant, their local boundedness is enough for continuity. The remaining question is whether a section can be obtained as a pointwise limit of continuous names. Unlike the integer-target case, the answer is positive and admits a fully explicit construction.

264.1 The regularization formula

Fix increasing finite sets $\Omega_k \subseteq \mathbb{N} \times \Gamma$, $k \geq 1$, with union $\mathbb{N} \times \Gamma$. For $t \in \mathbb{R}$ define two continuous functions

$$v_k(t) = \text{sgn}(t) \max(|t| - 2/k, 0), \quad w_k(t) = \min(1, k|t|/2), \quad (264.3)$$

where $v_k(0) = 0$. They have the crucial compatibility property

$$v_k(t) \neq 0 \implies w_k(t) = 1. \quad (264.4)$$

For every raw array X , legal or not, put

$$X_{n,\gamma}^{[k]} = \begin{cases} v_k(x_{n,\gamma}), & (n, \gamma) \in \Omega_k, \\ 0, & (n, \gamma) \notin \Omega_k, \end{cases} \quad (264.5)$$

$$b_i^{[k]}(X) = \max(\{h_i(\gamma)w_k(x_{n,\gamma}) : (n, \gamma) \in \Omega_k\} \cup \{0\}), \quad (264.6)$$

$$m_\gamma^{[k]}(X) = \max(\{nw_k(x_{n,\gamma}) : (n, \gamma) \in \Omega_k\} \cup \{0\}), \quad (264.7)$$

$$R_k(X) = (X^{[k]}, b^{[k]}(X), m^{[k]}(X)). \quad (264.8)$$

Each finite stage removes sufficiently small coefficients, instead of trying to make a discontinuous exact zero test on them. The bound ramp reaches its full value before a regularized coefficient can become nonzero.

Theorem 264.3 (Exact convergence of certified regularizations). *The maps*

$$R_k : \mathbb{R}^{\mathbb{N} \times \Gamma} \longrightarrow \text{Cert}_\Gamma^{\mathbb{R}}$$

are continuous and defined on the entire raw space. For every raw array X ,

$$X \in \mathcal{S}_\Gamma \iff (R_k(X))_{k \geq 1} \text{ converges in } \text{Cert}_\Gamma^{\mathbb{R}}. \quad (264.9)$$

On the legal domain the limit is

$$r_{\mathbb{R}}(X) = (X, b^*(X), m^*(X)), \quad (264.10)$$

with the least bounds from (263.4)–(263.5). In particular, $r_{\mathbb{R}}$ is a Baire-class-1 section. For each individual bound coordinate its approximants are eventually exactly equal to the limiting integer value.

Proof. Every coordinate of R_k is a continuous function of finitely many input coefficients: this is immediate for $X^{[k]}$, and follows from continuity of finite maxima for the bound coordinates. These coordinate maps combine into a continuous product map.

The array $X^{[k]}$ has finite total support, hence is strongly summable. Whenever $X_{n,\gamma}^{[k]} \neq 0$, compatibility (264.4) gives $w_k(x_{n,\gamma}) = 1$. Thus $b_i^{[k]} \geq h_i(\gamma)$ and $m_\gamma^{[k]} \geq n$, proving that $R_k(X)$ is a valid real-certified name for every input X .

For every fixed coefficient, the finite sets Ω_k eventually contain its position and $v_k(t) \rightarrow t$. Hence $X^{[k]} \rightarrow X$ in the product topology. Suppose now that X is legal. Every term

contributing to $b_i^{[k]}$ is at most b_i^* , since a zero input has ramp value zero and a nonzero input has height at most b_i^* . If $b_i^* > 0$, some active coefficient attains that integer maximum. Once its position is included and k is large enough, its ramp equals 1, giving $b_i^{[k]} = b_i^*$ forever after. If $b_i^* = 0$, equality holds at every stage. The same argument applies to $m_\gamma^{[k]}$ using the largest active row. Thus the names converge coordinatewise to (264.10), which belongs to the closed name space.

Conversely, if $R_k(X)$ converges in $\text{Cert}_\Gamma^\mathbb{R}$, continuity of the raw projection and $X^{[k]} \rightarrow X$ imply that its limit is a name for X . Exactness of the projection forces $X \in \mathcal{S}_\Gamma$.

There is also a direct divergence description. The ramps increase with k and Ω_k increases, so every bound coordinate is nondecreasing. If some $I_\gamma(X)$ is infinite, $m_\gamma^{[k]} \rightarrow \infty$. If incidence is finite but $U(X)$ is not well ordered, the height criterion supplies an i for which active heights are unbounded, and $b_i^{[k]} \rightarrow \infty$. This proves the same converse and identifies a diverging coordinate on every illegal input. $\square$

Corollary 264.4 (The exact selector comparison). *The same pointwise least-bound rule defines a Σ_2^0 -measurable, nowhere-continuous section into either name space. As a map into $\text{Cert}_\Gamma^\mathbb{N}$ it is not Baire class 1, and no alternative section is Baire class 1. As a map into $\text{Cert}_\Gamma^\mathbb{R}$ it is Baire class 1.*

This does not change the topology on the subset of integer names. What changes is the *allowed range of continuous approximants*: real bounds let approximating names pass through noninteger values while coefficients are soft-thresholded. Baire class 1 is sensitive to this codomain enlargement.

264.2 What the regularization does and does not compute

At every stage, even an illegal raw array produces a valid finite-support name. No finite stage certifies legality of the infinite input. Exact legality is encoded in boundedness, or convergence, of the entire sequence of certificate coordinates. This is consistent with the Π_3^0 lower bound and with nowhere-continuity of the limiting selector.

If the exponent enumeration and all required height values are available effectively, the finite-coordinate formulas are implementable using ordinary continuous real operations. Without those hypotheses, they remain explicit mathematical formulas but are not an algorithm for arbitrary presentations. In particular, choosing all relevant cuts of an arbitrary countable scattered order is not made effective by writing down (258.4).

[write] (Batch 96.) The second paragraph is stated without proof; [remark 270.2](#) [merge] proves it in a precise form (each stage is a computable map of represented spaces, uniformly in k). It does not say that the limit is computable, and it is not: the Hilbert's-tenth programme's review of this archive exhibits a computable strongly summable binary array over $\mathbb{Z}$ with no uniformly computable incidence certificate, integer or real (notes after [remark 270.2](#)).

For $\Gamma = \mathbb{Z}$, the support part uses only $h(\gamma) = \max(0, -\gamma)$. A sequence of progressively more negative active exponents makes the support bound diverge. Repeated contributions at one fixed exponent make its incidence bound diverge. These two failure modes are separated by different coordinates of the same certificate space.

265 The non-scattered naming barrier

The positive constructions have a sharp scope. In the non-scattered case, allowing arbitrary standard Borel names does not suffice to name all legal coefficient arrays while keeping coefficients Borel observable.

Definition 265.1 (Exact coefficient-observable names). For a set C of raw arrays, an exact standard Borel naming system is a standard Borel space P and a map $d : P \rightarrow$ the raw product such that $d[P] = C$ and each coordinate of d is Borel. Repetition of names is allowed. Since the product has countably many coordinates, coordinatewise Borel observability is equivalent to d being a Borel map.

Theorem 265.2 (An exact naming dichotomy). *Let Γ be countable and ordered abelian. Both $V(\Gamma)$ and $\mathcal{S}_\Gamma$ have exact standard Borel coefficient-observable naming systems if and only if Γ is scattered. In the scattered case the name spaces may be chosen closed Polish, with continuous decoders. In the non-scattered case there is not even a Borel relation*

$$\mathcal{R} \subseteq (\text{raw product}) \times Y,$$

with Y standard Borel, whose projection onto the raw product is exactly the legal set.

Proof. For strongly summable families in the scattered case use $\text{Cert}_\Gamma^\mathbb{R}$ or $\text{Cert}_\Gamma^\mathbb{N}$ and their raw projection. For single series omit the row-incidence bounds and impose only

$$(h_i(\gamma) - b_i)_+ a_\gamma = 0 \quad (i \in I, \gamma \in \Gamma)$$

on $\mathbb{R}^\Gamma \times [0, \infty)^I$. This is a closed Polish space projecting exactly onto $V(\Gamma)$ by the height criterion.

In the non-scattered case, the two legal sets are coanalytic-complete and hence not analytic. The image of a standard Borel space under a Borel map into a Polish space is analytic, so no decoder with the stated range exists. Likewise the projection of a Borel relation with standard Borel witness space is analytic, proving the final assertion. $\square$

[write] (Batch 96.) For $V(\Gamma)$ the equivalence is (i) $\Leftrightarrow$ (iii) of Part II's [theorem 20.2](#), whose naming systems are its coefficient-observable standard Borel parametrizations ([definition 17.1](#)); Part II's proof of (i) $\Rightarrow$ (iii) uses V_Γ itself as the Borel domain, while the proof here gives closed Polish name spaces with continuous decoders. The clauses for $\mathcal{S}_\Gamma$ and for Borel witness relations are new to the report.

The barrier permits several legitimate alternatives: a restricted collection of supports, a non-Borel set of valid names, a decoder without the stipulated coefficient observability, or a different representation problem. It excludes none of these by implication. In particular, a syntax whose *ambient* strings form a Polish space but whose validity predicate is coanalytic is not an exact standard Borel naming system in the theorem's sense.

A related obstruction follows directly from the convergence formulation. For a sequence of continuous maps into a complete metric space, its set of convergence points is Π^0_3 , by the Cauchy criterion

$$\forall r \exists N \forall p, q \geq N \quad d(f_p(x), f_q(x)) \leq 2^{-r}.$$

Therefore the exact regularization theorem cannot extend to a non-scattered full-support domain simply by choosing a more ingenious Polish target. The complexity of that domain is already too high.

266 Uniform hardness on one fixed discrete group

The fixed-order Π_3^0 upper bound should not be confused with a uniform procedure recognizing which exponent orders admit it. We now make that separation precise, even when the underlying group and its least positive element are fixed in advance.

Let

$$G = \bigoplus_{n \in \mathbb{N}} \mathbb{Z}e_n, \quad G' = G \oplus \mathbb{Z}u.$$

Both are fixed countable free abelian groups. Compatible strict linear orders on G' form a closed subspace of $2^{G' \times G'}$: linearity, transitivity, and translation invariance are countably many finite-coordinate conditions. Requiring the fixed element u to be least positive is also closed.

Lemma 266.1 (Scattered sums and convex limits). *An ordered sum of scattered orders over a scattered index order is scattered. An increasing union of convex scattered suborders is scattered.*

Proof. For an ordered sum, suppose $f : \mathbb{Q} \hookrightarrow \sum_{j \in J} L_j$. If two rational points land in the same block, every rational point between them lands there as well, producing a copy of $\mathbb{Q}$ in that block. Otherwise the block projection of f is injective and gives a copy of $\mathbb{Q}$ in J . Both alternatives contradict scatteredness.

For an increasing union of convex suborders, two images of distinct rational points lie together in some suborder. Convexity places the entire rational interval between them in that suborder, again a contradiction. $\square$

For a linear order L on $\mathbb{N}$, order G by its largest active L -coordinate: a nonzero vector is positive if its coefficient at that largest coordinate is positive. Finite supports guarantee that the largest coordinate exists, even when L is not well ordered. Denote the resulting ordered group by G_L . Finally order

$$G'_L = G_L \times_{\text{lex}} \mathbb{Z}u$$

with the G_L coordinate dominant, so u is least positive.

Lemma 266.2. *G_L and G'_L are scattered if and only if L is well ordered. The map from L to the order relation of G'_L is continuous.*

Proof. If L has a decreasing sequence $\ell_0 >_L \ell_1 >_L \dots$, the subgroup spanned by the corresponding basis elements has no least positive element. Indeed, below any positive element one can put a positive basis element with index sufficiently far down this sequence. This subgroup is densely ordered, so contains a copy of $\mathbb{Q}$. Thus neither G_L nor G'_L is scattered.

If L has ordinal type α , prove that G_L is scattered by induction on α . At a successor stage, the leading integer coefficient partitions the group into a $\mathbb{Z}$ -indexed ordered sum of copies of the previous group. At a limit stage it is the increasing union of the groups supported on shorter initial segments, and these subgroups are convex. Apply [lemma 266.1](#) in both cases; the zero group starts the induction. Then G'_L is a sum of copies of $\mathbb{Z}$ indexed by the scattered order G_L , and is scattered as well.

To compare two fixed vectors in G'_L , only the finitely many L -comparisons among coordinates in their support difference are needed. Hence every output order-relation coordinate is locally constant as a function of L , proving continuity. $\square$

Theorem 266.3 (A fixed-group uniformity barrier). *Within the compact space of compatible orders on the fixed group G' for which u is least positive, the set of scattered orders is Π_1^1 -complete.*

Proof. Non-scatteredness is analytic: a witness is a map from $\mathbb{Q}$ into G' that is injective and preserves all pairwise order comparisons. These are Borel conditions on the order and the countable witness map.

For hardness, start with the coanalytic-complete set of well-orders on $\mathbb{N}$. [Lemma 266.2](#) gives a continuous reduction to scatteredness, and every output has the prescribed least positive element u . This proves the claim. $\square$

Every scattered group in this family is noncyclic and has Π_3^0 -complete single-series validity, yet recognizing which groups lie in that family is coanalytic-complete. There is no total Borel classifier that accepts exactly the scattered orders. A construction promised a suitable decomposition or height table is a different problem and is not excluded. This is why the certificate theorems fix the exponent order and its auxiliary data before analyzing raw coefficient arrays.

[write] (Batch 96.) This makes precise, for ordered groups, the remark after Part II's [theorem 18.4](#) that the dichotomy is not a statement about the complexity of the class of all scattered orders or about finding Hausdorff decompositions uniformly. The local letters G_L, G'_L are not Part IX's G_L or Part XIII's G_L^{lo} .

267 Integration with ProveIt and a formalization route

267.1 The inspected interface

The repository snapshot used for this report is

c39974f12a45c8795575ab222f3b84b2901c394f

The selected implementation source is

Algebra/SurrealNumbers/Surreal/HahnSeries/StrongEvaluation.lean

It imports `Evaluation` and `Regroup`, defines `PowerSeriesSummable` through finite coefficient incidence, and proves `evaluate_powerSeriesSum` and `summable_evaluate_powerSeries`. In constructing a `Hahn SummableFamily`, it supplies the proof fields `isPWO_iUnion_support` and `finite_co_support` [156].

This source inspection establishes a genuine connection with the repository's implemented abstractions. It does not amount to rebuilding the repository, checking all its reports, or formally checking any theorem newly proved here. Under a linear exponent order, the partially well-ordered support condition specializes to the well-order condition used in this article.

Proposition 267.1 (The ordinary power-series interface). *On real arrays $(a_{i,n})_{i,n \in \mathbb{N}}$, the predicate*

$$\forall n \quad \{i : a_{i,n} \neq 0\} \text{ is finite}$$

is Π_3^0 -complete. Its coefficientwise finite sum is Baire class 1 and nowhere continuous on this domain. The integer- and real-bound constructions apply with no support heights, retaining all the selector conclusions.

Proof. The upper bound is $\forall n \exists M \forall i > M \ a_{i,n} = 0$; restricting to binary coefficients and transposing a matrix reduces $\text{Fin}^{\mathbb{N}}$. Ordinary power-series exponents lie in the well-order $\mathbb{N}$, so there is no additional union-support condition. Finite partial row sums give Baire class 1, and a unit contribution in an unobserved row gives nowhere-continuity. The certificate proofs use only the remaining incidence bounds. The legal space remains a real vector subspace, and the real-bound regularization uses (264.5), (264.7) with the support-bound coordinates omitted. $\square$

[write] (Batch 96.) What the proposition is about, checked in the repository (the file is unchanged from the pin to the placement; its last change is 54cf883d9). The Lean predicate is `Surreal.HahnSeries.PowerSeriesSummable`, defined at line 31 of `Algebra/SurrealNumbers/Surreal/HahnSeries/StrongEvaluation.lean` for a family $f : \iota \rightarrow \text{PowerSeries } R$ over any commutative ring R and any index type ι as $\forall n, (\text{fun } i \Rightarrow (f\ i).\text{coeff } n)).\text{HasFiniteSupport}$. The proposition concerns its instance $R = \mathbb{R}, \iota = \mathbb{N}$, read as a subset of $\mathbb{R}^{\mathbb{N} \times \mathbb{N}}$ through $a_{i,n} = (f\ i).\text{coeff } n$ with the product topology. Lean puts no topology on this set, and nothing in the proposition is formalized. The “coefficientwise finite sum” is `powerSeriesSum` (line 35 of the same file), defined with `finsum`, which is total: at a degree with infinitely many nonzero coefficients it returns 0, so it agrees with the proposition’s sum only on the predicate’s domain. The theorem `powerSeriesSummable_iff_exists_hahnFamily` of `Surreal/HahnSeries/PowerSeriesSummation.lean` proves the predicate equivalent to native Hahn summability over the exponent well-order $\mathbb{N}$, so the proposition is the analogue, for the exponent order $\mathbb{N}$, of [theorem 261.2](#), which is stated only for groups; the union-support condition is void there. The collection’s `FORMALIZATION.md` maps these declarations, as proved, to `thm:exact` of the `hahn-evaluation-at-omega` report; the real and complex instances are used in `Surreal/Surcomplex/StrongEvaluation.lean` and `Surreal/Surcomplex/PowerSeriesStrongHom.lean`. No Lean statement concerns the topology of these sets.

Thus the topological issue appears already in the repository’s source power-series summability predicate, before moving to more complicated Hahn exponents. Theorems about preservation of a sum are algebraic statements; continuity of an evaluator on a space of uncertified source families is a separate question.

267.2 Proof objects versus informative certificates

A proposition asserting existence of finite incidence does not itself expose a chosen row bound as observable representation data. Likewise, a proof of well-founded support is different from a supplied family of cut heights and bounds. The name spaces developed here make these witnesses into coordinates of the input. That extra information is precisely what is used in the continuity proofs.

The distinction should survive formalization. An abstract `SummableFamily` can be obtained by forgetting the bounds of a certified name, but reconstructing those bounds from a raw array is the discontinuous selection problem studied above. Erasing the bounds too early would erase the representation theorem’s content.

267.3 A modular formalization plan

A practical dependency order is as follows. These are proposed milestones, not claims of completed Lean files.

Order kernel. Formalize closure of well-ordered subsets under finite unions, the convex relation of well-ordered intervals, the reverse-well-ordered quotient argument, and the explicit embedding (260.1). These are ordinary countable order-theoretic arguments and do not require a surreal-number development.

Height interface. Use as data a countable family $h_i : L \rightarrow \mathbb{N}$, together with the specification that a support is well ordered exactly when each height is bounded on it. Prove the support, incidence, and certificate theorems from this interface first. Derive the interface from countable cuts separately; this isolates the perfect-set theorem used to obtain countability.

Coefficient topology. Prove closedness of the certificate equations and continuity of the finite regularization maps. Establish the exact convergence theorem in a product of real spaces. Most of these arguments are finite maxima, finite sums, product continuity, and coordinatewise convergence, independent of advanced descriptive set theory.

Descriptive classification. Add the explicit Fin , $\text{Fin}^{\mathbb{N}}$, and Kleene–Brouwer reductions. Expose coanalytic completeness and the analytic-image theorem as named dependencies until they are formally available. An informal citation must not be represented as an imported checked theorem.

Hahn integration. Construct an existing summable-family object from certified arrays, prove that its coefficients and its hsum agree with the decoding formulas, and then transfer the topology theorem to this interpretation. Keep the topology of raw arrays, the topology of names, and any valuation topology distinct in the statements.

This ordering yields useful reusable mathematics even before the full coanalytic lower-bound infrastructure is implemented.

[write] (Batch 96.) None of these milestones exists in the repository at the placement; Part II’s [section 27](#) proposes a parallel layered plan for its own results.

268 Further research questions and concrete next targets

The questions below are unresolved in this report. Unless separately attributed, this means proposed project questions, not a claim that no answer exists anywhere in the literature.

Question 268.1 (Certified multiplication and inversion). Construct closed name spaces for single Hahn series with jointly continuous multiplication and continuous inversion away from zero, and explicit transformations of support and coefficient-incidence certificates.

The difficulty is to bound the finitely many pairs contributing to a convolution coefficient and to expose the leading term needed by inversion. A successful construction should state exactly which additional information is required, rather than assuming that the summation certificates automatically solve the field-operation problem.

[write] (Batch 96.) This bears on Part II’s [question 28.6](#) (certified support languages closed under arithmetic); neither answers the other.

Question 268.2 (Effective compilation of height tables). For computably presented scattered orders equipped with a specified Hausdorff decomposition, determine the oracle complexity of producing cut heights satisfying (258.5). Which decompositions admit a uniform computable compilation?

The fixed-order theorem supplies existence. Theorem 266.3 prevents turning it into an unrestricted Borel recognition procedure without a promise or extra certificate. An effective theorem should name that promise explicitly.

[write] (Batch 96.) With question 268.9, this carries the clause of Part II's question 28.3 that source 25 leaves open: computing the least Borel level directly from an economical Hausdorff decomposition (note after that question).

Question 268.3 (The information content of the selector). Determine the Weihrauch or related represented-space degree of selecting incidence bounds and support bounds on natural computable presentations. How does allowing real-valued bounds change that degree, if at all?

The integer/real distinction changes Baire-class-1 realizability of a particular selector. That topological statement is not by itself a computational equivalence or separation; a precise representation of input reals and output bounds is required.

[write] (Batch 96.) The review observation after remark 270.2 shows that on computable inputs neither selector need be computable; it gives no degree.

Question 268.4 (Regrouping and double families). Give explicit certificate transformations for regrouping a jointly strongly summable doubly indexed family, and for admissible interchange of summation.

This connects directly with the repository's regrouping interface. It should separate a certificate for the *joint* family from separate certificates for individual rows or columns, which need not by themselves control the joint union support.

Question 268.5 (Restricted-support fields; merged with Part II's question 28.4 [write]). This is Part II's question 28.4, which source 03 asks for Borel subfields of $\mathbb{R}((t^{\mathbb{Q}}))$ containing the rational monomials. Source 25 asks it independently, for the useful truncation-closed subfields of full Hahn fields over dense exponent groups, including real closed examples, in terms of descriptive complexity, and adds: which support languages support continuous certified arithmetic?

The full-support coanalytic barrier does not answer this question. The target is an algebraically stable restriction, not just an arbitrary low-complexity subset of coefficient space.

[write] (Batch 96.) The question is printed once, as Part II's, with source 25's generalization and added clause here; a note after question 28.4 points back.

Question 268.6 (Sharp support-capacity bounds). Relate the countable height family, the ideal-isomorphism structure of $\text{WO}(L)$, and bounds on order types of supports in specified subfamilies.

The two nontrivial Borel Wadge degrees deliberately discard substantial order information. Recovering information needed by multiplication, generated supports, and algebraic closure requires finer invariants.

Question 268.7 (Partial orders instead of linear orders). For countable partially ordered exponent-like systems, classify the complexity of well-founded or partially well-ordered support predicates. Which parts of the cut-height method survive, and where can higher Borel complexity occur?

The interval equivalence, final-segment test, and rational-order obstruction all use linearity. A generalization must handle antichains as well as descending sequences, rather than substituting a partial order into the existing proofs.

Question 268.8 (Minimal coefficient observability; merged with Part II's [question 28.5](#) [write]). This is Part II's [question 28.5](#), asked independently by source 25 as: which selected coefficient maps or linear observations suffice to force the non-scattered naming barrier ([theorem 265.2](#))? Can useful exact models avoid it by observing strictly less than the whole coefficient sequence?

The proof uses countably many observations to form one Borel decoder. A minimal observation theorem would identify the exact point at which the tree reduction becomes visible.

[write] (Batch 96.) Printed once, as Part II's question, with source 25's wording here; a note after [question 28.5](#) points back.

Question 268.9 (Uniform families with promised witnesses). Describe the complexity of producing the real-certificate regularization uniformly from an order together with a valid scatteredness witness, such as an appropriate decomposition tree.

[Theorem 266.3](#) rules out classification without a promise, but does not settle uniform construction over a space of witnessed inputs. The witness-validity predicate must be included in any claimed complexity bound.

Question 268.10 (Formal proof and foundational strength). Formalize the exact support classification and identify the weakest standard subsystems needed for the countable-cut lemma and the naming dichotomy.

The elementary interval and reduction arguments can be separated from the perfect-set and analytic-image principles. This makes a reverse-mathematical or proof-theoretic dependency analysis a concrete, rather than purely philosophical, next step.

[write] (Batch 96.) Compare Part II's formalization question at the end of [section 28](#) and Part XVI's [question 222.9](#).

Question 268.11 (A closer arithmetic obstruction). For a fixed definable integer-part or omnific subring in a restricted Hahn model, compare the certificate topology with all Borel-compatible Polish topologies on unique representatives. Which discontinuities follow from support representation alone, and which require arithmetic induction?

This returns to Glazer's setting. It requires checking the arithmetic hypotheses of the chosen structure; the field and naming results here do not supply them automatically.

[write] (Batch 96.) For the finite principal-part ring $\mathcal{A}_{\text{fin}}$ of Part II, Glazer's obstruction is Part II's [theorem 25.2](#), and consequences of open induction that already forbid continuous addition there are identified by Part XIV (note after [question 28.7](#)).

Further questions and research [write]

(Batch 96.) Under the repository's rule that unproved claims are kept as questions, the write adds the following. Source 25's informal effectivity sentence in [section 264.2](#) was proved for

this report ([remark 270.2](#)); no statement of source 25 was found false. The part of Part II's [question 28.3](#) that source 25 does not answer stays with that question (dated note there) and is carried by [questions 268.2](#) and [268.9](#).

Question 268.12 (The attributions to Guevara–Uzcátegui and Hall–Knight–Lange [write]). Do Theorem I and Theorem 4.13 of [154] state, for every countable scattered linear order L , that $\text{WO}(L)$ is Borel and in fact $F_{\sigma\delta}$, as [section 256.2](#) credits them? Does [155] contain the classification of [theorem 259.6](#) or [theorem 260.1](#), or their Σ_2^0 criterion?

Source 25 credits the Π_3^0 ceiling to Guevara–Uzcátegui and read the abstract and metadata only of Hall–Knight–Lange (its ledger says the full text was inaccessible); it claims no priority and no solution of a problem of the latter paper. The write read neither paper. What is missing is a reading of both: the hypotheses of the cited theorems (ideals on countable sets and their orthogonals, versus fixed orders), and whether the forbidden-order criterion of [lemma 259.2](#) occurs there. The mathematics of this part does not depend on the answer, since [theorem 258.2](#) is proved independently.

269 Conclusion

The support problem admits a coarse classification that is much smaller than the hierarchy of countable scattered orders. The order $\omega^* \cdot \omega$ detects the exact jump from countable generation and F_σ validity to $F_{\sigma\delta}$ completeness. A rational suborder detects the jump out of the Borel hierarchy. This gives a direct answer to the previous project's Borel-level and Wadge-degree question.

For countable families, finite incidence creates a second independent source of complexity. Strong summation is nowhere continuous on raw arrays, although it is a pointwise limit of continuous finite sums. Closed certificate spaces restore continuity by making support and incidence bounds observable. Allowing real rather than discrete bounds then permits continuous finite-stage names to converge to a canonical certificate exactly on legal inputs. The non-scattered case admits no exact standard Borel naming system with all coefficients observable, while the uniform recognition problem remains coanalytic-complete even on a fixed discrete group.

These are precise deductions with explicit hypotheses and full arguments, not a blanket impossibility claim about surreal computation. Their most immediate use is to guide representations: distinguish raw arrays from legal objects, legal objects from names, and logical existence proofs from informative bounds.

270 Where Part XIX meets Parts II, V and XVII [merge]

This section was written for this report. It lists the results of Parts II and V that source 25 proves again or contains as special cases, specializes its classification to Archimedean exponent groups, proves the effectivity sentence of [section 264.2](#), and records two observations of the Hilbert's-tenth programme's review of the archive. The proofs use only results proved in source 25 and in those parts and classical facts named where they are used; they have not been independently reviewed.

Second routes and special cases. Source 25 did not read this report's article; each identification below is this report's.

Source 25	This report	Relation
lemma 257.2	proof of theorem 19.1 ; note after theorem 67.2	classical; third proof
theorem 258.2	theorem 18.4(i)	sharpened from Borel to Π_3^0 , different proof
lemma 259.4 , proposition 259.5	lemma 18.2 , theorem 18.4(ii) , lemma 20.1	same reduction, printed again
theorem 259.6 , cases $\mathbb{Z}, \mathbb{Z}_{\text{lex}}^d$	theorem 19.1(i), (ii)	special case
theorem 260.1 , case $\Gamma = \mathbb{Q}$	theorem 67.3 with $k = \mathbb{R}$	special case
corollary 260.3	corollary 20.4	second route, weaker hypothesis, completeness added
theorem 265.2 for $V(\Gamma)$	theorem 20.2 , (i) $\Leftrightarrow$ (iii)	second route; closed Polish names

[Theorem 259.6](#) answers the level and Wadge parts of [question 28.3](#) (note there). Part V's [theorem 67.3](#) also covers coefficient fields $k \neq \mathbb{R}$, and part (iii) of [theorem 19.1](#) (support order types) and Part II's analytic boundedness ([theorem 21.2](#)) have no counterpart in source 25.

Proposition 270.1. *Let Γ be a countable ordered abelian group.*

- (i) *With the Borel structures inherited from $\mathbb{R}^\Gamma$ and $\mathbb{R}^{\mathbb{N} \times \Gamma}$, the sets $V(\Gamma)$ and $\mathcal{S}_\Gamma$ are standard Borel spaces if and only if Γ is scattered.*
- (ii) *Let $\Gamma \leq \mathbb{R}$. If $\Gamma = \{0\}$, $V(\Gamma)$ is the whole space and $\mathcal{S}_\Gamma$ is Σ_2^0 -complete. If Γ is cyclic and nonzero, $V(\Gamma)$ is Σ_2^0 -complete and $\mathcal{S}_\Gamma$ is Π_3^0 -complete. Otherwise both are Π_1^1 -complete.*

Proof. (i) If Γ is scattered, both sets are Borel subsets of Polish spaces ([theorems 260.1](#) and [261.2](#)), hence standard Borel. If A is a subset of a Polish space X whose relative Borel structure is standard, the inclusion $A \rightarrow X$ is a Borel injection from a standard Borel space, so A is Borel by the Lusin–Souslin theorem [8]. For non-scattered Γ both sets are Π_1^1 -complete, hence not Borel, so neither is standard Borel.

(ii) The cases $\{0\}$ and $\mathbb{Z}u$ are rows of [theorems 260.1](#) and [261.2](#). If $\Gamma \leq \mathbb{R}$ has a least positive element u , then $\Gamma = \mathbb{Z}u$: for $\gamma > 0$ choose n with $nu \leq \gamma < (n+1)u$, which exists because $\mathbb{R}$ is Archimedean; then $0 \leq \gamma - nu < u$ forces $\gamma = nu$, and negative elements follow by symmetry. So a noncyclic $\Gamma \leq \mathbb{R}$ has no least positive element; by the first paragraph of the proof of [theorem 260.1](#) it is dense and contains a copy of $\mathbb{Q}$, and both sets are Π_1^1 -complete by [theorems 260.1](#) and [261.2](#). $\square$

For every countable $\Gamma \leq \mathbb{R}$, Part XVII's left-finite field $\mathcal{L}_\mathbb{R}^\Gamma$ has a Π_3^0 code domain, by the display (226.1) in the proof of [proposition 226.4](#). By part (ii), as soon as Γ is noncyclic, enlarging left-finite supports to all well-ordered supports moves the code domain from Π_3^0 to Π_1^1 -complete. This is the remark in Part XVII's [question 237.1](#) that the descriptive complexity of full Hahn support classes “need not match”, made exact; it bears on that question, which asks about automatic summation, without answering it.

Remark 270.2 (The effectivity sentence of [section 264.2](#)). Suppose an enumeration of Γ is fixed, the index set I of the final segments is $\mathbb{N}$ or an initial segment of it, the heights $h_i(\gamma)$ are computable in (i, γ) , and the windows Ω_k are given by a computable function returning a canonical list of each. Represent reals by fast Cauchy sequences of rationals and raw arrays by sequences of such names. Then each coordinate of $R_k(X)$ is computable from a name of X , uniformly in k and in the coordinate. Indeed, membership in Ω_k is decidable; $v_k(t) = \max(t - 2/k, 0) - \max(-t - 2/k, 0)$ and $w_k(t) = \min(1, k|t|/2)$ are compositions of

computable real functions; and $b_i^{[k]}$ and $m_\gamma^{[k]}$ are finite maxima, over the explicitly listed set Ω_k , of computable reals. This is the sentence of source 25 in a precise form. It says nothing about the limit $r_{\mathbb{R}}(X)$, and the next note shows that the limit need not be computable.

[write] (Batch 96.) The Hilbert’s-tenth programme read the complete manuscript and authenticated the archive members in commit 3abf23ba1 (Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/review_new_support_complexity_fb9f5884b.md, with .py and .json): it found no mathematical defect within the classical inputs source 25 names, ran none of the delivered programs, and did not recover Part II’s question (it is [question 28.3](#)). It adds two observations, credited to it, which the write re-derived and which sharpen, without contradicting, source 25’s non-claims. *Observation 1*: a computable legal array need not have computable certificates. Take $\Gamma = \mathbb{Z}$ with its computable height table, arrange that every machine halts, if at all, at a step $n \geq 1$, and let $X[n, e] = 1$ for $e \geq 0$ exactly when machine e first halts at step n , all other entries 0. Each entry is decidable uniformly, the union support lies in $\mathbb{N}$ and each exponent has at most one active row, so $X \in \mathcal{S}_{\mathbb{Z}}$; its strong sum at e is the halting characteristic. A uniformly computable integer incidence bound m_e would decide halting by simulation up to step m_e , and a uniformly computable real bound gives such an integer bound by rational approximation; hence neither exists, although the support bounds can all be 0 and the regularizations of [theorem 264.3](#) are computable and converge ([remark 270.2](#)). *Observation 2*: let $Y^{(e)}[n, 0] = 1$ exactly when machine e has halted by step n , all other entries 0. This uniformly computable family is strongly summable exactly when e never halts. A uniform procedure turning indices of computable arrays into finite existential Diophantine problems solvable exactly for the strongly summable arrays would make the set of nonhalting machines computably enumerable, which it is not. The review notes that neither observation excludes computation on already certified inputs, finite-support promises or restricted support languages; both bear on [questions 268.2, 268.3](#) and [268.9](#).

The Lean predicate. The note after [proposition 267.1](#) states exactly how that proposition relates to `Surreal.HahnSeries.PowerSeriesSummable`: it is a statement about the real, $\mathbb{N}$ -indexed instance of that predicate with the product topology, and nothing in this part is formalized.

Part XX

Borel flows on left-finite series fields

[write] This part is source 26, manuscript 03 of batch 96, *Borel Flows on Left-Finite Series Fields: A local-finiteness criterion, a rational-rank dichotomy, and failure of Lie closure* (24 pages, A4, dated 4 October 2026; “research manuscript prepared with ChatGPT for Vladimir Reshetnikov”). It arrived in commit fb9f5884b, as archive Borel_Flows_Glazer_ProveIt, and was placed, with the sources of Parts XVIII, XIX and XXI, in 54ece48ab. It is pinned to c39974f12, 38 commits before the placement, at which it read the Lean file `Algebra/BakerCampbellHausdorff/Lean/BCH/Formal/Trunc.lean` (blob 5f238ed2f, unchanged at the placement); this report’s article is the same at the pin and at the placement. Its file prefix in this report is 26-borel-flows-. All unmarked text in this part is source 26’s; its fourteen sections are Sections [271–284](#) in order, with statement and equation numbers otherwise unchanged (its Theorem 5.2 is the second statement of the fifth of them, [theorem 275.2](#)), its nine questions are Questions [283.1–283.9](#), and its two appendices are [appendices AO](#) and [AP](#). [Section 285](#)

[merge] was written for this report.

What this part claims, and its status

Let $\Gamma \neq \{0\}$ be a countable *divisible* subgroup of $\mathbb{R}$ and let $\mathcal{L}_{\mathbb{R}}^{\Gamma}$ be Part XVII's left-finite field with real coefficients and the coefficient Borel structure. A Borel field flow is a jointly Borel action $\mathbb{R} \times \mathcal{L}_{\mathbb{R}}^{\Gamma} \rightarrow \mathcal{L}_{\mathbb{R}}^{\Gamma}$ by unital field automorphisms; no continuity, fixing of $\mathbb{R}$ or of the valuation is assumed. Every such flow fixes $\mathbb{R}$ and the valuation ([theorem 274.4](#)), and differentiation at time zero is a bijection onto the real-linear valuation-nondecreasing derivations that are locally finite on every bounded exponent window, the coefficients of every orbit being entire functions of time ([theorem 275.2](#)). For strict derivations this is local nilpotence on windows, or escape of the iterated valuations ([theorem 276.2](#)); every strict derivation integrates exactly when $\dim_{\mathbb{Q}} \Gamma < \infty$ ([theorem 277.2](#)), and in finite rank every valuation-nondecreasing Borel derivation integrates ([corollary 277.3](#)). In infinite rank two integrable strict derivations can have a nonintegrable sum and bracket ([theorem 278.2](#)), and the product of their time-one maps is the time-one map of no Borel flow ([theorem 279.1](#)); the Borel time-one maps are characterized exactly, with a unique embedding flow ([theorem 279.5](#)); the integrable strict derivations form a Borel set of parameters ([theorem 280.2](#)).

[Section 285](#) [merge] proves that Borel time dependence of the images of basis monomials already makes an action a Borel flow, which at $\Gamma = \mathbb{Q}$ identifies source 26's answer with Part XVIII's; and it proves the three statements source 26 asserts with sketches only: the BCH transfer under a common positive gain, the surreal realization (by citation) and the remark that finite sampling cannot certify integrability, which needs infinite rational rank.

Unrefereed, AI-assisted, not formalized; priority is not established. The finite checks test finite algebraic identities only.

Questions answered. Source 26 answers Part XVII's [question 237.5](#) for countable divisible Γ , which it calls "the flow question in the earlier ProveIt research draft" (Question 13.5 of source 23): no analyticity has to be imposed, and the integrable derivations are characterized (dated note there). Part XVIII answers the same question in rational rank one, for $\mathbb{Z} \leq \Gamma \leq \mathbb{Q}$, divisible or not; at $\Gamma = \mathbb{Q}$ the two answers coincide ([corollary 285.2](#) [merge]). The question stays open for nondivisible Γ of rational rank at least two. Source 26 states that it solves no remaining numbered problem of Glazer's. Its questions bear on Part XVII's [questions 237.1](#), [237.2](#), [237.6](#) and [237.7](#) without answering them (notes after its questions).

What was already here, and what is new. Source 26 read this report's README and source 23 as a standalone PDF, not this report's article (its [appendix AP](#)). Its [lemma 272.1](#) re-proves, for divisible Γ , Part XVII's [proposition 226.2](#); its display ([272.2](#)) is ([226.1](#)); its [lemma 272.2](#) is a special case of [theorem 228.2](#); its [lemma 274.2](#) re-proves [remark 239.4](#) [merge]; its [corollary 275.3](#) re-proves the clause of [theorem 230.2](#) that Borel derivations vanish on $\mathbb{R}$; its [lemma 277.1](#) is the strict case of [theorem 230.2](#); its [corollary 276.3](#) and the joint Borelness in [theorem 276.2](#) are the exponential half of [theorem 234.2](#) with [corollary 234.3](#), which it credits; its [corollary 279.6](#) generalizes the logarithm half of [theorem 234.2](#); its [example 275.5](#) generalizes [proposition 239.3](#) [merge]; and its surreal map ([281.1](#)) is ([235.1](#)). Its [corollary 275.4](#) is Part XXI's [remark 294.3](#) in another category of actions, and the last clause of Part XVIII's [theorem 248.3](#); its [corollary 277.3](#) with [theorem 275.2](#) is, at $\Gamma = \mathbb{Q}$, Part XVIII's [theorems 248.3](#) and [248.8](#) ([corollary 285.2](#)). Each is flagged in a note. New to the report are

the finite-orbit-span formulation of the Baire argument ([theorem 273.1](#), not claimed as a new principle), automatic normalization of flows, the exact flow criterion, the strict criterion and the rational-rank dichotomy, the complete finite-rank criteria for generators and for time-one maps, the alternating shears and the nonclosure of integrable derivations and of time-one maps, the exact time-one criterion and the Borel integrability locus.

Labels and notation. Every label of source 26 takes the prefix `pma:bfl:`; its eighteen unlabelled numbered statements ([remarks 273.3, 274.5, 276.4, 278.3](#) and [280.3](#), [definitions 274.1, 275.1](#) and [276.1](#), [example 277.4](#) and the nine questions) are labelled `pma:bfl:n<number>` by their numbers in the source, as are its conclusion and appendices (`pma:bfl:n14`, `pma:bfl:nA`, `pma:bfl:nB`); labels written for this report have the form `pma:bfl:w-<name>`. The eighth block of the table in [section 1.3](#) lists the symbols. In short: source 26's field $\mathcal{L}_\Gamma$ is printed $\mathcal{L}_\mathbb{R}^\Gamma$, as in Part XVII (tempting false reading: Part XV's $\mathcal{L}^\Gamma$ is a surreal image), and its filtration $\mathcal{L}_{\Gamma, \geq a}$ is printed $\mathcal{L}_{\mathbb{R}, \geq a}^\Gamma$; its windows $V_{a,b}$ are printed $\text{Win}_{a,b}$, because V_Γ and V_L are groups in Parts II and XIII; its logarithmic values $d(\gamma)$ are printed $\delta(\gamma)$, Part XVII's letter, so its gain constant δ is printed ε , as in [theorem 234.2](#), and its gaps $\epsilon_n = \beta_{n+1} - \beta_n$ are printed κ_n ; its alternating derivations E, F , their bracket H and their shear flows P_s, Q_s are printed $D_e, D_o, D_{eo}, \Phi_s^e, \Phi_s^o$, because E and E_i are Part XVII's Euler derivations, H_α is Part X's group and P_θ Part XVI's cone; the two generic derivations of its [section 281](#) are printed D_1, D_2 for the same reason, and the coordinate spaces E_n in the proof of [theorem 273.1](#) are printed V_n ; its parameter space $\mathcal{P}$ and integrability locus $\mathcal{I}$ are printed $\mathfrak{P}_B$ and $\mathfrak{I}_B$, because $\mathcal{P}, \mathcal{P}_k$ and $\mathcal{I}_k$ are Part II's and Part V's fields and rings; its transcendental θ is printed $\mathfrak{D}$; its $\text{span}_\mathbb{R}$ macro is printed $\text{span}_\mathbb{R}$. The letters A (an automorphism, not Part II's $\mathcal{A}_{\text{fin}}$), W (a finite-dimensional subspace, not Part XVII's W_N), $B, C, S, \Sigma, \Delta, \beta_n, z_n$ and λ are local to this part; Φ_s, D, v and t (a positive infinitesimal) are Part XVII's. No normalization changed.

Files. Source 26's program, recorded run, build script, claim ledger and source record are shipped as `code/26-borel-flows-verify.py`, `data/26-borel-flows-verification_results.json`, `code/26-borel-flows-build.sh`, `26-borel-flows-CLAIM_LEDGER.md` and `26-borel-flows-SOURCES.md`; the build script names the unshipped `borel_flows.tex` and rewrites the record in place ([appendix E](#)). The write reran the program on a copy: 5,372 checks pass, also under `-O`, and the record equals the shipped one after removing carriage returns.

A corrected reference. [write] (Batch 96.) Lines 166–167 of the delivered manuscript read “Theorem~” and, on the next line, “`ef{thm:timeoneexact}`”: the characters `\r` of `\ref` were consumed as a carriage-return escape when the manuscript was produced, so its PDF prints “Theorem `efthm:timeoneexact`” (printed page 2, the PDF's third page), and an undefined-reference check cannot see it. The reference is restored in item 3 of [section 271](#) as a reference to [theorem 279.5](#). The Hilbert's-tenth programme's review found the same defect (next paragraph; it gives the lines as 167–168).

Review elsewhere in the repository. The Hilbert's-tenth programme reviewed the archive in commit `3d87c3d6b` (`Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/review_new_borel_flows_fb9f5884b.md`, with `.json` and `.py`): it read all 1,724 lines of the manuscript, authenticated the nine members and the eight delivered checksums, read the cited 143-line `Trunc.lean`, challenged the proofs of [Sections 272–281](#) and found no substantive error. It records that the review-side halting-height construction of

its review of source 24 (commit 8dffc7af0), which uses a nondivisible Γ , does not transfer to this divisible setting; that the manuscript gives no uniform effective producer of invariant window spaces, spectral data or gain certificates from arbitrary Borel codes; and that no ordinary-integer Diophantine compiler or operation count follows from it. It ran none of the delivered programs. Its findings are incorporated here; it is not a proof certification.

271 The research intersection and the precise advance

271.1 Why this topic

Glazer's *A Topological Tennenbaum Theorem* studies how descriptive regularity constrains arithmetic operations: under its hypotheses, Borel models of arithmetic on uncountable Polish spaces cannot make either addition or multiplication continuous [1]. The connection pursued here is this interaction between algebra and measurability, not an inference from a professional title.

The ProveIt repository contains a surreal-series program and a Baker–Campbell–Hausdorff (BCH) program [162, 164]. A prior research report, *Borel Regularity Forces Summability*, studies measurable operators and derivations on left-finite fields. Its Question 13.5 asks which derivations admit Borel one-parameter automorphism flows without a uniform positive gain in valuation, and whether analytic parameter regularity should be imposed [161]. That question provides our concrete target.

We settle it for *countable divisible* exponent subgroups of $\mathbb{R}$. This includes $\Gamma = \mathbb{Q}$, the usual exponent group of the real Levi–Civita field. The divisibility assumption is used to recover the order from the field structure. We do not silently extend the classification to arbitrary nondivisible exponent groups.

271.2 What is established

For a bounded exponent interval $[a, b]$, a series has only finitely many coefficients in that interval. Nevertheless, the space of all such truncations is usually countably infinite-dimensional. The decisive condition is not finite dimensionality of the whole window, but finite dimensionality of each cyclic orbit in that window.

The principal results are:

1. **Exact integration and automatic analyticity** (Theorem 275.2). All Borel flows correspond bijectively to valuation-nondecreasing derivations that are locally finite on every bounded window. Coefficientwise entire dependence follows automatically.
2. **A sharp strict-derivation threshold** (Theorems 276.2 and 277.2). Strict increase of valuation is sufficient for all derivations exactly in finite rational rank. At infinite rank, strictly increasing iterates may remain trapped below one fixed bound.
3. **Failure of the naive Lie picture** (Theorems 278.2 and 279.1). Two individually integrable strict derivations may have nonintegrable sum and commutator. Two time-one maps may have a product that is not any Borel time-one map. Theorem 279.5 then gives an exact, unique-embedding criterion for all Borel time-one maps.
4. **Descriptive complexity** (Theorem 280.2). Within an explicit standard Borel parameter space, the strict derivations that integrate form a Borel set.

These statements are proved below rather than inferred from finite experiments. They resolve the stated extension problem in a definite setting; we do not label them a confirmed historical breakthrough.

271.3 Relation to existing formal Lie theory

Exponential correspondences for derivations on generalized-series algebras already have a substantial theory. Bagayoko, Krapp, Kuhlmann, Panazzolo, and Serra develop formal summability conditions under which exponential and logarithm give such correspondences [159]. Bagayoko’s 2026 *A formal Lie correspondence* treats formally nilpotent Lie algebras and exponential groups in a more general categorical setting [160]. The present work does not introduce the formal exponential or the BCH correspondence. Its focus is the necessity forced by Borel time dependence and the extra obstruction imposed by the *left-finite support class*. A family admissible in a larger Hahn field need not remain in that class.

The uniform-positive-gain construction is already present in the earlier ProveIt draft [161, Section 10]. We reprove what is needed and extend beyond it. Our proofs do not depend on the correctness of that draft’s more general automatic-summability theorem. The repository’s research-report catalogue explicitly separates research drafts from formally checked statements [163].

[write] (Batch 96.) The “prior research report” and “earlier ProveIt draft” is source 23, printed as [Part XVII](#): its Question 13.5 is [question 237.5](#), its Section 10 is [section 234](#), its “more general automatic-summability theorem” is [theorem 228.2](#), and the “research-report catalogue” is this report’s README. Source 26 also answers the question in the setting it names (dated note after [question 237.5](#)). Theorem 279.5 in item 3 above is a restored reference (introduction of this part). The two cited 2024–2026 preprints were not re-read for this report.

272 Left-finite fields, Borel coding, and finite windows

272.1 The field and its order

Throughout the main theorems, $\Gamma \neq \{0\}$ is a countable $\mathbb{Q}$ -vector subspace of $\mathbb{R}$, with the inherited order. Define

$$\mathcal{L}_{\mathbb{R}}^{\Gamma} = \left\{ x = \sum_{\gamma \in \Gamma} x_{\gamma} t^{\gamma} : x_{\gamma} \in \mathbb{R}, \quad |\text{supp}(x) \cap (-\infty, b)| < \infty \text{ for every } b \in \mathbb{R} \right\}. \quad (272.1)$$

Here t is formal, and a monomial t^{γ} exists for every $\gamma \in \Gamma$. For $x \neq 0$ put

$$v(x) = \min \text{supp}(x), \quad v(0) = +\infty.$$

Every nonempty left-finite support has a least member: choose any member, and take the minimum of the nonempty finite part below a slightly larger bound. Infinite left-finite supports tend to $+\infty$.

Lemma 272.1 (Elementary field facts). *Coefficientwise addition and convolution multiplication make $\mathcal{L}_{\mathbb{R}}^{\Gamma}$ a field. It is ordered by the sign of the leading coefficient. Every positive element has a square root, and the nonzero squares are exactly the positive elements. The function v is a valuation with value group Γ .*

Proof. If S, T are nonempty left-finite supports with minima s_0, t_0 , then $s + t < b$ implies $s < b - t_0$ and $t < b - s_0$. Only finitely many pairs satisfy these inequalities. Thus convolution is well-defined and has left-finite support. The least exponent of a product is the sum of the least exponents, with the product of the nonzero leading coefficients.

Write $x = at^\gamma(1 + u)$, where $a \neq 0$ and $v(u) > 0$, allowing $u = 0$. The geometric series $\sum_{n \geq 0} (-u)^n$ is left-finite: below a fixed bound only finitely many n occur, and each contributing power is left-finite. It is an inverse of $1 + u$. This proves the field assertion.

The leading-sign rule is compatible with addition and multiplication. If $x > 0$, then $a > 0$, $\gamma/2 \in \Gamma$, and

$$\sqrt{x} = \sqrt{a} t^{\gamma/2} \sum_{n \geq 0} \binom{1/2}{n} u^n$$

is well-defined by the same positive-valuation estimate. Formal coefficient identities show that its square is x . A nonzero square has positive leading coefficient. The valuation statements follow from leading terms. $\square$

The same binomial construction gives a unique positive rational power of a positive element. Uniqueness follows from order: for positive $r < s$, $r^n < s^n$ for every positive integer n .

[write] (Batch 96.) The field and valuation assertions of [lemma 272.1](#) are Part XVII's [proposition 226.2](#), there for every countable $\Gamma \leq \mathbb{R}$; for $\Gamma = \mathbb{Q}$ and a countable ordered coefficient field they are Part V's [proposition 65.2](#), and with coefficients in $\mathbb{R}_{\text{alg}}$ Part XV's [proposition 196.2](#). The order and the square roots use divisibility, which source 26 assumes throughout.

The valuation topology has neighborhoods determined by agreement below an exponent cutoff. It is not the coefficient topology inherited from $\mathbb{R}^\Gamma$. For example, the constants are discrete in the valuation topology but have their ordinary topology in coefficient coding. We shall not apply a Polish-group theorem to the full valuation-topological additive group.

272.2 The standard Borel structure

Give $\mathcal{L}_{\mathbb{R}}^\Gamma$ the Borel structure inherited from its coefficient realization inside $\mathbb{R}^\Gamma$. It is a Borel subset, because left-finiteness is expressed by

$$\bigcap_{m \in \mathbb{Z}} \bigcup_{F \in [\Gamma_{< m}]^{< \omega}} \bigcap_{\gamma \in \Gamma_{< m} \setminus F} \{x : x_\gamma = 0\}. \quad (272.2)$$

All operations displayed here are countable. In particular this is a standard Borel space. The coordinate maps, valuation, truncations, addition, scalar multiplication, and field multiplication are Borel. For multiplication, each coordinate is an eventually finite sum over an enumeration of possible convolution pairs. Inversion is Borel as well, by partitioning according to the least exponent and applying the geometric formula. The order is Borel by partitioning according to the least nonzero coefficient of a difference.

[write] (Batch 96.) Display (272.2) is Part XVII's (226.1) with strict integer cutoffs, and the Borelness of the field operations is [proposition 226.4](#).

A family $(x_i)_{i \in I}$ is *left-finitely summable* when below each real bound there are only finitely many pairs (i, γ) with γ in that range and $(x_i)_\gamma \neq 0$. Its sum is defined by finite coefficient addition. The family of nonzero monomial terms of one element of $\mathcal{L}_{\mathbb{R}}^\Gamma$ has this property.

272.3 Windows are direct sums, not completed products

For real $a < b$ define

$$\mathcal{L}_{\mathbb{R}, \geq a}^\Gamma = \{x : v(x) \geq a\}, \quad \text{Win}_{a,b} = \mathcal{L}_{\mathbb{R}, \geq a}^\Gamma / \mathcal{L}_{\mathbb{R}, \geq b}^\Gamma \cong \mathbb{R}^{(\Gamma \cap [a,b])}. \quad (272.3)$$

The superscript parentheses mean finite support. We identify a quotient class with its finite truncation and equip it with the corresponding coefficient Borel structure. Integer endpoints suffice to test all our conditions. Each window is a union of countably many finite-dimensional coordinate subspaces, but need not itself be finite-dimensional.

Lemma 272.2 (Tail determination). *Suppose $T : \mathcal{L}_{\mathbb{R}}^\Gamma \rightarrow \mathcal{L}_{\mathbb{R}}^\Gamma$ is real-linear and $v(Tx) \geq v(x)$ for every x . Then T is Borel, is determined by its values on monomials, and preserves left-finitely summable families. It induces a Borel linear map $T_{a,b}$ on each window.*

Proof. The inequality makes every $\mathcal{L}_{\mathbb{R}, \geq a}^\Gamma$ invariant. For $x \in \mathcal{L}_{\mathbb{R}, \geq a}^\Gamma$, write $x = x_{<b} + r$ with $v(r) \geq b$ and $x_{<b}$ a finite sum. Then

$$(Tx)_{<b} = (Tx_{<b})_{<b}.$$

Thus every window map is an ordinary linear map on a countable-dimensional direct sum. Such a map is Borel: on each finite-dimensional coordinate subspace it is a finite-coordinate linear map, and these subspaces form a countable cover. Partition $\mathcal{L}_{\mathbb{R}}^\Gamma$ by an integer lower bound for its valuation, and use integer upper cutoffs to obtain Borelness of every coordinate of T .

For a summable family, only finitely many input terms begin below b ; the other images begin at or above b . Each of those finitely many images has finite support below b . The same tail calculation proves commutation with the sum and the asserted summability of the images. $\square$

[write] (Batch 96.) [Lemma 272.2](#) is a special case of Part XVII's [theorem 228.2](#): a real-linear valuation-nondecreasing map is valuation-continuous, so it is Borel, given by a cutoff-admissible matrix, and preserves left-finitely summable families. The direct proof above does not use divisibility.

273 A Baire-category lemma for countable-dimensional actions

The measurable regularity input can be isolated from field theory entirely. Here $\mathbb{R}^{(I)}$, for countable I , always has its coefficient Borel structure. No complete vector-space topology on that direct sum is assumed.

Theorem 273.1 (Finite orbit spans). *Let G be a connected Polish group, and let $\rho : G \rightarrow \text{GL}(V)$ be an algebraic representation on a countable-dimensional real vector space V . Suppose that for every $w \in V$ the orbit map $g \mapsto \rho_g w$ is Borel in a fixed countable-basis coding. Then each w lies in a finite-dimensional G -invariant subspace. On any finite-dimensional invariant subspace the representation is continuous.*

Proof. Fix w . Exhaust V by finite-dimensional coordinate spaces V_n . The Borel sets $\{g : \rho_g w \in V_n\}$ cover G , so at least one is nonmeagre. Among all finite-dimensional subspaces W for which

$$A_W = \{g : \rho_g w \in W\}$$

is nonmeagre, choose one of least dimension. Every such W is a Borel subset of V , since it is a linear subspace of a finite coordinate space. The Baire property provides a nonempty open set $U \subseteq G$ on which A_W is comeagre.

If $h \in UU^{-1}$, then $U \cap hU$ is nonempty and open. Both A_W and hA_W are comeagre there, so $A_W \cap hA_W$ is nonmeagre. For any g in this intersection,

$$\rho_g w \in W \cap \rho_h W.$$

Consequently the inverse image of $W \cap \rho_h W$ under the orbit map is nonmeagre. Minimality of $\dim W$ implies $\dim(W \cap \rho_h W) = \dim W$, hence $\rho_h W = W$.

The stabilizer of W is a subgroup containing the neighborhood UU^{-1} of the identity. It is therefore open; its complementary cosets are open as well. Connectedness of G makes the stabilizer all of G . Since A_W is nonempty, invariance now implies $w \in W$.

On a finite-dimensional invariant space, the matrix entries of ρ_g are Borel, being coordinates of finitely many orbit maps. A Borel homomorphism from a Polish group into a second-countable topological group is continuous. For completeness, choose a small symmetric neighborhood O in the target and cover the target by countably many translates of O . Some nonempty inverse image is nonmeagre. Its quotient with itself contains a neighborhood of the identity by the Baire-category argument used in Pettis' lemma: a Baire nonmeagre set is comeagre in some open set, whose sufficiently small translates intersect. The image of this neighborhood lies in $O^{-1}O$. Choosing O small enough for any prescribed target neighborhood proves continuity. $\square$

[write] (Batch 96.) The automatic-continuity step of the last paragraph is Part V's [lemma 61.2](#) with [lemma 61.1](#), stated there for Polish groups written additively; the same proof works in multiplicative notation, here with target the Polish group $\mathrm{GL}(W)$ of a finite-dimensional invariant space W . Source 26 gives the step as a sketch and calls it classical; it is [8, Theorem 9.10] (Banach–Pettis).

Corollary 273.2 (Borel one-parameter representations). *For $G = (\mathbb{R}, +)$ under the hypotheses of Theorem 273.1, there is a unique locally finite linear operator A on V such that $\rho_s w = e^{sA} w$. Every coordinate of every orbit extends to an entire function of a complex time variable. The union of the supports of $\rho_s w$, for all real s , is finite.*

Proof. On a finite-dimensional invariant subspace, a continuous matrix one-parameter group has the form e^{sA} . One elementary verification uses the convergent matrix logarithm near the identity. For sufficiently small s, t , commutativity and the local inverse identity give $\log \rho_{s+t} = \log \rho_s + \log \rho_t$. A continuous locally additive matrix-valued function is sA ; subdivision extends the exponential identity to every real s . Differentiation at zero shows compatibility on intersections of invariant subspaces and uniqueness of A .

The operator preserves each of these finite-dimensional spaces, so it is locally finite. Matrix exponentials are entire entrywise. A finite-dimensional subspace of $\mathbb{R}^{(I)}$ uses only the finite union of the supports of a basis, proving the final assertion. $\square$

Remark 273.3 (Why connectedness matters). The discrete group $\mathbb{Z}$ acts on $\mathbb{R}^{(\mathbb{Z})}$ by shifting the basis. Every orbit map is Borel, but the span of one basis orbit is infinite-dimensional. Thus the lemma is not a consequence of the countability of the coordinate set alone.

274 Every Borel field flow is automatically normalized

Definition 274.1. A Borel field flow on $\mathcal{L}_{\mathbb{R}}^{\Gamma}$ is a jointly coefficient-Borel map $\Phi : \mathbb{R} \times \mathcal{L}_{\mathbb{R}}^{\Gamma} \rightarrow \mathcal{L}_{\mathbb{R}}^{\Gamma}$ such that each Φ_s is a unital field automorphism, $\Phi_0 = \text{id}$, and $\Phi_{s+t} = \Phi_s \circ \Phi_t$. We do not initially assume that it fixes $\mathbb{R}$, preserves the valuation, or is differentiable.

Lemma 274.2 (Real coefficients are fixed). *Every coefficient-Borel unital field endomorphism of $\mathcal{L}_{\mathbb{R}}^{\Gamma}$ fixes $\mathbb{R}$ pointwise.*

Proof. For any $\eta \in \Gamma$, the map $r \mapsto [t^{\eta}] \Psi(r)$ is a Borel additive function from $\mathbb{R}$ to $\mathbb{R}$. Borel additive functions are continuous by the automatic-continuity argument in Theorem 273.1, and a continuous additive function is $r \mapsto c_{\eta} r$. Since $\Psi(1) = 1$, the constant c_{η} is 1 when $\eta = 0$ and 0 otherwise. Thus $\Psi(r) = r$. $\square$

[write] (Batch 96.) Ψ is the endomorphism of the statement. For Borel embeddings $\mathcal{L}_{\mathbb{R}}^{\Gamma} \rightarrow \mathcal{L}_{\mathbb{R}}^{\Lambda}$ and every countable Γ this is Part XVII's [remark 239.4](#) [merge], which deduces it from real-linearity ([theorem 228.2](#)); Part XVIII proves it again for its coordinate group ([theorem 242.4](#)).

Lemma 274.3 (Valuation action of an order automorphism). *An order-preserving field automorphism of $\mathcal{L}_{\mathbb{R}}^{\Gamma}$ induces on Γ an order-preserving additive automorphism. Every order-preserving additive automorphism of $\Gamma \subseteq \mathbb{R}$ is multiplication by a positive real scalar.*

Proof. Two nonzero elements have equal valuation exactly when their absolute values bound one another up to positive integer factors. Indeed, an element of valuation zero is bounded above and away from zero by ordinary positive real constants, whereas positive valuation makes it smaller than every positive rational. This Archimedean-comparison description is preserved by an ordered field automorphism. Multiplication and comparison therefore induce an ordered group automorphism on the value group.

Fix $\gamma_* > 0$ and let σ be an ordered group automorphism. For every $\gamma \in \Gamma$, the rational inequalities $q\gamma_* < \gamma$ are equivalent to $q\sigma(\gamma_*) < \sigma(\gamma)$. Equality of the rational cuts implies $\sigma(\gamma)/\sigma(\gamma_*) = \gamma/\gamma_*$. Thus $\sigma(\gamma) = c\gamma$ with $c = \sigma(\gamma_*)/\gamma_* > 0$. $\square$

Theorem 274.4 (Automatic normalization). *Every Borel field flow fixes $\mathbb{R}$ pointwise and satisfies*

$$v(\Phi_s x) = v(x) \quad (s \in \mathbb{R}, x \in \mathcal{L}_{\mathbb{R}}^{\Gamma}). \quad (274.1)$$

It induces a Borel real-linear one-parameter representation on each window.

Proof. Real coefficients are fixed by Lemma 274.2. By Lemma 272.1, positive elements are precisely nonzero squares, so every field automorphism preserves the order. Write the induced value action as $\gamma \mapsto c_s \gamma$ using Lemma 274.3. The map

$$s \mapsto c_s = \frac{v(\Phi_s(t^{\gamma_*}))}{\gamma_*}$$

is Borel and takes only countably many values. The group law gives $c_{s+t} = c_s c_t$. Therefore $s \mapsto \log c_s$ is a Borel additive real function and equals as for some $a \in \mathbb{R}$. Countability of its image forces $a = 0$. This proves (274.1). The induced window action is Borel because finite-truncation sections and projections are Borel. $\square$

Remark 274.5 (No hidden time regularity). The preceding argument rules out a value-scaling flow even if no continuity is assumed. Individual Borel automorphisms may scale the value group, but a jointly Borel real-parameter group cannot move it through a nontrivial countable family of scalings. This is one place where both the real time parameter and countability of Γ are essential.

[write] (Batch 96.) In rational rank one the scalings are multiplications by positive rationals, and the proof of Part XVIII's [theorem 248.8](#) shows without any regularity that a real-parameter group of them is trivial ($r(s) = r(s/n)^n$ for every n). For Γ of higher rank the scalars are real, and the argument above uses countability of the value group and Borelness in time instead. Only the coefficients of the images of one monomial enter it, which [proposition 285.1](#) [merge] uses.

275 The exact integration theorem

A derivation in this section is real-linear and satisfies $D(xy) = D(x)y + xD(y)$. Call it *valuation-nondecreasing* when $v(Dx) \geq v(x)$ for every x . It then induces $D_{a,b}$ on every window and is Borel by Lemma [272.2](#).

Definition 275.1 (Truncation-local finiteness). A valuation-nondecreasing derivation D is *truncation-locally finite* if for every $a < b$ and every $w \in \text{Win}_{a,b}$,

$$\dim_{\mathbb{R}} \text{span}_{\mathbb{R}} \{w, D_{a,b}w, D_{a,b}^2w, \dots\} < \infty. \quad (275.1)$$

We abbreviate this condition by TLF. Integer endpoints suffice, and it suffices to check monomial basis vectors in each window.

The last assertion follows because a finite sum of locally finite cyclic subspaces is again finite-dimensional and invariant.

Theorem 275.2 (Exact Borel flow–derivation correspondence). *For a nonzero countable divisible $\Gamma \subseteq \mathbb{R}$, differentiation at time zero gives a bijection between:*

- (i) all Borel field flows on $\mathcal{L}_{\mathbb{R}}^{\Gamma}$;
- (ii) all real-linear, valuation-nondecreasing derivations of $\mathcal{L}_{\mathbb{R}}^{\Gamma}$ satisfying TLF.

For every flow, every $x \in \mathcal{L}_{\mathbb{R}}^{\Gamma}$, and every $\eta \in \Gamma$, the function $s \mapsto [t^{\eta}] \Phi_s x$ extends to an entire function. The generator is characterized coefficientwise by

$$[t^{\eta}] Dx = \left. \frac{d}{ds} [t^{\eta}] \Phi_s x \right|_{s=0}. \quad (275.2)$$

The inverse construction is the compatible family of finite-dimensional matrix exponentials on the cyclic window spaces of D .

Proof. From a flow to a derivation. Theorem [274.4](#) gives real linearity and valuation preservation. Apply Corollary [273.2](#) to each window representation. This gives a locally finite generator $A_{a,b}$ there. Projection between windows and inclusion of a higher lower-cutoff window intertwine the representations; differentiating their finite-dimensional restrictions shows that they also intertwine the generators.

Fix x and choose an integer $a \leq v(x)$. The vectors $A_{a,b}(x \bmod \mathcal{L}_{\mathbb{R}, \geq b}^{\Gamma})$, for integer $b > a$, are compatible finite truncations. They define a unique series Dx : each bounded window has finite support, and all exponents are at least a . Compatibility when a is changed makes this definition independent of the choice. It also shows $v(Dx) \geq v(x)$ and real linearity. Equation (275.2) holds by construction.

To verify the Leibniz rule, take $x \in \mathcal{L}_{\mathbb{R}, \geq a}^{\Gamma}$, $y \in \mathcal{L}_{\mathbb{R}, \geq c}^{\Gamma}$, and a product cutoff b . Modulo $\mathcal{L}_{\mathbb{R}, \geq b}^{\Gamma}$, the product is determined by $x \bmod \mathcal{L}_{\mathbb{R}, \geq b-c}^{\Gamma}$ and $y \bmod \mathcal{L}_{\mathbb{R}, \geq b-a}^{\Gamma}$. The two orbits in

these windows each lie in a fixed finite-dimensional subspace. Consequently all coefficients relevant to their product are differentiable functions in a fixed finite set of coordinates. Differentiate $\Phi_s(xy) = \Phi_s(x)\Phi_s(y)$ at zero. The ordinary finite product rule gives

$$D(xy) \equiv D(x)y + xD(y) \pmod{\mathcal{L}_{\mathbb{R}, \geq b}^\Gamma}.$$

Since b is arbitrary, this is the required identity in $\mathcal{L}_{\mathbb{R}}^\Gamma$. Every induced generator is locally finite by Corollary 273.2, so D has TLF. Borelness follows from Lemma 272.2. That corollary also proves entire dependence of every orbit coefficient.

From a derivation to a flow. Let D satisfy (ii). On a fixed window and a fixed vector w , take the finite-dimensional invariant cyclic space $W = \text{span}_{\mathbb{R}}\{D_{a,b}^n w : n \geq 0\}$. Define

$$\phi_s^{a,b}(w) = \exp(sD_{a,b}|_W)w.$$

The definition is unchanged by passing to a larger finite-dimensional invariant subspace. It is therefore linear in w , and satisfies the group law and inverse law. Window projections and inclusions commute with this construction.

For $x \in \mathcal{L}_{\mathbb{R}, \geq a}^\Gamma$, the compatible vectors $\phi_s^{a,b}(x \bmod \mathcal{L}_{\mathbb{R}, \geq b}^\Gamma)$ define a unique series $\Phi_s x$. Each is finitely supported, and all have the common lower bound a ; hence the series lies in $\mathcal{L}_{\mathbb{R}}^\Gamma$, not just in an unrestricted coefficient product. Independence of a , linearity, and the group law follow from the corresponding window statements. The inverse Φ_{-s} shows that Φ_s is bijective.

Measurability. A linear endomorphism of a countable-dimensional direct sum is Borel. On each finite-dimensional cyclic space its exponential series converges in the ordinary real topology. Thus every coordinate of $\phi_s^{a,b}(w)$ is the pointwise limit of the Borel functions

$$(s, w) \mapsto \left[\sum_{n=0}^N \frac{s^n}{n!} D_{a,b}^n w \right]_\eta.$$

This proves joint Borelness of the window maps. A countable partition by an integer lower bound for x , followed by integer upper cutoffs, proves joint coefficient-Borelness of $(s, x) \mapsto \Phi_s x$.

Multiplicativity. Use the input and output windows from the first half of the proof. Choose finite-dimensional D -invariant subspaces X, Y containing the two input truncations. The span Z of their products in the product window is finite-dimensional and D -invariant, because D satisfies Leibniz. The vector $z(s) = \phi_s(x)\phi_s(y)$ in Z satisfies the finite-dimensional ordinary differential equation

$$z'(s) = D|_Z z(s), \quad z(0) = xy.$$

Uniqueness for this matrix equation gives $z(s) = \exp(sD|_Z)(xy)$. Thus $\Phi_s(xy) \equiv \Phi_s(x)\Phi_s(y) \pmod{\mathcal{L}_{\mathbb{R}, \geq b}^\Gamma}$. Again b is arbitrary. Since $D(1) = 0$, the maps are unital field automorphisms. They preserve the valuation: both Φ_s and its inverse are valuation-nondecreasing.

Inverse constructions and uniqueness. On every finite-dimensional cyclic window space, differentiating the matrix exponential recovers D . Conversely, the unique matrix one-parameter group with generator $A_{a,b}$ is $e^{sA_{a,b}}$. All windows separate elements, so the two constructions are inverse, and the flow with a given generator is unique. $\square$

[write] (Batch 96.) With corollary 275.3 this answers Part XVII's question 237.5 for countable divisible Γ (dated note there): coefficientwise analyticity need not be imposed, since every

orbit coefficient is entire, and the integrable derivations are exactly those of (ii). For $\Gamma = \mathbb{Q}$ the same classification is Part XVIII's [theorems 248.3](#) and [248.8](#), proved independently and under hypotheses neither of which contains the other ([corollary 285.2](#) [merge]); by [proposition 285.1](#) [merge] the joint Borelness assumed in [definition 274.1](#) follows from Borelness in time of the images of the monomials of one rational basis.

Corollary 275.3 (Answer for arbitrary Borel derivations). *Every coefficient-Borel field derivation of $\mathcal{L}_{\mathbb{R}}^{\Gamma}$ is real-linear. Such a derivation generates a Borel field flow if and only if it does not decrease valuation and satisfies TLF. Thus no separate analyticity hypothesis is needed in the divisible-exponent case of the prior draft's Question 13.5.*

Proof. For each output exponent, the restriction of the derivation to $\mathbb{R}$ is a Borel additive scalar function, hence a real multiple of its argument. The multiple is zero because $D(1) = 0$. Therefore $D(r) = 0$ for $r \in \mathbb{R}$, and Leibniz gives $D(rx) = rD(x)$. Apply Theorem [275.2](#). $\square$

[write] (Batch 96.) The first assertion is the clause of Part XVII's [theorem 230.2](#) that a Borel derivation vanishes on $\mathbb{R}$, there a consequence of [theorem 228.2](#), for every countable Γ ; Part XVIII ([theorem 242.2](#)) and Part XXI ([corollary 288.4](#)) prove it again in rational rank one. "The prior draft's Question 13.5" is [question 237.5](#).

Corollary 275.4 (Negative valuation gain is impossible). *If a real-linear derivation D has $v(Dx) < v(x)$ for some x , it is not the coefficientwise generator of any Borel field flow. In particular this holds for $D(t^q) = qt^{q-1}$ on $\mathcal{L}_{\mathbb{R}}^{\mathbb{Q}}$.*

Proof. The necessary inequality is part of Theorem [275.2](#). In the displayed example, already $D(t) = 1$ lowers valuation. $\square$

[write] (Batch 96.) The same conclusion, for coefficient-differentiable one-parameter actions with a Borel generator rather than for Borel flows, is Part XXI's [remark 294.3](#), and in rational rank one the last clause of Part XVIII's [theorem 248.3](#); the three are proved independently. The displayed example is $d/dt = t^{-1}E$, of gain -1 by Part XVII's ([231.3](#)).

Example 275.5 (Diagonal flows are analytic, not formally summable). Let $\lambda : \Gamma \rightarrow \mathbb{R}$ be additive. The derivation

$$D_{\lambda} \left(\sum x_{\gamma} t^{\gamma} \right) = \sum \lambda(\gamma) x_{\gamma} t^{\gamma}$$

is valuation-nondecreasing and satisfies TLF, since it leaves every finite support invariant. Its flow is

$$\Phi_s \left(\sum x_{\gamma} t^{\gamma} \right) = \sum e^{s\lambda(\gamma)} x_{\gamma} t^{\gamma}. \quad (275.3)$$

When $\lambda(\gamma) \neq 0$, infinitely many powers $D_{\lambda}^n(t^{\gamma})$ contribute to the same coefficient. The operator exponential therefore uses ordinary real convergence at that coefficient; its terms are not a left-finitely summable family. The main theorem intentionally includes this case. It must not be replaced by an assertion of formal summability for every Borel flow.

[write] (Batch 96.) For the coordinate maps $\lambda = \ell_i$ of a rational basis this is Part XVII's [proposition 239.3](#) [merge], whose proof, with the homomorphism $\gamma \mapsto e^{s\lambda(\gamma)}$ from Γ to $\mathbb{R}_{>0}$ in place of $e^{s\ell_i(\gamma)}$, checks (275.3) directly; part (iv) there is the failure of formal summability stated here. An additive λ on the divisible group Γ is $\mathbb{Q}$ -linear, so $D_{\lambda} = \sum_i \lambda(b_i) E_i$ in the sense of Part XVII's [theorem 231.1](#), a Borel derivation of gain zero (or the zero derivation).

276 Strict increase: local nilpotence versus uniform gain

Definition 276.1. A real-linear derivation is *strict* if $v(Dx) > v(x)$ for every nonzero x , with $v(0) = +\infty$ as before. It has *uniform positive gain* if there is a real $\varepsilon > 0$ such that $v(Dx) \geq v(x) + \varepsilon$ for all nonzero x .

Strictness permits $Dx = 0$. It does not assert one common positive gain. The distinction is essential at infinite rational rank.

Theorem 276.2 (Strict integration criterion). *For a strict real-linear derivation D of $\mathcal{L}_{\mathbb{R}}^{\Gamma}$, the following conditions are equivalent:*

- (i) D generates a Borel field flow.
- (ii) D satisfies TLF.
- (iii) Every window map $D_{a,b}$ is locally nilpotent.
- (iv) For every $x \in \mathcal{L}_{\mathbb{R}}^{\Gamma}$, $v(D^n x) \rightarrow +\infty$ as $n \rightarrow \infty$.
- (v) For every $\gamma \in \Gamma$, $v(D^n t^{\gamma}) \rightarrow +\infty$.

When these hold, the unique flow is the genuine left-finite formal sum

$$\Phi_s x = \sum_{n \geq 0} \frac{s^n}{n!} D^n x. \quad (276.1)$$

For fixed x and fixed exponent, its coefficient is a polynomial in s .

Proof. The equivalence of (i) and (ii) is Theorem 275.2. On a finite-dimensional $D_{a,b}$ -invariant space W , let S be the finite union of the supports of a basis. For a nonzero vector, applying $D_{a,b}$ either gives zero or strictly increases the least exponent. All iterates remain supported in S . Hence $D_{a,b}^{|S|}$ vanishes on W . This proves (ii) $\Rightarrow$ (iii); the reverse implication is immediate.

Condition (iii), applied to the truncation of a fixed x , says that for each b some power lies in $\mathcal{L}_{\mathbb{R}, \geq b}^{\Gamma}$. All subsequent powers remain there, since D preserves every filtration level. This is exactly (iv). The implication (iv) $\Rightarrow$ (v) is immediate. Conversely, truncate x below b to finitely many monomials. By (v), a sufficiently high common power sends every one of them into $\mathcal{L}_{\mathbb{R}, \geq b}^{\Gamma}$. The tail of x was already in $\mathcal{L}_{\mathbb{R}, \geq b}^{\Gamma}$ and stays there, proving (iv).

Under (iv), only finitely many powers in (276.1) contribute below any fixed bound. Each power itself has left-finite support. Thus this is a left-finitely summable family, with only finitely many terms at every coefficient. Its window realization is exactly the matrix exponential in Theorem 275.2, now a finite nilpotent exponential on each cyclic window space. $\square$

Corollary 276.3 (The safe uniform-gain case). *Every real-linear derivation with uniform positive gain generates a unique Borel field flow. If the gain is at least $\varepsilon > 0$, then modulo $\mathcal{L}_{\mathbb{R}, \geq b}^{\Gamma}$ the exponential on $x \neq 0$ only requires powers with $n\varepsilon < b - v(x)$.*

Proof. Induction gives $v(D^n x) \geq v(x) + n\varepsilon$. Apply Theorem 276.2. $\square$

[write] (Batch 96.) [Corollary 276.3](#), with the formula (276.1) and the joint Borelness of [theorem 275.2](#), is the exponential half of Part XVII's [theorem 234.2](#) with [corollary 234.3](#), which source 26 credits ("the earlier ProveIt draft, Section 10", that is [section 234](#)) and proves

again here; Part XVII's theorem does not need divisibility. [Theorem 276.2](#) extends it from a uniform gain to escape of the iterated valuations.

Remark 276.4 (Local nilpotence is on windows, not on the field). A nonzero derivation of a characteristic-zero field cannot be locally nilpotent on the entire field. Indeed, under global local nilpotence one could choose b with $Db \neq 0$ and $D^2b = 0$ by taking a suitable derivative of a nonconstant element. But $D^n(b^{-1}) = (-1)^n n! (Db)^n b^{-n-1} \neq 0$ for every n . Our nilpotence statements concern quotient *vector spaces* $\text{Win}_{a,b}$, not quotient fields or ideals of the whole field.

277 A sharp rational-rank dichotomy

277.1 Logarithmic monomial values

For any derivation D set

$$\delta(\gamma) = t^{-\gamma} D(t^\gamma). \quad (277.1)$$

Leibniz shows that $\delta : \Gamma \rightarrow \mathcal{L}_{\mathbb{R}}^\Gamma$ is additive, hence $\mathbb{Q}$ -linear. For strict D , every $\delta(\gamma)$ is zero or has positive valuation. Conversely, this observation permits explicit constructions.

Lemma 277.1 (Constructing strict derivations). *Let B be a $\mathbb{Q}$ -basis of Γ , and choose for each $\beta \in B$ an element $\delta_\beta \in \mathcal{L}_{\mathbb{R}}^\Gamma$ that is zero or has positive valuation. Extend $\beta \mapsto \delta_\beta$ $\mathbb{Q}$ -linearly to $\delta : \Gamma \rightarrow \mathcal{L}_{\mathbb{R}}^\Gamma$. Then*

$$D \left(\sum_{\gamma} x_{\gamma} t^{\gamma} \right) = \sum_{\gamma} x_{\gamma} t^{\gamma} \delta(\gamma) \quad (277.2)$$

defines a strict, Borel, real-linear derivation. Every strict real-linear derivation is obtained uniquely in this way.

Proof. A finite rational combination of the δ_β is zero or has positive valuation. Below a fixed bound b , only the finitely many input exponents $\gamma < b$ can contribute to (277.2), and each of the corresponding series has finite support there. Thus the sum is left-finitely summable. All nonzero output exponents exceed the corresponding input exponent. Its least exponent, when it exists, is strictly greater than the least input exponent. This proves strictness.

Additivity and real linearity are immediate. On monomials the Leibniz identity is exactly $\delta(\gamma + \eta) = \delta(\gamma) + \delta(\eta)$. For general series, truncate inputs sufficiently far to compute any fixed output range and apply the finite identity there. This proves Leibniz without an unjustified exchange of infinite sums. Borelness follows from Lemma 272.2. Conversely, any strict real-linear derivation satisfies that lemma and is therefore determined by its monomial values, giving precisely (277.2). $\square$

[write] (Batch 96.) Source 26's $d(\gamma)$ is printed $\delta(\gamma)$ because it is Part XVII's logarithmic value (230.2), and (277.2) is (230.3). [Lemma 277.1](#) is therefore the strict case of Part XVII's [theorem 230.2](#): values in $\mathfrak{m} \cup \{0\}$ have the common lower bound 0. By (230.4) the optimal gain of such a derivation is $\inf_{\gamma} v(\delta(\gamma)) \geq 0$, which can be 0: a strict derivation need not have uniform positive gain, which is the point of this section.

Theorem 277.2 (Finite rational rank is exactly the universal-integration regime). *For every nonzero countable divisible $\Gamma \subseteq \mathbb{R}$,*

$$\dim_{\mathbb{Q}} \Gamma < \infty \iff \text{every strict real-linear derivation of } \mathcal{L}_{\mathbb{R}}^\Gamma \text{ generates a Borel flow.} \quad (277.3)$$

In the finite-rank case, every nonzero strict derivation in fact has uniform positive gain.

Proof. Suppose $\beta_1, \dots, \beta_r$ is a basis and $D \neq 0$ is strict. Take the minimum of the finitely many finite numbers $v(\delta(\beta_i))$ for which $\delta(\beta_i) \neq 0$, and call it ε . It is positive. Rational linearity implies $v(\delta(\gamma)) \geq \varepsilon$ for every γ , using $v(0) = +\infty$. The termwise formula gives $v(Dx) \geq v(x) + \varepsilon$. Corollary 276.3 proves integrability. The zero derivation gives the identity flow.

For infinite rank, Lemma 277.5 and Proposition 277.7 below construct a strict derivation with bounded iterated monomial valuations. Theorem 276.2 then excludes a Borel flow. $\square$

[write] (Batch 96.) The finite-rank gain is Part XVII's (231.3), $\text{gain}(D) = \min_i v(a_i)$ with $a_i = \delta(\beta_i)$. Part XVII's example 232.3 is an infinite-rank strict derivation of gain at least 1, which integrates by corollary 276.3; the derivation of proposition 277.7 has gain 0 because its gaps κ_n tend to 0. The two examples do not conflict; together with section 231.2 they show three different effects of infinite rational rank (no common lower bound for Borelness, gain 0 without escape for integration, and bounded supports of the logarithmic values).

Corollary 277.3 (Complete finite-rank generator criterion). *If $\dim_{\mathbb{Q}} \Gamma < \infty$, a Borel derivation of $\mathcal{L}_{\mathbb{R}}^{\Gamma}$ generates a Borel field flow if and only if it does not decrease valuation.*

Proof. Necessity holds in every rank. For sufficiency choose a finite rational basis $\beta_1, \dots, \beta_r$. The logarithmic values $\delta(\beta_i)$ have nonnegative valuation. Let Δ be the union of their supports with zero removed. It is a left-finite set of positive exponents. Its additive monoid Σ , including zero, is left-finite: below a fixed positive bound only finitely many members of Δ are available, and the length of a sum is bounded using the positive minimum of Δ . The empty case is immediate.

Every $\delta(\gamma)$ is a rational linear combination of the basis values. Consequently $D(t^{\gamma})$ is supported in $\gamma + \Sigma$, and induction gives the same support bound for every $D^n(t^{\gamma})$. Its intersection with a bounded window is finite, independent of n . Thus each cyclic window space is finite-dimensional. Apply Theorem 275.2, using real linearity from Corollary 275.3. $\square$

[write] (Batch 96.) In Part XVII's normal form (231.4), $D = \sum_i a_i E_i$ generates a Borel field flow exactly when every a_i is 0 or has $v(a_i) \geq 0$, by (231.3). This extends Part XVII's theorem 234.2 and proposition 239.3 to every derivation of gain zero in finite rank. At $\Gamma = \mathbb{Q}$ it is the criterion of Part XVIII's theorem 248.3 (corollary 285.2 [merge]).

Example 277.4 (A zero-gain, non-diagonal flow). On $\mathcal{L}_{\mathbb{R}}^{\mathbb{Q}}$ the derivation $D(t^q) = qt^q(1+t)$ has gain zero. Its flow is

$$\Phi_s(t^q) = e^{qs} t^q (1 - (e^s - 1)t)^{-q}. \quad (277.4)$$

For $q = 1$, this is the fractional-linear substitution $t \mapsto e^s t / (1 - (e^s - 1)t)$. Direct composition gives the group law, and differentiating at zero gives $D(t) = t + t^2$. Positive rational powers and the tail argument extend the construction to the field. The coefficient at t^{q+n} is $e^{qs} \binom{-q}{n} (-(e^s - 1))^n$, a real exponential polynomial. The denominator always has leading coefficient one, so there is no finite-real-time singularity in this formal field. This is an explicit example outside the strict and diagonal subclasses.

[write] (Batch 96.) In Part XVII's notation $D = (1+t)E$, of gain $v(1+t) = 0$. The flow is the substitution $f \mapsto f \circ h_s$ with $h_s = e^s t / (1 - (e^s - 1)t) = e^s t(1 + u_s)$, $u_s \in \mathfrak{m}$, a Borel ordered-field automorphism by Part XVIII's theorem 242.4. The write checked the group law in the variable $T = t^{-1}$, where h_s acts by $T + 1 \mapsto e^{-s}(T + 1)$, and $\frac{d}{ds} h_s|_{s=0} = t + t^2$. It is not Part XVII's flow (234.4), $t \mapsto t/(1 - st)$, which is generated by tE and has positive gain.

277.2 A bounded independent chain in every infinite-rank group

Lemma 277.5. *If Γ has infinite rational rank, it contains a $\mathbb{Q}$ -linearly independent sequence $(\beta_n)_{n \geq 0}$ with*

$$1 < \beta_0 < \beta_1 < \cdots < 2, \quad \beta_n \longrightarrow 2.$$

It extends to a countable $\mathbb{Q}$ -basis of Γ .

Proof. Every nonzero divisible subgroup of $\mathbb{R}$ is dense, since it contains all rational multiples of one positive element. Choose $\beta_0 \in \Gamma \cap (1, 3/2)$. Given the first n independent members, their rational span W is proper. Choose $z \in \Gamma \setminus W$. The set $z + \mathbb{Q}\beta_0$ is dense and disjoint from W . Choose β_n from it in $(2 - 2^{-n}, 2 - 2^{-(n+1)})$. This gives the claimed independent increasing sequence. Enumerate Γ and adjoin any element not yet in the span to extend it to a countable basis. $\square$

Example 277.6 (One explicit exponent group). Fix any transcendental $\vartheta \in (0, 1)$ and set $\beta_n = 2 - \vartheta^{n+1}$. These numbers increase to 2 and are rationally independent: a finite relation would give a nonzero polynomial over $\mathbb{Q}$ vanishing at ϑ . Thus $\Gamma = \text{span}_{\mathbb{Q}}\{2 - \vartheta^{n+1} : n \geq 0\}$ is an explicit family of exponent groups for all the counterexamples below. No numerical approximation to ϑ is used in their proofs.

Proposition 277.7 (A strict derivative that cannot flow). *In every infinite-rank Γ as above, there is a strict Borel real-linear derivation D and monomials $z_n = t^{\beta_n}$ such that*

$$D(z_n) = z_{n+1}, \quad D^k(z_0) = z_k. \quad (277.5)$$

This derivation generates no Borel field flow.

Proof. Complete the β_n to a basis $B = \{\beta_n : n \geq 0\} \cup C$. Put $\kappa_n = \beta_{n+1} - \beta_n > 0$ and prescribe

$$\delta(\beta_n) = t^{\kappa_n}, \quad \delta(c) = 0 \quad (c \in C).$$

Lemma 277.1 gives the derivation. The displayed identities follow immediately, while $v(D^k z_0) = \beta_k < 2$ for every k . Theorem 276.2 excludes integration. $\square$

For $s \neq 0$ the tempting expression

$$\sum_{k \geq 0} \frac{s^k}{k!} z_k \quad (277.6)$$

is not an element of $\mathcal{L}_{\mathbb{R}}^{\Gamma}$: its support has infinitely many members below 2. The support is well ordered, so this *individual series* is a legitimate Hahn series in the larger field with arbitrary well-ordered support. That fact does not show that the derivation, or a flow, extends to the whole larger field. Such an extension would require a separate summability proof.

278 Alternating shears: addition and bracket both fail

Use the bounded independent chain from the preceding section. Split its successor derivative into two derivations, prescribed on basis monomials by

$$\begin{aligned} D_e(z_{2n}) &= z_{2n+1}, & D_e(z_{2n+1}) &= 0, & D_e(t^c) &= D_o(t^c) = 0 \quad (c \in C), \\ D_o(z_{2n}) &= 0, & D_o(z_{2n+1}) &= z_{2n+2}, \end{aligned} \quad (278.1)$$

These prescriptions use Lemma 277.1; both derivations are strict and Borel. They are not globally nilpotent derivations of the field.

278.1 Explicit flows for the two halves

For the unique basis expansion $\gamma = \sum_i q_i \beta_i + \sum_{c \in \mathbb{C}} r_c c$, with finitely many nonzero rational coordinates, set

$$\begin{aligned}\Phi_s^e(t^\gamma) &= t^\gamma \prod_{i \text{ even}} (1 + s t^{\kappa_i})^{q_i}, \\ \Phi_s^o(t^\gamma) &= t^\gamma \prod_{i \text{ odd}} (1 + s t^{\kappa_i})^{q_i}.\end{aligned}\tag{278.2}$$

Each product is finite, and rational powers mean their positive binomial expansions. Extend the maps real-linearly over the monomial terms of every left-finite series.

Proposition 278.1 (The shears really are Borel flows). *The maps $(s, x) \mapsto \Phi_s^e x$ and $(s, x) \mapsto \Phi_s^o x$ are Borel field flows with generators D_e and D_o , respectively. On basis monomials,*

$$\begin{aligned}\Phi_s^e(z_{2n}) &= z_{2n} + s z_{2n+1}, & \Phi_s^e(z_{2n+1}) &= z_{2n+1}, \\ \Phi_s^o(z_{2n}) &= z_{2n}, & \Phi_s^o(z_{2n+1}) &= z_{2n+1} + s z_{2n+2}.\end{aligned}\tag{278.3}$$

Proof. For a fixed input monomial, only finitely many positive gaps κ_i occur in (278.2). The semigroup they generate has finite intersection with each bounded interval: the number of factors counted with multiplicity is bounded by the interval length divided by their positive minimum. Thus every binomial product is left-finite, with leading term t^γ .

For a general input and a cutoff b , only finitely many input exponents begin below b , and each of their images contributes only finitely many terms below b . This proves that the extension is well-defined and valuation-preserving. The formulas are multiplicative on monomials by addition of rational basis coordinates. Truncating proves multiplicativity on all series. They are unital and fix real coefficients.

Equation (278.3) follows directly. For Φ^e , the odd basis monomials are fixed, so $\Phi_s^e \Phi_t^e(z_{2n}) = z_{2n} + (s+t)z_{2n+1}$; all other basis monomials are fixed. Multiplicative maps preserving positive rational powers agree on every t^γ once they agree on the basis monomials. Here positivity also follows directly from leading terms. The tail argument extends equality to all series. Hence $\Phi_s^e \Phi_t^e = \Phi_{s+t}^e$ and Φ_{-s}^e is the inverse. The proof for Φ^o is identical, with even and odd interchanged.

For fixed x and output cutoff, all potentially contributing exponent combinations form a finite set independent of s . Each resulting coefficient is a polynomial in s . Joint Borelness follows either by these finite formulas on a countable partition or by viewing the coordinate sums as limits of Borel partial sums. Differentiating the binomial products at zero gives exactly the logarithmic values that define D_e and D_o . This proves the generator assertions. $\square$

Theorem 278.2 (Integrable strict derivations are not Lie-closed). *If $\dim_{\mathbb{Q}} \Gamma = \aleph_0$, there are strict Borel real-linear derivations D_e, D_o generating Borel flows such that neither $D_e + D_o$ nor $[D_e, D_o] = D_e \circ D_o - D_o \circ D_e$ generates a Borel field flow.*

Proof. Use the preceding construction. The sum is exactly the derivation D of Proposition 277.7, so it does not integrate. Set $D_{eo} = [D_e, D_o]$. Directly from (278.1),

$$D_{eo}(z_{2n}) = -z_{2n+2}, \quad D_{eo}(z_{2n+1}) = z_{2n+3}.\tag{278.4}$$

A commutator of derivations is a derivation. It is strict here: each composition either vanishes or strictly increases valuation, and a difference cannot lower the minimum of the two output valuations. By induction,

$$D_{eo}^k(z_0) = (-1)^k z_{2k}, \quad v(D_{eo}^k z_0) = \beta_{2k} < 2.$$

The strict integration criterion excludes a Borel flow for D_{eo} . $\square$

Remark 278.3 (The precise failure). The set of all valuation-nondecreasing real derivations is closed under addition and Lie bracket. The set of strict ones is also so closed. What fails at infinite rank is closure of the *integrable* strict subclass. Individual local nilpotence on every window does not imply joint local nilpotence of the associative algebra generated by two operators. Alternating between them can traverse an infinite bounded chain that neither operator traverses by itself.

279 Two time-one maps whose product is not a time-one map

The preceding obstruction is not merely a failure of one proposed logarithm or of a particular BCH expansion.

Theorem 279.1 (Nonclosure of Borel time-one maps). *For infinite rational rank, let Φ^e, Φ^o be the flows above and put $A = \Phi_1^e \circ \Phi_1^o$. Although both factors are time-one maps of Borel field flows, A is not the time-one map of any Borel field flow on $\mathcal{L}_{\mathbb{R}}^{\Gamma}$.*

Proof. The basis action is

$$\begin{aligned} A(z_{2n}) &= z_{2n} + z_{2n+1}, \\ A(z_{2n+1}) &= z_{2n+1} + z_{2n+2} + z_{2n+3}. \end{aligned} \tag{279.1}$$

For $k \geq 1$, the finite vector $A^k z_0$ has largest index $2k - 1$, with coefficient 1. This follows by induction: the largest index is odd, its image creates the next odd index two places later, and every smaller index creates only smaller indices. No other term can cancel or reach the new largest index. The vector z_0 has largest index zero. Therefore

$$z_0, Az_0, A^2 z_0, \dots$$

are linearly independent. All their exponents lie in $(1, 2)$, so they remain linearly independent in the single window $\text{Win}_{1,2}$.

Suppose $A = \Phi_1$ for a Borel field flow. By automatic normalization this induces a Borel real-parameter representation on $\text{Win}_{1,2}$. Theorem 273.1 makes the span of the entire real orbit of z_0 finite-dimensional. Its integer suborbit contains the displayed independent sequence, a contradiction. $\square$

Corollary 279.2 (A practical necessary test). *If a Borel automorphism A is the time-one map of a Borel field flow, then it fixes the valuation and its induced action on each window is locally finite. In particular, a single monomial with an infinite-dimensional integer-orbit span in one window certifies nonembeddability into such a flow.*

[write] (Batch 96.) Source 26 gives no separate proof; it is the argument of the last paragraph of the proof of [theorem 279.1](#). If $A = \Phi_1$, then $v(Ax) = v(x)$ by [theorem 274.4](#), and in every window [theorem 273.1](#) puts each vector w in a finite-dimensional space invariant under every Φ_s , which contains $A^n w$ for all $n \geq 0$.

The theorem does not assert that A has no algebraic roots, nor that it cannot belong to a flow with nonmeasurable time dependence. It excludes all flows in the precisely defined jointly Borel category. This is stronger than observing that one chosen formal logarithm leaves $\mathcal{L}_{\mathbb{R}}^{\Gamma}$.

279.1 An exact converse and uniqueness for time-one maps

The necessary test can in fact be completed. The order excludes the rotational logarithm ambiguities familiar over complex vector spaces.

Lemma 279.3 (Real exponential-polynomial uniqueness). *Let $f(s) = \sum_{j=1}^r p_j(s)e^{a_j s}$, where the a_j are distinct real numbers and the p_j are real polynomials. If $f(n) = 0$ for every nonnegative integer n , then every p_j is zero.*

Proof. Order the a_j increasingly and divide the identity at n by $e^{a_r n}$. The lower-exponent terms tend to zero, so $p_r(n) \rightarrow 0$. A nonzero real polynomial cannot tend to zero along the positive integers. Thus $p_r = 0$, and descending induction proves the assertion. $\square$

Lemma 279.4 (Positive-spectrum powers on invariant finite spaces). *Let T be an invertible linear map on a finite-dimensional real vector space, with all eigenvalues positive real. There is a canonical one-parameter group $T^{[s]}$ with $T^{[1]} = T$, whose entries are real exponential polynomials. This construction commutes with invariant subspace restriction, quotient, and linear intertwining maps.*

Proof. On the generalized eigenspace for an eigenvalue $\lambda > 0$, write $T = \lambda(I + N)$ with N nilpotent, and define

$$T^{[s]} = \lambda^s \sum_{k \geq 0} \binom{s}{k} N^k. \quad (279.2)$$

The sum is finite. The generalized eigenspace decomposition is canonical, so this defines the operator without a basis choice. The binomial Vandermonde identity gives the group law; alternatively verify the polynomial identity first at nonnegative integer arguments, then at all real arguments. Entries are polynomials times $e^{s \log \lambda}$.

If a linear map L satisfies $LT = SL$, then $LT^n = S^n L$ for every nonnegative integer n . Each entry of $LT^{[s]} - S^{[s]}L$ is a real exponential polynomial. Apply Lemma 279.3. Restrictions and quotients are special cases. $\square$

Theorem 279.5 (Exact and unique time-one embedding). *Let A be a coefficient-Borel field automorphism of $\mathcal{L}_{\mathbb{R}}^{\Gamma}$. It is the time-one map of a Borel field flow if and only if:*

- (i) $v(Ax) = v(x)$ for every x ;
- (ii) in every window, each vector has a finite-dimensional span under $I, A, A^2, \dots$

When these conditions hold, the Borel field flow with time-one map A is unique. Its generator is obtained by taking the canonical real matrix logarithm on every finite-dimensional invariant window space.

Proof. Necessity is Corollary 279.2. For sufficiency, first note that A fixes $\mathbb{R}$ by Lemma 274.2, and is order-preserving by Lemma 272.1. Each finite-dimensional cyclic space W in a window is A -invariant. Injectivity of A and finite dimensionality imply $AW = W$.

A finite-dimensional subspace of a window has a basis with distinct least exponents, obtained by Gaussian elimination on its finite union of supports. The corresponding valuation flag has one-dimensional successive quotients. Since A preserves valuation, its matrix in this basis is triangular. Its diagonal entries are positive: they are the ratios of leading coefficients of a vector and its order-preserving image. In particular $A|_W$ has positive real spectrum.

Define $A^{[s]}$ on W by Lemma 279.4. Passing to sums of cyclic spaces proves consistency, linearity, and the group law on the whole window. The intertwining conclusion of that lemma gives compatibility between windows. For each series, these compatible finite truncations, all with its original lower bound, assemble a unique series in $\mathcal{L}_{\mathbb{R}}^{\Gamma}$. This gives maps Φ_s with $\Phi_0 = \text{id}$, $\Phi_1 = A$, and $\Phi_s \Phi_t = \Phi_{s+t}$.

For Borelness, enumerate a basis of a window and let W_n be the sum of the cyclic spaces of its first n basis vectors. These are finite-dimensional invariant spaces exhausting the window. On each $\mathbb{R} \times W_n$, the displayed matrix formula is continuous in time and in the finite-dimensional input coordinates. The W_n are Borel, so the assembled window map is Borel. Integer lower and upper cutoffs then give joint Borelness on $\mathbb{R} \times \mathcal{L}_{\mathbb{R}}^{\Gamma}$.

To prove multiplicativity, take lower bounds a, c for x, y and an output cutoff b . Choose finite-dimensional invariant spaces for the input truncations in $\text{Win}_{a,b-c}$ and $\text{Win}_{c,b-a}$. Their product span in $\text{Win}_{a+c,b}$ is finite-dimensional and invariant under A and A^{-1} . For every output coefficient, the difference

$$\Phi_s(xy) - \Phi_s(x)\Phi_s(y)$$

modulo this cutoff is a real exponential polynomial. For every nonnegative integer $s = n$, it is zero, because $\Phi_n = A^n$ is multiplicative. Lemma 279.3 proves that it is zero for all real s . The same argument, or $A1 = 1$, shows $\Phi_s 1 = 1$. Thus the maps are field automorphisms. They preserve all filtration levels, and their inverses do as well, proving exact valuation preservation.

For uniqueness, let Ψ be any Borel field flow with $\Psi_1 = A$. In each window, every w lies in a finite-dimensional Ψ -invariant space W by Theorem 273.1. All matrices $\Psi_s|_W$ preserve the valuation flag and hence are triangular in a common basis. Their infinitesimal generator is a real triangular matrix, so its exponential has real exponential-polynomial entries. The canonical powers of $A|_W$ have the same property. At all nonnegative integers they agree; Lemma 279.3 makes them equal for every real s . All windows separate series, proving uniqueness. Differentiating (279.2) is precisely the canonical real logarithm on each generalized eigenspace. $\square$

Corollary 279.6 (Tangent-to-identity logarithms). *Suppose A is a coefficient-Borel field automorphism and $U = A - \text{id}$ strictly increases valuation. Then A is a time-one map if and only if U is locally nilpotent on every window. In that case*

$$\Phi_s x = \sum_{n \geq 0} \binom{s}{n} U^n x, \quad Dx = \sum_{n \geq 1} \frac{(-1)^{n-1}}{n} U^n x \quad (279.3)$$

are left-finitely summable formulas for the unique flow and its generator.

Proof. Strictness of U implies valuation preservation of A . Local finiteness of A is equivalent to local finiteness of U , since each is a polynomial in the other. Strictness turns local finiteness of U into local nilpotence by the finite-support argument of Theorem 276.2. Apply Theorem 279.5; its finite-dimensional formula now has the sole eigenvalue 1 and is the binomial formula above. The derivative at zero gives the logarithm. $\square$

Corollary 279.7 (Finite-rank time-one criterion). *If $\dim_{\mathbb{Q}} \Gamma < \infty$, a Borel field automorphism is a time-one map of a Borel field flow if and only if it preserves the valuation. The embedding flow is unique.*

Proof. Only sufficiency remains to be shown. Write the images of finite basis monomials as $A(t^{\beta_i}) = a_i t^{\beta_i} (1 + u_i)$, with $a_i > 0$ and $v(u_i) > 0$, allowing $u_i = 0$. Let Δ be the finite union of the supports of the u_i , and let Σ be its additive monoid. As in Corollary 277.3, Σ is left-finite. Rational powers show that $A(t^\gamma)$ is supported in $\gamma + \Sigma$ for every γ . Iteration preserves this support bound. Hence each monomial has a finite-dimensional cyclic space in every window, and Theorem 279.5 applies. $\square$

[write] (Batch 96.) Corollary 279.6 generalizes the logarithm half of Part XVII's theorem 234.2: a uniformly tangent automorphism, with $\text{gain}(A - \text{id}) \geq \varepsilon > 0$, has $U = A - \text{id}$ locally nilpotent on every window, and (279.3) is (234.2); here only window-local nilpotence is required. Part XVII's proof that the logarithm is a derivation, by a polynomial identity vanishing on the natural numbers, is the same device as the exponential-polynomial uniqueness of lemma 279.3.

280 The integrability locus is Borel

Fix a countable $\mathbb{Q}$ -basis B of Γ . Let

$$\mathfrak{m} = \{u \in \mathcal{L}_{\mathbb{R}}^{\Gamma} : v(u) > 0\}, \quad \mathfrak{P}_B = \mathfrak{m}^B,$$

where $0 \in \mathfrak{m}$. This is a standard Borel space. By Lemma 277.1, its points parameterize exactly the strict real-linear derivations via their logarithmic basis values. No computability of the basis or of the ordered embedding is assumed.

Lemma 280.1 (Borel evaluation in parameters). *The evaluation map $(p, x) \mapsto D_p x$ from $\mathfrak{P}_B \times \mathcal{L}_{\mathbb{R}}^{\Gamma}$ to $\mathcal{L}_{\mathbb{R}}^{\Gamma}$ is Borel, and so is $(p, x) \mapsto D_p^n x$ for every fixed $n \in \mathbb{N}$.*

Proof. For fixed γ , its rational basis expansion is finite, so every coefficient of $\delta_p(\gamma)$ is a finite linear combination of coordinates of p . For an output exponent η ,

$$[t^\eta] D_p x = \sum_{\gamma \in \Gamma} x_\gamma [t^{\eta-\gamma}] \delta_p(\gamma).$$

Only input exponents $\gamma < \eta$ can contribute; among those in $\text{supp } x$ there are finitely many. Enumerate Γ and take partial sums: they are Borel and eventually stabilize for every (p, x) . The limit is Borel. Induction and composition prove the assertion for iterates. $\square$

[write] (Batch 96.) $\mathfrak{P}_B$ is the Borel subset of Part XVII's parameter space $\mathcal{B}_{\Gamma}$ (proposition 231.2) on which every parameter has positive valuation or is zero, and lemma 280.1 is the restriction of the evaluation map proved Borel there.

Theorem 280.2 (Borel integrability locus). *The subset $\mathfrak{I}_B \subseteq \mathfrak{P}_B$ consisting of strict derivations that generate Borel field flows is Borel. Explicitly,*

$$\mathfrak{I}_B = \bigcap_{\gamma \in \Gamma} \bigcap_{m \in \mathbb{Z}} \bigcup_{n \in \mathbb{N}} \left\{ p : v(D_p^n t^\gamma) \geq m \right\}. \quad (280.1)$$

On $\mathfrak{I}_B \times \mathbb{R} \times \mathcal{L}_{\mathbb{R}}^{\Gamma}$, the universal evaluation $(p, s, x) \mapsto \exp(s D_p) x$ is Borel.

Proof. For a strict derivation the valuation sequence of an iterated monomial is nondecreasing, including its possible value $+\infty$. Thus the right-hand side of (280.1) is exactly condition (v)

of Theorem 276.2. Its constituent sets are Borel by Lemma 280.1; all quantifiers range over countable sets. For the final assertion, the exponential partial sums are jointly Borel. On $\mathfrak{S}_B$, each coefficient eventually stabilizes for each (s, x) , and the limit belongs to $\mathcal{L}_{\mathbb{R}}^{\Gamma}$. Taking coordinate limits proves the assertion. $\square$

Remark 280.3 (Borel does not mean decidable). Formula (280.1) is a descriptive classification, not an algorithm deciding integrability from finite information. It quantifies over every monomial and every cutoff and asks for an unbounded iteration index. For any finite set of tested monomials, one can choose the bounded independent chain over their finite rational span and set the logarithmic derivative to zero on that span. The resulting nonintegrable derivation passes every escape test on those monomials. Thus finite monomial sampling alone cannot certify integrability over this unrestricted parameter space.

[write] (Batch 96.) Source 26 gives this argument as a sketch, and its last sentence is false in finite rational rank, where every strict derivation integrates (theorem 277.2), so $\mathfrak{S}_B = \mathfrak{P}_B$ and the constant answer “integrable” is correct; the bounded independent chain it uses exists only in infinite rank (lemma 277.5). Proposition 285.4 [merge] proves the remark under its implicit hypothesis $\dim_{\mathbb{Q}} \Gamma = \infty$.

281 Consequences for BCH, surreal realizations, and formalization

281.1 A safe BCH interface

The failure of Lie closure identifies exactly why an unrestricted appeal to a formal identity is insufficient. Before substituting derivations into a degree-completed free algebra, one must prove that the resulting operator series acts within the chosen support class.

A useful sufficient certificate is a *common uniform positive gain*. If D_1 and D_2 each have gain at least $\varepsilon > 0$, a word of length n in these operators increases valuation by at least $n\varepsilon$. There are finitely many words of each length, and on a fixed input below a fixed output cutoff all sufficiently long words vanish from consideration. Any prescribed formal noncommutative power series with finitely many words in each degree therefore evaluates there by a finite calculation. Universal degreewise identities, including the BCH identity, transfer through these finite calculations. The associated exponential correspondence is the established formal mechanism, not a new claim of this manuscript [159, 160].

In finite rational rank, Theorem 277.2 supplies such a common ε for any finite family of nonzero strict derivations, by taking the minimum of their individual gains. In the alternating construction, the gaps κ_n tend to zero, and there is no common positive uniform gain. More importantly, there is an explicit failure of joint local finiteness, not merely absence of that sufficient estimate.

The ProveIt module `BCH/Formal/Trunc.lean` constructs a finite-dimensional truncated regular representation of the free algebra. Its declarations include `rho`, `rho_apply_delta`, and `proj_eq_zero_of_rho_eq_zero` [164]. That is a natural finite identity interface. The additional obligation here is a proved support or window bound connecting the formal degree cutoff to the actual series-field action. Formal correctness of word coefficients alone does not discharge that obligation.

[write] (Batch 96.) The second paragraph is a sketch, which source 26 attributes to the established formal mechanism; proposition 285.3 [merge] proves it for left-finite fields, including the obligation named here (part (i) there), and concludes that under a common positive gain the product of two time-one maps is again a time-one map. The module is

Algebra/BakerCampbellHausdorff/Lean/BCH/Formal/Trunc.lean, unchanged at the placement (blob 5f238ed2f); the theorem that the homogeneous components of $\log(e^X e^Y)$ are Lie polynomials is BCH.bch_formula_formal in BCH/Main.lean of the same library.

281.2 Set-sized surreal interpretation

The constructions concern ordinary sets and ordinary real coefficients. There is no quantification over a set of all surreal numbers. Under the usual real-exponent normal-form realization, the map

$$\iota \left(\sum_{\gamma} x_{\gamma} t^{\gamma} \right) = \sum_{\gamma} x_{\gamma} \omega^{-\gamma} \quad (281.1)$$

realizes $\mathcal{L}_{\mathbb{R}}^{\Gamma}$ as a set-sized surreal subfield. The left-finite support on γ is well ordered; its negative is reverse well ordered, as required for the displayed normal form. Addition and multiplication agree by finite coefficient calculation below cutoffs and the normal-form arithmetic. This is the normal-form framework used by the repository's surreal project [162].

One may also read (281.1) purely conditionally: given a field embedding with these normal-form properties, transport every result along it. This version requires no unproved assertion about a particular Lean implementation. The Borel structure on the image is *the transported coefficient coding*; it is not asserted to be the Borel structure of an unspecified topology on the proper class **No**. No flow on this subfield is claimed to extend to all surreals.

[write] (Batch 96.) The map ι is Part XVII's ι_{Γ} (235.1), which for $\Gamma = \mathbb{Q}$ and coefficients in $\mathbb{R}_{\text{alg}}$ is Part V's (71.1) and Part XV's proposition 206.1. The first paragraph's sketch is completed by the normal-form theorem: every surreal number has a unique Conway normal form, and sums and products of normal forms are computed by the Hahn-series formulas with $\omega^x \omega^y = \omega^{x+y}$ ([91, Chapter 5]), so ι is an injective field homomorphism whose coefficient sums are finite by lemma 272.1; it also preserves the order, t and ω^{-1} being positive infinitesimals. The claim is therefore not moved to the further questions. Under ι the theorems of this part transfer to the image by transport of structure, as in Part XVII.

281.3 A feasible proof-assistant decomposition

The following sequence keeps the theorem boundary explicit. First define the subtype of coefficient functions satisfying left-finiteness, its field operations, and its valuation. Then prove that bounded-window quotients identify with finite-support functions. The mathematical tasks should be separated into three interfaces:

- (a) **Finite algebra.** Prove the product rule, the alternating basis formulas, the commutator formulas, and independence of the integer orbit. Sparse polynomial or finite-support linear algebra is sufficient.
- (b) **Support control.** Prove the tail lemma, compatibility of window exponentials, and the equivalence between local nilpotence and escape to infinity. This is where a finite test cannot replace a proof.
- (c) **Descriptive regularity.** Prove the Baire finite-orbit-span lemma, automatic scalar and valuation fixing, and the Borel coding of the assembled maps. These are separate from the BCH coefficients.

The final theorem should quantify over actual Borel action maps, not postulate an opaque

“flow” type whose axioms already contain the desired finite-support conclusion. The mathematical classification here is not reported as a new kernel-checked theorem. No Lean build was run for this manuscript.

[write] (Batch 96.) Nothing in this part is formalized, and docs/FORMALIZATION.md maps none of its labels. Part XVII’s [remark 239.5](#) records the one formal special case of Part XVII, which concerns primitives for $\Gamma = \mathbb{Z}$ and none of the flows here.

282 Verification and adversarial proof checks

The companion `verify.py` uses exact rational arithmetic in a sparse polynomial ring. It checks the chain, parity, and commutator identities on finite ranges, tests the Leibniz and shear group laws on deterministically generated examples, and independently checks the largest-index certificate for $A^k z_0$. It also checks a rational-binomial uniform-gain exponential identity. The delivered run completed 5,372 exact checks: 1,656 basis and chain checks, 1,200 polynomial derivation and shear-law checks, 1,760 finite time-one orbit checks, and 756 rational-binomial checks. Ordinary and optimized Python runs produced identical receipts. The output records all tested bounds.

[write] (Batch 96.) The program is shipped as `code/26-borel-flows-verify.py` and its record as `data/26-borel-flows-verification_results.json`. The write reran it on a copy with Python 3.14.4 (`verify.py -json` with an output path), with and without `-O`: 5,372 checks in the four stated groups, status PASS, and both records equal the shipped one after removing carriage returns. Its checks are raised exceptions, not assert statements, so `-O` does not disable them.

The finite algebra treats the z_i as independent variables. This is sound for the tested polynomial identities: rational independence of the β_i makes distinct integer exponent vectors correspond to distinct monomials $t^{\sum k_i \beta_i}$. The program never attempts to deduce that independence from floating-point samples.

The tests are *not* proofs of the general Baire theorem, of left-finiteness for arbitrary inputs, or of the infinite-rank conclusion. Those are established by the arguments in the preceding sections. The following checks are particularly important for reviewing them.

Strict increase is not divergence. The sequence $\beta_n \uparrow 2$ supplies a literal counterexample to substituting strict monotonicity for escape to $+\infty$.

A window is not finite-dimensional. There are infinitely many β_n in the one window $[1, 2)$. Finite dimensionality in the integration proof comes from the Baire lemma or the TLF hypothesis, not from truncation alone.

A coefficient derivative must be a field element. Differentiating coordinates separately can in general produce forbidden support. In Theorem [275.2](#), each window orbit lies in one finite-dimensional space with one fixed finite support. This proves that the derivative has left-finite support before it is called Dx .

Products require two compatible input cutoffs. To compute a product below b , use cutoffs $b - c$ and $b - a$ for inputs with lower bounds a, c . A single common cutoff b is not generally sufficient when one input has negative valuation.

There are two different exponentials. General TLF generators use ordinary matrix convergence on finite cyclic spaces. Strict integrable generators use a formally summable family with finitely many contributions below every bound. Example 275.5 prevents conflating them.

The time-one obstruction is intrinsic. The proof of Theorem 279.1 uses every possible Borel flow, not just the series $\log(\Phi_1^e \Phi_1^o)$. It therefore does not depend on choosing or computing a logarithm branch.

Hypotheses are not silently enlarged. All main results use a countable divisible exponent subgroup of $\mathbb{R}$, real coefficients, and the coefficient Borel structure. Nondivisible exponents, arbitrary Hahn supports, complex coefficient fields, and proper-class ambient structures require separate arguments.

283 Further research questions and concrete targets

The following are proposed questions after the proved results. Except for the specifically cited prior-draft question, no claim is made that they are already published open problems.

283.1 Effective time-one recognition and logarithm extraction

Question 283.1. For which symbolic classes of Borel field automorphisms can the two conditions in Theorem 279.5 be decided, and can their canonical generating derivations be computed?

The mathematical existence and uniqueness criterion is now exact, but it ranges over every monomial and every window. An effective version must produce finite invariant spaces and spectral data, or explicit nilpotence bounds in the tangent-to-identity case. It should separate computable descriptions from arbitrary Borel codes and should not confuse the existence of a canonical logarithm with an algorithm to obtain it.

[write] (Batch 96.) A companion of Part XVII's question 237.6 and of Part XVIII's question on effective normalizers (question 253.4). The Hilbert's-tenth review of source 26 (introduction of this part) records that the manuscript supplies no uniform effective producer of invariant window spaces or spectral data from arbitrary Borel codes; its review of source 24 proves a co-c.e.-completeness result for conjugacy over a nondivisible Γ , which it does not transfer to this setting.

283.2 Joint rather than individual local nilpotence

Question 283.2. Give a necessary-and-sufficient support criterion for a finite family of integrable strict derivations to admit a common BCH evaluation domain inside $\mathcal{L}_{\mathbb{R}}^{\Gamma}$.

The alternating example shows that testing each generator separately is insufficient. A useful sufficient condition is finite-dimensionality and nilpotence of every cyclic module for the associative algebra generated by the family in every window. The research task is to translate that condition into the logarithmic basis parameters, with cancellation handled exactly rather than replaced by a crude uniform gap.

[write] (Batch 96.) A common uniform positive gain is a sufficient condition, proved in [proposition 285.3](#) [merge]; the question asks for a necessary and sufficient one.

283.3 The exact descriptive complexity

Question 283.3. What is the least Borel complexity of the locus $\mathfrak{S}_B$ in natural coefficient topologies, and is it complete for an appropriate Borel class when Γ has infinite rational rank?

Theorem [280.2](#) proves Borelness but does not optimize a Borel rank. Tree-coded bounded chains might give lower bounds, while valuation-zero tests and the iteration map control upper bounds. Any answer must specify the topology on the parameter space, not just its standard Borel structure.

283.4 Zero-gain normal forms and resonances

Question 283.4. Can TLF for a valuation-nondecreasing derivation be characterized by a diagonal part plus a triangular remainder in terms of its logarithmic basis values?

Diagonal flows are always integrable, whereas a strictly increasing remainder can fail. Even a uniformly positive remainder can interact with a complicated diagonal part. Corollary [277.3](#) already settles existence for all finite-rank valuation-nondecreasing derivations. The remaining target is infinite rank, for instance finite-support logarithmic data on each basis element but with no common left-finite set of shifts. The desired result should characterize finite invariant supports or the cancellations that produce them, rather than impose a merely sufficient global gap.

283.5 Nondivisible exponent groups and other coefficient fields

Question 283.5. How much of the exact flow classification survives without divisibility of Γ , or with complex, non-Archimedean, or positive-characteristic coefficients?

Divisibility currently makes positivity field-definable by nonzero squares. Without it, the automatic order and value-group normalization needs replacement or an explicit order-preservation hypothesis. For complex coefficients, even the route forcing real scalar linearity has to be reformulated. These changes should not be treated as cosmetic.

[write] (Batch 96.) Its first clause is what remains of Part XVII's [question 237.5](#) (dated note there). Part XVIII answers the flow question for the nondivisible groups of rational rank one, $\mathbb{Z} \leq \Gamma \leq \mathbb{Q}$, for groups of ordered-field automorphisms, the explicit order hypothesis suggested here ([theorems 248.3](#) and [248.8](#)); nondivisible groups of rational rank at least two remain open. The coefficient clause is a companion of Part XVII's [question 237.2](#).

283.6 Larger support ideals and Hahn enlargements

Question 283.6. For which support ideals does Borel time dependence force finite orbit spans in appropriate quotients, and what is the smallest stable series-field enlargement containing the bounded-chain exponentials?

The bad series ([277.6](#)) is an individual Hahn series, but a field closed under all relevant substitutions and derivations may require many more supports. A useful target is a smallest

explicitly coded closure under the two alternating shears and the missing summed flow, with a proof that the operations are well-defined there.

[write] (Batch 96.) A companion of Part XVII's [question 237.1](#) and Part V's [question 73.5](#), which ask about support classes beyond left-finiteness for summability and for Polish topologies; Part XIX computes the descriptive complexity of the well-ordered supports themselves.

283.7 Effective integration certificates

Question 283.7. Which symbolic derivation languages admit computable certificates of window-local nilpotence or TLF?

A certificate could give a finite invariant support for each symbolic input and cutoff, together with a recurrence or nilpotence bound. The soundness proof is accessible to a proof assistant. Completeness and effective production of the certificates are separate issues, and should not be inferred from the Borel parameterization.

283.8 Other connected parameter groups

Question 283.8. Classify jointly Borel actions of connected Polish groups on $\mathcal{L}_{\mathbb{R}}^{\Gamma}$ by field automorphisms, using the finite-window representations of Theorem [273.1](#).

Finite-dimensional Lie groups give compatible local Lie representations on the windows. Compact connected groups are a particularly rigid candidate class: an order-preserving action on a finite-dimensional ordered vector space is constrained by its lexicographic flag. A complete classification would need to keep compatibility between these flags and multiplication.

[write] (Batch 96.) In rational rank one, Part XVIII's [corollary 248.9](#) shows that a homomorphism from $(\mathbb{R}^n, +)$ with Borel time coordinate factors through one real parameter, and Part XXI classifies the finite-dimensional Lie algebras of Borel derivations.

283.9 Foundational and formalization strength

Question 283.9. What weak set-theoretic or second-order-arithmetic assumptions suffice for coded versions of the finite-orbit-span lemma and the integration theorem, and can the full result be formalized without theorem-specific axioms?

The proof uses Baire category, countable coding, and an explicitly countable basis extension. Determining optimal foundational strength requires fixed codes for actions and exponents. This is a substantive connection to Glazer's interests, but does not itself answer his separate reverse-mathematical question.

[write] (Batch 96.) A companion of Part XVII's [question 237.7](#) and [question 237.9](#); Glazer's question is Question 1, for which Part VIII's claimed answer is not used here.

Further questions and research [write]

(Batch 96.) Under the repository's rule that unproved claims are kept as questions, the write examined the statements source 26 asserts with a sketch only. None had to be moved here. The transfer of BCH identities under a common positive gain ([section 281](#)) is proved as

[proposition 285.3](#) [merge]; the surreal realization (281.1) is the normal-form map of Parts V, XV and XVII (note after it); [remark 280.3](#) is proved as [proposition 285.4](#) [merge] under its implicit hypothesis of infinite rational rank, and is false without it (note after it); the automatic-continuity step of [theorem 273.1](#) is Part V's [lemma 61.2](#); and [corollary 279.2](#), printed without proof, is proved in the note after it. Source 26's own questions above stay open. One question arises from comparing [proposition 285.1](#) with Part XVIII.

Question 283.10 (Differentiable time in higher rank [write]). Let $\Gamma \leq \mathbb{R}$ be countable and divisible, of rational rank at least two, and let $s \mapsto \Phi_s$ be a homomorphism from $(\mathbb{R}, +)$ to the coefficient-Borel field automorphisms of $\mathcal{L}_{\mathbb{R}}^{\Gamma}$ such that every coefficient of $\Phi_s(t^{\beta})$, for β in a rational basis, is differentiable at $s = 0$. Is Φ a Borel field flow?

For $\Gamma = \mathbb{Q}$ the answer is yes by Part XVIII's [theorem 248.8](#), hypothesis (ii), whose proof uses the centralizer of one time map in rational rank one. In higher rank the value-group scalings are still trivial: continuity at $s = 0$ of the leading coefficient $[t^{\beta}] \Phi_s(t^{\beta})$ gives $c_s \beta \leq \beta$, hence $c_s = 1$ for small $|s|$ (apply it to s and $-s$), and then for all s by the group law. What is missing is Borelness, or at least the Baire property, of the orbit maps of the window representations, which [theorem 273.1](#) needs, or a substitute in higher rank for the rank-one centralizer description.

284 Conclusion

For left-finite real series with a countable divisible real exponent group, measurable time dependence is unexpectedly rigid. It forces scalar fixing, valuation fixing, finite orbit spans in every bounded window, and entire coefficient functions. This yields an exact existence-and-uniqueness theorem for flows without inserting analytic regularity into the definition.

The obstruction is equally concrete. An infinite rationally independent chain of exponents can increase forever below a fixed bound. Splitting its successor derivative into alternating shears produces two valid flows but destroys integrability under both addition and commutator. The corresponding time-one obstruction rules out every Borel embedding of their product into a real-parameter flow.

These results provide a definite answer to the prior report's flow question in the divisible-exponent setting. They also supply the missing support-side warning for using BCH identities in restricted surreal or Levi–Civita subfields: a correct formal identity does not, by itself, make its infinite evaluation stay in the chosen field.

285 Where Part XX meets Parts XVII, XVIII and XXI [merge]

This section was written for this report. It shows that a weak time hypothesis already gives source 26's joint Borelness, which identifies its answer to Part XVII's [question 237.5](#) with Part XVIII's at $\Gamma = \mathbb{Q}$, and it proves the three statements source 26 asserts with sketches only: the transfer of the BCH identity under a common positive gain ([section 281](#)), its surreal realization (by citation, in the note after (281.1)) and [remark 280.3](#). The proofs use only results proved in source 26 and in Part XVII and, for [proposition 285.3](#), the classical formal BCH theorem; they have not been independently reviewed. Throughout, $\Gamma \neq \{0\}$ is a countable divisible subgroup of $\mathbb{R}$, as in source 26.

Proposition 285.1 (Borel time on a basis suffices). *Let B be a $\mathbb{Q}$ -basis of Γ , and let $s \mapsto \Phi_s$ be a homomorphism from $(\mathbb{R}, +)$ to the group of coefficient-Borel unital field automorphisms of $\mathcal{L}_{\mathbb{R}}^{\Gamma}$. Suppose that for every $\beta \in B$ and every $\eta \in \Gamma$ the function $s \mapsto [t^{\eta}] \Phi_s(t^{\beta})$ is Borel. Then $(s, x) \mapsto \Phi_s x$ is jointly coefficient-Borel, so Φ is a Borel field flow, and [theorems 274.4](#) and [275.2](#) apply to it.*

Proof. Valuation. Each Φ_s fixes $\mathbb{R}$ by [lemma 274.2](#) and preserves the order, since the positive elements are the nonzero squares ([lemma 272.1](#)). By [lemma 274.3](#) there is $c_s > 0$ with $v(\Phi_s x) = c_s v(x)$ for $x \neq 0$, and the group law gives $c_{s+r} = c_s c_r$. Fix $\beta \in B$. For $\gamma \in \Gamma$,

$$\{s : v(\Phi_s(t^{\beta})) = \gamma\} = \{s : [t^{\gamma}] \Phi_s(t^{\beta}) \neq 0\} \cap \bigcap_{\eta \in \Gamma, \eta < \gamma} \{s : [t^{\eta}] \Phi_s(t^{\beta}) = 0\}$$

is Borel, so $s \mapsto c_s = v(\Phi_s(t^{\beta}))/\beta$ is Borel with countable range. As in the proof of [theorem 274.4](#), $s \mapsto \log c_s$ is a Borel additive function, hence $s \mapsto as$, and its countable range forces $a = 0$. So every Φ_s preserves the valuation.

Monomials. Let $\gamma = \sum_{\beta \in B} q_{\beta} \beta$, a finite sum with rational q_{β} . Then $t^{\gamma} = \prod_{\beta} (t^{\beta})^{q_{\beta}}$, where a rational power of a positive element is the unique positive one (the remark after [lemma 272.1](#)). Since Φ_s is multiplicative and maps positive elements to positive elements, it maps positive n th roots to positive n th roots, and $\Phi_s(t^{\gamma}) = \prod_{\beta} \Phi_s(t^{\beta})^{q_{\beta}}$. Write $\Phi_s(t^{\beta}) = a_{\beta}(s) t^{\beta} (1 + u_{\beta}(s))$ with $a_{\beta}(s) > 0$ and $v(u_{\beta}(s)) > 0$; then $a_{\beta}(s) = [t^{\beta}] \Phi_s(t^{\beta})$ and $[t^{\eta}] u_{\beta}(s) = [t^{\eta+\beta}] \Phi_s(t^{\beta}) / a_{\beta}(s)$ for $\eta > 0$ are Borel in s . For rational q and fixed η , the map $u \mapsto [t^{\eta}] (1 + u)^q$ on $\{u : v(u) > 0\}$ is the limit of the Borel maps $u \mapsto [t^{\eta}] \sum_{n \leq N} \binom{q}{n} u^n$ (multiplication is Borel by [proposition 226.4](#)), which are eventually constant in N because $v(u^n) \geq nv(u)$; so it is Borel. Hence $s \mapsto \Phi_s(t^{\beta})^q = a_{\beta}(s)^q t^{q\beta} (1 + u_{\beta}(s))^q$ is Borel into $\mathcal{L}_{\mathbb{R}}^{\Gamma}$, and so is the finite product $s \mapsto \Phi_s(t^{\gamma})$: every function $s \mapsto [t^{\eta}] \Phi_s(t^{\gamma})$ is Borel.

Every input. Each Φ_s is real-linear and preserves the valuation, so by [lemma 272.2](#) it commutes with the left-finitely summable sum $x = \sum_{\gamma} x_{\gamma} t^{\gamma}$. Since $v(\Phi_s(t^{\gamma})) = \gamma$,

$$[t^{\eta}] \Phi_s x = \sum_{\gamma \in \Gamma, \gamma \leq \eta} x_{\gamma} [t^{\eta}] \Phi_s(t^{\gamma}),$$

a sum with finitely many nonzero terms. Enumerate $\Gamma \cap (-\infty, \eta]$ as $\gamma_1, \gamma_2, \dots$; the partial sums over $\gamma_1, \dots, \gamma_N$ are Borel in (s, x) and eventually constant, so their limit is Borel. Thus every coefficient of $\Phi_s x$ is a Borel function of (s, x) . With $\Phi_0 = \text{id}$ and the group law, Φ is a Borel field flow. $\square$

The hypothesis concerns countably many coefficient functions of countably many orbits, and no time regularity of any other orbit is assumed; by [theorem 275.2](#) every coefficient of every orbit is then an entire function of time. For $\Gamma = \mathbb{Q}$ one may take $B = \{1\}$, and the hypothesis is that every coefficient of $\Phi_s(t)$ be Borel in s .

Corollary 285.2 (The answer at $\Gamma = \mathbb{Q}$). *Let $\Gamma = \mathbb{Q}$, and let $E(\sum_q f_q t^q) = \sum_q q f_q t^q$ be the Euler derivation.*

- (i) *A Borel derivation D of $\mathcal{L}_{\mathbb{R}}^{\mathbb{Q}}$ generates a Borel field flow if and only if $D = aE$ with $a = 0$ or $v(a) \geq 0$. The flow is unique, and every coefficient of every orbit is an entire function of time.*
- (ii) *If $s \mapsto \Phi_s$ is a homomorphism from $(\mathbb{R}, +)$ to the coefficient-Borel field automorphisms of $\mathcal{L}_{\mathbb{R}}^{\mathbb{Q}}$ and every coefficient of $\Phi_s(t)$ is a Borel function of s , then Φ is the flow of (i) for a unique a with $a = 0$ or $v(a) \geq 0$.*

Proof. By Part XVII's [theorem 231.1](#) in rational rank one, with the basis $\{1\}$, every Borel derivation is aE with $a = t^{-1}D(t)$, and $\text{gain}(aE) = v(a)$ by [\(231.3\)](#) ($+\infty$ for $a = 0$). So D is valuation-nondecreasing exactly when $a = 0$ or $v(a) \geq 0$, and (i) is [corollary 277.3](#) with the uniqueness and entire coefficients of [theorem 275.2](#). For (ii), [proposition 285.1](#) with $B = \{1\}$ makes Φ a Borel field flow; its generator D is a real-linear valuation-nondecreasing derivation, Borel by [lemma 272.2](#), so (i) applies, and $a = t^{-1}D(t)$ is determined by Φ through [\(275.2\)](#). $\square$

Part (i) is the criterion of Part XVIII's [theorem 248.3](#) at $\Gamma = \mathbb{Q}$, and part (ii) is its [theorem 248.8](#) under that theorem's hypothesis (i), at $\Gamma = \mathbb{Q}$: there the acting group consists of the Borel ordered-field automorphisms, and for divisible Γ every Borel field automorphism preserves the order. A coefficientwise real-analytic flow in Part XVIII's sense has continuous, hence Borel, coefficients of $\Phi_s(t)$, so by [proposition 285.1](#) it is a Borel field flow; conversely every Borel field flow has entire coefficient functions. At $\Gamma = \mathbb{Q}$ the two classes of flows therefore coincide, and the two theorems are the same statement, proved independently: Part XVIII through coordinate normal forms and centralizers, source 26 through finite orbit spans in windows. Away from $\Gamma = \mathbb{Q}$ neither contains the other. Part XVIII also covers the nondivisible groups $\mathbb{Z} \leq \Gamma \leq \mathbb{Q}$, accepts differentiability at time zero (its hypothesis (ii)) in place of Borelness, describes the coefficient functions as polynomials or exponential polynomials, and excludes negative gain even for curves without a group law; source 26 covers every countable divisible $\Gamma \leq \mathbb{R}$ of every rational rank and adds the window criterion, the time-one theory and the nonclosure theorems. [Proposition 285.1](#) removes one difference: for divisible Γ , Part XVIII's Borel-time hypothesis, imposed on a basis, already gives source 26's joint Borelness.

Part XXI's [remark 294.3](#) excludes negative gain for coefficient-differentiable actions with a Borel generator; [corollary 275.4](#) excludes it for Borel flows; the two categories of actions are different, and neither proof uses the other. Part XXI works in rational rank one; its higher-rank example [example 295.5](#) uses a nondivisible group, outside source 26's hypotheses.

Proposition 285.3 (BCH under a common positive gain). *Let D_1, D_2 be real-linear derivations of $\mathcal{L}_{\mathbb{R}}^{\Gamma}$ and $\varepsilon > 0$ with $v(D_i x) \geq v(x) + \varepsilon$ for $x \neq 0$ and $i = 1, 2$. Let $\mathbb{R}\langle\langle X_1, X_2 \rangle\rangle$ be the algebra of noncommutative formal power series, write $S = \sum_w c_w w$ over words w , and let $\text{ord } S$ be the least length of a word with $c_w \neq 0$.*

- (i) *The formula $\text{ev}(S)x = \sum_w c_w w(D_1, D_2)x$ defines a real-linear operator with $v(\text{ev}(S)x) \geq v(x) + \varepsilon \text{ord } S$. The map ev is a unital algebra homomorphism, and $\text{ev}(\sum_k S_k) = \sum_k \text{ev}(S_k)$ whenever $\text{ord } S_k \rightarrow \infty$.*
- (ii) *Let $Z = \log(e^{X_1} e^{X_2}) = \sum_{n \geq 1} Z_n$, with Z_n homogeneous of degree n . Then $\text{ev}(Z)$ is a real-linear derivation of gain at least ε , and $\exp(D_1) \exp(D_2) = \exp(\text{ev } Z)$, where $\exp T = \sum_n T^n / n!$. In particular the product of the time-one maps of the Borel flows of D_1 and D_2 is the time-one map of the Borel flow of $\text{ev}(Z)$.*

Proof. (i) For a word w of length n , induction gives $v(w(D_1, D_2)x) \geq v(x) + n\varepsilon$. Fix $x \neq 0$ and a cutoff b . Only the finitely many words of length less than $(b - v(x))/\varepsilon$ have images with coefficients below b , and each image is left-finite; so $(c_w w(D_1, D_2)x)_w$ is left-finitely summable, its sum lies in $\mathcal{L}_{\mathbb{R}}^{\Gamma}$, and the valuation bound holds. The operator is real-linear. For words, $\text{ev}(uw) = \text{ev}(u) \text{ev}(w)$ by definition. For series $S = \sum c_u u$ and $T = \sum d_w w$, each $u(D_1, D_2)$ is a composition of real-linear valuation-nondecreasing maps and so commutes with left-finitely summable sums ([lemma 272.2](#)); hence $\text{ev}(S) \text{ev}(T)x = \sum_u \sum_w c_u d_w (uw)(D_1, D_2)x$, and the double family is left-finitely summable by the same length bound, so it may be regrouped by the product words into $\text{ev}(ST)x$. If $\text{ord } S_k \rightarrow \infty$, each word occurs in only finitely many S_k ,

and the family $(c_w^{(k)} w(D_1, D_2)x)_{k,w}$ is left-finitely summable by the same bound; regrouping it by k or by w gives the last assertion.

(ii) Each Z_n is a Lie polynomial in X_1, X_2 (the formal BCH theorem). By (i) ev maps commutators to commutators, and a commutator of derivations is a derivation, so every $\text{ev}(Z_n)$ is a real-linear derivation, of gain at least $n\varepsilon$. By (i), $\text{ev}(Z)x = \sum_n \text{ev}(Z_n)x$. The families $(\text{ev}(Z_n)(x)y)_n$ and $(x \text{ev}(Z_n)(y))_n$ are left-finitely summable, and multiplication by a fixed element commutes with such sums (each coefficient of a product is a finite sum, and below a cutoff only finitely many members of the family contribute), so summing the Leibniz identities of the $\text{ev}(Z_n)$ gives that of $\text{ev}(Z)$; its gain is at least ε because $\text{ord } Z \geq 1$. In $\mathbb{R}\langle\langle X_1, X_2 \rangle\rangle$, $e^{X_1} e^{X_2} = \exp(Z)$, by the formal identity $\exp(\log(1+U)) = 1+U$ for $\text{ord } U \geq 1$. Apply ev and (i) to the exponential series, whose n th terms have order at least n : $\exp(D_1) \exp(D_2) = \exp(\text{ev } Z)$. By [theorem 276.2](#) and [corollary 276.3](#), $\exp(D_i)$ and $\exp(\text{ev } Z)$ are the time-one maps of the unique Borel flows of D_i and $\text{ev}(Z)$. $\square$

The proof does not use divisibility: for any countable $\Gamma \leq \mathbb{R}$, [lemma 272.2](#) holds as proved, and the flows are Part XVII's [theorem 234.2](#) and [corollary 234.3](#). The same argument applies to any finite family $D_1, \dots, D_k$ with a common positive gain and to every universal identity in $\mathbb{R}\langle\langle X_1, \dots, X_k \rangle\rangle$. In finite rational rank any finite family of strict derivations has a common positive gain ([theorem 277.2](#); the zero derivation has gain $+\infty$), so the time-one maps of strict derivations are closed under composition there; [theorem 279.1](#) shows that without a common gain this fails in infinite rank. The input that the homogeneous components of $\log(e^X e^Y)$ are Lie polynomials is formalized in Lean as `BCH.bch_formula_formal` (Algebra/BakerCampbellHausdorff/Lean/BCH/Main.lean, over every field of characteristic zero); the evaluation (i), which source 26 calls the additional obligation of a support bound, is proved here on paper only.

Proposition 285.4 (Finite monomial sampling certifies nothing in infinite rank). *Suppose $\dim_{\mathbb{Q}} \Gamma = \infty$, and let $G \subseteq \Gamma$ be finite. There is a strict Borel real-linear derivation D of $\mathcal{L}_{\mathbb{R}}^{\Gamma}$ that generates no Borel field flow and satisfies $D(t^{\gamma}) = 0$, hence $v(D^n t^{\gamma}) = +\infty$ for $n \geq 1$, for every γ in the rational span of G . If $\dim_{\mathbb{Q}} \Gamma < \infty$, every strict real-linear derivation generates a Borel field flow, so $\mathfrak{S}_B = \mathfrak{P}_B$ for every basis B .*

Proof. Let $W = \text{span}_{\mathbb{Q}} G$, a finite-dimensional subspace of Γ . Choose $\beta_0 < \beta_1 < \dots$ in Γ as in the proof of [lemma 277.5](#), with $W_n = W + \text{span}_{\mathbb{Q}}\{\beta_0, \dots, \beta_{n-1}\}$ in place of the span used there: W_n is finite-dimensional, hence proper; choose $z \in \Gamma \setminus W_n$; if $W_n \neq 0$, choose a positive $u \in W_n$, so that $z + \mathbb{Q}u$ is dense in $\mathbb{R}$ and disjoint from W_n , and choose β_n in it inside $(1, 3/2)$ for $n = 0$ and inside $(2 - 2^{-n}, 2 - 2^{-(n+1)})$ for $n \geq 1$; if $W_0 = 0$, choose $\beta_0 \in \Gamma \cap (1, 3/2)$, Γ being dense. Then $\beta_n \notin W_n$ for every n , so a basis B_W of W together with the β_n is independent; extend it by a set C to a $\mathbb{Q}$ -basis of Γ . Put $\kappa_n = \beta_{n+1} - \beta_n > 0$, $\delta(\beta_n) = t^{\kappa_n}$ and $\delta(b) = 0$ for $b \in B_W \cup C$. [Lemma 277.1](#) gives a strict Borel real-linear derivation D ; δ vanishes on W , so $D(t^{\gamma}) = t^{\gamma} \delta(\gamma) = 0$ for $\gamma \in W$. As in [proposition 277.7](#), $D^k(t^{\beta_0}) = t^{\beta_k}$ has valuation $\beta_k < 2$ for every k , so [theorem 276.2](#) excludes a Borel flow. In finite rank, [theorem 277.2](#) gives the last assertion. $\square$

This is source 26's [remark 280.3](#), with its implicit hypothesis made explicit: the remark's construction uses the bounded independent chain, which exists only in infinite rational rank, and in finite rank its conclusion fails, since every parameter then integrates. In the parameter space of [section 280](#), D is the point $(\delta(b))_{b \in B}$ for the fixed basis B ; it agrees on every monomial of W with the zero derivation, which integrates.

Part XXI

Two derivations, scalar coordinates and Lie algebras on left-finite fields

[write] This part is source 27, manuscript 07 of batch 96, *Two Derivations Remove the Borel Conjugacy Obstruction: Scalar normalizing coordinates, simultaneous normal forms, and finite-dimensional Lie algebras on left-finite series fields* (19 pages, A4, dated 4 October 2026; “Research manuscript prepared with ChatGPT”; its title page calls it “a research direction connecting Elliot Glazer’s work with the formal-series and Lie-theoretic programs in ProvelT”). It arrived in commit 78528873b as archive `Two_Derivations_Borel_Conjugacy` and was placed in 54ece48ab. It names no commit: its source record gives the blob identifiers a7ebc59ad of `Algebra/SurrealNumbers/README.md` and e24d8fc8d of `Algebra/BakerCampbellHausdorff/README.md`, which are the current blobs of those files from ad52ef11e (the commit that described the newly merged BCH library) to the placement, so it was written against ad52ef11e or a later commit. Its file prefix in this report is 27-two-derivations-. All unmarked text in this part is source 27’s; its thirteen sections are Sections 286–299 (numbered on from the preceding parts, with statement and equation numbers within a section unchanged: its Theorem 5.2 is [theorem 290.2](#)), its eight questions are Questions 298.1–298.8, and its appendix is [appendix AQ. Section 300](#) [merge] was written for this report.

What this part claims, and its status

Let $\mathbb{Z} \subseteq \Gamma \subseteq \mathbb{Q}$ and let $\mathcal{L}_{\mathbb{R}}^{\Gamma}$ be Part XVII's left-finite real field for this Γ . Every coefficient-Borel derivation is $D_a = aE$ (a re-proof of Part XVII in rank one), and the coefficient-Borel ordered-field automorphisms are the substitutions $t \mapsto ct^r(1+u)$ with $c > 0$, $r \in U_{\Gamma} = \{r \in \mathbb{Q}_{>0} : r\Gamma = \Gamma\}$ and $v(u) > 0$ ([theorem 289.1](#), Part XVIII's [theorem 242.4](#) by an independent proof). Every nonconstant scalar h has a canonical scalar H_h and a *unique* such automorphism carrying H_h to h ; all stabilizers of nonconstant scalars are trivial ([theorem 290.2](#) and [corollary 290.3](#)). Consequently simultaneous conjugacy of noncommuting ordered pairs of Borel derivations has the explicit Borel cross-section (D_A, D_{H_A}) , trivial stabilizers and unique, Borelly recoverable conjugators, also for countable ordered tuples ([theorem 291.2](#) and [corollary 291.3](#)), whereas single derivations and commuting pairs are not smooth when $\log U_{\Gamma}$ is dense ([theorem 292.2](#)). Every finite-dimensional Lie algebra of Borel derivations has dimension at most three, with ramification-sensitive affine and projective normal forms ([theorems 293.2](#) and [293.3](#); the classical pattern is credited); the valuation-nondecreasing ones have dimension at most two ([corollary 294.1](#)) and integrate to an explicit whole-field action of $\text{Aff}^+(\mathbb{R})$ ([theorem 294.4](#)).

[Section 300](#) [merge] makes source 27's undefined "Lie-group action" precise, and proves, from Part XVIII's automatic analyticity in time, that every action of a connected Lie group whose one-parameter subgroups move t Borelly in time (equivalently, every coefficient-differentiable action with Borel generators) has a Lie-algebra homomorphism as generator map and factors, on the universal cover, through source 27's affine action or through one analytic flow; it fixes the translation between the opposite conjugation conventions of Parts XVIII and XXI, and records that the affine BCH formula of [equation \(294.2\)](#) is a Lean theorem of the BCH project in its matrix form.

Unrefereed, AI-assisted; nothing of this part is formalized except that matrix identity and, for full Hahn series, the kernel of E (note in [section 297](#)); priority is not established. The finite checks test finite identities only.

Questions answered. Source 27 answers its two opening questions ([questions 286.1](#) and [286.2](#)) in the delimited form it states. It answers Part XVIII's [question 253.3](#) (source 24's Question 14.3, which it cites) at the infinitesimal level: [theorems 293.2](#), [293.3](#) and [294.4](#) and [corollary 294.1](#). [Theorem 300.3](#) [merge], which combines it with Part XVIII's [theorem 248.8](#), answers the group level of that question for actions whose one-parameter subgroups have Borel time dependence at the monomial t (in particular for jointly coefficient-Borel actions), and answers source 27's own [question 298.4](#); notes after both questions say what remains. It answers the simultaneous form of Part XVII's [question 237.4](#) for noncommuting ordered pairs in rational rank one, and shows that commuting pairs do not help ([theorems 291.2](#) and [292.2](#)); the single-derivation case is Part XVIII's. A reviewer deduction about effective presentations, credited in the note after [question 298.6](#), bears on Part XVII's [question 237.6](#) and Part XVIII's [question 253.4](#).

What was already here, and what is new. Source 27 read the surreal and BCH READMEs, not this report, its sources or any Lean file, and cites sources 23 and 24 (Parts XVII and XVIII) as manuscripts of "the user's collection". Its [lemma 287.2](#) is the rank-one case of Part XVII's [propositions 226.2](#) and [226.4](#); [lemmas 288.1](#) and [288.2](#) and [theorem 288.3](#) re-prove [lemmas 227.1](#) and [227.2](#) and [theorem 228.2](#) for one field with $\mathbb{Z} \subseteq \Gamma \subseteq \mathbb{Q}$; [corollary 288.4](#) is the rank-one case of [theorem 231.1](#) with the bracket [\(234.3\)](#); [theorem 289.1](#) is Part XVIII's [theorem 242.4](#)

(independent proof) and an analogue of Part XV's [theorem 198.2](#); [proposition 292.1](#) is the zero-gain case of Part XVIII's [theorems 245.1](#) and [247.2](#); [theorem 292.2](#) strengthens the nonsmoothness half of Part XVIII's [theorem 246.4](#); [Proposition 294.3](#) is printed as a numbered note, since Part XVIII's [theorem 248.3](#) and Part XX's [corollary 275.4](#) already prove it; the b -flow of [theorem 294.4](#) is Part XVII's (234.4) for $p = 1$; the affine BCH formula is the BCH project's Theorem 9.4; and the surreal map is Part XVII's (235.1). Each is flagged in a note. New to the report are the scalar-coordinate theorem and the Borel product decomposition, the Borel cross-section and unique witnesses for noncommuting pairs and tuples, the Baire-measurable and commuting-pair form of the nonsmoothness contrast, the valuation proof and ramification-sensitive refinement of the Lie classification with its Borel form, and the explicit global affine action.

Labels and notation. Every label of source 27 takes the prefix `pma:tdv:`; its unlabelled numbered statements are labelled `pma:tdv:n<number>` (so its Question 13.4 is `pma:tdv:n13.4`), and the labels written for this report, among them those of its conclusion and appendix, begin `pma:tdv:w-`. The eighth block of the table in [section 1.3](#) lists the symbols. In short: source 27's field K_Γ is printed $\mathcal{L}_\mathbb{R}^\Gamma$, as in Part XVII, because Part II's K_Γ is the full Hahn field and Part XVII's K_Γ is the surreal image of $\mathcal{L}_\mathbb{R}^\Gamma$ (tempting false reading: $K_\mathbb{Q}$ is not $\mathbb{R}((t^\mathbb{Q}))$). Source 27 uses H for three things: its canonical scalar H_h is printed H_h and the set of canonical scalars, its $\mathcal{H}_\Gamma$, is printed Can_Γ (tempting false reading: Part V's $\mathcal{H}_k$ and Part XV's $\mathcal{H}^\Gamma$ are Hahn fields); its $\mathfrak{sl}_2$ -triple e, f, H is printed X_+, X_-, X_0 (f is also its letter for series); and the subgroup $H = \log U_\Gamma$ in the proof of [theorem 292.2](#) is written out as $\log U_\Gamma$. Its space of noncommuting pairs $\mathcal{P}_\Gamma$ is printed NC_Γ , because Part V's $\mathcal{P}_k$ is the Puiseux field. Its macros `\Span` (`spanℝ`) and `\m` are the report's `spanℝ` and `m`. Source 27 writes conjugation in the opposite direction to Part XVIII; [remark 289.2](#) gives the translation. The letters $E, D_a, v, \text{lc}, \text{ct}, U_\Gamma, G_\Gamma$ agree with Part XVIII (and E, v, ct with Part XVII); Θ_y is Part XVIII's S_y ; $L_y, \Lambda, \Phi_s, \Phi_{a,b}, A, c(h), \kappa, N_h, X, Y$ are local (L_y is not Part XIV's field L_Γ , Λ is not Part XVII's target group, Φ_s is not Part XVII's $\exp(sD)$). No normalization changed.

Files. Source 27's program, its recorded run (text and JSON), its pinned dependency, its LaTeX build script and its source audit are shipped as `code/27-two-derivations-verify.py`, `data/27-two-derivations-verification_report.txt`, `data/27-two-derivations-verification_report.json`, `data/27-two-derivations-requirements.txt`, `code/27-two-derivations-build.sh` and `27-two-derivations-SOURCE_AUDIT.md`. The manuscript itself is printed here, not shipped, so the build script, which builds `two_derivations.tex` beside itself, cannot run from `code/`. The Hilbert's-tenth programme reviewed the archive in full ([remark 300.4](#)).

286 The research intersection and the questions answered

Main distinction. A *Borel cross-section* selects exactly one representative from each orbit in a measurably specified way. A mere normal-form theorem need not do this. Sections [290–291](#) construct an actual cross-section for noncommuting pairs. Section [292](#) explains why the analogous assertion for all single derivations is false. Here “smooth classification” means classification by Borel complete invariants, not differentiability in the series variable.

Glazer's *A Topological Tennenbaum Theorem* studies the interaction between algebraic structure and descriptive regularity: in the uncountable Polish Borel models of arithmetic considered there, neither addition nor multiplication is continuous [1]. That interaction, rather

than a professional-title inference, motivates the present project. ProveIt's surreal-number materials include normal-form series, strong summation, and differential constructions; its Baker–Campbell–Hausdorff project develops formal and analytic Lie identities [150, 168]. These provide a concrete meeting point: measurable classification of differential structures on a set-sized non-Archimedean field.

Two preceding research manuscripts in the user's collection establish the immediate project context. *Borel Regularity Forces Summability* investigates automatic continuity and Borel derivations. *A Divisibility Threshold for Borel Conjugacy* studies individual rank-one derivations and, in its Question 14.3, asks for the possible nonabelian infinitesimal Lie algebras and corresponding ordered actions [166, 167]. We do not repeat the individual-derivation classification. Instead we answer the following precisely delimited questions.

[write] (Batch 96.) The two manuscripts are this report's sources 23 and 24, printed as [Parts XVII](#) and [XVIII](#); source 24's Question 14.3 is [question 253.3](#).

Question 286.1 (Simultaneous classification). Can adding a second, noncommuting derivation remove the obstruction to Borel classification, and can the conjugating coordinate itself be recovered uniquely and Borelly?

Question 286.2 (Rank-one infinitesimal symmetry). Which finite-dimensional real Lie algebras of coefficient-Borel derivations occur on a left-finite rank-one series field, up to the allowed coordinates? Which nonabelian cases can act globally on the whole ordered field?

The first question has an affirmative answer for every $\mathbb{Z} \subseteq \Gamma \subseteq \mathbb{Q}$. The second has affine and projective normal forms, with an exponent invariant that records the ramification allowed by Γ . Only the positive affine cases survive the necessary valuation condition for whole-field differentiable actions. This settles the stated rank-one Lie-algebra subproblem, not a general classification of every Borel action of every connected Lie group.

[write] (Batch 96.) The last sentence is source 27's boundary; [theorem 300.3](#) [merge] goes further for actions with Borel time dependence along one-parameter subgroups.

286.1 Attribution and scope

Left-finite series and Levi–Civita analysis have an established literature [31]. Strongly additive automorphisms of generalized series fields are studied systematically by Kuhlmann and Serra [133]. Formal coordinate normalization and flow embeddings are also classical; power-log extensions are developed, for example, by Mardešić, Resman, Rolin, and Županović [151]. The familiar dimension-three bound and affine/projective pattern for vector fields on a line belong to Lie's theory; a recent account is Azad–Biswas–Shah [165, Section 3]. We do not present those patterns, or the elementary affine BCH formula, as newly discovered.

[write] (Batch 96.) Source 27 uses no lemma of Kuhlmann–Serra; Part XV's [remark 204.2](#) corrects Lemma 4.2.4 of their preprint, which is not involved here.

What is developed here is the explicit scalar-coordinate theorem on the *entire left-finite rank-one field*, the resulting Borel product decomposition and simultaneous cross-section, the unique witness formula, and the precise exponent-orbit classification of finite-dimensional subalgebras. The required automatic-regularity facts are reproved, so the arguments do not depend on trusting an earlier unrefereed manuscript. A targeted literature check establishes these antecedents but does not certify historical priority for the combined results.

All mathematics below is in ordinary set-sized structures, in ZFC. No result about every proper-class automorphism of the surreal numbers is asserted. No convergence of a formal

series as a real or complex analytic function is assumed. In particular, these are not analytic Stokes-classification theorems.

287 Left-finite fields and their Borel coding

Fix once and for all an additive subgroup

$$\mathbb{Z} \subseteq \Gamma \subseteq \mathbb{Q}.$$

The inclusion of 1 fixes the monomial t . Any nonzero subgroup of $\mathbb{Q}$ can be put in this form after rescaling its exponents.

Definition 287.1. A set $S \subseteq \Gamma$ is *left-finite* if $S \cap (-\infty, B]$ is finite for every real B . Let

$$\mathcal{L}_{\mathbb{R}}^{\Gamma} = \left\{ f = \sum_{q \in \Gamma} f_q t^q : f_q \in \mathbb{R}, \text{ supp}(f) \text{ is left-finite} \right\}.$$

For $f \neq 0$, put $v(f) = \min \text{supp}(f)$ and $\text{lc}(f) = f_{v(f)}$; set $v(0) = +\infty$. The order is $f > 0$ exactly when $\text{lc}(f) > 0$. Write $\text{ct}(f) = f_0$ and $\mathfrak{m} = \{f : v(f) > 0\}$.

[write] (Batch 96.) This is Part XVII's [definition 226.1](#) for $\mathbb{Z} \subseteq \Gamma \subseteq \mathbb{Q}$, with the order of Part XVII's [section 239](#), and Part XVIII's field.

Addition is coefficientwise and multiplication is convolution. For a fixed output exponent there are only finitely many nonzero convolution terms: if $v(f) = a$ and $v(g) = b$, an exponent of f contributing to the coefficient at q lies in $[a, q - b]$. Every nonzero series factors uniquely as

$$f = ct^q(1 + u), \quad c \in \mathbb{R}^{\times}, \quad q \in \Gamma, \quad u \in \mathfrak{m}. \quad (287.1)$$

The geometric series for $(1 + u)^{-1}$ gives inversion. Powers of a positive unit are defined by

$$(1 + u)^{\beta} = \sum_{n \geq 0} \binom{\beta}{n} u^n \quad (\beta \in \mathbb{R}). \quad (287.2)$$

At any bounded exponent cutoff only finitely many powers contribute, because $v(u) > 0$. Thus these sums remain left-finite. For positive f and rational β , the expression f^{β} is defined by (287.1) whenever $\beta v(f) \in \Gamma$. It is the unique positive rational power satisfying the corresponding integer-power equation.

Lemma 287.2 (Completeness and finite-cutoff calculus). *The field $\mathcal{L}_{\mathbb{R}}^{\Gamma}$ is complete for its valuation metric. A sequence of increments whose valuations tend to $+\infty$ has a sum in $\mathcal{L}_{\mathbb{R}}^{\Gamma}$. Addition, multiplication, inversion on nonzero inputs, the leading exponent and coefficient, and the admissible rational powers above are Borel in the coefficient coding.*

Proof. A valuation-Cauchy sequence eventually has exactly the same coefficients below any given cutoff. Those coefficients are the coefficients of one of its terms and therefore have finite support there. Taking the stabilized coefficients gives a left-finite limit. This proves both completeness and the assertion about sums.

Choose an enumeration (q_j) of Γ . The left-finite condition is

$$\bigcap_{m \in \mathbb{Z}} \bigcup_{N \geq 0} \bigcap_{j \geq N} \{f : q_j > m \text{ or } f_{q_j} = 0\},$$

so $\mathcal{L}_{\mathbb{R}}^{\Gamma}$ is a Borel subset of the Polish product $\mathbb{R}^{\Gamma}$ and is a standard Borel space. Each convolution coefficient is the pointwise eventually constant limit of finite sums. Leading-exponent fibers are specified by countably many zero tests. Inversion and powers use the Borel leading factorization and countably many finite algebraic operations followed by coefficientwise limits. All stated maps are therefore Borel. $\square$

[write] (Batch 96.) The rank-one case of Part XVII's [proposition 226.2](#) and of the field operations in [proposition 226.4](#); the Borelness of the admissible rational powers is source 27's addition.

The coefficient Borel structure must not be confused with the valuation topology. In the latter, different real constant coefficients are separated by disjoint small balls; the full field is not being treated as a Polish space in that topology. We use Polish spaces only for fixed-support coefficient slices in the next section.

Define the Euler derivation

$$E\left(\sum_q f_q t^q\right) = \sum_q q f_q t^q. \quad (287.3)$$

It is coefficient-Borel and valuation-continuous, and

$$\ker E = \mathbb{R}. \quad (287.4)$$

Multiplication gives derivations $D_a = aE$, for every $a \in \mathcal{L}_{\mathbb{R}}^{\Gamma}$.

[write] (Batch 96.) E is Part XVII's [\(231.1\)](#) in rank one with basis element 1. Part XVII writes D_{δ} for the derivation with logarithmic monomial values δ ; source 27's $D_a = aE$ is D_{δ} with $\delta(q) = qa$ (tempting false reading: the subscript of D_a is a field element, not a map on Γ).

288 Why all Borel derivations have one coefficient

We include the regularity argument because the absence of this hypothesis would permit unrelated algebraic derivations and invalidate the subsequent classification.

Lemma 288.1 (Polish coefficient slices). *Let $T : \mathcal{L}_{\mathbb{R}}^{\Gamma} \rightarrow \mathcal{L}_{\mathbb{R}}^{\Gamma}$ be additive and coefficient-Borel. Then T is real-linear. If $\Lambda = \{q_0 < q_1 < \dots\}$ is left-finite and unbounded above, each output coefficient of*

$$(x_n)_{n \geq 0} \mapsto T\left(\sum_n x_n t^{q_n}\right)$$

is a continuous real-linear functional on $\mathbb{R}^{\mathbb{N}}$. It therefore depends on only finitely many input coordinates.

Proof. For fixed f and output exponent r , the map $x \mapsto [t^r]T(xf)$ is a Borel additive real function. The Baire-category proof of the measurable Cauchy equation makes it continuous and hence real-linear. Applying this at every r shows $T(xf) = xT(f)$.

The slice map $\mathbb{R}^{\mathbb{N}} \rightarrow \mathcal{L}_{\mathbb{R}}^{\Gamma}$ is coefficient-continuous, so its composition with any output coordinate of T is a Borel additive homomorphism to $\mathbb{R}$. Such homomorphisms are continuous. For completeness, a countable cover by preimages of bounded intervals supplies a nonmeager Borel preimage; the Baire-category difference-set argument makes the homomorphism bounded on a neighborhood of zero, and additivity then gives continuity. A continuous linear functional on a product of real lines vanishes on all vectors zero in a fixed finite set of coordinates: otherwise arbitrary scaling in a free coordinate contradicts its boundedness on a basic neighborhood. This is the claimed finite dependence. $\square$

Lemma 288.2 (The generic-support argument). *For each slice Λ and each output cutoff B , only finitely many output coordinates at exponents $r \leq B$ are nonzero functionals on that slice. Consequently there is a finite input prefix beyond which every slice vector is sent to a series of valuation greater than B .*

Proof. If infinitely many such coordinate functionals were nonzero, each of their zero sets would be a closed nowhere dense hyperplane in $\mathbb{R}^{\mathbb{N}}$. By the Baire theorem there would be an input outside all those hyperplanes. Its image would have infinitely many nonzero coefficients below B , contrary to membership in $\mathcal{L}_{\mathbb{R}}^{\Gamma}$. The finitely many surviving functionals each depend on finitely many inputs by Lemma 288.1; take the union of those finite dependency sets. $\square$

Theorem 288.3 (Automatic valuation continuity). *Every additive coefficient-Borel map $T : \mathcal{L}_{\mathbb{R}}^{\Gamma} \rightarrow \mathcal{L}_{\mathbb{R}}^{\Gamma}$ is real-linear and valuation-continuous. It preserves every left-finite summable family. Conversely, every real-linear valuation-continuous map is coefficient-Borel.*

Proof. Only global continuity remains in the forward direction. If it failed, some B and series f_n would satisfy $v(f_n) > n$ but $v(Tf_n) \leq B$. The union of their supports is left-finite: below a fixed cutoff only finitely many f_n contribute, each with finite support. Put this union into a single increasing slice Λ (adjoining an unbounded sequence if necessary). For large n , the input f_n is zero on the finite prefix in Lemma 288.2, a contradiction.

Here a left-finite summable family means that its union of supports is left-finite and each exponent occurs in only finitely many summands. For such a family, only finitely many nonzero summands can have valuation below a fixed bound. There are only countably many nonzero summands, since each uses a monomial and each monomial is used finitely often. After enumeration their valuations tend to infinity. Continuity makes the image family have the same property and sends the partial sums to their sum. This is preservation of left-finite summation.

Conversely, let T be real-linear and valuation-continuous. The truncations $f_{\leq n} = \sum_{q \leq n} f_q t^q$ are finite and tend valuation-wise to f . For a fixed output coordinate, the expression $[t^r]T(f_{\leq n}) = \sum_{q \leq n} f_q [t^r]T(t^q)$ is Borel, being a pointwise finite sum represented by countable partial sums. Continuity makes these coordinates tend to those of $T(f)$, proving Borelness. $\square$

[write] (Batch 96.) A re-proof, for one field with $\mathbb{Z} \subseteq \Gamma \subseteq \mathbb{Q}$, of Part XVII's lemmas 227.1 and 227.2 and of the equivalence (i) $\Leftrightarrow$ (ii) and the summation clause of theorem 228.2, which treats maps between two countable exponent groups $\Gamma, \Lambda \leq \mathbb{R}$ and adds the matrix description and Baire class one. The slice Λ here is a set of exponents, not Part XVII's target group.

Corollary 288.4 (All Borel derivations). *Every coefficient-Borel field derivation of $\mathcal{L}_{\mathbb{R}}^{\Gamma}$ is uniquely $D_a = aE$. Moreover*

$$[D_a, D_b] = D_{aE(b) - bE(a)}, \quad (288.1)$$

and, for $a \neq 0$,

$$[D_a, D_b] \neq 0 \iff b/a \notin \mathbb{R}. \quad (288.2)$$

Proof. A Borel derivation is real-linear by Theorem 288.3, and kills 1, hence kills $\mathbb{R}$. Put $a = D(t)/t$. For $q = m/n \in \Gamma$, differentiating $(t^q)^n = t^m$ gives $D(t^q) = qat^q$. Summation preservation extends the formula to every series. The product rule gives (288.1); writing $b = ha$ gives its coefficient as $a^2E(h)$. Now use (287.4). $\square$

[write] (Batch 96.) The first assertion is the rank-one case of Part XVII's theorem 231.1 (and of Part XVIII's theorem 242.2); (288.1) is (234.3) in rank one; (288.2) is new to the report.

289 The exact coordinate group

Let G_Γ be the group of coefficient-Borel order-preserving field automorphisms of $\mathcal{L}_\mathbb{R}^\Gamma$. A Borel field automorphism fixes $\mathbb{R}$, by real-linearity and preservation of 1. Define

$$U_\Gamma = \{r \in \mathbb{Q}_{>0} : r\Gamma = \Gamma\}. \quad (289.1)$$

This is a countable multiplicative group. Examples are $U_\mathbb{Z} = \{1\}$, $U_\mathbb{Q} = \mathbb{Q}_{>0}$, and $U_{\mathbb{Z}[1/p]} = p^\mathbb{Z}$ for prime p .

Theorem 289.1 (Substitution description). *Every member of G_Γ is uniquely a substitution Θ_y with*

$$\Theta_y(t) = y = ct^r(1+u), \quad c > 0, \quad r \in U_\Gamma, \quad u \in \mathfrak{m}. \quad (289.2)$$

Conversely, every such substitution is an element of G_Γ . Evaluation, composition, and inversion are Borel in this parameterization.

Proof. The valuation ring consists exactly of elements bounded in absolute value by a positive real constant. Its maximal ideal consists of elements smaller in absolute value than every positive real. An order-preserving automorphism fixing $\mathbb{R}$ preserves both, and induces an order automorphism of the natural value group Γ . Every additive order automorphism of a subgroup of $\mathbb{Q}$ containing 1 is multiplication by its value r at 1, where $r\Gamma = \Gamma$. Thus $v(\Theta f) = rv(f)$.

The positive image of t has form (289.2). Uniqueness of positive rational powers forces $\Theta(t^q) = y^q$. Valuation continuity and finite truncations then force $\Theta(f) = \sum_q f_q y^q$.

Conversely, the displayed sum is left-finite: $v(y^q) = rq$, and below any cutoff only finitely many input monomials contribute. Each unit expansion is left-finite by (287.2). The binomial identities prove multiplicativity. The leading term shows that the map preserves order. Monomial substitutions $t \mapsto ct^r$ are invertible because $r \in U_\Gamma$. After factoring one off, it remains to invert $t \mapsto t(1+u)$. If S is this last substitution and $\Delta = S - \text{id}$, then

$$v(\Delta f) \geq v(f) + v(u) \quad (289.3)$$

when $u \neq 0$; the identity case is immediate. The convergent operator series $\sum_{n \geq 0} (-\Delta)^n$ is a two-sided linear inverse to S and therefore an inverse field homomorphism. This also proves surjectivity without assuming a fixed common denominator for all exponents.

All coefficients in evaluation are pointwise limits of finite algebraic expressions and admissible powers. The same applies to the inverse construction and to composition. Lemma 287.2 therefore gives the joint Borel assertions. $\square$

[write] (Batch 96.) This is Part XVIII's [theorem 242.4](#) (same statement, independent proof through the induced automorphism of the value group; Part XVIII adds that the order hypothesis is automatic when Γ is 2-divisible, so that for such Γ , G_Γ is the group of all Borel field automorphisms). It is the real-coefficient, Borel analogue for $\mathbb{Z} \subseteq \Gamma \subseteq \mathbb{Q}$ of Part XV's [theorems 198.1](#) and [198.2](#) (coefficients $\mathbb{R}_{\text{alg}}$, all embeddings of $\mathcal{L}^\Gamma$ for a $\mathbb{Q}$ -subspace Γ); "a Borel field automorphism fixes $\mathbb{R}$ " is Part XVII's [remark 239.4](#). The substitution Θ_y is Part XVIII's S_y , $f \mapsto f \circ y$.

Thus G_Γ itself has a concrete standard Borel coding: it is the Borel set of positive series y with $v(y) \in U_\Gamma$. We do not need to claim that this Borel group admits a compatible Polish group topology.

Put $L_y = E(y)/y$. It has positive leading constant r , so is nonzero. The chain rule gives the convenient conjugacy formula

$$\Theta_y^{-1} D_a \Theta_y = D_{\Theta_y^{-1}(aL_y)}. \quad (289.4)$$

Our group action on derivations is ordinary conjugation $g \cdot D = gDg^{-1}$. Formula (289.4) expresses passage *from* an original coordinate to its y -coordinate and fixes all orientation conventions used below.

Remark 289.2 (Orientation of conjugacy in Parts XVIII and XXI [write]). Part XVIII ([definition 243.1](#)) parametrizes conjugacy by a *coordinate*: h carries D_a to D_b when $S_h^{-1} D_a S_h = D_b$, where $S_h f = f \circ h$ and $S_h S_k = S_{k \circ h}$. Source 27 parametrizes it by the *automorphism* g acting on the left: g sends D to gDg^{-1} . Since $\Theta_y = S_y$, the two are inverse to each other:

$$gDg^{-1} = D' \iff \text{the coordinate } h = g^{-1}(t) \text{ carries } D \text{ to } D' \text{ in Part XVIII's sense,}$$

because $S_h = g^{-1}$. Thus (289.4) says that y carries D_a to $D_{\Theta_y^{-1}(aL_y)}$ in Part XVIII's sense, which is Part XVIII's equation $a E(h)/h = b(h)$ for $h = y$. Source 27's normalizing coordinate y_h ([definition 290.1](#)) carries a pair to its normal form in Part XVIII's sense; its conjugator $g = \Theta_{h_2} \Theta_{h_1}^{-1}$ from the first pair to the second in (291.5) corresponds to Part XVIII's coordinate $w = g^{-1}(t) = y_{h_2}^{\langle -1 \rangle} \circ y_{h_1}$, where $\langle -1 \rangle$ is compositional inversion, of the same shape as Part XVIII's witness $h_b^{-1} \circ M \circ h_a$ with $M = t$. In particular the exponents are reciprocal: the automorphisms g with $g(cE)g^{-1} = dE$ have $g(t) = Ct^{c/d}$ ([proposition 292.1](#)), while the coordinates carrying cE to dE in Part XVIII's sense are $C't^{d/c}$, as in the proof of [theorem 245.1](#); both describe the same relation $c/d \in U_\Gamma$.

290 A nonconstant scalar determines a unique normalizing coordinate

Choose an enumeration of $\Gamma \setminus \{0\}$. For $q \neq 0$, let $\kappa(q)$ be the first member of its U_Γ -orbit. Then

$$\kappa(rq) = \kappa(q) \quad (r \in U_\Gamma), \quad q/\kappa(q) \in U_\Gamma. \quad (290.1)$$

This is a function on a countable set and is automatically Borel. Its values have the same sign as their arguments. For $\Gamma = \mathbb{Q}$ choose $\kappa(q) = \text{sgn}(q)$; for $\Gamma = \mathbb{Z}$ choose $\kappa(q) = q$.

[write] (Batch 96.) For $\Gamma = \mathbb{Q}$ the orbits are $\mathbb{Q}_{>0}$ and $\mathbb{Q}_{<0}$, and $\kappa = \text{sgn}$ is the output of the definition only for an enumeration that lists 1 before every other positive rational and -1 before every other negative one; “choose” therefore means choosing such an enumeration, which is harmless because every result below depends on κ only through (290.1). For $\Gamma = \mathbb{Z}$, $U_\mathbb{Z} = \{1\}$, the orbits are points, and $\kappa(q) = q$ for every enumeration. Part XVIII's κ ([theorem 245.1](#)) is defined on positive exponents only and orbits β/r ; source 27's is signed. Both are complete invariants of the scaling relation on their domains.

Definition 290.1 (Scalar data). For $h \in \mathcal{L}_\mathbb{R}^\Gamma \setminus \mathbb{R}$, define

$$c(h) = \begin{cases} 0, & v(h) < 0, \\ \text{ct}(h), & v(h) \geq 0, \end{cases} \quad k(h) = h - c(h).$$

Put $q(h) = v(k(h)) \neq 0$, $\sigma(h) = \text{sgn } \text{lc}(k(h))$, and $\kappa = \kappa(q(h))$. Define the canonical scalar and its coordinate by

$$H_h = c(h) + \sigma(h)t^\kappa, \quad y_h = (\sigma(h)(h - c(h)))^{1/\kappa}. \quad (290.2)$$

Let $\Theta_h = \Theta_{y_h}$ and $N_h = \Theta_h^{-1}$.

The subtraction in the finite case is essential: a nonconstant finite series has a distinguished real standard part, while the constant coefficient of an infinite series is not preserved by all substitutions. In the infinite case we deliberately retain the whole series and set $c(h) = 0$.

Theorem 290.2 (Scalar-coordinate theorem). *The definitions above have the following properties.*

- (i) y_h is an admissible coordinate and $\Theta_h(H_h) = h$.
- (ii) Two nonconstant scalars are in the same G_Γ -orbit exactly when their canonical scalars H_h agree.
- (iii) The stabilizer in G_Γ of every nonconstant scalar is trivial. Consequently Θ_h is the unique automorphism sending H_h to h .
- (iv) All these assignments are Borel, and for $g \in G_\Gamma$,

$$H_{gh} = H_h, \quad \Theta_{gh} = g\Theta_h, \quad N_{gh}g = N_h. \quad (290.3)$$

Proof. Write $\sigma k = \alpha t^q(1+u)$ with $\alpha > 0$. Then

$$y_h = \alpha^{1/\kappa} t^{q/\kappa} (1+u)^{1/\kappa}.$$

Its exponent q/κ belongs to U_Γ , so Theorem 289.1 applies. The rational-power identity gives $\sigma y_h^\kappa = k$, proving (i).

An allowed automorphism preserves finiteness, the standard part of finite elements, the sign of a leading coefficient, and the value-group orbit of a nonzero leading exponent. These are exactly the data in H_h , proving the necessity in (ii). Sufficiency follows by applying the two constructed coordinates.

Suppose an automorphism fixes $H = c + \sigma t^\kappa$. It fixes $\mathbb{R}$, hence t^κ . As $\kappa \neq 0$ is rational, uniqueness of positive integer roots forces it to fix t . It then fixes every rational monomial and every valuation limit of finite sums, hence the whole field. Conjugating this argument by Θ_h gives (iii).

Borelness follows from Lemma 287.2, countability of the exponent choices, and Theorem 289.1. In (iv), both Θ_{gh} and $g\Theta_h$ send the same canonical scalar to gh ; uniqueness proves their equality. The inverse identity follows. $\square$

[write] (Batch 96.) In (i), y_h is defined because $U_\Gamma \subseteq \Gamma$ ($r = r \cdot 1$ for $r \in U_\Gamma$), so the exponent q/κ of the power lies in Γ . In (iii), when $\kappa < 0$ the fixed element t^κ gives the fixed element $t^{-\kappa} = (t^\kappa)^{-1}$, and t is the unique positive root of a fixed integer power of it.

Let Can_Γ be the set of canonical scalars: σt^κ for negative representatives κ , and $c + \sigma t^\kappa$ for positive representatives, where $c \in \mathbb{R}$ and $\sigma \in \{-1, 1\}$. It is a standard Borel space, consisting of countably many signed points and real lines.

Corollary 290.3 (A global Borel product decomposition). *The map*

$$\mathcal{L}_\mathbb{R}^\Gamma \setminus \mathbb{R} \longrightarrow \text{Can}_\Gamma \times G_\Gamma, \quad h \longmapsto (H_h, \Theta_h)$$

is a Borel bijection with Borel inverse $(H, g) \mapsto g(H)$. Under this bijection the G_Γ -action becomes left multiplication on the second factor.

Proof. Every canonical H normalizes to itself, with coordinate t . Thus the inverse assertions and equivariance follow directly from Theorem 290.2. $\square$

This is stronger than an orbit list: it provides a uniform, unique coordinate for each scalar, with no choice of a representative in an uncountable orbit left unresolved.

291 Simultaneous normal forms for noncommuting derivations

Code a derivation by its coefficient a . The space of noncommuting ordered pairs is the Borel set

$$\mathrm{NC}_\Gamma = \{(a, b) \in \mathcal{L}_\mathbb{R}^{\Gamma^2} : aE(b) - bE(a) \neq 0\}.$$

Both coefficients are nonzero on this set. Define

$$h = b/a, \quad j = D_a(h) = aE(h). \quad (291.1)$$

Then $h \notin \mathbb{R}$ and $j \neq 0$.

Lemma 291.1 (Replacing two derivations by two scalars). *The assignment $(a, b) \mapsto (h, j)$ is a Borel bijection*

$$\mathrm{NC}_\Gamma \longleftrightarrow (\mathcal{L}_\mathbb{R}^\Gamma \setminus \mathbb{R}) \times \mathcal{L}_\mathbb{R}^{\Gamma^\times},$$

with inverse

$$a = j/E(h), \quad b = hj/E(h). \quad (291.2)$$

It is equivariant: simultaneous conjugation of the derivations sends (h, j) to $(g(h), g(j))$.

Proof. The inverse follows from (287.4); all operations are Borel. If $D_b = hD_a$, then $gD_bg^{-1} = g(h)(gD_ag^{-1})$. Applying the transformed first derivation to $g(h)$ gives $g(D_ag^{-1}h) = g(j)$. $\square$

Theorem 291.2 (Explicit cross-section and unique conjugators). *For $(a, b) \in \mathrm{NC}_\Gamma$, let $H = H_h$, $\Theta = \Theta_h$, and define*

$$A = \frac{\Theta^{-1}(j)}{E(H)} = \Theta^{-1}\left(a \frac{E(y_h)}{y_h}\right). \quad (291.3)$$

Then $A \neq 0$ and the following conclusions hold.

(i) *The normalized ordered pair is*

$$\Theta^{-1}(D_a, D_b)\Theta = (D_A, D_{HA}). \quad (291.4)$$

(ii) *(H, A) is a Borel complete invariant. The pairs*

$$\{(D_A, D_{HA}) : H \in \mathrm{Can}_\Gamma, A \in \mathcal{L}_\mathbb{R}^{\Gamma^\times}\}$$

form an actual Borel cross-section.

(iii) *Every pair in NC_Γ has trivial stabilizer.*

(iv) *If two pairs have equal invariants, their unique simultaneous conjugator, from the first to the second, is*

$$g = \Theta_{h_2}\Theta_{h_1}^{-1}. \quad (291.5)$$

This witness depends Borelly on the two conjugate pairs.

Proof. After applying Θ^{-1} , the ratio of the derivations is H and the first derivation applied to H is $\Theta^{-1}(j)$. Lemma 291.1 gives precisely the coefficient A in (291.3), proving (i). The second formula there follows independently from (289.4).

Equivariance (290.3) and Lemma 291.1 show that both H and A are unchanged by conjugacy. Conversely, equal invariants yield the same normalized pair, hence a conjugacy. All formulas are Borel. A canonical pair already has ratio H , whose scalar normalizer is the identity, so

it normalizes to itself. The parameterization of the displayed cross-section is injective and Borel with inverse obtained by taking the ratio and first coefficient; the image is Borel. This proves (ii).

An automorphism stabilizing the ordered pair fixes its nonconstant ratio, so is the identity by Theorem 290.2. Formula (291.5) conjugates the two copies of the common normal pair. Triviality of the stabilizer gives uniqueness, and Borel group operations give (iv). $\square$

There is no restriction on the series A beyond being nonzero. Thus the answer is not a finite-dimensional moduli space: its improvement is exact Borel classification, not reduction of every invariant to finitely many real parameters. Since a standard Borel space can be coded by reals, (H, A) nevertheless is a complete invariant in the usual descriptive-set-theoretic sense.

[write] (Batch 96.) In Part XVIII's orientation the witness (291.5) is the coordinate $y_{h_2}^{\langle -1 \rangle} \circ y_{h_1}$ (remark 289.2). Unlike Part XVIII's theorem 245.2, which selects a witness only after two conjugate inputs are given, theorem 291.2 also gives a transversal; this is the sense in which a second, noncommuting derivation "removes the obstruction". For $\Gamma = \mathbb{Z}$ the invariant (H, A) is complete but its equality is not decidable from coefficient programs: see remark 300.4.

291.1 Ordered tuples and functorial scalar markers

Corollary 291.3 (Finite and countable tuples). *For every finite or countable ordered tuple of Borel derivations containing a noncommuting pair, simultaneous conjugacy is smooth and has a Borel cross-section. Its stabilizers are trivial, and conjugating witnesses are unique and Borel on the equivalence relation.*

Proof. Fix an enumeration of index pairs and take the first pair with nonzero commutator. This choice is Borel and invariant under simultaneous conjugacy. Use its ratio to normalize the entire tuple. Equality of the resulting ordered tuples is equivalent to conjugacy by exactly the same argument as in Theorem 291.2. Coordinatewise evaluation handles countable tuples. $\square$

The same mechanism applies to any Borel-coded structure carrying an *equivariantly definable nonconstant scalar*: that scalar supplies the unique coordinate, after which all remaining named data can be recorded without quotient ambiguity. This principle does not apply automatically to unordered tuples or to structures where the proposed scalar depends on an arbitrary choice of generators.

292 Why one derivation, or two commuting ones, can still be nonsmooth

The contrast can be proved on a very small family, without importing the full earlier single-derivation classification.

Proposition 292.1 (Euler conjugacy). *For $c, d \in \mathbb{R}^\times$, the derivations cE and dE are conjugate in G_Γ if and only if $c/d \in U_\Gamma$. Every conjugator sending cE to dE has coordinate*

$$g(t) = Ct^{c/d} \quad (C > 0).$$

Proof. The relation $dE g = g cE$, evaluated at t , gives $E(g(t))/g(t) = c/d$. Write $g(t) = Ct^r(1+u)$. The leading constant of its logarithmic derivative is r , so $r = c/d$. The remaining equality gives $E(1+u) = 0$, whence $u = 0$. The converse is immediate. $\square$

[write] (Batch 96.) The zero-gain stratum (245.2) of Part XVIII's [theorem 245.1](#) ($c_b/c_a \in U_\Gamma$), together with the zero-gain centralizer of [theorem 247.2](#); the exponent c/d here and c_b/c_a there are reciprocal because the orientations are opposite ([remark 289.2](#)).

Theorem 292.2 (A sharp nonsmoothness contrast). *If $\log U_\Gamma$ is dense in $\mathbb{R}$, conjugacy of all single Borel derivations on $\mathcal{L}_\mathbb{R}^\Gamma$ is not smooth. It has no complete invariant whose restriction to the positive Euler line is Baire measurable into a standard Borel space. The same obstruction remains for commuting ordered pairs. In particular, these assertions hold for $\Gamma = \mathbb{Q}$, whereas noncommuting pairs are smooth by Theorem 291.2.*

Proof. On $c > 0$, Proposition 292.1 gives multiplicative cosets of U_Γ . After logarithms, the relation is translation by the countable dense subgroup $\log U_\Gamma$.

An $\log U_\Gamma$ -invariant subset of $\mathbb{R}$ with the Baire property is either meager or comeager. Indeed, if it is nonmeager it is comeager in some nonempty open interval. Countably many $\log U_\Gamma$ -translates of that interval cover $\mathbb{R}$, and invariance makes the set comeager everywhere. Now suppose a Baire measurable complete invariant took values in a standard Borel space. Use a countable Borel family separating points in its target. Each inverse image is $\log U_\Gamma$ -invariant, hence meager or comeager. Away from a single meager exceptional set, every point has the same membership pattern in all these inverse images. The invariant is therefore constant on a comeager set, impossible because its fibers are the countable $\log U_\Gamma$ -cosets.

For $\Gamma = \mathbb{Q}$, the logarithms of positive rationals are dense. Finally, $cE \mapsto (cE, 2cE)$ preserves exactly the same conjugacy relation and its image consists of commuting pairs. $\square$

[write] (Batch 96.) By Part XVIII's [lemma 246.2](#), U_Γ is generated by the primes of infinite height in Γ , so $\log U_\Gamma$ is dense exactly when at least two primes are infinitely divisible in Γ ; with at most one it is discrete, and Part XVIII's [theorem 246.4](#) shows that conjugacy is then smooth. So [theorem 292.2](#) is the nonsmoothness half of that threshold (at least two infinitely divisible primes $\Rightarrow$ not smooth), proved again, with two additions: Baire measurable invariants on the Euler line are excluded, and commuting pairs inherit the obstruction. The proof's subgroup $\log U_\Gamma$ is written H in source 27.

The conclusion is a minimality statement in a specific sense: over $\mathcal{L}_\mathbb{R}^\mathbb{Q}$, one named derivation does not suffice for uniform Borel classification, and adding an arbitrary redundant commuting derivation does not help; two *noncommuting* named derivations do suffice. It is not a statement about computational running time or analytic normal forms.

293 Finite-dimensional Lie algebras: valuation spectrum and normal forms

We now address the nonabelian infinitesimal question. All Lie algebras are real Lie subalgebras of $\{D_a : a \in \mathcal{L}_\mathbb{R}^\Gamma\}$, with the usual commutator. The classical affine/projective pattern is not new; the proof here keeps track of the left-finite field, the allowed ramification, and Borel normalizers.

Lemma 293.1 (Finite valuation spectrum). *If $V \subseteq \mathcal{L}_{\mathbb{R}}^{\Gamma}$ is a d -dimensional real vector space, then*

$$\{v(f) : f \in V \setminus \{0\}\}$$

contains exactly d elements. There is a basis with pairwise distinct valuations, in increasing order.

Proof. The union of the supports of a finite basis is left-finite and contains the support of every vector of V . Its least exponent at which some vector is nonzero gives a nonzero leading-coefficient functional on V . Its kernel has dimension $d - 1$ and strictly larger least available exponent. Repeat. The process ends after d steps and gives a basis with distinct valuations. Every nonzero combination has the valuation of the first basis vector with nonzero coefficient, proving the exact count. $\square$

Theorem 293.2 (Dimension bound). *Every finite-dimensional real Lie algebra of coefficient-Borel derivations on $\mathcal{L}_{\mathbb{R}}^{\Gamma}$ has dimension at most three. Its coefficient valuation spectrum is one of the following types:*

$$\{p\}, \quad \{0, p\} \ (p \neq 0), \quad \{-p, 0, p\} \ (p > 0),$$

apart from the zero algebra. An abelian such Lie algebra has dimension at most one.

Proof. Let S be its finite coefficient valuation spectrum. If $p, q \in S$ are distinct, choose leading coefficients A, B at those valuations. The bracket coefficient has leading term

$$(q - p)ABt^{p+q}, \tag{293.1}$$

so $p + q \in S$.

There cannot be two positive members $0 < p < q$: repeated bracketing of a p -valuation vector with the q -valuation vector gives valuations $q + np$, all distinct. Explicitly the n th leading coefficient is

$$BA^n \prod_{j=0}^{n-1} (q + (j-1)p) \neq 0.$$

The negative case is the same with signs reversed. Thus S has at most one positive member, at most one negative member, and possibly zero. If both $p > 0$ and $q < 0$ occur, their sum can be neither p nor q ; it must be zero, so $q = -p$ and zero also occurs. Lemma 293.1 gives the dimension bound and the listed types. Finally, if $D_a \neq 0$ commutes with every element of the algebra, Corollary 288.4 makes every coefficient a real multiple of a . $\square$

[write] (Batch 96.) In the proof A, B are leading coefficients of two coefficient series, not the normal coefficient A of theorem 291.2. The product is nonzero because $q + (j-1)p \geq q - p > 0$ for $0 < p < q$; in the negative case one brackets with the member nearer to zero.

For $p \in \Gamma \setminus \{0\}$ set

$$\mathfrak{a}_p = \text{span}_{\mathbb{R}}\{E, t^p E\}, \tag{293.2}$$

and for $p > 0$ set

$$\mathfrak{s}_p = \text{span}_{\mathbb{R}}\{t^{-p} E, E, t^p E\}. \tag{293.3}$$

Theorem 293.3 (Ramification-sensitive Lie classification). *Every two-dimensional real Lie algebra of coefficient-Borel derivations on $\mathcal{L}_{\mathbb{R}}^{\Gamma}$ is conjugate to $\mathfrak{a}_{\kappa(p)}$, where $\{0, p\}$ is its coefficient valuation spectrum. Every three-dimensional one is conjugate to $\mathfrak{s}_{\kappa(p)}$, where its spectrum is $\{-p, 0, p\}$ with $p > 0$. Within either family, two normal forms are conjugate exactly when their positive scaling exponent orbits agree. In the affine family the sign of p is also preserved. Equivalently, equality of the chosen κ values is the complete invariant within each dimension.*

Proof. First suppose the dimension is two. Take a valuation-adapted basis X, Y with $v(X) = 0$, $v(Y) = p \neq 0$, where valuation refers to coefficients. The leading-bracket formula gives $[X, Y] = \lambda Y + \mu X$ with $\lambda \neq 0$. If $p > 0$, the bracket has no constant term, so $\mu = 0$. If $p < 0$, replace Y by $Y + (\mu/\lambda)X$, preserving its valuation and obtaining $[X, Y] = \lambda Y$. In both cases replace X by X/λ , so

$$[X, Y] = Y, \quad v(X) = 0, \quad v(Y) = p. \quad (293.4)$$

Write $X = D_a$, $Y = D_b$ and $h = b/a$. Then $v(h) = p \neq 0$ and $D_a(h) = h$. Its canonical scalar is $H = \sigma t^{\kappa(p)}$ with no translation term: if $p > 0$ its standard part is zero, and if $p < 0$ the infinite normalization uses $c = 0$ by definition. In the scalar coordinate, $X = D_A$, $Y = D_{HA}$ and the relation becomes $AE(H) = H$. Therefore

$$A = 1/\kappa(p).$$

The transformed subalgebra is exactly $\text{span}_{\mathbb{R}}\{E, t^{\kappa(p)}E\}$.

Now suppose the dimension is three, with spectrum $\{-p, 0, p\}$. The nonnegative-valuation part is a two-dimensional subalgebra, so the preceding argument conjugates it to $\text{span}_{\mathbb{R}}\{E, t^k E\}$, where $k = \kappa(p) > 0$. The remaining basis coefficient a has valuation $-k$. Since $[E, D_a] = D_{E(a)}$ lies in the algebra, comparison of leading terms gives

$$E(a) + ka \in \text{span}_{\mathbb{R}}\{1, t^k\}.$$

Coefficientwise, $(q+k)a_q = 0$ for $q \notin \{-k, 0, k\}$. Thus $a = A_- t^{-k} + A_0 + A_+ t^k$, with $A_- \neq 0$, and the whole algebra is $\mathfrak{s}_k$.

An allowed automorphism multiplies every coefficient valuation by the same $r \in U_{\Gamma}$: this follows from (289.4), since the logarithmic coordinate derivative has valuation zero. Hence the indicated exponent orbit is necessary. Monomial substitutions realize the orbit scaling and show sufficiency. $\square$

The projective algebra is explicitly $\mathfrak{sl}_2(\mathbb{R})$. For $p > 0$, put

$$X_+ = \frac{1}{p} t^p E, \quad X_- = -\frac{1}{p} t^{-p} E, \quad X_0 = \frac{2}{p} E.$$

Then $[X_0, X_+] = 2X_+$, $[X_0, X_-] = -2X_-$, and $[X_+, X_-] = X_0$. The affine algebra has basis $X = E/p$, $Y = t^p E/p$ with $[X, Y] = Y$. For $\Gamma = \mathbb{Q}$ the classification has just two affine coordinate types, according to the sign of p , and one projective type. For $\Gamma = \mathbb{Z}$, the nonzero integer p in the affine case and positive integer p in the projective case remain genuine invariants.

[write] (Batch 96.) Source 27 writes the $\mathfrak{sl}_2$ -triple as e, f, H ; it is printed X_+, X_-, X_0 because f denotes series and H the canonical scalar here. The brackets were rechecked: $[t^p E, t^{-p} E] = D_{t^p E(t^{-p}) - t^{-p} E(t^p)} = -2pE$ by (288.1). The statement “two affine types for $\Gamma = \mathbb{Q}$ ” uses $\kappa = \text{sgn}$, that is, the enumeration chosen in section 290; for another enumeration the two types are $\mathfrak{a}_{\kappa(1)}$ and $\mathfrak{a}_{\kappa(-1)}$, conjugate to $\mathfrak{a}_1$ and $\mathfrak{a}_{-1}$.

293.1 Borel meaning of the subalgebra classification

Proposition 293.4. *Code a finite-dimensional subalgebra by a finite ordered independent tuple of coefficients spanning it. On the two- and three-dimensional subalgebra loci, conjugacy of the spanned subalgebras is Borel and smooth, with the countable complete invariants in Theorem 293.3. A conjugating coordinate to the indicated standard subalgebra can be chosen Borelly from the input basis.*

Proof. Linear independence of d coefficients is equivalent to a nonzero $d \times d$ minor of their coefficient matrix, hence is a countable union of Borel conditions. Choose the first nonzero minor in a fixed enumeration. It computes coordinates in the span by finite real linear algebra. Membership of a given series in the span is then a countable coefficient equality test. This makes closure under the finitely many basis brackets Borel.

The adapted-basis construction in Lemma 293.1 chooses a leading exponent by Borel zero tests and eliminates it using finite linear algebra. The adjustments in the proof of Theorem 293.3 are Borel. Finally the scalar normalizer is Borel. The resulting *subalgebra* is the standard one determined by the invariant, even though a choice of input basis was used to construct a witness. $\square$

Unmarked subalgebras can have nontrivial stabilizers, so the uniqueness assertion for *ordered noncommuting pairs* must not be transferred to these unmarked objects.

294 Whole-field affine actions and the obstruction to projective actions

Call a derivation D_a *valuation-nondecreasing* when $v(a) \geq 0$, with zero included. Equivalently, $v(D_a f) \geq v(f)$ for every f .

Corollary 294.1. *A finite-dimensional Lie algebra consisting of valuation-nondecreasing Borel derivations has dimension at most two. Every nonabelian such algebra is conjugate to $\mathfrak{a}_p$ with $p > 0$.*

Proof. Its valuation spectrum has no negative member. Apply Theorems 293.2 and 293.3. $\square$

The nondecreasing hypothesis is forced by a natural minimal differentiability assumption on whole-field one-parameter actions.

Definition 294.2. A *coefficient-differentiable one-parameter action with Borel generator* is a homomorphism $s \mapsto \Phi_s$ from $(\mathbb{R}, +)$ to G_Γ such that, for each $f \in \mathcal{L}_\mathbb{R}^\Gamma$, every coefficient of $\Phi_s(f)$ is differentiable at $s = 0$, and these derivatives form a coefficient-Borel derivation $D(f)$. The last requirement is explicit; we do not infer interchange of arbitrary formal sums and parameter differentiation without it.

Remark 294.3 (Negative gain is impossible; source 27's Proposition 9.3, printed as a note [write]). Source 27 states: *The generator of such an action is valuation-nondecreasing. Consequently neither a negative affine algebra $\mathfrak{a}_p$ with $p < 0$ nor a projective algebra $\mathfrak{s}_p$ can be the effective infinitesimal image of a coefficient-differentiable Lie-group action in this sense on the whole field.*

The first sentence is already in this report twice, under other hypotheses. Part XVIII's [theorem 248.3](#) (last clause) proves it for every $\mathbb{Z} \subseteq \Gamma \subseteq \mathbb{Q}$ with less: if $v(a) < 0$, not even a coefficientwise differentiable curve of ordered Borel automorphisms through the identity has derivative D_a there, and only the constant coefficient of the images of $t^{-v(a)}$ is used, with no group law and no Borel generator. Since the generator in [definition 294.2](#) is Borel, hence some D_a by [corollary 288.4](#), that clause implies the first sentence. Part XX's [corollary 275.4](#) proves it for countable divisible $\Gamma \leq \mathbb{R}$ of any rational rank and jointly coefficient-Borel field flows, assumed neither ordered nor differentiable (their coefficients are entire by Part XX's [theorem 275.2](#)). For $\Gamma = \mathbb{Q}$ all three statements apply. Source 27's proof, a third route through the value group, is the following.

Proof (source 27). Let $r_s \in U_\Gamma$ be the value-group multiplier of Φ_s . The action law makes $s \mapsto r_s$ a homomorphism from the divisible additive group $\mathbb{R}$ into $\mathbb{Q}_{>0}$ under multiplication.

For every ordinary prime ℓ , the prime valuation $s \mapsto v_\ell(r_s) \in \mathbb{Z}$ is additive. Its value at any s is divisible by every positive integer, since $s = n(s/n)$; hence it is zero. All prime valuations vanish, so $r_s = 1$. Thus $\Phi_s(t)$ has valuation exactly 1 for every s . Every coefficient at exponent less than 1 is identically zero as a function of s and has zero derivative. Therefore $v(D(t)) \geq 1$. Writing $D = aE$ gives $v(a) \geq 0$. Applying this to every one-parameter subgroup proves the consequence. $\square$

The second sentence uses a “Lie-group action” and its “effective infinitesimal image”, which [definition 294.2](#) does not define: it covers one-parameter actions only, and the proof’s last line applies the first sentence to each one-parameter subgroup. [Definition 300.1](#) [merge] supplies the definition, under which the second sentence holds (without effectiveness) because every derivation in the infinitesimal image is valuation-nondecreasing, while $\mathfrak{a}_p$ ($p < 0$) and $\mathfrak{s}_p$ contain $t^p E$, respectively $t^{-p} E$, of negative coefficient valuation, and so does every conjugate of them, since conjugation multiplies coefficient valuations by some $r \in U_\Gamma$ (proof of [theorem 293.3](#)). That the infinitesimal image is a Lie algebra at all was not shown by source 27; [theorem 300.3](#) [merge] proves it for every action in the sense of [definition 300.1](#), and equally when the one-parameter subgroups only move t Borelly in time.

Theorem 294.4 (Explicit global affine integration). *For $p \in \Gamma_{>0}$ and $(a, b) \in \mathbb{R}_{>0} \times \mathbb{R}$, define*

$$\Phi_{a,b}(t) = a^{1/p} t(1 - bt^p)^{-1/p}. \quad (294.1)$$

These substitutions form a faithful action of $\text{Aff}^+(\mathbb{R})$, with multiplication

$$(a, b)(c, d) = (ac, b + ad),$$

by members of G_Γ . Every coefficient of $\Phi_{a,b}(f)$ is real-analytic in (a, b) for $a > 0$. Its infinitesimal image is $\mathfrak{a}_p$.

Proof. The factor in parentheses is a positive 1-unit, so the substitution is admissible. Put $u = t^{-p}$. Then

$$\Phi_{a,b}(u) = \frac{u - b}{a}.$$

Direct composition gives $\Phi_{a,b}\Phi_{c,d}(u) = (u - b - ad)/(ac)$, which is $\Phi_{ac,b+ad}(u)$. Since a nonconstant scalar has trivial stabilizer, equality on u proves the group law on the whole field. The same formula proves faithfulness.

For a monomial,

$$\Phi_{a,b}(t^q) = a^{q/p} t^q \sum_{n \geq 0} \frac{(q/p)_n}{n!} b^n t^{np}.$$

At a fixed output exponent, only finitely many input monomials and finitely many n contribute. Each coefficient is therefore a finite sum of analytic functions $a^{q/p} b^n$. Differentiating $\Phi_{e^s,0}$ and $\Phi_{1,b}$ at the identity yields E/p and $t^p E/p$, respectively. $\square$

[write] (Batch 96.) (a, b) is the affine map $x \mapsto ax + b$, that is, the matrix $\begin{pmatrix} a & b \\ 0 & 1 \end{pmatrix}$, and $(q/p)_n$ is the rising factorial. The time- b maps $\Phi_{1,b}$ are the flow of $t^p E/p$, which is Part XVIII’s example “positive gain without residue” in [section 249](#) with $\alpha = p$ and $c = 1/p$, and for $p = 1$ Part XVII’s [\(234.4\)](#), the surcomplex-field-automorphisms report’s `saut:thm:shiftflow`; the maps $\Phi_{e^s,0}$, $t \mapsto e^{s/p} t$, are the Euler flow of E/p (Part XVII’s [proposition 239.3](#)). New are the two-parameter action and its group law.

This is an exact global action on formal series. It is stronger than merely writing a formal exponential, but different from a local analytic action on a punctured real neighborhood. The infinite element $u = t^{-p}$ permits all real translations $u \mapsto u - b$ without losing positivity or its infinite leading term. Translating the infinitesimal t^p instead would lose that property, which is the geometric reason for the sign boundary.

294.1 A BCH application with its convergence qualification

For $[X, Y] = Y$, the affine matrix representation gives

$$\log(e^{sX}e^{uY}) = sX + \frac{su}{1-e^{-s}}Y, \quad (294.2)$$

with the continuous value uY when $s = 0$. Indeed, represent $X = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$ and $Y = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$. The exponential of $sX + vY$ has upper-right entry $v(e^s - 1)/s$, whereas $e^{sX}e^{uY}$ has upper-right entry $e^s u$. This gives (294.2). The scalar expansion is

$$\frac{s}{1-e^{-s}} = 1 + \frac{s}{2} + \frac{s^2}{12} - \frac{s^4}{720} + \frac{s^6}{30240} - \frac{s^8}{1209600} + \cdots$$

As a complex Taylor series it has radius 2π , determined by the nearest nonzero zeros of $1 - e^{-s}$. The real closed expression is meaningful for all real s . In the series field, zero-gain Euler flows are interpreted by the coefficientwise analytic action, not by an unjustified valuation-convergent operator exponential. This is the precise bridge to the BCH program [168]; the affine formula itself is classical.

Remark 294.5 (The affine BCH formula is a Lean theorem [write]). Source 27 read only the BCH project's README and calls (294.2) classical. It is identity (9.6) of Theorem 9.4 of that project's combined article (Algebra/BakerCampbellHausdorff/docs/combined, thm:eigen, eq:eigenbch), proved in Lean as `BCH.exp_mul_exp_of_lie_eq_smul` (Algebra/BakerCampbellHausdorff/Lean/BCH/Eigen.lean, line 219; the library was merged into this repository with its history in abb123637). Its setting and hypotheses are: a scalar field $\mathbb{K}$ with `[RCLike  $\mathbb{K}$ ]` (that is, $\mathbb{R}$ or $\mathbb{C}$); a complete normed unital $\mathbb{K}$ -algebra $\mathbb{A}$ (`[NormedRing  $\mathbb{A}$ ]`, `[NormedAlgebra  $\mathbb{K}$   $\mathbb{A}$ ]`, `[CompleteSpace  $\mathbb{A}$ ]`); elements $X, Y \in \mathbb{A}$ and $s \in \mathbb{K}$ with $[X, Y] = s \cdot Y$ for the ring commutator, and $s = 0 \vee e^{-s} \neq 1$. Its conclusion is $e^X e^Y = \exp(X + \beta(s)Y)$, where `BCH.beta` is $\beta(s) = s/(1 - e^{-s})$ with $\beta(0) = 1$. The match: take $\mathbb{K} = \mathbb{R}$, $\mathbb{A}$ the 2×2 real matrices with an operator norm, and apply the theorem to sX and uY ; then $[sX, uY] = su[X, Y] = s \cdot (uY)$ because $[X, Y] = Y$, the side condition holds for every real s (for real $s \neq 0$, $e^{-s} \neq 1$), and the conclusion $e^{sX}e^{uY} = \exp(sX + \frac{su}{1-e^{-s}}Y)$ is (294.2), including the value at $s = 0$. Two qualifications. The Lean theorem proves this exponential identity, not uniqueness of the logarithm; that $\log$ in (294.2) is well defined is the injectivity of $\exp$ on the affine algebra, which source 27's upper-right-entry computation supplies. And the Lean theorem concerns elements of a Banach algebra: it does not apply to the derivations $X = E/p$, $Y = t^p E/p$ as operators on $\mathcal{L}_{\mathbb{R}}^{\Gamma}$, which form no Banach algebra and of which E/p , of gain zero, has no valuation-convergent exponential (Part XVII's [proposition 239.3\(iv\)](#)); it reaches the action of [theorem 294.4](#) only through the group homomorphism $(a, b) \mapsto \Phi_{a,b}$, which is not formalized. Part XX's [section 281](#) gives the complementary operator-level criterion (a common uniform positive gain), which E/p does not meet. The library, `BCH.Eigen` included, was built in the root workspace on 4 October 2026 with no errors, warnings or sorry (commit 0aa065f26); it was not rebuilt for this write.

295 Examples and boundaries

Example 295.1 (A pair with a nontrivial but rational normalizer). In $\mathcal{L}_{\mathbb{R}}^{\mathbb{Q}}$, take

$$D_1 = E, \quad D_2 = \frac{t}{1-t}E.$$

The ratio $h = t/(1 - t)$ has canonical scalar $H = t$, and $\Theta_h(t) = t/(1 - t)$, with inverse $t \mapsto t/(1 + t)$. Now $j = E(h) = t/(1 - t)^2$, so

$$\Theta_h^{-1}(j) = t(1 + t), \quad A = 1 + t.$$

The canonical pair is therefore

$$((1 + t)E, t(1 + t)E).$$

This is an exact identity, not a truncation calculation.

Example 295.2 (The exponent group changes the canonical scalar). Let $h = 3 - t^2(1 + t)^2$. Over $\mathcal{L}_{\mathbb{R}}^{\mathbb{Z}}$, its data are $H = 3 - t^2$ and $y_h = t(1 + t)$. Over $\mathcal{L}_{\mathbb{R}}^{\mathbb{Q}}$, we can normalize exponent 2 to 1, obtaining $H = 3 - t$ and $y_h = t^2(1 + t)^2$. Both are admissible in their respective fields; the second is not an automorphism coordinate over $\mathcal{L}_{\mathbb{R}}^{\mathbb{Z}}$, since its value-group multiplier is 2.

Example 295.3 (Why infinite constant coefficients are not invariants). Let $h = t^{-2} + 7 + t$. Over $\mathcal{L}_{\mathbb{R}}^{\mathbb{Q}}$, its canonical scalar is $H = t^{-1}$ and $y_h = h^{-1}$. Over $\mathcal{L}_{\mathbb{R}}^{\mathbb{Z}}$ it is $H = t^{-2}$ with

$$y_h = t(1 + 7t^2 + t^3)^{-1/2}.$$

The real coefficient 7 is absorbed into the coordinate. Treating it as a standard-part invariant would give the wrong orbit classification.

[write] (Batch 96.) The data over $\mathcal{L}_{\mathbb{R}}^{\mathbb{Q}}$ in [examples 295.1 to 295.3](#) use $\kappa = \text{sgn}$ ([section 290](#)); they were rechecked by hand, and the placement's independent spot check confirmed $A = 1 + t$ in [example 295.1](#).

Example 295.4 (Unbounded denominators are genuinely allowed). The series

$$u = t \left(1 + \sum_{n \geq 1} t^{n+1/(n+1)} \right)$$

is an admissible coordinate in $\mathcal{L}_{\mathbb{R}}^{\mathbb{Q}}$. Its exponents have unbounded denominators, but its support is left-finite. All scalar and pair normalization theorems apply to series involving u ; none requires a single Puiseux denominator for the entire input.

Example 295.5 (Higher rational rank destroys scalar rigidity). Let $\Gamma = \mathbb{Z} + \mathbb{Z}\sqrt{2} \subset \mathbb{R}$, outside the hypotheses of the main classification, and form its left-finite field. For $C > 0$, the rule

$$t^{m+n\sqrt{2}} \mapsto C^n t^{m+n\sqrt{2}}$$

defines a coefficient-Borel ordered-field automorphism. It fixes the nonconstant scalar t but is nontrivial when $C \neq 1$. Thus a single scalar no longer determines the coordinate.

The dimension bound also fails dramatically. Define Euler operators $E_1(t^{m+n\sqrt{2}}) = mt^{m+n\sqrt{2}}$ and $E_2(t^{m+n\sqrt{2}}) = nt^{m+n\sqrt{2}}$. For every $N \geq 1$,

$$\text{span}_{\mathbb{R}}\{E_1, E_2, tE_2, \dots, t^N E_2\}$$

is a nonabelian Lie algebra of dimension $N + 2$, consisting entirely of valuation-nondecreasing Borel derivations. Its brackets are $[E_1, t^j E_2] = jt^j E_2$ and $[t^i E_2, t^j E_2] = 0$. The reason is that the constant field of E_2 now contains nonconstant series in t .

[write] (Batch 96.) The field is Part XVII's $\mathcal{L}_{\mathbb{R}}^{\Gamma}$ for this rank-two Γ ; E_1, E_2 are Part XVII's coordinate Euler derivations (231.1) for the basis $1, \sqrt{2}$, and the automorphism is the time-log C map $\Psi_{\log C}$ of the coefficientwise flow of E_2 in Part XVII's [proposition 239.3](#). That the constants of E_2 contain all series supported in $\mathbb{Z}$ is the observation in the commentary on Part XVII's [question 237.3](#); the failure of scalar rigidity and of the dimension bound is new here. Part XX ([Part XX](#)) gives the higher-rank counterpart on the flow side.

Remark 295.6 (Other excluded extensions). The proofs do not cover full Hahn supports, arbitrary non-Borel algebraic derivations, or unordered field automorphisms when sign characters can occur. They classify named ordered pairs, not pairs modulo arbitrary real linear recombination. Formal roots and inverse substitutions need not converge as analytic germs. Each change of category can change both the stabilizers and the equivalence relation.

296 Transfer to a set-sized surreal subfield

Using the usual Conway normal-form arithmetic, the assignment

$$\iota \left(\sum_{q \in \Gamma} a_q t^q \right) = \sum_{q \in \Gamma} a_q \omega^{-q} \quad (296.1)$$

is an ordered-field embedding of $\mathcal{L}_{\mathbb{R}}^{\Gamma}$ into the surreals. Left-finite support makes the exponents on the right reverse well ordered; normal-form addition and multiplication agree with the coefficient and convolution rules. This is exactly the kind of normal-form field interface appearing in the ProveIt surreal materials [150]. The image is set-sized, since $\mathcal{L}_{\mathbb{R}}^{\Gamma}$ is a subset of $\mathbb{R}^{\Gamma}$.

[write] (Batch 96.) (296.1) is Part XVII's ι_{Γ} , (235.1), for $\mathbb{Z} \subseteq \Gamma \subseteq \mathbb{Q}$, and Part V's map (71.1) with real coefficients; its image is Part XVII's K_{Γ} , which is why source 27's name K_{Γ} for the field itself is not used here. In the ω -convention $E(\omega^{-q}) = q\omega^{-q}$ agrees with the note after (235.1).

Transport the coefficient Borel structure, the group G_{Γ} , and the Euler operator through this particular embedding. The latter satisfies

$$E(\omega^{-q}) = q\omega^{-q}.$$

All theorems then transfer verbatim to that image. There is no claim that this transported derivation is a particular canonical surreal derivation, or that a subfield automorphism extends to all of **No**. Nor is a Borel structure imposed on the whole proper class **No**.

This restriction is mathematically useful rather than merely defensive: the classification supplies exact normalizing certificates for a concrete symbolic subfield that can be represented and reasoned about without quantifying over proper-class maps.

297 Verification artifacts and a formalization route

The companion program `verify.py` performs exact symbolic regression checks with rational coefficients. It checks scalar-coordinate identities, conjugacy and commutator covariance, the affine and projective Lie brackets, the affine substitution group law, the BCH scalar coefficients, and finite valuation-spectrum combinatorics. Its generated report records the actual number of assertions and the software version used. The included run passed 669 assertions with Python 3.13.5 and SymPy 1.14.0: 104 scalar/pair checks, 37 Lie checks, 15

affine/BCH checks, and 513 finite-spectrum checks. These checks test the formulas and orientation conventions; they are not substitutes for the infinite-support, Baire-category, or Borel proofs.

[write] In this report the program is `code/27-two-derivations-verify.py`, its pinned dependency `data/27-two-derivations-requirements.txt` (`sympy==1.14.0`) and its recorded run `data/27-two-derivations-verification_report.txt` and `.json`; by default the program writes `verification_report.txt` and `.json` beside itself, so it must be run on a copy or with `-output-dir`. The intake reran it on a copy (Python 3.13.5 and SymPy 1.14.0 through `uv`): 669 assertions pass in the same four groups, and the printed record equals the shipped one after removing the carriage returns that Windows text mode adds. Its checks raise exceptions rather than use `assert`, so they also run under `-O`.

The exact algebraic check behind a normalizing coordinate is particularly small. Given a proposed positive coordinate y , canonical scalar H , and coefficients a, b, A , verify

$$\Theta_y(H) = b/a, \quad \Theta_y(A) = aE(y)/y. \quad (297.1)$$

The second normalized coefficient is then forced to be HA . These equations are a natural proof-certificate interface: the search for y may be external, while a checker proves substitution, support admissibility, and the two identities.

A Lean development could be split into four layers. First formalize left-finite supports, convolution, completeness, and positive rational powers. Second formalize substitutions, the gain estimate (289.3), their inverses, and the chain rule. Third formalize the scalar-coordinate and simultaneous-conjugacy identities; this layer is independent of Baire category once derivations are explicitly aE and coordinates are explicitly admissible. Finally add standard Borel coding and the Polish-slice argument that makes the classification exhaustive for Borel derivations and automorphisms. The valuation-spectrum proof of the Lie dimension bound is an additional relatively compact algebraic target.

No Lean or Rocq proof of these new statements is included in this package. The repository's existing formalized theorems and its unrefereed reports are kept separate in its own documentation [150, 168]; the same distinction is maintained here.

[write] (Batch 96.) True of the new statements. Two pieces of this part are nevertheless formal elsewhere: the matrix form of (294.2) (remark 294.5), and the facts $\ker E = \mathbb{R}$ and $\text{ct}(Ef) = 0$ behind (287.4): `Surreal.EulerDerivation.euler_eq_zero_iff` and `coeff_zero_euler` in `Algebra/SurrealNumbers/Surreal/HahnSeries/EulerDerivation.lean` prove them for the Euler derivation of a full Hahn series ring, with an exponent map $\varphi : \Gamma \rightarrow R$ vanishing only at 0 (for the first) and coefficients without zero divisors, so for $\mathbb{R}((t^\Gamma)) \supseteq \mathcal{L}_{\mathbb{R}}^\Gamma$ with φ the inclusion; the restriction to the left-finite subfield is not formalized but immediate. Part XVII's remark 239.5 records the Lean form of the rank-one integration criterion for $\Gamma = \mathbb{Z}$. Source 27 read no Lean file. No label of this part is mapped in `docs/FORMALIZATION.md`.

298 Further research questions and concrete next targets

The following questions describe boundaries of this manuscript, not certified claims that every formulation is new to the literature.

298.1 Minimal scalar coordinates in higher rank

Question 298.1. For a countable finite-rank exponent group, how many equivariantly definable scalars suffice to make the ordered Borel automorphism action free and smooth on

a natural locus of tuples of derivations?

Example 295.5 rules out a naive one-scalar theorem. A first target is a rank-two field with two scalars whose leading exponents form a rational basis, together with a precise summability condition on the remaining coordinate corrections.

[write] (Batch 96.) A companion of Part XVII's question 237.3 and of Part XVIII's question 253.2 (single derivations in higher rank); none answers another.

298.2 Unordered automorphisms and sign characters

Question 298.2. For Γ not divisible by 2, can the ordered scalar cross-section be extended to all coefficient-Borel field automorphisms by finitely many sign parameters, or do new orbit phenomena appear?

The map $t \mapsto -t$ on $\mathbb{R}((t))$ is the elementary test case. The positive root uniqueness used in scalar rigidity must be replaced by a compatible analysis of roots and exponent-group sign characters.

[write] (Batch 96.) For 2-divisible Γ every Borel field automorphism is ordered (Part XVIII's theorem 242.4), so the restriction to non-2-divisible Γ is exact. Part XVIII's question 253.1 asks the same for single derivations.

298.3 Full Hahn support domains

Question 298.3. Which standard Borel subspaces of a full Hahn field admit an analogue of the scalar product decomposition in Corollary 290.3?

The obstruction is not merely the existence of formal powers: an arbitrary well-ordered support can contain infinitely many exponents below a finite cutoff, and the automatic-summability and inverse-substitution arguments must be reworked. Both the coding domain and the allowed notion of summation must be specified before proposing a generalization.

[write] (Batch 96.) The whole Hahn field is excluded as a coding domain: its coefficient codes in $k^{\mathbb{Q}}$ form a Π_1^1 -complete set (Part V's theorem 67.3, for countable k ; source 25, printed as Part XIX, computes the exact Borel and Wadge complexity of well-ordered supports for every countable exponent order), so the question concerns Borel subfamilies, as stated. A companion of Part XVII's question 237.1 and Part XVIII's question 253.6.

298.4 Joint regularity of Lie-group actions

Question 298.4. Does joint coefficient-Borel dependence of a connected finite-dimensional Lie-group action force sufficient coefficientwise differentiability to apply Remark 294.3, without assuming a generator?

The present paper provides the complete finite-dimensional infinitesimal classification and explicit affine integrations. It does not prove this regularity implication for arbitrary nonabelian group actions. Establishing it would turn the infinitesimal conclusion into a broader group-action classification.

[write] (Batch 96.) Answered for ordered automorphisms in rank one. Part XVIII's theorem 248.8, applied to each one-parameter subgroup, gives the regularity: joint coefficient-Borel dependence, indeed Borel dependence on s of the coefficients of $\rho(\exp sX)(t)$ alone,

makes every $s \mapsto \rho(\exp sX)$ an analytic flow with a Borel generator, to which [remark 294.3](#) applies ([theorem 300.3\(i\)](#) [merge]). [Theorem 300.3\(ii\)–\(iv\)](#) then supply the broader classification asked for: the generator map is a homomorphism of Lie algebras, its image has dimension at most two, and the action factors, on the universal cover, through source 27's affine action [theorem 294.4](#) or through one analytic flow. These are results of the write; an independent check made for this report found them correct (with editorial fixes, applied), and they are unrefereed. Open beyond them: unordered automorphisms ([question 298.2](#)), higher rational rank, where scalar rigidity and the dimension bound fail ([example 295.5](#), [question 298.1](#)), and actions without time regularity along one-parameter subgroups, which Part XVIII's Hamel-basis reparametrizations (remark after [corollary 248.9](#)) show to exist.

298.5 Forgetting the ordered generators

Question 298.5. How does classification change when a noncommuting pair is considered up to real linear recombination of its two generators, rather than as an ordered pair?

The ratio b/a then undergoes a fractional-linear change, so the unique scalar coordinate is no longer immediately canonical. The finite-dimensional subalgebra theorem solves the bracket-closed cases of dimension two and three; arbitrary two-generator infinite-dimensional Lie algebras remain a different problem.

298.6 Effective normalizers and certified tails

Question 298.6. For computably presented Γ and exact symbolic coefficient data, which support certificates make $\kappa(q)$, the scalar coordinate, and its inverse effectively computable to a requested valuation cutoff?

Borelness alone gives no terminating numerical equality test for arbitrary real coefficients. A practical interface should require certified leading terms and lower bounds for omitted tails. The equations ([297.1](#)) provide a minimal checker target.

[write] (Batch 96.) Part XVIII's [question 253.4](#) is the same question for single derivations, and Part XVII's [question 237.6](#) its analogue for cutoff moduli. The Hilbert's-tenth programme's review shows that computing the normalizer does not suffice for deciding conjugacy: on an explicit family of coefficient programs over $\mathbb{Z}$ the normalizer is the identity but equality of the invariant A is co-c.e.-complete ([remark 300.4](#)). Certified tails of the kind the commentary asks for, here a supplied support endpoint, remove that obstruction for polynomials.

298.7 Analytic germs and Stokes-sensitive subclasses

Question 298.7. For convergent or summable analytic subclasses of the formal field, when does the scalar coordinate preserve the analytic category, and which additional sectorial or Stokes data are needed for simultaneous analytic conjugacy?

An arbitrary coefficient A in the formal normal pair may encode divergent series. Formal Borel classification therefore cannot simply be relabeled analytic classification. The formal cross-section can instead serve as one stage of a sector-by-sector comparison with explicitly retained gluing data.

298.8 Differential operators and larger named structures

Question 298.8. Which higher-order differential operators, connections, or Lie-algebra representations contain an equivariantly recoverable nonconstant scalar that makes their orbit relation smooth by the same normalization principle?

A concrete starting point is a pair of first-order operators with nonzero zeroth-order terms. One must identify a scalar invariant of the entire coordinate action, rather than choose a coefficient that transforms with an extra Jacobian factor.

299 Conclusion

The central mechanism is simple but strong: in a left-finite rank-one series field, a nonconstant scalar has no nontrivial ordered Borel stabilizer and admits a unique canonical coordinate. Two noncommuting derivations supply such a scalar without an arbitrary choice, through their ratio. This turns an otherwise obstructed conjugacy problem into an explicit Borel cross-section with unique witnesses.

The same coordinate method and the leading-bracket valuation rule yield ramification-sensitive affine and projective Lie normal forms. Global whole-field differentiability excludes the negative-valuation directions, leaving exactly the positive affine nonabelian infinitesimal models. These results extend the earlier single-derivation program in a distinct direction while preserving a clear separation between proved formal statements, classical antecedents, computational checks, and unresolved extensions.

[write] (Batch 96.) “Leaving exactly the positive affine nonabelian infinitesimal models” is [corollary 294.1](#) for Lie algebras of valuation-nondecreasing derivations; that the infinitesimal image of a whole-field action is such a Lie algebra is [theorem 300.3](#) [merge], for every action in the sense of [definition 300.1](#).

300 Where Part XXI meets Parts XVII, XVIII and XX [merge]

This section was written for this report. It makes precise the Lie-group actions of source 27’s Proposition 9.3 ([remark 294.3](#)), proves with Part XVIII’s automatic analyticity in time that the actions of connected Lie groups with Borel time dependence are classified by source 27’s Lie algebras, and records a reviewer’s effective boundary for [theorem 291.2](#). The proofs use only results proved in source 27, in Part XVIII and standard Lie theory. They were checked once independently for this report ([theorem 300.3](#): correct, with editorial fixes that are applied here); they are unrefereed. Throughout, $\mathbb{Z} \subseteq \Gamma \subseteq \mathbb{Q}$, and a *coefficientwise real-analytic flow* is a homomorphism $s \mapsto \Phi_s$ from $(\mathbb{R}, +)$ to G_Γ all of whose coefficient functions $s \mapsto [t^\lambda] \Phi_s(f)$ are real-analytic, with generator D when their derivatives at 0 are the coefficients of $D(f)$ (Part XVIII’s [definition 248.1](#)). By Part XVIII’s [theorem 248.3](#), the generators of such flows are exactly the D_a with $a = 0$ or $v(a) \geq 0$, and the flow with a given generator is unique.

Definition 300.1 (Lie-group actions with Borel generators). Let G be a connected real Lie group with Lie algebra $\mathfrak{g}$. A *coefficient-differentiable action of G with Borel generators* is a group homomorphism $\rho : G \rightarrow G_\Gamma$ such that, for every $X \in \mathfrak{g}$, $s \mapsto \rho(\exp sX)$ is a coefficient-differentiable one-parameter action with Borel generator ([definition 294.2](#)). Its generator is written D_X , and $\{D_X : X \in \mathfrak{g}\}$ is the *infinitesimal image* of ρ .

Lemma 300.2 (Products of analytic flows). *Let $\Phi^1, \dots, \Phi^n$ be coefficientwise real-analytic flows in G_Γ whose time maps all have value-group multiplier 1, let $f \in \mathcal{L}_\mathbb{R}^\Gamma$, and put $F(s) = \Phi_{s_1}^1 \cdots \Phi_{s_n}^n(f)$ for $s \in \mathbb{R}^n$. Then every coefficient $s \mapsto [t^\lambda]F(s)$ is real-analytic on $\mathbb{R}^n$, and $\bigcup_{s \in \mathbb{R}^n} \text{supp } F(s)$ is left-finite.*

Proof. The second assertion follows from the first. A real-analytic function on the connected space $\mathbb{R}^n$ that is not identically zero vanishes on a closed set with empty interior; by Baire's theorem some s_0 lies outside the zero sets of the countably many coefficient functions that are not identically zero, and then $\text{supp } F(s_0)$ is the whole union.

The first assertion is proved by induction on n ; $n = 0$ is trivial. Put $G(s') = \Phi_{s_2}^2 \cdots \Phi_{s_n}^n(f)$ for $s' = (s_2, \dots, s_n)$. By induction its coefficients are real-analytic and its supports lie in one left-finite set S' . Write $y(s_1) = \Phi_{s_1}^1(t) = A(s_1)t(1 + u(s_1))$, with $A(s_1) > 0$ and $v(u(s_1)) > 0$, by the multiplier hypothesis. Here $A(s_1) = [t^1]y(s_1)$ and $[t^e]u(s_1) = [t^{1+e}]y(s_1)/A(s_1)$ are real-analytic, so (by the case $n = 1$ of the second assertion, which is Part XVIII's lemma 248.2) the supports of all $u(s_1)$ lie in a left-finite set $U \subseteq \Gamma_{>0}$, with least element $\mu > 0$ if $U \neq \emptyset$. For $q \in \Gamma$,

$$[t^\lambda]y(s_1)^q = A(s_1)^q \sum_{0 \leq m \leq (\lambda - q)/\mu} \binom{q}{m} [t^{\lambda - q}]u(s_1)^m$$

vanishes for $q > \lambda$ and is otherwise a finite polynomial expression in $A(s_1)^q$ and finitely many coefficients of $u(s_1)$, hence real-analytic in s_1 . By theorem 289.1, $\Phi_{s_1}^1(g) = \sum_q g_q y(s_1)^q$, so

$$[t^\lambda]F(s) = \sum_{q \in S', q \leq \lambda} [t^q]G(s') \cdot [t^\lambda]y(s_1)^q,$$

a finite sum of products of a real-analytic function of s' and one of s_1 , hence real-analytic on $\mathbb{R}^n$. $\square$

Theorem 300.3 (Lie-group actions with Borel time [merge]). *Let $\mathbb{Z} \subseteq \Gamma \subseteq \mathbb{Q}$, let G be a connected real Lie group with Lie algebra $\mathfrak{g}$ and let $\rho : G \rightarrow G_\Gamma$ be a group homomorphism such that, for every $X \in \mathfrak{g}$, every coefficient of $\rho(\exp sX)(t)$ is a Borel function of $s \in \mathbb{R}$. (This holds when $(g, f) \mapsto \rho(g)f$ is coefficient-Borel on $G \times \mathcal{L}_\mathbb{R}^\Gamma$.)*

- (i) *For every $X \in \mathfrak{g}$ there is a unique $a_X \in \mathcal{L}_\mathbb{R}^\Gamma$, with $a_X = 0$ or $v(a_X) \geq 0$, such that $s \mapsto \rho(\exp sX)$ is the coefficientwise real-analytic flow generated by $D_X = D_{a_X}$. So ρ is a coefficient-differentiable action with Borel generators (definition 300.1), its infinitesimal image consists of valuation-nondecreasing derivations, and every $\rho(g)$ has value-group multiplier 1. Conversely, every coefficient-differentiable action with Borel generators satisfies the hypothesis of the theorem, by hypothesis (ii) of Part XVIII's theorem 248.8; so (i)–(iv) classify exactly the actions of definition 300.1.*
- (ii) *The generator map $\psi : X \mapsto D_X$ is a homomorphism of real Lie algebras. Its image $\mathfrak{h}$ has dimension at most two: either $\mathfrak{h} = \mathbb{R}D_a$ with $a = 0$ or $v(a) \geq 0$, or $\mathfrak{h}$ is conjugate in G_Γ to $\mathfrak{a}_k$ for some $k \in \Gamma_{>0}$. In particular $\mathfrak{h}$ contains no derivation that lowers valuation and is never conjugate to $\mathfrak{a}_p$ with $p < 0$ or to $\mathfrak{s}_p$.*
- (iii) *If $X_1, \dots, X_n$ is a basis of $\mathfrak{g}$ and ρ' is another homomorphism satisfying the hypothesis of the theorem, with the same generators D_{X_i} , then $\rho' = \rho$.*
- (iv) *Let $\pi : \widetilde{G} \rightarrow G$ be the universal covering. If $\mathfrak{h} = \mathbb{R}D_a$ with $D_a \neq 0$, then $\rho \circ \pi = \Phi^a \circ \Lambda$, where Φ^a is the analytic flow of D_a and $\Lambda : \widetilde{G} \rightarrow (\mathbb{R}, +)$ is the homomorphism with $d\Lambda(X)D_a = D_X$. If $\mathfrak{h}$ is nonabelian, there are $\theta \in G_\Gamma$, $k \in \Gamma_{>0}$ and a homomorphism of Lie groups $\Psi : \widetilde{G} \rightarrow \text{Aff}^+(\mathbb{R})$*

with $\rho(\pi(x)) = \theta \Phi_{\Psi(x)} \theta^{-1}$ for all x , where $\Phi_{(a,b)}$ is the action of [theorem 294.4](#) for $p = k$. If $\mathfrak{h} = 0$, ρ is trivial; in particular every such action of a connected semisimple Lie group is trivial.

Proof. (i) For fixed X , $s \mapsto \rho(\exp sX)$ is a homomorphism from $(\mathbb{R}, +)$ to G_Γ satisfying hypothesis (i) of Part XVIII's [theorem 248.8](#), which gives the unique a_X and shows that the flow is the analytic flow of D_{a_X} from [theorem 248.3](#), with entire coefficient functions; their derivatives at 0 are the coefficients of $D_{a_X}f$, and D_{a_X} is coefficient-Borel ([theorem 288.3](#)), so [definition 294.2](#) holds. Since $v(a_X) \geq 0$ or $a_X = 0$, $v(D_X f) \geq v(f)$. The value-group multiplier of every $\rho(\exp sX)$ is 1 (the first step of the proof of [theorem 248.8](#), or of [remark 294.3](#)); as G is connected, every element is a finite product of such elements, and multipliers multiply.

(ii) *Linearity.* Fix a basis $X_1, \dots, X_n$ and put $\mu(s) = \exp(s_1 X_1) \cdots \exp(s_n X_n)$, so that $\rho(\mu(s))f = F(s)$ with $\Phi_{s_i}^i = \rho(\exp s_i X_i)$ in [lemma 300.2](#), whose hypotheses hold by (i). The map μ is real-analytic with differential $e_i \mapsto X_i$ at 0, hence a local analytic diffeomorphism there. For $X = \sum_i c_i X_i$ there are $\varepsilon > 0$ and a real-analytic $\sigma : (-\varepsilon, \varepsilon) \rightarrow \mathbb{R}^n$ with $\sigma(0) = 0$, $\sigma'(0) = c$ and $\exp sX = \mu(\sigma(s))$. Then $[t^\lambda] \rho(\exp sX)f = [t^\lambda] F(\sigma(s))$, and the chain rule at $s = 0$ gives $D_X f = \sum_i c_i \partial_i F(0)$, where $\partial_i F(0) = \frac{d}{ds} \rho(\exp sX_i)f|_{s=0} = D_{X_i}f$ coefficientwise. So $D_X = \sum_i c_i D_{X_i}$.

Brackets. Fix $X, Y \in \mathfrak{g}$, f and λ , write $g_s = \rho(\exp sX)$ and $y_s = g_s(t)$, and let $\text{Ad}(\exp sX)Y = \sum_i c_i(s)X_i$, with c_i real-analytic and $\sum_i c'_i(0)X_i = [X, Y]$. Since $g_s \rho(\exp uY) g_s^{-1} = \rho(\exp(u \text{Ad}(\exp sX)Y))$, the two sides have the same generator. On the right it is $\sum_i c_i(s)D_{X_i}$ by linearity. On the left, $w_u = \rho(\exp uY)(g_s^{-1}f)$ has real-analytic coefficients with supports in one left-finite set S ([lemma 300.2](#) with $n = 1$), so $[t^\lambda]g_s(w_u) = \sum_{q \in S, q \leq \lambda} [t^q]w_u [t^\lambda]y_s^q$ is a finite sum and its u -derivative at 0 is $[t^\lambda]g_s D_Y g_s^{-1}f$. Hence

$$g_s D_Y g_s^{-1}f = \sum_i c_i(s) D_{X_i}f \quad (s \in \mathbb{R}).$$

Differentiate at $s = 0$ coefficientwise. The right side gives $\sum_i c'_i(0)D_{X_i}f = D_{[X,Y]}f$. On the left, $g_s^{-1}f = \rho(\exp(-sX))f$ has real-analytic coefficients with supports in one left-finite set S_1 , and $D_Y = D_{a_Y}$ acts on such a family by finite sums, $[t^\nu]D_Y w = \sum_{q \in S_1, q \leq \nu - v(a_Y)} q w_q [t^{\nu-q}]a_Y$; so $w_s = D_Y g_s^{-1}f$ has real-analytic coefficients, supported in the left-finite set $S_1 + \text{supp } a_Y$, and $\frac{d}{ds} w_s|_0 = -D_Y D_X f$. Now $[t^\lambda]g_s(w_s) = \sum_{q \in S_1 + \text{supp } a_Y, q \leq \lambda} [t^q]w_s [t^\lambda]y_s^q$, and the product rule at $s = 0$, where $y_0 = t$, gives

$$[t^\lambda] \left(-D_Y D_X f \right) + \frac{d}{ds} [t^\lambda] g_s(w_0) \Big|_{s=0} = [t^\lambda] (-D_Y D_X f + D_X D_Y f).$$

So $[D_X, D_Y] = D_{[X,Y]}$, and ψ is a Lie algebra homomorphism.

Its image $\mathfrak{h}$ is a finite-dimensional Lie algebra of Borel derivations, valuation-nondecreasing by (i). By [theorem 293.2](#) an abelian $\mathfrak{h}$ has dimension at most one, and by [corollary 294.1](#) a nonabelian one has dimension two and is conjugate to $\mathfrak{a}_p$ with $p > 0$, which by [theorem 293.3](#) is $\mathfrak{a}_k$ with $k = \kappa(p) > 0$. The last assertion of (ii) is the argument at the end of [remark 294.3](#).

(iii) The map μ of (ii) is a local diffeomorphism at 0, so the subgroups $\exp(\mathbb{R}X_i)$ generate a neighbourhood of the identity and hence the connected group G . On each of them ρ' and ρ are coefficientwise real-analytic flows with the same generator (by (i) for both), so they agree by the uniqueness in Part XVIII's [theorem 248.3](#). Hence $\rho' = \rho$.

(iv) Apply (iii) to $\tilde{G}$, whose Lie algebra is $\mathfrak{g}$. If $\mathfrak{h} = \mathbb{R}D_a$ with $D_a \neq 0$, then $\psi = \lambda D_a$ for a linear form λ vanishing on $[\mathfrak{g}, \mathfrak{g}]$, which integrates to a homomorphism $\Lambda : \tilde{G} \rightarrow \mathbb{R}$ with $d\Lambda = \lambda$ because $\tilde{G}$ is simply connected. For each X , $s \mapsto \Phi_{s\lambda(X)}^a$ is the analytic flow generated by $\lambda(X)D_a = D_X$, so $\Phi^a \circ \Lambda$ and $\rho \circ \pi$ have the same generators and coincide by (iii).

If $\mathfrak{h}$ is nonabelian, choose $\theta \in G_\Gamma$ with $\theta^{-1}\mathfrak{h}\theta = \mathfrak{a}_k$ (by (ii)). In [theorem 294.4](#) the derivatives of $s \mapsto \Phi_{(e^s, 0)}$ and $b \mapsto \Phi_{(1, b)}$ at the identity are E/k and $t^k E/k$; these are the images of the matrices $\alpha = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$ and $\beta = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ of the Lie algebra $\mathfrak{aff}(1)$ of $\text{Aff}^+(\mathbb{R})$, and since every coefficient of $\Phi_{(a, b)}f$ is real-analytic in (a, b) , the chain rule makes $d\Phi : \mathfrak{aff}(1) \rightarrow \mathfrak{a}_k$, $Z \mapsto \frac{d}{ds}\Phi_{\exp sZ}|_{s=0}$, linear; it is bijective, and a Lie algebra isomorphism because $[\alpha, \beta] = \beta$ and $[E/k, t^k E/k] = t^k E/k$. So $\chi = (d\Phi)^{-1} \circ \theta^{-1}\psi\theta : \mathfrak{g} \rightarrow \mathfrak{aff}(1)$ is a homomorphism of Lie algebras, and it integrates to a homomorphism of Lie groups $\Psi : \tilde{G} \rightarrow \text{Aff}^+(\mathbb{R})$. For each X , $s \mapsto \theta\Phi_{\Psi(\exp sX)}\theta^{-1} = \theta\Phi_{\exp s\chi(X)}\theta^{-1}$ is a homomorphism into G_Γ with real-analytic coefficients (conjugation by the fixed θ keeps them real-analytic, as in the bracket step), and its generator is $\theta d\Phi(\chi(X))\theta^{-1} = D_X$. By (iii) it agrees with $\rho \circ \pi$.

If $\mathfrak{h} = 0$, every one-parameter subgroup acts trivially and ρ is trivial by (iii). If $\mathfrak{g}$ is semisimple, $\psi(\mathfrak{g})$ is a quotient of a semisimple Lie algebra, hence semisimple or zero, and of dimension at most two, hence solvable; so it is zero. $\square$

For $G = \mathbb{R}^n$ the abelian case of (iv) is Part XVIII's [corollary 248.9](#), proved there under joint Borel dependence of $\rho(x)(t)$ on $x \in \mathbb{R}^n$, which implies the line-wise hypothesis used here; so for $G = \mathbb{R}^n$ part (iv) is the stronger statement. For a nonabelian G , (iv) says that, up to the universal covering, conjugation and a homomorphism to $\text{Aff}^+(\mathbb{R})$, source 27's action is the only one: this answers [question 298.4](#) and the group level of Part XVIII's [question 253.3](#) for ordered automorphisms in rank one and under this time regularity, which by the converse in (i) is the same as being an action in the sense of [definition 300.1](#). The infinitesimal image of such an action is therefore a Lie algebra, which settles the gap noted after source 27's Proposition 9.3 ([remark 294.3](#)) under source 27's own hypotheses; without the time regularity, Part XVIII's Hamel-basis reparametrizations give homomorphisms $\mathbb{R} \rightarrow G_\Gamma$ whose generator does not exist, and nothing is claimed for them.

Remark 300.4 (A reviewer's effective boundary for [theorem 291.2](#) [write]). The Hilbert's-tenth programme reviewed the archive in full on 4 October 2026, in `Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/review_two_derivations_78528873b.md` (commit 556836e9b). It requests no correction of source 27, and adds a deduction it attributes to its own reviewers, not to source 27. Take $\Gamma = \mathbb{Z}$ and an effective enumeration of machines, and let $A_e = 1 + t^n$ if machine e first halts at step $n \geq 1$, and $A_e = 1$ if it never halts. Then e gives a total program for the coefficients of A_e , though not its degree. The pair $P_e = (D_{A_e}, D_{tA_e})$ is noncommuting, its bracket coefficient being $A_e E(tA_e) - tA_e E(A_e) = tA_e^2$; its ratio is t , whose canonical scalar over $\mathbb{Z}$ is $H = t$ with $y_t = t$, so P_e is already in normal form with invariant (t, A_e) . By [theorem 291.2\(ii\)](#), P_e is conjugate to (E, tE) exactly when $A_e = 1$, that is, when machine e never halts. So on this uniform family conjugacy to a fixed pair is co-c.e.-complete, although the normalizer is computable (it is the identity); in particular no uniform procedure turns these program indices into existential Diophantine conditions whose solvability is equivalent to conjugacy. The obstruction is exact equality of an invariant given by a coefficient program; for polynomials given as finite coefficient lists equality is decidable. This bears on [question 298.6](#), Part XVIII's [question 253.4](#) and Part XVII's [question 237.6](#); it is not a defect of the Borel theorem. The review's direct rigidity argument is the case $H = t$ of [theorem 290.2\(iii\)](#).

A Worked examples and an audit of the hypotheses (source 05)

A.1 A negative constant term is compatible with positivity

In $M_{\mathbb{R}}$, take

$$x = (\frac{3}{2}, -17), \quad y = (\frac{1}{4}, 6).$$

Both are positive, and

$$x + y = (\frac{7}{4}, -11).$$

Division of x by 5 gives

$$q_5(x) = (\frac{3}{10}, -4), \quad r_5(x) = 3,$$

because $-17 = 5(-4) + 3$. The quotient is positive because its real coordinate is positive. A definition that incorrectly demanded a nonnegative integer coordinate for every element would break the predecessor and division axioms.

A.2 A nonstandard finite-candidate calculation

Let $L = (\frac{1}{2}, -5)$, $U = (\frac{2}{3}, 0)$, and require

$$y \equiv (\frac{7}{13}, 2) \pmod{3G}.$$

The real coefficient of a congruence parameter is irrelevant because $\mathbb{R}$ is divisible. Among $L, L+1, L+2$, the candidate $L+1$ has integer coordinate $-4 \equiv 2 \pmod{3}$, so it is a witness. Every candidate lies below U , since its leading coefficient is $1/2 < 2/3$. No iteration a nonstandard number of times is used: the finite search is justified by full invariance on $3G$ -cosets.

A.3 A Baire convergence that order cannot detect continuously

Let e_j be the rational unit vector at coordinate j . Then

$$(e_j, 0) \rightarrow (0, 0), \quad (e_j, 0) > n\mathbf{1} \quad \text{for every fixed standard } n.$$

The initial coefficients eventually agree with zero to every prescribed finite length. In omnific notation this sequence is

$$\omega, \quad \omega^{1/2}, \quad \omega^{1/3}, \quad \dots,$$

which converges to zero in the *specified coefficient topology*, not in the arithmetic order topology. This is a concrete illustration, not a claim about an unspecified topology on surreal numbers.

A.4 Hypothesis checklist

Potential pitfall	Resolution in the proofs
Only the division axioms are checked	Theorem 4.9 , corollary 4.11 , and proposition 4.12 establish the complete theory and every additive induction instance.
An open ray is given an incomplete metric	Section 5 provides a compatible logarithmic metric on each open ray and a complete metric on the whole sum.

Potential pitfall	Resolution in the proofs
The intended topology is silently replaced by the order topology	Both topologies are explicitly distinguished; the arithmetic order topology is discrete.
Standard-shift invariance is used with a nonstandard quotient	Example 4.4 exhibits the failure; lemma 4.7 uses invariance on mG -cosets instead.
The all-zero rational branch is lost	The nonnegative-cone tree in (6.2) explicitly includes it.
Computing an output tag secretly requires deciding equality of streams	For a negative output integer coordinate, a negative input supplies a finite bound on the first positive coefficient.
A type-theoretic “computable model” is confused with a countable one	The presentation is Type-2 on all Baire sequences; not every element is computable, and order has only a limit decision procedure.
The omnific image is assumed closed under multiplication	It is only an additive fragment. Theorem 10.4 rules out even an alternative semiring multiplication on the same carrier.
A proper class is treated as a Polish space	Only explicit set-sized images and a sufficiently large set-sized birthday cutoff are used.
Existing repository proofs are credited with new theorems	The formalization roadmap distinguishes the inspected integer theorems from the additional nonstandard and topological obligations.
Numerical testing is mistaken for a proof	The exact test scope is documented in section 13 ; no formal verification claim is made for this article.
An affirmative proof is mistaken for certified historical priority	The source question and version are identified; independent priority and refereeing remain separate from the mathematical proof.

B Reproduction and the scope of the executable checks (source 04)

The package requires a standard LaTeX installation for the article and Python 3.10 or later for the companion code. The Python program uses only the standard library and exact fractions.Fraction arithmetic. From the package directory, run

```
python3 verification.py --output verification_results.json
pdflatex -interaction=nonstopmode -halt-on-error polish_presburger.tex
pdflatex -interaction=nonstopmode -halt-on-error polish_presburger.tex
```

A Makefile supplies equivalent targets. The delivered run used seed 20261003 and passed 75,079 assertions. Its breakdown is:

Check category	Assertions
Rational, integer, and pairing codes	6,468
Finite group and order identities	9,955
Exact quotient and remainder	2,151
Boundary witness transfer	21,037
Finite candidate comparison	1,200
External periodicity counterexample	2,321
Positive-pivot matrix tests	18,000
Monus integer-coordinate obstruction	400
Baire-code round trips	97
Baire quotient and remainder prefixes	9,700
Baire addition prefixes	3,750
Total	75,079

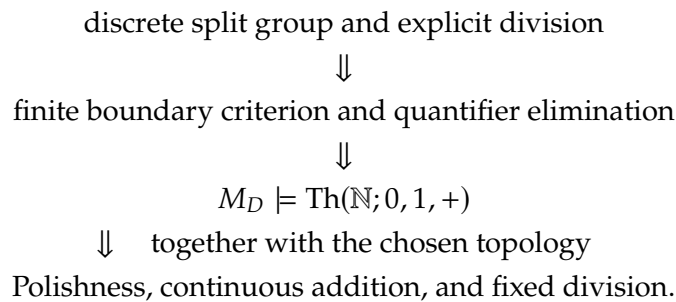
The finite group tests use rational coefficient vectors and signed integer coordinates. The boundary tests examine witnesses from a specified finite pool and compare them with the exact candidate set. They do not exhaust an infinite group. The matrix tests use finite initial pieces of matrices satisfying the positive-pivot conditions. The stream tests check finite prefixes of deterministic infinite oracles, including long zero runs. The monus tests verify exact coordinate obstructions along finite portions of the sequences used in the proof, not topological convergence by numerical approximation.

The model-theoretic, infinite-matrix, and topological statements rest on the proofs in the manuscript. Tests are useful for detecting sign errors, incorrect negative quotients, wrong coordinate offsets, and mismatched coding branches, but a successful run is not a substitute for those proofs.

[write] The commands above use source 04's delivery names. In this report its program is `code/04-polish-presburger-verification.py`, its recorded run data `data/04-polish-presburger-verification_results.json`, its Makefile `code/04-polish-presburger-Makefile` (which still names `polish_presburger.tex`, `verification.py` and `verification_results.json`), and its manuscript is merged into `this_article.tex`. The program's default output is `verification_results.json` in the working directory, so it is rerun with an explicit output path outside the report (README). The intake reran it on a copy with Python 3.14.4: 75,079 assertions passed, and the record was identical up to line endings.

C A compact proof-dependency map (source 04)

The answer can be read independently of the optional extensions:



For the real example, [section 5.4](#) completes the argument at that point. For literal Baire space, add only the two chart codings in [section 6.3](#). The computation of the monus locus,

the embedding classification, the growth-order obstruction, and the surreal realization are subsequent results; none is a hidden prerequisite for the affirmative construction.

D Source audit, contribution ledger, and reproducibility (source 03)

D.1 Repository material inspected

The repository branch was resolved to the commit printed in Section 16. The following guides were inspected in selected ranges at that revision:

```
Algebra/SurrealNumbers/README.md
Algebra/SurrealNumbers/docs/README.md
Algebra/SurrealNumbers/docs/
foundations-and-computation/omnific-notations/README.md
```

The first guide supplies the context of the omnific ring, its least positive element, and the floor boundary correction. The notations guide reports the existing Kleene–Brouwer support-validity classification and carefully states the limited scope of its finite checks. The present article does not certify all statements in either guide.

The external mathematical inputs are Newton–Puiseux, the classical Hahn-field construction, Hausdorff’s characterization of countable scattered orders, standard descriptive-set-theoretic facts, and Glazer’s Corollary 1. Analytic boundedness and the needed Kleene–Brouwer reduction are proved in the text. Shepherdson’s theorem is contextualized and its needed direction is also proved directly.

D.2 Contribution ledger

Statement	Status in this report
General coanalytic support hardness	Prior classical mechanism; already recorded for omnific notations in ProveIt. Reproved to make the reductions explicit.
Scattered/support dichotomy	Proved from Hausdorff theory and the fixed tree reduction; global priority not asserted.
Observable full-Hahn criterion	Proved, including Borel field operations on the scattered side and exclusion of even repeated observable names on the nonscattered side.
Exact $\mathbb{Z}^d$ Borel levels	Proved with explicit continuous reductions and a separate sharp support-order bound.
Uniform observable support bound	Classical analytic boundedness specialized to coefficient families; a proof is supplied.
Fixed-interval binary localization	Explicit application of the support reduction, not advertised as a new foundational lower-bound method.
Principal-part integer part and floor	Shepherdson-type classical model; negative-infinitesimal correction also already in ProveIt.

Statement	Status in this report
Exact cancellation loci	Proved for the stated Polish finite-normal-form topology, including closed-piece continuity and zero-factor cases.
No compatible continuous addition	Direct application of Glazer after all model and Borel hypotheses have been verified.
Formal verification	No Lean proof delivered. Python checks only finite exact-rational identities, order tests, floor cases, and finite embedding data.

D.3 Finite verification artifact

The archive contains `verify.py`, a Python 3.10+ standard-library-only program, and the actual result `verification.json`. The recorded run used Python 3.13.5, deterministic seed 2601003, and passed **84,388 assertions**. These include 6,000 ring-law checks, 2,000 closure checks, 2,000 positive-cone checks, 999 discreteness checks, 7,500 floor-interval checks, 63 negative-infinitesimal boundary checks, 25 cancellation checks, 65,792 rational Kleene–Brouwer embedding checks on 256 nodes, and 9 coherence and branch-approximation checks.

The finite embedding is an initial segment of one fixed midpoint construction: nodes are enumerated by the weight $\sum_i (s_i + 1)$, and each new node is inserted between its previously assigned neighbors. This checks the direction of the order reversal and the rational band. It does not test the well-foundedness of arbitrary infinite trees.

To reproduce:

```
python verify.py --output verification.json
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

The bibliography is internal, so no BibTeX run or network access is required. The finite checks are useful for catching sign, support, and boundary mistakes; they neither prove descriptive completeness nor replace the mathematical proofs or an independent review.

[write] In this report source 03’s program is `code/03-borel-presentations-verify.py`, its recorded run data `data/03-borel-presentations-verification.json`, and its provenance record `data/03-borel-presentations-provenance.json`; the record still describes 03’s own 23-page PDF, which is not shipped. Its manuscript is merged into `article.tex`. The intake reran the program on a copy with Python 3.14.4: 84,388 assertions passed, and the record differed from the delivered one only in the interpreter version and line endings.

E Provenance of this report

Sources of Parts I and II. Three manuscripts of batch 86 of the repository’s intake, all dated October 3, 2026, all AI-assisted (03: “prepared with ChatGPT”), arrived together in commit 31fdc6571 and were placed in commit 3d2177df4. Each inspected the repository at its own pin; the pins are 28 to 35 commits before the placement.

Source	Archive	Pin	Contributes
05	polish-presburger-glazer	a11efab09	base of Part I: sections 2 to 13 , its questions, section 15 and appendix A
04	Polish_Presburger_Glazer_ProveIt	3ad5f878c	every passage marked [04] in Part I ; sections 2.3, 4.3, 4.5, 5.2, 5.4, 6.3, 7.3, 8.1, 10.1, 11.2, 11.3 and 13.4 , its questions, appendices B and C
03	glazer-proveit-research	c5b4881fa	Part II except the notes and section 26 ; appendix D

Written for this report: [section 1](#) (except the quoted status statements), [section 13.5](#), the merged questions of [section 14.1](#), [section 26](#), the notes marked [write] or [merge], and these appendices.

Where the merge of Parts I and II chose.

- *Base.* Source 05, for the weaker hypotheses of its Polish criterion and Δ_2^0 theorem ([section 1.2](#)).
- *Printed once.* The main theorem (05's [theorem 3.1](#), with 04's statement as [remark 3.2](#)); the division algorithm ([proposition 4.2](#) = 04's Lemma 2.2); full Presburger arithmetic of the cone ([corollary 4.11](#) = 04's Corollary 3.5); the predecessor and truncated-subtraction loci ([proposition 8.2](#) and [theorem 8.3](#) = 04's Proposition 7.4 and Theorem 7.3); the omnific realization ([theorem 9.1](#) = 04's Proposition 10.1, same proof); the no-semiring theorem and its corollary ([theorem 10.4](#) and [corollary 10.5](#) = 04's Theorem 9.3 and Corollary 9.4, whose model-specific proof coincides with 05's); the real endomorphisms ([theorem 11.1](#) = 04's Theorem 8.2).
- *Second proofs from 04.* [Proposition 4.2](#) (uniqueness by coordinates), [lemma 4.7](#) (as [theorem 4.8](#)), [theorem 4.9](#) (as [theorem 4.10](#)), [theorem 6.1](#) (as [theorems 6.3](#) and [6.4](#)), [theorem 7.1](#) (as [theorem 7.2](#)), [proposition 8.2](#), [theorem 8.3](#) (with the nowhere-density clause), [theorem 10.4](#), [theorem 11.1](#).
- *Both printed.* The order obstructions [theorem 7.4](#) (05) and [theorem 7.5](#) (04), whose hypotheses are incomparable ([remark 7.7](#)); the counterexamples [example 4.4](#) (05) and [proposition 4.6](#) (04); the Polish criterion [theorem 5.2](#) (05) and the transfer theorem [theorem 5.4](#) (04), which alone gives continuous division for general D .
- *Questions.* Four questions asked by both 05 and 04 are merged ([section 14.1](#)); 05's question on canonical surreal cutoffs and topologies and 04's on coefficient topologies inside surreal arithmetic overlap only partly and are printed separately. 03's eight questions are printed as delivered. Part I has seventeen questions, Part II eight.
- *Not printed separately.* 04's title page, abstract (its non-claims are in [section 1.5](#)), reading routes, Definition 2.1 (05's setup), the opening paragraph of its Section 9 (Glazer's Corollary 1, in [section 1.1](#)), and the statements and proofs listed under "printed once"; 05's and 03's title pages and abstracts (their status statements are in [section 1.5](#)). The merged abstract replaces them.
- *Renamed symbols.* Listed in [section 1.3](#); no normalization was changed.
- *Added.* [Proposition 26.1](#) and the discussion in [section 26](#); the remark [remark 7.7](#); Glazer's speculation ([section 1.1](#)), which no source mentions; the notes citing the collection's

labels in [Part II](#).

Intake checks for Parts I and II. The intake ran the three delivered suites on copies (Python 3.14.4, standard library only): 03 passed 84,388 assertions in 11 groups, 04 passed 75,079 assertions in 13 categories, 05 passed 17,930 checks in 18 families. Their records agree with the shipped ones up to interpreter strings and line endings. The three delivered checksum manifests verified and were not shipped. It also checked by hand that $M_{\mathbb{R}}$ is a countable clopen sum of rays and a model of $\text{Th}(\mathbb{N}; 0, 1, +)$, the Polish clause of [theorem 11.2](#), [theorem 7.5](#), and the worked examples of [appendix A](#); no false statement was found. For this report the following references were checked against their publishers’ records on October 3, 2026: Glazer’s arXiv record (one version, no journal reference); Tserunyan’s notes (the file at the cited address is dated November 26, 2025); Paran and Vo (*Israel J. Math.* 273 (2026), no. 2, 979–1000, online December 11, 2025); Enayat, Hamkins and Wcisło (*Fund. Math.* 256 (2022), 171–193); Jeřábek (*Math. Logic Quart.* 65 (2019), no. 1, 108–115); Haase (*ACM SIGLOG News* 5(3) (2018), 67–82); Alvir and Rossegger (published in *J. Symb. Log.* 85 (2020), 1079–1101, added here). The other classical references were not re-checked.

Correction. [Section 26](#) corrects the placement record’s description of how 03’s topology restricts to the real model.

Sources of Parts III–V. Four further manuscripts of batch 86 arrived together in commit fb8414869 and were placed in commit 0bd0e5527, with manuscript 10 of the same arrival, which went to Algebra/SurrealNumbers/docs/surreal/discrete-initial-subgroups-and-omnific-normalization. They are numbered by archive time. Their pins are 33 to 43 commits before the placement; the repository files they inspected are unchanged between the pins and the placement. Sources 08 and 09 are AI-assisted by their own statement (“prepared with ChatGPT”, “prepared with AI assistance”); 06 and 07 describe themselves as research reports prepared for Vladimir Reshetnikov.

Source	Archive	Pin	Contributes
07	polish_presburger_research	0f95145cb	base of Part III : every unmarked passage, its questions, appendices G and H
06	polish_presburger_article	fa2f3e419	every passage marked [06] in Part III , among them sections 38 and 39 ; its twelve projects; appendices I and J
08	glazer_proveit_local_compactness	883e0b3b2	Part IV except the notes; appendix K ; staged code, data, sources and proof audit (prefix 08-local-compactness-)
09	Polish_Hahn_Puisseux_Levi_Civita_Research	4af956fe0	Part V except the notes; appendices L and M ; staged code, data and sources (prefix 09-hahn-polishability-)

Written for this report in Parts III–V: the introductions of the three parts, [section 42](#) with [proposition 42.1](#), the notes marked [write] or [merge], the additions to [section 1](#) (the paragraph on Parts III–V and Glazer’s speculation, the second block of the notation table, the status boxes), the note after [question 14.4](#), and the additions to these appendices.

Where the merge of [Part III](#) chose.

- *Base.* Source 07, for the weaker hypotheses of its realization theorem ([Part III](#)).
- *Printed again with a flag.* Results of Part I proved again by 06 or 07 are printed with their proofs, and each title names the statement of Part I it repeats (listed in the introduction of [Part III](#)). Source 06's [theorem 35.9](#) repeats 07's [theorems 35.1](#) and [35.4](#) with a different homeomorphism argument.
- *Both printed.* The two continuum theorems ([theorems 35.5](#) and [35.8](#)), whose constructions and refinements differ.
- *Placement.* Source 06's Sections 1 to 6 and 9 are printed as subsections of source 07's Sections 1, 2, 3, 6, 6, 8 and 9 respectively, its own subsections one level lower; its Sections 7 and 8 are sections of their own. Its twelve projects follow source 07's question with the same subject, the other five in [section 43.5](#).
- *Renamed.* Source 06's M_s, G_s, H_s ($s \in 2^{\mathbb{N}}$) are M'_s, G'_s, H'_s ; macros of all four sources are mapped to this report's ([section 1.3](#)); no normalization changed.
- *Labels added* to unlabelled statements so that notes can cite them: `pma:lc:q:remove1c`, `pma:lc:q:completion`, `pma:ser:q:cones`.
- *Not printed separately.* The four title pages and abstracts (their status statements are in [section 1.5](#); the merged abstract replaces them), source 07's table-of-contents material, and the section headings of 06 that were replaced by subsection headings. Source 07's figure, source 06's map of results and source 09's reading routes are printed.

Intake checks for Parts III–V. The suites of 08 and 09 were rerun on copies (Python 3.14.4, standard library only, about one second each): 08 passed 24,470 cases in four families, 09 passed 20,116 assertions in 29 families. Their records are JSON-equal to the shipped ones and differ only in line endings. Source 08's checksum manifest verified at placement (8 of 8) and was not shipped; 06, 07 and 09 ship none, and 06 and 07 ship no code. For this report the proofs of source 06's [theorem 39.1](#) and [theorems 38.1](#) and [38.2](#) and the oscillation table after [theorem 36.1](#) were checked by hand, with no error found, and the following records were checked on October 3, 2026: Block's arXiv record (v3 of June 16, 2026, the version cited by 06); Chen's arXiv record (one version, 2021, no journal reference). The arXiv record of Ehrlich and Kaplan lists no journal reference; the journal reference that source 07 gives was not re-checked. The other references of 06–09 were not re-checked.

Reviews elsewhere in the repository. The Hilbert's-tenth programme triaged the arrival fb8414869 for its own search in Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/recent_incoming_substrate_triage_20261003.md (commit 39e03adc5), after a README-only note on manuscripts 03–05 (review_recent_incoming_glazer_scope.md in the same directory, commit 0f3b760ae). Both are routing decisions, not proof audits; the later one notes, as [section 42](#) does, that source 06's cone topology is not automatically a locally compact signed-group topology.

Stale statements and addresses. Source 07's statement that the repository's text files never mention Glazer or Tennenbaum was true at its pin and is corrected after it ([appendix H](#)). Source 09 cites two repository files by unversioned addresses; the note in [appendix M](#) gives the pinned ones. Sources 08 and 09 cite sources 05 and 03 as “user-library” PDFs; in this report they are Parts I and II.

Sources of Parts VI–IX. Five manuscripts of batch 87 arrived in commit `fa0a0576e`, with a sixth (manuscript 02, written into `SetTheory/Cardinals/docs/reports/ordinals-and-order-types/measurable-box-games`), and were placed in commit `d151b39ca`. They are numbered by archive time in the batch and by file prefix in this report. Their pins are 44 to 51 commits before the placement. The repository files they inspected are unchanged between their pins and the placement, except that the Presburger and surreal READMEs gained the catalogue’s pointers to this report in `2b7b388ba`.

Source	Archive (batch 87 no.)	Pin	Contributes
12	Baire_Category_Arithmetic_Rigidity (03)	0f084afa9	base of Part VI : every unmarked passage, its questions; appendices N and O ; staged code, data and audits (prefix <code>12-baire-category-</code>)
10	Glazer_ProveIt_Baire_Rigidity (01)	6fef5383b	every passage marked [10] in Part VI , among them sections 79.1 , 79.3 and 79.4 and section 85 ; its twelve questions; appendices P and Q ; staged code, data and audit (prefix <code>10-baire-order-</code>)
13	polish_presburger_frontiers (04)	eab47e330	Part VII except the notes; appendices R and S ; staged code and data (prefix <code>13-beyond-split-</code>)
14	glazer_atr0_research (05)	6fef5383b	Part VIII except the notes and the status box; appendices T and U ; staged code, data and proof audit (prefix <code>14-atr0-tennenbaum-</code>)
15	polish_arithmetic_package (06)	0f084afa9	Part IX except the notes; appendix V ; staged build script and proof audit (prefix <code>15-polish-arithmetic-</code>)

Written for this report in Parts VI–IX: the introductions of the four parts, [section 90](#) with [proposition 90.1](#), the notes marked [write] or [merge] (in Parts VI–IX and at the questions and statements of Parts I and III–V that they answer or make stale), the status box of [Part VIII](#), the additions to [section 1](#) (the paragraph on Parts VI–IX and Question 1, the sources table, the third block of the notation table, the status boxes), and the additions to these appendices.

Where the merge of Part VI chose.

- *Base.* Source 12, for the weaker hypotheses of its cardinal, semiring and non-Polishability theorems and its second, Haar route ([Part VI](#)).
- *Printed once.* The principal theorem ([corollary 78.4](#) = source 10’s Theorem 5.1); the bounded-neighbourhood lemma ([lemma 77.2](#) = source 10’s Lemma 4.2); the open principal hull ([proposition 79.1](#) = source 10’s Proposition 4.3); the absence of an ordered-ring expansion ([corollary 81.1](#) = source 10’s Corollary 5.3); the three topologies on $\mathbb{Z} + t\mathbb{R}[t]$ (= source 10’s Examples 9.1, 9.3, 9.4); the completion ([proposition 86.2](#) = source 10’s Proposition 9.2). Source 10’s labels are second labels of these statements; its two displays `eq:localbound` and `eq:polynomialring` are not printed, and references to them point to source 12’s displays.
- *Second proofs from source 10.* [Lemma 77.2](#) (without Pettis’s lemma), [corollary 78.4](#) (through the principal-hull quotient embedding), [proposition 86.2](#).

- *Both printed.* The Polish corollaries ([corollary 78.5](#) and source 10's exact criterion [corollary 78.6](#)); the cardinal theorems ([theorems 78.3](#) and [79.13](#)); the spectrum theorems ([theorems 80.1](#) and [80.4](#)); the signed Presburger corollaries ([corollaries 81.3](#) and [81.4](#)); the omnific corollaries ([corollaries 86.3](#) and [86.4](#)).
- *Placement.* Source 10's Sections 1, 2, 3, 6, 7, 8, 10 and 12, and the last two subsections of its Section 13, are printed as subsections of source 12's Sections 1, 2, 5, 5, 5, 7, 6, 7, 11 and 12 respectively, their own subsections one level lower. Its Section 4 is divided between source 12's Sections 3 and 5, its Section 5 between Sections 4, 5 and 7, its Section 9 (examples) is merged into Section 10 and Proposition 9.2 into Section 11, and the first subsection of its Section 13 is printed in Section 11. Its Section 11 is a section of its own; its questions follow source 12's questions on the same subject; its conclusion follows source 12's.
- *Renamed.* Source 10's `\Fin` macro (printed `Fin`) and its $\mathcal{S}(G, e) \rightarrow \mathcal{U}(G)$; no normalization changed.
- *Labels added* to unlabelled statements: `pma:br:cor:signed-presburger`; and `pma:br:s10:sec:category`, `pma:br:sec:examples` for source 10's sections whose text is merged; `pma:br:sec:q10` for the subsection of source 10's further questions; and `pma:br:sec:meet`, `pma:br:prop:meet` for the merge section.

Where Parts VII–IX chose. Each is one source, printed whole with notes. The construction $\mathbb{R} \overrightarrow{\times} K$ with a countable nonsplit core is printed once in general ([theorem 94.1](#)); source 14's [proposition 113.2](#) and source 15's [theorem 122.2](#) keep their own proofs and are flagged in their titles. Source 14's Section 10 group D is printed K_* , the generic D of its nonsplit-topology question K , its Ψ as Φ_ω and its 2^ω as $2^\mathbb{N}$; source 13's `\MB` is printed M_ω as in its delivery, and its lexicographic sums, like those of sources 12, 14 and 15, as $\overrightarrow{\times}$. Source 15's Section 3 and its Theorems 3.6 and 8.1, special cases of [Part VI](#), are printed with flags. Labels were added to unlabelled statements of sources 13 (nine) and 14 (one, `pma:atr:subsec:inputs`). The `\Needspace` commands of sources 14 and 15 (page-break hints for a package this report does not load) were dropped.

Intake checks for Parts VI–IX. The four delivered suites were rerun on copies (Python 3.14.4, standard library only): source 10 passed 29,041 checks in seven families (seed 20261003), source 12 36,989 in fifteen (seed 20261003), source 13's diagnostics reported "all_passed": true, and source 14 printed "EXACT FINITE CHECKS: PASS"; each record equals the shipped one after removing the carriage returns that Windows text mode adds. Source 15 ships no executable checks. The checksum manifests of sources 10 (7 of 7), 12 (8 of 8) and 14 (8 of 8) verified at placement and were not shipped; source 13's `provenance.json` hashes (three archives and three repository blobs) and source 15's manifest hashes (its two deliverables and seven repository files at its pin) also verified. The intake read the main chains of sources 10, 12 and 15 and of source 14's [theorem 106.2](#) to [corollary 111.2](#), without finding a gap; that reading is not a review, and in particular it did not check source 14's declared inputs or its ATR_0 coding.

Files of the deliveries not shipped. Besides the PDFs, manuscripts, delivery READMEs and checksum manifests, two provenance files are not shipped because each contains a third party's personal profile address, which no tracked file carries: source 14's `SOURCE_AUDIT.txt` and source 15's `source_manifest.json`. Both survive in the archives of the arrival commit.

Their verified content, without that address, is as follows. Source 14’s audit records that the supplied profile page could not be retrieved and that the research overlap was grounded instead in Glazer’s paper and a public author biography; that the whole of Glazer’s paper was inspected, including Theorem 2, Corollary 1, the Optimization section and both questions, and that no later arXiv version or indexed solution of Question 1 was found (not an exhaustive search); that the Gaifman–Dimitracopoulos chapter is a contribution to monograph no. 30 of *L’Enseignement Mathématique*, not obtained, and that the statement used was checked against Chen, Müller and Yokoyama (*J. Symb. Log.* 90 (2025), p. 485 and Definition 5.2); the coded descriptive set theory of Simpson’s Sections V.3–V.4 and Theorem V.8.3; the repository paths inspected at the pin 6fef5383b (the Lean and Coq exponentiation modules, this report’s `article.tex`, the Hilbert’s-tenth scope note) and the three batch-86 archives at their arrival 31fdc6571; and the claim boundaries repeated in [appendix F](#). Source 15’s manifest records its pin 0f084afa9 with a clean working tree; the seven repository files it inspected with their blob identifiers and SHA-256 values (all recomputed and matching at the pin); its ten external references with what was inspected for each (Christensen–Kanovei–Reeken: abstract and record only; Morris: metadata, the structure theorem being checked on p. 68 of Gabrielyan–Morris); the sizes and SHA-256 values of its source and PDF (matching); and its validation summary (32 pages, eleven questions, fourteen references, no unresolved references, AI-assisted review passes and no proof-assistant certificate).

Reviews elsewhere in the repository (Parts VI–IX). The Hilbert’s-tenth programme reviewed the six archives of fa0a0576e for its own search in commit f15962acd: `Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/review_baire_polish_arithmetic_intake.md`, with a receipt (`review_baire_polish_arithmetic_intake.json`) and an independent audit (`review_baire_polish_arithmetic_intake_independent.md`). It authenticated every archive and member, read all READMEs and recorded spans of each manuscript, ran no code, and found no new universal polynomial or arithmetic schedule. It is not a proof review. Its findings bearing on this report are recorded where they apply: source 14’s internal-MRDP input and automatic-Borel argument were read, but the claimed ATR_0 resolution is not certified ([Part VIII](#)); source 13’s revision counts bound changes of output, not running time, coordinates read or arithmetic operations, with no known final stage, and its continuity procedure outputs an expanded coordinate language (notes in [Part VII](#)); sources 10 and 12 concern signed group topologies and grant no Polish group completion to a positive monoid; source 15’s classification and cardinality proofs were not reviewed there.

Stale statements and addresses (Parts VI–IX). Source 13 describes batch 86’s three manuscripts as packages in `docs/incoming` and cites them by addresses at its pin; they are Parts I and II here, and the archives, retired in 3d2177df4, remain at those addresses. Sources 13, 14 and 15 describe this report as it stood at their pins (Part I’s base text); sources 10 and 12 cite sources 08 and 09 as “user-library” files. Source 13 cites four repository files by unversioned addresses; the notes in its bibliography entries give the pinned ones. Notes after the statements concerned correct each of these.

Source of Part X. (Batch 89.) Batch 89 had five archives: four arrived in commit 7c0f2d9f9 and one in 8ea27d6c0, and all five were placed in 23adb85f9. Manuscript 02 is source 16 of this report; the other four went to `surreal-well-orders` (Part XV), `birthday-cutoffs-and-hereditary-sets` (Parts VI and VII) and the research-report collection’s `naming-elementary-embeddings`. Source 16’s pin 0f084afa9 is 70 commits before the

placement. Of the repository files it read, this report's `article.tex` has since become the merged report, the Presburger and surreal READMEs changed only by the catalogue's counts and pointers, and the batch-86 archives it compared were placed in `0bd0e5527`; the other files are unchanged. No two batch-89 manuscripts share more than 0.33% of their word 8-grams, and source 16 shares 1.24% with this report (Presburger phrasing).

Source	Archive (batch 89 no.)	Pin	Contributes
16	Glazer_ProveIt_Embedding s_and_Thresholds (02)	0f084afa9	Part X except its introduction, the notes and remark 133.2 ; appendices W and X ; staged code, data, source audit and proof review (prefix 16-presburger-embeddings-)

Written for this report in Part X: its introduction, the notes marked [write], [remark 133.2](#) [merge] with the note after it, the labels added, the additions to [section 1](#) (the paragraph on Part X, its sources table, the fourth block of the notation table, the status box), the notes at Part I's questions on coefficient groups, lattices, surreal cutoffs and embedding categories and the pointers after [questions 14.3](#) and [14.11](#), at the end of [section 11.3](#) and after [theorem 35.4](#), the four bibliography notes, the entry [96], and these paragraphs.

Where Part X chose.

- *One source, printed whole.* Source 16's twelve sections are Sections [127–138](#) and its two appendices [appendices W](#) and [X](#). Its title page, abstract and table of contents are not printed separately: its status paragraph is in [section 1.5](#), and the abstract of this report summarizes it.
- *Printed again with a flag.* Results of Parts I and III that source 16 proves again keep its proofs and name the earlier statement in their titles: [proposition 128.3](#) ([theorems 35.1](#) and [35.9](#)), [theorem 129.1](#) and [corollary 129.2](#) ([proposition 11.3](#) and [corollary 11.5](#)), [theorems 130.5](#) and [133.1](#) for $\alpha = \beta = \omega$ ([theorem 11.4](#)), [theorem 130.7](#) for $\alpha = \beta = \omega$ ([corollary 11.5](#)) and the first example of [section 136](#) ([example 11.6](#)). [Theorems 131.1](#) and [135.1](#) and [corollary 129.3](#) are flagged as extending [theorems 11.2](#), [11.7](#) and [35.4](#), and the omnific map ι_α is identified with [theorem 40.1](#) in a note. Source 16's quantifier-elimination appendix is printed as [appendix W](#), a fourth write-up of a classical theorem already printed three times.
- *Questions.* Source 16's twelve questions are printed in its own part, numbered with the report's shared counter; those that repeat questions of other parts carry notes pointing to them, and the questions they answer carry dated notes.
- *Renamed.* Source 16's D , D_{fin} , S_φ , its `\res` and its roman lex, and local letters in two proofs ([section 1.3](#)); its H_α is kept. No normalization changed. Its bibliography entries for Glazer, Enayat–Hamkins–Wcisło, Haase and Kechris are this report's; the others carry the prefix `s16:`.
- *Labels added* to unlabelled statements: `pma:emb:cor:orbitcount`, `pma:emb:ex:shifts`, `pma:emb:ex:stages` and the twelve `pma:emb:q:` labels; and the four Part I question labels listed in [section 1.2](#).

Intake checks for Part X. Source 16's suite was rerun on a copy (Python 3.14.4, standard library only): 11,243 exact checks in thirteen categories pass (seed 231113699), and the record equals the shipped one after removing the carriage returns that Windows text mode adds. Source 16 delivered no checksum manifest; the hashes of its build record (data/16-presburger-embeddings-BUILD_REPORT.json) verified, 8 of 8. The write read every proof of source 16 without finding a gap; that reading is not a review.

Review elsewhere in the repository (Part X). The Hilbert's-tenth programme routed the five batch-89 archives in commit 6aadaa4f2: Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/review_topology_surreal_arrivals_20261003.md, with a receipt (review_topology_surreal_arrivals_20261003.json). It authenticated every archive member, read bounded ranges of source 16 (its exclusion of multiplication and of ring closure, the dependence of the non-Borel shears on a functional annihilating the finite-support vectors, its questions on effective normal forms), ran no code and found no new Diophantine source. It is a routing decision, not a proof review, and it has no correctness finding.

Stale statements and files of the delivery (Part X). Source 16 describes this report as it stood at its pin (Part I's base text), whose "Research questions 5 and 8" are Part I's [questions 14.6](#) and [14.9](#), and cites the batch-86 archives in docs/incoming, retired since; its novelty statement omits source 04's results, which it had no way to see. Notes in [Part X](#) and in its bibliography entries correct each of these. Not shipped: its PDF, its manuscript (merged) and its delivery README; it delivered no checksum manifest. Its source audit, shipped as delivered, says that a supplied LinkedIn profile could not be retrieved (no address is given) and cites a public Epoch AI article as Glazer's author biography, the article this report cites as [\[77\]](#).

Sources of Parts XI and XII. (Batch 90.) Batch 90 had six archives, all of arrival commit a162e4386, placed in 12076b2e8. Manuscripts 01 and 03 are sources 17 and 18 of this report; manuscript 02 became Part II of the research-report collection's measurable-box-games, 04 the new report cantor-families-of-surreal-subfields, 05 Part VIII of birthday-cutoffs-and-hereditary-sets, and 06 a fourth source of discrete-initial-subgroups-and-omnific-normalization. Source 17's pin 7c0f2d9f9 is 122 commits and source 18's pin 2b7b388ba 135 commits before the placement; at both pins this report had Parts I–V. Source 18's source record lists six repository files with their blob identifiers at the pin; four are unchanged at the placement (the Presburger README, SignSequence.lean, SignSequenceTopology.lean and the foundations report), and this report and the birthday-cutoffs report have changed. Source 17's ListCoding README is unchanged. The placement measured at most 0.52% shared word 8-grams among the six batch-90 manuscripts; source 17 shares 1.23% with this report and 1.34% with source 14's manuscript (the shared scale and fusion construction), and source 18 at most 0.35% with any report.

Source	Archive (batch 90 no.)	Pin	Contributes
17	local_geometric_codes_research (01)	7c0f2d9f9	Part XI except its introduction and the notes; appendices Y and Z ; staged code, data, proof-status record and source audit (prefix 17-local-codes-)
18	glazer_proveit_topology_research (03)	2b7b388ba	Part XII except its introduction, the notes and propositions 155.6 and 155.7 ; appendices AA and AB ; staged build script and proof audit (prefix 18-polish-completions-)

Written for this report in Parts XI and XII: their introductions, the notes marked [write], [propositions 155.6](#) and [155.7](#) [merge] with the paragraph between them, the Division headings, the labels added, the additions to [section 1](#) (the paragraphs on Parts XI and XII in [section 1](#) and [sections 1.1](#), [1.2](#) and [1.4](#), the sources table, the fifth block of the notation table, the status boxes), the dated notes after [questions 14.3](#), [43.4](#), [57.2](#), [73.11](#), [88.4](#), [103.4](#), [115.2](#), [115.3](#), [126.2](#), [126.3](#), [138.9](#) and [138.10](#) and after [proposition 81.5](#) and [corollary 121.2](#), the bibliography notes, and these paragraphs.

Where Parts XI and XII chose.

- *One source each, printed whole.* Source 17's twelve sections are Sections [139–150](#) and its two appendices [appendices Y](#) and [Z](#); source 18's sixteen sections are Sections [151–166](#) and its two appendices [appendices AA](#) and [AB](#). After Appendix Z the appendix letters continue as AA and AB. Title pages, abstracts and tables of contents are not printed separately: the status statements are in [section 1.5](#), and the abstract of this report summarizes both parts.
- *Separate parts.* Source 17 is not merged into [Part VIII](#), although it proves several of its lemmas again: its hypotheses are incomparable with source 14's, and it keeps its own proofs. Source 18 is printed whole, Division II included, although that division answers no question of this report: a manuscript that continues a report goes to that report. Moving Division II to birthday-cutoffs-and-hereditary-sets was considered at placement and not done.
- *Divisions.* Source 18's three \parts are printed as unnumbered headings "Division I–III" with contents entries; its terminology appendix's "Part I" reads "Division I". Its "Research questions 1–12" share the report's counter and are Questions [165.1–165.12](#).
- *Printed again with a note.* Source 17's [lemmas 140.1](#) to [140.3](#), [143.1](#) and [143.3](#) are source 14's [lemmas 106.1](#), [108.1](#), [108.2](#), [109.1](#) and [110.1](#); its scales adapt [section 108](#); [corollary 146.1](#) is a second route to [theorem 104.1\(ii\)](#) for coded exponentiation. Source 18's [proposition 155.1](#) is a case of [proposition 34.3](#), [example 155.4](#) is Part I's $M_{\mathbb{R}}$, [example 155.5](#) is Part III's cone of [theorem 38.1](#), and [proposition 163.4\(ii\)–\(iii\)](#) restate `found:thm:discrete`. They are flagged in notes, not in titles, because both sources credit the earlier work or prove it in a different setting.
- *Renamed.* Source 17's \C, source 18's \dotminus and \cfрак ([section 1.3](#)); source 17's three \Needspace layout commands were dropped, and its unused macros \lex, \Cont and \ExpDom and source 18's unused \WF and \cf are not defined. No normalization changed. Their bibliography entries for Glazer, Enayat–Hamkins–Wcisło, Tserunyan,

Simpson and Blass are this report's ([1, 2, 9, 70, 71]); the others carry the prefixes s17: and s18:.

- *Added.* [Propositions 155.6](#) and [155.7](#) [merge], checked against [corollary 78.6](#) and [corollary 121.2](#); Glazer's own wording of the class-manifold question, checked against the Oberwolfach report [106]; the labels on eight questions of other parts.

Intake checks for Parts XI and XII. Source 17's program was rerun on a copy (Python 3.14.4, standard library only): status "passed", 14,953 assertions (1,692 beta-code, 900 scale, 12,180 buffer and 181 anchored assertions, seed 20261004), in under a second, and the record equals the shipped one after removing the carriage returns that Windows text mode adds. Source 17 delivered no checksum manifest; source 18's (six files) verified and is not shipped. Source 18 ships no executable checks, and its PDF was not rebuilt. The write read both manuscripts in full without finding a gap; that reading is not a review.

Source 18's source record. Not shipped, because its list of starting points gives a third party's professional-profile address. Its other content, verified at placement: the pin 2b7b388ba; Git blob identifiers of the six repository files named above; the two batch-87 archives `polish_arithmetic_package.zip` and `polish_presburger_frontiers.zip` with their members and SHA-256; the question it answers, source 15's "A usable group-completion criterion" near line 2008 of its manuscript ([question 126.3](#)), and at the pin `pma:lc:q:completion` and `pma:ser:q:cones`, with `pma:cp:prop:allborel` and `pma:cp:thm:general` as the credited prior results (its line references are exact at the pin); the scope of the formal sources it read; its primary sources with their addresses, as in its bibliography; and a statement that it does not establish priority.

Review elsewhere in the repository (Parts XI and XII). The Hilbert's-tenth programme read the six batch-90 archives at their arrival in commit bcc1a4438: `Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/review_polish_partx_c9bc70d8f.md`, with a receipt (`review_polish_partx_c9bc70d8f.json`). It hashed every archive member, read every delivery README, and read source 17's manuscript in selected spans (its lines 76–267, 401–437, 1287–1390, 1573–1916 and 2103–2255) and its proof-status record in full; source 18 it routed from its README only. It ran no code. It found no new arithmetic bound in source 17, stated that it does not certify source 17's ATR_0 argument, and recorded that source 18's group-completion claim "merits a separate future proof review rather than silently treating the old open-question language as a theorem defect". Accordingly the notes of this write at the old questions are dated additions, and nothing earlier was deleted. It is a routing review, not a proof review; no such proof review exists yet.

Stale statements and files of the delivery (Parts XI and XII). Both sources describe this report as it stood at their pins (Parts I–V) and cite the batch-87 archives in `docs/incoming`, retired since in d151b39ca; notes in Parts XI and XII and in their bibliography entries give the current places. Source 17's audit names `local_arithmetic_diagnostics.json` and its manuscript by their delivery names; its program writes its record beside itself when run without an argument. Source 18's build script builds `polish_completions_class_manifolds.tex`, which is not shipped. Not shipped: both PDFs, both manuscripts (merged), both delivery READMEs, source 18's checksum manifest and source record.

Sources of Parts XIII–XV. (Batch 92.) Batch 92 had six archives, numbered by archive time: manuscripts 01–05 arrived in 9dc8db274 and 06 in afd7ffabb. Manuscripts 01, 02 and 05 are sources 19, 20 and 21 of this report, placed in e24ce2ce0; manuscripts 03, 04 and 06 became Parts III and IV of the research-report collection’s measurable-box-games, placed in e38f368c2. Source 19’s pin a32e1b139 is 48 commits and the pin 9fe62865d of sources 20 and 21 52 commits before the placement. Source 19’s source manifest records the Git blobs of the two READMEs it read; source 20’s source record and source 21’s audit/sources.json those of the files they read; all match at the pins, and every file is unchanged at the placement except this report’s README, which had gained Parts X–XII. The placement measured at most 1.02% shared word 8-grams with this report (source 19, mostly Part IX and the bibliography), 0.60% (source 20) and 0.97% (source 21), at most 0.52% with the manuscript of any single earlier source (source 21 with source 09), and 0.11–0.14% among the three.

Source	Archive (batch 92 no.)	Pin	Contributes
19	One_Real_Dimension_Arithmetic (01)	a32e1b139	Part XIII except its introduction, the notes, remark 176.3 and section 182 ; appendices AC and AD ; staged checks, record, build script, claim list and source manifest (prefix 19-one-real-dimension-)
20	Glazer_ProveIt_Exponent_Threshold (02)	9fe62865d	Part XIV except its introduction and the notes; appendices AE and AF ; staged checks, record, build script, proof audit and source record (prefix 20-exponent-threshold-)
21	Levi_Civita_Rank_and_Self_Embeddings (05)	9fe62865d	Part XV except its introduction, the notes and remark 204.2 ; appendix AG ; staged checks, record, Makefile, build summary, source record and proof audit (prefix 21-levi-civita-rank-)

Written for this report in Parts XIII–XV: their introductions, the notes marked [write], [section 182](#) and [remark 176.3](#) [merge], the proof of source 20’s countable DIV + Root₂ example [merge], [remark 204.2](#), the labels added, the additions to [section 1](#) (the paragraphs on Parts XIII–XV in [section 1](#) and [sections 1.1](#), [1.2](#) and [1.4](#), the sources table, the sixth block of the notation table, the status boxes), the dated notes after [questions 14.12](#), [28.7](#), [43.2](#), [57.3](#), [73.5](#), [73.8](#), [103.5](#) and [126.4](#) to [126.6](#) and after [theorem 123.1](#), the bibliography notes, and these paragraphs.

Where Parts XIII–XV chose.

- *One source each, printed whole.* Source 19’s fifteen sections are Sections [167–181](#), source 20’s twelve Sections [183–194](#) and source 21’s fifteen Sections [195–209](#); their appendices are [appendices AC](#) to [AG](#). Title pages, abstracts, keywords and tables of contents are not printed separately; the status statements are in [section 1.5](#), source 20’s conventions paragraph opens [Part XIV](#), and the abstract of this report summarizes the three parts. Their “Research questions” share the report’s counter.
- *Separate parts.* Source 19 is not merged into Part IV or IX, although it re-proves results of Part IX, because its main theorems are new and its proofs are its own; source 21 is a part of this report rather than of the surreal self-embeddings report or a new report, because it names source 09 as its prior report and generalizes Part V’s field (alternatives

weighed at placement). Source 20 is a part of its own: it answers Part II's [question 28.7](#) and extends Part III's [section 42](#); its series field and floor are source 21's, mirrored, and are flagged in notes, not merged.

- *Printed again with a note.* Source 19's [lemmas 169.1, 174.1 and 177.1](#), [theorem 177.2](#), the carrier in [theorem 173.1](#), [proposition 173.3](#) and [lemma 173.2](#) and [appendix AC](#); source 20's [lemmas 184.3 and 186.2](#), [theorem 184.4](#), and [propositions 185.2, 187.2 and 190.1](#) and its cancellation example; source 21's [propositions 196.2, 196.3, 197.1, 205.2 and 206.1](#). Each is flagged in a note, not in its title, because the sources' proofs are their own and none of the three read this report's article.
- *Renamed.* Source 19's G_L , M_L , D_L , G_D , M_D , X and macros `\cfrac`, `\st`, `\repo`; source 20's macro `\PP`; source 21's fields $\mathcal{L}_\Gamma$, $\mathcal{L}_\Delta$, $\mathcal{H}_\Gamma$, $\mathcal{I}_\Gamma$, $\mathcal{Z}_\Gamma$ ([section 1.3](#)). No normalization changed. The bibliography entries for Glazer, Enayat–Hamkins–Wcisło, Jeřábek, Cooper, Blass, Kechris, Shamseddine–Berz and Ehrlich–Kaplan are this report's ([1, 2, 4, 58, 71, 8, 31, 5]); the others carry the prefixes s19:, s20: and s21:.
- *Added.* [Section 182](#) (the shear, the agreement of abstract and topological isomorphism, the classification of locally compact models with continuous partial subtraction, the prescribed-boundary corollary and the complexity remark), [remark 176.3](#), the proof of source 20's unproved countable example, and [remark 204.2](#), which checks source 21's correction against the preprint and adds a second counterexample; the labels on six questions of other parts.

Intake checks for Parts XIII–XV. The three delivered checksum manifests verified (8/8, 8/8 and 9/9 files) and are not shipped. The three programs were rerun on copies (Python 3.14.4, standard library only): source 19 passed 181,071 checks in 23 families (seed 20261004), also under -0, in about seven seconds; source 20 passed 13,034 assertions in 20 groups in about three seconds; source 21 passed 2,931 checks in 13 families in about 77 seconds. The records equal the shipped ones after removing carriage returns, except that source 20's records the interpreter version (3.14.4 for the recorded 3.13.5). The write read the three manuscripts in full, checked the arithmetic of source 19's [theorem 176.1](#), source 20's [proposition 188.3](#) and [theorem 186.4](#), and source 21's [theorems 199.2 and 200.2](#), and found three statements needing a note: source 20's unproved countable example (proved here), the implicit uncountability in [lemma 188.1](#), and source 21's unproved extension to larger coefficient fields (kept inside [question 208.5](#)); that reading is not a review.

The Kuhlmann–Serra preprint. The preprint arXiv:2107.03362v3 was read on 4 October 2026 (Definitions 2.1.1, 4.0.1, 4.0.3, 4.0.5, 4.2.3, Example 3.1.4, Lemma 4.2.4, Proposition 4.2.6 and Corollary 4.2.7) for [remark 204.2](#); the journal version was not.

Review elsewhere in the repository (Parts XIII–XV). The Hilbert's-tenth programme read the five archives of 9dc8db274 at README level in commit 3c25fbb55 (Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/review_latest_five_9dc8db274.md, with a manifest review_latest_five_9dc8db274.json): it authenticated all members by Git blob, SHA-256 and size, read the three delivery READMEs in full, ran no code, treated their claims as routing information, and noted that source 21's correction of another paper had not been checked there. It checked the placement in e663441d7 (review_polish_placement_e24ce2ce0.md and .json): all sixteen staged files match

archive members byte for byte, and the host article was then unchanged. Neither is a proof review; this report has checked the correction ([remark 204.2](#)).

Stale statements and files of the delivery (Parts XIII–XV). Source 19 cites sources 05 and 08 as Library files and does not know Parts IX and XII; source 20 describes this report as the 349-page report of its pin; source 21 cites source 09 as a Library file. Notes in the parts and in the bibliography give the current places. Source 19’s build script and program name `one_real_dimension.tex` and `checks/`; source 20’s build script builds `article.tex` beside itself and its program writes `verification_results.json` beside itself; source 21’s Makefile names `article.tex`, `code/check_identities.py` and `audit/`, and its build summary and proof audit describe its own PDF. Not shipped: the three PDFs, the three manuscripts (merged), the three delivery READMEs and the three checksum manifests.

Sources of Parts XVI and XVII. (Batch 93.) Batch 93 had six archives, numbered by archive time: manuscripts 01–05 arrived in 2faa3b37a and 06 in de37a66d1. Manuscripts 03 and 04 are sources 22 and 23 of this report, placed in 5e4d8eed8; the other four form batch 93A and go to other reports. Both are pinned to 3c25fbb55, 57 commits before the placement; at that commit this report had Parts I–X written. Source 23 also records the later commit 4f03443e0. The placement measured 0.65% shared word 8-grams of source 22 with this report (Part VI 0.21%), 0.10% with source 19, 0.14% of source 23 with this report and 0.24% with source 21, and at most 0.05% between the two.

Source	Archive (batch 93 no.)	Pin	Contributes
22	Ordinal_Rigidity_Borel_Hilbert_Orders (03)	3c25fbb55	Part XVI except its introduction, the notes, question 222.10 and section 224 ; appendices AH and AI ; staged program, record, build script, build record, proof review and source record (prefix 22-hilbert-orders-)
23	Borel_Summability_and_Derivations (04)	3c25fbb55	Part XVII except its introduction, the notes and section 239 ; appendices AJ and AK ; staged program, record and source and proof audit (prefix 23-borel-summability-)

Written for this report in Parts XVI and XVII: their introductions, the notes marked [write], [sections 224](#) and [239](#) [merge], [question 222.10](#), the labels added, the additions to [section 1](#) (the paragraphs on Parts XVI and XVII in [section 1](#) and [sections 1.1](#), [1.2](#) and [1.4](#), the sources table, the seventh block of the notation table, the status boxes), the dated notes after [questions 14.6](#), [14.14](#), [43.7](#), [57.4](#), [73.6](#), [138.7](#), [208.1](#) and [208.5](#) and [theorem 33.1](#) and after Part V’s remark on real coefficients, the bibliography notes, and these paragraphs.

Where Parts XVI and XVII chose.

- *One source each, printed whole.* Source 22’s fourteen sections are Sections [210–223](#), source 23’s fourteen Sections [225–238](#); their appendices are [appendices AH](#) to [AK](#). Title pages, abstracts and tables of contents are not printed; the status statements are in [section 1.5](#), source 23’s reading guide is omitted (its section references are this part’s), and the abstract of this report summarizes the two parts. Source 22’s “Research

questions” and source 23’s questions share the report’s counter; source 23’s \Needspace spacing command is dropped.

- *Separate parts.* Source 22 is not merged into Part III, although its flag theorem is Part III’s [theorem 33.1](#) for vector spaces, because its Hilbert classification and its proofs are its own; the identification is [proposition 224.1](#). Source 23 is a part of this report, not a new report or a part of the omnific-preserving-automorphisms report, because it shares Part V’s and Part XV’s field and Part V’s automatic-continuity spine (alternatives weighed at placement).
- *Printed again with a note.* Source 22’s [lemma 212.1](#) and [theorem 213.2](#), the Δ_2^0 half of [corollary 213.3](#), [proposition 213.4](#), the finite-dimensional case of [theorem 210.1](#), [lemmas 216.1](#) and [216.7](#), [theorem 218.1](#), the algebraic half of [theorem 218.3](#), [theorems 219.1](#) and [220.2](#); source 23’s [propositions 226.2](#) and [226.4](#) and [lemma 227.1](#), the logarithm proof of [theorem 234.2](#), the flow ([234.4](#)) and the surreal map ([235.1](#)). Each is flagged in a note, because neither source read this report’s article.
- *Renamed.* Source 22’s H_θ , $H_{\theta,\alpha}$, G_θ , M_θ , J_θ , E_α (and the flag H_α of its Theorem 1.2), its diagonal D and typewriter \repo; source 23’s L_Γ , F_S , H_N and \restr ([section 1.3](#)). No normalization changed; source 22’s equations, numbered consecutively in the source, are numbered by section. The bibliography entries for Glazer, Kechris, Pettis, Cooper, L’Innocente–Mantova and Shamseddine–Berz are this report’s ([1, 8, 45, 58, 72, 31]); the others carry the prefixes s22: and s23:.
- *Added.* [Section 224](#): the identification with Part III’s recursion, the L^p theorem for every nonatomic measure and every completely metrizable space with zero dual (proving source 22’s abstract), a convex cone that is not Borel (bearing on the priority question), the characterization of invariant orders asserted after [question 222.6](#), the analogues of Part X’s thresholds and the cone closure rank of Hilbert orders. [Section 239](#): the second route to source 23’s non-Polishability through Part V, the proof that Baire measurable slices suffice and of the Euler flows (both asserted by source 23 without proof), the Borel embeddings fixing $\mathbb{R}$ with their bearing on Part XV, and the Lean special case. [Question 222.10](#); the labels on three questions of other parts.

Intake checks for Parts XVI and XVII. The two delivered checksum manifests verified (9/9 and 6/6 files) and are not shipped; source 22’s build record repeats the SHA-256 of eight files, which match. Both programs were rerun on copies (Python 3.14.4, standard library only): source 22 passed 136,562 assertions in seven families (seed 231113699), also under -0, in about five seconds; source 23 passed 1,640 checks in 13 groups (seed 20261004) in about two seconds. The records equal the shipped ones after removing carriage returns, except that source 22’s records the interpreter version (3.14.4 for the recorded 3.13.5). The write read both manuscripts in full and checked the proofs used in [sections 224](#) and [239](#); no mathematical error was found. Five statements needed action under the repository’s rule on unproved claims: source 22’s abstract states the L^p theorem for nonatomic spaces but proves it for $L^p([0, 1])$ (proved in general, [proposition 224.2](#)); its priority against Küçük–Soyertem–Küçük is unchecked (kept as [question 222.10](#), with [remark 224.3](#)); its commentary on [question 222.6](#) asserts a characterization without proof (proved, [proposition 224.4](#)); source 23’s remark that Baire measurable slices suffice is a sketch (proved, [proposition 239.2](#)); and its claim that the Euler derivation has a coefficientwise flow is unproved (proved, [proposition 239.3](#)). That reading is not a review.

The pin is a commit. Source 22 calls 3c25fbb55 its “canonical tree identifier” and its source record “Pinned tree/commit”; source 23 calls it “a tree identifier, not a commit identifier”. It is a commit: `git cat-file -t` prints `commit`, and its tree is c72baba04d76f05c8b9c7c751777ad97c41b474b. The citations resolve, because GitHub’s tree/*sha* addresses accept commits. Notes in [appendices AI](#) and [AK](#) record the correction; the delivered files keep their wording.

The Küçük–Soyertem–Küçük abstract. For [question 222.10](#) the write read the abstract of the 2012 paper on 4 October 2026 at the Anadolu University record cited in its bibliography entry; the publisher’s page refused the request, and no full text was found.

Review elsewhere in the repository (Parts XVI and XVII). The Hilbert’s-tenth programme read the five archives of 2faa3b37a at README level in commit f78d3ab0f (Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/review_five_arrivals_2faa3b37a.md, with review_five_arrivals_2faa3b37a.json): it authenticated every member by SHA-256, read the two delivery READMEs in full, ran no code and certified no claim. It checked the placement in 929fe1e02 (review_polish_borel_placement_5e4d8eed8.md and .json): the nine staged files match archive members byte for byte, and the host article was then unchanged. Neither is a proof review.

Stale statements and files of the delivery (Parts XVI and XVII). Source 22 describes this report as the thirteen-manuscript, 379-page report of its pin and presents the canonical transfinite extraction as its key point; source 23 says no declaration of the repository verifies its theorems or their consequences, and cites the moving default branch. Notes give the current facts. Source 22’s build script runs `verify.py` and three `pdflatex` passes on `article.tex` beside itself; its program writes `verification_results.json` beside itself unless `-output` is given; its build record describes its own 22-page PDF and the delivery names. Source 23’s program prints its record, which its delivery README redirects to `verification_results.json`. Not shipped: the two PDFs, the two manuscripts (merged), the two delivery READMEs and the two checksum manifests.

Sources of Parts XVIII–XXI. (Batch 96.) Manuscripts 01, 02, 03 and 07 of batch 96 arrived in 62846e17a, 7be14aa84, fb9f5884b and 78528873b and are sources 24–27 of this report, placed together in 54ece48ab; the batch’s other manuscripts go to other reports. Source 24 is pinned to 715a716a3, the parent of its arrival commit, 39 commits before the placement, and sources 25 and 26 to c39974f12, 38 commits before it; at both pins this report had Parts I–XVII, and its article is unchanged from there to the placement. Source 27 records no commit but the blobs a7ebc59ad and e24d8fc8d of the surreal and BCH READMEs, current from ad52ef11e to the placement. Shared word 8-grams: sources 24 and 27 1.8% and 2.5% (boilerplate and bibliography), source 26 with 24 0.5% and with 27 0.6%, source 25 at most 0.41% with any of the three and 0.67% with this report, source 24 about 1% with source 23, source 26 0.1% with Part XVII.

Source	Archive (batch 96 no.)	Pin	Contributes
24	Borel_Conjugacy_Divisibility_Threshold (01)	715a716a3	Part XVIII except its introduction, the notes and section 255 ; appendices AL and AM ; staged program and record (prefix 24-borel-conjugacy-)
25	glazer_proveit_support_complexity (02)	c39974f12	Part XIX except its introduction, the notes, question 268.12 and section 270 ; appendix AN ; staged program, record, Makefile and source ledger (prefix 25-support-complexity-)
26	Borel_Flows_Glazer_Prove It (03)	c39974f12	Part XX except its introduction, the notes, question 283.10 and section 285 ; appendices AO and AP ; staged program, record, build script, claim ledger and source record (prefix 26-borel-flows-)
27	Two_Derivations_Borel_Conjugacy (07)	none; blobs a7ebc59ad, e24d8fc8d	Part XXI except its introduction, the notes, the numbered note remark 294.3 and section 300 ; appendix AQ ; staged program, two records, dependency pin, build script and source audit (prefix 27-two-derivations-)

Written for this report in Parts XVIII–XXI: their introductions, the notes marked [write], [sections 255, 270, 285](#) and [300](#) [merge], [questions 268.12](#) and [283.10](#), the remarks [remarks 289.2](#) and [294.5](#), the numbered note that prints source 27’s Proposition 9.3, the restored reference of source 26, the labels added, the additions to [section 1](#) (the paragraphs on Parts XVIII–XXI in [section 1](#) and [sections 1.1, 1.2](#) and [1.4](#), the sources table, the eighth block of the notation table, the status boxes), the dated notes after [theorems 18.4, 19.1, 20.2](#) and [67.3](#), [corollaries 20.4](#) and [234.3](#), [questions 28.3](#) to [28.6](#), [208.5](#), [237.1](#) and [237.3](#) to [237.6](#), and [proposition 239.3](#), the bibliography notes, and these paragraphs.

Where Parts XVIII–XXI chose.

- *One source each, printed whole.* Source 24’s fifteen sections are Sections [240–254](#), source 25’s fourteen Sections [256–269](#), source 26’s fourteen Sections [271–284](#) and source 27’s fourteen Sections [286–299](#); their appendices are [appendices AL](#) to [AQ](#). Title pages, abstracts, tables of contents and reading guides are not printed (the status statements are in [section 1.5](#)); source 25’s result box is a box, source 27’s “Main distinction” box the opening paragraph of its first section. Their questions share the report’s counter; source 25’s equations are numbered by section. Typography only: the boxed display of source 24’s first section uses smaller spaces and three of its tables ragged-right columns, two typewriter hashes in source 26’s Appendix B are set as paths, to fit the report’s margins.
- *Four parts, in manuscript order.* Source 24 continues Part XVII in rational rank one with its own spine (normal forms, invariants, threshold, flows); source 25 answers Part II’s [question 28.3](#) but its certificate and strong-summation theory is its own, so it is not merged into Part II; source 26 works in every rational rank with a different spine (finite orbit spans, the window criterion, the rank dichotomy, nonclosure); source 27 classifies pairs and Lie algebras. Results shared by these sources are printed in the earlier part and quoted in the later one: source 27’s Theorem 4.1 is [theorem 242.4](#); its Proposition 9.3

is the last clause of [theorem 248.3](#) and, at $\Gamma = \mathbb{Q}$, source 26's [corollary 275.4](#), so it is printed as the numbered note [remark 294.3](#) with source 27's own proof; at $\Gamma = \mathbb{Q}$ source 26's flow theorems and source 24's are the same statement under incomparable hypotheses elsewhere ([remark 255.2](#), [corollary 285.2](#)). Every reference between the four parts was checked after assembly.

- *Printed again with a note.* Source 24's [lemmas 241.2](#) and [242.1](#), [proposition 241.4](#), and [theorem 242.2](#), the integration obstruction after ([243.3](#)), [proposition 247.1](#), the logarithm in [lemma 248.6](#), a flow of its examples and its surreal map (Part XVII's, in rank one); source 25's [lemmas 257.2](#) and [259.4](#), [theorems 258.2](#) and [265.2](#), [proposition 259.5](#), and [corollary 260.3](#) (Part II's, [theorem 19.1](#) and [theorem 67.3](#) being special cases of its classification); source 26's field layer, Borel derivations, uniform-gain exponential and surreal map (Part XVII's); source 27's field layer, [theorem 288.3](#) and the zero-gain classification. Each is flagged in a note, because none of the four read this report's article.
- *Renamed.* Source 24's L_Γ , $P_\infty(\Gamma)$, its bookkeeping ℓ and F_S ; source 25's $\mathcal{H}_L$, C , E_k , E , A , B , $\mathcal{P}_n$ and I ; source 26's $\mathcal{L}_\Gamma$, $V_{a,b}$, $d(\gamma)$, δ , ϵ_n , E , F , H , P_s , Q_s , $\mathcal{P}$, I and θ ; source 27's K_Γ , its three H , $\mathcal{H}_\Gamma$, $\mathcal{P}_\Gamma$ and e , f ([section 1.3](#)). No normalization changed; sources 24 and 27 conjugate in opposite directions, which [remark 289.2](#) translates. The bibliography entries for Glazer, Kechris, Shamseddine–Berz, Kuhlmann–Serra, Rosendal–Suarez, Gonshor and the surreal README are this report's ([\[1, 8, 31, 133, 147, 91, 150\]](#)); the Mardešić–Resman–Rolin–Županović paper, cited by sources 24 and 27 under its journal and arXiv titles, is one entry ([\[151\]](#)); the others carry the prefixes s24:–s27:.
- *Added.* [Section 255](#): the time-one maps ([proposition 255.1](#)), the comparisons with Parts XX and XV ([remarks 255.2](#) and [255.3](#)), the reviewer's effectivity lemma ([remark 255.4](#)) and the check of the cited example ([remark 255.5](#)). [Section 270](#): a second-route table, the Archimedean specialization ([proposition 270.1](#)), a proof of an effectivity sentence ([remark 270.2](#)) and the review's two observations; [question 268.12](#). [Section 285](#): Borel time on a basis ([proposition 285.1](#)), the answer at $\Gamma = \mathbb{Q}$ ([corollary 285.2](#)), BCH under a common positive gain ([proposition 285.3](#), source 26's sketch), finite sampling ([proposition 285.4](#), source 26's Remark 10.3, with its scope corrected); [question 283.10](#). [Section 300](#): Lie-group actions ([definition 300.1](#), [lemma 300.2](#), and [theorem 300.3](#)) and the reviewer's effective boundary ([remark 300.4](#)); four labels on questions of Part II.

Intake checks for Parts XVIII–XXI. The four delivered checksum manifests verified at placement (5/5, 7/7, 8/8 and 9/9 files) and are not shipped. The programs were rerun on copies. Source 24's (Python 3.14.4, standard library, explicit output path): 143 checks in six families pass in about five seconds, also under -0. Source 25's (Python 3.14.4, standard library, seed 20261004): its printed record is produced in under a second, also under -0, but there only because its 17 checks are bare assert statements, which -0 removes. Source 26's (Python 3.14.4): 5,372 checks in four groups (1,656, 1,200, 1,760, 756) pass in about half a second, also under -0; its checks raise explicitly. Source 27's (Python 3.13.5 with SymPy 1.14.0, its pinned dependency): 669 assertions pass (104 scalar and pair, 37 Lie, 15 affine and BCH, 513 spectrum), checked explicitly. Every record equals the shipped one after removing carriage returns. The write read the four manuscripts in full and checked the proofs used in the four merge sections; no mathematical error was found, and the one statement found false as literally stated, source 26's Remark 10.3 in finite rational rank, is kept with its counterexample (note after it, and [proposition 285.4](#)). [Theorem 300.3](#) and [lemma 300.2](#) were checked once more by an independent reading for this report, which found them correct and asked for

four editorial fixes (the comparison with [corollary 248.9](#), the note after [question 253.3](#), the converse in part (i), and the hypotheses restated in the statement), all applied. These readings are not reviews.

Review elsewhere in the repository (Parts XVIII–XXI). The Hilbert’s-tenth programme read all four manuscripts in full and authenticated their archive members, in `Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/`: source 24 in `8dff7af0` (`review_new_borel_62846e17a.md`, with `.json` and `.py`), source 25 in `3abf23ba1` (`review_new_support_complexity_fb9f5884b.md`), source 26 in `3d87c3d6b` (`review_new_borel_flows_fb9f5884b.md`) and source 27 in `556836e9b` (`review_two_derivations_78528873b.md`). The file names of the reviews of sources 25 and 26 carry `fb9f5884b`, the snapshot they read; source 25 arrived in `7be14aa84`. They authenticated the members, ran no delivered program and found no correction; source 26’s review found the same malformed reference. Their deductions are printed with credit and re-checked by the write: the effectivity lemma [remark 255.4](#), two observations in [section 270](#), and the co-c.e.-complete invariant equality of [remark 300.4](#); the review of source 26 notes that the first does not transfer to divisible Γ .

Review remark 3 (The scope of the earlier reviews). The earlier wording, “read all four archives in full”, overstated those records. Each review explicitly says that its delivered PDF was hashed only, not read or rendered; source 27’s saved JSON was parsed as data but only selected line ranges were read. The complete mathematical manuscripts were read, and every archive member was authenticated. Likewise “found no correction” records the intake assessments at that time: source 26’s unqualified finite-sampling remark was missed and is false in finite rank, as the retained note after [remark 280.3](#) explains. [Proposition 285.4](#) proves the infinite-rank repair. These qualifications do not claim a new full-manuscript or PDF review.

Stale statements and files of the delivery (Parts XVIII–XXI). Source 24 describes Part XVII as a draft in the user’s library with no copy bundled; source 25 calls Part II a prior artifact “not bundled in this package”; source 26 calls source 23 an earlier draft and a prior report; source 27 cites sources 23 and 24 as manuscripts of the user’s collection. Notes give the current facts. Source 24’s program writes `verification_report.json` in the working directory unless `-output` is given, and its docstring names `borel_conjugacy.tex`. Source 25’s Makefile and Appendix A.4 name `article.tex`, `checks.py`, `python3` and `make`; its ledger names `glazer_proveit_article.pdf`. Source 26’s build script runs its program with `-json verification_results.json` and three `pdflatex` passes on `borel_flows.tex`, which is not shipped, overwriting the record in place; its source record names `borel_summability.pdf`, its claim ledger its own theorem numbers. Source 27’s program writes `verification_report.json` and `.txt` beside itself unless `-output-dir` is given; its build script builds the unshipped `two_derivations.tex`; its source audit uses its own section numbers and names sources 23 and 24 by delivery file names. Not shipped: the four PDFs, the four manuscripts (printed here), the four delivery READMEs and the four checksum manifests.

F Limitations and non-claims of the sources

Every limitation stated by a source is kept. They are grouped by source; the items written for the merge come last.

Source 05.

- 05.1 The construction answers Glazer’s Question 2 *as written*; independent bibliographic priority has not been established. The literature search verified the question in the source PDF and checked the arXiv record; it did not establish the absence of an earlier solution.
- 05.2 Classical Presburger elimination, the $\mathbb{Z}$ -group facts and elementary topological-group facts are not claimed as new; the lexicographic algebra is classical.
- 05.3 The manuscript is AI-assisted, not independently refereed and not Lean-verified; no new Lean or Rocq formalization is included; the proposed Lean modules are a plan, not existing files.
- 05.4 The executable tests are finite instances and stream prefixes; they do not quantify over arbitrary elements of $\mathbb{Q}^{\mathbb{N}}$, and finite prefixes do not decide equality of infinite streams.
- 05.5 No answer to Glazer’s Question 1 is claimed; no uncountable Polish model of Peano arithmetic; no continuous arithmetic order on these nondiscrete spaces; no terminating decision of equality of arbitrary infinite sequences (comparison is approximated with at most one change).
- 05.6 The omnific maps are additive and order-preserving, not ring embeddings; the topologies are specified coefficient topologies, not a canonical topology on $\mathbf{Oz}$; no topology on a proper class is used; all constructions are set-sized in ZFC.
- 05.7 Neither model admits any unital semiring multiplication with the given addition and unit; nothing contradicts Glazer’s obstruction for stronger arithmetic, and the no-semiring argument does not replace it.
- 05.8 The repository’s integer Cooper lemmas are correct on $\mathbf{Int}$; the existence of the integer eliminator does not certify the theorem for a new carrier; the original integer theorem need not be weakened or retracted.
- 05.9 The questions are proposed directions, not certified as open in the literature; the user’s repository was not modified.

Source 04.

- 04.1 The construction answers Question 2 as printed; historical priority has not been established; the literature check did not establish that the construction has never appeared elsewhere.
- 04.2 The manuscript is unrefereed; no new theorem is represented as machine-checked; Lean/Rocq certification is not supplied.
- 04.3 The classical $\mathbb{Z}$ -group characterization and quantifier elimination are not claimed as new.

- 04.4 The difference between one-step periodicity and residue invariance is a porting requirement, not a reported bug in the integer proof.
- 04.5 Baire coordinates encode the arithmetic structure and are not base digits; the stream API cannot decide totality of an arbitrary oracle, and partial, negative or noninteger entries are not supported.
- 04.6 The source audit was limited to the named files at its pin; no repository files were modified and no repository-wide build was performed; the formalization stages are proposals, not assertions about existing declarations.
- 04.7 The tests are finite, exact and partial; the model-theoretic, infinite-matrix and topological statements rest on the proofs.
- 04.8 No third-party font files or paper copies are included.

Source 03.

- 03.1 A research draft, not refereed and not a Lean proof; global priority and independent correctness review are not established.
- 03.2 The repository's coanalytic support obstruction and the negative-infinitesimal floor correction are credited, not claimed as new; the integer-part model is of Shepherdson type, not new; analytic boundedness and the Hausdorff characterization are classical.
- 03.3 The full-Hahn nonpresentation results need observable coefficients and coverage of the whole field; they do not exclude unobservable Borel presentations or Borel real closed subfields.
- 03.4 Only addition is shown impossible to make continuous; Glazer's multiplication obstruction (his Theorem 2) is not transferred, since only open induction is proved for $\mathcal{A}_{\text{fin}, \geq 0}$; continuous multiplication under another compatible topology is open.
- 03.5 The zero loci are not claimed minimal in cardinality, dimension or descriptive rank; countable closed-piece continuity is not identified with the Baire-one class.
- 03.6 The dichotomy is per fixed L , not about the class of scattered orders, nor uniform in a code of L ; the Hausdorff decomposition used need not be computable; the uniform support bound need not be computable from a code.
- 03.7 No general computability over arbitrary real coefficients is claimed.
- 03.8 All results are set-sized in ZFC; no class-sized Polish topology, global choice or large cardinal is used.
- 03.9 The Python checks are finite (they do not test well-foundedness of infinite trees, completeness, or the external theorems); the formalization plan is proposed only; no repository files were changed and the whole repository was not audited.

Source 07.

- 07.1 The construction answers Question 2 as printed; the claim that it is the first published resolution is not made. The literature search was targeted, not exhaustive; the precise place in the literature of the cone theorem, the Baire family and the rank computations is left to specialist review.

- 07.2 The lexicographic construction of a $\mathbb{Z}$ -group is classical; classical model theory and standard descriptive set theory are inputs; no historical priority for the formulation of [theorem 33.1](#) is needed or claimed, measurable orders on locally compact groups having been studied by Asmar and Montgomery-Smith.
- 07.3 No new result has been checked by Lean or Rocq; no Lean or Rocq build was run; the repository was used read-only; no new formal theorem is claimed; no publication priority or external peer review is claimed.
- 07.4 Only first-order induction is claimed, not second-order induction over external subsets.
- 07.5 No computable homeomorphism is claimed for an arbitrary presentation of a countable ordinal; no classification of the pure underlying orders is claimed; the oscillation-rank convention is fixed and not identified with every convention for difference-hierarchy ranks.
- 07.6 The omnific images are not claimed initial, closed under surreal multiplication or under surreal-analytic operations; the transported topology is not the surreal order topology; every model is set-sized and the class **No** is used only as a definable ambient class.
- 07.7 Glazer's Question 1 is not answered, and the metatheoretic strength of his theorem with multiplication is not determined.
- 07.8 The LinkedIn page supplied with the brief could not be read; the repository search covered `.tex`, `.md` and `.lean` files at the pin, not every branch or binary artifact.
- 07.9 The questions are proposed directions, not asserted to be historically new or unresolved in all of the literature.
- 07.10 No finite numerical experiment is offered as evidence; compilation and visual inspection verify the document, not the mathematics; the proof checks made during preparation are not external refereeing or proof-assistant certification.

Source 06.

- 06.1 No machine-checked certification and no unrestricted publication-priority claim is made; the literature search is a bounded bibliographic finding, not a proof of universal priority.
- 06.2 The lexicographic $\mathbb{Z}$ -group construction and its model theory are classical (Jeřábek); rational polynomial Presburger examples appear in lecture notes (Cook–Pitassi); the polynomial ring has classical precedents and occurs in the repository.
- 06.3 The computability claims are Type-2 claims about streams: no computable enumeration of the uncountable model and no decidability of equality of streams are asserted, so there is no conflict with Tennenbaum's theorem; the continuum family is not claimed computable.
- 06.4 The ordinal scales are classes of magnitude, not ω_1 -sequences in one order; there is no conflict with the obstruction of Enayat, Hamkins and Wcisło.
- 06.5 The failure of open induction concerns $\mathbb{Z} + t\mathbb{R}[t]$ only and does not transfer to **Oz**; the analogous topology on the signed ring is not a ring topology, and partial subtraction on the cone is not continuous.

- 06.6 The isolated-zero theorem does not imply that every standard numeral is isolated; it uses no induction inside the model, no standard division, exponentiation, saturation or integer-part theorem.
- 06.7 The surreal embeddings are additive; no closure under surreal multiplication is asserted; the topologies are transported, not inherited from a topology on the class **No**.
- 06.8 The formalization steps are proposed, not run; the repository reports are points of contact, not premises; no full-repository audit or Lean rebuild was made.
- 06.9 The projects are not certified as unsolved in the literature.
- 06.10 The proof reviews made during preparation are mathematical checks, not formal certification or peer review.

Source 08.

- 08.1 Bibliographic priority has not been established; no independently certified breakthrough is claimed; the manuscript is AI-assisted, unrefereed and not Lean-checked; a missing search result is not evidence of a first proof.
- 08.2 The compact-group obstruction, finite-dimensional separation and the locally compact abelian structure theorem are classical; the principal structure theorem is imported, not reproved; Presburger elimination is classical; Hewitt and Koshi's paper was not read in full.
- 08.3 A Hausdorff locally compact second-countable topology on the signed additive group is assumed; the result is not a theorem about every Polish presentation of a positive arithmetic model and does not replace Glazer's general theorem.
- 08.4 The real model is credited to the companion manuscript (source 05); the classification is of pointed topological additive groups, not of all orders or of all topological monoids.
- 08.5 No separable non-locally-compact counterexample is presented; the general Polish-group case and Glazer's Question 1 are not settled.
- 08.6 The transported examples are not effective; no topological-ring claim is made for the nonseparable example.
- 08.7 The finite tests cannot establish statements about all real points, arbitrary Borel sets, local compactness, Haar measure or infinite cardinalities; no Lean file is included as if checked, and a conditional formalization with an interface must not be advertised as unconditional.
- 08.8 The companion manuscripts were not independently audited; the repository guides identify interfaces only; no repository file was modified or built.
- 08.9 The proofs are in ZFC; no claim is made about the choice or reverse-mathematical strength needed; no topology, Haar measure or power set is applied to a proper class.

Source 09.

- 09.1 The proofs are conventional and unrefereed; nothing is Lean- or Rocq-verified; historical novelty is not established; the literature review was targeted and no absence-of-result argument is used as evidence.

- 09.2 Category theorems, Hahn and Puiseux constructions, Newton–Puiseux, real-closedness inputs, the Levi-Civita construction, the coanalytic support obstruction and the omnific floor rule are credited, not claimed as new.
- 09.3 No new answer to either of Glazer’s printed questions is claimed, and Glazer is not a collaborator or endorser; the LinkedIn page was not used.
- 09.4 The conventional real-coefficient Levi-Civita field is not claimed to be Polish.
- 09.5 The 20,116 finite checks are not proofs of the category, completeness, descriptive or real-closedness statements.
- 09.6 The repository’s formalization ledger could not be fetched; no claim-by-claim audit, local checkout or build was made; the pin is a locator.
- 09.7 The proofs are in ZFC with no choice-free assertion; no computability claim follows from Borel equivalence; no Polish topology on a proper class is used.
- 09.8 The questions are proposed next steps, not previously published open problems.

Source 12.

- 12.1 An AI-assisted, unrefereed manuscript, not an independently certified breakthrough and not a Lean- or Rocq-checked development; no such toolchain was run; historical priority is not established, and targeted searches do not prove absence from the literature.
- 12.2 The Pettis and Steinhaus difference principles are classical, and only the needed forms are proved; the bounded-neighbourhood lemma may be viewed as an application of classical regularity methods; Steinhaus is not credited with the locally compact generalization; Rothman is cited as background, and no theorem from it is assumed.
- 12.3 The theorems concern a topology on the signed additive group; they give no automatic Polish group completion of an arbitrary positive monoid and do not replace Glazer’s theorem under weaker hypotheses on the same objects.
- 12.4 Glazer’s ATR_0 question is not settled, and the earlier affirmative Presburger construction is not presented again as new.
- 12.5 “Countable” does not mean standard; the non-Polishability theorem does not exclude a Polish topology after the Borel structure and the regularity of the order are changed.
- 12.6 The pathological Polish presentation is a choice construction, not a computable coding, and its multiplication is not claimed to be Borel.
- 12.7 The finite checks, in $\mathbb{Z} + t\mathbb{Q}[t]$, do not certify Baire category, Borel sets, uncountable cardinalities, semiring completions or a theorem for every ordered ring.
- 12.8 No theorem treats the proper class **Oz** as a Polish space; the omnific corollary is set-sized and uses no unrefereed repository claim.
- 12.9 The repository was not built and its advertised audit results were not verified; reading its interfaces is not a check of the whole repository; the profile page supplied with the request was not available, and nothing personal is inferred from it.
- 12.10 The questions are proposed directions; except for the earlier project questions identified, their open status is not certified.

Source 10.

- 10.1 Unrefereed and AI-assisted; bibliographic priority is not established; no theorem is claimed Lean- or Rocq-checked, no new source file is labelled verified, and the repository's builds were not rerun.
- 10.2 The category mechanism is classical and its Polish version was already isolated by source 09; the divisible-plus-integer Presburger examples belong to the existing project constructions and are not claimed as new.
- 10.3 The theorem concerns signed additive-group topologies: it gives no Polish group completion of a positive arithmetic monoid and does not remove the Borel-multiplication hypothesis from Glazer's positive-model theorem; the comparison is between different hypothesis sets.
- 10.4 Glazer's Question 1 is not settled, and the affirmative Presburger examples are not presented as a new answer to Question 2.
- 10.5 "Uniqueness" in [corollary 78.6](#) is of the topology, not of the ring; countable nonstandard rings are not classified.
- 10.6 The transported example is chosen with vector-space bases, not by an algorithm, and gives no Borel multiplication or regular comparison.
- 10.7 Clause (4) of [theorem 80.4](#) concerns each hull separately, not the spectrum or the union of the hulls; the theorem is not about field self-embeddings of **No**.
- 10.8 Borel and continuous scalar maps are not identified outside the separable setting.
- 10.9 The omnific corollary is set-sized, with no Polish topology on a proper class; countable notation systems, incomplete presentation spaces, non-Baire comparisons and purely algebraic formalizations are not ruled out.
- 10.10 The exact tests do not verify category, uncountability, arbitrary orders, basis choices or any infinite theorem; the profile page supplied with the request was not used for personal details; the earlier project manuscripts were read as project documents, not refereed publications.
- 10.11 The questions are proposed targets, not certified open.

Source 13.

- 13.1 Conventional proofs with limited exact diagnostics; no first-publication priority, refereeing or new proof-assistant verification is claimed; an "open question" means one left open by a repository manuscript.
- 13.2 The affirmative existence construction is credited to the earlier manuscripts; the countable nonsplit algebra and the omnific criterion have prior repository counterparts and are not claimed as new; $\mathbb{Z}$ -groups, Presburger elimination, Polish-space facts and the ordered-group universality of **No** are classical.
- 13.3 Glazer's reverse-mathematical question is not resolved; arbitrary Polish topological monoids are not classified; the connected-universe question for monoid topologies is not settled.

- 13.4 Discontinuity sets need not be closed or Presburger-definable; the compiler's output is an external coordinate formula, an analysis tool and not a definability theorem; with arbitrary infinite parameters it is an oracle reduction, not a decision procedure.
- 13.5 A bound on revisions gives no stopping stage and no running-time bound; the exact finite calculation is not asserted for one fixed arbitrary parameter; decidability is not efficiency; the rank bound need not be the exact revision complexity.
- 13.6 The countable-core extension is topological; no computable presentation of the core is assumed.
- 13.7 The omnific criterion is additive and order-theoretic and set-sized; the monomial-shift embedding is not asserted to preserve multiplication or to be initial; the no-homomorphism result concerns **Oz**, not **No**.
- 13.8 The closed-translate theorem gives a necessary condition only.
- 13.9 `verify.py` is a finite diagnostic, not an implementation of quantifier elimination or a proof of the infinite theorems; the repository's Lean semantics over `List Int` is not a proof for nonstandard $\mathbb{Z}$ -groups; the profile page supplied with the request was inaccessible and not used.
- 13.10 The questions are not presented as solved, and are narrower than a theory of all Polish arithmetic.

Source 14.

- 14.1 The answer to Question 1 is claimed for the question as stated in Glazer's 2023 paper, under the coded-model convention of its Section 2; it is an upper bound only: no reversal, and no claim that ACA_0 fails or that the theorem is equivalent to ATR_0 .
- 14.2 Not mechanically verified and not externally refereed; the proof audits are in-session reviews; no exhaustive priority search (no later arXiv version or indexed solution was found).
- 14.3 The internal MRDP theorem is an established input; its original chapter was not obtained, the statement being corroborated by Chen, Müller and Yokoyama; analytic separation, coded Baire approximation and uniform perfect-set and order homogenization are declared inputs.
- 14.4 The standard- $\mathbb{N}$ Lean MRDP theorem does not prove the internal statement; no Lean or Coq file is a formalization of the obligations.
- 14.5 No global satisfaction predicate, set of all Borel functions or set of all models on an uncountable carrier is used.
- 14.6 "Cofinal" means unbounded in the arithmetic order, not dense, large or nonmeager.
- 14.7 Section 10 is ordinary set-theoretic mathematics, not formalized in ATR_0 ; nonsplit $\mathbb{Z}$ -groups are not a new discovery; preservation of the unit is essential (Φ_ω embeds into **Oz**); Φ is not a homeomorphism onto its image in the surreal order topology.
- 14.8 The finite checks prove no infinite theorem and do not simulate a nonstandard model.
- 14.9 The questions are proposed directions.

Source 15.

- 15.1 The basic affirmative answer to Question 2, the real and Baire constructions and the special multiplication obstructions are prior project material; quantifier elimination, automatic continuity, the structure theorem for locally compact abelian groups, the compact-subgroup and real-line order lemmas and finite-support techniques are classical; the probabilistic mechanism is Glazer's, extended, not independently invented.
- 15.2 The classification covers inherited group topologies only, not all Polish monoid topologies; the open real component need not be convex; local compactness of the coefficient group alone does not suffice.
- 15.3 The compact-open theorem concerns the topology at zero only: it does not rule out zero-dimensional locally compact presentations with isolated zero, nor make every numeral isolated.
- 15.4 The manifold boundary differs from the boundary in the ambient group; "initial" is meant arithmetically.
- 15.5 The omnific images are additive substructures, not subrings, and carry a transported coefficient topology, not the native order topology.
- 15.6 No Lean or Rocq build; AI-assisted review passes are not refereeing; publication priority is not certified; Glazer's Question 1 is not answered; "induction-free" refers to the modeled arithmetic, not to the metatheory.
- 15.7 The degree-stratum example carries no novelty claim; no finite computation is offered as a certificate, and none is shipped.
- 15.8 The questions are not all asserted to be previously published open problems.

Source 16.

- 16.1 AI-assisted, internally reviewed by two AI agents (record shipped as 16-presburger-embeddings-PROOF_REVIEW.md), not refereed and not a literature-certified breakthrough; no new Lean or Rocq formalization; its six proposed modules are a plan, and the repository's integer Presburger procedure is not a formalization of arbitrary $\mathbb{Z}$ -groups.
- 16.2 The classification and the thresholds concern the family M_α only; the lattices of arbitrary Polish presentations of arbitrary Presburger models are not classified.
- 16.3 The family M_α and its Baire-space carrier are prior work, and the positive construction is not a new answer to Glazer's Question 2; Glazer's Question 1 is not resolved.
- 16.4 Hahn-group matrix methods and strong additivity (Kuhlmann–Serra, Serra), formal infinite sums (Freni) and linearly compact duality (Castellano–Giordano Bruno) are credited and not claimed; Presburger quantifier elimination is classical, proved in the appendix only for completeness.
- 16.5 Historical priority is not established; the source audit was bounded, not exhaustive.
- 16.6 The finite checks (11,243 assertions) cover finite matrix identities, example normal forms, signs and standard division; they do not verify infinite products, ordinal recursion, Baire category, elementarity or the existence of the non-Borel functional, and no test confined to finite-support vectors can tell a wild shear from the identity.

- 16.7 Multiplication is absent; the omnific realization is additive and ordered, with a transported coefficient topology, without ring closure and without a topology on the proper class **No**.
- 16.8 The discontinuity statements concern the product topology (the shears are continuous for the valuation topology); Polishness is that of the inherited topology, and a nonclosed subspace may be abstractly isomorphic to a closed one.
- 16.9 The non-Borel constructions use choice; the questions are proposed directions, not certified open problems.

Source 17.

- 17.1 Prepared with ChatGPT; its reviews were in-session AI reviews, not external peer review; no publication priority, no proof-assistant certification, and no change to the repository is claimed. The source audit was targeted, not exhaustive.
- 17.2 It does not settle whether an uncountable Borel Polish $\mathsf{I}\Delta_0$ -model with separately or jointly continuous multiplication exists; the bare $\mathsf{I}\Delta_0$ question is open.
- 17.3 It does not identify the β -coded domain with any ordinary bounded-definition exponentiation; no recoding principle is assumed or proved; [proposition 149.2](#) isolates a sufficient criterion only.
- 17.4 Closure of the length domains D_q and $\mathcal{G}_M$ under successor, addition, multiplication or code concatenation is not established; bases, code parameters and endpoints $e_q(C)$ are not placed in N_M ; full elementarity and exponential closure of N_M are not claimed.
- 17.5 The variable-base endpoint relation is only shown analytic; the set P_E itself is not proved countable; no uniform effective Borel code or rank is asserted.
- 17.6 The ATR_0 bound is an upper bound for the local theorem only, with stated coding interfaces and Glazer's uniform perfect-set optimization as an input; the global countability results are ordinary ZFC theorems from the Harrington–Marker–Shelah theorem; no reversal, exact strength or global satisfaction predicate is claimed.
- 17.7 The polynomial cuts of [proposition 146.2](#) are not continuous-multiplication examples; the Borel PA-models are used through Enayat–Hamkins–Wcisło's Remark 17(a) and Glazer, and the Malitz–Mycielski–Reinhardt erratum was not inspected.
- 17.8 Glazer's total-exponentiation obstruction, source 14's argument and Glazer's countable cut are credited, not claimed; the perfect-set, partition and Borel-order theorems are inputs.
- 17.9 The finite diagnostics verify no nonstandard, uncountable, fusion or ATR_0 claim; its finite CRT codes do not prove internal code existence in $\mathsf{I}\Delta_0$.
- 17.10 The questions are research proposals, none resolved by its theorems.

Source 18.

- 18.1 AI-assisted review passes only; not refereed and not checked by a proof assistant; historical priority is not established; the completion criterion "should not be described as the resolution of a longstanding community-wide problem without further evidence".

- 18.2 Glazer's unrestricted higher-dimensional Global Choice manifold question is not resolved; its mechanisms are sufficient conditions, not shown to exhaust the routes from Global Choice to sethood, and Global Choice is not shown to supply a neighbourhood operator.
- 18.3 Glazer's Question 2 is not answered anew; the existing Presburger models are background; Borel measurability of partial subtraction is credited to this report's Part III.
- 18.4 The residue corollary assumes continuous addition and partial subtraction; existence of a connected Polish Presburger presentation is not settled.
- 18.5 No theorem for partially ordered cones; the topologies on a Presburger model, the resulting Polish groups, their locally compact factors and the interaction with multiplication are not classified.
- 18.6 Foundation is omitted, not negated; Global Choice, unrestricted Collection and class-length recursion are not assumed; its class theory is not to be identified with other notations in the literature; the ZFU results are context, not premises.
- 18.7 No fixed birthday cutoff for all manifolds or codings; the bound concerns points, not auxiliary parameters; quotient presentations whose points are proper classes are not covered.
- 18.8 The fine surreal topology results are comparisons, not hypotheses of the class-space theorems; that topology is not locally small.
- 18.9 The manifold cardinal bound (Nyikos, through Baillif–Gabard; Van Name), closure coding (YCor, Anschütz), the line with many origins and the long line are classical; the examples are hypothesis tests.
- 18.10 The repository's Lean sign-sequence files are not a formalization of the class-manifold theorem, and its Presburger formalization is not over arbitrary $\mathbb{Z}$ -groups; no finite experiment is offered as proof.
- 18.11 The repository comparison is scoped to the files it names; its questions are proposed projects.

Source 19.

- 19.1 It is a proof-based continuation, not a new claim to the affirmative construction for Glazer's Question 2, and it does not settle his Question 1.
- 19.2 It does not assume or prove that every locally compact Polish Presburger monoid has a locally compact group completion; its [lemma 173.2](#) is restricted to its own cones.
- 19.3 The normal form assumes a countable torsion-free pointed group (E, u) with $E/\mathbb{Z}u$ torsion-free and divisible, the product of the usual and the discrete topology, and $(0, u)$ least positive; the divisible quotient is not assumed to split, and only the rational-vector-space case uses a direct sum.
- 19.4 Borel completeness is for isomorphism of countable structures in an explicit standard Borel space of normal-form codes; it is not completeness for all analytic equivalence relations and says nothing about unspecified raw order codes.
- 19.5 The abstract automorphism formula concerns $\mathbb{R} \overrightarrow{\times} D \overrightarrow{\times} \mathbb{Z}$ for countable divisible ordered D ; the count 2^c needs $D \neq 0$, and arbitrary Hamel shears use choice.

- 19.6 The omnific map has a set-sized domain and finite supports; its topology is transported externally and is not a natural surreal order topology; its image is additive, not a subring; it is not a topological model of full arithmetic or a counterexample to Glazer's theorems.
- 19.7 No claim is made that all uncountable Polish Presburger models are nonrigid.
- 19.8 The finite checks test finite shadows of the proofs; the sign-rule check on $\mathbb{Q}^3$ is not a classification of the orders of $\mathbb{Q}^3$, and no check decides an infinite isomorphism problem, builds a Hamel basis or proves Borel completeness.
- 19.9 Historical priority of the new formulations is not established; the manuscript is unrefereed and not kernel-verified; Friedman–Stanley and Cooper elimination are classical inputs; no Lean file is supplied.
- 19.10 It uses the repository interfaces it read, not the truth of unrelated research claims.

Source 20.

- 20.1 It neither proves Glazer's Question 1 nor claims the repository's affirmative work on Question 2, and it settles no separately identified published conjecture.
- 20.2 The negative theorems need Borel multiplication, or at least the countably many scalar maps used; they do not exclude continuous multiplication with merely Borel addition, and do not say that multiplication is discontinuous when only addition is Borel.
- 20.3 The positive construction is the repository's polynomial cone, reparametrized; it gives no topological group on the signed ring and no continuous truncated subtraction.
- 20.4 The negative statements do not say that dense-exponent arithmetic has no standard Borel representation.
- 20.5 DIV is division with an order-bounded remainder, not a Euclidean norm; finite generation as an exponent group is not finite generation as a monoid of nonnegative exponents.
- 20.6 ZFC proofs give no reverse-mathematical bounds; Part VIII's ATR_0 claim is neither used nor certified.
- 20.7 All theorems concern set-sized cones; neither **No** nor **Oz** is declared a Polish space, and no topology on a proper class is used.
- 20.8 The finite checks test finite formulas, not Polish topologies, Borelness, fusion or measure.
- 20.9 Publication priority and independent review are not established; this report was not fully audited by source 20, and a failed keyword search is not evidence of novelty; no Lean or Rocq proof is supplied.

Source 21.

- 21.1 The field is left-finite over $\mathbb{R}_{\text{alg}}$; it is not the full Hahn field and not the real-coefficient Levi-Civita field, and a coefficient-fixing variant over a larger k is a different claim.
- 21.2 No measurability or continuity hypothesis is imposed on embeddings, and none is claimed to be needed.

- 21.3 No newly resolved printed question of Glazer's is claimed, and no uncountable Polish model of Peano arithmetic is produced.
- 21.4 No automatic extension to a full Hahn field, and no proper-class self-embedding or topology of **No**; the topology of the surreal image is the transported one.
- 21.5 The range map is not claimed continuous for the full Vietoris topology, and $S \subseteq T$ is not claimed to imply $F_S \subseteq F_T$.
- 21.6 The canonical integer part is not preserved by all embeddings, and a transported integer part of the image need not be one of the whole field.
- 21.7 The correction of Kuhlmann–Serra is version-specific (arXiv v3); the journal text was not audited, no other result of the paper is claimed to fail, and no external confirmation or search for errata is asserted.
- 21.8 The finite checks establish no infinite, completion, real-closedness, model-theoretic, basis or novelty claim.
- 21.9 It is unrefereed and AI-assisted; historical priority is not established; no Lean or Rocq proof is supplied, and the Lean interfaces it read are not a proof of its theorems.

Source 22.

- 22.1 It classifies the specified Hilbert family only, not all Polish Presburger models, and does not cover the repository's nonsplit or discrete-direction constructions; the $\aleph_1$ classification must not be extrapolated to them.
- 22.2 It does not claim the affirmative answer to Glazer's Question 2, which it credits to the repository; no contribution or endorsement by Glazer is implied.
- 22.3 The lexicographic example at index ω is not new (Küçük–Soyertem–Küçük; Tan); the 2012 paper was not read in full, so its priority must be compared before any exclusive claim, and no gap in it is claimed.
- 22.4 The oscillation rank is computed directly, with a fixed convention; no Kechris–Louveau comparison theorem is imported.
- 22.5 [Theorem 216.2](#) starts from a bounded operator; formal matrices without boundedness or invertibility are not covered, and not every algebraic order isomorphism is bounded ([example 216.6](#)).
- 22.6 The omnific realization is additive and set-sized with a transported topology, not the surreal order topology, not the coefficient topology and not a topology on the class **No**; no simplicity or initiality claim is made.
- 22.7 The no-ring theorem is narrower than Glazer's topological Tennenbaum theorem and does not replace it.
- 22.8 It neither relies on nor audits Part VIII's ATR_0 claim; its last question is not Glazer's Question 1.
- 22.9 The finite checks are not proofs (Baire category, Borel regularity, transfinite termination, infinite sums, surreal foundations and quantifier elimination are untested); there is no Lean or Rocq proof, no referee and no certified priority; the 379-page report was not read in full, so this is not a repository-wide nonduplication certificate.

Source 23.

- 23.1 “Borel” is the coefficient coding in $\mathbb{R}^\Gamma$, not a non-Archimedean measure theory or the Levi-Civita measure literature.
- 23.2 The results concern set-sized fields; no standard Borel structure is put on **No**; it is not claimed that every set-sized surreal subfield is left-finite, that the surreal exponential preserves K_Γ , or that a particular surreal derivation restricts to it.
- 23.3 It does not claim to solve a remaining question of Glazer’s paper; [proposition 229.1](#) solves neither, and there is no conflict with Glazer’s results.
- 23.4 It does not introduce strongly linear derivations (Kuhlmann–Matusinski); it derives summability from measurability.
- 23.5 Pettis’s lemma, Euler derivations, the residue obstruction and the filtered exponential method are not advertised as new.
- 23.6 The wild derivations use a transcendence basis (choice); the witness is explicit, but no computable global wild map is given.
- 23.7 For infinite rational rank, $\sum_i a_i E_i$ is not an unqualified operator sum.
- 23.8 Positive gain is sufficient, not necessary, for flows; valuation-formal and coefficientwise flows must not be conflated.
- 23.9 The cutoff (M2) exists but is not computed from a Borel code; a certified implementation needs a modulus.
- 23.10 Priority is not established (the search was limited); the manuscript is unrefereed and AI-assisted; there is no Lean development, its formalization section is a dependency plan, and the finite checks are not proofs.

Source 24.

- 24.1 AI-assisted and unrefereed; complete arguments are supplied, but “proved here” means that a proof is supplied, not that no earlier proof exists. Historical novelty of the assembled classification, the threshold and the automatic-regularity formulation is not established; specialist review is needed before any priority claim. Classical normal-form, filtered-exponential and dense-orbit mechanisms are credited, not claimed.
- 24.2 Glazer and the repository’s maintainer are not authors or endorsers; no remaining published question of Glazer’s is claimed solved.
- 24.3 The theorems concern formal series and their coefficient coding, not convergence in t ; coefficientwise analyticity in time is not analytic convergence in t , and nothing is claimed about Stokes or sectorial classification.
- 24.4 Left-finite is narrower than well ordered; only rational rank one ($\mathbb{Z} \leq \Gamma \leq \mathbb{Q}$) is treated, and only ordered automorphisms when Γ is not 2-divisible; all algebraic automorphisms are not classified.
- 24.5 The residue is an invariant of the log-free coordinate group only, not under logarithmic enlargements.
- 24.6 The Borel witness selection is not a transversal or an orbit-invariant normalization.

- 24.7 Borelness is not computability; no algorithm for arbitrary real coefficients or height presentations is given.
- 24.8 The surreal statement concerns one set-sized subfield; nothing is claimed about extensions to **No**, surreal exponentiation, simplicity or birthdays, or any Borel structure on a proper class.
- 24.9 The finite jet checks are regression tests; they prove nothing about unbounded supports, Borel uniformity, nonsmoothness or the no-flow obstruction. No Lean or Rocq development is supplied, and no repository declaration is claimed to verify the new theorems.
- 24.10 The ten questions are not all asserted to be previously unposed or open in every form.

Source 25.

- 25.1 All spaces are set-sized; exponent orders and index sets are countable; real coefficients carry their usual topology and raw arrays the product topology. Nothing is claimed about a valuation topology or a topology on the proper class **No**; complexity is boldface, with no computability of arbitrary order presentations.
- 25.2 The Π_3^0 ceiling for scattered orders is prior art (Guevara–Uzcátegui); “proved here” does not mean “first proved”, and historical priority is not established.
- 25.3 It does not claim to resolve an open problem of Hall, Knight and Lange, whose question is distinct.
- 25.4 It answers only the level and Wadge-degree parts of Part II’s Question 28.3: no effective classifier or certificate extraction from arbitrary decomposition codes, and no reclassification of ideal-isomorphism types.
- 25.5 Glazer’s theorem is motivation, not a lemma; Hahn fields are not, merely as ordered fields, his arithmetic models; certificate spaces are spaces of names, so the continuity results neither evade nor contradict his theorem.
- 25.6 [Corollary 260.3](#) is not an impossibility theorem for restricted-support (for example Puiseux) subfields.
- 25.7 The naming barrier excludes none of: restricted support collections, non-Borel sets of valid names, decoders without coefficient observability, other representation problems; [theorem 266.3](#) does not exclude constructions promised a decomposition or a height table.
- 25.8 Closed certificate spaces do not make validity decidable or height tables effective; certificates are infinite sequences, and no finite regularization stage certifies legality; Σ_2^0 -measurability is not Baire class 1.
- 25.9 The finite checks prove nothing infinitary; there is no peer review, no Lean verification and no rebuild of the repository; the formalization section is a proposed route; the LinkedIn page was not used.

Source 26.

- 26.1 Divisibility of Γ , real coefficients and the coefficient Borel structure are hypotheses; nothing is claimed for nondivisible exponent groups, arbitrary Hahn supports, complex or other coefficient fields, or proper-class ambient structures.
- 26.2 The formal exponential, the BCH correspondence and the uniform-gain case are prior mathematics (Bagayoko–Krapp–Kuhlmann–Panazzolo–Serra, Bagayoko, source 23); the finite-orbit-span argument is not claimed as a new Baire-category principle.
- 26.3 Strict increase of valuation is not escape; windows are not finite-dimensional; diagonal flows are not formally summable, and a Borel flow is not claimed to be a formal operator exponential.
- 26.4 [Theorem 279.1](#) does not exclude algebraic roots of the product, or flows with nonmeasurable time dependence.
- 26.5 The Borel integrability locus is a descriptive classification, not a decision procedure.
- 26.6 The surreal reading is set-sized and conditional on the stated realization; no flow is claimed to extend to **No**, and no Borel structure is put on **No**.
- 26.7 `Trunc.lean` is cited as an interface; no Lean build was run and the classification is not kernel-checked.
- 26.8 No peer review; priority is not established and the search was limited; the finite checks test finite identities only; the supplied LinkedIn page was not used.
- 26.9 No remaining numbered problem of Glazer’s is solved; neither Glazer nor the maintainer is an author or endorser.

Source 27.

- 27.1 AI-assisted and unrefereed; no Lean verification and no certified priority; neither Glazer nor the repository maintainer is an author or endorser.
- 27.2 The dimension-three bound, the affine/projective pattern and the affine BCH formula are classical (Lie; Azad–Biswas–Shah); the proposed contribution is the scalar-coordinate and Borel simultaneous-classification package, the unique witnesses and the ramification refinement.
- 27.3 Set-sized structures in ZFC; no claim about proper-class automorphisms of **No**, no Borel structure on **No**; the transported derivation is not claimed canonical, nor do subfield automorphisms extend to **No**.
- 27.4 Formal series only: no analytic convergence, not a Stokes classification; formal roots and inverse substitutions need not converge as germs.
- 27.5 Not covered: full Hahn supports, all surreals, non-Borel derivations, unordered automorphisms with sign characters, higher rational rank.
- 27.6 The Lie-action boundary assumes coefficient differentiability with a Borel generator; it does not classify all jointly Borel actions of all connected Lie groups. (Batch 96: [theorem 300.3](#) [merge] now classifies, for ordered automorphisms in rank one, the actions whose one-parameter subgroups have Borel time dependence, which include the jointly Borel ones.)

- 27.7 Uniqueness is for named ordered noncommuting pairs; it does not transfer to unmarked subalgebras, unordered tuples or pairs up to linear recombination.
- 27.8 The nonsmoothness contrast is descriptive, not a running-time statement.
- 27.9 The 669 checks are finite regression tests; they verify no Borel, infinite-support or novelty claim.
- 27.10 No named question of Glazer's is claimed solved; the supplied LinkedIn profile was not read and nothing relies on it; the earlier manuscripts are cited as antecedents and the facts needed from them are reproved.

The merge.

- M.1 The affirmative answer concerns Question 2 as printed. Glazer speculated the opposite; this report draws no conclusion about his Question 1 or about any variant of Question 2 with further hypotheses.
- M.2 [Proposition 26.1](#) and the observations of [section 26](#) are elementary consequences of the sources' theorems; they are not presented as new results about Glazer's problem.
- M.3 Citing a collection label does not certify that label: of those cited, only `odg:thm:floor` is formalized.
- M.4 Nothing in this report is formalized. The pointer from the formal project `Logic/PresburgerArithmetic` and the placement in the `Surreal` collection confer no formal status.
- M.5 The reading of Glazer's speculation for Parts III–V in [section 1.1](#) is editorial; no source of Parts III–V draws it, and nothing in them answers Question 1.
- M.6 [Proposition 42.1](#) and the observations of [section 42](#) are elementary consequences of the sources' theorems, not new results on Glazer's problem.
- M.7 The note after [question 14.4](#) re-scopes one sentence of that question; the rest of it, and every other question of the report except where a note says otherwise, stays open.
- M.8 (Batch 87.) Items M.1 and M.5 were written for Parts I–V. Part VIII prints source 14's claimed affirmative answer to Glazer's Question 1; this report has not reviewed that answer and draws no conclusion from it beyond the reading in [section 1.1](#).
- M.9 [Proposition 90.1](#) and the observations of [section 90](#) are elementary consequences of the sources' theorems.
- M.10 The identifications of source 14's K_* with G_{ζ_*} and of its omnific obstruction with `isg:cf:cor:unit` are this report's; source 14 does not make them (source 13 does for its core).
- M.11 Titles flagged "a case of" in Parts VIII and IX compare hypotheses only; the proofs of those sources are printed and are different.
- M.12 The notes that answer or re-scope questions of Parts I, III and IV record what the batch-87 sources prove; those answers are unrefereed and unformalized like the rest of the report. A question not named in such a note, and every part of a named question that a note does not say is answered, stays open.

- M.13 (Batch 89.) The notes of [Part X](#) that identify its results for $\alpha = \omega$ with source 04's and source 07's, and the statements there of what is new to the report, are this report's; source 16 does not make them.
- M.14 [Remark 133.2](#) was written for this report and has not been independently reviewed; it rests on Shelah's consistency theorem [96] as an input and does not decide which weak choice principle suffices for the discontinuous maps.
- M.15 The notes answering Part I's questions on coefficient groups (one clause), lattices and embedding categories record what source 16 proves for its family; those answers are unrefereed and unformalized, and the rest of each question stays open.
- M.16 (Batch 90.) The notes answering Questions 57.2, 88.3 (for Polish monoids), 88.4, 126.3 and the last sentence of 73.11, and superseding the note after [question 138.9](#), record what source 18 proves; that proof is unrefereed and not formalized, and the separate proof review the Hilbert's-tenth programme asked for has not been done. The connected-carrier questions stay open.
- M.17 [Propositions 155.6](#) and [155.7](#) were written for this report and have not been independently reviewed; they combine source 18's criterion with [corollary 78.6](#) and [corollary 121.2](#) and inherit their status.
- M.18 The identifications of source 17's lemmas with source 14's and of source 18's examples with Parts I and III are this report's. The reading of Glazer's manifold sentence with "connected" is source 18's; the comparison with his wording is this report's.
- M.19 The note after [question 115.2](#) records a partial answer, under hypotheses incomparable with source 14's; source 17's ATR_0 bound is printed as a claim.
- M.20 (Batch 92.) The notes answering Part II's [question 28.7](#), Part IV's [question 57.3](#) for $d = 1$, Part IX's [question 126.5](#) and [question 126.4](#), and in part [question 126.6](#), [question 73.8](#), [question 14.12](#), [question 103.5](#) and [question 43.2](#), record what sources 19, 20 and 21 prove, with [section 182](#); those proofs are unrefereed and not formalized, and every part of a question that a note does not say is answered stays open.
- M.21 [Propositions 182.1](#) to [182.3](#) and [corollary 182.4](#), [remarks 176.3](#) and [182.5](#), the proof of source 20's countable $\text{DIV} + \text{Root}_2$ example and the second counterexample in [remark 204.2](#) were written for this report and have not been independently reviewed; they inherit the status of the results they combine.
- M.22 The identifications of source 19's results with Parts I, IV, VII, IX and XII, of source 20's with Parts II, III and V, and of source 21's with Part V are this report's; none of the three sources read this report's article.
- M.23 [Remark 204.2](#) concerns the preprint arXiv:2107.03362v3 only; it does not say that the lemma fails for the maximal Hahn field or under further hypotheses, and the journal version was not checked.
- M.24 (Batch 93.) The notes answering Part I's [question 14.6](#) for separable Hilbert spaces, and in part [question 14.14](#), [question 43.7](#) and, in a variant, [question 138.7](#), record what sources 22 and 23 prove with [sections 224](#) and [239](#); those proofs are unrefereed and not formalized, and every part of a question that a note does not say is answered stays open.

- M.25 [Propositions 224.1, 224.2 and 224.4 to 224.6](#), [remark 224.3](#), [propositions 239.1 to 239.3](#) and [remarks 239.4 and 239.5](#) were written for this report and have not been independently reviewed. [Remark 224.3](#) uses the axiom of choice.
- M.26 The identifications of source 22's results with Parts I, III, IV, IX, X and XIII, and of source 23's with Parts V, XIV and XV and with other reports, are this report's; neither source read this report's article.
- M.27 The reading of the Küçük–Soyertem–Küçük abstract asserts nothing about that paper's theorems, whose hypotheses were not read.
- M.28 (Batch 96.) The notes answering Part XVII's [question 237.4](#) in rational rank one and [question 237.5](#) in rational rank one and for divisible Γ , Part II's [question 28.3](#) in part, and a case of Part XV's [question 208.5](#), record what sources 24–27 prove with the four merge sections; those proofs are unrefereed and not formalized, and every part of a question that a note does not say is answered stays open; in particular the decomposition clause of [question 28.3](#) stays open.
- M.29 [Proposition 255.1](#), [remarks 255.2 and 255.3](#), [proposition 270.1](#) and [remark 270.2](#), [propositions 285.1, 285.3 and 285.4](#) and [corollary 285.2](#) and [remarks 289.2 and 294.5](#) were written for this report and have not been independently reviewed. [Lemma 300.2](#) and [theorem 300.3](#) were written for this report and checked once by an independent reading made for it (correct, with editorial fixes applied); they are unrefereed. The lemmas of [remarks 255.4 and 300.4](#) and the two observations in [section 270](#) are the Hilbert's-tenth review's, re-checked by the write.
- M.30 The identifications of sources 24–27 with Parts II, V, XV and XVII and with each other are this report's; none of the four read this report's article. Source 25's statement about `PowerSeriesSummable` has no Lean counterpart, source 27's affine BCH identity is a Lean theorem only in its matrix form, and placement beside Lean files confers no formal status.
- M.31 Example 6.3 of Mardešić–Resman–Rolin–Županović was read in arXiv:1505.05929v2 only ([remark 255.5](#)); the preprints of Bagayoko–Krapp–Kuhlmann–Panazzolo–Serra, Bagayoko and Azad–Biswas–Shah and the papers of Guevara–Uzcátegui and Hall–Knight–Lange were not re-read ([question 268.12](#)).

G An explicit induction check after quantifier elimination (source 07)

For readers concerned that discrete order and division might conceal a missing induction argument, the standard quantifier-elimination input also yields a direct least-element proof.

Let G be any $\mathbb{Z}$ -group with positive cone M . After quantifier elimination, a unary definable set with parameters is a finite Boolean combination of linear inequalities and congruences. Division by fixed standard coefficients changes each inequality into an upper or lower bound on the variable. Taking a common standard modulus and partitioning at finitely many endpoints writes the set as a finite union of cells

$$I \cap \{x \in M : x \equiv r \pmod{d}\},$$

where I is an arithmetic interval, possibly with an infinite end. Empty cells may be discarded.

Every nonempty cell has a least element. Intersect with M and replace a strict finite lower bound by its successor; if there is no effective lower bound, use zero. This gives a closed lower endpoint $a \in M$. Choose the unique standard $t \in \{0, \dots, d-1\}$ such that $a + t \equiv r \pmod{d}$. Then $a + t$ is the first element of that residue class at or above a . If the cell is nonempty, $a + t$ also satisfies its upper bound, since it precedes any other candidate in the cell. Thus it is the cell's minimum. Taking the least of finitely many minima shows that every nonempty unary definable subset of M has a minimum.

Now let a definable $D \subseteq M$ contain zero and be closed under successor. If its complement is nonempty, choose its least element b . Then $b > 0$ and its predecessor $b - 1$ lies in D . Successor closure puts b in D , a contradiction. This proves every first-order induction instance with parameters. The only imported step is classical quantifier elimination, explicitly identified in Section 31; no topological assumption enters.

[write] This is a second route to the induction clause of [corollary 4.11](#); source 06's [appendix I](#) and source 08's [theorem 49.3](#) prove the elimination itself again.

H Source audit, proof dependencies, and scope (source 07)

H.1 Primary source checks

The supplied LinkedIn page could not be read directly during this investigation. The selection of topic rests on Glazer's named mathematical paper, whose full text was inspected, rather than on inferred biographical details. The exact wording and number of its Presburger question were checked on its final page.

The literature check covered the current arXiv record, the earlier Enayat–Hamkins–Wcisło paper and author page, and targeted searches combining Presburger arithmetic, Polish spaces, Glazer, and continuous addition. No later resolution of the cited question was located. This is a report of a limited search, not a certification that the problem remained open until the date of this manuscript.

General Baire presentations with low-complexity definable relations are already part of the background discussed in [2]. The additional content here is continuous addition together with explicit coordinate models, their ordinal and topological invariants, and their omnific realizations.

H.2 Repository inspection and duplication boundary

At the pinned snapshot, a search of the checked-out `.tex`, `.md`, and `.lean` files found no occurrence of “Glazer” or “Tennenbaum”. The initial-subgroup report does cite the earlier topological-arithmetic paper, but the inspected material does not give the Polish Presburger constructions used here. This statement concerns the searched text formats and snapshot; it is not a search of every historical branch or every binary artifact.

[write] *Correction of a stale statement.* The search result above was true at source 07's pin `0f95145cb`. It is no longer true. The word “Glazer” first occurs, in archive names, in the Hilbert's-tenth programme's scope note `Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/review_recent_incoming_glazer_scope.md` (commit `0f3b760ae`), and Glazer's paper, whose title names Tennenbaum, is discussed in this report since its placement commit `3d2177df4`. Part I there gives the real and Baire models of [sections 32](#) and [35](#) independently of source 07.

The exact formalization boundary was checked in:

- `Logic/PresburgerArithmetic/Lean/PresburgerArithmetic/Syntax.lean`: the interpretation `Formula.holds` uses ordinary integer valuations.
- `Logic/PresburgerArithmetic/Lean/PresburgerArithmetic/Decision.lean`: the elimination correctness theorem quantifies over `List Int`.
- `Logic/PresburgerArithmetic/Lean/PresburgerArithmetic/Audit.lean`: the file requests dependency reports for the existing integer theorems.

The repository was used read-only. No claim is made that this report has been merged into it or that any new formal theorem has been accepted by its kernel.

H.3 Dependency structure of the new arguments

Conclusion	Required ingredients
Affirmative real model	Lexicographic coordinate algebra; classical $\mathbb{Z}$ -group theorem; an explicit Polish topology
General cone regularity	Baire category; regularity of Borel sets; countable bases; closure and transfinite recursion
General Polish realization	Cone regularity; group division; classical Presburger quantifier elimination
Baire-space models	Countable products; direct G_δ cone computation; noncompact cylinders; Baire-space characterization
Ordinal nonisomorphism	Leading-coordinate comparison; group completion; Archimedean classes
Oscillation calculation	Finite-coordinate neighborhoods; relative closures; countable ordinal arithmetic
No ordered-semiring expansion	Distributivity; positivity; a cofinal family of standard multiples
Omnific realization	Set-sized Conway normal forms; coefficientwise addition and leading-term order

The real example does not depend on the general cone theorem, the transfinite Baire family, or surreal normal forms. The direct proof of the Baire family's sign-cone complexity likewise makes it independent of the general cone theorem. These independent routes reduce the chance that a single auxiliary claim could undermine the positive answer.

H.4 Review status

The coordinate model, the Borel-cone recursion, the transfinite Baire family, and the boundary oscillation calculation were subjected to separate mathematical proof checks during preparation. Those checks specifically examined induction, completeness versus the inherited metric, transfinite leading coordinates, limit stages, and the distinction between partial and truncated subtraction. They are independent reasoning checks, not external refereeing or proof-assistant certification.

No finite numerical experiment is offered as evidence for an infinite model-theoretic theorem. The deliverable consists of the proofs, identified classical inputs, exact source references, and a reproducible LaTeX source. Compilation and visual inspection verify the document, not the mathematics.

I A self-contained Presburger elimination argument (source 06)

This appendix proves the algebraic characterization used in the main constructions. It is included to make the induction claim fully checkable without relying only on a reference to a characterization theorem.

I.1 Division in a $\mathbb{Z}$ -group

Let G be a discretely ordered abelian group with least positive element 1 and divisible quotient $G/\mathbb{Z}$. Fix a positive standard integer n . For each $g \in G$, divisibility of the quotient gives $h \in G$ with $g - nh = k \cdot 1$ for some standard integer k . Write $k = nq + r$ in ordinary $\mathbb{Z}$, where $0 \leq r < n$. Then

$$g = n(h + q \cdot 1) + r \cdot 1.$$

This representation is unique: if two remainders r, r' in the specified finite interval differ by n times a group element, discreteness forces that element to be zero. Explicitly, a positive difference would be at least n , and a negative one at most $-n$, whereas $|r - r'| < n$.

Conversely, this division property makes $G/\mathbb{Z}$ divisible. Thus we can work with the ordered-group axioms, discreteness, and this schema of standard division algorithms.

I.2 Eliminating one quantified variable

Expand the language by predicates $D_n(u)$, for standard $n \geq 1$, with the interpretation $u \in nG$. Every group term is integer linear. To eliminate an existential quantifier, first put its quantifier-free matrix in disjunctive normal form and treat each conjunction separately. Split disequalities into the two strict inequalities and use discreteness to replace weak inequalities by strict inequalities with a shift of 1. Atoms not involving the quantified variable are retained as parameter conditions.

Suppose the variable is x . Let L be the least common multiple of the nonzero absolute coefficients of x in the conjunction, or 1 if there are none. Replace Lx by y and add $D_L(y)$. For an inequality or equality involving $ax + u$, multiply by the positive integer $L/|a|$; its coefficient of y becomes 1 or -1 . For a divisibility literal, use

$$D_n(ax + u) \iff D_{nm}(\text{sign}(a)y + mu), \quad m = L/|a|. \quad (\text{I.1})$$

This is valid because, in a torsion-free group, $v \in nG$ if and only if $mv \in nmG$. The same equivalence handles negated divisibility. Thus every occurrence of y has coefficient 1 or -1 .

If an equality specifies y , substitute its value. Otherwise the conjunction consists of finitely many lower bounds $a_i < y$, upper bounds $y < b_j$, and positive or negative congruence conditions, together with parameter-only literals.

Let N be a common multiple of all congruence moduli, including L , and split into the finitely many possibilities $y \equiv r \pmod{N}$ for $0 \leq r < N$. In each branch, a condition $D_n(\pm y + u)$ is equivalent to the parameter condition $D_n(\pm r + u)$. Only the one residue condition modulo N and the bounds now involve y .

If lower bounds are present, divide into finitely many cases choosing their greatest value a ; this is expressed by parameter comparisons. Among

$$a + 1, a + 2, \dots, a + N$$

there is exactly one member of the chosen residue class. It is the least member of that class above a . To justify minimality, if $y > a$ has the same residue and lies below that candidate, their positive difference is a multiple of N smaller than N , contradicting discreteness. Thus, when upper bounds are also present, existence is equivalent to the finite disjunction

$$\bigvee_{k=1}^N \left(D_N(a + k - r) \wedge \bigwedge_j (a + k < b_j) \right).$$

If there are no upper bounds, the candidate always works. If there are upper bounds but no lower bounds, a residue-class element exists among the N positions just below the least upper bound. If neither type of bound is present, the standard representative r itself works.

This eliminates one existential quantifier. Universal quantifiers are handled by negation, and induction on formulas gives quantifier elimination in the expanded language. The output consists of Boolean combinations of integer-linear comparisons and standard congruences.

I.3 Completeness and induction in the nonnegative cone

A closed quantifier-free formula in this language concerns only standard numerals and their congruences and comparisons. Its truth is the same in every $\mathbb{Z}$ -group and in ordinary $\mathbb{Z}$. Consequently every $\mathbb{Z}$ -group is elementarily equivalent to $\mathbb{Z}$.

The structure $G_{\geq 0}$ is interpreted in G by restricting quantifiers to $x \geq 0$. Its addition, constants, and order are the restrictions of those in G . Every sentence true in $(\mathbb{N}, 0, 1, +, <)$ therefore holds in the cone.

In particular, for every additive formula $\varphi(x, \bar{a})$ the sentence

$$\forall \bar{a} \left[(\varphi(0, \bar{a}) \wedge \forall x (\varphi(x, \bar{a}) \rightarrow \varphi(x + 1, \bar{a}))) \rightarrow \forall x \varphi(x, \bar{a}) \right]$$

holds. Parameters are covered because their universal closure is part of the sentence. Multiplication is absent from this language; the argument therefore makes no assertion about the polynomial model's formula $x^2 < t$.

[write] This is a second route to [theorem 4.9](#) and the induction clause of [corollary 4.11](#), by the finite search of [lemma 4.7](#).

J Provenance and verification record (source 06)

Mathematical dependencies. The main constructions use ordered-group algebra, elementary Presburger elimination, standard facts about Polish spaces, and explicit coordinate arguments. The isolated-zero theorem additionally uses analytic/coanalytic separation and fair product measure. Its tail zero-one step is proved in the text. The surreal realizations are explanatory embeddings and are not premises of any of the main topological theorems.

Independent scrutiny. The real and Baire models, the polynomial topology, the distinction between additive and open induction, and the isolated-zero argument were subjected to separate proof reviews during preparation. Those reviews specifically checked complete metrizable rather than mere separability, negative integer slices, the parameter version of induction, joint continuity, cancellation, and the tail-event invariance. The final proof omits an unnecessary Cantor-injectivity claim and never assumes continuous subtraction. These reviews are mathematical checks, not formal proof-assistant certification or external peer review.

Repository audit. The repository comparison used its path inventory and selected relevant files at the pinned commit. No full-repository theorem audit or Lean rebuild was performed. Existing exploratory reports are credited for their stated algebraic content, with the main results here proved independently.

Reproducibility. This source is self-contained and uses standard $\text{\LaTeX}$ packages. It can be compiled with `latexmk -pdf polish_presburger.tex`. No external image, private data, bibliography processor, or downloaded repository is needed to reproduce the PDF. The accompanying ZIP contains this source, its compiled PDF, and a short build-and-status readme.

[write] Source 06 shipped only its $\text{\LaTeX}$ source, its PDF and a short plain-text readme. None of them is shipped here (the source is merged into this report; the archive survives in commit fb8414869); there is no program to rerun.

K Contribution ledger, source audit, and reproducibility (source 08)

K.1 Contribution ledger

Statement or ingredient	Status in this manuscript
Compact-group order obstruction; finite-dimensional separation; LCA principal structure theorem	Classical material, credited in section 47 . The first two are proved in the needed form. The LCA principal structure theorem is an explicitly imported theorem.
Presburger quantifier elimination	Classical. A complete finite-witness argument is supplied to control the nonstandard-group transfer.
The real positive-cone example $M_{\mathbb{R}}$	Previously constructed in the companion manuscript [40] ; not claimed as new.
Integer-disjoint order-unit amplification and ordered direct sums	Derived and proved here, theorem 46.2 ; global priority unverified.
Arithmetic rigidity without multiplication regularity	Derived and proved here, theorem 44.1 ; global priority unverified. No induction assumption.
Pointed group classification; cone dimension threshold	Derived here from the classical structure inputs. Not a classification of all full orders or all topological monoids.
Uniform Δ_2^0 definability and continuous standard division	Derived for the classified presentations. Standard elimination and topological facts are not claimed as new.
Countable Borel-scalar subring; exact answer for $E = \mathbb{Z}$	Derived and proved here, theorem 53.4 ; does not assume order-compatible multiplication.
General Polish-group version; full order classification; Lean proof	Not established here. Listed as further research.

K.2 What was actually inspected

The repository snapshot used for the interfaces is

883e0b3b22263bd9d92fdbbf10376ca4b652b862.

The inspected guides were the root `README.md`, `Logic/PresburgerArithmetic/README.md`, and `Algebra/SurrealNumbers/README.md`. The first arithmetic guide identifies the syntax, normalization, Cooper, elimination, and decision layers and their stated audit scope. The surreal guide distinguishes constructed objects, mapped formal results, and unrefereed research reports. Those guides were used to identify interfaces, not as evidence that the new theorems in this manuscript have already been formalized. No repository files were modified, and no full repository build was run.

The Glazer source was the publicly available 2023 arXiv version of [1]; the supplied LinkedIn profile was not needed as mathematical evidence. Classical cross-checks used the primary papers or author notes listed below, including the measurable-order discussion in Section 3 of [22] and the LCA structural statement on page 4 of [27]. The bibliographic entry for [26] is recorded through its explicit discussion and citation in [22]; no claim of an independent full reading of that 1983 paper is made. The prior companion manuscripts were retrieved from the user's library.

K.3 Finite verification

The included `verify.py` uses Python's standard-library `fractions.Fraction`. Its output on the supplied version is:

Test family	Cases passed
Standard division in rational samples of $\mathbb{R}^d \times \mathbb{Z}$	7,595
Coset-invariant finite-witness tests	9,000
Polynomial amplification tuples	1,875
Lexicographic order triples	6,000
Total	24,470

The finite-witness tests compare the candidate construction with an independent oracle on rational lexicographic vector coordinates and integer intervals. The polynomial tests compute in $\mathbb{Q}[X]$, with $V = \mathbb{Q}X$ and u a positive rational multiple of X ; they check finite injectivity, the highest-coefficient sign rule, and lower-term domination. All comparisons are exact, with fixed random seeds for the sampled tests.

These checks can catch sign, indexing, and residue mistakes. They cannot establish any statement involving all real points, arbitrary Borel sets, local compactness, Haar measure, or infinite cardinalities. The mathematical proofs, not the tests, carry those conclusions.

K.4 Build instructions

The bundle contains the source and compiled PDF, this verification script, its JSON output, a source-provenance file, and a short build script. With a standard TeX Live installation including the New PX packages:

```
latexmk -pdf -interaction=nonstopmode -halt-on-error article.tex
python3 verify.py --output verification_results.json
```

The PDF should then be rendered and visually checked. No external font files, third-party source PDFs, or unexecuted Lean proof claims are included.

[write] In this report the program is `code/08-local-compactness-verify.py`, its recorded output `data/08-local-compactness-verification_results.json`, the build script `code/08-local-compactness-build.sh`, and the source-provenance file and proof audit `08-local-compactness-sources.md` and `08-local-compactness-proof_audit.md`. They are byte-identical to the delivery and still use the delivery names `verify.py`, `verification_results.json` and `article.tex`; the build script builds a file `article.tex` in its own directory and must not be run in the report directory. The intake reran the program on a copy with Python 3.14.4: the four families and the total of 24,470 cases agree with the shipped record, which differs from the new output only in line endings.

L Contribution ledger and proof dependencies (source 09)

Statement or ingredient	Status and dependency boundary
Pettis, automatic continuity, open mapping	Classical tools; short proofs are included in Section 61. Their use is not claimed as new.
Local order bounds and forced order topology	Proved here from the category tools, with Baire order and scalar hypotheses explicit. Priority of this formulation has not been certified.
Full Hahn classification	Central presentation-invariant deduction, Theorem 63.3; depends only on the order bridge and the explicit separated binary family, plus classical existence of Hahn fields.
Proper-subfield cover and Puiseux obstruction	Proved here, Theorems 64.1 and 64.3; the measurability of covering subfields is essential and is verified in the application.
Levi-Civita field and completeness	Classical construction; explicit proofs are supplied. The uniqueness and comparison with non-Polishable Puiseux arithmetic use the category bridge.
Exact code domains	Explicit FIN and row-finiteness reductions are given. The full Hahn coanalytic obstruction is credited to classical descriptive methods and the preceding project report, not reclaimed as new.
Real closedness	Classical Newton–Puiseux input plus a supplied proof that completion preserves algebraic closedness. Not a new algebraic theorem.
Polish hulls and exact cover number	Proved here using countable dense field cores, prefix stabilization, and the fixed-scale binary obstruction. The hull construction uses a standard completion mechanism; the local/global classification is explicitly developed in Section 69.

Statement or ingredient	Status and dependency boundary
Countable integer parts	A consequence of the forced order topology, with no measurability assumption on the integer part.
Floor formula	Existing omnific normal-form principle, proved here in the restricted field for clarity.
Floor regularity	Explicit continuous approximants and exact discontinuity sets for Levi-Civita and strengthened Polish hulls, Theorems 70.2 and 70.4; not a kernel check.
Open induction	Classical integer-part implication, proved directly for the constructed ring; no uncountable arithmetic model is asserted.
Executable verification	Finite rational arithmetic regression tests only. They do not verify the set-theoretic or topological quantifiers.

The logical core can be read as

$$\begin{aligned} &\text{Pettis} \implies \text{local order bounds} \\ &\implies \text{order-topology refinement} \\ &\implies \left\{ \begin{array}{l} \text{full Hahn classification,} \\ \text{countability of integer parts,} \\ \text{uniqueness of complete order presentations.} \end{array} \right. \end{aligned}$$

The proper-subfield-cover theorem is a separate branch from Pettis and automatic continuity. Descriptive completeness and real closedness are independent supplementary branches; neither is needed for the main Hahn classification.

M Source audit and reproducibility (source 09)

Research scope. The research-interest overlap was based on Glazer’s mathematical paper [1], not on a speculative interpretation of a social-media profile. The article does not imply that he reviewed, collaborated on, or endorsed this manuscript. The public LinkedIn page supplied in the request was not needed to substantiate the mathematical connection.

Repository scope. The root and surreal READMEs were inspected, together with repository metadata and the directory structure of the surreal implementation. The repository head recorded during the audit was

4af956fe015e043c5e163ee5b4d2cea23e6157ea

This pin is a reproducible repository locator, not a claim to have rebuilt the project at that commit. The main proofs here are self-contained and do not depend on assuming a repository theorem beyond the explicitly credited normal-form background. The large formalization ledger could not be fetched through the available connector; no claim-by-claim audit of that ledger is represented as completed.

Earlier reports. The Library copies of the previous Polish Presburger and Borel-support manuscripts were checked at the abstract/contribution level to avoid presenting the same results as new. The present article does not use their proposed theorems as unproved premises. References [43, 44] are unpublished project manuscripts, not independently validated literature.

Literature scope. The primary mathematical sources below were accessed during preparation. The review was targeted, not exhaustive. The sources establish the classical inputs and the research connection; the priority of the combined classification and its refinements remains open to specialist review. In particular, no absence-of-search-result argument is used as evidence of novelty.

Build and checks. The archive contains this self-contained `article.tex`, the compiled PDF, `verify.py`, its machine-readable output `verification.json`, a source record, and a build README. Compile with a standard TeX Live installation using

```
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
python3 verify.py --output verification.json
```

The Python program uses only the standard library. It records deterministic finite test counts and fails on a violated identity. No numerical tolerance is used: all tested coefficients and exponents are exact rational numbers. The PDF was also rendered for layout inspection. Font packages are referenced by the TeX source; no font files are distributed in the archive.

[write] In this report the program is `code/09-hahn-polishability-verify.py`, its recorded output `data/09-hahn-polishability-verification.json`, the build script `code/09-hahn-polishability-build.sh`, and the source record `09-hahn-polishability-SOURCES.md`; they are byte-identical to the delivery and use the delivery names. The intake reran the program on a copy with Python 3.14.4: 20,116 assertions in 29 families with seed 20261003, equal to the shipped record up to line endings. Source 09's bibliography and source record cite the surreal README and root README by unversioned `blob/main` addresses, beside the pin above; the pinned versions are <https://github.com/VladimirReshetnikov/ProveIt/blob/4af956fe015e043c5e163ee5b4d2cea23e6157ea/Algebra/SurrealNumbers/README.md> and the root README.md at the same commit. Both files are unchanged between the pin and the placement commit `0bd0e5527`.

N Reproduction and the scope of the exact checks (source 12)

[write] Source 12's checks are shipped as `code/12-baire-category-verify.py`, which writes `verification_results.json` beside itself; the recorded run is `data/12-baire-category-verification_results.json`. Run it on a copy (README). The build commands below name the delivered `article.tex`, whose text is merged into this report.

Run `python verify.py` with Python 3.10 or later. The script uses only the standard library and exact rational arithmetic. Its random seed is fixed at 20261003. The accompanying run completed **36,989 exact checks**, all passing.

The checks exercise leading-coefficient order, the least-positive-unit property, positivity and absolute values of products, distributivity, the separation inequality (87.1), difference-ring products, finite convolution locality, degree escape from principal convex subgroups, and the finite identity (86.2). Coefficients are rational and all supports are finite. The tested ring is $\mathbb{Z} + t\mathbb{Q}[t]$, not an approximation whose floating-point comparisons might obscure signs.

Check family	Count
Packing separation	2,399
Distinct-point unit separation	2,399
Convex-quotient degree escape	1,618
Sampled convex-quotient bounds	6,472
Finite geometric inverse identities	101
Ten further algebraic families, 2,400 each	24,000
Total	36,989

Some candidate pairs are equal, so the two separation checks have 2,399 instances rather than 2,400; equal pairs are correctly excluded by the theorem hypothesis. Similarly the quotient check excludes equal pairs. The JSON output records the full family breakdown.

These checks can detect implementation mistakes in the examples and sign errors in formulas. They cannot establish the statements quantified over all rings, all points of an uncountable space, or arbitrary Baire sets. Those statements rest on the proofs in the body of the article.

The source compiles with an ordinary LaTeX installation including `newpx`, `tcolorbox`, and `cleveref`. Run

```
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

A third pass may be used after substantial changes to the contents. The bibliography is embedded in the source, so no separate BibTeX step is needed.

O Provenance and contribution ledger (source 12)

The public research anchor is Glazer's 2023 arXiv paper. The supplied LinkedIn page was not available for inspection, so no employment, affiliation, or personal-biography claim is inferred from it. The research overlap is grounded in the paper's mathematical content.

The earlier local-compactness report was read from the user’s Library, including the exact research-question passage in its LaTeX source. It is an earlier AI-assisted project manuscript, not a peer-reviewed reference. Its role is to identify a concrete previously unanswered project question; the proof given here is independent of its main theorem. The Presburger companion is cited to avoid duplicating its construction or claiming it again as a new result.

The ProveIt interface review used the pinned repository revision stated in [section 87](#). The relevant README blob hashes are `a29a6b54c98ae5dbf5e3267a96f10dc5e1514a79` for Presburger arithmetic and `4f830b663aa28748468a9382829d0e4d90e7f8ba` for the surreal development. Reading those interfaces is not an independent check of the whole repository.

Targeted searches combined discrete ordered rings with Polish groups, Baire category, measurable order, and local compactness. They did not identify a source establishing the exact general theorem formulated here. Such a search does not prove absence from the literature or establish priority. In particular, the bounded-neighborhood lemma may be viewed as an application of classical regularity methods; its presentation here does not make those methods new.

Statement or ingredient	Status in this manuscript
Pettis and Steinhaus difference principles	Classical; needed forms proved explicitly
Bounded-neighborhood reflection	Proved here; no priority assertion
Packing and $c = d = R $	Principal established result
Removal of local compactness	Resolves the identified earlier project question
Ambient Haar-measurable version	Proved without hereditary restriction assumptions
Convex quotient and unit spectrum	Additional proved algebraic/topological consequences
Polish positive monoids in full generality	Not settled; group-completion boundary retained
Glazer’s ATR_0 question	Not settled
Lean/Rocq verification of this article	Not performed
Historical breakthrough or priority	Not certified

[write] “The earlier local-compactness report” is source 08, [Part IV](#); “the Presburger companion” is source 05, [Part I](#). The README blob hashes are those of `Logic/PresburgerArithmetic/README.md` and `Algebra/SurrealNumbers/README.md` at source 12’s pin `0f084afa9`.

P A semantic Presburger transfer argument (source 10)

This appendix justifies the abstract arithmetic language used in [section 81.3](#) without treating a decision procedure on ordinary integers as an automatic theorem about all ordered groups.

Let G be a $\mathbb{Z}$ -group. For a positive ordinary integer m , a predicate $P(x)$ will be called *coset-periodic modulo mG* if

$$x - y \in mG \implies (P(x) \leftrightarrow P(y)).$$

This is stronger than merely knowing $P(x + m) = P(x)$ and iterating the identity an ordinary finite number of times. The distinction matters in a nonstandard group: an arbitrary element of mG need not be an ordinary integer multiple of m .

Lemma P.1 (Finite interval reduction [10]). *Suppose P is coset-periodic modulo mG , and L, U are finite lists of elements of G . Then*

$$\exists x ((\forall \ell \in L \ell < x) \wedge (\forall u \in U x < u) \wedge P(x))$$

is equivalent to a disjunction of at most m candidate conditions, once the relevant finite extremum of the bounds is chosen. This is a logical reduction; an effective algorithm additionally requires effective comparison and evaluation of P .

Proof. If L is nonempty, put $b = \max L$ and test $b + 1 + r$ for ordinary $0 \leq r < m$. Given a witness x , discreteness gives $x \geq b + 1$. Divide $x - (b + 1)$ by m in G :

$$x - (b + 1) = mq + r, \quad 0 \leq r < m.$$

Necessarily $q \geq 0$: otherwise $q \leq -1$, so $mq + r \leq -m + (m - 1) < 0$. Thus $y = b + 1 + r \leq x$, and y lies above every lower bound and below every upper bound. Since $x - y \in mG$, $P(y)$ holds. The converse is immediate because every accepted candidate is a witness.

If L is empty and U is nonempty, use $b = \min U$ and test $b - 1 - r$, dividing $b - 1 - x$ for a given witness. This moves the witness upward without crossing an upper bound. If both lists are empty, test the standard residues $0, \dots, m - 1$, since every element of G is congruent to exactly one of them modulo mG . $\square$

To eliminate a quantified variable from a Presburger formula, first put the formula in disjunctive normal form. Terms are affine integer combinations of variables. Negated equalities and inequalities can be converted to finite combinations of strict bounds; discreteness permits the usual shifts by 1. Congruence conditions and their negations form a Boolean combination invariant on the cosets of a suitable mG .

For completeness, coefficients of the quantified variable can be normalized by choosing a positive ordinary common multiple k of their nonzero absolute values, substituting $y = kx$, and imposing $y \in kG$. An inequality $ax < t$ with $a > 0$ becomes $y < (k/a)t$ after multiplication by the positive integer k/a ; negative coefficients are handled by reversing the bound. A divisibility atom $d \mid ax + t$ becomes

$$dk \mid ay + kt.$$

Under $y = kx$ this equivalence follows by cancellation of k in the torsion-free group. The new congruence Boolean combination, including $k \mid y$, is invariant on cosets of mG for a common multiple of its ordinary moduli, even when the quotient elements are nonstandard. **Lemma P.1** now eliminates the existential quantifier. Recursion eliminates all quantifiers; universals are handled by negation. The finite maximum or minimum choice can itself be replaced by a finite disjunction specifying which listed bound is extremal; it does not add a new primitive operation to the language.

A resulting closed quantifier-free sentence has only ordinary integer constants. Their comparisons have their usual truth values, and an ordinary divisibility equation has a group solution exactly when it has an ordinary integer solution: any such quotient is ordinarily bounded and therefore an ordinary integer. Hence every $\mathbb{Z}$ -group has the same signed first-order arithmetic theory as $\mathbb{Z}$.

In the examples G_D , this proof also shows that every definable relation with parameters is Δ_2^0 . Integer affine maps are continuous, order comparisons are Δ_2^0 , and $mG_D = D \times m\mathbb{Z}$ is clopen. Finite Boolean combinations preserve Δ_2^0 . Standard quotient and remainder maps are continuous as well, directly from (81.4). These are explanatory consequences of the construction, not additional claims of historical novelty.

[write] [Lemma P.1](#) is source 08's [lemma 49.2](#) and the coset-invariant finite search of Part I ([lemma 4.7](#)), proved again.

Q Proof audit and reproducible checks (source 10)

Q.1 The shortest proof chain

The entire Polish obstruction has the following dependency chain:

$$\begin{aligned} \text{Baire group} + \text{Baire positive cone} &\implies \text{bounded neighborhood} \\ &\implies \text{open principal convex hull } H_a \\ &\implies |A/\mathbb{Z}1| \leq |A/H_a| \leq \aleph_0 \\ &\implies A \text{ countable} \implies A \text{ discrete.} \end{aligned}$$

The middle injection is algebraic. The first and last steps are category arguments. The countable quotient is topological. No step requires applying a continuity or measurability theorem to multiplication.

Q.2 Potential mistakes explicitly excluded

There is no claim that an order interval is a subgroup, that a principal convex subgroup is an ideal, or that an arbitrary scalar map is continuous. Ordinary finite bounds are not internal nonstandard bounds. The group topology includes continuous negation. Baire-property order is assumed for the given topology, not an unrelated transported one. Countability of a quotient is obtained from open cosets, not from an uncountable union of small fibers. The last use of Baire is essential: a countable metrizable group need not be discrete.

The quotient injection is well defined because the kernel is computed exactly, not merely bounded from above. Order preservation of the quotient uses convexity. The cardinal equalities for uncountable rings use $|A/\mathbb{Z}1| = |A|$, not an assumption that quotienting by a countable subgroup always preserves every finite or countable cardinal.

Q.3 Finite exact-arithmetic tests

The companion `verify_examples.py` uses rational coefficients and deterministic random inputs. It checks the principal-hull kernel identity in $\mathbb{Z} + X\mathbb{Q}[X]$ with $a = X^k$, leading-sign preservation for bounded-subgroup amplification with $H = \mathbb{Z} + \mathbb{Q}X$, $c = X^2$, $a = X^2 + 2$, and standard division/remainder identities in finite rational-coordinate models of G_D . It also checks the truncated geometric-inverse identity underlying [proposition 86.2](#).

The tests use exact rational arithmetic. They do not verify category, uncountability, arbitrary orders, vector-space basis choices, or any infinite theorem. They are regression checks for formulas and examples; the proofs in the article are the justification of the theorems. The generated JSON result records the seed, counts, and scope.

Q.4 Source and priority audit

The external research anchor is Glazer’s arXiv paper, with its theorem, corollary, and question numbering checked against the text. The LinkedIn URL supplied in the request was not used to infer unverified personal or current-employment details. The selected intersection is supported by the published paper itself.

The ProveIt snapshot inspected for this article is

6fef5383b126be343cccc9baf47081ef37ab5afe.

The relevant files are the Presburger and surreal READMEs listed in the bibliography. Their descriptions are attributed to the repository; this session did not independently rebuild or certify them. The earlier local-compactness and Hahn manuscripts were read as project documents, not as refereed publications.

The literature check did not establish that the main algebraic or topological deductions are new in the historical literature. Their status here is a proved contribution relative to the inspected project materials and a resolution of the explicitly posed project question. A submission should undergo independent proof review and a fuller priority search before making a stronger novelty claim.

[write] The “companion `verify_examples.py`” is shipped as `code/10-baire-order-verify_examples.py`, with the recorded run data `data/10-baire-order-verification_results.json`; source 10’s audit is `10-baire-order-SOURCE_AUDIT.md`. The “earlier local-compactness and Hahn manuscripts” are sources 08 and 09, [Parts IV](#) and [V](#).

R Source audit and contribution ledger (source 13)

The three incoming ZIPs were read at repository revision

eab47e3301e0ed2eb0e8eb09824b90a6e34af7b0.

Their bytes matched the archive identities recorded by the repository’s triage. The subsequent overlap check read the omnific-normalization article at

4f2d7897a114f65a30a1920fb03492d5f9d69d76.

The same omnific source blob was also verified at the earlier revision. The two retrieval revisions are recorded separately; this article does not imply that the changing repository was frozen under one revision throughout the investigation.

Source	What was used
Glazer, arXiv:2311.13699v1	The precise published Presburger question, the distinction between basic-operation continuity and stronger arithmetic, and the separate reverse-mathematical problem
Two incoming Polish Presburger manuscripts	Existing split constructions, their proof interfaces, and their explicit unresolved questions
Incoming Hahn/omnific manuscript	Scope comparison; no support-complexity result from that manuscript is required by a new proof here
Omnific-normalization article	Existing algebraic splitting, residue-image, and unital-omnific-embedding criteria, including a stronger initial-image result
Presburger Lean syntax and decision source	The actual ordinary-integer semantic boundary of the existing formal development
Jeřábek and Ehrlich–Kaplan	Classical model theory and the ordered-group embedding input, respectively

The exact omnific embeddability criterion is *not* claimed as a new result relative to Provett. Its earlier article proves a stronger initial-realization version and explicitly states the equivalence even for noninjective unit-preserving homomorphisms. We include a direct normal-form proof to keep the application self-contained. The new use is the construction and analysis of perfect Polish models outside that split class. Likewise, the ordinary algebraic residue splitting facts are background; the automatic topological consequence is proved here.

The central further results are the exact continuity compiler, the target-sensitive Baire classification, the dimension and exact finite-pattern revision theorems, and the complete-metric obstruction. These were developed against the questions in the inspected manuscripts and audited independently at their delicate proof steps. No search can certify that an equivalent statement has never appeared elsewhere; global historical priority is not asserted.

The source and status metadata are also recorded in `provenance.json`. The bibliography provides permanent paper identifiers and repository links. A repository theorem is not treated as kernel-checked solely because it occurs in a research report.

[write] The “three incoming ZIPs” are batch 86’s manuscripts 03, 04 and 05, now [Parts I](#) and [II](#); the “omnific-normalization article” is the `discrete-initial-subgroups-and-omnific-normalization` report. `provenance.json` is shipped as `data/13-beyond-split-provenance.json`; its archive hashes and blob identifiers were verified at placement.

S Executable diagnostics and reproduction (source 13)

The accompanying `verify.py` is a standard-library Python program. It uses exact rational arithmetic and includes finite checks of the directional feasibility calculation, the nonsplit core’s Chinese remainder division formula, and stream examples attaining the stated revision bounds. Its output is recorded in `verification_results.json`.

The program implements finite algebraic diagnostics. It is not a complete implementation of arbitrary Presburger quantifier elimination, a proof of the infinite-stream lower bound, a proof that a space is Polish, or a machine-checked verification of the theorems. The exact algorithms in the text include transformations beyond the diagnostic program’s scope. This boundary is deliberate and explicit.

To reproduce the checks, run

`python3 verify.py`

from the unpacked package. To compile the standalone article, run

```
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

No external figures, bibliography database, network request, or repository checkout is needed to build the PDF. The source is a single file. The delivered PDF was checked by compilation, text extraction, and visual page inspection.

[write] In this report the program is `code/13-beyond-split-verify.py`; it writes `verification_results.json` beside itself, so it must be run on a copy (README). The recorded output is `data/13-beyond-split-verification_results.json`. The `article.tex` named above is source 13's delivered manuscript, merged into this report.

T An optional measure version of the final contradiction (source 14)

The category argument is sufficient and avoids a measure-theoretic dependency. For comparison with the original probabilistic mechanism, the multiplicative events in (8.5) also have fair-coin measure zero.

Fix n, k , and let $B = \{n + 1, \dots, n + k\}$. No two distinct members of E_n agree outside B . To prove this, suppose r, s do, and let

$$p = \prod_{i \in r \setminus s} y_i, \quad q = \prod_{i \in s \setminus r} y_i.$$

Orient the pair so the largest differing index m belongs to $r \setminus s$. Repeated one-bit identities give

$$z_r = pu, \quad z_s = qu, \quad z_{r^*} = qv, \quad z_{s^*} = pv$$

for positive u, v . The estimate $Q_m^4 < y_m$, with $Q_m = \prod_{i < m} y_i$, gives

$$p^2 > y_m q^2 > y_n q^2.$$

Indeed $p \geq y_m$, $q \leq Q_m$, and $q^2 < y_m$. On the other hand, $r, s \in E_n$ implies

$$pu \leq y_n qv, \quad qu > pv.$$

Multiplying by q and p , respectively, and cancelling $v > 0$ gives $p^2 < y_n q^2$, a contradiction.

Thus the 2^k translates of E_n under flips in B are disjoint and have equal measure. The resulting bound is

$$\mu(E_n) \leq 2^{-k}.$$

Letting k increase gives $\mu(E_n) = 0$. Complement symmetry and the absence of ties imply $\mu(D) = 1/2$, contradicting the tail zero-one law. The non-strict finite-section bound is all that is needed.

Using direct inequalities with an inclusive upper endpoint, as in (8.5), avoids any need to identify an event using a floor of a ratio. This formulation also keeps every comparison inside the arithmetic semiring.

U Short checks on the elementary inequalities (source 14)

Three estimates support the whole construction.

Root scales. With $t_n \geq t_{n+1}^2$, $c \geq q_n t_n$, and $t_{n+1} \geq t_{n+2}^2$,

$$q_{n+1} \geq q_n t_{n+1} \geq q_n t_{n+2}^2.$$

Whenever $t_{n+2} > 8$, this exceeds $8q_n t_{n+2}$. The nonstandard construction guarantees this last hypothesis for every external n .

Finite-bit absorption. If $h_{i+1} > 8h_i$ and $k = \max(n, j) + 1$, then

$$h_k > h_n + 2h_j.$$

If $y_{i+1} > y_i^5$ and $y_i \geq 2$, the corresponding multiplicative estimate is

$$y_k > y_n y_j^2.$$

These are exactly the bounds needed for invariance under one flip.

Factorial obstruction. The identity $s_{n+1} = s_n + n!$ gives (10.3), and induction gives $n!b_n = b_1 + s_n$. The elementary bound

$$s_n \leq 2(n-1)! \qquad (n \geq 3)$$

follows by induction. It guarantees that a fixed proposed value of b_1 eventually produces a residue strictly between consecutive multiples of $n!$, disproving a retraction without any limiting argument inside the group.

V Claim and source audit (source 15)

Claim	Evidence or proof supplied	Status
Research overlap	Glazer’s topological arithmetic paper; inspected Presburger and omnific repository sources	Directly documented
Basic affirmative answer to Glazer’s Question 2	Earlier repository manuscript; coefficient construction recalled in Section 118	Prior background
Full Presburger theory	Explicit residue division plus classical $\mathbb{Z}$ -group quantifier elimination	Classical semantic dependency
Borel multiplier barrier	Complete automatic-continuity and Cantor tail-event proof	Extension of Glazer’s mechanism
Cardinality injection	Explicit map $a \mapsto ah + H$ and trivial-kernel argument	Proved here; priority not certified

Claim	Evidence or proof supplied	Status
Inherited locally compact classification	Cone reflection, classical LCA structure, measurable-order lemmas, rational-hull splitting	Full proof with named dependencies
Arithmetic boundary	Infinitesimal convex subgroup, endpoint identification, purity, and quantifier elimination	Proved here; every countable boundary realized
Continuum family	Finite condensation, Archimedean classes, and separable-space counting bound	Exact lower and upper bounds
Omnific realization	Finite normal-form map with sign and injectivity proofs	Set-sized mathematical construction
Formal implementation	Existing declarations inspected; roadmap given	No new Lean or Rocq certificate claimed
Historical breakthrough	Targeted comparison with retrieved primary sources and prior repository article	Not established or claimed

The source manifest accompanying this article records the pinned Git commit, the inspected repository paths and blob hashes, and the external reference URLs. It does not bundle copies of the external papers. The TeX file is self-contained and uses an inline bibliography. A `latexmk -pdf` invocation rebuilds the PDF with the indicated standard packages; alternatively, repeat `pdflatex` until the references are stable.

[write] “The source manifest accompanying this article” was delivered as `source_manifest.json` and is not shipped (it contains a third party’s personal profile address); [appendix E](#) summarizes its verified content. The “earlier repository manuscript” is [Part I](#).

W The precise quantifier-elimination input (source 16)

For completeness, this appendix records why the arithmetic argument is valid in every $\mathbb{Z}$ -group, rather than relying on calculations over the standard integers. The theorem itself is classical [3, 92].

Work in the group language with all congruence predicates. Put a quantifier-free formula into disjunctive normal form. Negated inequalities and equalities can be expressed using the linear order; negated congruences are finite disjunctions of the other standard residues. Thus it suffices to eliminate an existential variable from a conjunction of affine equalities, inequalities, and congruences.

Choose a positive common multiple L of the absolute values of all nonzero coefficients of the quantified variable x , taking $L = 1$ if there are none. Replace Lx by y and add $y \equiv_L 0$. After correcting signs, a constraint $ax < t$, $a > 0$, becomes $y < (L/a)t$. A congruence $m \mid (ax + s)$ becomes

$$(mL/a) \mid (y + (L/a)s).$$

This equivalence is valid in a torsion-free group: multiplying both the dividend and modulus by L/a preserves divisibility, with cancellation providing the reverse implication. Equalities

can be normalized in the same way; if one remains, substitute its specified value and check the other constraints.

In the equality-free case the conjunction has the form

$$\bigwedge_i \ell_i < y \quad \wedge \quad \bigwedge_j y < u_j \quad \wedge \quad P(y),$$

where P is a conjunction of congruences. Let N be a standard common multiple of their moduli, using $N = 1$ if there are none. Then $P(y)$ depends only on the complete congruence class of y modulo N . This is stronger than merely knowing $P(y) \leftrightarrow P(y + N)$ by a single standard shift.

If there is a lower bound, choose a maximal ℓ from the finite list. Any witness $y > \ell$ has the same residue as one of the finitely many candidates

$$\ell + 1, \ell + 2, \dots, \ell + N.$$

Choose the candidate strictly above ℓ with that residue. Standard division of the positive element $y - \ell$ shows the candidate is at most y . It therefore preserves every upper bound as well. The maximum can be selected in a finite disjunction of comparisons between the ℓ_i .

If there are upper bounds but no lower bounds, choose a minimal u and use the candidates $u - 1, \dots, u - N$. A witness can be replaced by the same-residue candidate at least as large as it and still strictly below u . If there are no bounds, standard division replaces any witness by one of $0, \dots, N - 1$. In every case existence is equivalent to a finite quantifier-free disjunction. Iterating eliminates all quantifiers.

Every closed quantifier-free sentence in the expanded language has the same truth value in every $\mathbb{Z}$ -group, because its terms are standard integer multiples of 1 and standard congruences have their prescribed residues. Thus the theory is complete and agrees with the standard integer group. Finally a formula over the positive cone is interpreted in the group by restricting every quantified variable to ≥ 0 . This proves the input used throughout the paper.

[write] This is the classical theorem printed as source 05's [theorem 4.9](#), source 04's [theorem 4.10](#) and source 06's [appendix I](#); it is printed again because it is source 16's, and its remark on invariance under complete congruence classes is the coset invariance of [lemma 4.7](#) and [theorem 4.8](#). "[92]" is source 05's delivered text, now [Part I](#).

X Claim ledger and source boundary (source 16)

Claim	Evidence and scope	Status
Polish Presburger cone construction	Section 128 ; existing ProveIt main and ordinal reports; classical $\mathbb{Z}$ -group theory	Prior construction, reproved
Unital homomorphisms are elementary embeddings	Theorem 129.1 ; explicit divisible-group and congruence argument	Proved here in general split form
Continuous matrix classification and closed image	Theorems 130.5 and 130.7 ; finite-dependence and finite-ancestry proofs	Proved specialization and extension
Closed-submodel normal forms, orbits, lattice	Theorems 131.1 , 131.3 , 132.1 and 132.4 ; finite-support dual separation	Proved for this family

Claim	Evidence and scope	Status
Automatic-continuity threshold	Theorems 133.1 and 133.3 ; rank argument and non-Borel shear	Exact theorem; related Hahn phenomena credited
Closed-image threshold	Theorems 134.1 and 134.2 ; terminal-tail quotient and shifted graph	Exact theorem for all countable ordinals
Self-copy core and clopen intersections	Theorems 135.1 and 135.2 ; fixed terminal ranks and finite coordinate approximations	Proved here
Omnific interpretation	Section 137 ; normal-form existence and fixed-support addition	Additive and ordered; topology transported
Finite example calculations	Included rational verification script and recorded output	Executed checks only
Proof-assistant verification	No new Lean or Rocq files supplied; modules proposed in section 137	Not claimed
Historical priority or major breakthrough	Bounded source and repository audit	Not certified

[write] For $\alpha = \beta = \omega$ the rows on the dictionary, the matrix classification, the closed image and automatic continuity were proved first by source 04 ([proposition 11.3](#), [theorem 11.4](#), and [corollary 11.5](#)), and the omnific interpretation is source 07’s [theorem 40.1](#); notes in [Part X](#) give the details.

Repository snapshot

The inspected repository snapshot is `0f084afa9be466593ffa6aff37749e78c0adc733`, with commit timestamp October 3, 2026, 19:50:54 Pacific time. The main relevant text is `Algebra/SurrealNumbers/docs/foundations-and-computation/polish-models-of-omnific-arithmetic/article.tex`. Its named research questions on coefficient endomorphisms and Polish elementary-submodel lattices determine the specific follow-up problem. The incoming ordinal-family source and other existing arithmetic archives were checked to avoid presenting their constructions as new. The archive’s source audit gives the exact paths and limits of this comparison.

[write] The source audit is shipped as `16-presburger-embeddings-SOURCE_AUDIT.md`. The archives it compared are now [Part III](#) (`polish_presburger_article`, source 06; `polish_presburger_research`, source 07), [Part IV](#) (`glazer_proveit_local_compactness`) and [Part V](#) (`Polish_Hahn_Puisseux_Levi_Civita_Research`), and the standard-cut manuscript `Glazer_ProveIt_Arithmetic_Research` is source 03 of `Algebra/SurrealNumbers/docs/surreal/discrete-initial-subgroups-and-omnific-normalization`; all were placed in `0bd0e5527`, after source 16’s pin.

Proof review

Independent mathematical reviews were used to challenge the statements before delivery. They particularly checked the distinction between a G_δ cone and a closed cone, inherited versus abstract Polishness, finite-support versus full-product codimension, the limit-coordinate inversion argument, and the different thresholds for map continuity and image closedness. This is ordinary mathematical review of an AI-assisted manuscript, not external refereeing or a formal proof certificate.

[write] The review record is shipped as 16-presburger-embeddings-PROOF_REVIEW.md; its reviewers are AI agents of the delivering session. The intake has not reviewed the proofs.

Y **A compact proof dependency ledger (source 17)**

The logical dependencies can be recorded without collapsing the object arithmetic into the metatheory.

Conclusion	Arithmetic input	External input
Borel decoder	Division with unique remainder	Borel operations and analytic separation
Local exponential laws	One geometric code; bounded induction	No regularity theorem
Separated grids	Integer roots and quotient estimates	Finite preservation of countability
Ordered Cantor maps	Borel order on code values	Perfect-set theorem and Galvin's pair theorem
Continuous limit injection	Distinct finite products	Complete metric and summable error bounds
One-bit identity	Cancellation and one local product law	Continuity of multipliers indexed in the selected interval
Contradiction	Ordered product growth	Borel comparison and category zero-one law
Countable core	Initial arithmetic intervals	Harrington-Marker-Shelah in ZFC
Countable code domains	Existence of each individual code	Local theorem and countable core
Borel certificate relation	Fixed recurrence	Countability of every possible valid index interval

Z **Reproducing the package (source 17)**

The package contains the complete LaTeX source, the compiled PDF, the arithmetic diagnostic program and its JSON results, a source audit, and a proof-status record. The source requires ordinary LaTeX packages and can be built with:

```
latexmk -pdf -interaction=nonstopmode \  
-halt-on-error local_geometric_codes.tex  
python3 verify_local_arithmetic.py \  
local_arithmetic_diagnostics.json
```

The diagnostic output is deterministic for the recorded seed. Build auxiliary files and temporary page images are not needed to reproduce the mathematical content.

[write] In this report the program is code/17-local-codes-verify_local_arithmetic.py and its record data/17-local-codes-local_arithmetic_diagnostics.json; the

manuscript `local_geometric_codes.tex` is merged into this part and not shipped. The command above would overwrite the record if run in place; the README gives a rerun on a copy.

AA Proof dependency and assumption ledger (source 18)

Result	Main inputs	Particular issue checked
Hausdorff completion	Cancellation, linear natural order, joint addition, partial subtraction	Original and induced topologies agree in both directions
Polish upgrade	Countable local base, invariant metric, two completely metrizable cone subspaces, Baire category	Subspace complete metrizability is not confused with metric completeness
Borel subtraction	Continuous injective F , Lusin–Souslin	Borel inverse does not imply continuous inverse
Closure coding	Hausdorffness, power sets, class-parameter Separation, class Replacement	No ranks, Choice, or choice of a limit point
Neighborhood operator	Local set neighborhoods and an ordinal exhaustion	All candidates lie inside one power set before their union
Clopen set hull	Set-valued closure, specified enlargement, tightness $\leq \kappa$, regular κ^+	Recursion has set length; no class recursion is used
Manifold size	First countability, connectedness, sethood theorem	Chart choice for the cardinal estimate occurs after sethood
Surreal transport	Bounded birthday stages and a specified class map	Images use Replacement; inverse images use injectivity

Table 13: The main dependencies. The first three rows use ordinary set topology; the remaining rows use the explicitly stated class theory.

AB Terminology and notation (source 18)

Polish space

A separable topological space admitting some compatible complete metric. A particular displayed metric need not be complete.

Natural order on a monoid

$x \leq y$ means $y = x + z$ for some monoid element z . Its antisymmetry and totality are assumptions in Division I (Sections 152–156).

Partial subtraction

The function $\delta(y, x) = z$ on $x \leq y$, where $x + z = y$. Cancellation makes it unique.

Set-sized or set

A carrier that is a set in the specified foundational theory, rather than merely an informal collection with local coordinate descriptions.

Ordinal exhaustion

A specified increasing ordinal-indexed family of sets covering a class. No continuity at limit indices is required.

Specified class function

A class whose graph assigns exactly one set output to each input; this is the form to which class Replacement applies.

Foundation omitted

The axiom is not among the assumptions. Its negation is not being asserted.

Fine surreal topology

The native order topology of the proper-class surreal ordered field, distinguished from ordinary real chart topology.

AC The model-uniform finite-witness argument (source 19)

For clarity, this appendix proves the Presburger fact used in [proposition 168.3](#). The method is classical Cooper elimination [58]; its relevance is that the proof uses cosets of mG , not only finitely many ordinary shifts.

[write] (Batch 92.) This is one more write-up of the classical elimination already proved in this report for nonstandard $\mathbb{Z}$ -groups: [lemma 4.7](#) and [theorem 4.9](#) (source 05) with source 04's second proof, source 06's [appendix I](#), source 08's [theorem 49.3](#) and source 16's [appendix W](#). It is kept because source 19's proofs cite it.

Let G be a discretely ordered torsion-free abelian group with least positive unit 1 , and suppose every element has a unique standard residue modulo every positive ordinary integer. We suppress the factor 1 when writing ordinary integers inside G .

Lemma AC.1 (Coset-invariant finite witnesses). *Fix $m \geq 1$, a predicate P satisfying*

$$x - y \in mG \implies (P(x) \leftrightarrow P(y)),$$

and finite lists of strict lower and upper bounds. If a lower bound is present, let ℓ be the largest. Then a witness to

$$P(x) \wedge \bigwedge_i (\ell_i < x) \wedge \bigwedge_j (x < u_j)$$

exists if and only if one of $\ell + 1, \dots, \ell + m$ is a witness. If no lower bound but an upper bound is present, with least upper bound u , it suffices to test $u - 1, \dots, u - m$. If neither is present, test $0, \dots, m - 1$.

Proof. Suppose first that $x > \ell$ is a witness. Choose $r \in \{1, \dots, m\}$ with $x - \ell - r \in mG$, writing $x - \ell = r + mg$. If $g < 0$, discreteness gives $g \leq -1$, whence $x - \ell \leq r - m \leq 0$, a contradiction. Thus $g \geq 0$ and $\ell + r \leq x$. This candidate is above every lower bound, below every upper bound, and congruent to x modulo mG , so it satisfies P .

With only upper bounds, choose $r \in \{1, \dots, m\}$ representing $u - x$ modulo mG . The same reasoning shows $x \leq u - r < u$; no lower bounds need to be retained. With no bounds, every mG -coset has a standard representative. The reverse directions are immediate. $\square$

Theorem AC.2 (Uniform quantifier elimination). *In the language of ordered additive groups with 1 and predicates $x \in nG$ for ordinary $n \geq 1$, every formula is equivalent, uniformly over these groups, to a quantifier-free formula.*

Proof. It suffices to eliminate one existential quantifier from a conjunction of literals, after putting a quantifier-free formula in disjunctive normal form. Negated comparisons and disequalities are split into strict inequalities using totality and discreteness; equality is retained when useful. Terms are integer affine.

Let N be a positive common multiple of the absolute values of all nonzero coefficients of the variable y to be eliminated. Substitute $x = Ny$ and add the congruence $x \in NG$. For an atom involving $ay + b$, with $a \neq 0$, put $k = N/|a|$. Multiplication by positive k turns a comparison into one involving

$$\text{sgn}(a)x + kb,$$

so its x -coefficient is 1 or -1 . For a congruence, the exact rule is

$$ay + b \in mG \iff \text{sgn}(a)x + kb \in mkG,$$

under $x = Ny$. The reverse direction follows by cancelling k in the torsion-free group. Atoms with $a = 0$ are independent of x . The substitution is existentially exact because $x \in NG$ gives a unique y with $x = Ny$.

Any surviving equality $x = t$ eliminates x directly by substitution. Otherwise we have finitely many strict lower and upper bounds and a Boolean combination of congruences with x -coefficient 1 or -1 . Choose m divisible by all their moduli. This congruence part is invariant on cosets of mG , so [lemma AC.1](#) eliminates x by a finite disjunction. The choice of a largest lower or least upper bound is also expressible by a finite disjunction; alternatively test all bound-plus-residue candidates and retain every original bound.

Iterating eliminates all existential quantifiers; universal quantification is handled by negation. The resulting construction is syntactic and works in every group satisfying the two stated arithmetic properties. $\square$

A ground quantifier-free formula compares only integer numerals and their ordinary residues, so it has the same truth value as in $\mathbb{Z}$. This proves completeness and the signed-group claim. Translating the nonnegative cone as a definable domain gives all of $\text{Th}(\mathbb{N}; 0, 1, +, <)$, including induction. Nothing requires an induction along the externally non-well-ordered set of nonstandard positive elements.

AD Dependency and claim ledger (source 19)

Result or input	Status and dependency
Presburger elimination	Classical; the required uniform argument is proved in appendix AC .
Order regularity	Elementary Baire-category proof in lemma 169.1 ; Baire-to- Δ_2^0 upgrade in corollary 170.6 .
Three-layer normal form	Proved in theorem 170.2 ; uses real completeness and the countable extension lemma.
Complete countable invariant	Proved in theorem 171.1 ; no assumption that an arbitrary countable extension splits.
Pair invariant in the split case	Proved in theorem 172.1 using rational vector-space complements.
Fixed analytic manifold	Proved in theorem 173.1 ; group-completion continuity is separately checked in lemma 173.2 .
Continuous coding map	Explicit construction and proof in theorem 174.3 .
Borel completeness	New reduction composed with the classical Friedman–Stanley theorem; explicit code space in section 175 .
Abstract/Borel automorphisms	Proved in theorem 176.1 ; arbitrary Hamel shears require ordinary choice.
Omnific realization	Finite normal forms only; no ring-closure or natural-topology claim.
Global historical novelty	Not established by the bounded source review. No independently certified breakthrough is claimed.
Lean/Rocq status	No new kernel-verified proof supplied. Repository interfaces were read, not rebuilt.

[write] The ledger is source 19’s. The results written for this report ([section 182](#), [remark 176.3](#)) are not in it; they are unrefereed and not formalized, like the rest of the part.

AE Source provenance and inspection limits (source 20)

The source review was performed on 4 October 2026. The public arXiv record of Glazer’s article displayed the single 2023 version at the time of inspection. Its full eight-page mathematical text was read, and the final page was inspected as an image. For Blass, the scanned pages 271–272 were inspected to verify the perfect-set partition input. The Enayat–Hamkins–Wcisło abstract and bibliographic record were inspected for historical context; their older “open” wording is not treated as a current status claim.

For ProveIt, the tree response supplied the pinned revision stated in [section 183](#). The root README was also read on the default branch, and the relevant surreal catalogue files and floor source were subsequently fetched at the pinned revision. The following paths were inspected:

```
README.md
Algebra/SurrealNumbers/docs/README.md
Algebra/SurrealNumbers/docs/foundations-and-computation/
polish-models-of-omnific-arithmetic/README.md
Algebra/SurrealNumbers/Surreal/Foundations/OmnificFloor.lean
```

The topology README was read in relevant contiguous ranges including its statement catalogue, limitations, existing polynomial example, and the status of its claimed reverse-mathematical extension. The first 150 lines of the floor file were read. Source inspection is

not a fresh build or kernel audit. No theorem from the unreviewed reverse-mathematical extension is used here.

Keyword searches were also made for the proposed intersection. They did not provide an exhaustive literature search, and several narrow queries returned irrelevant results. We therefore do not infer historical novelty from an absence of search hits. The repository’s catalogue descriptions are useful for avoiding obvious duplicate work, but cannot establish that no closely related statement occurs in an unread manuscript.

[write] (Batch 92.) At the placement e24ce2ce0 the root README, the surreal catalogue Algebra/SurrealNumbers/docs/README.md and OmnificFloor.lean were unchanged since the pin; this report and its README had grown by Parts X–XII. The catalogue commit cd514f61b has since changed the two READMEs. The declarations named exist, and the “claimed reverse-mathematical extension” is Part VIII. The “companion source ledger” is 20-exponent-threshold-SOURCES.md.

AF Proof ledger (source 20)

Claim	Evidence here	Attribution and status
Theorem 183.1: exact exponent threshold	Explicit positive topology; dense-grid contradiction	Proposed classification extension; no priority claim
Theorem 183.2: one-root obstruction	Countability cut, iterated roots, quotient grid estimates	Proposed weakening; not a claim about multiplication-only continuity
Theorem 186.4: separated-grid criterion	Full Cantor, metric, finite-flip and measure proof	Abstracted adaptation of Glazer’s method
Proposition 187.2: polynomial-cone topology	Direct chart and continuity proof	Already recorded in Provelt
Theorem 184.4: finite-support floor	Cutoff support estimate and sign-sensitive constant correction	Standard integer-part mechanism; related formal omnific floor exists
Theorem 185.1: exact k -root spectrum	Degree obstruction and binomial-series construction	Algebraic companion; novelty not separately established
Corollary 189.2: infinite root-tower failure	Closed finite-stage embeddings and dense-group theorem	Consequence of the classification
Proposition 190.1: Borel generic arithmetic	Countable support charts and polynomial cancellation loci	Extends an existing rational-support construction
New Lean or Rocq theorem	None in this package	No formal verification claimed
Finite diagnostics	Executed exact-arithmetic tests	Sanity checks, not infinite proof certificates

AG Reproducibility and the exact scope of the checks (source 21)

The package includes code/check_identities.py, using only Python’s standard library and exact fractions.Fraction arithmetic. Its recorded run passed 2,931 finite checks.

These cover rational binomial products and roots, the Neumann inverse in $\mathbb{Q}[t]/(t^{25})$, Catalan coefficients through degree 24, finite shift telescoping through $N = 200$, selected independent-chain identities, and the rational-function reflection identity.

The shift checks retain the final boundary monomial. Removing it would turn the example into a finite nilpotent shift and misleadingly produce a finite-dimensional inverse. The checked chain symbols are formal independent symbols; approximate rational numbers are not used as a substitute for rationally independent real exponents.

None of the checks establishes the infinitude, basis-extension, completeness, real-closedness, elementary-subfield, or historical novelty claims. Those require the proofs and explicitly identified classical inputs. The script is deliberately an independent finite sanity check, not a purported Lean or Rocq proof.

To reproduce the checks and typeset the article:

```
python3 code/check_identities.py
latexmk -pdf -interaction=nonstopmode -halt-on-error article.tex
```

The Makefile supplies these two targets. Repository and source versions are recorded in `audit/sources.json`. The exact-arithmetic check output is in `audit/checks.json`. Bibliography entries below identify the inspected preprint versions. Source PDFs and font files are not redistributed in the package.

[write] In this report the program is `code/21-levi-civita-rank-code-check_identities.py`, the Makefile `code/21-levi-civita-rank-Makefile`, and the files `audit/checks.json`, `audit/sources.json` and `audit/build-summary.json` are `data/21-levi-civita-rank-audit-checks.json`, `data/21-levi-civita-rank-audit-sources.json` and `data/21-levi-civita-rank-audit-build-summary.json`; the proof audit is `21-levi-civita-rank-audit-proof-audit.md`. The manuscript `article.tex` is merged into this part and not shipped, so the commands above refer to delivery names. The program prints its record; the Makefile's check target would write it to `audit/checks.json`. The intake reran it on a copy: 2,931 checks in 13 families pass, and the printed record equals the shipped one after removing the carriage returns that Windows text mode adds.

AH Separation without local convexity (source 22)

Here are details justifying the separation step used in Lemma 212.2, including its application to L^p with $0 < p < 1$.

Let C be a proper closed convex set with nonempty interior in a real Hausdorff topological vector space, and choose $y \notin C$ and $u_0 \in U = \text{int}(C)$. Put $A = U - u_0$. This is open and convex and contains zero, so it is absorbing. Its gauge

$$p_A(x) = \inf\{t > 0 : x \in tA\}$$

is a finite nonnegative sublinear function and $A = \{x : p_A(x) < 1\}$. The vector $z = y - u_0$ is outside A , so $p_A(z) \geq 1$. On the line $\mathbb{R}z$, the linear functional $\ell_0(tz) = tp_A(z)$ is dominated by p_A : for $t \geq 0$ use positive homogeneity, and for $t < 0$ use nonnegativity of the gauge. The algebraic Hahn–Banach theorem extends it to a real linear functional ℓ on E with $\ell \leq p_A$.

On the neighborhood $A \cap (-A)$, both ℓ and $-\ell$ are bounded above by 1. A linear functional bounded on a neighborhood of zero is continuous, since shrinking that neighborhood by a scalar gives arbitrarily small bounds. Also $\ell(z) = p_A(z) \geq 1$, so $\ell \neq 0$. For $u \in U$,

$$\ell(u - u_0) \leq p_A(u - u_0) < 1 \leq \ell(y - u_0),$$

and hence $\ell(u) < \ell(y)$.

Finally $\text{cl}(U) = C$. Indeed, for $c \in C$ and $0 < t < 1$, the point $(1 - t)c + tu_0$ is interior: adding a sufficiently small neighborhood scaled by t keeps it in C by convexity. These points tend to c . Thus ℓ weakly separates C from y . In the cone case, scaling shows $\ell(c) \leq 0$ for every $c \in C$; replacing ℓ by $-\ell$ gives the functional nonnegative on C required above.

The argument uses an open convex neighborhood arising from the set being separated; it does not assert that the topology has a base of convex neighborhoods. This is why it remains valid without local convexity.

[write] (Batch 93.) [Proposition 224.2](#) [merge] uses this argument, which needs no separability either.

AI Source audit and scope of the claims (source 22)

The repository snapshot inspected for this report is

3c25fbb5515748040decb853139ed39543885b90.

Its canonical tree identifier was obtained through the GitHub API.

[write] (Batch 93.) Correction: 3c25fbb55 is a *commit*, not a tree (`git cat-file -t` prints `commit`); its tree is c72baba04d76f05c8b9c7c751777ad97c41b474b. The commit is the Hilbert's-tenth programme's review commit of 4 October 2026, 10:49 ("Verify batch-90 file placement and route five later research arrivals"), 57 commits before the placement 5e4d8eed8. GitHub's `tree/sha` addresses accept a commit, so the citations resolve. The delivered 22-hilbert-orders-SOURCES.md says "Pinned tree/commit" and is kept byte-identical. The audit used the surreal collection's README, the full requested README range for `polish-models-of-omnific-arithmetic`, and repository searches for Glazer and the Hilbert/orthonormal-order intersection. The Polish report's catalogue identifies thirteen manuscripts, including constructions, Baire-category rigidity, nonsplit models, and elementary embeddings. It explicitly distinguishes research-draft proofs from formal verification and unreviewed claims. We have not read all 379 pages of that merged report and do not claim a complete repository-wide duplication audit.

[write] (Batch 93.) True at the pin: this report then had thirteen sources, Parts I–X, and 379 pages; Parts XI–XII were placed but not yet written. It now has twenty sources. The unread article contained the prior art listed in this part's introduction. The LinkedIn page is mentioned without an address; no personal data is shipped.

The direct LinkedIn page supplied in the request was not accessible in this session. The research connection instead rests on Glazer's own paper, whose stated problem and final questions were read, with its last page visually checked. No employment or biographical claim is inferred from the inaccessible profile.

Tan's five-page 2021 paper was read and its displayed construction and final page visually checked. Its naturally indexed Schauder-basis construction is credited rather than claimed as new. The full text of Küçük–Soyertem–Küçük (2012) could not be fetched; its abstract makes broad representation claims, so priority of our converse must be compared with that full text before submission. The Kechris–Louveau article was bibliographically verified through the authors' institutional repository; the rank computation here is proved directly, without claiming a review of that entire article. The opening normal-form description in L'Innocente–Mantova was read; its factorization theorems play no role in this report.

Claim family	Evidential status in this package
Lexicographic examples, Presburger quantifier elimination, surreal normal forms	Established inputs or constructions; sources credited.
Category and canonical flag theorem	Complete conventional proof here, including the nonlocally convex separation step.
Hilbert classification, ranks, embeddings	Complete proofs here; precise novelty relative to all earlier literature not certified.
Oscillation rank and ambient pairs	Exact derivative calculation with an explicitly fixed rank convention.
Arithmetic and omnific transfer	Complete deductions from the stated standard inputs; not a new claim to the original Presburger existence answer.
Finite computation	Reproducible exact tests only; no substitute for infinite or transfinite proofs.
Formal verification and priority	No proof-assistant verification, independent referee report, or priority certification supplied.

AJ Notation and dependency summary (source 23)

Notation	Meaning
Γ, Λ	Countable additive subgroups of the ordered real numbers.
$\mathcal{L}_{\mathbb{R}}^{\Gamma}$	Real-coefficient series with finite support below every real cutoff.
$v(f)$	Least exponent of a nonzero series; $v(0) = +\infty$.
τ_c	Subspace topology inherited from $\mathbb{R}^{\Gamma}$ with ordinary real coefficients.
τ_v	Topology defined by $v(f - g) > B$; constants are discrete.
Sl_S	Polish coefficient slice on a fixed left-finite support S .
$\delta(\gamma)$	Logarithmic monomial derivative $t^{-\gamma} D(t^{\gamma})$.
E_i	Euler derivation for the i th rational-basis coordinate.
$d_h(\gamma)$	Coefficient $[t^h]\delta(\gamma)$.
$\text{gain}(T)$	$\inf_{f \neq 0} (v(Tf) - v(f))$.
$\text{ct}(f)$	Constant coefficient $[t^0]f$.

The dependency spine is

left-finite field and coefficient coding $\longrightarrow$ Polish-slice functional lemma
 $\longrightarrow$ generic-support lemma $\longrightarrow$ matrix cutoff theorem
 $\longrightarrow$ uniform logarithmic bound $\longrightarrow$ derivation classification.

The coefficient-continuity theorem uses the finite-row corollary of the matrix theorem. The wild-derivation theorem uses the elementary lacunary lemma and algebraic extension. The exponential-logarithm theorem uses the bounded-gain estimates and valuation completeness. The surreal interpretation is a final transport and is not an input to the core proof.

AK Source and novelty record (source 23)

The initial repository inventory returned the Git tree identifier

3c25fbb5515748040decb853139ed39543885b90.

This is a tree identifier, not a commit identifier. The root and surreal-project READMEs were read from the public main branch on 4 October 2026; that branch continued to receive updates during preparation. No theorem from an unreviewed repository research report is assumed in the core proofs.

[write] (Batch 93.) Correction: 3c25fbb55 is a *commit*, not a tree (`git cat-file -t` prints `commit`); its tree is c72baba04d76f05c8b9c7c751777ad97c41b474b. It is the Hilbert’s-tenth programme’s review commit of 4 October 2026, 10:49, 57 commits before the placement 5e4d8eed8. The “later main-branch inspection” of the delivered 23-borel-summability-SOURCE_AND_PROOF_AUDIT.md is the merge commit 4f03443e0, whose tree d4b916d25 it gives correctly. The delivered audit, which repeats the tree reading, is kept byte-identical.

The literature search, conducted on 4 October 2026, included Glazer’s arithmetic paper, the Levi–Civita overview, generalized-series derivation papers, and an automatic-continuity survey. Searches combining Borel regularity, Levi–Civita fields, derivations, and left-finite supports did not establish whether the exact statements here already occur in print. Search coverage was limited and some keyword searches returned irrelevant results; absence of a located match is not evidence of novelty.

For a priority review, the most distinctive statements to compare with earlier work are Theorems 228.2, 230.2, and 232.1, including their precise support and topology hypotheses. The finite-rank Euler expression, the residue obstruction in a familiar Laurent-series normalization, and the basic filtered exponential method should not independently be advertised as new inventions.

AL Notation and dependency summary (source 24)

Symbol	Meaning
Γ	Fixed additive subgroup $\mathbb{Z} \subseteq \Gamma \subseteq \mathbb{Q}$.
$\mathcal{L}_{\mathbb{R}}^{\Gamma}$	Real-coefficient series with left-finite Γ -support.
v, ct	Least series exponent; constant-coefficient extraction.
$\mathfrak{m}$	Positive-valuation series, with zero included.
E, D_a	Euler derivation $t d/dt$; the derivation aE .
α, c, ρ	$v(a)$, leading coefficient of a , and $\text{ct}(1/a)$.
J	Euler integration after deleting the constant coefficient.
G_{Γ}	Borel ordered-field automorphisms in coefficient coding.
S_h	Substitution $f \mapsto f(h)$; $S_h S_k = S_{k \circ h}$.
U_{Γ}	Positive rational scalars preserving Γ bijectively.
$\omega_{\infty}(\Gamma)$	Primes having infinite height in Γ .
κ	A fixed enumeration-based representative of an exponent orbit.
θ	Full positive-gain residue invariant ρ/α .

[write] (Batch 96.) The table prints source 24's L_Γ , $\mathfrak{m}$ and $P_\infty(\Gamma)$ as $\mathcal{L}_\mathbb{R}^\Gamma$, $\mathfrak{m}$ and $\omega_\infty(\Gamma)$; the bookkeeping symbol of [section 243](#) (source 24's ℓ) is $\mathfrak{l}$, and the normal form of [theorem 244.1](#) is a^{nf} . Symbols of the whole report are in [section 1.3](#).

The normalization proof uses only support calculus, the Euler primitive, and unit contraction. The conjugacy theorem adds coordinate classification and residue covariance. The smoothness threshold adds elementary prime heights and the dense-translation Baire argument. The centralizer theorem uses the normal forms and the positive-gain exponential. Analytic-flow uniqueness adds the real-analytic support lemma. Automatic time analyticity instead uses logarithms of tangent automorphisms, linearization of nonunit dilations, and the additive Cauchy equation. No step in this dependency chain relies on the computational regression tests.

AM Source and novelty record (source 24)

The source review used Glazer's primary paper, the author offprint on Levi-Civita analysis, the generalized-series automorphism paper, the power-log normal-form paper, the automatic-continuity survey, the earlier user manuscript, and the ProveIt documentation. The LinkedIn page supplied in the request was not needed to infer mathematical interests after the primary paper had been identified. No employer, current position, or personal information is used in any mathematical claim.

The repository snapshot identifier returned during inspection was

715a716a3002b1e9f28daa2d81c009a392851b94.

The relevant path is Algebra/SurrealNumbers/README.md; the root README.md records the broader trust boundary. The existing research draft [\[152\]](#) is part of the user's Library and is not represented as a published article. No replacement copy of that earlier draft is bundled. The present paper re-proves its needed rank-one ingredients.

[write] (Batch 96.) 715a716a3 is a commit (`git cat-file -t` prints `commit`), the parent of the arrival commit 62846e17a and 39 commits before the placement 54ece48ab. The earlier draft is no longer only in a private library: it is source 23 of this report, printed as [Part XVII](#) in the write commit 9a8894d0a, which the pin already contains; the re-proved rank-one ingredients are flagged in notes. The page of a professional profile mentioned above was not used by the write either, and no address of it appears in any shipped file.

The proposed new emphasis is the exact Borel-uniform classification together with the prime-divisibility threshold and the compatibility of nonsmoothness with Borel witness selection. The normal-form shape, filtered exponentiation, and elementary dense-orbit obstruction have substantial antecedents. Specialist review is needed before making a historical priority claim or submitting the work as a claimed solution to a published open problem.

AN Proof audit and reproducibility (source 25)

AN.1 Contribution and dependency ledger

Statement	Status	Dependency or verification boundary
Scattered-support Borel dichotomy and Π_3^0 ceiling	Established prior art; independent proof included	Guevara–Uzcátegui; countable cuts and the perfect-set theorem.
F_σ iff B is absent; exact binary Wadge cases	Complete argument in this report; priority not certified	Convex well-ordered intervals, Fin and $\text{Fin}^{\mathbb{N}}$ reductions.
Real-code and ordered-group classification	Complete deductions here	Direct zero-coordinate formulas; least-positive-element argument.
Strong-family classification	Complete deduction here	Row-finiteness reduction and union-support cut test.
Nowhere-continuity and meagerness	Complete elementary proofs here	A unit coefficient inserted at an unobserved row.
Closed integer and real certificate spaces	Explicit constructions and proofs here	Height bounds, finite incidence, closed real equations.
Selector separation and exact regularization	Explicit formulas and full proofs here; priority not certified	Connectedness for the negative part; soft thresholds and finite maxima for the positive part.
No standard Borel full-support names in the non-scattered case	Deduction of classical analytic-image obstruction	Coanalytic-complete validity and Borel observability of every coefficient.
Uniform fixed-group hardness	Explicit continuous reduction here	Reverse lexicographic finite-support groups and completeness of well-orders.
Existing repository summation theorems	Selected source inspected	Pinned <code>StrongEvaluation.lean</code> ; not rebuilt in this task.
Finite Python checks	Reproducible finite illustrations	They do not certify infinite orders, Borel completeness, or Lean terms.

[write] (Batch 96.) “In this report” and “here” in the ledger mean source 25. Within this report the scattered-support Borel dichotomy is Part II’s [theorem 18.4](#), and the other re-proved statements are listed in [section 270](#).

AN.2 The specific prior question

The earlier report [158], dated 3 October 2026, has a section headed “Exact complexity beyond finite lexicographic rank” at Question 12.3. It asks for the least Borel level of $\text{WO}(L)$ for fixed countable scattered L and for the occurring Wadge degrees. Its supplied examples are the integer order and finite lexicographic powers. The present classification proves that these already exhaust the nontrivial Borel *degrees*, with the forbidden-order criterion identifying which one applies. It does not claim an effective algorithm from arbitrary decomposition codes, and it does not reclassify the much finer ideal-isomorphism types.

[write] (Batch 96.) In this report that question is Part II's [question 28.3](#), whose title is the section heading quoted above; its first sentence asks to compute the least level “directly from an economical Hausdorff decomposition”, the clause that stays open (note after the question).

AN.3 Checks provided with the article

The accompanying `checks.py` uses exact rational arithmetic and a fixed random seed. It checks the finite order-embedding pattern in [\(260.1\)](#), validates the real-certificate regularization for finite rational arrays over $\mathbb{Z}$, checks eventual exactness of the bound coordinates on those finite inputs, and exhibits the two diverging-bound patterns on finite windows of explicitly specified infinite arrays. It also checks that changing an unobserved row changes the chosen sum coefficient by exactly 1 while all declared observations remain unchanged.

Every finite support is well ordered. Therefore none of these finite tests can prove the infinitary classification or rule out a hidden flaw in its proofs. Their purpose is regression checking of formulas, inequality directions, inclusive row-bound conventions, and the finite algebra in the construction. The mathematical assertions rest on the proofs in the article, not on the number of tests passed.

[write] (Batch 96.) The program is shipped as `code/25-support-complexity-checks.py` and its record as `data/25-support-complexity-checks.txt`. Rerun on a copy with Python 3.14.4 (standard library, seed 20261004, under a second), it prints the shipped record up to carriage returns. Its checks are bare `assert` statements (17 of them) and nothing else stops it on failure: under `python -O`, which removes assertions, it prints the same “all passed” record while checking nothing (verified on the copy). Rerun it without `-O`. The program is shipped unchanged.

AN.4 Compilation and source scope

The package is self-contained for PDF production: `article.tex` contains the bibliography and uses standard TeX Live packages. Run `make` or `latexmk -pdf article.tex`; run `python3 checks.py` for the finite checks. The supplied PDF was compiled and rendered for visual inspection. No fonts, copied research papers, or external repository snapshot are bundled.

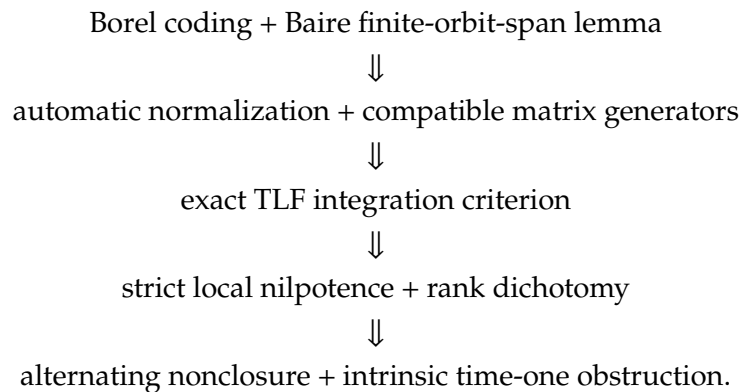
Public primary sources, selected repository files, and the earlier project report were inspected. The LinkedIn page supplied in the request was not used as a substitute for mathematical publications. The source search is not a complete priority search and the manuscript has not undergone external peer review. None of the new arguments is represented as a Lean-checked theorem.

[write] (Batch 96.) These instructions use the delivery names. In this report source 25's text is this part, built with the report; its manuscript, PDF, delivery README and checksum manifest are not shipped (they are in the arrival commit [7be14aa84](#)). The shipped `code/25-support-complexity-Makefile` names `article.tex`, `checks.py` and `python3: run` on a copy with the delivery names restored (the report's README gives the recipe), never in this directory, where its all target would build the report's own `article.tex`.

AO Notation and a dependency map (source 26)

Symbol	Meaning
Γ	Nonzero countable $\mathbb{Q}$ -vector subspace of ordered $\mathbb{R}$.
$\mathcal{L}_{\mathbb{R}}^{\Gamma}$	Real series with support finite below every real bound.
$v(x)$	Least exponent; $v(0) = +\infty$.
$\text{Win}_{a,b}$	$\mathcal{L}_{\mathbb{R}, \geq a}^{\Gamma} / \mathcal{L}_{\mathbb{R}, \geq b}^{\Gamma}$, identified with finite-support coefficients on $[a, b)$.
TLF	Local finiteness of the induced linear operator in every window.
$\delta(\gamma)$	Logarithmic monomial derivative $t^{-\gamma} D(t^{\gamma})$.
z_n, κ_n	t^{β_n} and $\beta_{n+1} - \beta_n$ for the bounded independent chain.
D_e, D_o	Alternating successor derivations.
Φ_s^e, Φ_s^o	Their explicitly constructed shear flows.
A	Product $\Phi_1^e \circ \Phi_1^o$, not any Borel time-one map.
$\mathfrak{P}_B, \mathfrak{S}_B$	Parameters of all strict derivations and of the integrable ones.

The logical spine is



The Borel-locus result uses the strict criterion and joint parameter evaluation. The finite verification script is outside this proof spine: it checks examples and identities, not the general logical implications.

[write] (Batch 96.) The symbols are printed as in [Part XX](#); source 26's $\mathcal{L}_{\Gamma}$, $V_{a,b}$, $d(\gamma)$, ϵ_n , E , F , P_s , Q_s , $\mathcal{P}$ and $\mathcal{I}$ are the $\mathcal{L}_{\mathbb{R}}^{\Gamma}$, $\text{Win}_{a,b}$, $\delta(\gamma)$, κ_n , D_e , D_o , Φ_s^e , Φ_s^o , $\mathfrak{P}_B$ and $\mathfrak{S}_B$ of the table (eighth block of [section 1.3](#)).

AP Source, scope, and novelty record (source 26)

The literature and repository review was performed on 4 October 2026. The supplied LinkedIn page was not fully accessible; the research intersection is grounded in Glazer's paper rather than in a guessed profile or current affiliation. The earlier standalone `borel_summability.pdf` was inspected for its stated results, Section 10 flow construction, and Question 13.5. The repository catalogue was inspected to locate that report as source 23, Part XVII of the merged Polish-models report; the full merged article was not audited.

The BCH source citation is pinned to commit `c39974f12a45c8795575ab222f3b84b2901c394f`; the inspected `Trunc.lean` blob has hash `5f238ed2f05b54e940e51f974465768c0dc23ba7`. The root and research-catalogue documentation was also read during the review. Repository documentation is evidence of the project interface, not a substitute for checking every claimed theorem or rebuilding it.

The search included generalized-series automorphism/derivation correspondences, the 2026 formal Lie correspondence, and combinations of Levi–Civita fields, derivations, and one-parameter automorphisms. The reviewed sources did not establish priority for the exact classification and bounded-chain package proved here. Absence of a match in this limited search is not a proof of novelty. The finite-orbit-span argument is elementary and is not claimed to be a newly discovered Baire-category principle.

The contribution advanced for specialist review is the combination of an exact Borel flow criterion with the all-infinite-rank obstruction, nonclosure of integrable strict derivations under sum and bracket, the exact and unique time-one embedding criterion, nonclosure of time-one maps, and the Borel integrability locus. The formal exponential mechanism and its uniform-gain special case are explicitly treated as prior mathematics. All new general claims remain subject to independent mathematical review and formalization.

[write] (Batch 96.) The pin `c39974f12` is a commit, as stated (`git cat-file -t` prints `commit; tree dbbe29a67`), of 4 October 2026, 15:49, 38 commits before the placement `54ece48ab`; `Trunc.lean` has the stated blob there and at the placement, and this report’s article did not change between the pin and the placement. The “merged Polish-models report” is this report; because source 26 did not read its article, its re-proofs of Part XVII’s results are flagged in notes, and [section 285](#) [merge] relates it to Parts XVII, XVIII and XXI. The standalone `borel_summability.pdf` is source 23’s delivered PDF, which this report does not ship. The 2024–2026 preprints [159, 160] were not re-read for this report; the priority statements are source 26’s.

AQ **Proof dependencies and source record (source 27)**

Result	Decisive input	Boundary
All Borel derivations are aE	Polish slices; generic support; rational powers	Left-finite, rank-one exponents
Scalar cross-section	Positive rational root; exponent-orbit representative	Ordered, real-fixing coordinates
Pair cross-section	Nonconstant ratio; scalar uniqueness	Named noncommuting pair
Single-derivation obstruction	Dense countable translation zero–one law	Dense $\log U_\Gamma$
Lie dimension bound	Finite valuation spectrum; leading bracket	Finite real dimension
Affine/projective refinement	Scalar normalization; coefficient comparison	Same coordinate group
Whole-field action boundary	Divisibility of time; rational prime valuations	Differentiability with Borel generator

The public sources were checked on 4 October 2026. The supplied LinkedIn profile was not available as a complete readable page; no mathematical attribution relies on its inaccessible contents. Glazer’s research connection is supported by his own paper. Repository context

was read through the GitHub connector from the Algebra/SurrealNumbers/README.md and Algebra/BakerCampbellHausdorff/README.md files. Their returned blob identifiers were, respectively, a7ebc59ad60d52ac4448e12ea8d2385227e34031 and e24d8fc8dbb4901bdaa9c588367e0b5ed2797451. These identify the inspected files, not a claim that all repository results were audited.

[write] (Batch 96.) Both blobs are the files' contents from ad52ef11e (4 October 2026, 14:57; the surreal README's blob dates from cd514f61b) through the placement 54ece48ab; source 27 names no commit, so this report treats ad52ef11e as its earliest possible pin. No address of the profile appears in any shipped file.

The two preceding manuscripts were retrieved from the user's file library. They are cited as project antecedents rather than refereed authorities. Every mathematical ingredient from them needed here has an independent proof in the present text. The classical Lie antecedent is explicitly separated from the proposed new Borel simultaneous-classification package.

[write] (Batch 96.) The two manuscripts are sources 23 and 24, printed as [Parts XVII](#) and [XVIII](#); the notes in [Part XXI](#) name the overlaps. The last row of the table is [remark 294.3](#); [theorem 300.3](#) [merge] replaces its differentiability hypothesis by Borel time dependence, through [Part XVIII's theorem 248.8](#).

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- [55] Vladimir Reshetnikov, *ProveIt: Surreal*, repository guide in Algebra/SurrealNumbers, same inspected snapshot. <https://github.com/VladimirReshetnikov/ProveIt/blob/6fef5383b126be343cccc9baf47081ef37ab5afe/Algebra/SurrealNumbers/README.md>. [write] Since the pin the file has gained only the catalogue’s pointer to this report (2b7b388ba).
- [56] *Local Compactness Forces Arithmetic Rigidity: Borel ordered groups, Presburger models, and multiplication without regularity assumptions*, AI-assisted research manuscript prepared for Vladimir Reshetnikov, 3 October 2026. Unrefereed project document; inspected source

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- [58] D. C. Cooper, *Theorem proving in arithmetic without multiplication*, in *Machine Intelligence* 7, 1972, pp. 91–99. Volume contents: <https://www.doc.ic.ac.uk/~shm/MI/mi7.html>. [write] Cited by sources 13, 15, 19 and 22; sources 15 and 19 give pp. 91–100, source 22 pp. 91–99. The page range was not re-checked.
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- [61] Vladimir Reshetnikov, Provelt, *PresburgerArithmetic/Decision.lean*, inspected source blob 9281d1504325e3a399370ce5d2f052909af8e991. <https://github.com/VladimirReshetnikov/ProveIt/blob/main/Logic/PresburgerArithmetic/Lean/PresburgerArithmetic/Decision.lean>. [write] The address is unversioned; the file is unchanged between source 13’s pin eab47e330 and the current state.
- [62] Vladimir Reshetnikov, Provelt, *Surreal*, project README. <https://github.com/VladimirReshetnikov/ProveIt/blob/main/Algebra/SurrealNumbers/README.md>. [write] The address is unversioned; the version at source 13’s pin is <https://github.com/VladimirReshetnikov/ProveIt/blob/eab47e3301e0ed2ebee8eb09824b90a6e34af7b0/Algebra/SurrealNumbers/README.md>.
- [63] *Polish Presburger Arithmetic*, incoming research manuscript in Provelt, archive Polish_Presburger_Glazer_ProveIt.zip, member polish_presburger.tex, 2026. https://github.com/VladimirReshetnikov/ProveIt/blob/eab47e3301e0ed2ebee8eb09824b90a6e34af7b0/docs/incoming/Polish_Presburger_Glazer_ProveIt.zip. [write] This is batch-86 manuscript 04, printed in Part I as the passages marked [04]. The archive was retired from docs/incoming in 3d2177df4 and remains at the cited address.
- [64] *Polish Presburger Arithmetic inside the Omnific Integers*, incoming research manuscript in Provelt, archive polish_presburger_glazer.zip, member polish_presburger_glazer.tex, 2026. https://github.com/VladimirReshetnikov/ProveIt/blob/eab47e3301e0ed2ebee8eb09824b90a6e34af7b0/docs/incoming/polish_presburger_glazer.zip. [write] This is batch-86 manuscript 05, the base of Part I; the archive remains at the cited address.
- [65] *Borel Presentations and Support Barriers in Omnific Arithmetic*, incoming research manuscript in Provelt, archive glazer_proveit_research.zip, member article.tex, 2026. https://github.com/VladimirReshetnikov/ProveIt/blob/eab47e3301e0ed2ebee8eb09824b90a6e34af7b0/docs/incoming/glazer_proveit_research.zip. [write] This is batch-86 manuscript 03, Part II; the archive remains at the cited address.
- [66] Provelt, *Incoming topology triage*, archive inventory and README-level scope note. <https://github.com/VladimirReshetnikov/ProveIt/blob/eab47e3301e0ed2ebee8eb0982>

- [4b90a6e34af7b0/Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/review_recent_incoming_glazer_scope.md](#). [write] A routing note of the Hilbert's-tenth programme, not a proof audit (appendix E).
- [67] Provelt, *Discrete Initial Groups and Omnific Normalization*, research article, source blob 79fa6c696e56e7ebe45944651ce489505a65e384. See Main Theorem E, the Presburger initiality criterion, and the corollary entitled "Unit-preserving omnific embeddings."
<https://github.com/VladimirReshetnikov/ProveIt/blob/4f2d7897a114f65a30a1920fb03492d5f9d69d76/Algebra/SurrealNumbers/docs/surreal/discrete-initial-subgroups-and-omnific-normalization/article.tex>. [write] This is the report discrete-initial-subgroups-and-omnific-normalization; its Presburger initiality criterion is `isg:cf:main:presburger`, the corollary is `isg:cf:cor:unit`, and the factorial family is `isg:cf:prop:family`.
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<https://mpla.math.uoa.gr/en/~cdimitr/publications/index.html>. [write] Source 14's author did not obtain this chapter (theorem 105.2).
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- [73] Vladimir Reshetnikov, Provelt repository, *Natural-number exponentiation is Diophantine*, Lean module `ExponentiationDiophantine.lean`, snapshot 6fef5383b126be343cccc9baf47081ef37ab5afe. [Permanent link to the Lean module](#).
- [74] Vladimir Reshetnikov, Provelt repository, Coq module `ExponentiationDiophantine.v`, same snapshot. [Permanent link to the Coq module](#).
- [75] Provelt research manuscript, *Polish Presburger Arithmetic inside the Omnific Integers*, `Algebra/SurrealNumbers/docs/foundations-and-computation/polish-models-of-omnific-arithmetic/article.tex`, same snapshot. [Permanent link to the inspected manuscript](#). Unformalized research source; its stated scope was audited. [write] This report, as it stood at source 14's pin: then the delivered text of batch-86 source 05, now the base of Part I.
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- [84] ProveIt research manuscript, prepared for Vladimir Reshetnikov. *Polish Presburger Arithmetic inside the Omnific Integers: An affirmative construction for Glazer’s Question 2, effective Baire-space models, and structural obstructions*. Dated 3 October 2026. [Pinned article.tex](#). Git blob `88f0244d0f1155cf44cd7d4915a3a5dfa3e50d03`. The current combined source, rather than stale README filenames, was used for the comparison. [write] This report, as it stood at source 15’s pin: then the delivered text of batch-86 source 05, now the base of [Part I](#). Parts III–V were placed later.
- [85] Vladimir Reshetnikov and ProveIt contributors. `OmnificIntegers.lean` and `OmnificPurelyInfiniteModule.lean`. Same pinned commit as above. [Pinned surreal foundations source directory](#).
- [86] Vladimir Reshetnikov and ProveIt contributors. `OmnificAlgebraDiscreteness.lean`. Same pinned commit as above. [Pinned topology source](#).
- [87] Salma Kuhlmann and Michele Serra, *Automorphisms of valued Hahn groups*, arXiv:2302.06290v4, February 26, 2026. [Versioned text](#). The Hahn sum/product distinction and scope of the matrix section are relevant to the attribution in this article.
- [88] Michele Serra, *Automorphism groups of Hahn groups and Hahn fields*, doctoral dissertation, University of Konstanz, 2021. See the discussion of strong additivity in Section 2.4 and the later higher-rank examples. [Repository PDF](#).
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- [93] Vladimir Reshetnikov, ProvelT research corpus, incoming archive docs/incoming/polish_presburger_article.zip, member polish_presburger/polish_presburger.tex, at the same snapshot. The theorem labelled thm:ordinals already constructs the countable-ordinal family. [Pinned archive](#). [write] Source 06 of this report, now in [Part III](#) (its thm:ordinals is [theorem 35.9](#)); the archive was retired from docs/incoming in 0bd0e5527 and remains at the pinned address.
- [94] Vladimir Reshetnikov, ProvelT, *Presburger arithmetic is decidable*, Logic/PresburgerArithmetic/README.md, same snapshot. [Pinned source](#). [write] The formal Presburger project's README ([section 1.4](#)).
- [95] Vladimir Reshetnikov, ProvelT, Algebra/SurrealNumbers/README.md, same snapshot. [Pinned source](#). [write] The Surreal project's README; unchanged at the placement except for the catalogue's pointers.
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- [102] Vladimir Reshetnikov and ProvelT contributors, *Polish models of omnific arithmetic*, research materials in Algebra/SurrealNumbers/docs/foundations-and-computation/polish-models-of-omnific-arithmetic/. Inspected snapshot: 7c0f2d9f92c3d51ec85703bed1022924a4b7359b. [Permanent repository directory link](#). Unrefereed research manuscripts; the source audit includes the more recent retained arrivals. [write] This report at source 17's pin: Parts I–V.
- [103] ProvelT research manuscript, *Diophantine Definability and Topological Arithmetic: An ATR_0 proof of Glazer's theorem, cofinal unary discontinuities, and nonsplit Presburger models*, October 2026. Retained arrival archive at commit fa0a0576e5cd7ff8a2978d6ad0aefd4d24511808. [Permanent archive link](#). Member glazer_atr0_research/glazer_atr0_topological_arithmetic.tex. Theorem 8.1 and Question 12.2 are the principal comparison points. This is prior unformalized project work. [write] Source 14 of this report, now [Part VIII](#): its Theorem 8.1 and Question 12.2 are [theorem 111.1](#) and [question 115.2](#). The archive was retired from docs/incoming in d151b39ca and remains at the pinned address.

- [104] Provelt contributors, arithmetic list coding and Diophantine exponentiation, `Logic/PeanoArithmetic/ListCoding/`. [Permanent repository directory link](#). These modules motivate a formalization route; they are not a formal verification of the present theorem. [write] The formal project `Logic/PeanoArithmetic/ListCoding` ([section 1.4](#)); its README is unchanged since the pin.
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- [110] *Polish Models of Omnific Arithmetic*. Merged AI-assisted research report in Provelt, October 2026. Snapshot identified in [section 166](#). <https://github.com/VladimirReshetnikov/ProveIt/tree/2b7b388ba81a3355b19a7c2d2fe797e92f572355/Algebra/SurrealNumbers/docs/foundations-and-computation/polish-models-of-omnific-arithmetic>. [write] This report at source 18’s pin: Parts I–V; its Parts III and IV are this report’s.
- [111] *Polish Arithmetic: Multiplication Obstructions and Local Compactness*. AI-assisted research manuscript, 4 October 2026 UTC. Member `polish_arithmetic/polish_arithmetic.tex`. https://github.com/VladimirReshetnikov/ProveIt/blob/2b7b388ba81a3355b19a7c2d2fe797e92f572355/docs/incoming/polish_arithmetic_package.zip. [write] Source 15, now [Part IX](#); its group-completion question is [question 126.3](#). The archive was retired from `docs/incoming` in `d151b39ca` and remains at the pinned address.
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- [113] *Foundations for Surreal and Surcomplex Mathematics*. Merged foundational audit in Provelt. Snapshot identified in [section 166](#). <https://github.com/VladimirReshetnikov/ProveIt/tree/2b7b388ba81a3355b19a7c2d2fe797e92f572355/Algebra/SurrealNumbers/docs/foundations-and-computation/foundations>. [write] Unchanged at the placement; its `found:thm:discrete` is cited after [proposition 163.4](#).
- [114] *Birthday Cutoffs and Hereditary Sets*. Research material in Provelt. Snapshot identified in [section 166](#). <https://github.com/VladimirReshetnikov/ProveIt/tree/2b7b388ba81a3355b19a7c2d2fe797e92f572355/Algebra/SurrealNumbers/docs/foundations-and-computation/birthday-cutoffs-and-hereditary-sets>. [write] Extended since the pin (batch 89).
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- <https://github.com/VladimirReshetnikov/ProveIt/blob/2b7b388ba81a3355b19a7c2d2fe797e92f572355/Algebra/SurrealNumbers/Surreal/Foundations/SignSequence.lean>.
- [116] Provelt. `SignSequenceTopology.lean`. Native order topology and small-subset results. Snapshot identified in [section 166](#). <https://github.com/VladimirReshetnikov/ProveIt/blob/2b7b388ba81a3355b19a7c2d2fe797e92f572355/Algebra/SurrealNumbers/Surreal/Foundations/SignSequenceTopology.lean>. [write] `SignSequence.lean` and `SignSequenceTopology.lean` are unchanged at the placement.
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- [118] YCor. Answer to *The cardinal of the closure of a set in a topological space*. MathOverflow, 29 October 2020. Original closure-pattern argument. <https://mathoverflow.net/questions/375156/the-cardinal-of-the-closure-of-a-set-in-a-topological-space>.
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- [126] Vladimir Reshetnikov, Provelt, `Logic/PresburgerArithmetic/README.md`, commit a32e1b139db2 (full hash in [section 167.2](#)), accessed 4 October 2026. [Pinned arithmetic guide](#).
- [127] Vladimir Reshetnikov, Provelt, `Algebra/SurrealNumbers/README.md`, same snapshot and access date. [Pinned surreal guide](#). [write] Both READMEs are unchanged at the placement e24ce2ce0.
- [128] Companion research manuscript prepared for Vladimir Reshetnikov, *Polish Presburger Arithmetic inside the Omnific Integers*, 3 October 2026. Unrefereed manuscript retrieved from the user’s library, `polish_presburger_glazer.pdf` and accompanying LaTeX source. [write] Source 05, now the base text of [Part I](#).

- [129] Companion research manuscript prepared with ChatGPT for Vladimir Reshetnikov, *Local Compactness Forces Arithmetic Rigidity: Borel ordered groups, Presburger models, and multiplication without regularity assumptions*, 3 October 2026. Unrefereed manuscript retrieved from the user's library as `article(20261004-024117).pdf`. Section 7 and Research Question 3 are the prior classification target; its other theorems are not assumed in this article. [write] Source 08, now [Part IV](#); its Section 7 is Section 50 ([theorem 50.1](#)) and its Research Question 3 is [question 57.3](#).
- [130] Vladimir Reshetnikov, ProveIt repository, *Polish Models of Omnific Arithmetic*, merged research-report catalogue, revision `9fe62865dc30d71256183466b744b98ebf7e64d6`, accessed 4 October 2026. Path: `Algebra/SurrealNumbers/docs/foundations-and-computation/polish-models-of-omnific-arithmetic/README.md`. In particular, its entries for Theorems 23.2, 38.1, 38.2, Section 42, and its limitations and review-status discussion. <https://github.com/VladimirReshetnikov/ProveIt/tree/9fe62865dc30d71256183466b744b98ebf7e64d6>. [write] This report at source 20's pin, with Parts I–IX; Theorems 23.2, 38.1, 38.2 and Section 42 keep their numbers.
- [131] Vladimir Reshetnikov, ProveIt repository, *The exact omnific integer part*, Lean source, `Algebra/SurrealNumbers/Surreal/Foundations/OmnificFloor.lean`, same pinned revision; lines 1–150 inspected. Blob: `d332b117ae170b1eebffa1982d9cf567ecf4915fe`. Declarations include `omnificFloor_spec` and `existsUnique_omnific_integerPart`.
- [132] Vladimir Reshetnikov, ProveIt repository, surreal research catalogue, `Algebra/SurrealNumbers/docs/README.md`, same pinned revision; its finite normal-form, omnific arithmetic, and proof-status descriptions. Blob: `adfd006467ec4f8ec9e92585e393739c6e5e7c37`.
- [133] Salma Kuhlmann and Michele Serra, *The automorphism group of a valued field of generalised formal power series*, *Journal of Algebra* **605** (2022), 339–376. DOI: [10.1016/j.jalgebra.2022.04.023](https://doi.org/10.1016/j.jalgebra.2022.04.023). Inspected text: arXiv:2107.03362v3, 11 April 2022, Definitions 4.0.3, 4.0.5, 4.2.3 and Lemma 4.2.4, printed pp. 20–24. <https://arxiv.org/abs/2107.03362>. The correction discussed here is to that preprint formulation. [write] Not the same paper as [87], the same authors' *Automorphisms of valued Hahn groups*. Lemma 4.2.4 is on printed page 24; see [remark 204.2](#). [write] Also cited by sources 24 and 27, as an antecedent only ([remark 204.2](#) is not involved).
- [134] Franz-Viktor Kuhlmann, *The defect*, in *Commutative Algebra—Noetherian and Non-Noetherian Perspectives*, Springer, New York, 2011, pp. 277–318. Inspected text: arXiv:1004.2135v1, especially Corollary 2.4 and §2. <https://arxiv.org/abs/1004.2135>.
- [135] *Polishability at the Hahn–Puisseux–Levi-Civita Boundary: Automatic order bounds, local Polish hulls, and countable omnific integer parts*. AI-assisted research manuscript prepared for Vladimir Reshetnikov, 3 October 2026. Prior project report in the user's Library, filename `article(20261004-024447).pdf`. Unrefereed; cited for provenance, not used as a theorem dependency. [write] Source 09, now [Part V](#).
- [136] Vladimir Reshetnikov, *ProveIt*, `Algebra/SurrealNumbers/README.md`, revision `9fe62865dc30d71256183466b744b98ebf7e64d6`. <https://github.com/VladimirReshetnikov/ProveIt/blob/9fe62865dc30d71256183466b744b98ebf7e64d6/Algebra/SurrealNumbers/README.md>.
- [137] Vladimir Reshetnikov, *ProveIt*, `Algebra/SurrealNumbers/Surreal/HahnSeries/CharacterTwist.lean`, revision `9fe62865dc30d71256183466b744b98ebf7e64d6`. <https://github.com/VladimirReshetnikov/ProveIt/blob/9fe62865dc30d71256183466b744b98ebf7e64d6/Algebra/SurrealNumbers/Surreal/HahnSeries/CharacterTwist.lean>.
- [138] Vladimir Reshetnikov, *ProveIt*, `Algebra/SurrealNumbers/Surreal/HahnSeries/BinomialRoots.lean`, revision `9fe62865dc30d71256183466b744b98ebf7e64d6`. <https://github.com/VladimirReshetnikov/ProveIt/blob/9fe62865dc30d71256183466b744b98ebf7e64d6/Algebra/SurrealNumbers/Surreal/HahnSeries/BinomialRoots.lean>.

- [44b98ebf7e64d6/Algebra/SurrealNumbers/Surreal/HahnSeries/BinomialRoots.lean](#). [write] The two Lean files are unchanged at the placement e24ce2ce0.
- [139] V. Reshetnikov et al., *Polish Models of Omnific Arithmetic*, Provelt research collection, catalogue and research-draft report, snapshot 3c25fbb5515748040dec853139ed39543885b90, inspected 4 October 2026. <https://github.com/VladimirReshetnikov/ProveIt/tree/3c25fbb5515748040dec853139ed39543885b90/Algebra/SurrealNumbers/docs/foundations-and-computation/polish-models-of-omnific-arithmetic>. [write] This report at source 22's pin (Parts I–X written); source 22 read its README. The “snapshot” is a commit.
- [140] V. Reshetnikov et al., *Hilbert Geometry at Surreal Scales*, Provelt research collection, catalogue and research-draft report, same snapshot. <https://github.com/VladimirReshetnikov/ProveIt/tree/3c25fbb5515748040dec853139ed39543885b90/Algebra/SurrealNumbers/docs/surcomplex/hahn-hilbert-geometry>.
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- [152] *Borel Regularity Forces Summability: Automatic continuity and measurable derivations on left-finite series fields*. Research manuscript prepared with ChatGPT, 4 October 2026. User's research collection, files `borel_summability.tex` and `borel_summability.pdf`; especially Questions 13.4 and 13.5. Unrefereed draft; no independent priority certification. [write] This is source 23 of this report, printed as [Part XVII](#); its Questions 13.4 and 13.5 are [questions 237.4](#) and [237.5](#).
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- [160] Vincent Bagayoko, *A formal Lie correspondence*, arXiv:2604.04224v1 (2026).
<https://arxiv.org/abs/2604.04224>. [write] Neither preprint was re-read for this report.
- [161] *Borel Regularity Forces Summability: Automatic continuity and measurable derivations on left-finite series fields*, AI-assisted research manuscript prepared with ChatGPT, 4 October 2026, 24 pages; standalone file `borel_summability.pdf`. Incorporated as source 23, Part XVII, in the report catalogued in [163]. In particular, Section 10 and Question 13.5. [write] Printed as [Part XVII](#); its Section 10 is [section 234](#) and its Question 13.5 is [question 237.5](#).
- [162] Vladimir Reshetnikov, *ProveIt*, repository and root documentation, reviewed 4 October 2026.
<https://github.com/VladimirReshetnikov/ProveIt>.
- [163] *Polish Models of Omnific Arithmetic*, research-report catalogue in ProveIt, reviewed 4 October 2026. Repository path: `Algebra/SurrealNumbers/docs/foundations-and-computation/polish-models-of-omnific-arithmetic/README.md`. The catalogue distinguishes unrefereed research manuscripts from formal results. [write] This report's README.
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<https://github.com/VladimirReshetnikov/ProveIt/blob/c39974f12a45c8795575ab222f3b84b2901c394f/Algebra/BakerCampbellHausdorff/Lean/BCH/Formal/Trunc.lean>. [write] The file has 143 lines at the pin; its blob is unchanged at the placement `54ece48ab`.
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<https://arxiv.org/abs/2605.24948>. [write] Not re-checked for this report.
- [166] *Borel Regularity Forces Summability: Automatic continuity and measurable derivations on left-finite series fields*, research manuscript prepared with ChatGPT, 4 October 2026. Prior project artifact `borel_summability.pdf`; unrefereed. [write] Source 23, printed as [Part XVII](#).
- [167] *A Divisibility Threshold for Borel Conjugacy: Explicit normal forms, measurable classification, and analytic flows on left-finite series fields*, research manuscript prepared with ChatGPT, 4 October 2026. Prior project artifact `borel_conjugacy.tex`; unrefereed. Question 14.3 supplies the nonabelian infinitesimal extension target. [write] Source 24, printed as [Part XVIII](#); its Question 14.3 is [question 253.3](#).
- [168] Vladimir Reshetnikov. *ProveIt: The Baker–Campbell–Hausdorff formula*, repository documentation, consulted 4 October 2026 (blob `e24d8fc8d` of `Algebra/BakerCampbellHausdorff/README.md`).
<https://github.com/VladimirReshetnikov/ProveIt/blob/main/Algebra/BakerCampbellHausdorff/README.md>.